

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

**ARMY MODELS UH-60A, UH-60L,
EH-60A, HH-60A AND HH-60L
HELICOPTERS**

**CHAPTER 7
MAINTENANCE INSTRUCTIONS**

*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19 and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

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HEADQUARTERS, DEPARTMENT OF THE ARMY

17 April 2006

WARNING SUMMARY

Personnel performing operations, procedures, and practices which are included or implied in this technical manual shall observe the following warnings. To disregard these warnings and precautionary information can cause serious injury, death, or an aborted mission.

WARNING

FIRST AID - refer to FM 4-25.11.

AC POWER - before applying ac power to the helicopter, make sure that the area around the stabilator is clear of personnel and equipment. Should the stabilator be in any position other than trailing edge fully down, it may reposition automatically to fully down when ac power is applied.

CARBON MONOXIDE - during cold weather operations when preheating is necessary, all personnel shall become acquainted with the various types of heaters and where they are to be used. Be careful of accumulations of carbon monoxide and damage to heat-sensitive equipment.

CONSUMABLE MATERIALS - observe all cautions and warnings on the containers when using consumables. When applicable wear necessary protective gear during handling and use. If a consumable is flammable or explosive, MAKE SURE consumable and its vapors are kept away from heat, spark, and flame. MAKE SURE helicopter is properly grounded and firefighting equipment is readily available for use.

SAFETY PINS - make sure all safety pins are installed in ejector racks prior to performing any maintenance. To disregard this warning can cause serious injury, death, or an aborted mission.

ELECTRICAL GROUNDING OF THE HELICOPTER - army regulations require the grounding of the helicopter when doing all fueling and defueling operations. Do not operate helicopter electrical switches, except those essential for servicing during fueling and defueling. Do not smoke or use flame during fueling and defueling operations.

HIGH VOLTAGE - there are dangerous voltages in the helicopter. Use extreme care when working with equipment having these voltages.

HYDRAULIC SYSTEMS - there are dangerous high hydraulic system pressures in the helicopter. Use extreme care when working on systems having this pressure.

USE OF FIRE EXTINGUISHER - when fire extinguishers are used in a confined area, ventilate immediately. Serious injury or death could result if the area is not ventilated or the user does not use a self-contained breathing device.

MAIN ROTOR PYLON SLIDING COVER - when opening and closing main pylon sliding cover, keep hands away from sharp edges near track, mating end, and especially ventilation blower inlet hole.

TECHNICAL ACETONE - is extremely flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

ALKALINE AQUEOUS CLEANER - alkaline cleaner is harmful to skin and eyes. Use adequate ventilation. Operators should wear protective gloves, aprons, and goggles. Upon contact with cleaning compound, wash affected area for at least 20 minutes with clean water. Seek medical attention.

CLEANING COMPOUND SOLVENT - cleaning Compound Solvent is combustible and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames or other sources of ignition.

ELECTRON - use electron in well ventilated area only. Avoid prolonged breathing of vapors. Avoid bodily contact. The use of chemical gloves and chemical splash goggles are required. Do not use near heat, spark or flame. This solvent is

reactive with acids and oxidizers; do not mix or cross-apply with other chemicals. Organic vapor respirator with dust and mist filter is recommended when solvent is spray applied. Keep containers closed between applications.

- Provide mechanical ventilation if used in confined spaces. Coordinate the use of this material with your supporting industrial hygiene and safety offices. Ensure you read and understand the material safety data sheet (MSDS) for electron prior to use.

ISOPROPYL ALCOHOL - isopropyl alcohol is a volatile and flammable liquid which must be kept away from flame and other ignition sources. It must be used only in well ventilated area. Personnel must wear protective gloves and eye protection. Excessive inhalation of vapors or contact with skin must be avoided. If contact occurs with eyes or skin, flush area thoroughly with water. If ingestion occurs, induce vomiting and call a doctor. Remove to fresh air if overexposed to vapor.

DS-108 - use in well ventilated area. Avoid breathing vapors. Organic vapor respirator with dust and mist filter is recommended. May be harmful by inhalation, ingestion, or skin absorption. Avoid bodily contact. Chemical safety glasses/goggles and chemical resistant gloves are required. Combustible, keep away from heat, spark, or flame. Vapors are heavier than air and may travel over a distance to an ignition source and then flash back. Do not mix or cross apply with other chemicals.

LOCKWIRE - in critical applications use lockwire instead of safety cable which is specifically prohibited from use in critical applications. The use of safety cable is restricted in these specific applications with a warning.

PURPOSE OF A WARNING - alerts personal to operation, procedure, practice, etc., which if not strictly observed, could result in personal injury or loss of life.

CHANGE
NO. 3

HEADQUARTERS DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 20 June 2008

TECHNICAL MANUAL
AVIATION UNIT AND INTERMEDIATE MAINTENANCE
FOR
ARMY MODELS
UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS
CHAPTER 7 MAINTENANCE INSTRUCTIONS

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TM 1-1520-237-23-7, 17 April 2006, is updated as follows:

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5. Remove old pages and insert new pages as indicated below.

remove pages

a and b
A and B

Insert Pages

a and b
A and B

6. Replace the following work packages with their revised version.

Work Package Number

WP 0534 00
WP 0538 00

Work Package Number

WP 0542 00
WP 0543 00
WP 0547 00
WP 0549 00
WP 0551 00
WP 0552 00
WP 0553 00
WP 0554 00
WP 0556 00
WP 0557 00
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WP 0586 00
WP 0587 00
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WP 0636 00
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WP 0649 00
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WP 0654 01
WP 0656 00
WP 0659 00

CHANGE
NO. 3

TM 1-1520-237-23-7

Work Package Number

WP 0670 00
WP 0679 00
WP 0680 00
WP 0681 00

By Order of the Secretary of the Army:

GEORGE W. CASEY, JR.
General, United States Army
Chief of Staff

Official:

A handwritten signature in black ink, reading "Joyce E. Morrow". The signature is written in a cursive, flowing style.

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0815615

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CHANGE
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TECHNICAL MANUAL
AVIATION UNIT AND INTERMEDIATE MAINTENANCE
FOR
ARMY MODELS
UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS
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Remove Pages

None
A and B
i thru iv
v/vi and Blank

Insert Pages

a and b
A and B
i thru iv
v/vi and Blank

6. Replace the following work packages with their revised version.

Work Package Number

WP 0534 00
WP 0538 00
WP 0542 00
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WP 0547 00
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WP 0587 00
WP 0590 00
WP 0597 00
WP 0603 00
WP 0613 00
WP 0616 00
WP 0623 00
WP 0624 00
WP 0629 00
WP 0630 00
WP 0653 00
WP 0656 00
WP 0667 00
WP 0669 00
WP 0678 00

By Order of the Secretary of the Army:

Official:



JOYCE E. MORROW
*Administrative Assistant to the
Secretary of the Army*
0733121

GEORGE W. CASEY, JR.
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TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE
FOR
ARMY MODELS
UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

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5. Remove old pages and insert new pages as indicated below.

Remove Pages

None
i through vi
v/vi Blank
Chapter 7 Title and Blank
Cover

Insert Pages

A and B
i through vi
v/vi Blank
Chapter 7 Title and Blank
Cover

6. Replace the following work packages with their revised version.

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WP 0534 00
WP 0538 00
WP 0546 00
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Work Package Number

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WP 0655 00
WP 0656 00
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WP 0678 00
WP 0683 00

7. Add the following new work packages.

Work Package Number

WP 0611 01
WP 0654 01

By Order of the Secretary of the Army:

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General, United States Army
Chief of Staff

Official:



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LIST OF EFFECTIVE PAGES/WORK PACKAGES

Dates of issue for original and changed pages / work packages are:

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Change	1	21 March 2007
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Change	3	20 June 2008

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 10 AND TOTAL NUMBER OF WORK PACKAGES IS 159 CONSISTING OF THE FOLLOWING:

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* Zero in this column indicates an original page or work package.

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WP 0629 00 (2 pgs)	2	WP 0655 00 (8 pgs)	1	WP 0682 00 (4 pgs)	0
WP 0630 00 (40 pgs)	3	WP 0656 00 (10 pgs)	3	WP 0683 00 (12 pgs)	1
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HEADQUARTERS,
DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 17 April 2006

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

Army Model Helicopters UH-60A, UH-60L, EH-60A, HH-60A, and HH-60L

CHAPTER 7 MAINTENANCE INSTRUCTIONS

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS.

You can improve this manual. If you find any mistakes or if you know of a way to improve these procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms), located in the back of this manual, directly to: Commander, US Army Aviation and Missile Command, ATTN:AMSAM-MMC-MA-NP, Redstone Arsenal, AL 35898-5000. A reply will be furnished to you. You may also send in your comments electronically to our E-mail address: 2028@redstone.army.mil or by fax 256-842-6546/DSN 788-6546. For the World Wide Web use: <https://amcom2028.redstone.army.mil>. Instructions for sending an electronic 2028 may be found at the back of this manual immediately preceding the hard copy 2028.

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CHAPTER 7
MAINTENANCE INSTRUCTIONS
FOR
ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A,
AND HH-60L
HELICOPTERS



UNIT LEVEL
ROTOR SYSTEM
SPINDLE INSPECTIONS

This WP supersedes WP 0534 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Depth Gage
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13

Primer Coating Item 243, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Abrasive, Item 80, WP 1803 00
Corrosion Resistant Coating Item 110, WP 1803 00
Corrosion Resistant Coating Item 111, WP 1803 00

References

TM 1-1520-265-23
WP 0595 00
WP 1803 00

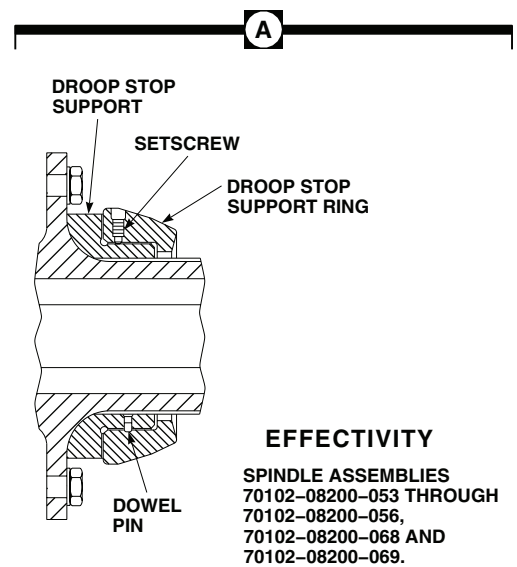
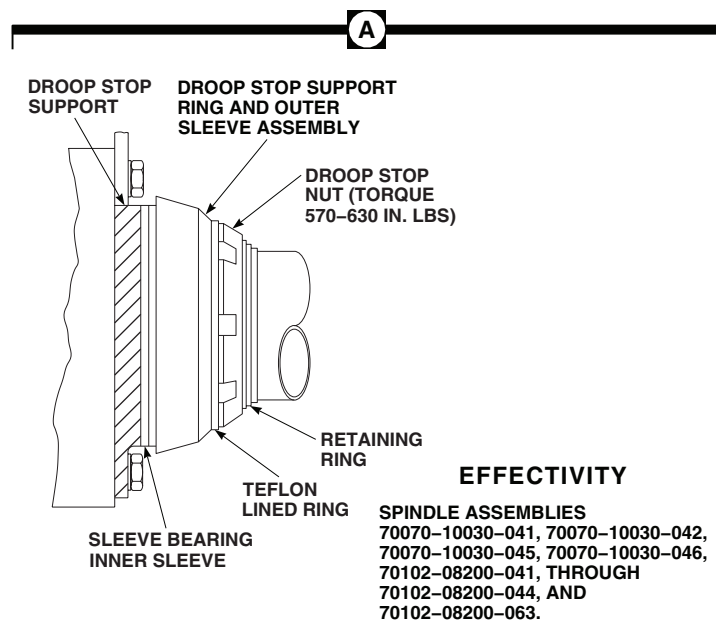
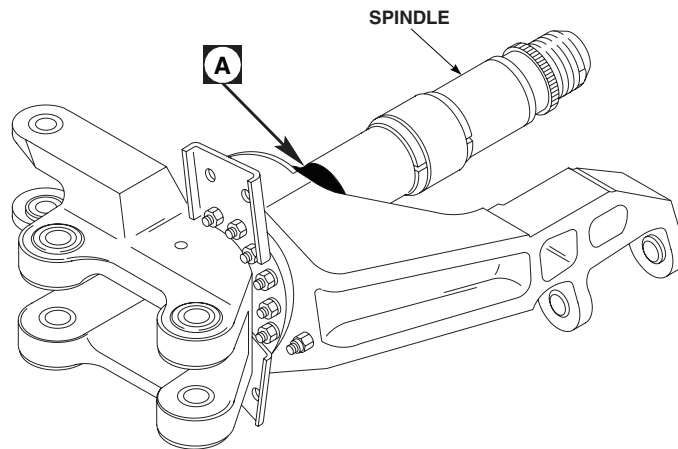
INSPECT SPINDLE

NOTE

The following inspection applies to spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063 only.

1. Visually check exposed portion of sleeve bearing (inner race of droop stop support ring) around entire circumference for cracks (Figure 1, Detail A). **IF A CRACK IS VISIBLE IN ONE LOCATION ONLY, REPLACE SPINDLE AT NEXT PERIODIC INSPECTION (PMD) OR UNTIL A SECOND CRACK IS FOUND. REPLACE SPINDLE IF MORE THAN ONE CRACK IS FOUND.**
2. If spindle must be replaced, a flight may be continued up to 10 hours under these conditions:
 - a. Flight is a one time flight to a site capable of replacing spindle assembly.
 - b. Continuance of flight is allowed to enable helicopter to return to a site where repairs can be made.

INSPECT SPINDLE - Continued



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Figure 1. Inspect Spindle.

SPINDLE LUG INSPECTION

WARNING

Injury to personnel and damage to equipment will result if cracks are allowed on spindle lug. **NO CRACKS ALLOWED ON SPINDLE LUG.**

NOTE

Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102 and main rotor blade solid pins, 70103-08108-041. In the following procedures both types of pins are referred to as main rotor blade pins, excepted where noted.

1. Check for any discrepancies in the spindle lug. Visually inspect exposed areas of lug for fractures which may appear as cracks. If lug is broken, any movement of the blade cuff or pin will cause displacement of the broken portion.
2. If any lugs are cracked or broken, **IMMEDIATELY REPLACE THE SPINDLE ASSEMBLY, MAIN ROTOR BLADE ASSEMBLY, AND THE MAIN ROTOR BLADE PINS.**

INSPECT DROOP STOP SUPPORT

1. Inspect droop stop supports for cracks (Figure 1, Detail A). Inspect upper and lower puller grooves of droop stop supports for cracks. **REPLACE BEARING AND SPINDLE ASSEMBLY IF SUPPORTS SHOW SIGNS OF CRACKS.** Send bearing and spindle assembly to depot.
2. Visually check upper portion (1 to 2 o'clock position, when looking inboard) and lower portion (6 to 9 o'clock position, when looking inboard) of droop stop support for cracks. **IF A CRACK IS VISIBLE IN ONE LOCATION ONLY, REPLACE SPINDLE AT NEXT PERIODIC INSPECTION (PMI).**
3. On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, do this:
 - a. Visually check exposed portion of sleeve bearing inner race (pressed on to droop stop support) around entire circumference for crack. **IF HELICOPTER IS IN A PERIODIC INSPECTION (PMI) AND A CRACK IS VISIBLE, REPLACE SPINDLE. IF HELICOPTER IS NOT IN A PERIODIC INSPECTION AND ONLY ONE CRACK IS VISIBLE, REPLACE SPINDLE AT NEXT PERIODIC INSPECTION (PMI). IF MORE THAN ONE CRACK IS VISIBLE, REPLACE SPINDLE.**
 - b. Check droop stop support ring sleeve bearing inner sleeve for nicks, gouges, and burrs. **DAMAGE SHALL NOT EXCEED 0.005-INCH DEEP.** Replace spindle assembly if damage is more than limits.
 - c. Check droop stop support ring sleeve bearing inner sleeve for corrosion pitting. **CORROSION PITTING SHALL NOT EXCEED 0.005-INCH.** Replace spindle assembly if damage is more than limits.
 - d. Inspect droop stop support for corrosion. If corrosion is found, do this:
 - (1) Remove corrosion from support using abrasive cloth, Item 80, WP 1803 00. **REPLACE SPINDLE IF CORROSION IS DEEPER THAN 0.035-INCH ACROSS ENTIRE CIRCUMFERENCE OF FIRST TWO THREADS. REPAIR OF REMAINING THREADS LIMITED TO 30% AT THREAD CIRCUMFERENCE.**

INSPECT DROOP STOP SUPPORT - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- (2) Clean repaired area using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
 - (3) Coat repaired area with corrosion resistant coating, Item 111, WP 1803 00.
 - (4) Coat repaired area with primer coating, Item 243, WP 1803 00.
4. On spindle assemblies, 70102-08200-053 through 70102-08200-056, 70102-08200-068, and 70102-08200-069, do this:
- a. Inspect the droop stop support for corrosion. If corrosion is found do the following:
 - (1) Remove corrosion from support using abrasive cloth, Item 80, WP 1803 00.
 - (2) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect rework area (TM 1-1520-265-23).
 - (3) After corrosion removal, touch up rework area with corrosion resistant coating, Item 110, WP 1803 00.
 - (4) **REPLACE SPINDLE ASSEMBLY IF CORROSION IS DEEPER THAN 0.020-INCH IN UNDERCUT AREA, ATTACHMENT FLANGE, OR BEARING SURFACE (Figure 3, Sheet 4).**
 - (5) **REPLACE SPINDLE ASSEMBLY IF CORROSION EXCEEDS 20% OF SURFACE AREA.**
 - (6) **ON ALL OTHER SURFACES, REPLACE SPINDLE IF CORROSION IS DEEPER THAN 0.050-INCH, OR EXCEEDS 20% OF SURFACE AREA.**
 - b. Inspect droop stop support for wear as follows:
 - (1) Wear on bottom half (3 - 9 o'clock position) of droop stop support up to 0.002-inch deep is acceptable (Figure 2). Wear 0.003 to 0.050-inch deep is also acceptable, provided that droop stop support ring is replaced with a new/unused droop stop support ring.
 - (2) **REPLACE SPINDLE IF WEAR ON SUPPORT IS GREATER THAN 0.050-INCH.**
 - (3) Blend scratches, nicks, and burrs on droop stop support using abrasive cloth, Item 80, WP 1803 00.
 - (4) Using fluorescent penetrant inspection kit, fluorescent penetrant inspect rework area (TM 1-1520-265-23).
 - (5) After rework, touch-up rework area with corrosion resistant coating, Item 110, WP 1803 00.

INSPECT TEFLON SLEEVE AND SHRINK TUBING

1. Inspect Teflon sleeve and shrink tubing for damage. Surface damage to Teflon sleeve and metal retaining strap is acceptable. Cracking of adhesive squeezed out from around Teflon sleeve is acceptable. Flight without shrink tubing is acceptable provided peeling of the Teflon sleeve from the spindle does not occur.
2. Inspect Teflon sleeve for axial movement along the spindle and rotation about the spindle centerline. Use caution not to damage or tear shrink tubing. If Teflon sleeve has axial and/or rotational movement, rebond teflon sleeve at next periodic maintenance inspection (PMI). If Teflon sleeve is disbonded, a crack in the Teflon sleeve greater than 0.5-inch in length is not acceptable.

INSPECT TEFLON SLEEVE AND SHRINK TUBING - Continued

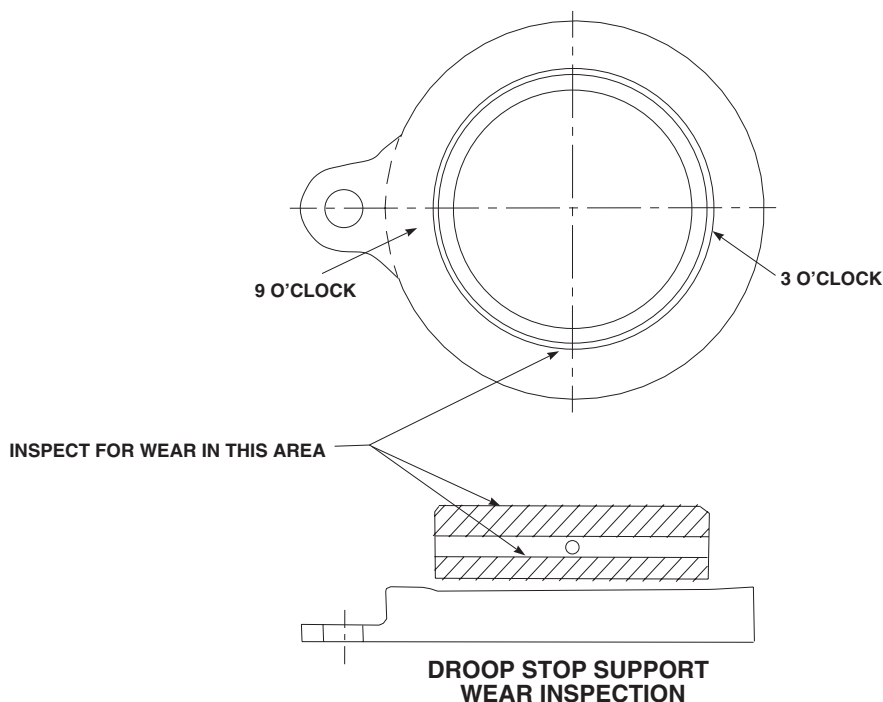
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Figure 2. Inspect Stop Support.

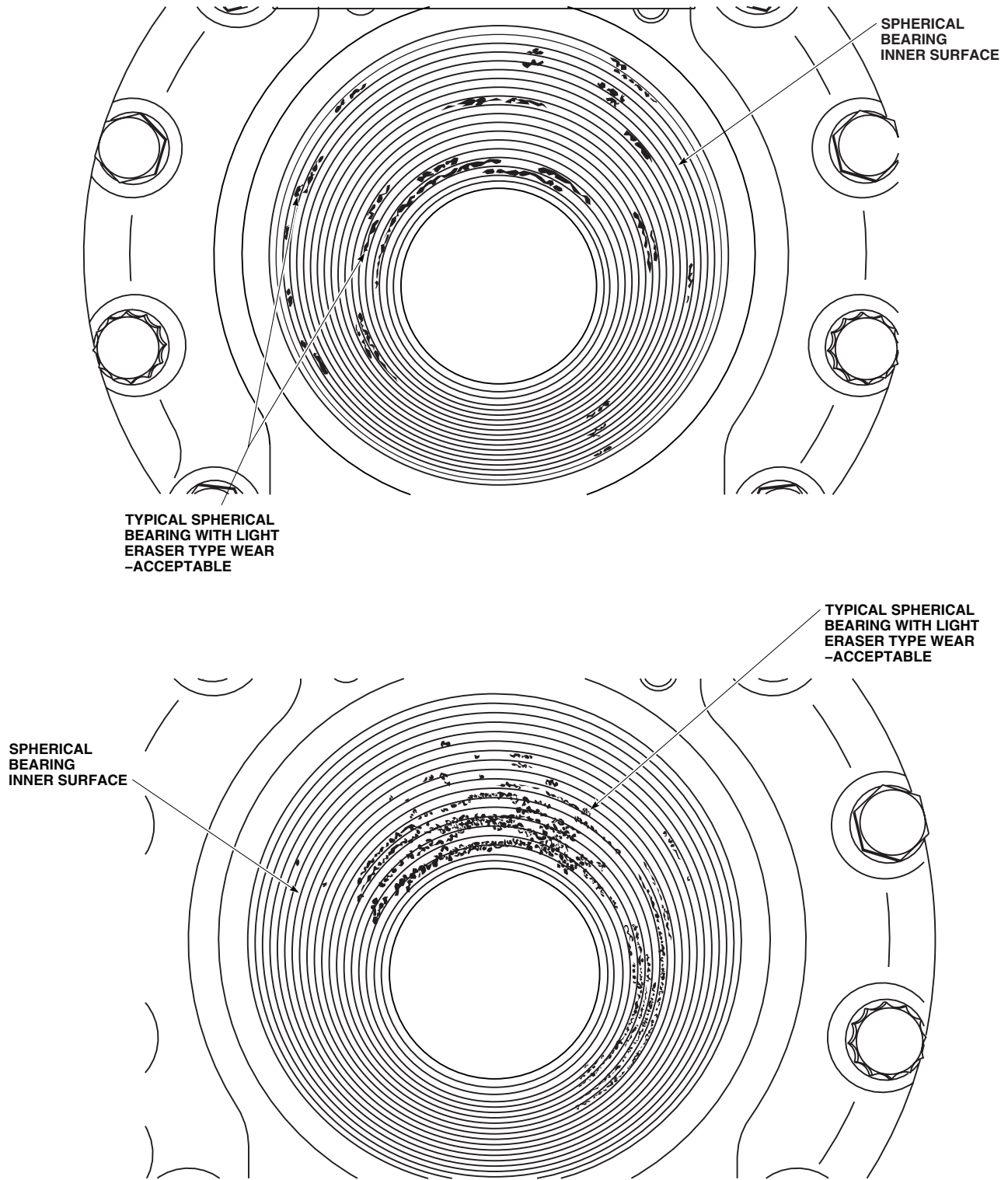
INSPECT SPINDLE ELASTOMERIC BEARING

1. Inspect droop stop supports for damage. Damage in the area around droop stop pin is blended out using a 1 inch abrasive wheel, Item 350, WP 1803 00. **REPLACE BEARING IF SURFACE DAMAGE EXCEEDS 0.030-INCH DEEP AFTER BLENDING OR IF CORNER DAMAGE EXCEEDS 0.050-INCH DEEP AFTER BLENDING.**

NOTE

- The spindle elastomeric bearing consists of a spherical bearing and a thrust bearing. Wear of either bearing is a gradual process that will take several hundreds of hours from first detection of rubber damage to point where bearing must be replaced. Visual comparison of the bearings with Figure 3 is the criterion for acceptance or rejection.
 - Do not disassemble bearings to inspect.
2. Types of wear shown include eraser-type wear, elastomer extrusion, and shim cracks. Description of wear is as follows:
 - a. Eraser-type wear is a pile-up of shredded rubber similar to the result of using a soft eraser.
 - b. Elastomer extrusion is the squeezing out of the elastomer from between the bearing laminations indicating debonding of elastomer. Overlap of elastomer on endplate OD from manufacturing process is not cause for rejection. Peeling of rubber mold lines running from inboard to outboard of bearing is not cause for rejection.
 - c. If spindle is installed on hub, use a flashlight and mirror to visually inspect outer surface of spherical bearing for wear, laminate separations, and cracked shims through the 2 inch diameter inspection hole.

INSPECT SPINDLE ELASTOMERIC BEARING - Continued

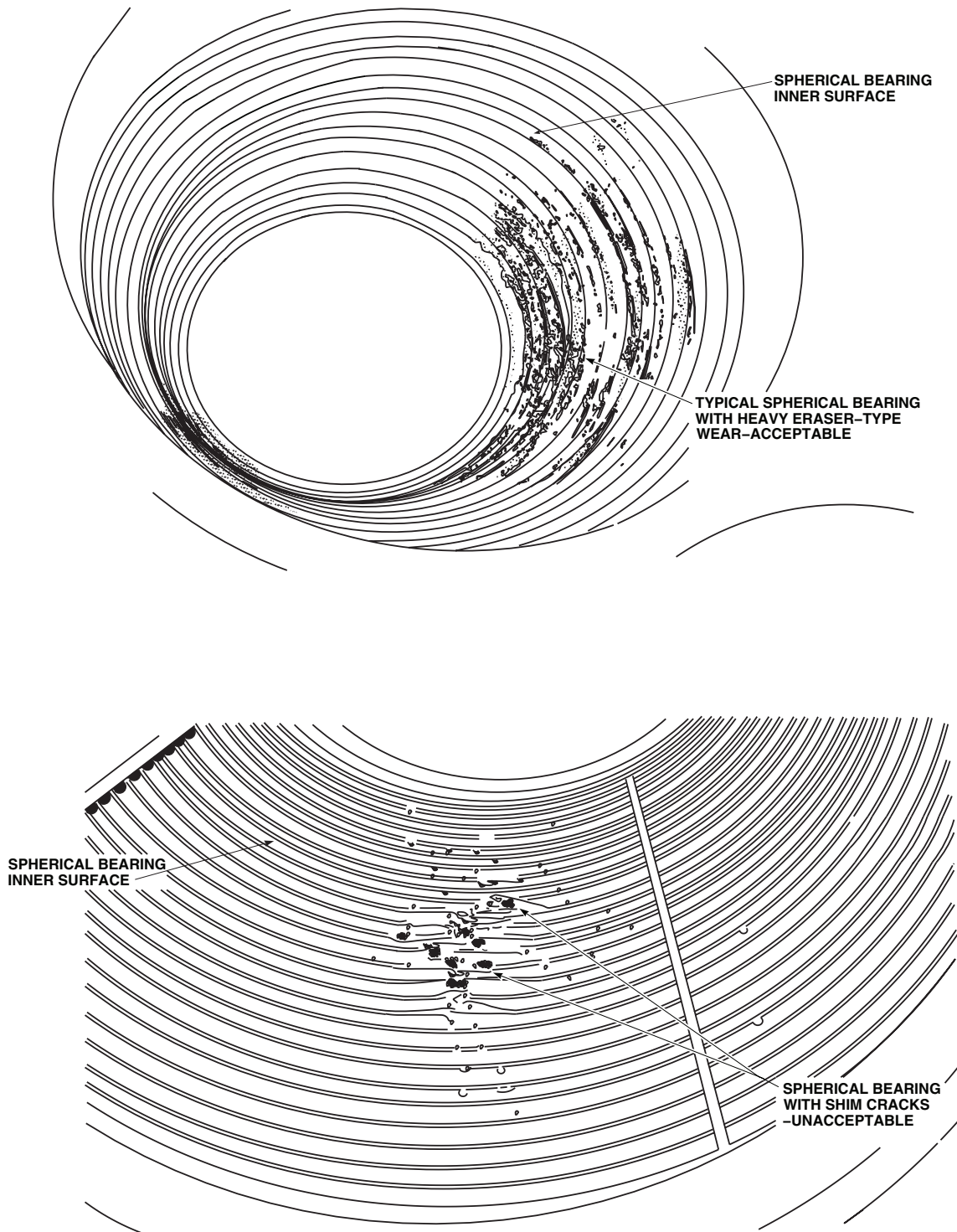


SPHERICAL BEARING INSPECTION

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Figure 3. Inspect Spindle Elastomeric Bearings. (Sheet 1 of 5)

INSPECT SPINDLE ELASTOMERIC BEARING - Continued



SPHERICAL BEARING INSPECTION

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Figure 3. Inspect Spindle Elastomeric Bearings. (Sheet 2 of 5)

INSPECT SPINDLE ELASTOMERIC BEARING - Continued

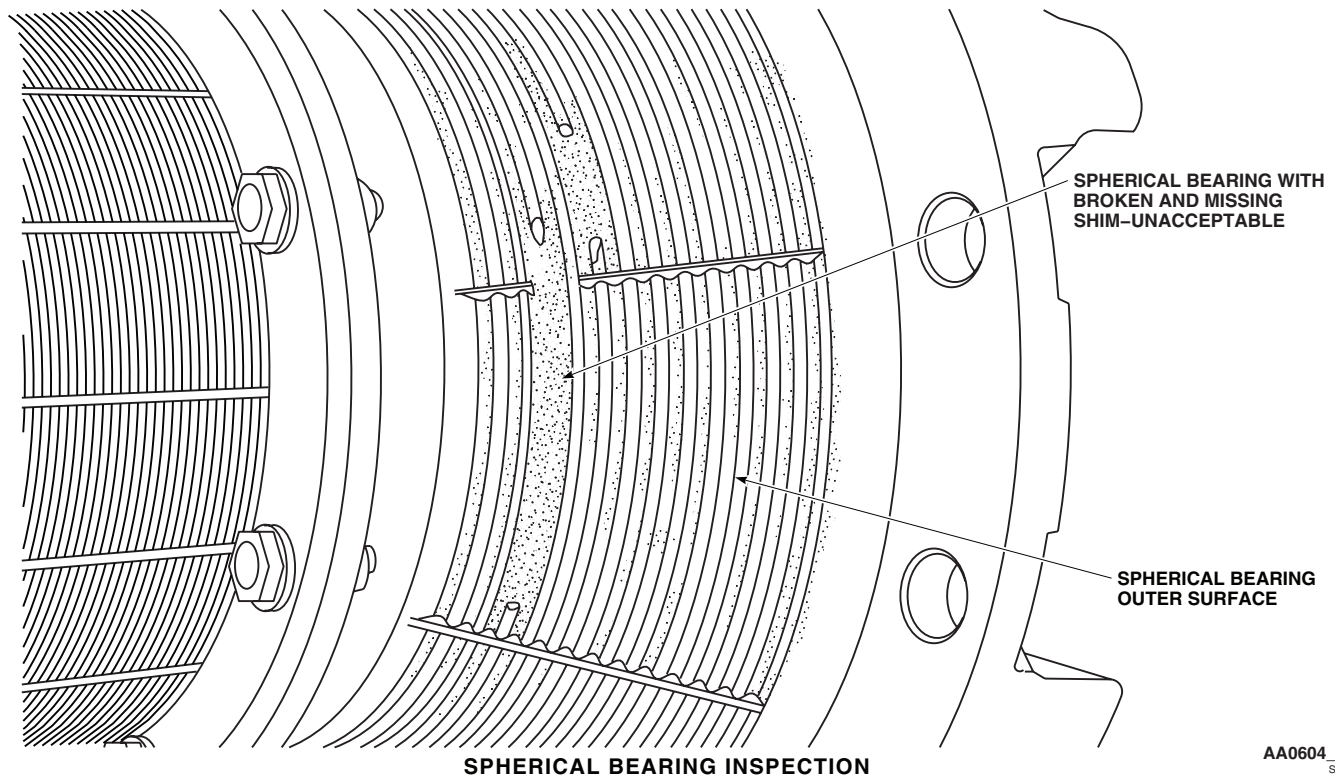


Figure 3. Inspect Spindle Elastomeric Bearings. (Sheet 3 of 5)

NOTE

Do not remove spindle at 40 hour special inspection to inspect inner surface of thrust bearing.

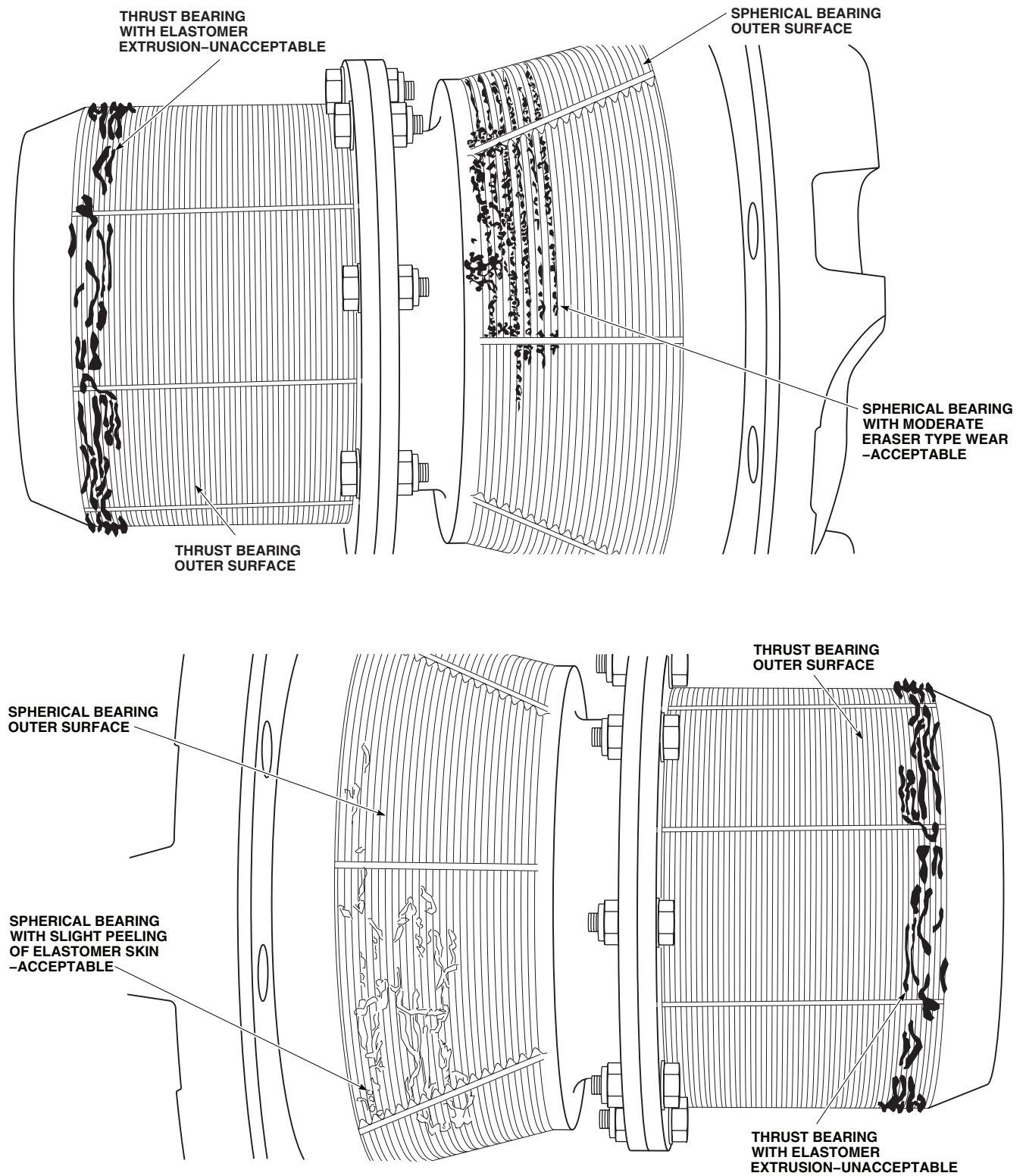
- d. **NO ELASTOMERIC EXTRUSION OF THRUST BEARING IS ALLOWED.** Elastomeric extrusion normally will be detectable on lower portion of thrust bearing. Replace thrust bearing within the next 30 flight hours if any elastomer extrusion is found (WP 0595 00).
- e. Shim cracks will appear in the edge of the metal shims. **NO CRACKS ALLOWED.** Replace spherical or thrust bearing if cracks are found (WP 0595 00).
- f. A white wax-like substance on rubber surface is normal and should not be removed.

NOTE

Both inner and outer surfaces of spherical bearing must be inspected. Outer surface of thrust bearing must be inspected.

- 3. Check elastomeric bearings for laminate separations by doing this:
 - a. Inspect between rubber layers and metal shims or endplate. Also, inspect within rubber layers themselves.

INSPECT SPINDLE ELASTOMERIC BEARING - Continued

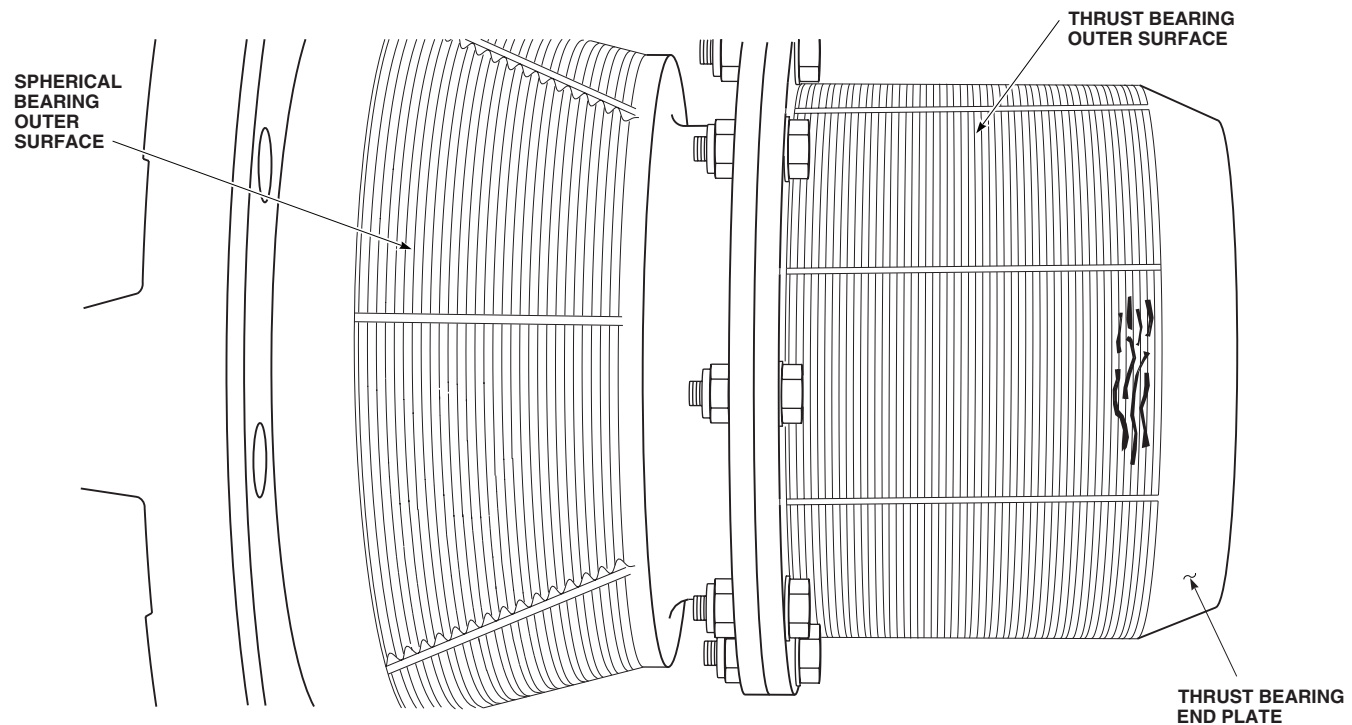


SPHERICAL AND THRUST BEARING INSPECTION

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Figure 3. Inspect Spindle Elastomeric Bearings. (Sheet 4 of 5)

INSPECT SPINDLE ELASTOMERIC BEARING - Continued



THRUST BEARING INSPECTION

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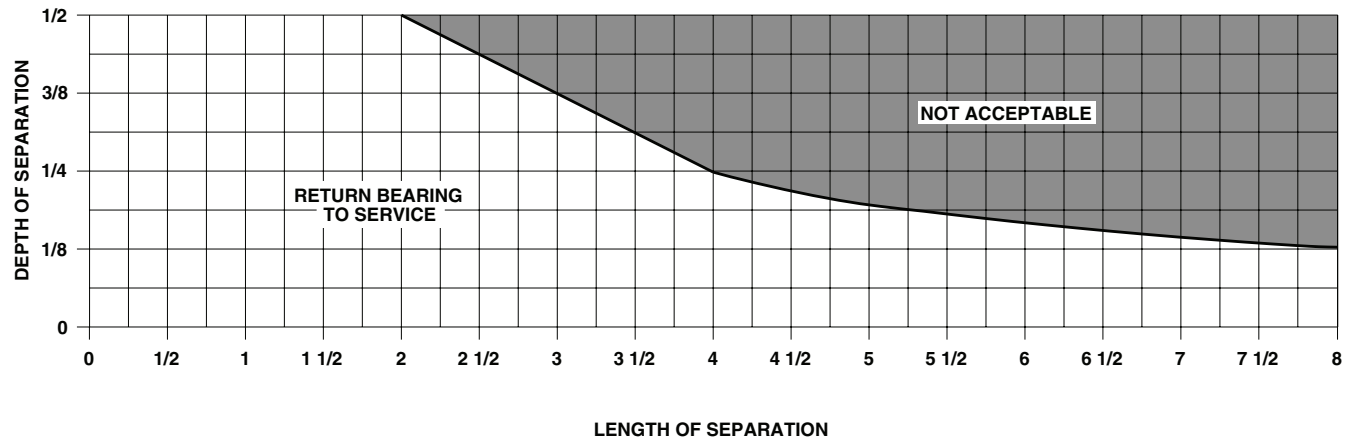
Figure 3. Inspect Spindle Elastomeric Bearings. (Sheet 5 of 5)

CAUTION

Damage to elastomeric bearing will result if feeler gage is forced between rubber layers. Do not force feeler gage when checking for laminate separations.

- b. Determine amount of laminate separations by probing with a feeler gage (0.005-inch). Carefully insert feeler gage in separation to measure depth and length of separation.
- c. Compare separations on spherical bearing with graph in Figure 4. Replace elastomeric spherical bearing if any separation extends into shaded area (WP 0595 00). **MULTIPLE SEPARATIONS OF 0.061-INCH DEEP OR GREATER ARE ALLOWED IN SPHERICAL BEARING IF THEY ARE AT LEAST 5 INCHES APART AROUND CIRCUMFERENCE, AND AT LEAST THREE METAL SHIMS APART.**

INSPECT SPINDLE ELASTOMERIC BEARING - Continued



NOTE

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

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Figure 4. Elastomeric Bearing Separation Limits.

- d. Check thrust bearing for separations. **UP TO 3/8-INCH DEEP SEPARATION ALLOWED AROUND ENTIRE CIRCUMFERENCE.** Replace thrust bearing beyond limits (WP 0595 00).
- e. If elastomeric spherical or thrust bearing has total delamination between rubber layer and metal shim endplate, replace spherical or thrust bearing (WP 0595 00).
4. Check elastomeric bearings for surface cracking or crazing of rubber. **UP TO 0.060-INCH DEEP ALLOWED AROUND ENTIRE CIRCUMFERENCE, BETWEEN ALL LAMINATES.** Small hairline surface cracking or crazing of rubber surface is acceptable. Replace spherical or thrust bearing beyond limits (WP 0595 00).
5. Examine edges of shims for cracks. **NO CRACKS ALLOWED.** Replace cracked spherical or thrust bearing (WP 0595 00).
6. If droop stop pounding is experienced, inspect droop stop ears on elastomeric bearing endplate for cracks. **NO CRACKS ALLOWED.** If cracks are found, replace elastomeric bearing assembly.
7. Inspect hub arm for signs of contact with spindle retention nut. **IF CONTACT IS FOUND, REMOVE AND REPLACE SPINDLE.**

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
SPINDLE HORN INSPECTION

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

TM 55-1500-345-23
WP 0381 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

INSPECTION**NOTE**

The following instructions apply to spindle horn attachment bolt heads with bolts installed only. Bolts will be discarded when removed.

1. Inspect spindle horn attachment boltheads for corner damage. **MAXIMUM CORNER DAMAGE SHALL NOT EXCEED 0.070 X 0.070-INCH.** Replace bolts if damage is more than limits.
2. Touch up finish on damaged areas if damage is within limits (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
DROOP STOP INSPECTION

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Goggles

References

TM 55-1500-345-23
WP 0381 00
WP 0551 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT**WARNING**

Injury to personnel may result if contact with beryllium occurs. The droop stop is made of beryllium copper metal, which can be a health hazard. The use of goggles is required. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

1. Visually check droop stop for cracks. **NO CRACKS ARE ALLOWED EXCEPT DROOP STOP CAM MAY HAVE SMALL CRACKS (UP TO 1/8-INCH) AT THE CORNER RADIUS OF IMPACT SURFACES.** Replace cam if damaged beyond limits (Figure 1, Detail A).
2. Check cam impact surfaces for deformation. **DEFORMATION UP TO 0.10-INCH IS ACCEPTABLE.** Replace cam if damaged beyond limits.
3. Check cam for corner damage. **CORNER DAMAGE UP TO 1/8-INCH IS ACCEPTABLE AFTER REMOVING SHARP EDGES WITH A MEDIUM CUT FILE.** Replace cam if damaged beyond limits.
4. Touch up reworked areas (WP 0381 00 and TM 55-1500-345-23).
5. Visually inspect droop stop spring and attaching hardware for signs of corrosion. **NONE ALLOWED.** Replace corroded parts.
6. If droop stop pounding is experienced, inspect droop stop ears on elastomeric bearing endplate for cracks. **NO CRACKS ALLOWED.** If cracks are found, replace elastomeric bearing assembly (WP 0551 00).

INSPECT - Continued

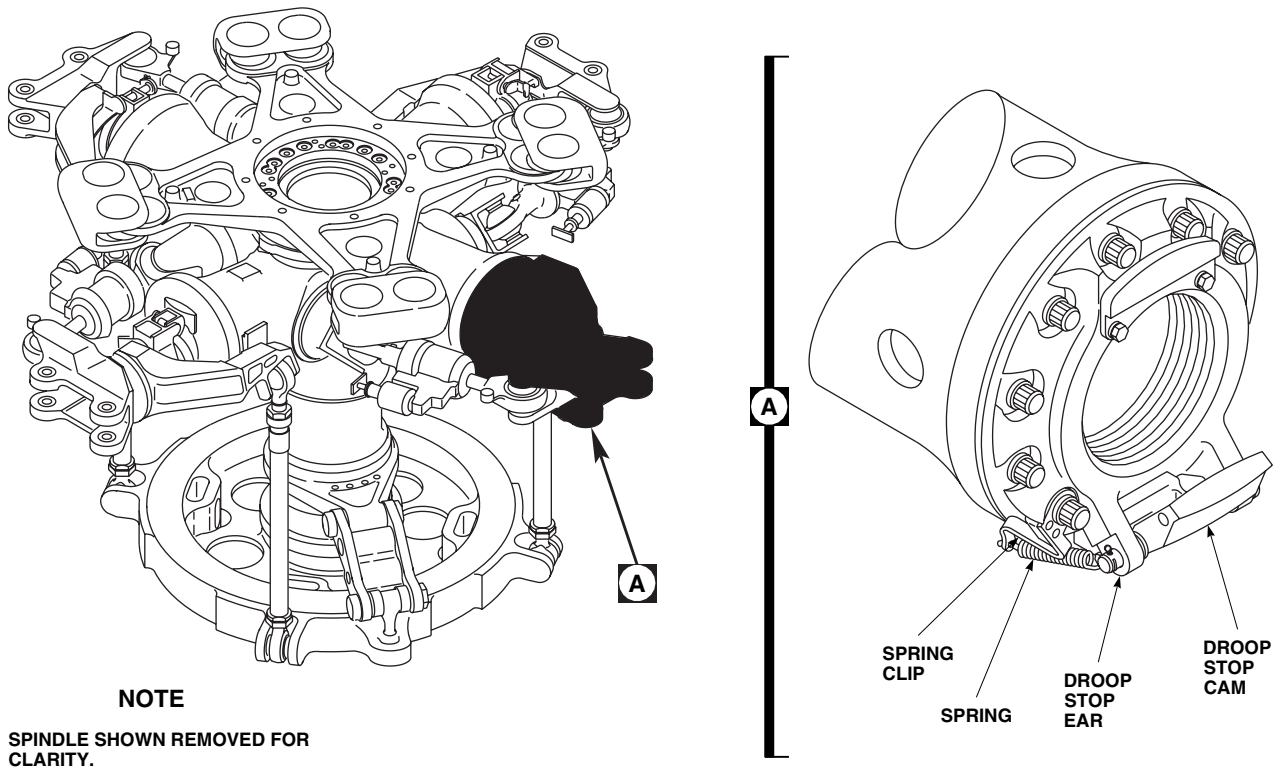


Figure 1. Inspect Droop Stop.

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
DAMPER INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Dial Indicator, MIL-I-18422
 Teflon Lined Spherical Bearing Plastic Feeler Gage,
 Locally-Made, Figure 209, WP 1805 00

References

WP 0558 00
 WP 0559 00
 WP 0561 00
 WP 0596 00
 WP 1803 00
 WP 1805 00

Material/Parts

Pad, Scouring, Item 205, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT DAMPER INDICATOR

1. Check for wear or scoring of gold band. As long as level of servicing can be seen, indicator is usable.
2. Check for sticky indicator piston as follows:

NOTE

Too much bleeding will result in fluid loss and damper will require servicing.

- a. Slightly crack open each of bleed plugs on damper, one at a time.
- b. If any trapped air or small amount of fluid is expelled, indicator is usable.
- c. If no fluid or air comes out and indicator piston does not move, repair indicator (WP 0596 00).
3. Inspect damper indicator for leakage. **IF LEAKAGE OF 1 DROP PER HOUR OR MORE IS VISIBLE, REPAIR/REPLACE INDICATOR.**

INSPECT DAMPER FOR LEAKAGE

1. If the damper indicator goes from the full indication (full gold band visible) to the refill indication (no gold visible) in less than 10 flight hours, inspect damper and indicator for leakage. Replace component as necessary (WP 0558 00 or WP 0559 00).
2. Inspect damper piston rod for continuous leakage. **IF LEAKAGE OF 1 DROP PER HOUR OR MORE IS VISIBLE REPLACE DAMPER (WP 0558 00).**
3. Inspect damper indicator for continuous leakage. **IF LEAKAGE OF 1 DROP PER HOUR OR MORE IS VISIBLE REPLACE INDICATOR (WP 0559 00).**
4. If there is leakage at damper cylinder head check valve, monitor damper indicator. If indicator goes from full indication (full gold band visible) to refill indication (no gold band visible) during next 10 flight hours, replace damper (WP 0558 00).

INSPECT DAMPER FOR LEAKAGE - Continued**INSPECT DAMPER BEARING****Inspect Bearing Surface****NOTE**

An accumulation of reddish-brown powder on ball bearing is normal for a Teflon bearing and is not cause for removal.

1. Visually inspect inboard and outboard damper bearings (Figure 1, Sheet 1). **IF EITHER BEARING HAS GOUGES GREATER THAN 0.040-INCH DEEP ON 1/2 OR MORE OF THE CIRCUMFERENCE AREA ON EDGE OF OUTER RACE, REPLACE BEARING (WP 0561 00) AND NYLON WASHER.**
2. Check bearing surface for corrosion by doing this:

CAUTION

Damage to Teflon liner will result if solvent is used to clean bearings. Do not use solvent to clean bearings.

- a. Remove minor surface corrosion with scouring pad, Item 205, WP 1803 00.
- b. Inspect spherical ball surface for corrosion pitting. **IF 5 OR MORE PITS EXCEED 0.030-INCH DIAMETER IN A 1/2-INCH DIAMETER CIRCLE OR ANY PIT IS LARGER THAN 0.050-INCH, REPLACE BEARING (WP 0561 00).**

Inspect For Wear - Feeler Gage Method**NOTE**

Check at least half of each spherical bearing closest to damper body.

1. Check inboard and outboard damper bearings by doing this:
 - a. With an assistant, position blade over nose. Engage gust lock. Grasp main rotor blade and apply load to blade in direction of full lead.
 - b. Push damper toward side of bearings to be inspected (Figure 1, Sheet 1).
 - c. Locally make Teflon lined spherical bearing plastic feeler gage (WP 1805 00).
 - d. Using 0.010-inch end of locally-made Teflon lined spherical bearing plastic feeler gage, probe both sides of bearing around entire circumference for clearance between ball and outer race (Figure 1, Sheet 2, Detail A).
 - e. If Teflon lined spherical bearing plastic feeler gage cannot be inserted to 0.250-inch mark, bearing is acceptable.
 - f. If Teflon lined spherical bearing plastic feeler gage can be inserted to 0.250-inch mark, or more, check clearance again using 0.015-inch end of feeler gage.
 - g. If 0.015-inch end of Teflon lined spherical bearing plastic feeler gage cannot be inserted to 0.250-inch mark between ball and outer race, bearing is acceptable. Repeat inspection every 15 flight hours. Annotate Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 with requirement for repeat inspection.
 - h. If 0.015-inch end of Teflon lined spherical bearing plastic feeler gage can be inserted to 0.250-inch mark between ball and outer race, inspect bearing for radial and axial play using dial indicator method.
 - i. Move damper in opposite direction and repeat inspection for opposite side of bearings.

INSPECT DAMPER BEARING - Continued

NOTE

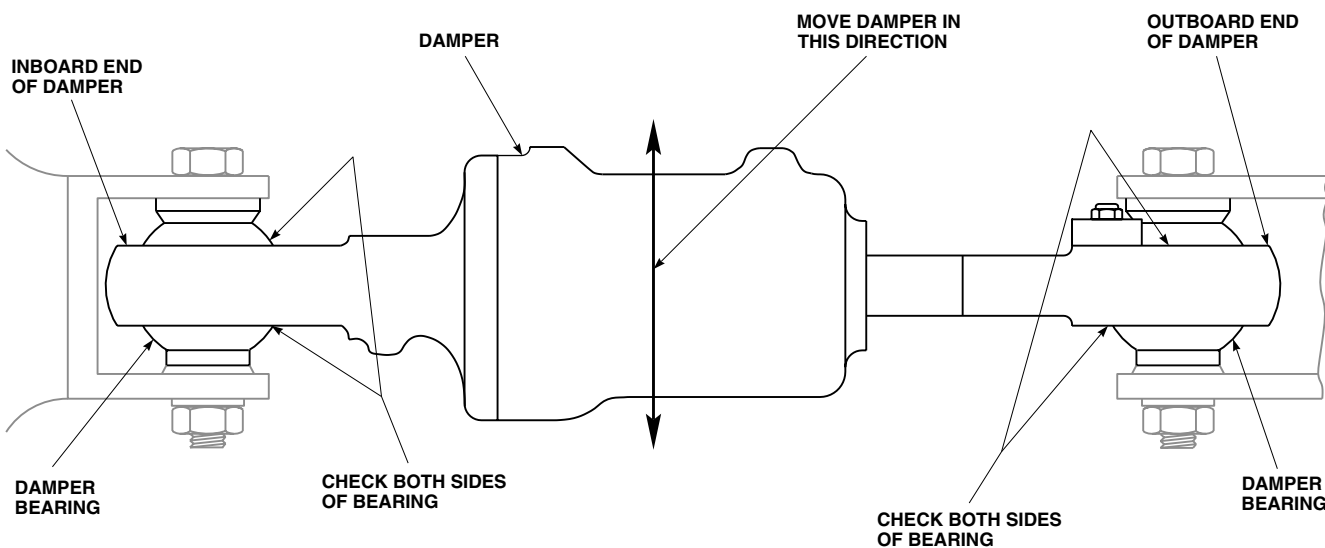
An accumulation of reddish-brown powder on the surface of the ball is normal for a Teflon bearing and is not cause for removal.

2. Check inboard and outboard bearings for evidence of metal-to-metal contact between ball and outer race. Metal-to-metal contact is present if worn or discolored spots are seen on load bearing surface of bearing ball, or if you hear a clicking sound when bearing ball is rotated in its outer race. **NONE ALLOWED.** Replace damper bearing if either bearing shows any metal-to-metal contact between ball and outer race (WP 0561 00).

NOTE

When operating helicopters with high gross weights or external sling load operation, contact of damper rod end bearing with nylon washer may occur.

3. If there is evidence of contact on lower outside or upper inside area of rod end bearing, remove one washer from counterweight bracket.
4. If there is evidence of contact on lower inside or upper outside area of rod end bearing, add one washer to counterweight bracket.



NOTE
INBOARD DAMPER BRACKET
ROTATED TO SHOW DAMPER
INSTALLATION.

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Figure 1. Inspect Damper Bearings. (Sheet 1 of 2)

INSPECT DAMPER BEARING - Continued

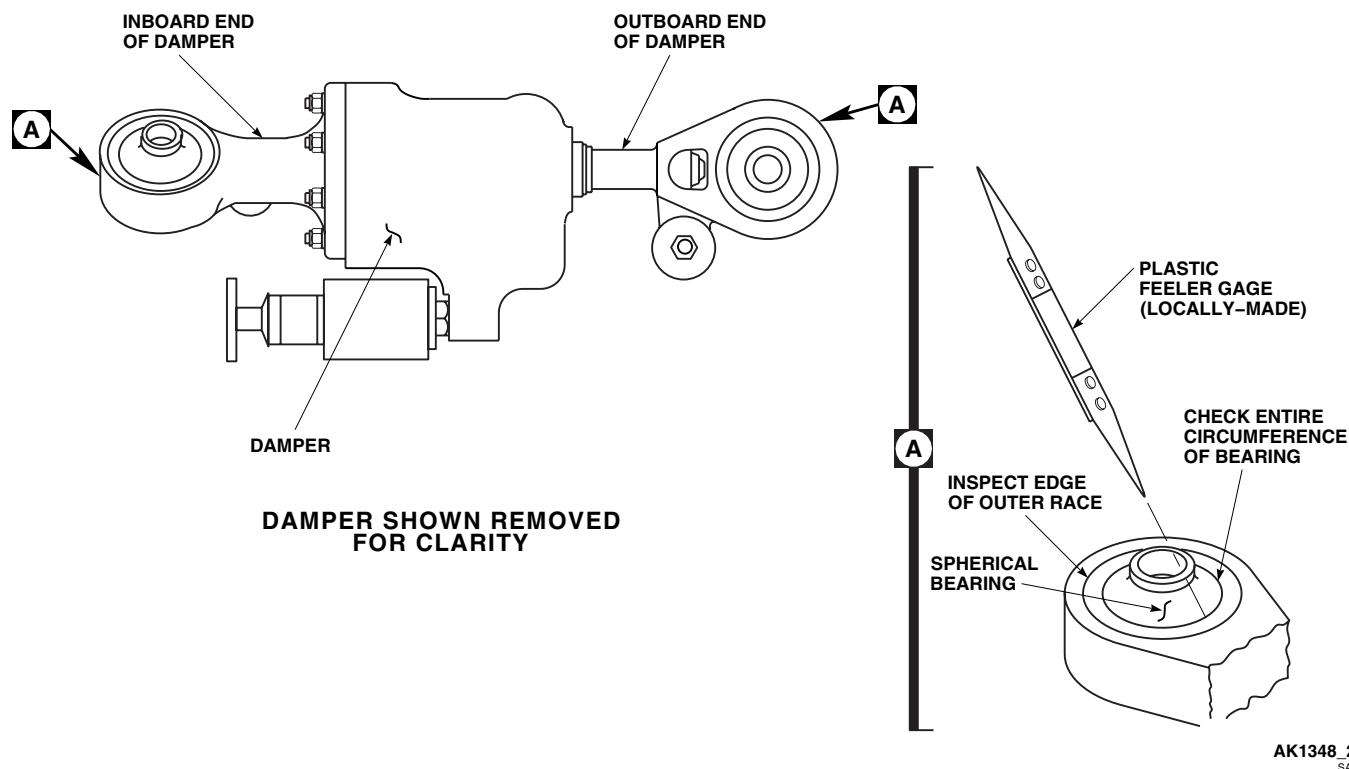


Figure 1. Inspect Damper Bearings. (Sheet 2 of 2)

Inspect For Radial and Axial Play

NOTE

The following inspection is not required until bearing has exceeded maximum tolerance. Refer to, Inspect For Wear-Feeler Gage Method, Step 1.h., in this work package.

1. Clamp dial indicator, to inboard end of damper (Figure 2).
2. Position blade over nose. Grasp main rotor blade and apply a load to blade in lead and lag direction. Do not pull blade down more than 6 inches. **MAXIMUM ALLOWABLE RADIAL PLAY MUST NOT BE MORE THAN 0.023-INCH.** If play is more, replace bearing (WP 0561 00).
3. Repeat inspection on outboard end of damper.
4. Clamp dial indicator to inboard end of damper, with indicator probe against head of bolt (Figure 2).
5. Hold damper from turning and move damper bearing in direction shown. **MAXIMUM ALLOWABLE AXIAL PLAY AS SHOWN ON DIAL INDICATOR MUST NOT BE MORE THAN 0.042-INCH.** If play is more, replace bearing (WP 0561 00).
6. Repeat inspection on outboard end of damper.
7. If damper bearings do not exceed maximum allowable radial and/or axial play, bearing is acceptable. Repeat inspection every 15 flight hours until bearing is replaced. Annotate Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 with requirement for repeat inspection.

INSPECT DAMPER BEARING - Continued

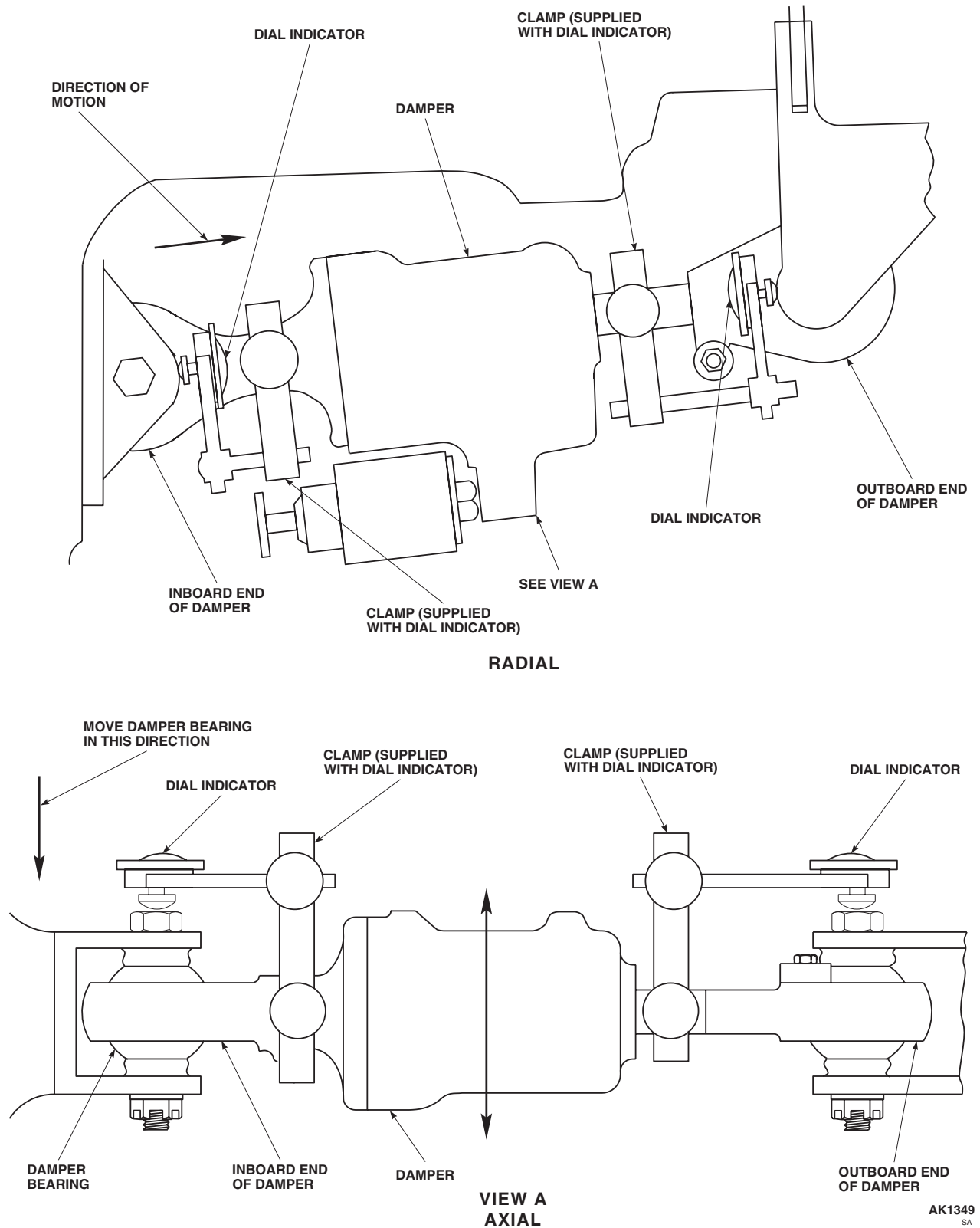


Figure 2. Inspect Damper Bearings.

END OF WORK PACKAGE

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UNIT LEVEL

ROTOR SYSTEM

PITCH CONTROL ROD INSPECTIONS

This WP supersedes WP 0538 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Depth Gage
 Elastomeric Rod End Bearing Plastic Feeler Gage,
 Locally-Made, Figure 232, WP 1805 00
 Gloves
 Goggles/Face Shield
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Teflon Lined Spherical Bearing Plastic Feeler Gage,
 Locally-Made, Figure 209, WP 1805 00

Material/Parts

Acetone, Technical Item 5, WP 1803 00
 Lubricant, Solid Film Item 180, WP 1803 00

Pad, Scouring, Item 205, WP 1803 00
 Towel, Machinery Wiping Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0563 00
 WP 0564 00
 WP 0565 00
 WP 1803 00
 WP 1805 00

INSPECT PITCH CONTROL ROD SPHERICAL AND ELASTOMERIC ROD END BEARING SURFACES

1. Check upper and lower spherical rod end bearing surface for corrosion by doing this:

CAUTION

Damage to Teflon liner will result if solvent is used to clean bearings. Do not use solvent to clean bearings.

- a. Remove surface corrosion with scouring pad, Item 205, WP 1803 00.
 - b. Inspect spherical ball surface for corrosion pitting. **IF FIVE OR MORE PITS EXCEED 0.030-INCH DIAMETER IN A 1/2-INCH DIAMETER CIRCLE, OR IF ANY PIT IS LARGER THAN 0.050-INCH, REPLACE BEARING (WP 0564 00).**
2. Inspect upper and lower spherical rod end bearing for gouges. **IF BEARING HAS GOUGES GREATER THAN 0.040-INCH DEEP ON 1/2 OR MORE OF THE CIRCUMFERENCE AREA ON EDGE OF OUTER RACE, REPLACE BEARING (WP 0564 00).**
 3. Inspect elastomeric rod end bearing surfaces as follows:

NOTE

- Elastomeric rod ends consist of laminated elastomer and steel shells. Wear of elastomer is a gradual process that will take several hundreds of hours from first detection of elastomer damage to point where rod end must be replaced.
 - A wax-like substance on rubber surface is normal and should not be removed.
- a. Check for a concentration or pileup of eraser-type wear. A concentration or pileup of eraser-type wear may indicate an elastomeric crack or separation may exist. Inspect rod end for elastomeric crack or separation if a

INSPECT PITCH CONTROL ROD SPHERICAL AND ELASTOMERIC ROD END BEARING SURFACES -
Continued

concentration or pileup of eraser-type wear is present. Refer to INSPECT PITCH CONTROL ROD ELASTOMERIC ROD END BEARING FEELER GAGE WEAR, in this work package.

NOTE

Peeling of elastomer mold lines which run radially across laminations is acceptable.

- b. Elastomeric extrusion is squeezing out of rubber from between bearing shims and inner or outer members, including local disbonding of elastomer. Inspect rod end for elastomeric crack or separation if an extrusion is detected. Refer to INSPECT PITCH CONTROL ROD ELASTOMERIC ROD END BEARING FEELER GAGE WEAR, in this work package.
- c. Visually check for cracking or disbonding of elastomer from shim inner or outer members. Cracking or disbonding of elastomer from shim inner or outer members will cause separation. Inspect rod end if cracking or disbonding of elastomer from shim inner or outer members is suspected. Refer to INSPECT PITCH CONTROL ROD ELASTOMERIC ROD END BEARING FEELER GAGE WEAR, in this work package.
- d. Visually check all elastomeric rod end bearings for signs of inner member separation using hand force, shaking control rod. **NO INNER MEMBER LOOSENESS OR EXPOSED METAL OF INNER BALL SURFACE OR MISSING ELASTOMER ALLOWED.** Replace elastomeric rod end if damage is beyond limits (WP 0565 00).

NOTE

Shims or steel shells alternate with elastomer layers of elastomeric rod end bearing and are composed of two 180 degree pieces (Figure 1). Within each shim layer, two pieces are separated by small gap measuring approximately 0.050-inch. These gaps are filled with elastomer during manufacture and are evident on surfaces as small lump of elastomer. Gaps becoming exposed over time is acceptable.

- e. Check for gaps or cracks in areas other than manufactured shim gaps. **NO GAPS OR CRACKS ALLOWED.** Replace elastomeric rod end with gaps or cracks (WP 0565 00).

INSPECT PITCH CONTROL ROD ELASTOMERIC ROD END BEARING FEELER GAGE WEAR

- 1. Inspect bearing surfaces. Refer to INSPECT PITCH CONTROL ROD SPHERICAL AND ELASTOMERIC ROD END BEARING SURFACES, in this work package.
- 2. Locally make elastomeric rod end bearing plastic feeler gage (WP 1805 00).
- 3. Check upper and lower elastomeric rod end bearings as follows:

INSPECT PITCH CONTROL ROD ELASTOMERIC ROD END BEARING FEELER GAGE WEAR - Continued**CAUTION**

- Damage to elastomeric rods end will result if metallic feeler gage is used. Elastomer will be damaged. Use only elastomeric rod end bearing plastic feeler gage when inspecting elastomeric rod end bearing.
- Damage to elastomeric rod ends will result if elastomeric rod end bearing plastic feeler gage is forced between elastomeric layers. Do not force elastomeric rod end bearing plastic feeler gage when checking for laminate separations.

NOTE

Elastomeric rod end bearing plastic feeler gage should only be used if possible separation exists based on visual inspection of bearing surfaces.

- a. On upper and lower elastomeric rod end bearings, check for elastomer/metal separations or disbonding of elastomer from shim inner or outer members in all areas carefully using elastomeric rod end bearing plastic feeler gage (0.005-inch) (Figure 1). If elastomeric rod end bearing plastic feeler gage can be inserted, separation has occurred. Measure depth of separations using elastomeric rod end bearing plastic feeler gage (0.005-inch). **SMALL HAIRLINE SURFACE CRACKING, MINOR DISBONDING AND CRAZING OF ELASTOMER UP TO 0.060-INCH IS ACCEPTABLE IN ALL AREAS OF ROD END BEARING.**
- b. On upper elastomeric rod end bearings, check for elastomer/metal separations or disbonding of elastomer from shim inner or outer members in Zone I carefully using elastomeric rod end bearing plastic feeler gage (0.005-inch) (Figure 1). If elastomeric rod end bearing plastic feeler gage can be inserted, separation has occurred. Measure depth of separations using elastomeric rod end bearing plastic feeler gage (0.005-inch). **SEPARATIONS UP TO 0.40-INCH IN ZONE I AND UP TO 0.060-INCH OUTSIDE ZONE I IS ACCEPTABLE.** Replace elastomeric rod end within next 10 flight hours if damage is beyond limits (WP 0565 00).
- c. On lower elastomeric rod end bearings, check for elastomer/metal separations or disbonding of elastomer from shim inner or outer members in Zone I carefully using elastomeric rod end bearing plastic feeler gage (0.005-inch) (Figure 1). If elastomeric rod end bearing plastic feeler gage can be inserted, separation has occurred. Measure depth of separations using elastomeric rod end bearing plastic feeler gage (0.005-inch). **SEPARATIONS UP TO 0.20-INCH IN ZONE I AND UP TO 0.060-INCH OUTSIDE ZONE I IS ACCEPTABLE.** Replace elastomeric rod end within next 10 flight hours if damage is beyond limits (WP 0565 00).

INSPECT PITCH CONTROL ROD SPHERICAL ROD END BEARING FEELER GAGE WEAR

1. Inspect bearing surfaces Refer to INSPECT PITCH CONTROL ROD SPHERICAL AND ELASTOMERIC ROD END BEARING SURFACES, in this work package.
2. Locally make Teflon lined spherical bearing plastic feeler gage (WP 1805 00).
3. Check upper and lower spherical rod end bearings as follows:
 - a. Apply hand force to control rod on opposite side of bearing that probing will be done (Figure 2, Sheet 1 or Figure 2, Sheet 2).

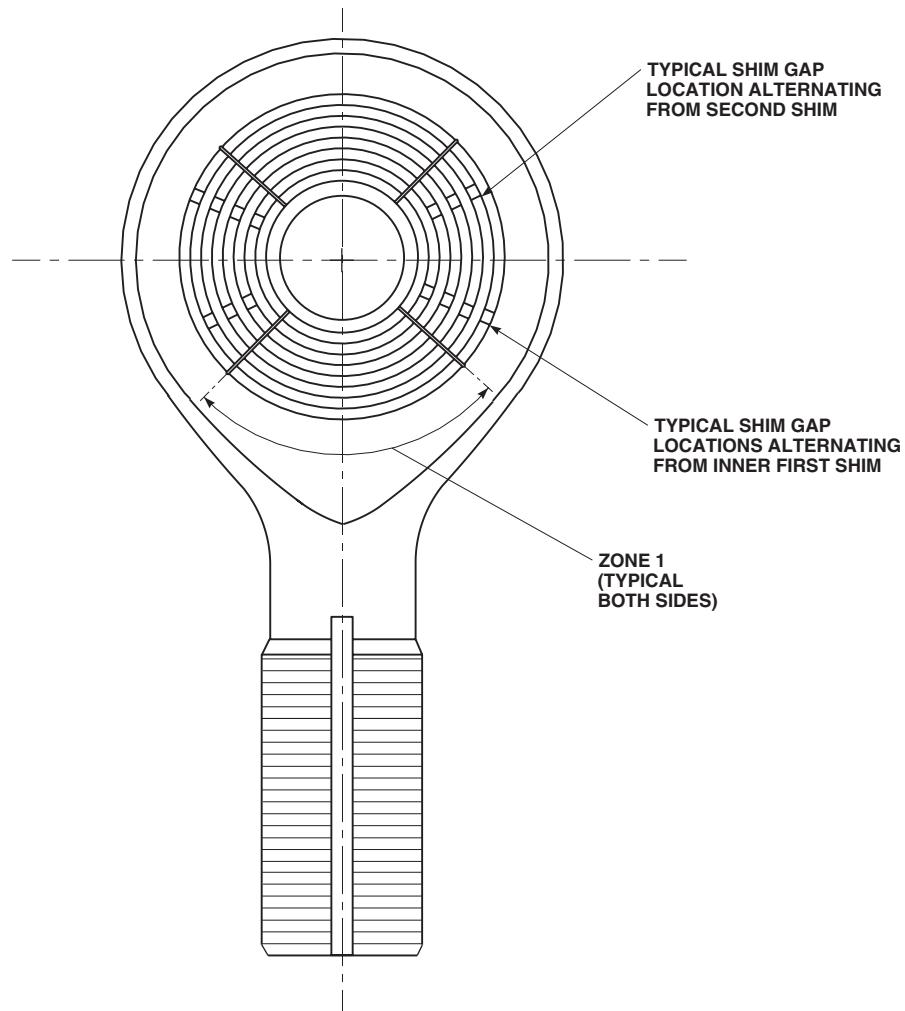
INSPECT PITCH CONTROL ROD SPHERICAL ROD END BEARING FEELER GAGE WEAR - ContinuedAA8115A
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Figure 1. Inspect Pitch Control Rod Elastomeric Rod End Bearings.

CAUTION

- Damage to spherical rod end bearing will result if metallic feeler gage is used. Teflon race will be damaged. Use only Teflon lined spherical bearing plastic feeler gage when inspecting spherical rod end bearing.
 - Damage to spherical rod end bearing will result if Teflon lined spherical bearing plastic feeler gage is not inserted carefully. Teflon race will be damaged. Use care when inserting Teflon lined spherical bearing plastic feeler gage.
- b. Using 0.010-inch end of locally-made Teflon lined spherical bearing plastic feeler gage, probe entire circumference of spherical rod end bearing.
- c. If feeler gage (0.010-inch) cannot be inserted to 0.250-inch mark, bearing is acceptable.

INSPECT PITCH CONTROL ROD SPHERICAL ROD END BEARING FEELER GAGE WEAR - Continued

- d. If feeler gage (0.010-inch) can be inserted to 0.250-inch mark, or more, probe entire circumference again using 0.015-inch end of Teflon lined spherical bearing plastic feeler gage.
- e. If 0.015-inch end of Teflon lined spherical bearing plastic feeler gage cannot be inserted to 0.250-inch mark between ball and outer race, spherical rod end bearing is acceptable. Repeat inspection every 15 flight hours.
- f. If 0.015-inch end of Teflon lined spherical bearing plastic feeler gage can be inserted between ball and outer race to 0.250-inch mark, do this:
 - (1) Disconnect upper or lower end of pitch control rod (WP 0563 00).

CAUTION

Damage to Teflon bearings will result if bearings are lubricated. Do not lubricate Teflon bearings.

NOTE

- An accumulation of reddish-brown powder on spherical rod end bearing ball is normal for a Teflon bearing.
 - Ratcheting indicates worn Teflon and metal-to-metal contact.
- (2) Check for ratcheting by rotating ball against outer race. **NONE ALLOWED.**
 - (3) If ratcheting exists, replace bearing and pitch control rod end (WP 0565 00).
 - (4) If ratcheting does not exist, replace rod end bearing only (WP 0564 00).
 - (5) Connect upper or lower end of pitch control rod (WP 0563 00).
4. Apply hand force to opposite side of pitch control rod and repeat step 3.
 5. Check upper and lower spherical rod end bearings for evidence of metal-to-metal contact between ball and outer race. Metal-to-metal contact is present if worn or discolored spots are seen on load bearing surface of bearing ball. **NONE ALLOWED.** Replace spherical rod end bearing if there is any metal-to-metal contact (WP 0565 00).

INSPECT PITCH CONTROL SPHERICAL AND ELSTOMERIC ROD END BEARING THREADS.

1. Inspect thread condition for corroded, flattened, stripped or crossed threads.
2. Replace components as required (WP 0565 00).
3. If bare metal is showing on threads, reapply solid film lubricant, Item 180, WP 1803 00, as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- a. Clean areas of bare metal using dry machinery wiping towel, Item 344, WP 1803 00 or equivalent dampened with technical acetone, Item 5, WP 1803 00. Allow technical acetone to dry thoroughly.
- b. At room temperature, spray exposed area with solid film lubricant, Item 180, WP 1803 00
- c. Apply multiple thin coats of solid film lubricant, Item 180, WP 1803 00, allowing 15 - 20 minutes drying time between coats.

INSPECT PITCH CONTROL SPHERICAL AND ELSTOMERIC ROD END BEARING THREADS. - Continued

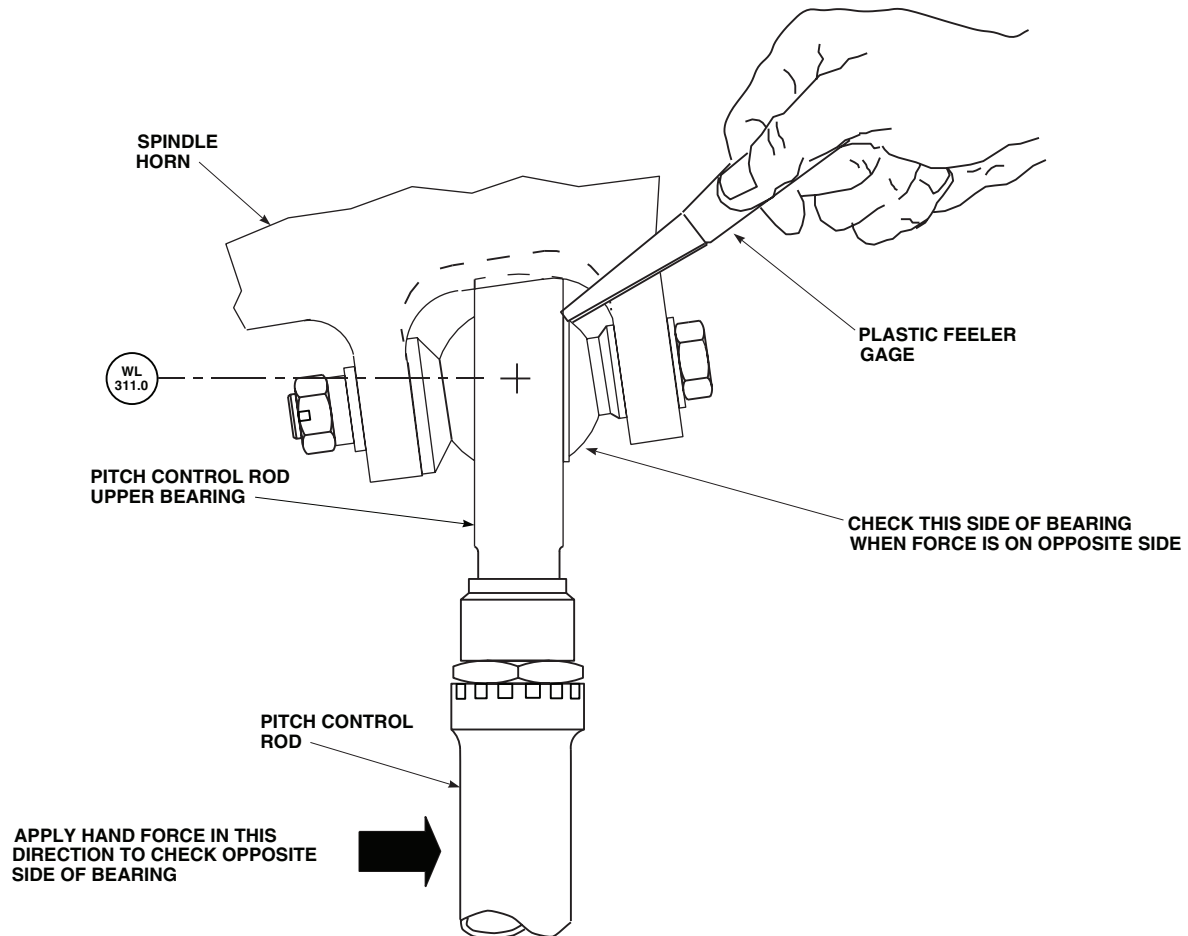
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Figure 2. Inspect Pitch Control Rod Spherical Rod End Bearings. (Sheet 1 of 2)

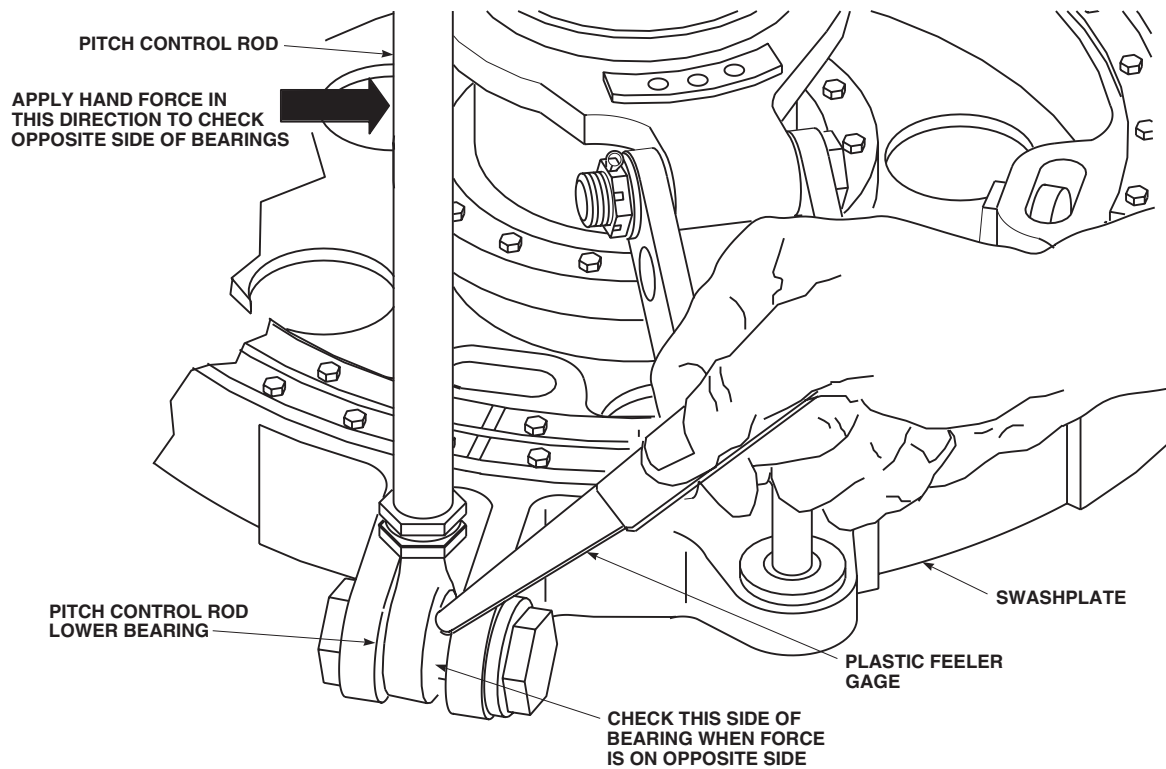
INSPECT PITCH CONTROL SPHERICAL AND ELSTOMERIC ROD END BEARING THREADS. - ContinuedFB3225
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Figure 2. Inspect Pitch Control Rod Spherical Rod End Bearings. (Sheet 2 of 2)

- d. Air cure solid film lubricant a minimum of 12 hours before using rod end.

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
SWASHPLATE INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Dial Indicator, MIL-I-18422
 Powertrain Repairer's Toolkit, SC 5180-99-B13

References

WP 0135 00
 WP 0563 00
 WP 0567 00
 WP 0568 00
 WP 0572 00
 WP 0598 00
 WP 1803 00

Material/Parts

Pad, Scouring, Item 205, WP 1803 00
 Tags

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

INSPECT SWASHPLATE LARGE SPHERICAL BEARING**NOTE**

- The large spherical bearing (uni-ball) is inspected by checking the condition of the spherical surface guide, and Teflon liners.
 - A reddish-brown powder on the ball is normal for a Teflon bearing and is not cause for removal.
 - The swashplate bearing may squeak when moved. Do not lubricate, since this is normal for this bearing.
1. Position swashplate to allow maximum exposure of spherical bearing.
 2. Check plated surfaces of bearing. If they show signs of metal-to-metal contact or peeling of plating greater than 1/4-inch in diameter in more than two places, replace spherical bearing (WP 0598 00).
 3. Inspect Teflon surfaces for debonding or separation from ball or outer race, or pieces of Teflon coming out of bearing. **NONE ALLOWED.** Replace spherical bearing if damage is found (WP 0598 00).

INSPECT SWASHPLATE SPHERICAL BEARING INNER RACE**NOTE**

Note the location of expandable pins for reinstallation.

1. Disconnect input connecting links (WP 0567 00).
2. Pull swashplate assembly toward you.
3. **REPLACE SWASHPLATE IF RATCHETING, CHATTERING, OR BINDING IS FELT IN FLIGHT CONTROLS.**

INSPECT SWASHPLATE SPHERICAL BEARING INNER RACE - Continued**CAUTION**

Damage to Teflon will result if feeler gage is forced between bearing race and swashplate guide. Do not force feeler gage between bearing race and swashplate guide.

4. Measure gap between inner race and guide by inserting feeler gage between swashplate guide and inner race of spherical bearing (Figure 1). **MAXIMUM GAP IS 0.028-INCH. NO METAL-TO-METAL CONTACT PERMITTED.**
5. Replace spherical bearing if either limit is exceeded (WP 0598 00).
6. Reconnect input connecting links (WP 0567 00).

INSPECT SWASHPLATE SPHERICAL BEARING OUTER RACE

1. Clamp dial indicator on lower pressure plate. Place dial indicator probe on swashplate guide as near as possible to inner ring of bolts. Probe must be at a 90 degree angle to main rotor shaft (Figure 1).
2. Disconnect input connecting links (WP 0567 00).
3. Insert feeler gage between bearing and swashplate guide to lock out play from inner race. Feeler gage must be located directly in line and above dial indicator probe.
4. Move swashplate assembly laterally by pushing and pulling on assembly (Figure 1).
5. To measure gap between outer race and swashplate, do this:
 - a. Read dial indicator when swashplate is pushed; then read dial indicator when swashplate is pulled.

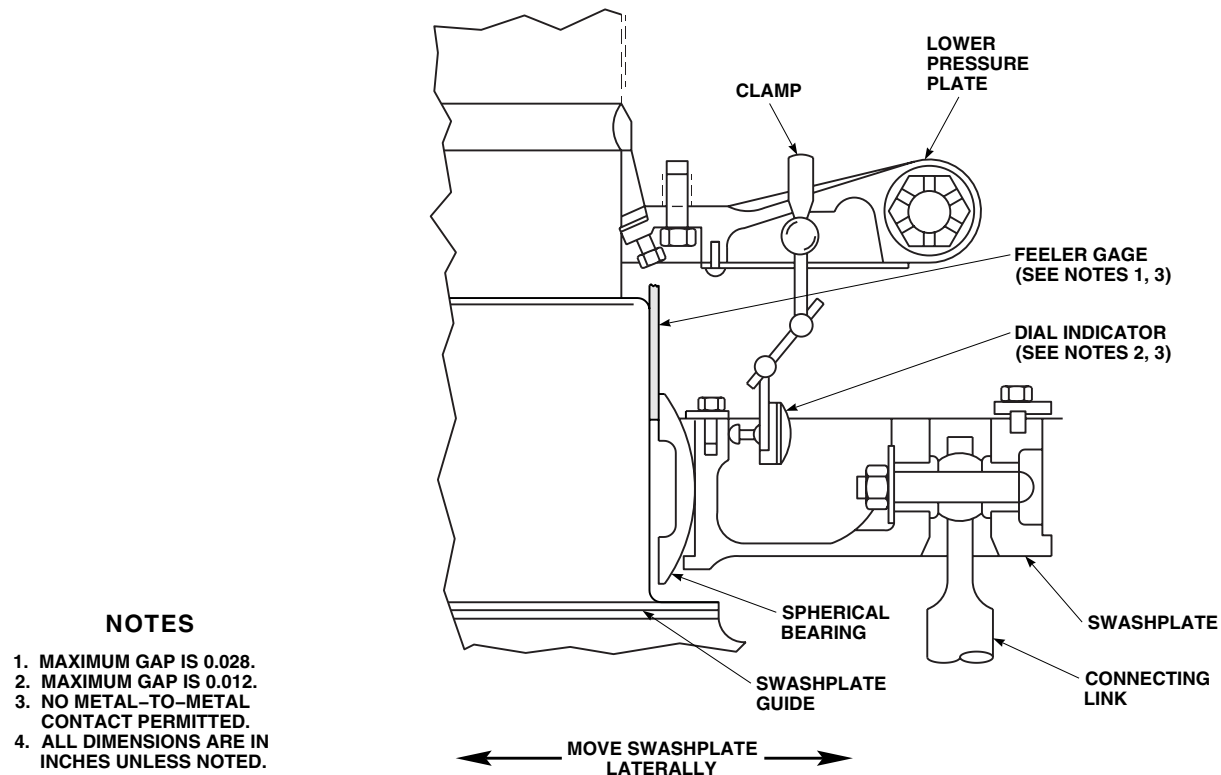


Figure 1. Inspect Swashplate Spherical Bearings.

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INSPECT SWASHPLATE SPHERICAL BEARING OUTER RACE - Continued

- b. The gap is the difference between the two readings. **MAXIMUM GAP IS 0.012-INCH. NO METAL-TO-METAL CONTACT PERMITTED.** Replace spherical bearing if either limit is exceeded (WP 0598 00).

6. Reconnect input connecting links (WP 0567 00).

INSPECT SWASHPLATE BALL BEARINGS BETWEEN STATIONARY AND ROTATING SWASHPLATE

1. Inspect swashplate extruded grease for metal particles. **NONE ALLOWED.** Replace swashplate with metal particles in extruded grease. Tag swashplate with reason for removal and send to depot.

NOTE

A newly lubricated swashplate will be more difficult to rotate. The following inspection should be performed prior to lubricating swashplate.

2. Disconnect pitch control rods (WP 0563 00) and scissors (WP 0572 00) from swashplate.
3. Check swashplate for smooth rotational motion by turning rotating swashplate by hand in direction of rotor head rotation (counterclockwise) through at least one complete rotation. **SWASHPLATE SHOULD ROTATE FREELY WITH NO SIGNS OF HESITATION, RATCHETING, OR UNEVEN MOVEMENT.** Repeat in opposite direction (Figure 2). If swashplate does not turn freely or has signs of hesitation, ratcheting, or uneven movement, replace swashplate assembly. Tag removed swashplate with reason for removal and return to depot.
4. Reconnect pitch control rod (WP 0563 00) and scissors (WP 0572 00) to swashplate.
5. Check ball bearings for wear as follows:
 - a. Turn on all hydraulic power.

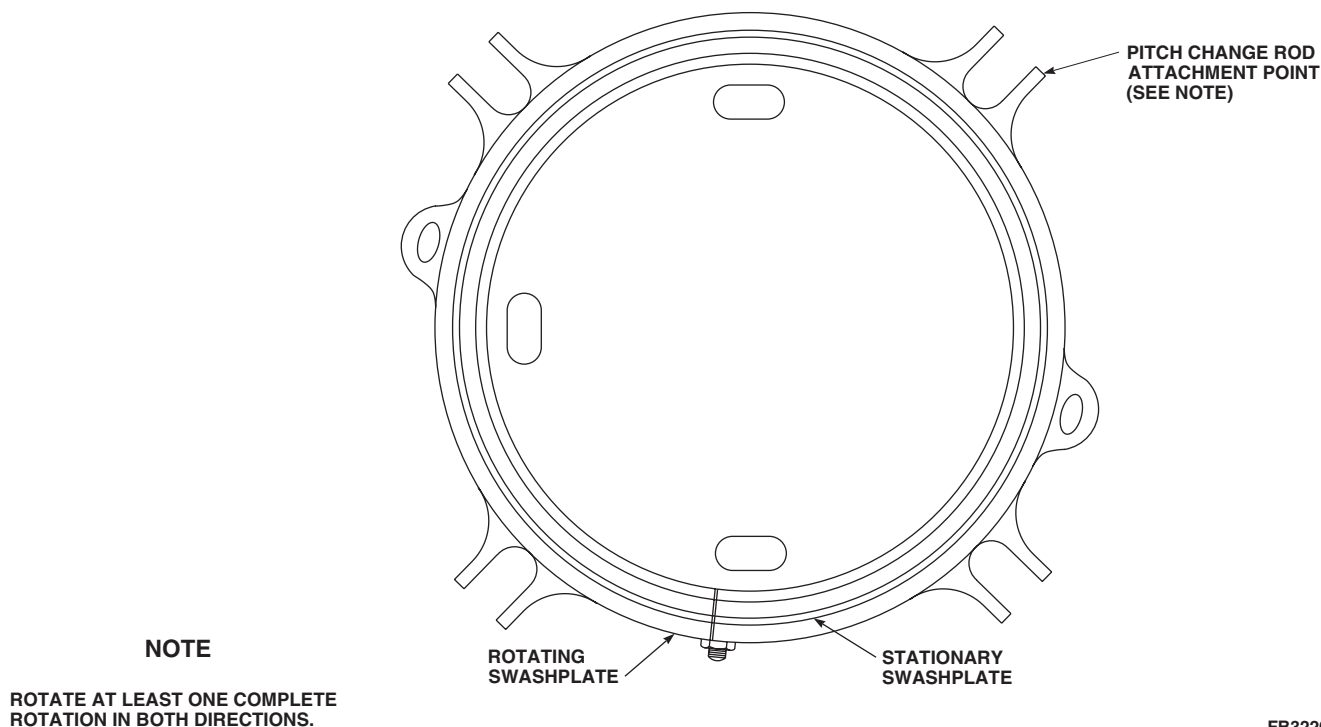


Figure 2. Inspect Swashplate Bearings.

INSPECT SWASHPLATE BALL BEARINGS BETWEEN STATIONARY AND ROTATING SWASHPLATE -
Continued

- b. Clamp dial indicator to rotating swashplate. Place dial indicator pointer on stationary swashplate. Connect dial indicator as close to bearing as possible.
- c. Move collective stick through its normal range (WP 0135 00). Measure vertical movement between rotating and stationary swashplates. **MAXIMUM RELATIVE VERTICAL MOTION IS 0.010-INCH.** Replace swashplate if limits are exceeded.
- d. Remove dial indicator.
- e. Turn off all hydraulic power.

INSPECT SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARING

1. Check bearing surface for corrosion as follows:
 - a. Disconnect lower link of rotating scissors (WP 0572 00).

CAUTION

Damage to Teflon lining will result if solvent comes in contact with bearings. Do not use solvent to clean bearings.

- b. Remove all surface corrosion with a scouring pad, Item 205, WP 1803 00.
 - c. Check spherical ball surface for corrosion pitting. **IF 5 OR MORE PITS ARE VISIBLE IN A 1/2-INCH DIAMETER CIRCLE OR ANY PIT IS LARGER THAN 0.050-INCH, REPLACE BEARING.**
 - d. Connect lower link of rotating scissors (WP 0572 00).
2. Check for radial play in the bearing connecting the lower scissors link to swashplate as follows:
 - a. Clamp dial indicator to scissors lower link.
 - b. Move link in direction shown and measure play (Figure 3). **MAXIMUM RADIAL PLAY IS 0.015-INCH.**
 - c. Replace bearing if play is more than limits (WP 0568 00).
 - d. Remove dial indicator.
 3. Check for axial play as follows:
 - a. Clamp dial indicator to scissors lower link.
 - b. Move link in direction shown (axial) and measure play (Figure 3). **MAXIMUM AXIAL PLAY IS 0.038-INCH.**
 - c. Replace bearing if play is more than limits (WP 0568 00).
 - d. Remove dial indicator.

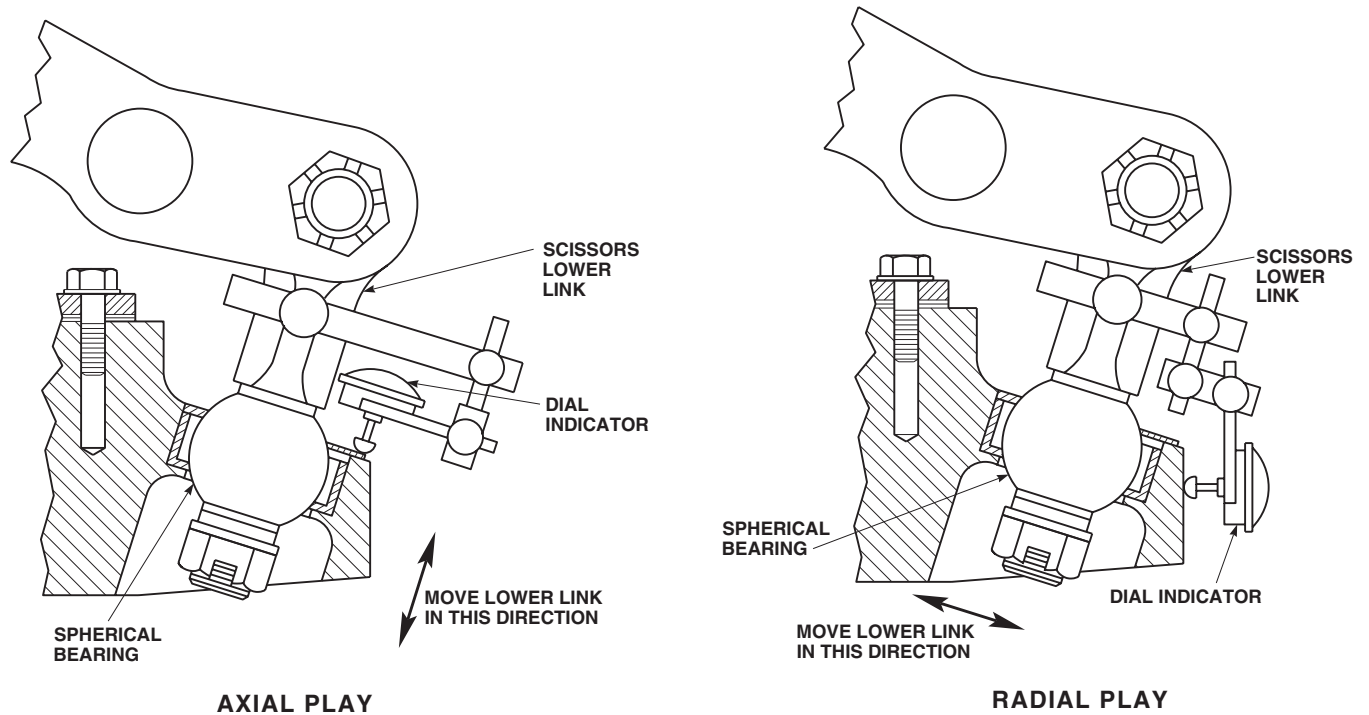
INSPECT SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARING - ContinuedAB0132
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Figure 3. Inspect Swashplate Scissors Attachment Spherical Bearings.

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
ROTATING SCISSORS INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Dial Indicator, MIL-I-18422
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

WP 0599 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT UPPER ROTATING SCISSORS BEARING

1. Rotate swashplate and scissors in direction of main rotor rotation. Make sure there is no load on rotating scissors upper link.
2. Clamp dial indicator, to pressure plate lug (Figure 1).
3. Move rotating scissors upper link up and down at indicator. **RADIAL PLAY AS SHOWN ON DIAL INDICATOR MUST NOT BE MORE THAN 0.012-INCH.** If play is more, replace bearings (WP 0599 00).
4. **GREATEST ALLOWABLE AXIAL CLEARANCE BETWEEN PRESSURE PLATE LUG AND SCISSORS LUG IS 0.014-INCH, MEASURED WITH A FEELER GAGE.** If clearance is more, measure height of outer bearing shoulder (WP 0599 00).

INSPECT LOWER ROTATING SCISSORS BEARING

1. Clamp dial indicator, to lower link (Figure 1, Detail A).
2. Move upper link up and down at dial indicator. **RADIAL PLAY AS SHOWN ON DIAL INDICATOR MUST NOT BE MORE THAN 0.012-INCH.** If play is more, replace bearings (WP 0599 00).
3. **GREATEST ALLOWABLE AXIAL CLEARANCE BETWEEN UPPER SCISSORS LUG AND LOWER SCISSORS LINK IS 0.014-INCH, MEASURED WITH A FEELER GAGE.** If clearance is more, measure height of outer bearing shoulder (WP 0599 00).

INSPECT LOWER ROTATING SCISSORS BEARING - Continued

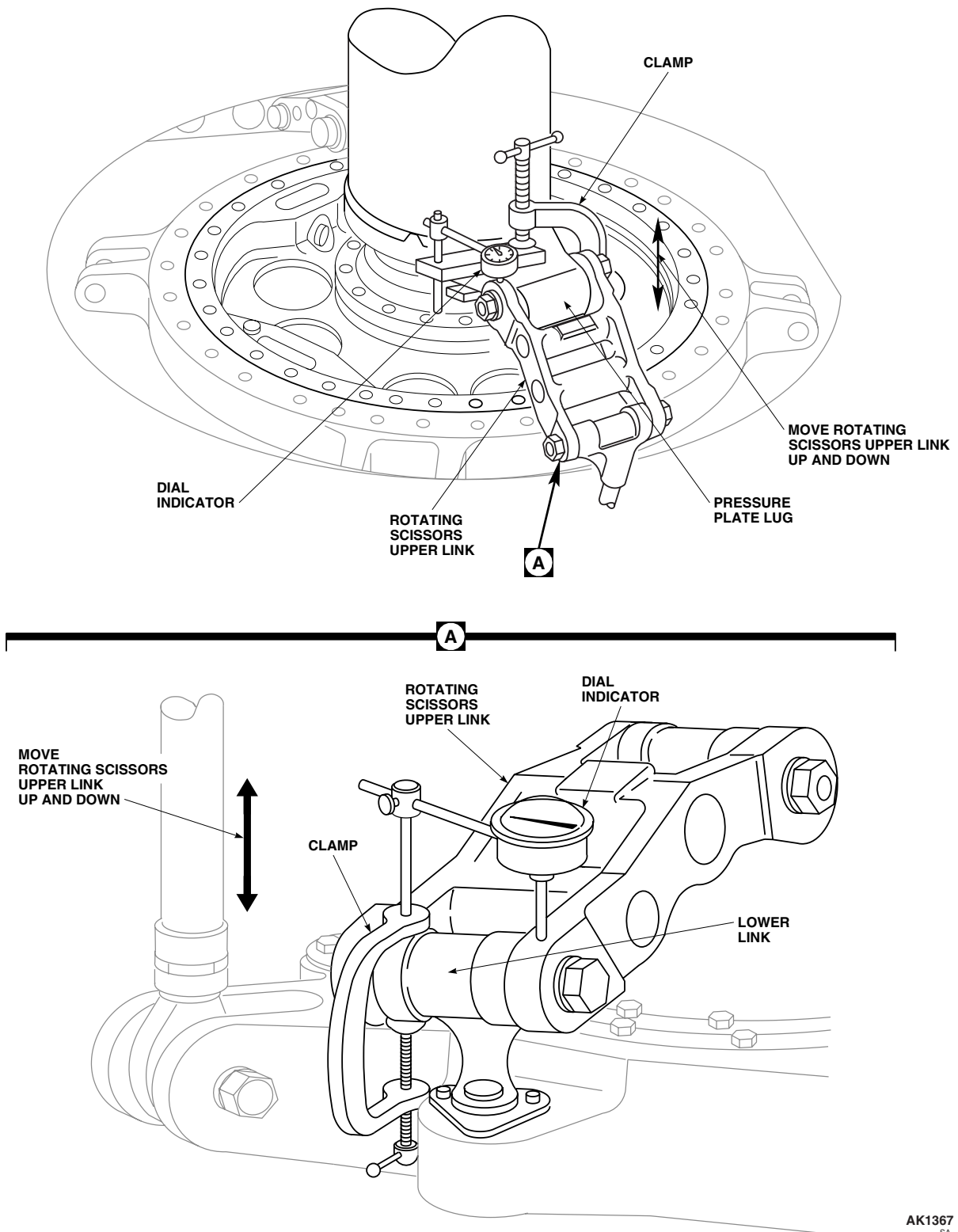


Figure 1. Inspect Upper/Lower Rotating Scissors Bearings.

END OF WORK PACKAGE

UNIT LEVEL**ROTOR SYSTEM**

MAIN ROTOR BLADE SPAR INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Checking and Filling Unit, S1670-15000-25
Checking Unit, S1670-15002-2
Free-Air Thermometer, MIL-T-1205C
Torque Wrench, 30 - 200 in. lbs.

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1624 00

WP 1803 00

Material/Parts

Leak Detection Fluid, Item 175, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

CHECK MAIN ROTOR BLADE SPAR PRESSURE/CHECKING UNIT

1. Check pressure in blade spar at intervals given in inspection requirements. Also check spar pressure whenever a new, overhauled, or stored blade is being placed in service. Check spar pressure as follows:

CHECK MAIN ROTOR BLADE SPAR PRESSURE/CHECKING UNIT - Continued**CAUTION**

Damage to blade will result if minimum pressures are not within limits. Do not attempt to pressurize blade spar if pressure is found to be below "MINIMUM WHEN CHECKING INSTALLED BLADE" limit of WP 1624 00.

NOTE

- Do not attempt to measure spar pressure with blade exposed to sunlight. Solar heating of blade can raise spar pressure by over 2 psi above free-air-temperature pressure, resulting in an inaccurate spar pressure measurement. Allow a minimum of 90 minutes after removing blade from sunlight, for blade spar temperature to stabilize, before measuring pressure in spar.
 - If spar pressure is below that given in "MINIMUM WHEN SERVICING BLADE" column of WP 1624 00, but above "MINIMUM WHEN CHECKING INSTALLED BLADE" pressure, increase spar pressure so it is between "MINIMUM WHEN SERVICING BLADE" and "MAXIMUM" pressures. Any blade with spar pressure found to be below "MINIMUM WHEN CHECKING INSTALLED BLADE" limit during normal checking shall be treated as though the pressure indicator had shown a full red (unsafe) condition.
 - Before measuring spar pressure, make sure that calibration expiration date on checking unit has not expired.
 - To make sure no pressure is lost from spar, connect coupling of checking unit to air valve before loosening air valve outer hex nut.
2. Remove cap from air valve in blade (Figure 1, Detail A). Connect coupling of checking unit, to air valve. Check for tight connection. Loosen air valve outer hex nut 2 1/2 turns while keeping inner hex nut and air valve adapter from turning. Note gage reading.

NOTE

Free-air thermometer should be in same lighting conditions as blade and should be within 10 feet of blade. Allow free-air thermometer to stabilize before taking final reading.

3. Using an accurate free-air thermometer, measure temperature in vicinity of blade being checked to nearest degree.
4. Record temperature and pressure.

WARNING

Injury to personnel and damage to equipment will result if blade with spar pressure below acceptable limits is reserviced and released for flight. Cause of pressure loss must be determined and corrected before releasing blade for flight. Remove blade and send it to AVIM level or depot.

NOTE

Make an entry on DA Form 2408-16 Aircraft Component Historical Record, stating reason for removal, whenever a blade is removed from service.

5. If spar pressure is within acceptable limits for temperature recorded above, as shown in WP 1624 00, blade can be continued in service.
6. **TORQUE AIR VALVE OUTER HEX NUT TO 50 - 70 INCH-POUNDS.** Disconnect checking unit from air

CHECK MAIN ROTOR BLADE SPAR PRESSURE/CHECKING UNIT - Continued

- valve. Install valve cap and tighten finger tight, plus 1/4 turn. Using nonelectrical wire, Item 357, WP 1803 00, lockwire air valve outer hex nut to air valve body (Figure 1, Detail A).
7. Check air valve for leakage, especially in stem area, by applying a liquid, such as leak detection fluid, Item 175, WP 1803 00, or a mild liquid detergent.
 8. When check is completed and before test liquid dries, clean area thoroughly with clean, fresh water to remove all traces of liquid; then dry blade using machinery wiping towel, Item 344, WP 1803 00.
 9. Test pressure indicator for proper operation (WP 1624 00).

CHECK MAIN ROTOR BLADE SPAR PRESSURE/CHECKING AND FILLING UNIT

1. Check pressure in blade spar at intervals given in inspection requirements. Also check spar pressure whenever a new, overhauled, or stored blade is being placed in service. Check spar pressure as follows:

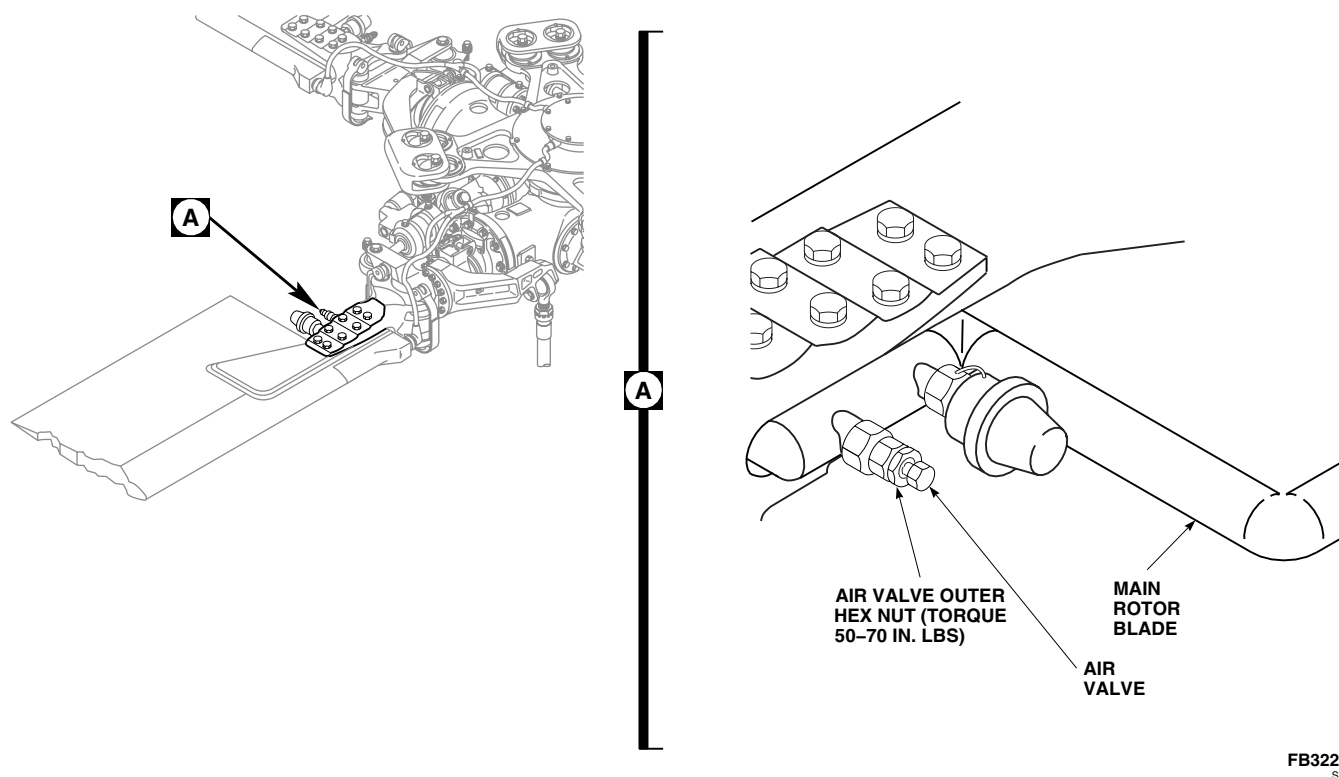


Figure 1. Check Main Rotor Blade Spar Pressure/Checking and Filling Unit.

CHECK MAIN ROTOR BLADE SPAR PRESSURE/CHECKING AND FILLING UNIT - Continued**CAUTION**

- Damage to blade will result if minimum pressures are not within limits. Do not attempt to pressurize blade spar if pressure is found to be below "MINIMUM WHEN CHECKING INSTALLED BLADE" limit of WP 1624 00.
- Damage to checking and filling unit will result if not purged of air. When using checking and filling unit to check spar pressure, it is mandatory that 20 foot hose line be purged of air before opening valve core control nut on blade air valve.

NOTE

- Do not attempt to measure spar pressure with blade exposed to sunlight. Solar heating of blade can raise spar pressure by over 2 psi above free-air temperature pressure, resulting in an inaccurate spar pressure measurement. Allow a minimum of 90 minutes after removing blade from sunlight, for blade spar temperature to stabilize, before measuring pressure in spar.
 - If spar pressure is below that given in "MINIMUM WHEN SERVICING BLADE" column of WP 1624 00, but above "MINIMUM WHEN CHECKING INSTALLED BLADE" pressure, increase spar pressure so it is between "MINIMUM WHEN SERVICING BLADE" and "MAXIMUM" pressures. Any blade with spar pressure found to be below "MINIMUM WHEN CHECKING INSTALLED BLADE" limit during normal checking shall be treated as though the pressure indicator had shown a full red (unsafe) condition.
2. Open cover of checking and filling unit. Remove adjustment handle from cylinder regulator. Remove regulator from bracket. Reinstall handle on regulator.
 3. Connect cylinder regulator of checking and filling unit to source of nitrogen at fitting on regulator. If necessary, use one of the adapters supplied with checking and filling unit to attach regulator to nitrogen container.
 4. Make certain station regulator, cylinder regulator, and station valve, next to station regulator, are closed. Turn station regulator shutoff knob and cylinder regulator adjustment handle counterclockwise to close. Turn station valve shutoff knob clockwise to close.
 5. Open valve on nitrogen bottle. Gage will now show apply pressure.
 6. Turn on flow of nitrogen at cylinder regulator and adjust pressure to between 30 and 40 psi, using adjustment handle on cylinder regulator.
 7. Remove cap from air valve in blade (Figure 1, Detail A). Just before connecting checking and filling unit coupling to air valve, open station valve shutoff knob and station regulator shutoff knob for a few seconds to purge air from lines. Close valves and connect coupling to air valve. Check for tight connection. Loosen air valve outer hex nut 2 1/2 turns while keeping inner hex nut and air valve adapter from turning. Note gage reading.

NOTE

Free-air thermometer should be in the same lighting condition as blade and should be within 10 feet of blade. Allow free-air thermometer to stabilize before taking final reading.

8. Using an accurate free-air thermometer, measure temperature in vicinity of blade being checked to nearest degree.
9. Record temperature and pressure.
10. If spar pressure is within acceptable limits for temperature recorded above, as shown in WP 1624 00, blade can be continued in service.

CHECK MAIN ROTOR BLADE SPAR PRESSURE/CHECKING AND FILLING UNIT - Continued**WARNING**

Injury to personnel and damage to equipment will result if blade with spar pressure below acceptable limits is reserviced and released for flight. Cause of pressure loss must be determined and corrected before releasing blade for flight. Remove the blade and forward to AVIM level or depot.

NOTE

Make an entry on DA Form 2408-16 Aircraft Component Historical Record, stating reason for removal, whenever a blade is removed from service.

11. If it is necessary to increase spar pressure, do this:
 - a. To increase spar pressure, open station valve by turning station valve shutoff knob counterclockwise. Using station regulator shutoff knob, gradually increase pressure until gage reaches required pressure that was found in WP 1624 00, for measured temperature. Allow spar to fill to this pressure.
 - b. Pressure in spar should be checked periodically by turning station valve shutoff knob fully clockwise and noting reading on gage at station regulator. When pressure in blade spar has reached specified limit, shut off nitrogen supply at station valve by turning station valve shutoff knob fully clockwise. Check gage at station regulator for a reading within limits specified in WP 1624 00, for measured blade spar temperature.
12. **TORQUE AIR VALVE OUTER HEX NUT TO 50 - 70 INCH-POUNDS.** Disconnect checking and filling unit from air valve. Install valve cap and tighten finger tight, plus 1/4 turn. Using nonelectrical wire, Item 357, WP 1803 00, lockwire air valve outer hex nut to air valve body (Figure 1, Detail A).
13. Check air valve for leakage, especially in stem area, by applying a liquid, such as leak detection fluid, Item 175, WP 1803 00, or a mild liquid detergent.
14. When check is completed and before test liquid dries, clean area thoroughly with clean, fresh water to remove all traces of liquid, then dry blade using machinery wiping towel, Item 344, WP 1803 00.
15. Shut off nitrogen supply to cylinder regulator. Open station valve shutoff knob to release pressure in line. Shut off station regulator and disconnect nitrogen supply from checking and filling unit.
16. Test pressure indicator for proper operation (WP 1624 00).

END OF WORK PACKAGE

UNIT LEVEL

ROTOR SYSTEM

MAIN ROTOR BLADE PIN INSPECTIONS

This WP supersedes WP 0542 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Adapter, 70700-20426-041
 Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Blade Clamp Assembly, 70700-20324-041,
 Depth Gage
 Dial Indicator, A-A-2348
 Gloves
 Goggles/Face Shield
 Heat Gun
 Hoist, Capacity to Support Main Rotor Blade (500 lb
 Minimum Capacity)
 Hoisting Sling, 70700-20320-042
 Maintenance Crane, 70700-20300-042 (Alternate for
 Hoist)
 Spring Scale, AAA-S-133
 Spring Tester, 719-40

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 77, WP 1803 00
 Gloves, Cotton, Item 386, WP 1803 00
 Lubricant, Solid Film, Item 180, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (4)

References

WP 1634 00
 WP 1803 00

INSPECT EXPANDABLE PIN

CAUTION

- Damage to main rotor blade will result if blade trim tab is bent or damaged. While working on main rotor blade, be careful not to bend or damage blade trim tab. Bending trim tab will cause blade out-of-track condition. Position a workstand under tip of blade for assistant to stand on to help guide blade clear of helicopter.
- Expandable pin will be damaged if disassembled. **DO NOT DISASSEMBLE EXPANDABLE PIN.**

NOTE

Verify expandable pin has been cleaned prior to inspection.

1. Release expandable pin handle from lower nut and open handle.
2. Loosen blade pin nut so that the thread protrusion past nut is at least one thread, but not more than two threads. Check nut breakaway torque. **BREAKAWAY TORQUE MUST NOT BE LESS THAN 32 INCH-POUNDS.** Replace expandable blade pin that fails breakaway torque.
3. Check expandable pin tension as follows:

INSPECT EXPANDABLE PIN - Continued**NOTE**

Make sure blade tension is off blade expandable pins during tension check. Use blade clamp assembly, hoist (or maintenance crane), hoisting sling and adapter as necessary.

- a. Using spring scale, place end of spring scale in handle nut hole at clip and pull.
- b. **KEEPING THE SPRING FORCE PERPENDICULAR TO THE LONG, RIVETED SECTION OF THE HANDLE. THE FORCE REQUIRED TO CLOSE HANDLE SHALL BE 25 POUNDS MINIMUM.**
- c. **IF FORCE REQUIRED TO CLOSE HANDLE IS NOT AT LEAST 25 POUNDS, DO THIS:**
 - (1) **TIGHTEN NUT 1/6 (ONE-SIXTH) TURN AND RECHECK CLOSING FORCE. REPEAT UNTIL MINIMUM OF 25 POUNDS CLOSING FORCE IS OBTAINED.**
 - (2) If proper tension cannot be obtained after adjustment of nut, replace expandable pin.
- d. Check expandable pin for a minimum of two threads extending beyond nut after adjusting tension. If a minimum of two threads are not exposed, do this:
 - (1) **REPEAT TENSION CHECK. KEEPING THE SPRING FORCE PERPENDICULAR TO THE LONG, RIVETED SECTION OF THE HANDLE. THE FORCE REQUIRED TO CLOSE HANDLE SHALL BE 25 POUNDS MINIMUM, WITH TWO THREADS EXTENDING BEYOND THE NUT. REPLACE IF CRITERIA IS NOT MET.**
4. Inspect expandable pin cam handle for cracks, nicks, dents, scratches, and corrosion. **NO CRACKS ALLOWED. IF CRACKED, REPLACE IMMEDIATELY.** Repair corrosion, nicks, dents, and scratches using abrasive cloth, Item 77, WP 1803 00, as follows:
 - a. **BLEND SURFACE DAMAGE ON CAM SURFACE OF HANDLE UP TO 0.002-INCH DEEP. TOTAL REPAIR AREA MUST NOT EXCEED 0.25-SQUARE INCHES.**
 - b. **BLEND SURFACE DAMAGE TO CAM HANDLE UP TO 0.020-INCH DEEP. TOTAL REPAIR AREA MUST NOT EXCEED 2 SQUARE INCHES.**
 - c. **BLEND EDGE DAMAGE TO CAM HANDLE UP TO 0.040-INCH DEEP. TOTAL LENGTH OF REPAIR MUST NOT EXCEED 3 INCHES.**
 - d. **BLEND SURFACE DAMAGE TO SPRING CLIP UP TO 0.005-INCH DEEP. NO LIMITS ON TOTAL REPAIR AREA.**
 - e. Replace expandable pin if damage or repairs are beyond limits.

WARNING**CSI**

Nicks, dents, scratches, and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.

5. Check outer segments OD for nicks, dents and scratches. **DAMAGE MUST NOT EXCEED 0.005-INCH.** Remove metal pickup using abrasive wheel, Item 350, WP 1803 00. Replace expandable pin if damage is beyond limits.
6. Check outer segments OD for condition of solid film lubricant. **UP TO 50% OF SOLID FILM LUBRICANT MISSING IS ACCEPTABLE.** If more than 50% of solid film lubricant is missing, do this:
 - a. Clean areas of bare metal using machinery wiping towel, Item 344, WP 1803 00, dampened with solvent cleaning compound, Item 71, WP 1803 00. Allow to dry.

INSPECT EXPANDABLE PIN - Continued

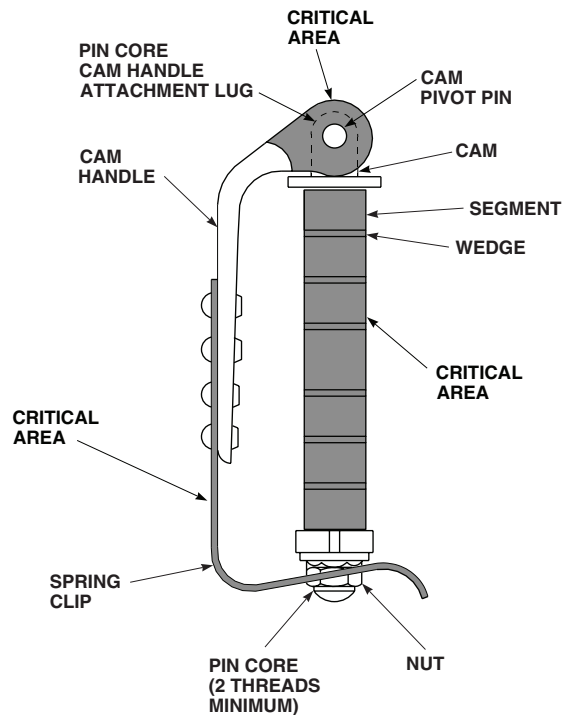
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Figure 1. Inspect Main Rotor Blade Expandable Pin.

NOTE

Solid film lubricant will not adhere to contaminated surfaces. Make sure surface remains clean while handling parts.

- b. Apply two thin coats of solid film lubricant, Item 180, WP 1803 00, to outer segment OD. Allow 15 to 20 minutes between coats. Total thickness of dry film lubricant to be 0.0003 to 0.0005-inches.
 - c. Air dry expandable pins for a minimum of 12 hours (alternate accelerated cure time is 30 minutes air dry plus 30 minutes at 150°F(66°C)).
 - d. Check expandable pin tension per step 2.
7. Check pivot pin for wear limits or movement as follows:

NOTE

If the pin becomes unseated when the handle is opened, or the pin was unseated prior to opening the handle, carefully raise the main rotor blade, using proper procedures, to remove any binding of the expandable blade pin and then reseal the pin.

- a. Using spring tester, check pivot pin security (axial play) by applying a measured forty (40) pounds of force to the inboard face of the pivot pin, in an outboard direction.
- b. Mount dial indicator to cam handle and zero out indicator on the inboard pivot pin face.
- c. Using spring tester, check pivot pin security (axial play) by applying a measured forty (40) pounds of force to outboard face of the pivot pin, in an inward direction.

INSPECT EXPANDABLE PIN - Continued

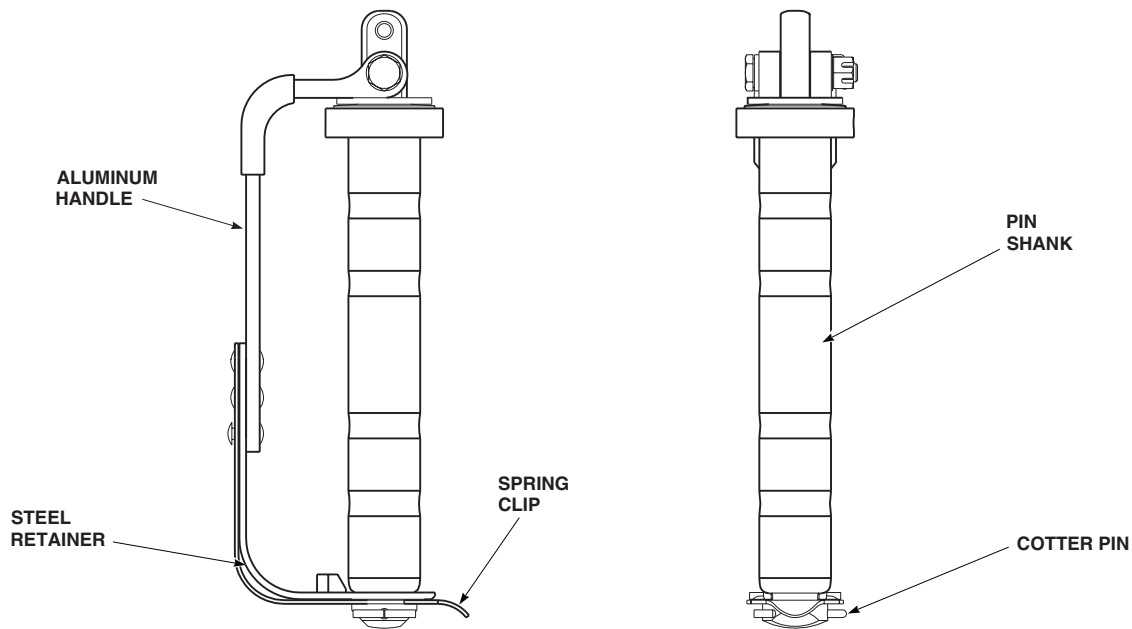
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Figure 2. Inspect Main Rotor Blade Solid Pin.

- d. Axial play of the cam pivot pin shall not exceed 0.025-inch in either direction. If axial play is less than 0.025-inch, inspection of the affected expandable pin is complete. Move handle to locked (closed) position as depicted in applicable maintenance procedure.

INSPECT SOLID PIN

CAUTION

Damage to main rotor blade will result if blade trim tab is bent or damaged. While working on main rotor blade, be careful not to bend or damage blade trim tab. Bending trim tab will cause blade out-of-track condition. Position a workstand under tip of blade for assistant to stand on to help guide blade clear of helicopter.

NOTE

- The following inspections apply only to main rotor blade solid pin, 70103-08108-041, that have been removed from the helicopter (Figure 2).
- Main rotor blade pins should be handled with cotton gloves, Item 386, WP 1803 00, following cleaning and during inspection. Do not touch pin shaft with bare hands.

1. Clean main rotor blade solid pins (WP 1634 00).
2. **QA** Inspect main rotor blade pin handle for cracks, nicks, dents, scratches, and corrosion (Figure 2). **NO CRACKS ALLOWED. IF CRACKED, REPLACE IMMEDIATELY.** Repair corrosion, nicks, dents, and scratches using abrasive cloth, Item 77, WP 1803 00, as follows:

INSPECT SOLID PIN - Continued

- a. Inspect handle for surface damage. **BLEND SURFACE DAMAGE TO HANDLE UP TO 0.020-INCH DEEP. TOTAL REPAIR AREA MUST NOT EXCEED 2 SQUARE INCHES.**
- b. Inspect handle for edge damage. **BLEND EDGE DAMAGE TO HANDLE UP TO 0.040-INCH DEEP. TOTAL LENGTH OF REPAIR MUST NOT EXCEED 3 INCHES.**
- c. Inspect steel spring clip and steel retainer for surface and edge damage. **BLEND SURFACE AND EDGE DAMAGE UP TO 0.005-INCH DEEP. NO LIMITS ON TOTAL REPAIR AREA.**
3. Inspect steel pin shank for wear. Measure pin diameter in worn areas. **PIN OUTSIDE DIAMETER MUST NOT BE LESS THAN 0.991-INCH.**
4. **QA** Inspect steel pin shank outside diameter for scoring or grooves. **BASE METAL AXIAL SCORING OR GROOVES UP TO 0.002-INCH DEEP ARE ACCEPTABLE. BLEND OUT BASE METAL AXIAL SCORING OR GROOVES 0.003 - 0.005-INCH DEEP. BLEND OUT BASE METAL CIRCUMFERENTIAL SCORING UP TO 0.005-INCH DEEP.**
5. **QA** Inspect condition of solid film lubricant. **UP TO 50% OF SOLID FILM LUBRICANT MISSING IS ACCEPTABLE.** If more than 50% of solid film lubricant is missing, do this:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- a. Clean areas of bare metal using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Allow to dry.

NOTE

- Solid film lubricant will not adhere to contaminated surfaces. Do not touch clean areas with bare hands. Use cotton gloves, Item 386, WP 1803 00.
 - Apply solid film lubricant, Item 180, WP 1803 00, where ambient temperatures are between 65 - 85 degrees F°(18 -30 degrees C°).
- b. Apply two thin coats of solid film lubricant, Item 180, WP 1803 00, to outer diameter of main rotor blade solid pin shaft. Allow 15 - 20 minutes between coats. Use depth gage to measure total thickness of dry film lubricant to be 0.0003 to 0.0005-inches.
 - c. Air dry main rotor blade solid pins for a minimum of 18 hours (alternate accelerated cure time is 6 to 9 hours at 120 to 150°F (49 - 66°C) using heat gun.

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
BIFILAR INSPECTION

This WP supersedes WP 0543 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Depth Gage
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Nonmetallic Scraper

Cloth, Abrasive, Item 77, WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Accelerator, Adhesive Curing, Item 3, WP 1803 00
 Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 33, WP 1803 00
 Brush, Acid Swabbing, Item 57, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 75, WP 1803 00

References

TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0381 00
 WP 0602 00
 WP 1803 00

INSPECT

NOTE

In the following steps, do all blending with abrasive cloth, Item 77, WP 1803 00, or an abrasive wheel, Item 350, WP 1803 00.

1. Check bifilar support arm for corrosion (Figure 1). **ON OPEN AREAS (NON-MATING SURFACES) REMOVE CORROSION PITS UP TO 0.050-INCH DEEP BY BLENDING OUT. NO TWO AREAS MAY BE CLOSER THAN 1 INCH APART. ON MATING SURFACES, REMOVE CORROSION PITS UP TO 0.010-INCH DEEP BY BLENDING OUT OVER AN AREA UP TO 10% OF MATING SURFACE AREA.** Replace bifilar if damage after repair is more than limits. Touch up reworked areas with corrosion resistant coating, Item 111, WP 1803 00. Apply primer coating, Item 243, WP 1803 00, and touch up paint (WP 0381 00 and TM 55-1500-345-23).
2. **IN BOLTHOLES, REMOVE CORROSION PITS UP TO 0.005-INCH DIAMETER.** Replace bifilar if damage after repair is more than limits. Touch up reworked area with corrosion resistant coating, Item 111, WP 1803 00. Apply primer coating, Item 243, WP 1803 00, to boltholes when installing bolts.
3. **CHECK BIFILAR SUPPORT ARM FOR NICKS AND GOUGES. CORNER DAMAGE UP TO 0.10-INCH MAY BE BLENDED OUT TO A 1/2-INCH RADIUS. MULTIPLE CORNER DAMAGE MUST BE AT LEAST 1 INCH APART.** Replace bifilar if damage after repair is more than limits. Touch up reworked area with corrosion resistant coating, Item 111, WP 1803 00. Touch up paint all reworked areas (WP 0381 00 and TM 55-1500-345-23).
4. **ON MATING SURFACES AND FLANGES, REMOVE DAMAGE UP TO 0.020-INCH. DAMAGED AREA MUST NOT BE OVER 10% OF SURFACE AREA.** Replace bifilar if damage after repair is more than limits.

INSPECT - Continued

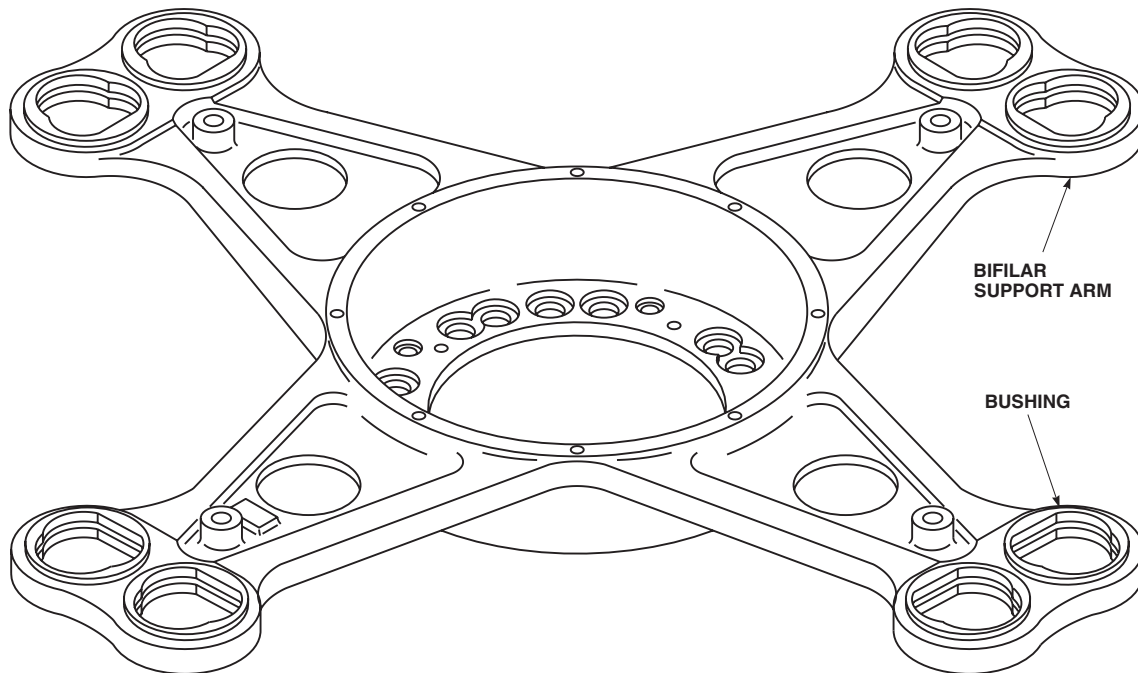
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Figure 1. Inspect Bifilar Support Arm.

Remove raised material around damage and blend out to 1/2-inch radius. After repairing area, visually inspect for cracks. **NO CRACKS ALLOWED.** Touch up reworked area with corrosion resistant coating, Item 111, WP 1803 00. Touch up paint all reworked areas (WP 0381 00 and TM 55-1500-345-23).

5. Visually inspect for cracks. **NONE ALLOWED.** If cracks are suspected, using fluorescent penetrant inspection kit, fluorescent penetrant inspect bushing ID for stress cracks along its edges (TM 1-1520-265-23). **NONE ALLOWED.** If cracks are present, replace bushings (WP 0602 00).

NOTE

A knife edge on bushing edge is not cause for rejection.

6. Check bifilar support arm and bifilar weight bushings for wear. **ON FLANGE FACES UP TO 0.030-INCH WEAR IS ALLOWED.** Original thickness of flange face on bushings, 70107-08202-106, -108, and -109 is 0.050-inch. On bushings 70107-08202-107, -110, and -111, original thickness of flange face is 0.0625-inch. If part number is not readable use 0.050-inch as original thickness. Replace bushings if wear is beyond limits (WP 0602 00).
7. Inspect weight for nicks, gouges, and cracks in AREA A, AREA B, and AREA C as follows (Figure 2):
 - a. **AREA A - NICKS AND GOUGES SHALL NOT EXCEED 0.150-INCH DEEP.**
 - b. **AREA B - CORNER NICKS AND GOUGES SHALL NOT EXCEED 0.10-INCH DEEP. SURFACE NICKS AND GOUGES SHALL NOT EXCEED 0.030-INCH DEEP. NO CRACKS ALLOWED.**
 - c. **AREA C - NO NICKS OR GOUGES ALLOWED.**
 - d. Blend nicks and gouges in AREA A and AREA B to a 1/2-inch radius using abrasive cloth, Item 77, WP 1803 00, or abrasive wheel, Item 350, WP 1803 00.

INSPECT - Continued

- e. Replace weight if damage is more than limits.

NOTE

Cracked paint in area C of weights with "UT" marked in area of part number is acceptable.

8. Inspect weight marked "UT" in area of part number for cracks in paint in AREA A and AREA B. Inspect weight without "UT" marked in area of part number for cracks in paint in AREA A, AREA B, and AREA C. If cracks in paint are found, do this:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

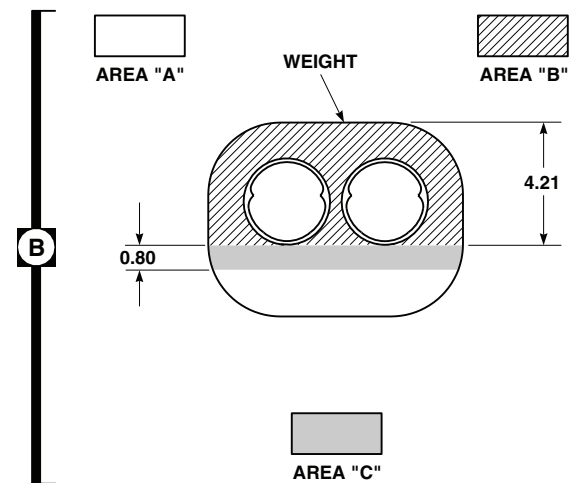
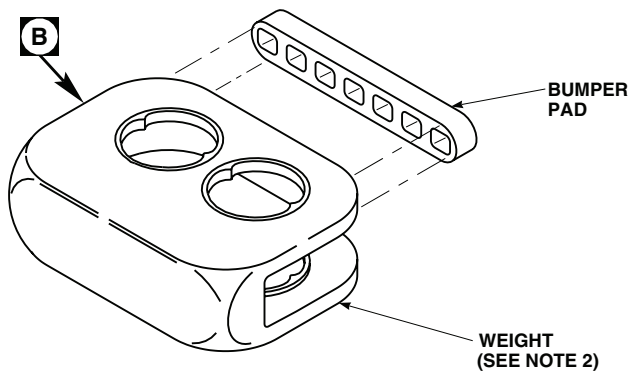
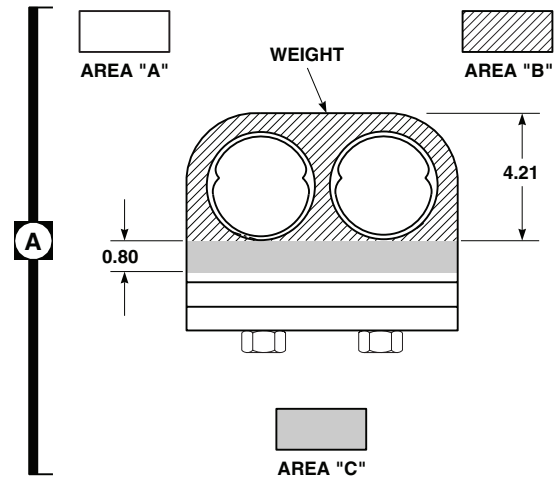
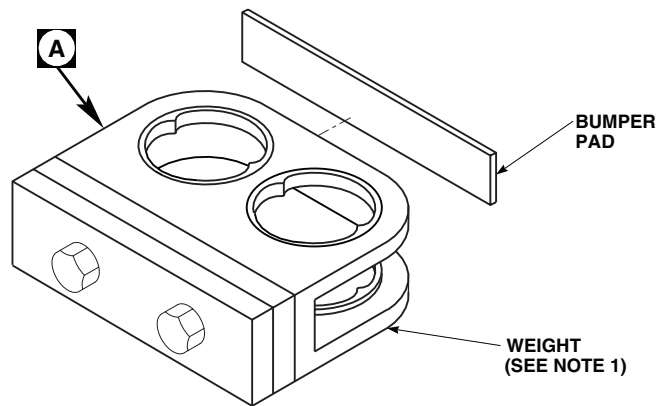
- a. Remove paint with technical acetone, Item 5, WP 1803 00.
 - b. Inspect AREA A and AREA B for cracks. **NONE ALLOWED.** Replace weight if cracks are found.
 - c. Inspect AREA C for bond separation. **NONE ALLOWED.** Replace weight if bond separation is found.
 - d. Touch up paint on weight as required (WP 0381 00 and TM 55-1500-345-23).
9. Check bumper pad for damage. **NONE ALLOWED.**
10. Remove damaged bumper pad using nonmetallic scraper (Figure 2). Clean pad area on weight by rubbing with abrasive cloth, Item 77, WP 1803 00, until surface is clean and bright.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- 11. Wipe contact area with technical acetone, Item 5, WP 1803 00.
- 12. Apply adhesive curing accelerator, Item 3, WP 1803 00, on new pad and bonding area on weight using acid swabbing brush, Item 57, WP 1803 00. Let primer dry at room temperature for 3 minutes.
- 13. Apply bead of adhesive, Item 33, WP 1803 00, to weight. Immediately place pad on pad area on weight. Apply pressure for 15 minutes before handling weight. Allow adhesive to cure for 24 hours before installing weight on bifilar.
- 14. Check pins for damage. **IN NON-CONTACT AREAS, REMOVE DAMAGE UP TO 0.020-INCH DEEP BY BLENDING OUT WITH ABRASIVE CLOTH, Item 75, WP 1803 00. CORNER DAMAGE UP TO 0.040-INCH MAY BE BLENDED OUT.** Replace pin if damage is more than limits.
- 15. Check pins for being out-of-round. **REPLACE PINS MORE THAN 0.002-INCH OUT-OF-ROUND.**
- 16. Check pin diameter for wear. **REPLACE PINS WITH MORE THAN 0.010-INCH WEAR.** See Table 1 for original diameter of pins.

INSPECT - Continued



NOTES

1. WEIGHT, 70107-08404-042.
2. WEIGHTS, 70107-08404-043 AND 70107-08404-044.
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 2. Inspect Bifilar Weights.

INSPECT - Continued

Table 1. Pin Diameters.

PART NUMBER	ORIGINAL DIAMETER
70107-08406-102	1.173
70107-08406-103	1.194
70107-08406-112	1.173
70107-08406-113	1.194
70107-08406-121	1.200
70107-08406-122	1.200

17. Visually check pins for cracks. If crack is suspected do a fluorescent penetrant inspection (TM 1-1520-265-23), as follows:
 - a. Clean pin using machinery wiping towel, Item 344, WP 1803 00, dampened with solvent cleaning compound, Item 71, WP 1803 00.
 - b. Check for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NONE ALLOWED.**
 - c. Replace pin if crack is found.
18. Check tapered washers for damage (Figure 3). **IN NON-CONTACT AREAS, REMOVE DAMAGE UP TO 0.020-INCH DEEP BY BLENDING OUT WITH ABRASIVE CLOTH, Item 75, WP 1803 00. CORNER DAMAGE UP TO 0.040-INCH MAY BE BLENDED OUT.** Replace washer if damaged beyond limits.
19. Check washers for wear. **IF MORE THAN 20% OF WASHER CONTACT SURFACES ARE PITTED, WORN, OR HAVE METAL PICKUP ON SURFACE, REPLACE WASHER. EVEN, CIRCULAR WEAR UP TO 0.020-INCH DEEP IS ACCEPTABLE IN CONTACT AREA BETWEEN BUSHINGS AND WASHERS.** Replace washers over limits.
20. Visually check washers for cracks. If crack is suspected do a fluorescent penetrant inspection (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked washers.

INSPECT - Continued

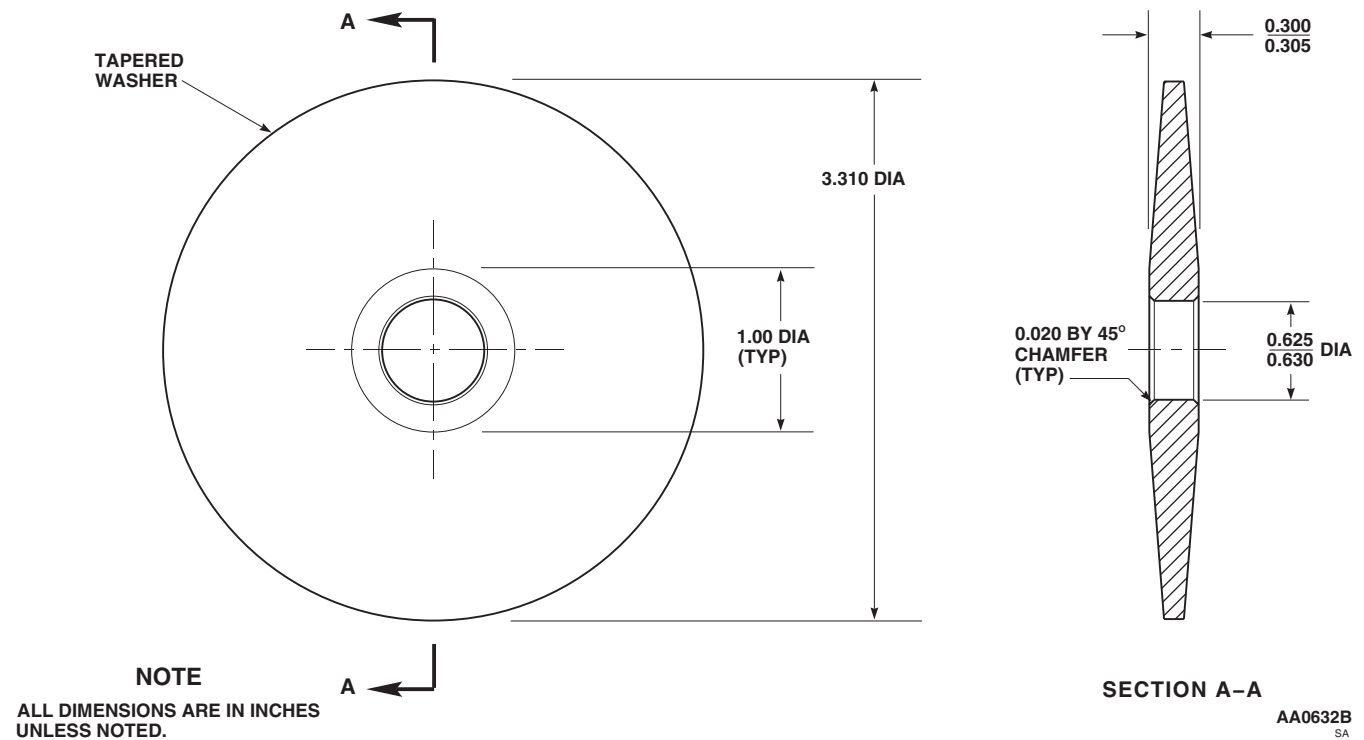


Figure 3. Check Bifilar Tapered Washers.

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
MAIN ROTOR BLADE INSPECTIONS
This WP supersedes WP 0544 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Drill Bit, No. 40
 File, Medium Cut
 Fluorescent Penetrant Inspection Kit, XMA101
 Grease Pencil
 Nonmetallic Scraper
 Punch, Soft Metal
 Soft Copper (or equivalent) 1 1/2 to 2 inch diameter x
 1/8 to 3/16-inch thick
 Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Adhesive, Item 11, WP 1803 00
 Cloth, Abrasive, Item 75, WP 1803 00
 Cloth, Abrasive, Item 77, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00

Sealing Compound, Item 257, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-265-23
 TM 1-6625-724-13&P
 WP 0082 00
 WP 0145 00
 WP 0576 00
 WP 0578 00
 WP 0584 00
 WP 1378 00
 WP 1486 00
 WP 1803 00

GENERAL**WARNING**

Injury to personnel and damage to equipment will result if the appearance red band, or portions of red band are evident. The cause for the appearance of red bands, or portions of red bands, in the BIM pressure indicator, shall be determined and corrected before flight (WP 0082 00). Discovery of spar failure, or inability to determine cause for appearance of red bands, will require removal and replacement of blade.

1. Damage to blade can be divided into two types, detectable and nondetectable.
2. See Table 1 and Figure 3, Sheet 1, for reference to damage limits and repair on main rotor blade.
3. Detectable damage can be seen and felt as a blister, or dent.
 - a. A blister can be seen as a raised bump above blade contour, indicating a separation between skin and core. 4 to 5 pounds of thumb pressure can move the skin against core without a crinkly sound.
 - b. A dent indicates a core crush. It can be felt and seen. 4 to 5 pounds of thumb pressure against dent will result in a crinkly sound.
4. Nondetectable damage can be found with coin-tapping, but cannot be seen or felt.
 - a. 4 to 5 pounds of thumb pressure will not make sound with nondetectable blisters.
 - b. 4 to 5 pounds of thumb pressure will make crinkly sound with nondetectable dents.
5. After repairing blade, record repair on Aircraft Component Historical Record DA Form 2408-16 by doing this:

GENERAL - Continued**NOTE**

Repairs inboard of Zone 1 (STA 60.0) do not need to be recorded. Repairs in this area will not affect chordwise or spanwise balance.

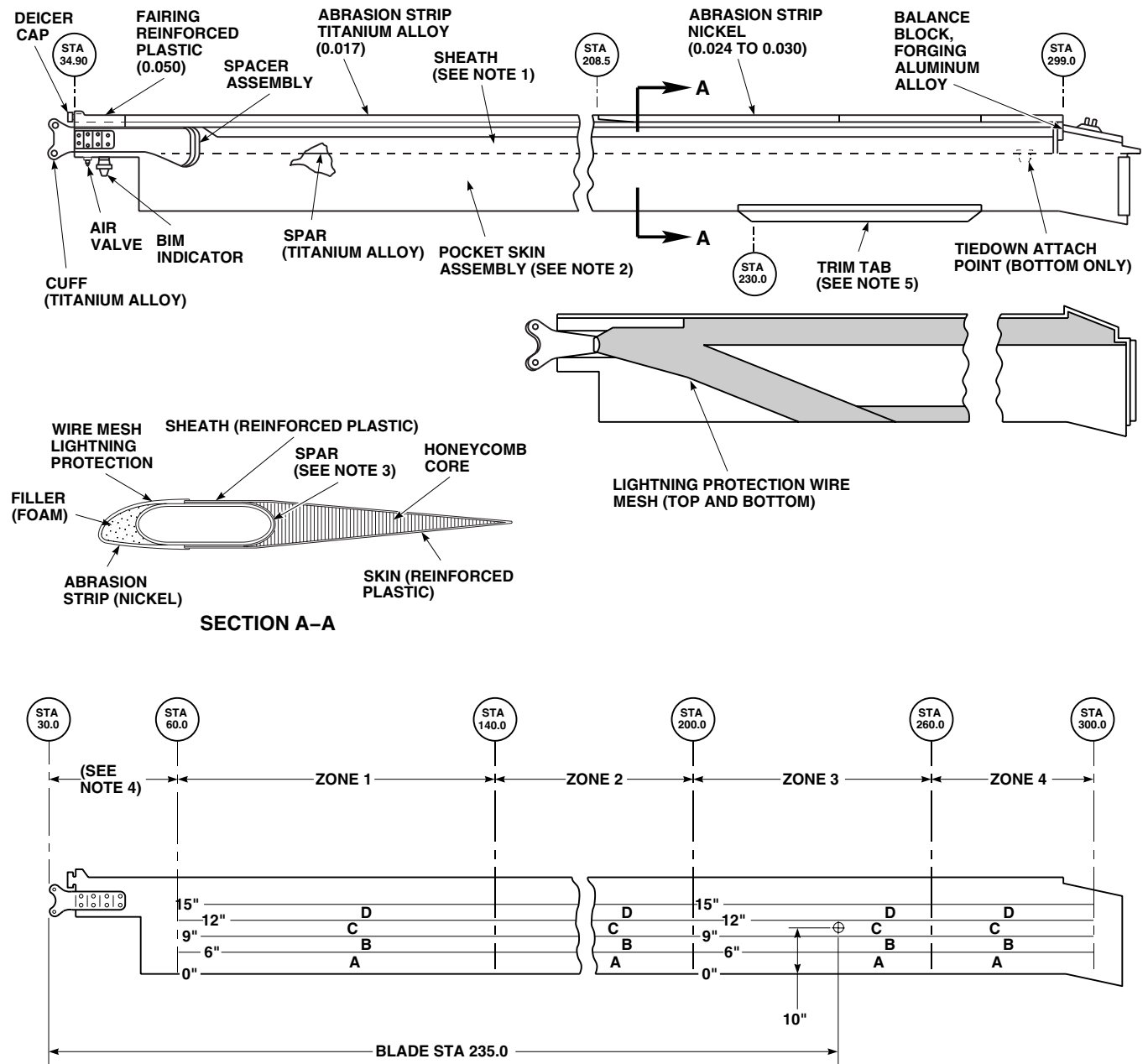
- a. Refer to Figure 1, Sheet 1, to find Zone of repair.
- b. Refer to Table 2 to find chordwise shim unbalance, and Table 3 to find spanwise weight unbalance.
- c. Record Date, Description of Damage, Zone, Repair kit Number, Chordwise Shim, and Spanwise Weight unbalances as shown in Table 4.
6. Repairs to blade can affect blade track and blade balance, causing main rotor vibration. The total number of repairs to the blade before adjusting weights will depend upon acceptable blade tracking and vibration levels (TM 1-6625-724-13&P).
7. If weights need to be adjusted, do this:
 - a. Add chordwise shim adjustment numbers for all repairs on blade (if weights have been adjusted previously, add all numbers from last new chordwise shim unbalance). Round this number down to nearest whole number. Record remaining fraction under new chordwise shim unbalance Table 2.
 - b. Remove tip cap (WP 0576 00).
 - c. Move whole number (obtained from above) of shims from rear chordwise studs, and install on front chordwise studs (WP 0576 00).
 - d. Add spanwise weight unbalances for all repairs on blade (if weights have been adjusted previously, add all numbers from last new spanwise weight unbalance).
 - e. Remove combination of shims from spanwise studs (WP 0576 00), whose weight comes closest to, but is not greater than, total spanwise weight unbalances.
 - f. Record difference between weight removed and total spanwise weight unbalances as new spanwise weight unbalance.
 - g. Install tip cap (WP 0576 00).

NOTE

New unbalance numbers are to be added to future repairs and balances.

- h. Track and balance main rotor blades (TM 1-6625-724-13&P).
8. Replace blade if damage is greater than maximum patch limits.
9. Do all repairs carefully according to instructions. Even small repairs affect blade track and balance.
10. Damaged area for patch repair must be at least 1 inch from shaded area (Figure 3, Sheet 1).
11. Damage not requiring patch may extend into shaded area.
12. Edge of new patch not to be closer than 1 inch from edge of previously installed patch on same side of blade.

GENERAL - Continued



NOTES

1. REINFORCED PLASTIC (0.090) INCORPORATES DEICING BLANKET.
2. COMPOSITE MADE FROM GLASS FABRIC AND GRAPHIC PLYS, BONDED WITH ADHESIVE RESIN.
3. INCORPORATES BIM BLANKET.
4. REPAIRS INBOARD OF ZONE 1 (STATION 60.0) DO NOT NEED TO BE RECORDED. REPAIRS IN THIS AREA WILL NOT AFFECT CHORDWISE OR SPANWISE BALANCE.
5. HELICOPTERS WITH BLADES SERIAL NOS A007-02550 TO A007-02900 WILL HAVE A NOTCH IN TRIM TAB.
6. ALL SHEET METAL CALLOUTS ARE 6061-T6 ALUMINUM ALLOY, QQ-A-250-11.
A -BASIC SKIN (0.063) REDUCED TO 0.033 TO 0.043 THICKNESS.
B -BASIC SKIN (0.063) REDUCED TO 0.018 TO 0.023 THICKNESS.
7. ON TIP CAP 70150-09107-049.
8. CRACKS ARE USUALLY FOUND AT HOLES 4 AND 5; 1/2-INCH OUTBOARD OF HOLE 6; BETWEEN HOLES 6 AND 7, 1 INCH AFT OF HOLE 6.
9. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

EXAMPLE

DAMAGE 10 INCHES TO FRONT OF TRAILING EDGE AT BLADE STA 235.0 IS IN ZONE 3C.

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Figure 1. Main Rotor Blade Inspection Areas. (Sheet 1 of 5)

GENERAL - Continued

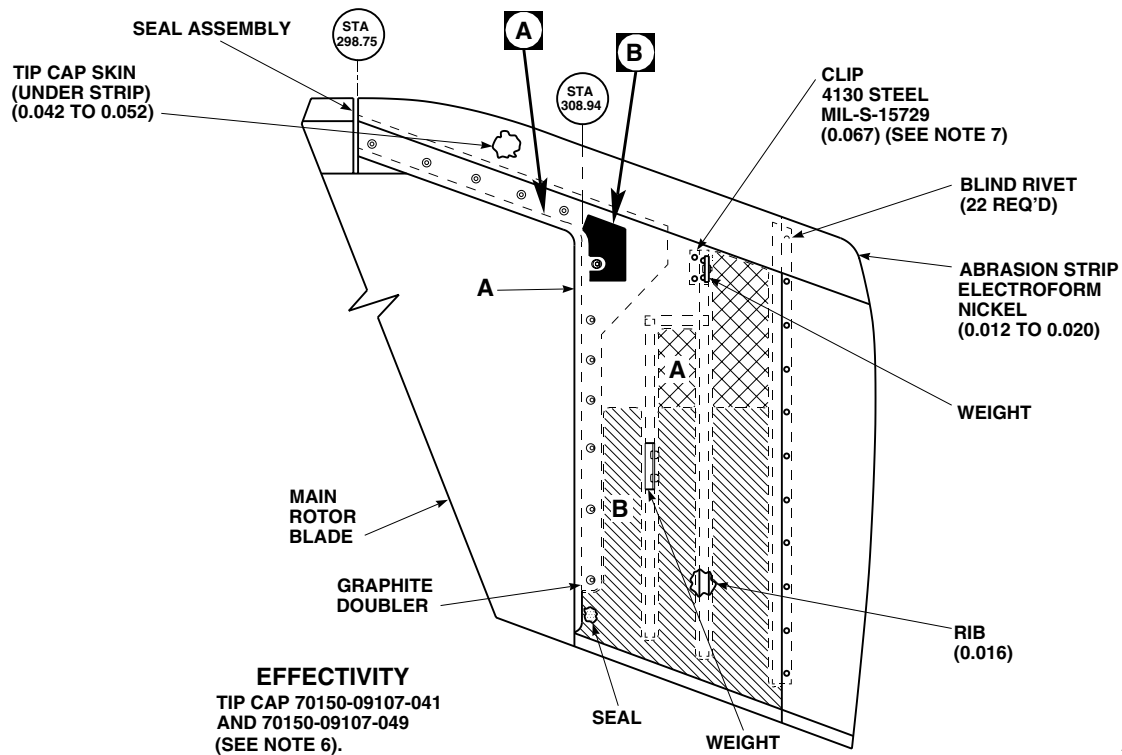
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Figure 1. Main Rotor Blade Inspection Areas. (Sheet 2 of 5)

CAUTION

Damage to blade skin will result if blade is sanded excessively. Heavy or careless sanding can produce enough damage to cause skin cracking. Sand reinforced plastic skin lightly and carefully.

NOTE

If bulk adhesive, Item 11, WP 1803 00, is used, mix 100 parts of "A" part to 22 parts of "B" part by weight. Weigh each part then place two parts in cup and stir until color of mixture is uniform. Pot life of mixture is 25 minutes at 75°F(24°C). Do not try to use all adhesive; use as necessary and discard leftover mix.

13. Blades damaged beyond repairable limits will be forwarded to depot.
14. Remove and install main rotor blades as required (WP 0576 00).

GENERAL - Continued

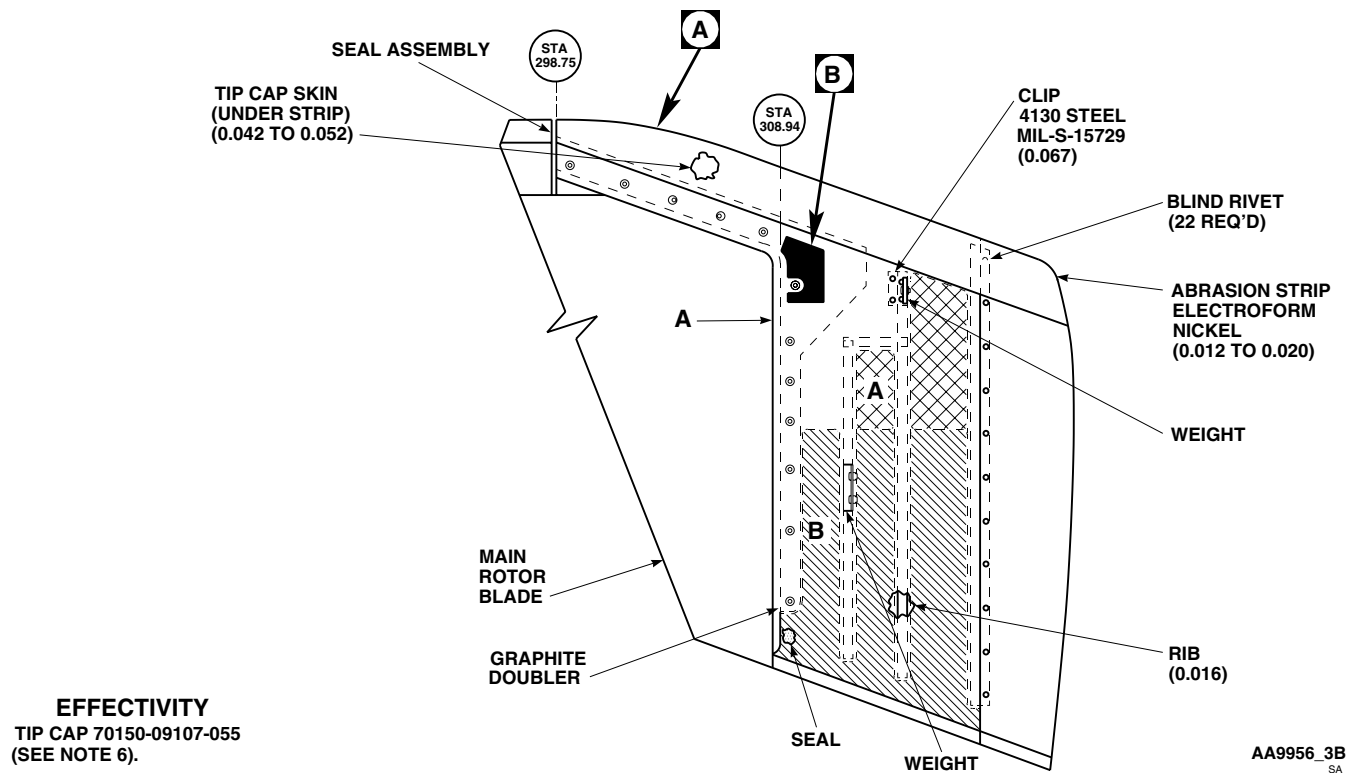


Figure 1. Main Rotor Blade Inspection Areas. (Sheet 3 of 5)

MAIN ROTOR BLADE COIN TAPPING INSPECTION METHOD

NOTE

- To verify suspected damage, use coin-tapping method to determine size and shape of disbonds/delaminations between skin and spar, skin and honeycomb core, trim tab laminations, and tip cap and abrasion strip bond areas.
 - Coin is smooth disc of soft copper, or equivalent metal, 1 1/2 to 2 inches in diameter and 1/8 to 3/16-inch thick. Round and smooth edge of coin to prevent damage to skin.
1. Hold coin loosely between thumb and finger, tap lightly along bond line or areas suspected of having disbonds/delaminations, and listen for variations in tapping sound. A sharp solid sound indicates a good bond and a dull dead sound indicates a bond separation.
 2. Tap across suspected area in horizontal, vertical, and diagonal lines and with grease pencil, mark points on skin where sound varies. Continue tapping and marking until outline of bond separation is completed.

INSPECT MAIN ROTOR BLADE SPAR REGION

1. Visually inspect skin for dents, gouges, scratches, and surface imperfections.
2. Using coin-tapping method, inspect skin over spar for disbonds/delaminations.
3. Check depth of dents and gouges to determine if damage has penetrated to spar.
4. Record location, dimensions and degree of damage, and surface area of disbonds/delaminations.
5. Spar region repaired by adhesive filling or skin patch (Figure 3, Sheet 1) (WP 0576 00).

INSPECT MAIN ROTOR BLADE SPAR REGION - Continued

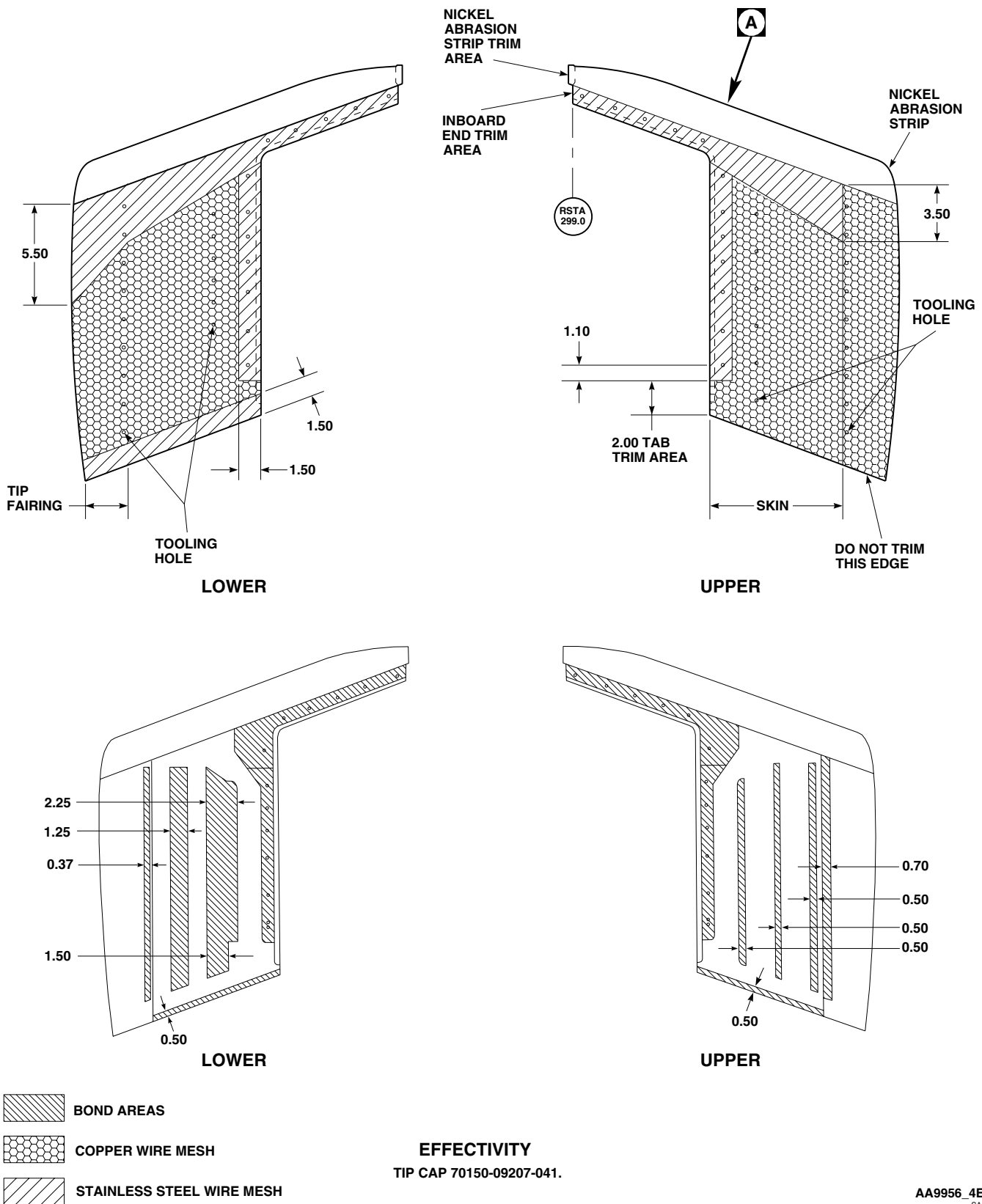


Figure 1. Main Rotor Blade Inspection Areas. (Sheet 4 of 5)

INSPECT MAIN ROTOR BLADE SPAR REGION - Continued

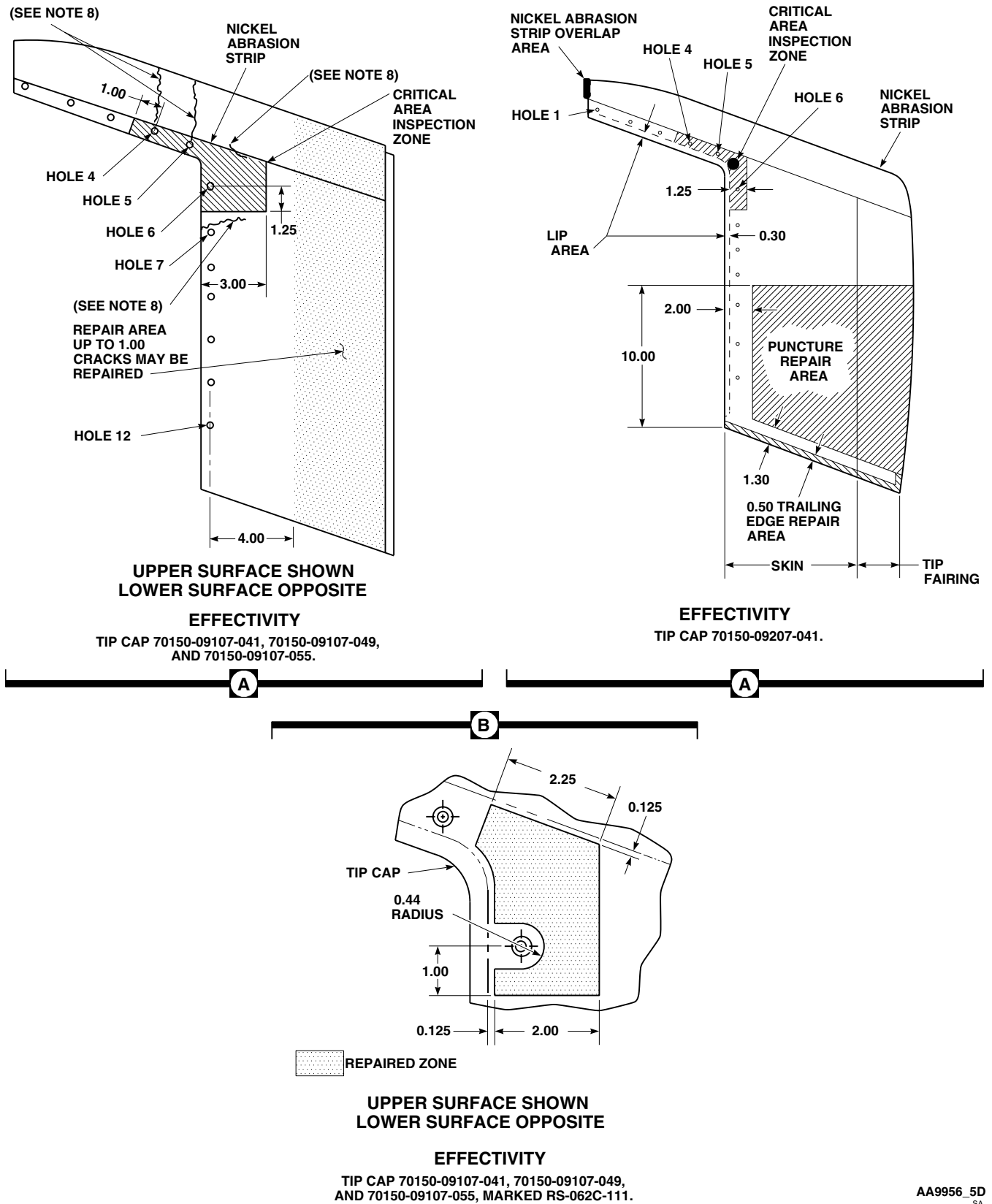


Figure 1. Main Rotor Blade Inspection Areas. (Sheet 5 of 5)

INSPECT MAIN ROTOR BLADE SPAR REGION - Continued

6. Area damage limited to 6 inch diameter (for 8 inch diameter patch).
7. Two or more damaged areas falling within 6 inch diameter will be combined and patched as one repair.
8. New damage occurring within previously repaired damage with total diameter greater than 6 inches, will not be repaired. Replace blade and send damaged blade to depot.

INSPECT MAIN ROTOR BLADE HONEYCOMB CORE REGION

1. Visually inspect reinforced plastic skin/honeycomb core area for dents, crushes, breaks, blisters, gouges and holes.
2. Using coin-tapping method, inspect skin and area around tiedown fitting for disbonds/delaminations.
3. Record location, dimensions and degree of damage, and surface area of disbonds/delaminations.
4. Damaged area not greater than 6 inch diameter repaired by skin or plug patch (Figure 3, Sheet 1) (WP 0576 00).
5. Diameter of routed hole for plug patch must be confined to honeycomb core region. Overlapping skin of plug patch may extend 1 inch beyond boundary of core.
6. Two or more damaged areas falling within 6 inch diameter will be combined and patched as one repair.
7. New damage occurring within previously repaired damage, with total diameter greater than 6 inches, will not be repaired. Replace blade and send damaged blade to depot.

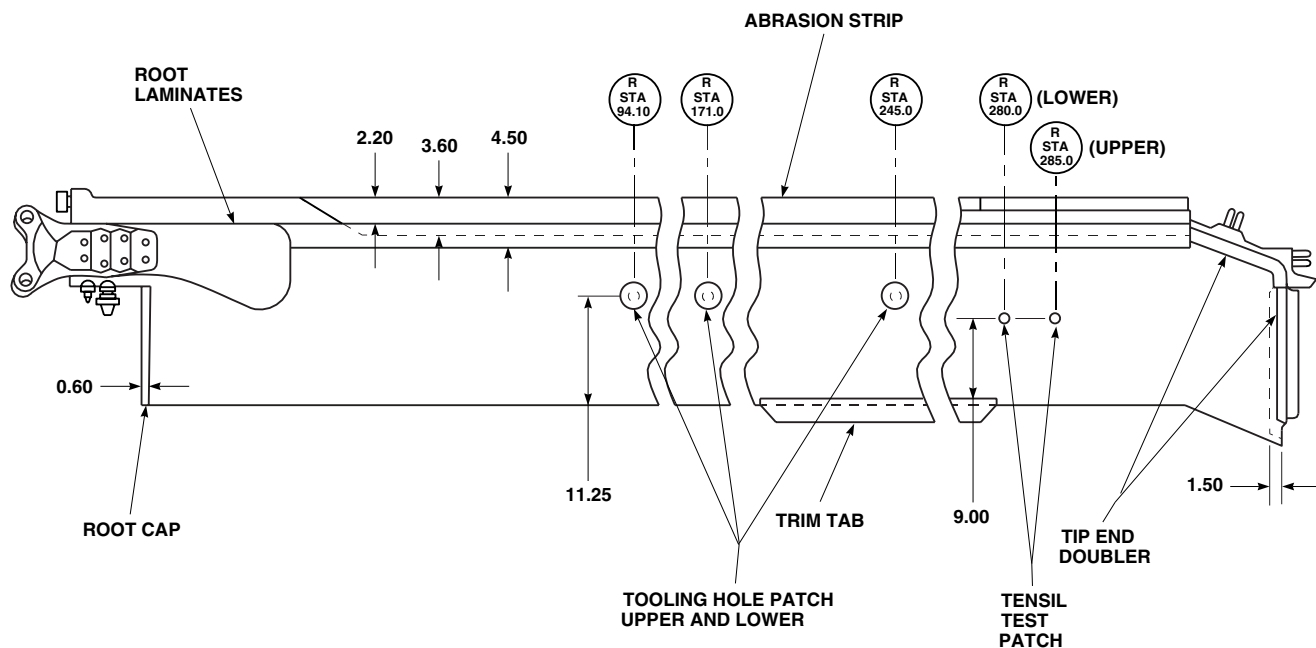
INSPECT MAIN ROTOR BLADE TRAILING EDGE REGION

1. Visually inspect for dents, gouges, and scratches, at top, bottom, and thickness of trailing edge.
2. Using coin-tapping method, inspect for disbonds/delaminations at trailing edge thickness. Check condition of bond between trailing edge skin and trim tab.
3. Record location, dimensions and degree of damage, and surface area of disbonds/delaminations.
4. Visually inspect edges of trim tab for cracks in paint or adhesive running spanwise at front edge or chordwise at the inboard and outboard ends of tab (Figure 2). If cracks are noted, check for bonding voids using coin-tapping method of inspection.
5. Trailing edge region repaired by trailing edge patch, adhesive filling or injection (Figure 3, Sheet 1) (WP 0576 00).
6. Skin patches from honeycomb core region may extend to trailing edge.
7. Two trailing edge patches must be no closer than 5 inches along trailing edge.

INSPECT MAIN ROTOR BLADE ABRASIVE STRIP

1. Visually inspect for erosion, dents, holes, cracks, and scratches.
2. Using coin-tapping method, inspect for bonding voids in the leading and trailing edges, top and bottom.
3. Visually inspect for spanwise cracks in paint 2.2-inches from leading edge of abrasion strip on upper surface and 3.6-inches from leading edge of abrasion strip on lower surface (Figure 2). If cracks are noted, check for bonding voids using coin-tapping method of inspection.
4. Visually inspect for spanwise cracks, small pits or voids in paint 4.5-inches from leading edge on upper and lower surface (Figure 2). If cracks are noted, check for bonding voids using coin-tapping method of inspection.

INSPECT MAIN ROTOR BLADE ABRASION STRIP - Continued

**NOTE**

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

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Figure 2. Main Rotor Blade Damage Inspection.

CAUTION

Damage to blade spar will result if damage to titanium abrasion strip affects deice heater mat wires under titanium abrasion strip. Damaged deice heater mat wires may cause an arc burn to spar, resulting in spar crack. Make sure to check blade deice system for proper operation whenever damage to titanium abrasion strip is found.

5. If installed, check blade deice system for proper operation (WP 0145 00).
6. Abrasion strip repaired by adhesive filling or injection (Figure 3, Sheet 1) (WP 0576 00).

CAUTION

Damage to blade spar will result if damage to titanium abrasion strip affects deice heater mat wires under titanium abrasion strip. Damaged deice heater mat wires may cause an arc burn to spar, resulting in spar crack. Make sure to check blade deice system for proper operation whenever damage to titanium abrasion strip is found.

7. Erosion through abrasion strip will not be repaired. Replace blade and send damaged blade to depot.

INSPECT MAIN ROTOR BLADE TIP CAP AND TIP CAP FAIRING**General****WARNING**

Injury to personnel and damage to equipment will result if the following inspection is not carefully performed. Tip cap could separate from the blade. If erosion protection coating or boots are applied, make sure they are completely removed from inspection Zone to inspect for cracks. Do not reapply erosion protection coating or boots to the inspection Zone.

NOTE

This inspection applies to tip caps, 70150-09107-041, 70150-09107-049, and 70150-09107-055 only.

1. If installed, remove polyurethane coating or boot in inspection Zone of tip cap (Figure 1, Sheet 5, Detail A) (WP 1486 00).
2. If installed, inspect/repair polyurethane coating or boot (WP 1378 00).
3. Visually inspect for secure attachment.

WARNING**CSI**

The tip cap assembly has critical surfaces which must not be damaged. Refer to Figure 1, Sheet 5, Detail A, for areas with critical characteristics.

4. Inspect skin for nicks, scratches, dents, scuffs, and edge damage in inspection Zone (Figure 1, Sheet 5, Detail A). Nicks, scratches, dents, scuffs, and edge damage may be repaired provided that damage does not exceed limits specified in Table 1.
5. Inspect skin for cracks using at least magnifying glass in inspection Zone (Figure 1, Sheet 5, Detail A). **NO CRACKS ALLOWED.** Replace cracked tip cap (WP 0576 00).
6. Inspect skin for cracks in area a minimum of 4 inches outboard from center on attachment holes No. 6 through No. 12 (Figure 1, Sheet 5, Detail A). Cracks up to 1 inch long may be repaired provided that damage does not exceed limits specified in Table 1.
7. Inspect Kevlar surface for wear and fraying. Repair surface using adhesive filler repair (WP 0576 00).
8. Inspect drain holes for elongation and wear. **OUTBOARD DRAIN HOLE ELONGATION SHALL NOT EXCEED 0.25-INCH VERTICALLY AND 0.50-INCH CHORDWISE. LOWER SURFACE DRAIN HOLE DIAMETER SHALL NOT EXCEED 0.25-INCH.**
9. Visually inspect tip cap attachment screws for corrosion. **NONE ALLOWED.** Replace corroded screws (WP 0576 00).

NOTE

Tip caps marked RS-062C-111, have been previously repaired at depot. See Figure 1, Sheet 5, Detail B, for repaired Zone.

10. On tip caps marked RS-062C-111, inspect for cracks progressing from previously stop drilled hole (Figure 1, Sheet 5, Detail B). **NONE ALLOWED.** Replace cracked tip cap (WP 0576 00).

INSPECT MAIN ROTOR BLADE TIP CAP AND TIP CAP FAIRING - Continued

11. Visually inspect for paint and adhesive cracks around tip doubler 1.5-inches inboard from joining edge of tip cap and blade and running parallel to edge of tip cap on upper and lower surface of blade. If cracks are noted, check for bonding voids using coin-tapping method of inspection (Figure 2).

Tip Cap Assembly, 70150-09107-041, 70150-09107-049, and 70150-09107-055

1. Erosion through tip cap abrasion strip requires tip cap replacement (WP 0576 00).
2. Holes in skin patched with reinforced plastic patches.
3. Fraying Kevlar repaired by adhesive filling (WP 0576 00).
4. Nicks, scratches, dents, and scuffs in inspection Zone (Figure 1, Sheet 5, Detail A). Nicks, scratches, dents, and scuffs may be repaired provided that damage does not exceed limits specified in Table 1.
5. Cracks in inspection Zone (Figure 1, Sheet 5, Detail A). **NO CRACKS ALLOWED.** Replace cracked tip cap (WP 0576 00).
6. Cracks in area a minimum of 4 inches outboard from center on attachment holes No. 6 through No. 12 (Figure 1, Sheet 5, Detail A). Cracks up to 1 inch long may be repaired provided that damage does not exceed limits specified in Table 1.
7. On tip cap, 70150-09107-055, scratches and gouges of any length in graphite doubler not exceeding 0.010-inch deep or penetrating into graphite core can be repaired by adhesive filling (WP 0576 00). Damage beyond these limits requires tip cap replacement (WP 0576 00).
8. Elongation of outboard drain hole and erosion of surface surrounding outboard drain hole patched with fiberglass patches using tip cap holes repair (WP 0576 00).
9. Elongation of lower surface drain hole requires tip cap replacement (WP 0576 00).

NOTE

Tip caps marked RS-062C-111, have been previously repaired at depot. See Figure 1, Sheet 5, Detail B, for repaired Zone.

10. On tip caps marked RS-062C-111, cracks progressing from previously stop drilled hole requires tip cap replacement (WP 0576 00).
11. On tip caps marked RS-062C-111, breaks or disbonding of internal graphite doubler requires tip cap replacement (WP 0576 00).

Tip Cap Assembly, 70150-09207-041

1. Erosion through tip cap abrasion strip requires tip cap replacement (Figure 3, Sheet 1) (WP 0576 00).
2. Holes in skin patched with reinforced plastic patches.
3. Fraying graphite repaired by adhesive filling (WP 0576 00).

WARNING**CSI**

The tip cap assembly has critical surfaces which must not be damaged. Refer to Figure 1, Sheet 5, Detail A for areas with critical characteristics.

4. Nicks, scratches, dents, and scuffs in inspection Zone (Figure 1, Sheet 5, Detail A). Nicks, scratches, dents, and scuffs may be repaired provided that damage does not exceed limits specified in Table 1.
5. Cracks in inspection Zone (Figure 1, Sheet 5, Detail A) . **NO CRACKS ALLOWED.** Replace cracked tip cap (WP 0576 00).

INSPECT MAIN ROTOR BLADE CUFF ASSEMBLY

1. Visually inspect for security of attaching hardware.

WARNING**CSI**

Nicks, dents, scratches, and or scuff marks within the critical characteristic area shall not exceed acceptance limits.

2. Visually inspect for nicks, scratches, and corrosion.
3. Visually inspect for cracks in blade attachment lugs.
4. Visually inspect attachment bolts and nuts for corrosion.
5. No cracks allowed on cuff assembly (Figure 3, Sheet 1). Replace blade and send damaged blade to depot.
6. Liner disbond is repaired at depot (spindle to cuff mating washers).

INSPECT MAIN ROTOR BLADE ROOT LAMINATES, ROOT CAP, AND ROOT FAIRING

1. Visually inspect edge of root laminates for cracks in paint. If cracks are noted inspect for disbonding/delamination using coin-tapping method of inspection (Figure 2).
2. Visually inspect edge of root cap for cracks in paint. If cracks are noted inspect for bonding voids using coin-tapping method of inspection (Figure 2).
3. No damage allowed to deicing system (Figure 3, Sheet 1). Replace blade and return damaged blade to depot.

NOTE

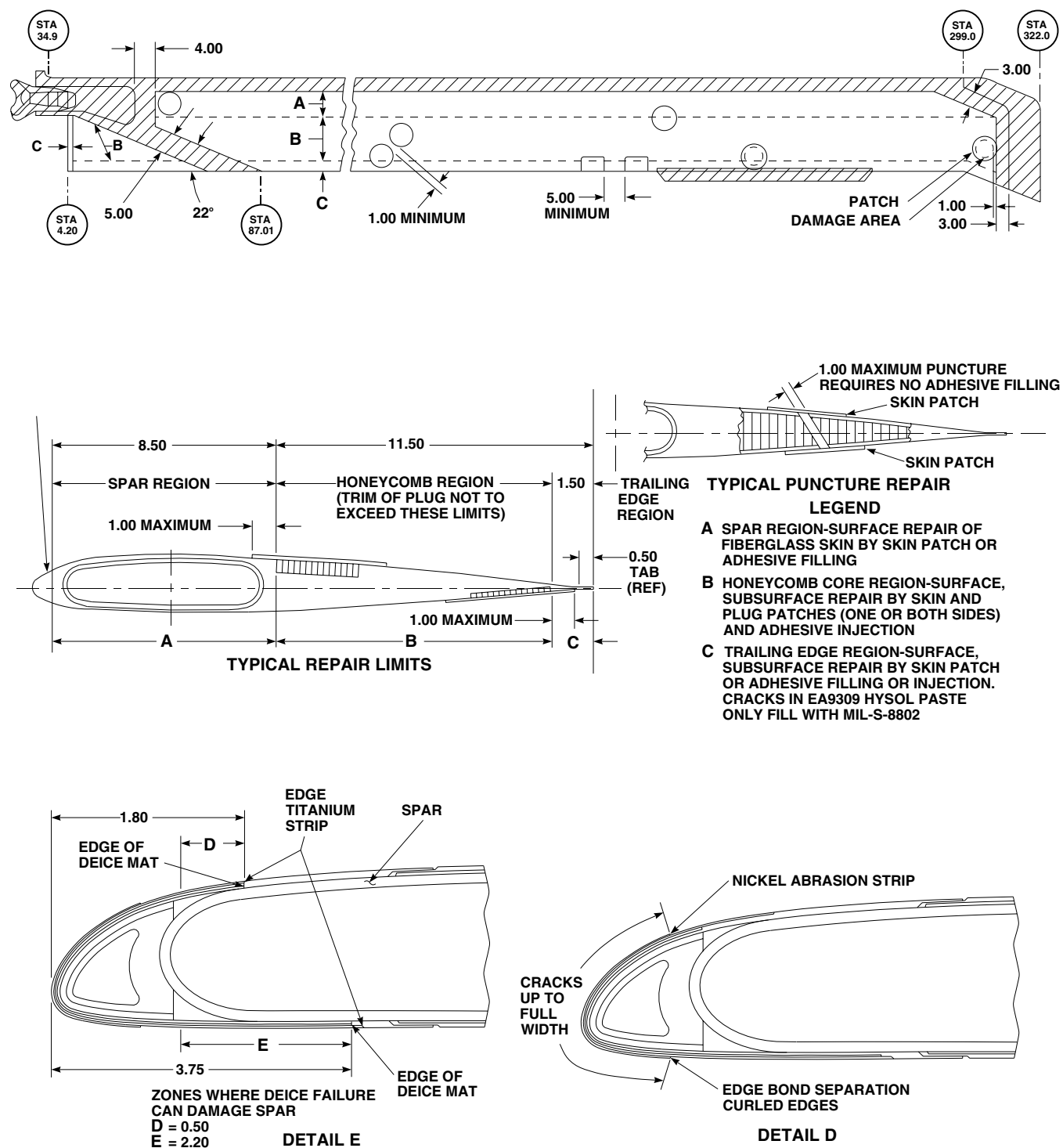
Sealing compound must be applied within first 50 flight-hours.

4. Seal cracks through skin and Hysol filler in shaded area (Figure 3, Sheet 2) with a layer of sealing compound, Item 257, WP 1803 00 using adhesive filler repair (WP 0576 00).
5. Cracks in paint around edge of root laminates are allowed provided that no disbonding/delamination is noted after coin-tapping inspection. Cracks in paint are to be carefully blended out and touched up using procedures in WP 0576 00. No cracks allowed in root laminates if coin-tapping reveals disbonding/delamination (Figure 3, Sheet 2). Replace blade and send damaged blade to depot.
6. Repair cracks in root end cap using procedures in WP 0576 00.

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES

Visually inspect tooling and tensile test patches for cracks in paint and slight contour changes. If cracks or contour changes are noted check for disbonding/delamination using coin-tapping method of inspection (Figure 2).

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

**NOTE**

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

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Figure 3. Main Rotor Blade Repair Limits. (Sheet 1 of 2)

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

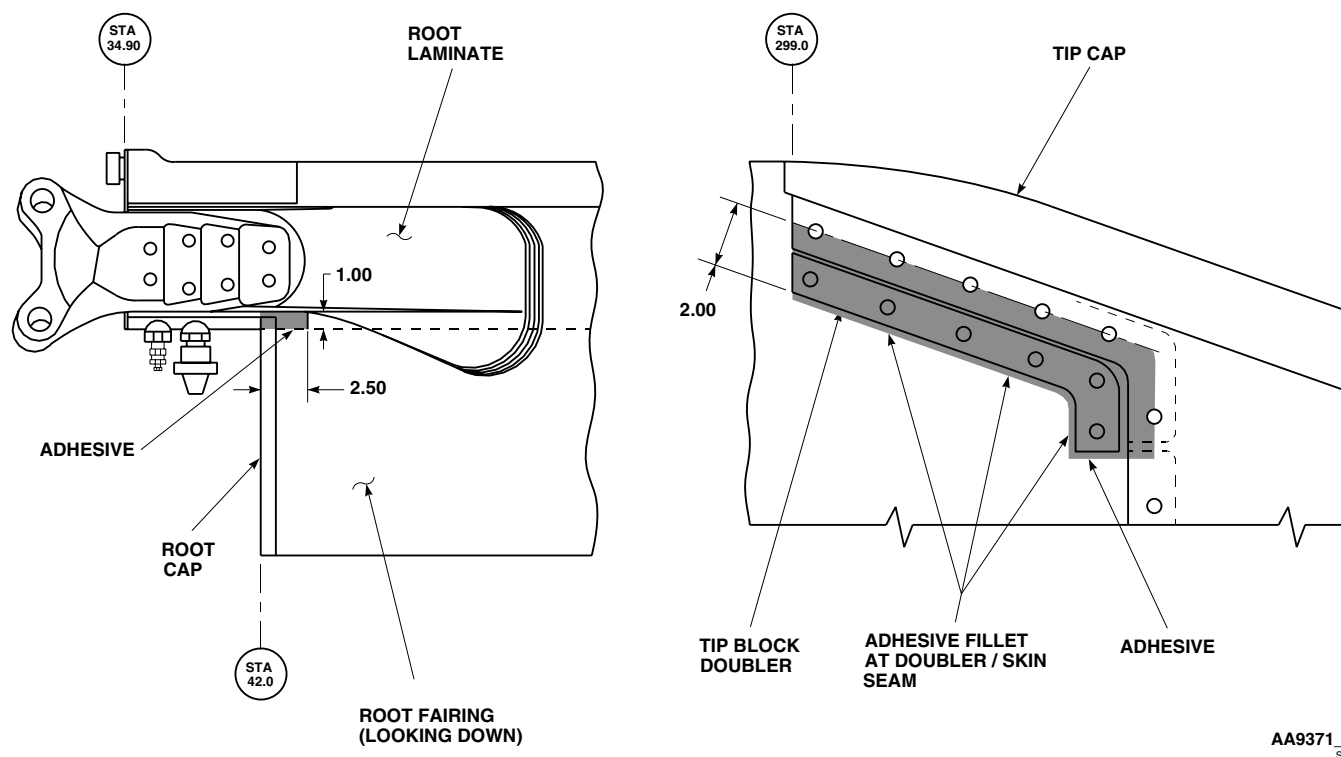


Figure 3. Main Rotor Blade Repair Limits. (Sheet 2 of 2)

Table 1. Main Rotor Blade Inspection/Repair Limits.

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Pocket Skin Over Spar Region	Scratches.			
	- - Less than 0.015-inch deep and any length.	No repair required. Touch up paint, as needed (WP 0576 00).	x	
	- - 0.015 to 0.030-inch deep and any length span-wise.	Fill scratch with adhesive (WP 0576 00).	x	
	- - Span wise only 0.015 to 0.050-inch deep (not through to spar) and not over 6 inches long.	Repair using skin patch repair (WP 0576 00).		x
	- - Deeper than 0.050-inch, or more than 6 inches long.	Not acceptable. Replace blade, send damaged blade to depot.		x
	Gouges and digs.			

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - 0.015 to 0.030-inch deep and 1 to 2 inch diameter.	Fill gouge with adhesive (WP 0576 00).		x
	- - 0.030 to 0.050-inch deep (not through to spar) and not over 6 inch long.	Repair using skin patch repair (WP 0576 00).		x
	- - Deeper than 0.050-inch or more than 6 inch diameter.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Tear/Rip (Skin torn and separated).			
	- - Not deeper than 0.040-inch and up to 2 inch diameter.	Repair using skin patch repair (WP 0576 00).		x
	More than 0.040-inch deep or longer than 2 inch diameter.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Bond Separations/ Delaminations.			
	- - Less than 2 inch diameter.	No repair required. Touch up paint (WP 0576 00). Daily inspection of area required.	x	
	- - More than 2 inch diameter, but not more than 6 inch diameter.	Repair using skin patch repair (WP 0576 00).		x
	- - More than 6 inch diameter.	Not acceptable. Replace blade. Send damaged blade to depot.		x
Receiver Assembly	Loose.	Repair per WP 0578 00.		x
Receiver Block	Cracked or Disbonded.	Not acceptable. Replace blade. Send damaged blade to depot.		x
BIM Indicator	Indication of spar pressure loss or any type of damage to indicator.	Refer to WP 0584 00.		

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Pocket Skin/ Honeycomb Core Region	Dents, Smooth Unbroken.			
	- - No deeper than 0.040-inch or larger than 3 inch diameter.	No repair required. Touch up paint, as needed (WP 0576 00).	x	
	- - No deeper than 0.040-inch and 3 to 6 inch diameter.	Trim out damage and patch using skin patch repair (WP 0576 00).		x
	- - Deeper than 0.040-inch up to 6 inch diameter.	Repair using plug patch repair (WP 0576 00).		x
	- - Deeper than 0.040-inch and greater than 6 inch diameter.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Gouges/Holes.			
	- - Up to 6 inch diameter using damage to core.	Repair using plug patch repair (WP 0576 00).		x
	- - Damage through blade but core damage less than 1 inch diameter.	Repair using skin patch, both sides (WP 0576 00).		x
	- - Damage through blade and core damage less than 6 inch diameter.	Repair using plug patch repair, both sides (WP 0576 00).		x
	- - Damage through blade and core damage greater than 6 inch diameter.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Tear/Rip (Skin damage only).			
	- - Up to 6 inch diameter.	Repair using skin patch repair (WP 0576 00).		x
	- - Greater than 6 inch diameter.	Not acceptable. Replace blade. Send damaged blade to depot.		x
Pocket Skin/ Honeycomb Core Region	Blisters/Bond Separations.			

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Pocket Skin Trailing Edge	- - Less than 2 inches diameter and maximum blistering 0.020-inch high.	No repair required. Touch up paint as needed (WP 0576 00).	x	
	- - More than 2 inches diameter but not more than 6 inch diameter.	Trim out damage and patch using skin patch repair (WP 0576 00).		x
	- - More than 6 inch diameter.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Dents, Smooth Unbroken.			
	- - No deeper than 0.040-inch	No repair required. Touch up paint as needed (WP 0576 00).	x	
	- - More than 0.040-inch deep and up to 1 inch diameter.	Fill damage using adhesive (WP 0576 00).		x
	- - Deeper than 0.040-inch and up to 3 inches spanwise and 2 inches chordwise.	Repair using trailing edge patch repair (WP 0576 00).		x
	- - Damage deeper than 0.040-inch and more than 3 inches spanwise and 2 inches chordwise.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Gouges/Scratches.			
	- - Up to 0.030-inch deep and 1 inch spanwise.	No repair required. Touch up paint as required (WP 0576 00).	x	
	- - Deeper than 0.030-inch and up to 3 inch spanwise.	Repair using trailing edge patch repair (WP 0576 00).		x
	End Gashes/Slits.			
	- - No more than 1/2-inch chordwise and 1 1/2-inch spanwise in trailing edge.	Repair using trailing edge patch repair (WP 0576 00).		x

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Pocket Skin Trail- ing Edge (Cont)	- - Through damage more than 1/2-inch chordwise and 1 1/2-inch spanwise.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Tear/Rip.			
	- - Up to 3 inches spanwise and 2 inches chordwise.	Repair using trailing edge patch repair (WP 0576 00).		x
	Bond Separations/Delaminations (in tab edge).			
Trim Tab	- - Area up to 1/4-inch x 3 inch spanwise.	No repair required. Touch up paint, as needed (WP 0576 00).	x	
	- - Area up to 1/2-inch x 6 inch spanwise.	Repair using adhesive injection repair (WP 0576 00).	x	
	Scratch/Gouge.			
	- - Up to full thickness of tab, full chordwise width of tab, and less than 3 inch spanwise.	Blend out using abrasive cloth, Item 83, WP 1803 00. Touch up Alodine and touch up paint area as needed (WP 0576 00).	x	
	End Gashes.			
	- - 1/4-inch or less through end of tab.	No repair required. Touch up paint as needed (WP 0576 00).	x	
	Bond Separations/Delaminations.			
	- - Up to 1/4-inch deep chordwise and less than 3 inch spanwise (in trailing edge).	No repair required. Monitor disbond.	x	
	- - Up to 1/2-inch deep chordwise, and less than 3 inch spanwise between trim tab and blade skins.	Repair using adhesive injection repair (WP 0576 00).		x

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - Between 1/4-inch and 1 inch deep chordwise and less than 4 inch spanwise (in trailing edge).	Repair using adhesive injection repair (WP 0576 00).		x
	NOTE Not all blades have slots in trim tab.			
	Cracks starting at front end of slot, running chordwise toward leading edge of trim tab.			
	- - Up to full width of tab.	No repair required.	x	
	Cracks not at slot, up to full chordwise width.	No repair required.		
	- - Cracks in spanwise direction up to 2 inches	Stop drill if rear of skin. No repair required if over skin.	x	
	Bends.			
	- - Not greater than 6 inch spanwise.	Straighten, using straight edge for alignment with adjacent areas of trim tab.	x	
Abrasion Strips-Nickel and Titanium	Cracks.			
	- - In nickel strips (outboard).	No repair required if the bond separation is within limits (See Bond Separation). Seal crack using abrasion strip repair (WP 0576 00).	x	
	- - In titanium strip (up to full width of strip).	No repair required. If installed, check blade deice system for proper operation (WP 0145 00).	x	
	Dents.			

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - In titanium strip up to 1/8-inch and not over spar.	No repair required. If installed, check blade deice system for proper operation (WP 0145 00).	x	
	- - Bond separation up to 1/2-inch beyond dent.	No repair required. Touch up paint as needed (WP 0576 00).	x	
	- - Cracks in dent.	If crack does not extend into reinforced plastic, no repair required. Seal crack using abrasion strip repair (WP 0576 00).	x	
	Bond Separation.			
	- - Edge voids in nickel strips up to 1 inch deep x 2 inches long.	Seals voids using abrasion strip repair (WP 0576 00).	x	
	- - Edge voids in nickel strips up to 1/2-inch deep x 4 inches long.	Seals voids using abrasion strip repair (WP 0576 00).	x	
	- - Internal voids in nickel strips up to 3 inches long or wide.	No repair required.	x	
	- - Multiple voids in nickel strips totaling up to 3 inches long or wide.	No repair required.	x	
	- - Edge or internal voids in nickel strips within 2 inches of inboard or outboard end of strip.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	- - Edge voids in titanium strip up to 1 inch deep x 6 inches long.	Repair using abrasion strip repair (WP 0576 00).	x	
	- - Internal voids in titanium strips up to 3 inches long or wide.	No repair required.	x	
	- - Edge Voids. One void up to 12 inches long spanwise per side or several totaling not more than 20 square inches.	Repair by adhesive injection (WP 0576 00).	x	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Abrasion Strips-Nickel	- - Edge Voids. One void greater than 12 inches long spanwise per side or several totaling more than 20 square inches.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	- - Internal voids up to 20 square inches (not closer than 0.5-inch to an edge).	Acceptable without repair. Total voided area cannot exceed 40 square inches per blade.	x	
	- - Internal voids up to 20 square inches (closer than 0.5-inch to an edge).	Repair by adhesive injection through the edge (WP 0576 00).	x	
	Gouges.			
	- - Up to full depth of nickel strip.	Blend out damage using medium cut file. Fill with adhesive (WP 0576 00).	x	
	- - Up to full thickness of titanium strip (0.020-inch).	No repair required. If installed, check blade deice system for proper operation (WP 0145 00).	x	
	Wear.			
	- - Up to full depth of strip nickel strip.	No repair until worn through to adhesive, then send damaged blade to depot.		x
	- - Up to full depth of titanium strip.	No repair until worn through to adhesive, then send damaged blade to depot.		x
	- - Rear 1/4-inch edge (both upper and lower) (See Note 1).	No repair until worn or corroded through to adhesive; then send damaged blade to depot.		x
Abrasion Strips-Nickel	Loose or curled edges.			
	- - Rear 1/4-inch edge (both upper and lower) (See Note 1).	Remove using nonmetallic scraper.	x	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Deice Fairing	Holes and Cracks.	If installed, check blade deice system for proper operation (WP 0145 00). Repair damage using corrosion removal repair (WP 0576 00).	x	
Root Laminates	Cracks.	Not acceptable. Send damaged blade to depot.		x
	Dents/Gouges.			
	- - No deeper than 0.050-inch x 4 inches long with no radiating cracks or bond separations.	Fill with adhesive (WP 0576 00).		x
	- - More than 0.050-inch deep or more than 4 inches long. Also if dent includes cracks or bond separation.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Bond Separation/Delaminations.			
	- - Up to 3 inches long and 1 inch wide.	Repair using tip cap-to-blade sealing repair (WP 0576 00).	x	
Root End Fairing	- - More than 3 inches long or wider than 1 inch	Not acceptable. Replace blade. Send damaged blade to depot.		x
	Cracks (If deicing system not affected).	Patch using bond repair (WP 0576 00).	x	
	- - Dents/Gouges (If deicing system not affected).	Fill with adhesive (WP 0576 00).	x	
	- - Damage affecting deicing system.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	- - Bond separations up to 6 inches long.	Repair using bond repair (WP 0576 00).	x	
	- - Loose or missing sealant at root fairing-to-spar joint.	Reseal joint using sealing compound, Item 257, WP 1803 00.	x	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Cuff Assembly	Cracks.	Not acceptable. Replace blade. Send damaged blade to depot.		x
	WARNING CSI			
	Nicks, dents, scratches, and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.			
	Nicks, Scratches, Corrosions..			
	- - No deeper than 0.005-inch and covering more than 10% of surface area.	Blend out damage to 1/2-inch radius using abrasive cloth, Item 77, WP 1803 00, or equivalent. Fluorescent penetrant inspect blended area of cuff assembly (TM 1-1520-265-23). Touch up paint (WP 0576 00).	x	
Blade Cuff Attach Bolts, Nuts	Surface corrosion and pits.	Remove all sealing compound from area around corrosion.	x	
	- No greater than 0.005-inch in base metal.	Remove corrosion by blending using abrasive cloth, Item 77, WP 1803 00.	x	
	Bushing movement.	Return to correct location by driving into position using soft metal punch.	x	
Tip Cap, 70150-09107-041, 70150-09107-049, and 70150-09107-055	Dents, smooth unbroken.			
	- - Less than 1/8-inch deep.	No repair required. Touch up paint as needed (WP 0576 00).	x	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - More than 1/8-inch deep.	Remove tip cap and inspect for internal damage. If no damage is found, and dent can be straightened, reuse tip cap.	x	
WARNING				
CSI				
Nicks, dents, scratches, and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.				
	Nicks, Scratches, Dents, Scuffs.			
	- - Less than 0.020-inch deep and 1 inch long in inspection Zone (Figure 1, Sheet 5, Detail A).	Repair using adhesive filler repair (WP 0576 00).	x	
	More than 0.020-inch deep and 1 inch long in inspection Zone (Figure 1, Sheet 5, Detail A).	Not acceptable. Replace tip cap (WP 0576 00).		x
	Edge Damage.			
	Less than 0.040-inch deep in inspection Zone (Figure 1, Sheet 5, Detail A).	Repair using adhesive filler repair (WP 0576 00).	x	
	More than 0.040-inch deep in inspection Zone (Figure 1, Sheet 5, Detail A).	Not acceptable. Replace tip cap (WP 0576 00).	x	
	Holes.			
	- - Less than 1/2-inch diameter with no internal damage.	Patch using tip cap hole repair (WP 0576 00).		x
	- - More than 1 1/2-inch diameter or with internal damage.	Not acceptable. Replace tip cap (WP 0576 00).		x

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	Cracks.			
	- - Skin crack less than 1 inch long in areas other than attachment holes, 4 inches outboard from center of attachment holes No. 6 through No. 12.	Stop-drill using No. 40 (0.098-inch diameter) drill and patch using tip cap skin and fairing crack repair (WP 0576 00).		X
	- - Skin crack more than 1 inch long in areas other than attachment holes.	Not acceptable. Replace tip cap (WP 0576 00). Visually inspect all tip cap attachment areas of blade assembly for cracking or other damage.		X
	- - Skin crack progressing from previously stop drilled hole.	Not acceptable. Replace tip cap (WP 0576 00).		X
	- - Attachment hole crack.	Not acceptable. Replace tip cap (WP 0576 00). If tip cap is replaced due to cracks, visually inspect all tip cap attachment areas of blade assembly for cracking, or other damage.	X	
	- - Tip rib crack up to 1 inch long.	Stop-drill using No. 40 drill.	X	
	- - Tip rib crack greater than 1 inch long.	Repair using tip cap rib cracks repair (WP 0576 00).	X	
	- - Abrasive strip cracks.	No repair required if disbond area is less than 1 square inch. Seal crack, as needed using abrasion strip repair (WP 0576 00). Repair disbond area 1 square inch to 3 square inch bond repair (WP 0576 00).	X	
	- - Balance weight attachment blocks and hardware cracks.	Not acceptable. Replace tip cap (WP 0576 00).		X

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - Edge cracks in balance shim weights.	No repair required if crack less than 1/8-inch long. If longer cracks are found, replace tip cap (WP 0576 00).		x
	Corrosion.			
	- - Pitting in thick portion of skin no more than 0.020-inch deep.	Remove corrosion using corrosion removal repair (WP 0576 00). Touch up paint (WP 0576 00).	x	
	- - Pitting in thin portion of skin (Figure 2) no more than 0.010-inch deep.	Remove corrosion using corrosion removal repair (WP 0576 00). Touch up paint (WP 0576 00).	x	
	- - Pitting deeper than 0.020-inch in thick portion of skin, or 0.010-inch in thin portion of skin.	Replace tip cap (WP 0576 00).		x
	- - Localized erosion/corrosion adjacent to No. 4 and No. 5 screws on lower side up to 0.030-inch deep.	Remove corrosion using corrosion removal repair (WP 0576 00). Touch up paint (WP 0576 00). Apply sealing compound using tip cap-to-blade sealing repair (WP 0576 00).	x	
	- - Rib corrosion no deeper than 0.005-inch.	Remove corrosion using corrosion removal repair (WP 0576 00). Touch up paint (WP 0576 00).	x	
	- - Corrosion of screws, studs, and mounting hardware.	Replace corroded hardware (WP 0576 00).	x	
	- - Corrosion of aluminum tip weight blocks no deeper than 0.005-inch.	Remove corrosion using corrosion removal repair (WP 0576 00). Touch up paint (WP 0576 00).	x	
	Bond Separations.			
	- - Trailing edge separation up to 1/4 x 1 inch.	No repair required. Touch up paint (WP 0576 00).	x	
	Bond Separation.			

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - Trailing edge separation more than 1/4 x 1 inch and up to 4 inches long.	Repair using bond repair (WP 0576 00).	x	
	- - Rib separation less than 2 inches long.	No repair required.	x	
	- - Rib separation more than 2 inches long.	Repair using bond repair (WP 0576 00).		x
	Erosion.			
	- - Through nickel-abrasion strip.	Replace tip cap (WP 0576 00).		x
	- - Local erosion (small isolated areas).	No repair required if erosion is less than 1/3 material thickness. Touch up paint (WP 0576 00).	x	
	Loose or missing sealing compound at tip cap-to-blade joint.	Replace sealing compound using tip cap-to-blade sealing repair (WP 0576 00).	x	
	Separations in a 2 inches wide area behind centerline of first six tip cap screws (Figure 1, Sheet 2).			
	- - Separations less than 1/16-inch wide.	No repair required.	x	
	- - Separations parallel to tip cap/blade caulking line.	No repair required.	x	
	- - Separations not past tip cap leading edge.	No repair required.	x	
	- - Separations beyond limits listed above.	Not acceptable. Replace blade.		x
	Elongated or worn lower surface drain hole.			
	- - Lower surface drain hole worn more than 0.25-inch in diameter.	Replace tip cap (WP 0576 00).	x	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Tip Cap Fairing (Kevlar Reinforced Plastic)	Graphite doubler.			
	- - Scratches and gouges less than 0.010-inch deep of any length not penetrating into graphite core.	Repair using adhesive filler repair (WP 0576 00).		x
	- - Scratches and gouges deeper than 0.010-inch deep or penetrating into graphite core.	Replace tip cap (WP 0576 00).		x
	- - Broken or disbonded.	Replace tip cap (WP 0576 00).		x
	- - Delaminated or disbonded greater than 1/2-inch long.	Not acceptable. Replace tip cap (WP 0576 00).		x
	- - Missing pieces less than 1/4-square inch.	Repair tip cap using adhesive filler repair (WP 0576 00).		x
	- - Missing pieces greater than 1/4-square inch.	Not acceptable. Replace tip cap (WP 0576 00).		x
	Cracks.			
	- - Cracks less than 2 inches long.	Repair using tip cap skin and fairing crack repair (WP 0576 00).		x
	- - Crack in attachment holes.	Replace tip cap (WP 0576 00).		x
	- - Bond Separation at edges or trailing edge.	Repair using adhesive injection repair (WP 0576 00).		x
	Dents/Gouges.			
	- - Up to 0.020-inch deep.	Fill with adhesive (WP 0576 00).		x
	- - Damage preventing secure attachment of tip cap fairing.	Replace tip cap (WP 0576 00).		x

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	Elongated worn or outboard drain hole.			
	- - Outboard drain hole worn more than 0.25-inch vertically and 0.25-inch chordwise.	Apply fiberglass patch up to 2 inches in diameter and 1 to 3 plies thick using skin patch repair (WP 0576 00). After patch is cured reopen drain hole using 0.190 drill.		x
	Worn or frayed Kevlar surface.	Lightly sand frayed Kevlar. Fill with adhesive (WP 0576 00).	x	
	Missing or broken screws.			
	- - No missing or broken screws in positions 4, 5, and 6, both top and bottom locations (Figure 1, Sheet 5, Detail A).	Replace screws (WP 0576 00).	x	
	- - No more than one screw total in positions 1, 2, and 3, either top or bottom locations (one screw total).	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (WP 0576 00).	x	
	- - No more than one screw per set in positions 7 thru 12, both top and bottom locations (two screws total).	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (WP 0576 00).	x	
	- - No more than two screws total per tip cap.	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (WP 0576 00).	x	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Tip Cap, 70150-09207-041	Cracks.			
	- - Within inspection Zone (Figure 1, Sheet 5, Detail A).	Not acceptable. Replace tip cap (WP 0576 00). If tip cap is replaced due to cracks, inspect tip cap attachment area of blade for cracks and damage.		X
	- - Within lip area (Figure 1, Sheet 5, Detail A).	No limit. Inject with adhesive (WP 0576 00) or trim away loose material.	X	
	- - All other areas.	No through cracks allowed. Replace tip cap with through cracks (WP 0576 00).		X
	Delaminations or dis-bonds.			
	- - Within inspection Zone (Figure 1, Sheet 5, Detail A).	Not acceptable. Replace tip cap (WP 0576 00). If tip cap is replaced due to delaminations or dis-bonds, inspect tip cap attachment area of blade for cracks and damage.		X
	- - Within lip area.	No limit. Inject with adhesive (WP 0576 00) or trim away loose material.	X	
	- - Internal rib to skin up to 1 inch long.	Inject with adhesive (WP 0576 00).		X
	- - Trailing edge up to 1 inch long.	Inject with adhesive (WP 0576 00).	X	
	- - Trailing edge greater than 1 inch long up to 4 inches long.	Repair using 4 ply wet lay up skin patch repair (WP 0576 00).		X
	- - Trailing edge greater than 4 inches long.	Not acceptable. Replace tip cap (WP 0576 00).		X
	- - All other areas up to 1/8-inch deep.	No repair required.	X	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - All other areas greater than 1/8-inch deep.	Inject with adhesive (WP 0576 00).		x
	Surface damage, raised or frayed fibers, tears, or surface delaminations.			
	- - Up to 0.015-inch deep.	Repair using adhesive filler repair (WP 0576 00).	x	
	- - Greater than 0.015-inch deep up to 0.030-inch deep.	Repair using 2 ply wet lay up skin patch repair (WP 0576 00).		x
	- - Greater than 0.030-inch deep.	Not acceptable. Replace tip cap (WP 0576 00).		x
WARNING				
CSI				
Nicks, dents, scratches, and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.				
	Scratches or abrasions, no limit on length or number.			
	- - Up to 0.005-inch deep.	No repair required.	x	
	- - Greater than 0.005-inch deep up to 0.010-inch deep.	Touch up paint (WP 0576 00).	x	
	- - Greater than 0.010-inch deep, up to 0.015-inch deep.	Repair using adhesive filler repair (WP 0576 00).	x	
Tip Cap, 70150-09207-041	Erosion. (See Figure 1 for definition of stainless steel and copper wire mesh areas.)			
	- - Erosion which exposes stainless steel or copper wire mesh.	Touch up paint (WP 0576 00).	x	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - Erosion into stainless steel wire mesh, but not completely through.	Repair using adhesive filler repair (WP 0576 00). Touch up paint (WP 0576 00).	x	
	- - Erosion completely through stainless steel wire mesh in local areas, including area adjacent to attachment holes (less than 0.25-square inch each).	Remove any loose or raised mesh by lightly hand sanding with abrasive cloth, Item 75, WP 1803 00.	x	
CAUTION				
Sand only to remove loose or raised mesh, do not remove excessive material from adjacent areas. Excessive sanding may damage tip cap. Repair all exposed edges using adhesive filler repair (WP 0576 00). Touch up paint (WP 0576 00). Install mission kit, 70073-15000, if flight will be continued in highly erosive environment.				
	- - Erosion completely through stainless steel wire mesh in large areas, (greater than 0.25-square inch each).	Not acceptable. Replace tip cap (WP 0576 00).		x
	- - Erosion completely through copper wire mesh, no limit on areas.	Remove any loose or raised mesh by lightly hand sanding with abrasive cloth, Item 75, WP 1803 00.	x	
CAUTION				
Sand only to remove loose or raised mesh, do not remove excessive material from adjacent areas. Excessive sanding may damage tip cap. Repair all exposed edges using adhesive filler repair (WP 0576 00). Touch up paint (WP 0576 00).				
	- - Erosion into composite material with no cracking, disbonding, or delamination.	Touch up paint (WP 0576 00).	x	
	- - Erosion with associated cracking, disbonding, or delamination.	See specific limits for damage to composite material within this table.	x	
	Cracks, torn areas, and missing pieces. in trailing edge repair area.			

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - Up to 0.50-inch wide and 2 inches long.	Repair using 4 ply wet lay up skin patch repair (WP 0576 00).		X
	- - Greater than 0.50-inch wide and 2 inches long.	Not acceptable. Replace tip cap (WP 0576 00).		X
	Holes.			
	- - Punctures through skin in puncture repair area only up to 1 inch diameter (Figure 1, Sheet 5, Detail A).	Repair using 3 inch diameter patch or 4 ply wet lay up skin patch repair (WP 0576 00).		X
	- - Punctures through skin in puncture repair area only greater than 1 inch diameter (Figure 1, Sheet 5, Detail A).	Not acceptable. Replace tip cap (WP 0576 00).		X
	Attachment hole elongation at holes 4, 5, and 6, upper and lower.			
	- - Up to 0.030-inch.	No repair required.	X	
	- - Up to 0.040-inch.	No repair required provided no more than 3 holes are elongated between 0.030 to 0.040-inch.	X	
	- - Greater than 0.040-inch.	Not acceptable. Replace tip cap (WP 0576 00). If tip cap is replaced due to attachment hole elongation, inspect tip cap attachment area of blade for cracks and damage.		X
	Delaminations, disbonds, and other damage around attachment holes 4, 5, and 6, upper and lower.			
	- - Up to 0.015-inch deep, up to 5/8-inch diameter around hole.	Inject with adhesive (WP 0576 00).	X	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
	- - Greater than 0.015-inch deep, or greater than 5/8-inch diameter around hole.	Not acceptable. Replace tip cap (WP 0576 00). If tip cap is replaced due to attachment hole delaminations, disbonds, and other damage, inspect tip cap attachment area of blade for cracks and damage.		x
	Attachment hole elongation at attachment holes 1 through 3 and 7 through 12, upper and lower.			
	- - Up to 0.040-inch.	No repair required.	x	
	- - Greater than 0.040-inch.	Not acceptable. Replace tip cap (WP 0576 00). If tip cap is replaced due to attachment hole elongation, inspect tip cap attachment area of blade for cracks and damage.		x
	Delaminations, disbonds, and other damage around attachment holes 1 through 3 and 7 through 12, upper and lower.			
	- - Up to 0.030-inch deep, up to 0.60-inch diameter around hole.	Inject with adhesive (WP 0576 00).	x	
	- - Greater than 0.030-inch deep, up to 0.60-inch diameter around hole.	Not acceptable. Replace tip cap (WP 0576 00). If tip cap is replaced due to attachment hole delaminations, disbonds, and other damage, inspect tip cap attachment area of blade for cracks and damage.		x
	Missing or broken screws.			
	- - No missing or broken screws in positions 4, 5, and 6, both top and bottom locations (Figure 1, Sheet 5, Detail A).	Replace screws (WP 0576 00).	x	

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Tip Cap Nickel Abrasion Strip	- - No more than one screw total in positions 1, 2, and 3, either top or bottom locations (one screw total).	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (WP 0576 00).	x	
	- - No more than one screw per set in positions 7 through 12, both top and bottom locations (two screws total).	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (WP 0576 00).	x	
	- - No more than two screws total per tip cap.	Check that torque on remaining screws is no less than 25 inch-pounds (check in tightening direction only). Replace screws at next available opportunity (WP 0576 00).	x	
	Erosion.			
	- - Through nickel strip.	Not acceptable. Replace tip cap (WP 0576 00).		x
	Cracks.			
	- - Overlap area at inboard end.	Remove loose pieces. Blend sharp edges. Seal using sealing compound, Item 257, WP 1803 00.	x	
	- - All other areas.	Acceptable provided no pieces are missing.	x	
	Disbonds.			
	- - Edge disbonds up to 0.50-inch chordwise x 3 inches spanwise.	Inject with adhesive (WP 0576 00).	x	
	- - Edge disbonds greater than 0.50-inch chordwise x 3 inches spanwise.	Not acceptable. Replace tip cap (WP 0576 00).		x

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued

Table 1. Main Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Tip Block	- - Internal disbonds more than 0.50-inch from any edge up to 6 square inches.	No repair required.	x	
	- - Internal disbonds more than 0.50-inch from any edge greater than 6 square inches.	Not acceptable. Replace tip cap (WP 0576 00).		x
	- - Stripped or missing inserts.	Repair using tip block insert repair (WP 0576 00).	x	
Tip Block Doubler	Cracks.			
	- - Visible in adhesive fillet.	Coin tap doubler to insure bond integrity. If bond is good, no repair needed.	x	
	Bond Separation.			
	- - Up to 2 square inches.	Repair using adhesive injection repair (WP 0576 00).		x
	- - Greater than 2 square inches.	Return blade to depot for repair.		x
Aluminum Lightning Protection Wire Mesh	Corrosion (Note 2).			
	- - Less than 10 pinholes per square foot.	No repair required.		
	- - More than 10 pinholes per square foot.	Repair using aluminum wire mesh lightning repair (WP 0576 00).		x
NOTES				
1. If helicopter is not operated in high-erosion (desert sand) environment, 1/4-inch dimension may be increased to 3/4-inch.				
2. Lightning mesh is nonstructural. Corrosion is a cosmetic problem. Repair corrosion only if blade has already been removed from helicopter.				

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued**Table 2. Number of Chordwise Shim Weights to be Moved from Rear Chordwise Stud to Front Chordwise Stud.**

SPANWISE CHORDWISE	0 INCHES TO 6 INCHES A	6 INCHES TO 9 INCHES B	9 INCHES TO 12 INCHES C	12 INCHES TO 15 INCHES D	KIT NO.
Zone 1 Rotor STA 60.0 to STA 140.0 (SEE NOTE)	0.4	0.3	0.2	0.1	-014, -020
	0.3	0.2	0.2	0.1	-018, -019
	0.2	0.1	0.1	0.0	-013, -015, -016, -017
	0.2	-	-	-	-021
	0.1	0.0	0.0	0.0	-012
Zone 2 Rotor STA 140.0 to STA 200.0	0.7	0.6	0.4	0.2	-014, -020
	0.5	0.4	0.3	0.1	-018, -019
	0.3	0.2	0.1	0.1	-013, -015, -016, -017
	0.3	-	-	-	-021
	0.1	0.1	0.0	0.0	-012
Zone 3 Rotor STA 200.0 to STA 260.0	0.9	0.8	0.5	0.2	-014, -020
	0.7	0.6	0.4	0.1	-018, -019
	0.4	0.3	0.2	0.1	-013, -015, -016, -017
	0.4	-	-	-	-021
	0.1	0.1	0.1	0.0	-012
Zone 4 Rotor STA 260.0 to STA 300.0	1.2	0.9	0.6	0.3	-014, -020
	0.8	0.7	0.4	0.2	-018, -019
	0.5	0.3	0.2	0.1	-013, -015, -016, -017

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued**Table 2. Number of Chordwise Shim Weights to be Moved from Rear Chordwise Stud to Front Chordwise Stud. - Continued**

SPANWISE CHORDWISE	0 INCHES TO 6 INCHES A	6 INCHES TO 9 INCHES B	9 INCHES TO 12 INCHES C	12 INCHES TO 15 INCHES D	KIT NO.
	0.5	-	-	-	-021
	0.2	0.1	0.1	0.0	-012
EXAMPLE					
Chordwise Shim Determination					
Given: Zone 3C and Kit No. -018					
Find: Chordwise Shim Unbalance = 0.4 shim					
NOTE					
Repairs inboard of Zone 1 (STA 60.0) do not need to be recorded. Repairs in this area will not affect chordwise balance.					

Table 3. Spanwise Weight Unbalance With the Addition of Patches (LBS).

Zone Kit No.	-014, -020	-018, -019	-013, -015, -016, -017, -021	-012
Zone 1 Rotor STA 60.0 to STA 140.0 (SEE NOTE)	0.036	0.026	0.013	0.004
Zone 2 Rotor STA 140.0 to STA 200.0	0.061	0.044	0.022	0.007
Zone 3 Rotor STA 200.0 to STA 260.0	0.083	0.060	0.030	0.010
Zone 4 Rotor STA 260.0 to STA 300.0	0.101	0.073	0.036	0.012
EXAMPLE				
Chordwise Shim Determination				
Given: Zone 3 and Kit No. -019				
Find: Spanwise Shim Unbalance = 0.060 lbs				

INSPECT MAIN ROTOR BLADE TOOLING AND TENSILE TEST PATCHES - Continued**Table 3. Spanwise Weight Unbalance With the Addition of Patches (LBS). - Continued**

Zone Kit No.	-014, -020	-018, -019	-013, -015, -016, -017, -021	-012
NOTE				
Repairs inboard of Zone 1 (STA 60.0) do not need to be recorded. Repairs in this area will not affect chordwise balance.				

Table 4. Blade Damage Repair Historical Record Entries.

DATE	DESCRIPTION OF DAMAGE	ZONE	REPAIR KIT NO.	CHORDWISE UNBALANCE	SPANWISE WGT UNBALANCE
4/19/91	Damage 1 inch x 5 inches x 1/2-inch deep (bottom)	3C	-019	+ 4 shims	+ 0.060 lbs
8/21/91	Skin tear 5 inches long (top)	3A	-014	+ 9 shims	+ 0.083 lbs
1/20/92	Skin disbond 1 inch x 4 inches (top)	4A	-014	+ 1.2 shims	+ 0.101 lbs
1/20/92	Blade Tip Wgt. Adjustment Chordwise - 2 shim adjusted Spanwise -0.224 lbs removed			New + 0.5 shims	New + 0.020 lbs

INSPECT MAIN ROTOR BLADE DEICE FAIRING

Blade deice fairing repaired by adhesive injection or filling (Figure 1, Sheet 1) (WP 0576 00).

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
TAIL ROTOR INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Dial Indicator
Elastomeric Rod End Bearing Plastic Feeler Gage,
Locally-Made, Figure 232, WP 1805 00
Teflon Lined Spherical Bearing Plastic Feeler Gage,
Locally-Made, Figure 209, WP 1805 00

References

WP 0546 00
WP 0585 00
WP 0586 00
WP 0606 00
WP 1805 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT TAIL ROTOR BLADE

1. Grasp tail rotor blade tip at mid width. Apply light hand force in flatwise direction toward or away from tail pylon for 3 to 4 inches. Observe opposite blade for equal deflection at blade tip. Repeat for adjacent tail rotor blade.
2. If movement of an equal deflection is observed in opposite blade, do this:
 - a. Remove tail rotor blades (WP 0585 00).
 - b. Inspect nylon surfaces of tail rotor blade spars and retention plate shims for wear and heat damage (WP 0546 00). **NONE ALLOWED.** Replace damaged tail rotor blade and retention plate shims.
 - c. Inspect tail rotor blade spars for evidence of interlaminar cracking. Pay particular attention to leading and trailing edges of graphite spar in area from STA 8.0 through STA 0.0 to STA 8.0 on opposite end. **NONE ALLOWED.** Replace damaged tail rotor blade.
 - d. Check screws that hold nylon washers to make sure they are hand tight.
 - e. Install tail rotor blades (WP 0585 00).

INSPECT TAIL ROTOR PITCH CONTROL ROD**Inspect Spherical Rod End Bearing For Wear - Feeler Gage Method**

1. Locally make Teflon lined spherical bearing plastic feeler gage (WP 1805 00).
2. Apply hand pressure to tail rotor blades, and twist each blade in direction of reduced pitch.
3. Check spherical rod end bearings on both rod ends by doing this:

INSPECT TAIL ROTOR PITCH CONTROL ROD - Continued

CAUTION

- Damage to spherical rod end bearing will result if metallic feeler gage is used. Teflon race will be damaged. Use only Teflon lined spherical bearing plastic feeler gage when inspecting spherical rod end bearings.
 - Damage to spherical rod end bearing will result if Teflon lined spherical bearing plastic feeler gage is not inserted carefully. Teflon race will be damaged. Use care when inserting Teflon lined spherical bearing plastic feeler gage.
- a. Using locally-made Teflon lined spherical bearing plastic feeler gage (0.010-inch), probe both sides of bearing around entire circumference for clearance between ball and outer race (Figure 1, Detail A).
 - b. If Teflon lined spherical bearing plastic feeler gage (0.010-inch) cannot be inserted to 0.125-inch mark, bearing is acceptable. Recheck in 40 flight hours.
 - c. If Teflon lined spherical bearing plastic feeler gage (0.010-inch) can be inserted to 0.125-inch mark, or more, check clearance again using Teflon lined spherical bearing plastic feeler gage (0.015-inch).
 - d. If Teflon lined spherical bearing plastic feeler gage (0.015-inch) cannot be inserted to 0.125-inch mark between ball and outer race, bearing is acceptable. Recheck in 15 flight hours.
 - e. **IF TEFLON LINED SPHERICAL BEARING PLASTIC FEELER GAGE (0.015-INCH) CAN BE INSERTED TO 0.125-INCH MARK BETWEEN BALL AND OUTER RACE, REPLACE PITCH CONTROL ROD END (WP 0586 00).**

NOTE

An accumulation of reddish-brown powder on bearing ball is normal for a Teflon bearing and not cause for removal.

4. Check spherical rod end bearings on both rod ends for evidence of metal-to-metal contact between ball and outer race. **NONE ALLOWED.** Metal-to-metal contact is present if worn or discolored spots are seen on bearing ball, or if you hear a clicking sound when bearing ball is rotated in its outer race. Replace pitch control rod end bearings if metal-to-metal contact is present (WP 0606 00).
5. Check spherical rod end bearings for play. **REPLACE BEARINGS IF RADIAL PLAY (MEASURE IN DIRECTION OF LOAD), IS GREATER THAN 0.012-INCH, OR IF AXIAL PLAY (MEASURE IN DIRECTION OF BOLT) IS GREATER THAN 0.023-INCH (WP 0606 00).**

Inspect Elastomeric Rod End Bearing Surface**NOTE**

- Elastomeric rod ends consist of laminated elastomer and steel shells. Wear of elastomer is a gradual process that will take several hundreds of hours from first detection of elastomer damage to point where rod end must be replaced.
- Eraser-type wear is a pileup of shredded rubber similar to the result of using a soft eraser. Eraser-type wear is acceptable.
- A wax-like substance on rubber surface is normal and should not be removed.

1. Inspect elastomeric rod end bearing surface as follows:
 - a. Check for a concentration or pileup of eraser-type wear. A concentration or pileup of eraser-type wear may indicate an elastomer crack or separation may exist. Inspect rod end for elastomeric crack or separation if a concentration or pileup of eraser-type wear is present. Refer to, INSPECT ELECTROMERIC ROD END BEARNING FOR WEAR-FEELER GAGE METHOD, in this work package.

INSPECT TAIL ROTOR PITCH CONTROL ROD - Continued**NOTE**

Peeling of elastomer mold lines which run radially across laminations is acceptable.

- b. Elastomeric extrusion is squeezing out of rubber from between bearing shims and inner or outer members, including local disbonding of elastomer. Inspect rod end for elastomeric crack or separation if an extrusion is detected. Refer to, INSPECT ELECTROMERIC ROD END BEARING FOR WEAR-FEELER GAGE METHOD, in this work package.
- c. Visually check for cracking or disbonding of elastomer from shim inner or outer members. Cracking or disbonding of elastomer from shim inner or outer members will cause separation. Inspect rod end if cracking or disbonding of elastomer from shim inner or outer members is suspected Refer to, INSPECT ELECTROMERIC ROD END BEARING FOR WEAR-FEELER GAGE METHOD, in this work package.
- d. Visually check all elastomeric rod end bearings for signs if inner member separation using hand force, shaking control rod. **NO INNER MEMBER LOOSENESS OR EXPOSED METAL OF INNER BALL SURFACE OR MISSING ELASTOMER ALLOWED.** Replace elastomeric rod end if damage is beyond limits (WP 0586 00).

NOTE

Shims or steel shells alternate with elastomer layers of elastomeric rod end bearing and are composed of two 180 degree pieces. Within each shim layer, two pieces are separated by small gap measuring approximately 0.050-inch. These gaps are filled with elastomer during manufacture and are evident on surface as small lump of elastomer. Gaps becoming exposed over time is acceptable.

- e. Check for gaps or cracks in area other than manufactured shim gaps. **NO GAPS OR CRACKS ALLOWED.** Replace elastomeric rod end with gaps or cracks (WP 0586 00).

Inspect Elastomeric Rod End Bearing For Wear - Feeler Gage Method

1. Inspect bearing surfaces Refer to, INSPECT ELASTOMERIC ROD END BEARING SURFACE, in this work package.
2. Locally make elastomeric rod end bearing plastic feeler gage (WP 1805 00).
3. Check elastomeric rod end bearings as follows:

CAUTION

- Damage to elastomeric rod end bearing will result if metallic feeler gage is used. Elastomer will be damaged. Use only elastomeric rod end bearing plastic feeler gage when inspecting elastomeric rod end bearing.
- Damage to elastomeric rod end bearings will result if elastomeric rod end bearing plastic feeler gage is forced between elastomeric layers. Do not force elastomeric rod end bearing plastic feeler gage when checking for laminate separations.

NOTE

Elastomeric rod end bearing plastic feeler gage should only be used if possible separation exists based on visual inspection of bearing surfaces.

- a. Check for elastomeric/metal separations or disbonding of elastomer from shim inner or outer members in all areas carefully using elastomeric rod end bearing plastic feeler gage (0.005-inch). If elastomeric rod end bearing plastic feeler gage can be inserted, separation has occurred. Measure depth of separations using elastomeric rod

INSPECT TAIL ROTOR PITCH CONTROL ROD - Continued

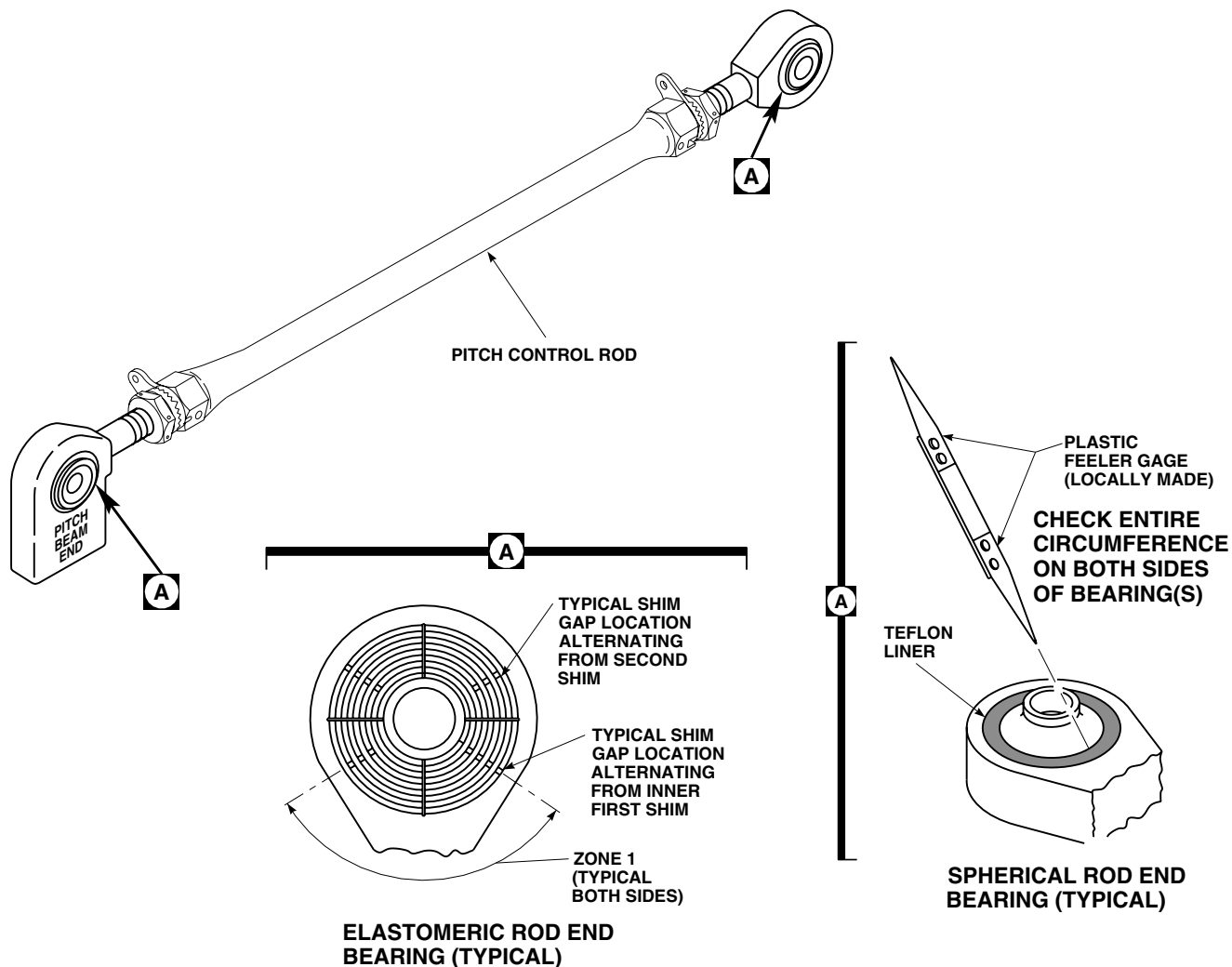
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Figure 1. Inspect Tail Rotor Pitch Control Rod Bearings.

end bearing plastic feeler gage (0.005-inch). **SMALL HAIRLINE SURFACE CRACKING, MINOR DISBONDING AND CRAZING OF ELASTOMER UP TO 0.060-INCH ARE ACCEPTABLE IN ALL AREAS OF ROD END BEARING.**

- b. Check for elastomeric/metal separations or disbonding of elastomer from shim inner or outer members in Zone I carefully using elastomeric rod end bearing plastic feeler gage (0.005-inch). If elastomeric rod end bearing plastic feeler gage can be inserted, separation has occurred. Measure depth of separations using elastomeric rod end bearing plastic feeler gage (0.005-inch). **SEPARATIONS UP TO 0.20-INCH IN ZONE I AND UP TO 0.060-INCH OUTSIDE ZONE I ARE ACCEPTABLE.** Replace elastomeric rod end if damage is beyond limits (WP 0586 00).

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
TAIL ROTOR BLADE INSPECTIONS
This WP supersedes WP 0546 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Bit, No. 40
Fluorescent Penetrant Inspection Kit, XMA101
Soft Copper (or equivalent) 1 1/2 to 2 inch diameter x
1/8 to 3/16-inch thick

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
Adhesive, Item 11, WP 1803 00
Cloth, Abrasive, Item 75, WP 1803 00

Cloth, Abrasive, Item 81, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0585 00
WP 0610 00
WP 1379 00
WP 1803 00

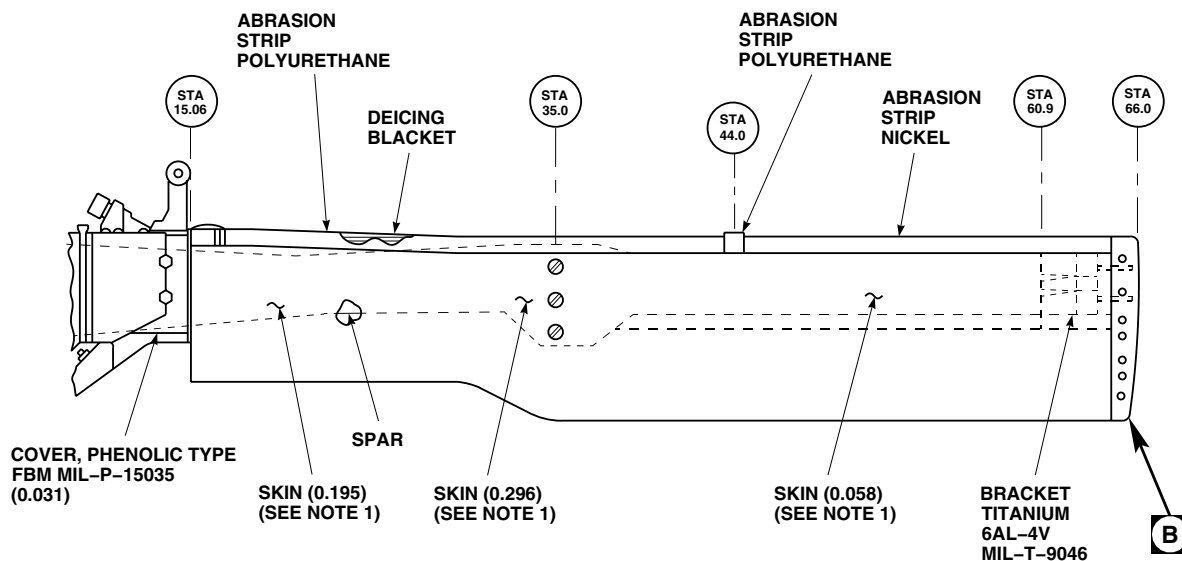
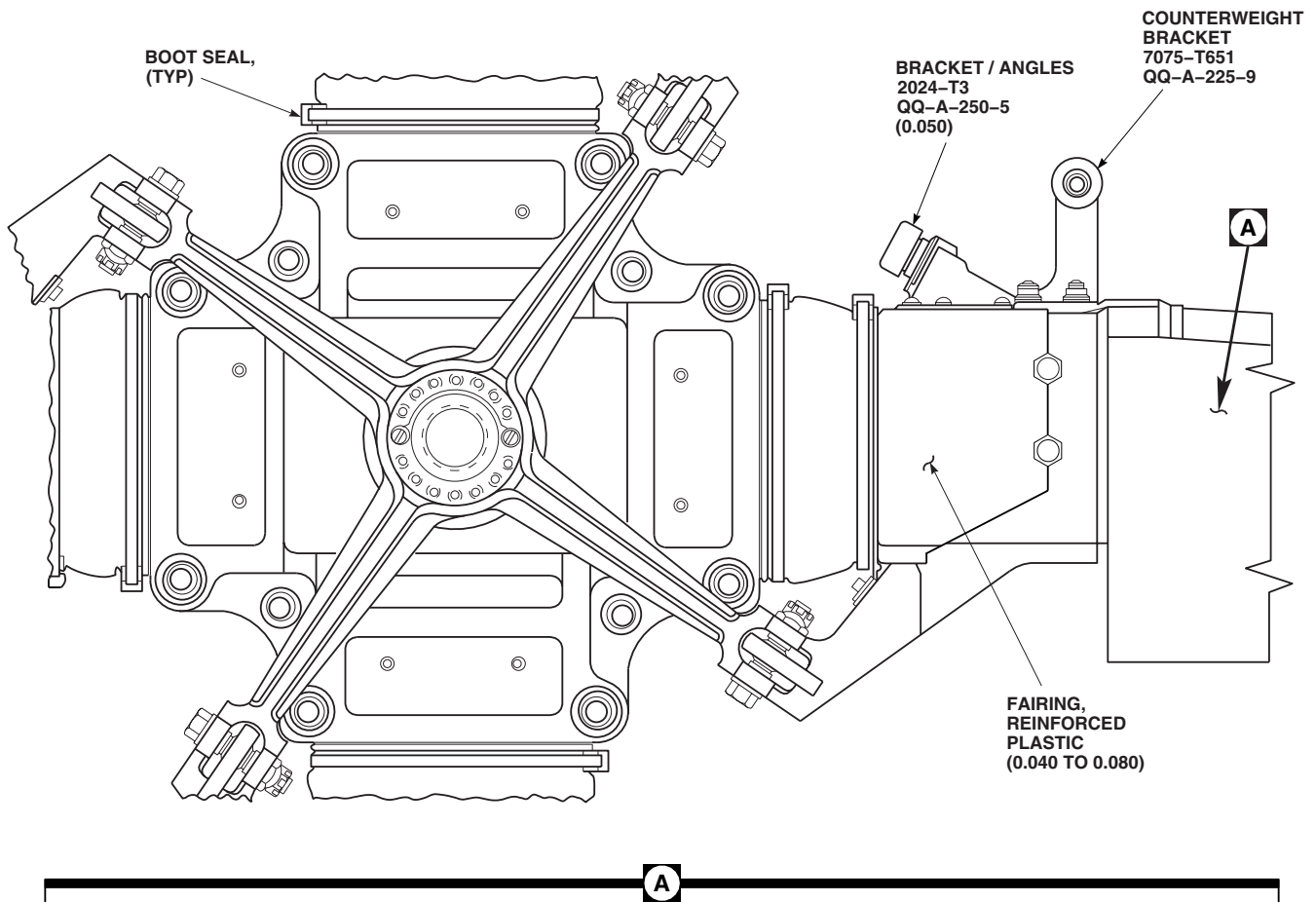
GENERAL

1. Damage to blade can be divided into two types; detectable and nondetectable.
2. Detectable damage can be seen and felt as a blister or dent.
 - a. A blister can be seen as a raised bump above blade contour, indicating a separation between skin and core. 4 to 5 pounds of thumb pressure can move the skin against core without crinkly sound.
 - b. A dent indicates a core crush. It can be felt and seen. 4 to 5 pounds of thumb pressure against dent will result in a crinkly sound.
3. Nondetectable damage can be found with coin-tapping, but cannot be seen or felt.
 - a. 4 to 5 pounds of thumb pressure will not make sound with nondetectable blisters.
 - b. 4 to 5 pounds of thumb pressure will make crinkly sound with nondetectable dents.
 - c. Use scale if necessary to find out what 4 to 5 pounds of pressure feels like.

NOTE

- To verify suspected damage, use coin-tapping method to determine size and shape of any possible bond separations between skin and honeycomb core, skin and control horn, and skin and abrasion strip bond areas.
 - "Coin" is smooth disc of soft copper, or equivalent metal, 1 1/2 to 2 inches in diameter and 1/8 to 3/16-inch thick. Round and smooth edge of coin to prevent damage to skin.
 - Coin tapping should be done by experienced personnel only.
4. Hold coin loosely between thumb and finger and tap lightly along bond line or areas suspected of having bonding voids, and listen for variations in tapping sound. A sharp solid sound indicates a good bond and a dull dead sound indicates a bond separation (Figure 1, Sheet 1, Detail A).
 5. Tap across suspected area in horizontal, vertical, and diagonal lines. Mark points on skin with grease pencil where sound varies. Continue tapping and marking until outline of bond separation is complete.

GENERAL - Continued



NOTES

1. VARYING THICKNESS OF COMPOSITE SKIN MADE FROM GLASS FIBER AND GRAPHITE PLYS BONDED WITH ADHESIVE RESINS.
2. TIP CAP 70101-31131-041 HAS ABRASION STRIP BONDED AS SHOWN. TIP CAP 70101-31131-042 HAS LEADING EDGE ELECTRO-PLATED NICKEL (0.050 MAXIMUM).
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Tail Rotor Blade Inspection Areas. (Sheet 1 of 2)

GENERAL - Continued

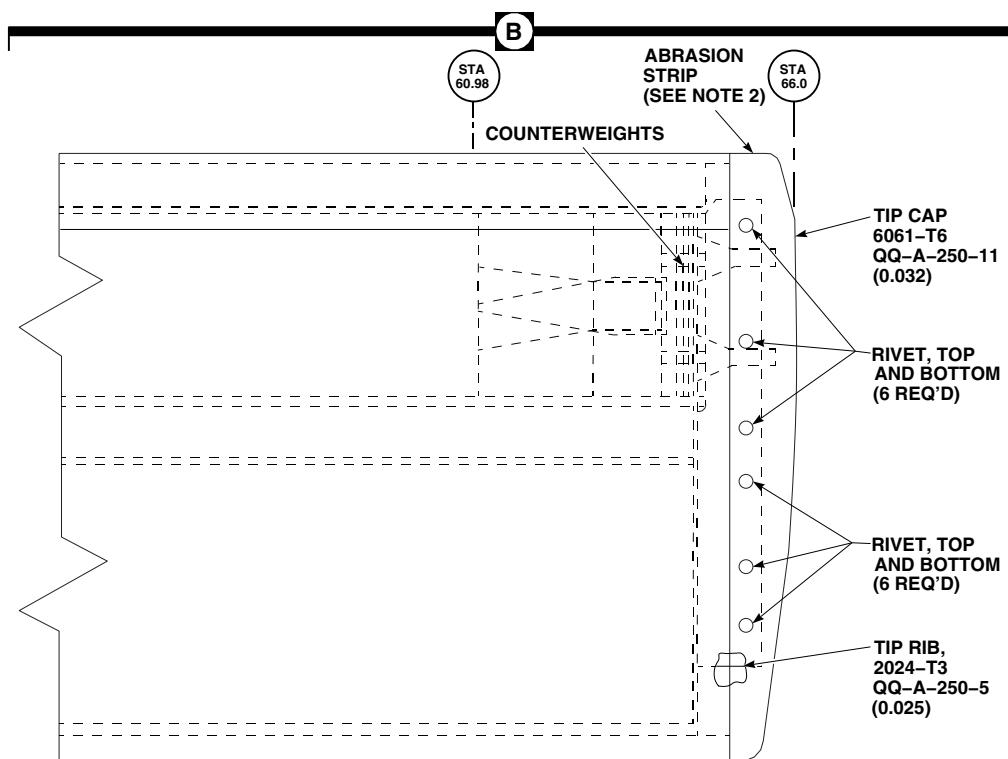
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Figure 1. Tail Rotor Blade Inspection Areas. (Sheet 2 of 2)

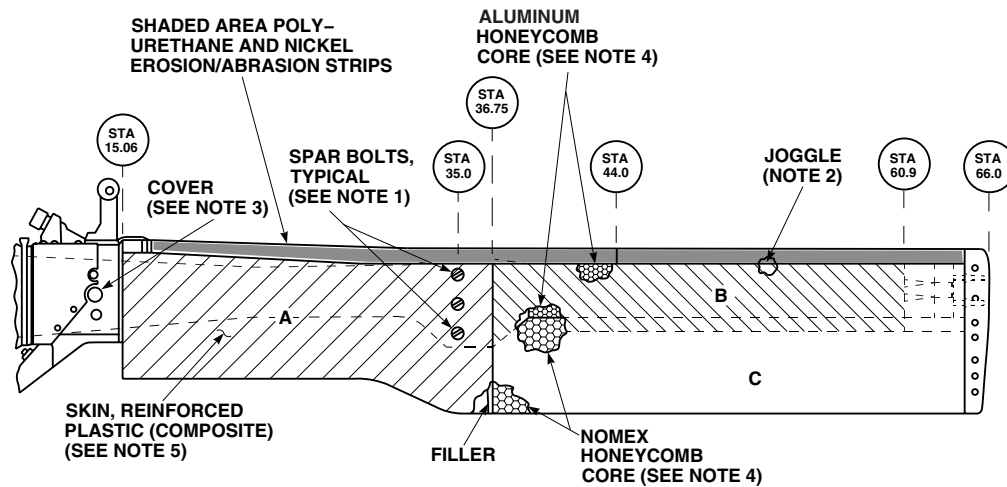
6. Visually inspect tip cap for cracks, dents, gouges, and security of attachment (Figure 1, Sheet 2, Detail B).
7. Visually inspect nickel abrasion strip for cracks, dents, and gouges.
8. Visually inspect polyurethane abrasion strip for gouges, wear, and loose bonding.
9. If installed, inspect/repair polyurethane tape (Erosion Protection Kit) (WP 1379 00).
10. Visually inspect pocket skin for holes, dents, cracks, and gouges.
11. Using coin-tapping method, inspect abrasion strip and skin for bond separations particularly in the area where skin is bonded to horn, and around three spar bolts at center of blade.
12. Using either a fluorescent penetrant inspection kit, or an eddy current inspection using a 90 degree shielded probe, inspect center spar plug for cracks or disbonding from the spar (TM 1-1520-265-23).

WARNING**CSI**

Bonding of skin along trailing edge of tail rotor blade is a critical characteristic. Damage shall not exceed limits given in Table 1.

13. See Table 1 and Figure 2 for reference to damage limits and repair procedures on tail rotor blade.
14. Record all damage and repairs to blade.

GENERAL - Continued



NOTES

1. SPAR BOLT THREADS AND TOP OF BOLTS ARE SEALED WITH EA9309.3NA ADHESIVE. COVER BOLT HEADS EVEN WITH BLADE CONTOUR.
2. JOGGLE DEPRESSION FILLED TO BLADE CONTOUR WITH PR-1826, B-2 SEALANT.
3. BOND COVER WITH EA9309.3NA ADHESIVE.
4. CORE DAMAGE NOT REPAIRABLE.
5. SKIN DAMAGE (AREAS A, B, C) LIMITED TO ADHESIVE FILLER OR ADHESIVE INJECTION REPAIR.

AREA A - COMPOSITE SKIN ONLY.
 B - COMPOSITE SKIN/ALUMINUM HONEYCOMB CORE.
 C - COMPOSITE SKIN/NOMEX HONEYCOMB CORE.

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Figure 2. Tail Rotor Blade Repair Limits.

TAIL ROTOR BLADE HORN FAIRING-TO-SPAR CLEARANCE

1. No delamination or splintering of spar allowed. Corner damage up to a maximum depth of 0.070-inch is allowable in the area of the inboard end of the horn fairing (approximate blade STA 9.0).
2. Repairs to blade can affect blade track and balance, causing tail rotor vibration. Due to nature of track and balance requirements, only minor repairs are made to the blade. The total number of repairs will depend upon acceptable blade tracking and vibration levels.
3. Do all repairs carefully and according to instructions.

CAUTION

Damage to blade skin will result if blade is sanded excessively. Heavy or careless sanding can damage and crack the skin. Sand reinforced plastic skin lightly and carefully.

4. Replace blade if damage is greater than established repair limits. Forward damaged blade to depot.
5. Remove and install tail rotor blades (WP 0585 00).

TAIL ROTOR BLADE EROSION PROTECTION STRIPS

If installed, inspect erosion protection strips (WP 1379 00).

TAIL ROTOR BLADE POCKET

1. Area A, Figure 2, consists of laminate skin with no filler. Area B consists of laminate skin bonded to aluminum honeycomb core which is bonded to the spar. Area C consists of laminate skin bonded to Nomex fiber honeycomb core. Repairs to blade pocket in all three areas is limited to adhesive filler and adhesive injection repairs.
2. Cracks or holes in skin will not be repaired. Replace blade.
3. Dents or gouges in skin, within limits listed in Table 1 are repairable using adhesive filler repair procedure.
4. Core damage greater than dent limits will not be repaired. Replace blade (WP 0585 00).
5. Bond separations between skin and core (Area B and Area C) are limited to 1/2- square inch in area.
6. Abrasion strips bonded to leading edge of pocket are repairable until complete wear through occurs.
7. If installed, inspect/repair polyurethane tape (Erosion Protection Kit) (WP 1379 00).
8. Bond separation between pocket and tip rib is repairable using adhesive injection repair procedure.

Table 1. Tail Rotor Blade Inspection/Repair Limits.

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Blade Pocket Skin	Cracks.	Unacceptable. Remove and replace blade (WP 0585 00).		x
	Dents.			
	- - Unbroken skin up to 0.040-inch deep x 2 inches diameter.	No repair required. Touch up paint as required (WP 0585 00).	x	
	- - Skin broken within dent or dent deeper than above limits.	Unacceptable. Remove and replace blade (WP 0585 00).		x
	Bond Separation.			
	- - Between skin and tip rib.	Repair using adhesive injection repair (WP 0585 00).	x	
	- - Skin to core up to 1/2-square inch area.	No repair required. Touch up paint as required (WP 0585 00).	x	
	- - Larger than 1/2-square inch.	Unacceptable. Remove and replace blade (WP 0585 00).		x

TAIL ROTOR BLADE POCKET - Continued

Table 1. Tail Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
WARNING				
CSI				
Bonding of skin along trailing edge of tail rotor blade is a critical characteristic. Damage shall not exceed the following limits.				
	- - Between skin plies at trailing edge: inboard of STA 39.0, voids up to 0.10-inch chordwise and 0.050-inch spanwise; outboard of STA 39.0, voids up to 0.25-inch chordwise and 0.50-inch spanwise.	Repair using adhesive injection repair (WP 0585 00).		X
	- - Skin separation between STA 19.0 and STA 40.0, disbond up to 6 inches.	Repair using trailing edge disbond repair (WP 0585 00).		X
	Core Damage (Nomex Core).			
	- - Greater than Dent limits.	Unacceptable. Remove and replace blade (WP 0585 00).		X
	Core Damage (Aluminum Core).			
	- - Greater than Dent limits.	Unacceptable. Remove and replace blade (WP 0585 00).		X
	Gouges.			
	- - Up to 0.030-inch deep x 3 inches long in Area A and Area B (Figure 2).	Repair using adhesive filler repair (WP 0585 00).	X	
	- - Up to 0.005-inch deep x 2 inches long in Area C (Figure 2).	Repair using adhesive filler repair (WP 0585 00).	X	

TAIL ROTOR BLADE POCKET - Continued

Table 1. Tail Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Nickel Abrasion Strip - Blade	- - Greater than above limits.	Unacceptable. Remove and replace blade (WP 0585 00).		x
	Holes (Skin or Core).	Unacceptable. Remove and replace blade (WP 0585 00).		x
	Erosion.			
	- - Up to one ply thickness (0.006-inch deep).	No repair required. Touch up paint (WP 0585 00).	x	
	- - Deeper than one ply thickness.	Unacceptable. Remove and replace blade (WP 0585 00).		x
	Cracks.			
	- - 3 or less per strip.	Acceptable if bond separations are within limits. Refer to BOND SEPARATIONS, in this table.) Seal cracks using abrasion strip repair (WP 0585 00).	x	
	- - More than 3 per strip.	Unacceptable. Remove and replace blade (WP 0585 00).		x
	Dents.			
	- - Up to 1/16-inch deep x 1 inch across and bond separation less than 1/2-inch beyond edge of dent.	No repair required. Touch up paint (WP 0585 00).	x	
	- - Greater than above limits.	Unacceptable. Replace blade (WP 0585 00).		x
	Bond Separations.			
	- - Up to 1/8 x 1/2-inch area.	Seal open voids using abrasion strip repair (WP 0585 00).	x	

TAIL ROTOR BLADE POCKET - Continued

Table 1. Tail Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Polyurethane Erosion Strip	- - Up to 1/2 x 2 inch area.	Repair using adhesive injection repair (WP 0585 00).	x	
	- - Greater than above limits.	Unacceptable. Replace blade (WP 0585 00).		x
	Gouges.	Acceptable within thickness of strip. Blend raised metal with abrasive cloth, Item 81, WP 1803 00, and touch up paint (WP 0585 00).	x	
	Erosion (Wear and Corrosion).	Acceptable until repair limits of 0.007-inch deep for large areas, or up to 0.012-inch deep, for no more than 1 square inch are exceeded. Apply thin coat of adhesive, Item 11, WP 1803 00, at point where nickel erosion strip and composite skin/aluminum honeycomb core mates. Touch up paint (WP 0585 00). Erosion greater than above requires replacement of blade (WP 0585 00).	x	
	Loose bond.	Acceptable with limit if repaired using polyurethane erosion strip repair (WP 0585 00).	x	
	Gouges.			
	- - Up to 1/8-inch wide.	Repair using adhesive filler repair (WP 0585 00).	x	
	- - More than 1/8-inch wide.	Repair using polyurethane erosion strip repair (WP 0585 00).		x

TAIL ROTOR BLADE POCKET - Continued

Table 1. Tail Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Tip Cap	Erosion.	Acceptable until worn through, then repair using polyurethane erosion strip repair (WP 0585 00).		x
	Delaminations.			
	- - Less than 4 square inches.	Repair using polyurethane erosion strip repair (WP 0585 00).		x
	- - Greater than 4 square inches.	Replace polyurethane erosion strip (WP 0585 00).		x
	Cracks.			
	- - Up to 1/4-inch long.	Stop-drill using No. 40 drill. (No more than two cracks allowed at attachment rivet holes). Touch up paint (WP 0585 00).	x	
	- - More than 1/4-inch long, or more than two cracks at attachment holes.	Replace tip cap (WP 0585 00).	x	
	Dents.			
	- - With no cracks or gouges.	No repair required. Touch up paint (WP 0585 00).	x	
	- - With cracks up 1/4-inch long.	Stop-drill with No. 40 drill and touch up paint (WP 0585 00).	x	
	- - With cracks more than 1/4-inch long.	Replace tip cap (WP 0585 00).	x	
	Gouges and Erosion/Corrosion.			
	- - Up to 0.005-inch deep (after cleanup).	Blend out with abrasive mat, Item 2, WP 1803 00, and touch up paint (WP 0585 00).	x	

TAIL ROTOR BLADE POCKET - Continued

Table 1. Tail Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Tip Cap Abrasion Strip	- - Deeper than 0.005-inch (after cleanup).	Replace tip cap (WP 0585 00).	x	
	Bond Separation.			
	- - Up to 1/8 x 1/2-inch area.	Seal voids using abrasion strip repair (WP 0585 00).	x	
	- - More than 1/8 x 1/2-inch area.	Repair using adhesive injection repair (WP 0585 00).	x	
	- - Erosion (Wear and Corrosion).	Acceptable until worn through, then replace tip cap (WP 0585 00).	x	
Aluminum Lightning Protection Wire Mesh	Cracks.	Acceptable if bond separation within limits. (Refer to Bond Separation).	x	
	Corrosion (see note).			
	- - Less than 10 pinholes per square-foot.	No Repair Required.		
	- - Greater than 10 pinholes per square-foot.	Repair using aluminum wire mesh lightning protection repair (WP 0585 00).		x
NOTE				
Lightning mesh is nonstructural. Corrosion is strictly a cosmetic problem. Repair corrosion only if blade has already been removed from helicopter.				
Spar Center Plug	Scratches, Gouges, and Corrosion.			

TAIL ROTOR BLADE POCKET - Continued

Table 1. Tail Rotor Blade Inspection/Repair Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	ON ACFT	OFF ACFT
Boot Support Assembly	- - Top and bottom surface up to 0.060-inch deep.	Blend out with abrasive cloth, Item 75, WP 1803 00. Touch up paint (WP 0585 00).	x	
	- - On the 2.525-inch diameter hole surface up to 0.020-inch deep, 50% of area. No single blend larger than 1 square inch.	Blend out with abrasive cloth, Item 75, WP 1803 00. Touch up paint (WP 0585 00).	x	
	Cracks and Disbonding	No cracks or disbonding allowed. Remove and replace blade (WP 0585 00).		x
	- - Crack in adhesive fillet (any length, boot support firmly attached).	Repair using adhesive injection repair (WP 0585 00).		x
	- - Crack in boot support.	Replace boot support (WP 0610 00).		x
	- - Disbond of boot support to spar.	Rebond boot support (WP 0610 00).		x

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**MAIN ROTOR HEAD AND SHAFT EXTENSION**

This WP supersedes WP 0547 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Adapter, 70700-20421-041
 Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Air Transport Set, 70700-70002-012
 Aluminum, 0.032-inch x 1 inch x 4 inch
 Aluminum, 0.060-inch x 1 inch x 4 inch
 Depth Gage
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Guide Rope
 Jacking Bolts, NAS566-41
 Jacking Bolts, NAS629-26
 Main Rotor Shaft Impact Tool, Locally-Made,
 Figure 255, WP 1805 00
 Mallet, GGG-H-33
 Nonmetallic Scraper
 Pencil, Grease
 Tags
 Thread Protection, Cardboard
 Torque Wrench, Dial, 0 - 75 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.
 Torque Wrench, 300 - 2500 in. lbs.
 Torque Wrench, 0 - 300 ft. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Antiseize Compound, Item 49, WP 1803 00
 Cloth, Abrasive, Item 78, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Grease, Aircraft, Item 132, WP 1803 00
 Grease, Molybdenum Disulfide, Item 138,
 WP 1803 00

Lubricating Oil, General Purpose, Item 185,
 WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Sealing Compound, Item 267, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Sealing Compound, Item 275, WP 1803 00
 Tags
 Tape, Pressure Sensitive, Adhesive, Item 322,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Assistant - (1)
 UH-60 Helicopter Repairer MOS 15T (2)

References

TM 1-1500-204-23
 TM 1-1500-344-23
 TM 1-1520-237-MTF
 TM 1-1520-265-23
 TM 1-6625-724-13&P
 TM 55-1500-345-23
 WP 0381 00
 WP 0548 00
 WP 0549 00
 WP 0550 00
 WP 0563 00
 WP 0573 00
 WP 0576 00
 WP 0636 00
 WP 1037 00
 WP 1220 00
 WP 1590 00
 WP 1691 00
 WP 1803 00
 WP 1805 00

PREPARE TO REMOVE MAIN ROTOR HEAD

NOTE

Main rotor head weighs about 875 pounds.

1. If maintenance crane will be used, assemble and install crane (WP 1590 00).
2. Remove main rotor blades (WP 0576 00). Put blade pins back into same main rotor spindle holes.
3. Disconnect pitch control rod at swashplate (WP 0563 00). Keep attaching hardware with swashplate.
4. If deice kit is installed, disconnect and remove main rotor blade deice distributor, slipring assembly, and tube (WP 1220 00).
5. Remove bolts, washers, and bifilar cover (Figure 1, Sheet 1).

REMOVE MAIN ROTOR HEAD

WARNING

CSI

- Exposed surfaces of main rotor shaft and shaft extension are critical surfaces which must not be damaged during and after removal of main rotor head. Use extreme caution when removing and provide protective covering over critical surfaces after removal of main rotor head.
- Exposed surfaces of main rotor shaft and shaft extension are critical surfaces which must not be damaged while breaking torque. Use protective covering as required.

CAUTION

Damage to main rotor head assembly will result if metallic scrapers are used to clean sealing compound from components. Do not use metallic items to remove sealant.

1. Clean any sealant from pressure plate, shaft extension, bifilar, and bifilar cover using nonmetallic scraper.
2. Remove bolts holding lockring to main rotor shaft nut (Figure 1, Sheet 2). Remove lockring. Inspect lockring for missing rollpins from bushings or damaged bushings. Repair as necessary (TM 1-1500-204-23).
3. Gradually loosen bolt tension, at main rotor shaft nut, in 1/4-turn increments in sequence, as numbered on shaft nut. Repeat sequence several times as necessary. Leave bolts loosely screwed into main rotor shaft nut.
4. Remove lock.
5. Remove main rotor shaft nut and thrust washer.
6. Inspect main rotor shaft nut for corrosion and repair as follows:
 - a. Remove light corrosion (TM 1-1500-344-23).
 - b. Remove remaining corrosion on outside diameter and end of shaft nut by blending with abrasive cloth, Item 82, WP 1803 00. Do not blend or chase threads. **MINIMUM SPHERICAL OUTSIDE DIAMETER IS 7.290-INCHES. MINIMUM THICKNESS IS 1.240-INCHES.**
 - c. If corrosion cannot be blended out within limits, replace nut.
 - d. Apply 2 coats of epoxy primer coating kit, Item 120, WP 1803 00.
 - e. Touch up paint reworked areas (WP 0381 00 and TM 55-1500-345-23).

REMOVE MAIN ROTOR HEAD - Continued

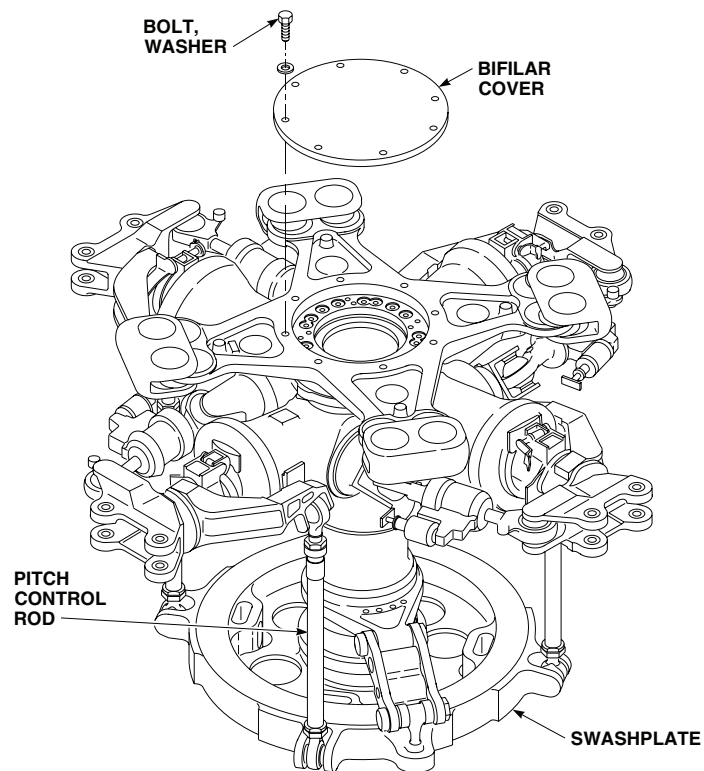
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 1 of 13)

7. Inspect thrust washer for corner damage, corrosion pitting, nicks, and scratches. Repair washer using abrasive cloth, Item 78, WP 1803 00, as follows:
 - a. **CORNER DAMAGE MAY BE BLENDED OUT UP TO 0.003-INCH DEEP AROUND ENTIRE CIRCUMFERENCE OF WASHER. IF CORNER DAMAGE IS BEYOND 0.003-INCH, ONE SIDE OF WASHER MAY BE BLENDED TO A MAXIMUM DEPTH OF 0.010-INCH, OVER 10% OF WASHER CIRCUMFERENCE.** Replace washer if repair exceeds limits, or if both sides of washer were blended beyond the initial limit of 0.003-inch.
 - b. **CORROSION PITTING DAMAGE ON BOTH SIDES OF WASHER MAY BE BLENDED OUT TO A MAXIMUM DEPTH OF 0.003-INCH, OVER 10% OF WASHER SURFACE AREA. IF DAMAGE IS BEYOND 0.003-INCH, ONE SIDE OF WASHER MAY BE BLENDED TO A MAXIMUM DEPTH OF 0.010-INCH, OVER 10% OF WASHER SURFACE AREA.** Replace washer if repair exceeds limits, or if both sides of washer were blended beyond the initial limit of 0.003-inch.
 - c. **NICKS, GOUGES, AND SCRATCHES ON BOTH SIDES OF WASHER MAY BE BLENDED OUT TO A MAXIMUM DEPTH OF 0.003-INCH, OVER 10% OF WASHER SURFACE AREA. IF DAMAGE IS BEYOND 0.003-INCH, ONE SIDE OF WASHER MAY BE BLENDED TO A MAXIMUM DEPTH OF 0.010-INCH, OVER 10% OF WASHER SURFACE AREA.** Replace washer if repair exceeds limits, or if both sides of washer were blended beyond initial limit of 0.003-inch.
 - d. Check washer for warping. **WARPING MUST NOT EXCEED 0.005-INCH.** Replace washer if warping is beyond limits.
8. Slowly loosen bolts at lower pressure plate in 1/4-turn increments, in sequence numbered on lower pressure plate. Repeat this procedure until bolts are loose. Do not completely remove bolts until split cones are free from shaft extension (Figure 1, Sheet 2).

REMOVE MAIN ROTOR HEAD - Continued

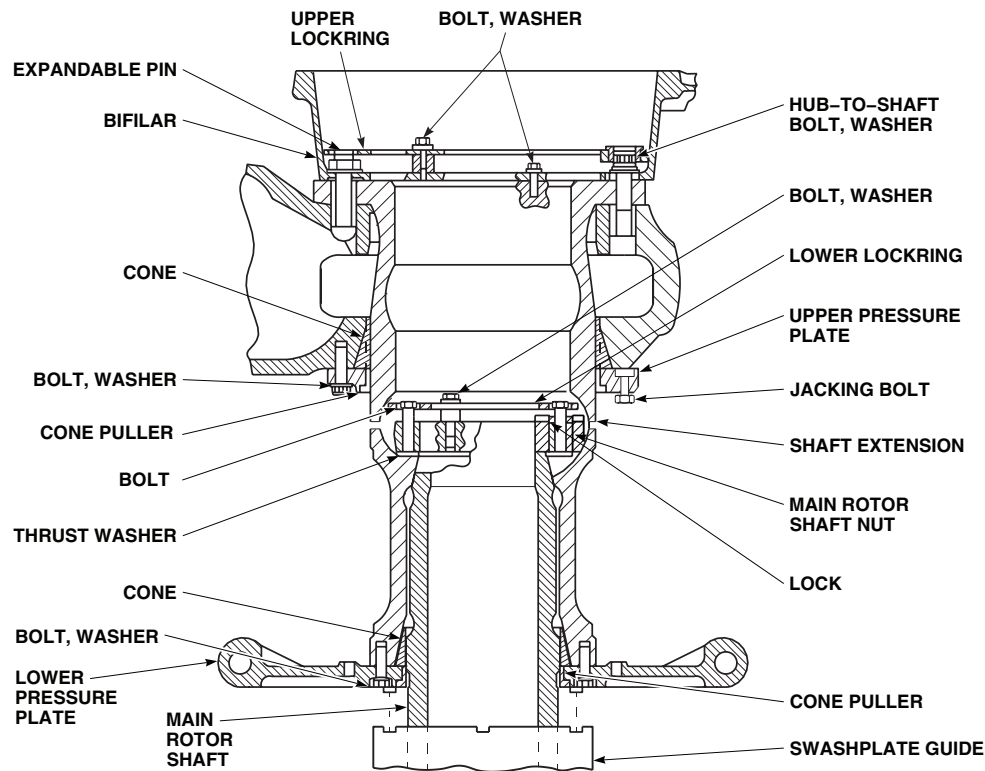
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 2 of 13)

9. Inspect upper and lower pressure plate (WP 0548 00).
10. Install jacking bolts, NAS566-41, in lower pressure plate.
11. Turn in jacking bolts until cones are free from shaft extension. Make sure all six jacking bolts are turned in evenly, and that split cones are free.
12. Remove all bolts and washers from lower pressure plate (Figure 1, Sheet 3). Slide lower pressure plate down and rest it on swashplate.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

13. Clean all bolts that have primer on them using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
14. Remove split cones (Figure 1, Sheet 3).
15. Inspect split cones for nicks and burrs. **NICKS AND BURRS SHALL NOT EXCEED 0.010-INCH DEEP.** Replace split cones if not within limits.
16. Inspect split cones for corrosion pits. **CORROSION PITS SHALL NOT EXCEED 0.010-INCH DEEP.** Replace split cones if not within limits.

REMOVE MAIN ROTOR HEAD - Continued

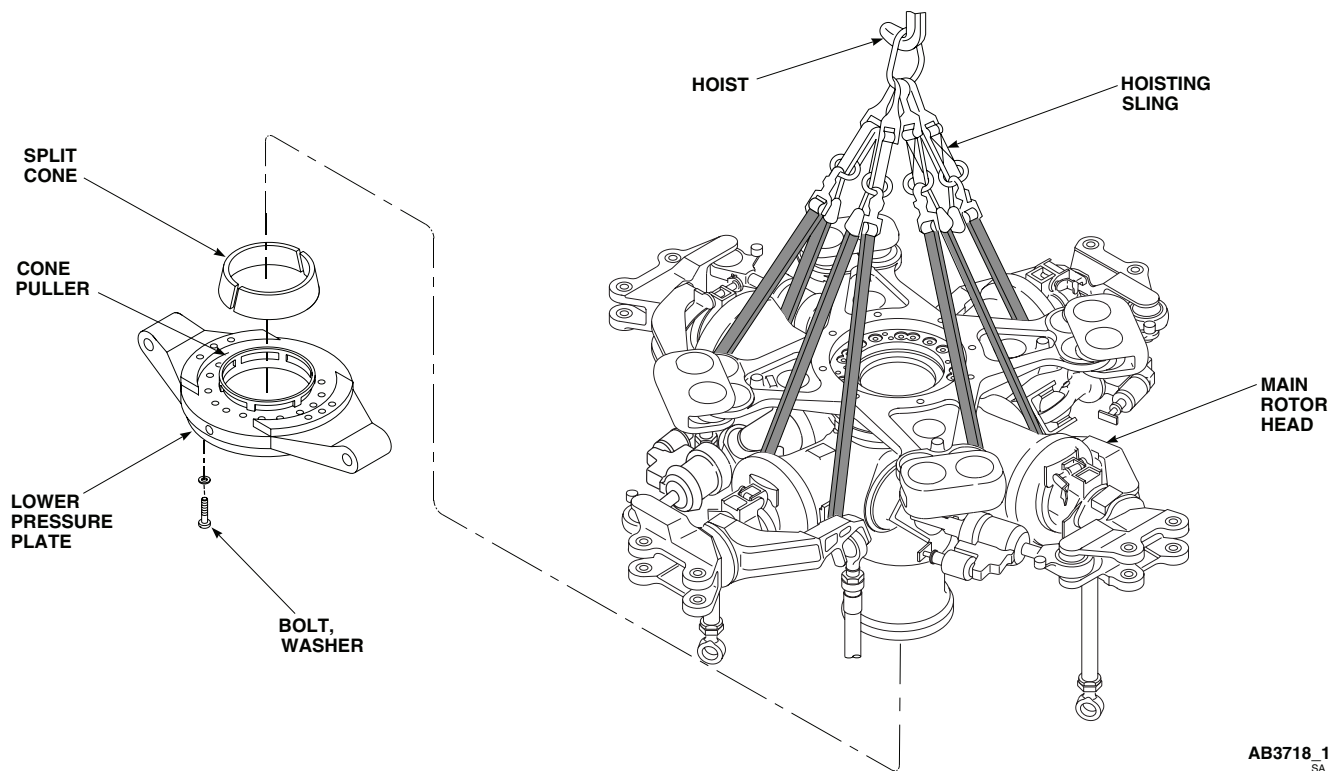
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 3 of 13)

17. Check split cones for fretting. **FRETTING UP TO 0.005-INCH DEEP OVER 50% OF CONTACTING SURFACE IS ACCEPTABLE AFTER REMOVAL OF METAL PICKUP USING ABRASIVE WHEEL, Item 350, WP 1803 00.** Replace split cones if not within limits.
18. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect split cones for fractures or cracks (TM 1-1520-265-23). **NO FRACTURES OR CRACKS ALLOWED.** Replace fractured or cracked split cones.
19. Tag split cones and place in bag for reinstallation.
20. Protect main rotor shaft threads with a wrap such as cardboard or machinery towels, Item 344, WP 1803 00, and secure with pressure sensitive adhesive tape, Item 322, WP 1803 00.
21. Place straps of hoisting sling around each arm of main rotor hub (Figure 1, Sheet 3).

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
 - Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.
22. Attach hoisting sling to maintenance crane or a hoist with capacity to lift main rotor head.

REMOVE MAIN ROTOR HEAD - Continued

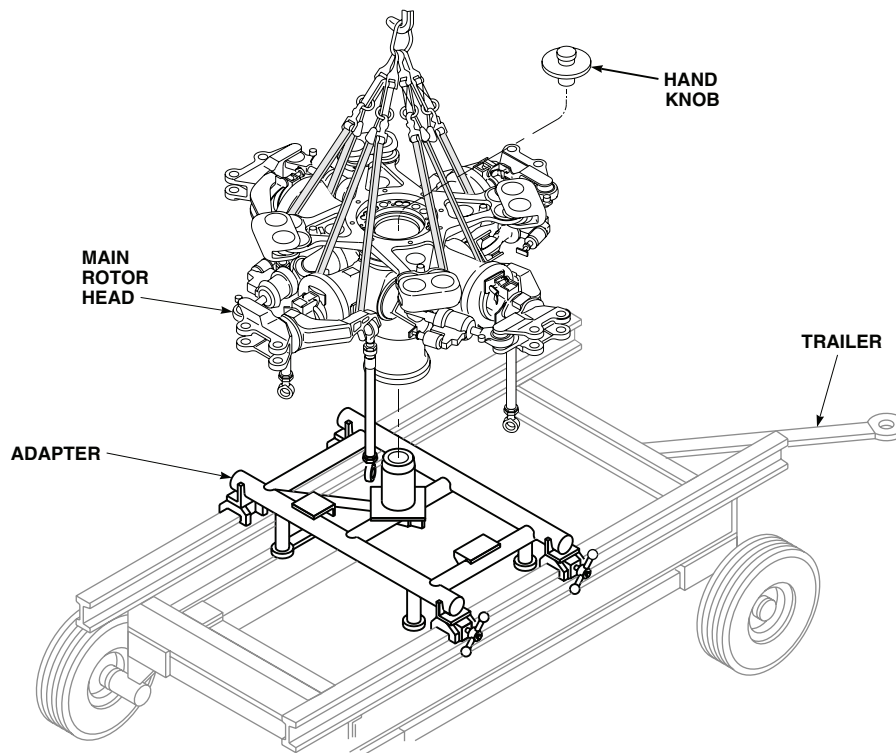
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 4 of 13)

23. Adjust hoisting sling straps before lifting, for even distribution of weight and so rotor head can be lifted straight up. If using maintenance crane, make sure straps are adjusted so main rotor shaft extension will clear main shaft when head is lifted.
24. Remove any slack from hoisting sling using hoist.
25. Attach guide rope.
26. Lift rotor head clear of helicopter.
27. If rotor head becomes seized to main rotor shaft at shaft extension, do this:
 - a. Apply light coat of general purpose lubricating oil, Item 185, WP 1803 00, around split cone seat area.
 - b. Apply tension to hoisting sling. If rotor head fails to become detached, relieve tension on hoisting sling. Locally make main rotor shaft impact tool (WP 1805 00).
 - c. Position impact tool on main rotor shaft with steel rod of impact tool protruding out of shaft extension. Make sure slot in aluminum bottom plate of impact tool lines up with tang on main rotor shaft.

WARNING

Injury to personnel and damage to equipment will result if excessive tension is used and personnel and equipment is not clear of rotor head when applying tension to hoisting sling and impacting main rotor shaft. Rotor head may break free of main rotor shaft suddenly. Do not apply excessive tension to hoisting sling and make sure personnel and equipment are clear from main rotor head when performing this procedure.

- d. Apply tension to hoisting sling.

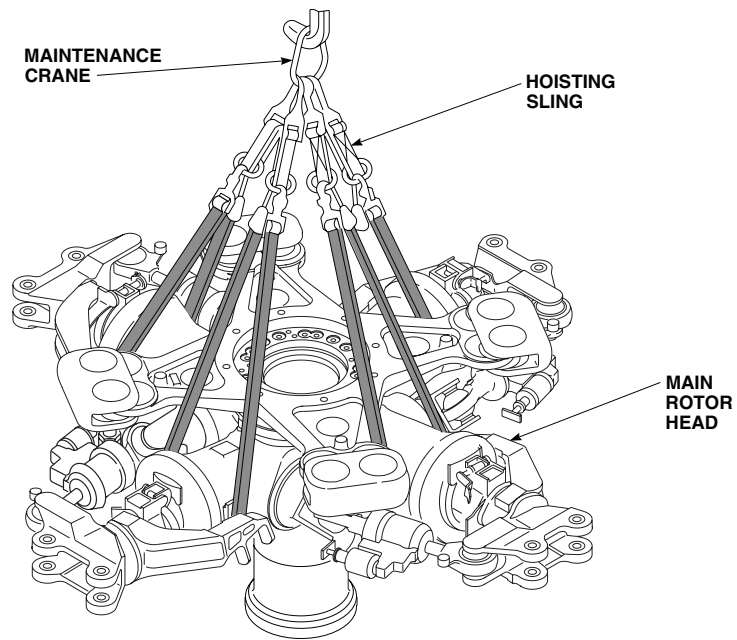
REMOVE MAIN ROTOR HEAD - ContinuedAK2010
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 5 of 13)

- e. Hold weight of impact tool one foot above bottom plate of impact tool. Drop weight onto bottom plate. Repeat until rotor head is free from shaft extension.
- 28. Place main rotor head on adapter (Figure 1, Sheet 4).
- 29. Attach main rotor head to adapter with hand knob.
- 30. Remove bolts, washers, shims, and nuts securing rotating scissors to lower pressure plate.
- 31. Inspect lower pressure plate bolts, rotating scissors bolts and lower pitch control rod bolts (WP 0550 00).
- 32. Remove hoisting sling from main rotor head.

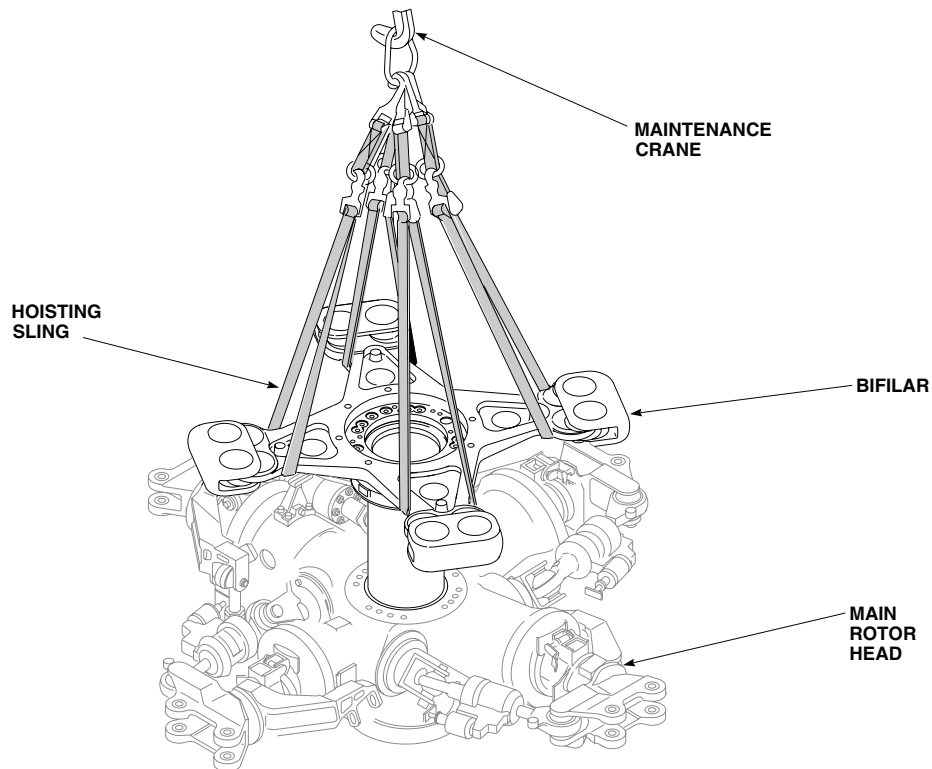
REMOVE MAIN ROTOR HEAD - ContinuedAK2012A
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 6 of 13)

REMOVE SHAFT EXTENSION**WARNING****CSI**

Exposed surfaces of main rotor shaft and shaft extension are critical surfaces which must not be damaged while lowering main rotor hub. Use protective covering as required.

CAUTION

Damage to main rotor head assembly will result if metallic items are used to clean off sealant. Do not use metallic items to remove sealant.

NOTE

- Place index marks on all rotor head components with grease pencil, to make sure parts are positioned correctly during buildup.
 - Do not completely remove bolts until cones are free from shaft extension.
1. Clean any sealant from upper pressure plate and shaft extension using nonmetallic scraper.
 2. Gradually loosen bolts in upper pressure plate in sequence as numbered on pressure plate (Figure 1, Sheet 2). Do not completely remove bolts. Repeat several times as necessary until all bolts are loose.

REMOVE SHAFT EXTENSION - Continued

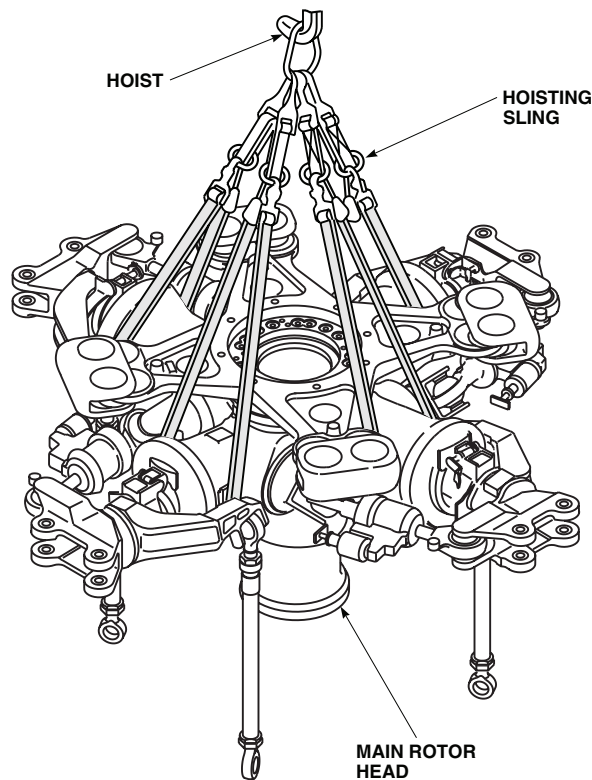
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 7 of 13)

3. Install four jacking bolts, NAS629-26, in upper pressure plate.
4. Turn in jacking bolts until cones are free from hub. Make sure jacking bolts are turned in evenly, and cones are free.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Remove upper pressure plate bolts, washers, and jacking bolts. Clean bolts with technical acetone, Item 5, WP 1803 00.
6. Install hub guide, on shaft extension below main rotor hub, and close latch.
7. Lower upper pressure plate from hub and remove split cone. Make sure serial numbers match.
8. Inspect split cones for nicks and burrs. **NICKS AND BURRS SHALL NOT EXCEED 0.010-INCH DEEP.** Replace split cones if not within limits.
9. Inspect split cones for corrosion pits. **CORROSION PITS SHALL NOT EXCEED 0.010-INCH DEEP.** Replace split cones if not within limits.

REMOVE SHAFT EXTENSION - Continued

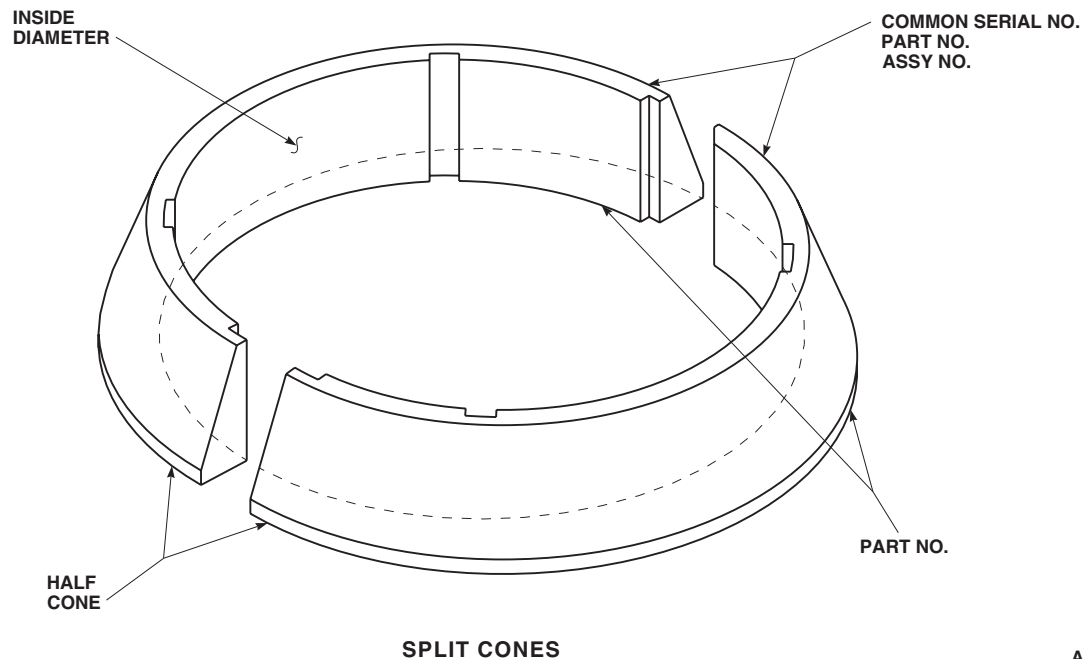
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 8 of 13)

10. Check split cones for fretting. **FRETTING UP TO 0.005-INCH DEEP OVER 50% OF CONTACTING SURFACE IS ACCEPTABLE AFTER REMOVAL OF METAL PICKUP USING ABRASIVE WHEEL, Item 350, WP 1803 00.** Replace split cones if not within limits.
11. Fluorescent penetrant inspect split cones for fractures or cracks using fluorescent penetrant inspection kit. **NO FRACTURES OR CRACKS ALLOWED.** Replace fractured or cracked split cones.
12. Place split cones in bag for reinstallation and tag.
13. Raise upper pressure plate to hub and loosely install bolts.
14. Remove straps from hoisting sling.

CAUTION

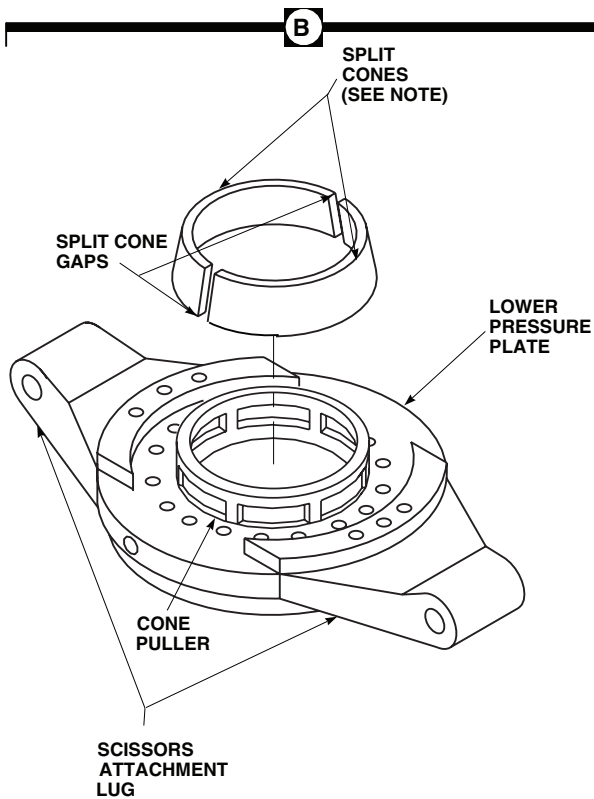
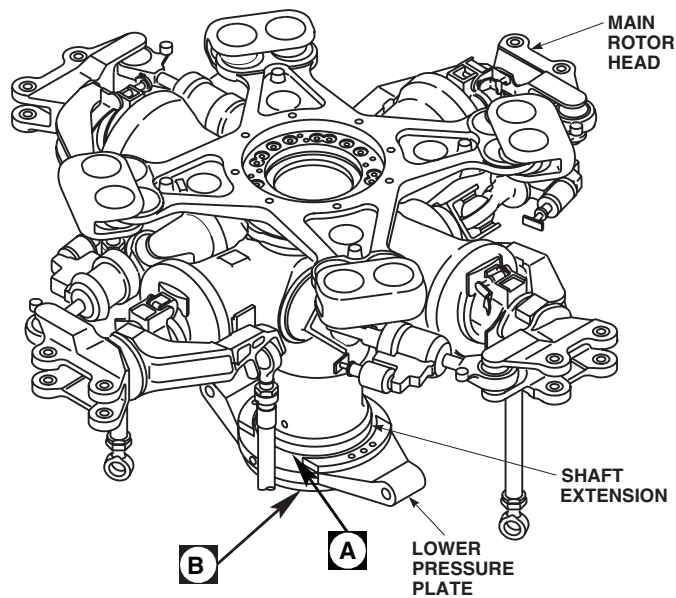
Damage to main rotor head will result if hoisting sling is not properly positioned when hoisting rotor hub. Hoisting sling must be put securely around rotor hub arms.

NOTE

Straps must be adjusted to full length or hub will not completely lower and rest on adapter.

15. Fasten straps around main rotor hub arms. Connect hoist to lifting eye of hoisting sling (Figure 1, Sheet 5).
16. Take up slack until hoisting sling is supporting weight of main rotor hub.

REMOVE SHAFT EXTENSION - Continued



NOTE
SPLIT CONE GAPS SHOULD BE
POSITIONED AS CLOSE TO 90
DEGREES FROM SCISSORS
ATTACHMENT LUG AS POSSIBLE.

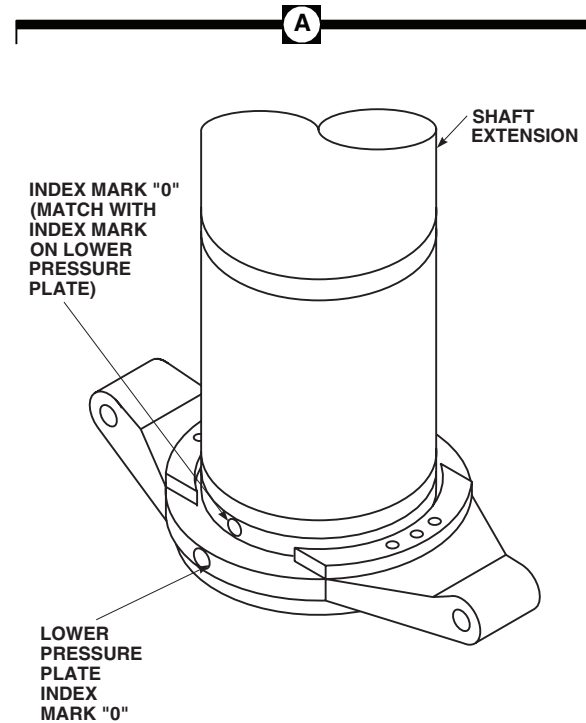


Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 9 of 13)

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REMOVE SHAFT EXTENSION - Continued

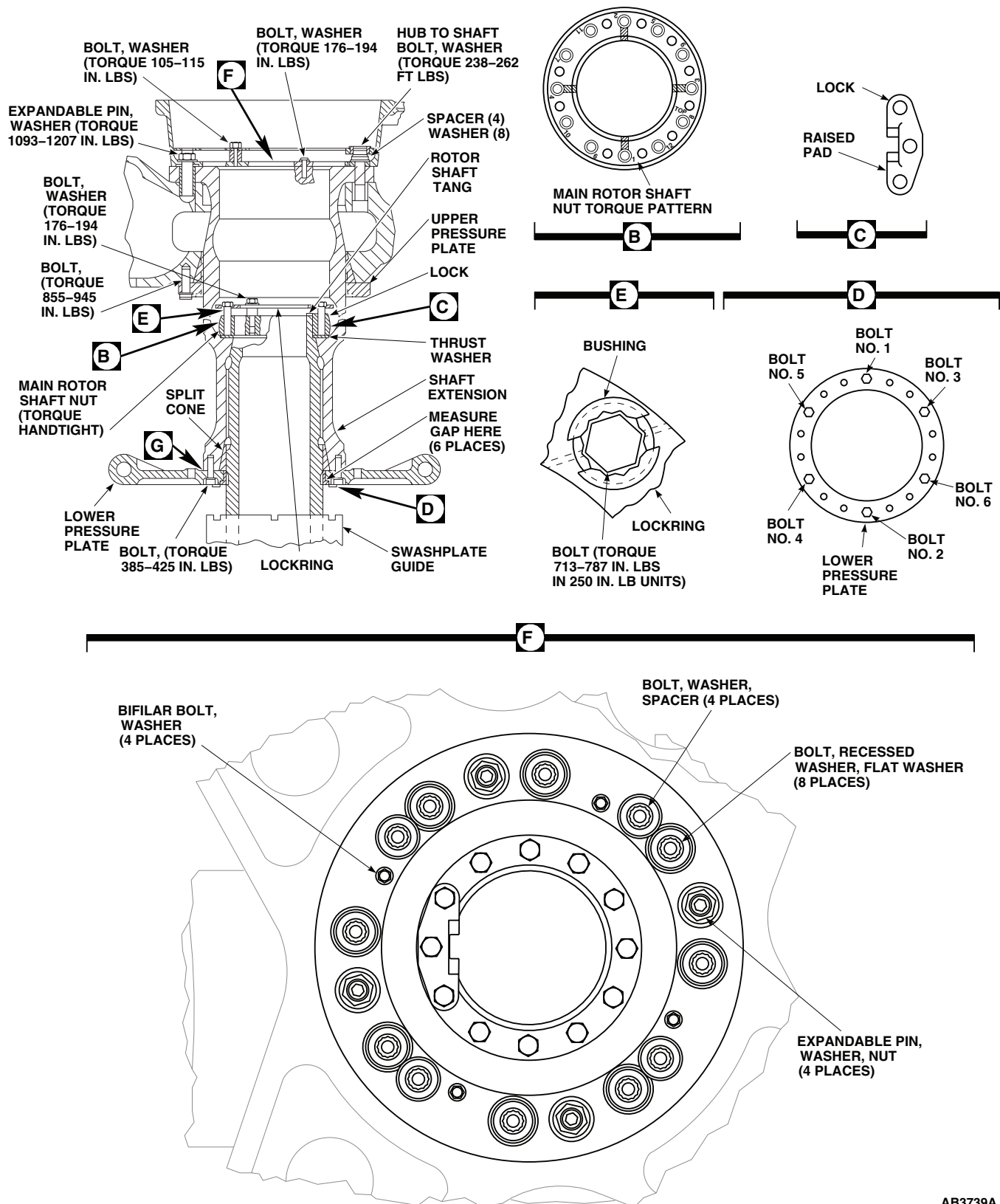
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 10 of 13)

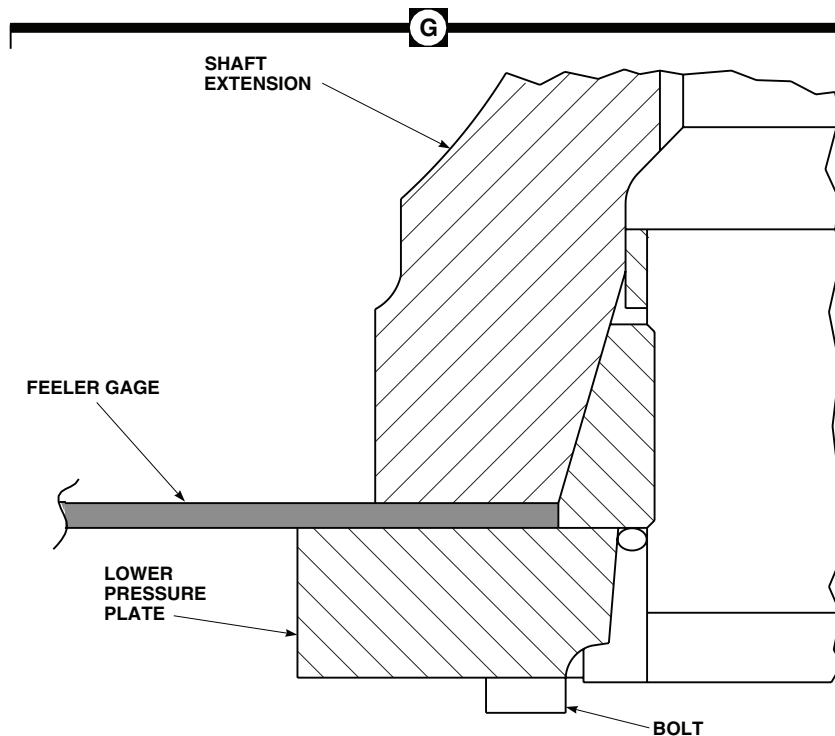
REMOVE SHAFT EXTENSION - ContinuedAK1326 2
SA

Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 11 of 13)

17. If necessary, hold expandable pins from turning with a 5/16 Allen wrench (Figure 1, Sheet 2). Loosen four expandable pins by loosening nuts until top of nuts are flush with top of studs, and then lightly tap with plastic or leather mallet. Use special wrench, to loosen nuts.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when removing hub to shaft extension bolts. After hub to shaft extension bolts are removed, entire hub and blades are supported only by hoisting sling straps and hoist. Always keep area underneath hub clear while lowering.

NOTE

Note location of washers and spacers for reinstallation on bolts.

18. Loosen and remove 12 hub to shaft extension bolts.

CAUTION

Damage to hub will result if not protected when being placed on adapter. Place protective padding on adapter prior to lowering hub.

19. Place protective padding on top of adapter. Slowly lower hub until it rests on top of adapter.
 20. Lower hoisting sling to allow slack in straps, and remove hoisting sling from main rotor hub.
 21. Fasten straps of hoisting sling to arms of bifilar (Figure 1, Sheet 6).

REMOVE SHAFT EXTENSION - Continued

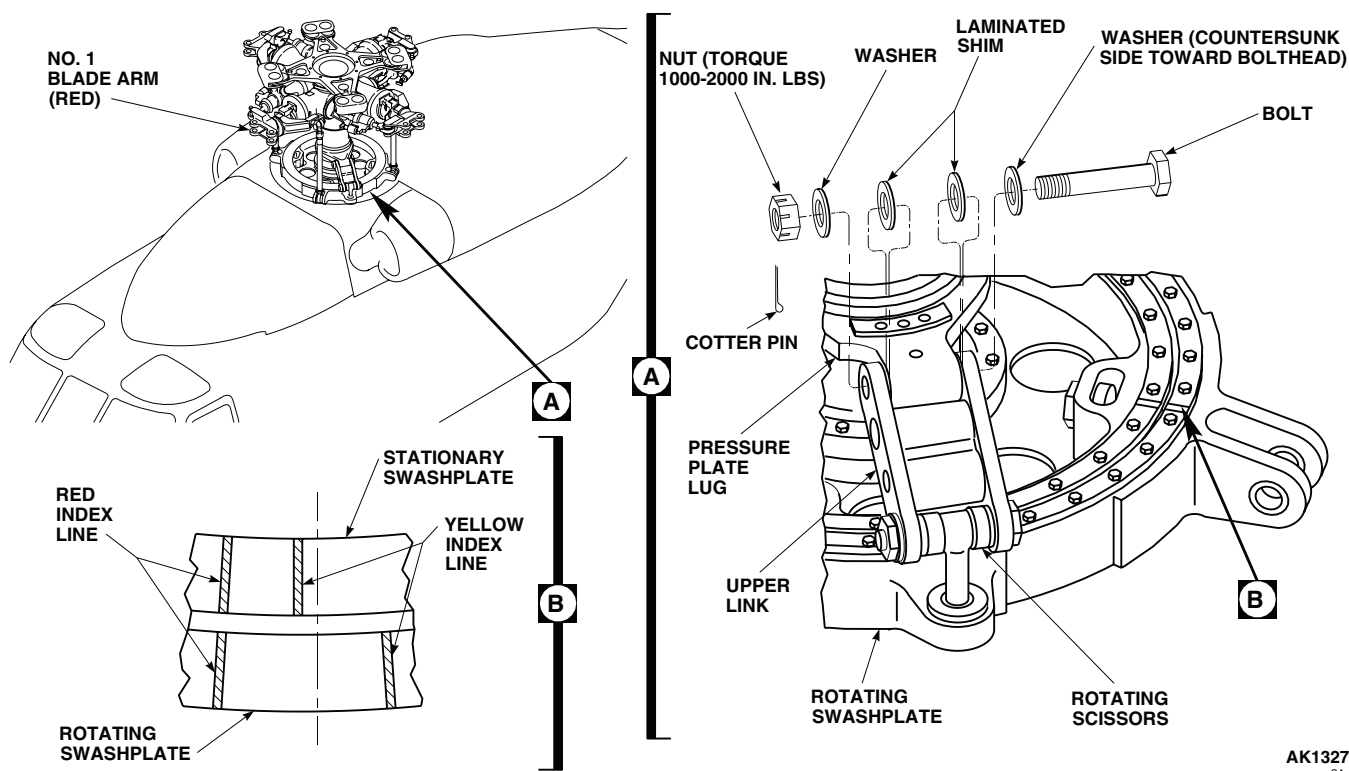


Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 12 of 13)

22. Shorten straps as much as possible after connecting to bifilar.

NOTE

- Hub guide will come up with bifilar and shaft extension.
- Attach guide rope to component before lowering bifilar and shaft extension. Have assistant hold rope to help guide part.

23. Raise and remove bifilar and shaft extension with hoist.

WARNING**CSI**

Exposed surfaces of main rotor shaft extension are critical surfaces which must not be damaged during removal. Use extreme caution when removing, and provide protective covering after removal.

24. Remove bifilar from shaft extension (WP 0573 00).

INSPECT

1. Check for voids in bonded liners and sleeves on main shaft extension. **EDGE VOIDS OF UP TO 25% OF LINER OR SLEEVE WIDTH ARE ACCEPTABLE. VOIDS UP TO 5% OF TOTAL LINER OR SLEEVE AREA ARE ACCEPTABLE.** Replace shaft extension if liners or sleeves are not within limits.

INSPECT - Continued

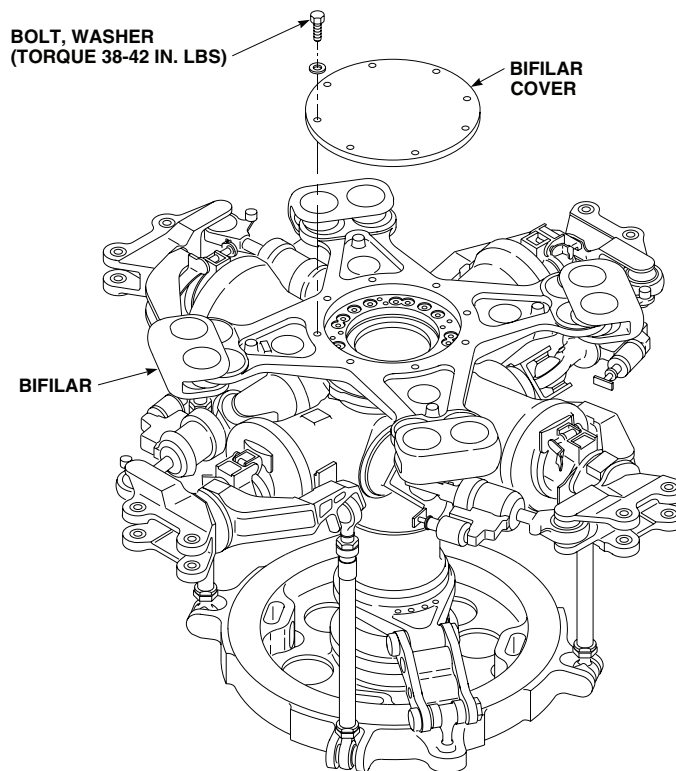
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Figure 1. Replace Main Rotor Head and Shaft Extension. (Sheet 13 of 13)

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean bolts with technical acetone, Item 5, WP 1803 00.
3. Inspect upper pressure plate bolts and four hub to shaft extension bolts (WP 0550 00).

WARNING**CSI**

The shaft extension surfaces are critical surfaces which must not be damaged during installation. After removal of protective covering, use extreme caution when installing shaft extension.

4. Check shaft extension surfaces for damage. **NO DAMAGE ALLOWED.**
5. Inspect/repair main rotor shaft (WP 0636 00).

INSTALL SHAFT EXTENSION

1. Install bifilar on new shaft extension (WP 0573 00).

INSTALL SHAFT EXTENSION - Continued**WARNING****CSI**

Exposed surfaces of shaft extension are critical surfaces which must not be damaged while during installation. Use protective covering as required.

2. Attach hoisting sling to hoist.
3. Attach hoisting sling to bifilar (Figure 1, Sheet 6).
4. Shorten straps as much as possible after attaching to bifilar.

WARNING

Injury to personnel and damage to equipment will result if caution is not taken when hoisting shaft extension and bifilar. Keep area clear when hoisting shaft extension and bifilar.

5. Raise shaft and bifilar and position above hub. Do not lower into hub yet.

NOTE

Make sure hub guide is installed on shaft extension. Prevent shaft extension from turning during lowering, to keep index marks "0" lined up as accurately as possible.

6. **QA** Line up index mark on bifilar arm with No. 1 (red) blade arm. Slowly lower shaft extension and bifilar into hub.
7. **QA** Install hand knob on adapter (Figure 1, Sheet 4).
8. Lower hoisting sling, to allow slack in straps, and remove hoisting sling from bifilar.
9. Fasten straps of hoisting sling around arms of hub (Figure 1, Sheet 5).

WARNING

- Always keep area under main rotor head clear while raising main rotor head, until shaft to hub bolts and expandable bolts are installed.
 - Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
 - Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.
10. Connect hoisting sling to hoist and slowly lift main rotor hub.
 11. **QA** Watch that expansion bolts line up with holes in hub. Stop hoisting when head is firmly against upper flange of shaft extension.

INSTALL SHAFT EXTENSION - Continued**NOTE**

When installing expandable pins, hold each pin from turning initially using 5/16 Allen socket wrench. Do not use Allen socket wrench when applying final torque requirement.

12. **QA** Install four expandable pins. Make sure expandable pins are fully seated. **TORQUE NUTS TO 238 - 262 INCH-POUNDS USING SPECIAL WRENCH (Figure 1, Sheet 10).**
13. Coat shank, threads, and area under head of hub-to-shaft bolts with primer coating, Item 243, WP 1803 00.
14. **QA** Install 12 hub-to-shaft bolts, washers, and four spacers while primer coating is still wet. Use 1.250-inch OD washers under all 12 bolts and 1.500-inch OD washers under 8 bolts (bolts without spacers) (Figure 1, Sheet 10, Detail F). **TORQUE TO 950 - 1050 INCH-POUNDS. STABILIZE TORQUE AT 950 - 1050 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**
15. **QA** Release torque on all four expandable pins.
16. **QA TORQUE 12 HUB-TO-SHAFT BOLTS TO 238 - 262 FOOT-POUNDS IN SEQUENCE. STABILIZE TORQUE AT 238 - 262 FOOT-POUNDS BY TURNING EACH BOLT IN A TIGHTENING DIRECTION UNTIL NO MOTION IS FELT (Figure 1, Sheet 10).**
17. **QA TORQUE NUTS ON FOUR EXPANDABLE PINS TO 1093 - 1207 INCH-POUNDS, USING SPECIAL WRENCH.**
18. Remove hoisting sling from hub arms.
19. Remove hub guide from shaft extension.

WARNING**CSI**

Verification of proper installation and torque of bolt, and presence and proper installation of lockring is a critical characteristic.

20. **QA** Coat shank, threads, and area under head of lockring bolts with primer coating, Item 243, WP 1803 00.
21. **QA** Make sure lockring is properly seated over boltheads. Install upper lockring over hub to shaft bolts, and attach with bolts and washers while primer coating is still wet. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
22. Using aluminum:
 - a. Make four shims measuring 0.060-inch thick x 1 inch wide x 4 inches long.
 - b. Make four shims measuring 0.032-inch thick x 1 inch wide x 4 inches long.

NOTE

Feeler gages may be substituted for aluminum shims.

23. Before torquing upper pressure plate attachment bolts, check threaded inserts as follows:
 - a. **QA** Hold pressure plate up and thread bolts until heads are 0.030-inch away from surface of pressure plate.
 - b. **QA** Using dial type torque wrench, unthread each bolt separately. **MINIMUM BREAKAWAY TORQUE IS 18 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 150 INCH-POUNDS.** If torque is not within these limits, replace insert (WP 0550 00).

INSTALL SHAFT EXTENSION - Continued**NOTE**

The upper pressure plate has an offset bolthole used for alignment. Make sure offset bolthole is properly aligned prior to installation.

24. Install upper pressure plate as follows:

- a. Remove upper pressure plate bolts and washers (Figure 1, Sheet 10).
- b. Wipe inside diameter of upper split cones with machinery wiping towel, Item 344, WP 1803 00.
- c. **QA** Lightly coat inside diameter of upper cones, and upper part of shaft extension where upper cones seat, with molybdenum disulfide grease, Item 138, WP 1803 00, or aircraft grease, Item 132, WP 1803 00. Outside tapered surface of all cones shall be left dry.
- d. **QA** Position upper split cones on upper pressure plate with equal gap between cones.
- e. **QA** Lift up upper pressure plate and cones to seat cones in hub.

CAUTION

Damage to upper pressure plate and hub will result if wrong length attachment bolts are used. Make sure correct length bolts are used to install upper pressure plate.

- f. **QA** Coat shank, threads, and area under head of pressure plate bolts with primer coating, Item 243, WP 1803 00.
- g. **QA** Install four bolts and washers while primer coating is still wet in holes numbered 3, 4, 5, and 6 to hold pressure plate in place.
- h. **QA** Install remaining bolts and washers in pressure plate while primer coating is still wet.

NOTE

The purpose of the shims is to keep the pressure plate level during installation. Drag on the shims may or may not be reached prior to step 24. l.

- i. Insert 0.092-inch thickness of shims at four places, equally spaced between pressure plate and bottom of hub.
- j. Tighten bolts in sequence numbered on pressure plate while checking for equal drag on shims. Shims should keep plate and hub parallel.
- k. Remove 0.032-inch thick shims. Tighten bolts in sequence while checking for equal drag on shims.
- l. **QA** Remove all shims. **TORQUE BOLTS TO 550 INCH-POUNDS, FOLLOWING NUMBERED SEQUENCE ON PRESSURE PLATE.**
- m. **QA TORQUE BOLTS TO 855 - 945 INCH-POUNDS IN 200 INCH-POUND UNITS, FOLLOWING NUMBERED SEQUENCE ON PRESSURE PLATE. STABILIZE TORQUE AT 855 - 945 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**
- n. **QA** Measure gap between pressure plate and hub at four equally spaced places 90 degrees from each other. **ALL FOUR GAPS MUST BE WITHIN 0.030-INCH OF EACH OTHER. IF NOT, REMOVE BOLTS AND RETORQUE.**

25. Seal areas around upper pressure plate bolts with sealing compound, Item 273, WP 1803 00.

INSTALL MAIN ROTOR HEAD

1. Place a strap of hoisting sling around each arm of main rotor hub (Figure 1, Sheet 7).
2. Attach hoisting sling to hoist or maintenance crane.

INSTALL MAIN ROTOR HEAD - Continued

3. Attach guide rope.
4. Adjust hoisting sling straps before lifting, for even distribution of weight.
5. Wipe main rotor shaft and rotor head mating surfaces clean using a clean, dry machinery wiping towel, Item 344, WP 1803 00.
6. Inspect main rotor shaft thread for crossed, stripped, or flattened threads. **NONE ALLOWED.** Replace main module if threads are damaged.

NOTE

- Split cones are a matched set and cannot be replaced individually.
 - Serial number, part number, and assembly numbers may be found on either upper or lower surface of cone half.
7. Check serial number on split cones to make sure they are a matched set. If numbers do not match, replace split cones (Figure 1, Sheet 7).
 8. Wipe cones clean using machinery wiping towel, Item 344, WP 1803 00.
 9. Check cone pullers for corrosion. **CORROSION UP TO 0.020-INCH DEEP CAN BE BLENDED OUT. REPLACE CONE PULLERS IF CORROSION IS DEEPER THAN 0.020-INCH DEEP.**
 10. Touch up finish on reworked areas using epoxy primer coating kit, Item 120, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
 11. **QA** Coat inside diameter of split cones lightly with molybdenum disulfide grease, Item 138, WP 1803 00, or aircraft grease, Item 132, WP 1803 00. Do not put grease on outside tapered surface of split cones. Outside surface shall be left dry.
 12. **QA** Coat main rotor shaft splines lightly with molybdenum disulfide grease, Item 138, WP 1803 00, or aircraft grease, Item 132, WP 1803 00.
 13. Check lower pressure plate attachment bolt threaded inserts in shaft extension as follows:
 - a. Remove countersunk washers from bolts (Figure 1, Sheet 10).
 - b. **QA** Thread in bolts until head of bolts are 0.580-inch away from bottom of shaft extension.
 - c. **QA** Using dial type torque wrench, unthread each bolt individually. **MINIMUM BREAKAWAY TORQUE IS 9.5 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 80 INCH-POUNDS.**
 - d. If torque is not within limits, check condition of bolt for corrosion and wear. If corrosion or wear is evident, replace bolt, and clean insert.
 - e. If breakaway or removal torque is still not acceptable, replace insert (WP 0549 00).
 - f. Remove bolts from shaft extension.
 - g. Install countersunk washers on bolts with countersunk side of washer facing bolt head.

INSTALL MAIN ROTOR HEAD - Continued**WARNING****CSI**

Exposed surfaces of main rotor shaft and shaft extension are critical surfaces which must not be damaged during installation of main rotor head. After removal of protective covering, use extreme caution when installing main rotor head.

14. If cone puller is loose, apply a coat of sealing compound, Item 275, WP 1803 00, to outside diameter and flange face of puller. Make sure cone puller is positioned to allow split cones gap to be 90 degrees away from scissors attachment points.
15. **QA** Place lower pressure plate, split cones and cone puller over main rotor shaft. Position split cones gap to be 90 degrees to scissors attachment points (Figure 1, Sheet 9). Lower pressure plate onto main transmission.
16. Check main rotor shaft for thread protection. If there is none, tape cardboard or machinery wiping towels, Item 344, WP 1803 00, around threads with pressure sensitive adhesive tape, Item 322, WP 1803 00.

NOTE

Main rotor shaft thrust washer, nut, and bolts must be installed and torqued while primer coating is wet.

17. Apply a light coat of epoxy primer coating, Item 120, WP 1803 00, to thread relief and exposed area of tapered cone seat of the main rotor shaft. **DO NOT APPLY PRIMER TO THREADS OF SPLINES ON MAIN ROTOR SHAFT.** Allow epoxy primer to dry.
18. Hoist main rotor head to top of main rotor shaft.
19. **QA** Carefully lower main rotor head down main rotor shaft. Lower head so that No. 1 blade arm (red) is in the left front position.
20. Remove thread protection from main rotor shaft.
21. Lightly coat threads of main rotor shaft nut with antiseize compound, Item 49, WP 1803 00.
22. Apply sealing compound, Item 267, WP 1803 00, to thread relief up to the bottom thread of the main rotor shaft. **DO NOT APPLY SEALING COMPOUND TO TAPERED CONE.**

NOTE

- If one side of thrust washer was repaired, washer must be installed with repaired side facing down. Refer to, INSPECT, in this work package.
 - If both sides of thrust washer were repaired, washer must be installed with side of deepest repair facing down. Refer to, INSPECT in this work package.
23. **QA** Install thrust washer within 15 minutes of sealing compound application on main rotor shaft. Make sure thrust washer is properly seated and centered in the shaft extension (Figure 1, Sheet 10). Spread sealing compound over exposed area of tapered cone seat and up to thrust washer. Apply additional sealing compound, Item 267, WP 1803 00, as required.
 24. **QA** Install main rotor shaft nut. Make sure painted white stripes are facing up (Figure 1, Sheet 10, Detail B). **HANDTIGHTEN NUT, THEN BACK OFF NUT UP TO 1/4-TURN TO LINE UP NEAREST WHITE STRIPE ON NUT WITH TANG ON ROTOR SHAFT.**

INSTALL MAIN ROTOR HEAD - Continued**WARNING****CSI**

Verification of proper hardware, lockring installation, and torque of bolts is a critical characteristic.

NOTE

Raised pads on lock shall face up.

25. **QA** Place lock on top of nut and line up notch in lock with tang of rotor shaft.
26. Coat shank, threads, and area under head of shaft nut bolts with primer coating, Item 243, WP 1803 00.
27. **QA** Install bolts on nut while primer coating is still wet. **TIGHTEN BOLTS UNTIL A SLIGHT RESISTANCE IS FELT AS BOLTS CONTACT THRUST WASHER. TORQUE BOLTS TO 713 - 787 INCH-POUNDS IN 250 INCH-POUND STEPS, FOLLOWING NUMBERED SEQUENCE ON NUT. STABILIZE TORQUE ON ALL BOLTS AT EACH 250 INCH-POUND STEP BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT. STABILIZE TORQUE AT 713 - 787 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT (Figure 1, Sheet 10, Detail E).**
28. **QA** Check that tang on rotor shaft is flush to above tang on lock.
29. Wipe off any primer coating that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.
30. Touch up paint on bolts as necessary (WP 0381 00 and TM 55-1500-345-23).

NOTE

Lockring must be installed over bolts on main rotor shaft nut before corrosion preventive compound dries.

31. Apply corrosion preventive compound, Item 106, WP 1803 00, to main rotor shaft nut and to area inside shaft extension.
32. Coat shank, threads, and area under head of lockring bolts with primer coating, Item 243, WP 1803 00.

CAUTION

Damage to lockring will result if it is not seated properly prior to torquing bolts. Lockring must sit flat on main rotor shaft nut. Turn bushings in lockring to allow lockring to seat over bolts.

33. **QA** Install lockring over bolts on main rotor shaft nut (Figure 1, Sheet 10, Detail E). Make sure two shorter spacers of lockring are resting on lock. Install bolts and washers while primer coating is still wet. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS** (Figure 1, Sheet 10). Wipe off any primer coating that has seeped out from under boltheads using machinery wiping towels, Item 344, WP 1803 00.
34. **QA** Line up "0" index marks on lower pressure plate and shaft extension (Figure 1, Sheet 9).

INSTALL MAIN ROTOR HEAD - Continued**NOTE**

Due to installation of cone pullers the 90 degree positioning of split cone gaps from scissors attachment lugs may not be accurately attained. Split cone gaps should be positioned as close to 90 degrees from scissors attachment lugs as possible.

35. Install split cones on lower pressure plate with gaps (typical two) positioned approximately 90 degrees away from scissors attachment lugs (Figure 1, Sheet 9, Detail B).
36. Torque bolts on lower pressure plate as follows:
 - a. Coat shank, threads, and area under head of pressure plate bolts with primer coating, Item 243, WP 1803 00.
 - b. **QA** Install bolts and washers in two holes while primer coating is still wet, 90 degrees from cone slots.
TIGHTEN BOLTS FINGERTIGHT.
 - c. **QA** Install bolts and washers in third hole on each side of first two bolts. **TIGHTEN BOLTS FINGERTIGHT.**
 - d. **QA TIGHTEN THESE SIX BOLTS IN REVERSE SEQUENCE AS MARKED ON LOWER PRESSURE PLATE TO 150 INCH-POUNDS. MAKE SURE TORQUE HAS STABILIZED.**

NOTE

Number sequence on pressure plate may have to be shifted so that largest gap can be torqued first. For example, if largest gap is at bolt hole No. 17, torque sequence is: 17, 18, 1, 2, etc.

- e. **QA** Using feeler gage, measure gap between lower pressure plate and shaft extension at each of six bolts. **GAPS MUST MEASURE WITHIN 0.025-INCH OF EACH OTHER.** If gaps are more than 0.025-inch of each other, loosen bolts and repeat step 35. d and step 35. e (Figure 1, Sheet 11, Detail G).
- f. **QA STARTING AT BOLT WHERE GAP IS LARGEST, TORQUE THESE SIX BOLTS TO 285 - 315 INCH-POUNDS.**
- g. Using feeler gage, measure gap between lower pressure plate and shaft extension at each of six bolts.
- h. **QA** Install remaining bolts and washers. **STARTING AT BOLT WHERE GAP IS LARGEST, TORQUE ALL 18 BOLTS, IN SEQUENCE, TO 285 - 315 INCH-POUNDS, IN 150 INCH-POUND STEPS. STABILIZE TORQUE AT EACH 150 INCH-POUND STEP.**
- i. Using feeler gage, measure gap between lower pressure plate and shaft extension at each bolt.
- j. **QA STARTING AT BOLT WHERE GAP IS LARGEST, TORQUE ALL 18 BOLTS IN SEQUENCE, TO 385 - 425 INCH-POUNDS. MAKE SURE TORQUE HAS STABILIZED AT FINAL TORQUE.**
- k. Recheck torque on all 18 bolts. Torque should be equal on all bolts.
- l. **QA** Measure gap between pressure plate and shaft extension. **SMALLEST GAP ALLOWED IS 0.005-INCH. A GAP DIFFERENCE OF NOT OVER 0.035-INCH IS ALLOWED BETWEEN SMALLEST AND LARGEST GAP AREAS.**
- m. If above gaps are not met, lower pressure plate and repeat procedure.
- n. Wipe off any primer coating that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.
- o. Touch up paint on boltheads as necessary (WP 0381 00 and TM 55-1500-345-23).

INSTALL MAIN ROTOR HEAD - Continued**NOTE**

Pressure plate torque check is required between 9 to 11 flight hours.

p. To ease maintenance during torque check, do this:

- (1) Apply 2 coats of corrosion preventive compound, Item 106, WP 1803 00, between pressure plate and shaft extension and around locking ring bolt heads.
- (2) Inspect area where corrosion preventive compound, Item 106, WP 1803 00, every three days and reapply as required. After each torque check reapply two coats of corrosion preventive compound, Item 106, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- (3) After torque check has stabilized, remove corrosion preventive compound using technical acetone, Item 5, WP 1803 00.
 - (4) Seal gap between pressure plate and shaft extension, and areas around lockring and pressure plate bolt heads with sealing compound, Item 273, WP 1803 00.
37. Make sure gust lock is disengaged.
 38. Turn swashplate until red index lines on rotating and stationary swashplate line up (Figure 1, Sheet 12, Detail B).
 39. Turn main rotor hub until rotating scissors line up with lugs on pressure plate.
 40. Turn hub and swashplate together until No. 1 blade arm (red) is positioned over left front of helicopter (Figure 1, Sheet 12, Detail B).
 41. Engage gust lock.
 42. Make sure red index lines are lined up, scissors are lined up with pressure plate lugs, and No. 1 blade arm is correctly positioned.
 43. Temporarily install rotating scissors upper link to pressure plate lug using bolt and washers. Do not install nut at this time (Figure 1, Sheet 12, Detail A).
 44. Using feeler gage, measure gap between upper link and pressure plate lug, on both sides of lug. Record measurement.
 45. Peel laminated shims equally on both sides of link to obtain 0.002 to 0.004-inch overall gap on each side of lug.

WARNING**CSI**

Verification of proper hardware installation, torque, and safety installation are critical characteristics.

46. **QA** Connect upper link to lower pressure plate lug using bolt, washers, shims and nut. Make sure countersunk side of washer faces bolt head, and bolt head is in direction of rotation.
47. **QA TORQUE NUT TO 1000 - 2000 INCH-POUNDS.** Install cotter pin. Apply torque stripe to nut and bolt only (WP 0381 00 and TM 55-1500-345-23).

INSTALL MAIN ROTOR HEAD - Continued

48. Disengage gust lock.
49. Connect pitch control lower rod end to swashplate (WP 0563 00).
50. If helicopter is equipped, install main rotor blade deice tube and distributor and slipping assembly (WP 1220 00).
51. Coat shank, threads, and area under head of bifilar bolts with primer coating, Item 243, WP 1803 00.
52. **QA** Place bifilar cover over bifilar and install bolts and washers while primer coating is still wet (Figure 1, Sheet 13). **TORQUE BOLTS TO 38 - 42 INCH-POUNDS.**
53. Wipe off any primer coating that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.
54. Seal edge of bifilar cover and areas around bifilar boltheads with sealing compound, Item 273, WP 1803 00.
55. Touch up paint on boltheads as necessary (WP 0381 00 and TM 55-1500-345-23).
56. Install main rotor blades (WP 0576 00).
57. Perform main rotor rig check (WP 1037 00).
58. Make sure area is clean and free of foreign objects.
59. Track and balance main rotor blades (TM 1-6625-724-13&P).
60. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure main rotor head torque check is shown as due 9 to 11 flight hours after installation (WP 1691 00).
61. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ROTOR SYSTEM

MAIN ROTOR HUB

This WP supersedes WP 0548 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Fluorescent Penetrant Inspection Kit, XMA 101
Nonmetallic Scraper
Main Rotor Hub Flat Plate Tool, Item 272,
WP 1805 00
Powertrain Repairer's Toolkit, SC 5180-99-B13
Syringe, 215955

Parting Compound, Foundry Mold, Item 212,
WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00
WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Alcohol, Denatured, Item 48, WP 1803 00
Cloth, Abrasive, Item 79, WP 1803 00
Paper, Abrasive, Item 210, WP 1803 00

INSPECT/REPAIR

NOTE

- In the following steps, do all blending using abrasive cloth, Item 79, WP 1803 00. Blend out all reworked areas to a 1/2-inch radius.
 - Record all repairs on aircraft component historical record, DA Form 2408-16.
 - Record all component reduction in life limits due to repair, on DA Form 2410 I/A/W Section III repair/overhaul/gain data.
1. Check History Recorder, Component, Module Record, DA Form 2408-16 to make sure area has not been previously reworked.

WARNING

CSI

Inspection holes at hub Zone I of inspection area (Figure 1, Detail C) are a critical characteristic area of the hub.

2. Check hub Zone I for nicks and gouges (Figure 1, Detail C).
 - a. Check inspection holes for damage penetrating through paint into base metal. **MAXIMUM ALLOWABLE BASE METAL DAMAGE IS 0.0003-INCH.** Replace hub if damage penetrating through paint into base metal is beyond limits.
 - b. Visually check hub Zone I for cracks. **NO CRACKS ALLOWED.** Replace cracked hub.

INSPECT/REPAIR - Continued

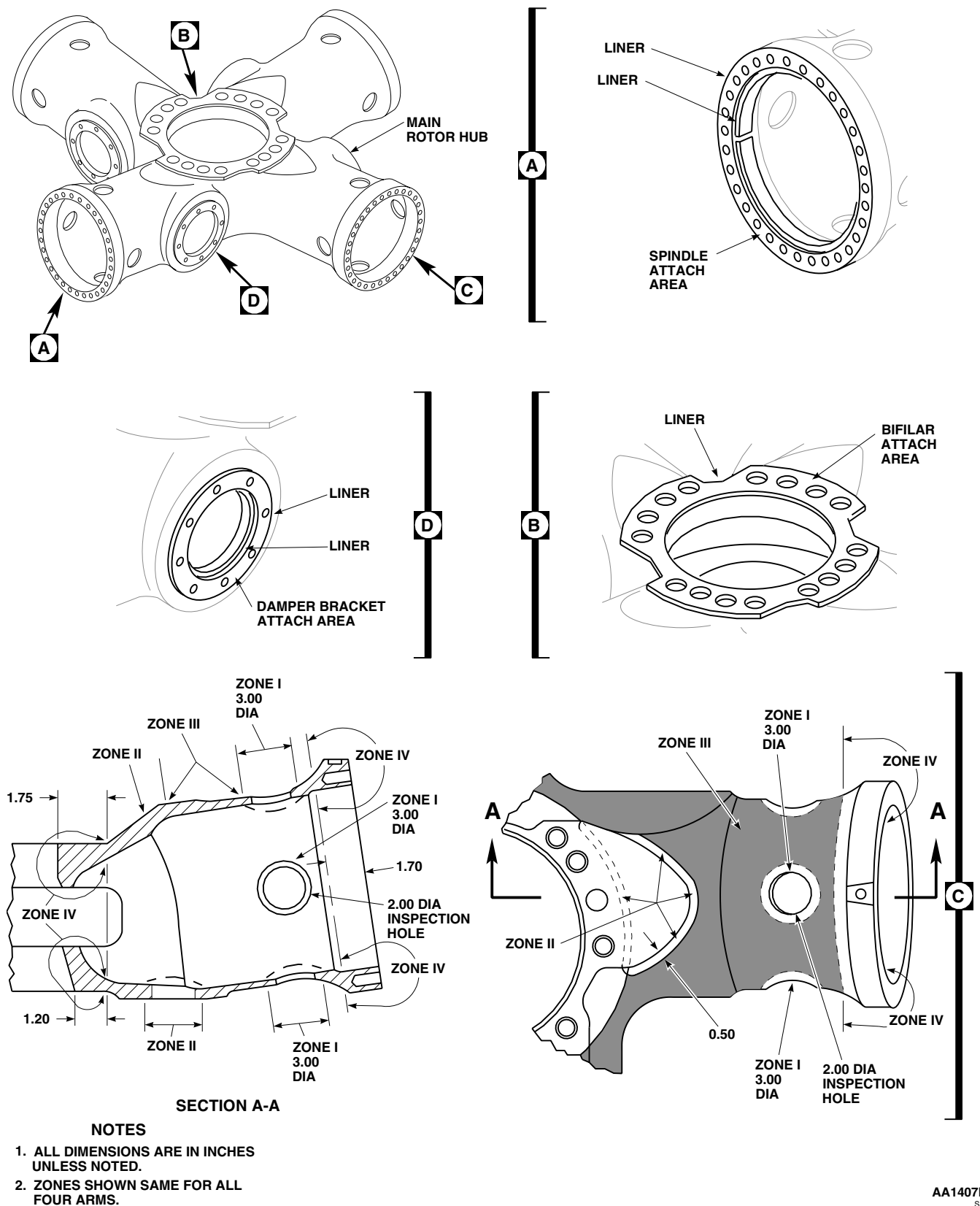


Figure 1. Inspect/Repair Main Rotor Hub.

INSPECT/REPAIR - Continued

- c. If cracks in hub Zone I are suspected, fluorescent penetrant inspect, Type I, Method C, area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked hub.
3. Check hub Zone II for nicks and gouges.
 - a. **DAMAGE UP TO 0.010-INCH DEEP CAN BE BLENDED OUT. DAMAGED AREA MUST BE LESS THAN 15% OF AREA. REPLACE REPAIRED HUB AT THE NEXT PMI INSPECTION REQUIRING ROTOR SYSTEM DISASSEMBLY, NOT TO EXCEED 720 FLIGHT HOURS.**
 - b. Fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked hub immediately.
4. Check hub Zone III for nicks and gouges.
 - a. **DAMAGE UP TO 0.020-INCH DEEP, 0.500-INCH WIDE AND LONG CAN BE BLENDED OUT. DAMAGE MUST BE LESS THAN 15% OF CIRCUMFERENCE OF PART. REPLACE REPAIRED HUB WITHIN 2000 FLIGHT HOURS OF REPAIR.**
 - b. Fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked hub.
5. Check hub Zone IV and upper and lower pressure plate for nicks and gouges.
 - a. **CORNER DAMAGE UP TO 0.010-INCH DEEP CAN BE BLENDED OUT, EXCEPT FOR LUG AREAS. IN LUG AREAS, DAMAGE UP TO 0.050-INCH IS ALLOWED.**
 - b. **ON OPEN AREAS (NONMATING SURFACES) SURFACE DAMAGE UP TO 0.040-INCH DEEP CAN BE BLENDED OUT. DAMAGED AREA MUST BE LESS THAN 20% OF CIRCUMFERENCE.**
 - c. **ON MATING AREAS, SURFACE DAMAGE UP TO 0.020-INCH DEEP CAN BE BLENDED OUT. DAMAGED AREA MUST BE LESS THAN 10% OF SURFACE AREA AFTER BLENDED OUT AND REMOVING ANY METAL PICKUP.**
 - d. **SURFACE DAMAGE ON BOTTOM OF HUB DUE TO JACKING BOLTS BEARING ON HUB SURFACE IS ACCEPTABLE.**
 - e. Fluorescent penetrant, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked hub.
6. Touch up finish all reworked areas (WP 0381 00 and TM 55-1500-345-23).

NOTE

Fretting on bonded liners is considered normal wear and is acceptable.

7. Check cone seat surfaces on shaft extension and hub Zone IV for fretting. **FRETTING UP TO 0.002-INCH DEEP OVER ENTIRE CONTACTING SURFACE IS ACCEPTABLE AS LONG AS THERE IS NO METAL PICKUP THAT CAN NOT BE REMOVED WITH ABRASIVE WHEEL, Item 350, WP 1803 00.**
8. **VISUALLY INSPECT LINERS FOR EDGE VOIDS (Figure 1). VOIDS UP TO 25% OF LINERS WIDTH OR AREA ARE ACCEPTABLE.**
9. **VOIDS GREATER THAN 25% BUT LESS THAN 50% OF WIDTH OR AREA OF BONDED EXTERNAL SPINDLE, DAMPER BRACKET, OR BIFILAR LINERS, REQUIRE REPAIR (REPAIR BONDED LINER). VOIDS GREATER THAN 50% OF WIDTH OR AREA OF BONDED EXTERNAL SPINDLE, DAMPER BRACKET, AND BIFILAR LINERS REQUIRE REPLACEMENT (ROTOR HUB LINER REPLACEMENT). VOIDS GREATER THAN 25% OF WIDTH OR AREA OF INTERNAL SPINDLE AND DAMPER BRACKET LINER ARE CAUSE FOR HUB REPLACEMENT.**
10. Record time, type, and location of rework in History Recorder, Component, Module Record, DA Form 2408-16.

REPAIR BONDED LINER**WARNING**

Denatured alcohol is flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/faceshield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flame, sparks, or other sources of ignition.

1. Clean void areas using denatured alcohol, Item 48, WP 1803 00. Inject denatured alcohol into void areas using syringe.
2. If possible, sand voided surfaces, using abrasive paper, Item 210, WP 1803 00. Clean again with denatured alcohol, Item 48, WP 1803 00, and allow to dry.
3. Mix adhesive, Item 11, WP 1803 00, following manufacturer's directions. Inject mixed adhesive into void areas, using syringe.
4. Bond liner in place by using a main rotor hub flat plate tool, Item 272, WP 1805 00, with a thin film of foundry mold parting compound, Item 212, WP 1803 00, against liner. Install plate using the components (spindle, damper bracket, or bifilar) bolts and washers.
5. Allow to cure for 2 to 3 days at room temperature.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****MAIN ROTOR HUB THREADED INSERTS**

This WP supersedes WP 0549 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13
Soldering Iron, 1102-0004
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Sealing Compound, Item 275, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Primer, Sealing Compound, Item 245, WP 1803 00

REMOVE**CAUTION**

Temperatures above 240°F(116°C) may degrade bond integrity of adjacent bonded liners. After removal of inserts, inspect bonded liners. Refer to, INSPECT, in this work package. For repair, refer to, INSTALL, in this work package.

NOTE

If it is difficult to remove inserts, 79698-820, between spindle and hub, bring temperature of insert up to 300°F(149°C) to 450°F(232°C) by using soldering iron inserted into insert. Try to remove inserts periodically during temperature increase.

Remove damaged insert using hex bit from screw driver bit set (Figure 1).

INSPECT**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean any old sealant from hole using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Dry area thoroughly.
2. Check threaded hole where new insert will be installed as follows:

INSPECT - Continued

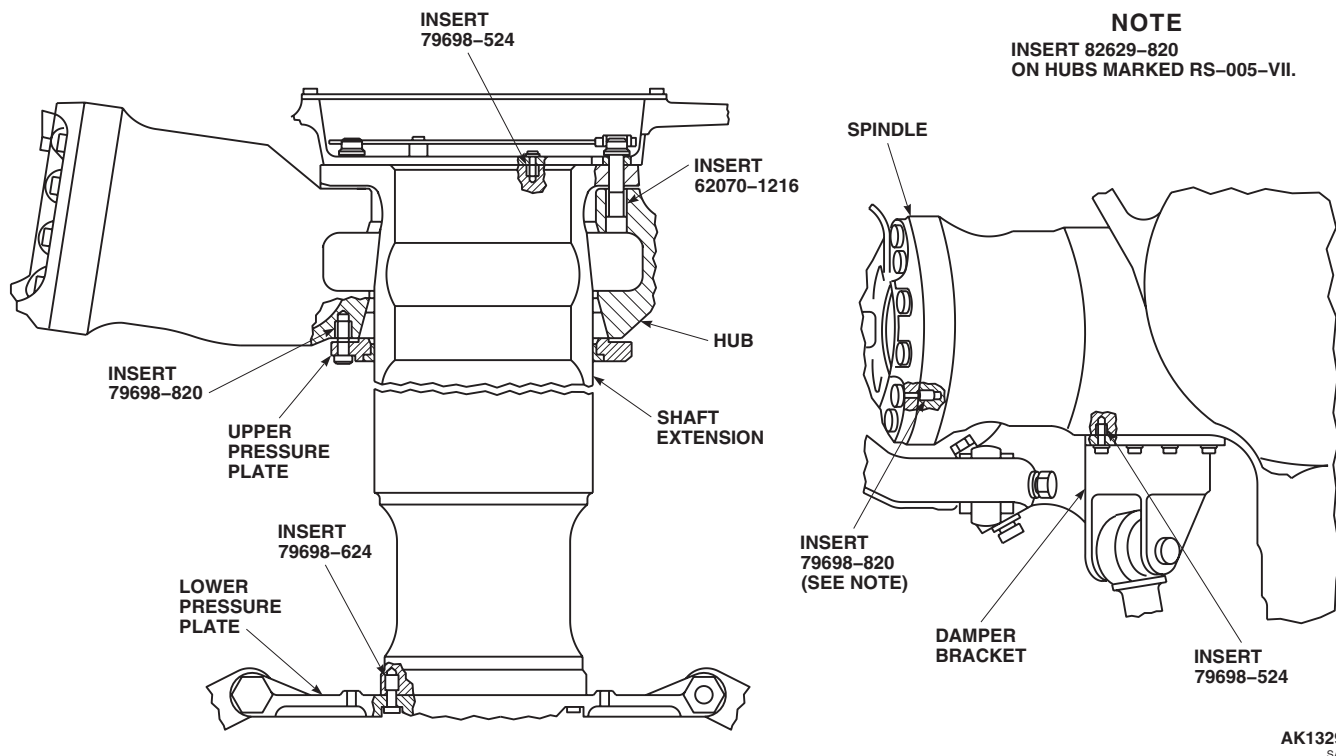


Figure 1. Replace Main Rotor Hub Threaded Inserts.

- a. Install insert into threaded hole using hex bit up to insert locking feature. **INSERT MUST THREAD IN BY HAND WITHOUT USING TORQUE WRENCH.**
- b. Replace hub if insert will not thread in by hand.
- c. Remove insert from threaded hole.

INSTALL

1. Grind 5/16-inch hex bit to fit insert, 79698-524, and 3/8-inch hex bit to fit insert, 79698-624.
2. Treat outside threads of replacement insert with sealing compound primer, Item 245, WP 1803 00. Apply a thin coat of sealing compound, Item 275, WP 1803 00, to first two threads of insert only.
3. **QA** Install insert in housing with hex bit from screwdriver bit set and torque wrench. See Table 1 for correct torque limits and depth of insert.

INSTALL - Continued**Table 1. Main Rotor Head Inserts.**

INSERT PART NO.	MINIMUM TORQUE WHEN INSTALLING INSERT	MAXIMUM TORQUE WHEN INSTALLING INSERT	INSTALLATION DEPTH (INCHES)	INSERT LOCATION	REMOVAL/ INSTALLATION TOOL KEY SIZE
79698-524	66 inch-pounds	500 inch-pounds	0.000 to 0.031 below face of hub (NOTE 3)	Damper bracket to hub	0.264-inch
79698-524	66 inch-pounds	483 inch-pounds	0.000 to 0.031 below shaft extension surface	Bifilar to shaft extension	0.264-inch
79698-820 (NOTE 2)	165 inch-pounds	790 inch-pounds (NOTE 1)	0.000 to 0.031 below face of hub (NOTE 3)	Spindle to hub	7/16-inch
82629-820	165 inch-pounds	790 inch-pounds (NOTE 1)	0.000 to 0.031 below face of hub (NOTE 3)	Spindle to hub	7/16-inch
79698-820	165 inch-pounds	790 inch-pounds (NOTE 1)	0.000 to 0.031 below hub surface	Upper pressure plate to hub	7/16-inch
62070-1216	432 inch-pounds	1200 inch-pounds	0.000 to 0.045 below face of hub (NOTE 3)	Hub to shaft	5/8-inch
79698-624	90 inch-pounds	660 inch-pounds	0.000 to 0.031 below surface of shaft extension	Lower pressure plate to shaft extension	0.328-inch
79698-524	66 inch-pounds	483 inch-pounds	0.000 to 0.031 below face of bearing	Antiflap stop to bearing assembly	0.264-inch
NOTES					
1. If proper torque cannot be achieved, up to 2 inserts which are not next to each other, may be installed up to 0.030-inch below installation depth at each location.					
2. On hubs marked RS-005-VII, replace insert with insert, 82629-820.					
3. If insert depth is measured from liner surface, add actual liner thickness to installation depth.					
4. If needed, clean hub's thread using 5/8 in. NF bit.					

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****MAIN ROTOR HEAD BOLTS**

This WP supersedes WP 0550 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Fluorescent Penetrant Inspection Kit, XMA 101
Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00

References

TM 1-1520-265-23
WP 0381 00
WP 1803 00

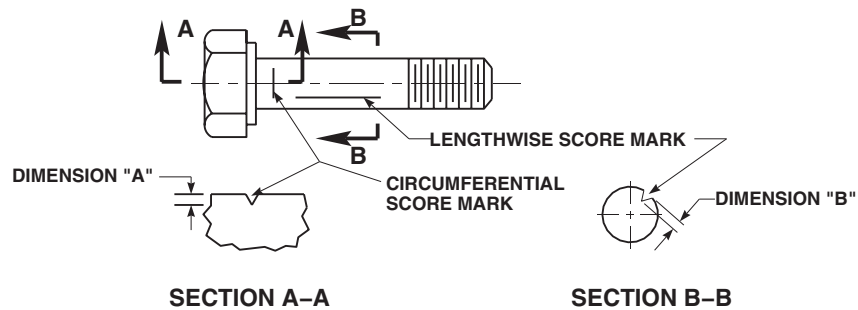
INSPECT TITANIUM AND STEEL BOLTS**WARNING****CSI**

On bolts, 70103-08801-102/-103, the shank outer diameter is a critical dimension and must not measure below minimum allowable diameter after rework as reflected in Figure 1.

NOTE

- The following instructions apply to bolts used to attach damper, pitch control rods and rotating scissors.
 - Whenever one of these bolts is removed it should be checked as follows before it is reinstalled.
1. Clean any oil or grease from bolts using cleaning compound solvent, Item 71, WP 1803 00.
 2. Check bolts for any material displaced above normal surface of bolt. Remove any displaced material using abrasive cloth, Item 82, WP 1803 00.
 3. Check bolts for scoring or grooves. Lengthwise score marks or grooves may be smoothed out, but can be left as is, if within limits. Circumferential score marks must be blended out. See Figure 1 for limits on depth of scoring. Blend out as follows:

INSPECT TITANIUM AND STEEL BOLTS - Continued



BOLT PART NUMBER	LOCATION	DIMENSION "A" MAXIMUM	DIMENSION "B" MAXIMUM	MINIMUM ALLOWABLE DIAMETER AFTER REWORK
70103-28702-101	DAMPER (BOTH ENDS)	0.005	0.005	0.7435
70103-08801-103	PITCH CONTROL ROD (TOP)	0.002	0.003	0.7480
70103-08801-102	PITCH CONTROL ROD (BOTTOM)	0.001	0.003	0.6230
SS5011-16-80A	SCISSORS (SCISSORS TO PRESSURE PLATE)	0.002	0.003	0.9945
SS5011-12-81H	SCISSORS (UPPER LINK TO LOWER LINK)	0.002	0.003	0.7445
70103-08114-101	DAMPER (BOTH ENDS)	0.005	0.005	0.7435
SS5032-12-66	PITCH CONTROL ROD (TOP)	0.002	0.003	0.7442
SS5032-10-59	PITCH CONTROL ROD (BOTTOM)	0.001	0.003	0.6202
NAS1956-9	HORN	0.001	0.001	0.3716
NAS1956-45	HORN	0.001	0.001	0.3716
70102-08117-102	HORN	0.001	0.001	0.3716
SS5021-10-34A	BIFILAR (WEIGHT TO SUPPORT)	0.001	0.003	0.6170
SS5211-12-26	BIFILAR SUPPORT LOCKRING	0.002	0.002	0.7420

**NOTE**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Inspect/Repair Main Rotor Head Bolts.

INSPECT TITANIUM AND STEEL BOLTS - Continued**WARNING****CSI**

Fluorescent penetrant inspection is a critical characteristic.

NOTE

- On steel bolts, 70103-08114-101 and 70103-28702-101, magnetic particle inspect (TM 1-1520-265-23), before and after rework.
 - On bolts, 70102-08117-102, corner damage may be permitted up to 0.070 by 0.070-inch. Prime and paint damaged areas as necessary (WP 0381 00 and TM 1-1520-345-23). If damage exceeds these limits, remove and replace bolts.
 - If, after rework, bolt diameter is less than specified in figure, discard bolt.
 - If, after rework, repaired area is over 30% of bolt shank area, discard bolt.
- a. Before reworking bolt, fluorescent penetrant inspect, Type I, Method C, and check for cracks (TM 1-1520-265-23), using fluorescent penetrant inspection kit. **NO CRACKS ALLOWED.** Replace cracked bolts.
 - b. Blend out scratches using abrasive wheel, Item 350, WP 1803 00. Blend area smooth with abrasive cloth, Item 82, WP 1803 00.
 - c. After reworking bolt, fluorescent penetrant inspect, Type I, Method C, for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked bolts.

INSPECT PLATED BOLTS**NOTE**

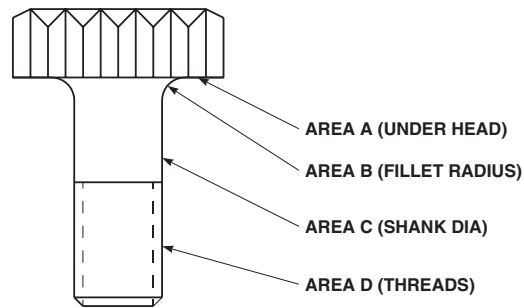
- The following instructions apply to both cadmium and silver-plated bolts. Silver-plated bolts are identified by an S after basic part number.
 - Whenever one of these bolts is removed it should be checked as follows before it is reinstalled.
1. Clean any oil or grease from bolts using cleaning compound solvent, Item 71, WP 1803 00.
 2. Check bolts for corrosion pitting. **REPLACE BOLTS WITH CORROSION PITTING GREATER THAN 0.030-INCH ON EXPOSED BOLT HEAD AREA.** Refer to Figure 2 for corrosion limits on other areas of bolt.

INSPECT NON-PLATED BOLTS**NOTE**

Whenever one of these bolts is removed it should be checked as follows before it is reinstalled.

1. Clean any oil or grease from bolts using cleaning compound solvent, Item 71, WP 1803 00.
2. Check bolts for corrosion pitting. **REPLACE BOLTS WITH CORROSION PITTING GREATER THAN 0.030-INCH ON EXPOSED BOLT HEAD AREA.** Refer to Figure 2 for corrosion limits on other areas of bolt.

INSPECT NON-PLATED BOLTS - Continued



BOLT PART NO.	LOCATION	PIT LIMIT AREA A	PIT LIMIT AREA B	PIT LIMIT AREA C	PIT LIMIT AREA D
76683-5-7 OR 80489-5-7	BIFILAR SUPPORT	0.003	0.001	0.004	0.001
76683-5-8 OR 80489-5-8	DAMPER BRACKET	0.003	0.001	0.004	NONE
76683-6-10 OR 80489-6-10	LOWER PRESSURE PLATE	0.003	0.001	0.004	0.001
76683-8-10 OR 80489-8-10	SPINDLE ELASTOMERIC BEARING PLATE	0.003	0.001	0.004	NONE
76683-8-12 OR 80489-8-12	UPPER PRESSURE PLATE	0.003	0.001	0.004	0.001
NAS1956-9	HORN	0.001	NONE	0.002	NONE
NAS1956-45	HORN	0.001	NONE	0.002	NONE
70102-08117-102	HORN	0.001	NONE	0.002	NONE
SS5211-12-26	BIFILAR SUPPORT LOCKRING	0.002	NONE	0.004	NONE

**NOTE**

ALL DIMENSIONS ARE IN
INCHES UNLESS NOTED.

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Figure 2. Inspect/Repair Main Rotor Head Bolts.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**SPINDLE**

This WP supersedes WP 0551 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Brush, Soft Bristle
 Cotton Gloves
 Depth Micrometer, GGG-C-105
 Dial Indicator, MIL-I-18422
 Fluorescent Penetrant Inspection Kit, XMA 101
 Gloves
 Goggles/Face Shield
 Heat Gun, LFT500
 Inside Micrometer Caliper, GGG-C-105
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Strap Wrench, GGG-W-651
 Torque Wrench, Dial, 0 - 600 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.
 Torque Wrench, 300 - 2500 in. lbs.
 Vernier Caliper, GGG-C-111
 Wood, 2 inch x 4 inch x 3 feet

Material/Parts

Accelerator, Adhesive Curing, Item 3, WP 1803 00
 Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 11, WP 1803 00
 Adhesive, Item 33, WP 1803 00
 Adhesive Primer, Item 23, WP 1803 00
 Cloth, Abrasive, Item 77, WP 1803 00
 Cloth, Abrasive, Item 80, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Detergent, General Purpose, Item 115, WP 1803 00
 Etching Solution, Item 121, WP 1803 00
 Insulation Sleeving, Electrical, Item 158, WP 1803 00
 Lubricant, Solid Film, Item 180, WP 1803 00

Primer Coating, Item 243, WP 1803 00
 Remover, Paint, Item 250, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Tags
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

DA Form 2408-16
 TM 1-6625-724-13&P
 TM 55-1500-345-23
 WP 0381 00
 WP 0534 00
 WP 0535 00
 WP 0548 00
 WP 0549 00
 WP 0550 00
 WP 0554 00
 WP 0555 00
 WP 0558 00
 WP 0563 00
 WP 0576 00
 WP 0593 00
 WP 0594 00
 WP 0595 00
 WP 1691 00
 WP 1703 00
 WP 1726 00
 WP 1803 00

REMOVE**NOTE**

One main rotor spindle assembly weighs about 65 pounds.

1. Turn main rotor so spindle can be removed easily.

REMOVE - Continued

2. Lock gust lock.
3. Remove main rotor blade (WP 0576 00).

WARNING**CSI**

The pitch control rod attachment areas, and critical areas from horn lugs to elbow, are critical surfaces which must not be damaged during and/or after removal of horn and/or control rod. Use extreme caution when removing and provide protective covering after removal from horn and attaching hardware. If damaged, inspect horn (WP 0535 00).

CAUTION

Damage to spindle horn will result if allowed to come in contact with droop stop pin or cotter pin. Do not allow spindle horn to come in contact with droop stop pin or cotter pin.

NOTE

It may be necessary to move spindle horn slightly up and down, for easy removal of pitch control rod bolt.

4. Remove pitch control rod (WP 0563 00).
5. Disconnect damper from spindle bracket (WP 0558 00).
6. Remove antifiap stop from spindle. Move bonding jumper aside (WP 0554 00).
7. Remove droop stop (WP 0555 00).

CAUTION

Damage to main rotor spindle will result if metallic items are used to remove sealant. Do not use metallic items to remove sealant.

8. Clean sealing compound from spindle bolt heads using nonmetallic scraper.
9. Remove bolts and washers holding spindle to main rotor (Figure 1). Leave one upper bolt in until hoisting sling is attached. Keep two clips that attach to droop stop springs with springs. There may not be enough wrench clearance to remove bolt behind horn and antifiap assembly. If necessary, insert a piece of wood 2 inch x 4 inch x about 3 feet long between blade cuff lugs, and turn spindle.
10. Place hoisting sling around spindle and adjust hoisting sling straps. Attach hoisting sling to hoist. Take up slack.
11. Remove remaining bolt from spindle.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

12. Clean primer from bolts using machinery wiping towel, Item 344, WP 1803 00, with technical acetone, Item 5, WP 1803 00.

REMOVE - Continued

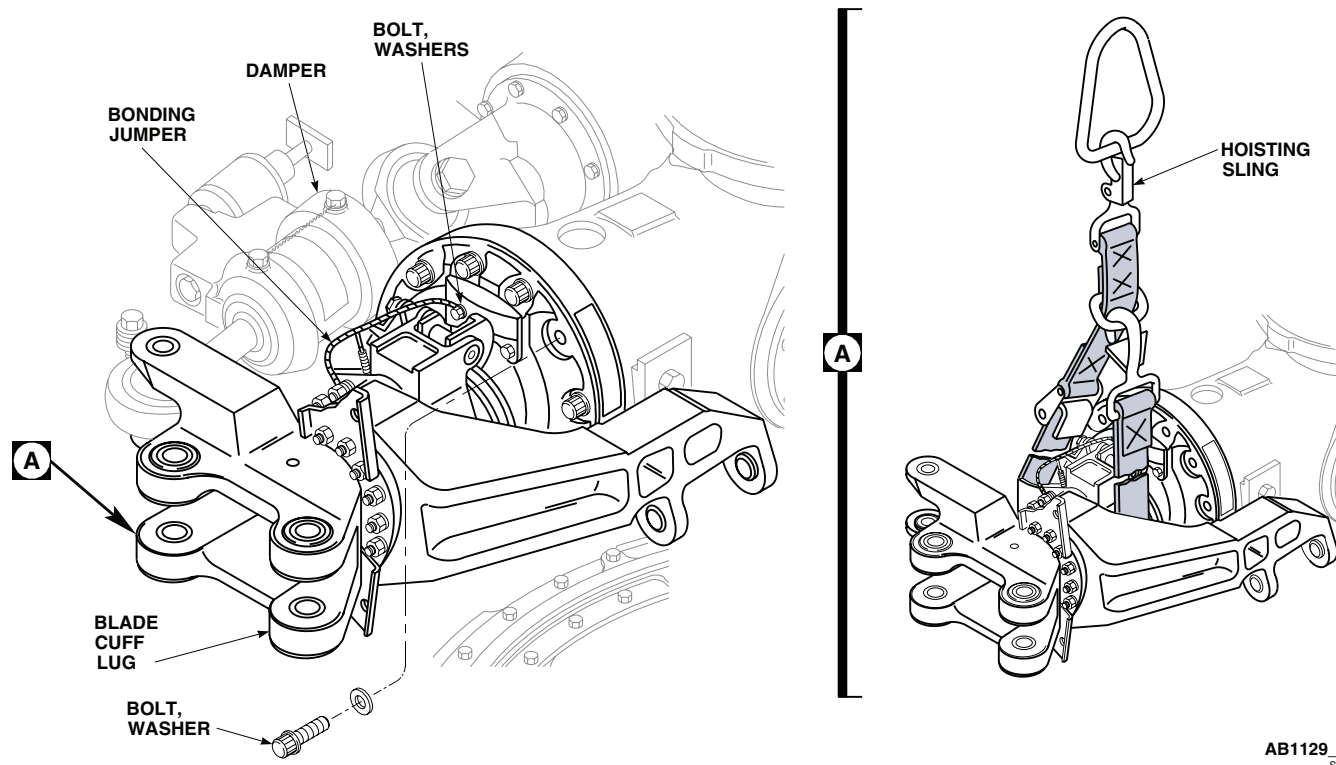


Figure 1. Remove Spindle.

CAUTION

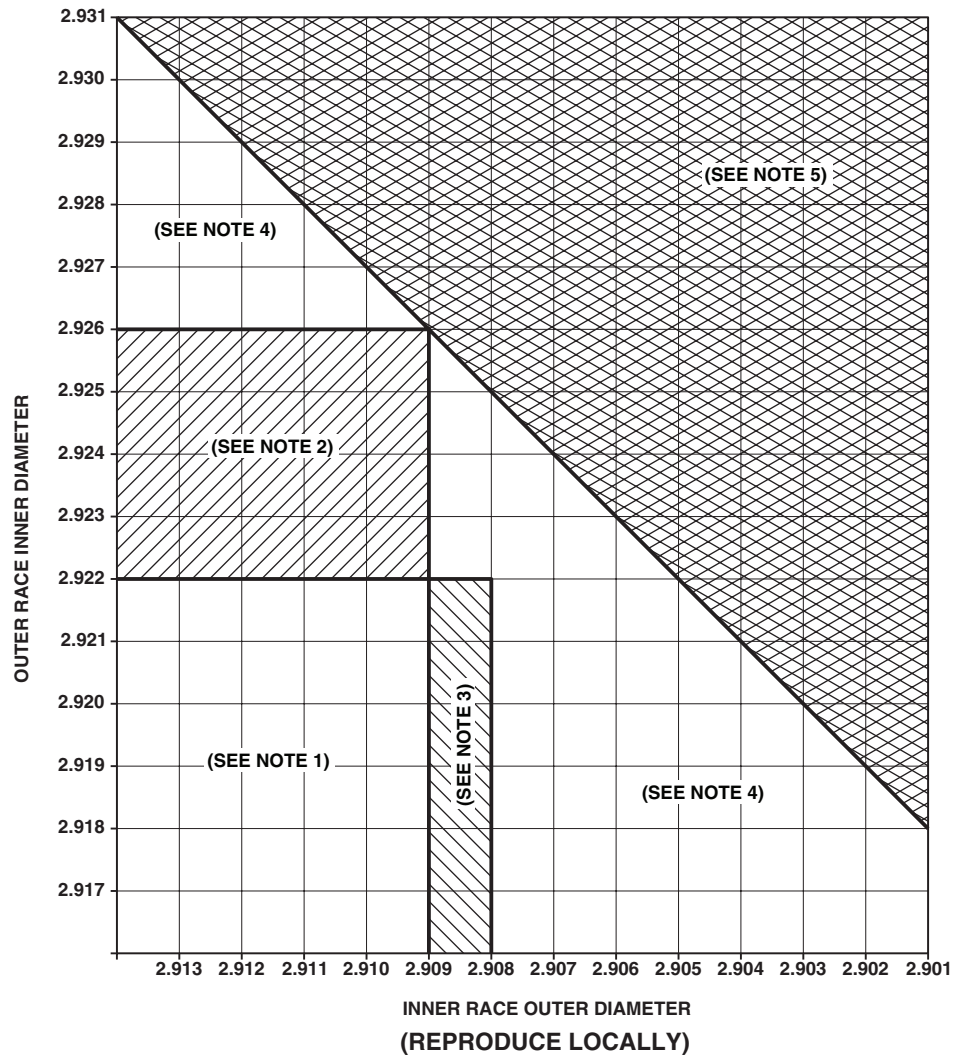
Damage to hub will result if metallic tools are used to pry spindle from hub. Do not pry spindle from hub with metal tools.

13. Slide spindle from hub.
14. Move spindle clear of helicopter onto workstand.
15. Install protective covering to spindle threads, undercut and shank.
16. Remove hoisting sling.

DISASSEMBLE

1. Measure movement between spindle spline and thrust bearing spline at spindle nut locking bolts. **MAXIMUM MOVEMENT IS 0.010-INCH.** Replace spindle if movement is beyond limits.
2. If replacement spindle is to be installed and previous spindle has balance weights installed, remove washers, balance weights, and nuts from spindle (Figure 2, Sheet 2). Keep these parts for installation in new spindle. Tag parts.
3. If replacement spindle is to be installed, remove bonding jumper from spindle.
4. Remove bolts and washers holding spindle nut to thrust bearing (Figure 2, Sheet 3, Detail B). Remove spindle nut. It may be necessary to use a strap wrench to loosen nut.
5. Slide elastomeric bearing assembly from spindle.

DISASSEMBLE - Continued



NOTES

1. REASSEMBLE SPINDLE AND ELASTOMERIC BEARING AND RETURN TO SERVICE.
2. RETURN ELASTOMERIC BEARING ASSEMBLY TO HIGHER LEVEL MAINTENANCE TO REPLACE SLEEVE BEARING OUTER SLEEVE.
3. RETURN SPINDLE AND LINER ASSEMBLY TO HIGHER LEVEL MAINTENANCE TO REPLACE SLEEVE BEARING INNER SLEEVE.
4. RETURN ELASTOMERIC BEARING ASSEMBLY AND SPINDLE AND LINER ASSEMBLY TO HIGHER LEVEL MAINTENANCE TO REPLACE SLEEVE BEARING INNER AND OUTER SLEEVES.
5. RETURN ELASTOMERIC BEARING ASSEMBLY TO HIGHER LEVEL MAINTENANCE TO REPLACE SLEEVE BEARING OUTER SLEEVE. SCRAP SPINDLE AND LINER ASSEMBLY AND SPINDLE NUT.
6. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 2. Disassemble/Assemble Spindle. (Sheet 1 of 5)

DISASSEMBLE - Continued

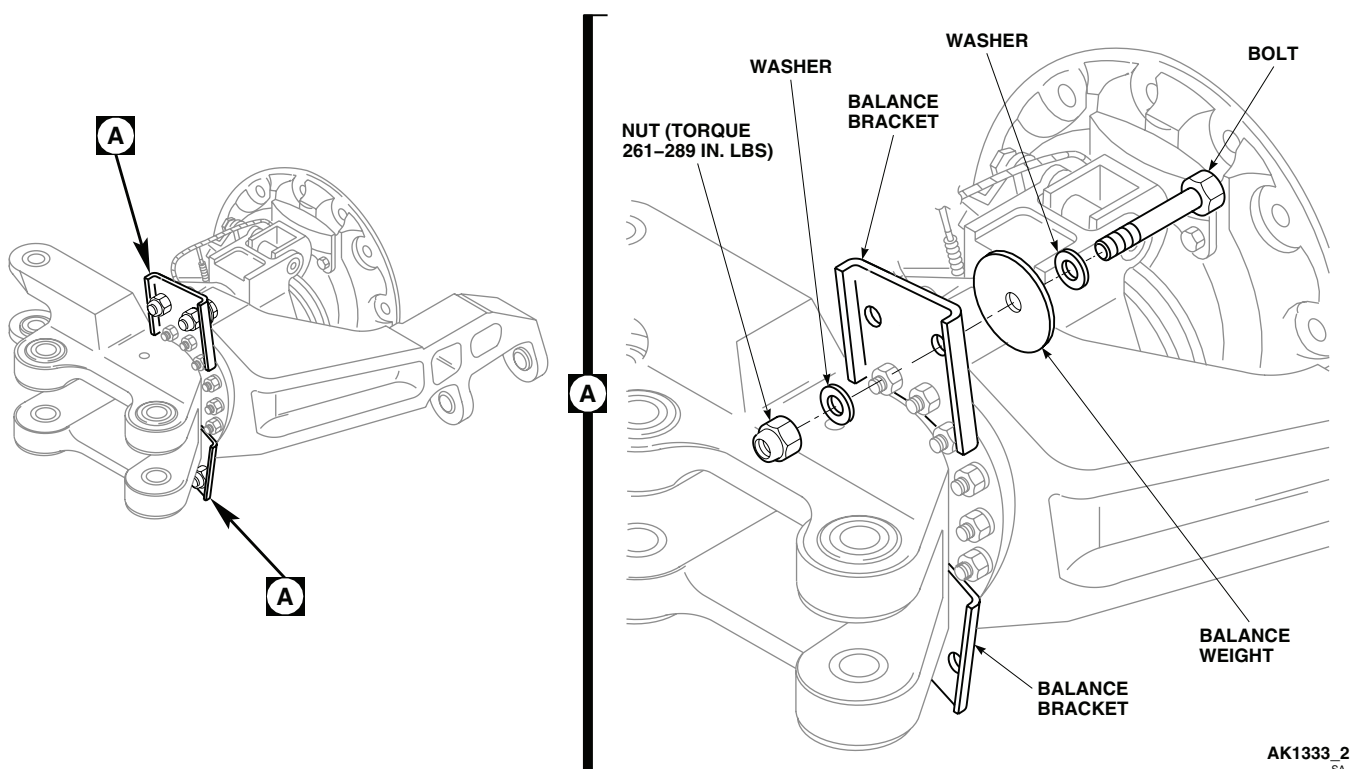


Figure 2. Disassemble/Assemble Spindle. (Sheet 2 of 5)

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

6. Clean primer from bolts using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
7. On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, disassemble spindle as follows (Figure 2, Sheet 3, Detail C):
 - a. Using dial indicator, measure radial play between droop stop ring and inner race of sleeve bearing. **MAXIMUM RADIAL PLAY IS 0.030-INCH.** Replace sleeve bearing outer sleeve if radial play is beyond limits (WP 0593 00).
 - b. Using dial indicator, measure axial play between droop stop ring and inner race of sleeve bearing. **MAXIMUM AXIAL PLAY IS 0.020-INCH.** Replace sleeve bearing outer sleeve and Teflon lined ring if axial play is more than limits (WP 0593 00 00).
 - c. Visually inspect exposed sleeve bearing Teflon wear surfaces. **NO MORE THAN 30% OF THE CIRCUMFERENCE OF EITHER WEAR SURFACE MISSING ALLOWED.** Replace sleeve bearing outer sleeve and/or Teflon lined ring having more than 30% of the circumference of wear surface missing (WP 0593 00).

DISASSEMBLE - Continued

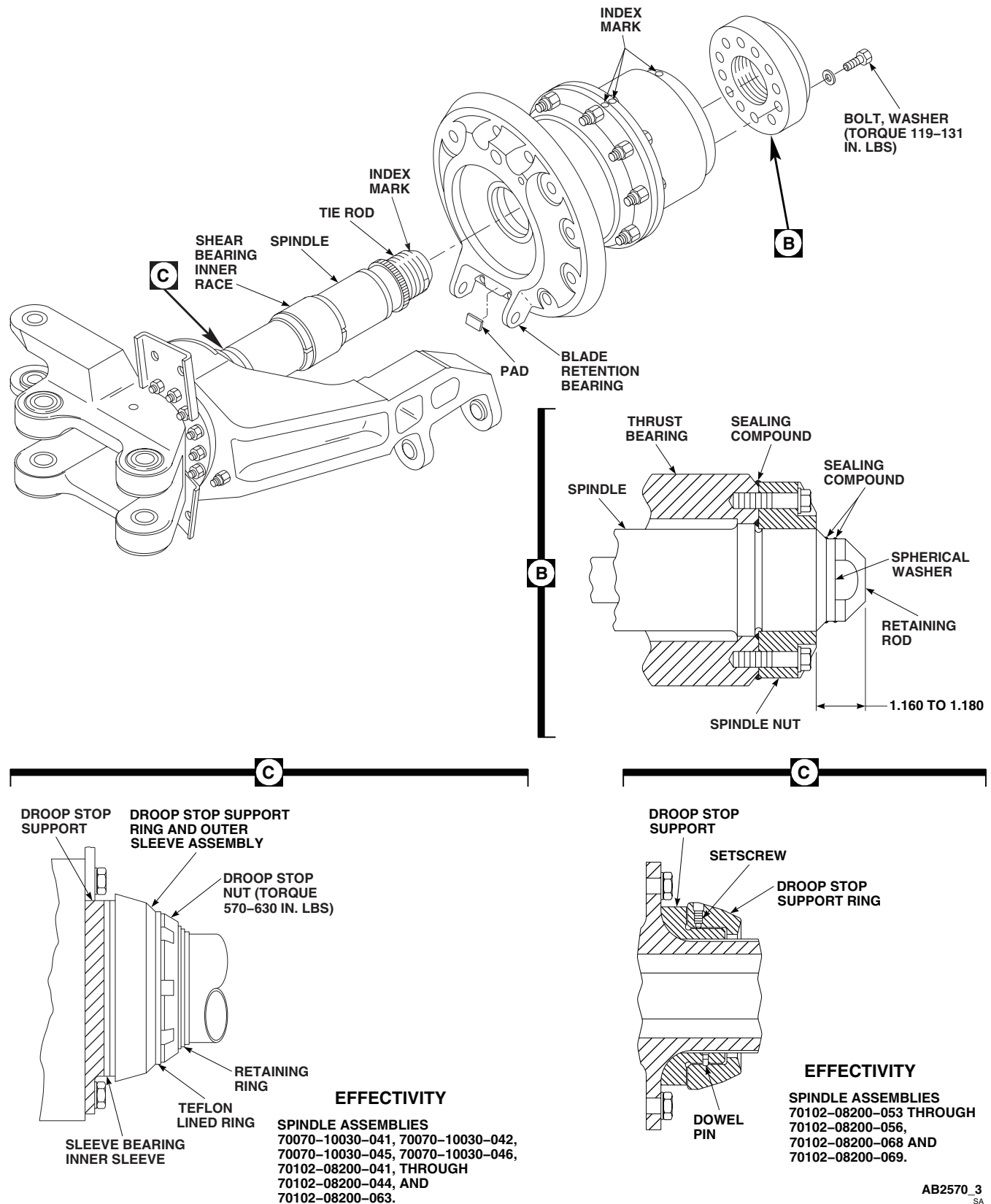


Figure 2. Disassemble/Assemble Spindle. (Sheet 3 of 5)

DISASSEMBLE - Continued

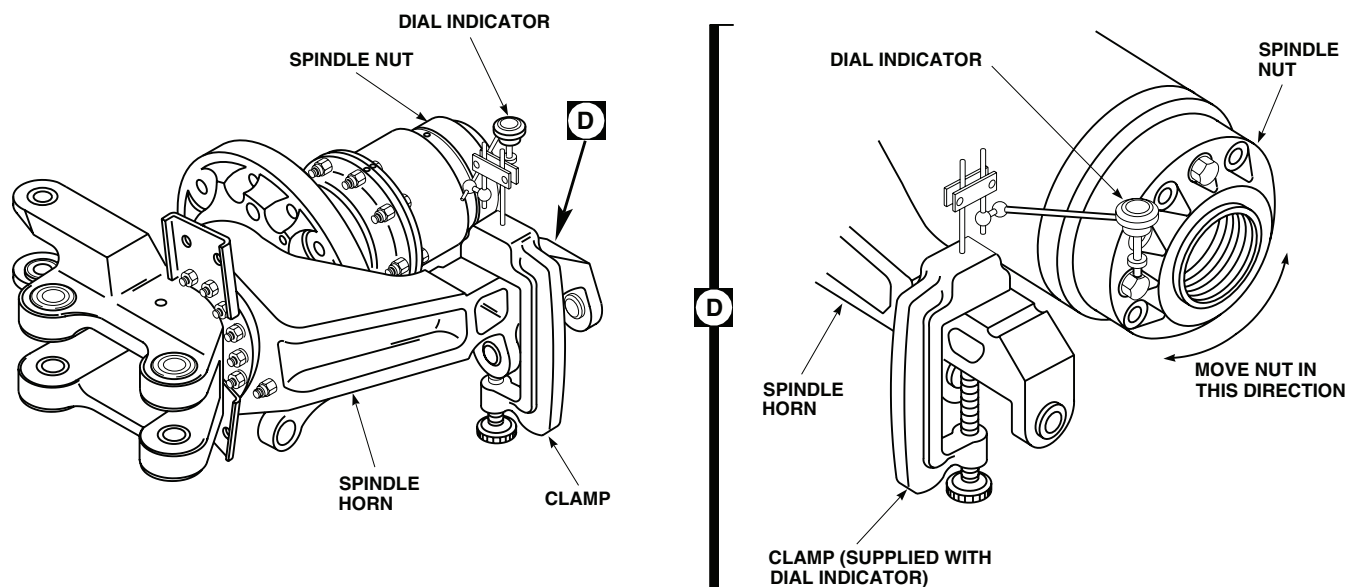
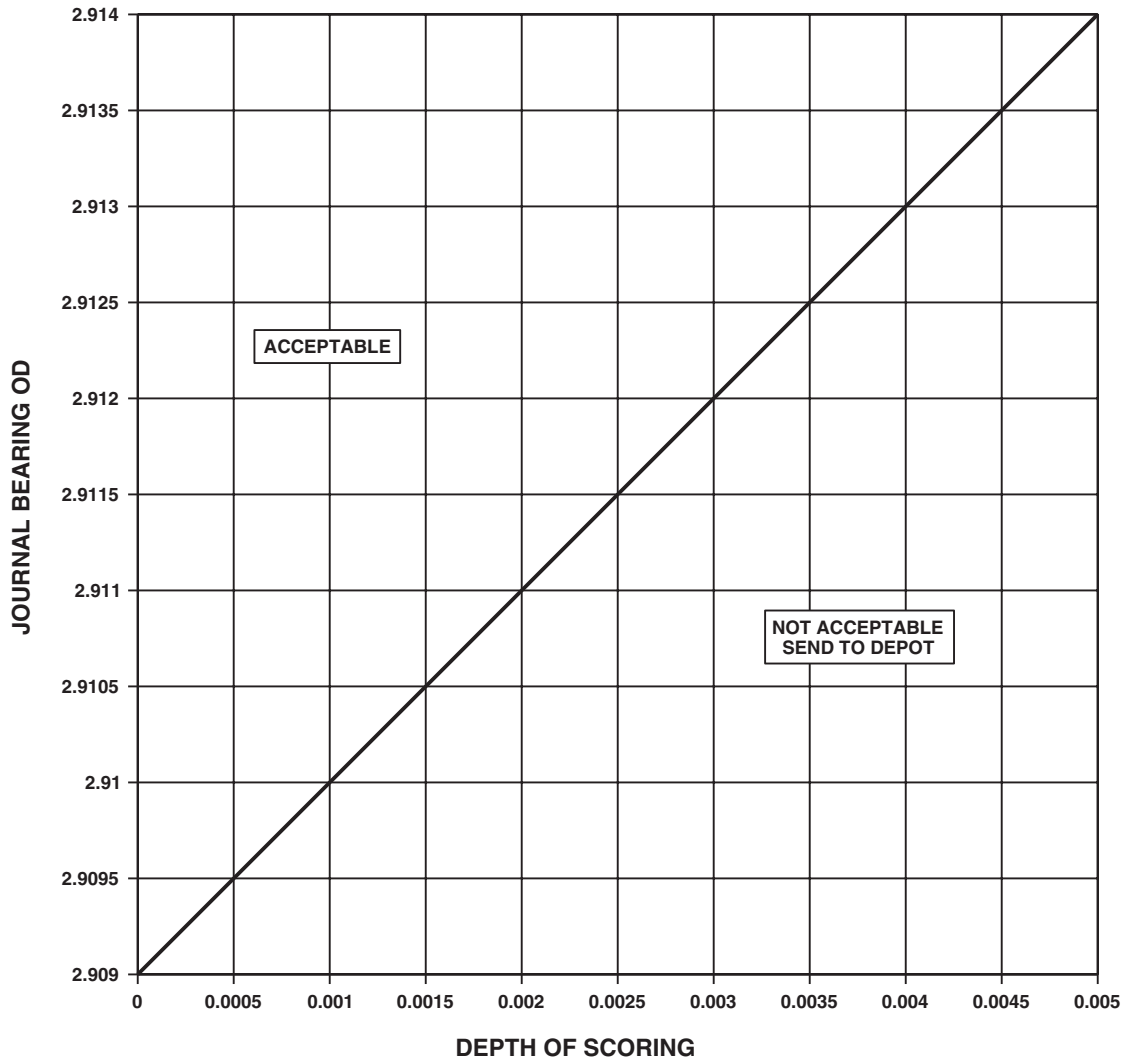
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Figure 2. Disassemble/Assemble Spindle. (Sheet 4 of 5)

- d. Visually inspect exposed sleeve bearing Teflon wear surfaces for metal to metal contact. **NO METAL TO METAL CONTACT ALLOWED.** Replace sleeve bearing outer sleeve and/or Teflon lined ring if metal to metal contact is found (WP 0593 00).
- e. Remove retaining ring, nut, Teflon lined ring, and droop stop support ring and outer sleeve assembly from spindle.
8. On spindle assemblies, 70102-08200-053 through 70102-08200-056, 70102-08200-068, and 70102-08200-069, disassemble spindle as follows (Figure 2, Sheet 3, Detail C):
 - a. Using dial indicator, measure radial play in vertical direction between droop stop ring and droop stop support. **MAXIMUM RADIAL PLAY IS 0.025-INCH.** Replace droop stop ring if radial play is beyond limits.
 - b. Visually inspect exposed droop stop ring Teflon wear surface as follows:
 - (1) Move droop stop ring inboard to expose edge of Teflon liner.
 - (2) With droop stop ring moved inboard, rotate 360 degrees and check condition of Teflon liner. **NO METAL TO METAL CONTACT ALLOWED.** Replace droop stop ring if metal to metal contact is found.
 - (3) Using feeler gage, measure gap between droop stop support ring outboard Teflon surface and inboard surface of droop stop support. **MAXIMUM GAP SHALL NOT EXCEED 0.090-INCH.** Replace droop stop support ring if gap exceeds limits.
 - c. Remove droop stop ring as follows:
 - (1) Remove sealant to expose setscrew in droop stop support ring using nonmetallic scraper.
 - (2) Back out setscrew and rotate droop stop support ring while pulling slightly to remove ring.

DISASSEMBLE - Continued

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SCORE DEPTH VS BEARING DIAMETER



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Figure 2. Disassemble/Assemble Spindle. (Sheet 5 of 5)

INSPECT SPINDLE SHAFT**WARNING****CSI**

Thread relief and first seven full threads from thread relief on spindle shaft are critical characteristics of the part and must not contain any nicks, dents, scratches, or scuff marks. Make a detailed inspection of this area of spindle.

1. Make a detailed inspection of spindle critical areas. **REPLACE SPINDLE IF THERE IS ANY NICKS, DENTS, SCRATCHES, OR SCUFF MARKS TO THREADS, THREAD RELIEF, RADIUS AT ANTIFLAP BRACKET ASSEMBLY MOUNTING PAD, UNDERCUT AREA NEXT TO THREADS, OR IF SPINDLE NUT WILL NOT TURN FREELY WHEN YOU THREAD IT ON.**
2. Check spindle threads for condition of solid film lubricant, Item 180, WP 1803 00.
 - a. **IF SPINDLE THREADS ARE MISSING MORE THAN 50% SOLID FILM LUBRICANT, REPLACE SPINDLE.**
 - b. **IF SPINDLE THREADS ARE MISSING SOLID FILM LUBRICANT NOT EXCEEDING 50% OF SURFACE AREA AND SPINDLE HAS PREVIOUSLY BEEN TOUCHED-UP, REPLACE SPINDLE.**
3. If metal is showing on spindle, and spindle has not been touched up previously, reapply solid film lubricant, Item 180, WP 1803 00, up to 50% of surface area as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- a. Clean areas of bare metal using dry machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Allow to dry thoroughly.

NOTE

Solid film lubricant will not adhere to contaminated surfaces. Do not touch cleaned areas with bare hands. Use clean cotton gloves.

- b. At room temperature, spray exposed area with solid film lubricant, Item 180, WP 1803 00. Apply multiple thin coats allowing 15 to 20 minutes drying time between coats.
- c. Air cure solid film lubricant a minimum of 12 hours before using spindle.

WARNING**CSI**

100% coverage of solid film lubricant is a critical characteristic.

4. Visually inspect spindle for condition of solid film lubricant. **REPLACE SPINDLE IF COATING IS NOT SLATE BLACK AND UNIFORM IN COLOR, IF COATING HAS ANY RUNS, BLISTERS, ROUGHNESS OR SEPARATION OF INGREDIENTS, OR IF COATING HAS ANY OTHER SURFACE IMPERFECTIONS.**
5. If solid film lubricant repair is successful, reinstall part. Note component serial number, percentage of area touched up, and total time on part, on Aircraft Component Historical Record, DA Form 2408-16.

INSPECT SPINDLE SHAFT - Continued

6. Visually check that white torque stripe on retaining rod and nut is unbroken. If torque stripe is broken indicating a potential torque loss, replace spindle.
7. Check retaining rod for damage and corrosion as follows:
 - a. Remove sealing compound from around retaining rod head and washer using nonmetallic scraper.
 - b. Blend any damage or corrosion found with a 1 inch diameter abrasive wheel Item 350, WP 1803 00. **REPLACE SPINDLE IF DAMAGE OR CORROSION TO THE RETAINING ROD HEAD IS DEEPER THAN 0.010-INCH AFTER BLENDING OR IF DAMAGE AREA EXCEEDS 0.050-INCH DIAMETER OR DAMAGE COVERS THREE OR MORE 0.050-INCH DIAMETER LOCATIONS, OR COMES CLOSER THAN 0.150-INCH TO WASHER CONTACT FACE. REPLACE SPINDLE IF CORROSION DAMAGE ON THE WASHER IS GREATER THAN 0.015-INCH DEEP OR COVERS MORE THAN 25% OF THE WASHER CIRCUMFERENCE.**
 - c. Coat blended area with primer coating, Item 243, WP 1803 00. Touch up paint on tie rod as necessary (WP 0381 00 and TM 55-1500-345-23). Reapply sealing compound, Item 273, WP 1803 00, around tie rod and washer.
8. Check cylindrical part of spindle for surface damage. Blend any damage found with 1 inch diameter abrasive wheel, Item 350, WP 1803 00. **REPLACE SPINDLE IF DAMAGE IS DEEPER THAN 0.010-INCH AFTER BLENDING OR IF DAMAGE GOES MORE THAN 1/4 CIRCUMFERENCE OF SPINDLE.**

NOTE

Crazing of the adhesive in the slits of the spindle shank liners is acceptable.

9. Check the space consisting of a sleeve bearing (approximately 2 1/2-inches long) and the thin metal retaining strap (band) or the plastic shrink tubing located between the inner race and the droop stop ring assembly (Figure 2, Sheet 3). On spindles, 70102-08200-043 and 70102-08200-044, inspect shrink tubing positioned on bonded Teflon sleeve as follows:
 - a. Inspect shrink tubing for signs of axial movement, displacement, or incorrect installation.
 - b. Grasp inboard end of shrink tubing, and apply an axial force toward blade end. Use caution not to damage or tear shrink tubing. Shrink tubing must not move beyond limits of channel in sleeve bearing. Rotational movement is acceptable.
 - c. **IF SHRINK TUBING HAS BEEN INSTALLED WITH EDGE OF TUBING BEYOND CHANNEL IN SLEEVE BEARING, SHIFTED POSITION, OR MOVES BEYOND CHANNEL LIMITS IN SLEEVE BEARING, REPLACE SHRINK TUBING AT NEXT 500 HOUR INSPECTION.**
10. Any surface damage to the Teflon sleeve and metal retaining strap is acceptable. Cracking of adhesive squeeze out from around Teflon sleeve is acceptable.
11. **REBOND THE METAL BAND IF IT BECOMES DISBONDED BY LIGHTLY SANDING (IF POSSIBLE), CLEAN WITH DRY CLOTH AND APPLY ADHESIVE, Item 11, WP 1803 00.**
12. **RESHRINK THE SHRINK TUBING WITH A HEAT GUN IF THE TUBING BECOMES LOOSE (ONE TIME ONLY). FLIGHT WITHOUT THE SHRINK TUBING IS ACCEPTABLE PROVIDED THE TEFLON SLEEVE DOES NOT START PEELING FROM THE SPINDLE.**

INSPECT SPINDLE CUFF LUG BUSHINGS AND LUG AREA**WARNING****CSI**

Nicks, dents, scratches, and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.

1. Check ID of spindle cuff lug bushings for wear. **MAXIMUM ID IS 1.003-INCHES.** Check for proper seating of bushings flush to 0.003-inch below surface. Replace bushing if beyond limits (WP 0594 00).

WARNING

Spindle cuff lug bushings are made of beryllium copper metal, which can be a health hazard. The use of goggles is required. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

2. Repair spindle cuff lug bushings using abrasive cloth, Item 82, WP 1803 00, as follows:
 - a. **BLEND NICKS, GOUGES, OR SCORING UP TO 0.005-INCH DEEP. TOTAL BLENDED AREA SHALL NOT EXCEED 25% OF BORE AREA.**
 - b. **BLEND CORROSION NO MORE THAN A MAXIMUM BUSHING ID OF 1.003-INCHES.**
 - c. Replace spindle cuff lug bushings if repairs are more than limits (WP 0594 00).
3. Check lug area for surface damage. Blend any damage found with a 1 inch diameter abrasive wheel, Item 350, WP 1803 00. **REPLACE SPINDLE IF DAMAGE IS DEEPER THAN 0.020-INCH AFTER BLENDED.**
4. Check corners for surface damage. Blend any damage found with a 1 inch diameter abrasive wheel, Item 350, WP 1803 00. **REPLACE SPINDLE IF DAMAGE IS DEEPER THAN 0.050-INCH AFTER BLENDED.**

INSPECT DAMPER SHOULDER BUSHINGS AND LUG AREA

1. Measure wear on damper shoulder bushings where outboard end of damper attaches by doing this:
 - a. Measure distance between shoulder of each bushing. **REPLACE SPINDLE IF DISTANCE IS GREATER THAN 2.509-INCHES.**
 - b. Check ID of bushing for wear. **REPLACE SPINDLE IF ID OF EITHER BUSHING IS GREATER THAN 0.7513-INCH.**
2. Check for corrosion on damper shoulder bushings. **REPLACE SPINDLE IF CORROSION ON CHAMFER OF BUSHING IS GREATER THAN 0.040-INCH DEEP.** Blend out corrosion with abrasive cloth, Item 77, WP 1803 00. Coat blended area with primer coating, Item 243, WP 1803 00. Touch up paint.
3. Check lug area for surface damage. Blend any damage found with abrasive wheel, Item 350, WP 1803 00. Blend ratio is 10:1 length or width to depth. **MAXIMUM REWORK IS 30% OF THE AREA WITHIN THE LUG CIRCUMFERENCE. MINIMUM ACCEPTABLE THICKNESS OF THE LUG IS 0.475-INCH.**

INSPECT ELASTOMERIC BEARING AND SPINDLE SLEEVE BEARING**NOTE**

At least six intersections will be plotted. If any intersection falls into replace area of graph, replace component as applicable.

1. Check sleeve bearing on spindle and inside elastomeric bearing for wear by doing this:
 - a. Measure ID of outer race 1/2-inch in from both ends and at center at two places: one place at 12/6-O'clock position and one place at 3/9-O'clock position.
 - b. Measure inner race OD 3/4-inch in from both ends and at center at two places: one place at 12/6-O'clock position, and one place at 3/9-O'clock position.
 - c. Enter each pair (inner and outer race) of measurements taken on spindle on graph (Figure 2, Sheet 1) using all clock positions, and both ends and center.
 - d. Find where intersection of readings is. Replace or keep sleeve bearing in elastomeric bearing or spindle in use as determined by where intersection is on graph.
 - e. Measure bearing in any other position where it seems to be worn. Measure mating sleeve bearings in same position. Enter these measurements on graph (Figure 2, Sheet 1). Replace or keep sleeve bearing in elastomeric bearing or spindle in use as determined by where intersection is on graph.
 - f. If outer race ID minus inner race OD is 0.017-inch or greater, remove spindle and spindle nut. Return as scrap.

NOTE

Some Teflon protrusion between the races is not cause for rejection.

2. Check sleeve bearing on spindle. **REPLACE SPINDLE AND/OR BEARING IF:**
 - a. **BEARING IS CRACKED, OR SLEEVE BEARING INNER RACE SHOWS SIGNS OF MOVEMENT ON SPINDLE.** If sleeve bearing inner race displacement occurs, flight may be continued up to 10 hours after confirmation of a slipped sleeve bearing inner race under the following conditions only:
 - (1) **THE SLEEVE BEARING INNER RACE HAS NOT DISENGAGED FROM THE OUTER RACE.**
 - (2) **THE SLEEVE BEARING INNER RACE HAS NOT MOVED BEYOND THE INBOARD CONTACT FACE OF THE TEFLON SLEEVE OR BROKEN THE INBOARD TIEDOWN STRAP.**
 - (3) **THE FLIGHT IS ONE TIME FLIGHT TO A SITE CAPABLE OF REMOVING/REPLACING THE SPINDLE ASSEMBLY.**
 - (4) **THE CONTINUANCE OF FLIGHT IS ALLOWED TO ENABLE THE HELICOPTER TO RETURN TO A SITE WHERE REPAIRS MAY BE COMPLETED.**
 - b. **SLEEVE BEARING SHOWS SIGNS OF METAL-TO-METAL WEAR.**
 - c. **DAMAGE ON ONE END OF BEARING SLEEVE EXCEEDS 3/8-INCH OR DAMAGE IS ON BOTH ENDS.**
3. Check sleeve bearing on spindle for corrosion as follows:
 - a. Clean surface using detergent cleaning compound, Item 115, WP 1803 00, and water.
 - b. Visually inspect sleeve bearing for corrosion pitting. **REPLACE SPINDLE IF 5 OR MORE PITS EXCEED 0.030-INCH OR GREATER IN DIAMETER IN A 1/2-INCH DIAMETER CIRCLE OR IF ANY PIT EXCEEDS 0.050-INCH IN DIAMETER.**

INSPECT ELASTOMERIC BEARING AND SPINDLE SLEEVE BEARING - Continued

4. Check location of sleeve bearing on spindle. **BEARING MUST NOT EXTEND BEYOND ENDS OF BONDED LINER.** Replace spindle if bearing extends beyond ends of liner.
5. Inspect sleeve bearing inner race for circumferential wear or scoring. **NO RAISED METAL ALLOWED.** Inspect OD of inner race in area of wear. Measure depth of score marks on inner race. Find intersection of measurements on graph (Figure 2, Sheet 5). Disposition as indicated.
6. If sleeve bearing becomes loose on spindle, rebond as follows:
 - a. Remove shrink tubing and Teflon sleeve.
 - b. Remove remaining adhesive from bonding areas of spindle and Teflon sleeve using paint remover, Item 250, WP 1803 00, and nonmetallic scraper.

WARNING

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- c. Clean bonding areas using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
- d. Apply adhesive primer, Item 23, WP 1803 00, to bonding area on spindle. Allow to cure at room temperature for 1 hour.
- e. Apply etching solution, Item 121, WP 1803 00, to electrical insulation sleeving, Item 158, WP 1803 00, and sleeve bearing bonding surfaces. Allow surfaces to turn brown in color and remove using machinery towel, Item 344, WP 1803 00.
- f. Apply adhesive, Item 11, WP 1803 00, to spindle bonding area and inside diameter of sleeve bearing.
- g. Install sleeve bearing butted against liner.

CAUTION

Damage to droop stop support ring will result if adhesive is applied to support ring.
Do not apply adhesive to droop stop support ring.

- h. Apply adhesive, Item 11, WP 1803 00, to outside diameter of sleeve bearing.
- i. Install electrical insulation sleeving, Item 158, WP 1803 00, centered over sleeve bearing. Preshrink sleeving in place using heat gun at 300°F(149°C) to 325°F(165°C).
- j. Cure adhesive at 70°F(22°C) to 80°F(24°C) for 24 hours or 140°F(60°C) to 160°F(71°C) for 1 hour.

NOTE

Polishing of inner race (smooth finish) is acceptable if uniform. Reddish-brown powder on surface of either race is normal wear and not cause for replacement.

7. Check sleeve bearing in elastomeric bearing. **REPLACE SLEEVE BEARING IF TEFLON SLEEVE BEARING SHOWS SIGNS OF TEARS, RIPS, OR MISSING TEFLON MORE THAN ONE LAYER OF TEFLON DEEP (WP 0595 00).**
8. Check sleeve bearing in elastomeric bearing at 1:30/4:30-o'clock position and at 7:30/10:30-o'clock position for visible metal. **NONE ALLOWED.** Replace sleeve bearing having visible metal (WP 0595 00).

INSPECT SPINDLE - GENERAL

1. Check rest of spindle for damage. Blend out damage with a 1 inch diameter abrasive wheel, Item 350, WP 1803 00. **REPLACE SPINDLE IF DAMAGE IS GREATER THAN 0.040-INCH AFTER REPAIR.**
2. Check spindle for cracks. Cracking is allowed only on corner radius of droop stop cam and anti flap cam contact surfaces. **REPLACE SPINDLE IF CRACKS ON CORNER RADIUS OF DROOP STOP CAM AND ANTI-FLAP CAM CONTACT SURFACES ARE GREATER THAN 1/8-INCH LONG.**
3. Check anti flap bracket for cracks. **IF BRACKETS ARE CRACKED DUE TO HIGH WINDS, CHECK PER WP 1726 00. REPLACE ANTI FLAP BRACKET, IF ANY CRACKS ARE FOUND. REPLACE SPINDLE IF THERE ARE ANY CRACKS IN OTHER AREAS OF SPINDLE.**
4. Check for corrosion. Blend any corrosion with abrasive cloth, Item 77, WP 1803 00. **REPLACE SPINDLE IF CORROSION ON MATING SURFACES COVERS MORE THAN 10% OF SURFACE, OR IS DEEPER THAN 0.010-INCH AFTER REMOVAL. REPLACE SPINDLE IF CORROSION ON OPEN AREAS IS DEEPER THAN 0.030-INCH AFTER REMOVAL.**

INSPECT SPINDLE NUT

1. Check spindle nut threads for condition of solid film lubricant, Item 180, WP 1803 00.
 - a. **IF SPINDLE NUT IS MISSING MORE THAN 50% SOLID FILM LUBRICANT, REPLACE SPINDLE NUT.**
 - b. **IF SPINDLE NUT IS MISSING SOLID FILM LUBRICANT NOT EXCEEDING 50% OF SURFACE AREA AND SPINDLE NUT HAS PREVIOUSLY BEEN TOUCHED UP, REPLACE SPINDLE NUT.**
2. If metal is showing on spindle nut, and spindle nut has not been touched up previously, reapply solid film lubricant, Item 180, WP 1803 00, up to 50% of surface area as follows:

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- a. Clean areas of bare metal using dry machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Allow to dry thoroughly.

NOTE

Solid film lubricant will not adhere to contaminated surfaces. Do not touch cleaned areas with bare hands. Use clean cotton gloves.

- b. At room temperature, spray exposed area with solid film lubricant, Item 180, WP 1803 00. Apply multiple thin coats allowing 15 to 20 minutes drying time between coats.
- c. Air cure solid film lubricant a minimum of 12 hours before using spindle nut.
3. Visually inspect spindle nut for condition of solid film lubricant. **REPLACE SPINDLE NUT, IF COATING IS NOT SLATE BLACK AND UNIFORM IN COLOR, IF COATING HAS ANY RUNS, BLISTERS, ROUGHNESS OR SEPARATION OF INGREDIENTS, OR IF COATING HAS ANY OTHER SURFACE IMPERFECTIONS.**

INSPECT SPINDLE NUT - Continued

4. After solid film lubricant has cured, firmly press a strip of pressure sensitive adhesive tape, Item 334, WP 1803 00, onto the surface of the touched up area. Remove pressure sensitive adhesive tape with an abrupt motion. A uniform deposit of powdery material clinging to pressure sensitive adhesive tape is acceptable. **REPLACE SPINDLE NUT, IF SOLID FILM LUBRICANT FLAKES OR LEAVES ANY BARE METAL EXPOSED.**
5. If solid film lubricant repair is successful, reinstall part. Note component serial number percentage of area touched up and total time on part, on Aircraft Component Historical Record, DA Form 2408-16.

INSPECT DROOP STOP SUPPORT RING AND NUT

1. On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, do this:
 - a. Check droop stop support ring sleeve bearing outer sleeve for nicks, gouges, and burrs. **DAMAGE SHALL NOT EXCEED 0.005-INCH DEEP.** Replace sleeve bearing if damage is more than limits (WP 0593 00).
 - b. Check droop stop support ring sleeve bearing outer sleeve for corrosion pitting. **CORROSION PITTING SHALL NOT EXCEED 0.005-INCH.** Replace sleeve bearing if damage is more than limits (WP 0593 00).
 - c. Check mating surface of droop stop support ring for nicks, gouges, and scratches. **DAMAGE SHALL NOT BE MORE THAN 0.002-INCH DEEP.** Replace droop stop support ring if damage is beyond limits.
 - d. Check corners of droop stop support ring for nicks, gouges, and scratches. Blend out damage in not less than 0.50-inch radius using abrasive cloth, Item 80, WP 1803 00. **DAMAGE AFTER REPAIR SHALL NOT BE MORE THAN 0.020-INCH DEEP.** Replace droop stop support ring if damage after repair is more than limits.
 - e. Inspect droop stop ring nut for cracks and corrosion.
 - f. **IF RADIAL CRACK IS FOUND, NUT MUST BE REPLACED.**
 - g. **IF CIRCUMFERENTIAL CRACK ON LUG IS MORE THAN 3/4 OF CIRCUMFERENTIAL WIDTH OF LUG, REPLACE NUT.** If replacement nut is not available, do this:
 - (1) Remove outer portion of damaged lug by breaking remaining section with pliers.
 - (2) Clean fractured area with abrasive cloth, Item 80, WP 1803 00.
 - (3) Touch up reworked area with corrosion resistant coating, Item 111, WP 1803 00.
 - (4) Coat reworked area with primer coating, Item 243, WP 1803 00.
 - (5) Replace nut within 500 flight hours.
 - h. If corrosion is found on nut, do this:
 - (1) Remove corrosion using abrasive cloth, Item 80, WP 1803 00. **REPLACE NUT IF CORROSION IS DEEPER THAN 0.035-INCH AFTER REMOVAL.**

WARNING

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- (2) Clean repaired area using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
 - i. Coat repaired area with primer coating, Item 243, WP 1803 00.
2. Inspect droop stop nut retaining ring for damage. Replace retaining ring if damage is found.

INSPECT DROOP STOP SUPPORT RING AND NUT - Continued

3. On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, disassemble spindle as follows (Figure 2, Sheet 3, Detail C):
 - a. Using dial indicator, measure radial play between droop stop ring and inner race of sleeve bearing. **MAXIMUM RADIAL PLAY IS 0.030-INCH.** Replace sleeve bearing outer sleeve if radial play is beyond limits (WP 0593 00).
 - b. Using dial indicator, measure axial play between droop stop ring and inner race of sleeve bearing. **MAXIMUM AXIAL PLAY IS 0.020-INCH.** Replace sleeve bearing outer sleeve and Teflon lined ring if axial play is more than limits (WP 0593 00 00).
 - c. Visually inspect exposed sleeve bearing Teflon wear surfaces. **NO MORE THAN 30% OF THE CIRCUMFERENCE OF EITHER WEAR SURFACE MISSING ALLOWED.** Replace sleeve bearing outer sleeve and/or Teflon lined ring having more than 30% of the circumference of wear surface missing (WP 0593 00).
 - d. Visually inspect exposed sleeve bearing Teflon wear surfaces for metal to metal contact. **NO METAL TO METAL CONTACT ALLOWED.** Replace sleeve bearing outer sleeve and/or Teflon lined ring if metal to metal contact is found (WP 0593 00).
 - e. Remove retaining ring, nut, Teflon lined ring, and droop stop support ring and outer sleeve assembly from spindle.

INSPECT ELASTOMERIC BEARING

1. Inspect elastomeric bearing (WP 0534 00).
2. Check rubber pad on elastomeric bearing for damage or disbond. If pad is damaged, replace it. To bond pad to elastomeric bearing, do this:

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- a. Clean area to be bonded with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
- b. Lightly scuff bonding surface of pad with abrasive cloth, Item 80, WP 1803 00.
- c. Apply adhesive acclerator, Item 3, WP 1803 00, to bonding area of pad and elastomeric bearing. Let accelerator dry at room temperature for at least 3 minutes.
- d. Apply a bead of adhesive, Item 33, WP 1803 00, to pad and install on elastomeric bearing.
- e. Clamp pad to elastomeric bearing, for good contact, and let dry for at least 15 minutes.

ASSEMBLE**WARNING****CSI**

The spindle thrust bearing retaining nut, spindle threads, thread undercut area next to threads, radius at antifiap mounting bracket and thread relief are critical surfaces which must not be damaged during assembly. After removal of protective covering from spindle use extreme caution when assembling spindle.

CAUTION

- If any oil, grease, or other fluids are spilled or accidentally get on the bearings, do not use a solvent to remove them. Use only warm water, mild detergent and a soft bristle brush to clean the elastomeric bearings. After cleaning, rinse with clear water and then dry.
- To prevent damage to Teflon bearing outer race, use extreme care when sliding thrust bearing assembly onto spindle.

NOTE

A white wax-like substance on the rubber surface is normal and should not be removed. If some of this white coating comes off during cleaning, it is not cause for rejection of bearing.

1. **QA** On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, install droop stop support ring and outer sleeve assembly as follows (Figure 2, Sheet 3, Detail C):
 - a. **QA** Install droop stop support ring and outer sleeve assembly on spindle.
 - b. **QA** Install Teflon lined ring with Teflon lining facing droop stop support ring and outer sleeve assembly.
 - c. Coat threads of droop stop support and face of droop stop nut with primer coating, Item 243, WP 1803 00.
 - d. **QA** Thread droop stop nut on droop stop support while primer coating is still wet. **TORQUE NUT TO 570 - 630 INCH-POUNDS.**
 - e. Wipe off excess primer coating from support and nut using machinery wiping towel, Item 344, WP 1803 00.
 - f. **QA** Install retaining ring.
 - g. Seal around droop stop nut with sealing compound, Item 273, WP 1803 00.
2. **QA** On spindle assemblies, 70102-08200-053 through 70102-08200-056, 70102-08200-068, and 70102-08200-069, install droop stop support ring as follows (Figure 2, Sheet 3, Detail C):
 - a. **QA** Position droop stop support ring on spindle. Push ring into support, rotating slightly to line up slot and dowel pin, until fully seated.
 - b. **QA** Rotate ring until setscrew hole faces 180 degrees away from dowel pin.
 - c. **QA** Holding ring against support, install setscrew until ring lifts off support.
 - d. **QA** Back out setscrew one full turn.
 - e. Fill setscrew hole with sealing compound, Item 273, WP 1803 00. Wipe off excess sealing compound from ring using machinery wiping towel, Item 344, WP 1803 00.

ASSEMBLE - Continued**WARNING**

Injury to personnel and damage to equipment will result if elastomeric bearing assembly is installed without sleeve bearing outer race installed in ID of spherical bearing. Make sure sleeve bearing outer race is installed in spherical bearing prior to installing elastomeric bearing assembly.

NOTE

Sleeve bearing outer race is 2.25-inches long with Teflon fabric lining on ID.

3. Visually inspect elastomeric bearing assembly for presence of sleeve bearing outer race in ID of spherical bearing. If sleeve bearing outer race is not installed, install per WP 0595 00 .
4. **QA** Line up wide tooth spline on thrust bearing with wide tooth on spindle, and slide bearing assembly on spindle.
5. **QA INSTALL SPINDLE NUT SO THAT DISTANCE FROM INBOARD END OF SPINDLE NUT TO TIE ROD END FACE IS 1.160 TO 1.180-INCHES (Figure 2, Sheet 3, Detail B).**
6. Coat shanks, threads and area under heads of spindle nut attachment bolts with primer coating, Item 243, WP 1803 00.
7. **QA** Install bolts and washers that hold spindle nut to thrust bearing while primer coating is still wet. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts (Figure 2, Sheet 3).
8. Wipe off any primer coating that has seeped from under boltheads.
9. Attach dial indicator to horn (Figure 2, Sheet 4, Detail D).
10. Measure movement between spindle spline and thrust bearing spline at spindle nut locking bolts. **MAXIMUM MOVEMENT IS 0.010-INCH.** Replace spindle if movement is beyond limits.
11. **QA** Verify that distance from inboard end of spindle nut to tie rod end face is 1.160 - 1.180-inches (Figure 2, Sheet 3, Detail B).
12. Seal boltheads and gap between thrust bearing and spindle nut with sealing compound, Item 273, WP 1803 00.

INSTALL**NOTE**

Before installing new spindle assembly onto hub, make sure that preservative grease is removed from mating flanges, using warm, soapy water and clean cloth.

1. Check depth of inserts (WP 0549 00).
2. Inspect and repair hub liner in spindle attach area as required (WP 0548 00).
3. Inspect bolts attaching elastomeric bearing endplate to hub (WP 0550 00).
4. Check breakaway torque of spindle attachment bolts as follows:
 - a. Thread in bolts until boltheads are 0.610-inch away from surface of hub.
 - b. Measure torque required to breakaway and remove bolts using a dial-type torque wrench. **MINIMUM BREAK-AWAY TORQUE IS 14 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 150 INCH-POUNDS.**
5. If breakaway or removal torque of spindle attachment bolts is not within limits, do this:
 - a. Inspect bolt threads for evidence of corrosion or wear. **NONE ALLOWED.** Replace bolts if corrosion or wear is found.

INSTALL - Continued**WARNING**

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- b. Inspect insert for cleanness. Clean insert using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
- c. Repeat step 4. If breakaway or removal torque is still not within limits, replace insert (WP 0549 00).
6. **QA** If replacement spindle is being installed and previous spindle had balance weights, remove hardware and balance washers. Make sure balance washers are installed in original location on replacement spindle.
7. Inspect and treat balance washers and bolts (WP 1703 00).
8. **QA** Install weights as follows (Figure 2, Sheet 2, Detail A):
 - a. Remove tag.
 - b. Slide attaching washer and weights on bolts.
 - c. **QA** Install bolt in original location on bracket (with bolthead facing center of main rotor).
 - d. **QA** Secure with washer and nut.
 - e. **QA TORQUE NUT TO 261 - 289 INCH-POUNDS.**
 - f. Apply a bead of sealing compound, Item 273, WP 1803 00, around boltheads and nuts.
9. If replacement spindle is being installed, install bonding jumper (WP 0554 00).

WARNING

The spindle has critical areas that must not be damaged during installation. Nicks, dents, scratches, and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.

CAUTION

To prevent damage to elastomeric bearings, wipe off grease from exterior surfaces of bearings, using warm, soapy water and clean cloth.

10. Remove protective covering from spindle.
11. Attach hoisting sling around spindle and adjust hoisting sling straps. Attach hoisting sling to hoist.
12. **QA** Slide spindle in main rotor head, with horn facing direction of rotation (Figure 3, Sheet 1). Line up offset hole in spindle with offset hole in hub.
13. Coat shank, threads, and area under head of spindle attachment bolts with primer coating, Item 243, WP 1803 00.

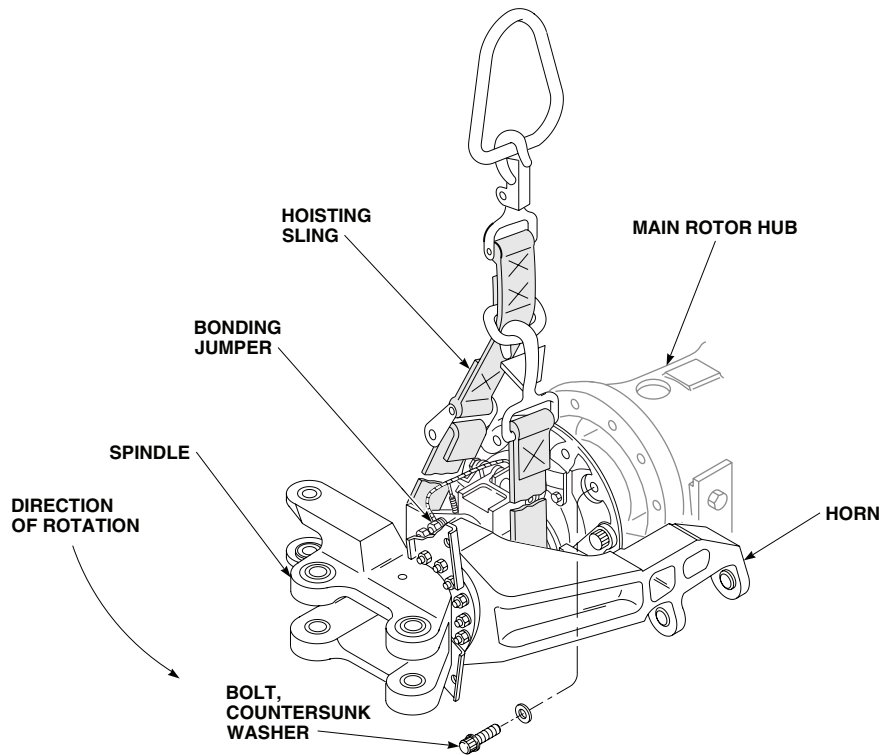
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Figure 3. Install Spindle. (Sheet 1 of 2)

WARNING**CSI**

Verification of proper hardware and spring clip installation and torque of bolts are critical characteristics.

NOTE

When installing the droop stop spring clips and blade deice harness brackets, do not use a washer underneath the bolt head.

14. **QA** Install bolts, washers, and blade deice harness bracket around upper half of spindle while primer coating is still wet (Figure 3, Sheet 2).
15. Remove hoisting sling.
16. **QA** Install remaining bolts, washers, and blade deice harness bracket while primer coating is still wet. Droop stop spring clips (no washers) will go on lower outer bolts. There may not be enough wrench clearance to install bolt behind horn and antiflap assembly. If necessary, insert a piece of wood 2 inch x 4 inch x about 3 feet long, between blade cuff lugs and turn spindle. Make sure top blade deice bracket is positioned vertically to prevent blade deice harness from chaffing on damper. **TORQUE ALL BOLTS TO 910 - 1010 INCH-POUNDS IN CRISSCROSS SEQUENCE.** Apply sealing compound, Item 273, WP 1803 00 around boltheads
17. Install droop stop (WP 0555 00).
18. **QA** Install spring from spring clip to droop stop.

INSTALL - Continued

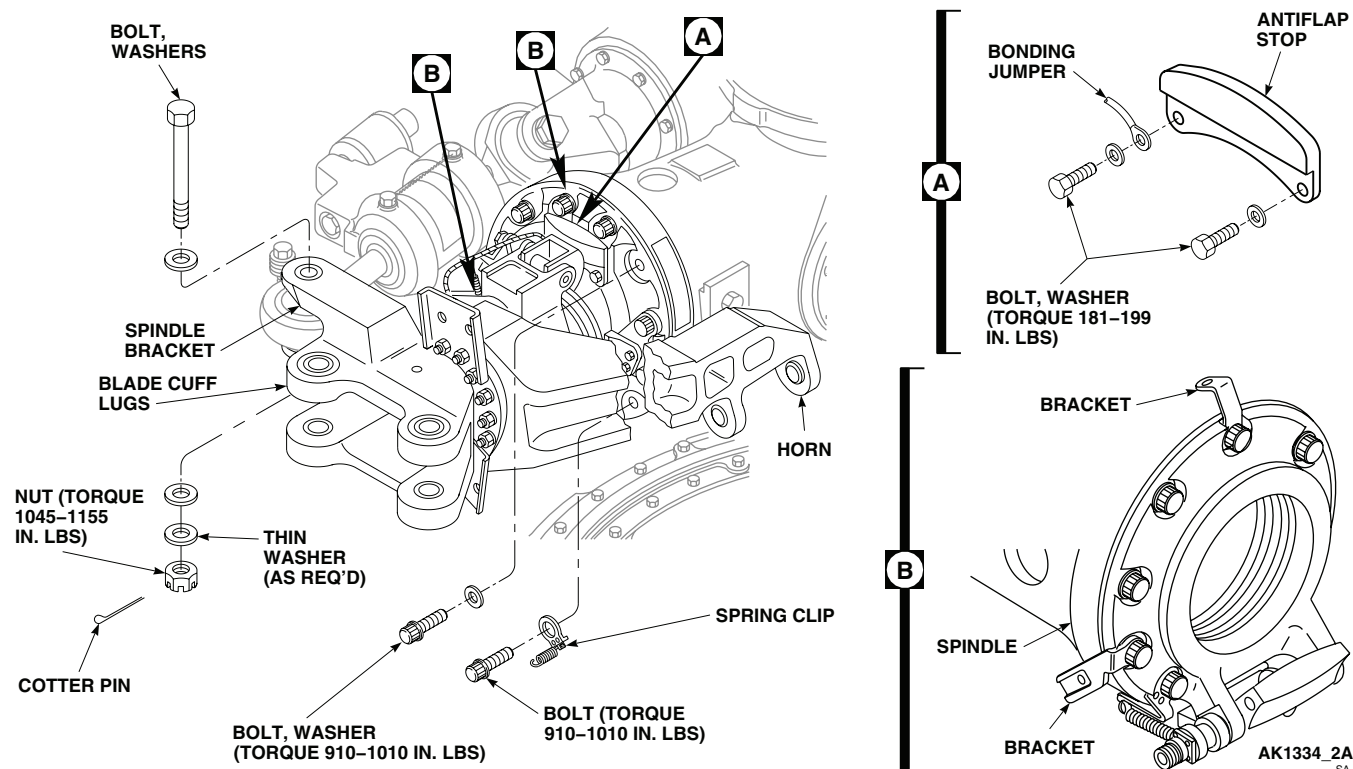


Figure 3. Install Spindle. (Sheet 2 of 2)

19. Install antiflap stop (WP 0554 00).

20. Install damper (WP 0558 00).

WARNING**CSI**

The pitch control rod attachment areas, and critical areas from horn lugs to elbow, are critical surfaces which must not be damaged during installation of horn and/or control rod. Use extreme caution when installing horn and attaching hardware.

CAUTION

Damage to pitch control rod and spindle horn will result if nylon washer is not installed properly. Make sure nylon washer is on opposite side of bearing snapping.

NOTE

- It may be necessary to gently rock spindle horn, to make it easy to line up pitch control rod bolthole.
- Make sure pitch control rod bolthead faces outboard (WP 0563 00).

21. Place upper end of pitch control rod in spindle horn. Install pitch control rod (WP 0563 00).

INSTALL - Continued

22. Touch up paint exposed boltheads, nuts, and threads as necessary (WP 0381 00 and TM 55-1500-345-23).
23. Install main rotor blade on spindle (WP 0576 00).
24. Check area for cleanness and foreign material.
25. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure spindle attachment bolts and damper attachment bolt retorque is shown due 9 to 11 flight hours after installation (WP 1691 00).
26. Check for free rotation of the antifiap cam. If the cam contacts the antifiap stop with main rotor blades installed and rotor in static position, do the following:
 - a. Run up helicopter to 100% Nr in flat pitch for 15 minutes.
 - b. Shut down helicopter and reinspect for cam contact. If cam contact continues, replace elastomeric bearing assembly with a bearing assembly, 70102-08100-044.
27. Track and balance main rotor blades (TM 1-6625-724-13&P).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****SPINDLE HORN**

This WP supersedes WP 0552 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Fluorescent Penetrant Inspection Kit, XMA 101
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13
Spanner Set, AN8515-1
Torque Adapter, Locally-Made, Figure 205,
WP 1805 00
Torque Wrench, Dial, 0 - 600 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.
Vernier Caliper, GGG-C-111

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Abrasive, Item 77, WP 1803 00
Cloth, Abrasive, Item 79, WP 1803 00

Cloth, Abrasive, Item 80, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

DA Form 2408-16
TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 0555 00
WP 0563 00
WP 1691 00
WP 1803 00
WP 1805 00

REMOVE

1. Turn main rotor so horn can be easily removed.
2. Engage gust lock.

WARNING**CSI**

The pitch control rod attachment areas, and critical areas from horn lugs to elbow are critical surfaces which must not be damaged during and after removal of horn and/or control rod. Use extreme caution when removing and provide protective covering for horn and attaching hardware after removal.

3. Disconnect pitch control rod (WP 0563 00).
4. Remove droop stop (WP 0555 00).

CAUTION

Damage to main rotor spindle horn will result if metallic scraper is used to clean horn mounting bolts. Do not use metallic items to remove sealing compound.

5. Clean sealing compound from spindle horn bolts using nonmetallic scraper.

REMOVE - Continued**NOTE**

Removal of droop stop support ring is not required on spindle assemblies, 70102-08200-053 through 70102-08200-056.

6. On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, do this:
 - a. Remove retaining ring from droop stop nut (Figure 1, Detail A).
 - b. Back off droop stop nut using locally-made spanner wrench.
 - c. Slide Teflon lined ring away from droop stop support ring.
 - d. Slide droop stop support ring and outer sleeve assembly off inner sleeve far enough to allow antifiap assembly bolts to be removed.
7. Remove bolts, washers, nuts, and upper and lower bracket (Figure 2, Sheet 1).
8. Remove remaining bolts, washers, and nuts securing horn. Discard old bolts.
9. Remove horn.
10. Using a nonmetallic scraper, remove all sealing compound from spindle flange.

WARNING

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11. Clean primer coating from bolts and brackets with technical acetone, Item 5, WP 1803 00.

INSPECT/REPAIR

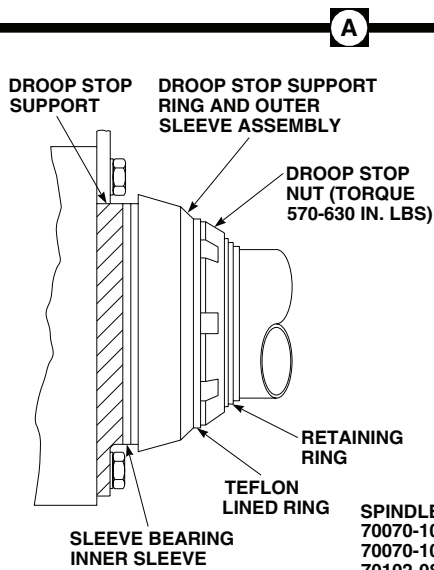
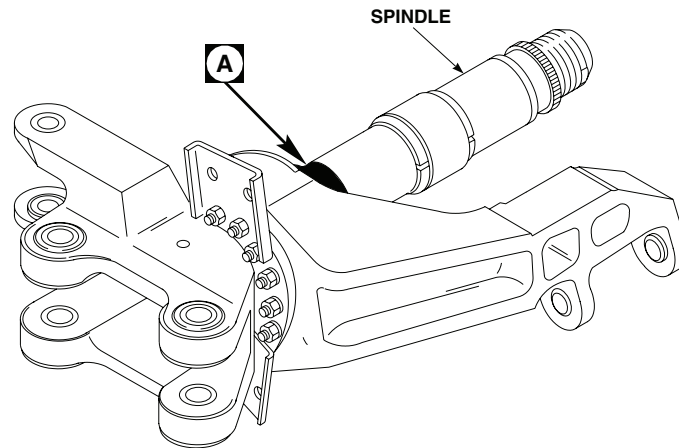
1. Check balance weight brackets for surface damage or corrosion. Blend damage using abrasive cloth, Item 80, WP 1803 00. **REPLACE BRACKET IF DAMAGE OR CORROSION IS DEEPER THAN 0.004-INCH TOTAL THICKNESS AFTER BLENDING.** Coat blended area with primer coating, Item 243, WP 1803 00, and touch up paint as necessary (Figure 2, Sheet 1) (WP 0381 00 and TM 55-1500-345-23).
2. Check balance weight brackets for cracks. **NO CRACKS ALLOWED.** Replace cracked brackets.

WARNING**CSI**

Shaded area of horn (Figure 2, Sheet 2) is a critical surface. No repairs are allowed in this area.

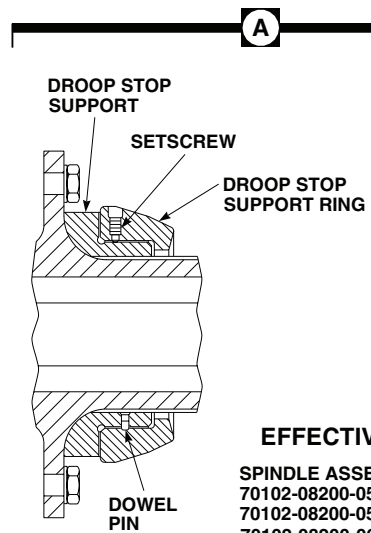
3. Visually inspect shaded area of horn (Figure 2, Sheet 2). Damage in paint or protective finishes not penetrating into base metal may be repaired with touch up paint (WP 0381 00 and TM 55-1500-345-23). **REPLACE HORN IF DAMAGE PENETRATES INTO BASE METAL.**
4. Check Aircraft Component Historical Record, DA Form 2408-16, to see if any inspection areas on horn have been previously reworked.
5. Inspect horn AREA A for scratches, nicks, and gouges (Figure 2, Sheet 2) and blend out all rework areas to a 0.030-inch radius using abrasive cloth, Item 79, WP 1803 00, as follows:

INSPECT/REPAIR - Continued



EFFECTIVITY

SPINDLE ASSEMBLIES
70070-10030-041, 70070-10030-042,
70070-10030-045, 70070-10030-046,
70102-08200-041, THROUGH
70102-08200-044, AND
70102-08200-063.



EFFECTIVITY

SPINDLE ASSEMBLIES
70102-08200-053 THROUGH
70102-08200-056,
70102-08200-068 AND
70102-08200-069.

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SA

Figure 1. Remove Spindle Horn.

INSPECT/REPAIR - Continued

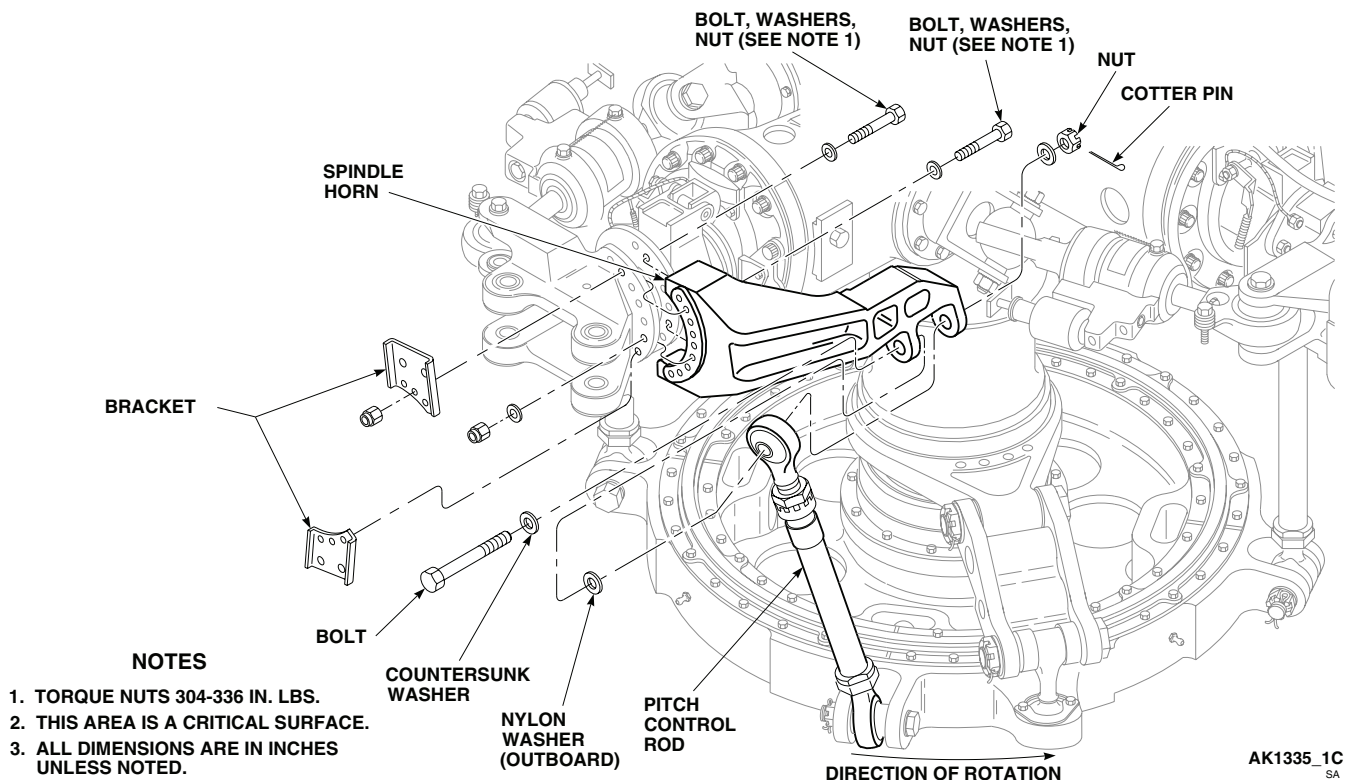
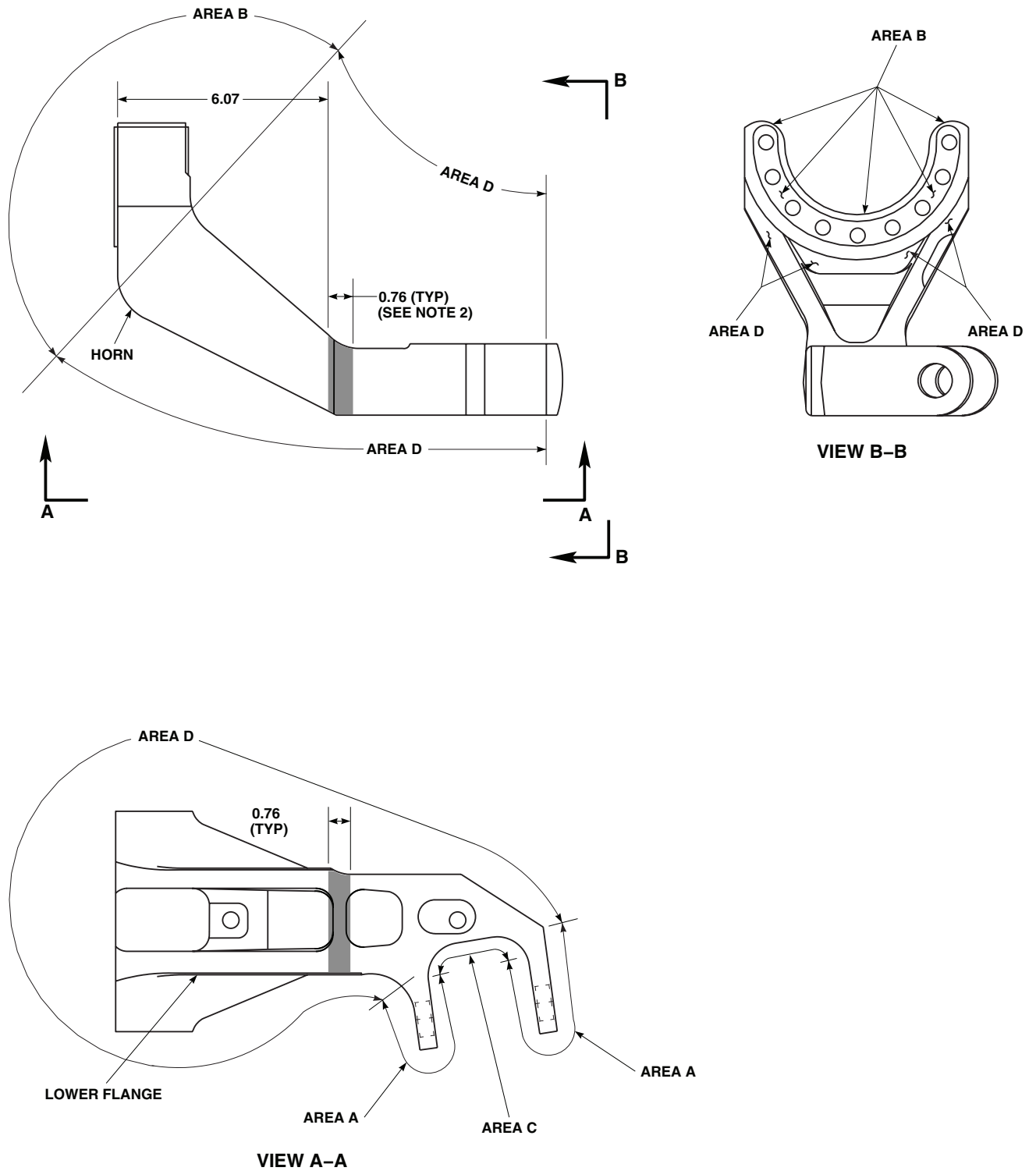


Figure 2. Replace Spindle Horn. (Sheet 1 of 2)

- a. **DAMAGE TO LUG SURFACE UP TO 0.010-INCH DEEP CAN BE BLENDED OUT. TOTAL BLEND AREA NOT TO EXCEED 1 SQUARE INCH.** Replace horn if damage after blending exceeds limits.
- b. **DAMAGE TO LUG EDGE UP TO 0.020-INCH CAN BE BLENDED OUT. TOTAL REPAIR LENGTH TO LUG EDGE NOT TO EXCEED 0.075-INCH.**
- c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace horn if any cracks are found.
6. Inspect horn AREA B for scratches, nicks, and gouges (Figure 2, Sheet 2) and blend out all rework areas to a 0.030-inch radius using abrasive cloth, Item 79, WP 1803 00, as follows:
 - a. **DAMAGE TO ATTACHMENT FLANGE UP TO 0.010-INCH DEEP CAN BE BLENDED. TOTAL LENGTH OF REPAIR NOT TO EXCEED 3 INCHES.** Replace horn if damage after blending exceeds limits.
 - b. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace horn if any cracks are found.
7. Inspect horn AREA C for scratches, nicks, and gouges (Figure 2, Sheet 2) and blend out all rework areas to a 0.030-inch radius using abrasive cloth, Item 79, WP 1803 00, as follows:
 - a. **DAMAGE IN RADIUS BETWEEN HORN LUGS UP TO 0.010-INCH DEEP CAN BE BLENDED. TOTAL BLEND AREA MUST NOT EXCEED 1.500-SQUARE INCHES.** Replace horn if damage after blending exceeds limits.
 - b. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace horn if any cracks are found.

INSPECT/REPAIR - Continued



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SA

Figure 2. Replace Spindle Horn. (Sheet 2 of 2)

INSPECT/REPAIR - Continued**WARNING****CSI**

Nicks, dents, scratches, and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.

8. Inspect horn AREA D for scratches, nicks, and gouges (Figure 2, Sheet 2) and blend out all rework areas to a 0.030-inch radius using abrasive cloth, Item 79, WP 1803 00, as follows:
 - a. **DAMAGE TO "I" BEAM AND RECTANGULAR SECTION UP TO 0.050-INCH DEEP CAN BE BLENDED. TOTAL REPAIR WIDTH AT ANY CROSS SECTION MUST NOT EXCEED 0.600-INCH.** Replace horn if damage after blending exceeds limits.
 - b. **TOTAL BLEND AREA ALONG LOWER FLANGE MUST NOT EXCEED 2 SQUARE INCHES.** Replace horn if damage after blending exceeds limits.
 - c. Using fluorescent penetrant inspection kit, fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace horn if any cracks are found.
9. Check for corrosion. Remove any corrosion with abrasive cloth, Item 77, WP 1803 00. **REPLACE HORN IF CORROSION IS DEEPER THAN 0.030-INCH ON OPEN AREAS, OR DEEPER THAN 0.010-INCH ON MATING AREAS AFTER REMOVAL.**
10. Measure distance between shoulders of bushing where pitch control rod attaches to horn using inside micrometer caliper. **REPLACE HORN IF DISTANCE IS GREATER THAN 2.509-INCHES.**
11. Check silver-plated bushing for wear. **REPLACE HORN IF DEPTH OF WEAR IS DEEPER THAN 0.005-INCH IN SHOULDER AREA OR 0.040-INCH IN CHAMFER AREA.**
12. Check anti flap attachment lugs for damage. Blend damage in areas using a 1 inch abrasive wheel, Item 350, WP 1803 00. **REPLACE BEARING ASSEMBLY IF DAMAGE EXCEEDS 0.015-INCH DEEP AFTER BLENDING.**
13. Check anti flap attachment lugs for cracks. **NONE ALLOWED.** Replace spindle assembly if cracks are found.
14. Touch up paint reworked areas (WP 0381 00 and TM 55-1500-345-23).
15. Enter amount of rework and area information on Aircraft Component Historical Record, DA Form 2408-16.

INSTALL

1. Locally make spindle horn nut torque adapter (WP 1805 00).
2. Coat shank, threads, area under boltheads and exposed portion of each bracket at mounting holes with primer coating, Item 243, WP 1803 00.
3. **QA** Place horn on spindle while primer coating is still wet.

WARNING**CSI**

Torque requirement for spindle to horn attachment nuts is a critical characteristic of the assembly. Verification of the required torque is required.

4. **QA** Install bolts, washers, and nuts. Install upper bracket on three bolts at top of and lower bracket on three bolts at bottom of spindle. **USING LOCALLY-MADE TORQUE ADAPTER, TORQUE ALL NUTS TO 304 - 336 INCH POUNDS. BOLTS MUST EXTEND AT LEAST 0.040-INCH BEYOND NUT (Figure 2, Sheet 1).**

INSTALL - Continued

5. Wipe off any primer coating that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.
6. **QA** On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, secure droop stop support ring (Figure 1, Detail A).

WARNING**CSI**

The pitch control rod attachment areas on horn lugs are critical surfaces which must not be damaged during installation of horn and/or control rod. After removal of protective covering, use extreme caution when installing horn and attaching hardware.

NOTE

- Make sure pitch control rod bolthead faces outboard (Figure 2, Sheet 1).
 - It may help to gently rock horn to make it easy to line up pitch control rod bolthole.
7. Place upper end of pitch control rod in spindle horn. Install bolt, without primer coating, washers, and nut (WP 0563 00).
 8. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure horn attachment nuts torque check is shown as due in 9 to 11 flight hours (WP 1691 00).
 9. Touch up paint on exposed boltheads, nuts and threads as necessary (WP 0381 00 and TM 55-1500-345-23).
 10. Seal areas around joint of washer and nut to bracket and threads of bolts to nuts with sealing compound, Item 273, WP 1803 00.
 11. Install droop stop (WP 0555 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****SPINDLE BONDED WASHER REPAIR**

This WP supersedes WP 0553 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
C-Clamp, GGG-C-406
Compression Plates, Locally-Made, Figure 44,
WP 1805 00
Gloves
Goggles/Face Shield
Heat Gun, LFT500
Inside Micrometer Caliper, GGG-C-105
Locating Tool, Locally-Made, Figure 43, WP 1805 00
Nonmetallic Scraper
Vernier Caliper, GGG-C-111

Cloth, Abrasive, Item 80, WP 1803 00
Parting Compound, Foundry Mold, Item 212,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1803 00
WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00

REPLACE

1. Locally make two sizing blocks to be used as compression plates (WP 1805 00).
2. Locally make locating tool (WP 1805 00).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

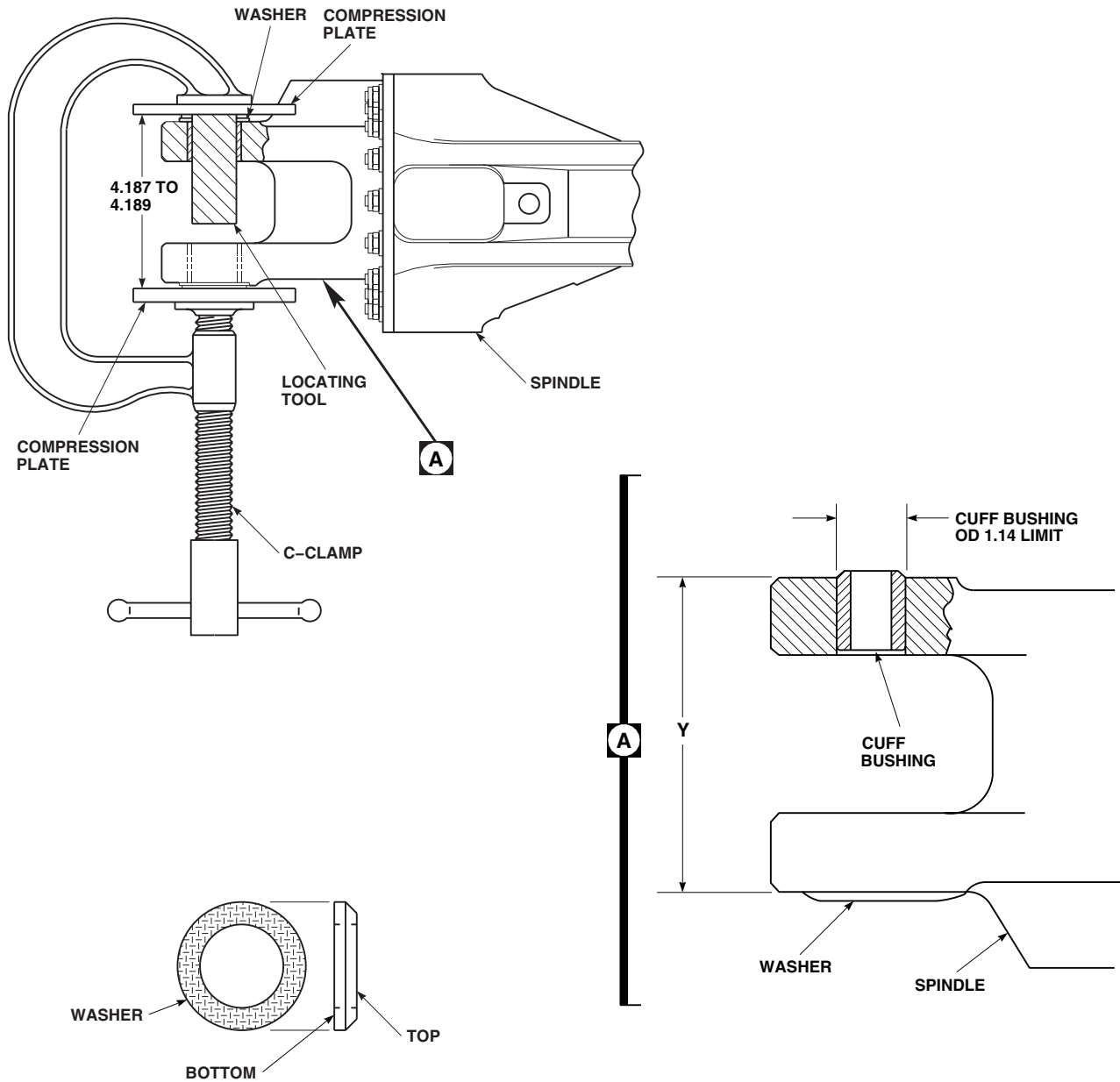
3. Clean spindle surface with nonmetallic scraper and technical acetone, Item 5, WP 1803 00.

NOTE

Only one washer from each expandable pin hole can be missing or unserviceable. If both washers are missing or unserviceable from one expandable pin hole, replace spindle.

4. If original washer is unserviceable or missing, do this:
 - a. Using vernier caliper, measure OD of cuff bushing (Figure 1, Detail A). **IF CUFF BUSHING OD IS GREATER THAN 1.14-INCHES, REPLACEMENT WASHERS CANNOT BE USED.** Replace spindle if cuff bushing OD is more than limits.

REPLACE - Continued



NOTE

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

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Figure 1. Replace Spindle Bonded Washer.

REPLACE - Continued

- b. Measure from existing washer to spindle surface that washer will be bonded to (Figure 1). Record as dimension Y.

Table 1. Washer Thickness.

Washer, 70102-08010-105, is 0.079 to 0.085-inch thick.	Washer, 70102-08010-110, is 0.062 to 0.064-inch thick.
Washer, 70102-08010-106, is 0.050 to 0.052-inch thick.	Washer, 70102-08010-111, is 0.065 to 0.067-inch thick.
Washer, 70102-08010-107, is 0.053 to 0.055-inch thick.	Washer, 70102-08010-112, is 0.068 to 0.070-inch thick.
Washer, 70102-08010-108, is 0.056 to 0.058-inch thick.	Washer, 70102-08010-113, is 0.071 to 0.073-inch thick.
Washer, 70102-08010-109, is 0.059 to 0.061-inch thick.	Washer, 70102-08010-114, is 0.074 to 0.076-inch thick.

- c. Subtract dimension Y from 4.181. The total will be replacement washer thickness required to obtain a distance of 4.187 to 4.189-inches between inner faces of compression plates. **REPLACEMENT WASHER THICKNESS SHALL NOT BE MORE THAN THE THICKEST WASHER AVAILABLE IN TABLE 1.** Replace spindle if thickness is not within limits.
- d. Select washer thickness required to obtain a distance of 4.187 to 4.189-inches between inner faces of compression plates.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean bottom surface of washer with abrasive cloth, Item 80, WP 1803 00. Wipe clean with clean machinery wiping towel, Item 344, WP 1803 00, soaked with technical acetone, Item 5, WP 1803 00.
6. Spray thin film of foundry mold parting compound, Item 212, WP 1803 00, onto areas on compression plates and tool that might touch adhesive, Item 11, WP 1803 00.
7. Mix adhesive, Item 11, WP 1803 00, using manufacturer's instructions. Apply thin coat to spindle on area where washer will be placed. Make sure to cover entire area.
8. Apply thin coat of adhesive, Item 11, WP 1803 00, to bottom of washer.
9. Place locating tool through bolt hole in spindle. This will keep washer centered over hole until C-clamp can hold it in place (Figure 1).
10. **QA** Place washer over locating tool, bottom side towards spindle, and press onto spindle.
11. **QA** Place compression plates on washers.

REPLACE - Continued

12. Clamp compression plates with C-clamp, applying pressure until distance between inner faces of compression plates is 4.187 to 4.189-inches.

CAUTION

Damage to elastomeric bearing will result if it is exposed to direct heat from heat gun.
Do not expose bearing to direct heat greater than 148°F(65°C).

13. Using heat gun, cure adhesive for 45 minutes at 190°F(87°C) to 210°F(99°C) or 2 hours at 140°F(60°C) to 160°F(71°C).
14. If heat gun is not available, allow adhesive to cure for 12 hours at room temperature.
15. If using adhesive, Item 11, WP 1803 00, cure for 24 hours at 70°F(21°C) to 85°F(20°C) or for 1 hour at 140°F(60°C) to 160°F(71°C).
16. **QA** Remove locating tool, C-clamp, and compression plate.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**ANTIFLAP ASSEMBLY**

This WP supersedes WP 0554 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Blade Clamp Assembly, 70700-20324-041
 Droop Stop Nut Wrench, Locally-Made, Figure 168,
 WP 1805 00
 Gloves
 Goggles/Face Shield
 Nonmetallic Drift
 Nonmetallic Scraper
 Rubber Mallet, GGG-H-0033
 Spanner Wrench, 1520-BH5E-001, Locally-Made,
 Figure 52, WP 1805 00
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00

Insulation Sheet, Electrical, Item 153, WP 1803 00
 Primer, Adhesive, Item 240, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Sealing Compound, Item 275, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
 WP 0381 00
 WP 0549 00
 WP 1691 00
 WP 1803 00
 WP 1805 00

REMOVE

1. Locally make droop stop nut wrench, if required (WP 1805 00).
2. Turn main rotor head until antiflap assembly is easy to reach.
3. Install blade clamp on main rotor blade balance strips. Fasten lock.
4. Secure antiflap cam in flight position.
5. Attach hoist or maintenance crane to blade clamp and lift blade until droop stop ring is off droop stop cam.

NOTE

Removal of droop stop support ring is not required on spindle assemblies, 70102-08200-053 through 70102-08200-056.

6. On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, do this:
 - a. Remove retaining ring from droop stop nut (Figure 1, Sheet 1, Detail A).
 - b. Back off droop stop nut using locally-made spanner wrench (WP 1805 00).
 - c. Slide Teflon lined ring away from droop stop support ring.
 - d. Slide droop stop support ring and outer sleeve assembly off inner sleeve far enough to allow antiflap assembly bolts to be removed.

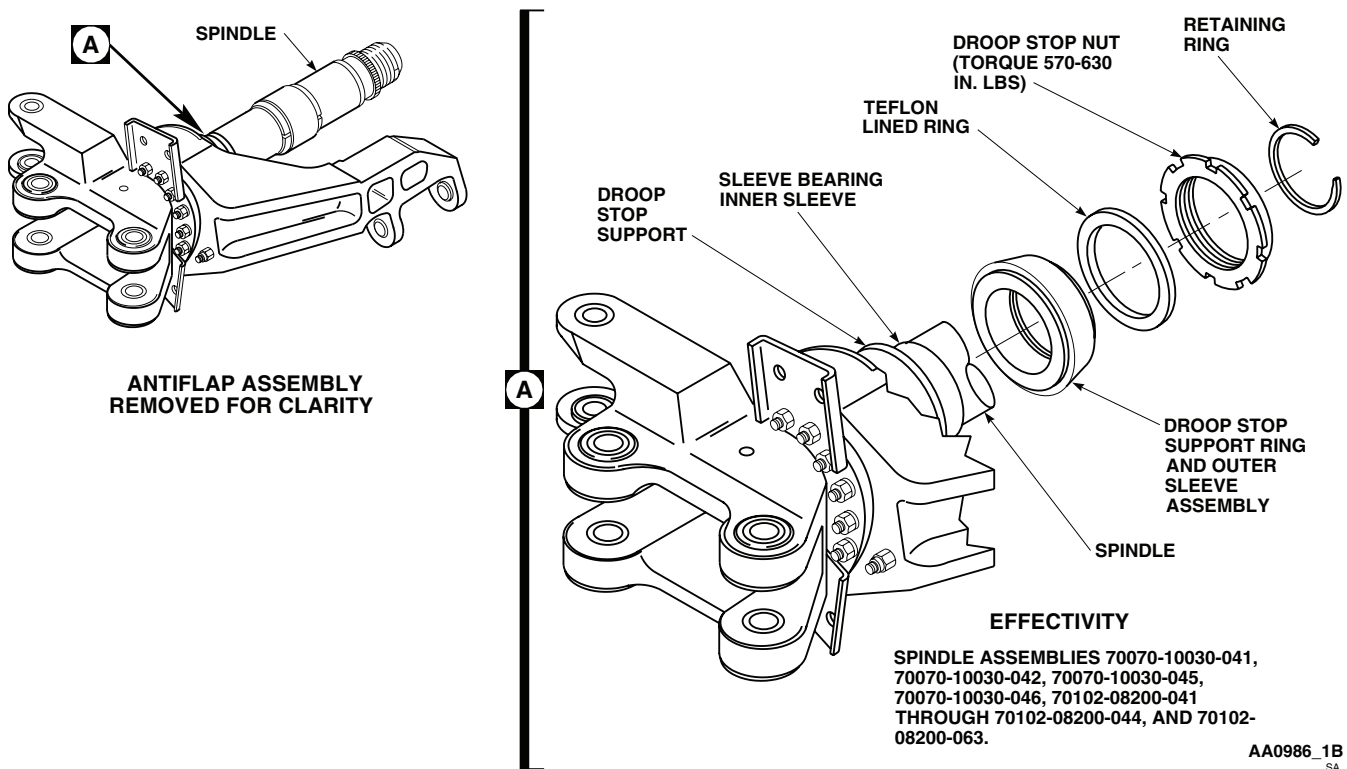
REMOVE - Continued

Figure 1. Replace Antiflap Assembly. (Sheet 1 of 4)

7. Remove spring and allow antiflap cam to swing down.
8. If stop is damaged, remove it.

CAUTION

Damage to antiflap assembly will result if metallic scraper is used to clean mounting bolts. Do not use metallic items to remove sealant.

9. Clean sealing compound from antiflap assembly boltheads using nonmetallic scraper.
10. Remove bolts, washers, nuts, and antiflap assembly from spindle (Figure 1, Sheet 2). Discard long bolts.

DISASSEMBLE**NOTE**

It is not necessary to remove antiflap assembly from spindle to remove shaft from antiflap cam.

1. Remove nut, washer, and spring retainer from antiflap shaft (Figure 1, Sheet 3, Detail B).
2. Using a pin punch, gently tap out spring pin until free of antiflap cam and shaft.
3. Slide shaft out of antiflap assembly and remove cam.

DISASSEMBLE - Continued

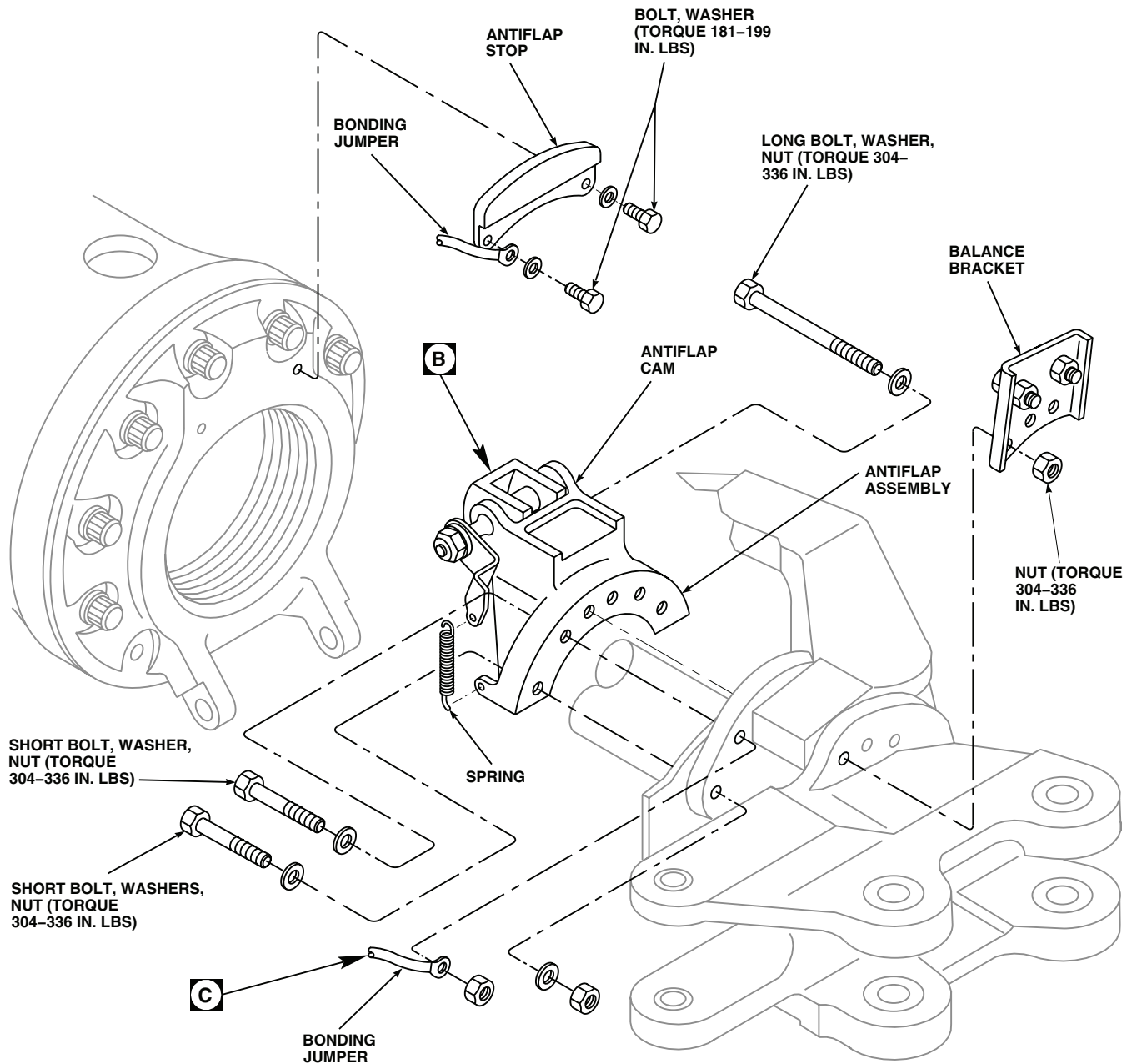
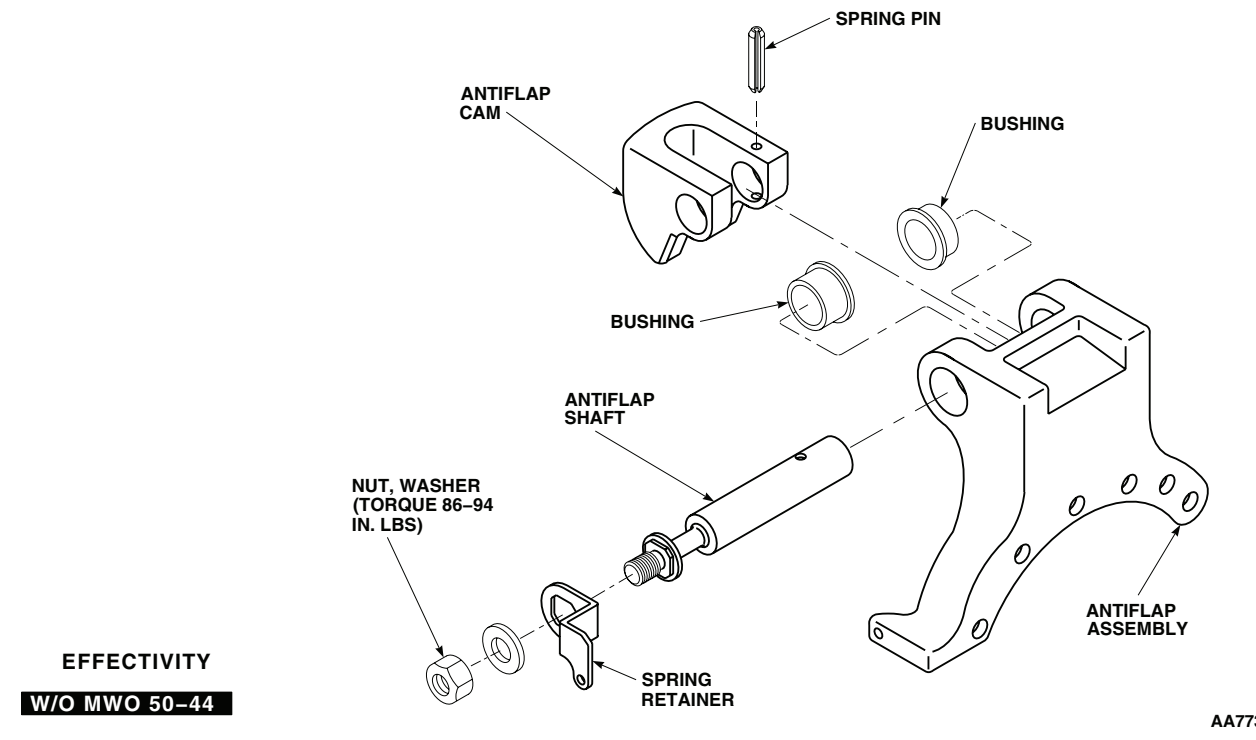
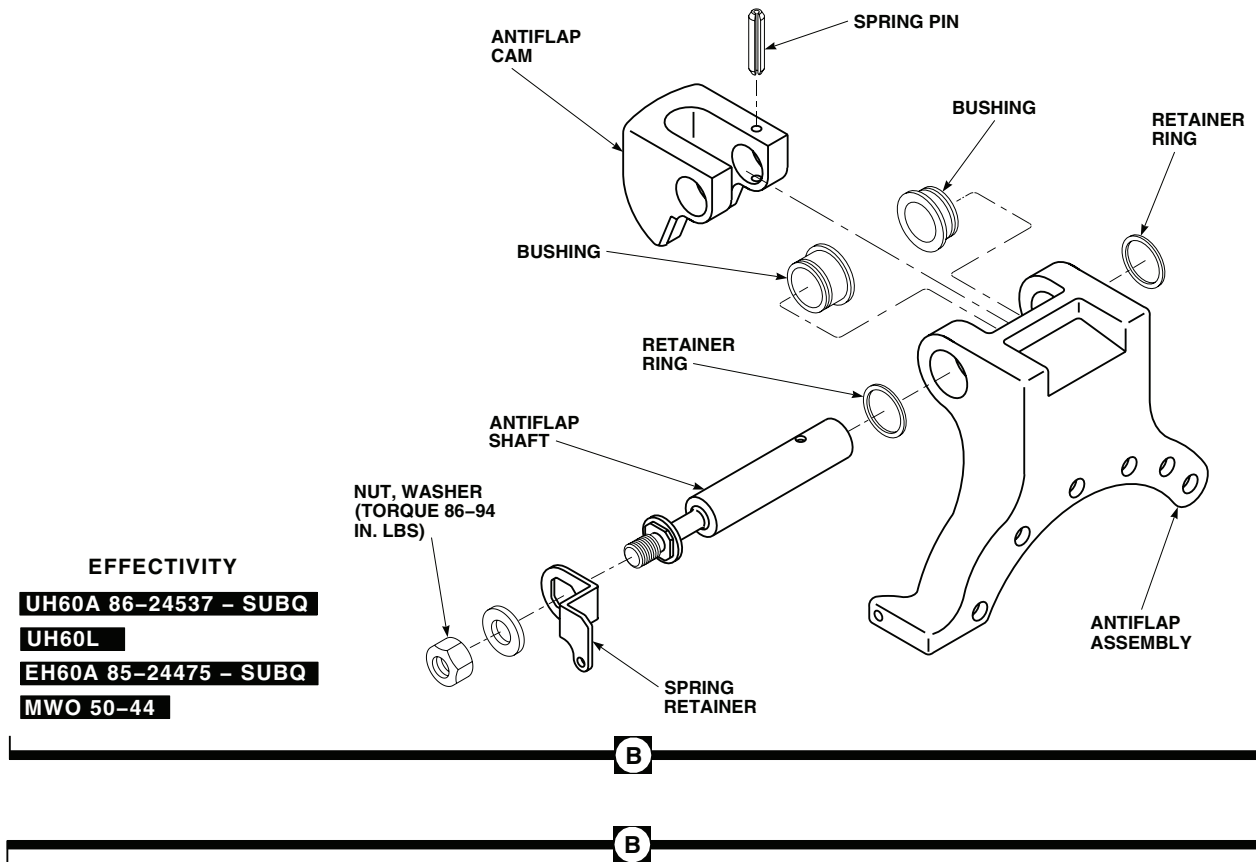
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Figure 1. Replace Antiflap Assembly. (Sheet 2 of 4)

DISASSEMBLE - Continued



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Figure 1. Replace Antiflap Assembly. (Sheet 3 of 4)

DISASSEMBLE - Continued

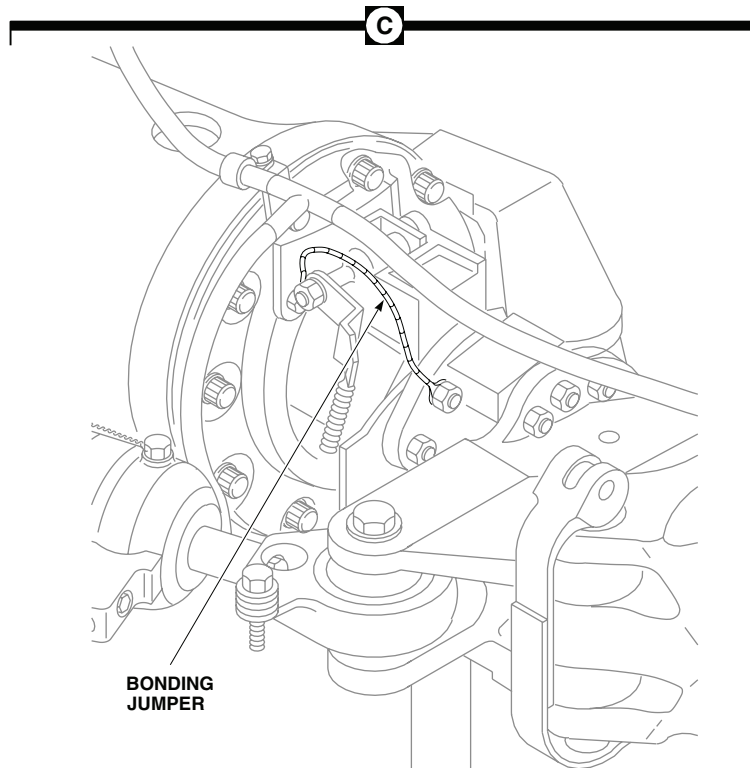
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Figure 1. Replace Antiflap Assembly. (Sheet 4 of 4)

INSPECT/REPAIR

1. Inspect threaded inserts in elastomeric spherical bearing that antiflap stop bolts go into as follows (Figure 1, Sheet 2):
 - a. Thread new bolt into insert until bolt reaches insert locking feature. **BOLT SHOULD THREAD DOWN SMOOTHLY WITH NO EVIDENCE OF STRIPPED OR CROSSED THREADS. MAXIMUM TORQUE REQUIRED TO THREAD BOLTS IS 20 INCH-POUNDS.**
 - b. Replace insert if not within limits (WP 0549 00).
2. Visually check antiflap for cracks. **NO CRACKS ARE ALLOWED EXCEPT ANTIPLAP CAM MAY HAVE SMALL CRACKS (UP TO 1/8-INCH) AT THE CORNER RADIUS OF IMPACT SURFACES.** Replace anti-flap if damage is beyond limits.
3. Check antiflap bracket for nicks, gouges, burrs, or corrosion. **BLENDING OUT OF NICKS, GOUGES, BURRS, OR CORROSION UP TO 0.020-INCH IS ACCEPTABLE. BLENDING OF CORNERS AND SURFACES UP TO 0.030-INCH IS ACCEPTABLE.** Replace antiflap bracket if damage is beyond limits.
4. Check cam impact surfaces for deformation. **DEFORMATION UP TO 0.10-INCH IS ACCEPTABLE.** Replace cam if deformation is beyond limits.
5. Check cam for corner damage (nicks, gouges, etc.). **CORNER DAMAGE UP TO 1/8-INCH IS ACCEPTABLE AFTER REMOVING SHARP EDGES WITH A MEDIUM CUT FILE.** Replace cam if corner damage is beyond limits.
6. Touch up reworked areas (WP 0381 00 and TM 55-1500-345-23).

INSPECT/REPAIR - Continued

7. Visually inspect antifiap stop attachment bolthead and antifiap stop spring for signs of corrosion. **NONE ALLOWED.** Replace corroded bolts and springs.
8. Inspect antifiap bushings for displacement.

CAUTION

Damage to antifiap assembly will result if metallic tools are used to clean or remove bushings. Do not use metallic items to remove sealing compound or bushings.

9. **W/O MWO 50-44** If bushings are displaced do this:
 - a. Clean sealing compound from antifiap bushing using nonmetallic scraper.
 - b. Remove bushing using nonmetallic drift and rubber mallet (Figure 1, Sheet 3, Detail B).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- c. Clean outside diameter of bushing and inside diameter of hole in antifiap assembly using machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe outside diameter of bushing using abrasive cloth, Item 82, WP 1803 00. Wipe outside diameter of bushing and antifiap assembly dry using machinery wiping towel, Item 344, WP 1803 00.

NOTE

Do not allow sealing compound to contact inside diameter of bushing or bushing face.

- d. Apply sealing compound, Item 275, WP 1803 00, to outside diameter of bushing.
- e. Place bushing in antifiap assembly (Figure 1, Sheet 3, Detail B).
- f. **UH-60A 86-24537 - SUBQ UH-60L EH-60A 85-24475 - SUBQ MWO 50-44** Install retainer ring on grooved bushings, using needlenose pliers.
- g. Apply adhesive primer, Item 240, WP 1803 00, around outside diameter of bushing flange.

ASSEMBLE

1. **QA** Install shaft through bushings in antifiap assembly and holes in cam (Figure 1, Sheet 3, Detail B).
2. Line up spring pin hole in cam with hole in shaft. Press or tap spring pin into hole until flush with cam.

NOTE

Do not reuse self-locking nut.

3. **QA** Install spring retainer, washer, and nut on shaft. Make sure retainer is correctly seated on square end of shaft with small hole for spring facing down. **TORQUE SHAFT NUT TO 86 - 94 INCH-POUNDS.**
4. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire spring pin to cam.

INSTALL

1. Coat shank, threads, and area under head of all bolts with primer coating, Item 243, WP 1803 00.

INSTALL - Continued**WARNING****CSI**

Torque requirement for spindle to horn attachment nuts is a critical characteristic of the assembly. Verification of the required torque is required.

2. **QA** Place antifiap assembly on spindle. Install long bolts, washers, and nuts as shown while primer is still wet. **TORQUE NUTS TO 304 - 336 INCH-POUNDS.** Install short bolts, washers, nuts, and bonding jumper, while primer is still wet (Figure 1, Sheet 2). Omit washer between bonding jumper and nut. Orientate bonding jumper to 45 degrees up position (Figure 1, Sheet 4, Detail C). **TORQUE NUTS TO 304 - 336 INCH-POUNDS.**
3. **QA** Install spring.
4. Coat shank, threads, and area under head of antifiap stop bolts with primer coating, Item 243, WP 1803 00.
5. **QA** Install antifiap stop and bonding jumper with bolts and washers while primer is still wet (Figure 1, Sheet 2). Orientate bonding jumper to 30 to 45 degrees up position (Figure 1, Sheet 4, Detail C). **TORQUE BOLTS TO 181 - 199 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts. Cover nonelectrical wire with electrical insulation sheet, Item 153, WP 1803 00.
6. Wipe off any primer that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.
7. Seal areas around boltheads and nuts with sealing compound, Item 273, WP 1803 00.
8. **QA** On spindle assemblies, 70070-10030-041, 70070-10030-042, 70070-10030-045, 70070-10030-046, 70102-08200-041 through 70102-08200-044, and 70102-08200-063, do this:
 - a. Apply wet primer coating, Item 243, WP 1803 00, to threads of droop stop support and to face of droop stop nut.
 - b. **QA** Slide droop stop support ring and outer sleeve assembly onto inner sleeve.
 - c. **QA** Slide Teflon lined ring against droop stop support ring.
 - d. **QA** Install droop stop nut with locally-made spanner wrench. **TORQUE NUT TO 570 - 630 INCH-POUNDS.**
 - e. Wipe off excess primer from under nut.
 - f. **QA** Install retaining ring behind droop stop nut.
 - g. Seal area around droop stop nut with sealing compound, Item 273, WP 1803 00.
9. Lower blade until droop stop ring is resting on droop stop cam. Remove hoist and blade clamp.
10. Touch up paint on bolts and exposed threads, as necessary (WP 0381 00 and TM 55-1500-345-23).
11. Check area for cleanness and foreign material.
12. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure antifiap assembly to spindle attachment bolts torque check is shown due at 9 to 11 flight hours after installation (WP 1691 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****DROOP STOP**

This WP supersedes WP 0555 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Aspirator, GGG-M-125/6
Blade Clamp, 70700-20324-041
Goggles
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0536 00
WP 0556 00
WP 1220 00
WP 1803 00

Material/Parts

Strap Tiedown, Electrical Item 301, WP 1803 00

REMOVE**WARNING**

The droop stop is made of beryllium copper metal, which can be a health hazard. The use of goggles is required. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

1. Turn main rotor head so droop stop is easy to reach.

CAUTION

To prevent damage to the antifiap cam and antifiap stop, secure the antifiap in the flight position prior to lifting the blade.

2. Push down on antifiap cam lever (flight position) and secure to horn with electrical tiedown strap, Item 301, WP 1803 00.
3. Install blade clamp on main rotor blade on balance strips. Fasten lock.
4. Attach hoist to blade clamp and lift blade until spindle is off droop stop cam.
5. Disconnect springs from spring pins (Figure 1, Sheet 1 and Figure 1, Sheet 2).
6. If deice kit is installed, remove electrical connector from heater pin (WP 1220 00).
7. Remove cotter pin, slide out pin and remove droop stop cam.
8. If necessary, remove clips and springs from spindle.
9. On droop stop cam, 70105-08151-041, remove nuts, washers, bushings, and spring pins from cam (Figure 1, Sheet 1).
10. On droop stop cam, 70105-08151-043 or 70105-08151-045, remove nuts, washers, and spring pins from cam (Figure 1, Sheet 2).

REMOVE - Continued

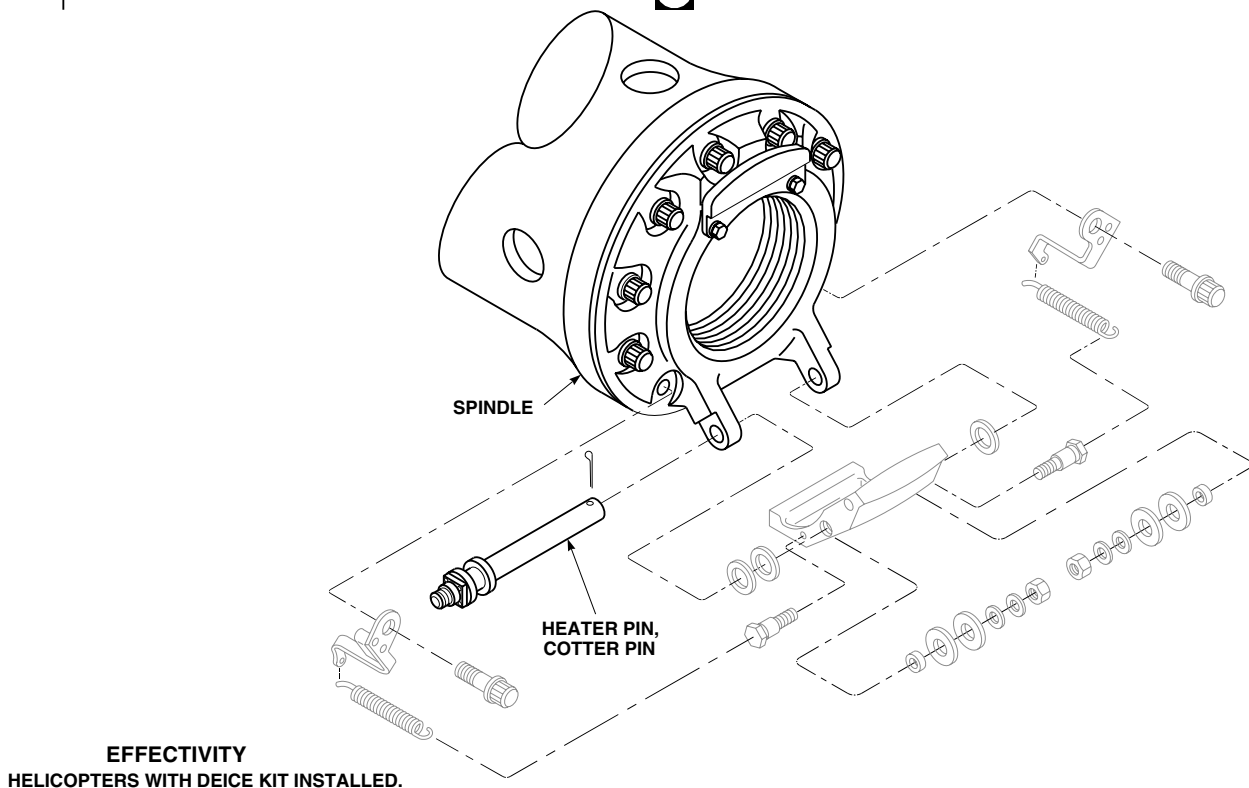
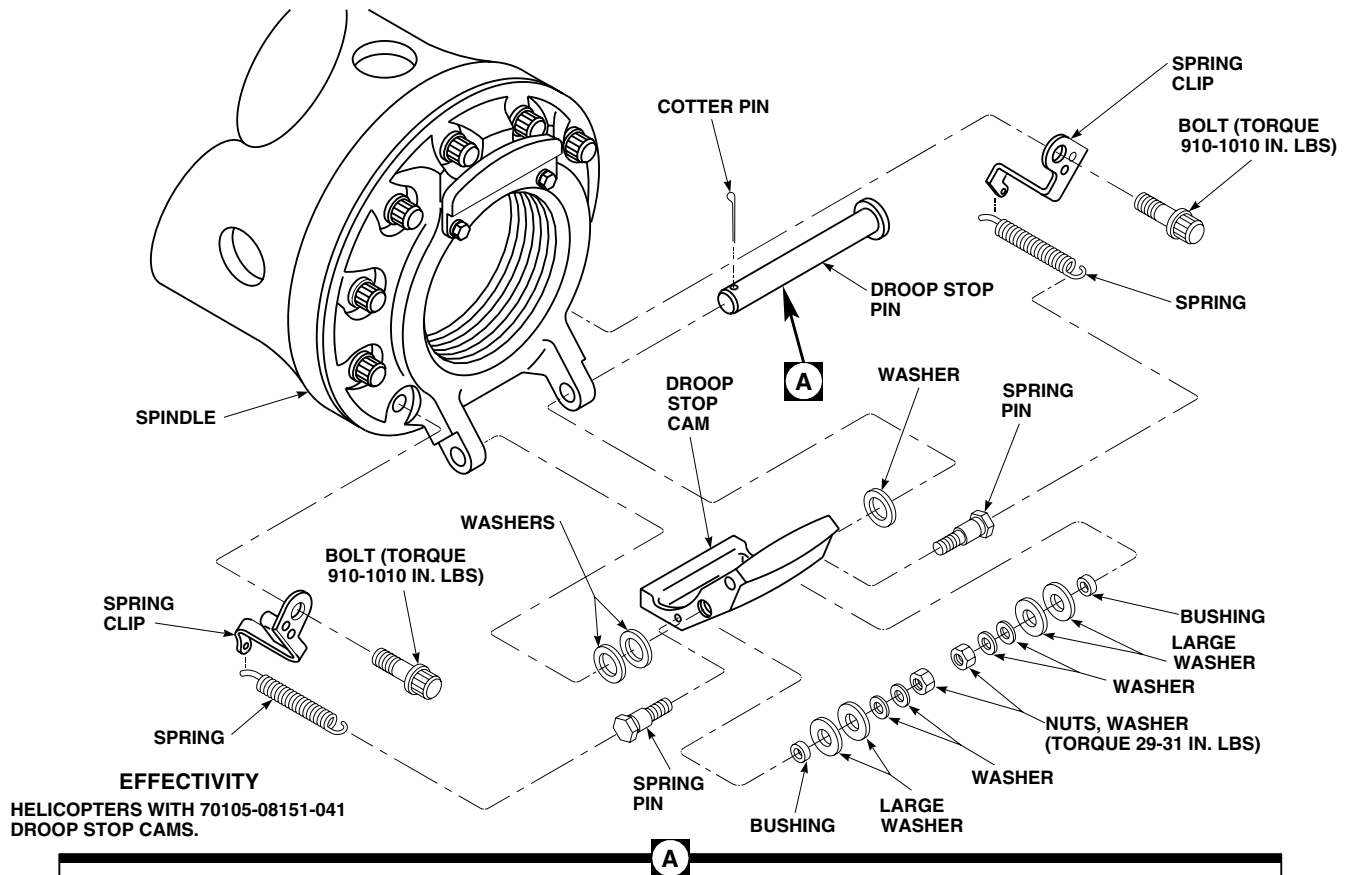
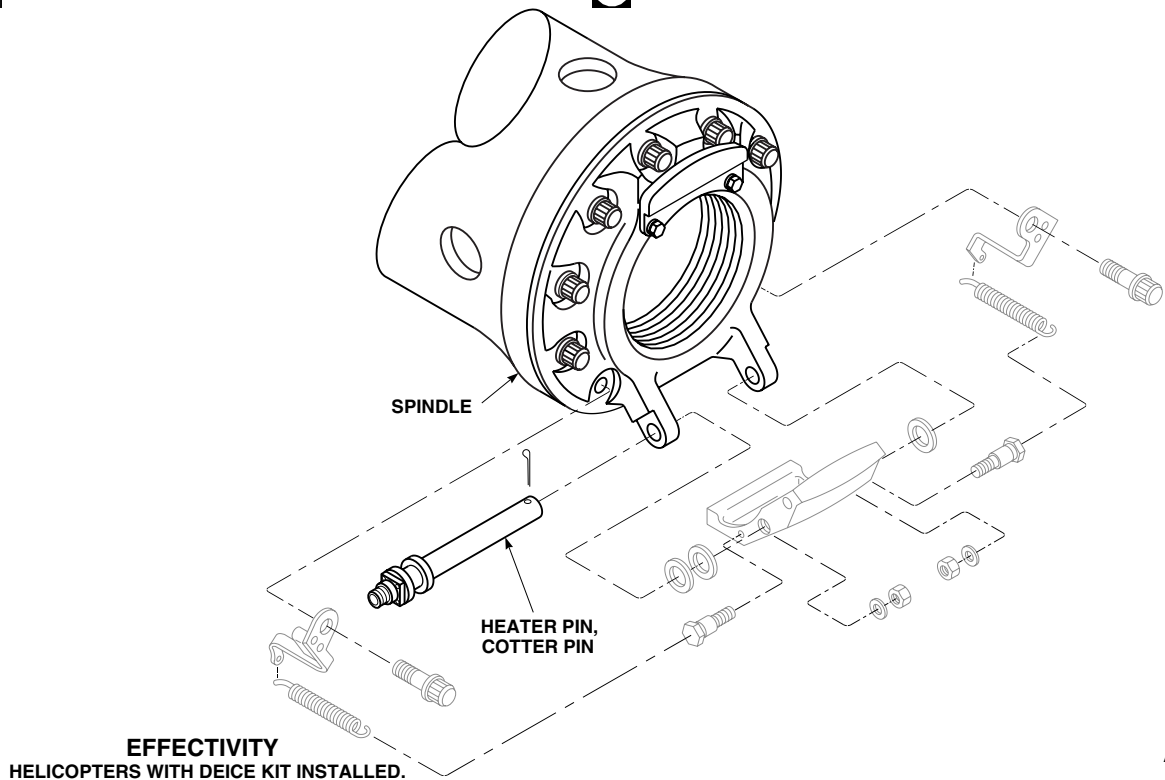
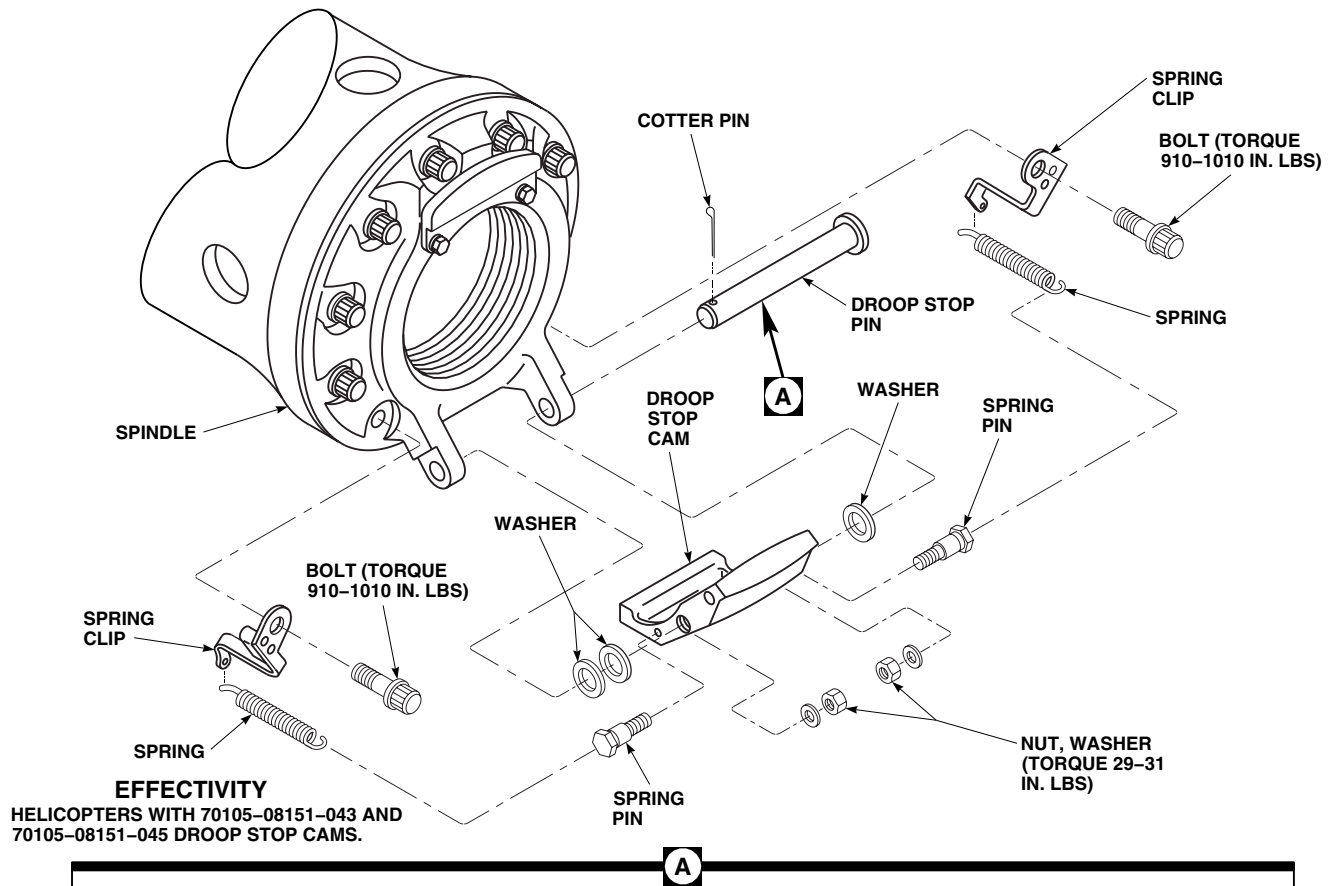
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Figure 1. Replace Droop Stop. (Sheet 1 of 2)

REMOVE - Continued



AA0109_2A
SA

Figure 1. Replace Droop Stop. (Sheet 2 of 2)

REMOVE - Continued

11. Inspect droop stop (WP 0536 00).
12. Inspect droop stop bushings (WP 0556 00).

INSTALL

1. **QA** On droop stop cam, 70105-08151-041, install nuts, washers, bushings, and spring pins on sides of droop stop cam (Figure 1, Sheet 1). **TORQUE NUTS TO 29 - 31 INCH-POUNDS.**
2. **QA** On droop stop cam, 70105-08151-043 or 70105-08151-045, install nuts, washers, and spring pins on sides of droop stop cam (Figure 1, Sheet 2). **TORQUE NUTS TO 29 - 31 INCH-POUNDS.**

WARNING**CSI**

Verification of proper hardware and spring clip installation and torque of bolts are critical characteristics.

3. **QA** If removed, place clips on spindle and install bolts. **TORQUE BOLTS TO 910 - 1010 INCH-POUNDS.** Connect springs to spring clips.
4. **QA** Install droop stop cam, washers, and pin on spindle. Install cotter pin.
5. **QA** If deice kit is installed, reinstall pin and electrical connector to heater pin (WP 1220 00).
6. **QA** Connect springs to spring pins.
7. Lower blade. Remove blade clamp.
8. **QA** Remove electrical tiedown strap from antifiap cam.
9. Check area for cleanness and foreign material.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**DROOP STOP BUSHINGS**

This WP supersedes WP 0556 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Arbor Press, OO-P-636
 Drift
 Droop Stop Bushing Replacement Tool, Figure 273,
 WP 1805 00
 Gloves
 Goggles/Face Shield
 Inside Micrometer Caliper
 Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00

Cloth, Cheesecloth, Item 85, WP 1803 00
 Sealing Compound, Item 284, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0551 00
 WP 0555 00
 WP 0595 00
 WP 1803 00
 WP 1805 00

INSPECT BUSHINGS**WARNING**

The droop stop is made of beryllium copper metal, which can be a health hazard. The use of goggles is required. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

1. Check droop stop bushings for nicks and gouges. **NICKS AND GOUGES ON SHOULDER OF BUSHINGS MAY BE BLENDED UP TO 0.020-INCH DEEP AND UP TO 25% OVERALL DAMAGE, USING ABRASIVE CLOTH, Item 82, WP 1803 00.**
2. Check ID of bushings. **ID OF EITHER BUSHING MUST NOT BE MORE THAN 0.762-INCH.** Replace any droop stop bushing that is over above limits.
3. Check droop stop bushings for displacement or disbonding. **NO DISPLACEMENT OR DISBONDING ALLOWED.** If displacement or disbonding has occurred, do this:
 - a. Remove displaced or disbonded droop stop bushing. Refer to, REMOVE, in this work package.
 - b. Check droop stop bushing OD. **DROOP STOP BUSHING OD SHALL NOT BE LESS THAN 0.8755-INCH.** Replace bushing if beyond limits.
 - c. Inspect/repair spherical bearing lug bore. Refer to, INSPECT LUG BORE, in this work package.
 - d. Install droop stop bushing. Refer to INSTALL, in this work package.

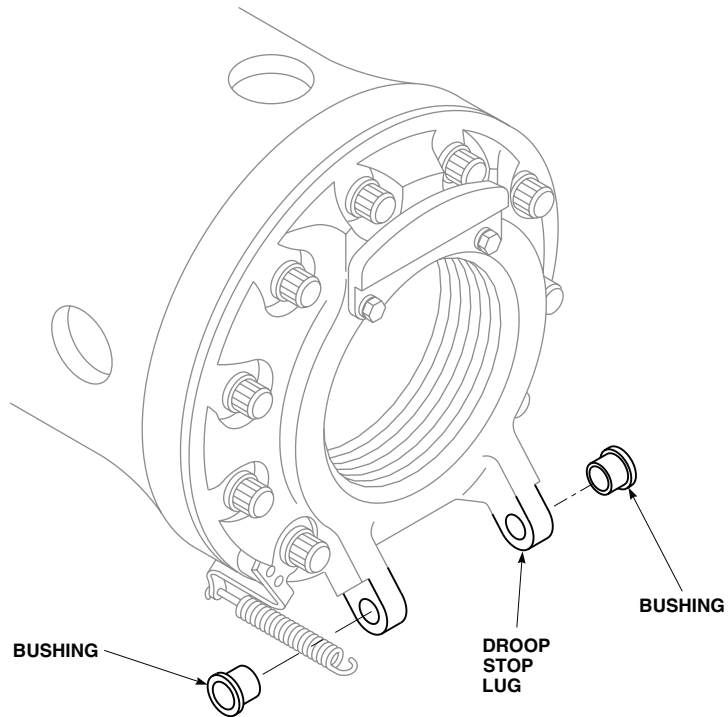
INSPECT BUSHINGS - ContinuedAK1342
SA

Figure 1. Replace Droop Stop Bushings.

REMOVE**WARNING**

The droop stop is made of beryllium copper metal, which can be a health hazard. The use of goggles is required. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

1. Locally make droop stop bushing replacement tool, Figure 273, WP 1805 00.
2. Remove droop stop (WP 0555 00).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Apply technical acetone, Item 5, WP 1803 00, on old bushings to soften adhesive.
4. Press old bushings from droop stop lugs using droop stop bushing replacement tool, Figure 273, WP 1805 00 (Figure 1).

REMOVE - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean hole in bearing endplate with nonmetallic scraper, and cheesecloth cloth, Item 85, WP 1803 00, soaked in technical acetone, Item 5, WP 1803 00.

INSPECT LUG BORE

1. Check lug bore for scoring. Remove scoring using abrasive cloth, Item 82, WP 1803 00.
2. Using inside micrometer caliper, check ID of lug bore in two places minimum. **AVERAGE LUG BORE DIAMETER MUST NOT BE MORE THAN 0.8760-INCH.** If lug bore is worn beyond limits, remove spindle (WP 0551 00), and replace spherical bearing (WP 0595 00).

INSTALL

1. Coat new bushing with sealing compound, Item 284, WP 1803 00.
2. **QA** Using droop stop bushing replacement tool, Figure 273, WP 1805 00, press bushing into droop stop lugs (Figure 1).
3. Wipe extra adhesive off bushing and bearing endplate using machinery wiping towel, Item 344, WP 1803 00. Let cure for 24 hours.
4. Install droop stop (WP 0555 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****DROOP STOP PAD**

This WP supersedes WP 0557 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Aspirator, GGG-M-125/6
Gloves
Goggles/Face Shield

Cloth, Cleaning, Item 86, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Cloth, Abrasive, Item 75, WP 1803 00

References

TM 55-1500-345-23
WP 0381 00
WP 0555 00
WP 1803 00

REMOVE

1. Remove droop stop cam assembly (WP 0555 00).
2. Remove old pad (Figure 1).

INSTALL

1. Clean bond areas on cam and new wear pad with abrasive cloth, Item 75, WP 1803 00.

WARNING

- Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.
 - The droop stop is made of beryllium copper metal, which can be a health hazard. The use of goggles is required. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.
2. Wipe roughed areas on cam and pad with cleaning cloth, Item 86, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
 3. Apply adhesive, Item 14, WP 1803 00, on cam and pad.
 4. Allow adhesive to dry for 10 minutes, until most of solvent has evaporated and adhesive is tacky.
 5. **QA** Put pad on cam so that surfaces to be bonded are joined together firmly, making sure entire mating surfaces are in direct contact (Figure 1). Do not move or disturb bonded surfaces after mating.
 6. Allow bond to cure at room temperature for at least 24 hours. Greatest strength is obtained in about seven days.
 7. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

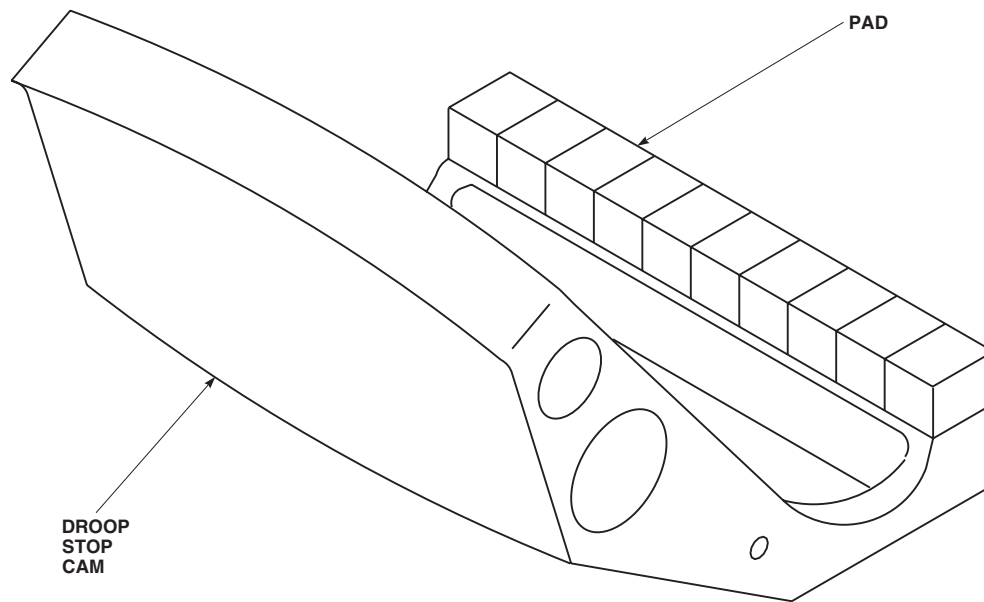
INSTALL - ContinuedAA7739
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Figure 1. Replace Droop Stop Pad.

8. Install droop stop cam (WP 0555 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****DAMPER**

This WP supersedes WP 0558 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Depth Gage
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 150 - 1000 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Material/Parts

Cloth, Abrasive, Item 77, WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00
Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Primer Coating, Item 243, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Sealing Compound, Item 258, WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Pneudraulics Repairer MOS 15H (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0381 00
WP 0537 00
WP 0550 00
WP 0551 00
WP 0560 00
WP 0562 00
WP 1610 00
WP 1691 00
WP 1803 00

REMOVE

1. Turn main rotor until damper is easy to reach.
2. Engage gust lock.
3. Remove sealant from around damper bolts and nuts using nonmetallic scraper.

NOTE

If necessary, when removing inboard and outboard bolts, lightly tap them with a rawhide mallet and a nonmetallic drift.

4. Remove bolt, washers, and nut securing damper to spindle (Figure 1).
5. Remove bolt, washers, and nut securing damper to damper bracket. Remove damper.

NOTE

- Keep restraining cable for possible installation on new damper.
 - Restraining cables are not required on dampers with part numbers, 70106-08100-044, 70106-08100-046, or marked, RS-001C-V1.
6. If necessary to remove restraining cable from damper cylinder head, do this:
 - a. Note location and position of cable clip attached to damper cylinder head stud.
 - b. Remove nut securing cable clip to damper cylinder head stud. Remove cable.

REMOVE - Continued

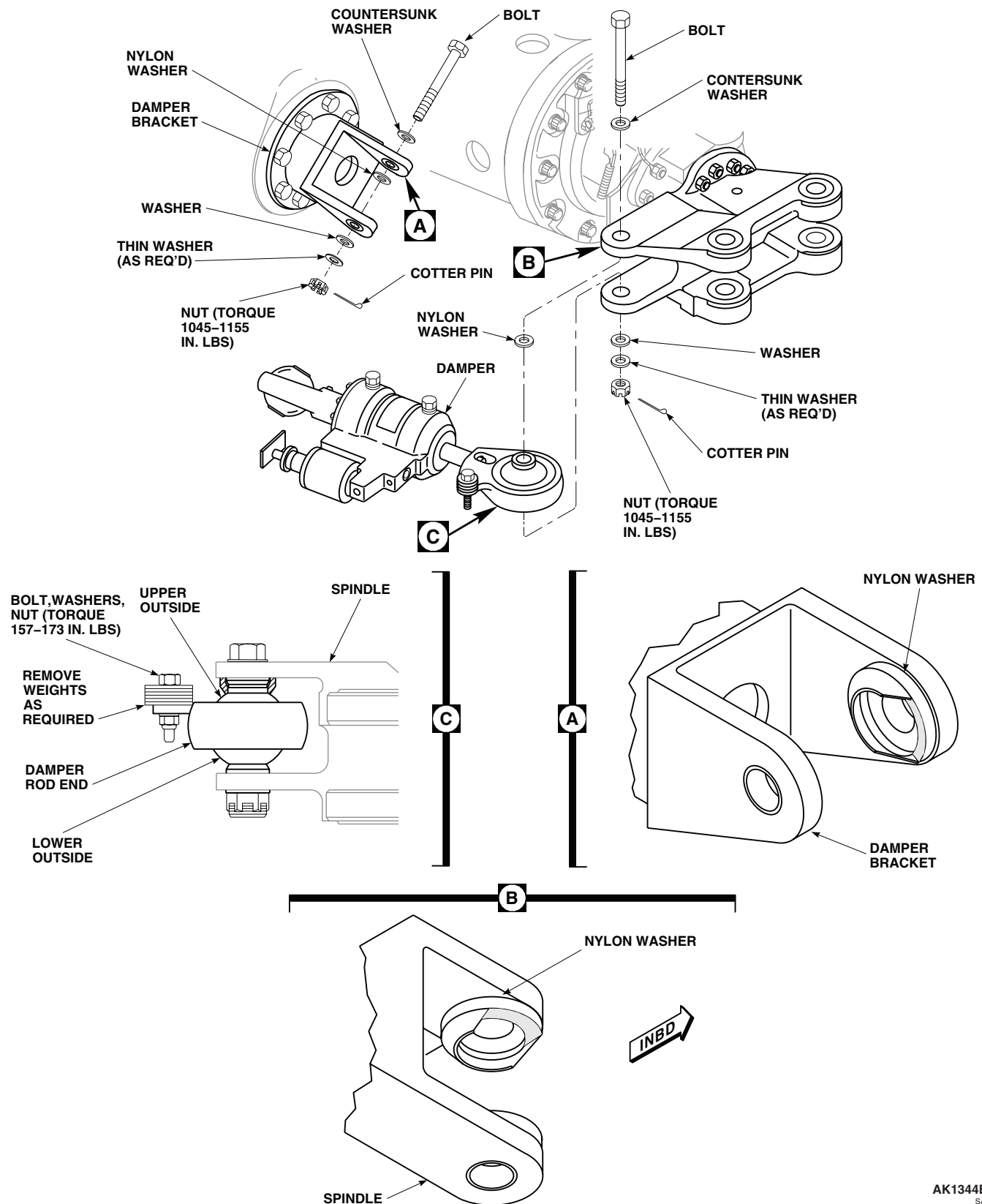


Figure 1. Replace Damper.

INSPECT

1. Check shoulder bushings for wear (WP 0551 00). If wear is detected, do this:
 - a. Remove bolt and nut.
 - b. Remove one washer from counterweight bracket if wear is on lower outside, or upper inside area of damper spindle attachment lug washer (Figure 1, Detail C).
 - c. Add one washer to counterweight bracket if wear is on lower inside, or upper outside area of damper spindle attachment lug washer (Figure 1, Detail C).
 - d. Install bolt and nut. **TORQUE NUT TO 157 - 173 INCH-POUNDS.**
2. Inspect damper bracket (WP 0560 00).
3. Inspect damper bearings as required (WP 0537 00).
4. Inspect damper. Refer to, INSPECT/REPAIR DAMPER, in this work package.
5. Inspect bolts attaching damper to damper bracket and spindle (WP 0550 00).

INSTALL

1. **QA** If replacement damper is filled with hydraulic preservative fluid, drain damper, refill with petroleum base hydraulic fluid, Item 145, WP 1803 00, and bleed (WP 0562 00).
2. **QA** Install nylon washer on damper bracket and spindle, with bearing retaining ring on damper facing away from nylon washer. Position slot on washer toward centerline of damper body (Figure 1, Detail A and Detail B).
3. Apply a bead of sealing compound, Item 273, WP 1803 00, around edge of nylon washer and lug of bracket or spindle.

NOTE

Restraining cables are not required on dampers with part numbers, 70106-08100-044, 70106-08100-046, or marked, RS-001C-V1.

4. **QA** If required, install restraining cable on damper cylinder head as follows:
 - a. **QA** Apply primer coating, Item 243, WP 1803 00, to restraining cable clip. Install clip on damper stud while primer is still wet.
 - b. Make sure bend in cable clip faces away from damper assembly.
 - c. **QA** Fasten cable clip to damper stud with nut. **TORQUE NUT TO 167 - 183 INCH-POUNDS.** Touch up paint on nut with primer coating, Item 243, WP 1803 00.
5. **QA** Place damper between damper bracket and spindle. Make sure damper is positioned with word TOP facing up and out (Figure 1).

NOTE

If necessary, when installing inboard and outboard bolts, lightly tap them with a rawhide mallet.

6. **QA** Coat bolt threads and shank with sealing compound, Item 258, WP 1803 00. Install bolt and countersunk washer at damper bracket end. Install countersunk washer with countersunk side facing bolthead.
7. **QA** If restraining cable is installed, position clip on bolt with bend toward damper assembly.

INSTALL - Continued**NOTE**

Two thick washers may be used to align cotter pin hole and nut if required.

8. **QA** Install washer against damper bracket. Install thin washers, as required to align cotter pin hole, and nut. **TORQUE NUT TO 1045 - 1155 INCH-POUNDS.** Install cotter pin.
9. **QA** Coat bolt threads and shank with sealing compound, Item 258, WP 1803 00. Install bolt and countersunk washer at spindle end. Install countersunk washer with countersunk side facing bolthead.
10. Install washer against spindle. Install thin washers, as required to align cotter pin hole, and nut. **TORQUE NUT TO 1045 - 1155 INCH-POUNDS.** Install cotter pin.
11. Touch up paint on boltheads, nuts, and exposed threads as necessary (WP 0381 00 and TM 55-1500-345-23).
12. Seal areas around boltheads and nuts with sealing compound, Item 273, WP 1803 00.
13. Service damper using instructions in WP 1610 00.
14. Check area for cleanness and foreign material.
15. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure damper attachment bolts re-torque check is shown as due 9 to 11 flight hours after installation (WP 1691 00).

INSPECT/REPAIR DAMPER**NOTE**

In the following steps, do all blending with abrasive wheel, Item 350, WP 1803 00, abrasive cloth, Item 77, WP 1803 00, or a medium cut file.

1. Check damper for corrosion. **ON OPEN AREAS (NONMATING SURFACES) REMOVE CORROSION PITS UP TO 0.030-INCH DEEP ON NO MORE THAN 10% OF ANY CIRCUMFERENCE BY BLENDING OUT.** Replace damper if damaged beyond limits.
2. **ON MATING SURFACES REMOVE CORROSION PITS UP TO 0.010-INCH DEEP ON NO MORE THAN 10% OF THE SURFACE BY BLENDING OUT.** Replace damper if damaged beyond limits.
3. Touch up aluminum areas that were reworked with corrosion resistant coating, Item 111, and apply primer and paint (WP 0381 00 and TM 55-1500-345-23). On other than aluminum apply primer and paint (WP 0381 00 and TM 55-1500-345-23).
4. Check damper for nicks and gouges. **CORNER DAMAGE UP TO 0.060-INCH DEEP MAY BY BLENDED OUT TO A 1/2-INCH RADIUS.** Replace damper if damaged beyond limits.
5. **SURFACE DAMAGE UP TO 0.040-INCH DEEP AND LESS THAN 10% OF THE CIRCUMFERENCE MAY BE BLENDED OUT TO A 1/2-INCH RADIUS.** Replace damper if damaged beyond limits.
6. Visually check reworked areas for cracks.
7. Touch up aluminum areas that were reworked with corrosion resistant coating, Item 111, WP 1803 00, and apply primer and paint (WP 0381 00 and TM 55-1500-345-23). On other than aluminum apply primer and paint (WP 0381 00 and TM 55-1500-345-23).
8. Check piston rod surface for damage. If any damage can be felt with the fingernail, replace damper.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****DAMPER INDICATOR**

This WP supersedes WP 0559 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 150 - 1000 in. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Insulation Sheet, Electrical, Item 153, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

References

WP 1610 00
WP 1803 00

REMOVE

1. Engage gust lock.
2. Loosen bleed plugs enough to relieve pressure, so indicator bottoms at empty position (handle in). Catch any overflow of fluid with machinery wiping towel, Item 344, WP 1803 00.
3. Tighten plugs handtight.

NOTE

Have new indicator ready to install, otherwise insert plug to stop fluid loss.

4. Remove indicator and packing (Figure 1). Discard packing.

INSTALL

1. **QA** Install packing on indicator (Figure 1).

WARNING

Use of safety cable is not allowed in this application. Use only nonelectrical wire as specified.

2. **QA** Make sure handle is all the way in and install indicator in damper. **TORQUE JAMNUT TO 285 - 315 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire indicator to damper, covering nonelectrical wire with electrical insulation sheet, Item 153, WP 1803 00.
3. Service damper (WP 1610 00).

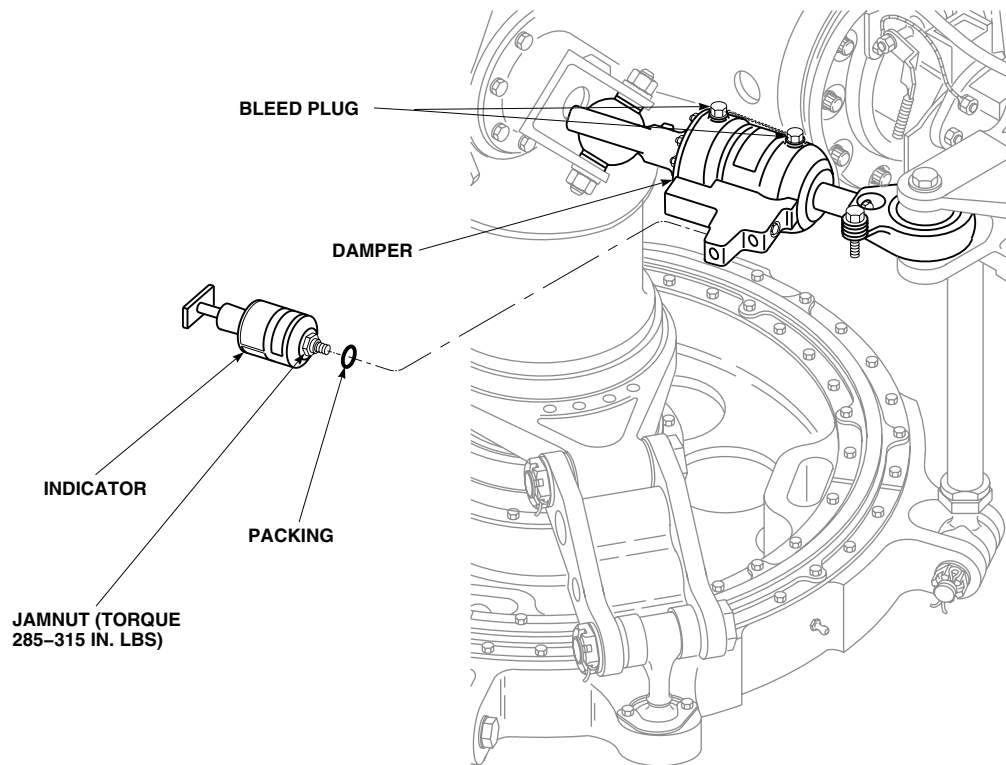
INSTALL - Continued**AK1346**
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Figure 1. Replace Damper Indicator.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****DAMPER BRACKET**

This WP supersedes WP 0560 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Depth Gage
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, Dial, 0 - 75 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Abrasive, Item 77, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Sealing Compound, Item 273, WP 1803 00
Sealing Compound, Item 258, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0381 00
WP 0549 00
WP 0550 00
WP 1691 00
WP 1803 00

REMOVE

1. Turn main rotor until damper bracket is easy to reach.
2. Engage gust lock.

NOTE

Restraining cables are not required on dampers, 70106-08100-044, 70106-08100-046, or dampers marked, RS-001C-VI.

3. Using nonmetallic scraper remove sealant from damper attachment bolt and nut and damper bracket attachment bolts and nuts.
4. Remove bolt, washers, restraining cable (if installed), and nut securing damper to damper bracket and swing damper out of way (Figure 1, Detail A).
5. Remove bolts, washers, and damper bracket from main rotor hub.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Clean primer off bolts using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
7. Clean sealing compound off damper bracket using nonmetallic scraper.

Technical drawing of a damper bracket assembly. The left side shows a perspective view of the engine block with the damper bracket (A) highlighted in black. The right side shows a top-down view of the damper bracket (B) with dimensions: 2.509 MAXIMUM for the outer diameter and 0.7513 MAXIMUM for the inner diameter. The chamfer is also indicated.



Change 3

INSPECT**NOTE**

On hub assemblies with part number beginning with 70070-10046, a bolt is not installed at the 11 o'clock position to attach the damper bracket.

1. Inspect bolts attaching damper bracket to hub (WP 0550 00).
2. Check distance between shoulder bushings (Figure 1, Detail B). **REPLACE BRACKET IF DISTANCE IS GREATER THAN 2.509-INCHES.**
3. Check bushing for wear. **REPLACE BRACKET IF ID OF BUSHING IS GREATER THAN 0.7513-INCH.**
4. Check for corrosion on bushing. **REPLACE DAMPER BRACKET IF CORROSION ON CHAMFER OF BUSHING IS GREATER THAN 0.040-INCH DEEP.** Blend out corrosion with abrasive cloth, Item 77, WP 1803 00. Coat blended area with primer coating, Item 243, WP 1803 00. Touch up paint (WP 0381 00 and TM 55-1500-345-23).
5. Check bracket for cracks. **NO CRACKS ALLOWED.** Replace cracked bracket.
6. Check bracket for nicks and gouges. Blend out damage with abrasive cloth, Item 77, WP 1803 00. **REPLACE BRACKET IF:**
 - a. **DAMAGE AFTER BLENDING SHALL NOT BE GREATER THAN 0.050-INCH DEEP.**
 - b. **DAMAGE ON MATING AREAS AFTER BLENDING SHALL NOT BE GREATER THAN 0.020-INCH DEEP, OR COVER MORE THAN 10% OF SURFACE AREA.**

INSTALL**NOTE**

On hub assemblies with part number beginning with 70070-10046, a bolt is not installed at the 11 o'clock position to attach the damper bracket.

1. Before torquing damper bracket attachment bolts, check threaded inserts as follows:
 - a. **QA** Hold bracket up against hub and thread in bolts and washers until surface of washers are 0.030-inch away from surface of damper bracket.
 - b. **QA** Using dial type torque wrench, unthread each bolt. **MINIMUM BREAKAWAY TORQUE IS 6.5 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 60 INCH-POUNDS.**
 - c. If torque is not within limits, do this:
 - (1) Check bolt for corrosion and wear. **NONE ALLOWED.** Replace corroded or worn bolts.
 - (2) Repeat step 1. If breakaway or removal torque is still not within limits, replace insert (WP 0549 00).
2. Coat shank, threads and area under head of damper bracket bolts with primer coating, Item 243, WP 1803 00.
3. **QA** Place damper bracket on main rotor hub with word "TOP" facing up (Figure 1, Detail A).
4. **QA** Install bolts and washers while primer is still wet.
5. **QA TORQUE BOLTS TO 214 - 236 INCH-POUNDS.**
6. Wipe off any primer that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.

INSTALL - Continued**NOTE**

To prevent excessive wear of bolts, install enough washers under nut so that at least half of cotter pin will be inside castellations of nut when cotter pin is installed.

7. **QA** Check to make sure nylon washer is installed on damper bracket.
8. **QA** Place damper in damper bracket. Install bolt and washer at damper bracket end. Install countersunk washer with countersunk side facing bolthead. Coat bolt threads and shank with sealing compound, Item 258, WP 1803 00.
9. **QA** If restraining cable is required, position cable clip on bolt with bend toward damper assembly.
10. **QA** Install thin washers, as required to align cotter pin hole, and nut. **TORQUE NUT TO 1045 - 1155 INCH-POUNDS.** Install cotter pin.
11. Touch up paint on boltheads, nuts, and exposed threads (WP 0381 00 and TM 55-1500-345-23).
12. Seal gap between damper bracket and hub, and areas around boltheads and nuts with sealing compound, Item 273, WP 1803 00.
13. Check area for cleanness and foreign material.
14. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure damper attachment bolts torque check is shown as due 9 to 11 flight hours after installation (WP 1691 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****DAMPER ROD END BEARING**

This WP supersedes WP 0561 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Bearing Installer/Remover, 70700-77306-041
Gloves
Goggles/Face Shield
Inside Micrometer Caliper, 0 - 4 inches
Powertrain Repairer's Toolkit, SC 5180-99-B13

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00
Cloth, Abrasive, Item 75, WP 1803 00

References

WP 0558 00
WP 1803 00

NOTE

The following instructions apply to either bearing, except where indicated.

REMOVE

1. Disconnect damper from spindle or damper bracket as required (WP 0558 00).
2. Remove retaining ring.
3. Using bearing installer/remover, 70700-77306-041, remove bearing from rod end by pressing from retaining ring side of bearing (Figure 1).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer coating off rod end bore or cylinder headliner with technical acetone, Item 5, WP 1803 00. Do not allow primer coating to dry.

INSPECT/REPAIR

1. Check rod end bore for metal pickup as follows:
 - a. Remove metal pickup by blending out bore using abrasive cloth, Item 75, WP 1803 00.

INSPECT/REPAIR - Continued

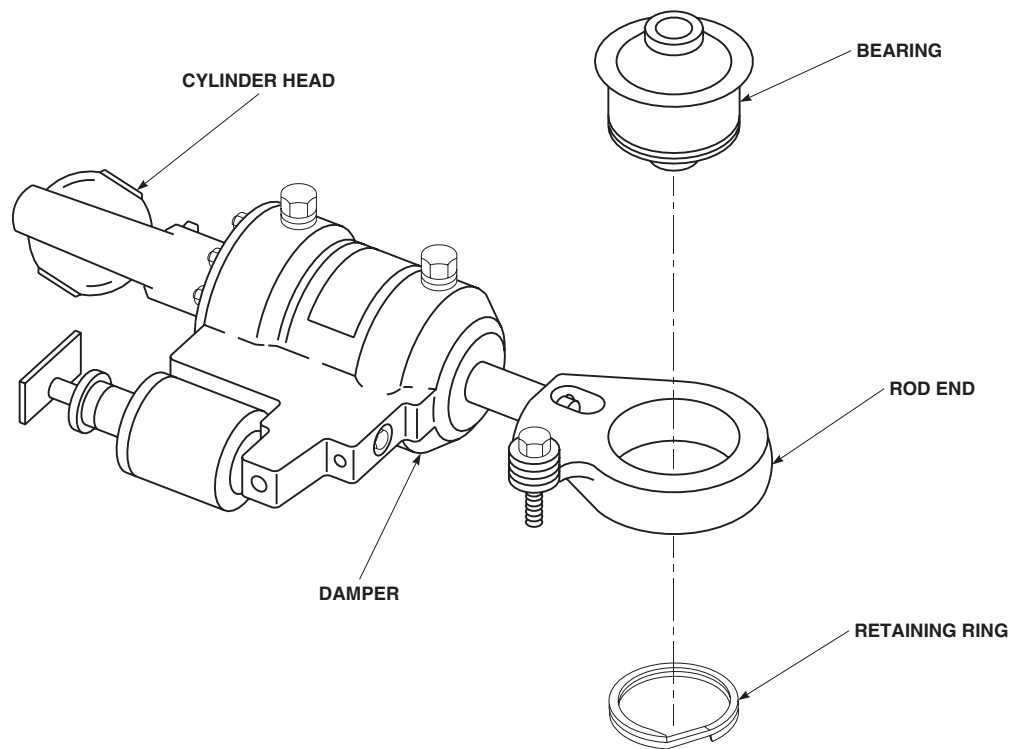
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Figure 1. Replace Damper Rod End Bearing.

NOTE

Bore dimensions are different at each end.

- b. Measure rod end bore diameter in two places 90 degrees apart. **ID OF ROD END BORE MUST NOT AVERAGE MORE THAN 2.749-INCHES AFTER REWORK.**
 - c. Measure rod end bore diameter in two places 90 degrees apart. **ID OF CYLINDER HEAD END BORE (LINERS INSTALLED) MUST NOT AVERAGE MORE THAN 2.748-INCHES AFTER REWORK.**
2. Visually check for security of the bonded liner in the cylinder head rod end. **NO DISBONDING ALLOWED.**

INSTALL**WARNING**

Injury to personnel will result if protective gloves are not worn when handling carbon dioxide or chilled parts. Wear protective gloves when handling carbon dioxide or chilled parts.

NOTE

Bearings to remain packaged during chilling process.

1. If available, place new bearing in carbon dioxide, Item 64, WP 1803 00, for at least 30 minutes or freezer for at least 1 hour before installation.

INSTALL - Continued**NOTE**

Make sure bearings are pressed into bore from side with chamfered edge.

2. Coat rod end bore or cylinder head liner with primer coating, Item 243, WP 1803 00. Do not allow primer coating to dry.
3. **QA** Using bearing installer/remover, 70700-77306-041, install bearing into damper (Figure 1). Wipe off extra primer coating using machinery wiping towel, Item 344, WP 1803 00.

NOTE

Make sure bearing is fully seated in rod end bore.

4. **QA** Apply a coat of epoxy primer coating kit, Item 120, WP 1803 00, to bottom side of lug area of inboard cylinder head and retaining ring. Install retaining ring on the side of bearing marked retaining ring side. If bearing is not marked, install retaining ring on opposite side of chamfered shoulder.
5. Install damper to spindle or damper bracket as required (WP 0558 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****BLEED DAMPER (DAMPER REMOVED)**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 2-Gallon
Hydraulic Repairer's Toolkit, SC 5180-99-A03
Torque Wrench, 30 - 200 in. lbs.

Reducer, AN919-6D
Selector Valve, AN6293-6
Towel, Machinery Wiping, Item 344, WP 1803 00
Urethane Tubing, 70652-02251-174
Wire, Nonelectrical, Item 357, WP 1803 00

Material/Parts

Globe Valve, S-138-4BP
Hose, MS8005E-204A
Hydraulic Fluid Dispenser AF-5
Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Insulation Sheet, Electrical, Item 153, WP 1803 00
Reducer, AN919-2D

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0558 00
WP 1610 00
WP 1803 00

BLEED PROCEDURE

1. Locally make a bleed unit per Table 1.
 - a. Connect tubing, fittings, fluid dispenser, and valves as shown in Figure 1.
 - b. Secure tubing to fittings with tiedown straps.

NOTE

- Do not connect damper until tubing is bled.
 - Make sure that port numbers on selector valve faceplate are lined up, with port numbers stamped on selector valve housing.
- c. Prepare hydraulic fluid dispenser for bleeding. Service dispenser with at least 4 gallons of petroleum base hydraulic fluid, Item 145, WP 1803 00.
 - d. Place 2 gallon container under drain line.
 - e. Bleed lines and fittings of air as follows: Open bleed valve. Place selector valve to cylinder 1. Open fluid dispenser shutoff valve and purge line No. 2 of air. Close shutoff valve and connect line No. 2 to damper. Place selector valve to cylinder 2. Open shutoff valve and purge line No. 4 of air. Close shutoff valve and connect line No. 4 to damper. Make sure damper is positioned with bleed ports up.
2. Make sure damper is properly installed in bleeding unit with bleed ports up, all lines bled, and fluid dispenser serviced.
 3. Open bleed valve. Place selector valve to cylinder 2. Open fluid dispenser shutoff valve until damper piston is fully extended. Check for air in line No. 2. Leave selector valve in cylinder 2 position until line No. 2 is purged of all air bubbles. Place selector valve to cylinder 1. Piston should fully retract. Check for air in line No. 4. Leave selector valve in cylinder 1 position until line No. 4 is purged of all air bubbles.

BLEED PROCEDURE - Continued

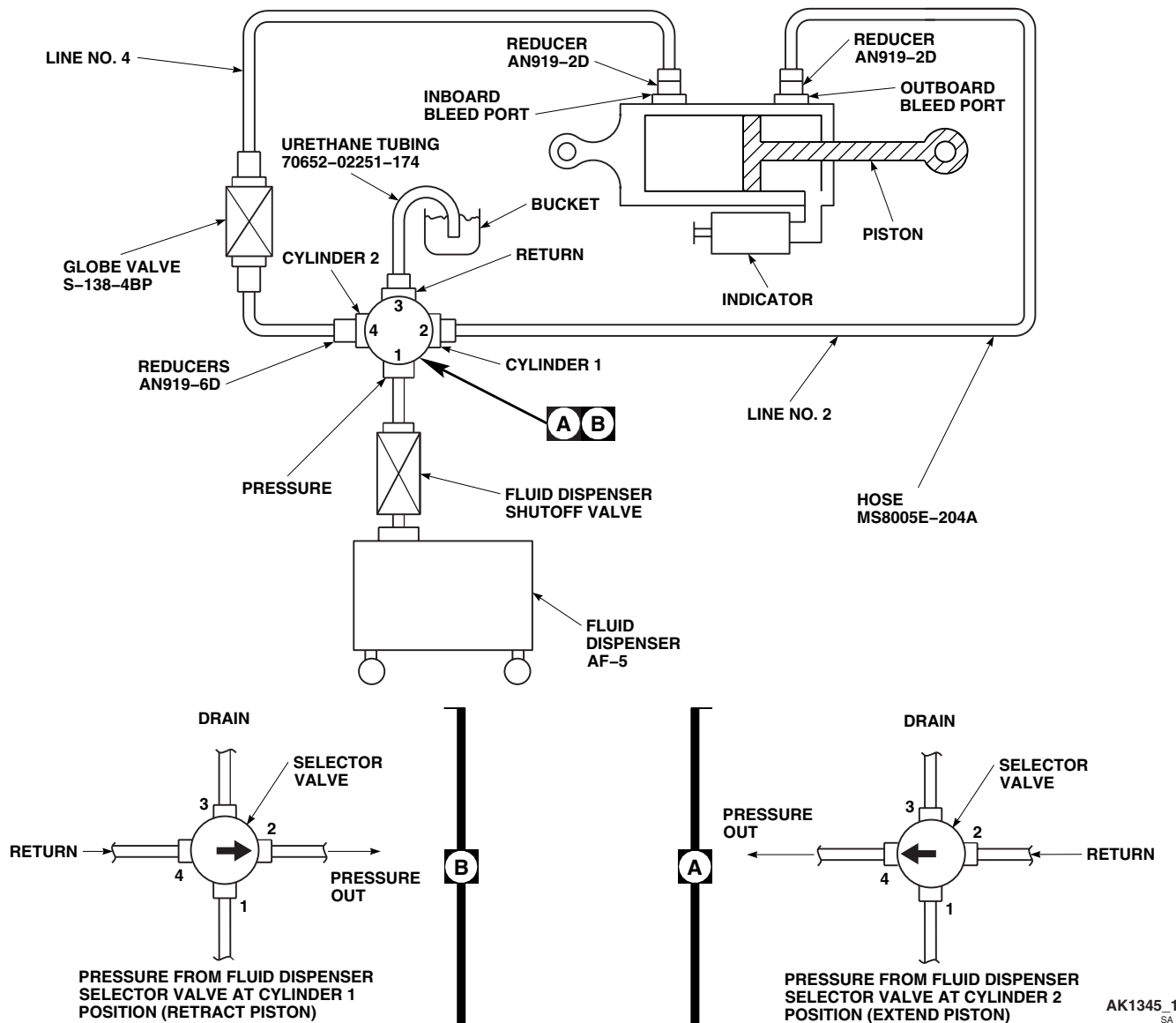


Figure 1. Bleed Setup.

4. Repeat step 3. until no more air bubbles appear in drain line. Leave damper piston in retracted position (selector valve at cylinder 1). Close fluid dispenser shutoff valve.
5. Stand damper on inboard end. Make sure selector valve is in cylinder 1 position. Open fluid dispenser shutoff valve and close bleed valve. Allow damper indicator to fill with fluid (indicator rod fully extended). Close shutoff valve and place selector valve to cylinder 2. Allow indicator to bleed down (indicator rod will retract) and check for air bubbles in line No. 2. Press in on indicator to aid in bleeding.
6. If air bubbles are seen, purge line No. 2 of air by opening bleed valve and fluid dispenser shutoff valve.
7. Close bleed valve and place selector valve to cylinder 1. Allow damper indicator to fill with fluid (indicator rod fully extended). Close shutoff valve and place selector valve to cylinder 2. As indicator bleeds down, tilt damper

BLEED PROCEDURE - Continued

about 45 degrees in the direction that will allow air to flow to outboard bleed port (Figure 2). Check for air bubbles in line No. 2. Press in on indicator to aid in bleeding. If air bubbles are seen, purge line No. 2 of air by opening bleed valve and fluid dispenser shutoff valve. When air bubbles are gone, close shutoff valve and place selector valve to port 2.

8. Repeat step 7. until no more air bubbles appear.

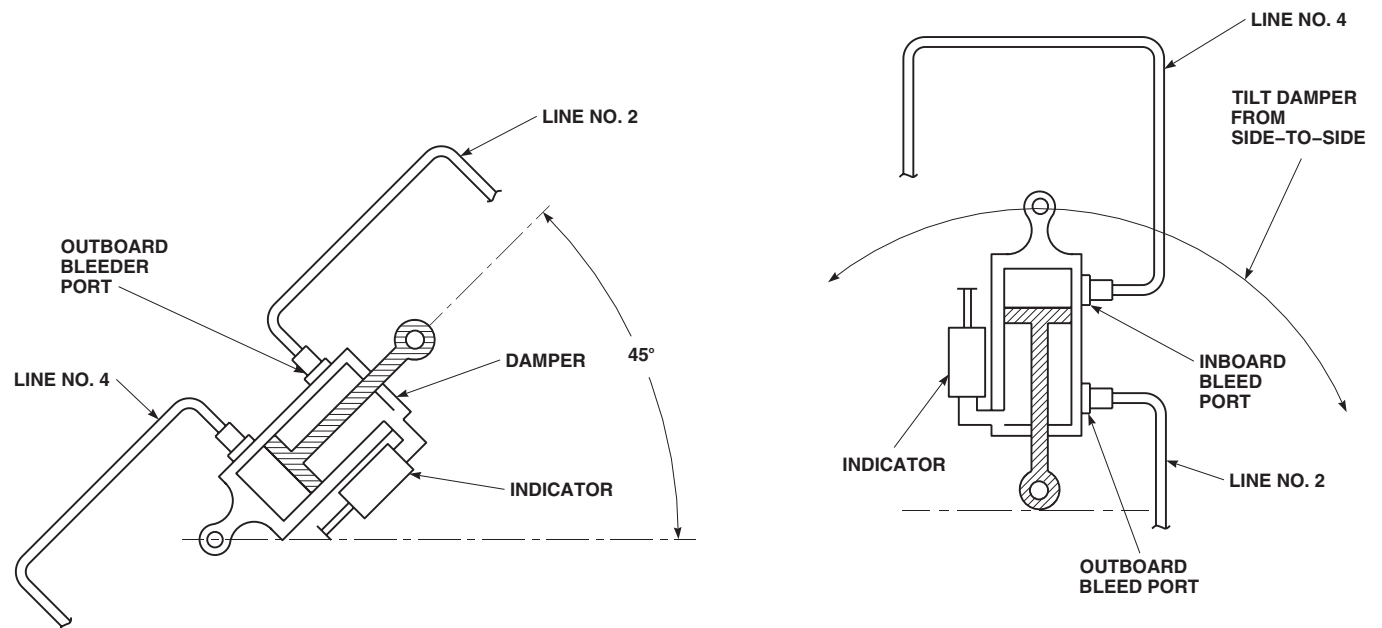
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Figure 2. Bleed Positions.

BLEED PROCEDURE - Continued**Table 1. Locally Made Bleed Unit.**

ITEM	P/N	QTY
Container, 2-gallon		1
Globe Valve	S-138-4BP	1
Hose	MS8005E-204A	3
Hydraulic Fluid Dispenser	AF-5	1
Reducer	AN919-6D	4
Reducer	AN919-2D	2
Selector Valve	AN6293-6	1
Urethane Tubing	70652-02251-174	2 feet

9. Open fluid dispenser shutoff valve and close bleed valve. Allow damper indicator to fill with fluid (indicator rod fully extended). Close shutoff valve. Hold damper in upright position with inboard bleed port up. Open bleed valve and allow indicator to bleed down (indicator rod will retract). As indicator bleeds down tilt damper from side-to-side and watch for air bubbles in line No. 4. If air bubbles are seen, purge line No. 4 of air by opening fluid dispenser shutoff valve (selector valve still at cylinder 1 and bleed valve still open). When all air bubbles are gone, close shutoff valve.
10. Turn damper 180 degrees so it rests on inboard end. Repeat above step until no more air bubbles appear.
11. Lay damper down horizontally with bleed ports up. Place selector valve to cylinder 2. Open fluid dispenser shutoff valve until damper piston is fully extended. Check for air in line No. 2. Leave selector valve in cylinder 2 position until line No. 2 is purged of all air bubbles. Place selector valve to cylinder 1. Piston should fully retract. Check for air in line No. 4. Leave selector valve in cylinder 1 position until line No. 4 is purged of all air bubbles.
12. Repeat step 11. until air bubbles no longer appear in lines. Leave damper piston in retracted position (selector valve at cylinder 1).

NOTE

When installing clamp on indicator, be careful not to score or scratch gold band on indicator. If necessary, pad clamp.

13. With selector valve still at cylinder 1, close bleed valve. Allow damper indicator to fill with fluid (indicator rod extended). Install hose clamp on indicator rod to prevent rod from retracting. Close fluid dispenser shutoff valve.
14. Disconnect tubing from damper. Remove fittings from bleed ports. Install packings lubricated with petroleum base hydraulic fluid, Item 145, WP 1803 00, on bleed plugs and install bleed plugs fingertight. Remove clamp.
15. Crack bleed plug on outboard port. If air is present it can be heard as a hissing sound. When hissing stops, tighten bleed plug. Check gold band on indicator. If full gold band is not visible, repeat entire bleeding procedure. If gold band is visible, tighten bleed plug.
16. Repeat step 15., except crack inboard bleed plug.

BLEED PROCEDURE - Continued

17. **QA TORQUE BOTH BLEED PLUGS TO 65 - 75 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bleed plugs to each other and indicator. Cover nonelectrical wire with electrical insulation sheet, Item 153, WP 1803 00.
18. Clean damper of any spilled hydraulic fluid using machinery wiping towel, Item 344, WP 1803 00.
19. Install damper (WP 0558 00).
20. Check damper as follows:
 - a. Position blade over nose.
 - b. Grasp blade at tip and move toward lead position. Rapidly change blade to lag position. Blade tip should move 3 to 9 inches. Do not pull blade down more than 6 inches.
 - c. Have assistant observe damper indicator as direction of blade is changed. If movement of damper is 1/8-inch or less, damper has been properly bled and serviced. If damper indicator moves more than 1/8-inch, service damper on helicopter using instructions in WP 1610 00. After servicing, repeat damper checkout. If damper indicator still moves more than 1/8-inch, remove damper and repeat entire bleeding procedure.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**PITCH CONTROL RODS**

This WP supersedes WP 0563 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Depth Gage
 Fluorescent Penetrant Inspection Kit, XMA101
 Pitch Control Link Tool, Locally-Made, Figure 259, WP 1805 00
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Torque Adapter, Wrench, Box, 70700-20210-101
 Torque Wrench, 150 - 1000 in. lbs.
 Torque Wrench, 300 - 2500 in. lbs.

Material/Parts

Adhesive, Item 25, WP 1803 00
 Cloth, Abrasive, Item 77, WP 1803 00
 Corrosion Preventive Compound Item 103, WP 1803 00
 Corrosion Preventive Compound Item 106, WP 1803 00
 Lubricant, Solid Film, Item 180, WP 1803 00
 Lubricating Oil, Molybdenum Disulfide, Item 189, WP 1803 00

Sealing Compound, Item 273, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-237-MTF
 TM 1-1520-265-23
 TM 1-6625-724-13&P
 TM 55-1500-345-23
 WP 0381 00
 WP 0538 00
 WP 0550 00
 WP 0552 00
 WP 0564 00
 WP 0565 00
 WP 1803 00
 WP 1805 00

REMOVE

1. Engage gust lock.
2. Locally make pitch control link tool (WP 1805 00).

CAUTION

- Damage to swashplate and horn lug areas will result if extreme caution is not used during disconnecting of pitch control rod end. Use extreme caution when disconnecting pitch control rod end to prevent damage to swashplate and horn lug areas.
- Pitch control link upper rod ends are left hand threaded. Loosen pitch control link upper jamnuts by turning clockwise. Excessive force applied in the counterclockwise direction may damage pitch control rod end threads, pitch control rod link threads or anti-corrosive coatings.

NOTE

It may be necessary to gently rock main rotor blade to remove a binding pitch control rod bolt.

3. To remove bolt, from upper and lower pitch control rod and swashplate (Figure 1), do this:

REMOVE - Continued

- a. Loosen nut on bolt. Depress retaining pawl on bolt with thumb and continue backing off nut until castellations clear retaining pawl. Release pawl and remove nut from bolt.
- b. Depress retaining pawl on bolt with thumb and remove flat washer from bolt.

NOTE

Retaining pawl on bolt will make bolt hard to remove once inside bearing. Use needed force to pull bolt through rod end.

- c. Depress retaining pawl on bolt with thumb and start removing upper and lower bolts from rod ends. Use force needed to pull bolts from rod ends.
- d. Depress retaining pawl on bolt with thumb and remove flat washers, if installed, and countersunk washer from upper bolt.
- e. Depress retaining pawl on bolt with thumb and remove spacer and countersunk washer from lower bolt.
4. Measure dimension X on removed pitch control rod (Figure 1). Note number of exposed threads on lower rod end.
5. Remove upper and lower rod ends, lock key, and jamnuts from pitch control link (WP 0565 00).

INSPECT

1. Inspect bolts attaching pitch control rod (WP 0550 00). Apply molybdenum sulfide lubricating oil, Item 189, WP 1803 00, as necessary. Apply or touch up solid film lubricant, Item 180, WP 1803 00, as necessary.
2. Inspect pitch control rod attaching washers for cracks. **NO CRACKS ALLOWED.** Replace cracked washers.
3. Inspect bushings on swashplate lugs (WP 0564 00).
4. Inspect bushings on spindle horn (WP 0552 00).
5. Inspect spacer for corrosion pitting. **REPLACE SPACER IF CORROSION PITS ARE DEEPER THAN 0.003-INCH ON ID, OR 0.005-INCH ON OD.**
6. Inspect spacer for wear. **MAXIMUM WEAR IS 0.020-INCH.** Replace spacer if wear exceeds limits.
7. Inspect spacer for nicks, scratches, or gouges. **REPLACE SPACER IF NICKS, SCRATCHES, OR GOUGES PENETRATE THROUGH SILVER PLATING.**
8. Inspect bearings (WP 0538 00).

NOTE

On control rods with Teflon lined bearings, be sure movement is not between spherical bearing and outer race before replacing components.

9. Inspect for looseness between pitch control link, jamnuts and rod ends by moving control rod assembly in all directions. **NO LOOSENESS IS ALLOWED.**
10. Inspect jamnuts, rod ends and pitch control link for corroded, flattened, stripped, or crossed threads. Replace components as required (WP 0538 00). If bare metal is visible, reapply solid film lubricant, Item 180, WP 1803 00.

INSTALL

1. Install upper and lower rod end, lock key, and jamnut on pitch control link (WP 0565 00).

INSTALL - Continued

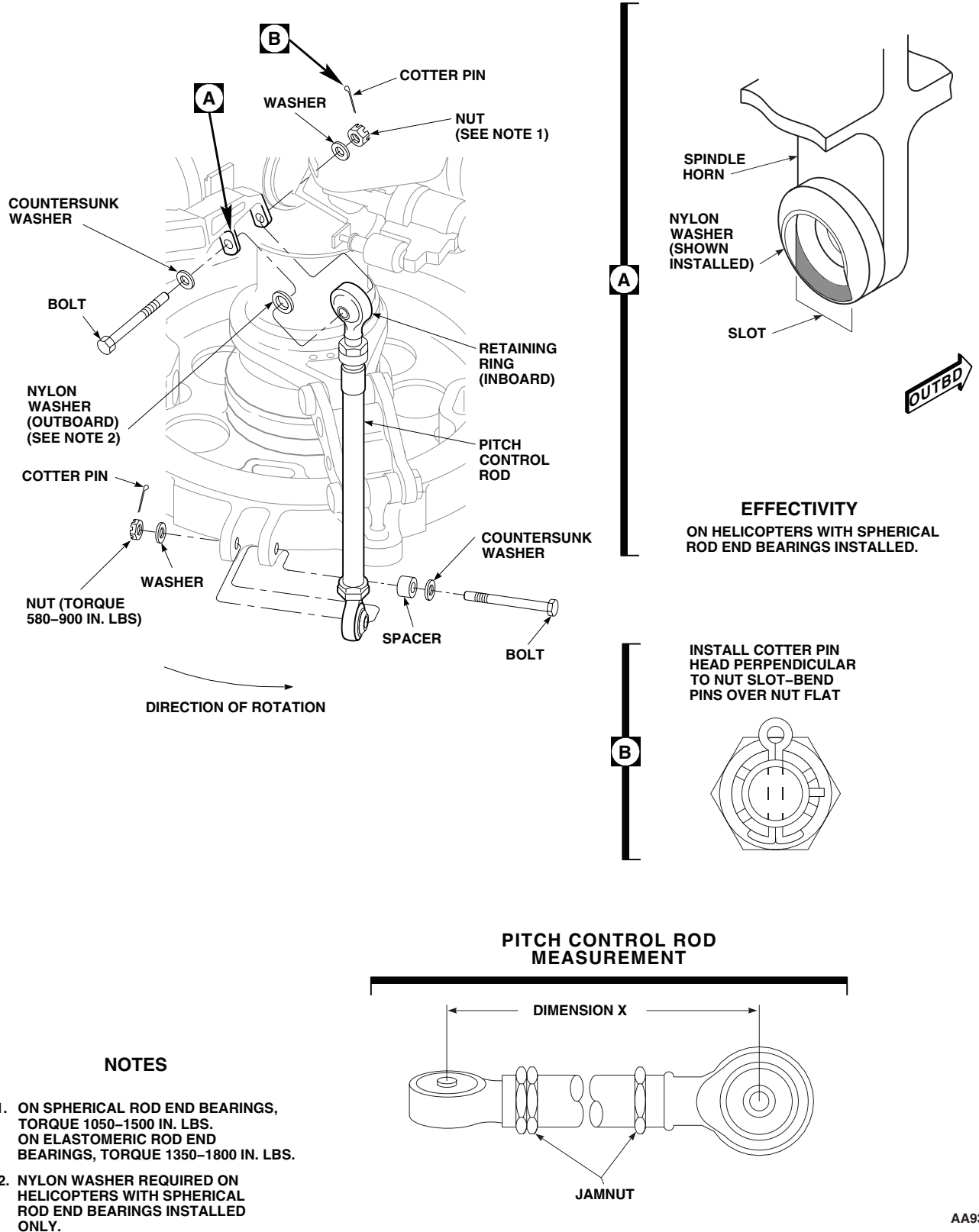


Figure 1. Replace Pitch Control Rods.

INSTALL - Continued**NOTE**

If recorded dimensions and angle are not available, set length of rod to 25.22-inches with 0.43-inch of thread exposure from face of jamnut to end of threads on lower rod end only.

2. Apply corrosion preventive compound, Item 106, WP 1803 00, on upper and lower rod threads and jamnuts. Adjust replacement pitch control rod to length of dimension X measured on removed control rod (Figure 1). Make sure lower rod end has same amount of exposed threads as noted on removal. Hand tighten jamnuts now.

NOTE

Nylon washer is required on helicopters with spherical rod end bearings installed only. Do not install nylon washers if elastomeric rod end bearings are installed.

3. **QA** On helicopters with spherical rod end bearings installed, do this:

CAUTION

Damage to pitch control rod and spindle horn will result if nylon washer is installed incorrectly. Make sure nylon washer is installed on side opposite bearing retaining ring.

- a. **QA** Install nylon washer on outboard lug of spindle horn. Position slot on washer toward swashplate with slot in line with centerline of control rod (Figure 1, Detail A). Bond washer in place with sealing compound, Item 273, WP 1803 00.
- b. Apply a bead of sealing compound, Item 273, WP 1803 00, around edge of nylon washer and lug of spindle horn.

WARNING**CSI**

Verification of proper hardware installation, torque and safety installation of bolts are critical characteristics.

NOTE

Make sure pitch control rod bolthead faces outboard.

4. **QA** Install pitch control rod upper rod end on spindle horn with locking key outboard and snapping, if installed, on inboard side. Install bolt as follows:
 - a. Apply molybdenum disulfide lubricating oil, Item 189, WP 1803 00, on bolt.
 - b. **QA** Depress retaining pawl on bolt with thumb and install countersunk washer on bolt with countersunk side of washer facing bolthead.
 - c. **QA** Depress retaining pawl on bolt with thumb and install bolt and countersunk washer through pitch control rod and spindle horn, with pawl in 6 o'clock position.
 - d. **QA** Depress retaining pawl on bolt with thumb and install washer and nut. Do not torque upper nut now.

INSTALL - Continued**WARNING****CSI**

Verification of proper hardware installation, torque and safety installation of bolts are critical characteristics.

CAUTION

Damage to swashplate and horn lug areas will result if extreme caution is not used during installation of pitch control rod. Use extreme caution when installing pitch control rod to prevent damage to swashplate and horn lug areas.

NOTE

- It may be necessary to gently rock or pitch main rotor blade or spindle up to allow lower rod end to swing over swashplate lug from side to side and be inserted into clevis from the top, and to line up pitch control rod for bolt insertion.
 - On pitch control rods which have the lower control rod end bearing retained by a retaining ring, make sure that the retaining ring is installed in direction of rotation toward leading edge of blade.
 - Make sure pitch control rod bolthead faces direction of rotation.
5. **QA** Install pitch control lower rod end in swashplate and install bolt by doing this:
 - a. Apply molybdenum disulfide lubricating oil, Item 189, WP 1803 00, on bolt.
 - b. **QA** Depress retaining pawl on bolt with thumb and install countersunk washer on bolt with countersunk side of washer facing bolthead. Install spacer on bolt.
 - c. **QA** Depress retaining pawl on bolt with thumb and install bolt and washer through swashplate lugs and pitch control rod end bearing with pawl in 6 o'clock position.
 - d. **QA** Depress retaining pawl on bolt with thumb and install washer and nut. Do not torque lower nut now.
 6. **QA** Use torque adapter on upper nut. **ON SPHERICAL ROD END BEARINGS, TORQUE UPPER NUT TO 1050 - 1500 INCH-POUNDS. ON ELASTOMERIC ROD END BEARINGS, TORQUE UPPER NUT TO 1350 - 1800 INCH-POUNDS.** Install cotter pin in castellated nut with head perpendicular to nut slot, and bend pins over hex flats (Figure 1, Detail B).
 7. **QA TORQUE LOWER NUT TO 580 - 900 INCH-POUNDS.** Install cotter pin in castellated nut with head perpendicular to nut slot, and bend pins over hex flats (Figure 1, Detail B).
 8. Check to make sure both upper and lower rod ends are parallel to their respective attachment lugs at the same time. Rotate pitch control rod by hand to make sure there is no metal-to-metal contact on lower rod end attachment lugs. Make small adjustments as necessary. Make sure that lower pitch control rod end has no more than 3/4-inch exposed threads. Tighten jamnuts as follows:

NOTE

To prevent improper torque of upper jamnut on pitch control rod, make sure torque wrench is not in contact with lock key when torquing.

- a. **QA** Lock key in notch by tightening upper jamnut. **TORQUE TO 650 - 750 INCH-POUNDS.**
- b. **QA TORQUE LOWER NUT TO 650 - 750 INCH-POUNDS.**

INSTALL - Continued**WARNING**

The use of safety cable is not allowed in this application. Use only nonelectrical wire as specified.

- c. Using nonelectrical wire, Item 357, WP 1803 00, lockwire lower jamnut to barrel and then back to lower jamnut.

WARNING

The use of safety cable is not allowed in this application. Use only nonelectrical wire as specified.

- d. Using nonelectrical wire, Item 357, WP 1803 00, lockwire from upper jamnut to tang and then back to opposite side of upper jamnut.
9. Touch up paint on nut and bolthead as necessary (WP 0381 00 and TM 55-1500-345-23). Do not allow paint or sealing compound to contact retaining pawl area of bolt.
10. Track and balance main rotor blades (TM 1-6625-724-13&P).

NOTE

A maintenance test flight is not required if only one end of pitch control rod was disconnected at a time and no adjustments were made.

11. Do a maintenance test flight (TM 1-1520-237-MTF).
12. Seal interface between upper rod end and top of upper jamnut only with sealing compound, Item 273, WP 1803 00. **DO NOT APPLY SEALANT TO UPPER JAMNUT WRENCH FLATS. DO NOT APPLY SEALANT TO LOWER ROD END OR JAMNUTS.**

INSPECT/REPAIR PITCH CONTROL RODS

1. Check pitch control rod for nicks and gouges and repair as follows:

NOTE

Damage should be blended out using a abrasive wheel, Item 350, WP 1803 00, abrasive cloth, Item 77, WP 1803 00, and/or a medium cut file. After blending out, check reworked area for cracks.

- a. **CORNER DAMAGE UP TO 0.050-INCH DEEP MAY BE BLENDED OUT TO A 1/2-INCH RADIUS. MULTIPLE CORNER DAMAGE MUST BE AT LEAST 1 INCH APART.**
- b. **SURFACE DAMAGE IN ROUND PORTION OF ROD END BETWEEN THREAD AREA AND LUG UP TO 0.030-INCH DEEP MAY BE BLENDED OUT TO A 1/2-INCH RADIUS. DAMAGED AREA MUST BE LESS THAN 25% OF THE CIRCUMFERENCE.**
- c. **SURFACE DAMAGE IN LUG PORTION OF ROD ENDS UP TO 0.040-INCH DEEP ACROSS ANY SINGLE SURFACE MAY BE BLENDED OUT TO A 1/2-INCH RADIUS. MULTIPLE DAMAGE MUST BE AT LEAST 1 INCH APART.**
- d. **SURFACE DAMAGE IN LINK PORTION OF CONTROL ROD UP TO 0.060-INCH DEEP MAY BE BLENDED OUT TO A 1/2-INCH RADIUS. DAMAGED AREA MUST NOT BE MORE THAN 25% OF LINK CIRCUMFERENCE.**
- e. Replace control rod if damaged beyond limits.

INSPECT/REPAIR PITCH CONTROL RODS - Continued

2. Fluorescent penetrant inspect reworked areas for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23).
3. Touch up, prime, and paint reworked areas (WP 0381 00 and TM 55-1500-345-23).
4. Inspect threads of jamnuts for corrosion using a magnifying glass. **NO CORROSION ALLOWED.** Replace jamnuts if corroded.
5. Inspect threads of rod ends, link, and jamnuts for stripped or crossed threads. **NO DAMAGED THREADS ALLOWED.** Replace any damaged parts.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
PITCH CONTROL ROD END SPHERICAL BEARINGS
This WP supersedes WP 0564 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Bearing Installer/Remover, 65700-40030-041
Bearing Installer/Remover, 70700-77306-041
Gloves
Goggles/Face Shield
Inside Micrometer Caliper, GGG-C-105
Powertrain Repairer's Toolkit, SC 5180-99-B13

Cloth, Abrasive, Item 75, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0563 00
WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00

REMOVE**WARNING****CSI**

- Upper and lower end of pitch control rods are critical characteristic surfaces which must not be damaged during and after removal. Use extreme caution when removing or disconnecting rod end and provide protective covering after removal.
- Verification of proper hardware installation, torque and safety installation of bolts are critical characteristics.

CAUTION

Damage to swashplate and horn lug areas will result if extreme caution is not used when disconnecting pitch control rod end. Use extreme caution when disconnecting pitch control rod end to prevent damage to swashplate and horn lug areas.

NOTE

- The following procedures apply to either spherical bearing, except where indicated.
- The following procedures apply to spherical rod end bearings only.
- Pitch control rod does not have to be removed if only one spherical bearing is being replaced. Disconnect pitch control rod end with spherical bearing being replaced only.
- A maintenance test flight is not required if only one end of pitch control rod was disconnected at a time and no adjustments were made.

1. Disconnect pitch control rod upper or lower rod end (WP 0563 00).

REMOVE - Continued

2. Remove retaining ring (Figure 1).
3. Using bearing installer/remover, 70700-77306-041, on upper rod end, or bearing remover/installer, 65700-40030-041, on lower rod end, remove bearing from rod end by pressing from retaining ring side of bearing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer out of rod end bore using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Do not allow primer to dry in bore.

INSPECT/REPAIR

1. Check upper rod end bore for metal pickup. Remove metal pickup by blending out bore using abrasive cloth, Item 75, WP 1803 00.

NOTE

Bore dimensions for upper and lower rod ends are different.

2. Measure upper rod end bore diameter in two places 90 degrees apart and take average of these two measurements. **INSIDE DIAMETER MUST NOT AVERAGE MORE THAN 2.749-INCHES AFTER REWORK.** Replace rod end if beyond limits.
3. Check lower rod end bore for metal pickup. Remove metal pickup by blending out bore using abrasive cloth, Item 75, WP 1803 00.
4. Measure lower rod end bore diameter in two places 90 degrees apart and take average of these two measurements. **INSIDE DIAMETER MUST NOT AVERAGE MORE THAN 1.8747-INCHES AFTER REWORK.** Replace rod end beyond limits.

INSTALL**WARNING**

Injury to personnel will result if protective gloves are not worn when handling dry ice or chilled parts. Wear protective gloves when handling dry ice or chilled parts.

NOTE

Bearings to remain packaged during chilling process.

1. If available, place new bearing in carbon dioxide, Item 64, WP 1803 00, for at least 30 minutes or freezer for at least 1 hour before installation.

NOTE

Make sure upper control rod end bearings are pressed into bore so that retaining ring is installed opposite key slot.

2. Apply primer coating, Item 243, WP 1803 00, to inside of rod end bore. Do not allow primer to dry.

INSTALL - Continued

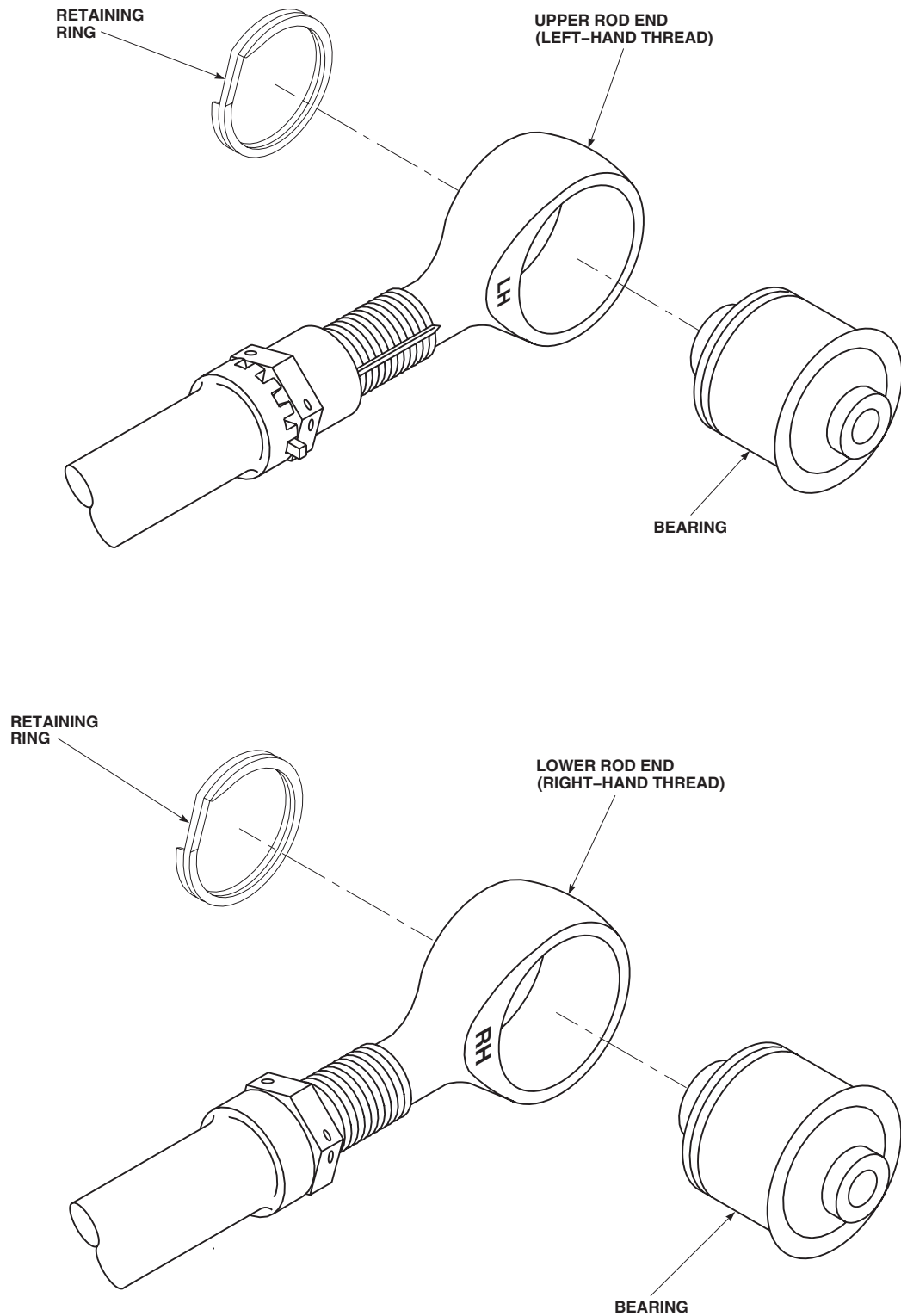


Figure 1. Replace Pitch Control Rod End Spherical Bearings.

INSTALL - Continued

3. **QA** Using bearing remove/installer, 70700-77306-041, on upper rod end, or bearing remover/installer, 65700-40030-041, on lower rod end, install bearing into rod end (Figure 1). Wipe off extra primer using machinery wiping towel, Item 344, WP 1803 00.

NOTE

Make sure bearing is fully seated in rod end bore.

4. **QA** Install retaining ring.
5. Connect pitch control rod upper or lower end (WP 0563 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****PITCH CONTROL ROD UPPER AND LOWER ROD ENDS**

This WP supersedes WP 0565 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Compound, Corrosion Preventive Item 106,
WP 1803 00
Sealing Compound, Item 273, WP 1803 00

References

TM 1-6625-724-13&P
WP 0563 00
WP 1803 00

REMOVE**NOTE**

- If spherical rod end bearing is replaced with elastomeric rod end bearing, both upper and lower rod end bearings on that pitch control rod must be replaced as well as both upper and lower ends of opposing pitch control rod.
- Reidentify rod assembly from P/N 70101-08000-046 to P/N 70101-08000-048 after installing elastomeric rod end bearings.

1. Remove pitch control rod (WP 0563 00).
2. Measure and record length of control rod bolthole to bolthole.

CAUTION

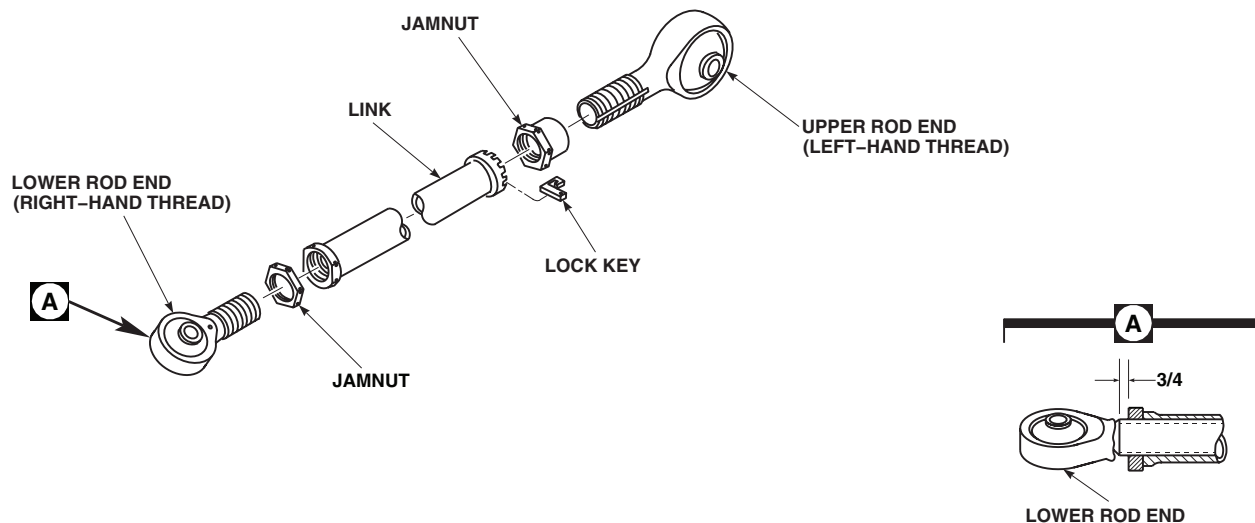
Damage to pitch control rod will result if metallic scrapers are used to clean sealing compound around rod end and jamnuts. Do not use metallic items to remove sealant.

3. Clean any sealant from the pitch control rod end and jamnuts using a nonmetallic scraper.

CAUTION

Pitch control link upper rod ends are left hand threaded. Loosen pitch control link upper jamnuts by turning clockwise. Excessive force applied in the counter-clockwise direction may damage pitch control rod end threads, pitch control rod link threads or anti-corrosive coatings.

4. Loosen jamnut and remove rod end (Figure 1).
5. Remove lock key from upper rod end.

REMOVE - Continued**NOTE**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Replace Pitch Control Rod Upper And Lower Rod Ends.

INSTALL**NOTE**

If the shelf life for SB7110-101 and SB7111-101 has exceeded 60 months, do not install. If the shelf life is less than 60 months prior to installation, serviceability will be in accordance with the applicable technical manual.

1. **QA** Install upper or lower rod end, lock key on upper rod end, and jamnut. Adjust rod length to recorded dimension and angle. Engage lock key on upper rod end. Tighten jamnut fingertight (Figure 1).
2. **QA** Make sure lower rod end has no more than 3/4-inch thread exposure from face of jamnut to end of threads (Figure 1, Detail A).
3. Apply corrosion preventive compound, (Item 106, WP 1803 00), on upper and lower rod threads and jamnuts.

NOTE

- If recorded dimension is not available, set length of rod to 25.22-inches, and with 0.43-inch of thread exposure from face of jamnut to end of threads on lower rod end only.
- Jamnut is torqued and lockwired during installation of pitch control rod (WP 0563 00).

4. Install pitch control rod (WP 0563 00).
5. Track and balance main rotor blades (TM 1-6625-724-13&P).

INSTALL - Continued

6. Seal interface between upper rod end and top of upper jamnut only with sealing compound, Item 273, WP 1803 00.
DO NOT APPLY SEALANT TO UPPER JAMNUT WRENCH FLATS. DO NOT APPLY SEALANT TO LOWER ROD END OR JAMNUT.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ROTOR SYSTEM

PITCH CONTROL ROD ADJUSTMENT

This WP supersedes WP 0566 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 150 - 1000 in. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Sealing Compound, Item 273, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

References

TM 1-1520-237-MTF
TM 1-6625-724-13&P
WP 1803 00

ADJUST

CAUTION

Pitch control link upper rod ends are left hand threaded. Loosen pitch control link upper jamnuts by turning clockwise. Excessive force applied in the counter-clockwise direction may damage pitch control rod end threads, pitch control rod link threads or anti-corrosive coatings.

NOTE

- For every three notches on link there will be a 1% Nr change.
- For purpose of adjusting autorotation rpm, all four pitch control rods must be adjusted in equal amounts.
- A one notch change to a pitch control rod will affect track by approximately 1/4-inch.

1. To adjust pitch control rods, do this:

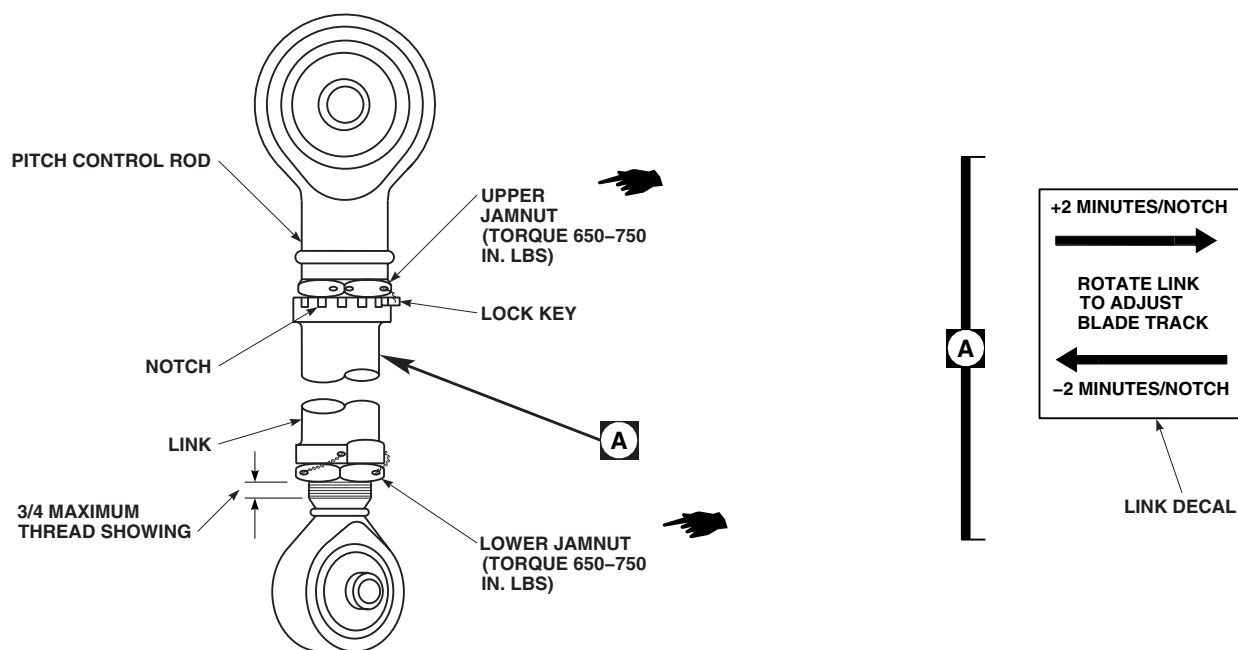
- a. Loosen jamnuts and slide lock key upward enough to allow link to be free to rotate (Figure 1).
- b. To decrease rpm, turn link in +2 minutes (counterclockwise) direction.
- c. To increase rpm, turn link in -2 minutes (clockwise) direction.
- d. **QA** Check for no more than 3/4-inch of threads showing on lower end of pitch control rod. If exposed thread is more than 3/4-inch, pitch control rod has been improperly adjusted. Recheck adjustment.

NOTE

To prevent improper torque of upper jamnut on pitch control rod, make sure torque wrench is not in contact with lock key when torquing.

2. **QA** Secure lock key in notch by tightening upper jamnut. **TORQUE NUT TO 650 - 750 INCH-POUNDS.**

ADJUST - Continued

**NOTE**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Adjust Pitch Control Rod.

3. **QA** Separately tighten lower jamnut. **TORQUE NUT TO 650 - 750 INCH-POUNDS.**
4. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire lower jamnut to barrel and then back to lower jamnut.
5. **QA** Install nonelectrical wire, Item 357, WP 1803 00, from upper jamnut to key, and then back to opposite side of upper jamnut.
6. Check area for cleanness and foreign material.
7. Track and balance main rotor blades (TM 1-6625-724-13&P).
8. Perform maintenance test flight (TM 1-1520-237-MTF).
9. Seal interface between upper rod end and top of upper jam nut only with sealing compound, Item 273, WP 1803 00. **DO NOT APPLY SEALANT TO UPPER JAMNUT WRENCH FLATS. DO NOT APPLY SEALANT TO LOWER ROD END OR JAMNUT.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**SWASHPLATE**

This WP supersedes WP 0567 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Grease Pencil
 Powertrain Repairer's Toolkit, SC 5180-99-A13
 Syringe, Z15955
 Torque Adapter Set
 Torque Wrench, 150 - 1000 in. lbs.
 Torque Wrench, 300 - 2500 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 11, WP 1803 00
 Alcohol, Denatured, Item 48, WP 1803 00
 Brush, Artist's, Item 58, WP 1803 00
 Cloth, Abrasive, Item 75, WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Grease, General Purpose, Item 136, WP 1803 00
 Polyurethane, Coating, Item 226, WP 1803 00
 Polyurethane, Coating, Item 233, WP 1803 00
 Polyurethane, Coating, Item 236, WP 1803 00

Tape, Pressure Sensitive, Adhesive, Item 325, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
 TM 1-1520-237-MTF
 TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0381 00
 WP 0539 00
 WP 0547 00
 WP 0569 00
 WP 0572 00
 WP 0576 00
 WP 1037 00
 WP 1590 00
 WP 1661 00
 WP 1803 00

NOTE

Swashplate weighs about 180 pounds.

REMOVE

1. If maintenance crane will be used, assemble and erect crane (WP 1590 00).
2. Remove main rotor blades (WP 0576 00).
3. Remove main rotor head (WP 0547 00). Reinstall bolts, washers, spacers, and nuts that secure pitch control rod to swashplate, back in swashplate.

REMOVE - Continued**CAUTION**

Damage to main rotor shaft can result when removing lower pressure plate. Use care when removing lower pressure plate.

4. Place straps of hoisting sling around bolts in pitch control rod attachment points on swashplate (Figure 1, Sheet 1).

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

5. Attach hoisting sling to maintenance crane or hoist.
6. Adjust lengths of hoisting sling straps for even distribution of weight when hoisting.
7. Insert 1/4-inch Allen wrench into center pin of expandable pin holding connecting links. While holding Allen wrench, remove nut and washers (Figure 1, Sheet 1). Do not allow pin to turn.
8. Raise hoist enough to carry weight of swashplate.
9. Remove expandable pins (Figure 1, Sheet 2).

WARNING**CSI**

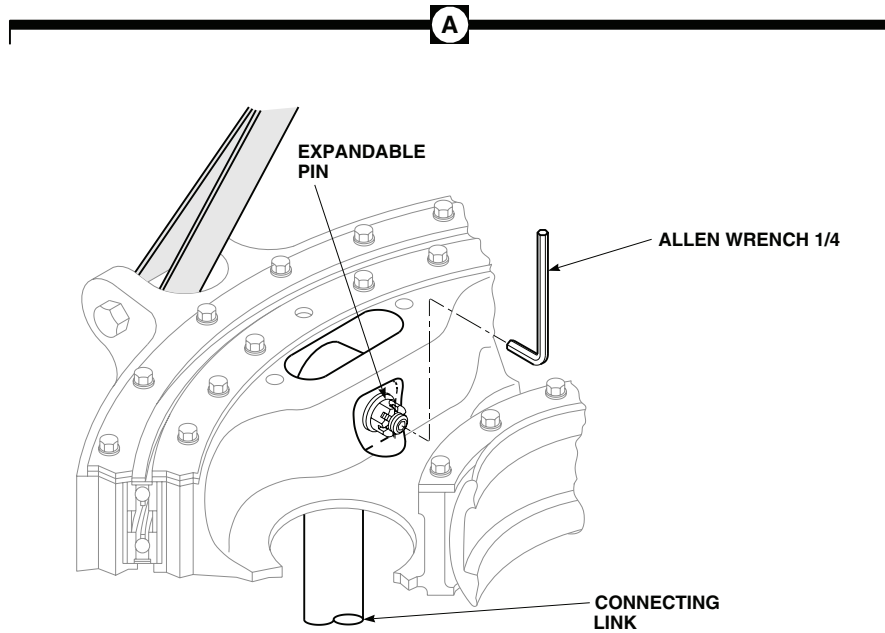
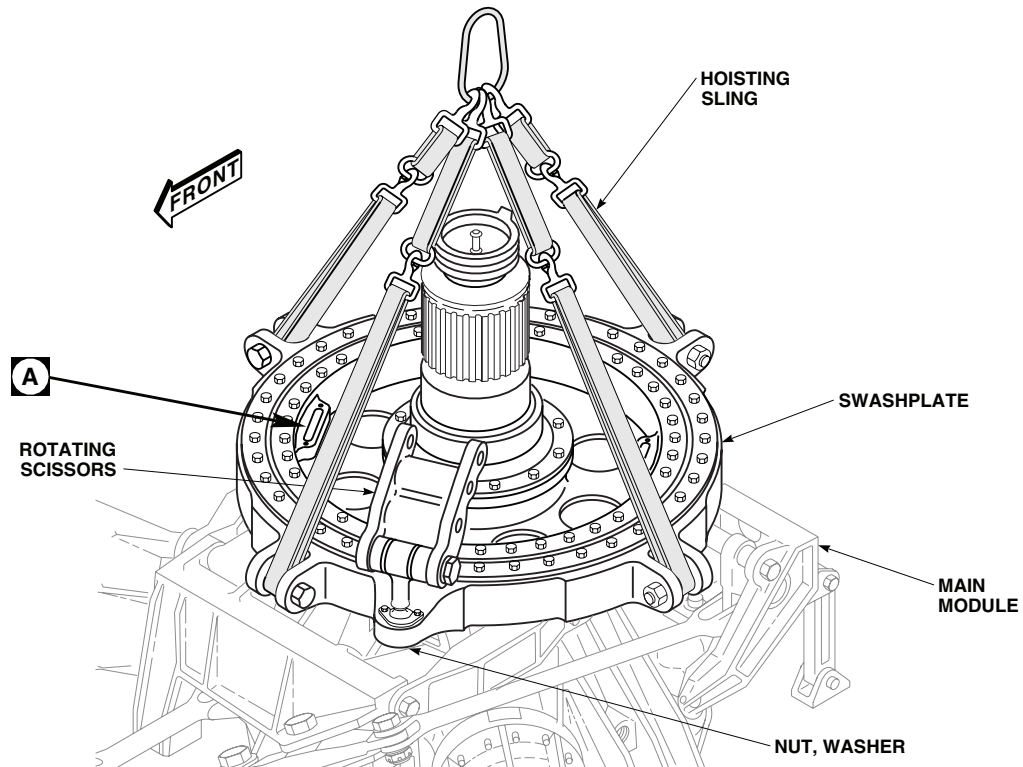
Exposed surfaces of main rotor shaft and connecting link lug areas are critical surfaces which must not be damaged during and after removal of swashplate. Use extreme caution when removing and provide protective covering over critical surfaces after removal of swashplate.

10. Remove swashplate from helicopter, taking care not to damage main rotor shaft and connecting links.
11. Inspect expandable pins (WP 0569 00).
12. Reinstall expandable pins in swashplate. Use only enough torque to hold pin.
13. Place swashplate on work platform or in shipping container.
14. Remove rotating scissors (WP 0572 00).
15. Remove hoisting sling.

INSPECT

Inspect swashplate (WP 0539 00). Refer to, INSPECT/REPAIR SWASHPLATE, in this work package.

INSPECT - Continued

**NOTE**

ALL DIMENSIONS ARE IN
INCHES UNLESS NOTED.

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Figure 1. Replace Swashplate. (Sheet 1 of 2)

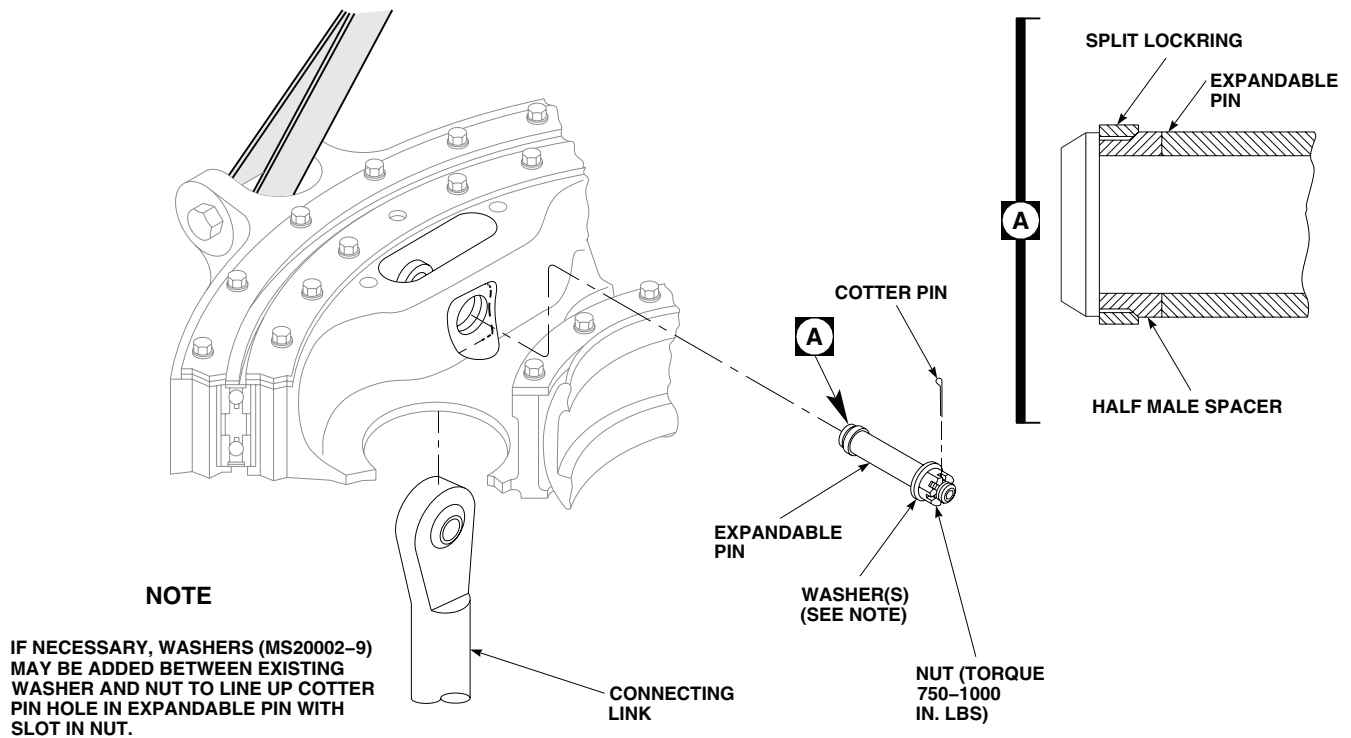
INSPECT - ContinuedAK1358A
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Figure 1. Replace Swashplate. (Sheet 2 of 2)

INSTALL**NOTE**

If new or replacement swashplate was built-up more than 12 months prior to installation date, new or replacement swashplate must be lubricated prior to installation.

1. **QA** Check new or replacement swashplate build up date. If new or replacement swashplate was built-up more than 12 months prior to installation, lubricate swashplate (WP 1661 00).
2. Connect straps of hoisting sling around bolts in pitch control rod attachment points on swashplate. Adjust length of hoisting sling straps to evenly distribute weight when hoisting (Figure 1, Sheet 1).
3. Attach hoisting sling to maintenance crane or hoist.
4. Install rotating scissors (WP 0572 00).

WARNING**CSI**

Exposed surfaces of main rotor shaft and connecting link lug areas are critical surfaces which must not be damaged during installation of swashplate. After removal of protective covering from shaft and links, use extreme caution when installing swashplate.

5. **QA** Lift and place swashplate onto main rotor shaft, taking care not to damage shaft.
6. **QA** Lower swashplate and guide connecting links into position (Figure 1, Sheet 2).

INSTALL - Continued

7. Back off nut of expandable pin until split locking segments are loose on pin (Figure 1, Sheet 1).

WARNING**CSI**

- Verification of proper hardware installation and torque of nut is a critical characteristic.
- Verification of correct installation of expandable pin split lock ring and half male spacer is a critical characteristic.

NOTE

- If necessary, add washer between existing washer and nut, to line up cotter pin hole in expandable pin with slot on nut.
 - A light coat of general purpose grease, Item 136, WP 1803 00, may be used to aid in inserting and removing the expandable pin.
8. **QA** Insert expandable pin through stationary swashplate and connecting link of control rod end. Make sure washer under nut is flush with inner side of swashplate.
9. Tighten nut on expandable pin handtight.
10. **QA** Insert Allen wrench into center on pin. **HOLD ALLEN WRENCH AND TORQUE NUT TO 750 - 1000 INCH-POUNDS USING TORQUE ADAPTER.** Do not allow pin to turn.
11. **QA CHECK THAT CLEARANCE UNDER WASHER IS NOT MORE THAN 0.002-INCH.** This means that bolt is locked in place.
12. **QA** Install cotter pin.
13. Remove hoisting sling.
14. Install main rotor head (WP 0547 00).
15. Check swashplate ball bearings for wear as follows:
- a. Turn on all hydraulic power.
 - b. Clamp dial indicator to rotating swashplate, place dial indicator pointer on stationary swashplate. Connect dial indicator as close to bearing as possible.
 - c. Slowly move collective stick through normal range. Measure vertical movement between rotating and stationary swashplates. **MAXIMUM RELATIVE VERTICAL MOTION IS 0.010 - INCH.** Replace swashplate if limits are exceeded.
 - d. Remove dial indicator.
 - e. Turn off all hydraulic power.
16. Install main rotor blades (WP 0576 00).
17. Perform main rotor rig check (WP 1037 00).
18. Check area for cleanness and foreign material.

INSTALL - Continued**NOTE**

After the installation of a new or relubricated swashplate, considerable grease may be displaced from the duplex ball bearing between stationary and rotating swashplate during normal helicopter operation.

19. Perform maintenance test flight (TM 1-1520-237-MTF).

INSPECT/REPAIR**NOTE**

In the following steps, do all blending with an abrasive wheel, Item 350, WP 1803 00, abrasive cloth, Item 75, WP 1803 00, or a medium cut file.

1. Visually inspect swashplate grease shield for cracks. **NO MORE THAN 15 CRACKS ALLOWED IN GREASE SHIELD, AND CRACKS MAY NOT BE CLOSER THAN 1 INCH APART.**
2. If crack is found in grease shield, do this:
 - a. Turn main rotor head until red index marks on rotating and stationary swashplates line up. Engage gust lock or rotor brake (Figure 2).
 - b. Remove bolts and washers holding inner bearing retainer assembly to swashplate. Slide retainer assembly to one side.
 - c. Stop drill cracks (TM 1-1500-204-23).
 - d. Install inner bearing retainer assembly with bolts and washers, making sure red index marks line up. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**
3. If crack is suspected in any other part of swashplate, do this:
 - a. Clean and strip suspected area (TM 1-1520-265-23). If necessary, protect bearings with pressure sensitive adhesive tape, Item 325, WP 1803 00

WARNING**CSI**

Fluorescent penetrant inspection is a critical characteristic.

- b. Do a fluorescent penetrant inspection, Type I, using fluorescent penetrant inspection kit and check for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.**
 - c. If crack is found, replace swashplate.
 - d. If crack is not found, touch up stripped area with primer and paint (WP 0381 00 and TM 55-1500-345-23).
4. Check corners of swashplate for nicks and gouges. **DAMAGE TO CORNERS SHALL NOT EXCEED 0.010-INCH DEEP.** Blend damage out to 1/2-inch radius. Remove any raised material around damaged area. Touch up paint repaired area (WP 0381 00 and TM 55-1500-345-23).

INSPECT/REPAIR - Continued

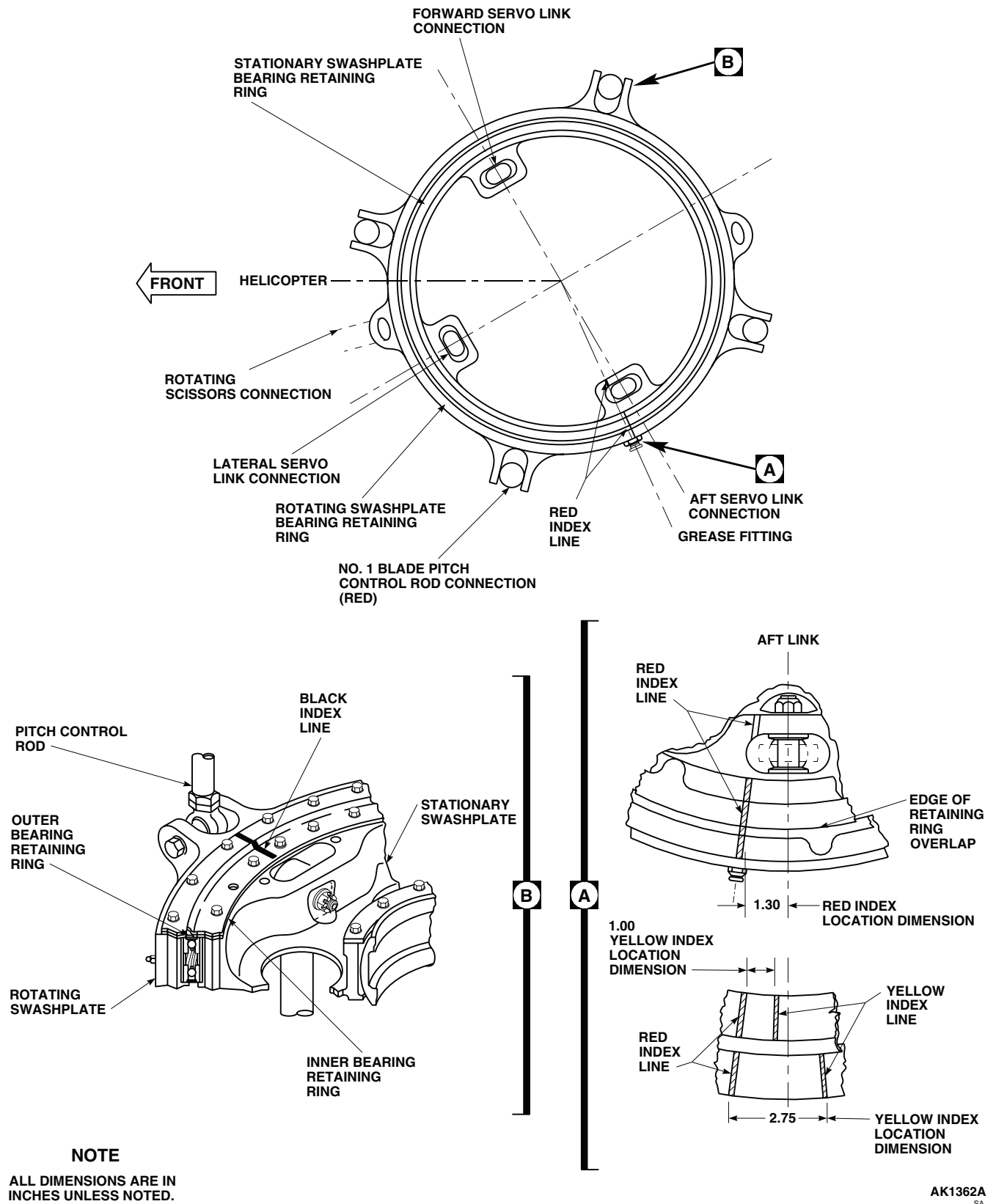


Figure 2. Inspect/Repair Swashplate.

INSPECT/REPAIR - Continued

5. Check nonmating surfaces of swashplate for nicks and gouges. **DAMAGE TO NONMATING SURFACES, EXCEPT PITCH CONTROL ROD LUGS SURFACES, SHALL NOT EXCEED 0.050-INCH DEEP. DAMAGE TO PITCH CONTROL ROD SURFACES SHALL NOT EXCEED 0.030-INCH DEEP.** Blend damage out to 1/2-inch radius. Remove any raised material around damaged area. Touch up paint repaired area (WP 0381 00 and TM 55-1500-345-23).
6. Check mating surface of swashplate for nicks and gouges. **DAMAGE SHALL NOT EXCEED 0.020-INCH DEEP ON NO MORE THAN 10% OF MATING AREA.** Blend damage out to 1/2-inch radius. Remove any raised material around damaged area. Touch up paint repaired area (WP 0381 00 and TM 55-1500-345-23).
7. Check bushings in control rod attachment lugs for corrosion pitting. **REPLACE SWASHPLATE IF CORROSION PITS ARE DEEPER THAN 0.005-INCH IN SHOULDER AREA, OR 0.040-INCH IN CHAMFER AREA.**
8. Visually check swashplate for voids or disbonding of shield on retainer assembly. If disbond or void is over 5 inches, do the following:
 - a. Turn main rotor head until red index marks on rotating and stationary swashplates line up (Figure 2, Detail A).
 - b. Remove bolts and washers securing retainer assembly to stationary swashplate.
 - c. Raise retainer assembly and cover swashplate.

CAUTION

Damage to swashplate will result if solvents or adhesives enter grease. Provide adequate protection to the swashplate while using solvents and adhesives.

- d. Clean disbonded area by injecting denatured alcohol, Item 48, WP 1803 00, into area using syringe.
 - e. If possible, sand area using abrasive cloth, Item 75, WP 1803 00. Clean again with denatured alcohol, Item 48, WP 1803 00, and allow to dry.
 - f. Mix adhesive, Item 11, WP 1803 00, following manufacturer's directions. Inject mixed adhesive into void areas using syringe.
 - g. Apply pressure on rebonded area using clamps.
 - h. Allow to cure for 24 hours at room temperature.
 - i. Install retainer assembly by aligning the red index marks. Secure with bolts and washers. Make sure that all nine tabs of shield receive a bolt. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**
9. If either or both red blade fold index marks on bearing retaining rings are missing, or lines cannot be matched, do this:
- a. Turn rotating swashplate so that the scissors connection is approximately between aircraft centerline and lateral servo link on stationary swashplate (Blade No. 1 on left front side).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- b. Thoroughly clean areas on retaining rings and aft servo link boss, near grease fitting, using a clean machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe area again, do not allow solvent to dry on parts.

INSPECT/REPAIR - Continued

- c. Locate and mark centerline of aft servo link on stationary swashplate bearing retaining ring with grease pencil (Figure 2).
 - d. Locate and mark centerline of grease fitting on rotating swashplate bearing retaining ring with grease pencil. Turn rotating swashplate until the measured distance between the aft servo link centerline, and the grease fitting centerline equal 1.30-inches on the overlap edge of bearing retaining ring (Figure 2, Detail A). Engage main rotor gust lock or rotor brake.
 - e. Make a guide on both sides of pencil centerline, using pressure sensitive adhesive tape, Item 325, WP 1803 00, to form a line approximately 0.10-inch wide from outer bearing retainer ring edge to inner servo link boss. Remove grease pencil marks, using previous cleaning instructions.
 - f. Paint red index mark between guides, using polyurethane coating, Item 226, WP 1803 00, and artist's brush, Item 58, WP 1803 00.
 - g. Remove guide tape when paint has dried.
 - h. Check area for cleanness and foreign material.
 - i. Disengage main rotor gust lock or rotor brake.
10. If either or both yellow blade fold index marks on both rotating and stationary bearing retaining rings are missing, or lines cannot be matched, do this:
- a. Align red index lines on both rotating and stationary bearing retaining rings. Engage main rotor gust lock or rotor brake.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- b. Thoroughly clean area between red index mark and aft servo link on both bearing retaining rings, using a clean machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe area again, do not allow solvent to dry on parts.
 - c. On inside diameter of stationary bearing retaining ring, measure 1 inch from center of red index line, and mark yellow index centerline, using a grease pencil (Figure 2, Detail A).
 - d. On outside diameter of rotating bearing retaining ring, measure 2.75-inches from center of red index line, and mark yellow index centerline, using a grease pencil.
 - e. Make a guide on both sides of both pencil centerlines using pressure sensitive adhesive tape, Item 325, WP 1803 00, to form a line approximately 0.10-inch wide across each bearing retainer ring (Figure 2, Detail A). Remove grease pencil marks, using previous cleaning instructions.
 - f. Paint both yellow index lines between guides, using polyurethane coating, Item 236, WP 1803 00, and artist's brush, Item 58, WP 1803 00.
 - g. Remove guide tape when paint has dried.
 - h. Check area for cleanness and foreign material.
 - i. Disengage main rotor gust lock or rotor brake.
11. If any black rigging index marks on bearing retaining rings are missing, or lines cannot be matched (total 4 each retaining ring), do this:

INSPECT/REPAIR - Continued

- a. Turn rotating swashplate to align the pitch control rod centerlines with the servo link centerlines. Engage main rotor gust lock or rotor brake.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- b. Thoroughly clean area to be painted on both retaining rings behind pitch control rods (4 places), using a clean machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe area again, do not allow solvent to dry on parts.
 - c. Carefully mark centerlines of each pitch control rod across both bearing retaining rings, using a grease pencil (Figure 2, Detail B).
 - d. Make a guide on both sides of pencil centerlines using pressure sensitive adhesive tape, Item 325, WP 1803 00, to form a line approximately 0.12-inch wide across both bearing retaining rings. Remove grease pencil marks, using previous cleaning instructions.
 - e. Paint black index lines between guides, using polyurethane coating, Item 233, WP 1803 00, and artist's brush, Item 58, WP 1803 00.
 - f. Remove guide tape when paint has dried.
 - g. Check area for cleanness and foreign material.
 - h. Disengage main rotor gust lock or rotor brake.
12. Touch up aluminum areas that were reworked, with corrosion resistant coating, Item 111, WP 1803 00, and apply primer and paint (WP 0381 00 and TM 55-1500-345-23). On parts other than aluminum, apply primer and paint (WP 0381 00 and TM 55-1500-345-23).
 13. Check swashplate for corrosion.
 14. Measure thickness of the part being inspected at the annealed surface where corrosion is found.
 15. **ON OPEN AREAS (NONMATING SURFACES) REMOVE CORROSION PITS UP TO 0.050-INCH DEEP. NO TWO REWORKED AREAS MAY BE CLOSER TOGETHER THAN THE THICKNESS OF THE PART.** Replace swashplate if damaged beyond limits.
 16. **ON MATING SURFACES REMOVE CORROSION PITS UP TO 0.010-INCH DEEP ON NO MORE THAN 10% OF THE SURFACE BY BLENDING OUT.** Replace swashplate if damaged beyond limits.
 17. Touch up aluminum areas that were reworked, with corrosion resistant coating, Item 111, WP 1803 00, and apply primer and paint (WP 0381 00 and TM 55-1500-345-23). On parts other than aluminum, apply primer and paint (WP 0381 00 and TM 55-1500-345-23).
 18. Inspect blade track bracket on rotating swashplate for general condition and security. Replace damaged or bent bracket. **TORQUE BOLTS TO 105 - 115 INCH POUNDS.**
 19. Inspect magnetic pickup bracket on stationary swashplate for general condition and security. Replace damaged or bent bracket. **TORQUE BOLTS TO 105 - 115 INCH POUNDS.**
 20. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM**

SWASHPLATE SCISSORS ATTACHMENT SPHERICAL BEARING

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13
Spherical Bearing Remover/Installer, Locally-Made,
Figure 195, WP 1805 00
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Carbon Dioxide, Item 64, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0571 00
WP 0572 00
WP 1803 00
WP 1805 00

REMOVE

1. Locally make spherical bearing remover/installer (WP 1805 00).
2. Disconnect lower link of rotating scissors (WP 0572 00).
3. Remove bolts, washers, and retainer from swashplate (Figure 1).
4. Using tool A, tool B, and tool C of locally-made spherical bearing remover/installer, remove spherical bearing as follows:
 - a. Position tool A above swashplate to receive spherical bearing (Figure 1, Detail A).
 - b. Position tool B below swashplate to push spherical bearing.
 - c. Install tool C through tool A, tool B, and spherical bearing.
 - d. Tighten tool C until spherical bearing is pushed out of swashplate. Remove tool A, tool B, tool C, and spherical bearing.
5. Inspect swashplate bearing bore as follows (Figure 1, Detail C):
 - a. Inspect inside diameter of bushing. **INSIDE DIAMETER OF BUSHING NOT TO EXCEED 1.7485-INCH. NO REPAIR ALLOWED.**
 - b. Inspect condition of well flange. **NO SEVERE DAMAGE (0.100-INCH) ALLOWED. NO REPAIR ALLOWED.**
 - c. Inspect scissors attachment surface for servicability. **NICKS, SCRATCHES AND GOUGES DEEPER THAN 0.020-INCH NOT ALLOWED. NO REPAIR ALLOWED.**

INSTALL

1. Using tool C, tool D, and tool E of locally-made spherical bearing remover/installer, install spherical bearing as follows:

INSTALL - Continued

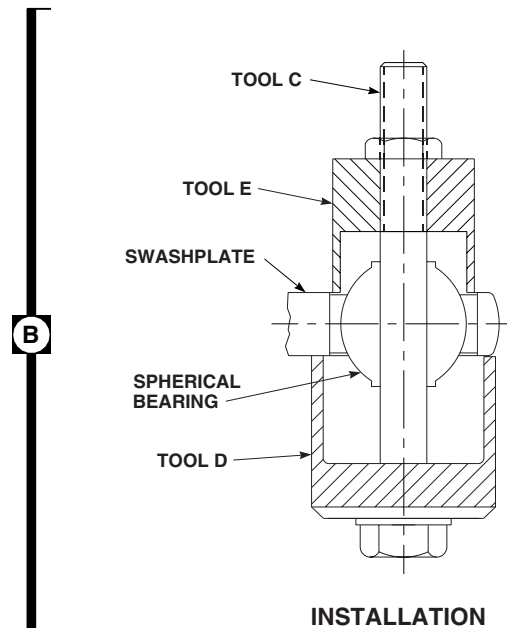
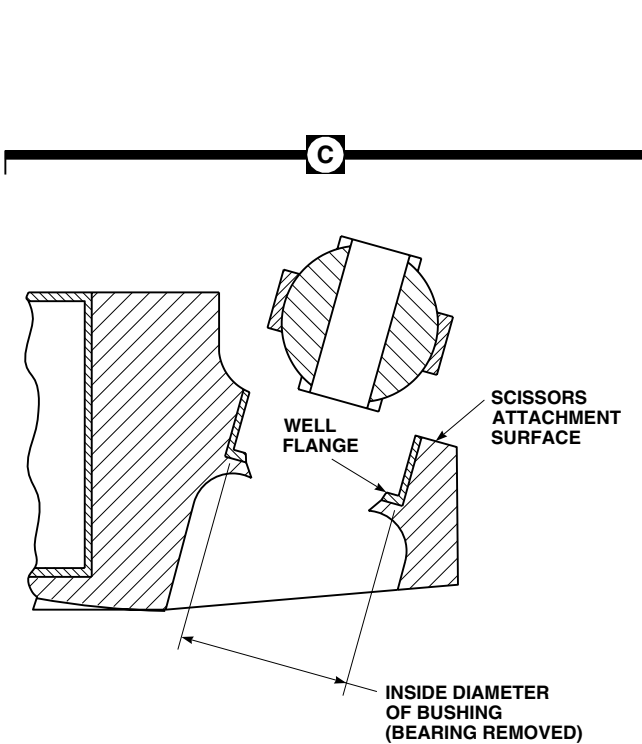
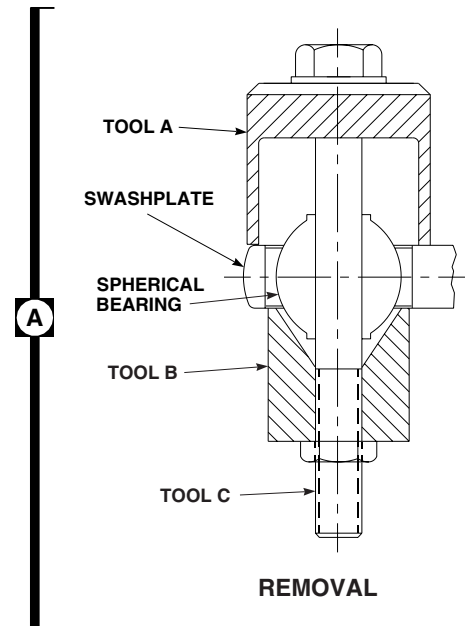
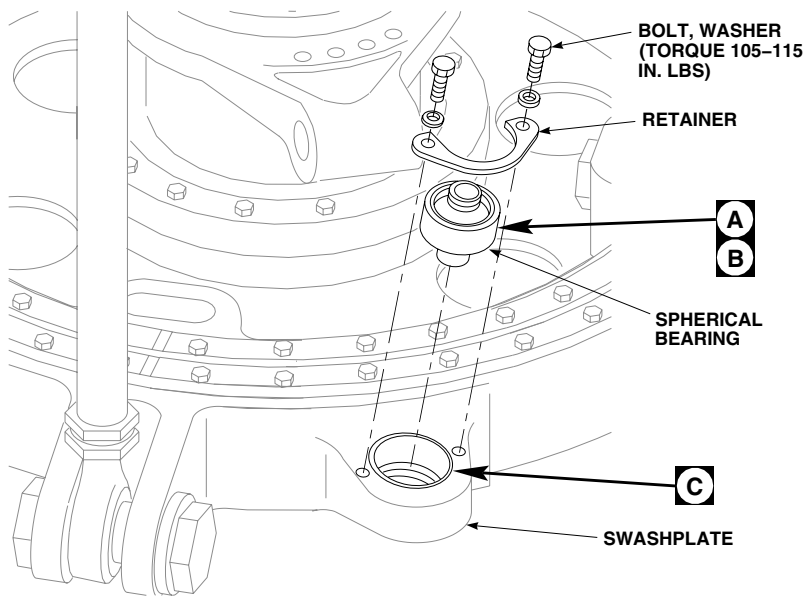


Figure 1. Replace Swashplate Scissors Attachment Spherical Bearing.

INSTALL - Continued**WARNING**

Injury to personnel will result if dry ice or chilled parts are handled without protective gloves. Use protective gloves when handling dry ice or chilled parts.

- a. Freeze replacement spherical bearing in carbon dioxide, Item 64, WP 1803 00, for a minimum of 30 minutes.
 - b. Remove spherical bearing from dry ice.
 - c. Apply a thin coat of primer coating, Item 243, WP 1803 00, to the mating surfaces of the spherical bearing and swashplate bearing bore.
 - d. Position spherical bearing above swashplate bearing bore.
 - e. Position tool E above swashplate and spherical bearing to push spherical bearing into swashplate bearing bore (Figure 1, Detail B).
 - f. Position tool D below swashplate to provide surface for tool C to pull against.
 - g. Install tool C through tool E, tool D, and spherical bearing.
 - h. **QA** Tighten tool C until spherical bearing is bottomed into swashplate bearing bore. Remove tool C, tool D, and tool E.
2. Check retainer inserts as follows:
 - a. Install bolts in insert until bottom of bolthead is 0.150-inch from lug. Do not install retainer or washers.
 - b. Check breakaway torque for each bolt. **BREAKAWAY TORQUE SHALL BE 3.5 - 30 INCH-POUNDS.**
 - c. If breakaway torque is not within limits, replace bolt and clean insert. Recheck breakaway torque.
 - d. If breakaway torque is still not within limits replace insert (WP 0571 00).
 3. **QA** Install retainer with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
 4. Connect rotating scissors (WP 0572 00).
 5. Check area for cleanness and foreign material.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
SWASHPLATE EXPANDABLE PIN REPAIR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00

References

WP 1803 00

INSPECT/REPAIR

1. Clean expandable pin with cleaning compound solvent, Item 71, WP 1803 00.
2. Remove surface corrosion with abrasive mat, Item 2, WP 1803 00.
3. Visually inspect pin collar and threads for corrosion, fretting, and scoring damage.
4. **REPLACE EXPANDABLE PIN IF THERE IS ANY EVIDENCE OF PITTING, SCORING, FRETTING OR 25% OF DRY FILM LUBRICANT HAS BEEN REMOVED FROM OUTSIDE DIAMETER OF PIN BUSHING ASSEMBLY.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
SWASHPLATE SPHERICAL BEARING RETAINER INSERTS

INITIAL SETUP:**Tools and Special Tools**

Depth Micrometer, GGG-C-105
Fluorescent Penetrant Inspection Kit, XMA101
Insert Kit, K4-314-314L
Powertrain Repairer's Toolkit, SC 5180-99-B13
Swage Tool, SR31W-A
Thread Gage, 599

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

TM 1-1520-265-23
WP 0598 00
WP 1803 00

Material/Parts

Primer Coating, Item 243, WP 1803 00

INSPECT

1. Check condition of inserts threads as follows:
 - a. Visually inspect for looseness. **NO LOOSENESS ALLOWED.** Replace loose inserts.
 - b. Using a thread gage, 5/16 - 24UNF-3A, check for crossed, stripped, or flattened threads. **NO DAMAGE ALLOWED.** Replace damaged inserts.

WARNING**CSI**

Fluorescent penetrant inspection is a critical characteristic.

2. Fluorescent penetrant inspect insert holes for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked swashplate.

REMOVE

1. Remove retainer and shim, if required (WP 0598 00).
2. Using insert kit, K4-314-314L, remove inserts as follows:
 - a. Counterbore 0.110-inch deep by 0.374 to 0.379-inch diameter with removal tool.
 - b. Back out insert with drive wrench (Figure 1).
3. Send swashplate to depot if insert cannot be removed.

INSTALL

1. Coat insert with primer coating, Item 243, WP 1803 00.
2. Using insert kit, K4-314-314L, install inserts as follows:
 - a. **QA** Thread insert 0.015 to 0.025-inch below surface of swashplate using drive wrench (Figure 1).

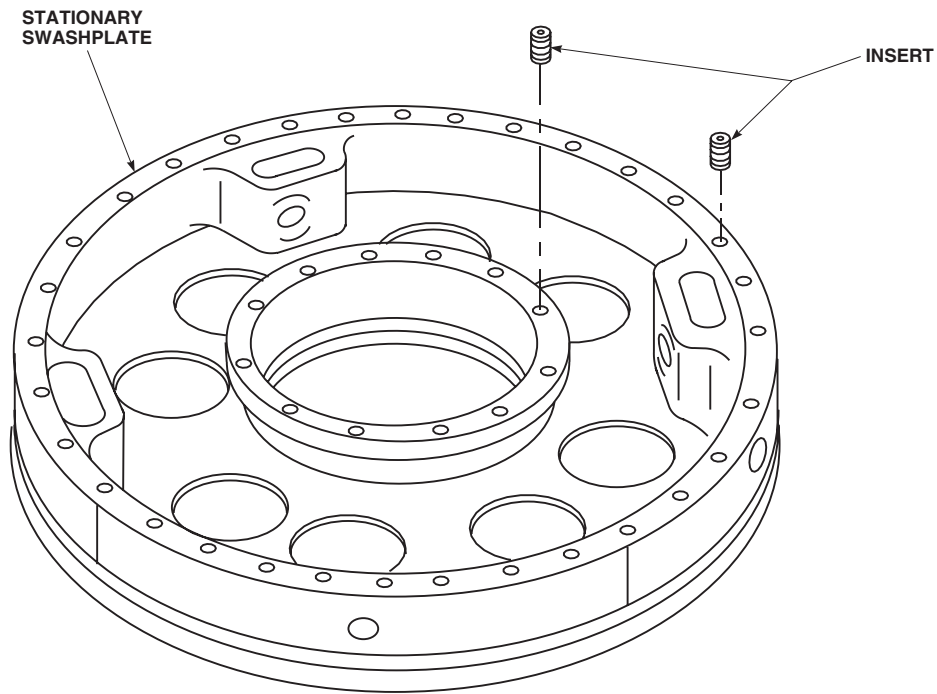
INSTALL - ContinuedAK1364
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Figure 1. Replace Swashplate Spherical Bearing Retainer Inserts.

- b. **QA** Place swage tool in insert and apply enough downward force to bottom nylon protective shoulder on surface of swashplate.
3. Install retainer and shim, if required (WP 0598 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM**

SWASHPLATE SCISSORS ATTACHMENT RETAINER INSERTS

INITIAL SETUP:**Tools and Special Tools**

Depth Micrometer, GGG-C-105
Insert Kit, K4-258-258L
Powertrain Repairer's Toolkit, SC 5180-99-B13
Swage Tool, SR25S-A
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

WP 1803 00

Material/Parts

Primer Coating, Item 243, WP 1803 00

REMOVE

1. Remove bolts and washers attaching retainer to rotating swashplate.
2. Using insert kit, K4-258-258L, remove inserts as follows:
 - a. Counterbore 0.095-inch deep by 0.311 to 0.316-inch diameter with removal tool.
 - b. Back out and remove insert from swashplate (Figure 1).

INSTALL

1. Coat insert with primer coating, Item 243, WP 1803 00.
2. Using insert kit, K4-258-258L, install inserts as follows:
 - a. **QA** Thread insert 0.015 to 0.025-inch below surface of swashplate (Figure 1).
 - b. **QA** Place swage tool in insert and apply enough downward force to bottom nylon protective shoulders on surface of swashplate.
3. Check threaded inserts as follows:
 - a. Install bolts in inserts until bottom of bolthead is 0.150-inch from lug. Do not install retainer or washers.
 - b. Check breakaway torque for each bolt. **BREAKAWAY TORQUE SHALL BE 3.5 - 30 INCH-POUNDS.**
 - c. If breakaway torque is not within limits, replace bolt and clean insert. Recheck breakaway torque.
 - d. If breakaway torque is still not within limits, replace insert.
4. Install retainer on rotating swashplate with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

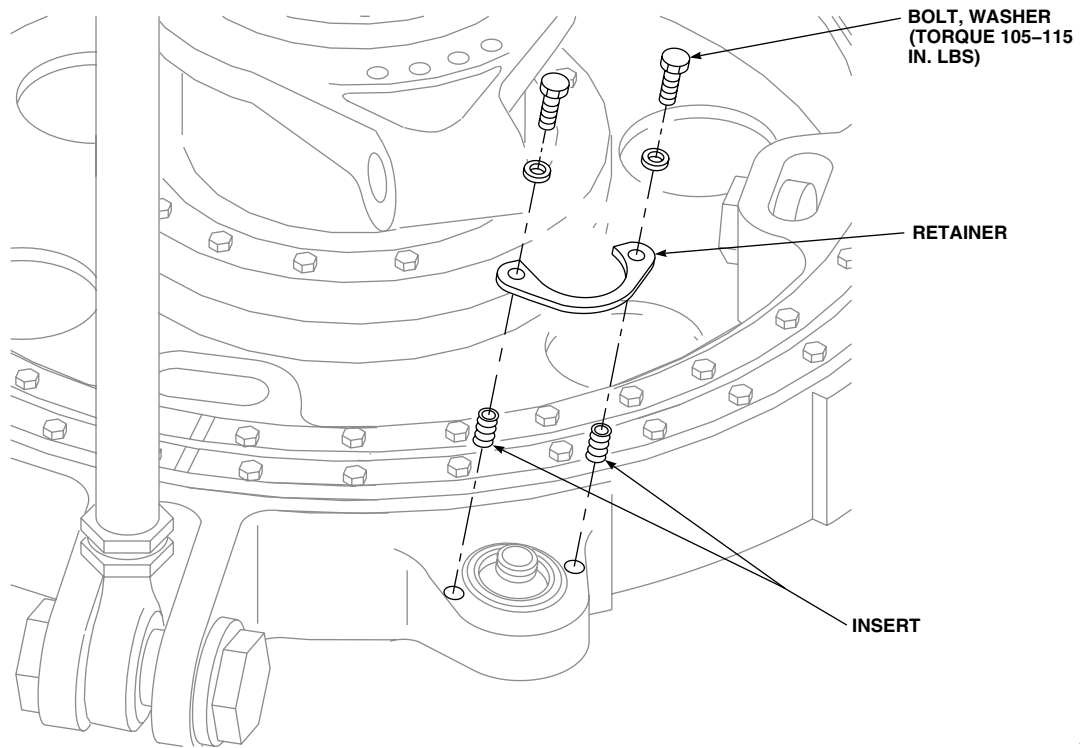
INSTALL - ContinuedAK1365B
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Figure 1. Replace Swashplate Scissors Attachment Retainer Inserts.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****ROTATING SCISSORS**

This WP supersedes WP 0572 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Depth Gage
Fluorescent Penetrant Inspection Kit, XMA101
Medium Cut File
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 300 - 2500 in. lbs.

Tape, Pressure Sensitive, Adhesive, Item 334,
WP 1803 00

Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cloth, Abrasive, Item 77, WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00
Lubricant, Solid Film, Item 180, WP 1803 00
Polyurethane, Coating, Item 227, WP 1803 00
Polyurethane, Coating, Item 233, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Remover, Paint, Item 250, WP 1803 00

References

TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 0540 00
WP 0550 00
WP 0599 00
WP 1803 00

REMOVE

1. Inspect upper and lower bearings for play (WP 0540 00).
2. Turn main rotor head so rotating scissors are easy to reach.
3. Engage gust lock.
4. Remove bolt, washers, shims, nut, and separate upper scissors link from lower scissors link (Figure 1, Detail B).
5. Remove bolt, washers, shims, and nut from upper link and pressure plate lug (Figure 1).
6. Remove washer, and nut from lower link. Remove scissors from rotating swashplate.

INSPECT/REPAIR**NOTE**

In the following steps, do all blending with abrasive wheel, Item 350, WP 1803 00, 180 grit abrasive cloth, Item 77, WP 1803 00, or a medium cut file, as necessary.

1. Inspect bolt attaching upper link to pressure plate (WP 0550 00).
2. Visually inspect scissors for cracks. If crack is suspected, do a fluorescent penetrant inspection of scissors as follows:
 - a. Clean and strip suspected area using paint remover, Item 250, WP 1803 00 (TM 55-1500-345-23).
 - b. Do a fluorescent penetrant inspection, and check for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23).

INSPECT/REPAIR - Continued

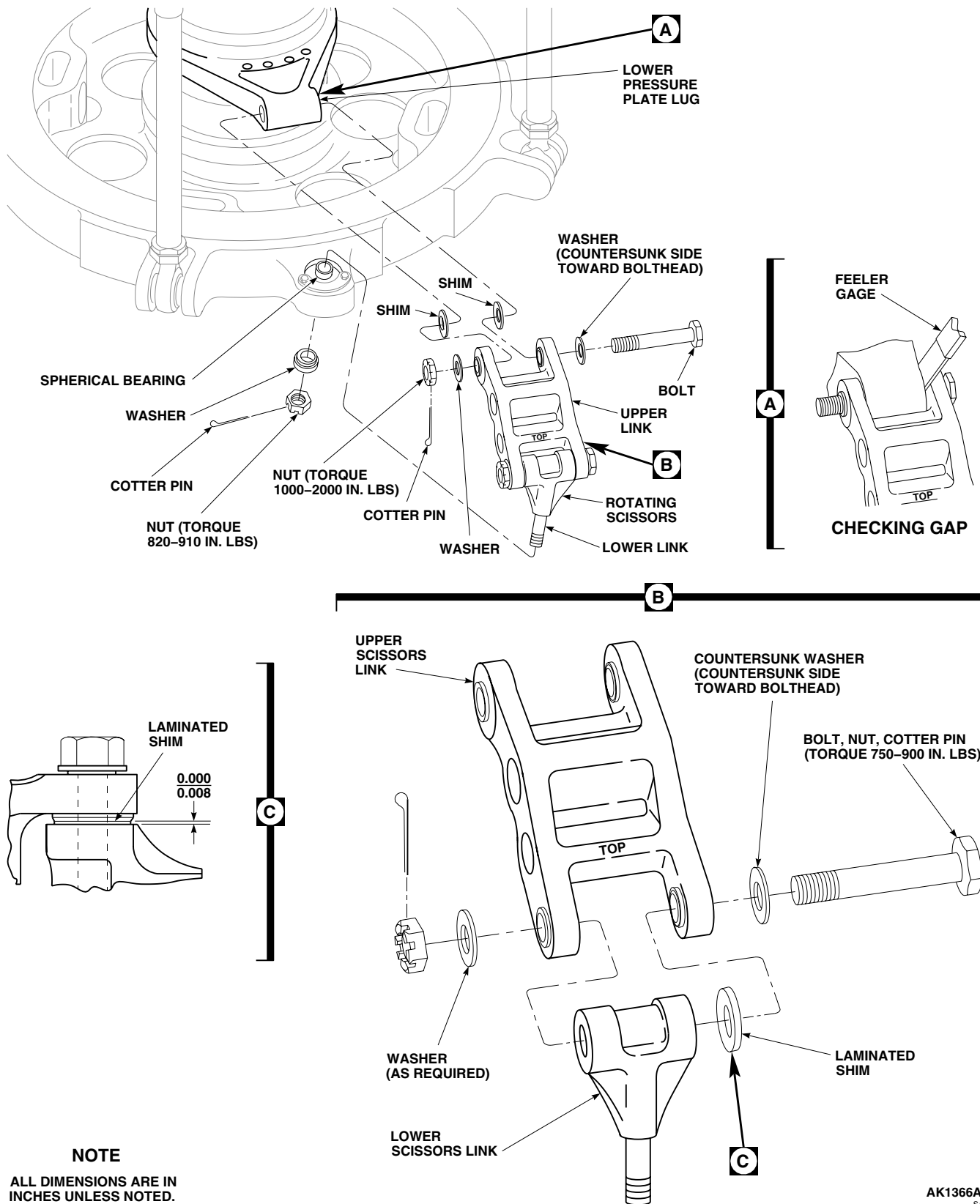


Figure 1. Replace Rotating Scissors.

INSPECT/REPAIR - Continued

- c. If crack is found, replace scissors.
- d. If crack is not found, touch up stripped area with primer and paint (WP 0381 00 and TM 55-1500-345-23).
- 3. Check scissors for nicks and gouges. **CORNER DAMAGE UP TO 0.10-INCH DEEP, EXCEPT IN LUG AREA WHERE LIMIT IS 0.050-INCH, MAY BE BLENDED OUT TO A 1/2-INCH RADIUS. NO TWO REWORKED AREAS MAY BE CLOSER TOGETHER THAN 1 INCH.** Replace scissors if damaged beyond limits.
- 4. **ON OPEN AREAS (NONMATING SURFACES) SURFACE DAMAGE UP TO 0.050-INCH DEEP, EXCEPT LUG AREA WHERE LIMIT IS 0.030-INCH, MAY BE BLENDED OUT TO A 1/2-INCH RADIUS.** Replace scissors if damaged beyond limits.
- 5. **ON MATING SURFACES, REMOVE DAMAGE UP TO 0.020-INCH ON NO MORE THAN 10% OF THE MATING AREA.** Remove any raised material around damaged area and blend out to a 1/2 - inch radius. Replace scissors if damaged beyond limits.
- 6. Touch up aluminum areas that were reworked, with corrosion resistant coating, Item 111, WP 1803 00, and apply primer and paint (WP 0381 00 and TM 55-1500-345-23).
- 7. Check scissors for corrosion.
- 8. **ON OPEN AREAS (NONMATING SURFACES), REMOVE CORROSION PITS UP TO 0.050-INCH DEEP. NO TWO REWORKED AREAS MAY BE CLOSER TOGETHER THAN 1 INCH.** Replace scissors if damaged beyond limits.
- 9. **ON MATING SURFACES, REMOVE CORROSION PITS UP TO 0.010-INCH DEEP ON NO MORE THAN 10% OF SURFACES, BY BLENDING OUT.** Replace scissors if damaged beyond limits.
- 10. Touch up aluminum areas that were reworked, with corrosion resistant coating, Item 111, WP 1803 00, and apply primer and paint (WP 0381 00 and TM 55-1500-345-23).
- 11. Paint both upper and lower links as follows (WP 0381 00 and TM 55-1500-345-23).
 - a. Mask bearing bores with pressure sensitive adhesive tape, Item 334, WP 1803 00.
 - b. Apply one coat of primer coating, Item 243, WP 1803 00, to surface.
 - c. Apply two coats of polyurethane coating, Item 227, WP 1803 00.
 - d. Stencil "top" and "bottom" on top and bottom surface of upper link, using polyurethane coating, Item 233, WP 1803 00.
 - e. Remove masking tape from bearing bores.

INSTALL

- 1. Loosely assemble upper link to lower link with bolt.
- 2. Using feeler gage measure gap between links (Figure 1, Detail A). **PEEL LAMINATED SHIM TO MAKE 0.000 to 0.008-INCH TOTAL CLEARANCE BETWEEN LINKS (Figure 1, Detail C).**

INSTALL - Continued**WARNING****CSI**

Verification of proper hardware installation, torque, and safety installation is a critical characteristic.

3. **QA** Assemble upper link to lower link with bolt, countersunk washer, shims, washers (as required), and nut. Make sure countersunk side of washer faces bolthead. Lubricate threads of bolt with solid film lubricant, Item 180, WP 1803 00. **TORQUE NUT TO 750 - 900 INCH-POUNDS.** Install cotter pin. Apply torque stripe (Figure 1, Detail B) (WP 0381 00 and TM 1-1520-345-23).

WARNING**CSI**

Verification of proper hardware installation, torque, and safety installation is a critical characteristic.

4. **QA** Install lower link in spherical bearing of rotating swashplate and fasten with nut and washer. Place shoulder of washer facing spherical bearing as shown (Figure 1). **TORQUE NUT TO 820 - 910 INCH-POUNDS.** Install cotter pin. Apply torque stripe (WP 0381 00 and TM 1-1520-345-23).
5. If same scissors are to be installed on same pressure plate, use same shim stackup. If existing shim stackup is insufficient to allow 0.002 to 0.004-inch overall gap on each side of each lug, inspect outer shoulder bearing height (WP 0599 00). **IF MINIMUM HEIGHT IS LESS THAN 0.086-INCH, REPLACE BEARING. IF MINIMUM HEIGHT IS 0.086-INCH OR GREATER, RESHIM TO 0.002 TO 0.004-INCH OVERALL GAP ON EACH SIDE OF LUG.**
6. If new scissors or pressure plate are being installed, do this:
 - a. Loosely connect upper link to pressure plate lug with bolt. Bolthead shall face direction of rotation.
 - b. Measure gap between upper link and each side of pressure plate lug with a feeler gage (Figure 1, Detail A).
 - c. **QA PEEL AND ADJUST LAMINATED SHIMS EQUALLY ON BOTH SIDES OF LINK AS REQUIRED TO ALLOW 0.002 TO 0.004-INCH OVERALL GAP ON EACH SIDE OF EACH LUG.**
7. Remove bolt.

NOTE

Install bolt with bolt head facing direction of rotation.

8. **QA** Assemble upper link to lower pressure plate lug with bolt, countersunk washer, shims, washers (as required), and nut. Make sure countersunk side of washer faces bolthead. Lubricate bolt threads with solid film lubricant, Item 180, WP 1803 00. **TORQUE NUT TO 1000 - 2000 INCH-POUNDS.** Install cotter pin (Figure 1).
9. Disengage gust lock.
10. Check area for cleanness and foreign material.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**BIFILAR**

This WP supersedes WP 0573 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Depth Micrometer, GGG-C-105
 Drive Tool, REA048D
 Driver Wrench, SR25W-A
 Extension, 3-inch, GGG-W-641
 Handle Socket Wrench, GGG-W-641
 Lockring Removal Tool, SR25R
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Socket, 13/16-inch, GGG-W-641
 Swage Tool, SR25S-A
 Torque Wrench, Dial, 0 - 75 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.
 Torque Wrench, 300 - 2500 in. lbs.
 Torque Wrench, 0 - 300 ft. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cloth, Abrasive, Item 75, WP 1803 00

Corrosion Preventive Compound, Item 106, WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Paper, Abrasive, Item 208, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 Assistant - (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
 WP 0381 00
 WP 0550 00
 WP 0574 00
 WP 1220 00
 WP 1590 00
 WP 1660 00
 WP 1691 00
 WP 1803 00

REPLACE BIFILAR**Remove****NOTE**

Bifilar weighs about 165 pounds.

1. If maintenance crane will be used, assemble and install crane (WP 1590 00).
2. Engage gust lock.
3. If helicopter is equipped with blade deice system, remove main rotor blade deice slipping assembly (WP 1220 00).
4. If installed, remove bolts, washers, and bifilar cover (Figure 1, Sheet 1).
5. If required, remove bifilar weights (WP 0574 00).
6. Install straps of hoisting sling, on support arms of bifilar.

REPLACE BIFILAR - Continued

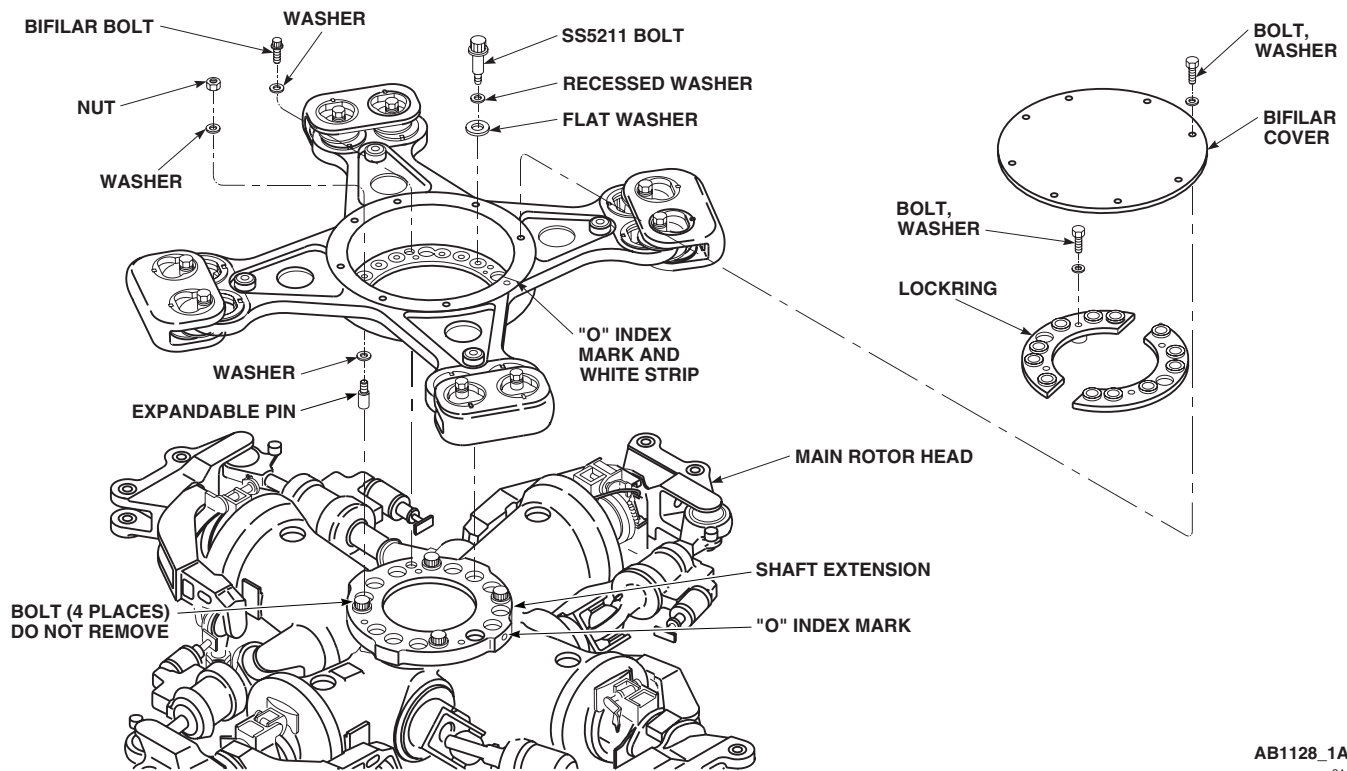
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Figure 1. Replace Bifilar. (Sheet 1 of 2)

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

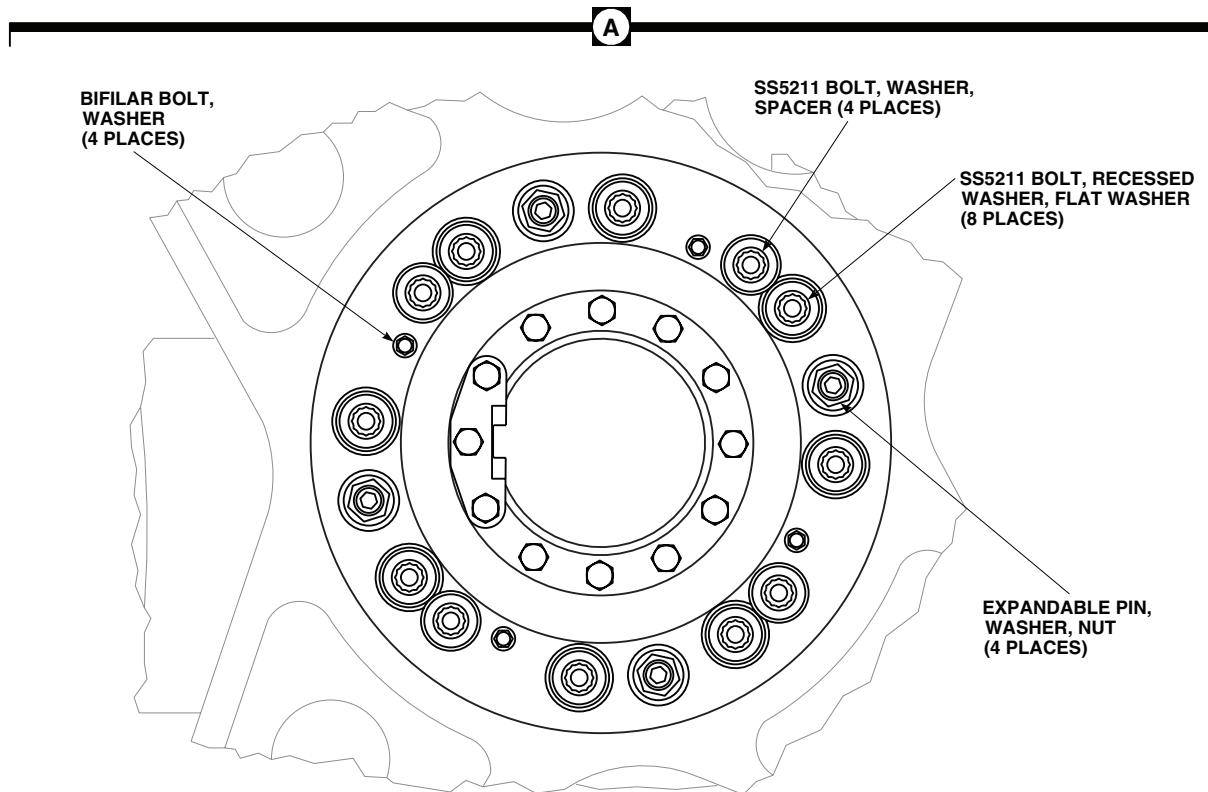
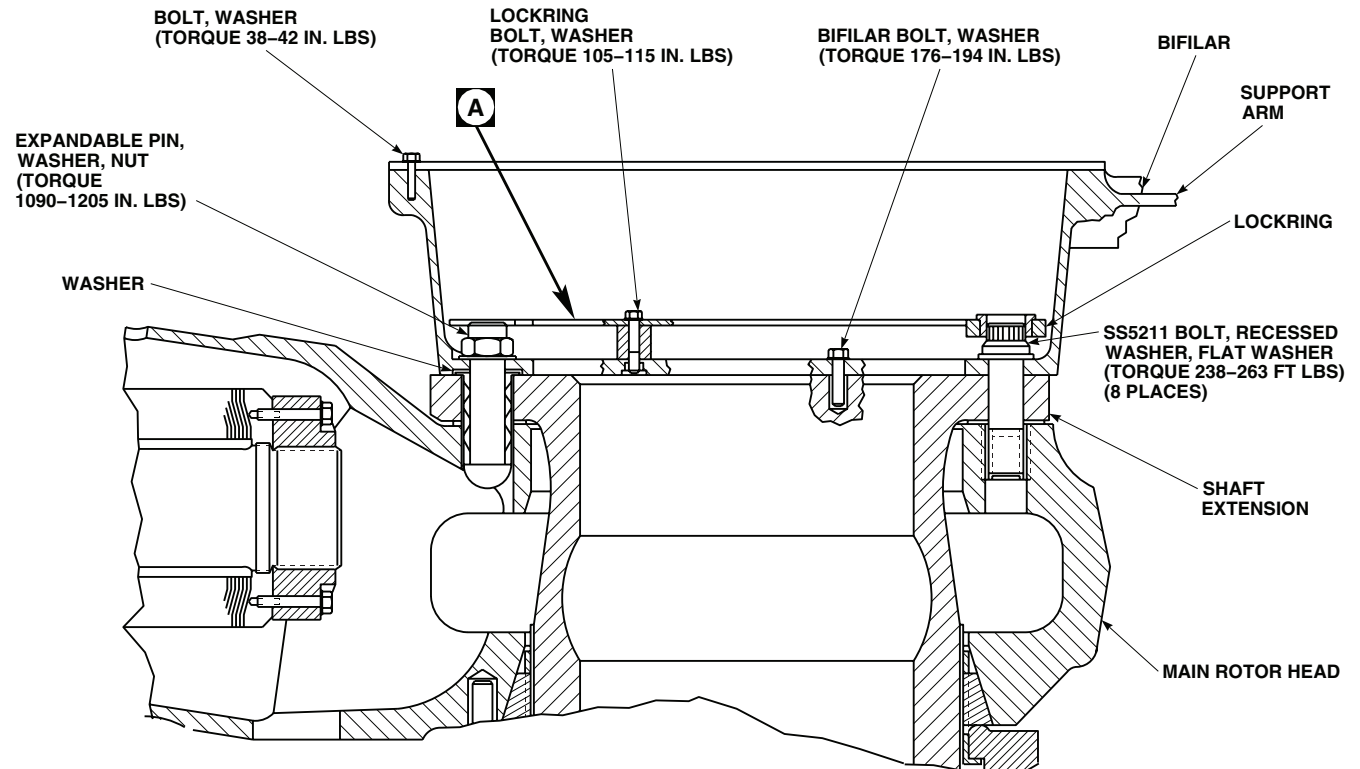
7. Attach hoisting sling to maintenance crane or to a hoist with enough capacity to lift bifilar. Adjust lengths of hoisting sling straps to evenly distribute weight when hoisting.
8. Remove bolts, washers, and lockring from bifilar (Figure 1, Sheet 1).
9. Loosen expandable pins, do not remove nuts. Use hex wrench to prevent pin from turning while loosening nut.
10. Remove four bifilar bolts and washers from bifilar.

WARNING

To avoid damage to equipment, and personnel injury, do not remove four bolts that have spacers under bolthead.

11. Remove SS5211-12-26 bolt, recessed washer and flat washer (8 places) attaching bifilar to main rotor head using handle socket wrench extension and 13/16-inch socket.

REPLACE BIFILAR - Continued



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Figure 1. Replace Bifilar. (Sheet 2 of 2)

REPLACE BIFILAR - Continued**WARNING**

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

12. Clean primer off bolts using machinery towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
13. Lift bifilar from main rotor head with expandable pins still in and place on suitable work area or in shipping container.

CAUTION

Damage to main rotor head and bifilar will result if metallic scrapers are used when cleaning rotor head components. Do not use metallic items to remove sealing compound.

14. Clean sealing compound from bifilar and rotor head using a nonmetallic scraper.
15. Remove expandable pins.
16. Remove hoisting sling from bifilar.
17. Inspect bolts attaching bifilar to main rotor head (WP 0550 00).

Inspect/Repair

1. Inspect bifilar cover/blade deice distributor (if installed) mating surface of bifilar for corrosion or pitting and repair as follows:
 - a. **REMOVE CORROSION PITTING UP TO 0.010-INCH DEEP USING ABRASIVE CLOTH, Item 75, WP 1803 00. UP TO 100% OF REPAIR AREA MAY BE BLENDED PROVIDED A MAXIMUM "H" DIMENSION OF 4.400-INCHES IS MAINTAINED (Figure 2, Detail A).**
 - b. **REMOVE CORROSION PITTING UP TO 0.020-INCH DEEP USING ABRASIVE PAPER, Item 208, WP 1803 00. MAINTAIN A 10:1 LENGTH TO DEPTH RATIO. UP TO 25% OF REPAIR AREA MAY BE BLENDED PROVIDED A MAXIMUM "H" DIMENSION OF 4.390-INCHES IS MAINTAINED (Figure 2, Detail A). A MAXIMUM REPAIR OF 0.015-INCH DEEP IS ALLOWED WITHIN 0.500-INCH RADIUS OF ATTACHMENT HOLES.**
 - c. **REMOVE CORROSION PITTING UP TO 0.030-INCH DEEP USING ABRASIVE PAPER, Item 208, WP 1803 00. MAINTAIN A 10:1 LENGTH TO DEPTH RATIO. UP TO 5% OF REPAIR AREA MAY BE BLENDED PROVIDED A MAXIMUM "H" DIMENSION OF 4.380-INCHES IS MAINTAINED (Figure 2, Detail A). A MAXIMUM REPAIR OF 0.015-INCH DEEP IS ALLOWED WITHIN 0.500-INCH RADIUS OF ATTACHMENT HOLES.**
 - d. **REPAIR STEP 1. A. MAY BE COMBINED WITH REPAIR STEP 1. B. OR STEP 1. C. UP TO THE MAXIMUM LIMITS FOR EACH TYPE OF REPAIR.**
 - e. Replace bifilar if repairs are more than limits.
2. Touch up repaired areas using corrosion resistant coating, Item 111, WP 1803 00.

Install

1. Install straps of hoisting sling on support arms of bifilar. Adjust length of hoisting sling straps to evenly distribute weight when hoisting.

REPLACE BIFILAR - Continued

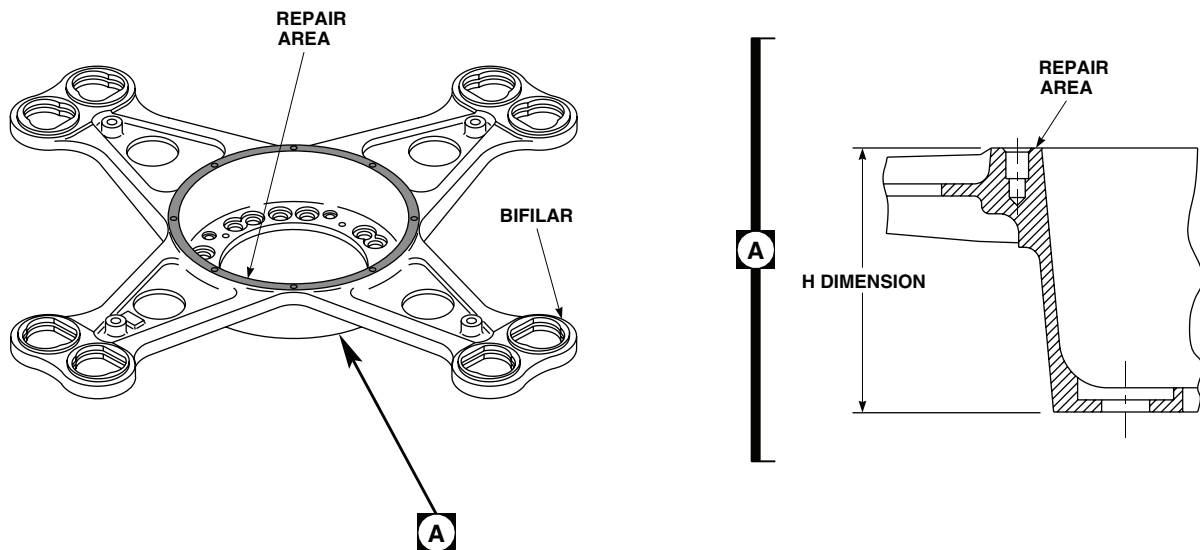
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Figure 2. Inspect/Repair Bifilar.

2. Connect hoisting sling to hoist.
3. Remove nuts from expandable pins and place washers that go between shaft extension and bifilar on pins. Insert top of expandable pin up through holes in bifilar. Install second washers on pins. Loosely install nuts (Figure 1, Sheet 1).
4. Hoist bifilar over main rotor head.
5. **QA** Line up "O" index mark (white stripe) on bifilar to "O" index mark on shaft extension. Lower bifilar and guide expandable pins into holes (Figure 1, Sheet 2).

NOTE

Do not use hex wrench to torque expandable pins.

6. **QA** Tighten expandable pins. **TORQUE NUTS TO 238 - 262 INCH-POUNDS.** Use hex wrench to stop pin from turning.
7. Coat shank, threads, and area under heads of SS5211-12-26 bolts with primer coating, Item 243, WP 1803 00.
8. **QA** Install SS5211-12-26 bolts, recessed washers, and flat washers (8 places) holding bifilar to shaft extension while primer is still wet (Figure 1, Sheet 2). Install bolts with extension and socket. **TORQUE BOLTS TO 950 - 1050 INCH-POUNDS. STABILIZE TORQUE AT 950 - 1050 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**
9. Break torque on expandable pins.

REPLACE BIFILAR - Continued

10. **QA TORQUE EIGHT SS5211-12-26 BOLTS HOLDING BIFILAR TO SHAFT TO 238 - 263 FOOT-POUNDS. STABILIZE TORQUE AT 238 - 263 FOOT-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**
11. Wipe off any primer that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.
12. **QA TORQUE NUTS ON EXPANDABLE PINS TO 1093 - 1207 INCH-POUNDS. STABILIZE TORQUE AT 1090 - 1205 INCH-POUNDS BY TURNING EACH BOLT IN TIGHTENING DIRECTION UNTIL NO MOTION IS FELT.**
13. Before torquing bifilar bolts, check threaded inserts that these bolts go into, as follows:
 - a. **QA** Thread in bifilar bolts until head of bolts are 0.030-inch away from surface of bifilar.
 - b. **QA** Using dial type torque wrench, unthread each bolt individually. **MINIMUM BREAKAWAY TORQUE IS 6.5 INCH-POUNDS. MAXIMUM REMOVAL TORQUE IS 60 INCH-POUNDS.**
14. If breakaway or removal torque is not within limits, do this:
 - a. Inspect bolts for evidence of corrosion or wear. **NONE ALLOWED.** Replace bolts if corrosion or wear is found.

WARNING

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- b. Inspect insert for cleanness. Clean insert using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
 - c. Repeat step 13. If breakaway or removal torque is still not within limits, replace insert (WP 0550 00).
15. Coat shank, threads, and area under head of bifilar bolts with primer coating, Item 243, WP 1803 00.
16. **QA** Install bifilar bolts and washers while primer is still wet (Figure 1, Sheet 2). **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**
17. Wipe off any primer that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.
18. Seal around all boltheads with sealing compound, Item 273, WP 1803 00.
19. Coat shank, threads, and area under head of locking bolts with primer coating, Item 243, WP 1803 00.
20. **QA** Install locking over bolts while primer is still wet.

CAUTION

Damage to locking will result if it is not seated properly prior to torquing holddown bolts. Make sure locking is properly seated over boltheads. If necessary, turn bushings to allow locking to seat over boltheads. Lockring must sit flat before torquing hold-down bolts.

21. **QA TORQUE LOCKRING HOLDDOWN BOLTS TO 105 - 115 INCH-POUNDS.**
22. Wipe off any primer that has seeped from under boltheads using machinery wiping towel, Item 344, WP 1803 00.
23. Touch up paint on boltheads as necessary (WP 0381 00 and TM 55-1500-345-23).
24. Seal areas around boltheads with sealing compound, Item 273, WP 1803 00.

REPLACE BIFILAR - Continued

25. Spray corrosion preventive compound, Item 106, WP 1803 00, inside shaft extension and on main rotor shaft nut before installing bifilar cover or main rotor blade deice slipping assembly (if installed).
26. If helicopter is equipped with blade deice system, install main rotor blade deice slipping assembly (WP 1220 00).
27. If bifilar cover is installed, coat shank, threads, and area under head of bifilar cover bolts with primer coating, Item 243, WP 1803 00.
28. Check area for cleanness and if required, install cover, gasket side down on top of bifilar. Install bolts and washers while primer is still wet (Figure 1, Sheet 2). **TORQUE BOLTS TO 38 - 42 INCH-POUNDS.**
29. Wipe off any primer that has seeped from under boltheads.
30. Touch up paint on boltheads as necessary (WP 0381 00 and TM 55-1500-345-23).
31. Seal areas around bifilar boltheads, and gap between bifilar and shaft extension with sealing compound, Item 273, WP 1803 00.
32. If required, install bifilar weights (WP 0574 00).
33. Lubricate bifilar (WP 1660 00).
34. Check area for cleanness and foreign material.
35. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure bifilar to shaft extension bolts and expandable pins torque check is due 9 to 11 flight hours after installation (WP 1691 00).

REPLACE INSERT**Remove**

1. Drill out insert lockring to 0.082-inch using removal tool (Figure 3).
2. Remove remainder of insert with drive tool.
3. Clear insert hole and threads of any foreign material and chips.

Install

1. **QA** Install insert in hole with driver wrench, until top of insert is 0.015 to 0.025-inch below surface (Figure 3).
2. **QA** Swage insert with swage tool, until shoulder of tool touches surface of bifilar.

REPLACE CLINCH NUT**Remove**

Remove clinch nut from bifilar using clinch nut drive tool (Figure 4).

Install

1. Apply epoxy primer coating kit, Item 120, WP 1803 00, to mating surface of clinch nut and bore of bifilar.
2. **QA** Install clinch nut into bifilar while primer is still wet using drive tool.

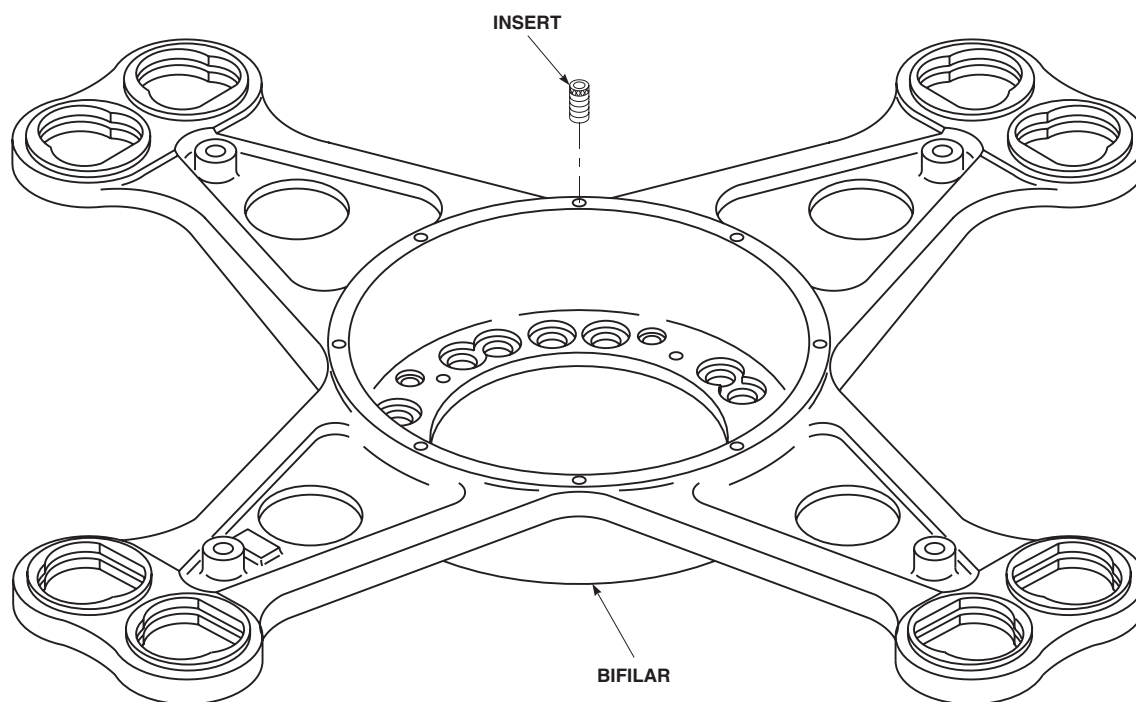
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Figure 3. Replace Insert.

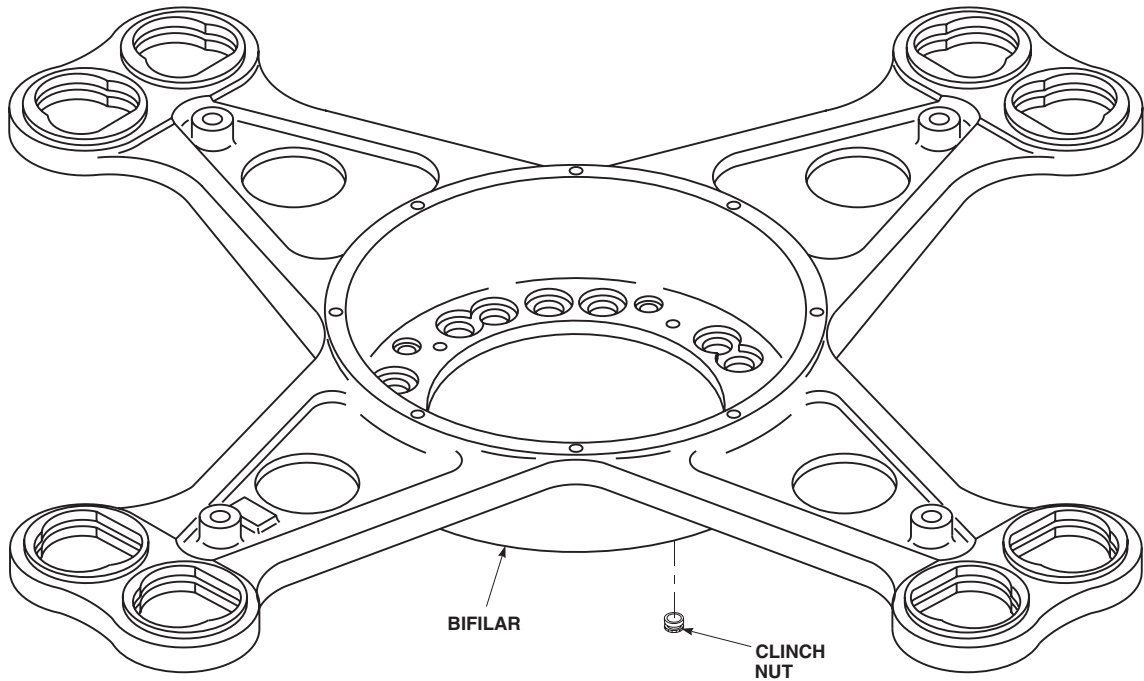
REPLACE CLINCH NUT - ContinuedAA2160
SA

Figure 4. Replace Clinch Nut.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****BIFILAR WEIGHT**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 300 - 2500 in. lbs.

References

WP 0543 00
WP 1660 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn main rotor head so bifilar weight is easy to reach.
2. Engage gust lock.
3. Remove bolts, washers, tapered washers, pins, nuts, and weight from bifilar (Figure 1).

INSPECT/REPAIR

Inspect/repair bifilar weights (WP 0543 00).

INSTALL

1. **QA** Place weight, tapered washers and pins on support. Install bolt, countersunk washer, washers, and nut. Make sure countersunk side of washer faces bolthead (Figure 1). **TORQUE NUT TO 1333 - 1367 INCH-POUNDS.**
2. Lubricate bifilar (WP 1660 00).
3. Check area for cleanness and foreign material.

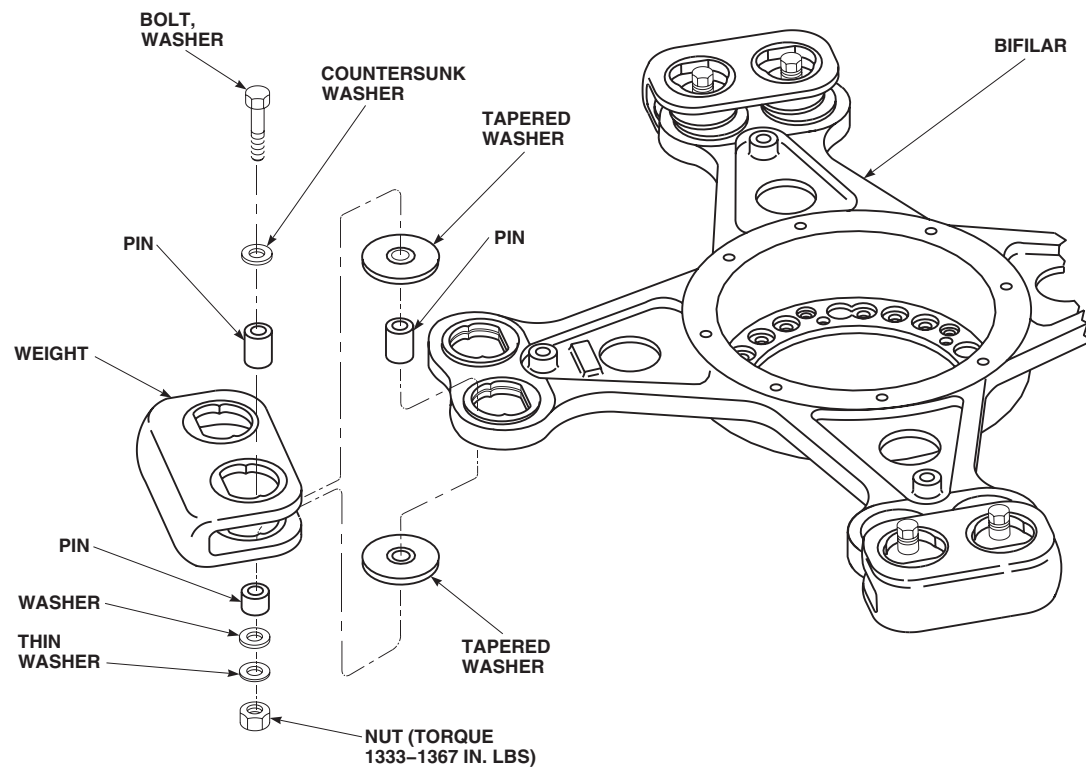
INSTALL - ContinuedAK1370
SA

Figure 1. Replace Bifilar Weight.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****BIFILAR COVER GASKET**

This WP supersedes WP 0575 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Cloth, Cheesecloth, Item 85, WP 1803 00

Paper, Abrasive, Item 211, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00

References

WP 1803 00

REMOVE

1. Remove old gasket with nonmetallic scraper (Figure 1).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Remove remaining residue from cover with cheesecloth cloth, Item 85, WP 1803 00, soaked in technical acetone, Item 5, WP 1803 00.

INSTALL

1. Thoroughly scuff up surface of gasket and cover to be mated with abrasive paper, Item 211, WP 1803 00 (Figure 1).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean area with technical acetone, Item 5, WP 1803 00.
3. Apply adhesive, Item 14, WP 1803 00, to mating surface of cover and gasket.

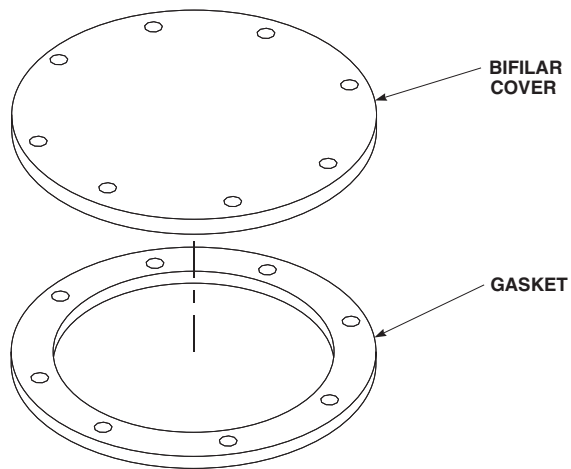
INSTALL - ContinuedAB2685
SA

Figure 1. Replace Bifilar Cover Gasket.

NOTE

Allow adhesive to air-dry for 10 minutes until most of the solvent has evaporated and adhesive is very tacky.

4. **QA** Install gasket on cover.
5. Lay cover assembly on a flat surface with gasket side down.
6. Press down firmly on cover assembly to make sure that cover and gasket mate.
7. Allow adhesive to cure for 24 hours.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****MAIN ROTOR BLADE**

This WP supersedes WP 0576 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Adapter, 70700-20426-041
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Blade Alignment Pin
Blade Clamp Assembly, 70700-20324-047
Blade Tieddown Ropes (2)
C-Clamp, GGG-C-406
Cutter, GGG-C-771
DC Ammeter
DC Voltmeter
Digital Low-Resistance Ohmmeter, Model, DLRO 247000
Drill Bit, No. 40
Driving Tool, R3605-D
Gloves
Goggles/Face Shield
Flexible Metal Spatula (or equivalent)
Heater Mat Resistance Test Set, Locally Made, TGN55-01052-041, Figure 261, WP 1805 00
Hoist (500 lb minimum capacity)
Main Rotor Blade Solid Pin Removal Tool, 70700-25450-041, Locally-Made, Figure 274, WP 1805 00
Nonmetallic Scraper
Pencil, Soft
Plastic Straight Edge
Plastic Tool
Pressure/Heat Pack
Rotary Repair Kit, 70072-15001-011
Screw Extractor Set (or equivalent), GGG-E-936
Shot Bags (weights or equivalent)
Spring Scale, AAA-S-133
Squeegee
Tap, SRW19T
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Trailing Edge Repair Kit, 70072-15001-021
Wire Brush, HB178

Wood Strips
Wrench, R1105-W

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive, Item 13, WP 1803 00
Alcohol, Denatured, Item 48, WP 1803 00
Applicator, Disposable, Item 53, WP 1803 00
Brush, Paint, Item 61, WP 1803 00
Cloth, Abrasive, Item 75, WP 1803 00
Cloth, Abrasive, Item 78, WP 1803 00
Cloth, Abrasive, Item 80, WP 1803 00
Cloth, Abrasive, Item 81, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Abrasive, Item 84, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Cloth, Satin, Item 90, WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00
Epoxy Coating Kit, Item 118, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Heavy Paper (or equivalent protective covering)
Lacquer, Item 171, WP 1803 00
Padding,
Paper, Abrasive, Item 208, WP 1803 00
Paper, Abrasive, Item 211, WP 1803 00
Polyurethane Coating, Item 227, WP 1803 00
Polyurethane Coating, Item 233, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Primer, Pressure, Item 374, WP 1803 00
Protective Caps and Plugs,
Repair Kit, Rotary B, Item 253, WP 1803 00
Sealing Compound, Item 258, WP 1803 00
Sealing Compound, Item 265, WP 1803 00
Shot Bags (weights or equivalent)
Stirring Stick, Beverage, Item 300, WP 1803 00
Strap, Tiedown, Electrical, Item 301, WP 1803 00
Strap, Tiedown, Electrical, Item 306, WP 1803 00

Tape, Pressure Sensitive, Adhesive, Item 325, WP 1803 00	TM 55-1500-345-23 WP 0145 00
Tape, Pressure Sensitive, Adhesive, Item 333, WP 1803 00	WP 0381 00 WP 0542 00
Towel, Machinery Wiping, Item 344, WP 1803 00	WP 0544 00
Wire, Nonelectrical, Item 357, WP 1803 00	WP 0553 00 WP 0554 00
Personnel Required	WP 1220 00
Aircraft Structural Repairer MOS 15G (1)	WP 1590 00
Assistant - (3)	WP 1624 00
UH-60 Helicopter Repairer MOS 15T (4)	WP 1634 00 WP 1690 00
References	WP 1803 00 WP 1805 00
TM 1-1500-204-23	
TM 1-1520-237-MTF	
TM 1-6625-724-13&P	

REMOVE

1. If maintenance crane will be used, assemble and erect crane (WP 1590 00).
2. Turn off all electrical and hydraulic power.
3. Disengage gust lock.

NOTE

It is recommended that blade being removed be positioned left or right of helicopter centerline.

4. Manually position blade to be removed (Figure 1).
5. Engage gust lock.

NOTE

Blade pretrack number will be used to adjust the pitch control rod during installation of replacement blade.

6. Note and record blade's pretrack number and location.
7. If blade deice system is installed, disconnect blade deice electrical connector (WP 1220 00).
8. Disconnect antifiap spring (WP 0554 00)

CAUTION

To prevent damage to antifiap cam and antifiap stop attaching bolts, antifiap cam must be secured whenever main rotor blades are removed.

9. Rotate antifiap cam down and secure cam bracket using electrical tiedown strap, Item 301, WP 1803 00

REMOVE - Continued**WARNING**

Blade clamp may open during removal causing injury to personnel and damage to equipment. If maintenance sling is used, one leg of maintenance sling should be fastened loosely around blade clamp in case blade clamp opens. Install safety strap over blade clamp before removing blade.

10. Place blade clamp assembly between lifting stripes and lock to blade (Figure 1, Detail A). Install safety strap.
11. Attach guide lines to blade cuff and blade tip. Position assistants at blade tip and cuff to help guide blade.

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

CAUTION

While working on main rotor blade, be careful not to bend or damage blade trim tab. Bending trim tab will cause blade out of track condition.

NOTE

To prevent undue binding of hoisting hook and blade clamp hoisting ring, hook must fit ring with sufficient clearance.

12. Attach blade clamp to maintenance crane or hoist. Use maintenance sling, to attach blade clamp assembly to any hoist other than the maintenance crane assembly.

WARNING**CSI**

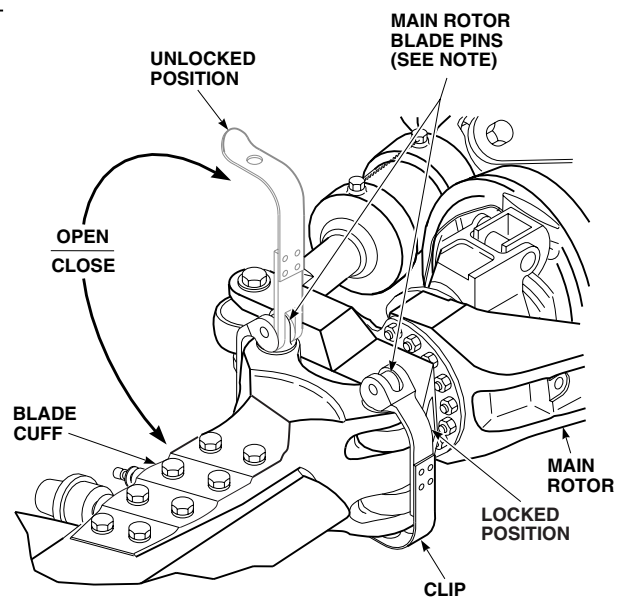
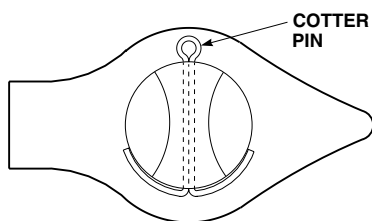
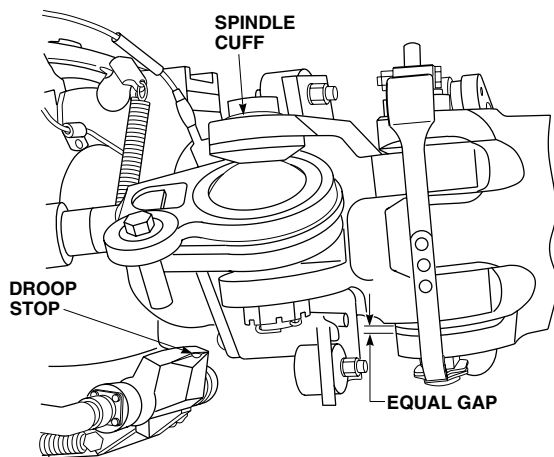
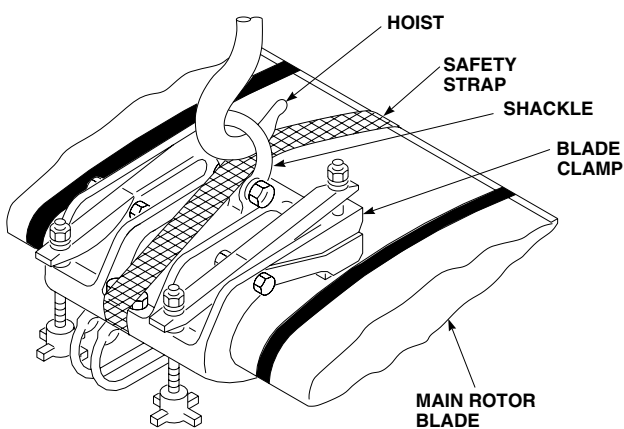
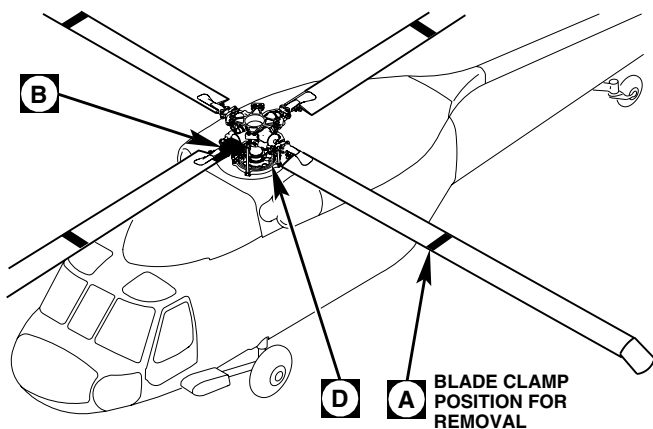
The outside lug surface of the blade cuffs are critical surfaces which must not be damaged during and after blade removal. Use extreme caution when removing and provide protective covering after removal.

NOTE

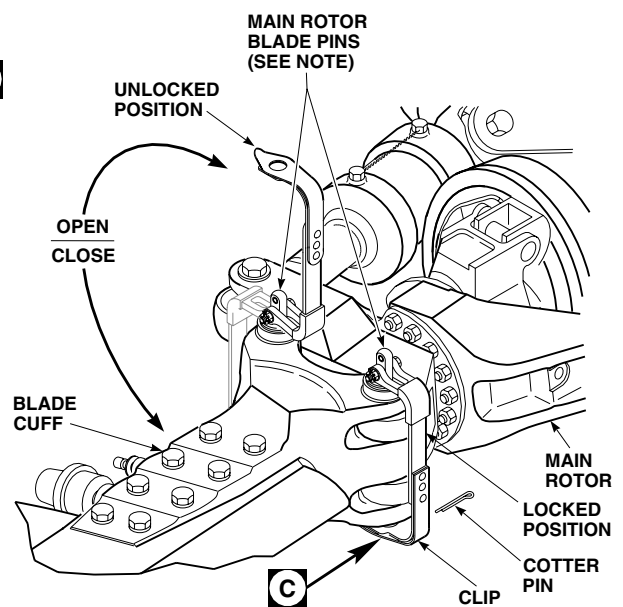
Main Rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102, and main rotor blade solid pins, 70103-08108-041. In the following procedures both types of pins are referred to as main rotor blade pins, except where noted.

13. Relieve load on main rotor blade pins by lifting blade to obtain an equal gap between blade cuff washers and spindle cuff washers (Figure 1).

REMOVE - Continued

**EFFECTIVITY**

HELICOPTERS WITH 70103-08107-102
MAIN ROTOR BLADE EXPANDABLE PINS.

**EFFECTIVITY**

HELICOPTERS WITH 70103-08108-041
MAIN ROTOR BLADE SOLID PINS.

NOTE

USE TWO EXPANDABLE PINS ON ANY BLADE OR TWO SOLID PINS ON ANY BLADE AS LONG AS BOTH PINS ON ANY ONE BLADE ARE THE SAME. DO NOT INTERMIX EXPANDABLE AND SOLID MAIN ROTOR BLADE PINS ON THE SAME BLADE.

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Figure 1. Replace Main Rotor Blade.

REMOVE - Continued**CAUTION**

Damage to main rotor blade expandable pin, 70103-08107-102, and cuff will result if pin is removed before lower ring is fully compressed. Pin will jam cuff. Make sure lower ring is fully compressed before attempting to remove main rotor blade expandable pin.

14. Unlock and open main rotor blade pin handles as follows:

- a. On helicopters with main rotor blade expandable pins, 70103-08107-102, unlock and fully open handles on both expandable pins. Check lower ring to see if it is fully compressed by rotating it around pin shaft. If lower ring is not compressed, or shows evidence of dirt or contaminants, clean out lower ring before removing (WP 1634 00).
- b. On helicopters with main rotor blade solid pins, 70103-08108-041, remove cotter pin, unlock and fully open handles on both pins.

WARNING**CSI**

- Main rotor blade pins contain critical surfaces which must not be damaged during and after removal of main rotor blade. Use extreme caution when removing and provide protective covering for pins after removal, if not immediately reinstalled in spindle.
- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.

NOTE

- If bonded washers are loose, rebond washers (WP 0553 00).
- On helicopters with 70103-08107-102 main rotor blade expandable pins, make sure expandable pins are installed in the same hole from which they were removed. Note and record pin hole and blade location of all expandable pins removed.

15. Remove both pins from blade cuff and main rotor sleeve as follows:

- a. With blade lifted to relieve pressure on main rotor blade pin, and while applying upward force on main rotor blade pin, gently lead and lag main rotor blade until main rotor blade pin begins to move up.
- b. Stop lead and lag of main rotor blade once up and down movement of main rotor blade pin is achieved.
- c. While upward force is applied on main rotor blade pin, lower or raise main rotor blade as necessary until pressure is relieved and main rotor blade pin can be removed.
- d. Repeat steps 15. a. through 15. c. until main rotor blade pin can be removed.
- e. On helicopters using main rotor blade solid pins, 70103-08108-041, if main rotor blade solid pin cannot be removed, Refer to REMOVE - DAMAGED/STUCK MAIN ROTOR BLADE SOLID PIN, in this WP.

REMOVE - Continued**NOTE**

On helicopters using 70103-08107-102 main rotor blade expandable pins, if main rotor blade expandable pin has to be disassembled to remove, pin replacement is required.

16. If 701103-08107-102 main rotor blade expandable pin still cannot be removed, loosen the expandable pin nut without completely removing it and repeat step 15. a through 15. c. If pin had to be disassembled to remove pin, replace pin.
17. Remove blade.
18. Lower blade and place on adapter, or padding material.
19. Remove blade clamp and maintenance sling, if used, from main rotor blade.
20. Remove guidelines if installed.

CAUTION

70103-08107-102 main rotor blade expandable pins must be installed in main rotor spindle in same holes from which they were removed.

21. Install main rotor pins in main rotor spindle.
22. If more than one blade is to be removed, repeat entire procedure for remaining blades. If more than one blade is to be removed, repeat entire procedure for remaining blades.

REMOVE DAMAGED/STUCK MAIN ROTOR BLADE SOLID PIN

1. Release load on main rotor blade solid pin using overhead crane and blade clamp.
2. Unlock spring clip from main rotor blade solid pin and unlock handle.
3. Remove main rotor blade solid pin using locally-made solid pin removal tool, 70700-25450-041, (Figure 2) as follows:
 - a. Locally made main rotor blade solid pin removal tool (WP 1805 00).
 - b. Extend rod by backing off nut. Extend rod to bottom of removal tool.
 - d. Remove T-pin from rod.
 - e. Place tool over main rotor blade solid pin. Align hole in top lug of main rotor blade pin with slot in tool. Insert T-pin fully through slot and main rotor blade pin.
 - f. Handtighten nut until bottom of tool sits flush on blade cuff.
 - g. Support tool by the handle and using a 7/8-inch closed end box wrench, extract main rotor blade solid pin from lugs of blade and blade cuff.
 - h. Disengage T-pin from main rotor blade solid pin.

REMOVE DAMAGED/STUCK MAIN ROTOR BLADE SOLID PIN - Continued

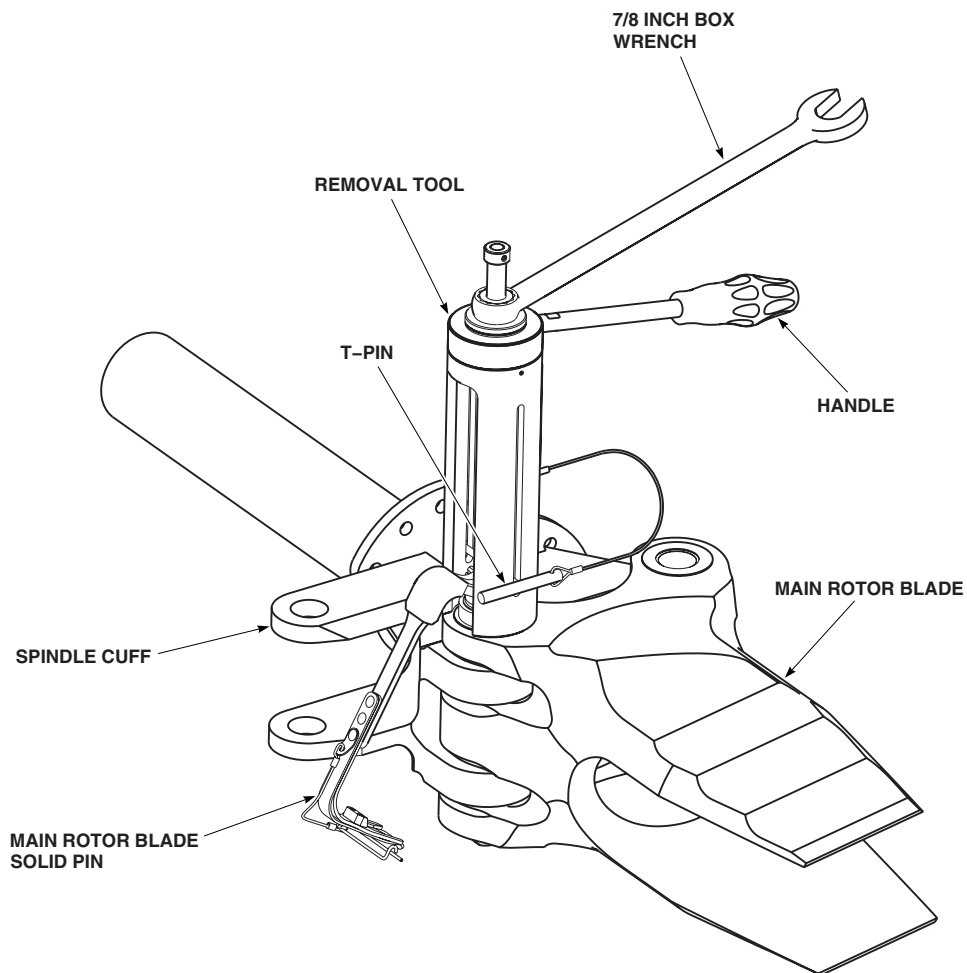
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Figure 2. Remove Damaged/Stuck Main Rotor Blade Solid Pin.

DEICE HEATER MAT FAILURE TEST**NOTE**

- The procedures contained in this deice heater mat failure test are intended for checkout of main rotor blades, 70150-09100, that have been removed from aircraft for suspected heater mat resistance faults.
- Main rotor blades that are installed on aircraft can be checked with a digital low resistance ohmmeter in accordance with WP 0145 00. For resistance checks in WP 0145 00 to be accurate, power must have been applied to main rotor blade under test during system operational check a maximum of one (1) hour prior to taking resistance readings.
- Resistance test for main rotor blade deice heater mats using heater mat resistance test set, TGN55-01052-041. This procedure requires use of an external dc voltmeter and a dc ammeter.
- Test points TP1, TP2, and TP3 on the heater mat resistance test box are not used during the deice heater mat failure test.

1. Locally make heater mat resistance test set (WP 1805 00).
2. Make sure the controls on the heater mat resistance test set, TGN55-01052-041, are in the following positions:
 - a. CB1 POWER SUPPLY circuit breaker - Disengaged.
 - b. CB2 115 vac circuit breaker - Disengaged.
 - c. HEATING ELEMENTS (S1) switch - Set to 1-2 position.
 - d. S2 switch - Open.
 - e. S3 switch - Open.
 - f. POWER SUPPLY ADJUSTMENT - Fully counterclockwise position.
3. Connect heater mat resistance test set, TGN55-01052-041, to 115 vac 60 Hz electrical power source.
4. Connect electrical connector, M83723/86W2212N, on heater mat resistance test set, TGN55-01052-041, to main rotor blade to be tested.
5. Connect dc voltmeter, set to the 20 vdc range, to VOLTMETER jacks TP4 (+) and TP5 (-) on heater mat resistance test set, TGN55-01052-041.
6. Connect dc ammeter, set to the 2 amp dc range, to AMMETER jacks TP6 (-) and TP7 (+) on heater mat resistance test set, TGN55-01052-041.
7. Engage CB2 115 vac circuit breaker on heater mat resistance test set, TGN55-01052-041. Verify that the ac POWER ON IND. lamp illuminates.
8. Engage CB1 POWER SUPPLY circuit breaker on heater mat resistance test set, TGN55-01052-041. Verify that dc voltmeter displays between 11.9 to 12.1 vdc reading. Allow heater mat resistance test set, TGN55-01052-041, to warm-up for at least 15 minutes before continuing.
9. Set S2 and S3 switches to the CLOSE position on heater mat resistance test set, TGN55-01052-041.

DEICE HEATER MAT FAILURE TEST - Continued**CAUTION**

If less than 6.0 vdc reading is displayed on the dc voltmeter, do not proceed further, to prevent possible damage to the main rotor blade and/or heater mat resistance test set, TGN55-01052-041.

10. Verify that the dc voltmeter displays greater than 6.0 vdc reading. If less than 6.0 vdc reading is displayed on the dc voltmeter, do not proceed further, to prevent possible damage to the main rotor blade and/or heater mat resistance test set, TGN55-01052-041.
11. Verify that the HEATING ELEMENTS (S1) switch is set in the 1-2 position.
12. Adjust POWER SUPPLY ADJUSTMENT on heater mat resistance test set, TGN55-01052-041, slowly clockwise until 1.0 AMP dc reading is displayed on dc ammeter.
13. The dc voltmeter will now correspond directly to a resistance reading. Verify that the dc voltmeter displays a reading between 7 vdc to 10 vdc.
14. Return POWER SUPPLY ADJUSTMENT on heater mat resistance test set, TGN55-01052-041, fully to the counterclockwise position.
15. Return S2 and S3 switches to the OPEN position on heater mat resistance test set, TGN55-01052-041.
16. Repeat step 8. through step 14. with the HEATING ELEMENTS (S1) switch in each of the following positions:
 - a. 1-3
 - b. 2-3
 - c. 4-5
 - d. 4-6
 - e. 5-6
 - f. 7-8
 - g. 7-9
 - h. 8-9
 - i. 10-11
 - j. 10-12
 - k. 11-12
17. Disengage CB2 115 vac circuit breaker on heater mat resistance test set, TGN55-01052-041. Verify that the ac POWER ON IND. lamp is not illuminated.
18. Disengage CB1 POWER SUPPLY circuit breaker on heater mat resistance test set, TGN55-01052-041.
19. Disconnect dc voltmeter, from VOLTMETER jacks TP4 (+) and TP5 (-) on heater mat resistance test set, TGN55-01052-041.
20. Disconnect dc ammeter from AMMETER jacks TP6 (-) and TP7 (+) on heater mat resistance test set, TGN55-01052-041.
21. Disconnect heater mat resistance test set, TGN55-01052-041, from main rotor blade.

INSPECT**NOTE**

Inspect main rotor to determine if blade has six to eight lug cuffs. Retirement time is different for these two cuffs.

1. Inspect main rotor blade cuff (WP 0544 00).
2. Inspect main rotor blade abrasion strip (WP 0544 00).
3. Inspect main rotor blade skin (WP 0544 00).
4. Clean main rotor blade pins (WP 1634 00).
5. Inspect main rotor blade pins (WP 0542 00).

INSTALL**NOTE**

- Main rotor blades may be secured by two different kinds of pins, main rotor blade expandable pins, 70103-08107-102, and main rotor blade solid pins, 70103-08108-041. In the following procedure both types of pins are referred to as main rotor blade pins, except where noted.
- Blades with erosion protection kits installed must not be mixed with blades that do not have erosion protection kits installed. Main rotor head balance and vibration levels will be adversely effected.

1. Test operation of pressure indicator (WP 1624 00).
2. When installing a new, overhauled, or stored blade from the supply system, check and service blade (WP 1624 00).

NOTE

It is recommended that rotor head be positioned to install blade left or right of helicopter center line.

3. Manually position rotor head to accept blade to be installed and engage gust lock.
4. Check trim tab for possible damage. Also, do not fold, bend or adjust trim tab in any way.

WARNING

- Injury to personnel and damage to equipment will result if blade clamp opens during blade installation. Make sure safety strap is installed over blade clamp and through shackle prior to installing blade.
 - Make sure clamp is positioned within lifting stripes on blade. Blade will be out of balance and swing erratically when lifted if blade clamp is not positioned correctly.
5. Place blade clamp assembly, 70700-20324-047, between lifting stripes on main rotor blade. Install safety strap over blade clamp and through shackle (Figure 1, Detail A).
 6. Lock blade clamp.

INSTALL - Continued**WARNING****CSI**

Main rotor blade pins contain critical surfaces which must not be damaged during blade removal. Use extreme caution when removing and provide protective covering, if not immediately reinstalled in spindle.

NOTE

70103-08107-102, main rotor blade expandable pins must be installed in main rotor spindle in same holes from which they are removed.

7. Remove main rotor blade pins from main rotor spindle.
8. Make sure that leading edge of blade faces direction of rotation. Attach hoist or maintenance crane to blade clamp. Attach guide lines to blade cuff and blade tip. Position assistants at blade cuff and blade tip to help guide blade.
9. Hoist blade into place.
10. Line up blade cuff with main rotor spindle.

WARNING**CSI**

- The outside surface of blade cuffs are critical surfaces which must not be damaged during blade installation. Extreme care shall be used when installing main rotor blades.
- Do not intermix expandable pins, with solid pins on the same blade. Intermixing expandable and solid main rotor blade pins on the same blade may cause fatigue damage to spindle, cuff, and blade attachment pins. Failure to comply could result in injury to personnel or damage to equipment.
- Each blade must be secured by two of the same type of main rotor blade pin. (Example: Use two main rotor blade expandable pins, 70103-08107-102 OR two main rotor blade solid pins, 70103-08108-041, on any one blade.) Failure to comply could result in injury to personnel or damage to equipment.
- It is permissible to use expandable pins on any blade or solid pins on any blade as long as both pins on any one blade are the same. Failure to comply could result in injury to personnel or damage to equipment.

CAUTION

On helicopters with 70103-08107-102 main rotor expandable pins, make sure expandable pins are installed in the same hole noted and recorded during remove procedure.

NOTE

In the following procedures both types of pins are referred to as main rotor blade pins, except where noted.

11. **QA** Insert main rotor blade pins through blade cuff lugs and main rotor spindle. Align pin handle with decal on cuff. Do not hit or force main rotor blade pins into position. Gently lead/lag blade to obtain proper alignment for main rotor blade pin to properly seat.

INSTALL - Continued**NOTE**

Make sure blade tension is off main rotor blade expandable pins during tension check.
Have blades holders position blade by lowering or raising blade as necessary to relieve tension.

12. If installing 70103-08107-102 main rotor blade expandable pins, perform expandable pin tension check (WP 0542 00).
13. **QA** Close and lock main rotor blade pin handles as follows:
 - a. On helicopters with main rotor blade expandable pins, 70103-08107-102, insert pin fully through lugs on main rotor spindle and blade cuff. Close handle, making sure spring handle of pin snaps over fully engages hex on locknut. Visually check for expansion of locking on end of pin. Repeat for other main rotor blade expandable pin (Figure 1, Detail B).
 - b. On helicopters with main rotor blade solid pins, 70103-08108-041, insert pin fully through lugs on main rotor spindle and blade cuff. Close handle, making sure retaining clip fork engages groove in pin and that bottom of pin protrudes through hole in spring clip. Install cotter pin from side of pin closest to main rotor hub (Figure 1, Detail B and C). Repeat for other main rotor blade solid pin.
14. **QA** Cut and remove tiedown strap holding antifiap cam ito bracket.
15. Lower blade. Remove safety strap and unlock and remove blade clamp from blade.
16. If blade deice system is installed, connect blade deice electrical connector (WP 1220 00).
17. Connect antifiap spring (WP 0554 00).
18. **QA** Perfrom an adjustment of main rotor blade, Refer to ADJUST in this work package.

ADJUST**NOTE**

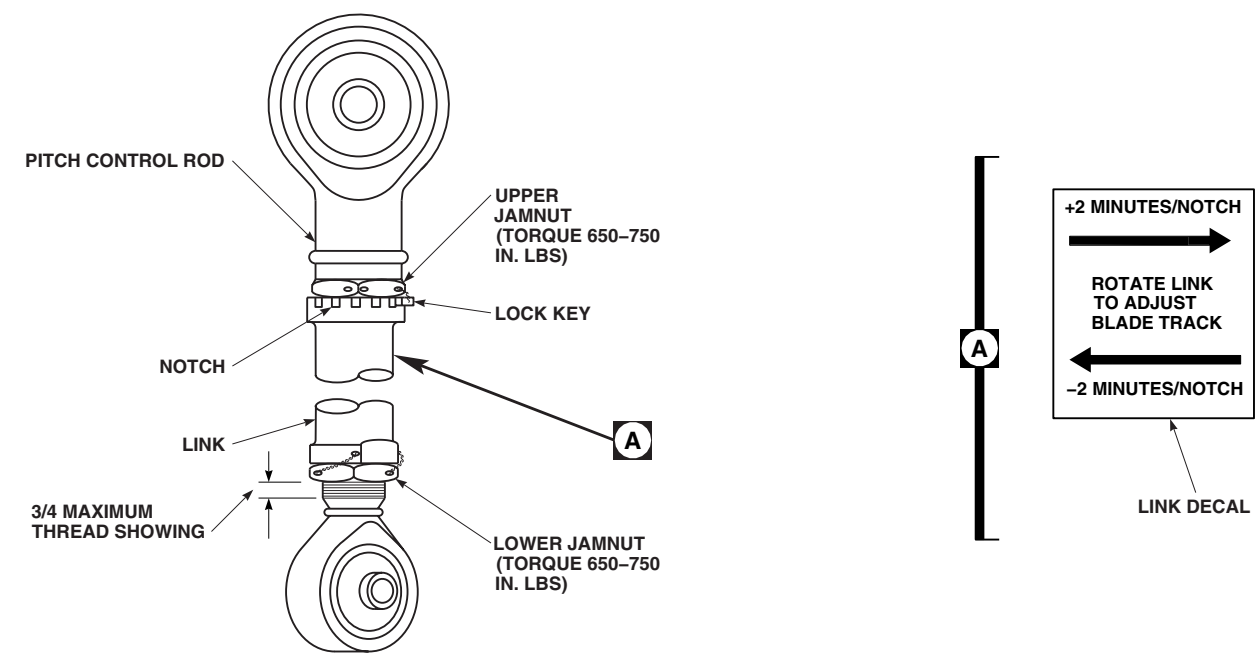
A one notch change to a pitch control rod will affect track by approximately 1/4-inch.

1. Adjust pitch control rod as follows:
 - a. Compare pretrack number of removed (old) blade with number on blade now installed.

NOTE

- Pretrack numbers are stenciled on bottom of blade. Adjustment is not necessary if numbers are alike.
 - Each notch on pitch control rod equals 2 minutes.
- b. Subtract pretrack number of installed (new) blade from number of removed (old) blade.
 - c. If difference between removed (old) blade and installed (new) blade is a plus (+) number, loosen locknuts on rod and turn in -2 MINUTES/NOTCH direction (Figure 2, Detail A) (Table 1 and Table 2).
 - d. If difference between old blade and new blade is a minus (-) number, loosen locknuts on rod and turn in +2 MINUTES/NOTCH direction (Figure 2, Detail A) (Table 1 and Table 2).

ADJUST - Continued



NOTE
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 3. Adjust.

Table 1. Examples.

EXAMPLES	TIME	REQUISITE
Example 1.	+8 MINUTES	removed (old) blade
subtract	+4 MINUTES	installed (new) blade
Total	+4 MINUTES	
Example 2.	+8 MINUTES	removed (old) blade
subtract	-4 MINUTES	installed (new) blade
Total	+12 MINUTES	
Example 3.	-2 MINUTES	removed (old) blade
subtract	+10 MINUTES	installed (new) blade

ADJUST - Continued**Table 1. Examples. - Continued**

EXAMPLES	TIME	REQUISITE
Total	-12 MINUTES	
NOTES		
1. Example 1 shows old blade at +8 and new blade at +4; therefore adjust control rod 2 notches in -2 MINUTES/NOTCH direction.		
2. Example 2 shows old blade at +8 and new blade at -4; therefore adjust control rod 6 notches in -2 MINUTES/NOTCH direction.		
3. Example 3 shows old blade at -2 and new blade at +10; therefore adjust control rod 6 notches in +2 MINUTES/NOTCH direction.		

- e. Check for no more than 3/4-inch of threads showing on lower end of pitch control rod as shown. If exposed thread is more than 3/4-inch, pitch control rod has been improperly adjusted. Recheck adjustment.
- f. **QA** Lock key in notch by tightening upper locknut. **TORQUE NUT TO 650 - 750 INCH-POUNDS.**
- g. **QA TORQUE LOWER LOCKNUT TO 650 - 750 INCH-POUNDS.**
- h. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire locknuts in both directions (double safety).
- 2. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire from upper locknut to lock key and then back to opposite side of upper locknut.
- 3. If more than one blade is to be installed, disengage gust lock before turning rotor head, and repeat entire procedure.
- 4. Check area for cleanness and foreign material.
- 5. If different 70103-08107-102 main rotor blade expandable pins were installed, check main rotor balance (TM 1-6625-724-13&P).
- 6. If different blade was installed, check main rotor balance (TM 1-6625-724-13&P).

Table 2. Examples.

MINUTES SHOWN ON BLADE	EXAMPLE 1	EXAMPLE 2	EXAMPLE 3
+12			
+10			***NB
+8	*OB	**OB	
+6			
+4	*NB		
+2			

ADJUST - Continued**Table 2. Examples. - Continued**

MINUTES SHOWN ON BLADE	EXAMPLE 1	EXAMPLE 2	EXAMPLE 3
0			
-2			***OB
-4		**NB	
-6			
-8			
-10			
-12			
NOTES			
1. OB = old blade, NB = new blade			
2. *Example 1 shows old blade at +8 and new blade at +4; therefore adjust control rod 2 notches in -2 MINUTES/NOTCH direction.			
3. **Example 2 shows old blade at +8 and new blade at -4; therefore adjust control rod 6 notches in -2 MINUTES/NOTCH direction.			
4. ***Example 3 shows old blade at -2 and new blade at +10, therefore adjust control rod 6 notches in +2 MINUTES/NOTCH direction.			

NOTE

A maintenance test flight is not required when in flight characteristics are normal and blades have been removed for inspection or shipment, have not been abused or damaged, require no adjustments or replacement, and are reinstalled in their original positions and locations.

7. Do a maintenance test flight (TM 1-1520-237-MTF) if:
 - a. Blades or components have been abused or damaged in operational or non-operational conditions.
 - b. In-flight characteristics are abnormal requiring adjustments to blades or components to correct.
 - c. Different blade or components are installed.

ADHESIVE FILLER REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
- Get medical evaluation for skin or throat irritations and breathing problems.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wear protective cotton and plastic gloves when handling solvents or adhesives.
- Wash hands and face before eating or smoking.

1. Mask area around damage 1/2-inch larger than damage outline.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in prolonged contact with bonded areas or exposed laminate fibers. Do not allow solvent to seep into bonded area or contact exposed laminate fibers.

2. Remove paint inside masked area using clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
3. On repairs to leading edge fairing, slightly undercut foam, leaving cavity slightly larger than hole in fairing. Remove all loose foam from hole.

CAUTION

Damage to blade skin will result if blade is sanded excessively. Sand only until yellow primer is removed.

4. Sand yellow primer from repair area using abrasive cloth, Item 81, WP 1803 00.

ADHESIVE FILLER REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Wipe repair area with clean, machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.
6. Mix package of two part adhesive, Item 11, WP 1803 00. Empty tube of curing agent into cup of adhesive resin. Stir until mixture is free of streaks and color is uniform. Pot life of mixture is 25 minutes at 75°F (24°C).
7. Using beverage stirring stick, Item 300, WP 1803 00, tongue depressor (or equivalent), fill in area with adhesive mixture.

NOTE

To reduce sanding, remove extra adhesive from repair area.

8. With wood or plastic straight edge bridging repair spanwise, scrape off extra adhesive in chordwise direction. Make sure level of adhesive is even with contour of skin.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

9. Wipe area outside repair with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
10. Cover repair with plastic film peel ply large enough to cover bladder area.

NOTE

- Do not use pressure/heat pack on leading edge fairing repair.
 - Alternate curing time without pressure/heat pack is 24 hours at 75°F (24°C).
11. Place pressure/heat pack on blade so that rubber bladder is over repair and hinges are toward trailing edge. Close fixture and engage toggle bolts. Tighten toggle bolts until stops are contacted (Figure 3, Sheet 1).

NOTE

Relief valve on pump prevents pressure over 5 psi.

12. Actuate handpump until gage reads 4 psi.
13. Disconnect air pump hose from air valve.
14. Connect pressure/heat pack to 110 vac electrical power source.
15. Maintain heat and pressure for 1 hour.
16. Disconnect electrical power and depress plunger in valve stem, on top of pack assembly, to release air pressure.
17. Remove pack assembly from blade.

ADHESIVE FILLER REPAIR - Continued

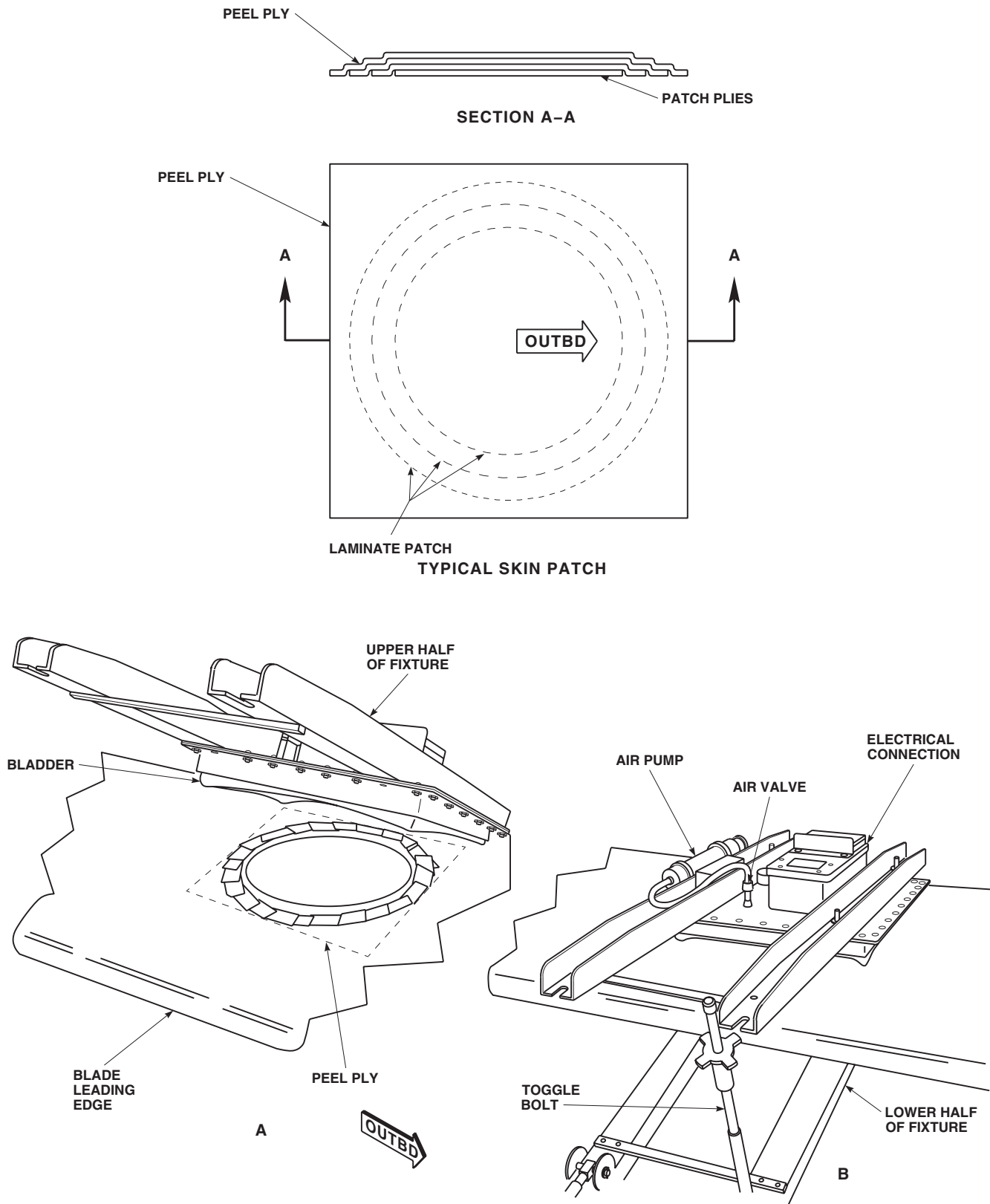


Figure 4. Repair Procedures. (Sheet 1 of 9)

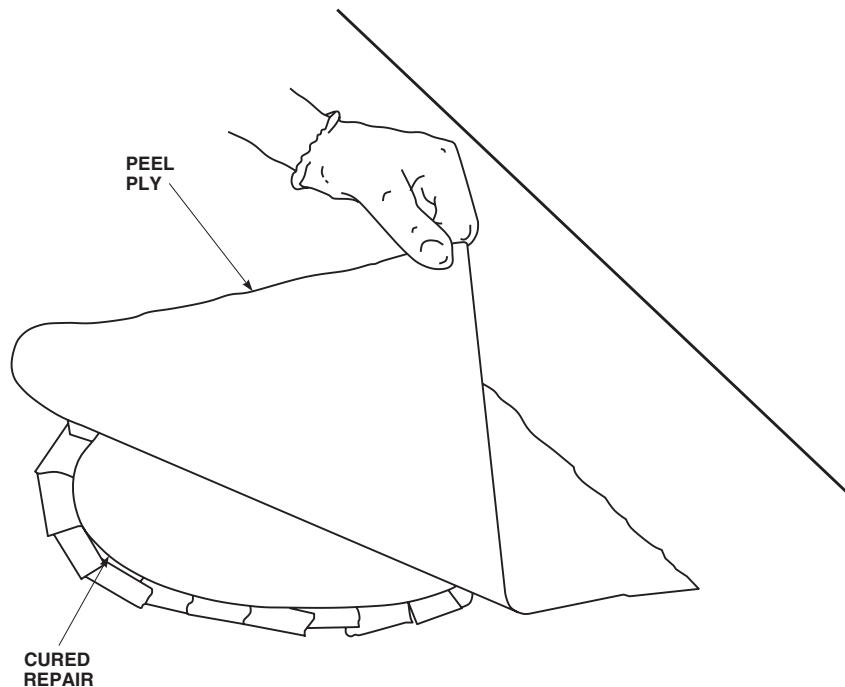
ADHESIVE FILLER REPAIR - ContinuedAK1391
SA

Figure 4. Repair Procedures. (Sheet 2 of 9)

18. Remove peel ply and masking tape from repair.

CAUTION

Damage to blade skin will result if blade is sanded excessively. Use only enough pressure to lightly sand adhesive from blade.

19. **QA** Lightly sand edges of adhesive to blend repair to surrounding surface, using abrasive cloth, Item 75, WP 1803 00. Leave 0.062-inch bead of adhesive over damaged area when repairing leading edges fairing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

20. Wipe area clean using clean, machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean, machinery wiping towel, Item 344, WP 1803 00.
21. Touch up paint finish. Refer to, PAINT TOUCHUP, in this work package.

ADHESIVE FILLER REPAIR - Continued

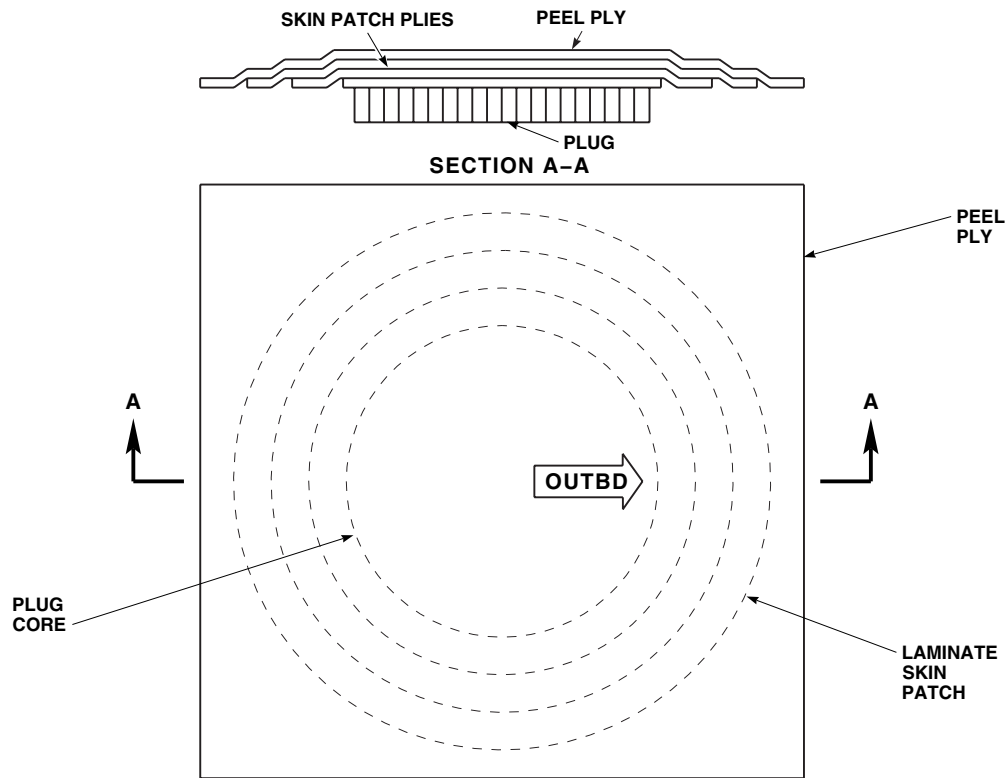


Figure 4. Repair Procedures. (Sheet 3 of 9)

AK1392
SA

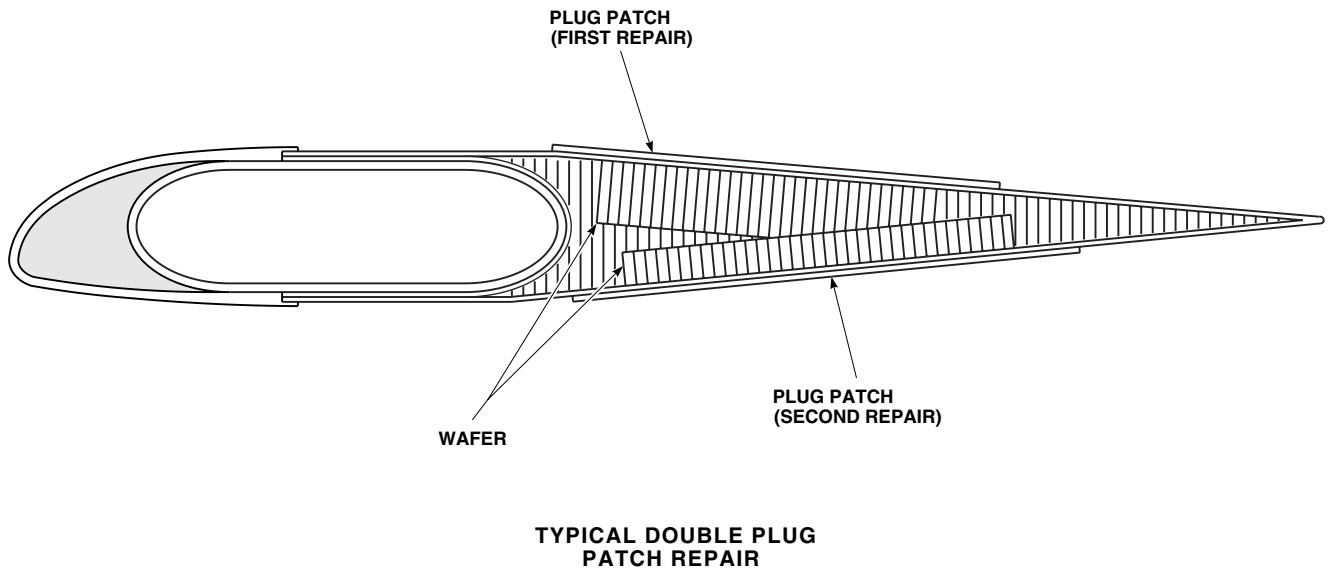
ADHESIVE FILLER REPAIR - ContinuedAK1393
SA

Figure 4. Repair Procedures. (Sheet 4 of 9)

ADHESIVE FILLER REPAIR - Continued

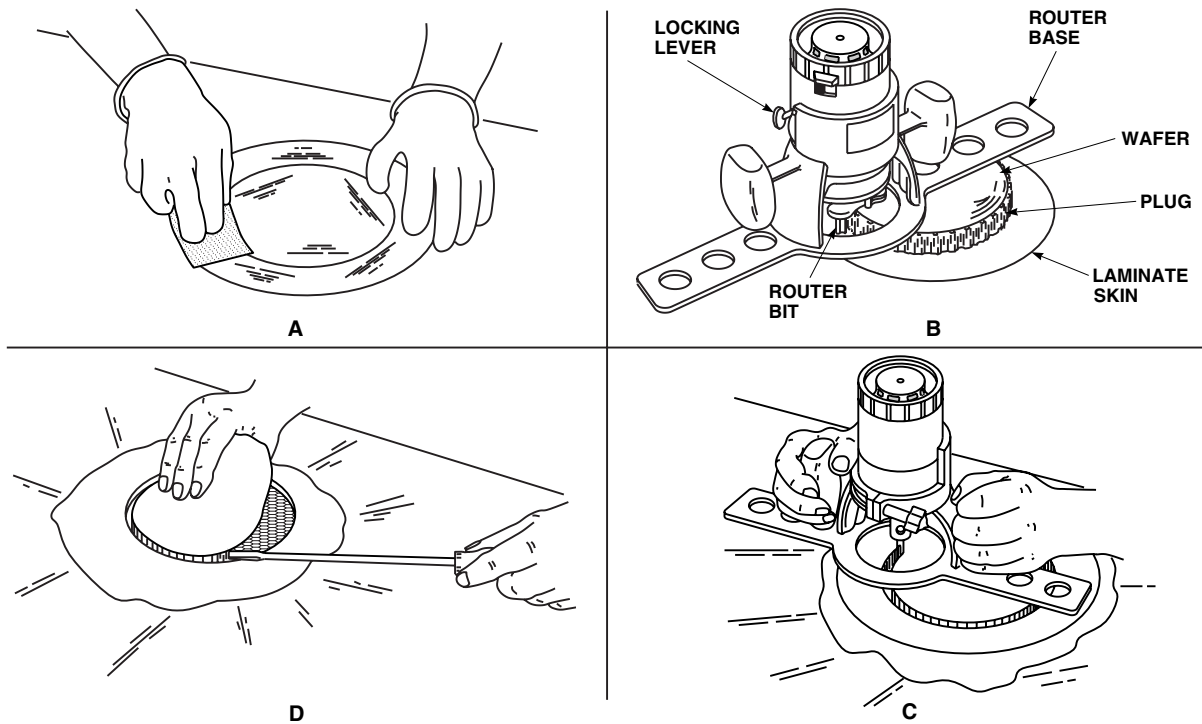
AK1394A
SA

Figure 4. Repair Procedures. (Sheet 5 of 9)

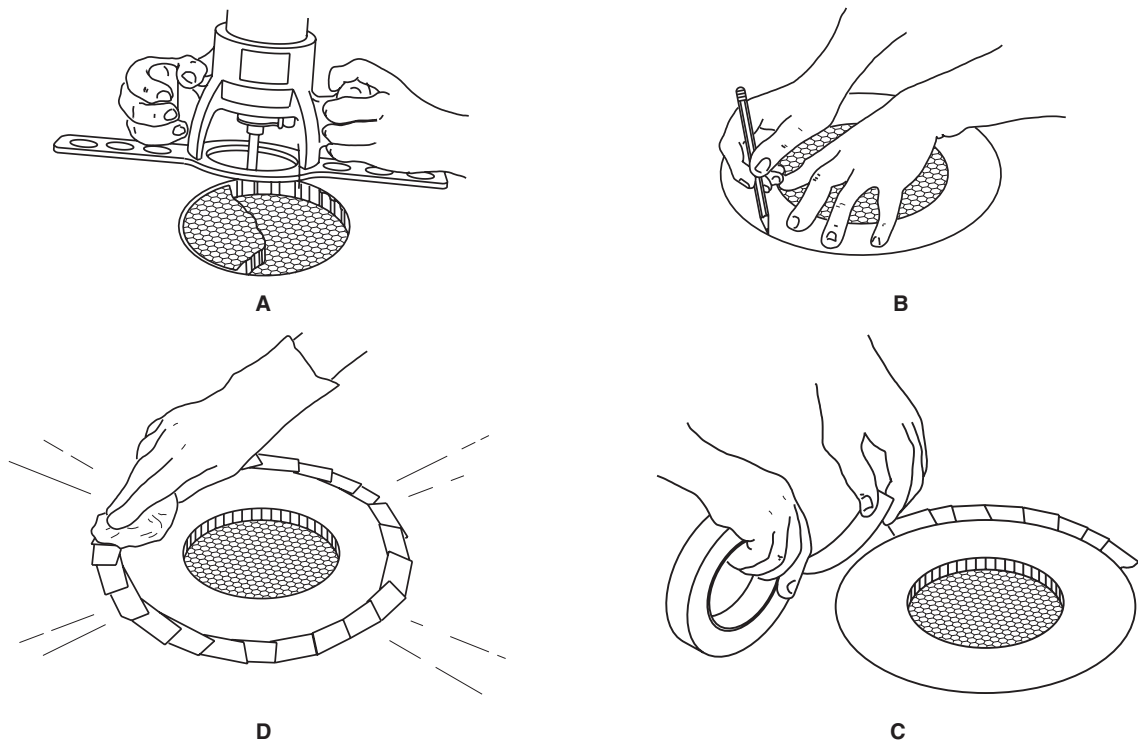
ADHESIVE FILLER REPAIR - ContinuedAK1395
SA

Figure 4. Repair Procedures. (Sheet 6 of 9)

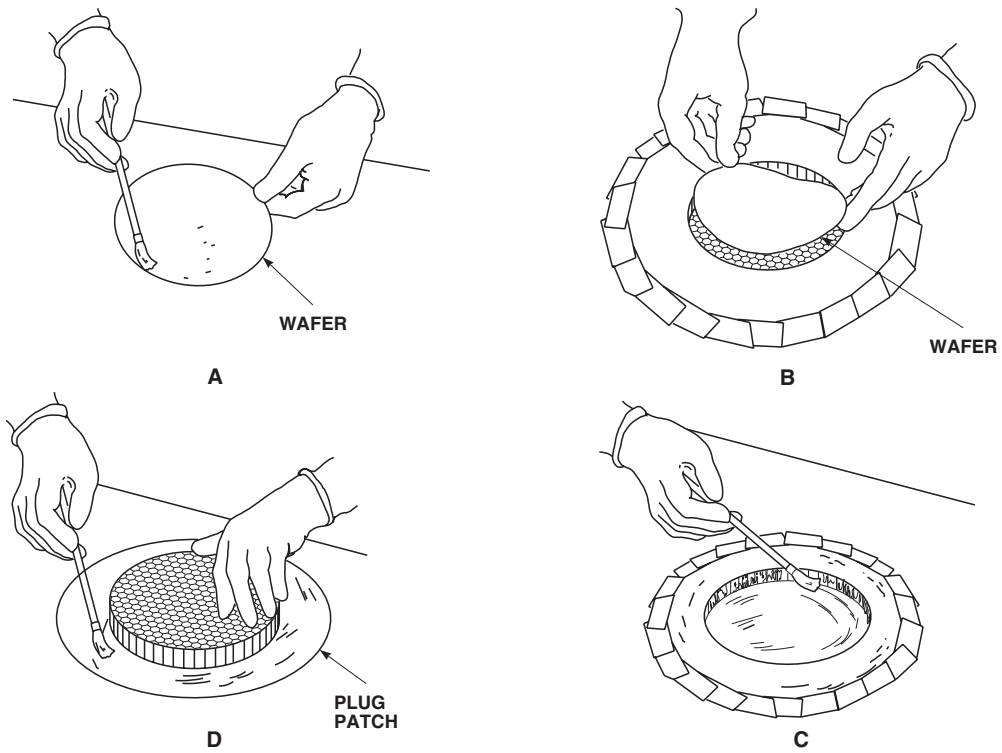
ADHESIVE FILLER REPAIR - ContinuedAK1396
SA

Figure 4. Repair Procedures. (Sheet 7 of 9)

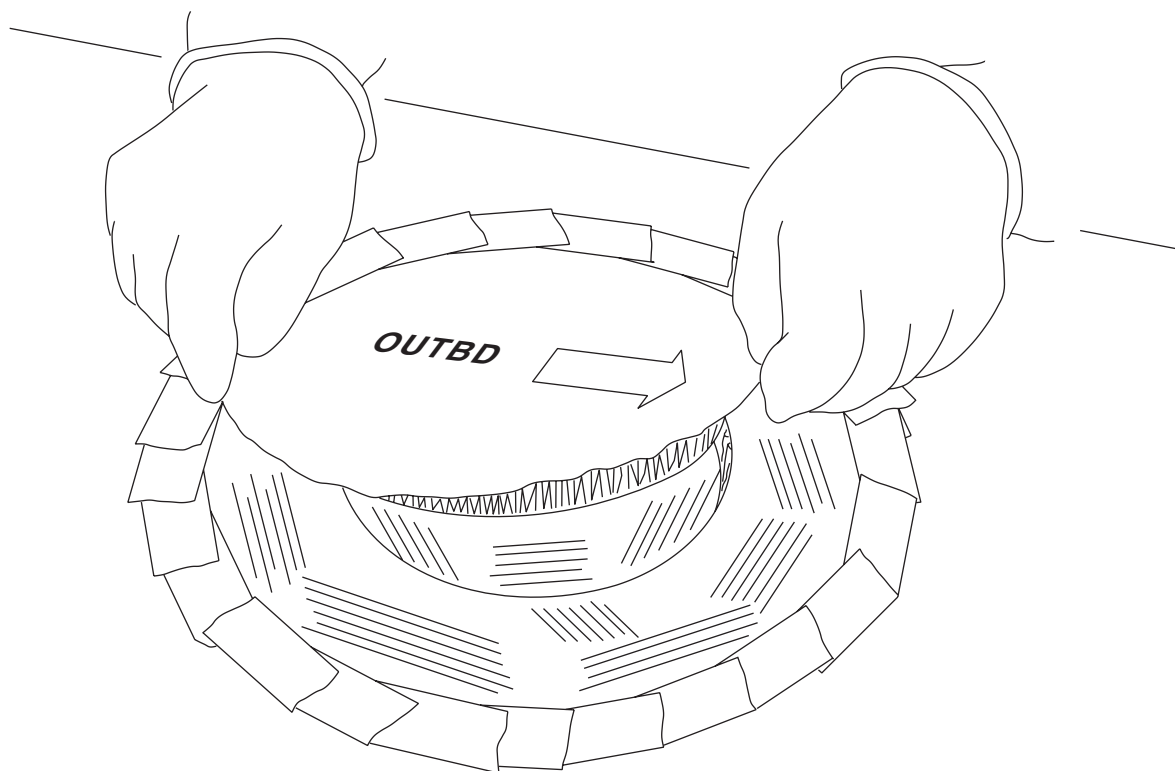
ADHESIVE FILLER REPAIR - ContinuedAK1397
SA

Figure 4. Repair Procedures. (Sheet 8 of 9)

ADHESIVE FILLER REPAIR - Continued

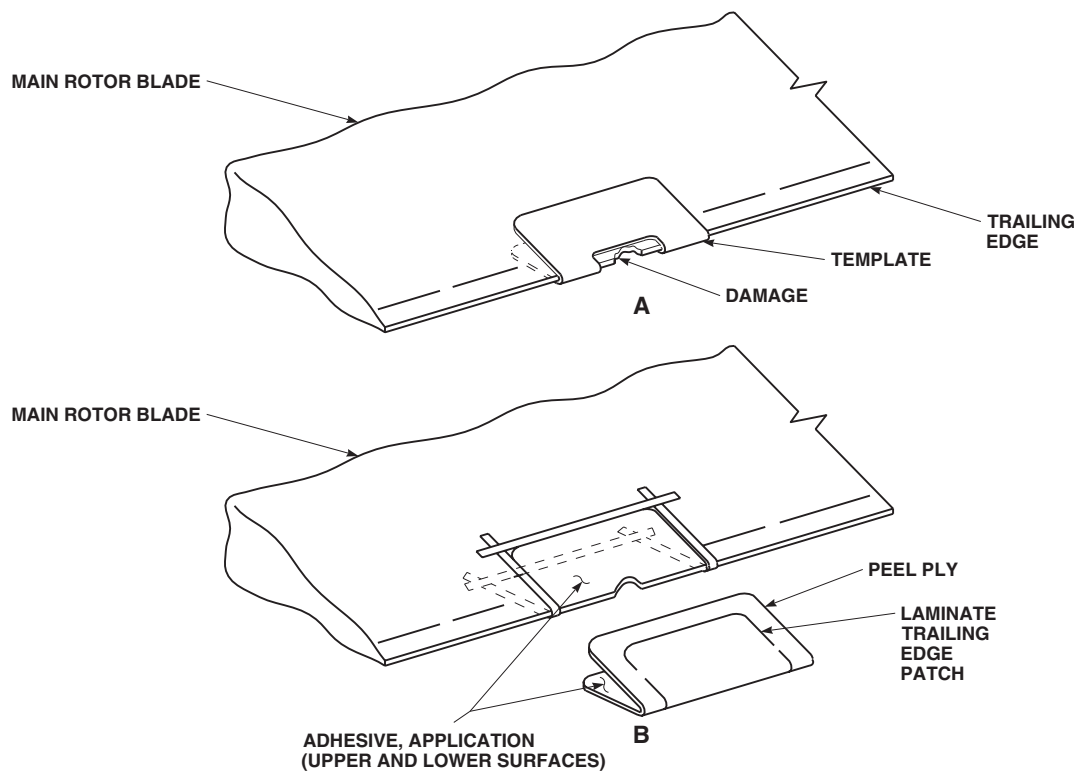
AK1398
SA

Figure 4. Repair Procedures. (Sheet 9 of 9)

ADHESIVE INJECTION REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
 - Get medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
1. Position blade horizontally, with leading edge 2 to 3 inches higher than trailing edge.
 2. Clean out delamination with denatured alcohol, Item 48, WP 1803 00.
 3. Mix two part adhesive, Item 11, WP 1803 00. Empty tube of curing agent into cup of resin. Stir until color of mixture is uniform. Pot life of mixture is 25 minutes at 75°F (24°C).

CAUTION

Damage to blade skin will result if excessive force is used to open up separation in blade. Do not apply too much pressure with plastic tool.

4. Carefully open up separation with clean thin plastic tool, or similar tool, and apply adhesive to slot. Use small disposable applicator, Item 53, WP 1803 00, (or equivalent) to force adhesive into separation.
5. Remove tool and finish filling slot with adhesive.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Wipe extra adhesive from tab with clean, machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
7. Cover repair with plastic film peel ply.
8. Clamp flat strips of wood to sides of tabs with light but firm pressure.

ADHESIVE INJECTION REPAIR - Continued

9. Allow adhesive to cure for 1 hour at 160°F (71°C) or 24 hours at 75°F (24°C).
10. Remove clamping material.
11. **QA** Remove peel ply and sand lightly to smooth out repair.
12. Wipe area clean with cheesecloth cloth, Item 85, WP 1803 00, moistened with denatured alcohol, Item 48, WP 1803 00.
13. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.

SKIN PATCH REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
 - Get medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
1. This repair applies to 3 inch, 5 inch, and 8 inch patches (rotary repair kit part numbers, 70072-15001-012 through 70072-15001-014, in that order) in spar and honeycomb regions. Use smallest patch size that will allow 1 inch overlap of cleaned up damage.
 2. Use skin patch when damage does not extend through skin to spar (spar region) or when core damage is less than 1 inch diameter (honeycomb region). See repair limits in WP 0544 00.
 3. Use skin patch on two sides if damage is through both skins but core damage is less than 1 inch diameter (honeycomb region).
 4. If core damage is greater than 1 inch diameter, use plug patch (Procedure No. 4).

NOTE

- Part numbers referenced are all component parts of rotary b repair kit, Item 253, WP 1803 00.
 - Skin patch may straddle area between spar and honeycomb regions.
5. Select correct template from fiberglass repair kit. Use 5 inch, 7 inch, or 10 inch template for 3 inch, 5 inch, or 8 inch patch, respectively.

SKIN PATCH REPAIR - Continued

6. Center template over damage and with pencil, draw circle around damage. Circle should be 1 inch larger than damage, on all sides.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in prolonged contact with bonded areas or exposed laminate fibers. Do not allow solvent to seep into bonded area or contact exposed laminate fibers or core material.

7. Remove paint inside circle using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

CAUTION

Damage to blade skin will result if blade is sanded excessively. Sand only until yellow primer is removed.

8. Sand yellow primer from repair area using abrasive cloth, Item 81, WP 1803 00.

NOTE

When damage results from rips, tears, or holes less than 1 inch diameter, remove debris using vacuum cleaner.

9. Use correct template from fiberglass repair kit and draw circle 1/2-inch larger than patch to be applied.
10. Mask area outside circle with pressure sensitive adhesive tape, Item 325, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

11. Clean skin area inside masked circle with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.
12. Protect area to keep surface clean.

NOTE

Use 32 gram package of pre-measured adhesive resin and curing agent. Mix full amount. Use as needed, and discard leftover mixture.

13. Mix package of two part adhesive, Item 11, WP 1803 00. Empty tube of curing agent into cup of resin and stir with beverage stirring stick, Item 300, WP 1803 00, until color of mixture is uniform. Pot life of mixture is 25 minutes at 75°F (24°C).

SKIN PATCH REPAIR - Continued**NOTE**

If core is damaged less than 1 inch diameter, smear adhesive on frayed ends of honeycomb material. Routing out core is not necessary for this repair (WP 0544 00). Do not pack adhesive into cells of honeycomb. Added adhesive weight will affect blade balance.

14. Using clean 1 inch paint brush, Item 61, WP 1803 00, apply light, even coat of adhesive mixture to repair surface inside masked circle.

NOTE

Skin patches are made from several plies of material (WP 0544 00). A peel ply on the outer surface keeps adhesive from contacting pressure/heat pack during curing. An arrow is stenciled on peel ply for correct positioning of patch. Patches are packaged in plastic to protect bonding surface. Do not remove peel ply from patch until repair is cured.

15. Carefully remove patch from plastic wrapping. Do not handle inner (bonding) surface.
16. Using same disposable applicator, Item 53, WP 1803 00, (or equivalent), apply thin even coat of adhesive mixture to patch bonding surface.

NOTE

Patch must overlap damage by 1 inch on all sides.

17. Position patch on blade skin with arrow pointing outboard. Move patch back and forth with light hand pressure to seat patch and work out air bubbles in adhesive.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

18. Carefully lift edges of peel ply and wipe excess adhesive from around patch, using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
19. Check patch for correct position and tape down four corners to prevent movement.
20. Place pressure/heat pack (Figure 3, Sheet 1) on blade so that rubber bladder is over repair and hinges are toward trailing edge. Close fixture and engage toggle bolts. Tighten bolts until metal skirt around bladder is 1/8 - inch from skin.
21. Work handpump until gage reads 4 psi.
22. Disconnect air pump hose from air valve.
23. Connect fixture to 110 vac electrical power source.
24. Maintain heat and pressure for one hour.
25. Disconnect electrical source and press plunger in valve stem on top of fixture, to release air pressure.
26. Remove fixture from blade.
27. Allow blade to cool to room temperature.

SKIN PATCH REPAIR - Continued

28. **QA** Remove peel ply and masking tape from repair area (Figure 3, Sheet 2).

CAUTION

Damage to blade skin will result if blade is sanded excessively. Use only enough force to blend edges of patch into blade.

29. **QA** Lightly sand edges of patch to blend with surrounding skin, using abrasive cloth, Item 81, WP 1803 00.
30. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.

PLUG PATCH REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
 - Get medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
1. This repair uses 3 inch and 6 inch plug patches (rotary repair kit part numbers, 70072-15001-015 through 70072-15001-020, in that order) (Figure 3, Sheet 3) for repairing damage in honeycomb core region (WP 0544 00). Use size that allows 1 inch skin overlap of cleaned up damage.
 2. **QA** Holes routed for plug patch must be within dimensions allowed in WP 0544 00.
 3. Overlapping plug patch skin may extend 1 inch into spar or trailing edge region (WP 0544 00).
 4. For damage through both skins and core damage greater than 1 inch diameter, use plug patch on both sides (Figure 3, Sheet 4). Complete repair on one side of blade before starting repair on reverse side.

PLUG PATCH REPAIR - Continued**NOTE**

- Use rotary repair kit.
 - Plug patches 3 inches and 6 inches in diameter are made from prefabricated skin patch and Nomex honeycomb plug. See Table 3 for patch size and plug patch repair kit part number. A peel ply on the outer surface keeps adhesive from contacting pressure/heat pack during curing. An arrow is stenciled on peel ply for correct positioning of plug. A reinforced plastic wafer provides bonding surface for honeycomb core and plug core.
5. Select proper template from plug patch repair kit. Use 3 inch to 7 inch template for 3 inch plug, and 6 inch to 10 inch template for 6 inch plug.
 6. Center template over damage and with soft pencil draw two circles on blade skin around damage. Outer circle will be 2 inches in diameter larger than inner circle.
 7. Inner circle will be diameter of routed hole (Figure 3, Sheet 3). Depth will be 1/4, 1/2, or 7/8-inch, depending upon core thickness at point of damage. Limits in Table 4 show how close the cavity depth can come to trailing edge without breaking through opposite skin.

Table 3. Plug Patch Repair.

KIT PART NO.	DIAMETER	THICKNESS
70072-15001- 015	3 inch	1/4-inch
70072-15001- 016	3 inch	1/2-inch
70072-15001- 017	3 inch	7/8-inch
70072-15001- 018	6 inch	1/4-inch
70072-15001- 019	6 inch	1/2-inch
70072-15001- 020	6 inch	7/8-inch

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in prolonged contact with bonded joints, exposed laminate fibers or core. Do not allow solvent to seep into bonded joints or contact exposed laminate fibers or core.

8. Remove paint from inside both circles, using clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
9. Redraw inner circle.

PLUG PATCH REPAIR - Continued**Table 4. Plug Patch Repair.**

Plug Depth	Closest Distance to Trailing Edge
1/4-inch	1 1/2-inches
1/2-inch	3 inches
7/8-inch	5 1/2-inches

CAUTION

Damage to blade skin will result if blade is sanded excessively. Sand only until yellow primer is removed.

10. Sand yellow primer from area between both circles, using abrasive cloth, Item 81, WP 1803 00 (Figure 3, Sheet 5).

NOTE

- To use plug and wafer as gage for setting router bit depth, place plug patch on flat surface with plug facing up. Place wafer on plug. Set router base on wafer with router bit along side of plug. Adjust base until router bit touches inner surface of patch. Tighten locking lever to lock base (Figure 3, Sheet 5).
- When patching both sides of blade, start with larger diameter repair.

11. Redraw inner circle for routing operation.
12. Insert router bit in router collet and set depth of cut to match depth of required plug patch (including wafer).

NOTE

Ensure there is no contact between trailing blade pin handle and spindle lug after pin isw closed. Reposition blade pin as necessary to remove contact.

13. Start router and hold with both hands. Rest edge of base on blade with router bit parallel to skin. Tilt router forward towards vertical, bringing router bit in contact with skin near center of inner circle. Keep router base in contact with blade skin.
14. Move router slowly towards pencil line of inner circle, then carefully follow line until complete circle is cut in skin (Figure 3, Sheet 5, Detail C). Shut off router and remove from cut.
15. Using flat-bladed screwdriver (or equivalent), lift edge of cut skin, inside circle, and peel from core (Figure 3, Sheet 5).
16. Using same depth of cut and 1/2 - inch diameter end mill, reinsert router and with base spanning cut, route out core material using back-and-forth motion (Figure 3, Sheet 6).
17. Shut off router and remove from blade.

NOTE

To make sure plug patch will fit routed hole, check diameters of plug and hole.

18. Clean up debris in cavity. Remove dust and small particles with vacuum cleaner.

PLUG PATCH REPAIR - Continued

19. Mark circle on skin 2 1/2-inches larger than routed hole (Figure 3, Sheet 6). (Kit template for 3 inch plug has 5 1/2-inch OD and template for 6 inch plug has 8 1/2-inch OD).
20. Mask outside this circle with pressure sensitive adhesive tape, Item 325, WP 1803 00 (Figure 3, Sheet 6).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

21. Clean skin area to be bonded with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00 (Figure 3, Sheet 6). Wipe dry before solvent evaporates.
22. Protect area to keep surface clean.

NOTE

The amount of mixed adhesive required will depend on plug size and depth (See Table 5). Each 32 gram package is pre-measured with curing agent and resin. Mix full package and use as required. Discard leftover mixture.

23. Mix one or more packages, as required, of two part adhesive, Item 11, WP 1803 00. Empty tube of curing agent into cup of resin and stir with beverage stirring stick, Item 300, WP 1803 00, until color of mixture is uniform. Pot life of mixture is 25 minutes at 75°F (24°C).

Table 5. Plug Patch Repair.

PLUG DIAMETER	PLUG DEPTH	QTY
3 inch	1/4-inch	1/3 Pkg
	1/2-inch	1/3 Pkg
	7/8-inch	2/3 Pkg
6 inch	1/4-inch	1 Pkg
	1/2-inch	1 1/4 Pkg
	7/8-inch	2 Pkgs

24. Using clean 1 inch paint brush, Item 61, WP 1803 00, apply liberal coat of mixed adhesive to one side of wafer (Figure 3, Sheet 5, Detail A).
25. Place wafer in cavity with adhesive side down (Figure 3, Sheet 7, Detail B), when repairing top side of blade, or place wafer on plug side of plug patch when repairing bottom side of blade.

PLUG PATCH REPAIR - Continued**CAUTION**

Damage to blade will result if honeycombs are packed with adhesive. Added adhesive weight will affect blade balance. Do not pack honeycomb cells with adhesive.

26. Apply heavy coat of adhesive, Item 11, WP 1803 00, to exposed side of wafer and wall of core cavity (Figure 3, Sheet 7). Apply thin even coat of adhesive, Item 11, WP 1803 00, to skin area to be bonded.
27. Remove plug patch from plastic bag and apply light coat of adhesive, Item 11, WP 1803 00, to mating surfaces of laminate skin patch and plug wall (Figure 3, Sheet 7). Do not apply adhesive to plug end.
28. Position plug patch in cavity with stenciled arrow pointing outboard (Figure 3, Sheet 8) and press firmly in place. With firm hand-pressure, work out air bubbles and extra adhesive from skin patch overlapping blade skin.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

29. Carefully lift edges of peel ply and wipe squeezed out adhesive from around skin patch, using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
30. Check patch for correct position and tape down four corners to prevent movement.
31. Place pressure/heat pack (Figure 3, Sheet 1) on blade with rubber bladder over repair and hinges toward trailing edge. Close fixture and engage toggle bolts. Tighten bolts until metal skirt around bladder is 1/8-inch from skin.
32. Work handpump until gage reads 4 psi.
33. Disconnect air pump hose from air valve.
34. Connect fixture to 110 vac electrical power source.
35. For 3 inch plug, maintain heat and pressure for one hour.
36. For 6 inch plug, maintain heat and pressure for one hour and 15 minutes.
37. Disconnect electrical source and press plunger in valve stem on top of fixture to release air pressure.
38. Remove fixture from blade.
39. Allow blade to cool to room temperature.
40. **QA** Remove peel ply and masking tape from repair area (Figure 3, Sheet 2).

CAUTION

Damage to blade skin will occur if blade is sanded excessively. Use only enough force to blend patch into blade.

41. Lightly sand edges of patch to blend with surrounding skin, using abrasive cloth, Item 81, WP 1803 00.

PLUG PATCH REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

42. Wipe area clean using clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.

NOTE

If lower side of blade requires plug patch (Figure 3, Sheet 4), repeat this procedure for lower side.

43. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.

TRAILING EDGE PATCH REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
 - Get medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
 - This repair applies to trailing edge region (WP 0544 00).
 - Tab groove damage greater than 0.060-inch deep and up to 3 inches long spanwise can be repaired.
 - Trailing edge gashes up to 1/2-inch chordwise (from trailing edge) and 1 1/2-inches spanwise can be repaired.
 - Skin tears no greater than 2 inches chordwise and 3 inches spanwise can be repaired.
 - Damage is repaired using 3 by 5 inch vee-shaped doubler patch bonded to both sides of trailing edge. Patch must overlap damage, on all sides, by at least 1 inch.
 - Use trailing edge repair kit Part No. 70072-15001-021.
 - Trailing edge patch is made from piles of impregnated glass cloth fabric. A peel ply on the outer surface keeps adhesive from contacting pressure/heat pack during curing. Patch is packaged in plastic to protect bonding surface.
1. Turn blade to workable position and support blade to prevent movement and droop.
 2. Locate work stand and adjust to safe, comfortable working height.
 3. Center kit template spanwise over damaged area and with soft pencil draw line around template on upper and lower surface of blade (Figure 3, Sheet 9, View A).

TRAILING EDGE PATCH REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints.

4. Remove paint from inside template guide lines using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

CAUTION

Damage to blade will result if blade is sanded excessively. Sand only until yellow primer is removed.

5. Sand yellow primer from inside guidelines using abrasive cloth, Item 81, WP 1803 00. Sand off damaged material that rises above blade contour. Do not sand undamaged skin fibers (Figure 3, Sheet 9).
6. Clean up sanding dust and small particles using vacuum cleaner or low-pressure air (5 to 10 psi).
7. Using same template, remark guide lines.
8. Mask around guide lines using pressure sensitive adhesive tape, Item 325, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

9. Clean skin to be bonded using clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.

NOTE

Use 32 gram package of pre-measured curing agent and adhesive resin. Mix full amount and use as required. Discard leftover mixture.

10. Mix package of two part adhesive, Item 11, WP 1803 00. Empty tube of curing agent into cup of resin and stir with beverage stirring stick, Item 300, WP 1803 00, until color of mixture is uniform. Pot life of mixture is 25 minutes at 75°F (24°C).
11. Using clean 1 inch paint brush, Item 61, WP 1803 00, apply light, even coat of adhesive mixture to area of trailing edge inside guide lines and to inside of doubler patch (Figure 3, Sheet 9).
12. Center patch within guide lines and press in place. To seat patch, move it back and forth with light hand pressure.
13. Push patch firmly against trailing edge and work out air bubbles and extra adhesive, working from center to edge of patch.

TRAILING EDGE PATCH REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

14. Carefully lift edges of peel ply and wipe squeezed out adhesive from around patch, using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
15. Check patch for correct position and tape down four corners to prevent movement.

NOTE

Use pressure/heat pack curing fixture (Figure 3, Sheet 3).

16. Place pressure/heat pack fixture on blade so that bladder and opposing pad are over repair area. Close pack assembly and engage toggle bolts. Tighten bolts until metal skirt around bladder is 1/8-inch from blade skin.
17. Work handpump until gage reads 4 psi.
18. Disconnect air pump hose from air valve.
19. Connect fixture to 110 vac electrical power source. (Thermostat will keep curing temperature of 160°F (71°C)).
20. Maintain heat and pressure for 1 hour.
21. Disconnect electrical source and press plunger in valve stem on top of fixture to release air pressure.
22. Remove fixture from blade.
23. Allow blade to cool to room temperature.
24. **QA** Remove peel ply and masking tape from repair area.
25. **QA** Carefully sand edges of patch to blend with surrounding skin, using abrasive cloth, Item 81, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

26. Wipe area clean using clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.
27. Touch up paint finish. Refer to, PAINT TOUCHUP, in this work package.

REPLACE TIP CAP

1. Turn blade to workable position and support it to prevent movement or droop.
2. Locate workstand and adjust to safe, comfortable, working height.
3. Remove screws from first six holes (from leading edge), top and bottom, and store separately.
4. Remove remaining screws from rear six holes, top and bottom, and store separately.
5. With sharp nonmetallic scraper, scrape out old adhesive from joint of tip cap and blade.

REPLACE TIP CAP - Continued**CAUTION**

Damage to blade and tip cap will result if tip cap is pried off without removing sealing compound. Use flexible metal spatula (or equivalent) to cut through sealing compound that bonds tip cap to tip block.

6. Remove tip cap.
7. Inspect interior lower airfoil skin of tip cap in area where spanwise weights are located for missing paint or indication of contact by spanwise weights. If evidence of contact is found, repair spanwise weights. Refer to, TIP CAP WEIGHT REPAIR, in this work package.
8. Inspect tip cap for damage (WP 0544 00).
9. Remove broken screws using screw extractor set (or equivalent). Center punch broken screw before drilling screw extractor hole.
10. Inspect inserts in first six holes (from leading edge) for damage (stripped, corroded, loose, or broken). If damaged, replace inserts, MS124735, per TM 1-1500-204-23. Replace inserts. Refer to, TIP BLOCK INSERT, RZA12487, REPAIR, in this work package.
11. Inspect tip weights. Refer to, TIP CAP WEIGHT REPAIR, in this work package.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints.

12. Using nonmetallic scraper, scrape/clean old sealing compound from blade tip end and tip cap (if tip cap is to be reused). Scrape/clean old sealing compound from screws and attachment holes. Clean area with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
13. Trial fit replacement tip cap on rotor blade. If abrasion strip binds, or does not fit, file binding end so that tip cap will fit without putting pressure on abrasion strip. **EDGE OF ABRASION STRIP MUST BE WITHIN 0.005 to 0.090-INCH OF ROTOR BLADE AS LONG AS SEAL CAN BE MAINTAINED WITH SEALING COMPOUND.**

REPLACE TIP CAP - Continued**WARNING****CSI**

Verification of tip cap screw torque is a critical characteristic.

CAUTION

Different screws are used on blades, 70150-09100-041 and 70150-09100-043. Screws holding front section and rear section of tip cap are also different. Refer to Figure 4, Sheet 1 and Sheet 2, before installing screws.

NOTE

Tip cap, 70150-09107-055, uses pan head screws, NAS 7203P5, at holes 4 and 5 on bottom surface of tip cap (Figure 4, Sheet 2).

14. **QA** Position tip cap on blade and install screws, wet with sealing compound, Item 258, WP 1803 00. Refer to Figure 4 for part number of screws. **TORQUE SCREWS TO 29 - 31 INCH-POUNDS.**
15. Make an entry on Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 showing tip cap screw torque stabilization check is due 9 to 11 flight hours after installation (WP 1690 00).
16. Apply a bead of sealing compound, Item 265, WP 1803 00, to the gap where the tip cap butts to the blade. Refer to, TIP CAP-TO-BLADE SEALING, in this work package.
17. If tip cap abrasion strip was filed, fill area with sealing compound, Item 265, WP 1803 00, to form flush contour.
18. Finish sealing tip cap-to-blade joint. Refer to, TIP CAP-TO-BLADE SEALING, in this work package.

TIP CAP-TO-BLADE SEALING**WARNING**

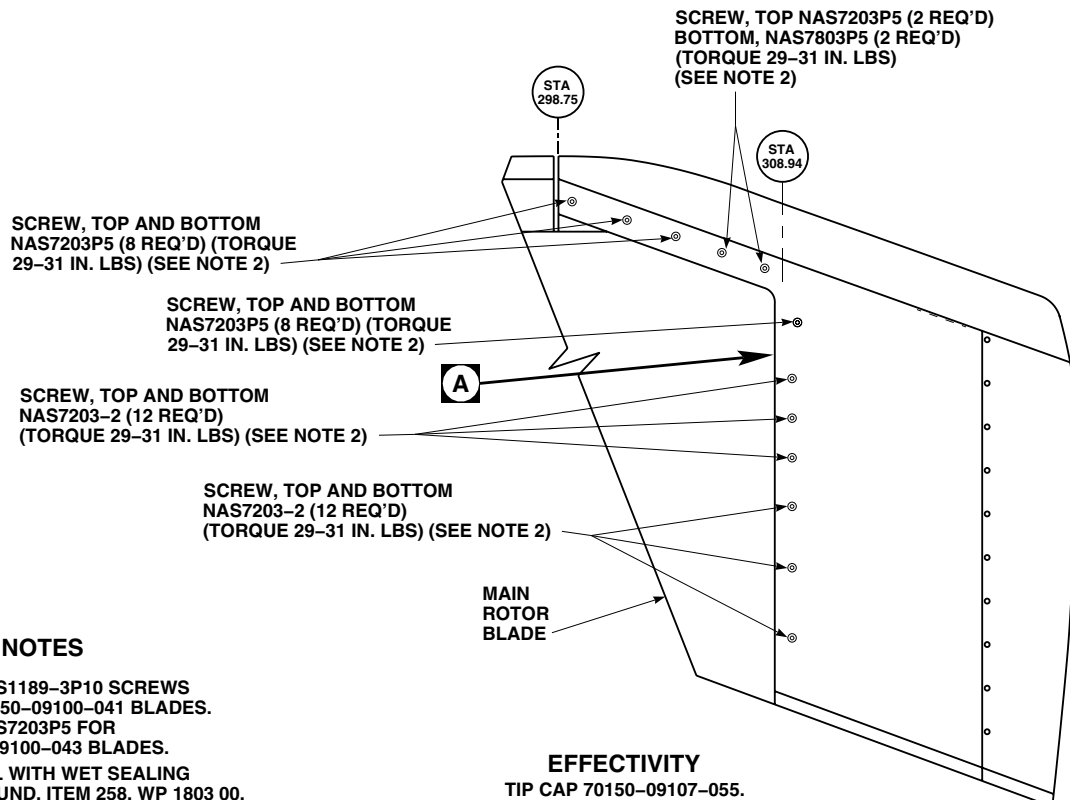
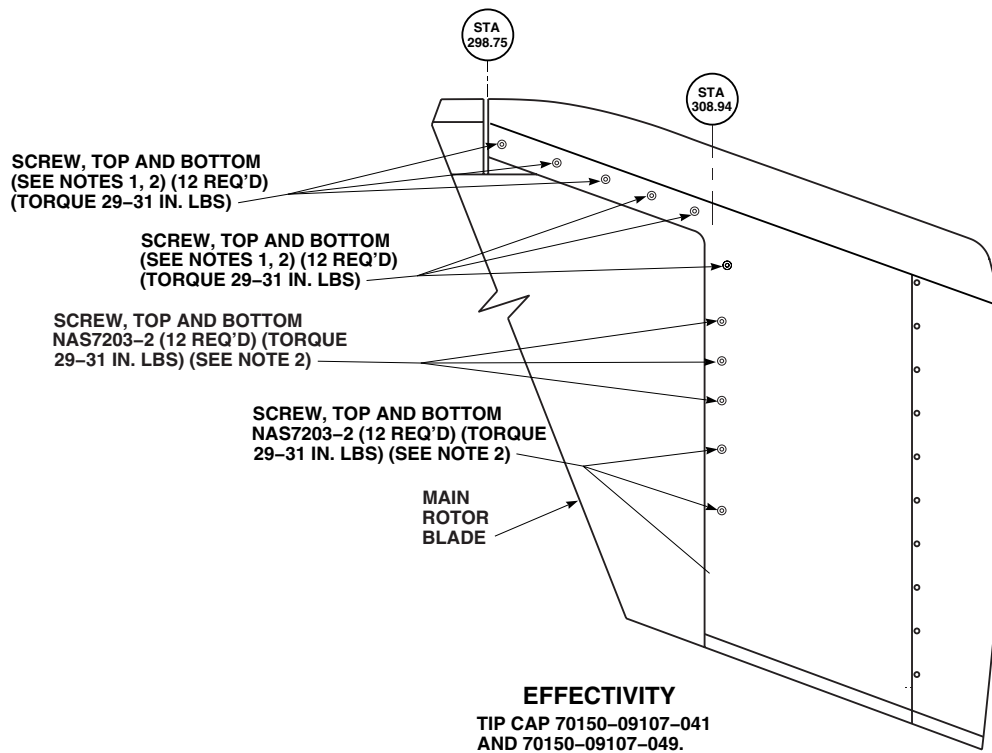
Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Blade will be damaged if solvent seeps into bonded joints or contact with reinforced plastic is prolonged. Bonded joints will separate or plastic will dissolve. Do not allow solvent to seep into bonded joints or remain in contact with reinforced plastic.

1. Clean metal and reinforced plastic surfaces thoroughly with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
2. Apply sealing compound, Item 265, WP 1803 00, into all joints to be sealed.
3. **QA** Fair compound even with tip cap surface, using nonmetallic scraper.
4. **QA** Place test specimen of compound next to repair. Repair compound is cured when test specimen is cured.
5. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.
6. Track and balance main rotor blades (TM 1-6625-724-13&P).

TIP CAP-TO-BLADE SEALING - Continued



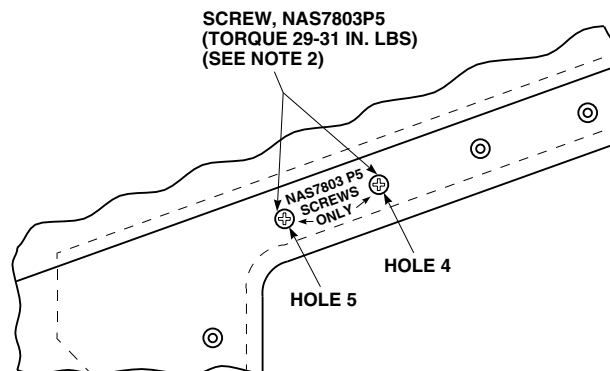
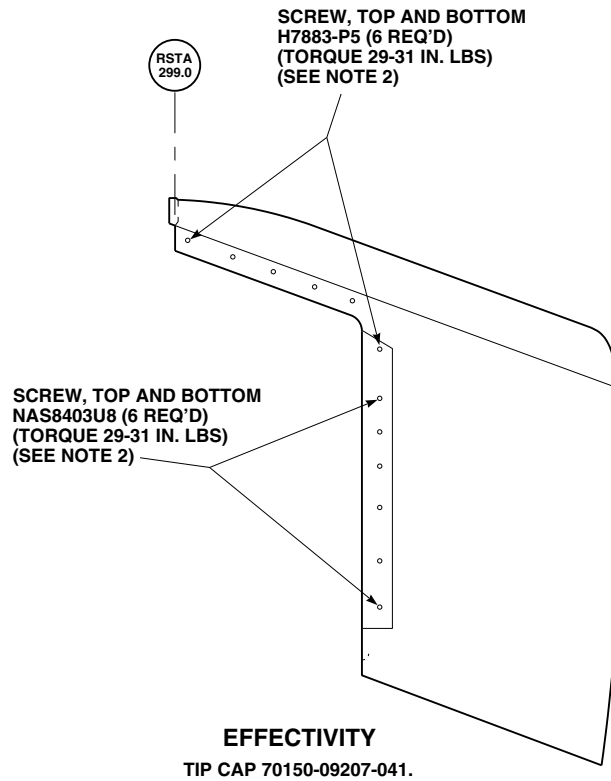
NOTES

1. USE NAS1189-3P10 SCREWS FOR 70150-09100-041 BLADES. USE NAS7203P5 FOR 70150-09100-043 BLADES.
2. INSTALL WITH WET SEALING COMPOUND, ITEM 258, WP 1803 00, OR PRIMER COATING, ITEM 243, WP 1803 00.

AB0173_1B
SA

Figure 5. Replace Tip Cap. (Sheet 1 of 2)

TIP CAP-TO-BLADE SEALING - Continued



BOTTOM VIEW

EFFECTIVITY
TIP CAP 70150-09107-055.

AB0173_2B
SA

Figure 5. Replace Tip Cap. (Sheet 2 of 2)

BOND REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

CAUTION

Blade will be damaged if solvent seeps into bonded joints or contact with reinforced plastic is prolonged. Bonded joints will separate or plastic will dissolve. Do not allow solvent to seep into bonded joints or remain in contact with reinforced plastic.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
 - Get medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
1. Clean bond separation by first wetting separation with technical acetone, Item 5, WP 1803 00, and while wet, blow out with clean filtered forced air (5 to 10 psi).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Carefully insert clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, between separation. Clean with back-and-forth motion. Change machinery towel, as required, until machinery towel remains clean.
3. Prepare adhesive, Item 11, WP 1803 00, per manufacturer's instructions.

NOTE

Pot life of mixture is 25 minutes at 75°F (24°C).

4. Apply mixed adhesive inside separation, using thin disposable applicator, Item 53, WP 1803 00, (or equivalent). Make sure both surfaces are coated.
5. Apply enough pressure to keep surfaces mated, using C-clamps, bungee (elastic) cord, or tape.

BOND REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Clean up squeezed-out adhesive with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
7. Cure adhesive for 24 hours at 75°F (24°C) or 1 hour at 160°F (71°C).
8. **QA** Remove clamps, cord or tape from repair.
9. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.

CORROSION REMOVAL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to blade will result if solvent or chemical conversion solution is allowed to come in prolonged contact with bonded joints. Do not allow solvent or chemical conversion material solution to seep into bonded joints.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
 - Get medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
1. Remove paint from corroded area, using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
 2. Lightly sand aluminum surface with abrasive cloth, Item 84, WP 1803 00 and abrasive cloth, Item 80, WP 1803 00, until corrosion is removed.

CORROSION REMOVAL - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Wipe area clean with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean, machinery wiping towel, Item 344, WP 1803 00.
4. Apply corrosion resistant coating, Item 111, WP 1803 00, to reworked surface (WP 0381 00 and TM 55-1500-345-23).
5. Allow chemical conversion material to air-dry.
6. Prime and paint reworked surface. Refer to, PAINT TOUCHUP, in this work package.1

TIP CAP HOLES REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
 - Get medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
1. Holes up to 1/2-inch diameter or length in tip cap, with no internal damage, may be patched using 2 inch diameter reinforced plastic patch.
 2. Smooth edges of hole, using fine-cut file or abrasive paper, Item 211, WP 1803 00.
 3. Polish edges using abrasive cloth, Item 75, WP 1803 00.
 4. Outline area around damage with pressure sensitive adhesive tape, Item 325, WP 1803 00, 1/4-inch larger than patch size.

TIP CAP HOLES REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean off paint inside masked area using clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe surface dry before solvent evaporates.
6. Prepare adhesive, Item 11, WP 1803 00, per manufacturer's instructions. Refer to Procedure No. 3.

NOTE

Pot life of mixture is 25 minutes at 75°F (24°C).

7. On clean smooth surface, lay up three 2 inch diameter plies of glass fabric saturated with mixed adhesive. Squeegee out extra adhesive.
8. Apply light even coating of mixed adhesive on repair area inside masking tape.
9. Center prepared patch over hole and press firmly into place.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

10. Clean up squeezed-out adhesive with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
11. Cover patch with plastic film peel ply and place felt or rubber mat over film.
12. Place shot bag (weights or equivalent) on pad to keep patch in good contact while curing.
13. Cure adhesive for 24 hours at 75°F (24°C) or 1 hour at 160°F (71°C).
14. **QA** Remove weights, pad, and peel off film.
15. **QA** Lightly sand patch edges to blend with surrounding skin.
16. Remove masking tape and touch up paint. Refer to, PAINT TOUCHUP, in this work package.

TIP CAP SKIN AND FAIRING CRACK REPAIR**NOTE**

- Cracks up to 2 inches long in Kevlar fairing are patched.
- Cracks in aluminum skin up to 1 inch long are stop-drilled and patched.

1. Stop-drill cracks in aluminum skin only, using No. 40 (0.098-inch diameter) drill.
2. Prepare surface and patch over cracks with glass fabric patches. Refer to, TIP CAP HOLES REPAIR, in this work package.

ABRASION STRIP REPAIR

1. **QA** Cracks or negligible voids in abrasion strips on main rotor blade and tip cap, within limits described in WP 0544 00, are sealed to prevent moisture from entering bond line and causing further damage. Apply smooth even coat of mixed adhesive, Item 11, WP 1803 00, over crack or void. For applicable instructions. Refer to, BOND REPAIR, in this work package.
2. Bond separations listed as reparable in WP 0544 00 are bonded using adhesive, Item 13, WP 1803 00. Refer to, BOND REPAIR, in this work package.

TIP CAP WEIGHT REPAIR

1. Visually inspect weights for discoloration (slight corrosion). Remove any discoloration with abrasive cloth, Item 82, WP 1803 00.
2. Inspect for corrosion pitting. If corrosion pitting is found, repair as follows:

NOTE

Under each weight retaining nut is one or more washers used for fine-tuning blade balance. Record exact location of each weight and number, and exact location of washers under each nut, for reinstallation.

- a. Remove nuts, washers, and weights from blade (Figure 5, Sheet 1, Detail A).
 - b. Inspect weights for corrosion pits and repair using abrasive cloth, Item 75, WP 1803 00, as follows:
 - (1) **REMOVE CORROSION PITS FROM WEIGHTS UP TO 0.005-INCH.**
 - (2) Replace weights if corrosion pits are more than limits.
 - (3) Apply coat of epoxy primer coating kit, Item 120, WP 1803 00, to weights. Refer to, PAINT TOUCHUP, in this work package.
 - c. Inspect tip block studs for corrosion pits and repair using wire brush as follows:
 - (1) **REMOVE CORROSION PITS FROM MINOR DIAMETER OF THREADS INBOARD OF RETENTION NUTS UP TO 0.005-INCH DEEP.**
 - (2) **REMOVE CORROSION PITS FROM MAJOR DIAMETER OF THREADS UP TO 0.010-INCH DEEP.**
 - (3) Replace blade if corrosion pits are more than limits.
 - (4) Apply epoxy primer coating kit, Item 120, WP 1803 00, to repaired threads. Refer to, PAINT TOUCHUP, in this work package.
 - d. Remove light corrosion from washers and nuts by brushing lightly with wire brush.
 - e. Reprime tip block studs, bolts, nuts, washers and weights with primer coating, Item 243, WP 1803 00, and while primer is still wet, install weights, washers and nuts in exact location and stack up as recorded when removed.
 - f. **TORQUE NUTS ON FRONT AND REAR CHORDWISE WEIGHTS TO 156 - 173 INCH-POUNDS (Figure 5, Sheet 1, Detail A).**
 - g. **WHILE PUSHING SPANWISE WEIGHTS TOWARD TOP OF BLADE, TORQUE NUTS TO 262 - 288 INCH-POUNDS.**
3. **QA** Check spanwise weights for protrusion below bottom edge of tip block using straight edge (Figure 5, Sheet 1, Detail B and Figure 5, Sheet 1, Detail C). If weights protrude below the bottom edge of tip block, do this:
 - a. Loosen nuts securing spanwise weights.

TIP CAP WEIGHT REPAIR - Continued

- b. Push spanwise weights toward top of blade and check for protrusion below bottom edge of tip block using straight edge.
- c. If spanwise weights cannot be positioned to eliminate protrusion below bottom edge of tip block, do this:
 - (1) Mark spanwise weights which protrude below bottom edge of tip block.
 - (2) Remove nuts, washers, and spanwise weights from blade.
 - (3) Remove material from protruding spanwise weights using medium cut file.
 - (4) Install spanwise weights on blade, push weights toward top of blade, and check for protrusion below bottom edge of tip block using straight edge.
 - (5) Repeat step (3) and step (4) until all protruding spanwise weights are eliminated.
- d. Apply epoxy primer coating kit, Item 120, WP 1803 00, to reworked areas of weights.
- e. **QA** Install spanwise weights, washers, and nuts in exact location and stackup as when removed. **WHILE PUSH-ING SPANWISE WEIGHTS TOWARD TOP OF BLADE, TORQUE NUTS TO 262 - 288 INCH-POUNDS.**

TIP CAP RIB CRACKS REPAIR**NOTE**

Repairs on tip cap ribs will not affect weight or balance of tip cap.

1. **QA** Clip ribs as shown in (Figure 5, Sheet 1).
2. Touch up exposed clipped edges with primer coating, Item 243, WP 1803 00.

TIP BLOCK INSERT, RZA12487, REPAIR

1. Thread pilot from cutter into insert and slide cutter over pilot (Figure 5, Sheet 2). Cut serrations from top of insert and lockring. Do not allow cutter to remove metal from bottom of lockring counterbore.
2. Remove cutter and pilot. Lightly tap screw extractor into insert and unscrew insert from hole. Remainder of lockring will lift out as insert is removed.
3. If lockring does not come out with insert, remove by carefully prying out or collapsing it with a center punch.
4. Clear insert hole and threads of any foreign material and chips.
5. Using fingers only, start new insert into hole to make sure proper engagement of threads.
6. Using wrench, screw insert into hole until top is 0.005 to 0.010-inch below surface of tip block (Figure 5, Sheet 2).
7. **QA** Place lockring over top of insert. Line up lockring outer serrations with original serrations in sides of counterbore.
8. Using driving tool, drive lockring down flush with surface of tip block or not more than 0.010-inch below surface (Figure 5, Sheet 2).
9. Apply a coat of primer coating, Item 243, WP 1803 00, to seal lockring, insert, and counterbore.

TIP BLOCK INSERT, RZA12487, REPAIR - Continued

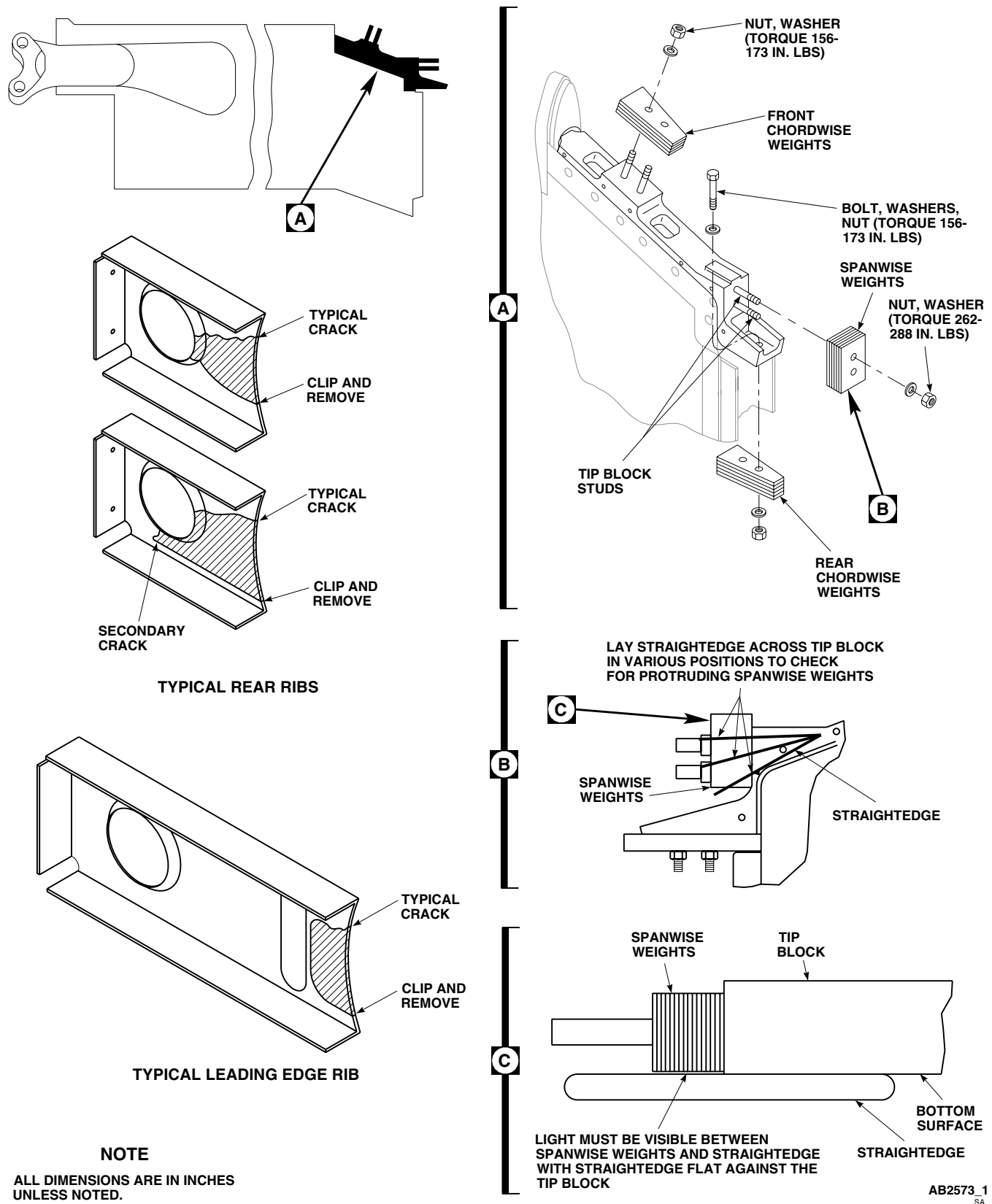


Figure 6. Tip Cap Repair Procedures. (Sheet 1 of 2)

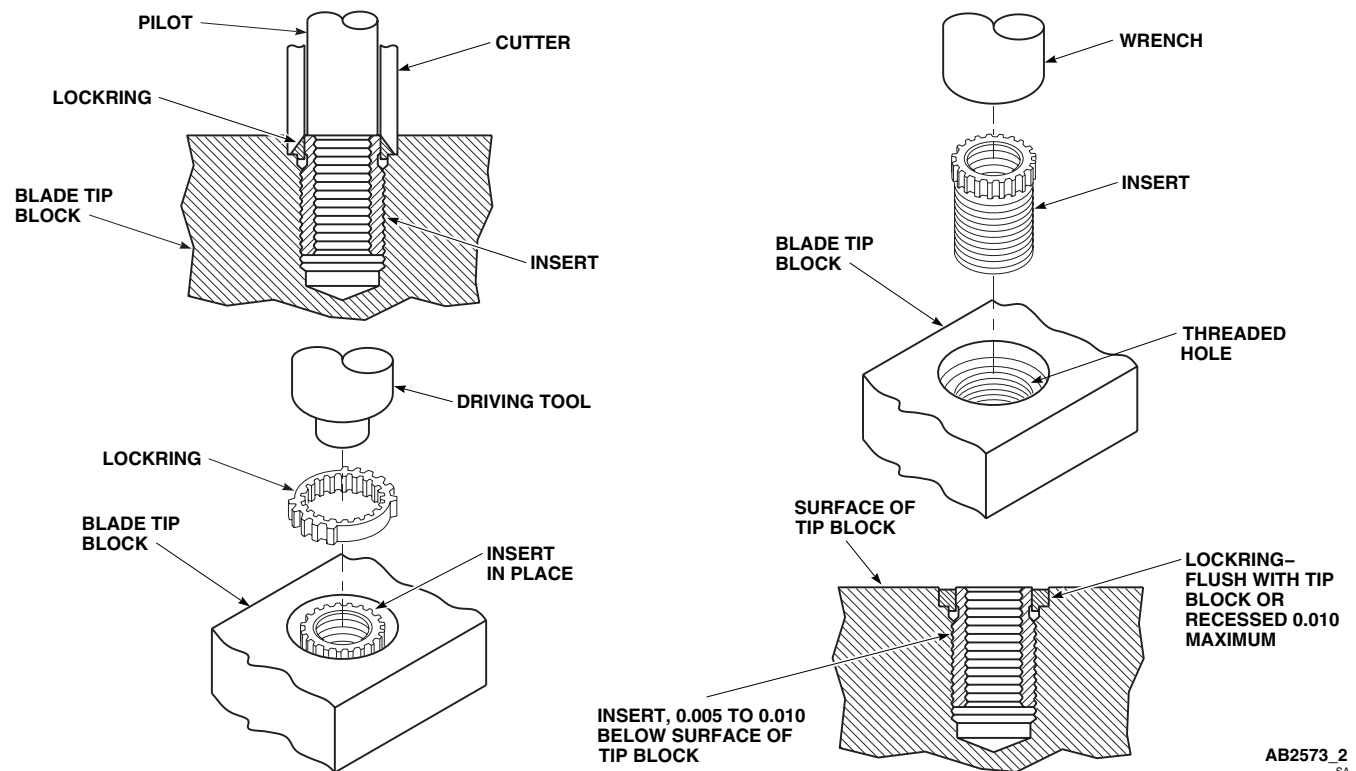
TIP BLOCK INSERT, RZA12487, REPAIR - Continued

Figure 6. Tip Cap Repair Procedures. (Sheet 2 of 2)

ALUMINUM WIRE MESH LIGHTNING PROTECTION REPAIR**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
- Get medical evaluation for skin or throat irritations and breathing problems.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wear protective cotton and plastic gloves when handling solvents or adhesives.
- Wash hands and face before eating or smoking.

1. Mask area around corrosion 1/2-inch larger than corrosion outline.

ALUMINUM WIRE MESH LIGHTNING PROTECTION REPAIR - Continued**CAUTION**

Damage to blade will result if blade is sanded excessively. Use only enough force to remove paint, primer, and corrosion. Do not sand through underlying fiberglass skin plies.

NOTE

- Sand only in immediate areas of pin holes.
- Up to 20% of chordwise width of wire mesh may be removed without affecting lightning protection.

2. Sand away paint, primer, and corrosion.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in prolonged contact with bonded joints or inside of blade. Do not allow solvent to seep into bonded joints or inside of blade.

3. Wipe sanded area with a clean, cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.
4. Treat exposed wire mesh with corrosion resistant coating, Item 111, WP 1803 00.
5. **QA** Prime repaired area with epoxy primer coating kit, Item 120, WP 1803 00. Refer to, PAINT TOUCHUP, in this work package.
6. Paint repaired area with epoxy coating kit, Item 118, WP 1803 00.
7. Apply another light coat of epoxy primer coating kit, Item 120, WP 1803 00. Refer to, PAINT TOUCHUP, in this work package.
8. Apply top coat of touchup paint. Refer to, PAINT TOUCHUP, in this work package.

ROOT END CAP REPAIR

WARNING

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
 - Get medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
1. Cracks up to 2 1/2-inch length in root end cap, with no other damage, may be patched using reinforced plastic patch exceeding the size of the crack by 1/2-inch in each direction.
 2. Outline area around damage with pressure sensitive adhesive tape, Item 333, WP 1803 00, 1/4-inch larger than patch size.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean area inside tape using machinery wiping towel, Item 344, WP 1803 00, moisten with technical acetone, Item 5, WP 1803 00.

CAUTION

Damage to blade will result if blade is sanded excessively. Sand only until yellow primer is removed.

4. Sand paint and yellow primer from inside tape using abrasive cloth, Item 78, WP 1803 00. **DO NOT SAND SKIN FIBERS.**

ROOT END CAP REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Wipe repair area with clean, machinery wiping towel, Item 344, WP 1803 00, moisten with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.
6. Prepare adhesive, Item 11, WP 1803 00, per manufacture's instructions.

NOTE

Pot life of mixture is 25 minutes at 75°F (24°C).

7. On clean smooth surface, lay 1 ply of satin cloth, Item 90, WP 1803 00. Saturate with mixed adhesive. Squeegee out extra adhesive. Trim to size needed.
8. Apply light even coating of mixed adhesive on repair area inside masking tape.
9. Center prepared patch over crack and press firmly into place.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

10. Clean up excess adhesive with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
11. Cover patch with plastic film peel ply and apply vacuum bag to area.
12. Cure adhesive for 24 hours at 75°F (24°C) or 1 hour at 160°F (71°C).
13. Remove vacuum bag and peel off film.
14. Lightly sand patch edges to blend with surrounding skin.
15. Remove masking tape and touch up paint. Refer to, PAINT TOUCHUP, in this work package.

PAINT TOUCHUP**WARNING**

- Volatile and toxic fumes occur when using solvents or mixing and using adhesives, causing both a fire and health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush them with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat and skin.
- Get medical evaluation for skin or throat irritations and breathing problems.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wear protective cotton and plastic gloves when handling solvents or adhesives.
- Wash hands and face before eating or smoking.

1. Refer to Table 6 for paint requirements.

CAUTION

Rotor blade imbalance will result if excessive paint is applied to rotor blade. The weight of the paint, although slight has large effects on rotor blade balance. Restrict the application of paint to avoid inducing balance problems

NOTE

Paint may be missing from rotor blades due to erosion. There is no limit to the areas of erosion that can be reprimed and repainted if the paint is missing.

2. When repainting eroded or repaired areas of a blade, apply primer and paint to those areas only. Spot-spray or touch up with small brush. Apply just enough paint for good coverage.

PAINT TOUCHUP - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Cloths wet with solvent should not be left lying on blade.

3. Wipe surface to be painted with clean, machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

CAUTION

Damage to blade will occur if blade is sanded excessively. Use only enough force to remove resin gloss. Removal of paint to allow cosmetic painting is not allowed.

4. Lightly sand surface to be painted to remove resin gloss, using abrasive paper, Item 208, WP 1803 00, or equivalent.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Rewipe surface with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and wipe dry before solvent evaporates.

Table 6. Paint Touchup.

AREA OR DETAILS	BLADE PART NO.	PAINT COLOR, TYPE, AND SPECIFICATION	NUMBER OF COATS
Blade, Overall	Blade Part No. 70150-09100-041	Epoxy Primer Coating Kit, Item 120, WP 1803 00	One
		Lacquer, Item 171, WP 1803 00	Two
Data Markings and Lift Line Stripes	Blade Part No. 70150-09100-041	Lacquer, Item 171, WP 1803 00	One
Blade, Overall	Blade Part No. 70150-09100-043	Epoxy Primer Coating Kit, Item 120, WP 1803 00	One
		Polyurethane Coating, Item 233, WP 1803 00	Two

PAINT TOUCHUP - Continued**Table 6. Paint Touchup. - Continued**

AREA OR DETAILS	BLADE PART NO.	PAINT COLOR, TYPE, AND SPECIFICATION	NUMBER OF COATS
Data Markings and Lift Line Stripes	Blade Part No. 70150-09100-043	Polyurethane Coating, Item 227, WP 1803 00	One

6. Mask around touchup area to reduce cleanup, using pressure sensitive adhesive tape, Item 325, WP 1803 00, and heavy paper (or equivalent protective covering).

NOTE

Do not apply epoxy primer coating kit when temperature is below 50°F (10°C). Recommended temperature is 75°F (24°C) to 86°F (30°C). Lower temperature will cause longer drying time. Apply primer by brush or spray directly over cleaned reinforced plastic surfaces.

7. Epoxy primer coating kit, Item 120, WP 1803 00, consists of two components packaged separately. Add component II to component I. Do not reverse mixing, or mix components from different manufacturers.
8. Pour required amount of component I (base) into clean container. Add equal amount, by volume, of component II (converter). Stir mixture slowly and thoroughly for at least 10 minutes.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

9. Thin mixture for spraying by adding one part, by volume, thinner to two parts, by volume, of mixed primer. Thinner is mixture of equal amounts of technical acetone, Item 5, WP 1803 00.
10. Mix thoroughly, strain mixture, and allow to stand 1 hour before using. Stir again just before using. Pot life of mixture is 8 hours at 75°F (24°C).
11. Spray light even coat of primer on prepared surface and allow to dry 1 hour at room temperature 75°F (24°C).

NOTE

- Minor feathering during repair painting into existing painted areas is acceptable.
 - When repainting erosion damage on the forward edge of the blade, a 3 inch wide feathering into painted areas is acceptable on both the upper and lower blade surfaces along the entire length of the blade. Maintain an even coverage along the rotor blade to evenly distribute the weight of the paint.
 - Cosmetic repainting of blades is not allowed.
12. **QA** Apply top coat of proper paint over primer. Refer to Table 6 for paint requirements.
13. Remove masking tape from repair area.
14. Lightly sand repair area using abrasive cloth, Item 78, WP 1803 00, to produce a uniform surface finish.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM**

MAIN ROTOR BLADE CUFF DECAL

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Wedge
Pencil, Grease
Rawhide Mallet, E-2200-D

Cloth, Cheesecloth, Item 85, WP 1803 00
Remover, Paint, Item 252, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (4)

Material/Parts

Adhesive, Item 14, WP 1803 00
Alcohol, Denatured, Item 48, WP 1803 00

References

WP 1803 00

REMOVE

1. Turn main rotor head until blade cuff is easy to reach.
2. Mark position of decal on blade cuff using grease pencil (Figure 1).
3. Using a nonmetallic wedge and rawhide mallet, remove old decal from cuff.

INSTALL

1. Remove old sealing compound from blade cuff using paint remover, Item 252, WP 1803 00. Clean blade cuff and back of new decal with denatured alcohol, Item 48, WP 1803 00, and cheesecloth cloth, Item 85, WP 1803 00. Dry areas cleaned.
2. Apply adhesive, Item 14, WP 1803 00, (sticky side) to decal. Smooth out tape to remove any trapped air.
3. **QA** Remove backing paper from tape and install decal on cuff in same position as decal just removed (Figure 1).
4. Apply firm pressure to decal to get a good contact of adhesive tape to decal and cuff.
5. Seal all edges of decal with sealing compound, Item 273, WP 1803 00. Allow to cure at room temperature for 24 hours.
6. Check area for cleanness and foreign material.

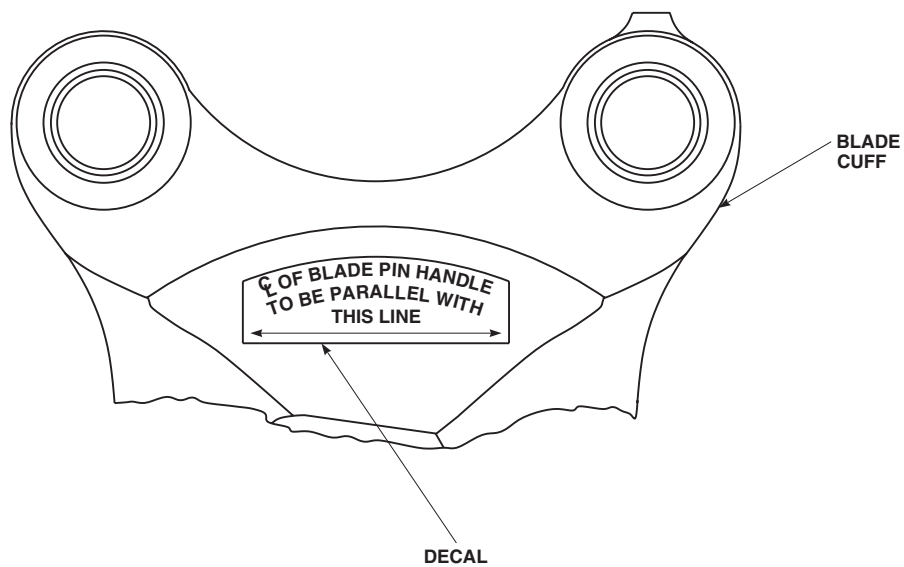
INSTALL - ContinuedAK1377
SA

Figure 1. Replace Main Rotor Blade Cuff Decal.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
MAIN ROTOR BLADE RECEIVER ASSEMBLY
This WP supersedes WP 0578 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Installation Tool, 70150-09148-041T046
Torque Wrench, 30 - 200 in. lbs.

Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (4)

References

TM 1-1500-204-23
WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

REMOVE

1. Using installation tool, remove receiver assembly from main rotor blade (Figure 1, Detail A).

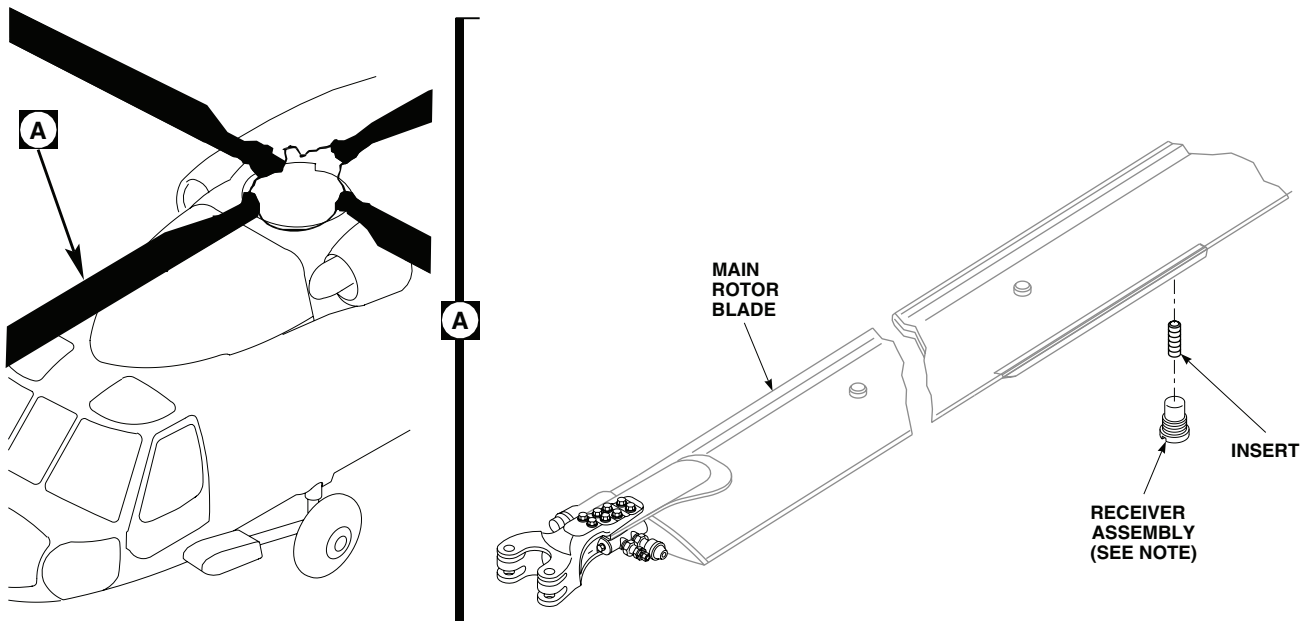
WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean area using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

INSPECT

1. Check receiver block on main rotor blade for cracks. **NO CRACKS ALLOWED.** Replace main rotor blade with cracked receiver block.
2. Check receiver block on main rotor blade for disbonding from blade. **NO DISBONDING ALLOWED.** Replace main rotor blade with disbonded receiver block.
3. Visually check threaded insert for damage or missing threads. **NO DAMAGED OR MISSING THREADS ALLOWED.** If damaged or missing threads are found, do this:
 - a. Remove threaded insert (TM 1-1500-204-23) (Figure 1, Detail A).
 - b. Inspect receiver block threads for damage. **NO DAMAGE ALLOWED.** Replace main rotor blade if damage is found.
 - c. Install threaded insert into receiver block (TM 1-1500-204-23).

INSPECT - Continued**NOTE**

TORQUE UNTIL FLUSH WITH MAIN ROTOR BLADE OUTSIDE SURFACE.

AA9528
SA

Figure 1. Replace Main Rotor Blade Receiver Assembly.

INSTALL

1. **QA** Using installation tool, install receiver assembly into main rotor blade (Figure 1, Detail A). **MINIMUM RUNNING TORQUE IS 70 INCH-POUNDS.** Inspect threaded insert if minimum running torque is not met. Refer to, INSPECT, in this work package.
2. Final torque receiver assembly until flush with main rotor blade outside surface.
3. Apply sealing compound, Item 273, WP 1803 00, to OD of receiver assembly.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
MAIN ROTOR BLADE TRIM TAB ADJUSTMENT

INITIAL SETUP:

References

TM 1-6625-724-13&P

Refer to TM 1-6625-724-13&P.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
MAIN ROTOR BLADE TRACK AND BALANCE

INITIAL SETUP:

References

TM 1-6625-724-13&P

Refer to TM 1-6625-724-13&P.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM

BLADE INDICATION METHOD (BIM®) PRESSURE INDICATOR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Nonmetallic Scraper
 Pin Arm, AN8514-4
 Spanner Wrench, AN8514-1
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Towel, Machinery Wiping, Item 344, WP 1803 00

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0582 00

WP 0583 00

WP 1624 00

WP 1803 00

Material/Parts

Leak Detection Fluid, Item 175, WP 1803 00
 Petrolatum, Item 216, WP 1803 00
 Sealing Compound, Item 257, WP 1803 00

REPLACE BLADE INDICATION METHOD (BIM®) PRESSURE INDICATOR
Remove**NOTE**

If any repair is done because the spar pressure is below limits, a whirl test is required.

1. Release pressure in blade spar by removing cap from air valve and loosening outer hex nut 2 1/2 turns while keeping inner hex nut and air valve adapter from turning (Figure 1).
2. Install wrench on body only and remove pressure indicator and packing using adjustable spanner wrench with pin arm installed. Do not remove adapter in which indicator was installed.
3. Immediately install plug in opening in blade, to prevent air from entering blade spar, and install cap over opening in indicator.

Inspect

Perform a torque check on indicator adapter. **TORQUE SHOULD BE 176 - 194 INCH-POUNDS.**

Install

1. **QA** Lubricate new packing with petrolatum, Item 216, WP 1803 00. Install packing on pressure indicator (Figure 1).
2. Remove cap from replacement pressure indicator and remove plug from opening in blade.
3. **QA** Immediately install pressure indicator and new packing (Figure 1). **INSTALL WRENCH ON BODY ONLY AND TORQUE INDICATOR TO 105 - 115 INCH-POUNDS USING ADJUSTABLE SPANNER WRENCH WITH PIN ARM INSTALLED.**
4. Service blade spar to normal pressure (WP 1624 00).
5. Check indicator packing for leakage by applying a test liquid such as leak detection fluid, Item 175, WP 1803 00, or a mild liquid detergent over and around base of indicator and watching for bubbles to form.

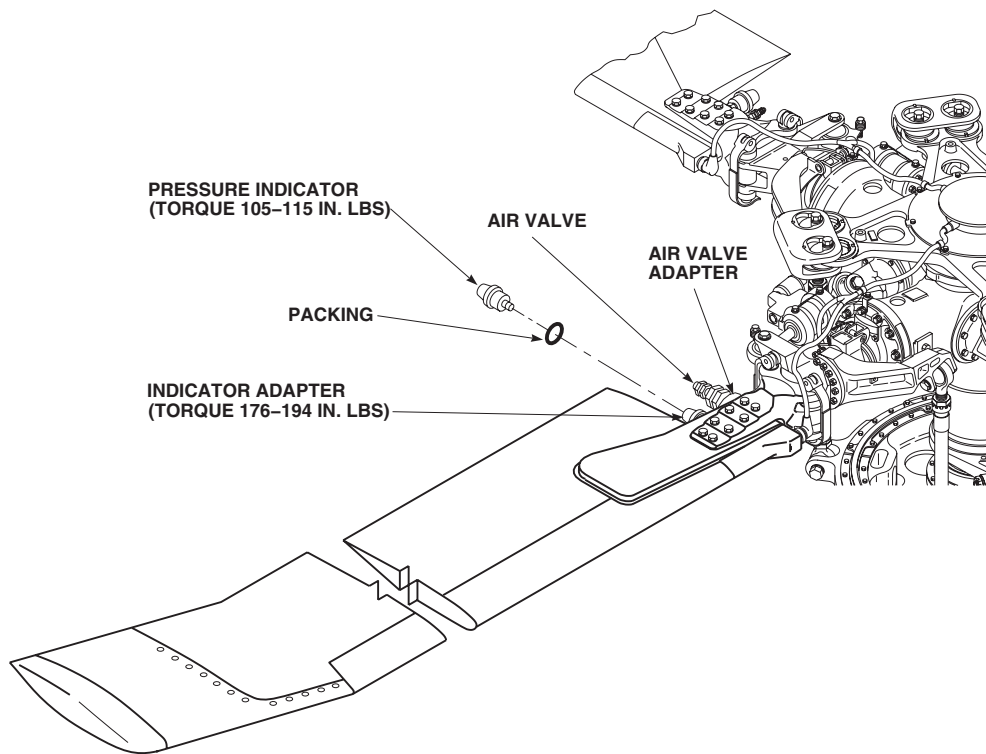
REPLACE BLADE INDICATION METHOD (BIM®) PRESSURE INDICATOR - ContinuedAA9368
SA

Figure 1. Replace Blade Indication Method (BIM®) Pressure Indicator.

6. When check is completed and before test liquid dries, clean area thoroughly with clean, fresh water to remove all traces of liquid; then dry blade using machinery wiping towel, Item 344, WP 1803 00.
7. Test indicator for proper operation (WP 1624 00).
8. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire indicator to indicator adapter and to air valve adapter.

REPLACE BLADE INDICATION METHOD BIM® PRESSURE INDICATOR LEVER**Remove**

Remove screw holding manual test lever to base of indicator (Figure 2).

Install

1. **QA** Remove lever. Place manual test lever on base of indicator and install screw (Figure 2).
2. Test indicator (WP 1624 00).
3. Check area for cleanness and foreign material.

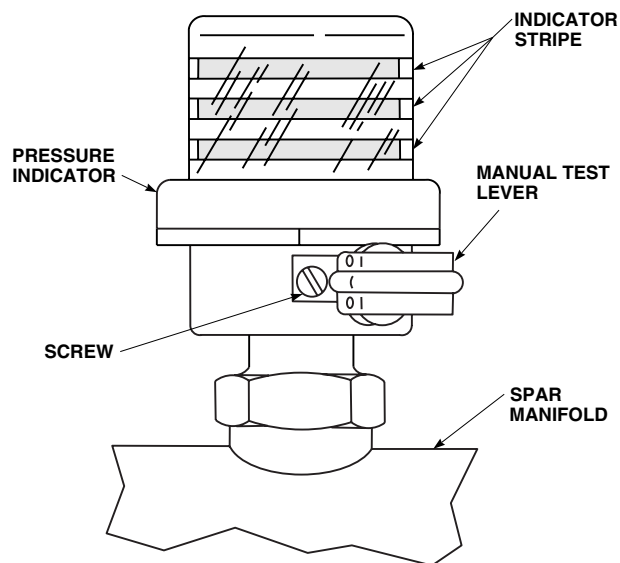
REPLACE BLADE INDICATION METHOD BIM® PRESSURE INDICATOR LEVER - ContinuedAK1384
SA

Figure 2. Replace Blade Indication Method (BIM®) Pressure Indicator Lever.

REPLACE BLADE INDICATION METHOD BIM® PRESSURE INDICATOR/AIR VALVE ADAPTER**Remove**

1. Remove pressure indicator. Refer to, REPLACE BLADE INDICATION METHOD (BIM) PRESSURE INDICATOR, in this work package. Remove air valve (WP 0582 00).
2. Remove adapter (Figure 3, Detail A).
3. Remove and discard packing.

Clean/Inspect**CAUTION**

Damage to blade spar will result if metallic tools are used to remove sealing compound from adapter cavity. Use only nonmetallic tools to remove sealing compound.

NOTE

All traces of sealing compound do not need to be removed from cavity. Only enough sealing compound should be removed to provide good seal.

1. Using locally made nonmetallic scraper, remove enough sealing compound from adapter cavity to provide good seal.

REPLACE BLADE INDICATION METHOD BIM® PRESSURE INDICATOR/AIR VALVE ADAPTER - Continued

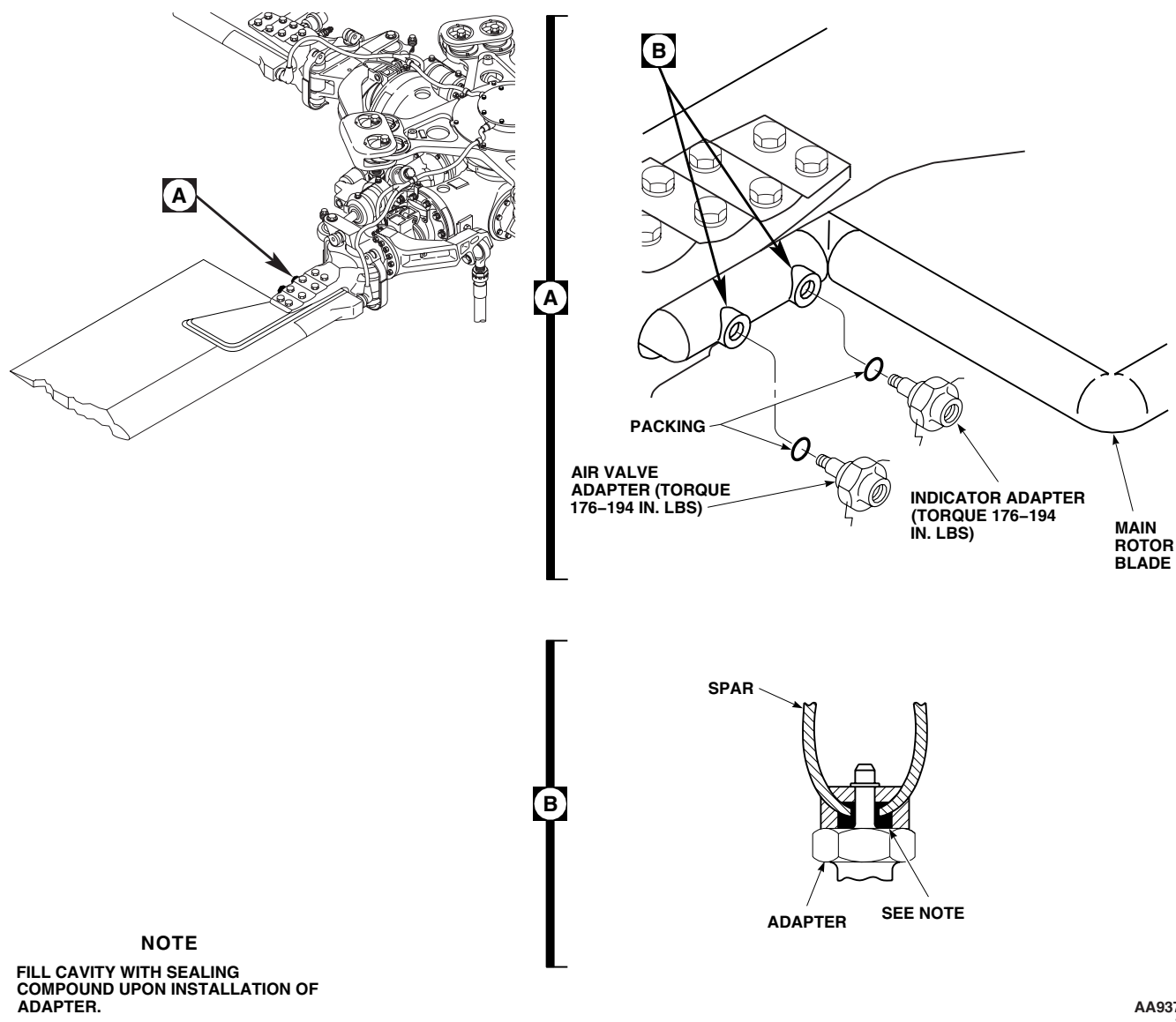


Figure 3. Replace Blade Indication Method (BIM®) Pressure Indicator/Air Valve Adapter.

CAUTION

Damage to blade will result if debris is not removed from housing prior to installing new pressure indicator/air valve adapter. Make sure all debris is removed from housing.

2. Vacuum debris from housing.

Install

1. Fill cavity of housing with sealing compound, Item 257, WP 1803 00 (Figure 3, Detail B).
2. **QA** Install adapter in housing (Figure 3, Detail A). **TORQUE ADAPTER TO 176 - 194 INCH-POUNDS.**
3. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire adapter to other adapter.

REPLACE BLADE INDICATION METHOD BIM® PRESSURE INDICATOR/AIR VALVE ADAPTER - Continued

4. Install pressure indicator. Refer to, REPLACE BLADE INDICATION METHOD (BIM) PRESSURE INDICATOR, in this work package. Remove air valve (WP 0582 00).
5. If required, do a main rotor blade One Hour Whirl Test (One Hour) (WP 0583 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM**

MAIN ROTOR BLADE AIR VALVE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Leak Detection Fluid, Item 175, WP 1803 00
Packing, MS29512-07
Towel, Machinery Wiping, Item 344, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

References

WP 0583 00
WP 0584 00
WP 1624 00
WP 1803 00

REMOVE**NOTE**

If any repair is done because the spar pressure is below limits, a whirl test is required.

1. Release pressure in blade spar by removing cap from air valve and loosening outer hex nut 2 1/2 turns while keeping inner hex nut and air valve adapter from turning (Figure 1, Detail A).

NOTE

- Have new replacement air valve ready for installation prior to removing old air valve.
 - Make sure that only packing, (MS29512-07), is installed on new replacement air valve. Packing which comes with new replacement air valve, is not to be used.
2. Remove air valve and packing keeping air valve adapter from turning. Discard old packing. Do not remove adapter in which valve was installed. Immediately install replacement air valve and packing, (MS29512-07), to prevent air from entering blade spar.

INSPECT

Perform torque check on air valve adapter. **TORQUE SHALL BE 176 - 194 INCH-POUNDS.**

INSTALL**NOTE**

Make sure that only packing, (MS29512-07), is installed on new replacement air valve. Packing which comes with new replacement air valve, is not to be used.

1. **QA** Install new air valve and packing, (MS29512-07). **TORQUE AIR VALVE TO 105 - 115 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire valve to adapter and valve adapter to pressure indicator adapter (Figure 1, Detail A).

INSTALL - Continued

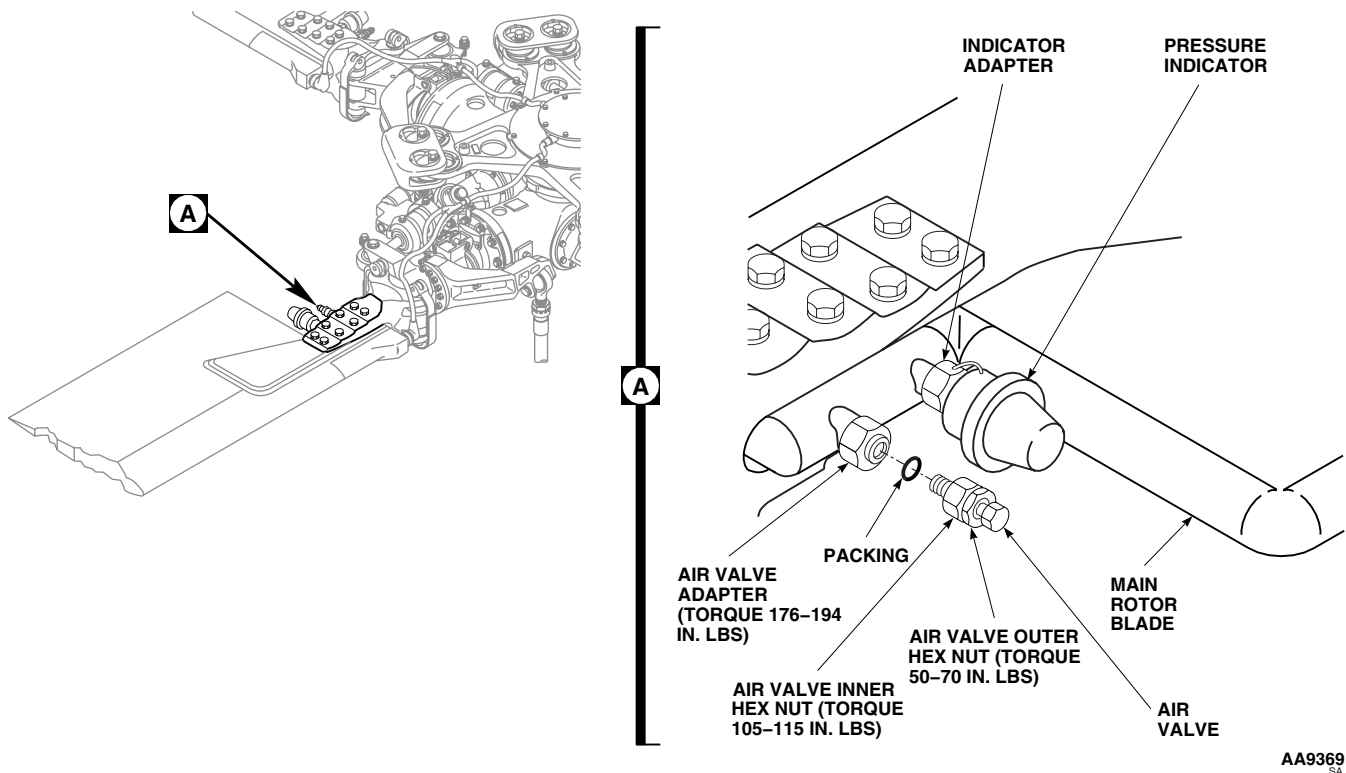


Figure 1. Replace Main Rotor Blade Air Valve.

2. **IF BLADE SPAR WILL NOT BE SERVICED IMMEDIATELY, TORQUE AIR VALVE OUTER HEX NUT TO 50 - 70 INCH-POUNDS TO PREVENT AIR AND/OR MOISTURE FROM ENTERING SPAR.**
3. If air valve was replaced because of an unsafe indication on pressure indicator, check valve for leakage (WP 0584 00).
4. If air valve was replaced because of an unsafe indication on pressure indicator, do a spar leakage test (WP 0584 00).

NOTE

Make sure air valve outer hex nut is lockwired to base of valve when servicing is finished.

5. Service blade spar to normal pressure (WP 1624 00).
6. If, when air valve was replaced, spar pressure was within acceptable limits, check air valve for leakage by applying a test liquid, such as leak detection fluid, Item 175, WP 1803 00, or a mild liquid detergent over and around entire air valve and watching for bubbles to form.
7. When check is completed and before test liquid dries, clean area thoroughly with clean, fresh water to remove all traces of liquid; then dry blade using machinery wiping towel, Item 344, WP 1803 00.
8. If required, do a main rotor blade One Hour Whirl Test (WP 0583 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
MAIN ROTOR BLADE WHIRL TEST (ONE HOUR)
This WP supersedes WP 0583 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

TM 1-1520-237-10

WP 0576 00

WP 1624 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

MAIN ROTOR BLADE WHIRL TEST**NOTE**

- Whirl test is a final check on the integrity of a rotor blade spar that has had a loss of spar pressure. This test is made after all other corrective actions have been taken.
- Maximum pressure in spar not to exceed 18 psig.

1. Check pressure indicator for proper operation (WP 1624 00).
2. Check spar pressure to see that it is within limits shown in Table 1 (WP 1624 00). Pressure must be at "MAXIMUM WHEN SERVICING BLADE" pressure for actual measured spar temperature. Service to that pressure if required. If pressure is within limits, record pressure to nearest 0.1 psi, record temperature within 1°F (-17°C), and proceed with test.
3. If pressure is not within limits, service blade to "MAXIMUM WHEN SERVICING BLADE" pressure for actual measured spar temperature. Record pressure to nearest 0.1 psi, and temperature within 1°F (-17°C), and proceed with test.
4. Install main rotor blade, if required (WP 0576 00).
5. With qualified personnel at controls, ground run helicopter for one half-hour period at 100% RPM, low collective stick, and neutral cyclic stick and directional control (TM 1-1520-237-10).
6. At the end of one half-hour period, check pressure indicator. **PRESSURE INDICATOR STRIPES SHALL BE YELLOW.** If indication is not within limits, send blade to AVIM for repair.
7. Repeat step 5.
8. At the end of one half-hour period, measure spar pressure and temperature. Corrected spar pressure is "MAXIMUM WHEN SERVICING BLADE" pressure obtained in Table 1 at the actual measured spar temperature at the end of test.
9. Subtract measured pressure from corrected pressure. **PRESSURE DIFFERENCE SHALL NOT BE GREATER THAN 0.25 PSI.** If pressure is not within limits, send blade to AVIM for repair.

MAIN ROTOR BLADE WHIRL TEST - Continued**Table 1. Example.**

	PRESSURE	TEMPERATURE	
	(psi)	°F	°C
Initial Reading:	18.0	73.0	22.8
Reading after 24 hours:	17.0	63.0	17.2
Temperature change is 10.0°F(5.6°C) decrease correction to be added to reading: $10.0 \times 0.07 = 0.70$ psi (for °F) or $5.6 \times 0.126 = 0.70$ psi (for °C) Therefore the actual corrected pressure in the spar is: $17.0 + 0.70 = 17.7$ psi NOTE For an increase in temperature, the same procedure is followed, but the correction is subtracted.			

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****MAIN ROTOR BLADE SPAR LEAKAGE TEST**

This WP supersedes WP 0584 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts

Leak Detecting Fluid, Item 175, WP 1803 00

Nitrogen, Item 199, WP 1803 00

Towel, Machinery Wiping, Item 344, WP 1803 00

References

WP 0082 00

WP 0583 00

WP 1624 00

WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

TEST COMPONENTS FOR LEAKAGE**NOTE**

- Perform these procedures following an unsafe BIM® indication.
 - The procedures outlined in this task are to be used with and in the order of, the instructions in the AVIM procedures in WP 0082 00.
 - This procedure contains instructions for testing for leakage at the pressure indicator valve and packing, air valve, root end plate seal, and cuff/spar bolts.
 - Maximum pressure in spar not to exceed 18 psig.
1. Pressurize blade spar to maximum pressure with nitrogen, Item 199, WP 1803 00, using procedures given in WP 1624 00.
 2. Make sure that air valve outer hex nut on air valve is torqued and lockwired correctly (WP 1624 00).

NOTE

- A significant leak should be readily determined by this test. However, a slow leak may be more difficult to find. Should there be any doubt that the source of leak has been found, send blade to depot for further tests.
 - To prevent false indications of leakage, make certain that bubbles formed during application of liquid are allowed to settle out.
3. Using test liquid such as leak detecting fluid, Item 175, WP 1803 00, or alternate, check the following areas for leakage:
 - a. Spread test liquid over and around base of pressure indicator and around test valve under handle.
 - b. Spread test liquid over and around air valve, especially stem area.
 - c. Spread test liquid over and around entire edge of root endplate.

TEST COMPONENTS FOR LEAKAGE - Continued

- d. Spread test liquid over and around each bolt head and nut that secures cuff to spar.
4. Watch carefully at all of these areas for bubbles to form.
5. Clean area thoroughly with clean, fresh water to remove all traces of liquid. Dry blade using machinery wiping towel, Item 344, WP 1803 00.
6. If no leakage is found, do one hour whirl test (WP 0583 00).
7. Service blade to "MAXIMUM WHEN SERVICING BLADE" pressure for actual measured spar temperature (WP 1624 00).
8. If reparable leakage is found, repeat step 1. through step 7. after completion of repair.
9. If nonreparable leakage is found, forward the damaged blade to depot. A blade component record card entry should be made whenever a blade is removed from service stating reason for removal.
10. If a blade is to be forwarded to depot it must have with it a statement that describes accurately the specific maintenance action, or actions that were performed. Statement should include spar pressure that existed when blade was inducted and when it left and specific reason for forwarding blade to depot. All data, including identification of original problem, should also be with blade.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ROTOR SYSTEM

TAIL ROTOR BLADES

This WP supersedes WP 0585 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Airframe Repairer's Toolkit, SC 5180-99-B02
 Caliper Set, Outside Micrometer, GGG-C-105
 C-Clamps (4 inch), GGG-C-406
 Countersink
 Drill Bit, No. 50
 Drill Bit, No. 60
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Heat Gun, LFT 500
 Micrometer GGG-C-105
 Plastic Tool, Thin
 Pneumatic Blind Riveter, BP-350RP
 Pressure Strips, Wooden or Metal
 Single Cut Mill File, GGG-F-325
 Spring Scale, AAS-S133
 Tail Rotor Blade Installation Checking Shim "A",
 Locally-Made, Figure 208, WP 1805 00
 Tail Rotor Blade Installation Checking Shim "B",
 Figure 112, WP 1805 00
 Tail Rotor Maintenance Crane, (Alternate for Hoist)
 70700-20413-041
 Toothpick
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, Dial, 0 - 75 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.
 Wooden Wedges

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 11, WP 1803 00
 Adhesive, Item 15, WP 1803 00
 Adhesive, Item 47, WP 1803 00
 Adhesive Primer, Item 23, WP 1803 00
 Alcohol, Denatured, Item 48, WP 1803 00
 Applicator, Disposable, Item 53, WP 1803 00
 Brush, Acid Swabbing, Item 57, WP 1803 00
 Cloth, Abrasive, Item 78, WP 1803 00
 Cloth, Abrasive, Item 80, WP 1803 00
 Cloth, Abrasive, Item 81, WP 1803 00

Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00
 Cloth, Cheesecloth, Item 85, WP 1803 00
 Cloth, Satin, Item 90, WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Epoxy Coating Kit, Item 118, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Isopropyl, Alcohol Technical, Item 170, WP 1803 00
 Lacquer, Item 171, WP 1803 00
 Lubricant, Solid Film, Item 180, WP 1803 00
 Paper, Abrasive, Item 209, WP 1803 00
 Paper, Abrasive, Item 211, WP 1803 00
 Plastic Film
 Plastic Sheet, Item 221, WP 1803 00
 Polyurethane Coating, Item 230, WP 1803 00
 Polyurethane Coating, Item 233, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Shim, 70102-11109-101
 Shim, 70102-11109-102
 Strap Tiedown, Electrical, Item 310, WP 1803 00
 Stirring Stick, Beverage, Item 300, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 222,
 WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 232,
 WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 325,
 WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 333,
 WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 334,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
 Assistant - (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

MIL-C-46168
 MIL-L-19538

TM 1-1520-237-MTF	WP 0590 00
TM 1-1520-265-23	WP 0605 00
TM 1-6625-724-13&P	WP 0607 00
TM 55-1500-345-23	WP 0608 00
WP 0136 00	WP 0609 00
WP 0378 00	WP 0610 00
WP 0381 00	WP 1039 00
WP 0545 00	WP 1246 00
WP 0546 00	WP 1590 00
WP 0586 00	WP 1692 00
WP 0587 00	WP 1803 00
WP 0588 00	WP 1805 00

REMOVE

1. Remove pitch control rods (WP 0586 00).
2. Remove pitch beam (WP 0587 00).
3. If tail rotor maintenance crane will be used, erect crane (WP 1590 00).
4. Protect pitch control shaft threads with pressure sensitive adhesive tape, Item 325, WP 1803 00.
5. If blade deice system is installed, disconnect blade deice electrical connector (WP 1246 00).

CAUTION

Damage to outboard retention plate will result if wrenches are used to remove bolts.
Use deep well socket when removing outboard retention plate bolts.

NOTE

- If tail rotor blades being removed are to be reinstalled on same tail gear box, and color bands have worn off, use a grease pencil and put index marks on blades, retention plates, pitch control rods, and pitch beam so they are reinstalled in the same clockwise and inboard/outboard position, to minimize rebalancing of tail rotor.
 - If balance weights are installed with bolt through both retention plates, note location of hole on outboard retention plate. Keep all washers, bolt, and nut together for that specific hole. Do not remove balance weights installed with screw and nut in one retention plate, or on tail rotor blade.
6. Remove 12 bolts attaching outboard retention plate to inboard retention plate. If installed, remove balance bolt, washers, and nut. Remove outboard retention plate (Figure 1, Sheet 1).

NOTE

Keep bolts, washers, and nuts for reinstallation.

7. Using hoisting sling, remove outboard and inboard tail rotor blade individually as follows:

REMOVE - Continued

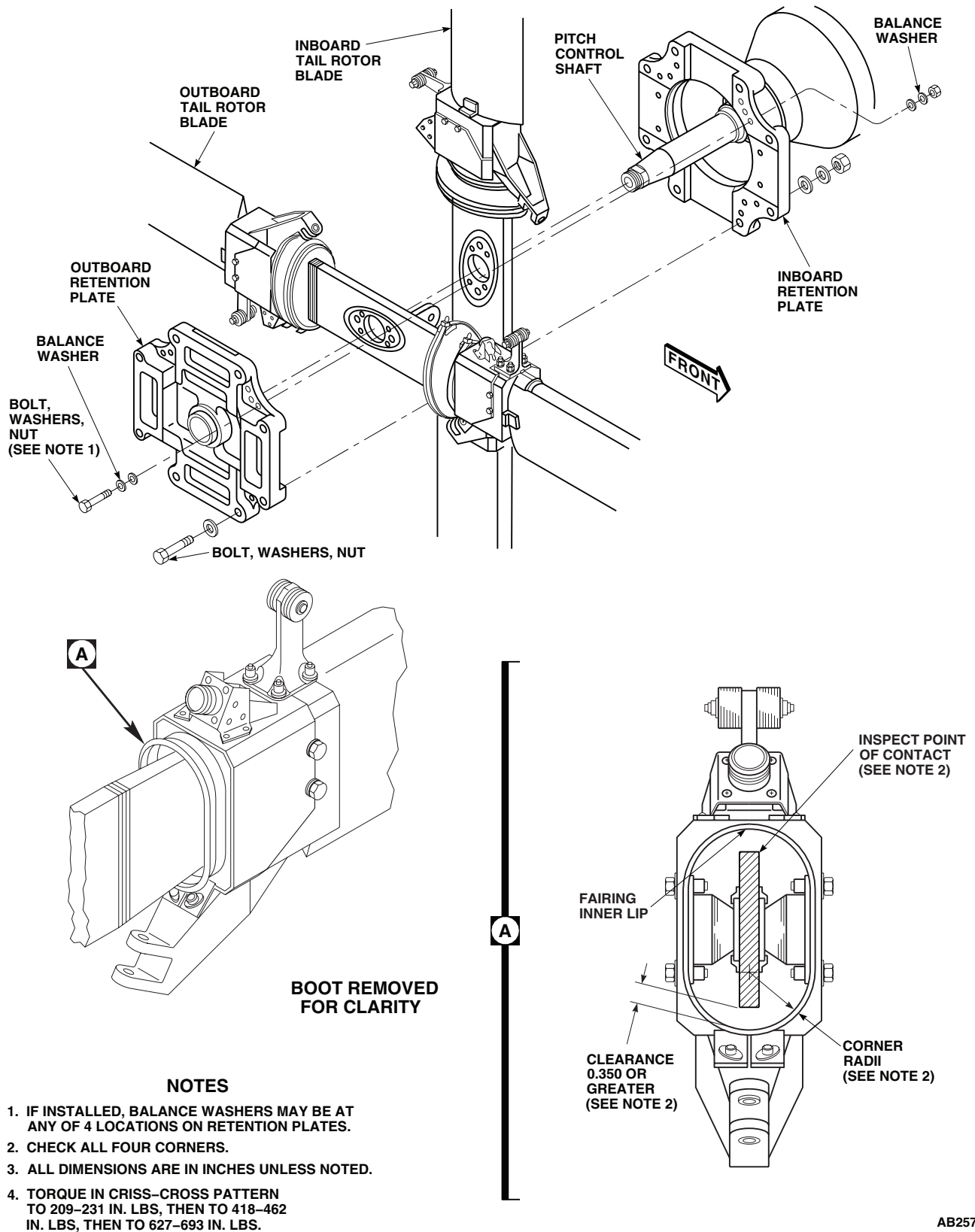


Figure 1. Replace Tail Rotor Blades. (Sheet 1 of 3)

REMOVE - Continued

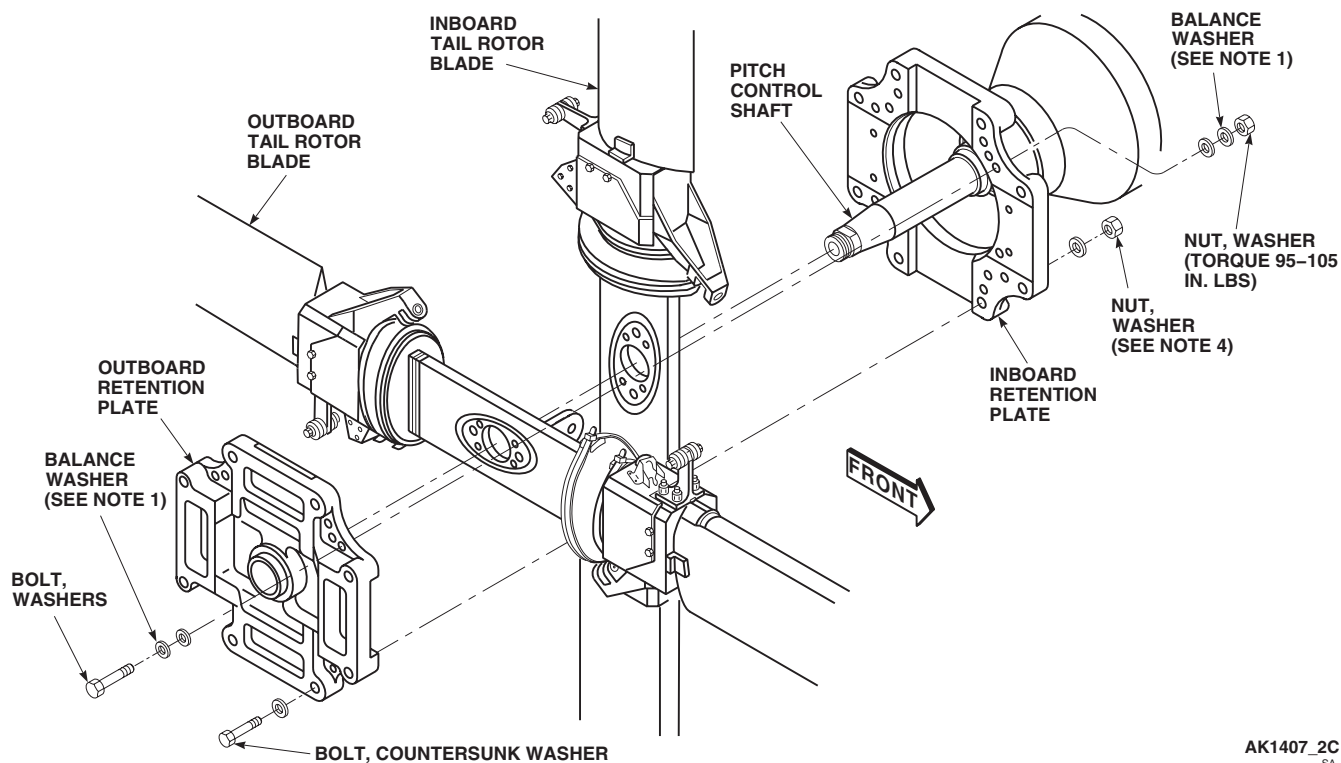


Figure 1. Replace Tail Rotor Blades. (Sheet 2 of 3)

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

a. Attach hoisting sling to tail rotor maintenance crane or to overhead hoist, with enough capacity to support blade.

CAUTION

Damage to tail rotor blade and pitch control shaft will result if excessive force is used to remove blades. Guide tail rotor blade carefully over pitch control shaft.

b. Remove tail rotor blade from pitch control shaft and lower blade on adapter.

8. Remove hoisting sling.

9. Place hoisting sling straps around inboard tail rotor blade. Attach hoisting sling to hoist and adjust straps. Remove and lower tail rotor blade on adapter.

INSPECT TAIL ROTOR BOOT

Inspect tail rotor boot for cuts, tears, breaks, and dry rot. **NONE ALLOWED.** Replace damaged boot (WP 0588 00).

INSPECT TAIL ROTOR BOOT - Continued

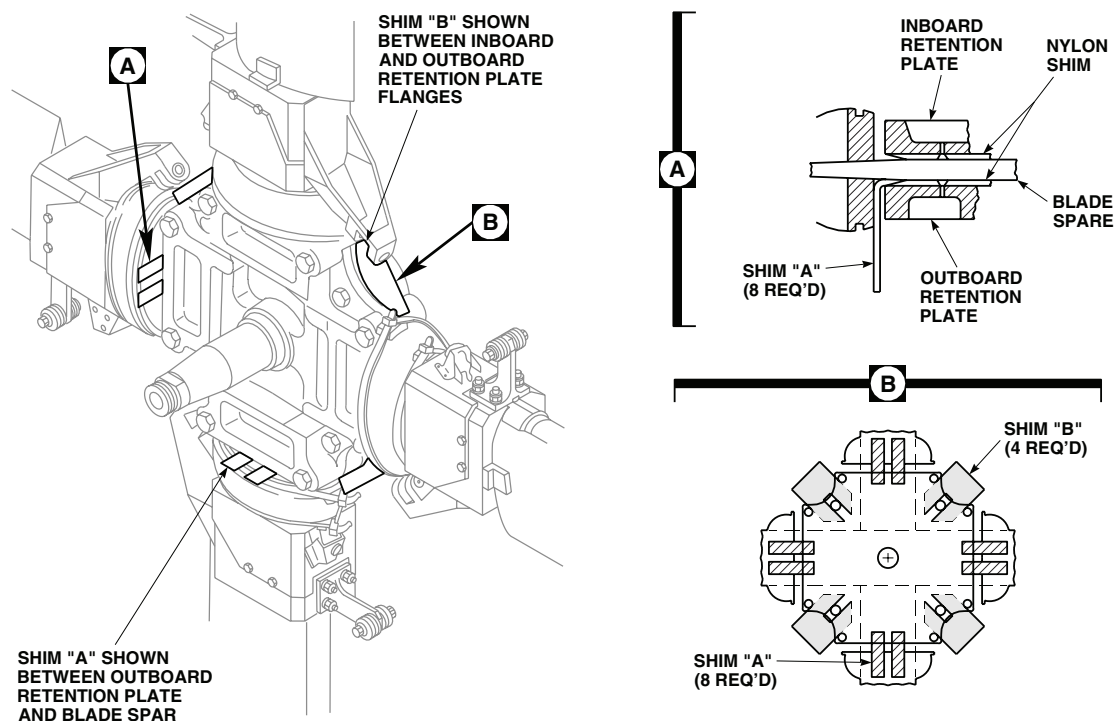
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Figure 1. Replace Tail Rotor Blades. (Sheet 3 of 3)

NOTE

Certain work performed on tail rotor system requires a balance check to be completed prior to flight. This includes outboard retention plate bolts removed or loosened, folding of tail rotor blades, tail rotor blades removed, skin repairs to tail rotor blades, and any other work accomplished to tail rotor system which may affect tail rotor balance.

1. Do a tail rotor balance check (TM 1-6625-724-13&P).

INSPECT TAIL ROTOR BLADE DEICE BRACKET ASSEMBLY

Inspect blade deice bracket assembly for cracks. **NONE ALLOWED.** Replace cracked blade deice bracket assembly (WP 0590 00).

INSPECT TAIL ROTOR BLADE SPAR

1. Remove one of the tail rotor boot ties and roll back tail rotor boot (WP 0588 00).
2. Measure clearance between fairing inner ridge of the tie wrap channel and spar.
3. Record minimum clearance at leading and trailing edges. **MINIMUM CLEARANCE MUST NOT BE LESS THAN 0.350-INCH.**
4. If clearance between fairing inner ridge and spar is less than the minimum acceptable limits, rework fairing (WP 0608 00).

INSPECT TAIL ROTOR BLADE BOOT SUPPORTS

1. Inspect boot supports for displacement from their proper position. **NO DISPLACEMENT ALLOWED.** Replace tail rotor blade boot supports if displacement is found (WP 0610 00).
2. Inspect boot supports for looseness using hand pressure. **NO LOOSENESS ALLOWED.** Replace tail rotor blade boot supports if looseness is found (WP 0610 00).
3. Inspect inboard and outboard boot support adhesive fillets adjacent to spar for cracks. Repair cracks in adhesive per adhesive injection repair instructions (ADHESIVE INJECTION REPAIR).
4. Apply sealing compound, Item 273, WP 1803 00, to inboard and outboard adhesive fillets as follows:
 - a. Clean inboard and outboard adhesive fillets using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical isopropyl alcohol, Item 170, WP 1803 00.
 - b. Apply sealing compound, Item 273, WP 1803 00, to adhesive fillets and edge area (Figure 2).
 - c. Make sure sealing compound beads do not interfere with retention plate during tail rotor blade installation.

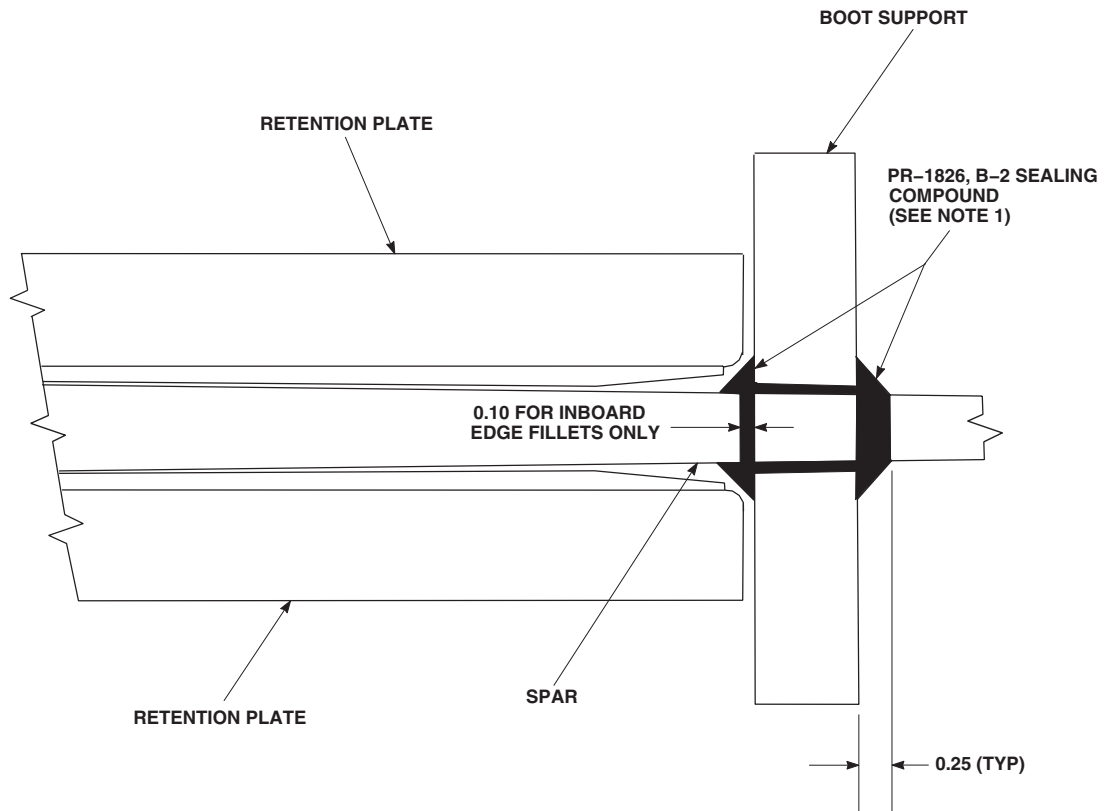
INSPECT TAIL ROTOR PIVOT BEARING

1. Using flashlight and mirror, check pivot bearing for delaminations, and bearing retainer for disbond as follows:
 - a. Check pivot bearings for separations within rubber layers. **IF SEPARATION IS MORE THAN 1/8-INCH DEEP AROUND 1/4 OF CIRCUMFERENCE, REPLACE PIVOT BEARING (WP 0607 00).**
 - b. Check pivot bearings for eraser type extrusion. **NO EXTRUSIONS ALLOWED.** Replace pivot bearing if extrusions are found (WP 0607 00).
 - c. Check pivot bearings for rubber to shim separation. **NO RUBBER TO SHIM SEPARATIONS ALLOWED. BENT SHIMS ARE ACCEPTABLE.** Replace pivot bearings if shim separations are found (WP 0607 00).
 - d. **NO DISBONDING OF RETAINERS ALLOWED.** If any retainer is disbonded (four retainers per blade), replace retainer (WP 0609 00).
 - e. If no visible disbond of retainer or delamination of bearing is detected, press directly on tail rotor horn over pivot bearing attachment bolts on each side of blade. Check for looseness of horn relative to spar. **NO LOOSENESS ALLOWED.** If any looseness is found, remove pivot bearings and inspect pivot bearings and retainers (WP 0607 00).
2. Using flashlight and mirror, inspect pivot bearing base plate for cracks. Typical cracks run spanwise from near edge of base plate towards tip of blade. **NO CRACKS ALLOWED.** If crack is suspected, remove and inspect pivot bearing (WP 0607 00).
3. Using flashlight and mirror check inside surfaces of pitch horn, pivot bearing retainer, and pivot bearing base plates for missing or damaged primer. **NONE ALLOWED.** If primer is damaged or missing, remove pivot bearing and replace primer (WP 0607 00).
4. Secure tail rotor boot (WP 0588 00).

INSPECT PITCH CHANGE HORN

1. Visually inspect pitch change horn for cracks. If cracks are suspected, inspect pitch change horn using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace blade if cracks are found.
2. Check pitch change horn for nicks and gouges. Remove raised edges and blend out damage with abrasive wheel, Item 350, WP 1803 00. **REPLACE TAIL ROTOR BLADE IF:**

INSPECT PITCH CHANGE HORN - Continued



NOTES

1. MAKE SURE SEALING COMPOUND BEAD DOES NOT INTERFERE WITH RETENTION PLATES.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 2. Application of Sealing Compound to Tail Rotor Boot Support.

INSPECT PITCH CHANGE HORN - Continued**NOTE**

Torque tube attachment area is where horn attaches to blade.

- a. **CORNER DAMAGE IN HORN AFTER BLENDING IS GREATER THAN 0.020-INCH DEEP OR 0.010-INCH DEEP IN LUG AND TORQUE TUBE ATTACHMENT AREAS.**
 - b. **SURFACE DAMAGE IN HORN AFTER BLENDING IS GREATER THAN 0.010-INCH DEEP.**
 - c. **NICKS AND GOUGES IN HORN MATING SURFACES ARE GREATER THAN 0.010-INCH DEEP, OR COVER MORE THAN 10% OF MATING SURFACE.**
3. Check pitch change horn for corrosion. **REPLACE BLADES IF HORN HAS CORROSION OVER 0.010-INCH DEEP, OR IF CORRODED AREAS ARE LESS THAN 1 INCH APART.**
 4. Check fairing for edge disbonds. Rebond edges with sealing compound, Item 273, WP 1803 00.
 5. Check bushing bore and flange surface for corrosion. Remove corrosion, inspect and replace bushings as necessary (WP 0605 00).
 6. Refer to WP 0545 00 for damage limits for other components of tail rotor system.
 7. Inspect tail rotor blade for damage (WP 0546 00).

INSPECT/REPAIR RETENTION PLATE BOLTS

1. Visually inspect solid film lubricant on outboard retention plate bolt. Bolt shank must be completely covered with lubricant. **NO SIGNS OF BASE METAL ALLOWED.** If dry film lubricant is missing, do this:
 - a. Check bolt shank for any material displaced above surface of bolt. Remove displaced material using abrasive cloth, Item 82, WP 1803 00.
 - b. Check bolt shank diameter using outside micrometer. **MINIMUM BOLT SHANK DIAMETER IS 0.4970-INCH.** Replace bolt if shank diameter is less than limits.
 - c. If dry film lubricant is missing and bolt shank diameter is within limits, using fluorescent penetrant inspection kit, fluorescent penetrant inspect bolts for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked bolts.
2. Blend out nicks and gouges smoothly to 1/2-inch radius, using abrasive paper, Item 211, WP 1803 00. Polish out sanding marks using abrasive paper, Item 211, WP 1803 00.
3. Remove corrosion by lightly sanding surface using abrasive paper, Item 211, WP 1803 00, until corrosion is removed.
4. If outboard retention plate bolt shank is missing solid film lubricant or has been reworked and is still within acceptable limits, reapply solid film lubricant, Item 180, WP 1803 00, as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- a. Clean areas of exposed base metal using dry machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Allow to dry thoroughly.
- b. At room temperature, spray exposed area with solid film lubricant, Item 180, WP 1803 00. Apply multiple thin coats allowing 15 to 20 minutes drying time between coats.

INSPECT/REPAIR RETENTION PLATE BOLTS - Continued

- c. Allow solid film lubricant to air-cure at least 12 hours before using bolt.
- d. **REPLACE BOLT IF COATING IS NOT SLATE BLACK AND UNIFORM IN COLOR. IF COATING HAS ANY RUNS, BLISTERS, ROUGHNESS OR SEPARATION OF INGREDIENTS, OR IF COATING HAS ANY OTHER SURFACE IMPERFECTIONS.**
5. After solid film lubricant has cured, firmly press a strip of pressure sensitive adhesive tape, Item 325, WP 1803 00, onto surface of touched-up area. Remove masking tape with an abrupt motion. A uniform deposit of powdery material clinging to tape is acceptable. **REPLACE BOLT IF SOLID FILM LUBRICANT FLAKES OR LEAVES ANY BARE METAL EXPOSED.**
6. If solid film lubricant repair is successful, reinstall bolts.

INSTALL**NOTE**

- If tail rotor blades being installed are the same ones that were removed, make certain they are installed in the same clockwise and inboard/outboard position, to minimize rebalancing of tail rotor.
 - To prevent additional tail rotor balancing, match color bands or index marks on blades, retention plates, pitch control rods, and pitch beam during installation.
 - Tail rotor blades, 70101-11200-042 through 70101-11200-044, are individually interchangeable with each other.
 - Blades with erosion protection kits installed must not be mixed with blades that do not have erosion protection kits installed. Tail rotor balance and vibration levels will be adversely effected.
 - When shim "A" is made from 0.001-inch thick stock, use 70102-11109-101 as shim "B". When shim "A" is made from 0.002-inch thick stock, use 70102-11109-102 as shim "B".
1. Locally make tail rotor blade installation checking shims "A" (WP 1805 00) and "B" (WP 1805 00).
 2. Visually check pitch control shaft and make sure dust plug is installed. If dust plug is not installed do the following:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- a. Clean dust plug opening using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
- b. Using acid swabbing brush, Item 57, WP 1803 00, apply a coat of adhesive, Item 15, WP 1803 00, around dust plug.
- c. Install dust plug in opening on pitch control shaft. Make sure plug is seated firmly against pitch control shaft.
- d. Apply a bead of sealing compound, Item 273, WP 1803 00, around edges of dust plug. Allow to dry.
3. Place hoisting sling straps around inboard tail rotor blade. Attach hoisting sling to hoist and adjust straps.
4. Make sure pitch control shaft threads are protected with pressure sensitive adhesive tape, Item 325, WP 1803 00.

INSTALL - Continued**CAUTION**

Damage to blade will result if blade is dragged across shaft. Guide blade carefully over shaft.

NOTE

During assembly, make sure all blade leading edges are facing proper direction.

5. **QA** Lift tail rotor blade into place. Place inboard blade into deep slot of inboard retention plate (Figure 1, Sheet 2).
6. Remove hoisting sling.
7. On work table, install outboard blade into deep slot of outboard retention plate. Lightly tap blade into place using a rawhide or rubber mallet until blade is seated.
8. Inspect spar area for delamination and damage to blade.
9. Place tiedown strap, Item 310, WP 1803 00, through installation bolt holes of retention plate and around blade to help secure assembly while hoisting.
10. Place hoisting sling straps around outboard tail rotor blade. Attach hoisting sling to hoist and adjust straps.

CAUTION

Damage to blade will result if blade is dragged across shaft. Glide blade carefully over shaft.

11. **QA** Lift tail rotor blade and outboard retention plate assembly into place.
12. Remove tiedown strap from outboard blade and retention plate.
13. Remove hoisting sling.
14. **QA** Visually inspect outboard retention plate and make sure dust seal is installed. Line up index marks on inboard and outboard retention plates. Outboard blade should line up with deep slot in outboard retention plate.
15. If a new or different inboard retention plate, outboard retention plate, tail rotor blade, or gear box has been installed, do tail rotor clamp-up check procedure as follows:
 - a. Temporarily install shims "A" (8 places) between outboard retention plate and blade spars (Figure 1, Sheet 3, detail A).
 - b. Temporarily install shims "B" (4 places) between outboard and inboard retention plate flanges. Make sure shims "B" do not contact tail rotor blades.

CAUTION

Damage to outboard retention plate will result if wrenches are used to install bolts. Use deep well socket when installing outboard retention plate bolts.

NOTE

Refer to WP 1246 00 for shimming slipring, rotor, and stator.

- c. **QA** With all shims in place, install bolts, countersunk washers, washers, and nuts holding retention plate halves together. Make sure countersunk side of washer faces bolthead. Snug up nuts evenly (Figure 1, Sheet 2).
- d. **QA TORQUE NUTS IN CRISSCROSS PATTERN TO 209 - 231 INCH-POUNDS.**
- e. **TORQUE NUTS IN CRISSCROSS PATTERN TO 418 - 462 INCH-POUNDS.**

INSTALL - Continued

- f. **QA TORQUE NUTS IN CRISSCROSS PATTERN TO A FINAL TORQUE OF 627 - 693 INCH-POUNDS.**
- g. **QA** Check that 0.080 to 0.125-inch of bolt extends beyond nut. Add or subtract washers under nut, as needed to correct dimension.
- h. Shims "A" should remain in place when normal finger-force is applied to remove them.
- i. If shims "A" still do not remain in place, do the following:
 - (1) Inspect tail rotor blade thickness in area of nylon wrap for **0.719-INCH MINIMUM**.
 - (2) Inspect thickness of nylon pads of inboard and outboard retention plates for **0.123-INCH MINIMUM**.
 - (3) Inspect retention plate spar channels with nylon pads removed for **0.821-INCH MAXIMUM DEPTH (DEEP CHANNEL) AND 0.141-INCH MAXIMUM DEPTH (SHALLOW CHANNEL)**.
 - (4) Replace any component, which exceeds above requirements, and repeat tail rotor blade clamp-up procedure.
- j. If shims do remain in place, loosen bolts, washers, and nuts holding retention plates together and remove shims "A" (8 places), and shims "B" (4 places). Proceed with normal tail rotor installation.
- 16. Install bolts, countersunk washers, washers and nuts holding retention plate halves together. Make sure countersunk side of washer faces bolthead.
- 17. Check friction torque on self-locking nuts by doing this:
 - a. Install nuts on bolts finger tight.
 - b. Using dial indicating torque wrench turn nut in tightening direction. **TORQUE MUST BE GREATER THAN 18 INCH-POUNDS BEFORE MORE THAN 0.080 TO 0.125-INCH OF BOLT EXTENDS BEYOND NUT.**
 - c. If torque is 18 inch-pounds or less, replace nut.
- 18. **QA TORQUE NUTS IN CRISSCROSS PATTERN TO 209 - 231 INCH-POUNDS.**
- 19. **QA TORQUE NUTS IN CRISSCROSS PATTERN TO 418 - 462 INCH-POUNDS.**
- 20. **QA TORQUE NUTS IN CRISSCROSS PATTERN TO A FINAL TORQUE OF 627 - 693 INCH-POUNDS.**
- 21. **QA** Check that 0.080 to 0.125-inch of bolt extends beyond nut. Add or subtract washers under nut as needed to get correct dimension.

NOTE

Balance washers must be stacked on both sides of retention plates.

- 22. **QA** If necessary, install balance washers in assigned holes with bolt, washers, and nut. Make sure washer, AN960C416, is between balance washer and retention plate on both sides. **TORQUE NUT TO 95 - 105 INCH-POUNDS.**
- 23. Install pitch beam (WP 0587 00).
- 24. Install pitch control rods (WP 0586 00).
- 25. Connect blade deice electrical connectors (WP 1246 00).
- 26. Do a tail rotor rig check (WP 1039 00).
- 27. Check area for cleanness and foreign material.
- 28. Do an operational check of tail rotor flight control system (WP 0136 00).

INSTALL - Continued**NOTE**

Certain work performed on the tail rotor system requires a balance check to be completed prior to flight. This includes outboard retention plate bolts removed or loosened, folding of tail rotor blades, tail rotor blades removed, skin repairs to tail rotor blades, and any other work accomplished to tail rotor system which may have an affect on tail rotor balance.

29. Do a tail rotor balance check (TM 1-6625-724-13&P).
30. Do a maintenance test flight (TM 1-1520-237-MTF).

WARNING**CSI**

Verification of torque stabilization and final stabilization torque are critical characteristics.

NOTE

Inboard retention plate nut torque must be stabilized prior to performing outboard retention plate bolt and tail rotor pitch beam retaining nut torque check.

31. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1, to make sure outboard retention plate nut and pitch beam retaining nut torque checks are shown as due 9 to 11 flight hours after installation (WP 1692 00).

ADHESIVE FILLER REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents, or mixing and using adhesives. These fumes are both a fire and health hazard.
- Provide adequate ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush out eyes with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Seek medical evaluation for skin or throat irritations and breathing problems.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wear protective cotton and plastic gloves when handling solvents or adhesives.
- Wash hands and face before eating or smoking.

1. Mask area around damage 1/2-inch larger than damage outline.

ADHESIVE FILLER REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in prolonged contact with bonded joints or inside of blade. Do not allow solvent to seep into bonded joints or inside of blade.

2. Remove paint inside masked area, using clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

CAUTION

Damage to blade will result if blade is sanded excessively. Sand only until yellow primer is removed.

3. Hand sand yellow primer from repair area using abrasive cloth, Item 81, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. Wipe sanded area clean, with clean, machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.
5. Mix package of 2 part adhesive, Item 11, WP 1803 00. Empty tube of curing agent into cup of adhesive resin. Stir until mixture is free of streaks and color is uniform. Pot life of mixture is 25 minutes at 75°F (24°C).
6. Fill in damage with adhesive mixture, using beverage stirring stick, Item 300, WP 1803 00, tongue depressor, or equivalent.

NOTE

To reduce sanding, remove extra adhesive from repair area.

7. **QA** With wood or plastic straight edge bridging repair, scrape off extra adhesive in chordwise direction. Make sure level of adhesive is even with contour of skin.

ADHESIVE FILLER REPAIR - Continued**WARNING**

- Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.
- Wipe area outside repair with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

8. Allow adhesive to cure for 24 hours at 75°F (24°C). Remove tape from repair.

CAUTION

Damage to blade will result if blade is sanded excessively. Heavy sanding will damage blade skin. Use only enough force to blend repair into blade.

9. Lightly sand edges of adhesive to blend repair to surrounding surface, using abrasive cloth, Item 81, WP 1803 00.
10. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.

ADHESIVE INJECTION REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents, or mixing and using adhesives. These fumes are both a fire and health hazard.
- Provide adequate ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush out eyes with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Seek medical evaluation for skin or throat irritations and breathing problems.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wear protective cotton and plastic gloves when handling solvents or adhesives.
- Wash hands and face before eating or smoking.

1. Turn blade horizontally with leading edge 2 to 3 inches higher than trailing edge.
2. Clean out bond separation with denatured alcohol, Item 48, WP 1803 00.
3. Mix package of two part adhesive, Item 11, WP 1803 00. Empty tube of curing agent into cup of adhesive resin. Stir until mixture is free of streaks and color is uniform. Pot life of mixture is 25 minutes at 75°F (24°C).

ADHESIVE INJECTION REPAIR - Continued**CAUTION**

Damage to blade skin will result if excessive force is used to expose damage. Use only enough pressure to slowly open up separation.

4. Carefully open up separation with clean thin plastic tool, or similar tool and apply adhesive mixture to slot. Use small disposable applicator, Item 53, WP 1803 00 (or equivalent) to force adhesive into separation.
5. Remove tool and finish filling slot with adhesive.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

6. Wipe excess adhesive from repair with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.
7. Cover repair with plastic film.
8. Clamp flat pressure strips of wood to sides of repair with light firm pressure.
9. Allow adhesive to cure for 8 hours at 75°F (24°C), or using heat lamps, 1 hour at 160°F (71°C).
10. **QA** Remove clamps, wood pressure strips, and plastic film.
11. Sand lightly to smooth edges of adhesive, using abrasive cloth, Item 81, WP 1803 00.
12. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.

POLYURETHANE EROSION STRIP REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents, or mixing and using adhesives. These fumes are both a fire and health hazard.
- Provide adequate ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush out eyes with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Seek medical evaluation for skin or throat irritations and breathing problems.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wear protective cotton and plastic gloves when handling solvents or adhesives.
- Wash hands and face before eating or smoking.

1. Use this procedure as an alternate method to repair damage to tail rotor blade polyurethane strip.

CAUTION

Any repair to polyurethane strip on one blade must be counterbalanced by same sized patch on opposite blade in same location. Any full length replacement must be same length and width as supplemental polyurethane strip on opposite blade and same distance from STA 15.06.

NOTE

- Replacement of opposite blades polyurethane strip is only required if it has previously been repaired. Replacement can be performed with blade assembly installed or removed.
- The supplemental polyurethane erosion strip is applied over the existing blade erosion strip from rotor stations 31.50–44.50 if there is unacceptable damage to existing strip.

2. For replacement of supplemental polyurethane strip, remove strip and mask blades as follows:
 - a. Remove the entire supplemental polyurethane strip and adhesive.

CAUTION

Do not remove adhesive filler when removing supplemental polyurethane strip and adhesive.

- b. Mask to area 1/4-inch beyond outboard end of existing strip.
- c. Mask to raised portion of deice blanket.
- d. Mask area chordwise 1/4-inch beyond edges of existing strip.

POLYURETHANE EROSION STRIP REPAIR - Continued

3. For partial repairs, remove polyurethane strip and mask blades as follows:
 - a. Remove all loose or torn pieces of polyurethane strip.
 - b. Mask area at least 1 inch beyond either side (spanwise) of damage.
 - c. Mask area chordwise 1/4-inch beyond edges of existing strip.
 - d. Mask opposite blade in same locations.
4. Fill in cavities, or areas worn below contour of leading edge skin per Adhesive Filler Repair. Sand area smooth to original contour using abrasive paper, Item 209, WP 1803 00.
5. Lightly sand polyurethane strip on blade where patch will overlap with abrasive paper, Item 209, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

6. Clean areas with technical acetone, Item 5, WP 1803 00. Wipe dampened area with dry machinery wiping towel, Item 344, WP 1803 00, before acetone dries.
7. Make flat pattern for new strip. **EDGES OF NEW STRIP SHALL BE AT LEAST EVEN WITH, BUT NOT MORE THAN 1/8-INCH GREATER THAN EXISTING STRIP. CORNERS SHALL BE ROUNDED TO 1/2-INCH RADIUS.**

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

8. Repeat cleaning with Acetone (Item 5, WP 1803 00) and wiping dry until cloth is clean.
9. Make flat pattern for new strip. Edges of new strip shall be at least even with, but not more than 1/8-inch greater than existing strip (not to exceed 2.87-inch total width). Corners shall be rounded to 1/2-inch radius.
10. Apply a thin coat of adhesive primer, (Item 23, WP 1803 00), to the entire area to be covered with pressure sensitive adhesive tape. Let the adhesion promoter dry for 10 minutes prior to application of the pressure sensitive tape. When it is dry it will not appear glossy. Pressure sensitive tape must be applied within one hour after adhesion promoter application.
11. Cut pressure sensitive tape, (Item 222, WP 1803 00, Item 232 WP 1803 00, Item 333, WP 1803 00 or Item 334, WP 1803 00) to flat pattern size.
12. Prepare a wetting solution as follows: To a pint sized spray bottle, add two drops of liquid dishwashing detergent, 1/2-cup isopropyl alcohol (Item 170, WP 1803 00), and fill the bottle with clean water. (The actual mixture is 75% water, 25% isopropyl alcohol, and two drops of liquid dishwashing detergent per one liter of solution).
13. Remove poly liner from adhesive tape. Liberally spray the adhesive with the wetting solution prepared in step 12. Also spray the surface of the part to which the pressure sensitive adhesive tape is to be applied.
14. Flip the pressure sensitive tape down and into position.

POLYURETHANE EROSION STRIP REPAIR - Continued

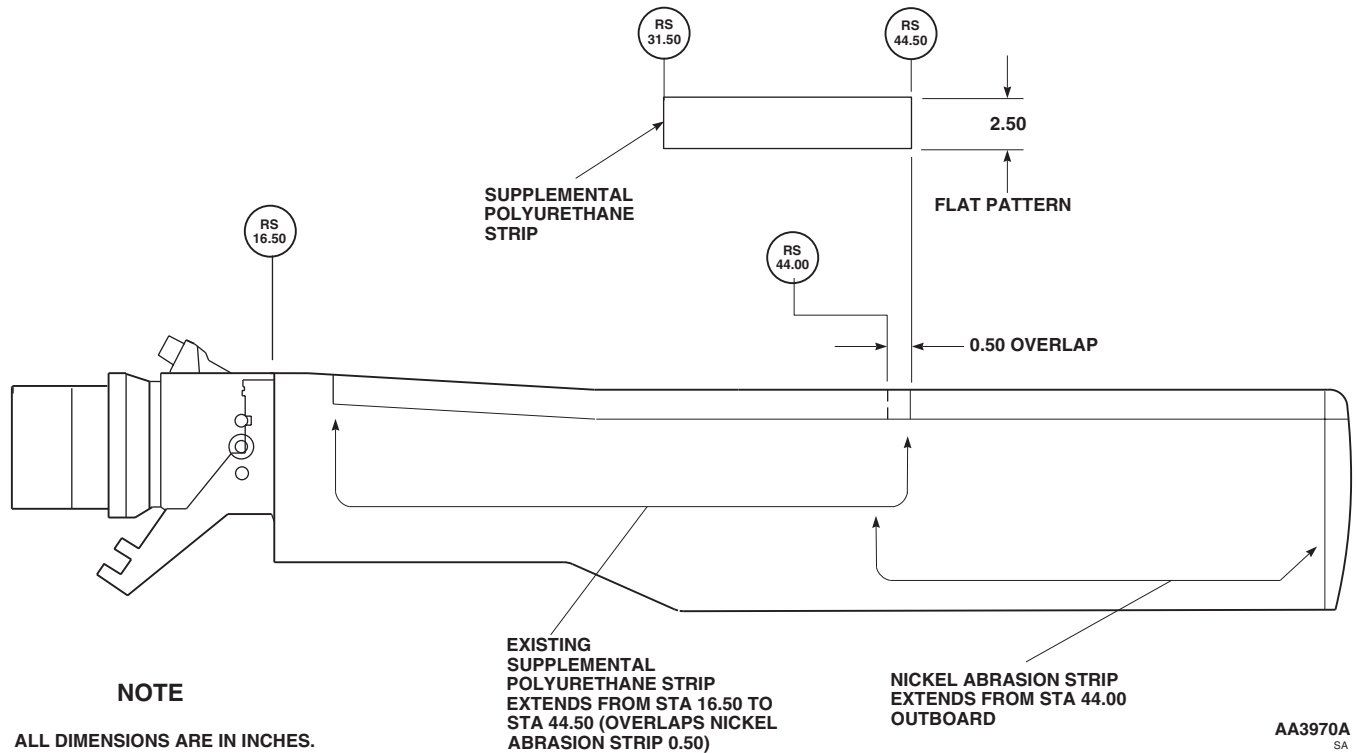


Figure 3. Polyurethane Erosion Strip Repair.

15. Using a plastic squeegee, start at the middle of the tape, and remove the wetting solution from beneath the pressure sensitive tape by using long overlapping strokes. If air/fluid bubbles become trapped, pull the tape back far enough to release the air/fluid, re-spray wetting solution and re-apply using the squeegee. Use a clean rag to mop up the excess wetting solution at the edges to prevent the solution from wicking back under the tape.
16. Squeegee any remaining fluid from the blade until all fluid and bubbles are gone. Use a clean rag to mop up the excess wetting solution at the edges to prevent the wetting solution from wicking back under the tape.
17. **QA** Seal edges of tape with coat of adhesive (Item 47, WP 1803 00). Allow adhesive to dry for 20 to 30-minutes. Apply a second coat over the first.
18. Allow repair to cure at least 8-hours at (70°F/21° C).
19. Touch up paint as required. Refer to, PAINT TOUCHUP, in this work package.

REPLACE TIP CAP

1. Turn blade to safe, comfortable working position.

CAUTION

Damage to blade will result if blade tip attachment rivets are drilled out carelessly.
Holes will be enlarged. Use care in drilling out tip cap attachment rivets.

2. Carefully drill out rivet heads and remove twelve rivets attaching tip cap to tip rib, using 1/8-inch diameter drill and 3/32-inch diameter drive pin punch.

REPLACE TIP CAP - Continued

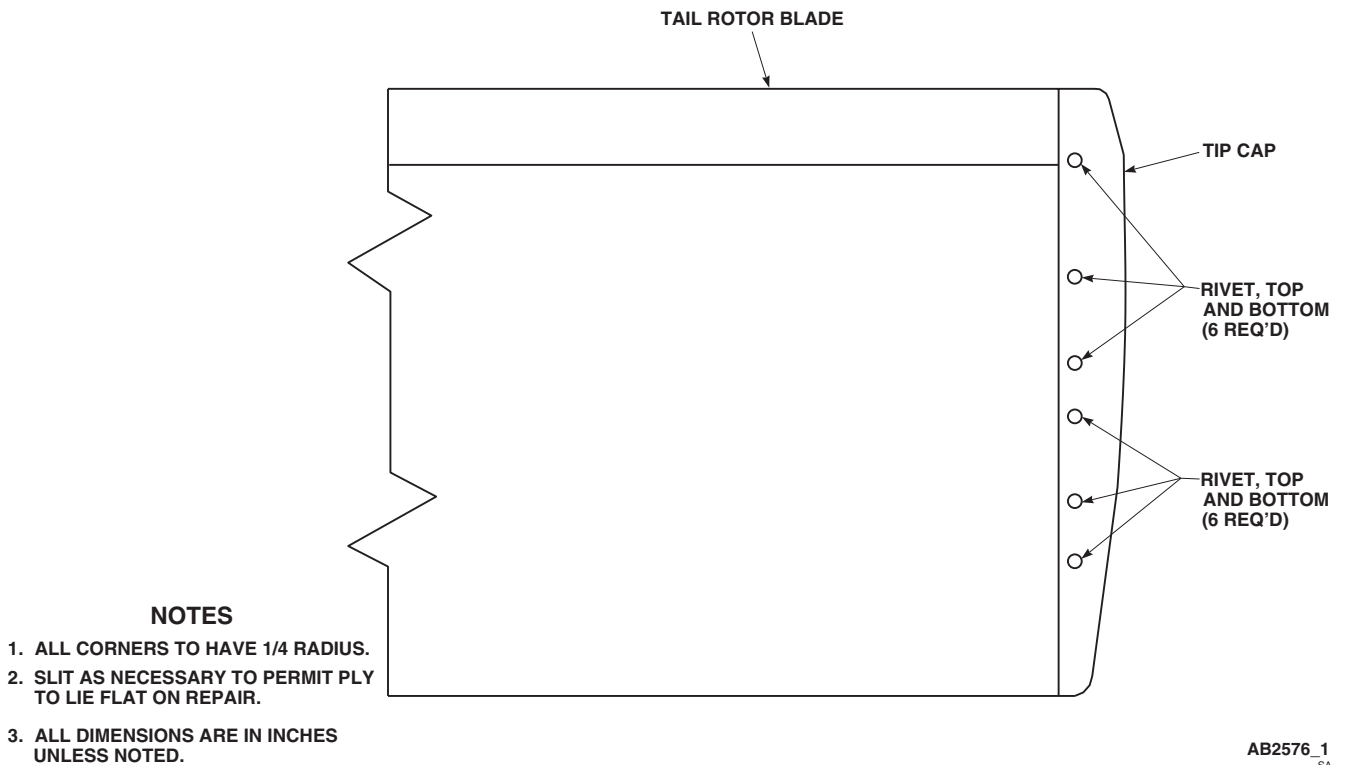


Figure 4. Repair Procedures. (Sheet 1 of 2)

CAUTION

Damage to blade will result if counterweights on blade tip are disturbed when removing tip cap. Blade balance will be lost. Do not disturb counterweights on tip of blade.

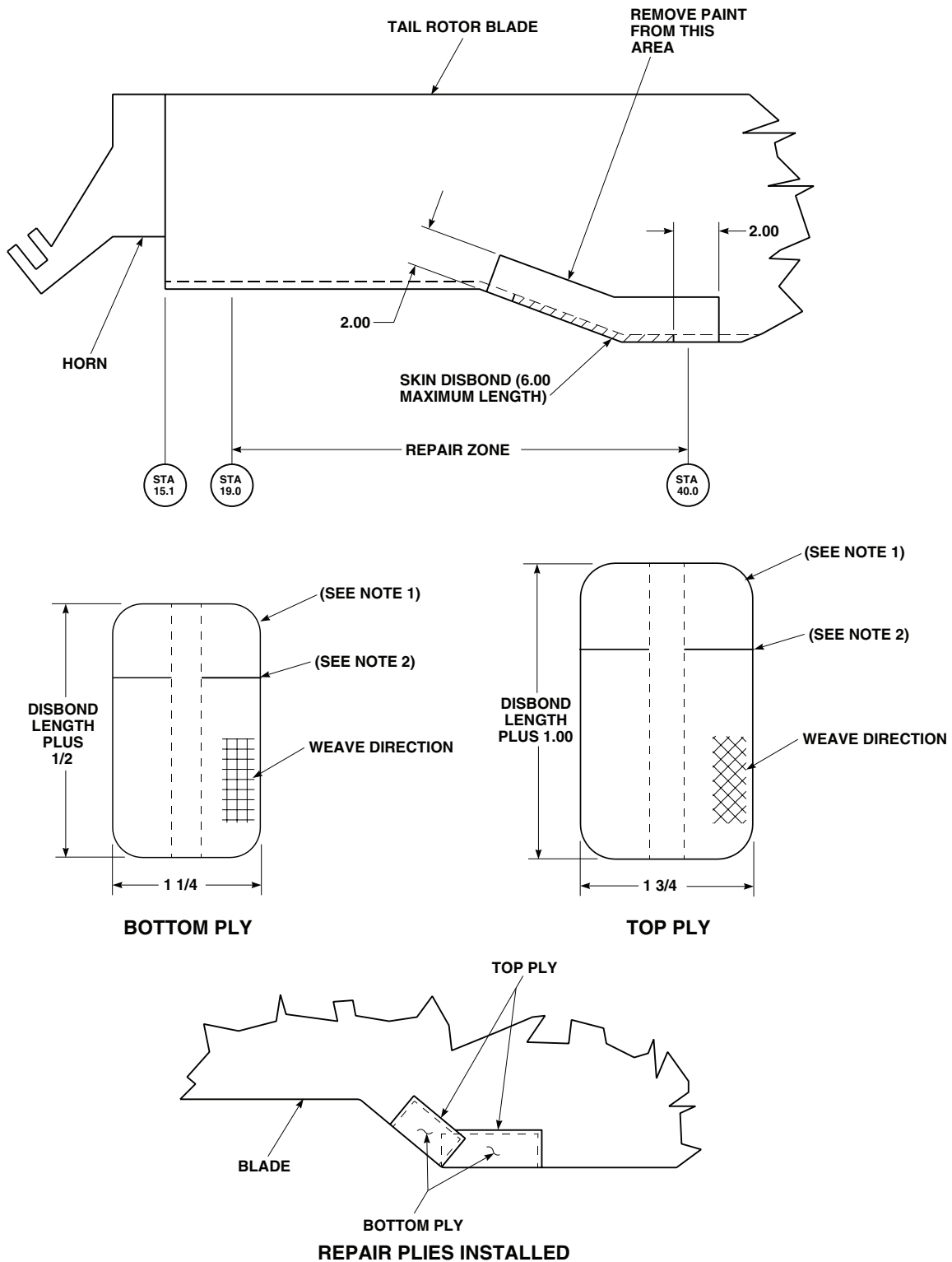
3. Carefully remove tip cap and clean out rivet debris from tip end of blade.
4. When installing new tip cap, carefully open pilot holes to full size, using (0.1285-inch diameter) drill. Use care when drilling through nickel abrasion cap.
5. Countersink all holes to 0.225-inch diameter, using 100 degree countersink.
6. Deburr holes.
7. Treat holes and countersunk areas with corrosion resistant coating, Item 111, WP 1803 00. Refer to PAINT TOUCHUP, in this work package.
8. Position tip cap on blade and hold in place using 1/8-inch diameter sheet metal hole holders in two top and two bottom holes.

NOTE

When installing a new tip cap on 70101-11200 blade, position tip cap on blade. Match drill second hole from leading edge of tip cap into tip rib, both top and bottom.

9. **QA** Lightly coat holes with primer coating, Item 243, WP 1803 00, and while primer is wet, install required rivets using pneumatic blind riveter.
10. **QA** Install twelve blind rivets in holes, top and bottom, from leading edge (Figure 4, Sheet 1).

REPLACE TIP CAP - Continued



AB2576.2
SA

Figure 4. Repair Procedures. (Sheet 2 of 2)

REPLACE TIP CAP - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

11. Wipe off excess primer from around rivets with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
12. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.

ABRASION STRIP REPAIR

1. Cracks or negligible voids in abrasion strips on tail rotor blade and tip cap, within limits described in WP 0545 00, are sealed to prevent moisture from entering bond line and causing further damage.
2. **QA** Apply smooth even coat of mixed adhesive, Item 11, WP 1803 00, over crack or void. Refer to, ADHESIVE INJECTION REPAIR, in this work package.

ALUMINUM WIRE MESH LIGHTNING PROTECTION REPAIR**WARNING**

- Volatile and toxic fumes occur when using solvents, or mixing and using adhesives. These fumes are both a fire and health hazard.
- Provide adequate ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush out eyes with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
 - Seek medical evaluation for skin or throat irritations and breathing problems.
 - Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
 - Post NO SMOKING signs in area and keep materials away from spark or flame.
 - Wear protective cotton and plastic gloves when handling solvents or adhesives.
 - Wash hands and face before eating or smoking.
1. Mask area around corrosion 1/2-inch larger than corrosion outline.

ALUMINUM WIRE MESH LIGHTNING PROTECTION REPAIR - Continued**CAUTION**

Damage to blade will result if blade is sanded excessively. Heavy sanding will damage blade. Do not sand through underlying fiberglass skin plies. Use only enough force to remove paint, primer and corrosion.

NOTE

- Sand only in immediate areas of pin holes.
- Up to 20% of chordwise width of wire mesh may be removed without affecting lightning protection.

2. Sand away paint, primer, and corrosion.

WARNING

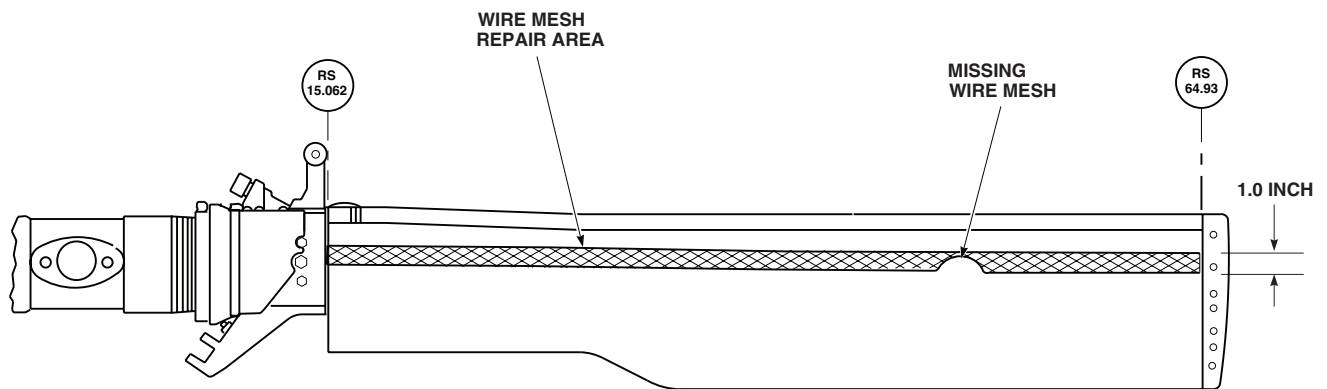
Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

CAUTION

Damage to blade will result if solvent is allowed to come in contact with bonded joints or inside of blade. Do not allow solvent to seep into bonded joints or inside of blade.

3. Wipe sanded area with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry before solvent evaporates.
4. **QA** If more than 20% of chordwise width of lightning wire mesh has been removed after repair, replace tail rotor blade.
5. **QA** Treat exposed wire mesh with corrosion resistant coating, Item 111, WP 1803 00.
6. Apply epoxy primer coating kit, Item 120, WP 1803 00, to repaired area. Refer to, PAINT TOUCHUP, in this work package.
7. Paint repaired area with epoxy coating kit, Item 118, WP 1803 00.
8. Apply another light coat of epoxy primer coating kit, Item 120, WP 1803 00. Refer to, PAINT TOUCHUP, in this work package.
9. Apply top coat of touchup paint. Refer to, PAINT TOUCHUP, in this work package.

ALUMINUM WIRE MESH LIGHTNING PROTECTION REPAIR - Continued



70101-11200-044
WIRE MESH REPLACEMENT AREA

AA1849
SA

Figure 5. Wire Mesh Replacement Area. (Sheet 1 of 2)

TRAILING EDGE DISBOND REPAIR

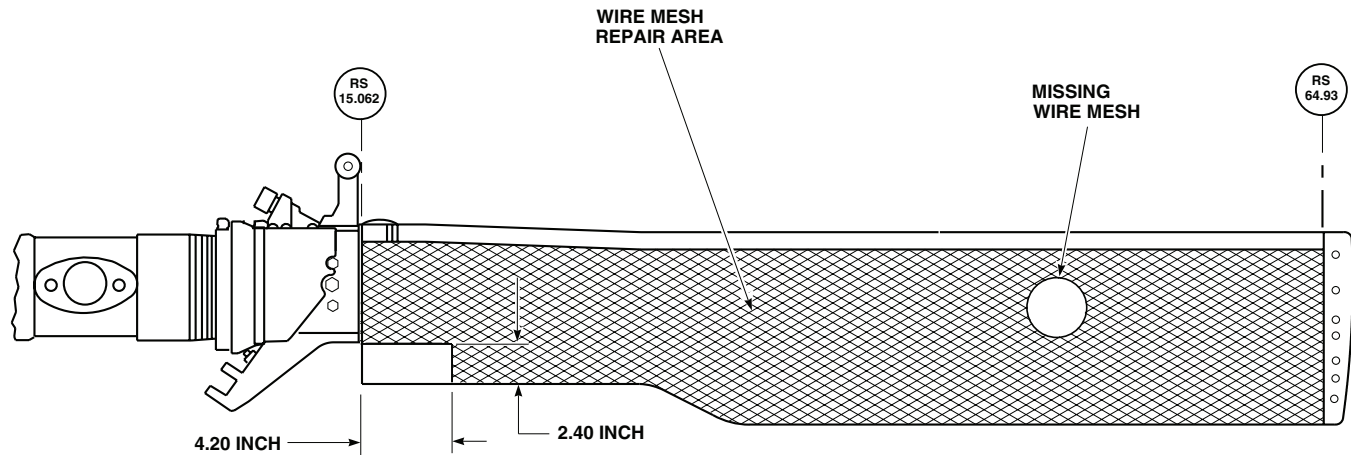
WARNING

- Volatile and toxic fumes occur when using solvents, or mixing and using adhesives. These fumes are both a fire and health hazard.
- Provide adequate ventilation and protective clothing, including eye shield, when using solvents and adhesive mixtures. Avoid breathing vapors and skin contact with these materials as much as possible. Wash contacted skin with soap and water. If materials contact eyes, flush out eyes with clean water and get immediate medical attention.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Seek medical evaluation for skin or throat irritations and breathing problems.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wear protective cotton and plastic gloves when handling solvents or adhesives.
- Wash hands and face before eating or smoking.

TRAILING EDGE DISBOND REPAIR - Continued



70101-31000-043
WIRE MESH REPLACEMENT AREA

AA1850
 SA

Figure 5. Wire Mesh Replacement Area. (Sheet 2 of 2)

- If damage is present on both blade ends, larger damaged area determines size of repairs applied to both ends.
- Repair shall be duplicated on opposite end of blade to maintain blade balance.

1. Clamp trailing edge just beyond each end of disbond, using C-clamps.

CAUTION

Damage to blade will result if skins are forced apart during repair. Length of disbond will be extended. Do not force skins apart during repair. Use care when inserting wedges.

2. Insert small wooden wedges, to spread skins apart for access.
3. Sand bond surfaces of both skins to remove old adhesive, using abrasive paper, Item 209, WP 1803 00, wrapped around a thin tool to penetrate opening.
4. Sand into end of split so that adhesive can form uniform bond line.

TRAILING EDGE DISBOND REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

5. Wipe bond surfaces using small pieces of cheesecloth cloth, Item 85, WP 1803 00, wrapped around toothpick or small tool, and moistened with technical acetone, Item 5, WP 1803 00. Repeat until cheesecloth remains clean.
6. Wipe surfaces with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical isopropyl alcohol, Item 170, WP 1803 00.
7. Mix adhesive, Item 11, WP 1803 00 (WP 0378 00).

CAUTION

Damage to blade will result if adhesive is allowed to run inside of blade. Do not allow adhesive to run inside blade.

8. Apply adhesive to entire length of disbond with thin tool or toothpick, moving wedges as required.
9. Remove wedges and use metal or wooden pressure strips, clamped with C-clamps to apply pressure enough (10 to 15 psi) for intimate contact at bond line.
10. Cure adhesive for 24 hours at 70°F (22°C) or for 2 hours at 140°F (60°C) to 160°F (71°C) using heat gun. Remove all C-clamps and pressure strips.
11. Remove excess adhesive squeezeout and paint from blade repair area (Figure 4, Sheet 2), using abrasive cloth, Item 80, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

12. Wipe repair area with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
13. Cut repair plies from satin cloth, Item 90, WP 1803 00 (Figure 4, Sheet 2).
14. Mix adhesive, Item 11, WP 1803 00 (WP 0378 00).
15. Place repair plies on flat surface covered with plastic sheet, Item 221, WP 1803 00.
16. Saturate repair plies with adhesive, using acid swabbing brush, Item 57, or beverage stirring stick, Item 300, WP 1803 00.
17. Apply adhesive to sanded surface of blade in repair area.
18. Apply bottom ply, centering it on disbond repair. Slit ply at blade transition area if necessary (Figure 4, Sheet 2).
19. Apply top ply, slitting if necessary, overlapping bottom ply (Figure 4, Sheet 2).
20. Cover repair with plastic sheet, Item 221, WP 1803 00.
21. Work out bubbles and excess adhesive using beverage stirring stick, Item 300, WP 1803 00, as a squeegee.

TRAILING EDGE DISBOND REPAIR - Continued

22. Cure repair for 24 hours at 70°F (22°C) or for 2 hours at 140°F (60°C) to 160°F (71°C) using heat gun.
23. **QA** Remove plastic sheet.
24. **QA** Remove excess adhesive and feather edge of patch, using abrasive cloth, Item 80, WP 1803 00 and abrasive cloth, Item 83, WP 1803 00.
25. Touch up paint. Refer to, PAINT TOUCHUP, in this work package.

PAINT TOUCHUP**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

NOTE

- Use eye, skin, and breathing protection when sanding reinforced plastic laminates. Resulting dust is abrasive and an irritant to eyes, nose, throat, and skin.
- Seek medical evaluation for skin or throat irritations and breathing problems.
- Use properly grounded and explosion proof electrical equipment in areas where combustible vapors or materials may be present.
- Post NO SMOKING signs in area and keep materials away from spark or flame.
- Wear protective cotton and plastic gloves when handling solvents or adhesives.
- Wash hands and face before eating or smoking.

1. Refer to Table 1 for touch up paint requirements.

PAINT TOUCHUP - Continued

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

CAUTION

- Damage to blade will result if large portions of blade are repainted. Blade balance will be lost. Do not repaint entire blade, or large areas of the blade. Spot-spray or touch up with small brush all repaired areas. Apply just enough paint for good coverage.
- Damage to blade will result if solvent is allowed to come in prolonged contact with bonded joints. Do not allow solvent to seep into bonded joints. Cloths wet with solvent should not be left lying on blade.

NOTE

Do not apply any paint on nickel-abrasion strip, including tip cap.

2. Wipe surface to be painted with clean machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

CAUTION

Damage to blade will result if blade is sanded excessively. Use only enough force to remove resin gloss.

3. Lightly sand surface to be painted to remove resin gloss, using abrasive cloth, Item 78, WP 1803 00, or equivalent.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. Rewipe surface with clean, machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and wipe dry before solvent evaporates.
5. Mask around touch up area to reduce cleanup, using pressure sensitive adhesive tape, Item 325, WP 1803 00, heavy paper, or equivalent protective covering.

Table 1. Paint Touchup.

AREA OR DETAILS	BLADE EFFECTIVITY	PAINT COLOR, TYPE AND SPECIFICATION	NUMBER OF COATS
Blade, overall	Data Plate Marked Paint, per MIL-L-19538	Epoxy Primer Coating Kit, Item 120, WP 1803 00	One
		Lacquer, Item 171, WP 1803 00	Two

PAINT TOUCHUP - Continued**Table 1. Paint Touchup. - Continued**

AREA OR DETAILS	BLADE EFFECTIVITY	PAINT COLOR, TYPE AND SPECIFICATION	NUMBER OF COATS
Data Markings		Lacquer, Item 171, WP 1803 00	One
Blade, overall	Data Plate Marked Paint, per MIL-C-46168	Epoxy Primer Coating Kit, Item 120, WP 1803 00	One
		Polyurethane Coating, Item 233, WP 1803 00	Two
Data Markings		Polyurethane Coating, Item 230, WP 1803 00	One

NOTE

- Do not apply epoxy primer coating kit when temperature is below 50°F (10°C). Recommended temperature is 75°F (24°C) to 86°F (30°C). Lower temperature will cause longer drying time. Apply primer by brush or spray directly over cleaned reinforced plastic surfaces.
- Aluminum surfaces which have finish removed must be treated with corrosion resistant coating, Item 111, WP 1803 00, before primer coating is applied. Refer to (WP 0381 00 and TM 55-1500-345-23).

6. Epoxy primer coating kit, Item 120, WP 1803 00, consists of two components packaged separately. Add component II to component I. Do not reverse mixing, or mix components from different manufacturers.
7. Pour required amount of component I (base) into clean container. Add equal amount, by volume, of component II (converter). Stir mixture slowly and thoroughly for at least 10 minutes.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

8. Thin mixture for spraying by adding one part, by volume, thinner to two parts, by volume, of mixed primer. Thinner is mixture of equal amounts of technical acetone, Item 5, WP 1803 00.
9. Mix thoroughly, strain mixture, and allow to stand 1 hour before using. Stir again just before using. Pot life of mixture is 8 hours at 75°F (24°C).
10. Spray light even coat of primer on prepared surface and allow to dry 1 hour at room temperature 75°F (24°C).
11. **QA** Apply top coat of proper paint over primer. Refer to Table 1 for paint requirements.
12. Remove masking from repair area.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**TAIL ROTOR PITCH CONTROL RODS**

This WP supersedes WP 0586 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Caliper Set, Outside Micrometer, GGG-C-105
 Depth Gage
 Fluorescent Penetrant Inspection Kit, XMA101
 Micrometer, GGG-C-105
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Tape, Pressure Sensitive, Adhesive, Item 325,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Antiseize Compound, Item 49, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Corrosion Preventive Compound, Item 105,
 WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Pad, Scouring, Item 205, WP 1803 00

References

TM 1-1520-237-MTF
 TM 1-1520-265-23
 WP 0545 00
 WP 0606 00
 WP 1039 00
 WP 1803 00

REMOVE

1. Engage gust lock.
2. Install tail rotor servo rigging pin.
3. To remove bolts do this:
 - a. Loosen nut on bolt. Depress retaining pawl on bolt using thumb and continue backing off nut until castellations clear retaining pawl. Release pawl and remove nut from bolt.
 - b. Remove washer from bolt by depressing pawl and sliding washer from bolt.

NOTE

Retaining pawl on bolt will make bolt hard to remove once inside rod end. Use needed force to pull bolt through rod end.

- c. Depress retaining pawl on bolts and remove bolts and pitch control rod.
- d. Depress pawl and remove bonding jumper and countersunk washer from bolt.
4. Inspect spherical or elastomeric rod ends (WP 0545 00).

REMOVE SPHERICAL AND ELASTOMERIC ROD END

1. Loosen locknut securing rod end to control rod.
2. Remove rod end from pitch control rod.

REMOVE SPHERICAL AND ELASTOMERIC ROD END - Continued

3. Remove locknut and locking devices from rod end.
4. If required, replace spherical rod end bearing (WP 0606 00).
5. Inspect/repair pitch control rod, rod ends, and attaching hardware (Replace Tail Rotor Pitch Control Rods).

INSPECT/REPAIR

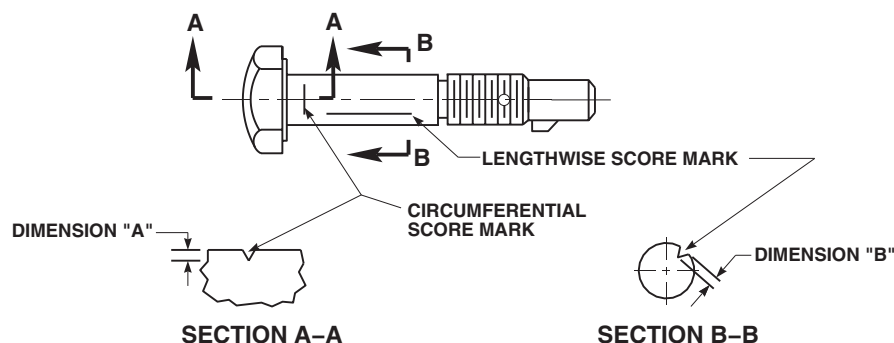
1. Check pitch control rod attaching washers for cracks. **NO CRACKS ALLOWED.** Replace cracked washers.
2. Check spherical rod end bearings for corrosion by doing this:
 - a. Check spherical bearing surface for corrosion pitting. **IF 5 OR MORE PITS ARE VISIBLE IN A 1/2-INCH DIAMETER CIRCLE OR ANY PIT IS LARGER THAN 0.050-INCH, REPLACE SPHERICAL BEARING.**

CAUTION

Damage to Teflon liner will result if solvent is used to clean bearings. Do not use solvent to clean bearings.

- b. Remove surface corrosion with a scouring pad, Item 205, WP 1803 00.
3. Check spherical and elastomeric rod end threads for damage. **REPLACE ROD OR ROD ENDS WITH NICKS OR GOUGES IN THREAD AREA, OR ANY DAMAGE TO THREADS THAT PREVENTS FREE MOVEMENT OF ROD ENDS OR LOCKNUTS.**
4. Check pitch control rods for nicks and gouges. Blend out damage.
5. **REPLACE PITCH CONTROL ROD IF:**
 - a. **CORNER DAMAGE AFTER BLENDING IS GREATER THAN 0.035-INCH DEEP.**
 - b. **SURFACE DAMAGE, AFTER BLENDING, IS DEEPER THAN 0.020-INCH OR COVERS MORE THAN 1/4 OF CIRCUMFERENCE OF ROD BETWEEN LUG AND THREADS.**
 - c. **SURFACE DAMAGE AFTER BLENDING IN LUG PORTION OF ROD ENDS IS GREATER THAN 0.030-INCH DEEP.**
6. Check pitch control rod for cracks. **NO CRACKS ALLOWED.** Replace cracked rod.
7. Inspect pitch control rod end and jamnut for corrosion pits and repair using scouring pad, Item 205, WP 1803 00, as follows:
 - a. **REMOVE CORROSION PITS FROM ROD END UP TO 0.005-INCH DEEP.**
 - b. Replace rod end if corrosion pits are more than limits.
 - c. Remove corrosion pits from jamnut. **REPLACE JAMNUT IF 5 OR MORE PITS ARE FOUND ON ANY SINGLE FLAT.**
 - d. Mask rod end bearing and threads with pressure sensitive adhesive tape, Item 325, WP 1803 00.
 - e. Apply 2 coats of epoxy primer coating kit, Item 120, WP 1803 00, to rod end and jamnut.
8. Check pitch control attachment bolts as follows:
 - a. Clean any oil or grease from bolts using solvent cleaning compound, Item 71, WP 1803 00.
 - b. Check bolts for any material displaced above normal surface of bolt. Remove any displaced material using abrasive cloth, Item 82, WP 1803 00.
 - c. Check bolts for scoring or grooves. Lengthwise score marks or grooves may be smoothed out, but can be left as is, if within limits. Circumferential score marks must be blended out. See Figure 1 for limits on depth of scoring. Blend out as follows:

INSPECT/REPAIR - Continued



BOLT PART NUMBER	LOCATION	DIMENSION "A" MAXIMUM	DIMENSION "B" MAXIMUM	MINIMUM ALLOWABLE DIAMETER AFTER REWORK
70103-08801-101	PITCH CONTROL ROD (BOTH ENDS)	0.004	0.004	0.3655

NOTE

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Inspect Tail Rotor Pitch Control Rod Bolt.

WARNING**CSI**

Fluorescent penetrant inspection is a critical characteristic.

- (1) Before reworking bolt do a fluorescent penetrant inspection, Type I, and check for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23). Discard cracked bolts.
- (2) Blend out scratches using abrasive wheel, Item 350, WP 1803 00. Blend area smooth with abrasive cloth, Item 82, WP 1803 00.

WARNING**CSI**

Minimum shank diameter is a critical characteristic.

- (3) **IF, AFTER REWORK, USING A MICROMETER, BOLT DIAMETER IS LESS THAN SPECIFIED IN FIGURE 1, DISCARD BOLT.**
- (4) **IF, AFTER REWORK, REPAIR AREA IS OVER 30% OF BOLT SHANK SURFACE AREA, DISCARD BOLT.**
- (5) **IF, AFTER REWORK, REPAIR AREA IS OVER 30% OF SHANK CIRCUMFERENCE, DISCARD BOLT.**

INSPECT/REPAIR - Continued**WARNING****CSI**

Fluorescent penetrant inspection is a critical characteristic.

- (6) After reworking bolt, using fluorescent penetrant inspection kit, do a fluorescent penetrant inspection, Type I, and check for cracks (TM 1-1520-265-23). **NONE ALLOWED.** Discard cracked bolts.
9. Check bonding jumper for broken strands. **NONE ALLOWED. CRACKS AND CHIPS IN PLASTIC COATING ARE ACCEPTABLE.** Replace bonding jumper if beyond limits.

INSTALL SPHERICAL AND ELASTOMERIC ROD END**NOTE**

- If the shelf life for SB7112-101 and SB7112-102 has exceeded 60 months, do not install. If the shelf life is less than 60 months prior to installation, serviceability will be in accordance with the applicable technical manual.
 - If spherical rod end bearing is replaced with elastomeric rod end bearing, both rod end bearings on that pitch control rod must be replaced as well as both ends of opposing pitch control rod.
1. Remove pitch control rod. Refer to, REMOVE, in this work package.
 2. Remove pitch control rod end. Refer to, REMOVE SPHERICAL AND ELASTOMERIC ROD END, in this work package.
 3. Apply light coat of antiseize compound, Item 49, WP 1803 00, to threads of pitch control rod end.
 4. Install locknuts and locking devices onto pitch control rod end (Figure 2).
 5. Thread replacement pitch control rod ends into pitch control rod. Make sure each end is threaded in the same amount until a measurement of 5.564 to 6.804-inches measured from center of each rod end is obtained.
 6. Adjust pitch control rod. Refer to, ADJUST, in this work package.

ADJUST

1. Temporarily install pitch control rod to pitch beam and tail rotor blade using bolt, countersunk washer, washer and nut. Do not torque nuts now.
2. Adjust replacement pitch control rod (WP 1039 00).

WARNING

The use of safety cable is not allowed in this application. Use only nonelectrical wire as specified.

3. **QA** Make sure locking device is in proper position. **TORQUE LOCKNUTS TO 119 - 131 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire from locknut to locking device then back to opposite side of locknut.

INSTALL

1. Make sure replacement pitch control rod has been adjusted. Refer to, ADJUST, in this work package.

INSTALL - Continued

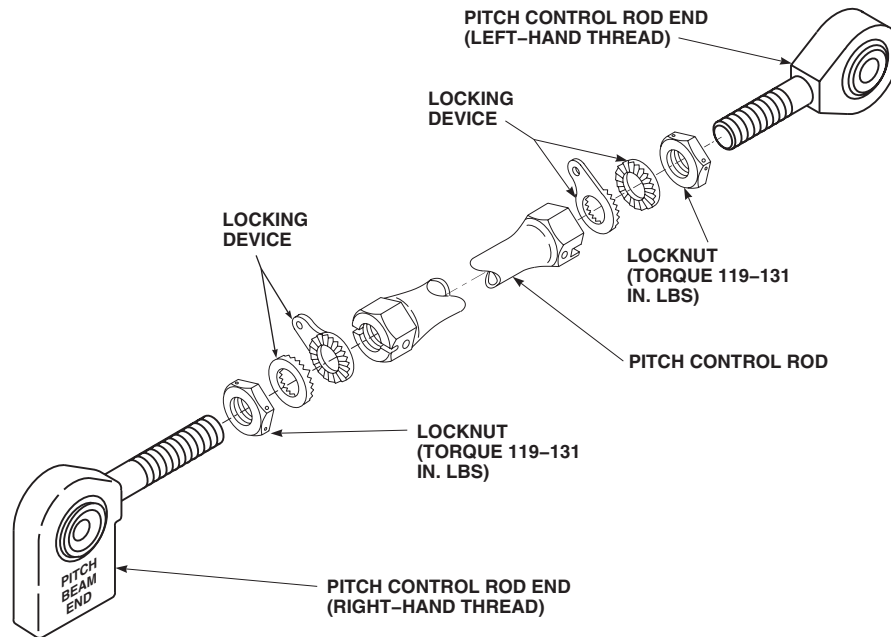
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Figure 2. Install Spherical and Elastomeric Rod End.

WARNING**CSI**

Verification of proper hardware, bonding jumper, cotter pin installation and torque is a critical characteristic.

2. **QA** Install bolt through pitch beam and tail rotor blade horn by doing this:

CAUTION

Damage to bolthead will result if countersunk washer is not installed correctly. When installing countersunk washers, make sure to install the washer with the countersunk side facing the bolthead.

NOTE

Bending of bonding jumper terminal 10° is allowed to prevent interference between bonding jumper and pitch beam or tail rotor blade horn.

- a. **QA** Press pawl and install one washer and bonding jumper on bolt, making sure countersunk side of washer is next to head of bolt.

INSTALL - Continued

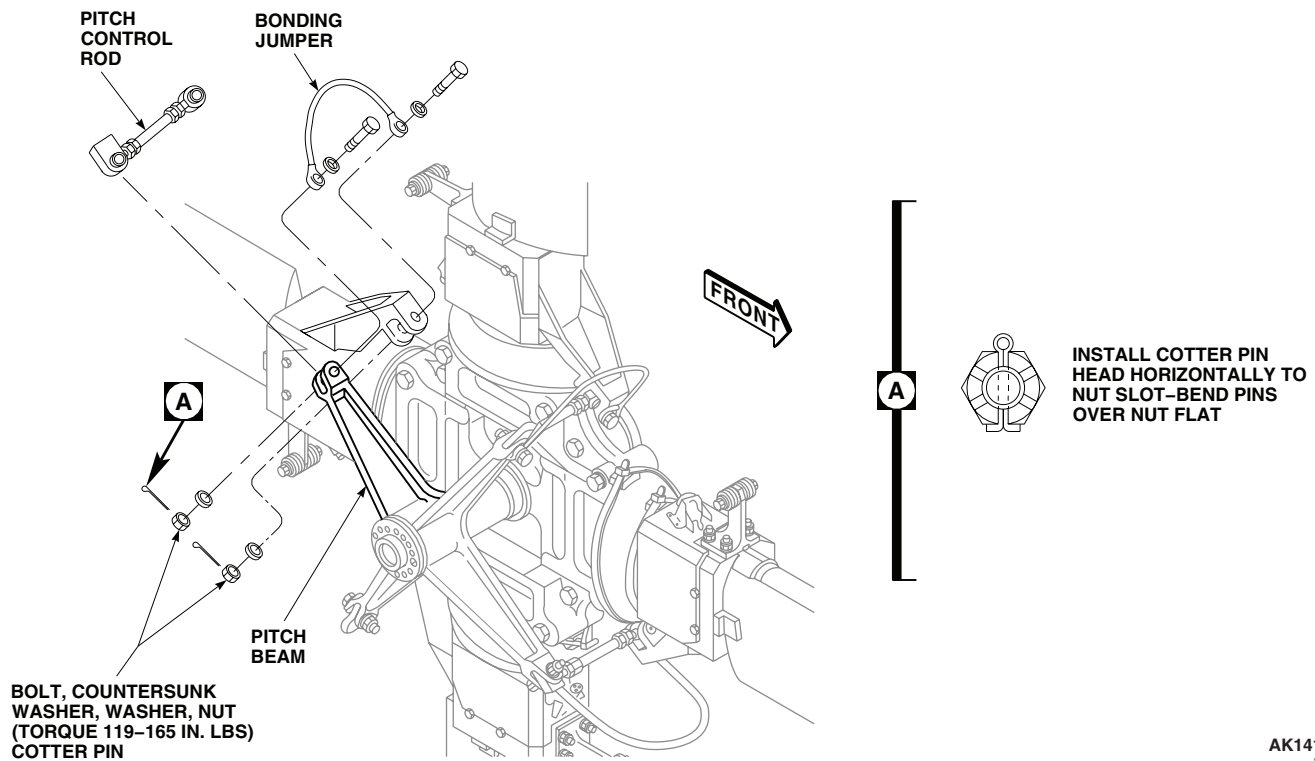


Figure 3. Replace Tail Rotor Pitch Control Rod.

CAUTION

Loss of torque on pitch control rod bolts will result if bonding jumper is not bent 10 degrees before installation. Make sure bonding jumper is bent 10 degrees before installation.

NOTE

Paint or dirt under bonding jumper washer will prevent proper bonding of parts. Make sure area of pitch beam and on horn where bonding jumper is connected is completely clean and free of paint or contaminants.

- b. **QA** If not already done, bend bonding jumper 10° to eliminate chafing against pitch beam and tail rotor blade horn.
- c. Apply a coating of corrosion preventive compound, Item 105, WP 1803 00, on inside diameter of pitch beam bushings and on tail rotor horn bushings at control rod connection.

NOTE

If installing pitch control rod with elastomeric rod end bearings, install with rod end bearing marked with yellow dot at pitch beam end.

- d. **QA** Line up rod end bearing with tail rotor blade horn and install bolt with head facing away from direction of rotation with pawl facing outboard (Figure 3).

INSTALL - Continued

- e. Before installing nut, wipe excess corrosion preventive compound from bolt threads using machinery wiping towel, Item 344, WP 1803 00.
 - f. **QA** Press pawl and install one washer on bolt and install nut. Do not torque nut now.
 - g. **QA** Repeat step 2. a through step 2. f to install pitch beam end of control rod. Center control rod ends within ears of pitch beam.
 - h. **QA** Align bonding jumper 90° from pitch control rod.
 - i. **QA TORQUE NUT TO 119 - 165 INCH-POUNDS.** Install cotter pins in castellated nut, with head horizontal to nut slot and bend pins over hex flats (Figure 3, Detail A).
- 3. **QA** Remove tail rotor servo rigging pin.
 - 4. Check area for cleanness and foreign material.
 - 5. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****TAIL ROTOR PITCH BEAM**

This WP supersedes WP 0587 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Caliper Set, Outside Micrometer, GGG-C-105
Depth Gage
Fluorescent Penetrant Inspection Kit, XMA101
Gloves
Goggles/Face Shield
Micrometer, GGG-C-105
Pitch Beam Puller, 65700-10048-041
Rawhide Mallet, E-2200-D
Spanner Wrench, AN8515
Torque Wrench, 300 - 2500 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Corrosion Preventive Compound, Item 105,
WP 1803 00
Corrosion Preventive Compound, Item 107,
WP 1803 00

Corrosion Resistant Coating, Item 111, WP 1803 00
Insulation Sleeving, Electrical, Item 156, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 0585 00
WP 0586 00
WP 1039 00
WP 1148 00
WP 1692 00
WP 1803 00

REMOVE

1. Engage gust lock.
2. Remove bolts from pitch control rods at pitch beam (WP 0586 00).
3. Remove screws from pitch beam retaining nut (Figure 1, Sheet 1, Detail A).
4. Remove retaining nut using spanner wrench. Remove washer (Figure 1, Sheet 1, Detail B).

CAUTION

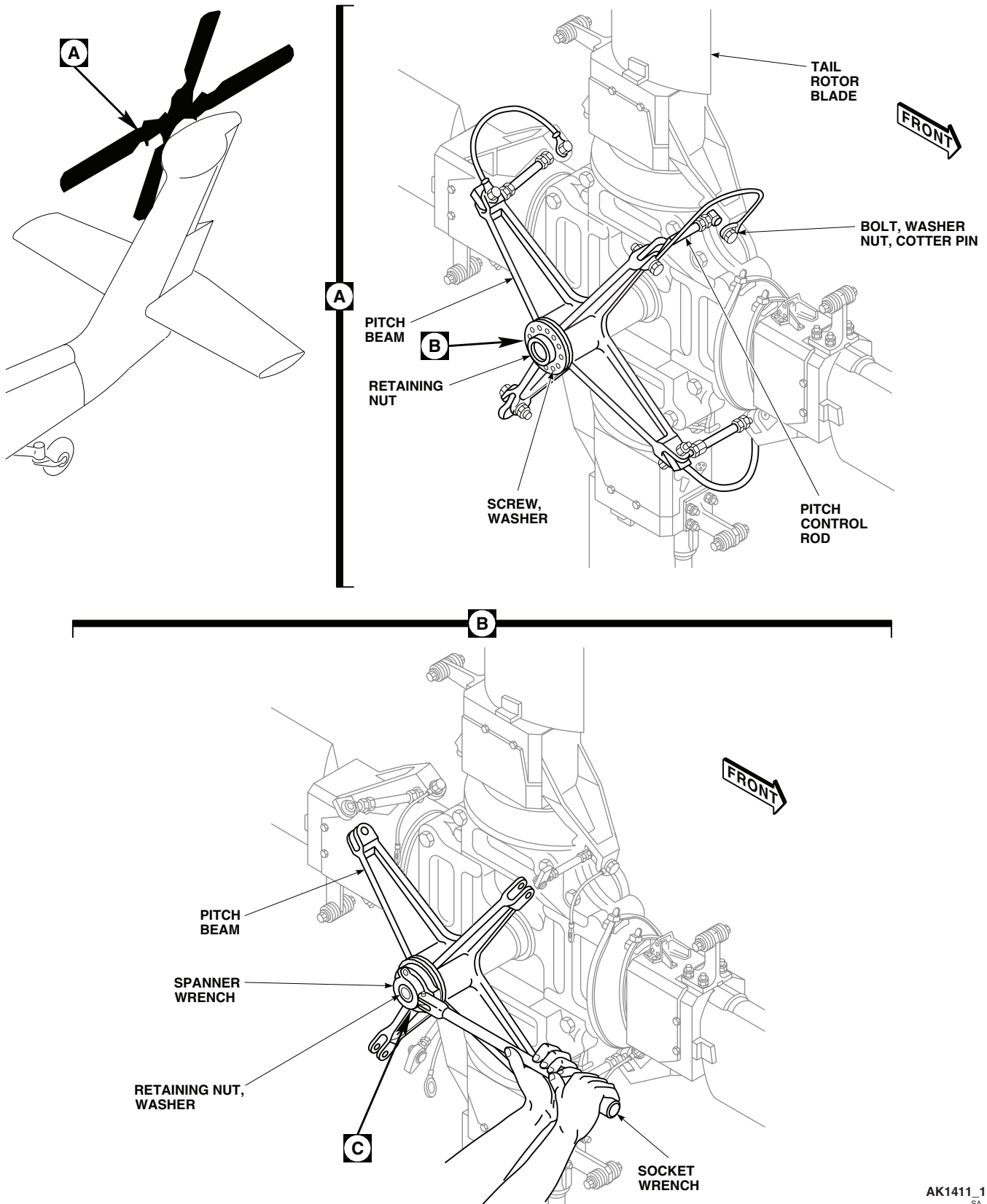
Damage to pitch change shaft dust plug will result if excessive force is used to remove retaining nut and pitch beam. Use care when removing retaining nut and pitch beam.

5. Remove pitch beam using pitch beam puller. Tap with rawhide mallet on inboard side, if necessary.

INSPECT

1. Visually inspect pitch beam washer for cracks. **NO CRACKS ALLOWED.** Replace washer if cracks are found.
2. Check surfaces of pitch beam washer which come in contact with pitch beam retaining nut for nicks, gouges, corrosion, or fretting. Remove damage using abrasive wheel, Item 350, WP 1803 00. **REPLACE PITCH BEAM WASHER IF DAMAGE AFTER BLENDING EXCEEDS 0.010-INCH DEEP OR IS MORE THAN 15% OF SURFACE.**

INSPECT - Continued



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Figure 1. Replace Tail Rotor Pitch Beam. (Sheet 1 of 2)

INSPECT - Continued

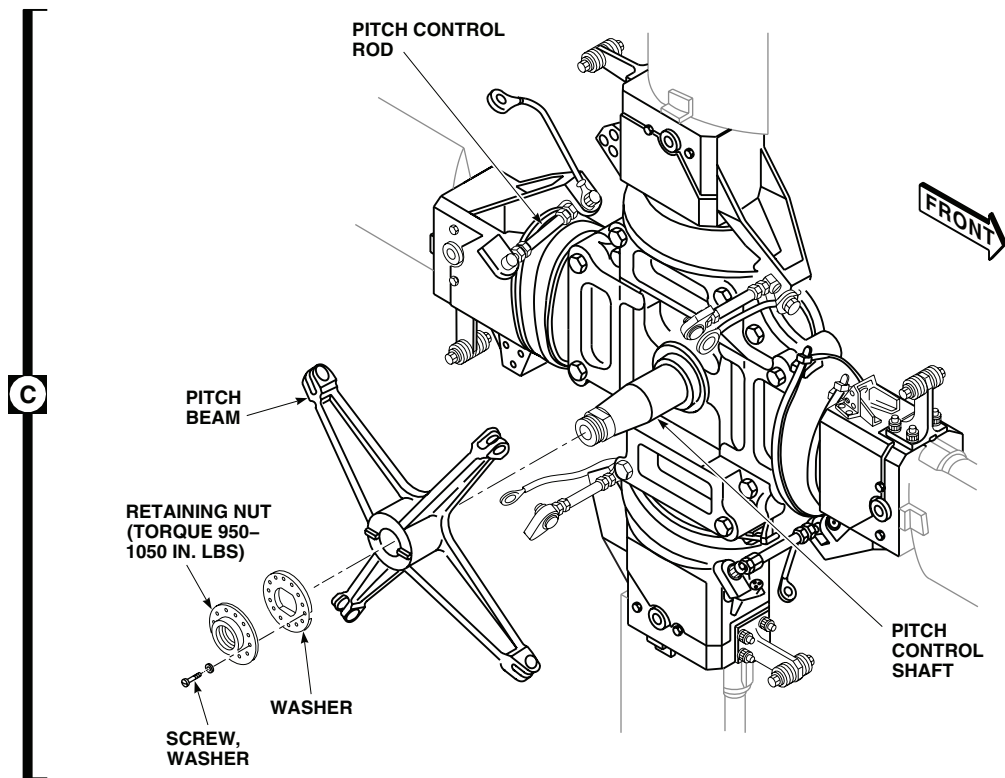
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Figure 1. Replace Tail Rotor Pitch Beam. (Sheet 2 of 2)

3. Visually inspect pitch beam retaining nut for cracks. **NO CRACKS ALLOWED.** Replace nut if cracks are found.
4. Visually check pitch beam retaining nut for thread face wear or chafing. **NO DAMAGE TO THREADS ALLOWED.** Replace nut with damaged threads.
5. Check surfaces of pitch beam retaining nut which come in contact with pitch beam washer for nicks, gouges, corrosion, or fretting. Remove damage using abrasive wheel, Item 350, WP 1803 00. **REPLACE PITCH BEAM NUT IF DAMAGE AFTER BLENDING EXCEEDS 0.010-INCH DEEP OR IS MORE THAN 15% OF SURFACE.**
6. Visually check pitch beam for cracks. **NO CRACKS ALLOWED.** If a crack is suspected, check using fluorescent penetrant inspection kit (TM 1-1520-265-23). **REPLACE PITCH BEAM IF CRACKED.**
7. Check for nicks, gouges, and fretting. Remove raised edges and blend out damage using abrasive wheel, Item 350, WP 1803 00.
8. Replace pitch beam if:
 - a. **CORNER DAMAGE AFTER BLENDING IS GREATER THAN 0.100-INCH DEEP, OR 0.020-INCH DEEP IN LUG AREA. MULTIPLE DAMAGE MUST BE 1/2-INCH APART.**
 - b. **SURFACE DAMAGE AFTER BLENDING IS GREATER THAN 0.050-INCH DEEP, OR 0.020-INCH DEEP IN LUG AREA. MULTIPLE DAMAGE MUST BE AT LEAST 1 INCH APART.**
 - c. **NO RADIAL OR AXIAL PLAY ALLOWED ON PITCH BEAM TO PITCH CONTROL SHAFT INSTALLATION.**
9. Check for corrosion (TM 1-1500-344-23). **REPLACE PITCH BEAM IF CORROSION IS:**

INSPECT - Continued

- a. **DEEPER THAN 0.010-INCH OR COVERS MORE THAN 10% OF SURFACE AREA IN BUSHING BORES AND ON PITCH CONTROL SHAFT MATING SURFACES.**
- b. **DEEPER THAN 0.030-INCH IN OPEN SECTIONS.**
- c. **DEEPER THAN 0.020-INCH IN LUG AREAS.**

NOTE

Do not use any type of scotch-brite pad.

- d. Remove corrosion and/or any discoloration that is less than limits, using abrasive wheel, Item 350, WP 1803 00.
10. Check flat surfaces of pitch beam which come in contact with pitch beam washer for nicks, gouges, corrosion, or fretting. Remove damage using abrasive wheel, Item 350, WP 1803 00. **REPLACE PITCH BEAM IF DAMAGE AFTER BLENDING EXCEEDS 0.010-INCH DEEP OR IS MORE THAN 15% OF SURFACE.**
11. Check pitch beam for cracks if a repair has been made or if a crack is suspected using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked pitch beam.
12. Apply corrosion resistant coating, Item 111, WP 1803 00, to repaired areas.
13. Touch up paint on pitch beam (WP 0381 00 and TM 55-1500-345-23).
14. Inspect exposed portion of pitch control shaft (WP 1148 00).
15. Check pitch control shaft to make sure dust plug is installed. If dust plug is not installed, install dust plug (WP 0585 00).

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean OD of pitch control shaft, conical seat of pitch beam, washer, and retaining nut using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, or cleaning compound solvent, Item 71, WP 1803 00.

WARNING

Improper torque of pitch beam retaining nut will result if corrosion preventive compound is applied to pitch beam retaining nut or washer mating surface. Do not apply corrosion preventive compound to pitch beam retaining nut or washer mating surfaces.

NOTE

Corrosion preventive compound, Item 107, WP 1803 00, is preferred for use on pitch control shaft mating surface. If corrosion preventive compound, Item 107, WP 1803 00, is not available, corrosion preventive compound, Item 105, WP 1803 00, may be used as an alternate.

2. Remove thread protection from shaft.

INSTALL - Continued**CAUTION**

Damage to pitch control shaft dust plug will result if excessive force is used to install pitch beam and retaining nut. Use care when installing pitch beam and retaining nut.

3. Lightly coat pitch control shaft tapered surface with corrosion preventive compound, Item 107, WP 1803 00, or corrosion preventive compound, Item 105, WP 1803 00.
4. **QA** Slide pitch beam on pitch control shaft (Figure 1, Sheet 2). Wipe off any excess corrosion preventive compound on pitch control shaft and pitch beam using machinery wiping towel, Item 344, WP 1803 00.
5. **QA** Install washer. Line up pitch beam lands with slots in washer. Turn pitch beam as necessary. Install retaining nut. **TORQUE NUT TO 950 - 1050 INCH-POUNDS USING SPANNER WRENCH. DO NOT EXCEED 1050 INCH-POUNDS TORQUE LIMIT WHEN ATTEMPTING TO ALIGN HOLES IN NUT AND WASHER.**
6. If holes in retaining nut do not align with holes in washer when 1050 inch-pounds is reached, do this:
 - a. Remove retaining nut, washer, and pitch beam. Check pitch beam conical seat and washer mating surface, retaining nut, washer, and pitch control shaft tapered surface for damage or debris which may prevent alignment of holes. Replace components as required.
 - b. Reinstall pitch beam.

WARNING

Aviation safety action message, UH-60-91-01, requires the use of a steel washer, 70103-11015-106. All aluminum washers, 70103-11015-101/-104, are not to be installed on the aircraft and are to be purged from the supply system.

- c. Install washer 180 degrees from original position.
- d. Install retaining nut. **TORQUE NUT TO 950 - 1050 INCH-POUNDS USING SPANNER WRENCH.**
7. **QA** Install screws and washers to retaining nut. **TIGHTEN SCREWS SNUGLY, DO NOT TORQUE.**
8. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire, screws. Install electrical insulation sleeving, Item 156, WP 1803 00, to nonelectrical wire to prevent chafing.
9. Apply corrosion preventive compound, Item 107, WP 1803 00, or corrosion preventive compound, Item 105, WP 1803 00, to exterior surfaces of tail rotor pitch beam retaining nut and washer.
10. Install pitch control rods to pitch beam (WP 0586 00).
11. Do tail rotor rig check (WP 1039 00).
12. Check area for cleanness and foreign material.
13. Check DA Form 2408-13-1 to make sure pitch beam retaining nut torque check is shown as due 9 to 11 flight hours after installation (WP 1692 00).
14. After torque has stabilized, do this:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- a. Clean corrosion preventive compound from retaining nut, washer and protruding end of shaft outside retaining

INSTALL - Continued

nut with a clean machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00. Then wipe this area with a clean machinery wiping towel, Item 344, WP 1803 00 dampened with technical acetone, Item 5, WP 1803 00.

- b. Apply a thin coat of corrosion preventive compound, Item 107, WP 1803 00, to pitch beam retaining nut, washer and protruding end of shaft outside retaining nut. Allow to dry and apply a second thin coat.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**TAIL ROTOR BOOT**

This WP supersedes WP 0588 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Tail Rotor Blade Boot Installation Tool, Locally-Made,
 Figure 207, WP 1805 00

Tape, Insulation, Electrical, Item 316, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Strap, Tiedown, Electrical Components, Item 310,
 WP 1803 00

References

TM 1-6625-724-13&P
 WP 0586 00
 WP 1803 00
 WP 1805 00

REMOVE

1. Remove tiedown straps and clips from boot (Figure 1).
2. Disconnect pitch control rod from horn (WP 0586 00).
3. If replacing boot, cut off old boot.
4. Remove tape from support and fairing if unserviceable (Figure 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean support and fairing where tape was removed, with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

INSPECT

Inspect tail rotor boot for cuts, tears, breaks, and dry rot. **NONE ALLOWED.** Replace damaged boot.

INSTALL

1. Locally make tail rotor blade boot installation tool (WP 1805 00).
2. Apply electrical insulation tape, Item 316, WP 1803 00, to ends of support and fairing (Figure 1, Detail A).
3. Coat mating surface on support with sealing compound, Item 273, WP 1803 00.

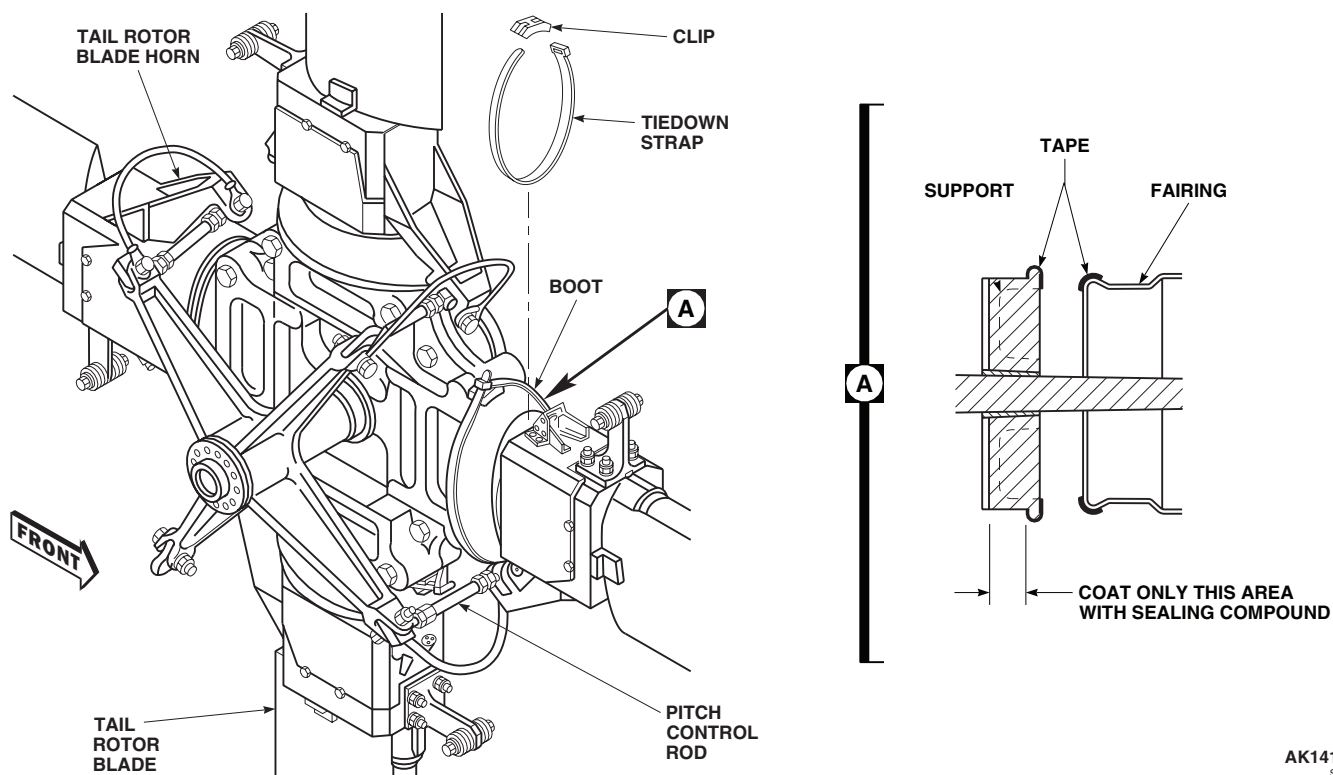
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Figure 1. Replace Tail Rotor Boot.

4. **QA** Lubricate leading edge and trailing edge of blade with water. Stretch boot over blade. Make sure egg-shaped end of boot goes on first.
5. **QA** With locally-made tail rotor blade boot installation tool, slide boot along spar to horn. Stretch boot over horn. Make sure both ends of boot are seated in grooves.
6. **QA** Place clips on boot at leading edge and secure boot and clips with electrical components tiedown straps, Item 310, WP 1803 00. Using spring scale, apply 10 to 20 pounds tension to end of tiedown straps (Figure 1).
7. Connect pitch control rod (WP 0586 00).

NOTE

A maintenance test flight is not required for replacement of tail rotor boot.

8. Do a tail rotor blade balance check (TM 1-6625-724-13&P).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****OUTBOARD RETENTION PLATE REPAIR**

This WP supersedes WP 0589 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Caliper Set, Outside Micrometer, GGG-C-105
Depth Gage
Micrometer, GGG-C-105
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 0 - 600 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Alcohol, Denatured, Item 48, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Sealing Compound, Item 282, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 327,
WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 55-1500-345-23
WP 0381 00
WP 0585 00
WP 1803 00

INSPECT/REPAIR**NOTE**

- Damage limits are the same for both inboard and outboard retention plates.
 - Nylon shim inspection is the same for inboard and outboard retention plates.
1. Check retention plate for cracks. **NO CRACKS ALLOWED.** Replace cracked retention plate.
 2. Visually inspect outboard retention plate for nicks, gouges, fretting, and scoring.
 3. Remove raised material and blend out damage using abrasive wheel, Item 350, WP 1803 00. **REPLACE RETENTION PLATE IF:**
 - a. **CORNER DAMAGE AFTER BLENDING IS GREATER THAN 0.100-INCH DEEP, OR 0.040-INCH DEEP IN BOLTHOLE AREAS. MULTIPLE DAMAGE MUST BE AT LEAST 1 INCH APART.**
 - b. **SURFACE DAMAGE AFTER BLENDING IS GREATER THAN 0.040-INCH DEEP IN OPEN AREAS, OR GREATER THAN 0.020-INCH DEEP IN BOLTHOLE AREAS. MULTIPLE DAMAGE MUST BE AT LEAST 1 INCH APART.**
 - c. **SURFACE DAMAGE IN MATING AREAS AFTER BLENDING IS GREATER THAN 0.020-INCH DEEP, OR COVERS MORE THAN 10% OF MATING AREA.**
 - d. **FRETTING ON AREAS THAT MATE WITH INBOARD RETENTION PLATE AFTER BLENDING IS GREATER THAN 0.010-INCH DEEP, OR COVERS MORE THAN 10% OF MATING AREA.**
 4. Touch up paint on repaired areas (WP 0381 00 and TM 55-1500-345-23).
 5. Remove screws and nylon shims from outboard and inboard retention plates (Figure 1).

INSPECT/REPAIR - Continued

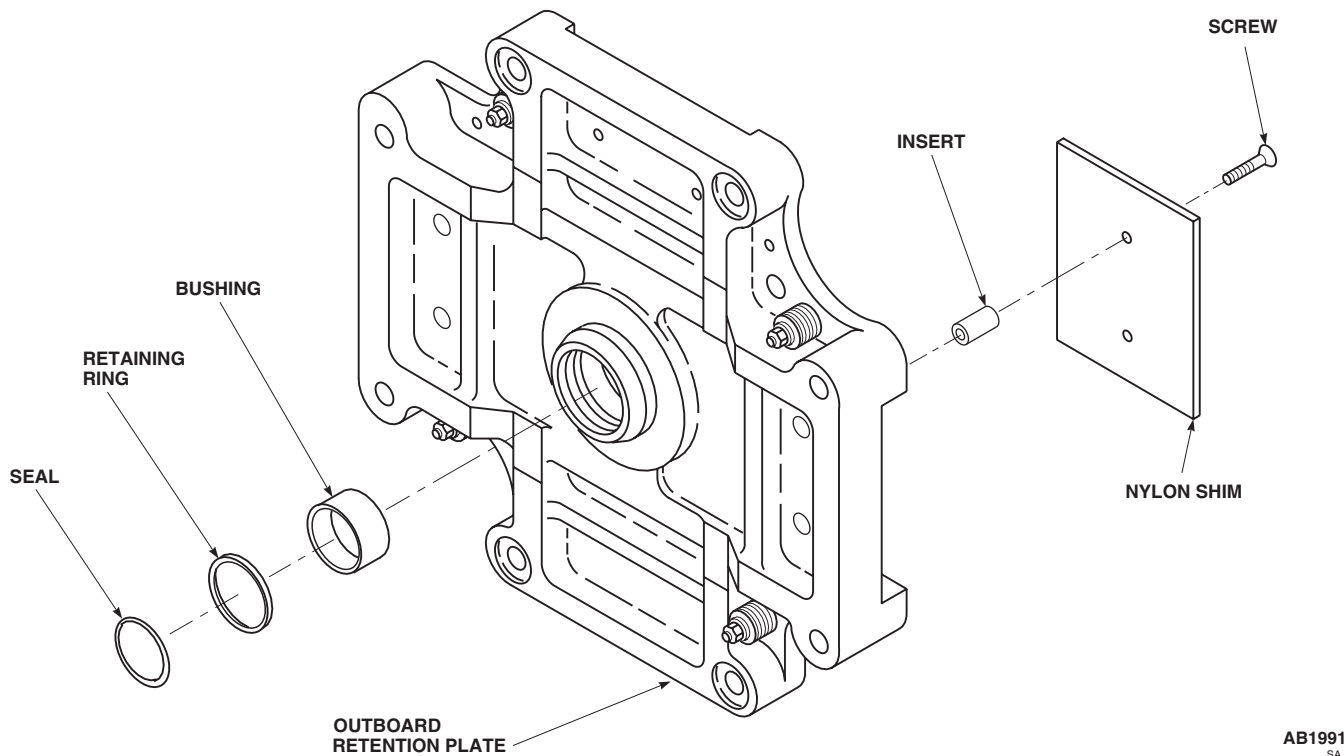


Figure 1. Inspect/Repair Outboard Retention Plate.

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6. Inspect nylon shims one inch in from edge of each corner using micrometer. **SHIM THICKNESS MUST BE 0.123 TO 0.126-INCH.** Replace nylon shims not within limits or if surface is not flat.
7. If new nylon shims are being installed inspect per above step. Do not use shim that is not within limits.
8. Check threaded inserts as follows:
 - a. Install screws in inserts and check drag torque required to install screws. **DRAG TORQUE SHALL BE 2 - 18 INCH-POUNDS.**
 - b. If drag torque is not within limits, replace screw and clean insert. Recheck drag torque. If drag torque is still not within limits, replace insert as follows:
 - (1) Remove insert using 5/32 Allen wrench.
 - (2) Clean tapped hole in outboard retention plate with denatured alcohol, Item 48, WP 1803 00. Blow dry cleaned area using clean, dry compressed air.
 - (3) Inspect tapped hole in retention plate for stripped, crossed, or damaged threads. **NONE ALLOWED.**
 - (4) Clean outer threads of insert with denatured alcohol, Item 48, WP 1803 00. blow dry cleaned area using clean, dry compressed air.
 - (5) Apply a thin coat of sealing compound, Item 282, WP 1803 00, to first two threads of insert.
 - (6) **USING 5/32 ALLEN WRENCH, INSTALL INSERT FLUSH TO 0.025-INCH BELOW THE SURFACE OF THE TAPPED HOLE, TO A DRIVING TORQUE OF 25 - 300 INCH-POUNDS.**
9. Install nylon shims and screws (Figure 1). **TIGHTEN SCREWS HANDTIGHT.**

INSPECT/REPAIR - Continued

10. Remove seal from retention plate.
11. Visually inspect retaining ring and end of bushing for corrosion pitting (TM 1-1500-344-23). **IF CORROSION PITTING IS DEEPER THAN 0.003-INCH ON RETAINING RING, OR 0.020-INCH ON END OF BUSHING, REPLACE RETAINING RING AND/OR BUSHING.**
12. Visually inspect inside diameter of bushing. Surface will appear bronze speckled under normal wear. **REPLACE BUSHING IF 70% OF INSIDE DIAMETER APPEARS BRONZE, BRONZE APPEARS SMEARED, OR BRONZE LAYER IS CHIPPED AND STEEL BACKING IS EXPOSED.**
13. If corrosion pitting is less than above limits do the following:
 - a. Remove outboard retention plate (WP 0585 00).
 - b. Remove retaining ring from pilot.
 - c. Using abrasive cloth, Item 82, WP 1803 00, remove corrosion from retaining ring using limits in Table 1.
 - d. Remove corrosion from bushing as follows:
 - (1) Insert a dry cleaning cloth, Item 86, WP 1803 00, inside bushing.

Table 1. Retaining Ring Rework Limits.

Total Thickness - 0.049-inch	Width - 0.148-inch
After Cleaning - 0.040-inch	Width - 0.142-inch
Total Diameter of ring - not less than 2.670 -inches	

- (2) Using abrasive cloth, Item 82, WP 1803 00, remove corrosion from top of bushing only. Do not use abrasive cloth on bushing ID.
 - (3) Remove cloth from inside bushing, being careful not to imbed any grit on ID of Teflon-impregnated bushing. Wipe bushing clean with dry cleaning cloth, Item 86, WP 1803 00.
14. Mask bushing ID with pressure sensitive adhesive tape, Item 327, WP 1803 00.
15. Apply primer coating, Item 243, WP 1803 00, to top of bushing and retaining ring groove in pilot.
16. Remove tape from bushing.
17. Apply primer coating, Item 243, WP 1803 00, to retaining ring and install retaining ring in pilot groove while primer is still wet.

NOTE

Make sure that seal is installed with flange facing outboard.

18. Install seal.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****TAIL ROTOR BLADE DEICE BRACKET**

This WP supersedes WP 0590 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

References

TM 1-1500-204-23
TM 1-6625-724-13&P
WP 0588 00
WP 1803 00

REMOVE

1. Using nonmetallic scraper, remove sealing compound from blade electrical connector and tail rotor blade deice bracket base mounting surfaces.
2. Remove screws, washers, and nuts securing tail rotor blade deice electrical connector to tail rotor blade deice bracket (Figure 1).
3. Remove cap from tail rotor blade deice electrical connector.
4. Remove tail rotor blade deice electrical connector through hole in tail rotor blade deice bracket assembly.
5. Remove screws and tail rotor blade deice bracket from fairing.

INSPECT

1. Inspect nutplates for damage and secure attachment to fairing.
2. Replace damaged or loose nutplates as follows:
 - a. Cut tail rotor boot tie and roll back tail rotor boot (WP 0588 00).
 - b. Replace nutplates (TM 1-1500-204-23).
 - c. Secure tail rotor boot (WP 0588 00).

INSTALL

1. **QA** Install tail rotor blade deice bracket to fairing with screws (Figure 1).
2. **QA** Install tail rotor blade deice electrical connector through hole in tail rotor blade deice bracket.

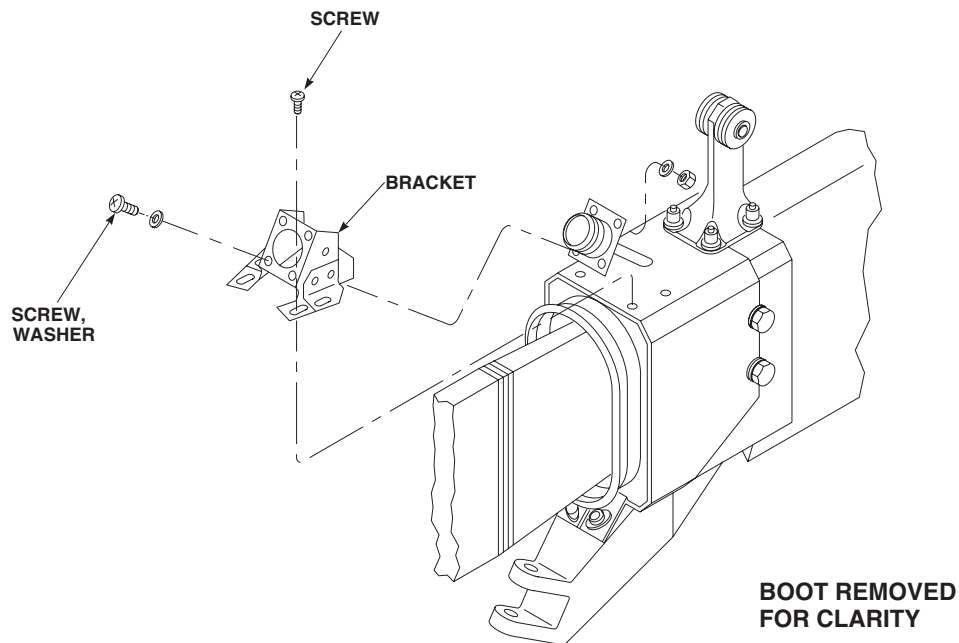
INSTALL - ContinuedAA9740_1C
SA

Figure 1. Replace Tail Rotor Blade Deice Bracket.

CAUTION

Correct hardware must be used or failure of the tail rotor blade deice bracket may occur.

NOTE

If tail rotor deice slipring rotor electrical harnesses are not to be installed, substitute one MS35265-16 screw in place of NAS600-7P screw.

3. **QA** If installed apply epoxy primer coating kit, Item 120, WP 1803 00, to screws, washers, and nuts. Secure tail rotor blade deice electrical connector to tail rotor blade deice bracket with screws, washers, and nuts while primer is still wet.
4. If tail rotor deice slipring rotor electrical harness are not to be installed. Install cap on electrical connector and lock-wire using nonelectrical wire, Item 357, WP 1803 00.
5. Apply sealing compound, Item 273, WP 1803 00, to edges of tail rotor blade deice bracket to fairing mounting surfaces and tail rotor blade deice electrical connector to tail rotor blade deice bracket mating surfaces.

INSTALL - Continued**NOTE**

A maintenance test flight is not required for replacement of tail rotor blade deice bracket.

6. Do a tail rotor balance check (TM 1-6625-724-13&P).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ROTOR SYSTEM

TAIL ROTOR BALANCE

INITIAL SETUP:

References

TM 1-6625-724-13&P

Refer to TM 1-6625-724-13&P.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
MAIN ROTOR SHAFT NUT (AVIM)

INITIAL SETUP:
Tools and Special Tools

Drift, 7/16-Inch Diameter
 Fluorescent Penetrant Inspection Kit, XMA101
 Inside Micrometer Caliper, GGG-C-105
 Magnaflux
 Micrometer, GGG-C-105
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Telescopic Gage Set, GGG-G-17

Sealing Compound, Item 289, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 334,
 WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

TM 1-1520-265-23
 WP 1803 00

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
 Lacquer, Item 172, WP 1803 00

REMOVE

Using arbor press and 7/16-inch diameter drift, press insert out from top of nut (Figure 1, Detail A).

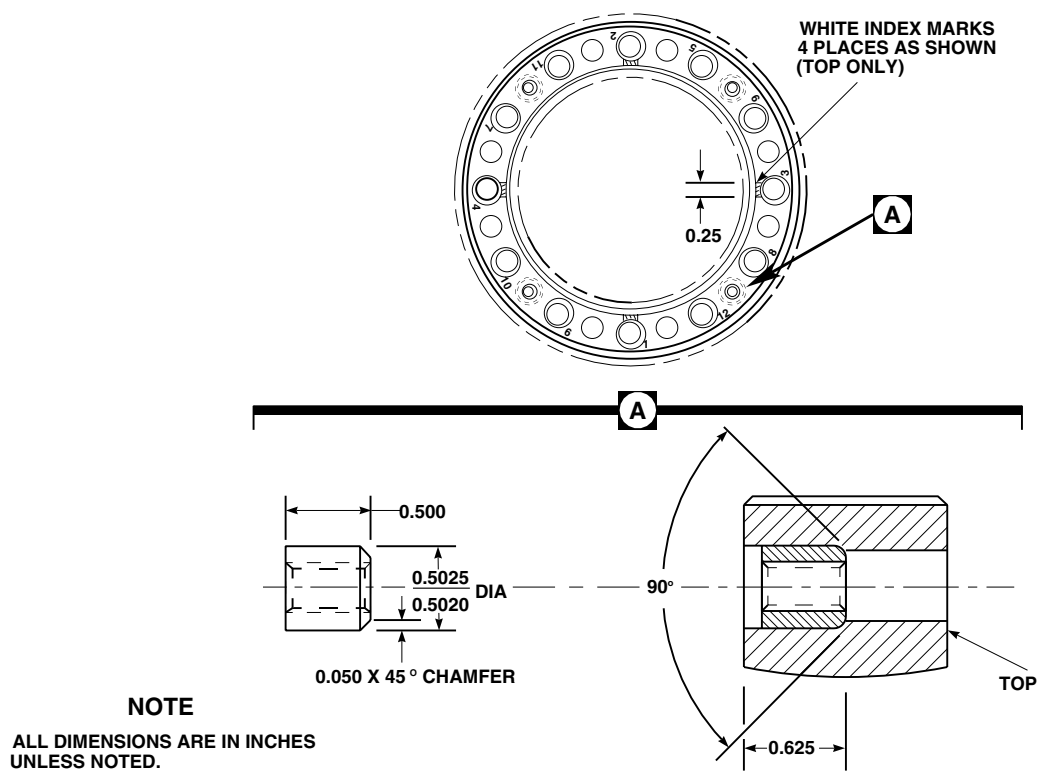


Figure 1. Replace Main Rotor Shaft Nut Insert (AVIM).

INSPECT

1. Magnetic particle inspect nut, using Magnaflux (TM 1-1520-265-23).
2. Measure ID of hole where insert is to be installed. **ID MUST BE NO GREATER THAN 0.501-INCH.** Replace main rotor shaft nut if hole ID is greater than limits.

INSTALL

1. Apply sealing compound, Item 289, WP 1803 00, following manufacturer's instructions.

NOTE

Before installing insert, make sure chamfer on insert is facing bottom of nut.

2. **QA** Using arbor press and 7/16-inch diameter drift, press insert in from bottom of nut.
3. Mask threads and inserts with pressure sensitive adhesive tape, Item 334, WP 1803 00.
4. Apply one coat of solvent cleaning compound, Item 71, WP 1803 00, to nut.
5. Apply two coats of lacquer, Item 172, WP 1803 00, to nut.
6. Paint index marks on nut with lacquer, Item 172, WP 1803 00.
7. **QA** Paint numbers indicating torque sequence with lacquer, Item 172, WP 1803 00, over existing stencil marks.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
SPINDLE SLEEVE BEARING OUTER SLEEVE (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

REMOVE**NOTE**

The following procedure applies to droop stop support ring, 70105-08056-101, only.

Place droop stop support ring and outer sleeve assembly in arbor press and carefully remove outer sleeve (Figure 1).

INSTALL

1. Heat droop stop support ring in an oven not over 200°F(93°C) for 30 minutes.
2. **QA** Using old outer sleeve as a drift, carefully press outer sleeve into droop stop support ring (Figure 1).

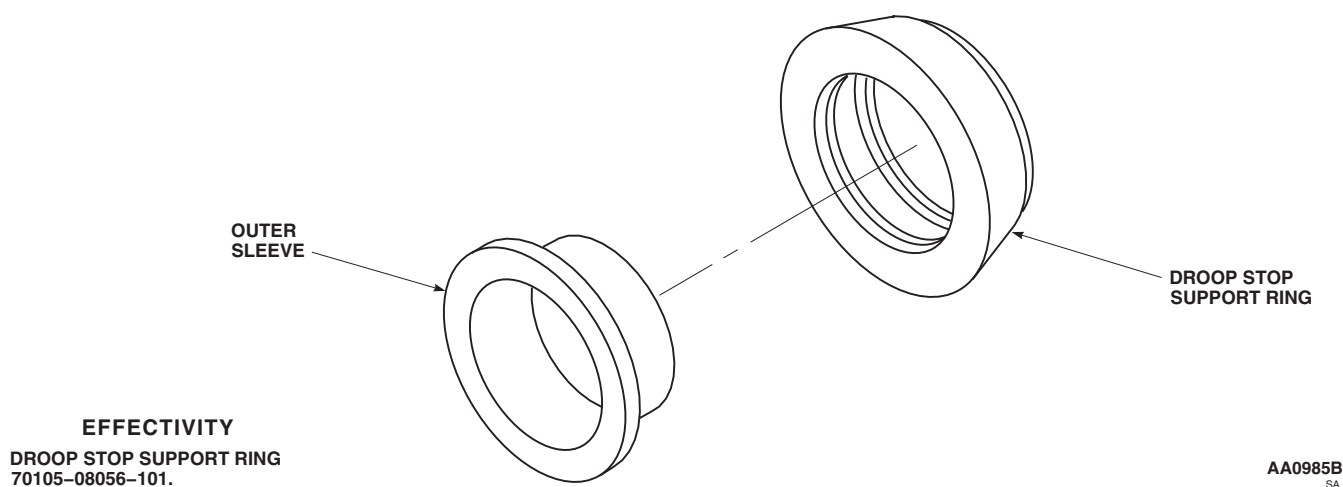


Figure 1. Replace Spindle Sleeve Bearing Outer Sleeve (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ROTOR SYSTEM

SPINDLE CUFF LUG BUSHINGS (AVIM)

This WP supersedes WP 0594 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Blade/Spindle Cuff Bushing Installation Tool, Locally-Made, Figure 161, WP 1805 00
Inside Micrometer Caliper, GGG-C-105
Powertrain Repairer's Toolkit, SC 5180-99-B13

Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

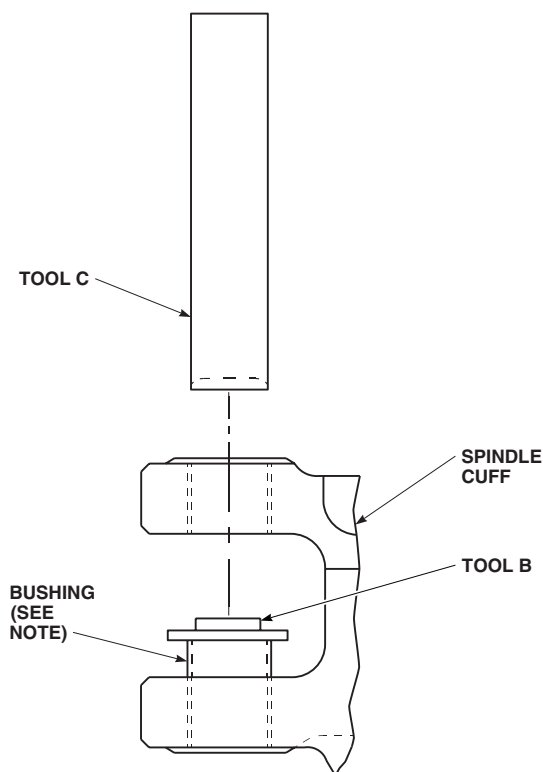
WP 1803 00
WP 1805 00

Material/Parts

Carbon Dioxide, Item 64, WP 1803 00
Primer Coating, Item 243, WP 1803 00

REMOVE

1. Locally make blade/spindle cuff bushing installation tool A, tool B, and tool C (WP 1805 00).
2. Using tool A and arbor press, press out bushing.



NOTE
INSTALL FLUSH TO 0.003
INCH BELOW SPINDLE
CUFF SURFACE.

AA1144
SA

Figure 1. Replace Spindle Cuff Lug Bushings (AVIM).

REMOVE - Continued**INSPECT****WARNING****CSI**

Minimum press fit is a critical characteristic. Spindle bore must not exceed 1.125 and bushing O. D. must not be less than 1.1255.

1. Check bores for scoring or corrosion. **NONE ALLOWED.** Replace spindle if damage is found.
2. Check ID of bushing bore. **MAXIMUM ID IS 1.125-INCHES.** Replace spindle if ID is more than limits.

INSTALL**WARNING**

Injury to personnel will result if protective gloves are not worn when handling carbon dioxide or chilled parts. Wear protective gloves when handling carbon dioxide or chilled parts.

1. Place new bushings in carbon dioxide, Item 64, WP 1803 00, for at least 3 hours before installation.
2. Coat bushing outer diameter and bushing bore with primer coating, Item 243, WP 1803 00.
3. **QA** Press in bushings while primer coating is still wet flush to 0.003-inch below surface using tool B and tool C (Figure 1).
4. Apply bead of sealing compound, Item 273, WP 1803 00, to edges of bushing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**SPINDLE ELASTOMERIC BEARING (AVIM)**

This WP supersedes WP 0595 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Gloves
 Goggles/Face Shield
 Mallet, Rawhide
 Micrometer, GGG-C-105
 Nonmetallic Scraper, Locally-Made
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Sleeve Bearing Removal/Installation Tool, Locally-Made, Figure 198, WP 1805 00
 Soft Bristle Brush
 Telescoping Gage
 Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Detergent, General Purpose, Item 115, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00

Corrosion Resistant Coating, Item 111, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

WP 0534 00
 WP 0536 00
 WP 0554 00
 WP 0557 00
 WP 1803 00
 WP 1805 00

DISASSEMBLE

1. Locally make sleeve bearing removal/installation tool (WP 1805 00).
2. Remove sealing compound from bolt heads using nonmetallic scraper.
3. Remove bolts, washers, and nuts securing elastomeric spherical and thrust bearings together (Figure 1). Discard nuts.
4. Separate spherical bearing from thrust bearing.

NOTE

- Shims, SS52C3000Z3237, between elastomeric bearing sleeve bearing and thrust bearing or shims, 70102-08119-101, between spherical and thrust bearing endplates will no longer be required.
 - If shims, 70102-08119-101, are found between spherical bearing and thrust bearing endplates, bolts securing bearings must be replaced with shorter bolts.
5. If shims are found between spherical bearing, sleeve bearing and thrust bearing, discard shims.
 6. If shims are found between spherical bearing and thrust bearing endplates, discard shims and bolts.

CLEAN**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to bearings may result if any oil, grease, or other fluids are spilled or accidentally get on bearings, do not use solvent to remove them. Use only warm water, mild detergent, and soft bristle brush to clean elastomeric bearings. After cleaning, rinse with clear water and then dry.

1. Clean spherical and thrust bearing titanium and aluminum endplates and bolts and washers using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
2. Clean spherical and thrust bearings using machinery wiping towel, Item 344, WP 1803 00, detergent, general purpose, Item 115, WP 1803 00, and a soft bristle brush.

INSPECT/REPAIR

1. Inspect spherical and thrust bearings (WP 0534 00).
2. Inspect droop stop lug bushings (WP 0536 00).
3. Inspect droop stop chafe pad (WP 0557 00).
4. Inspect antifiap stop threaded inserts (WP 0554 00).
5. Inspect spherical bearing endplates as follows:
 - a. Inspect spherical bearing endplates for cracks. **NO CRACKS ALLOWED.** Replace spherical bearing with cracked endplates.
 - b. Inspect outboard titanium endplate for corner and surface damage, corrosion pitting, nicks, and scratches. **MAXIMUM DAMAGE MUST NOT EXCEED 0.050-INCH DEEP . A MINIMUM DISTANCE BETWEEN MULTIPLE DAMAGE MUST NOT EXCEED DEPTH OF DAMAGE. DAMAGE MUST NOT EXCEED 20% OF TOTAL SURFACE AREA.**
 - c. Inspect inboard aluminum endplate and bearing seat area (3.2465 to 3.2485-inch diameter) for corner and surface damage, corrosion pitting, nicks and scratches. **MAXIMUM DAMAGE MUST NOT EXCEED 0.050-INCH DEEP . A MINIMUM DISTANCE BETWEEN MULTIPLE DAMAGE MUST NOT EXCEED DEPTH OF DAMAGE. DAMAGE MUST NOT EXCEED 20% OF TOTAL SURFACE AREA.** Replace spherical bearing if damaged beyond limits.
 - d. **NO REPAIR ALLOWED IN 0.255 TO 0.260-INCH HOLES.**
6. Repair spherical bearing endplates as follows:
 - a. Blend out reparable areas using abrasive cloth, Item 82, WP 1803 00. Polish reworked area using abrasive cloth, Item 82, WP 1803 00, 0.001 to 0.002-inch below damage depth. Blend to 0.005-inch radius.
 - b. Apply corrosion resistant coating, Item 111, WP 1803 00, to repaired inboard aluminum endplate surfaces.
7. Inspect thrust bearing endplates as follows:

INSPECT/REPAIR - Continued

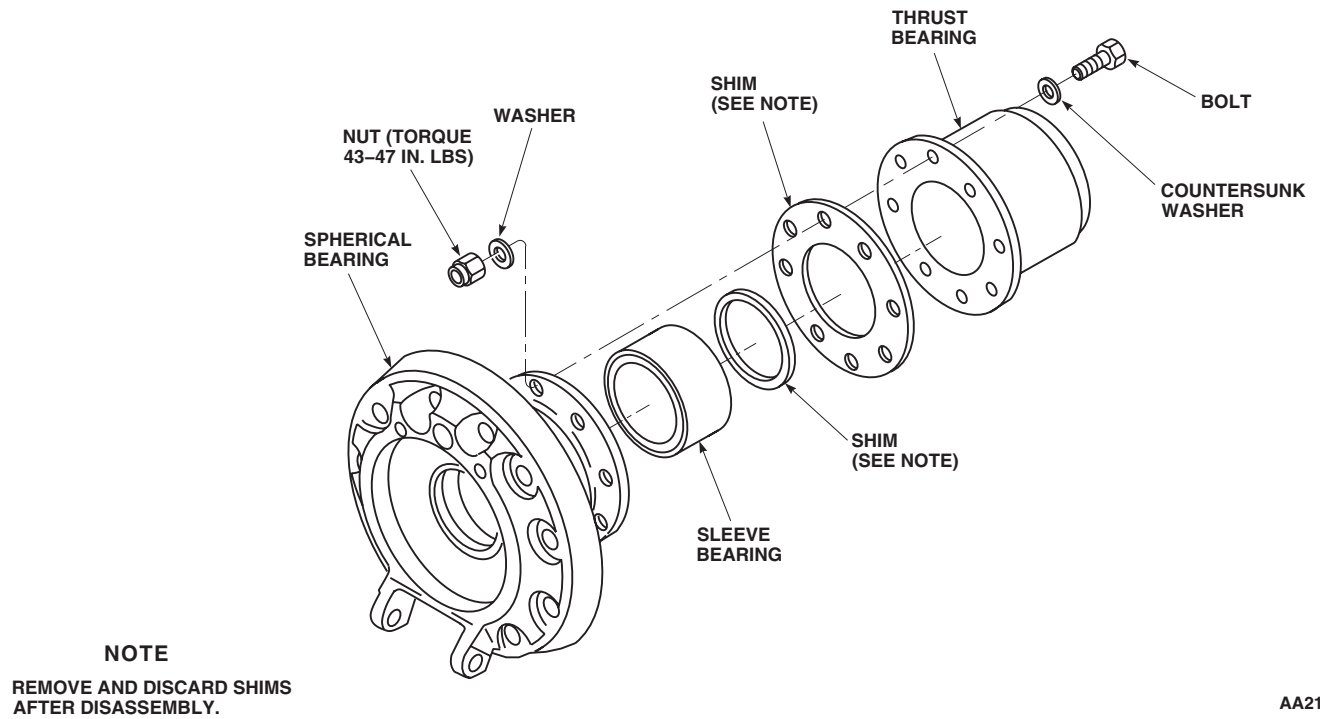


Figure 1. Repair Spindle Elastomeric Bearing (AVIM).

- a. Inspect thrust bearing endplates for cracks. **NO CRACKS ALLOWED.** Replace thrust bearing with cracked endplates.
- b. Inspect endplates for corner and surface damage, corrosion pitting, nicks, and scratches. **MAXIMUM DAMAGE MUST NOT EXCEED 0.050-INCH DEEP. A MINIMUM DISTANCE BETWEEN MULTIPLE DAMAGE MUST NOT EXCEED DEPTH OF DAMAGE. DAMAGE MUST NOT EXCEED 20% OF SURFACE AREA.** Replace thrust bearing if damaged beyond limits.
- c. **NO REPAIR ALLOWED IN 0.255 TO 0.260-INCH HOLES OR IN SPLINE AREAS.**
8. Repair thrust bearing endplates as follows:
 - a. Blend out reparable areas using abrasive cloth, Item 82, WP 1803 00. Polish reworked area using abrasive cloth, Item 82, WP 1803 00, 0.001 to 0.002-inch below damage depth. Blend to 0.005-inch radius.
 - b. Apply corrosion resistant coating, Item 111, WP 1803 00, to repaired outboard aluminum endplate surfaces.
9. Measure sleeve bearing ID 0.050-inch from each end of ID in 4 places, at 12/6 o'clock position and at 3/9 o'clock position. **AVERAGE MAXIMUM ID IS 2.922-INCHES. NO MORE THAN 0.014-INCH OUT OF ROUND ALLOWED.** Replace sleeve bearing if not within limits.
10. Inspect sleeve bearing Teflon liner for disbonding. **NO MORE THAN 0.250-INCH EDGE DISBOND ALLOWED. MULTIPLE EDGE DISBONDS MUST NOT BE CLOSER THAN 1 INCH APART.** Replace sleeve bearing if not within limits.
11. Inspect sleeve bearing Teflon liner for scoring (either circumferential or longitudinal), nicks, or gouges. Measure damage and outer race inside diameter and then chart results on graph (Figure 2). Replace outer sleeve bearing if depth of damage extends into the **NOT ACCEPTABLE** area of graph.

INSPECT/REPAIR - Continued

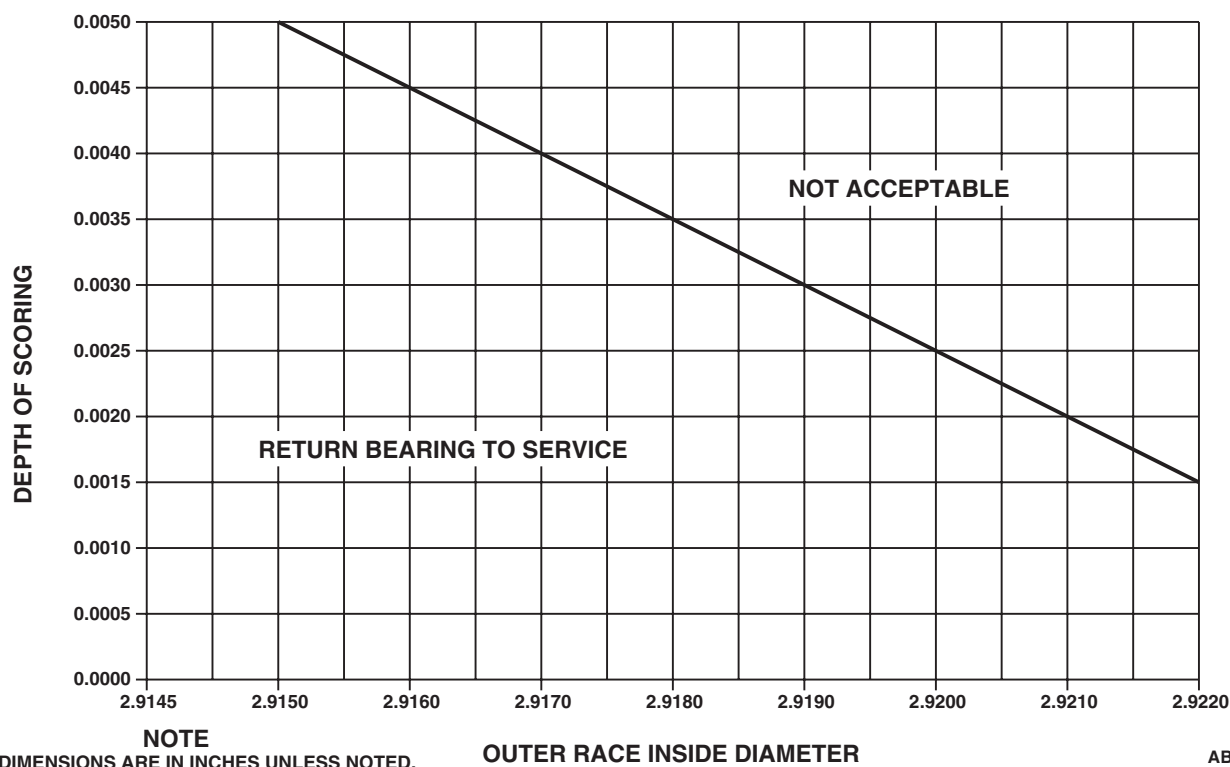


Figure 2. Inspect Spindle Elastomeric Outer Sleeve Bearing (AVIM).

12. To remove sleeve bearing from spherical bearing, do this:

- Using locally-made sleeve bearing removal/installation tool (WP 1805 00), support spherical bearing in arbor press with droop stop lug facing up.
- Using locally-made sleeve bearing removal/installation tool (WP 1805 00), remove sleeve bearing from spherical bearing. Discard sleeve bearing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- Clean sleeve bearing aluminum seat in spherical bearing using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

ASSEMBLE**WARNING**

Injury to personnel and damage to equipment will result if spherical and thrust bearings are assembled without installing sleeve bearing outer race into spherical bearing. Make sure sleeve bearing outer race is installed into spherical bearing prior to assembling spherical and thrust bearings.

NOTE

Sleeve bearing outer race is 2.25-inches long with Teflon fabric lining on ID.

1. Check spherical bearing for sleeve bearing installation. If sleeve bearing is not installed, install sleeve bearing into spherical bearing as follows:

WARNING

Injury to personnel will result if carbon dioxide or chilled parts come in contact with skin. Use protective gloves when handling carbon dioxide or chilled parts.

- a. Freeze sleeve bearing in carbon dioxide, Item 64, WP 1803 00, for 1 hour minimum.
 - b. Heat spherical bearing in oven at 140°F(60°C) to 160°F(71°C) for 20 minutes maximum.
 - c. Position spherical bearing support tool in arbor press.
 - d. Have rawhide mallet and spherical bearing installation tool ready at arbor press to install sleeve bearing.
 - e. Remove spherical bearing from oven and coat sleeve bearing bore with primer coating, Item 243, WP 1803 00.
 - f. Position spherical bearing into spherical bearing support tool droop stop lug down.
 - g. Remove sleeve bearing from carbon dioxide and carefully position into spherical bearing bore. Tap sleeve bearing evenly to align bearing and prevent cocking.
 - h. **QA** Using spherical bearing installation tool and arbor press, press sleeve bearing into spherical bearing 1 3/4 to 2 inches. Remove spherical bearing installation tool and spherical bearing from support tool and check leading edge of sleeve bearing for burrs and debris pickup. Clean burrs and pickup and continue pressing sleeve bearing into spherical bearing until bottomed.
 - i. **QA** Using 0.002-inch feeler gage, check to make sure sleeve bearing is fully seated in spherical bearing bore.
 - j. Allow assembly to return to room temperature.
 - k. Measure sleeve bearing ID 0.050-inch from each end of ID in 4 places, at 12/6 o'clock position and at 3/9 o'clock position. **AVERAGE MINIMUM ID IS 2.9145-INCHES.** Replace sleeve bearing if not within limits.
2. Apply thin coat of epoxy primer coating kit, Item 120, WP 1803 00, to mating surfaces of spherical and thrust bearings and to bolts.
 3. Line up three index marks on spherical and thrust bearings. Install bolts, and countersunk washers, with countersunk side of washer facing bolt head, with bolt head facing thrust bearing side.
 4. **QA** Install washers and nuts on bolts (Figure 1). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.** Wipe excess primer from bolts and bearing mating surfaces using machinery wiping towel, Item 344, WP 1803 00.
 5. Apply sealing compound, Item 273, WP 1803 00, to edges of bolt heads and nuts.
 6. Touch up exposed surfaces of bolts, washers, and nuts with primer coating, Item 243, WP 1803 00.

ASSEMBLE - Continued

7. **QA** If shim, 70102-08119-101, was removed, reidentify elastomeric bearing assembly to part No. 70102-08100-044.
Adjust bolt grip length to provide proper assembly grip length and sufficient threads safety past the nut.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****DAMPER INDICATOR (AVIM)**

INITIAL SETUP:**Tools and Special Tools**

Damper Indicator Cradle, Locally-Made, Figure 164,
WP 1805 00
Damper Indicator Insertion Tool, Locally-Made, Figure
165, WP 1805 00
Damper Indicator Installation Tool, Locally-Made,
Figure 166, WP 1805 00
Drill Bit, No. 52
Hydraulic Repairer's Toolkit, SC 5180-99-A03

Grease, Aircraft and Instrument, Item 133,
WP 1803 00
Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)

References

WP 1803 00
WP 1805 00

Material/Parts

Adhesive, Item 38, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00

DISASSEMBLE

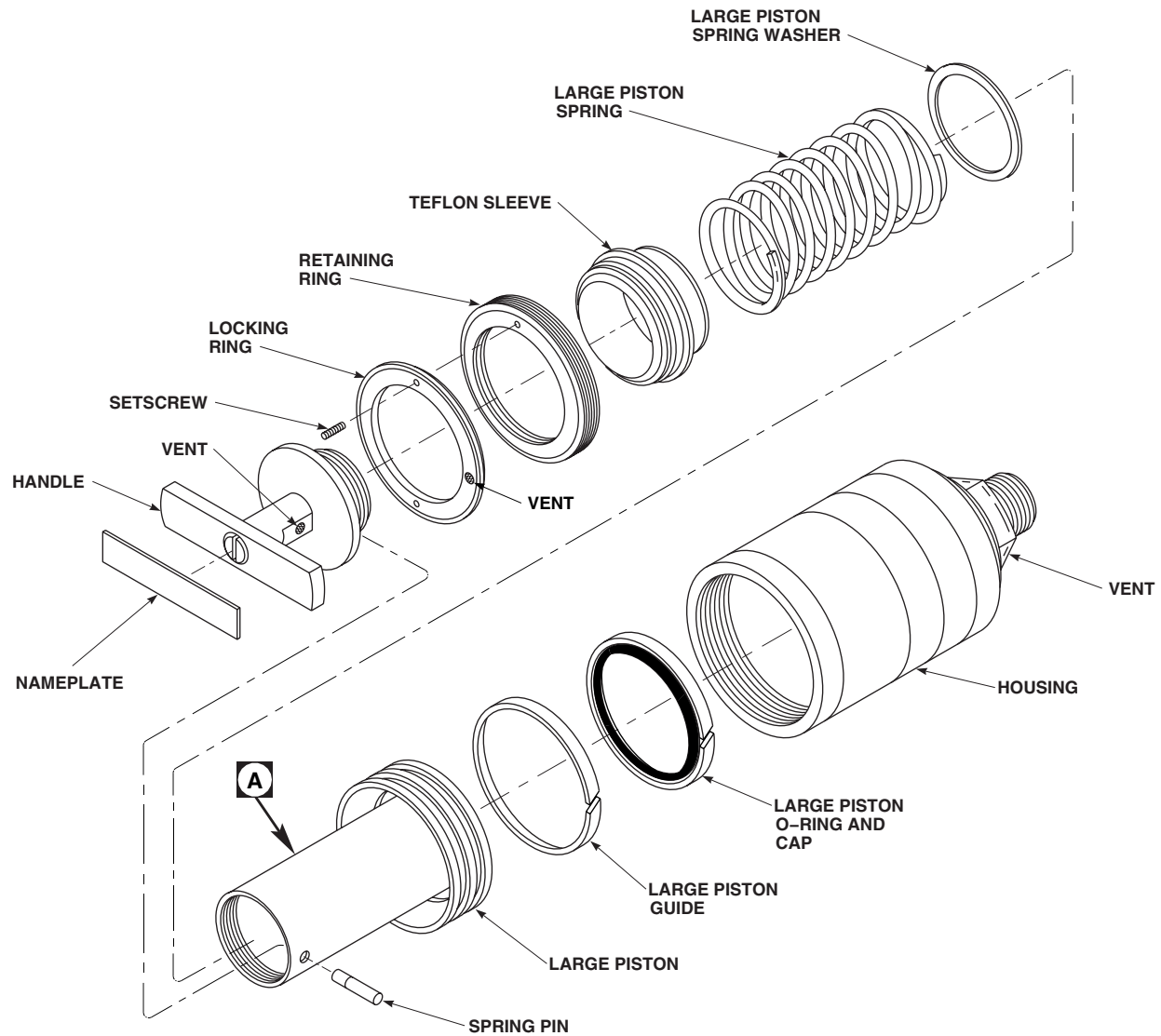
1. Locally make damper indicator cradle, insertion tool, and installation tool (WP 1805 00).
2. Manually pull handle to extend piston and position in locally-made damper indicator cradle (WP 1805 00).
3. Tap spring pin into handle using pin punch (Figure 1, Sheet 1 and Figure 1, Sheet 2). Remove indicator from cradle.
4. Mark housing in line with vent on the locking ring.
5. Using an Allen wrench, remove two setscrews from locking ring.
6. On damper indicator, 9018-1, do this:
 - a. Using an adjustable spanner wrench, unscrew locking ring from housing and slide it over the handle (Figure 1, Sheet 1).
 - b. Using same wrench, unscrew retaining ring and slide it over the handle.
 - c. Grasping the handle, slowly remove large piston from housing.

WARNING

Injury to personnel and damage to equipment will result if care is not taken when removing handle. Piston springs are compressed. Take care when removing handle.

- d. Slowly unscrew handle from piston.
- e. Remove spring pin from inside handle.
- f. Remove large piston spring and Teflon sleeve.
- g. Remove small piston spring from inside large piston.
- h. Push small piston out of large piston.

DISASSEMBLE - Continued

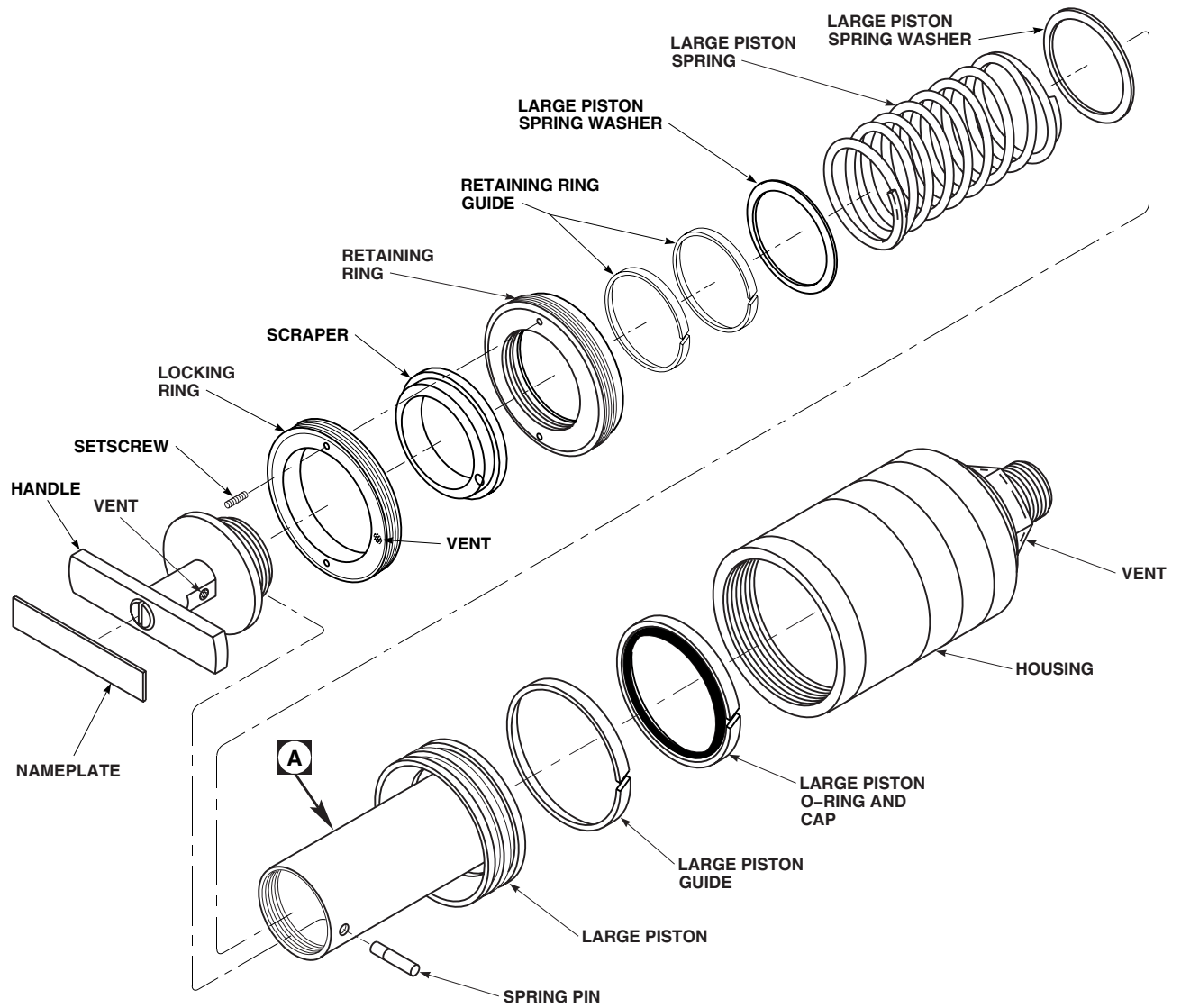


EFFECTIVITY
 DAMPER INDICATOR,
 9018-1.

AB2571_1
 SA

Figure 1. Repair Damper Indicator (AVIM). (Sheet 1 of 3)

DISASSEMBLE - Continued



EFFECTIVITY
DAMPER INDICATOR,
9018-2.

AB2571_2
 SA

Figure 1. Repair Damper Indicator (AVIM). (Sheet 2 of 3)

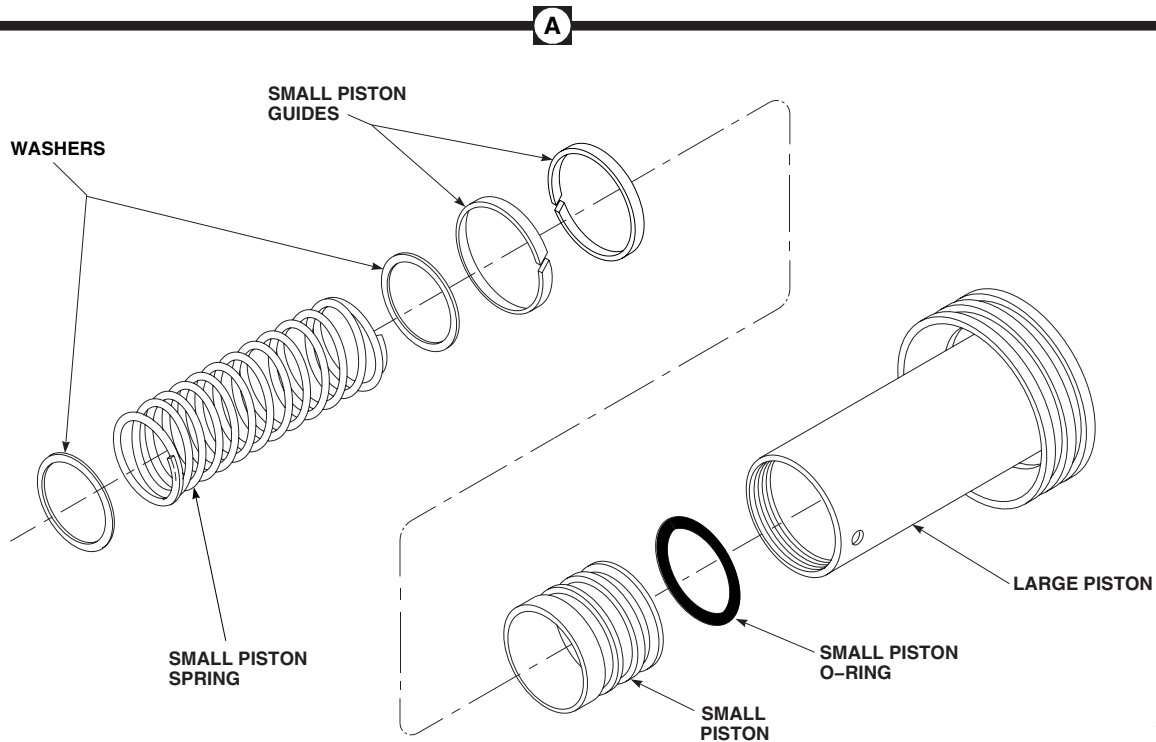
DISASSEMBLE - ContinuedAB3726_3
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Figure 1. Repair Damper Indicator (AVIM). (Sheet 3 of 3)

- i. Remove cap, packings and piston guides from large and small pistons (Figure 1, Sheet 3, Detail A). Discard packings.
 - j. Remove spring washers from large and small pistons, including small spring washer from underside of the handle.
7. On damper indicator, 9018-2, do this (Figure 1, Sheet 2):

WARNING

Injury to personnel and damage to equipment will result if care is not taken when removing handle. Piston springs are compressed. Take care when removing handle.

- a. Slowly unscrew handle from piston.
- b. Remove spring pin from inside handle.
- c. Remove small piston spring and washers from inside large piston (Figure 1, Sheet 3, Detail A).
- d. Using an adjustable spanner wrench, unscrew locking ring and scraper assembly from housing (Figure 1, Sheet 2). Separate scraper and locking ring.
- e. Using same wrench, unscrew retaining ring from housing. Remove guides from retaining ring.
- f. Remove large piston spring and washers.
- g. Slowly remove large piston from housing.
- h. Push small piston out of large piston.

DISASSEMBLE - Continued

- i. Remove cap, packings, and piston guides from large and small pistons. Discard packings.
8. Clean all parts in cleaning compound solvent, Item 71, WP 1803 00, and dry with compressed air.

INSPECT

1. Inspect exterior and interior of large and small pistons for nicks, scratches, gouges, and general condition. Pay particular attention to piston guide grooves and lands. Also inspect servicing marks on exterior of large piston. If black or gold band anodizing is worn through to aluminum, replace large piston.
2. Inspect exterior and interior of housing. If damaged, replace housing.
3. Inspect handle, locking ring and retaining ring for general condition. Pay particular attention for cracks in the locking ring.
4. Inspect large and small springs for corrosion and cracks.
5. Inspect stainless steel mesh in the three vent holes for integrity and condition.
6. Inspect for any damaged, loose or missing labels and decals on exterior of housing.

ASSEMBLE

1. If new large piston or new handle is being used, do this:
 - a. Thread handle into large piston (Figure 1, Sheet 1 and Figure 1, Sheet 2).
 - b. If spring pin holes do not line up, drill hole in handle in line with existing hole in large piston using No. 52 drill bit. Deburr hole.
 - c. Unthread handle from large piston and clean any metal particles from threads using cleaning compound solvent, Item 71, WP 1803 00.
 - d. **QA** Thread retaining ring and locking ring to housing. Remove rings and clean any metal particles from threads using cleaning compound solvent, Item 71, WP 1803 00.
2. Coat all internal part with petroleum base hydraulic fluid, Item 145, WP 1803 00, before assembly.

NOTE

Make sure that small cut on each piston guide is 180 degrees apart from each other.

3. **QA** On small piston install new piston guides and O-ring.
4. **QA** On large piston install new guide, O-ring, and cap using locally-made damper indicator installation tool (WP 1805 00).
5. **QA** Install new spring washers in large and small pistons, including small spring washer in underside of handle.
6. **QA** Install small piston into bore of large piston until it bottoms. Make sure packing and piston guides remain in their proper position.
7. On damper indicator, 9018-1, do this (Figure 1, Sheet 1):
 - a. Apply a thin coat of aircraft and instrument grease, Item 133, WP 1803 00, to threads of handle and large piston.
 - b. **QA** Place small piston spring into bore of large piston. Make sure spring seats properly inside small piston.
 - c. **QA** Place large piston spring over large piston. Place new Teflon sleeve in top of large spring.
 - d. Place handle into Teflon sleeve. Push down on handle, compressing both springs, and screw into large piston.

ASSEMBLE - Continued

- e. Install large piston into housing using locally-made damper indicator insertion tool (WP 1805 00). Make sure packing and piston guide remain in their proper position.
 - f. Apply a thin coat of aircraft and instrument grease, Item 133, WP 1803 00, to threads of housing, retaining ring, and locking ring.
 - g. Screw retaining ring into housing until piston no longer feels loose in housing (approximately 3 full turns).
 - h. Slip locking ring over handle. Screw locking ring into housing until it contacts retaining ring.
 - i. **QA** Continue to tighten locking ring until vent aligns with mark on housing.
8. On damper indicator, 9018-2, do this (Figure 1, Sheet 2):
- a. Install large piston into housing using locally-made damper indicator insertion tool (WP 1805 00). Make sure packing and piston guide remain in their proper position.
 - b. Apply a thin coat of aircraft and instrument grease, Item 133, WP 1803 00, to threads of housing, retaining ring, and locking ring.
 - c. **QA** Place large piston spring over large piston.
 - d. **QA** Install guides into retaining ring. Place washer and retaining ring on top of large spring.
 - e. Push down on retaining ring and screw into housing until top of retaining ring is approximately 1/4-inch below top of housing. Make sure retaining ring guides remain in their proper position.
 - f. **QA** Install scraper into locking ring.
 - g. Screw locking ring and scraper assembly into housing until it contacts retaining ring.
 - h. **QA** Continue to tighten locking ring until vent aligns with marks on housing.
 - i. Apply a thin coat of aircraft and instrument grease, Item 133, WP 1803 00, to threads of handle and large piston.
 - j. Place small piston spring into bore of large piston. Place washer and handle on top of spring.
 - k. **QA** Push down on handle, compressing spring, and screw into large piston.
9. Manually pull handle to extend piston and position in damper indicator cradle.
10. **QA** Install spring pin into large piston and handle using pin punch. Remove indicator from cradle.
11. **QA** Apply adhesive, Item 38, WP 1803 00, to setscrews and tighten them into the locking ring using an Allen wrench. Do not over tighten.

TEST

1. Connect hydraulic servicing unit to damper indicator.

CAUTION

Damage to piston may result if hydraulic pressure exceeds 100 psi. Do not exceed 100 psi hydraulic pressure.

- 2. Slowly, apply hydraulic pressure, and fully extend piston. Check valve in bottom of housing should release fluid before 100 psi.
- 3. Allow damper indicator to sit under pressure for 30 minutes. Check for leakage. **NONE ALLOWED.**
- 4. If indicator leaks, repeat repair procedures. Pay particular attention to installation of new packings.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**DAMPER SEALS (AVIM)**

This WP supersedes WP 0597 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Hydraulic Repairer's Toolkit, SC 5180-99-A03
 Hydraulic Test Stand, MIL-T-7612D
 Main Rotor Damper Holding Fixture, Locally-Made,
 Figure 223, WP 1805 00
 Main Rotor Damper Piston Rod Insertion Tool,
 Locally-Made, Figure 225, WP 1805 00
 Main Rotor Damper Piston Rod Wrench, Locally-
 Made, Figure 226, WP 1805 00
 Main Rotor Damper Seal Expansion Tool, Locally-
 Made, Figure 224, WP 1805 00
 Main Rotor Damper Sleeve Bearing Installation/
 Removal Tool, Locally-Made, Figure 222,
 WP 1805 00
 Packing, MS28778-6
 Plug, AN814-6D
 Tool Kit, Insert Installation, CUTA1870104C
 Torque Wrench, 150 - 1000 in. lbs.
 Torque Wrench, 300 - 2500 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Brush, Acid Swabbing, Item 57, WP 1803 00

Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 75, WP 1803 00
 Cloth, Abrasive, Item 79, WP 1803 00
 Cloth, Abrasive, Item 80, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Corrosion Removing Compound, Item 108,
 WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Hydraulic Fluid, Petroleum Base, Item 145,
 WP 1803 00
 Remover, Paint, Item 250, WP 1803 00
 Tags
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)

References

TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0381 00
 WP 0537 00
 WP 0559 00
 WP 0562 00
 WP 1803 00
 WP 1805 00

WARNING

Injury to personnel may result if contact with beryllium occurs. The bearing sleeve is made of beryllium copper metal, which can be a health hazard. The use of goggles is required. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

DISASSEMBLE

1. Locally make main rotor damper sleeve bearing installation/removal tool, holding fixture, seal expansion tool, piston rod insertion tool, and piston rod wrench (WP 1805 00).
2. Remove damper indicator (WP 0559 00). Plug port.

DISASSEMBLE - Continued**WARNING**

Injury to personnel will result if pressure is not released from damper prior to disassembly. Damper may have residual pressure. Relieve pressure from damper prior to disassembly.

CAUTION

Damage to damper may result if contaminated with dirt and debris. Damper assembly/disassembly can only be done in clean-room environment. Care must be taken during disassembly to avoid introducing dirt or other debris which will contaminate damper assembly.

NOTE

Any damper assembly removed from helicopter due to vibration problem shall be forwarded to depot level facility for repair. Tag damper assembly with tag stating "DAMPER REQUIRES FORCED VELOCITY TESTING".

3. Slowly remove plug from damper assembly. Remove and discard packing. Drain petroleum base hydraulic fluid from damper assembly into container.

NOTE

Note relationship between cylinder head and cylinder assembly. Components shall be reassembled in the same position.

4. Remove nuts, washers and cylinder head (Figure 1).
5. Remove packing and packing retainer from cylinder head. Discard packing.
6. Remove packing and ring from bushing in cylinder head. Discard packing.
7. Remove retaining ring and sleeve bearing from cylinder head. Remove and discard packing from cylinder head (Figure 1, Detail A).
8. Remove retaining ring from rod end on piston rod assembly (Figure 1).

CAUTION

Damage to cylinder assembly will result if piston rod wrench comes in contact with interior of cylinder. Use extreme caution when inserting piston rod wrench into cylinder assembly.

9. Install locally-made piston rod wrench into cylinder assembly until it engages piston rod assembly. Place cylinder assembly into locally-made holding fixture.
10. Using locally-made piston rod wrench, break torque to loosen piston rod assembly from rod end assembly.
11. Remove barrel nut and rod end assembly from piston rod assembly.
12. Remove cylinder assembly from locally-made holding fixture. Remove locally-made piston rod wrench.

CAUTION

Damage to cylinder assembly will result if piston rod assembly comes in contact with interior of cylinder. Use extreme caution when removing piston rod assembly from cylinder assembly.

13. Remove piston rod assembly from cylinder assembly.

DISASSEMBLE - Continued

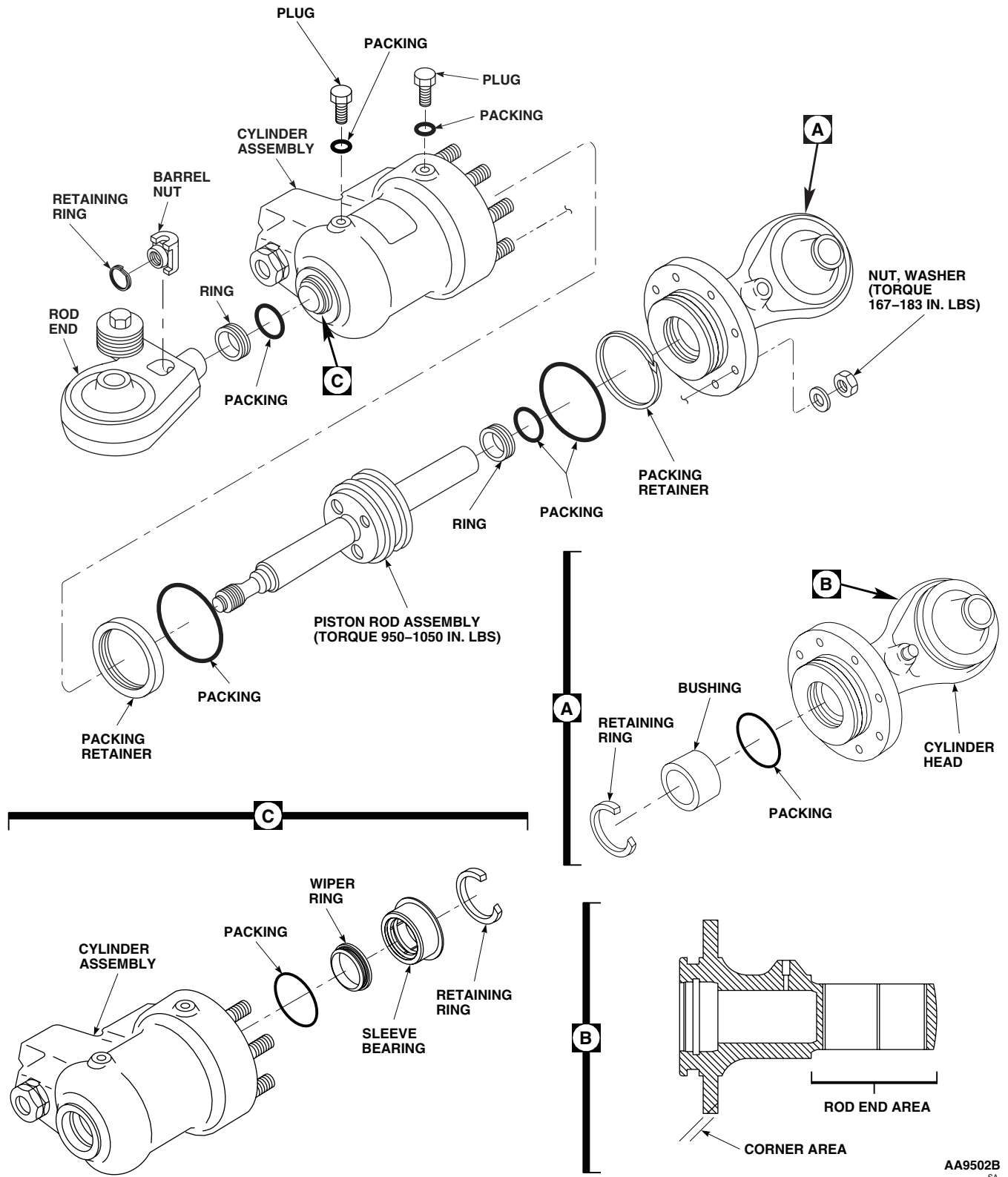


Figure 1. Replace Damper Seals (AVIM).

DISASSEMBLE - Continued

14. Remove and discard packing and packing retainer from piston rod assembly.
15. Remove and discard packing and ring from sleeve bearing in cylinder assembly.
16. Remove and discard wiper ring from sleeve bearing (Figure 1, Detail B).
17. Remove retaining ring from cylinder assembly. Using locally-made main rotor damper sleeve bearing installation/removal tool and arbor press, press out sleeve bearing from cylinder assembly.
18. Remove and discard packing from cylinder assembly.

CLEAN

1. Clean all parts using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
2. Wipe all parts dry using clean machinery towel, Item 344, WP 1803 00.

INSPECT/REPAIR

1. Check cylinder head bushing for burrs, scratches, nicks, or corrosion. **NONE ALLOWED.** Replace damaged bushing.
2. Inspect cylinder head bushing ID. **MAXIMUM ID IS 1.000-INCH.** Replace bushing if ID is beyond limits.
3. Inspect cylinder head bushing OD. **MAXIMUM OD IS 1.438-INCHES.** Replace bushing if OD is beyond limits.
4. Visually inspect cylinder head bushing for cracks using 3X magnifier. If a crack is suspected. Verify using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked cylinder bushing.
5. Inspect damper cylinder head rod end bearing (WP 0537 00).
6. Inspect cylinder head open areas for nicks, gouges, and corrosion. **DAMAGE SHALL NOT EXCEED 0.040-INCH DEEP OR 15% OF CIRCUMFERENCE ON CIRCULAR FEATURES. DAMAGE SHALL NOT EXCEED 1.000-INCH AXIAL LENGTH (EACH REWORK). DAMAGE SHALL NOT EXCEED 1.000-INCH DIAMETER. DAMAGE SHALL NOT BE CLOSER THAN 1.000-INCH TO OTHER REWORK.** Replace cylinder head if damage is more than limits.
7. Inspect cylinder head rod end area for nicks, gouges, burrs, or corrosion. **NONE ALLOWED.** Replace cylinder head if damage is more than limits.
8. Inspect corner areas of cylinder head for nicks, gouges, and corrosion. **CORNER DAMAGE SHALL NOT EXCEED 0.060-INCH DEEP OR 30% OF CIRCUMFERENCE ON CIRCULAR FEATURES TOTAL. CORNER DAMAGE SHALL NOT EXCEED 1.000-INCH IN LENGTH ON LINEAR FEATURES EACH REWORK. CORNER DAMAGE SHALL NOT BE CLOSER THAN 1.000-INCH TO OTHER REWORKS.** Replace cylinder head if damage is more than limits.
9. Inspect mating surface of cylinder head for nicks, gouges, and corrosion. **DAMAGE SHALL NOT EXCEED 0.010-INCH DEEP OR 10% OF SURFACE AREA. MATING PART SHALL NOT BE REWORKED IN SAME AREA.** Replace cylinder head if damage is more than limits.
10. Inspect cylinder head bushing bore diameter. **MAXIMUM DIAMETER SHALL NOT EXCEED 1.438-INCHES. OUT OF ROUND SHALL NOT EXCEED 0.001-INCH.** Replace cylinder head if damage is more than limits.
11. Inspect cylinder head piston rod bore diameter. **MAXIMUM DIAMETER IS 1.070-INCH.** Replace cylinder head if damage is more than limits.
12. Visually inspect cylinder head case for cracks using 3X magnifier. If a crack is suspected. Verify using eddy current inspection unit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked cylinder head.

INSPECT/REPAIR - Continued

13. Check for missing or damaged protective coating. Repair per step 14.
14. Repair cylinder head as follows:
 - a. Clean damaged areas using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
 - b. Remove paint from corroded area using paint remover, Item 250, WP 1803 00. Rinse part thoroughly with water to remove paint remover.
 - c. Apply corrosion removing compound, Item 108, WP 1803 00, to corroded area using acid swabbing brush, Item 57, WP 1803 00. On heavy corrosion, rub area with abrasive cloth, Item 79, WP 1803 00, while solvent cleaning compound is in corroded area. Do not allow solvent cleaning compound to remain on area for more than 10 minutes.
 - d. Remove compound from area using machinery wiping towel, Item 344, WP 1803 00, while rinsing with clean water. Repeat step 14. c. and step 14. d. until all corrosion is removed.
 - e. Blend out area using abrasive cloth, Item 79, WP 1803 00. Do not exceed repair limits.
 - f. Remove gouges and dents on external surfaces with abrasive wheel, Item 350, WP 1803 00. Blend to 0.001 to 0.002-inch deeper than original damage.
 - g. Fluorescent penetrant inspect repaired area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked cylinder head.
 - h. Apply corrosion resistant coating, Item 111, WP 1803 00, to exposed metal.
 - i. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
15. Inspect damper assembly rod end bearing (WP 0537 00).
16. Inspect outer rod end lug areas for nicks, gouges, and corrosion. **CORROSION SHALL NOT EXCEED 0.002-INCH DEEP OR 10% OF TOTAL SURFACE AREA.** Replace rod end if damage is more than limits.
17. Inspect rod end lug faces on both sides of rod end for nicks, gouges, and corrosion. **NONE ALLOWED.** Replace damaged rod end.
18. Inspect remaining areas of rod end for nicks, gouges, and corrosion. **DAMAGE SHALL NOT EXCEED 0.010-INCH DEEP OR 10% OF TOTAL SURFACE.** Replace rod end if damage is more than limits.
19. Check plated areas of damper piston rod assembly for scratches, nicks, burrs, or corrosion. **NONE ALLOWED.** Replace damaged damper piston rod assembly.
20. Inspect damper piston rod assembly shaft diameter. **DIAMETER SHALL NOT BE LESS THAN 0.996-INCH.** Replace damper piston rod assembly if diameter is not within limits.
21. Inspect damper piston rod assembly piston diameter. **DIAMETER SHALL NOT BE LESS THAN 2.970-INCHES.** Replace damper piston rod assembly if diameter is not within limits.
22. Check damper piston rod assembly threads for crossed, striped, or flattened threads. **NONE ALLOWED.** Replace damaged damper piston rod assembly.
23. Check damper piston rod assembly for missing or damaged protective coating (chrome plating). **NONE ALLOWED.** Replace damaged damper piston rod assembly.
24. Inspect damper piston rod assembly packing groove diameter. **DIAMETER SHALL NOT BE LESS THAN 2.621-INCHES.** Replace damper piston rod assembly if diameter is not within limits.
25. Visually inspect cylinder dampener piston rod assembly for cracks using 3X magnifier. If a crack is suspected. Verify using magnetic particle inspection (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked piston rod assembly.

INSPECT/REPAIR - Continued

26. Repair damper piston rod assembly as follows:
 - a. Polish minor axial scoring on 0.996 to 0.998-inch chrome plated diameter not deeper than 0.0002-inch using abrasive cloth, Item 75, WP 1803 00, and abrasive cloth, Item 82, WP 1803 00.
 - b. **IF SCORING IS DEEPER THAN 0.0002-INCH, REPLACE DAMPER ROD ASSEMBLY.**
27. Check cylinder assembly sleeve bearing for scratches, nicks, burrs, or corrosion. **NONE ALLOWED.** Replace damaged sleeve bearing.
28. Inspect cylinder assembly sleeve bearing OD. **OD SHALL NOT BE LESS THAN 1.438-INCH.** Replace cylinder assembly sleeve bearing if diameter is not within limits.
29. Inspect cylinder assembly sleeve bearing ID. **ID SHALL NOT BE MORE THAN 1.000-INCH.** Replace cylinder assembly sleeve bearing if diameter is not within limits.
30. Visually inspect cylinder assembly sleeve bearing for cracks using a 3X magnifier. If a crack is suspected. Verify using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked cylinder assembly sleeve bearing.
31. Check open areas of cylinder assembly for nicks, gouges, and corrosion. **DAMAGE SHALL NOT EXCEED 0.040-INCH DEEP. WALL THICKNESS SHALL NOT BE LESS THAN 0.275-INCH IN ALL OPEN AREAS EXCEPT CENTER OF FORGING. DAMAGE IN CENTER OF FORGING SHALL NOT EXCEED 1.000-INCH WIDE OR BE MORE THAN 0.020-INCH DEEP. DAMAGE SHALL NOT BE MORE THAN 15% OF CIRCUMFERENCE ON CIRCULAR FEATURES AND BE NO MORE THAN 1.000-INCH AXIAL LENGTH. EACH REWORK SHALL NOT BE CLOSER THAN 1.000-INCH TO OTHER REWORKS. DAMAGE SHALL NOT EXCEED 1.000-INCH DIAMETER. DAMAGE SHALL NOT BE CLOSER THAN 1.000-INCH TO OTHER DAMAGE.** Replace cylinder assembly if damage is more than limits.
32. Check corner areas of cylinder assembly for nicks, gouges, and corrosion. **CORNER DAMAGE SHALL NOT EXCEED 0.060-INCH DEEP. CORNER DAMAGE ON CORNER OF CYLINDER ADJACENT TO CYLINDER HEAD STUDS SHALL NOT EXCEED 0.040-INCH DEEP. DAMAGE SHALL NOT BE MORE THAN 30% OF CIRCUMFERENCE ON CIRCULAR FEATURES AND SHALL NOT BE MORE THAN 1.000-INCH LENGTH ON LINEAR FEATURES. EACH REWORK SHALL NOT BE CLOSER THAN 1.000-INCH TO OTHER REWORKS.** Replace cylinder assembly if damage is more than limits.
33. Check mating surfaces of cylinder assembly for damage. **DAMAGE SHALL NOT EXCEED 0.010-INCH DEEP. DAMAGE SHALL NOT EXCEED 10% OF SURFACE AREAS. MATING PART SHALL NOT BE REWORKED IN SAME AREA.** Replace cylinder assembly if damage is more than limits.
34. Inspect cylinder assembly ID. **ID SHALL NOT EXCEED 2.997-INCHES. NO MISSING OR DAMAGED CHROME PLATING OR PITTING ALLOWED.** Replace cylinder assembly if damage is more than limits.
35. Inspect cylinder assembly bushing bore diameter. **DIAMETER SHALL NOT EXCEED 1.438-INCHES.** Replace cylinder assembly if diameter is more than limits.
36. Check cylinder assembly exposed threads for crossed, striped, or flattened threads. **NONE ALLOWED.** Replace cylinder assembly with damaged threads.
37. Visually inspect cylinder assembly for cracks using a 3X magnifier. If a crack is suspected. Verify using eddy current inspection unit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked cylinder assembly.
38. Check cylinder assembly exterior for missing or damaged protective coating. Repair per step 39.
39. Repair cylinder assembly external surface damage as follows:
 - a. Clean area using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.

INSPECT/REPAIR - Continued

- b. Remove paint from corroded area using paint remover, Item 250, WP 1803 00. Rinse part thoroughly with water to remove paint remover.
- c. Apply corrosion removing compound, Item 108, WP 1803 00, to corroded area using acid swabbing brush, Item 57, WP 1803 00. On heavy corrosion, rub area with abrasive cloth, Item 80, WP 1803 00, while corrosion removing compound is in corroded area. Do not allow corrosion removing compound to remain on area for more than 10 minutes.
- d. Remove corrosion removing compound from area using machinery wiping towel, Item 344, WP 1803 00, while rinsing with clean water. Repeat step 39. c. and step 39. d. until all corrosion is removed.
- e. Blend out area using abrasive cloth, Item 80, WP 1803 00. Do not exceed repair limits.
- f. Remove nicks, dents, and gouges on external surfaces with abrasive wheel, Item 350, WP 1803 00. Blend to 0.001 to 0.002-inch deeper than original damage.
- g. Fluorescent penetrant inspection repaired areas for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked cylinder assembly.
- h. Fluorescent penetrant inspect cylinder assembly for cracks (TM 1-1520-265-23). **NONE ALLOWED.** Replace cracked cylinder assembly.
- i. Apply corrosion resistant coating, Item 111, WP 1803 00, to exposed metal.
- j. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

ASSEMBLE

1. Coat all internal parts with petroleum base hydraulic fluid, Item 145, WP 1803 00, before assembly.

NOTE

After installation, make sure that seals and packings are fully seated in packing grooves and are not twisted.

2. **QA** Install packing into groove in cylinder assembly (Figure 1, Detail B).

CAUTION

Damage to packing will result if sleeve bearing is forced into cylinder assembly. Use extreme caution when installing sleeve bearing.

3. **QA** Using locally-made sleeve bearing installation tool and arbor press, install sleeve bearing into cylinder assembly. Secure with retaining ring.
4. **QA** Install packing and ring into sleeve bearing (Figure 1).
5. Coat OD of locally-made seal expansion tool and ID of sleeve bearing with clean petroleum base hydraulic fluid, Item 145, WP 1803 00.
6. Insert locally-made seal expansion tool into sleeve bearing bore with twisting motion until ring and packing come in contact with largest OD of seal expansion tool. Remove seal expansion tool.
7. **QA** Install wiper ring into sleeve bearing (Figure 1, Detail B).
8. Coat OD of locally-made seal expansion tool and ID of bushing with clean petroleum base hydraulic fluid, Item 145, WP 1803 00.
9. Insert locally-made seal expansion tool into sleeve bearing bore with twisting motion until wiper ring comes in contact with largest OD of seal expansion tool. Remove seal expansion tool.
10. **QA** Install packing and packing retainer on piston rod assembly (Figure 1).

ASSEMBLE - Continued

11. Attach locally-made piston rod insertion tool to cylinder assembly. Coat ID of piston rod insertion tool with clean petroleum base hydraulic fluid, Item 145, WP 1803 00.

CAUTION

- Damage to cylinder assembly, ring and packing or wiper ring will result if piston rod is forced during installation. Use extreme caution when installing piston rod assembly.
 - Damage to cylinder assembly will result if piston rod assembly comes in contact with interior of cylinder. Use extreme caution when installing piston rod assembly into cylinder assembly.
12. Coat shank (threaded side) of piston rod assembly and ID of cylinder assembly with clean petroleum base hydraulic fluid, Item 145, WP 1803 00. Carefully insert piston rod assembly (threaded end) into cylinder assembly.
 13. Remove locally-made piston rod assembly insertion tool.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

14. Using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, thoroughly clean and dry threaded section of piston rod assembly. Remove all traces of petroleum base hydraulic fluid.
15. **QA** Install rod end assembly and barrel nut to piston rod assembly.

CAUTION

Damage to cylinder assembly will result if piston rod wrench comes in contact with interior of cylinder. Use extreme caution when inserting piston rod wrench into cylinder assembly.

16. **QA** Install locally-made piston rod wrench into cylinder assembly until it engages piston rod assembly. Place cylinder assembly with piston rod wrench into locally-made holding fixture. **TORQUE PISTON ASSEMBLY TO 950 - 1050 INCH-POUNDS.**
17. Remove cylinder assembly from locally-made holding fixture. Remove locally-made piston rod wrench. Install barrel nut retaining ring on end of piston assembly.
18. **QA** Install packing into cylinder head assembly (Figure 1, Detail A).

CAUTION

Damage to packing will result if sleeve bearing is forced into cylinder head assembly. Use extreme caution when installing sleeve bearing.

19. Using locally-made sleeve bearing installation tool and arbor press, install sleeve bearing into cylinder head assembly. Secure with retaining ring.
20. **QA** Insert packing and ring into sleeve bearing (Figure 1).
21. Coat OD of locally-made seal expansion tool and ID of sleeve bearing with clean petroleum base hydraulic fluid, Item 145, WP 1803 00.

ASSEMBLE - Continued

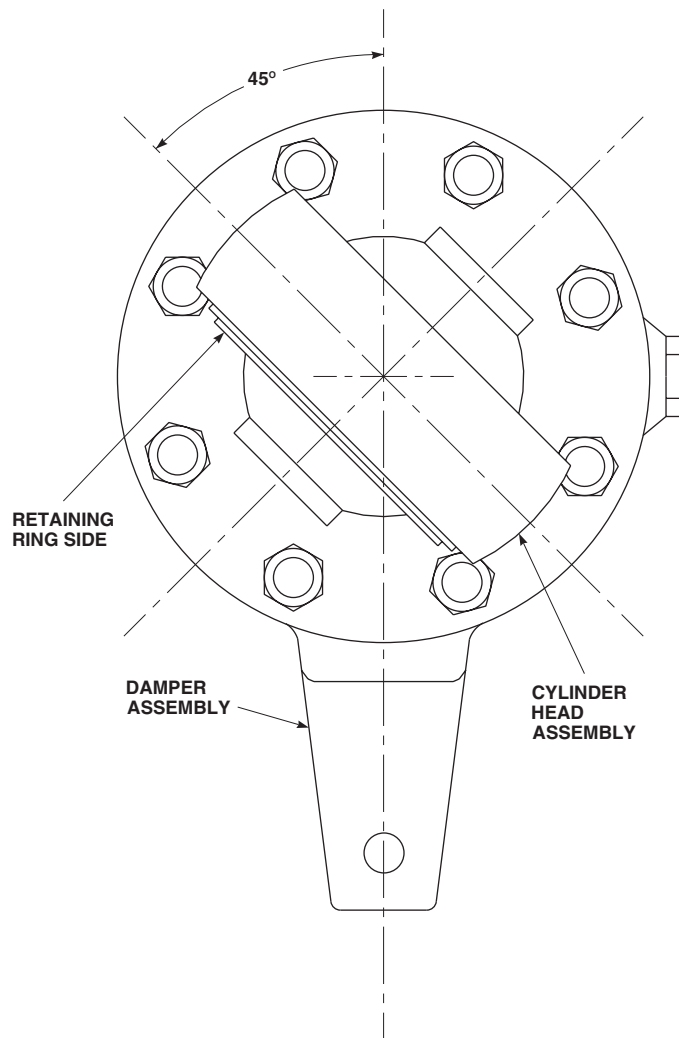
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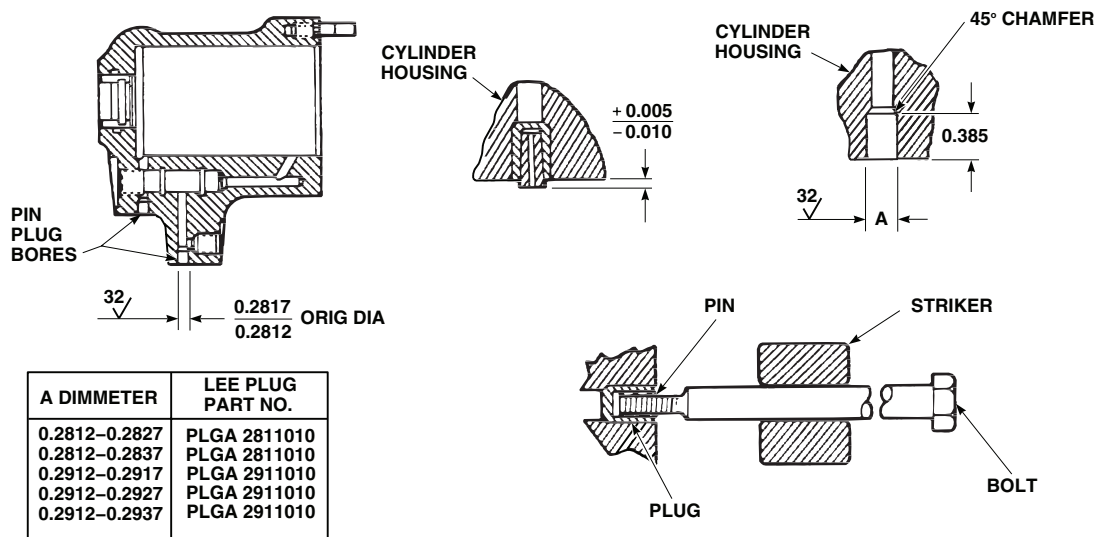
Figure 2. Replace Damper Seals (AVIM).

22. Insert locally-made seal expansion tool into sleeve bearing bore with twisting motion until ring and packing come in contact with largest OD of seal expansion tool.
23. Remove locally-made seal expansion tool.
24. **QA** Install packing and packing retainer in cylinder head assembly OD groove.

NOTE

Make sure cylinder head assembly and cylinder are aligned as shown in Figure 2.

25. Install cylinder head assembly with nuts and washers. **TORQUE NUTS TO 167 - 183 INCH-POUNDS.**
26. Remove plug and install damper indicator (WP 0559 00).

ASSEMBLE - Continued**NOTES**

1. ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.
2. SYMBOL:
✓ MICROINCH SURFACE FINISH

REMOVALAA6338
SA

Figure 3. Replacement of Pin Plugs.

TEST

Bleed damper (WP 0562 00) and check for leaks.

REPLACE CHECK VALVE

1. Replace check valve using appropriate body tools, pin extraction/installation tool, and striker from insert installation tool kit, as follows:
 - a. Installation method.
 - (1) Thread insert body onto body tool.
 - (2) Push insert firmly into reamed hole until seated against 45 degree shoulder. Remove body tool.
 - (3) Thread pin completely onto pin installation tool.
 - (4) Drive pin into body until pin is flush with body within +/- 0.005 inch.
 - b. Extraction method.
 - (1) Slide striker onto pin extraction tool and thread completely into pin.
 - (2) Strike head of pin extraction tool with striker until pin is removed.
 - (3) Repeat steps b(1) and b(2) using body bolt and striker to remove insert body.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
SWASHPLATE SPHERICAL BEARING (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Powertrain Repairer's Toolkit, SC 5180-99-B013
Torque Wrench, 150 - 1000 in. lbs.

Primer Coating, Item 243, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

Material/Parts

Cloth, Abrasive, Item 77, WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00

References

WP 1803 00

REMOVE

1. Remove bolts and washers attaching retainer to stationary swashplate (Figure 1).
2. Remove retainer and shim from swashplate.
3. Remove spherical bearing from swashplate.

INSPECT/REPAIR**NOTE**

Repair to stationary swashplate inner bore is limited to a maximum of 10 repairs with any repair no closer than 0.50-inch.

1. Inspect inner bore of stationary swashplate for corrosion or damage. Blend corrosion or damage using abrasive cloth, Item 77, WP 1803 00. **MAXIMUM ALLOWABLE DAMAGE AFTER BLENDING IS 0.010-INCH IN DEPTH AND 1 INCH IN DIAMETER.**
2. Touch up repaired area using corrosion resistant coating, Item 111, WP 1803 00.

INSTALL**NOTE**

- Do not lubricate bearings. Bearings are installed dry.
- Spherical bearing and outer races are a matched set. Make sure all parts have same serial number.

1. **QA** Line up etched "V" on bearing outer races (Figure 1, Detail A).

CAUTION

Swashplate and/or spherical bearing will be damaged if bearing is cocked during installation. Make sure spherical bearing is correctly aligned during installation.

2. **QA** Install spherical bearing in stationary swashplate using hand pressure, or tapping lightly with rubber mallet.
3. Measure gap between spherical bearing and swashplate (Figure 1, Detail B). **PEEL SHIM TO ALLOW 0.003 TO 0.006-INCH CLAMP-UP ON SPHERICAL BEARING OUTER RACE.**

INSTALL - Continued

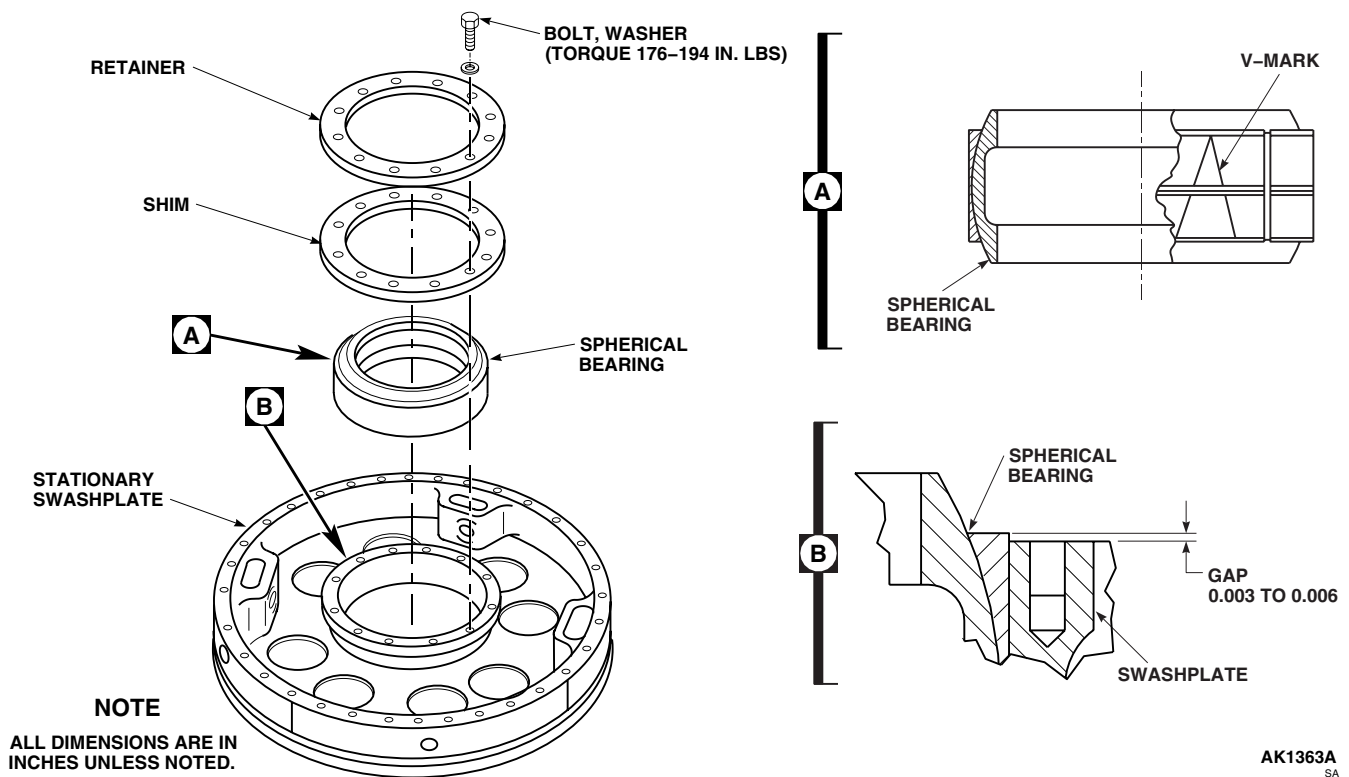


Figure 1. Replace Swashplate Spherical Bearing (AVIM).

4. Coat shank, threads, and under heads of bolts with primer coating, Item 243, WP 1803 00.
5. **QA** Install shim and retainer on swashplate with bolts and washers while primer coating is still wet. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**
6. Check installed bearing for freedom of movement.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
ROTATING SCISSORS BEARING (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Carbon Dioxide, Item 64, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Tags

References

WP 0572 00
WP 1803 00

REMOVE

1. Remove upper and lower scissor link assemblies (WP 0572 00).

NOTE

Bearings are a matched set and cannot be replaced individually. If same bearings are to be reinstalled, tag bearings for correct reassembly.

2. Remove inner sleeve bearings from upper scissors link.

INSPECT

1. Measure height of outer bearing shoulder (Figure 1, Detail A). **MINIMUM HEIGHT IS 0.086-INCH.**
2. Replace outer bearing if height is less than minimum.

DISASSEMBLE

Press out bearings from upper scissors link using arbor press (Figure 1, Detail A).

INSPECT/REPAIR

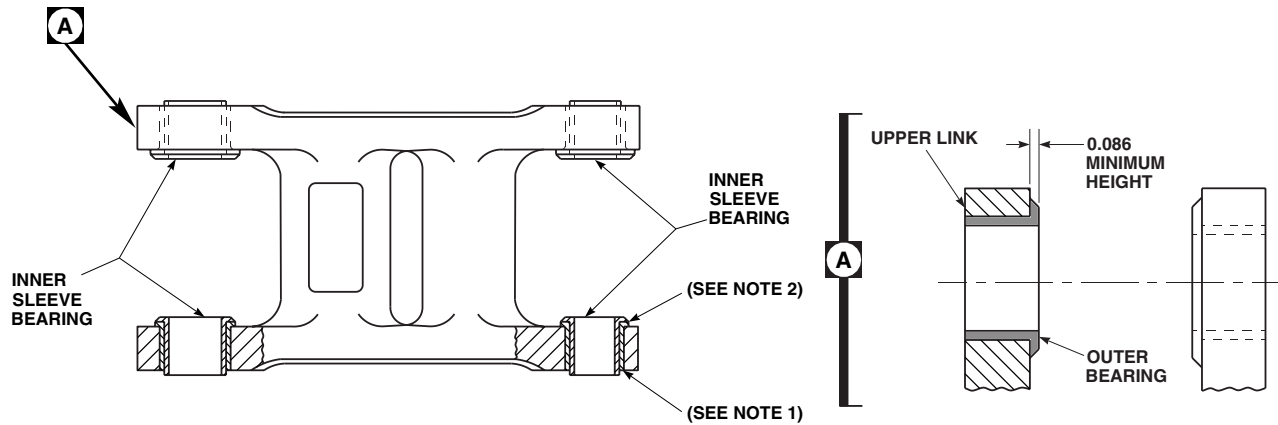
1. Blend out any wear or damage in bearing bores with abrasive cloth, Item 82, WP 1803 00.
2. **REPLACE UPPER SCISSORS IF DAMAGE IS DEEPER THAN 0.010-INCH AFTER BLENDING, OR IF BLENDED AREA IS MORE THAN 25% OF BORE AREA.**

ASSEMBLE**WARNING**

Injury to personnel will result if carbon dioxide or chilled parts are handled without protective gloves. Use protective gloves when handling carbon dioxide or chilled parts.

1. Freeze replacement bearings in carbon dioxide, Item 64, WP 1803 00, or a freezer at least -65°F(-54°C), for a minimum of 30 minutes.

ASSEMBLE - Continued



NOTES

1. FILL "V" GROOVE OF ALL SLEEVE BEARINGS WITH SEALING COMPOUND.
2. FORM FILLET OF SEALING COMPOUND ON ALL OUTER BEARINGS.
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2577
SA

Figure 1. Replace Rotating Scissors Bearing (AVIM).

NOTE

- Make sure shoulder of bearing is seated against flange of scissors.
 - Bearings are matched sets. Make sure serial numbers on inner and outer race are same.
2. **QA** Press sleeve bearings into bores using arbor press (Figure 1).

NOTE

Make sure sealing compound does not get in bearing. Bearing should rotate freely after installation.

3. Seal exposed edges of outer bearing with sealing compound, Item 273, WP 1803 00.

INSTALL

Install upper and lower scissor link assemblies (WP 0572 00). **QA check**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ROTOR SYSTEM

ROTATING SCISSORS LOWER LINK BUSHING (AVIM)

This WP supersedes WP 0600 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Hand Reamer Set, Adjustable, GGG-R-180

Primer Coating, Item 243, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

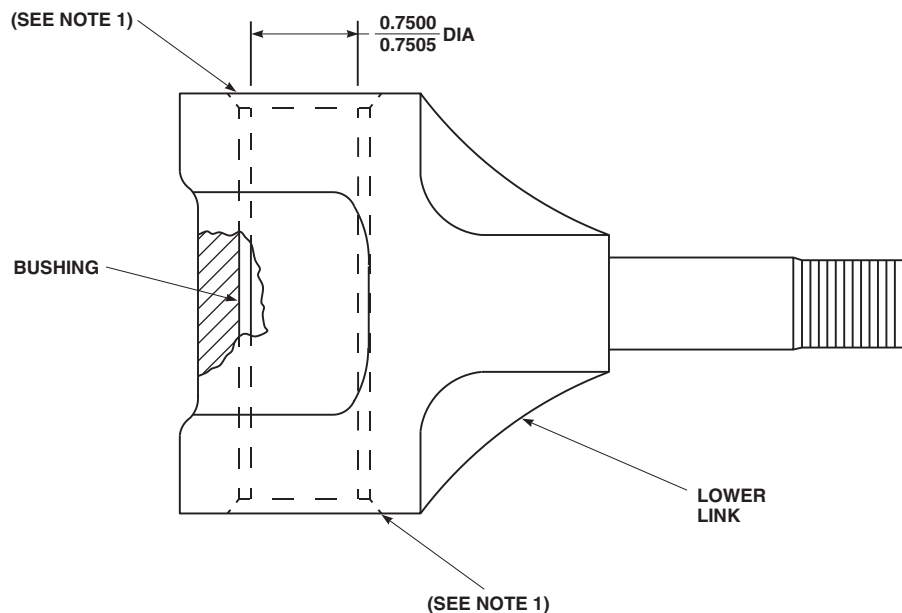
Acetone, Technical, Item 5, WP 1803 00
 Carbon Dioxide, Item 64, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00

References

TM 1-1520-265-23
 WP 1803 00

REMOVE

1. Press out bushing from lower link using arbor press (Figure 1).

**NOTES**

1. FILL "V" GROOVE OF BUSHING WITH SEALING COMPOUND.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2578
SA

Figure 1. Replace Rotating Scissors Lower Link Bushing (AVIM).

REMOVE - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean link bore using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
3. Fluorescent penetrant inspect link for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked link.
4. Check link bore for scoring and corrosion. Blend out scoring and corrosion using abrasive cloth, Item 82, WP 1803 00.

INSTALL**WARNING**

Injury to personnel will result if carbon dioxide or chilled parts are handled without protective gloves. Use protective gloves when handling carbon dioxide or chilled parts.

1. Freeze replacement bushing in carbon dioxide, Item 64, WP 1803 00, for a minimum of 30 minutes.
2. Remove bushing from carbon dioxide.
3. **QA** Coat outside of bushing with primer coating, Item 243, WP 1803 00. Press bushing into link bore using arbor press while primer coating is still wet (Figure 1).
4. Allow parts to return to room temperature.
5. **QA** Ream bushing to 0.7500 to 0.7505-inch using adjustable hand reamer set.
6. Apply sealing compound, Item 273, WP 1803 00, to fill V-groove of bushing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****BIFILAR ARM BUSHINGS (AVIM)**

This WP supersedes WP 0601 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Bifilar Bushing Replacement Tool, Locally-Made,
Figure 213, WP 1805 00
Drift
Flat Washer, MS15795-830
Gloves
Goggles/Face Shield
Installer Washer, Locally-Made, 2763-6, Figure 213,
WP 1805 00
Mallet
Nonmetallic Scraper
Nut, AN320-20

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Sealing Compound, Item 267, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0574 00
WP 1803 00
WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00

REMOVE

1. Locally make bifilar bushing replacement tools (WP 1805 00).
2. Remove bifilar weights (WP 0574 00).
3. Using drift and mallet, remove machine key, if installed (Figure 1, Detail A).

NOTE

- Note bushing part number on top bushing to make sure correct bushing replacement.
 - To make sure bushing alignment is maintained, only replace one set of bushings at a time.
4. Select set of pulling tabs, 2763-2 or 2763-3, that best fits groove inside support assembly between the two bushings.
 5. Install threaded portion of threaded shaft, 2763-1, through bushings from bottom side of support arm with flange of threaded shaft pressed against bottom side of pulling tabs.
 6. Place pulling washer, 2763-4, on threaded shaft along with one flat washer, MS15795-830, and nut, AN320-20. Tighten nut to sandwich pulling tabs between flange of threaded shaft and pulling washer.

WARNING

Injury to personnel will result if care is not taken when handling carbon dioxide or chilled parts. Wear protective gloves when handling carbon dioxide or chilled parts.

7. Freeze bushings by placing carbon dioxide, Item 64, WP 1803 00, down inside support arm.

REMOVE - Continued

8. Place puller disk, 2763-5, down over threaded shaft along with one flat washer and nut.
9. Tighten nut down in puller disk until top bushing is removed.
10. Remove nut, flat washer, and puller disk from threaded shaft. Leave pulling tabs for lower bushing removal.
11. Place threaded shaft with pulling tabs still secured to threaded shaft down through support arm from top, resting pulling tabs on lower bushing.
12. Install puller disk up through threaded shaft over lower bushing flange.
13. Align cut out on puller disk with support assembly. Secure with one flat washer and nut.
14. Tighten nut against puller disk until lower bushing is removed.
15. Remove tools from bifilar.

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

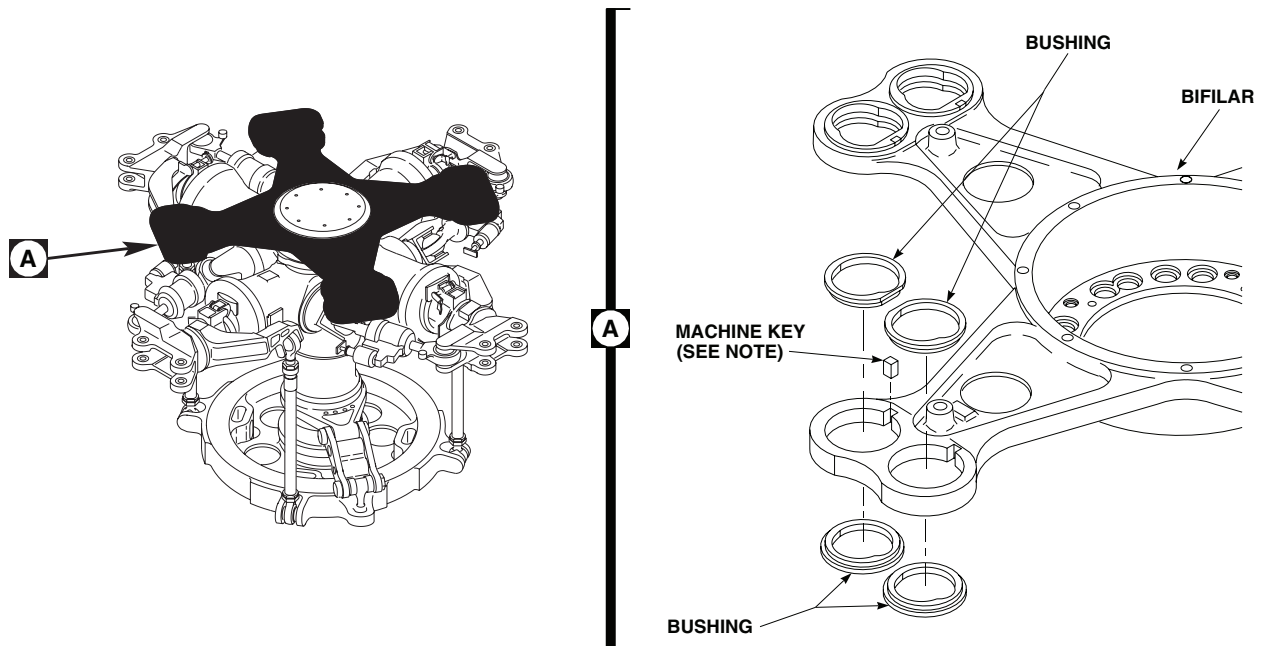
1. Clean all sealing compound from bifilar with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.

WARNING

Injury to personnel will result if care is not taken when handling carbon dioxide or chilled parts. Wear protective glove when handling carbon dioxide or chilled parts.

2. Freeze bushings in carbon dioxide, Item 64, WP 1803 00.
3. Coat bushing bore with epoxy primer coating kit, Item 120, WP 1803 00.
4. **QA** Line up top bushing flange flats with alignment bar, 2763-7, on top of bifilar arm while primer is still wet. If installed, make sure machine key slots are aligned (Figure 1, Detail A).
5. **QA** Using threaded shaft, 2763-1, install one installer washer, 2763-6, and one flat washer, MS15795-830, and secure with nut, AN320-20.
6. Insert threaded end of threaded shaft up through bifilar arm, lining up installer washer flat with alignment bar.
7. **QA** Place other installer washer, 2763-6, down over threaded shaft, and on top of new bushing, lining up flats of installer washer and bushing with alignment bar on top. Secure in place with one flat washer and nut.
8. Tighten nut against installer washer until top bushing has been installed in bifilar arm.
9. Remove nut, flat washer, and installer washer from threaded shaft and remove threaded shaft from bifilar arm. Leave other installing washer secured on threaded shaft.
10. With alignment bar still in place on bifilar arm and installer washer still secured on threaded shaft, place bushing over threaded end of threaded shaft with flange against installer washer.
11. Insert threaded end of threaded shaft up through bifilar arm lining up flats on installer washer and bushing with alignment bar.

INSTALL - Continued

**NOTE**

BIFILAR, 70108-08401-042, ONLY.

AA8904A
SA

Figure 1. Replace Bifilar Arm Bushings (AVIM).

12. Install other installer washer in threaded shaft from top of bifilar arm, lining up flat of installing washer with alignment bar. If installed, make sure machine key slot is aligned.
13. **QA** Install flat washer and nut.
14. Tighten nut until lower bushing has been installed in bifilar arm. Allow bushing to return to room temperature.
15. Remove tools from bifilar arm.
16. **QA** Install machine key, if required.
17. Seal mating surfaces of bushing flanges and bifilar arm with sealing compound, Item 267, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****BIFILAR ARM WEIGHT BUSHINGS (AVIM)**

This WP supersedes WP 0602 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Bifilar Bushing Replacement Tool, Locally-Made,
Figure 213, WP 1805 00
Drift
Gloves
Goggles/Face Shield
Mallet
Nonmetallic Scraper

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0574 00
WP 1803 00
WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00

REMOVE

1. Locally make bifilar bushing replacement tools (WP 1805 00).

NOTE

- Note bushing part number on top bushing to make sure correct bushing replacement.
- To make sure bushing alignment is maintained, only replace one set of bushings at a time.

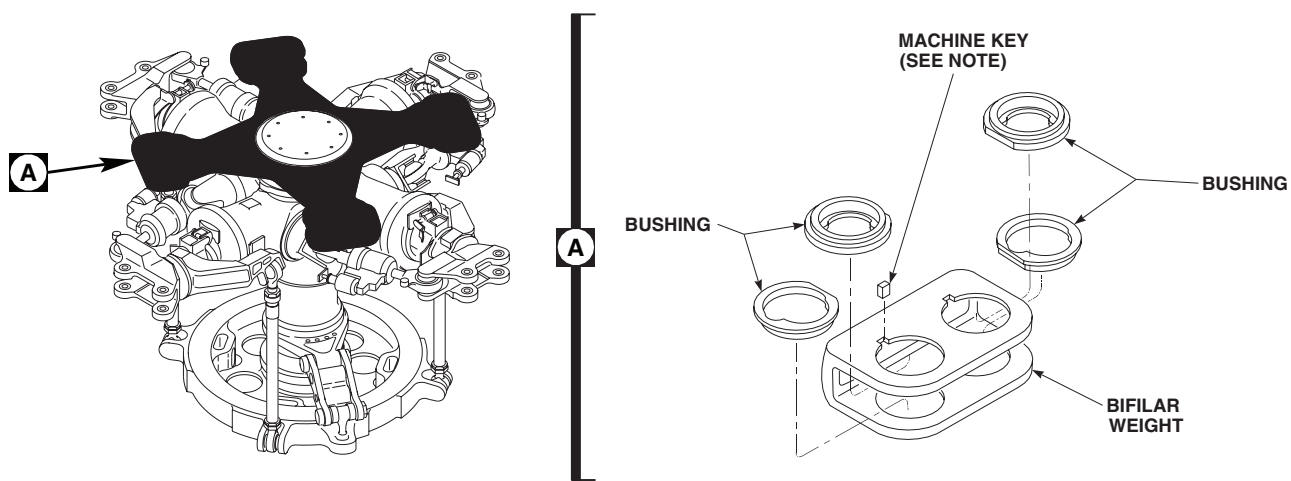
2. Remove bifilar weights (WP 0574 00).
3. Using drift and mallet, remove machine key, if installed (Figure 1, Detail A).
4. Using drift and mallet, remove one upper and one lower bushing as a set (Figure 1, Detail A).

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean all sealing compound from bifilar weight with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.

INSTALL - Continued

**NOTE**

BIFILAR WEIGHTS, 70107-08404-042
AND 70107-08404-043, ONLY.

AA8905A
SA

Figure 1. Replace Bifilar Arm Weight Bushings (AVIM).

WARNING

Injury to personnel will result if care is not taken when handling carbon dioxide or chilled parts. Wear protective gloves when handling carbon dioxide or chilled parts.

2. Freeze bushings in carbon dioxide, Item 64, WP 1803 00.
3. Coat bushing bore with epoxy primer coating kit, Item 120, WP 1803 00.
4. Place guide bar, 2803-1, inside bifilar weight and against flat portion of upper and lower flanges on existing bushings in bifilar weight (Figure 1, Detail A).
5. Line up top bushing flange flats with alignment bar on top of bifilar weight while primer is still wet. If installed, make sure machine key slots are aligned.
6. **QA** Using arbor press, place press plate, 2803-2, on top of bushing flange and press bushing into bifilar weight using alignment bar as guide.
7. **QA** Install bottom bushing in same way as top bushing. Allow bushing to return to room temperature.
8. **QA** Install machine key, if required.
9. Seal mating surfaces of bushing flanges and bifilar weight with sealing compound, Item 273, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****MAIN ROTOR BLADE CUFF BUSHINGS (AVIM)**

This WP supersedes WP 0603 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Arbor Press, 00-P-636
Blade/Spindle Cuff Bushing Installation Tool, Locally-Made, Figure 161, WP 1805 00
Gloves
Goggles/Face Shield
Inside Micrometer Caliper,
Powertrain Repairer's Toolkit, SC 5180-99-B13

Cloth, Abrasive, Item 82, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

WP 1803 00
WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00

INSPECT BUSHING

1. Inspect ID of blade cuff lug bushings for wear. **MAXIMUM ID IS 1.003-INCHES.** Replace bushing if wear is more than limits.
2. Inspect center lug bushings for displacement. **BUSHING SHOULD BE RECESSED 0.010 TO 0.030 INCHES FROM THE LUG SURFACE.** Replace bushing if protruding past center lugs causing interference with mating spindle lugs.

WARNING

Injury to personnel may result if contact with beryllium occurs. Main rotor blade cuff bushings are made of beryllium copper metal, which can be a health hazard. The use of goggles is required. Avoid breathing dust while reworking damaged areas. Use a dust respirator if necessary. Grinding is prohibited. Keep beryllium copper out of open wounds. Clean up visible dust piles and wash hands and arms after maintenance.

NOTE

This inspection is for both 6 and 8 lug main rotor blade cuffs.

3. Repair blade cuff lug bushings using abrasive cloth, Item 82, WP 1803 00, as follows:
 - a. **BLEND NICKS, GOUGES, OR SCORING ON ID OF BUSHING, UP TO 0.005-INCH DEEP. TOTAL BLENDED AREA SHALL NOT EXCEED 25% OF BORE AREA.**
 - b. **BLEND CORROSION ON ID OF BUSHING NO MORE THAN A MAXIMUM BUSHING ID OF 1.003-INCHES.**
 - c. **BLEND NICKS AND GOUGES ON SHOULDER OF BUSHING UP TO 0.010-INCH DEEP. CORNER DAMAGE SHALL NOT EXCEED 0.030-INCH.**
 - d. **BLEND NICKS AND GOUGES ON FACE OF BUSHING UP TO 0.010-INCH DEEP. BLENDED AREA SHALL NOT EXCEED 25% OF BUSHING FACE AREA.**

INSPECT BUSHING - Continued**NOTE**

Examine main rotor blade in vicinity of blade part number for reworking or repair identification data RS-007-1, RS-007L-1 or RS70100-007L-1. This data indicates a reworked or repaired lug bore resulting in oversized lug bushings. **IF BLADE EXHIBITS ANY OF THESE CONDITIONS, AND THE BUSHING EXCEED WEAR LIMITS OR REPAIR LIMITS, REPLACE MAIN ROTOR BLADE.**

- e. Replace bushing if repairs are more than limits.

REMOVE

1. Locally make blade/spindle cuff bushing installation tool (WP 1805 00).
2. Using tool A, press out outer lug bushings.
3. Using tool C and tool D, press out center lug bushings.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean lug bores using machinery wiping towel, Item 344, WP 1803 00, with technical acetone, Item 5, WP 1803 00.

INSPECT BUSHING BORE

1. Check bores for scoring and corrosion. Remove corrosion and blend out scoring with abrasive cloth, Item 82, WP 1803 00. **REPLACE BLADE IF DAMAGE IS DEEPER THAN 0.010-INCH AFTER BLENDING, OR IF BLENDED AREA IS GREATER THAN 25% OF BORE AREA.**
2. Check diameter of bushing bore. **REPLACE BLADE IF BORE IS GREATER THAN 1.125-INCHES AFTER BLENDING.**
3. Inspect blade cuff for cracks between attachment boltholes and leading and trailing edges of cuff on upper and lower surfaces. **NO CRACKS ALLOWED.** Replace blade with cracked cuff.

INSTALL**WARNING**

Personnel injury will result if protective gloves are not worn when handling carbon dioxide or chilled parts. Wear approved insulated gloves when handling carbon dioxide or chilled parts.

1. Place new bushings in carbon dioxide, Item 64, WP 1803 00, for at least 3 hours before installation.
2. Coat bushing outer diameter and bushing bore with primer coating, Item 243, WP 1803 00.

CAUTION

Use care when installing bushings in blade cuff or damage to blade lugs may result.

3. **QA** Press in shoulder bushings while primer coating is still wet. Use tool B, tool C, and arbor press, to press in shoulder bushings (Figure 1). Make sure shoulder of bushing is firmly seated against blade lug.

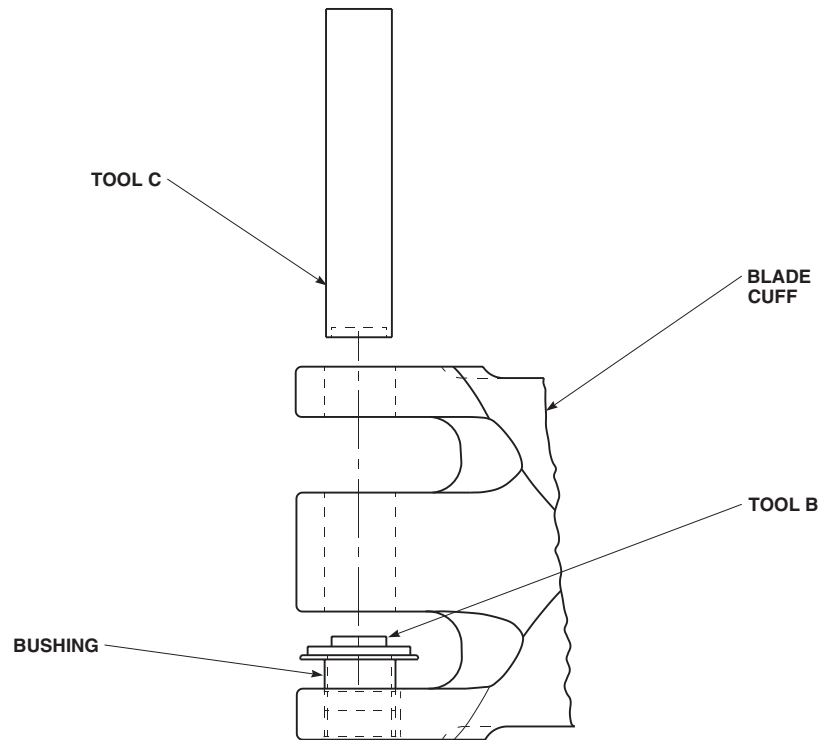
INSTALL - ContinuedAK1403
SA

Figure 1. Replace Main Rotor Blade Cuff Bushings (AVIM).

4. Press in center lug bushings while primer coating is still wet. Use tool B, tool C, and arbor press, to press in center lug bushings (Figure 1). Make sure bushings are recessed a minimum of 0.010 inches, but no more than 0.030 inches.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ROTOR SYSTEM

TAIL ROTOR PITCH BEAM BUSHINGS (AVIM)

INITIAL SETUP:

Tools and Special Tools

Fluorescent Penetrant Inspection Kit, XMA101
Pitch Beam/Horn Bushing Sizing Block, Locally-Made,
Figure 228, WP 1805 00
Powertrain Repairer's Toolkit, SC 5180-99-B13
Reamer Set, GGG-R-00180
Vernier Caliper, GGG-C-111

Material/Parts

Coating Compound, Metal, Item 92, WP 1803 00
Lacquer, Item 172, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Remover, Paint, Item 250, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00
WP 1805 00

REMOVE

1. Using arbor press, press out bushings from pitch beam (Figure 1).

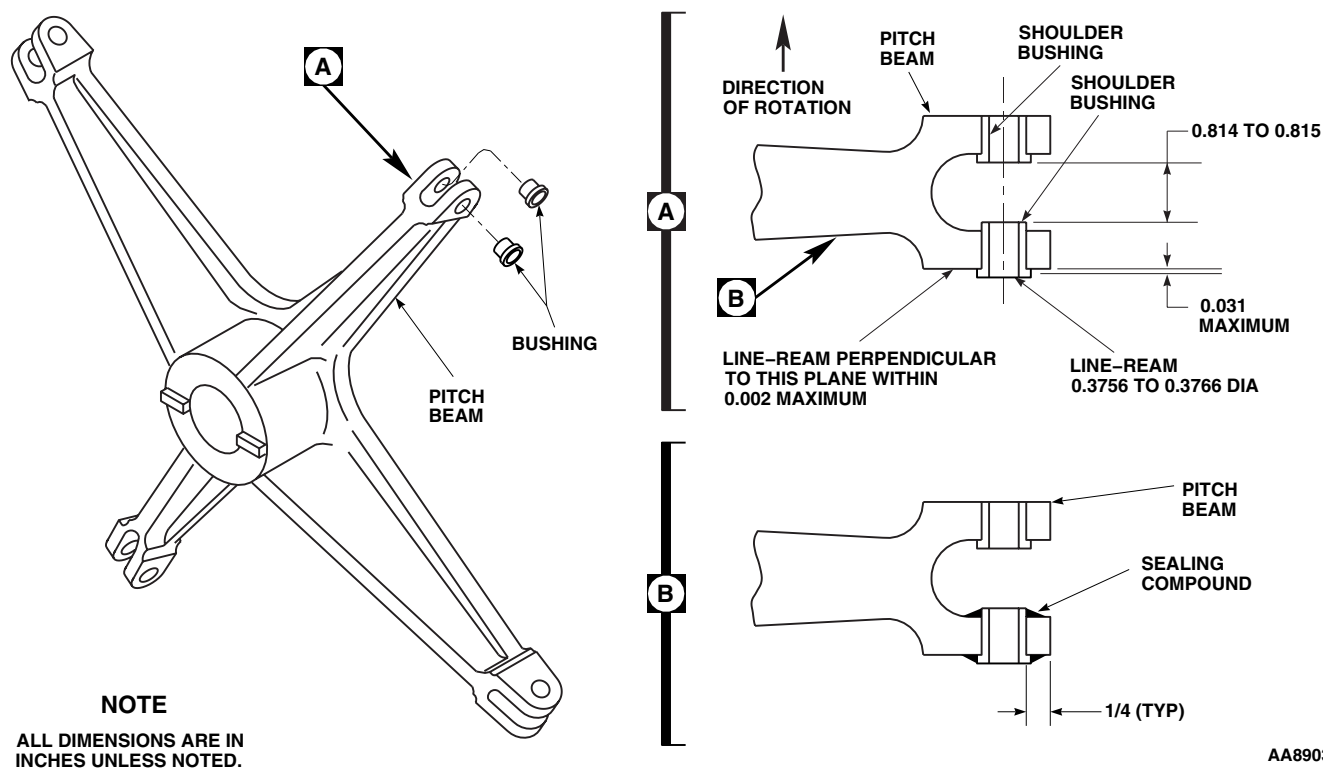


Figure 1. Replace Tail Rotor Pitch Beam Bushings (AVIM).

REMOVE - Continued

2. Remove paint from pitch beam bushing boss area, using paint remover, Item 250, WP 1803 00. Allow to dry.

INSPECT

1. Inspect bushing boss area for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23).
2. Inspect bushing hole in pitch beam for pits and corrosion. **NONE ALLOWED.** Replace pitch beam if corrosion or pits are found.
3. Check diameter of bushing hole in pitch beam using vernier caliper (Figure 1, Detail A). **MAXIMUM DIAMETER ALLOWED IS 0.5001-INCH.** Replace pitch beam if diameter exceeds limits.
4. **IF PITCH BEAM BUSHING IS MARKED RS-028-I NEXT TO PART NUMBER, REPLACE PITCH BEAM.**

INSTALL

1. Locally make pitch beam/horn bushing sizing block, 0.814 to 0.815-inch (WP 1805 00).
2. Apply metal coating compound, Item 92, WP 1803 00, to bare metal area of the sizing block.
3. Prime bushings and bushing holes with primer coating, Item 243, WP 1803 00.

NOTE

Bushing shoulder on the outside of the pitch beam will not seat on lug surface when distance between bushings is correct.

4. **QA** Using arbor press, install bushings while primer coating is still wet. Use pitch beam/horn bushing sizing block to make sure distance between bushings is 0.814 to 0.815-inch (Figure 1, Detail A).
5. **QA** Using reamer set, line-ream bushings to 0.3756 to 0.3766-inch.
6. Touch up with lacquer, Item 172, WP 1803 00 (WP 0381 00 and TM 1-1520-345-23).
7. Apply sealing compound, Item 273, WP 1803 00, around bushings (Figure 1, Detail B).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****TAIL ROTOR PITCH HORN BUSHING (AVIM)**

This WP supersedes WP 0605 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Arbor Press, 00-P-636
Gloves
Goggles/Face Shield
Inside Micrometer Caliper, GGG-C-105
Pitch Beam/Horn Bushing Sizing Block, Locally-Made,
Figure Figure 228, WP 1805 00
Powertrain Repairer's Toolkit, SC 5180-99-B13
Reamer Set, GGG-R-00180

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00

Pad, Scouring, Item 205, WP 1803 00
Paper, Abrasive, Item 211, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (2)

References

WP 0585 00
WP 1803 00
WP 1805 00

INSPECT/REPAIR

1. Check for corrosion of flange surfaces. **CORROSION UP TO 0.005-INCH DEEP AND NOT COVERING MORE THAN 20% OF AREA MAY BE HAND-POLISHED USING ABRASIVE PAPER, Item 211, WP 1803 00, OR SCOURING PAD, Item 205, WP 1803 00.**
2. Corrosion of ID may be blended using abrasive paper, Item 211, WP 1803 00, or scouring pad, Item 205, WP 1803 00. **ID MUST NOT EXCEED 0.3766-INCH (Figure 1, Detail A).**
3. If ID of pitch horn bushing is over 0.3766-inch, replace pitch horn bushing.
4. **REPLACE PITCH HORN BUSHINGS IF CORROSION PITS ARE STILL FOUND IN BORE, OR ON FLANGE.**
5. If pits are found, remove pitch horn bushings and check pitch horn bushing holes in pitch horn lug.

REMOVE

Press out pitch horn bushings from pitch horn lug using pusher and arbor press.

INSPECT

Check tail rotor blade pitch horn bushing holes. **REPLACE TAIL ROTOR BLADES IF THERE IS ANY CORROSION DAMAGE, OR IF HOLE IS LARGER THAN 0.5001-INCH.**

INSTALL

1. Locally make pitch beam/horn bushing sizing block, 0.814 to 0.815-inch (WP 1805 00).

INSTALL - Continued

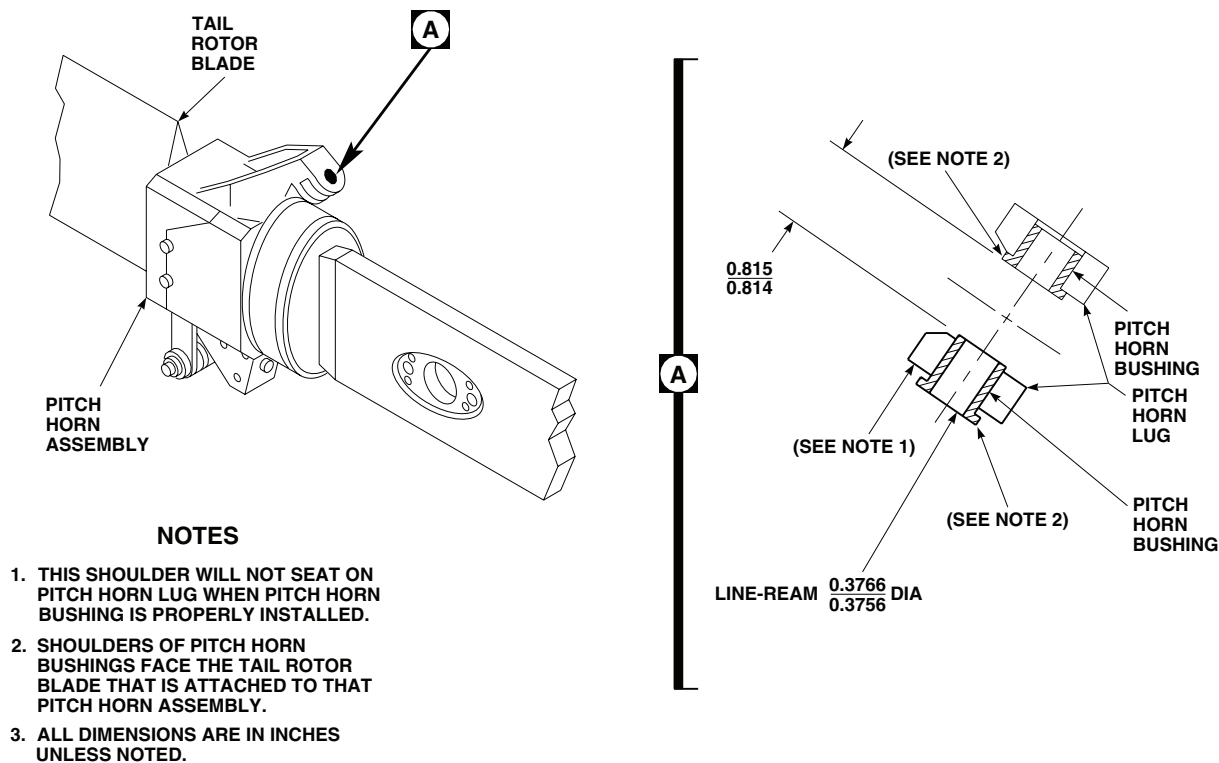


Figure 1. Replace Tail Rotor Pitch Horn Bushing (AVIM).

WARNING

Injury to personnel will result if carbon dioxide or chilled parts are handled without protective gloves. Use protective gloves when handling carbon dioxide or chilled parts.

2. Freeze replacement pitch horn bushings in carbon dioxide, Item 64, WP 1803 00, for at least 30 minutes.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean pitch horn bushing holes in pitch horn lug with technical acetone, Item 5, WP 1803 00.
4. **QA** Coat pitch horn bushings with primer coating, Item 243, WP 1803 00. Install pitch horn bushings while primer coating is still wet, using pitch beam/horn bushing sizing block to make sure distance between pitch horn bushings is 0.814 to 0.815-inch (Figure 1, Detail A).

NOTE

Pitch horn bushing shoulder on outside of pitch horn lug will not seat on lug surface when distance between pitch horn bushings is correct.

5. **QA** Using reamer set, line-ream pitch horn bushings to 0.3756 to 0.3766-inch (Figure 1, Detail A).

INSTALL - Continued

6. Seal edges of pitch horn bushings with sealing compound, Item 273, WP 1803 00.
7. Mark tail rotor data plate "RS-006A-II". Make sure data plate is marked so that this number will not be added to part number by mistake.
8. Touch up paint area as required (WP 0585 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**TAIL ROTOR PITCH CONTROL ROD BEARINGS (AVIM)****This WP supersedes WP 0606 00, dated 17 April 2006.****INITIAL SETUP:****Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
 Bearing Pusher/Puller, 70700-77454-041
 Bearing Staking Tool Set, 70700-77353-041
 Gloves
 Goggles/Face Shield
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Spinning Tool, STD-1850-458-017
 Torque Wrench, 0 - 30 in. lbs.
 Vernier Caliper, GGG-C-111

Cloth, Abrasive, Item 75, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Cheesecloth, Item 85, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Petrolatum, Item 216, WP 1803 00
 Primer Coating, Item 243, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00

References

TM 55-1500-322-24
 WP 1803 00

REMOVE**NOTE**

Removal and installation of bearing at either rod end is the same.

1. Remove bearing using arbor press and bearing pusher/puller, 70700-77454-041.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean primer coating residue from rod end bore and chamfer using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.
3. Make sure that rod end bore, chamfers, and adjacent surfaces are clean and dry.

INSPECT/REPAIR

1. Remove minor scoring or metal pickup from rod end bore using abrasive cloth, Item 75, WP 1803 00. Blend rod end bore using abrasive cloth, Item 82, WP 1803 00.
2. Using vernier caliper, measure rod end bore. **MAXIMUM ID SHALL NOT EXCEED 0.9994-INCH.** Replace rod end if ID after repair is beyond limits.

INSTALL**NOTE**

If the shelf life for SB7002-048 has exceeded 60 months, do not install. If the shelf life is less than 60 months prior to installation, serviceability will be in accordance with the applicable technical manual.

1. Apply a light coat of primer coating, Item 243, WP 1803 00, to inside of rod end bore and to outside surface of outer race of bearing.
2. **QA** Install bearing using arbor press and bearing pusher/puller, 70700-77454-041.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean primer coating, if any, from bearing V-grooves, using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry.
4. Using bearing staking tool set, 70700-77353-041, position rod end with bearing on drill press base plate, with bearing staking tool.

CAUTION

Damage to bearing will result if petrolatum is allowed to come in contact with bearing or exposed edge of fabric liner. If petrolatum gets on these areas, remove petrolatum immediately.

5. Lubricate V-groove of bearing and spinning tool with petrolatum, Item 216, WP 1803 00 (Figure 1, Detail A).

CAUTION

Damage to bearing will result if bearing staking tool is used improperly. Do not allow bearing staking tool to make contact with ball side of groove as binding may result.

6. **QA** Swage V-groove lip over chamfer in rod end bore using bearing staking tool (TM 55-1500-322-24).
7. Clean petrolatum from V-groove with cheesecloth cloth, Item 85, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00.
8. **QA** Inspect swag by inserting 0.005-inch feeler gage under staked lip of bearing. **NO INSERTION ALLOWED.** Reswage bearing if not within limits.
9. Proof test both sides of bearing using 100 pounds.
10. Do breakaway torque check of bearing ball as follows:
 - a. Rotate ball 360 degrees clockwise and 360 degrees counterclockwise.
 - b. Misalign bearing ball in at least 2 directions to full degree permitted by clearance.
 - c. **NO-LOAD ROTATIONAL BREAKAWAY TORQUE SHALL NOT EXCEED 10 INCH-POUNDS.**
 - d. Replace bearing if limits are exceeded.
11. Touch up bearing surfaces with corrosion preventive compound, Item 106, WP 1803 00.

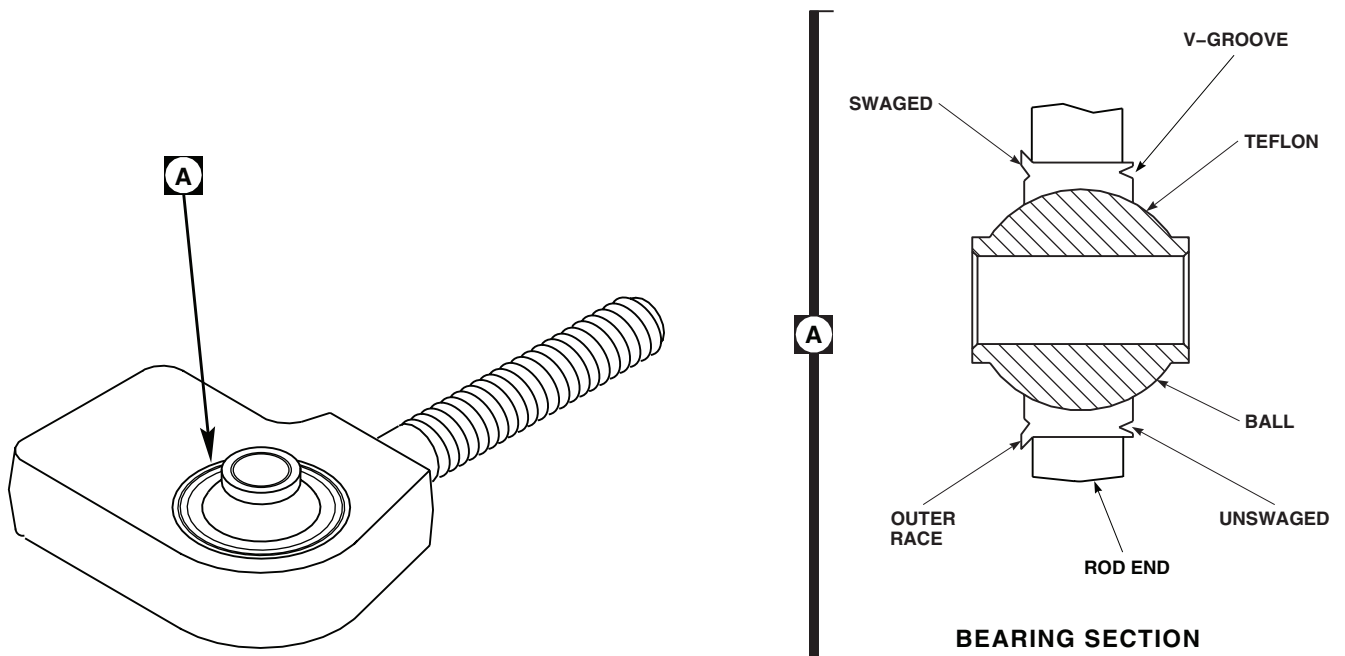
INSTALL - ContinuedAK1416
SA

Figure 1. Replace Tail Rotor Pitch Control Rod Bearings (AVIM).

12. Touch up rod end surfaces with cleaning compound solvent, Item 71, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
TAIL ROTOR BLADE PIVOT BEARING (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Compression Screw, Y-28446-1-4
Nonmetallic Scraper
Pivot Bearing Replacement Wrench, 1420-BH-013,
Locally-Made, Figure 51, WP 1805 00
Torque Wrench, 30 - 200 in. lbs.
Vernier Caliper, GGG-C-111

Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00
Isopropyl, Alcohol Technical, Item 170, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0588 00
WP 0609 00
WP 1803 00
WP 1805 00

Material/Parts

Adhesive, Item 17, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00

NOTE

If any one pivot bearing is removed or requires replacement, its mating bearing must be removed in order to take measurements for calculating shim thickness.

REMOVE

1. Locally make pivot bearing replacement wrench (WP 1805 00).
2. Cut tiedown strap on outboard end of tail rotor blade boot. Pull boot back, over opposite end (Figure 1, Sheet 2).
3. Remove screws and washers securing fairing to horn.
4. Remove screws securing deice bracket to fairing.
5. Using a nonmetallic scraper, remove bead of sealing compound from around edge of fairing.
6. Loosen bolts securing fairing and pivot bearing to blade. Use locally-made pivot bearing replacement wrench, to hold nut.
7. Carefully remove fairing from tail rotor blade.
8. Remove both bolts holding pivot bearing to each side of blade using locally-made pivot bearing replacement wrench.
9. Clean sealing compound from boltheads.
10. Using Allen wrench, remove compression screw from new pivot bearing. Install screw in old bearing. Turn compression screw clockwise and compress bearing enough to remove it. Carefully remove bearing by pulling it straight out. Remove shim and retain.
11. Repeat procedure to remove mating pivot bearing.

REMOVE - Continued

12. Clean old sealing compound from fairing and horn using nonmetallic scraper.

INSPECT

1. Inspect all four pivot bearing retainers for disbonding from spar. **NO DISBONDING ALLOWED.** Replace retainers if disbonding is found (WP 0609 00).

NOTE

A white wax-like substance on the rubber surface is normal and should not be removed.

2. Wear of elastomeric pivot bearings is a gradual process that will take several hundreds of hours from first sign of rubber damage to the point where bearings must be replaced. Compare bearings with bearings in Figure 1, Sheet 1 to find out whether they need replacement.
3. Bearings may fail by developing separations between rubber layers and metal shims or end plates, or within the rubber layers themselves. **REPLACE BEARING IF SEPARATION WITHIN RUBBER LAYERS IS GREATER THAN 1/8-INCH DEEP AROUND 1/4 OF CIRCUMFERENCE OF PIVOT BEARING.**
4. Check pivot bearings for eraser type extrusions (Figure 1, Sheet 1). **NO EXTRUSIONS ALLOWED.** Replace damaged pivot bearings.
5. Check pivot bearings for rubber to shim separations (Figure 1, Sheet 1). **NO RUBBER TO SHIM SEPARATIONS ALLOWED. BENT SHIMS ARE ACCEPTABLE.** Replace damaged pivot bearings.
6. Check pivot bearing base plate for cracks. **NO CRACKS ALLOWED.** Replace pivot bearings with cracked base plate.
7. Using flashlight and mirror check inside surfaces of pitch horn, pivot bearing retainer, and pivot bearing base plates for missing or damaged primer. **NONE ALLOWED.** If primer is missing or damaged, replace primer. Refer to, REPAIR, in this work package.

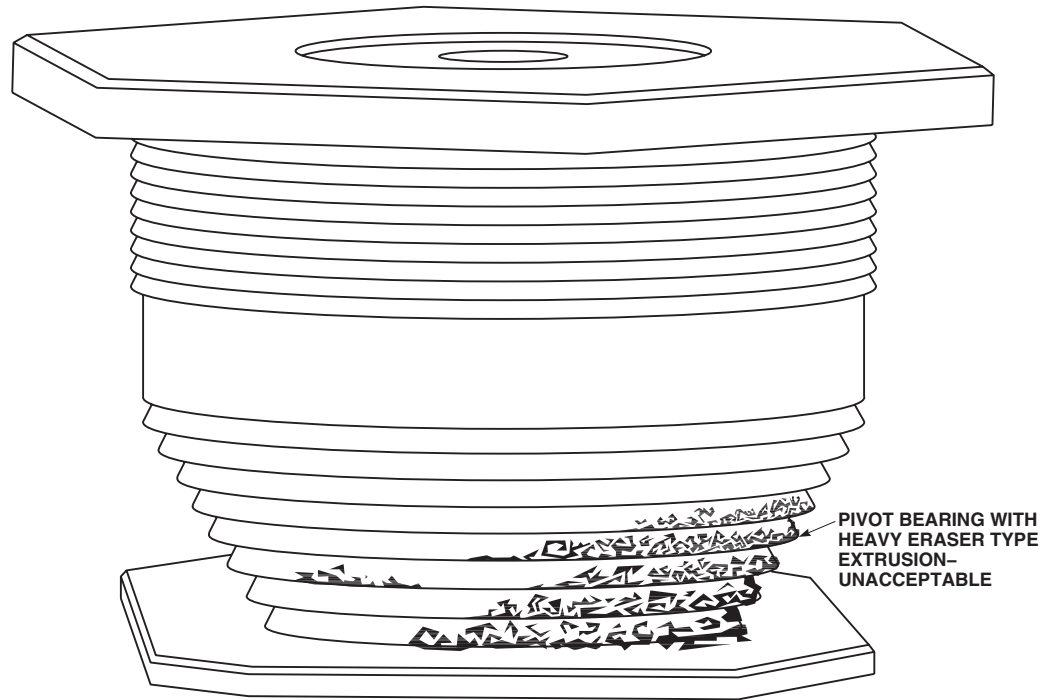
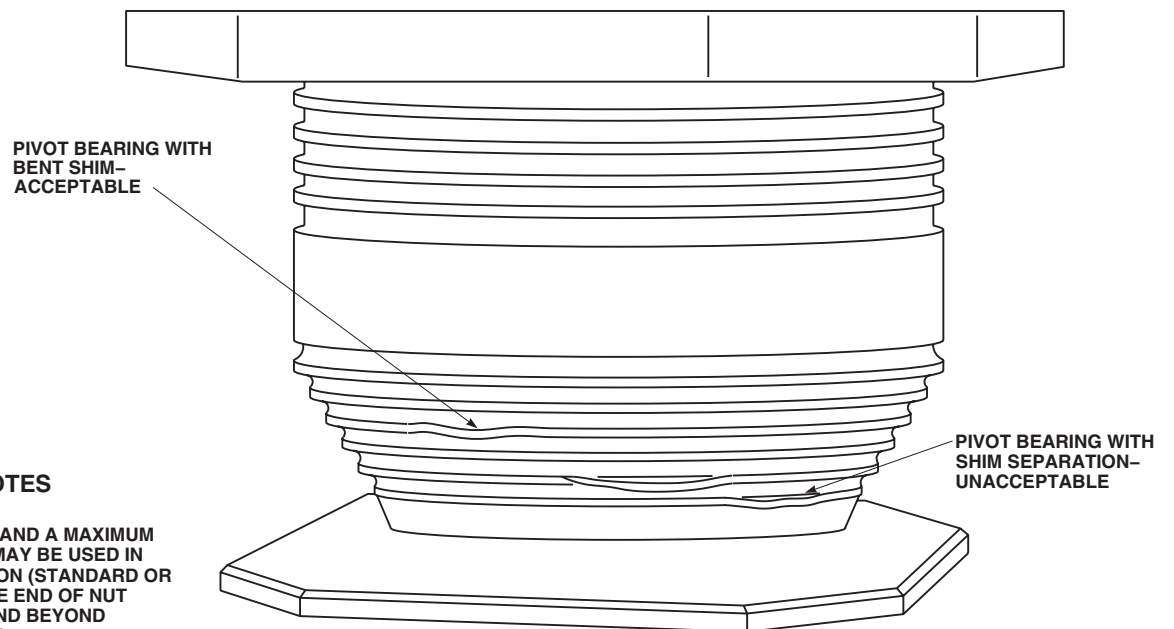
REPAIR

1. Lightly sand surfaces of horn that mate with pivot bearing shims using abrasive cloth, Item 83, WP 1803 00.
2. Clean sanded surface with a clean cheesecloth cloth, Item 85, WP 1803 00, moistened with technical isopropyl alcohol, Item 170, WP 1803 00.
3. Treat bare aluminum with corrosion resistant coating, Item 111, WP 1803 00, before applying primer.
4. Prime untreated surfaces of horn with cleaning compound solvent, Item 71, WP 1803 00.

INSTALL**NOTE**

- If the shelf life for SB7301-103 has exceeded 60 months, do not install. If the shelf life is less than 60 months prior to installation, serviceability will be in accordance with the applicable technical manual.
- The following procedure applies to both mating pivot bearings.

1. Calculate required thickness of new shim as follows:
 - a. Measure and record inside dimension of horn (dimension A) (Figure 2).
 - b. Measure and record thickness of spar, including both retainer seats (dimension B).

INSTALL - Continued**TAIL ROTOR PIVOT BEARING INSPECTION****NOTES**

1. A MINIMUM OF 1 AND A MAXIMUM OF 2 WASHERS MAY BE USED IN ANY COMBINATION (STANDARD OR THIN) TO ENSURE END OF NUT DOES NOT EXTEND BEYOND CROWN OF BOLT.
2. ALL DIMENSION ARE IN INCHES UNLESS NOTED.

TAIL ROTOR PIVOT BEARING INSPECTIONAB3727
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Figure 1. Replace Tail Rotor Blade Pivot Bearing (AVIM). (Sheet 1 of 2)

INSTALL - Continued

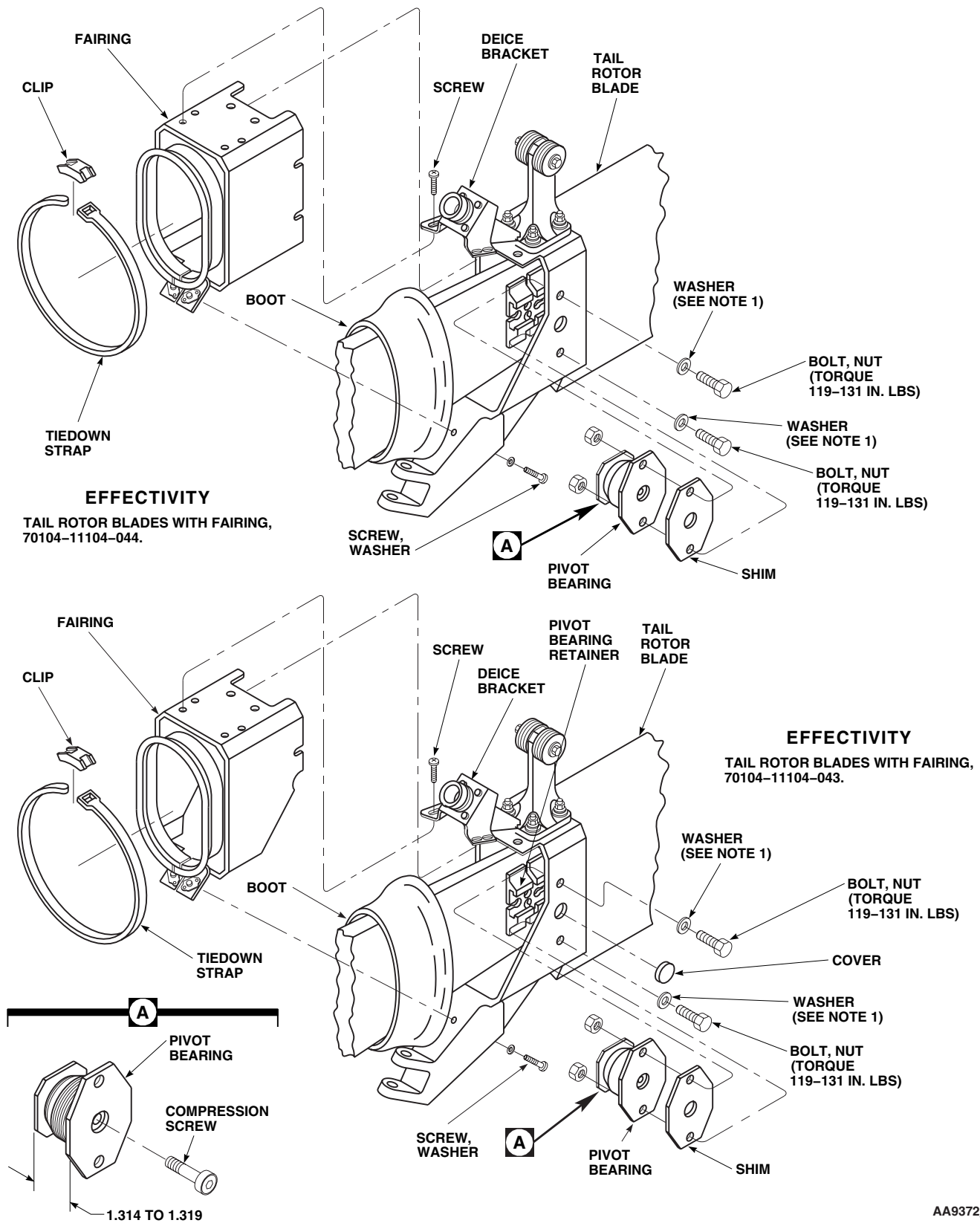
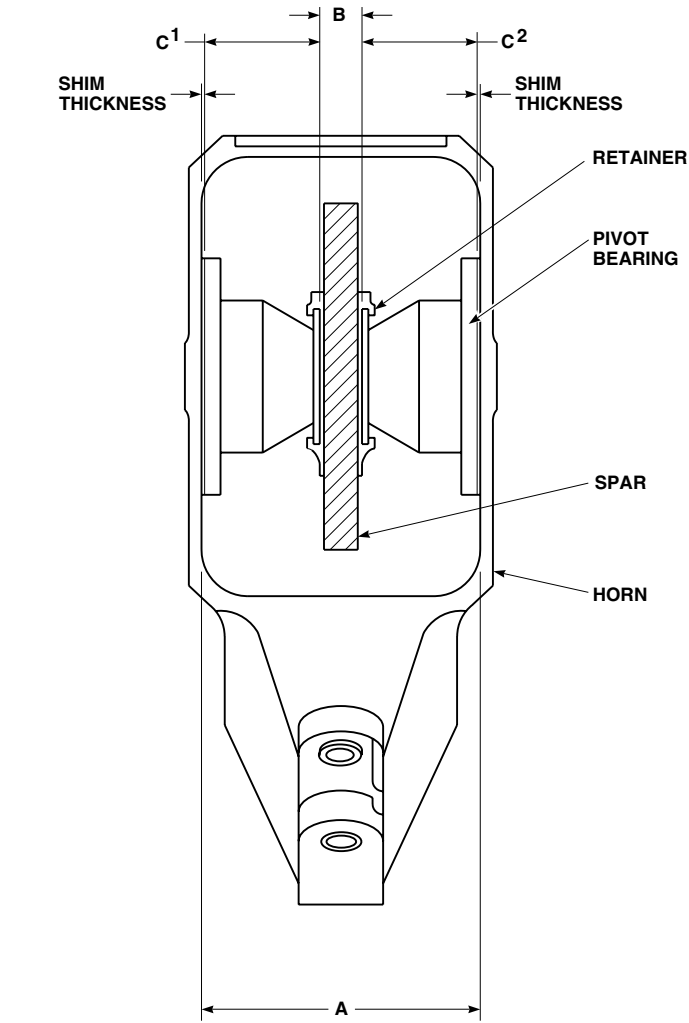


Figure 1. Replace Tail Rotor Blade Pivot Bearing (AVIM). (Sheet 2 of 2)

INSTALL - Continued



DIMENSION	MEASUREMENT	EQUATIONS
A		$B + C^1 + C^2 - 0.066 = D$ $(A - D) \div 2 = \text{SHIM THICKNESS}$
B		
C ¹		
C ²		
D		

NOTE
ALL DIMENSIONS ARE IN
INCHES UNLESS NOTED.

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Figure 2. Replace Tail Rotor Blade Pivot Bearing Retainer (AVIM).

INSTALL - Continued

- c. Measure and record free height (compression screw removed) of pivot bearings (dimensions C^1 and C^2). Take measurements in two places, either side of bearing center hole. Average of two measurements will determine dimensions.
 - d. Calculate dimension D using equation, $B + C^1 + C^2 - 0.066 = D$.
 - e. Calculate shim thickness using equation, $(A - D)$ divided by 2 = Shim Thickness.
 - f. Peel each shim to thickness calculated in step e.
2. **QA** Install compression screw into pivot bearing (Figure 1, Sheet 2, Detail A). **COMPRESS NEW PIVOT BEARING TO A HEIGHT OF 1.314 TO 1.319-INCHES.**

NOTE

All aluminum surfaces of pivot bearing, shims, and retainer must be primed prior to installation.

3. **QA** Place bearing and shim in horn while cleaning compound solvent, Item 71, WP 1803 00, is still wet. Line up centerhole and boltholes in bearing and shim with holes in horn. Make sure bearing is properly engaged to bearing retainer.
4. Loosely install bolts, washers, and nuts through blade and bearing (Figure 1, Sheet 2). Expand bearing by turning compression screw counterclockwise until screw is removed.
5. Place fairing on horn, and slide cutouts in fairing under washers and boltheads. Line up screwholes and install both screws and washers holding fairing to pitch change horn. Apply bead of sealing compound, Item 273, WP 1803 00, around edge of fairing and horn mating surfaces to form watertight seal.
6. **QA** Torque bolts securing bearing as follows:
 - a. Hold nuts with locally-made pivot bearing replacement wrench.
 - b. **QA TORQUE BOLTS SECURING BEARING TO 119 - 131 INCH-POUNDS.**

NOTE

- A minimum of 1 and a maximum of 2 washers may be used in any combination.
- At least 1 washer must be installed under head of bolt.
- Thread protrusion beyond end of nut is not required.

- c. Check that end of nut does not extend beyond crown of bolt. Remove or install washers as required. Repeat step 5.
7. **QA** Install deice bracket on tail rotor blade with screws.
8. Apply sealing compound, Item 273, WP 1803 00, to completely cover heads of bolts securing bearing to horn.
9. Apply a bead of sealing compound, Item 273, WP 1803 00, around edges of deice bracket and boltheads to form a watertight seal.
10. **QA** On tail rotor blades with fairing, 70104-11104-043, installed, bond cover to pivot bearing hole using adhesive, Item 17, WP 1803 00.
11. Install outboard edge of boot on fairing (WP 0588 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM**

TAIL ROTOR BLADE SPAR AND HORN FAIRING (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Bit, No. 50
Nonmetallic Scraper

References

WP 0607 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

1. Inspect for equal spacing on each side of fairing. If clearance between the fairing and spar is not equal, repositioning of the fairing should be completed before any measurements or rework of the fairing.
2. Remove horn fairing (WP 0607 00).
3. Check for delaminations and/or splinters on spar in area near horn fairing inner lip. If any evidence of delamination and/or splintering is found, blade assembly should be rejected and horn fairing should not be reworked.
4. Check for grooves at spar corners in area near horn fairing inner lip. If no grooves are found, rework horn fairing. If a groove is found measure its depth as follows:
 - a. Place a No. 50 drill bit (0.070-inch diameter) into damage notch and place a straight edge in contact notch and place a straight edge in contact with surfaces directly adjacent to damage, holding straight edge perpendicular to drill center line.
 - b. **IF STRAIGHT EDGE TOUCHES ADJACENT SURFACE BUT NOT DRILL BIT, DAMAGE EXCEEDS 0.070-INCH AND BLADE ASSEMBLY SHOULD BE REJECTED AND HORN FAIRING SHOULD NOT BE REWORKED.**
 - c. **IF DAMAGE DOES NOT EXCEED 0.070-INCH BLADE ASSEMBLY IS ACCEPTABLE AND HORN FAIRING SHOULD BE REWORKED.**
5. Using a single cut mill file or sanding block remove material from fairing inner lip at all four corner radii. Continue until inner lip is 0.060 to 0.100-inch outboard of the inside fairing surface under the boot. Tiedown strap area as measured with a straight edge.
6. If spar inspection has revealed no delaminations and/or splinters and no grooves deeper than 0.070-inch, clean old sealing compound from fairing and horn using nonmetallic scraper.
7. Install horn fairing and outboard edge of boot on fairing (WP 0607 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
TAIL ROTOR BLADE PIVOT BEARING RETAINER (AVIM)
This WP supersedes WP 0609 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Heat Gun, LFT500
 Pivot Bearing Replacement Wrench, 1420-BH-013,
 Locally-Made, Figure 51, WP 1805 00
 Torque Wrench, 30 - 200 in. lbs.
 Vernier Caliper, GGG-C-111

Cloth, Cheesecloth, Item 85, WP 1803 00
 Mat, Reinforcing, Fibrous Glass, Item 196,
 WP 1803 00
 Paper, Abrasive, Item 211, WP 1803 00
 Primer Coating Item 243, WP 1803 00
 Stirring Stick, Beverage, Item 300, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 17, WP 1803 00
 Alcohol, Denatured, Item 48, WP 1803 00
 Brush, Acid Swabbing, Item 57, WP 1803 00
 Cloth, Abrasive, Item 80, WP 1803 00

References

WP 0379 00
 WP 0585 00
 WP 0607 00
 WP 1803 00
 WP 1805 00

INSPECT

1. Remove tiedown straps and boots from both ends of tail rotor blade.

NOTE

If pivot bearing retainer becomes disbonded, remove horn fairing, pivot bearings, shims, and disbonded retainer (WP 0607 00).

2. Inspect all four pivot bearing retainers for disbonding from blade. **NO DISBONDING ALLOWED.**

REMOVE

Pivot bearing retainer disbond can cause trailing edge disbond. Make sure to inspect trailing edge for damage prior to repair of pivot bearing retainers (WP 0585 00).

INSTALL**NOTE**

If the shelf life for SB7301-103 has exceeded 60 months, do not install. If the shelf life is less than 60 months prior to installation, serviceability will be in accordance with the applicable technical manual.

1. Locally make pivot bearing replacement wrench (WP 1805 00).

INSTALL - Continued**CAUTION**

Damage to graphite plies in spar will result if care is not taken when removing residual adhesive from blade. Use care in smoothing and removing residual adhesive from blade.

2. Sand adhesive remaining on blade, using abrasive cloth, Item 80, WP 1803 00. Continue to sand only until a smooth bond surface is attained.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Wipe bond surface with clean cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Wipe bond surface with clean cheesecloth cloth, Item 85, WP 1803 00, moistened with denatured alcohol, Item 48, WP 1803 00.
5. Prepare bond surface on pivot bearing retainer, as follows:

WARNING

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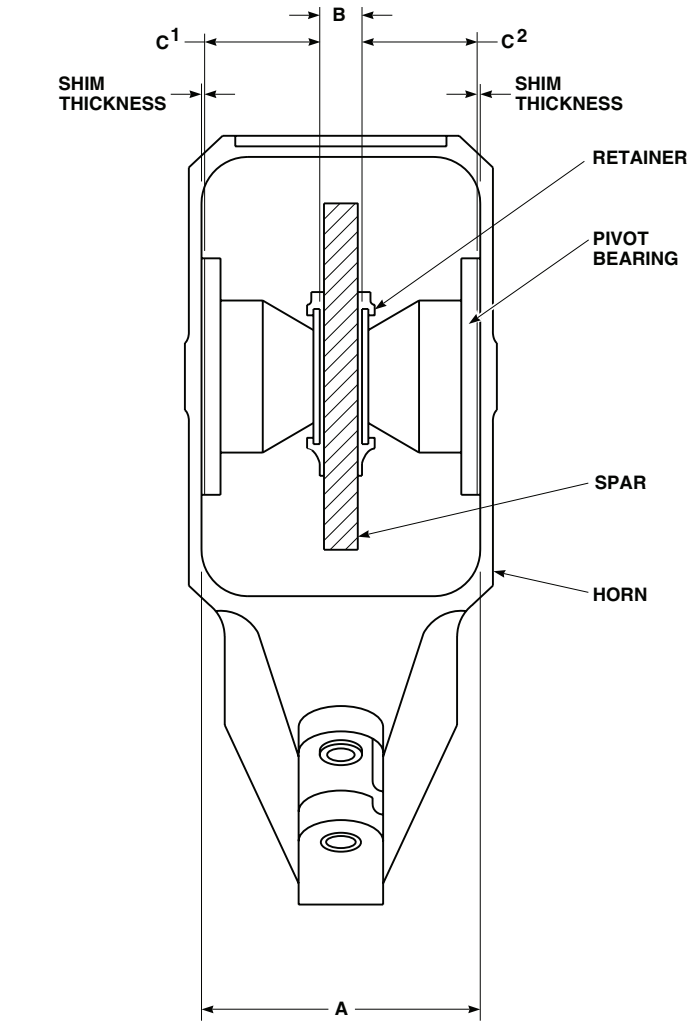
- a. Clean rubber surface, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
- b. Scuff rubber surface, using abrasive paper, Item 211, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- c. Wipe rubber with cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
 - d. Wipe rubber with clean cheesecloth cloth, Item 85, WP 1803 00, moistened with denatured alcohol, Item 48, WP 1803 00.
6. Calculate required thickness of new shim as follows:
 - a. Measure and record inside dimension of horn (dimension A) (Figure 1).
 - b. Measure and record thickness of spar, including both retainer seats (dimension B).

INSTALL - Continued



DIMENSION	MEASUREMENT	EQUATIONS
A		$B + C^1 + C^2 - 0.066 = D$ $(A - D) \div 2 = \text{SHIM THICKNESS}$
B		
C ¹		
C ²		
D		

NOTE
ALL DIMENSIONS ARE IN
INCHES UNLESS NOTED.

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Figure 1. Replace Tail Rotor Blade Pivot Bearing Retainer (AVIM).

INSTALL - Continued

- c. Measure and record free height (compression screw removed) of pivot bearings (dimension C^1 and dimension C^2). Take measurements in 2 places, either side of bearing center hole. Average of 2 measurements will determine dimensions.
 - d. Calculate dimension D using equation, $B + C^1 + C^2 - 0.066 = D$.
 - e. Calculate shim thickness using equation, $(A - D)$ divided by 2 = Shim Thickness.
 - f. Peel each shim to thickness calculated in step e.
7. Install compression screw in pivot bearings.

NOTE

Keep pivot bearing from twisting during compression, and align top and bottom plates when compressed.

8. **QA** Compress pivot bearing as required to allow installation of bearing and shim into blade (WP 0607 00).
9. **QA** Install pivot bearing retainer on pivot bearing, making sure it is fully seated.
10. Mix adhesive, Item 17, WP 1803 00 (WP 0379 00).

NOTE

If both retainers are disbonded, completely bond only one side at a time. Bond opposite side only after first side is cured.

11. Apply even coat of adhesive, Item 17, WP 1803 00, to bond surface of spar and rubber bond surface of retainer, using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00.
12. Embed fibrous glass reinforcing mat, Item 196, WP 1803 00, in wet adhesive.
13. Push horn to deflect it away from spar.

CAUTION

Poor bonding will result if adhesive is smeared during installation of pivot bearing retainer. Do not smear adhesive during installation of pivot bearing retainer.

14. **QA** Install pivot bearing, retainer, and shim in horn, fastening with bolts, washers, and nuts.
15. **QA** Hold nuts with locally-made pivot bearing replacement wrench. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.**
16. **QA** Compress opposite pivot bearing to as required to allow installation of bearing and shim into blade. Install bearing with shim.
17. **QA** If opposite retainer is also disbonded, install retainer on pivot bearing and install assembly with bolts, washers, and nuts. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.**
18. Release and remove compression screw on lower side (not being bonded) to provide bonding pressure against blade.
19. Cure adhesive for 24 hours at 70°F(22°C) or for 2 hours at 140°F(60°C) to 160°F(71°C) using heat gun.
20. Repeat procedures for each disbonded pivot bearing retainer on blade.

INSTALL - Continued**CAUTION**

Damage to rubber will result if contacted by primer coating. Prevent primer coating from contacting rubber parts of pivot bearing.

21. Coat pivot bearing end plates, retainers, shims, and all attaching hardware with primer coating, Item 243, WP 1803 00.
22. **QA** Install shims, pivot bearings, and horn fairing (WP 0607 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ROTOR SYSTEM****TAIL ROTOR BLADE BOOT SUPPORT (AVIM)****This WP supersedes WP 0610 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Boot Support Locator, Locally-Made, Figure 256,
WP 1805 00
Gloves
Goggles/Face Shield
Heat Gun, LFT500
Torque Wrench, 0 - 30 in. lbs.

Alcohol, Denatured, Item 48, WP 1803 00
Cloth, Abrasive, Item 80, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Stirring Stick, Beverage, Item 300, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00

References

WP 1803 00
WP 1805 00

REMOVE

1. Remove sealing compound coating screws. Remove screws and boot support assemblies (Figure 1).

CAUTION

Damage to tail rotor blade spar will result if sanding penetrates graphite plies on spar surface. Do not sand through graphite plies on spar surface.

2. Sand off adhesive down to wire mesh using abrasive cloth, Item 80, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean bond areas using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry using clean cheesecloth cloth, Item 85, WP 1803 00.
4. Trial fit new boot supports to spar. Install screws. **BOOT SUPPORTS SHALL PROVIDE CLAMPING TO SPAR WITHOUT BOTTOMING SCREWS.** Replace tail rotor blade if spar thickness does not provide boot support clamping to spar without bottoming screws.

INSTALL

1. Locally make boot support locator (WP 1805 00).

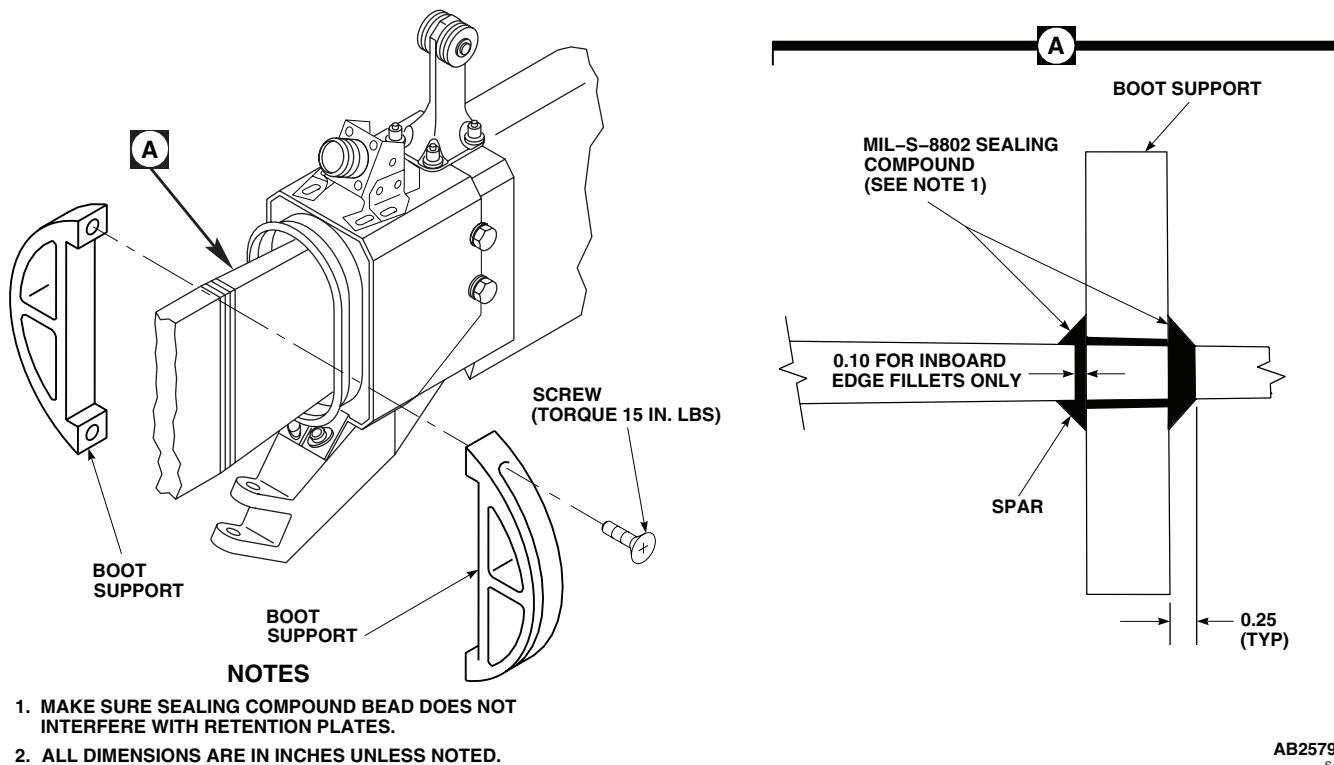
INSTALL - Continued

Figure 1. Replace Tail Rotor Blade Boot Support (AVIM).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Wipe bonding surface of spar using cheesecloth cloth, Item 85, WP 1803 00, saturated with technical acetone, Item 5, WP 1803 00. Rewipe bonding surface of spar using clean cheesecloth cloth, Item 85, WP 1803 00, dampened with denatured alcohol, Item 48, WP 1803 00. Wipe dry using clean cheesecloth cloth, Item 85, WP 1803 00.

CAUTION

Damage to tail rotor blade spar will result if sanding penetrates graphite plies on spar surface. Do not sand through graphite plies on spar surface.

3. Lightly abrade bonding surfaces of tail rotor blade spar and boot supports using abrasive cloth, Item 80, WP 1803 00. Remove resin dust using clean cheesecloth cloth, Item 85, WP 1803 00.
4. Check tail rotor blade spar for no waterbreak condition. Repeat step 2. until no waterbreak condition is obtained.
5. Dry tail rotor blade spar at 130°F(54°C) to 150°F(66°C) using heat gun for at least 30 minutes.
6. Mix adhesive, Item 11, WP 1803 00 (WP 0379 00).
7. Apply mixed adhesive, Item 11, WP 1803 00, to boot support and spar bonding surfaces.

INSTALL - Continued

8. Using locally-made boot support locator (WP 1805 00), position boot supports on tail rotor blade at STA 7.18.
9. Using mixed adhesive, Item 11, WP 1803 00, fill gap between boot supports and tail rotor blade spar using beverage stirring stick, Item 300, WP 1803 00.
10. **QA** Install screws securing boot support. **TORQUE SCREWS TO 3 INCH-POUNDS.**
11. Cure adhesive for 24 hours at 70°F(21°C) to 85°F(29°C), or 2 hours at 140°F(60°C) to 160°F(71°C), or 45 minutes at 190°F(88°C) to 210°F(99°C).
12. **QA RETORQUE SCREWS TO 15 INCH-POUNDS.**
13. Fill screw holes with sealing compound, Item 273, WP 1803 00.
14. Apply sealing compound, Item 273, WP 1803 00, to adhesive fillets and edge area (Figure 1, Detail A).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM**OUTBOARD RETENTION PLATE (AVIM)**

This WP supersedes WP 0611 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Arbor Press
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Powertrain Repairer's Toolkit, SC 5180-99-B13

Primer Coating, Item 243, WP 1803 00
 Tape, Pressure Sensitive Adhesive, Item 327,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00
 Lacquer, Item 172, WP 1803 00

References

TM 1-1520-265-23
 WP 1803 00

NOTE

Do not remove balance weights (washers) from retention plate.

DISASSEMBLE

Using an arbor press, press out pilot bushing from retention plate; then press out bushing from pilot (Figure 1).

INSPECT

1. Visually check retention plate for cracks, nicks, wear, and corrosion.
2. Check tail rotor pilot bushing for nicks, gouges and scratches on surface (Figure 1).
3. **DAMAGE UP TO 0.010-INCH DEEP COVERING LESS THAN 20% OF SURFACE AND OUTSIDE OF 0.10 RADIUS AREA, BLEND OUT DAMAGE USING ABRASIVE CLOTH, Item 83, WP 1803 00 (Figure 1, Detail A).** Replace pilot bushing if damaged beyond limits.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. Remove all paint on inspection surface using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00 (Figure 1, Detail A).

INSPECT - Continued

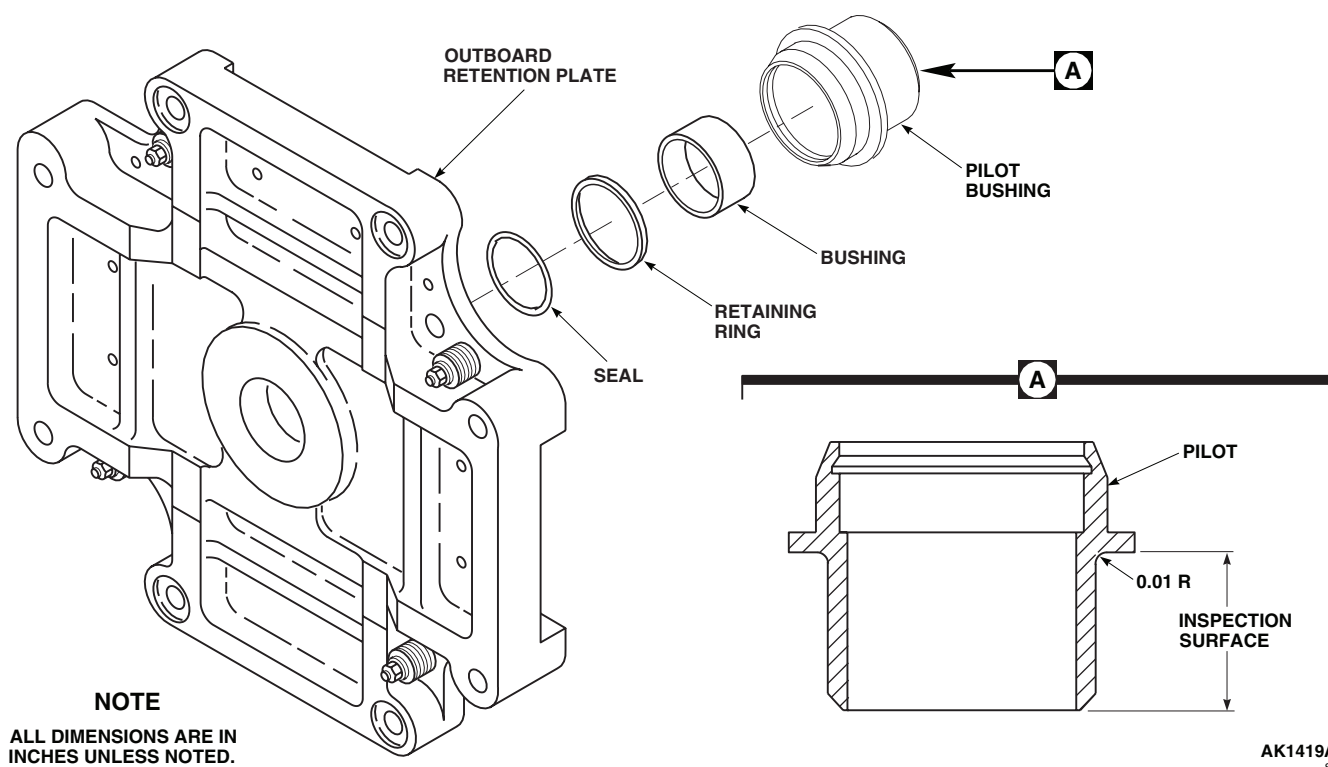


Figure 1. Repair Outboard Retention Plate (AVIM).

NOTE

Do not repaint pilot bushing.

5. Fluorescent penetrant inspect reworked areas for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23).

ASSEMBLE

1. Coat outboard surface of pilot bushing OD with primer coating, Item 243, WP 1803 00.
2. **QA** While primer coating is still wet, press in pilot bushing using an arbor press. Be careful not to cock pilot.
3. **QA** Check to make sure shoulder on pilot bushing is fully seated on retention plate all around.
4. Mask ID of bushing with pressure sensitive adhesive tape, Item 327, WP 1803 00. Coat OD of bushing with primer coating, Item 243, WP 1803 00, and press in bushing.
5. **QA** Apply primer coating, Item 243, WP 1803 00, to top of bushing, retaining ring groove in pilot bushing, and retaining ring. Remove pressure sensitive adhesive tape. Install retaining ring while primer coating is still wet.

NOTE

Make sure that seal is installed with flange facing outward.

6. **QA** Install seal.
7. If necessary, touch up plate with lacquer, Item 172, WP 1803 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ROTOR SYSTEM
ROTOR HUB LINER REPLACEMENT (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Powertrain Repairer's Toolkit, SC 5180-99-B13
Main Rotor Hub Flate Plate Tool, Item 272,
WP 1805 00

Parting Compound, Foundry Mold, Item 212,
WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00

References

WP 1803 00
WP 1805 00

ROTOR HUB LINER REPLACEMENT**WARNING**

Acetone is flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/faceshield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flame, sparks, or other sources of ignition.

CAUTION

All operation connected with bonding area shall be done (as best as possible) in a dirt and dust free environment.

1. Peel liner from rotor hub. **DO NOT NICK OR GOUGE HUB BOND AREA.**
2. Clean bonding surface of hub using cheesecloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. A nonmetallic scraper may be used to aid in removing old adhesive.
3. Clean bonding surfaces of new liner (external spindle, damper bracket, or bifilar) with cheesecloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Mix adhesive, Item 11, WP 1803 00, following manufactures instructions. Lightly apply adhesive to original bond area on hub and new liner. **DO NOT GET ADHESIVE ON BOLT HOLES.**
5. Bond liner (external spindle, damper bracket, or bifilar) in place by using a main rotor hub flat plate tool, Item 272, WP 1805 00, with a thin film of foundry mold parting compound, Item 212, WP 1803 00, against liner. Install plate using the components (spindle, damper bracket, or bifilar) bolts and washers. Clean off excessive adhesive with a dry cheesecloth, Item 85, WP 1803 00.
6. Allow to cure for 2 to 3 days at room temperature.

END OF WORK PACKAGE

UNIT LEVEL

DRIVE SYSTEM

MAIN TRANSMISSION MOUNTING BOLT INSPECTION

This WP supersedes WP 0612 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Mounting Bolt Adapter Tool, Locally-Made, Figure 46,
 WP 1805 00
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 0 - 300 ft. lbs.

Polyurethane Coating, Item 231, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Tape, Pressure Sensitive Adhesive, Item 326,
 WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
 WP 0259 00
 WP 0260 00
 WP 0261 00
 WP 0261 01
 WP 1803 00
 WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Antiseize Compound, Item 49, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00

INSPECTION

1. Open main rotor pylon sliding cover.
2. Open oil cooler compartment access doors.
3. Remove transmission drip pan (WP 0259 00, WP 0260 00, WP 0261 00, or WP 0261 01).

CAUTION

Damage to main transmission will result if metal scrapers are used on surfaces. Do not use metallic scrapers to remove sealant or other debris.

4. Remove sealing compound from main gear box mounting bolts using nonmetallic scraper.

WARNING

CSI

Verification of minimum torque of bolts is a critical characteristic.

5. **CHECK EIGHT MAIN TRANSMISSION MOUNTING BOLTS FOR TORQUE OF 154 - 170 FOOT-POUNDS.** Use mounting bolt adapter tool (WP 1805 00) to check torque on two front mounting bolts. Use TM 1-1500-204-23 to figure proper torque. Torque value will change due to added length of adapter.
6. Inspect barrel-nut(s) as follows:
 - a. On bench, install mounting bolt into barrel-nut until two threads are exposed beyond nut.

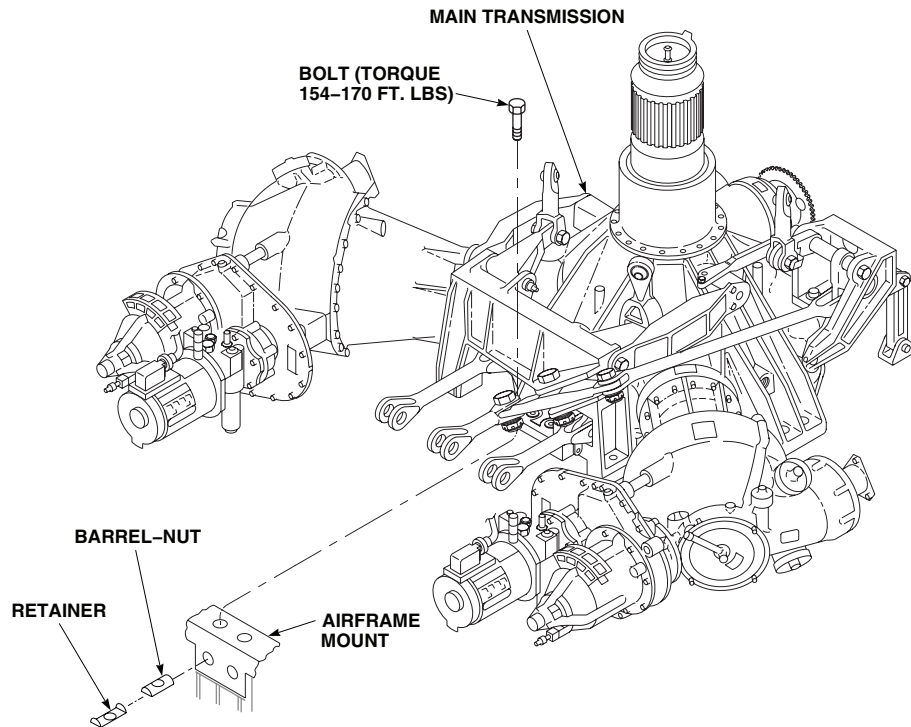
INSPECTION - ContinuedAK2004
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Figure 1. Check Main Transmission Mounting Bolts.

- b. Using torque wrench, back out bolt. **IF BREAKAWAY TORQUE IS LESS THAN 32 INCH-POUNDS, DISCARD NUT AND REPLACE WITH NEW ONE.**

CAUTION

Damage to recess in frame or beam will result if barrel-nut is incorrectly installed. To prevent damage to recess, make sure nut is installed with barrel (round side) toward bolthead.

NOTE

Make sure that only square barrel-nuts are used.

7. Install serviceable barrel-nut and retainer (Figure 1).

CAUTION

Damage to equipment will result if epoxy primer coating kit is applied to threads of mounting bolts. Epoxy primer coating kit will make removal of bolts difficult and could damage barrel-nuts. Do not allow epoxy primer coating kit to contact bolt threads.

8. Apply a coating of epoxy primer coating kit, Item 120, WP 1803 00, to shank of mounting bolts.

INSPECTION - Continued**WARNING****CSI**

Verification of minimum torque of bolts is a critical characteristic.

9. Put a light coat of antiseize compound, Item 49, WP 1803 00, on mounting bolt threads. Install bolt. **TORQUE TO 154 - 170 FOOT-POUNDS.** Use locally-made mounting bolt adapter tool (Figure 46, WP 1805 00) to check torque on two front mounting bolts. Use TM 1-1500-204-23 to figure proper torque. Torque value will change due to added length of adapter.
10. Mask off adjacent areas using pressure sensitive adhesive tape, Item 326, WP 1803 00, to prevent overspray.
11. Apply corrosion preventive compound, Item 106, WP 1803 00, to entire area where sealing compound was removed.
12. Allow to dry for 10 to 30 minutes and apply a second coat of corrosion preventive compound, Item 106, WP 1803 00.
13. If any main transmission mounting bolt is replaced, a torque check is required 9 to 11 flight hours after replacement (WP 1689 00).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

14. After torque has stabilized, remove corrosion preventive compound using technical acetone, Item 5, WP 1803 00.
15. Apply sealing compound, Item 273, WP 1803 00, to joints around bolt heads, nuts and washers. Allow to dry.
16. Apply epoxy primer coating kit, Item 120, WP 1803 00, to joints around boltheads, nuts and washers.
17. Apply one coat of polyurethane coating, Item 231, WP 1803 00, to joints around boltheads, nuts and washers. Allow to dry.
18. Apply two coats of corrosion preventive compound, Item 106, WP 1803 00, to gear box attachment feet. Allow to dry between coats.
19. Make sure area is clean and free of foreign material.
20. Install transmission drip pan (WP 0259 00, WP 0260 00, WP 0261 00, or WP 0261 01).
21. Close oil cooler compartment access doors.
22. Close main rotor pylon sliding cover.

END OF WORK PACKAGE

UNIT LEVEL
DRIVE SYSTEM
MAIN TRANSMISSION DOWEL PIN INSPECTION
This WP supersedes WP 0613 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Depth Gage, Dial Indicating, G43J

References

WP 0630 00
 WP 1138 00
 WP 1139 00
 WP 1140 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

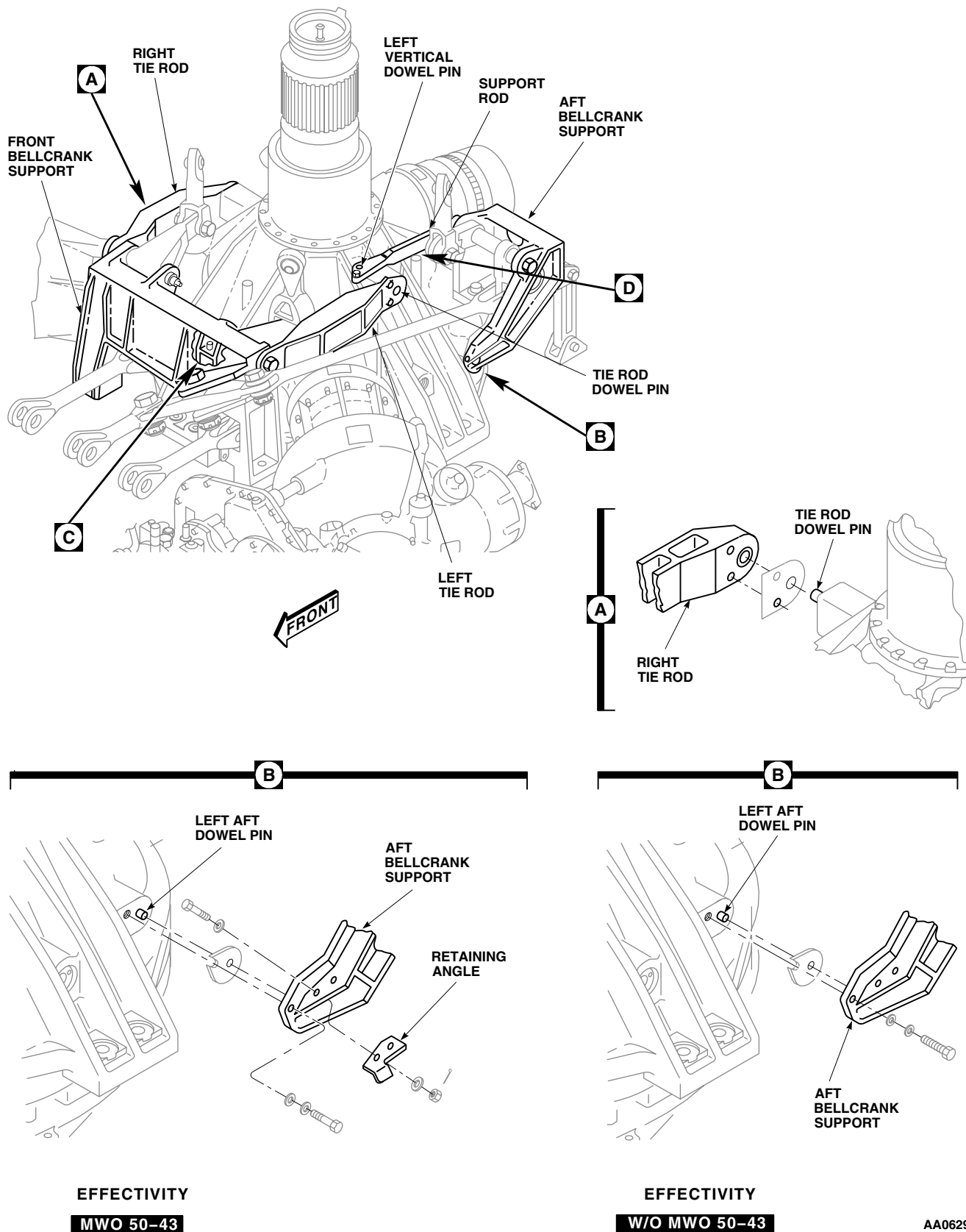
INSPECT

1. Open main rotor pylon sliding cover, No. 1 engine cowling, left oil cooler access door, and No. 2 engine cowling.
2. **W/O MWO 50-43** Inspect main transmission flight control mount dowel pins as follows (Figure 1, Sheet 1, and Figure 1, Sheet 2): 

NOTE

- On transmissions, 70351-08100-053 and 70351-08100-054, with slippage stripe, axial movement will reveal an unpainted portion of the pin. Radial movement will indicate a break in the black stripe between pin head and adjoining white painted surface.
 - On dowel pins with sealing compound, axial movement will be indicated by a bulging or bubbling of sealant surface. Radial movement will be indicated by clockwise or counterclockwise ripples on cracking of sealant surface.
- a. Inspect right and left tie rod dowel pins for evidence of axial or radial movement. **NONE ALLOWED. DOWEL PINS MUST BE FLUSH OR BELOW TIE ROD.** Replace main transmission if movement is found (WP 0630 00).
 - b. Inspect left vertical dowel pin for axial or radial movement. **NONE ALLOWED. DOWEL PIN MUST BE WITHIN 0.060 - INCH OF SUPPORT ROD SURFACE.** Replace main transmission if movement is found (WP 0630 00).
 - c. Inspect left aft dowel pin for evidence of axial or radial movement. **NONE ALLOWED. DOWEL PIN MUST BE APPROXIMATELY FLUSH WITH AFT BELLCRANK SUPPORT BRIDGE SURFACE.** Replace main transmission if movement is found (WP 0630 00).
 - d. Inspect forward dowel pin for evidence of axial or radial movement. **NONE ALLOWED.** Replace main transmission if movement is found (WP 0630 00).
3. **MWO 50-43** Inspect main transmission flight control mount dowel pins as follows: 
 - a. Inspect right and left tie rod attachment bolts for damage and security (Figure 1, Sheet 1, Detail A). **NO DAMAGE OR LOOSENESS ALLOWED.** If tie rod attachment bolts are damaged or loose, do this:
 - (1) Remove right or left tie rod (WP 1138 00).

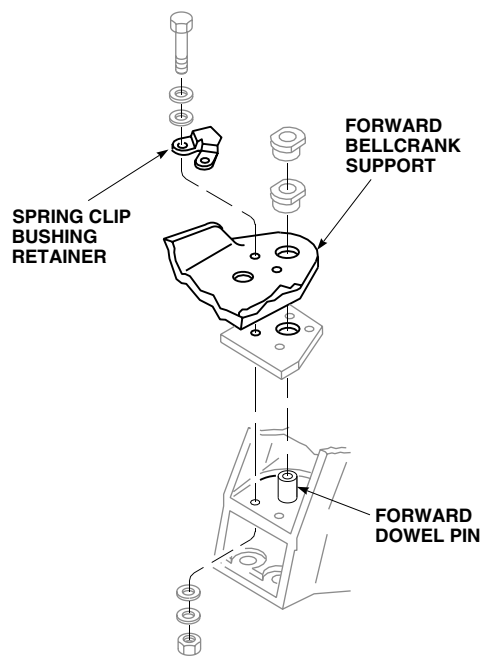
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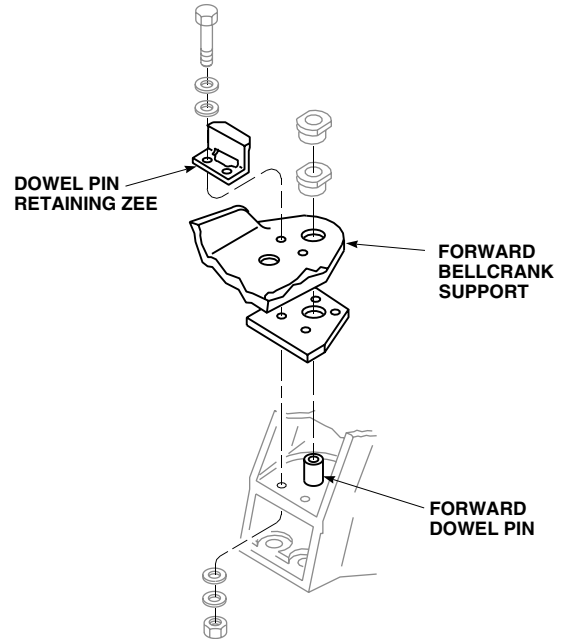
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Figure 1. Inspect Main Transmission Dowel Pins. (Sheet 1 of 2)

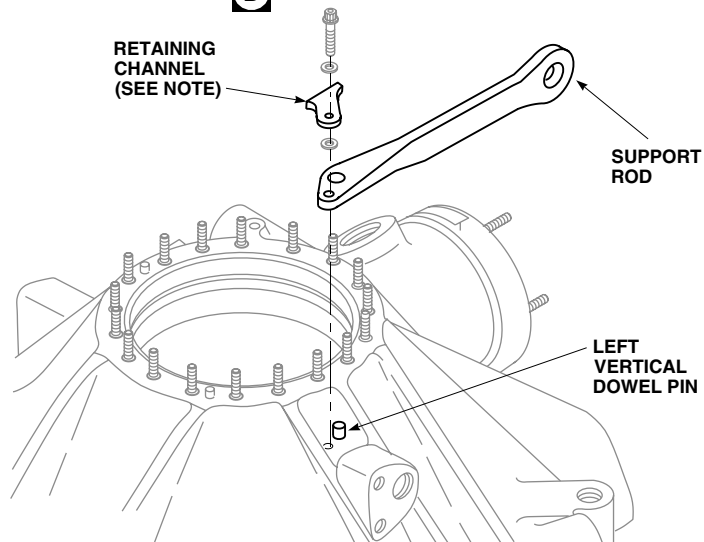
INSPECT - Continued



EFFECTIVITY
W/O MWO 50-43



EFFECTIVITY
MWO 50-43



NOTE
MWO 50-43

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Figure 1. Inspect Main Transmission Dowel Pins. (Sheet 2 of 2)

INSPECT - Continued

- (2) Inspect tie rod dowel pin for axial or radial movement. **NONE ALLOWED.** Replace main transmission if movement is found (WP 0630 00).
- (3) Install right or left tie rod (WP 1138 00).
- b. Inspect left vertical dowel pin retaining channel for deformation and attachment bolts for damage and security (Figure 1, Sheet 2, Detail D). **NO DEFORMATION OF RETAINING CHANNEL ALLOWED. NO DAMAGE OR LOOSENESS OF ATTACHMENT BOLTS ALLOWED.** If dowel pin retaining channel is deformed or attachment bolts are damaged or loose, do this:
 - (1) Remove left vertical dowel pin retaining channel from support rod (WP 0630 00).
 - (2) Inspect left vertical dowel pin for axial or radial movement. **NONE ALLOWED. DOWEL PIN MUST BE WITHIN 0.060-INCH OF SUPPORT ROD SURFACE.** Replace main transmission if movement is found (WP 0630 00).
 - (3) Install left vertical dowel pin retaining channel on support rod (WP 0630 00).
- c. Inspect left aft dowel pin retaining angle for deformation and attachment bolts for damage and security (Figure 1, Sheet 1, Detail B). **NO DEFORMATION OF RETAINING ANGLE ALLOWED. NO DAMAGE OR LOOSENESS OF ATTACHMENT BOLTS ALLOWED.** If dowel pin retaining angle is deformed or attachment bolts are damaged or loose, do this:
 - (1) Remove left aft dowel pin retaining angle from aft bellcrank support (WP 1140 00).
 - (2) Inspect left aft dowel pin for axial or radial movement. **NONE ALLOWED. DOWEL PIN MUST PROTRUDE APPROXIMATELY 1/8-INCH ABOVE AFT BELLCRANK SUPPORT BRIDGE SURFACE.** Replace main transmission if movement is found (WP 0630 00).
 - (3) Install left aft dowel pin retaining angle on aft bellcrank support (WP 1140 00).
- d. Inspect forward dowel pin retaining zee for deformation and attachment bolts for damage and security (Figure 1, Sheet 2, Detail C). **NO DEFORMATION OF RETAINING ZEE ALLOWED. NO DAMAGE OR LOOSENESS OF ATTACHMENT BOLTS ALLOWED.** If dowel pin retaining zee is deformed or attachment bolts are damaged or loose, do this:
 - (1) Remove forward dowel pin retaining zee (WP 1139 00).
 - (2) Inspect forward dowel pin for evidence of axial or radial movement. **NONE ALLOWED.** Replace main transmission if movement is found (WP 0630 00).
 - (3) Install forward dowel pin retaining zee (WP 1139 00).
4. Close No. 2 engine cowling, left oil cooler access door, No. 1 engine cowling, and main rotor pylon sliding cover.

END OF WORK PACKAGE

UNIT LEVEL**DRIVE SYSTEM**

MAIN ROTOR SHAFT INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

WP 0634 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT MAIN ROTOR SHAFT SWASHPLATE GUIDE AND SHAFT SEAL LEAKAGE**NOTE**

- Checking amount of shaft seal leakage can prevent premature seal replacement, before leakage is over recommended limits.
 - Allow at least 2 hours after flight for oil to drain back into sump before checking level.
1. Measure seal leakage by observing drop in oil level on main transmission dipstick.
 2. **IF LEAKAGE IS MORE THAN 1/16-INCH PER FLIGHT HOUR REPLACE UPPER SHAFT SEAL (WP 0634 00).**

INSPECT MAIN ROTOR SHAFT

Axial and radial movement of main rotor shaft is not cause for replacement of transmission. There is no requirement to inspect for radial or axial movement. If excessive axial and/or radial movement is suspected, contact AMCOM engineering for instructions.

END OF WORK PACKAGE

UNIT LEVEL**DRIVE SYSTEM****MAIN TRANSMISSION FUNCTIONAL TEST**

This WP supersedes WP 0615 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

TM 1-1520-237-MTF
WP 0616 00
WP 1607 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

FUNCTIONAL TEST

1. A functional test is required to make sure a replacement main transmission or a replacement module and its components are installed properly and that the lubrication system is working properly. Make these checks:

WARNING

Injury to personnel and damage to equipment will result if checks are not performed.
These checks are mandatory and must be conducted for safety of flight.

2. Prepare helicopter for runup. Make sure helicopter is serviced.

CAUTION

Damage to equipment will result if instrument readings are not carefully monitored. If any abnormal readings are indicated, shut down helicopter immediately and investigate according to WP 0616 00.

3. With qualified personnel, conduct a normal preflight warmup. Inspect main transmission for leakage. Check main gear box oil temperature and oil pressure for normal operation limits. Record gage readings when they stabilize.
4. With qualified personnel, if all checks are normal, hover helicopter several feet above ground for 30 minutes. Continuously check oil pressure, oil temperature, and chip indicator for abnormal conditions.
5. After landing, inspect main transmission assembly for leakage.
6. Inspect input and main module chip detectors screens for chips.

NOTE

If chips are found go to , WP 0616 00. Allow about 2 hours for oil to drain back before checking.

7. Check oil level on dipstick and service main transmission if required (WP 1607 00).

FUNCTIONAL TEST - Continued**NOTE**

Maintenance test flight not required after functional test for accessory module, oil cooler, or any seal replacement.

8. Perform maintenance test flight after functional test when replacing input or main modules (TM 1-1520-237-MTF).

END OF WORK PACKAGE

UNIT LEVEL

DRIVE SYSTEM

MAIN TRANSMISSION SERVICEABILITY INSPECTION

This WP supersedes WP 0616 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Container, 1-Gallon
 Powertrain Repairer's Toolkit, SC 5180-99-B13

WP 0643 00

WP 0644 00

WP 0645 00

WP 0646 00

WP 0647 00

Material/Parts

Cloth, Cheesecloth, Item 85, WP 1803 00

WP 0675 00

WP 0677 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

WP 1607 00

WP 1803 00

References

TB 43-0211
 WP 0617 00

NOTE

AOAP samples results are considered advisory in nature and cannot be used as a sole basis for determining airworthiness of a transmission or gearbox. If an AOAP lab reports an anomalous oil sample result and/or recommends removal of an otherwise normal functioning gearbox, contact the logistics assistance representative or liaison engineer for assistance.

OIL FILTER BYPASS BUTTON EXTENDED AND/OR PARTICLES IN OIL SCREENS

1. If oil filter bypass button is extended check maintenance logbook to determine if first stage filter element has been replaced more than three times within the last 200 flight hours with no oil changes or main transmission maintenance that caused oil to be drained, exposed to a nonpurified container, or exposed to dust.
 - a. If filter element has been replaced more than three times within the last 200 flight hours and the main transmission oil has not been subjected to the possible contamination factors listed in step 1., replace filter element. If bypass button continues to pop, perform a serviceability inspection. If no contamination/debris is found during inspection, replace filter bypass mechanism.
 - b. If first stage filter element has been replaced more than three times within the last 200 flight hours and the transmission oil has been subjected to the possible contamination factors listed in step 1., go to step 2. and continue serviceability check.
 - c. If filter element has not been replaced more than three times within the last 200 flight hours and main transmission has not been subjected to possible contamination factors listed in step 1., replace filter element.

OIL FILTER BYPASS BUTTON EXTENDED AND/OR PARTICLES IN OIL SCREENS - Continued**NOTE**

Only the main module and input module chip detectors have chip detector screens. No other screens require inspection. All main transmission chip detectors require inspection.

2. If 3 micron main transmission oil filter elements (70351-08143-104 or 70351-38801-101) are installed do this:
 - a. Check chip detectors and chip detector oil screens for chips. If no chips are found in chip detectors or oil screens, serviceability check is complete. Replace main transmission oil first stage filter element. Check 75 micron second stage filter element for contamination. Replace secondary filter only if contamination is found (WP 0647 00).
 - b. If chips are found on chip detectors or oil screens, serviceability check must be performed.

NOTE

Do the following procedure if particles are found on any chip detector or screen, or if oil filter bypass button is extended and 3 micron filters are not being used.

3. If chips are found in chip detectors see WP 0617 00.
4. Remove, clean, and reinstall oil strainer screen and chip detector of suspected module. Collect any debris or particles that are found, and package and label them for comparison after gear box runup.
5. If oil filter bypass button is extended, disconnect oil cooler inlet line (WP 0677 00). Drain oil through clean cheesecloth cloth, Item 85, WP 1803 00. If any metal particles are seen, do this:
 - a. Replace oil cooler radiator (WP 0675 00).
 - b. Drain main transmission oil through clean cheesecloth cloth, Item 85, WP 1803 00.
 - c. Remove, clean, and reinstall chip detectors, sleeves, and oil screens from both accessory modules (WP 0643 00 and WP 0644 00).
 - d. Remove, clean, and reinstall chip detectors, strainers, and screens for both input modules (WP 0645 00).
 - e. Remove, clean, and reinstall main module chip detector/temperature indicator and strainer from sump (WP 0646 00).
6. If no metal particles are present, reconnect oil cooler inlet line (WP 0677 00).
7. Replace main transmission first stage (3 micron) oil filter element. Check second stage oil filter element for contamination. Replace second stage oil filter element if contamination is found (WP 0647 00).
8. Service main transmission (WP 1607 00).
9. Ground run helicopter for 30 minutes and then hover for 30 minutes. Reinspect chip detector and screen of affected module. If particles were found in oil cooler inlet line, reinspect all oil screens.
10. Return transmission to service if no particles are found during check.

NOTE

If module is being replaced due to an increase in particles, make sure all foreign material from chip detectors, screens, and/or filters is carefully packaged and returned with module.

11. If amount of particles has increased, replace affected module.
12. If amount of particles has decreased, do this:
 - a. Enclose chips in a 1 gallon container.

OIL FILTER BYPASS BUTTON EXTENDED AND/OR PARTICLES IN OIL SCREENS - Continued

- b. Obtain an oil sample from gear box (TB 43-0211).
 - c. Submit chips and oil sample together to AOAP laboratory.
13. Service main transmission, if necessary (WP 1607 00).
14. Return helicopter to service.

END OF WORK PACKAGE

UNIT LEVEL**DRIVE SYSTEM****MAIN TRANSMISSION CHIP DETECTOR PARTICLE INSPECTION**

This WP supersedes WP 0617 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

WP 0616 00
WP 0839 00
WP 1803 00

Material/Parts

Cloth, Cheesecloth, Item 85, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

MAIN TRANSMISSION CHIP DETECTOR PARTICLE INSPECTION

1. When the CHIP INPUT MDL-LH, CHIP ACCESS MDL-LH, CHIP MAIN MDL SUMP, CHIP INPUT MDL-RH, or CHIP ACCESS MDL-RH light on the caution/advisory panel goes on, it indicates which chip detector has been activated.
2. The chip detectors and screens, in the main transmission, isolate a problem to a particular module, and avoid contaminating the lubrication system. If particles are found, it does not necessarily mean the affected module must be replaced. The quantity, size, type, and source of the particles, along with the history of the gear box, must be considered. Also, gear box time, previous malfunctions, operating conditions, temperatures, and pressure should be considered.
3. Particles found may be metallic/magnetic (steel), metallic/nonmagnetic (lead, cadmium, aluminum, magnesium, bronze, or silver), or nonmetallic (wood, paint, paper, lint, rubber, or phenolic).

CAUTION

Main gear box filter normally keeps oil clean. Metallic black oil discoloration found in main gear box is indicative of a high rate of fretting/wear. Damage to gear box may result. Perform oil analysis prior to next flight and contact your AMCOM LAR.

NOTE

Visual inspection may be enough to determine nature of contamination material. If identification of material is required, contact your supporting Army Oil Analysis Lab for assistance.

4. If particles are easily identified as fragments from gears, nuts, bearings, etc., replace module. If replacement of module is questionable, contact the AMCOM LAR. If 10 or more particles are found, or if any one particle is larger than 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long, replace the module. When a module is replaced due to contamination, foreign material from screens and filters must be carefully packaged and returned with the module. Also, gear box oil must be drained and strained with cheesecloth cloth, Item 85, WP 1803 00. Any particles found in oil should be combined with particles from screens and filters for evaluation and returned with module if necessary. Also, package cheesecloth cloth, Item 85, WP 1803 00, put oil in sealed container, and return both with module.

MAIN TRANSMISSION CHIP DETECTOR PARTICLE INSPECTION - Continued

5. For appropriate action after determining particle material, size, and quantity, refer to Table 1.

Table 1. Particle Classification.

MATERIAL	TYPE-QUANTITY SIZE	ACTION REQUIRED
Metallic/Magnetic (Steel)	Fuzz, fine hair-like particles.	Acceptable. Return gear box to service. No serviceability check is required.
	No more than 10 flakes, none over 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.0125)-inch long.	Do a serviceability check (WP 0616 00). Record findings in helicopter logbook. Refer to Table 2.
	Splinters or granular particles.	Do a serviceability check (WP 0616 00).
	Any quantity of chips beyond the above dimensions.	Replace module.
	Flathead No. 8 machine screws.	align="left"Remove screws. Replace generator (WP 0839 00).
	Pieces of small coil spring. Overall length of unstretched spring from lower main rotor shaft seal (approximately 9 inches).	Performance of shaft seal will not be seriously affected by loss of spring. Measure unstretched spring (9 inches) to make sure entire spring has been removed. Continue serviceability check.
Metallic/Nonmagnetic (Lead, Cadmium, Silver, Aluminum, Magnesium, Bronze)	Thin flakes not over 0.050 inch in diameter to 1/32 (0.032) by 1/8 (0.125) inch. Quantity of 10 or less flakes.	Retain flakes for future reference. Return gearbox to service. No serviceability check required. If previous chips have been experienced, refer to Table 2..
	Quantity of 11-24 flakes.	Retain flakes. Conduct a serviceability check (WP 0616 00). Contact AMCOM or Sikorsky Flight Service Representative for possible laboratory analysis of flakes. Record findings on Log card. Refer to Table 2.
	Any quantity of flakes that appear to be an aluminum material with a shim like appearance and main transmission gearbox was reported to have experienced a drop in pressure of 10 or more PSI.	Replace Main Module. Submit QDR.

MAIN TRANSMISSION CHIP DETECTOR PARTICLE INSPECTION - Continued

Table 1. Particle Classification. - Continued

MATERIAL	TYPE-QUANTITY SIZE	ACTION REQUIRED
Nonmetallic (Baked Resin/ Paint, Phenolic, Dust, Lint, Wood)	Any quantity of flakes that appear to be an aluminum material that have a shim like appearance but main transmission did not experience a pressure drop of 10 or more PSI.	Retain flakes. Contact AMCOM or Sikorsky Field Service Representative.
	More than 25 flakes.	Replace affected module. Submit QDR.
	Flakes or particles not over 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long. Quantity not to be over 10 flakes.	Acceptable. Return gear box to service. No serviceability check is required.
	Less than a total area of 1/2 (0.50)-square inch.	Do a serviceability check (WP 0616 00). Refer to Table 2.
Nonmetallic (Rubber)	More than a total area of 1/2 (0.50)-square inch.	If contamination is sufficient to partially block oil strainer, replace module.
	More than an area of 1/4 (0.250)-square inch.	Conduct visual leakage inspection of all external oil seal areas and gear box housing attachment joints. Evaluate indicator oil pressure readings. Remove gear box if the low oil pressure warning light has been illuminated and if leakage source cannot be corrected or identified.
NOTE		
If continuing chip indications are being found, the limitations in Table 2 apply.		

6. For appropriate action after determining particle material, type-quantity size, time interval, and No. of indications, refer to Table 2.

MAIN TRANSMISSION CHIP DETECTOR PARTICLE INSPECTION - Continued

Table 2. Particle Classification-Continuing Indications.

MATERIAL	TYPE-QUANTITY SIZE	TIME INTERVAL	NO. OF INDICATIONS	ACTION REQUIRED
Metallic/ Magnetic (Steel)	More than 2 flakes or granular particles over 0.005-inch thick- ness or 0.020- inch diameter noted at each indication.	10 hour period	Three	Replace module.
		25 hour period	Five	Replace module.
		50 hour period	Nine	Replace module.
Metallic/ Nonmagnetic (Lead, Cadmium, Silver, Aluminum, Magnesium, Bronze)	More than 10 flakes or granular particles over 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long or larger than 0.050-inch diameter.	10 hour period	Three	Replace module.
NOTE				
To make use of time interval and number of indications, chip light history must be entered in helicopter historical records. Entries on the Historical Record For Aircraft, DA FORM 2408-15 and Aircraft Component Historical Record, DA FORM 2408-16, identifying time interval, number of indications, dates, component hours, size, and type of particles, must be made.				

END OF WORK PACKAGE

UNIT LEVEL**DRIVE SYSTEM**

MAIN TRANSMISSION LUBRICATION SYSTEM INSPECTION

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

WP 0616 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT OIL FILTER

Inspect bypass indicator button on filter bowl. If button is popped, refer to WP 0616 00.

END OF WORK PACKAGE

UNIT LEVEL
DRIVE SYSTEM
INPUT MODULE INSPECTIONS

This WP supersedes WP 0619 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Depth Gage, Dial Indicating, 643J
Dial Indicator, MIL-I-18422
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0640 00
WP 0653 00
WP 0654 00
WP 1803 00

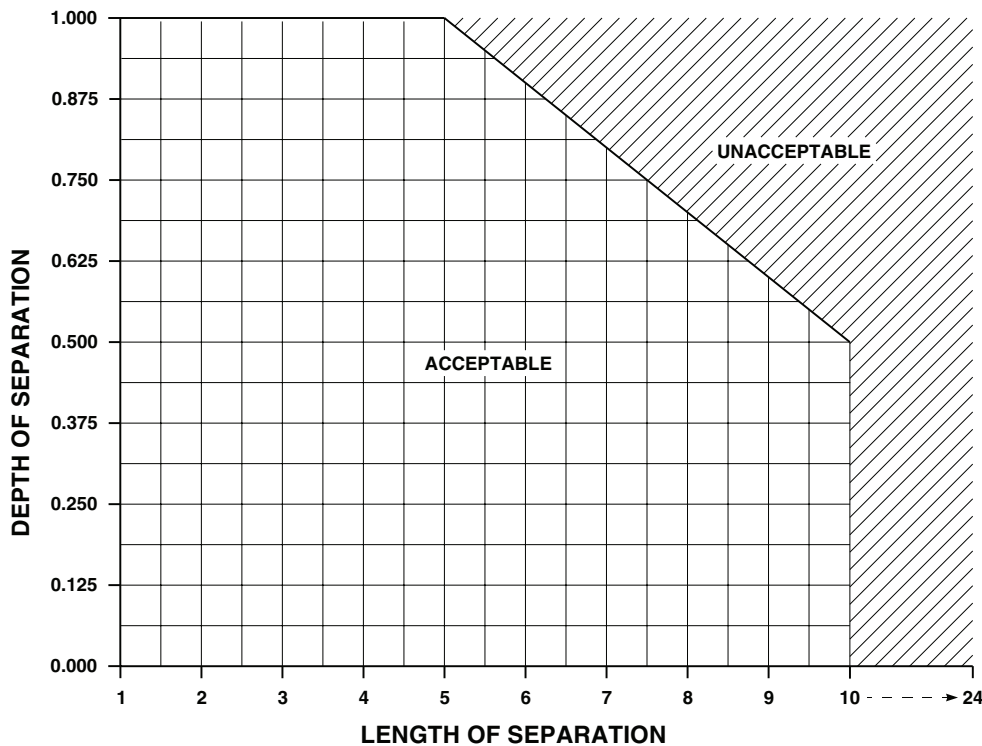
Material/Parts

Cloth, Abrasive, Item 78, WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00

INSPECT ELASTOMERIC GIMBAL

1. Inspect elastomeric gimbal, 70361-08001-101, as follows:
 - a. Check eight Allen-head bolts for tightness using fingers only.
 - b. If 1 to 4 bolts are loose, inspect for bond separation using a 0.005-inch feeler gage (Figure 1).
 - c. If more than 4 bolts are loose, replace gimbal (WP 0653 00).
 - d. Check gimbal for nicks, scratches, or gouges. **DAMAGE UP TO 0.015-INCH DEEP CAN BE BLENDED OUT. DAMAGE MUST BE LESS THAN 15% OF CIRCUMFERENCE OF GIMBAL.**
 - e. Blend out damage using abrasive cloth, Item 78, WP 1803 00, or abrasive wheel, Item 350, WP 1803 00, as needed. Blend out all reworked areas to a 1/2-inch radius. Touch up reworked areas with corrosion resistant coating, Item 111, WP 1803 00.
2. Inspect elastomeric gimbal, 70361-08001-102, as follows:
 - a. Visually inspect for loose or broken bolts.
 - b. If 1 to 4 bolts are loose, **RETORQUE LOOSE BOLTS TO 57 - 63 INCH-POUNDS.**
 - c. If any bolts are broken, replace gimbal (WP 0653 00).
 - d. If loose bolts are found, inspect for bond separation using 0.005-inch feeler gage (Figure 1).
 - e. Check gimbal for nicks, scratches, or gouges. **DAMAGE UP TO 0.015-INCH DEEP CAN BE BLENDED OUT. DAMAGE MUST BE LESS THAN 15% OF CIRCUMFERENCE OF GIMBAL.**
 - f. Blend out damage using abrasive cloth, Item 78, WP 1803 00, or abrasive wheel, Item 350, WP 1803 00, as needed. Blend out all reworked areas to a 1/2-inch radius. Touch up reworked areas with corrosion resistant coating, Item 111, WP 1803 00.
3. Inspect elastomeric gimbal, 70361-08001-103 70361-08015-102 or, as follows:
 - a. Visually inspect for loose bolts. **RETORQUE ANY LOOSE BOLT TO 81 - 89 INCH-POUNDS.**

INSPECT ELASTOMERIC GIMBAL - Continued



NOTES

1. ONE INCH DEPTH EQUALS TOTAL ELASTOMERIC BOND SEPARATION.
2. BOND SEPARATION 1 INCH DEEP WITHIN A 5 INCH RADIUS IS ACCEPTABLE.
3. BOND SEPARATIONS OVER 10 INCHES IN LENGTH ARE UNACCEPTABLE.
4. ALL DIMENSIONS IN INCHES.

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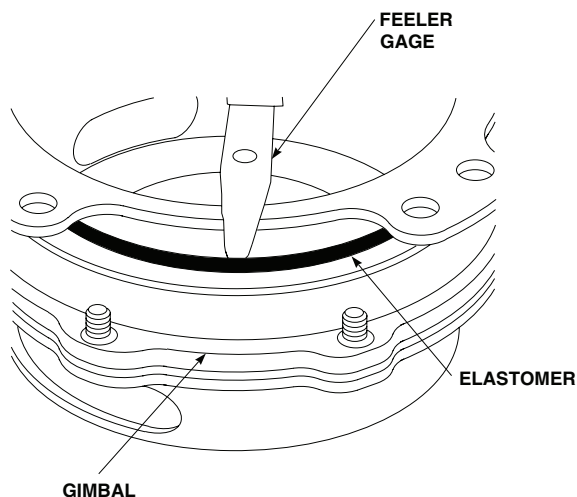
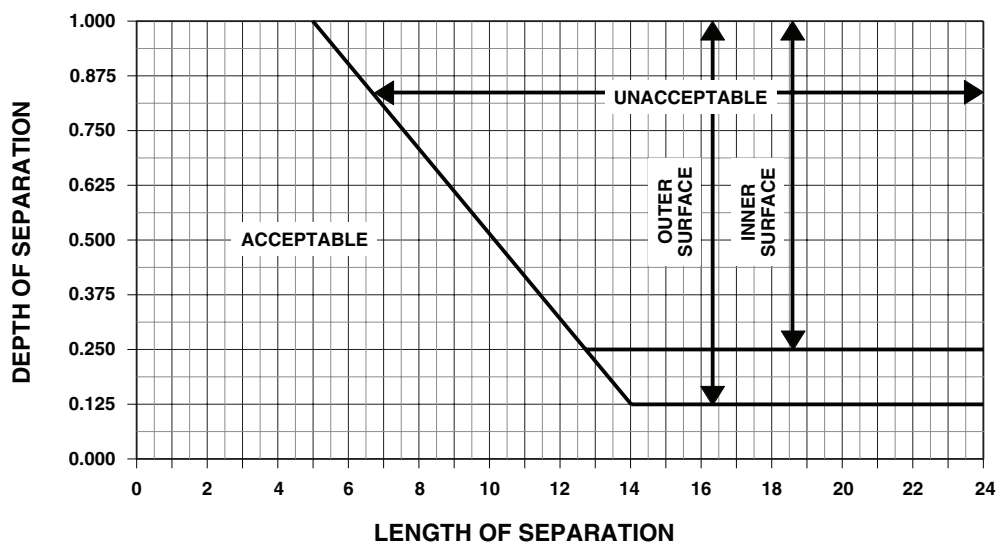
Figure 1. Inspect Elastomeric Gimbal (P/N 70361-08001-101, 70361-08001-102, 70361-08001-103).

- b. If any bolts are broken, replace gimbal (WP 0653 00).
 - c. For elastomeric gimbal 70361-08001-103, if loose bolts are found, inspect for bond separation using a 0.005-inch feeler gage (Figure 1).
 - d. For elastomeric gimbal 70361-08015-102, if loose bolts are found, inspect for bond separation. Using a 0.005-inch feeler gage, check for separation between inner ring and elastomer and outer ring and elastomer (Figure 2).
 - e. Check gimbal for nicks, scratches or gouges. **DAMAGE UP TO 0.015-INCH DEEP CAN BE BLENDED OUT. DAMAGE MUST BE LESS THAN 15% OF CIRCUMFERENCE OF GIMBAL.**
 - f. Blend out damage using abrasive cloth, Item 78, WP 1803 00, or a rubber abrasive wheel, Item 350, WP 1803 00, as needed. Blend out all reworked areas to a 1/2-inch radius. Touch up reworked areas with corrosion resistant coating, Item 111, WP 1803 00.
4. Check Drag Torque on forward support tube attachment nut plates. Thread bolt into nut plate so that 1-2 threads extend past each nut plate. Using a torque wrench, check breakaway torque. Breakaway torque must not be less than 6.5-in-lbs. Replace nut plates as required (WP 0654 00). This inspection shall be required for all elastomeric gimbals, regardless of part number.

INSPECT INPUT MODULE/QUILL SHAFT

1. Check input module housing surface for paint discoloration. Discoloration from original aluminized gray to a flat tone of gray is acceptable. **IF ANY PAINT HAS DISCOLORED TO A BROWN TONE, INDICATING POSSIBLE EXCESSIVE HEAT OR OIL LEAKAGE, REPLACE MODULE.**
2. Visually inspect module housing surface for nicks, gouges, scratches, and corrosion (WP 0640 00).

INSPECT INPUT MODULE/QUILL SHAFT - Continued



NOTES

1. ONE INCH DEPTH EQUALS TOTAL ELASTOMERIC BOND SEPARATION.
2. BOND SEPARATION 1 INCH DEEP WITHIN A 5 INCH RADIUS IS ACCEPTABLE.
3. ALL DIMENSIONS IN INCHES.
4. PLUNGING A METAL OR RIGID FEELER GAGE INTO THE ELASTOMER WILL DAMAGE IT.
5. INSPECT FOR ELASTOMERIC DEBONDING BETWEEN THE OUTER RING AND THE MIDDLE RING.

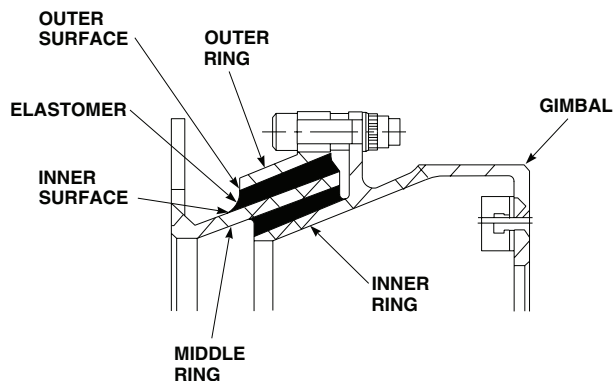
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Figure 2. Inspect Elastomeric Gimbal (P/N 70361-08015-102).

INSPECT INPUT MODULE/QUILL SHAFT - Continued

3. Inspect for axial movement of input pinion. Refer to, INSPECT INPUT PINION, in this work package.
4. Visually inspect quill shaft and splines for damage. **NONE ALLOWED.** Replace quill shaft if damaged.

INSPECT INPUT SEAL LEAKAGE**NOTE**

- Checking amount of input seal leakage can prevent premature seal replacement, before leakage is over recommended limits. Allow at least 2 hours after flight for oil to drain back into sump before checking level.
- Allow at least 2 hours after flight for oil to drain back into sump before checking level.

Measure seal leakage by observing drop in oil level on main transmission dipstick. **IF LEAKAGE IS MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT, REPLACE INPUT SEAL.**

INSPECT INPUT PINION**NOTE**

This procedure is to be performed whenever input flange, input module, elastomeric gimbal, output shaft, or engine is removed or replaced.

1. Press input flange forward toward input module to remove any axial play in pinion shaft.
2. Move input flange to rear and measure axial play using dial indicator. **MAXIMUM ALLOWABLE AXIAL PLAY IS 0.030-INCH.** Replace input module if limit is exceeded.

END OF WORK PACKAGE

UNIT LEVEL**DRIVE SYSTEM**

ACCESSORY MODULE INSPECTION

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

WP 0656 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT ACCESSORY MODULE LEAKAGES

1. Visually inspect accessory module for oil leaks, paying close attention to area adjacent generator mounting pad. **NO OIL LEAKS ALLOWED.** Replace leaking accessory module.
2. Inspect accessory module oil return screen per WP 0656 00 if any of the following conditions exist:
 - a. Increase of main gear box oil temperature above 212°F(100°C).
 - b. Main gear box oil pressure fluctuates below 30 psi.
 - c. Accessory module chip detector light on.
 - d. Accessory and/or input module housing paint discolored from aluminized gray to brown.

END OF WORK PACKAGE

UNIT LEVEL
DRIVE SYSTEM
INTERMEDIATE GEAR BOX INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 0 - 600 ft. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00

References

WP 0660 00
WP 0661 00
WP 1803 00

INSPECT INTERMEDIATE GEAR BOX INPUT SEAL LEAKAGE**NOTE**

- Checking amount of input seal leakage can prevent premature seal replacement, before leakage is over recommended limits.
- Allow at least 45 minutes after flight for oil to drain back into bottom of gear box before checking level.
- When flight longer than 3 hours is planned, total leakage cannot be greater than 5/100-inch per flight hour (no more than 1/2-inch drop in level over a 10 hour flight).

Measure seal leakage by observing drop in oil level on intermediate gear box sight gage. IF LEAKAGE IS MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT, REPLACE INPUT SEAL.

INSPECT INTERMEDIATE GEAR BOX OUTPUT SEAL LEAKAGE**NOTE**

- Checking amount of output seal leakage can prevent premature seal replacement before leakage is over recommended limits.
- Allow at least 45 minutes after flight for oil to drain back into bottom of gear box before checking level.
- When flight longer than 3 hours is planned, total leakage cannot be greater than 5/100-inch per flight hour (no more than 1/2-inch drop in level over a 10 hour flight).

Measure seal leakage by observing drop in oil level on intermediate gear box sight gage. IF LEAKAGE IS MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT, REPLACE OUTPUT SEAL.

INSPECT INTERMEDIATE GEAR BOX SIGHT PLUG

1. Check sight plug for cracks, crazing, and oil staining.

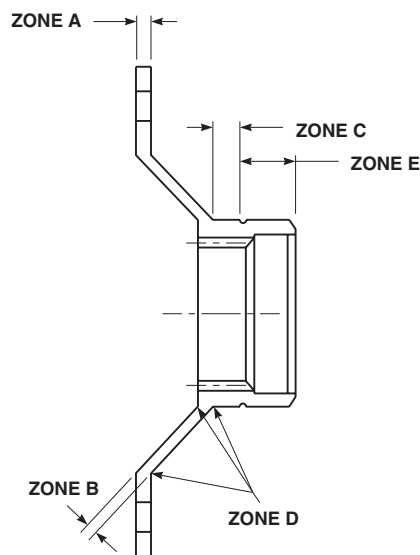
INSPECT INTERMEDIATE GEAR BOX SIGHT PLUG - Continued

2. **REPLACE SIGHT PLUG IF OIL LEVEL CANNOT BE SEEN THROUGH LENS DUE TO STAINING OR CRAZING.**
3. **REPLACE SIGHT PLUG IF IT LEAKS.**

INSPECT INTERMEDIATE GEAR BOX FLANGE**NOTE**

Rework of flange may be done on helicopter.

1. Remove damaged input flange (WP 0660 00) and/or output flange if necessary (WP 0661 00).
2. Inspect chrome plated seal journal area of flange for nicks or scratches. **NONE ALLOWED.** Replace flange if damage is found. Wear pattern caused by seal is not cause for replacement unless wear causes seal to leak.
3. Fine scratches may be blended smooth using abrasive cloth, Item 82, WP 1803 00.
4. Check present dimension of damaged area (Figure 1).
5. Check depth of damaged area to determine whether flange minimum dimension after repair can be maintained.
6. If damage can be repaired, blend smooth using abrasive cloth, Item 83, WP 1803 00. Remove only what is necessary, so flange will stay within limits.
7. Blend reworked area to remove all traces of abrasion marks and mark surface smoothness using abrasive cloth, Item 82, WP 1803 00.

**NOTES**

1. NO SCRATCHES OR NICKS ALLOWED IN CHROME PLATED LINER.
2. NO REMOVAL OF MATERIAL ALLOWED IN WASHER CONTACT AREA.
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

	ZONE	DIMENSION BEFORE REPAIR	MINIMUM DIMENSION AFTER REPAIR
LUG	A	0.175 (± 0.005)	0.160 (SEE NOTE 2)
WEB	B	0.105 (± 0.005)	0.090
OD	C	2.270 (± 0.010)	2.250
RADIUS	D	0.125 (± 0.010)	0.105
LINER	E	2.2505 (± 0.0015)	2.248 (SEE NOTE 1)

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Figure 1. Inspect Intermediate Gear Box Flange.

INSPECT INTERMEDIATE GEAR BOX FLANGE - Continued

8. Repair of nicks, scratches, or scoring shall be done by polishing as follows:
 - a. Polishing shall be done only in damaged area. Do not polish circumference unless damage extends around it.
 - b. Polish with blend radius of 1/2-inch or greater.

END OF WORK PACKAGE

UNIT LEVEL**DRIVE SYSTEM**

INTERMEDIATE GEAR BOX CHIP DETECTOR PARTICLE INSPECTION

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts

Cloth, Cheesecloth, Item 85, WP 1803 00

References

WP 0623 00

WP 0659 00

WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECTION

1. When the CHIP INT XMNS light on the caution/advisory panel goes on, it indicates the chip detector sensor in the intermediate gear box has activated.
2. Metal particles found on the chip detector may indicate an internal part of the gear box has malfunctioned. If particles are found it does not necessarily mean the gear box must be replaced. The quantity, size, type, and source of the particles, along with the history of the gear box, must be considered. Also gear box time, previous malfunctions, and operating conditions should be considered.
3. Particles found may be metallic/magnetic (steel), metallic/nonmagnetic (lead, cadmium, aluminum, magnesium, bronze, or silver), or nonmetallic (wood, paint, paper, lint, rubber, or phenolic).

NOTE

- Visual inspection may be enough to determine nature of contamination material. If identification of material is required, contact your supporting Army Oil Analysis Lab for assistance.
 - Intermediate and tail rotor gear boxes do not have oil filters and discoloration is possible.
4. If particles are easily identified as fragments from gears, nuts, bearings, etc., replace intermediate gear box (WP 0659 00). If replacement of intermediate gear box is questionable, contact the AMCOM LAR. If 10 or more particles are found, or if any one particle is larger than 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long, replace gear box. When a intermediate gear box is replaced due to contamination, foreign material from screens and filters must be carefully packaged and returned with the gear box. Also, intermediate gear box oil must be drained and strained with cheesecloth cloth, Item 85, WP 1803 00. Any particles found in oil should be combined with particles from screens and filters for evaluation and returned with module if necessary. Also, package cheesecloth cloth, Item 85, WP 1803 00, put oil in sealed container, and return both with module.
 5. For appropriate action after determining particle material, size, and quantity, refer to Table 1.

INSPECTION - Continued

Table 1. Particle Classification.

MATERIAL	TYPE-QUANTITY SIZE	ACTION REQUIRED
Metallic/Magnetic (Steel)	Fuzz, fine hair-like particles.	Acceptable. Return intermediate gear box to service. No serviceability check is required.
	No more than 10 flakes, none over 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long.	Do a serviceability check (WP 0620 00). Record findings in helicopter logbook. Refer to Table 2.
	Splinters or granular particles.	Do a serviceability check (WP 0620 00).
	Any quantity of chips beyond the above dimensions.	Replace gear box.
Metallic/Nonmagnetic (Lead, Cadmium, Silver, Aluminum, Magnesium, Bronze)	Thin or particle flakes not over 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long. Quantity not to be over 10 flakes.	Acceptable. Return intermediate gear box to service. No serviceability check is required.
	More than 10 flakes, but less than 25.	Do a serviceability check (WP 0620 00). If possible, contact analytical laboratory to determine material. Record findings in helicopter logbook. Refer to Table 2.
	More than 25 flakes.	Replace gear box.
Nonmetallic (Baked Resin/Paint, Phenolic, Dust, Lint, Wood)	Flakes or particles not over 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long. Quantity not to be over 10 flakes.	Acceptable. Return intermediate gear box to service. No serviceability check is required.
	Less than a total area of 1/2 (0.50)-square inch.	Do a serviceability check (WP 0620 00). Refer to Table 2.
	More than a total area of 1/2 (0.50)-square inch.	If contamination is sufficient to partially block oil strainer, replace gear box.
Nonmetallic (Rubber)	More than an area of 1/4 (0.250)-square inch.	Conduct visual leakage inspection of all external oil seal areas and intermediate gear box housing attachment joints. Evaluate indicator oil pressure readings. Remove intermediate gear box if the low oil pressure warning light has been illuminated and if leakage source cannot be corrected or identified.
NOTE		
If continuing chip indications are being found, the limitations in Table 2 apply.		

INSPECTION - Continued

6. For appropriate action after determining particle material, type-quantity size, time interval, and No. of indications, refer to Table 2.

Table 2. Particle Classification-Continuing Indications.

MATERIAL	TYPE-QUANTITY SIZE	TIME INTERVAL	NO. OF INDICATIONS	ACTION REQUIRED
Metallic/Magnetic (Steel)	More than 2 flakes or granular particles over 0.005-inch thick- ness or 0.020- inch diameter noted at each indication.	10 hour period	Three	Replace gear box.
		25 hour period	Five	Replace gear box.
		50 hour period	Nine	Replace gear box.
Metallic/ Nonmagnetic (Lead, Cadmium, Silver, Aluminum, Magnesium, Bronze)	More than 10 flakes or granular particles over 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long or larger than 0.050-inch diameter.	10 hour period	Three	Replace gear box.
NOTE				
To make use of time interval and number of indications, chip light history must be entered in helicopter historical records. Entries on the Historical Record For Aircraft, DA FORM 2408-15 and Aircraft Component Historical Record, DA FORM 2408-16, identifying time interval, number of indications, dates, component hours, size, and type of particles, must be made.				

END OF WORK PACKAGE

UNIT LEVEL
DRIVE SYSTEM
INTERMEDIATE GEAR BOX SERVICEABILITY INSPECTION

This WP supersedes WP 0623 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 2-Quart
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

TB 43-0211
WP 0622 00
WP 0659 00
WP 1608 00
WP 1803 00

Material/Parts

Cloth, Cheesecloth, Item 85, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

NOTE

Do this procedure if high metal count is detected during AOAP sampling, or if particles were found on chip detector.

1. Drain intermediate gear box oil through clean cheesecloth cloth, Item 85, WP 1803 00, into an empty 2 quart container.
2. Inspect and clean chip detector (WP 0622 00).
3. Install chip detector.
4. Service intermediate gear box (WP 1608 00).
5. Ground run helicopter for 30 minutes and hover for 30 minutes. Reinspect chip detector.
6. If amount of metal particles has increased, replace gear box (WP 0659 00).
7. If amount of particles has decreased, do this:
 - a. Enclose chips in a 2 quart container.
 - b. Obtain a sample of oil from gear box (TB 43-0211).
 - c. Submit chips and oil together to AOAP laboratory.
8. Service intermediate gear box, if required (WP 1608 00).

END OF WORK PACKAGE

UNIT LEVEL
DRIVE SYSTEM
TAIL ROTOR DRIVE SHAFT INSPECTION
This WP supersedes WP 0624 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0668 00

WP 0669 00

WP 0678 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0666 00

WP 0667 00

INSPECT**NOTE**

The inspections in step 1. and step 2. are performed during the Periodic Inspection (PMI) when the flexible coupling is disconnected.

1. Visually inspect drive shaft for cracks. **NONE ALLOWED.** Replace drive shaft if cracks are found.

NOTE

Light surface corrosion on inside collars of fasteners used to connect drive shaft tube to flange and stub shaft is not cause for rejection. Replace drive shaft if corrosion between drive shaft tube and flange or between drive shaft tube and stub shaft is present or if fasteners are loose.

2. Visually inspect drive shaft for corrosion between drive shaft tube and stub shaft, and drive shaft tube and flange. Corrosion between drive shaft tube and stub shaft, and drive shaft tube and flange is identified by flaking or lifting of paint, crack, or presence of white powder on shaft. **NO CORROSION ALLOWED.** Replace drive shaft if corrosion is found (WP 0666 00, WP 0667 00, WP 0668 00, or WP 0669 00).
3. Inspect section II, III, and IV tail rotor drive shaft flanges boltholes for elongation. **NONE ALLOWED.** Replace tail rotor drive shaft if boltholes are elongated.
4. Inspect oil cooler and section II tail rotor drive shaft flanges boltholes for elongation. **NONE ALLOWED.** If forward flange boltholes are elongated, replace oil cooler (WP 0678 00) or section II tail rotor drive shaft. If aft flange boltholes are elongated, replace flange (WP 0667 00).
5. Check balance weights for security. Check integrity of rivets used to secure balance weight to shaft **NO LOOSE-NESS OF BALANCE WEIGHTS OR RIVETS ALLOWED.** Replace tail rotor drive shaft if balance weights or rivets are loose.
6. Replace tail rotor drive shaft if rivets are not intact.

END OF WORK PACKAGE

UNIT LEVEL

DRIVE SYSTEM

TAIL ROTOR DRIVE SHAFT VISCOUS DAMPER INSPECTION

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

WP 0672 00
WP 0673 00
WP 1611 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

NOTE

Viscous damper bearing tube assembly allows movement of bearing liner relative to support outer housing. However, movement of viscous damper tube within support outer housing is not allowed. Radial and axial movement of bearing inside liner is allowed to accommodate shaft deflections during flight.

1. Visually inspect bearing assembly, tube assembly, tube assembly housing, and bearing mount area for indications of excessive temperature.
2. Visually inspect viscous dampers for evidence of disbonding between liner and tube assembly. If disbonding is evident do the following:

CAUTION

Damage to tube assembly will result if feeler gage is not used carefully. Use care when inserting feeler gage.

- a. Using 0.002-inch feeler gage, carefully insert and measure depth of separation on both sides of drive shaft assembly (Figure 1).
 - b. **MAXIMUM ALLOWABLE SEPARATION IS 0.200-INCH DEEP FOR 40% OF THE CIRCUMFERENCE ON ONE OR BOTH SIDES.** Replace liner and tube assembly if separation exceeds limit (WP 0673 00).
3. Inspect both sides of bearing to be sure that bearing seals are intact and properly positioned.

NOTE

- A small amount of grease purged from viscous damper bearing assembly, with no accompanying discoloration or damage to bearing, is not cause for replacement.
 - An assembly compound is used between viscous damper bearing liner and bearing. This compound darkens with age and appears like burned grease. Compound will appear as a thin bead at outside diameter of bearing. Compound may or may not be visible. Do not replace bearing if compound is visible.
4. Inspect viscous damper bearing assembly for grease seepage. If a large amount of grease is showing, do this:

INSPECT - Continued

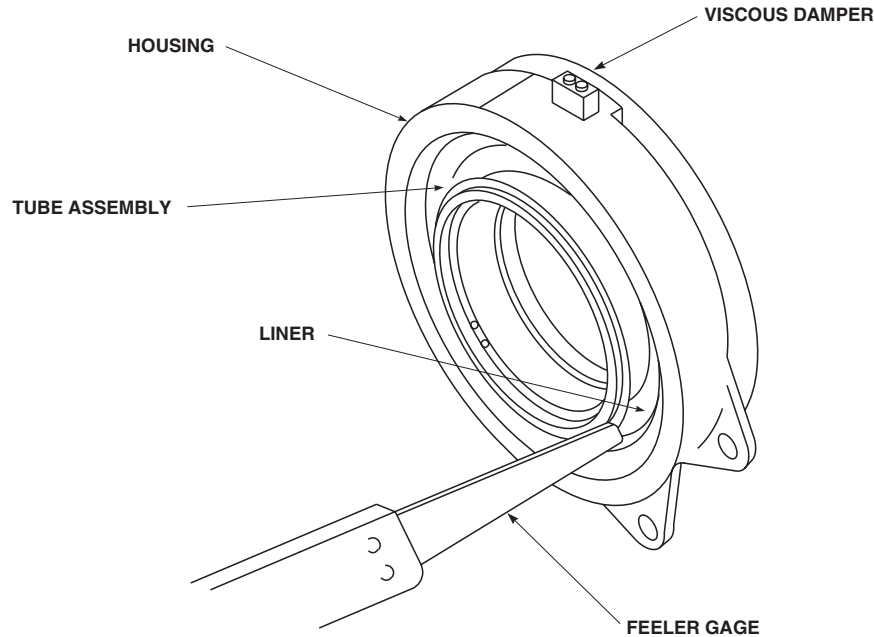
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Figure 1. Inspect Tail Rotor Drive Shaft Viscous Damper.

- a. Disconnect applicable tail rotor drive shaft.
 - b. Rotate bearing by hand. **BEARING MUST ROTATE SMOOTHLY.** Replace bearing that ratchets or feels rough when rotated (WP 0672 00).
 - c. Replace bearings that show a large amount of grease seepage accompanied by heat discoloration.
 - d. Reconnect tail rotor drive shaft.
5. Check viscous dampers for leakage. **NONE ALLOWED.** Replace viscous damper bearing tube assembly if leakage is suspected. If viscous damper tube is transparent, check for leakage as follows:
- a. Holding flashlight behind viscous dampers, check for air bubbles in viscous damper tubes.
 - b. **DAMPERS WITH NO AIR BUBBLE SHOULD BE CHECKED FOR FLUID.**
 - c. **DAMPERS WITH AIR BUBBLES LESS THAN 1 INCH LONG MAY BE CONTINUED IN USE WITHOUT RESERVICING.**
 - d. **DAMPERS WITH AIR BUBBLE 1 INCH TO 2 INCHES LONG SHALL BE SERVICED (WP 1611 00).**

INSPECT - Continued

- e. **DAMPERS WITH AIR BUBBLE OVER 2 INCHES LONG SHALL BE REPLACED (WP 0673 00).**

END OF WORK PACKAGE

UNIT LEVEL
DRIVE SYSTEM
OIL COOLER INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Borescope Kit, 6650-T700-003
Caliper Set, GGG-C-105
Depth Gage, Dial Indicating, 643J
Powertrain Repairer's Toolkit, SC 5180-99-B13
Puller Kit, Mechanical, GGG-P-781
Spring Scale
Torque Wrench, 0 - 600 ft. lbs.

Material/Parts

Corrosion Preventive Compound, Item 106,
WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00

Strap, Tiedown, Electrical, Item 310, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

PAM 738-751
TM 1-1500-344-23
WP 0671 00
WP 0672 00
WP 0675 00
WP 0678 00
WP 1803 00

TORQUE CHECK AND INSPECT OIL COOLER EXIT FLANGE RETAINING NUT**WARNING**

Injury to personnel and damage to equipment will result if tail rotor is not secured when drive shaft is disconnected. Secure tail rotor to prevent shafts and rotor from turning.

NOTE

This procedure applies only to helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed.

1. Remove oil cooler (WP 0678 00).
2. Remove radiator from oil cooler duct (WP 0675 00).
3. Match mark oil cooler drive shaft input flange with axial fan drive shaft exit flange prior to disconnecting shafts.

NOTE

It is not required to completely remove oil cooler drive shaft from oil cooler duct since this would require removing the viscous damper bearing support. Disconnecting oil cooler drive shaft will provide access to fan exit flange retaining nut.

4. Disconnect oil cooler drive shaft from axial fan exit flange and slide shaft aft through oil cooler duct (WP 0678 00).

TORQUE CHECK AND INSPECT OIL COOLER EXIT FLANGE RETAINING NUT - Continued**WARNING**

Injury to personnel and damage to equipment will result if fan exit flange retaining nut is not torqued properly. Loosening or removal of retaining nut may cause improper torque to be present and is not allowed. Only inspect and repair surfaces that do not require removal or loosening of fan exit flange retaining nut.

5. Visually inspect area around axial fan exit flange retaining nut, inside diameter of axial fan drive shaft, and inside diameter of oil cooler drive shaft for red oxide residue. **NONE ALLOWED.** Red oxide residue may be an indication of fretting of internal splines. Replace oil cooler if evidence of fretting or wear (red oxide) is indicated. Submit CAT I EIR per PAM 738-751.
6. Inspect exit flange, retaining nut, and mating surface of oil cooler drive shaft for corrosion. **REPLACE FAN IF PITS ARE 0.030-INCH DEEP, OR IF PITTING COVERS MORE THAN 10% OF OIL COOLER DRIVE SHAFT MATING SURFACE.**
7. Inspect axial fan shaft ID (WP 0678 00).
8. Remove corrosion from exposed surfaces of fan exit flange and/or retaining nut (TM 1-1500-344-23).
9. Remove cotter pin from axial fan drive shaft exit flange retaining nut.

WARNING**CSI**

- Verification of minimum torque of axial fan exit flange retaining nut is a critical characteristic.
- Injury to personnel and damage to equipment will result if fan exit flange retaining nut is not torqued properly. Loosening or removal of retaining nut may cause improper torque to be present and is not allowed. Check torque of fan exit flange retaining nut in tightening direction only.

NOTE

Torque reactor will be attached to axial fan input flange to react against torquing of axial fan exit flange retaining nut.

10. Check and record torque of axial fan exit flange retaining nut using torque wrench and torque adapter. Torque is to be checked in a tightening direction, while holding axial fan shaft with torque reactor, to the following specified torque or until nut moves indicating looseness, whichever occurs first. **TORQUE SHALL BE 214 - 236 FOOT-POUNDS.**
11. **IF RETAINING NUT TORQUE IS 214 - 236 FOOT-POUNDS, AND NO RED OXIDE RESIDUE IS EVIDENT,** install new cotter pin. If it is necessary to align cotter pin hole, turn nut in tightening direction. **DO NOT EXCEED MAXIMUM TORQUE. IF MAXIMUM TORQUE IS EXCEEDED AND DOES NOT ALIGN, REMOVE NUT AND REPLACE IT WITH A NEW NUT AND REPEAT PROCEDURE.** Torque check is complete.

WARNING**CSI**

Torque stabilization is a critical characteristic.

12. **IF RETAINING NUT TORQUE IS 180 - 213 FOOT-POUNDS, AND NO RED OXIDE RESIDUE IS EVIDENT, TORQUE THE EXIT FLANGE NUT AS FOLLOWS: THE TORQUE RANGE IS 214 - 236**

TORQUE CHECK AND INSPECT OIL COOLER EXIT FLANGE RETAINING NUT - Continued

FOOT-POUNDS. TORQUE RETAINING NUT TO 214 FOOT-POUNDS. Install cotter pin. To align cotter pin hole, turn nut in tightening direction. **DO NOT EXCEED MAXIMUM TORQUE. IF MAXIMUM TORQUE IS EXCEEDED AND DOES NOT ALIGN, REMOVE NUT AND REPLACE IT WITH A NEW NUT AND REPEAT PROCEDURE.** Torque check is complete.

13. **IF RETAINING NUT TORQUE IS LESS THAN 180 FOOT-POUNDS, REPLACE OIL COOLER (WP 0678 00).**
14. Apply epoxy primer coating kit, Item 120, WP 1803 00, to exposed surfaces of retaining nut.
15. Apply corrosion preventive compound, Item 106, WP 1803 00, to retaining nut and area around nut. Do not apply compound to flange mating surfaces.

CAUTION

Damage to oil cooler will result if match marks on flanges are not aligned correctly prior to installation. Make sure match marks on oil cooler drive shaft input flange line up with corresponding match marks on axial fan drive shaft exit flange.

16. Connect oil cooler drive shaft input flange to axial fan drive shaft exit flange making sure to line up match marks on flanges (WP 0678 00).
17. Install radiator on oil cooler duct (WP 0675 00).
18. Install oil cooler (WP 0678 00).

INSPECT OIL COOLER VISCOUS DAMPER BEARING**NOTE**

If oil cooler viscous damper bearing fails inspection and is to be replaced, a drive shaft runout check must be performed per WP 0678 00 prior to replacing bearing.

1. Turn main rotor blades as required to gain access to oil cooler.
2. Engage gust lock.
3. Open oil cooler access doors.

NOTE

- Movement of the viscous damper bearing tube assembly is not cause for replacement of bearing.
 - Grease purged from viscous damper bearing assembly, with no accompanying discoloration or indication of damage to bearing, is not cause for replacement.
4. On helicopters with sleeve adapter, 70361-03001-041, remove access panel at rear of oil cooler duct. Using inspection mirror and flashlight, inspect area inside of duct for debris. Use flexible borescope to inspect area at forward end of viscous damper bearing and between viscous damper bearing assembly and duct.
 5. On helicopters with sleeve adapter, 70361-03011-105, using slots in sleeve, inspect inside of duct for debris located at the forward end of viscous damper bearing, and between viscous damper bearing assembly and duct.
 6. If debris is present at bottom of duct, but no debris is present at forward end of viscous damper bearing or between bearing assembly and duct, clean out debris. Inspection is complete.
 7. If debris is present at forward end of viscous damper bearing or between bearing and duct, clean out debris as follows:

INSPECT OIL COOLER VISCOUS DAMPER BEARING - Continued

- a. On helicopters with sleeve adapter, 70361-03011-105, clean out debris through sleeve slots using scribe or equivalent.
 - b. On helicopters with sleeve adapter, 70361-03011-041, remove oil cooler assembly and viscous damper bearing assembly (WP 0678 00). Clean out debris from inside duct.
 - c. Check bearing for evidence of overheating (temperature discoloration) and debris damage. Pay particular attention to bearing seal. **NONE ALLOWED.** If any evidence of overheating or debris damage is found, replace bearing (WP 0672 00).
 - d. If removed, reinstall viscous damper bearing support assembly and oil cooler assembly (WP 0678 00).
8. Position a flashlight inside of duct, pointing to rear at the viscous damper bearing.
 9. Inspect bearing assembly for proper fluid level, indicated by bubbles in the fluid.
 10. Inspect viscous damper bearing for fluid leakage.
 11. If viscous damper bearings have any of the above deficiencies, replace viscous damper support assembly (WP 0671 00).
 12. On helicopters with sleeve adapter, 70361-03011-041, reinstall access panel at rear of oil cooler duct.
 13. Close oil cooler access doors.

INSPECT OIL COOLER SPLINE WEAR**NOTE**

This procedure applies only to helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed.

1. Open section II and section III tail rotor drive shaft covers.
2. Open oil cooler access doors.

WARNING

Injury to personnel and damage to equipment will result if spline wear inspection is not performed accurately. Excessive spline wear will cause loss of tail rotor drive. Make sure spline wear inspection is performed accurately.

3. Engage gust lock.

WARNING

Injury to personnel and damage to equipment will result if spline wear inspection is performed improperly by rotating tail rotor in normal direction of rotation. Tail rotor must be rotated in direction opposite normal rotation to simulate in-flight loads. Failure to rotate tail rotor in direction opposite normal rotation will result in false spline wear readings. If tail rotor is rotated in the wrong direction, the entire inspection procedure must be repeated.

4. Apply a 20 pound load for 5 seconds to tail rotor blade leading edge using electrical tiedown strap, Item 310, WP 1803 00, and spring scale. Make sure tail rotor is rotated in direction opposite normal rotation.
5. Release gust lock with load applied.

INSPECT OIL COOLER SPLINE WEAR - Continued

6. Slowly rotate tail rotor blade in opposite direction of rotation until index mark on tail rotor drive shaft lines up with spline wear index bracket slot on No. 3 viscous damper. Use a 6 inch ruler or similar straight edge to line up bracket slot with black index mark line (Figure 1, Detail B).
7. If index mark on tail rotor drive shaft rotates past spline wear index bracket slot on No. 3 viscous damper, continue to rotate in opposite direction of normal rotation until index mark is aligned with slot.
8. If tail rotor is inadvertently rotated in direction of normal rotation, repeat step 3. through step 7.
9. Check position of black painted index blade on oil cooler. **BLACK PAINTED INDEX BLADE MUST BE WITHIN BLACK WEAR INDICATOR RANGE** (Figure 1, Detail C).
10. If any portion of black painted index blade is not within black wear indicator range, do this:
 - a. Repeat step 3. through step 9.

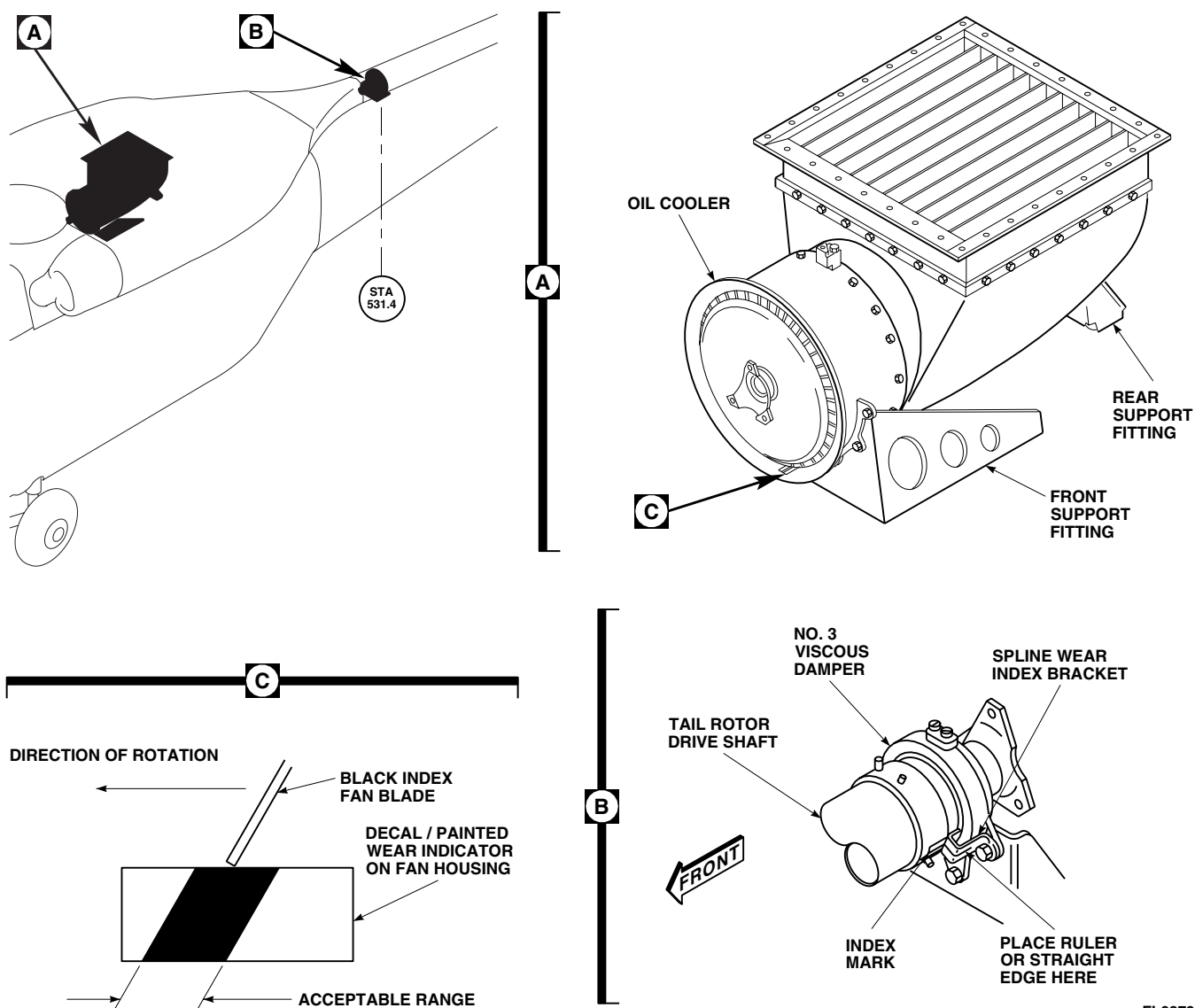
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Figure 1. Inspect Oil Cooler Spline Wear.

INSPECT OIL COOLER SPLINE WEAR - Continued

- b. If any portion of black painted index blade is still not within black wear indicator range, replace oil cooler (WP 0678 00). Submit CAT I EIR per PAM 738-751.
- 11. Check area for cleanness and foreign material.
- 12. Close oil cooler access doors.
- 13. Close tail rotor drive shaft covers.

END OF WORK PACKAGE

UNIT LEVEL
DRIVE SYSTEM
TAIL GEAR BOX INSPECTIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

References

WP 0621 00
WP 0682 00
WP 0683 00
WP 0684 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

INSPECT TAIL GEAR BOX INPUT SEAL LEAKAGE**NOTE**

- Checking amount of input seal leakage can prevent premature seal replacement before leakage is over recommended limits.
- When flight longer than 3 hours is planned (range extension kit installed) total leakage cannot be greater than 5/100-inch per flight hour (no more than 1/2-inch drop in level over a 10 hour flight).

Measure seal leakage by observing drop in oil level on tail gear box sight gage. **IF LEAKAGE IS MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT, REPLACE INPUT SEAL (WP 0682 00).**

INSPECT TAIL GEAR BOX INNER RETENTION PLATE SEAL LEAKAGE**NOTE**

- Checking amount of seal leakage can prevent premature seal replacement before leakage is over recommended allowable limits. Leakage is gaged by measuring the drop in oil level in a given period, as seen on sight plug. Total leakage must not lower the oil level over 1/8-inch in any one flight.
- When flight longer than 3 hours is planned, total leakage cannot be greater than 5/100-inch per flight hour (no more than 1/2-inch drop in level over a 10 hour flight).

Measure seal leakage by observing drop in oil level on tail gear box sight gage. **IF LEAKAGE IS MORE THAN 1/8-INCH AFTER ANY ONE FLIGHT, REPLACE SEAL (WP 0683 00).**

INSPECT TAIL GEAR BOX SIGHT PLUG

1. Check sight plug for cracks, crazing, and oil staining.
2. **REPLACE SIGHT PLUG IF OIL LEVEL CANNOT BE SEEN THROUGH LENS DUE TO STAINING OR CRAZING (WP 0684 00).**
3. **REPLACE SIGHT PLUG IF IT LEAKS (WP 0684 00).**

INSPECT TAIL GEAR BOX SIGHT PLUG - Continued**INSPECT TAIL GEAR BOX FLANGE**

Inspection of tail gear box flange is identical to intermediate gear box flange. Refer to WP 0621 00.

INSPECT TAIL GEAR BOX INPUT PLUG SEAL

If any leakage is present at outer end of shaft, replace seal (WP 0682 00).

END OF WORK PACKAGE

UNIT LEVEL

DRIVE SYSTEM

TAIL GEAR BOX CHIP DETECTOR PARTICLE INSPECTION

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts

Cloth, Cheesecloth, Item 85, WP 1803 00

References

WP 0629 00

WP 0681 00

WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

TAIL GEAR BOX CHIP DETECTOR PARTICLE INSPECTION

1. When the CHIP TAIL XMNS light on the caution/advisory panel goes on, it indicates the chip detector/temperature sensor in the tail gear box has activated.
2. Metal particles found on the chip detector/temperature sensor may indicate an internal part of the gear box has malfunctioned. If particles are found it does not necessarily mean the gear box must be replaced. The quantity, size, type, and source of the particles, along with the history of the gear box, must be considered. Also, gear box time, previous malfunctions, and operating conditions should be considered.
3. Particles found may be metallic/magnetic (steel), metallic/nonmagnetic (lead, cadmium, aluminum, magnesium, bronze, or silver), or nonmetallic (wood, paint, paper, lint, rubber, or phenolic).

NOTE

- Visual inspection may be enough to determine nature of contamination material. If identification of material is required, contact your supporting Army Oil Analysis Lab for assistance.
 - Intermediate and tail rotor gear boxes do not have oil filters and discoloration is possible.
4. If particles are easily identified as fragments from gears, nuts, bearings, etc., replace tail rotor gear box (WP 0681 00). If replacement of tail rotor gear box is questionable, contact the AMCOM LAR. If 10 or more particles are found, or if any one particle is larger than 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long, replace tail rotor gear box. When a tail rotor gear box is replaced due to contamination, foreign material from screens and filters must be carefully packaged and returned with the tail rotor gear box. Also, tail rotor gear box oil must be drained and strained with cheesecloth cloth, Item 85, WP 1803 00. Any particles found in oil should be combined with particles from screens and filters for evaluation and returned with module if necessary. Also, package cheesecloth cloth, Item 85, WP 1803 00, put oil in sealed container, and return both with module.
 5. For appropriate action after determining particle material, size, and quantity, refer to Table 1.

TAIL GEAR BOX CHIP DETECTOR PARTICLE INSPECTION - Continued

Table 1. Particle Classification.

MATERIAL	TYPE-QUANTITY SIZE	ACTION REQUIRED
Metallic/Magnetic (Steel)	Fuzz, fine hair-like particles.	Acceptable. Return tail rotor gear box to service. No serviceability check is required.
	No more than 10 flakes, none over 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long.	Do a serviceability check (WP 0629 00). Record findings in helicopter logbook. Refer to Table 2.
	Splinters or granular particles.	Do a serviceability check (WP 0629 00).
	Any quantity of chips beyond the above dimensions.	Replace tail rotor gear box.
Metallic/Nonmagnetic (Lead, Cadmium, Silver, Aluminum, Magnesium, Bronze)	Thin or particle flakes not over 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long. Quantity not to be over 10 flakes.	Acceptable. Return tail rotor gear box to service. No serviceability check is required.
	More than 10 flakes, but less than 25.	Do a serviceability check (WP 0629 00). If possible, contact analytical laboratory to determine material. Record findings in helicopter logbook. Refer to Table 2.
	More than 25 flakes.	Replace tail rotor gear box.
Nonmetallic (Baked Resin/Paint, Phenolic, Dust, Lint, Wood)	Flakes or particles not over 0.050-inch diameter or 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long. Quantity not to be over 10 flakes.	Acceptable. Return tail rotor gear box to service. No serviceability check is required.
	Less than a total area of 1/2 (0.50)-square inch.	Do a serviceability check (WP 0629 00). Refer to Table 2.
	More than a total area of 1/2 (0.50)-square inch.	If contamination is sufficient to partially block oil strainer, replace tail rotor gear box.

TAIL GEAR BOX CHIP DETECTOR PARTICLE INSPECTION - Continued

Table 1. Particle Classification. - Continued

MATERIAL	TYPE-QUANTITY SIZE	ACTION REQUIRED
Nonmetallic (Rubber)	More than an area of 1/4 (0.250)-square inch.	Conduct visual leakage inspection of all external oil seal areas and tail rotor gear box housing attachment joints. Evaluate indicator oil pressure readings. Remove tail rotor gear box if the low oil pressure warning light has been illuminated and if leakage source cannot be corrected or identified.
NOTE If continuing chip indications are being found, the limitations in Table 2 apply.		

6. For appropriate action after determining particle material, type-quantity size, time interval, and No. of indications, refer to Table 2.

Table 2. Particle Classification-Continuing Indications.

MATERIAL	TYPE-QUANTITY SIZE	TIME INTERVAL	NO. OF INDICATIONS	ACTION REQUIRED
Metallic/Magnetic (Steel)	More than 2 flakes or granular particles over 0.005-inch thickness or 0.020-inch diameter noted at each indication.	10 hour period	Three	Replace tail rotor gear box.
		25 hour period	Five	Replace tail rotor gear box.
		50 hour period	Nine	Replace tail rotor gear box.
Metallic/Nonmagnetic (Lead, Cadmium, Silver, Aluminum, Magnesium, Bronze)	More than 10 flakes or granular particles over 1/32 (0.032)-inch thick x 1/8 (0.125)-inch long or larger than 0.050-inch diameter.	10 hour period	Three	Replace tail rotor gear box.
NOTE				

TAIL GEAR BOX CHIP DETECTOR PARTICLE INSPECTION - Continued**Table 2. Particle Classification-Continuing Indications. - Continued**

MATERIAL	TYPE-QUANTITY SIZE	TIME INTERVAL	NO. OF INDICATIONS	ACTION REQUIRED
To make use of time interval and number of indications, chip light history must be entered in helicopter historical records. Entries on the Aircraft Component Historical Record, DA FORM 2408-5 and Aircraft Component Historical Record, DA FORM 2408-16, identifying time interval, number of indications, dates, component hours, size, and type of particles, must be made.				

END OF WORK PACKAGE

UNIT LEVEL
DRIVE SYSTEM
TAIL GEAR BOX SERVICEABILITY INSPECTION
This WP supersedes WP 0629 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 2-Quart

Material/Parts

Cloth, Cheesecloth, Item 85, WP 1803 00
Cloth, Cleaning, Item 86, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TB 43-0211
WP 0681 00
WP 0688 00
WP 1148 00
WP 1609 00
WP 1803 00

INSPECT**NOTE**

Do this procedure if high metal count is detected during AOAP sampling, or if particles were found on chip detector.

1. Drain tail gear box oil through clean cheesecloth cloth, Item 85, WP 1803 00, into an empty 2 quart container.
2. Remove, inspect and clean chip detector/temperature sensor (WP 0688 00).
3. Install chip detector/temperature sensor (WP 0688 00).

CAUTION

Damage to equipment will result if foreign objects enter tail gear box. Foreign objects may enter tail gear box when tail rotor servo is removed. Make sure that no foreign objects enter tail gear box while performing step 4 through step 6.

NOTE

Step 4 through step 6 are only required if excessive particle counts are reported in AOAP samples.

4. Remove tail rotor servo (WP 1148 00).
5. Check inside of tail gear box, especially inside diameter of input bevel pinion, for buildup of sludge or debris accumulation. Remove any sludge/debris accumulation using a dry cleaning cloth, Item 86, WP 1803 00.
6. Flush tail rotor tail gear box (WP 1609 00).
7. Install tail rotor servo (WP 1148 00).
8. Service tail gear box (WP 1609 00).
9. Ground run helicopter for 30 minutes and hover for 30 minutes. Reinspect chip detector/temperature sensor.
10. If amount of metal particles has increased, replace tail gear box (WP 0681 00).

INSPECT - Continued

11. If amount of particles has decreased, do this:
 - a. Enclose chips in a 2 quart container.
 - b. Obtain a sample of oil from tail gear box (TB 43-0211).
 - c. Submit chips and oil together to AOAP laboratory.
12. Service tail gear box, if required (WP 1609 00).
13. Return helicopter to service.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

DRIVE SYSTEM

MAIN TRANSMISSION

This WP supersedes WP 0630 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Adapter, 70700-20420-041,
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Brass Drift
Container, 1-Quart
Container, 10-Gallon
Gloves
Goggles/Face Shield
Fluorescent Penetrant Inspection Kit, XMA101
Mounting Bolt Adapter Tool, 1420-BH-008, Locally-
Made, Figure 46, WP 1805 00
Nonmetallic Scraper
Powertrain Repairers's Toolkit, SC 5180-99-B13
Rawhide Mallet
Strap, Guide
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.
Torque Wrench, 0 - 600 ft lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Antiseize Compound, Item 49, WP 1803 00
Brush, Paint, Item 61, WP 1803 00
Corrosion Preventive Compound, Item 101,
WP 1803 00
Corrosion Preventive Compound, Item 106,
WP 1803 00
Corrosion Preventive Compound, Item 107,
WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00
Epoxy Coating Kit, Item 118, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Lubricant, Solid Film, Item 180, WP 1803 00
Molybdenum Disulfide, Item 197, WP 1803 00
Polyurethane Coating, Item 231, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Sealing Compound, Item 258, WP 1803 00
Tags
Tape, Lacing and Tying, Item 321, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 326,
WP 1803 00

Towel, Machinery Wiping, Item 344, WP 1803 00
Wire, Nonelectrical, Item 354, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1520-265-23
TM 55-1500-345-23
WP 0085 00
WP 0259 00
WP 0260 00
WP 0261 00
WP 0261 01
WP 0301 00
WP 0302 00
WP 0308 00
WP 0381 00
WP 0486 00
WP 0506 00
WP 0547 00
WP 0567 00
WP 0576 00
WP 0613 00
WP 0615 00
WP 0637 00
WP 0653 00
WP 0666 00
WP 0677 00
WP 0839 00
WP 1019 00
WP 1021 00
WP 1022 00
WP 1023 00
WP 1024 00
WP 1026 00
WP 1029 00
WP 1037 00
WP 1125 00

INITIAL SETUP: - Continued

WP 1126 00	WP 1590 00
WP 1127 00	WP 1607 00
WP 1128 00	WP 1689 00
WP 1129 00	WP 1703 00
WP 1199 00	WP 1803 00
WP 1220 00	WP 1805 00
WP 1272 00	

REMOVE**NOTE**

- **UH-60A** Main transmission with all equipment (input modules, accessory modules, generators and hydraulic pumps) installed, weighs about 1180 pounds. Main transmission with all equipment, main rotor head, swashplate and flight controls installed, weighs about 2312 pounds. Main transmission weighs approximately 860 pounds. ■
- **UH-60L** Transmission with all equipment (input modules, accessory modules, generators, and hydraulic pumps) installed, weighs approximately 1395 pounds. Main transmission with all equipment, main rotor head, and swashplate and flight controls installed, weighs about 2448 pounds. Main transmission alone weighs approximately 860 pounds. ■
- If an overhead hoist capable of lifting main transmission with main rotor head and swashplate installed is available, it is easier to remove transmission with head installed and then remove head while on ground. If maintenance crane will be used, main rotor head and swashplate must be removed first.

1. Make sure helicopter wheels are chocked.
2. Turn off all electrical and hydraulic power.
3. Open engine cowlings.
4. Open main rotor pylon sliding cover.

CAUTION

Damage to maintenance crane and helicopter will occur if maintenance crane is used to remove transmission with main rotor head and swashplate installed. Use sling and overhead hoist to remove main transmission with main rotor head and swashplate installed.

5. If maintenance crane will be used, do this:
 - a. Assemble and erect crane (WP 1590 00).
 - b. Remove main rotor blades (WP 0576 00).
 - c. Remove main rotor head (WP 0547 00).
 - d. Remove swashplate (WP 0567 00).
 - e. **UH-60L** Remove fire extinguisher line from right firewall at STA 370.0 (WP 1199 00). ■
6. If overhead hoist and sling will be used, do this:
 - a. If deice kit is installed, remove main rotor blade deice slinging assembly (WP 1220 00).

REMOVE - Continued

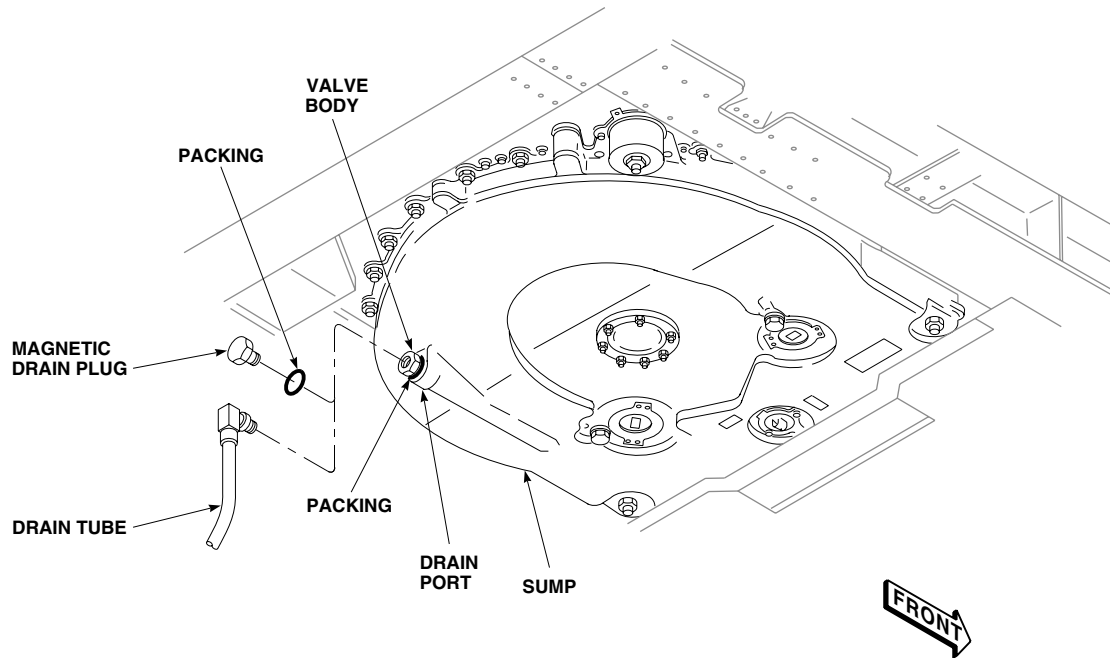
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Figure 1. Drain Gear Box.

- b. Remove main rotor blades (WP 0576 00).
- c. **UH-60L** Remove fire extinguisher line from right firewall (WP 1199 00).
- 7. Remove transmission drip pan (WP 0259 00, WP 0260 00, WP 0261 00, or WP 0261 01).
- 8. Remove magnetic drain plug from valve body (Figure 1).

NOTE

If main transmission is to be removed and reinstalled to facilitate other maintenance, draining oil from transmission is not required. Ensure all lines and ports are capped to prevent oil contamination or debris entering the transmission.

- 9. Install drain tube by turning clockwise until locked in position. Drain oil into 10 gallon container.
- 10. Remove drain tube from drain port, by turning counterclockwise and pulling out. Install new packing on magnetic drain plug.
- 11. If valve body is damaged, remove by turning counterclockwise and pulling out.
- 12. Reinstall new valve body with new packing.
- 13. Install magnetic drain plug. Using nonelectrical wire, Item 357, WP 1803 00, lockwire plug to valve body.
- 14. Remove oil cooler compartment access doors (WP 0308 00).
- 15. Remove transmission covers (WP 0302 00).
- 16. Remove both left and right air inlet fairings (WP 0301 00).

REMOVE - Continued

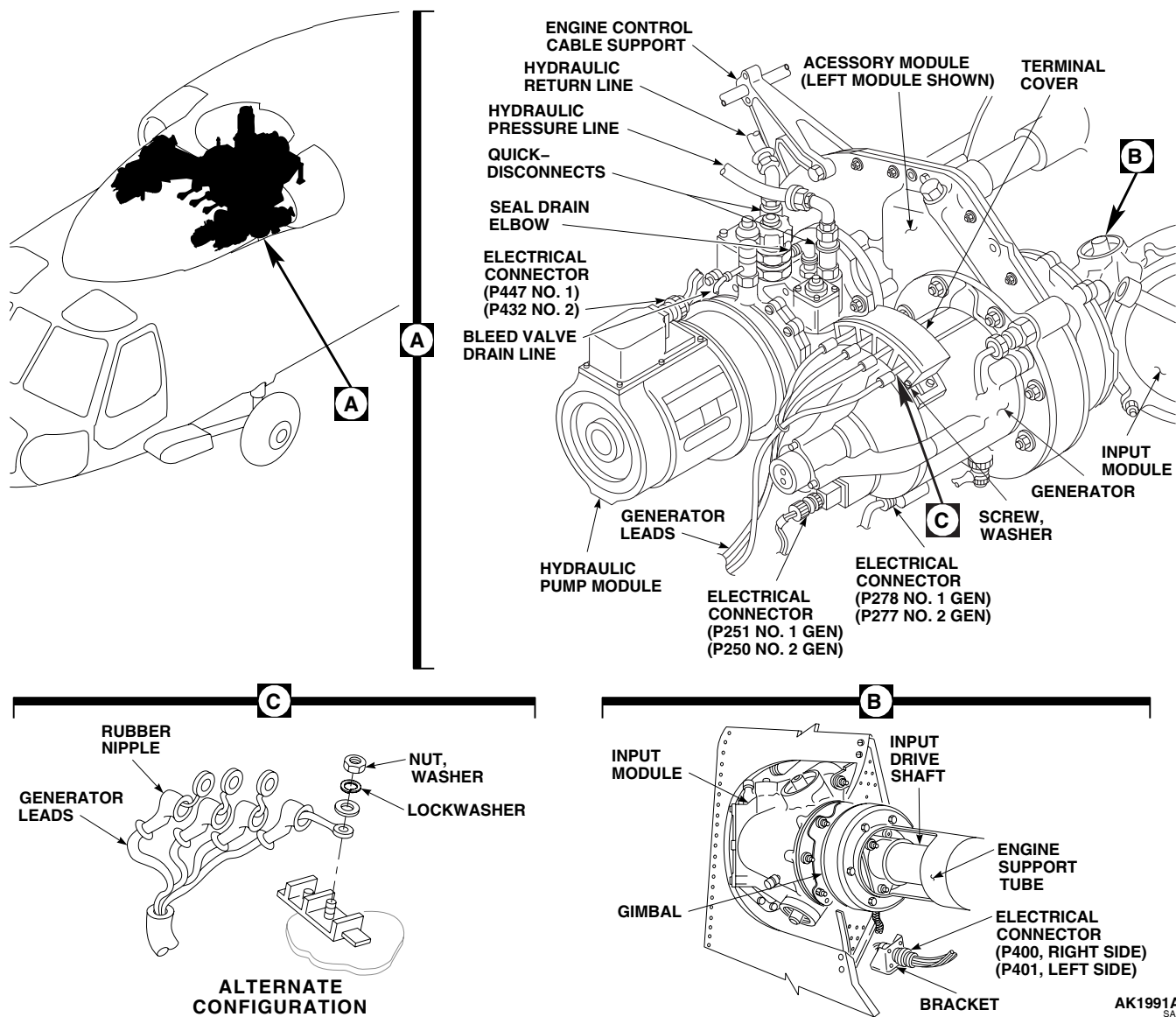


Figure 2. Disconnect Lines and Wiring.

17. Disconnect engine output shaft from input module flange (WP 0486 00).
18. Disconnect forward engine support tube from gimbal on input module (WP 0486 00).
19. Disconnect hydraulic pressure line and hydraulic return line at quick-disconnects (Figure 2, Detail A).
20. Disconnect electrical connector, P447, No. 1 or P432, No. 2 from pump modules.
21. Disconnect seal drain from seal drain elbow.
22. Disconnect bleed valve drain line from bleed valve.
23. Unclamp generator leads from generator.
24. Remove screws and slide terminal cover back from generator terminals, or pull rubber nipples back from terminals, as required, to expose terminal connections (Figure 2, Detail C).

REMOVE - Continued

25. Remove nuts and washers holding generator leads to terminals. Remove and tag leads.
26. Disconnect electrical connectors, P251 and P278 from No. 1 or P250 and P277 from No. 2 generator (Figure 2, Detail A).
27. If terminal cover was installed, pull wires through cover and remove cover.
28. Disconnect main transmission harness electrical connectors (P400 right side/P401 left side) from brackets under forward engine supports (Figure 2, Detail B).
29. Disconnect engine control cable supports, and clip from accessory modules. Keep bolts, washers, and nuts with supports (Figure 2, Detail A).
30. Disconnect collective bias tube rod ends from mixer (WP 0506 00). Reposition cables away from main transmission components.
31. Remove bleed-air tube (WP 1272 00).
32. Remove section I tail rotor drive shaft (WP 0666 00).
33. Remove gust lock lever at rear end of main gear box (WP 0637 00) (Figure 3, Detail A).
34. Place a 1 quart container under oil line to catch fluid trapped in lines.
35. Disconnect oil cooler lines from main transmission. Cap or plug all lines and fittings (WP 0677 00).
36. Unclamp oil drain line. Disconnect oil drain line from scupper (Figure 3, Detail B).

CAUTION

- Damage to maintenance crane and helicopter will occur if maintenance crane is used to remove main transmission with main rotor head and swashplate installed. Use sling and overhead hoist to remove main transmission with main rotor head and swashplate installed.
 - Damage to main rotor head components will result if sling is not properly installed. Make sure sling is not put around damper, damper reservoir, or other parts of rotor head, other than main rotor head hub arms.
37. Remove main transmission with main rotor head installed as follows:
 - a. Put sling straps around hub arms in four places (Figure 4).
 - b. Attach sling to hoist with enough capacity to lift main transmission and main rotor head.
 - c. Adjust sling straps before lifting, for even distribution of weight.

CAUTION

UH-60L Damage to sling will result if sling contacts studs on tail takeoff housing. Studs will cut into sling. Make sure sling does not contact studs. ■

38. Remove main transmission with main rotor head and swashplate removed as follows:
 - a. Attach sling straps to input modules and around tail takeoff housing (Figure 4).
 - b. Attach sling to maintenance crane or a hoist with enough capacity to lift main transmission.
 - c. Adjust sling straps before lifting, for even distribution of weight.

REMOVE - Continued

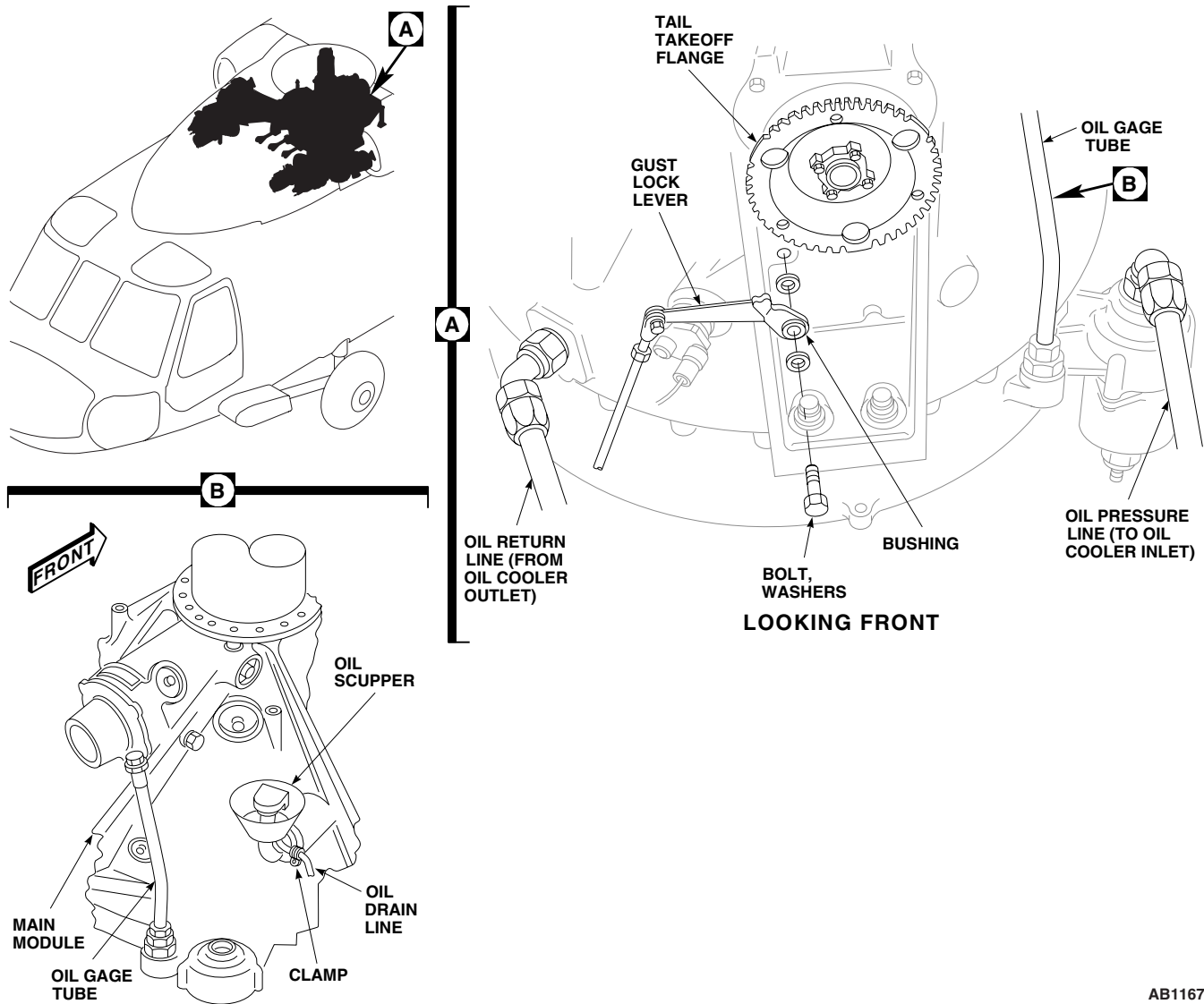
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Figure 3. Disconnect Oil Lines from Transmission.

WARNING**CSI**

The upper wall perimeter of the main transmission module housing is a critical area of the transmission. These surfaces must not be damaged during and after removal of the transmission. Use extreme caution when removing transmission and provide protective covering to these surfaces during removal or installation.

39. Remove bolts connecting control rods to primary servos. Press in on hole in bolthead to release retaining balls so bolts can be removed.

REMOVE - Continued

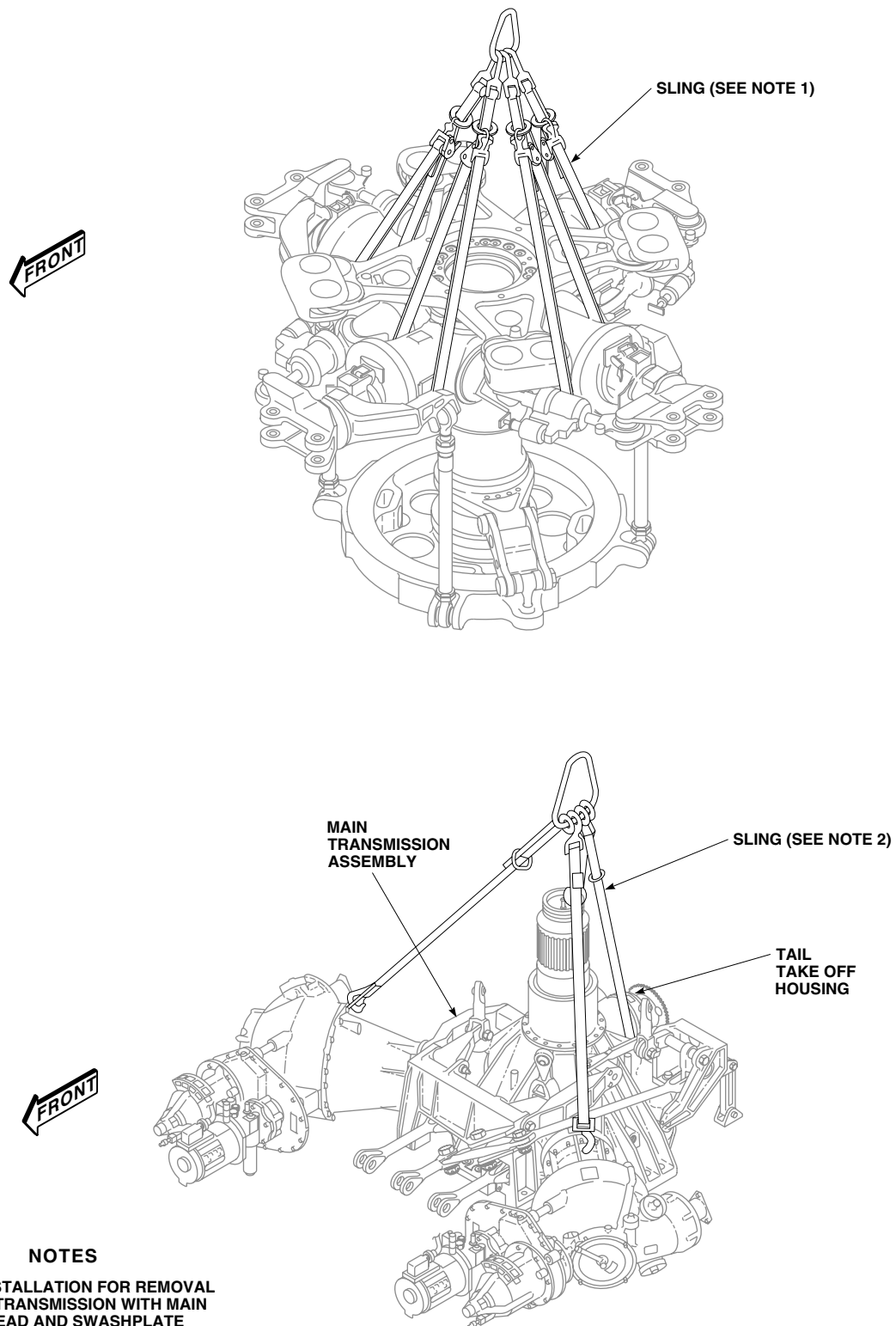


Figure 4. Attach Sling.

REMOVE - Continued**WARNING****CSI**

Upper and lower ends of tie rod at rear bellcrank support are critical characteristic surfaces which must not be damaged during and after removal of main transmission. Use extreme caution when removing or disconnecting and provide protective covering after removal.

40. Remove bolt, washers, and nut from lower end of aft tie rod (Figure 5).
41. Swing aft tie rod out. Remove bolts and washers securing support fitting to deck. Remove shims.
42. Temporarily install bolt, washers, and nut on lower end of aft tie rod through support fitting.
43. Remove aft bellcrank support mounting bolts and spacers from support legs on transmission deck.
44. Remove bolts and washers securing forward bellcrank support to airframe.
45. Remove main transmission mounting bolts. Locally make mounting bolt adapter tool (WP 1805 00), if necessary, to remove two front bolts.

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

CAUTION

Damage to helicopter will result if all connections to main transmission are not removed prior to hoisting. Make sure all attaching lines and electrical connections are removed and area is clear before hoisting.

NOTE

- Attach guide straps to main transmission.
- To help guide main transmission clear of helicopter, place assistant on each side of transmission.

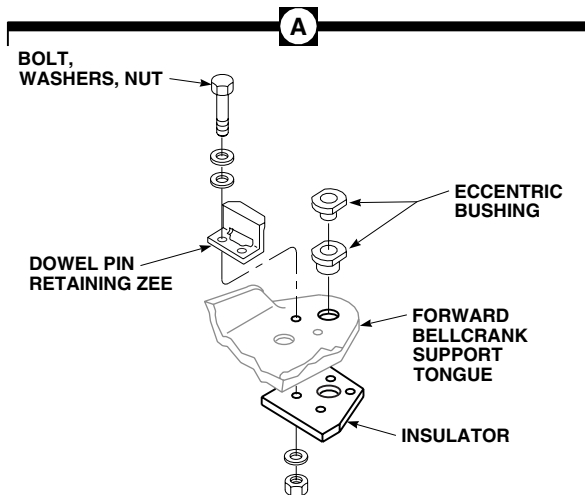
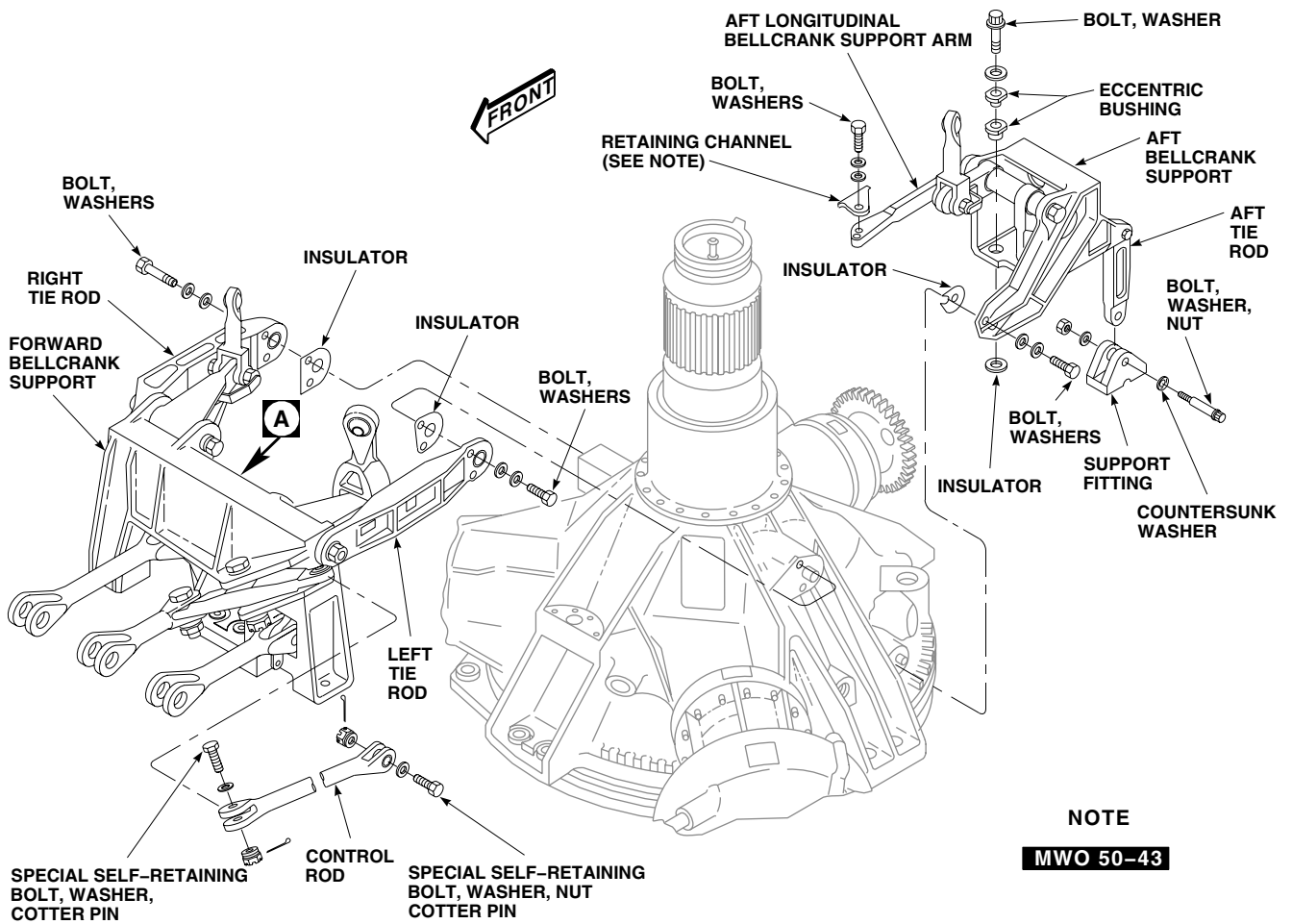
46. Hoist main transmission clear of helicopter.
47. Lower main transmission onto adapter (Figure 6).
48. Install bolts, washers, and nuts holding transmission to adapter.

NOTE

If transmission is to be reinstalled, removal of components is not required.

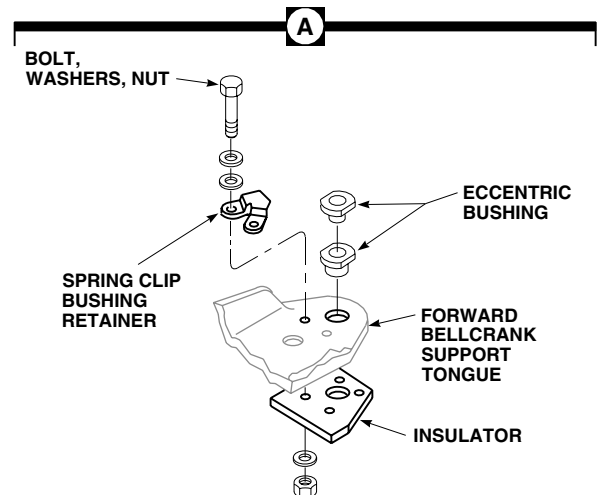
49. If main transmission is to be replaced, remove main rotor head (WP 0547 00).
50. If main transmission is to be replaced, remove swashplate (WP 0567 00).

REMOVE - Continued



EFFECTIVITY

MWO 50-43



EFFECTIVITY

W/O MWO 50-43

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Figure 5. Remove Bellcrank Supports.

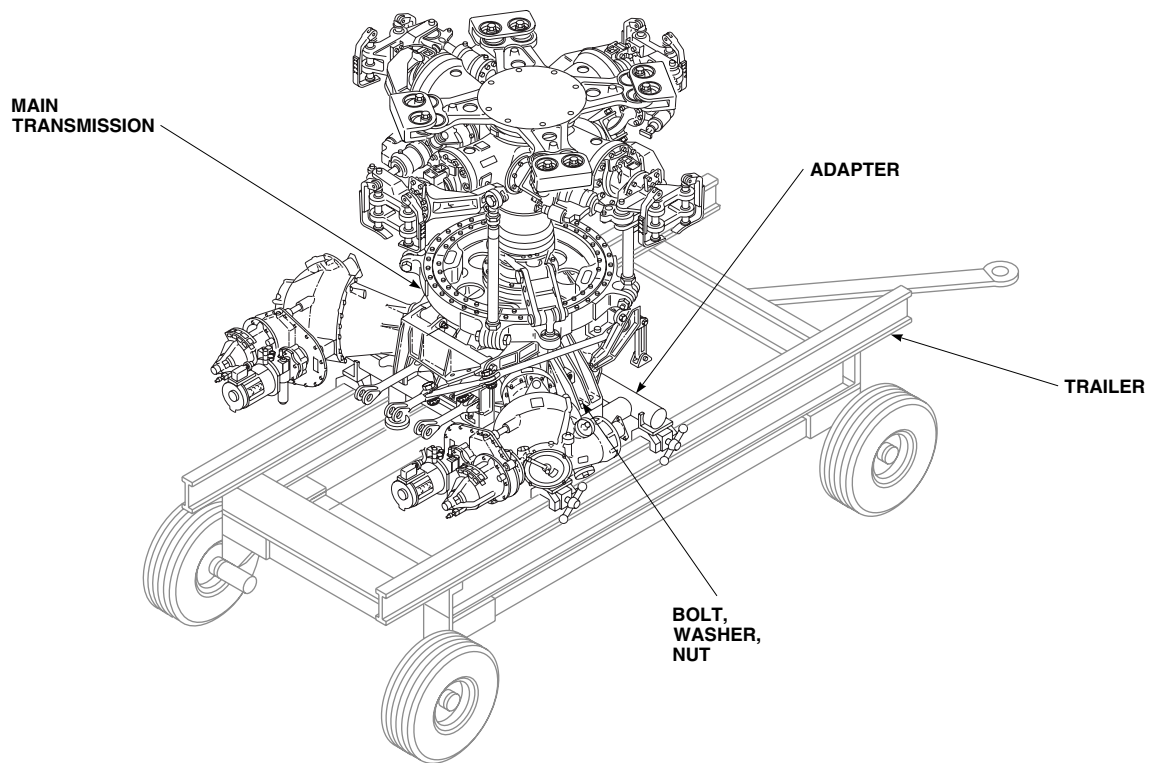
REMOVE - ContinuedAK1994
SA

Figure 6. Place on Adapter.

51. If main transmission is to be replaced, disconnect electrical connectors, remove clamp assemblies and lacing tape securing wiring to transmission. Remove wiring harness. To aid installation, note harness routing and location of clamps (Figure 8, Sheet 1, Figure 8, Sheet 2, Figure 8, Sheet 3, and Figure 8, Sheet 4).

MAIN TRANSMISSION BUILDUP**WARNING****CSI**

All self retaining bolts, bellcranks, tie rods, connecting links, control rods, bellcrank supports and support fitting contain critical surfaces which must not be damaged during and after removal, or disconnection. Use extreme caution when removing, and provide protective covering after removal.

NOTE

If new transmission is to be installed, remove bellcrank supports.

1. Remove bolts, washers, and nuts securing forward, aft, and lateral control rods to primary servos and main transmission (Figure 5). Remove control rods.
2. Disconnect and remove control rod between walking beam and aft bellcrank. To remove bolts, apply pressure using 1/8-inch Allen wrench to release pin located in center of bolthead and pull out bolt (Figure 5).
3. Remove bolts and washers securing left and right tie rods to main transmission. Remove tie rods.

MAIN TRANSMISSION BUILDUP - Continued

4. Remove bolts and washers securing forward bellcrank support tongue to main transmission. Remove eccentric bushings and bushing retainer from support.
5. Loosen nuts holding left and right tie rods enough so that tie rods can be slipped off dowel pins on main transmission.
6. Remove forward bellcrank support with control rods, bellcranks, and walking beam still attached. Remove insulators and eccentric bushings.
7. Remove bolt and washers securing aft longitudinal bellcrank support arm to main transmission.
8. **MWO 50-43** Remove retaining channel from support arm. ■
9. Remove bolt, washers, and nut securing aft tie rod to support fitting.
10. Remove bolts, and washers securing aft bellcrank support to main transmission.
11. Remove aft bellcrank support arm from aft bellcrank assembly. Remove aft bellcrank assembly with insulator and eccentric bushings (Figure 5, Detail A).

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
 - Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.
12. Clean primer from bolts, insulators, and areas on transmission that mate with tie rods, forward bellcrank support, and support rod with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
 13. Clean sealing compound from transmission with nonmetallic scraper.
 14. Visually inspect left and right tie rods, aft bellcrank, and support fitting lug surface areas (WP 1022 00, WP 1024 00, and WP 1026 00).
 15. Visually inspect entire surface of long control rod (WP 1023 00).
 16. Visually inspect forward bellcrank support lug areas, top flange, radiuses and support lug surfaces (WP 1021 00).
 17. Visually inspect forward bellcrank support bushings for damaged or worn cadmium plating. If cadmium plating is damaged or worn apply solid film lubricant, Item 180, WP 1803 00, for corrosion protection until part is replaced.
 18. Inspect support mount dowel pins (WP 0613 00).
 19. Visually inspect aft bellcrank support lug areas, top flange and radius (WP 1024 00).
 20. Visually inspect aft bellcrank support bushings for damaged or worn cadmium plating. If cadmium plating is damaged or worn apply solid film lubricant, Item 180, WP 1803 00, for corrosion protection until part is replaced.
 21. Visually inspect special self-retaining bolts (WP 1019 00).
 22. Visually inspect aft longitudinal bellcrank support arm lug and surface areas (WP 1029 00).
 23. Visually inspect transmission and cabin area at STA 308.0 through STA 402.0.

MAIN TRANSMISSION BUILDUP - Continued

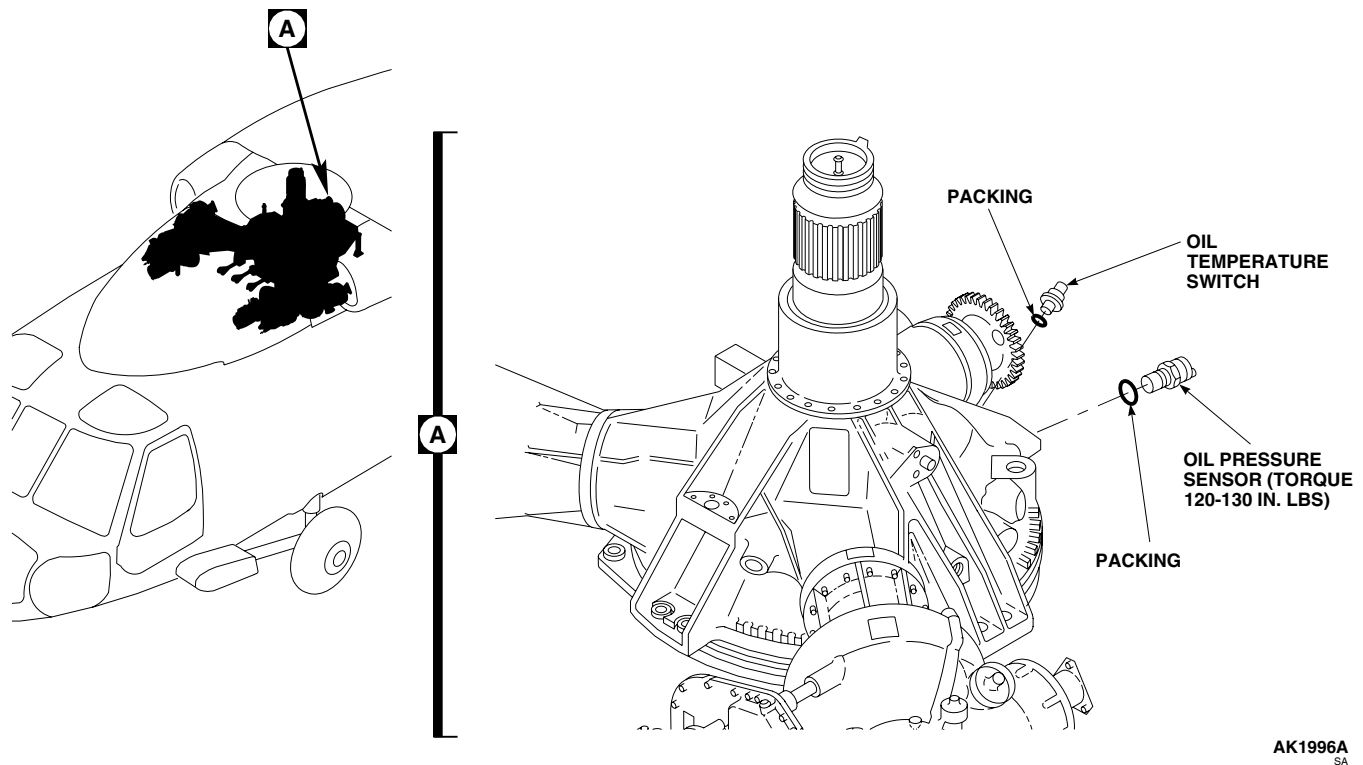


Figure 7. Replace Oil Temperature Switch and Oil Pressure Sensor.

NOTE

If new main module is being installed, the following items must be removed from old module and installed on new module.

24. Remove left and right input modules from main module (WP 0653 00).
25. **QA** Install left and right input modules on replacement main module (WP 0653 00).
26. Remove oil pressure sensor from main module (Figure 7, Detail A).
27. **QA** Replace oil pressure sensor packing.
28. **QA** Install sensor in replacement main module. **TORQUE TO 120 - 130 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire sensor.
29. Seal area around mating flange with sealing compound, Item 273, WP 1803 00.
30. Remove oil temperature switch from main module (Figure 7, Detail A).
31. **QA** Replace oil temperature switch packing.
32. **QA** Install switch in replacement main module. Using nonelectrical wire, Item 357, WP 1803 00, lockwire switch.
33. Seal area around mating flange with sealing compound, Item 273, WP 1803 00.
34. **QA** Install wiring harness on main transmission with clamps, screws, washers, nuts, spacers and lacing and tying tape, Item 321, WP 1803 00 (Figure 8).

MAIN TRANSMISSION BUILDUP - Continued

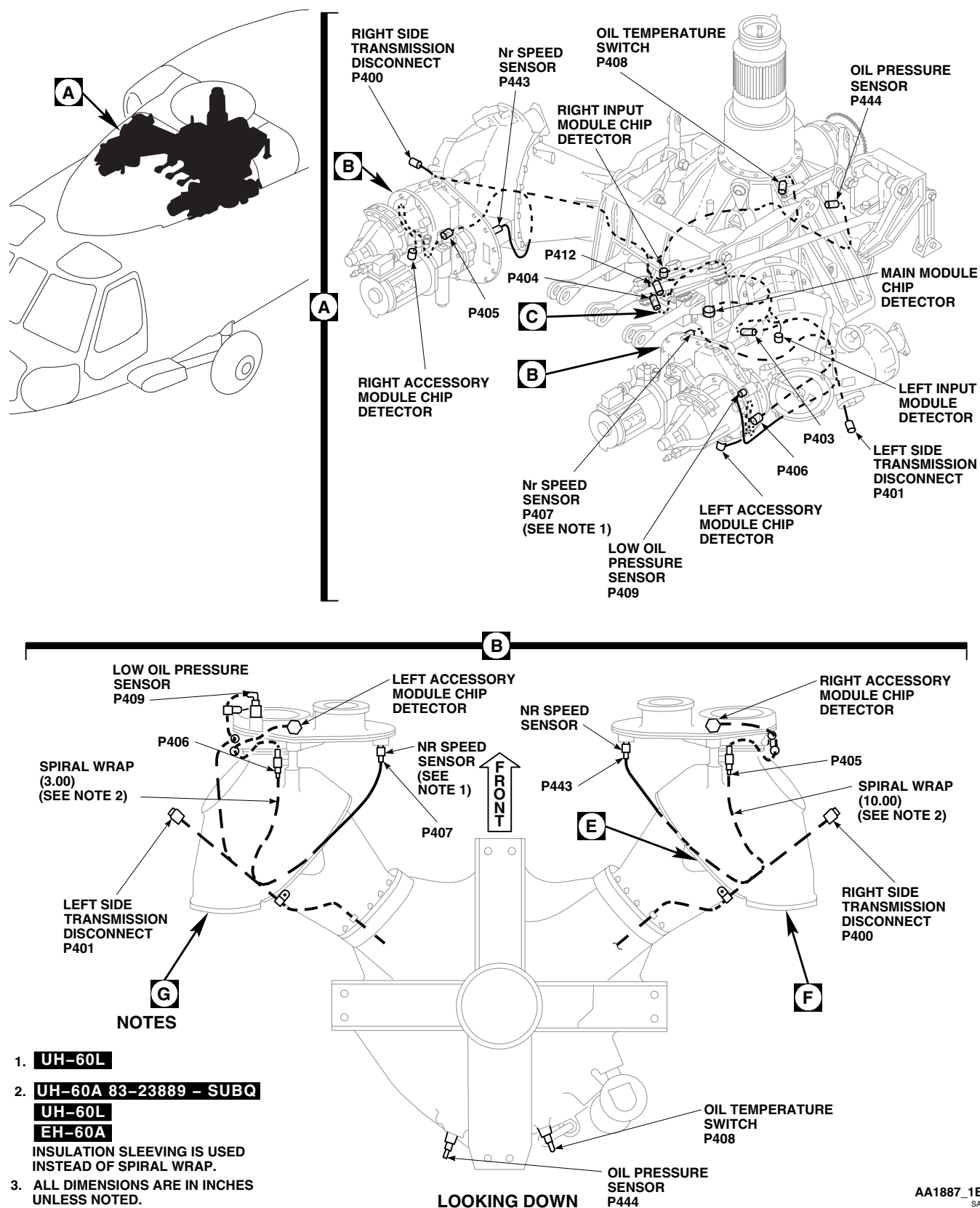


Figure 8. Install Wiring Harness. (Sheet 1 of 4)

MAIN TRANSMISSION BUILDUP - Continued

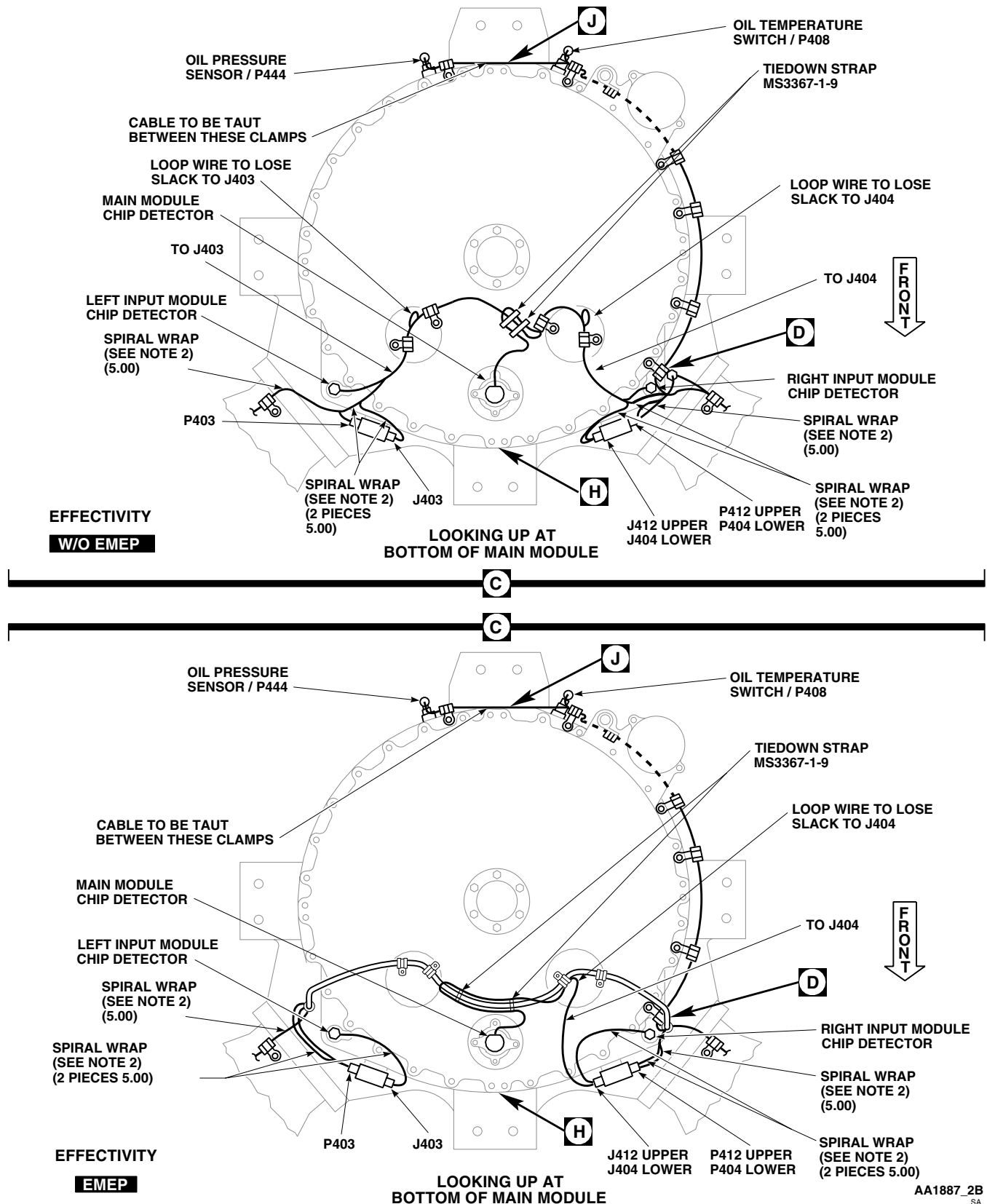
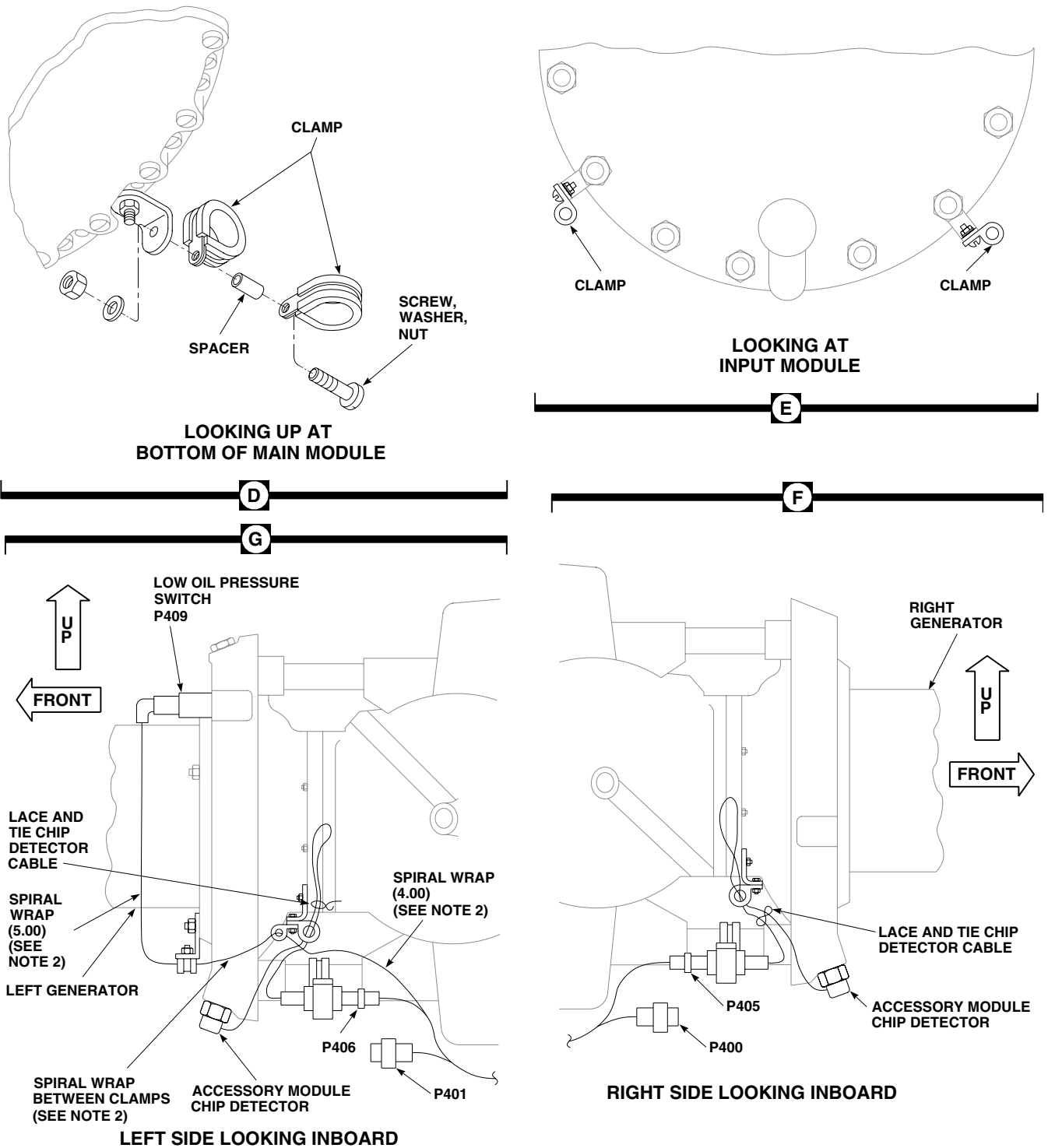


Figure 8. Install Wiring Harness. (Sheet 2 of 4)

MAIN TRANSMISSION BUILDUP - Continued



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Figure 8. Install Wiring Harness. (Sheet 3 of 4)

MAIN TRANSMISSION BUILDUP - Continued

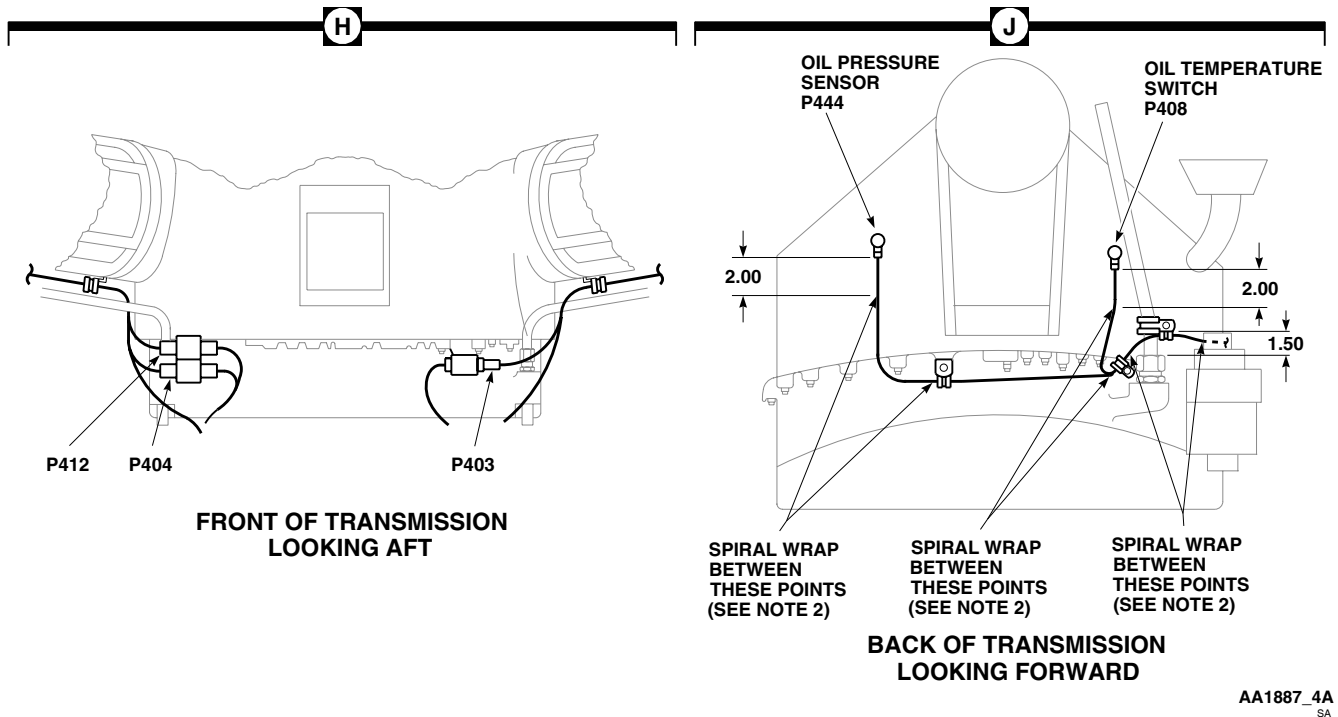


Figure 8. Install Wiring Harness. (Sheet 4 of 4)

PREPARE TO INSTALL

1. Make sure helicopter wheels are chocked.
2. Turn off all electrical and hydraulic power.
3. **QA** Check main transmission to make sure all parts removed from old module have been installed on replacement module, except for bellcrank supports. Main transmission must be installed prior to installing bellcrank supports.

NOTE

Replace bolts, NAS630-35, with bolts, SS5209-10-34. If bolts, SS5209-10-34, are not available, replace with new bolts, NAS630-35.

4. Check main transmission mounting bolts and barrel-nuts as follows:
 - a. Check bolt threaded area for damage.
 - b. Check bolt for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace bolt if crack is found.
 - c. Install bolt into barrel-nut until two threads are exposed beyond nut.
 - d. Using torque wrench back out bolt. **IF BREAKAWAY TORQUE IS LESS THAN 32 INCH-POUNDS, REPLACE NUT.**

PREPARE TO INSTALL - Continued**CAUTION**

- Damage to maintenance crane and helicopter will occur if maintenance crane is used to install main transmission with main rotor head and swashplate installed. Use sling and overhead hoist to install main transmission with main rotor head and swashplate installed.
 - Damage to main rotor head components will occur if sling is not properly installed. Make sure sling is not put around damper, damper reservoir, or other parts of rotor head, other than main rotor head hub arms.
5. Install main transmission with main rotor head installed as follows:
 - a. Put sling straps around hub arms in four places (Figure 4).
 - b. Attach sling to hoist with enough capacity to lift main transmission and main rotor head.
 - c. Adjust sling straps before lifting, for even distribution of weight.

CAUTION

UH-60L Damage to sling will result if sling contacts studs on tail takeoff housing. Studs will cut into sling. Make sure sling does not contact studs. ■

6. Install main transmission with main rotor head and swashplate removed as follows:
 - a. Attach sling straps to input modules and around tail takeoff housing (Figure 4).
 - b. Attach sling to maintenance crane or a hoist with enough capacity to lift main transmission.
 - c. Adjust sling straps before lifting, for even distribution of weight.

INSTALL**CAUTION**

Damage to recess in frame or beam will result if barrel nut is incorrectly installed. To prevent damage to recess, make sure nut is installed with barrel (round side) toward bolthead.

NOTE

Make sure that only square barrel-nuts are used.

1. Before hoisting main transmission, line up barrel-nuts as follows:
 - a. Screw mounting bolts into each barrel-nut two to three threads.
 - b. Line up mounting bolts perpendicular to airframe mounting surface.
 - c. Carefully remove bolts so as not to disturb position of barrel-nuts.
2. Clean airframe mounting surfaces and put a light coating of molybdenum disulfide, Item 197, WP 1803 00, on main transmission and bellcrank support airframe mounting surfaces.

INSTALL - Continued**CAUTION**

- Damage to main transmission and adjacent components will result if transmission is allowed to swing into components during installation. Carefully guide main transmission into place to clear all components on helicopter.
- Damage to maintenance crane and helicopter will occur if maintenance crane is used to install transmission with main rotor head and swashplate installed. Use sling and overhead hoist to install main transmission with main rotor head and swashplate installed.

NOTE

- **UH-60A EH-60A** Main transmission with all equipment (input modules, accessory modules, generators, and pumps) installed weighs about 1200 pounds. Main transmission with all equipment, main rotor head, and swashplate installed weighs about 2400 pounds. Main module alone weighs about 750 pounds. ■
- **UH-60L** A Main transmission with all equipment (input modules, accessory modules, generators, and pumps) installed weighs about 1365 pounds. Main transmission with all equipment, main rotor head, and swashplate installed weighs about 2565 pounds. Main module alone weighs about 915 pounds. ■

3. If not installed, attach guide straps to transmission.
4. Remove bolts, washers, and nuts holding main transmission to adapter (Figure 6).

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from adapter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from adapter. Do not allow hook assembly to contact pulley.
- Injury to personnel will result if sling or hoist allows main transmission to fall while personnel are working under main transmission. Do not work directly under main transmission when transmission is supported only by sling and hoist.

NOTE

Do not move adapter and trailer from under main transmission while applying sealing compound.

5. Hoist main transmission enough to gain access to underside of mounting feet.
6. Apply a thin brush coat of sealing compound, Item 273, WP 1803 00, using paint brush, Item 61, WP 1803 00, to underside of main transmission mounting feet.
7. Hoist main transmission and lower into position on helicopter while sealing compound is still wet.
8. Apply a small bead of sealing compound, Item 273, WP 1803 00, around inner rim of each mounting bolt hole on top side of mounting feet.
9. Apply a coat of antiseize compound, Item 49, WP 1803 00, to threads of bolt.

INSTALL - Continued

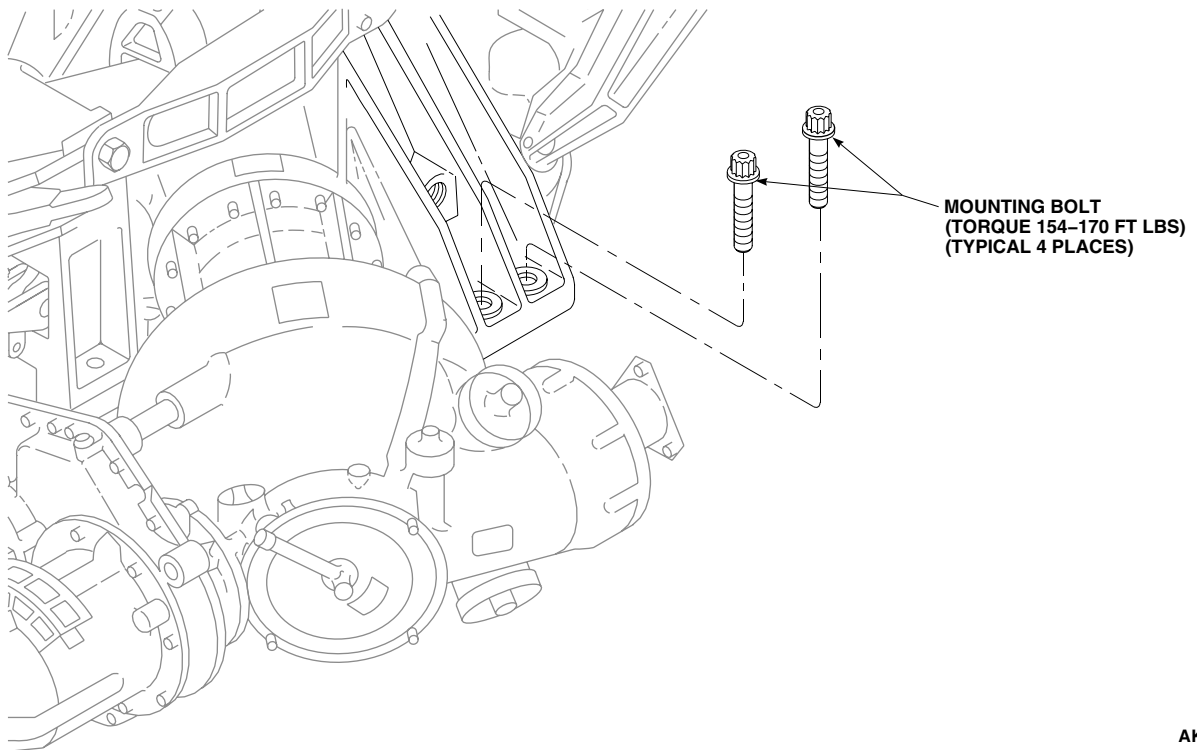
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Figure 9. Install on Mounts.

CAUTION

Damage to equipment will result if epoxy primer coating is applied to threads of mounting bolts. Epoxy primer coating will make removal of bolts difficult and could damage barrel-nuts. Do not allow epoxy primer coating to contact bolt threads.

10. Apply a coating of epoxy primer coating kit, Item 120, WP 1803 00, to shank of mounting bolts.

WARNING**CSI**

Verification of minimum torque of bolts is a critical characteristic.

NOTE

- It may be easier to line up front mounting holes first, insert bolts, and then line up rear holes and insert bolts.
 - Replace bolts, NAS630-35, with bolts, SS5209-10-34. If bolts, SS5209-10-34, are not available, replace with new bolts, NAS630-35.
11. **QA** Install mounting bolts while primer is still wet. **TORQUE ALL EIGHT MAIN TRANSMISSION MOUNTING BOLTS TO 154 - 170 FOOT-POUNDS.** If bellcrank support is installed on main transmission, locally make mounting bolt adapter tool (WP 1805 00), if necessary, to torque two front bolts. Use TM 1-1500-204-23 to figure proper torque. Torque value will change due to added length of adapter (Figure 9).

INSTALL - Continued

12. Remove sling.
13. Remove guide straps.
14. Form a fillet with squeezed out sealing compound, Item 273, WP 1803 00, around base of mounting feet.
15. Mask off adjacent areas using pressure sensitive adhesive tape, Item 326, WP 1803 00, to prevent overspray.
16. Apply corrosion preventive compound, Item 106, WP 1803 00, to entire area of mounting feet.
17. Allow to dry for 10 to 30 minutes and apply a second coat of corrosion preventive compound, Item 106, WP 1803 00.
18. **QA** Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure main transmission mounting bolt torque stabilization check is shown as due 9 to 11 flight hours after installation (WP 1689 00).

WARNING**CSI**

All self retaining bolts between servo and swashplate bellcranks, tie rods, connecting links, control rods, bellcrank supports, and support fittings contain critical surfaces which must not be damaged during installation. After removal of protective covering, use extreme caution when installing critical parts.

NOTE

To remove or install bolts that have retaining balls, insert 1/8-inch diameter Allen wrench in hole in bolthead. Push in on pin to release retaining balls at nut end of bolt. Keep pressure on pin while removing or installing bolt.

19. If tie rods are bolted to forward bellcrank support, remove nut and then remove tie rod from forward bellcrank support (Figure 12, Sheet 3, Detail E and Detail F). Tie rods contain eccentric bushings which must be lined up after bellcrank support is installed.

CAUTION

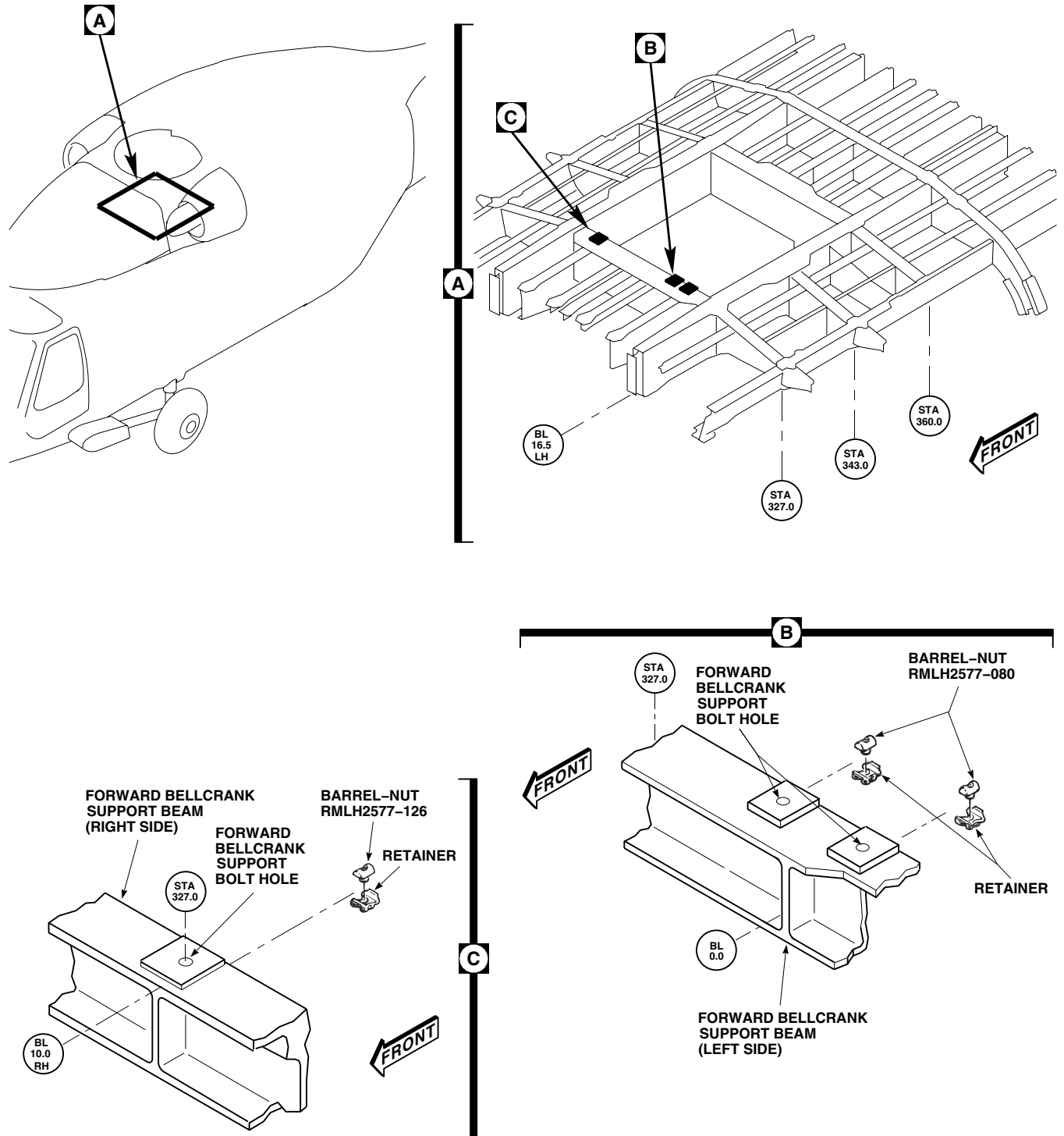
Damage to recess in frame or beam will result if barrel-nuts are incorrectly installed. To prevent damage to recess, make sure nuts are installed with barrel (round side) toward bolthead.

NOTE

Make sure that only square barrel-nuts are used.

20. Check barrel-nuts that forward bellcrank support bolts thread into as follows (Figure 10, Detail B and Detail C):
 - a. Install bolt into barrel-nut until two threads extend beyond nut.
 - b. Back bolt out of barrel-nut, noting torque necessary to start bolt turning out of nut.
 - c. **REPLACE RIGHT SIDE BARREL-NUT IF BREAKAWAY TORQUE IS LESS THAN 50 INCH-POUNDS.**
 - d. **REPLACE LEFT SIDE BARREL-NUTS IF BREAKAWAY TORQUE IS LESS THAN 18 INCH-POUNDS.**
 - e. If barrel-nut was replaced, recheck breakaway torque. **IF BREAKAWAY TORQUE IS STILL TOO LOW, REPLACE BOLT.**

INSTALL - Continued



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Figure 10. Replace Barrel-Nuts.

INSTALL - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

21. Clean support tongue mounting pad on main transmission using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe with clean towel.
22. Apply a coat of sealing compound, Item 258, WP 1803 00, to dowel pin and entire bridge forward mount pad on main transmission. Apply fillet around base of dowel pin. Fill bolt holes with sealing compound. Make sure there are no gaps in coating.
23. Apply a thin coat of sealing compound, Item 258, WP 1803 00, to bottom of insulator.
24. **QA** Install insulator over dowel pin and press onto support tongue mounting pad. Align holes in insulator with holes in support tongue mounting pad.
25. Apply a thin coat of sealing compound, Item 258, WP 1803 00, to entire lower surface of forward bellcrank support tongue.
26. Apply a thin coat of sealing compound, Item 258, WP 1803 00, to the entire upper surface of insulator.
27. Coat bottom of forward bellcrank support with sealing compound, Item 258, WP 1803 00.
28. Make a fillet around forward bellcrank support locator pin using sealing compound, Item 258, WP 1803 00.
29. **QA** Carefully place forward bellcrank support on airframe mounting surface and place forward bellcrank support tongue onto dowel pin on support tongue mounting pad on main transmission.
30. Put a light coat of antiseize compound, Item 49, WP 1803 00 on bolt threads.
31. Apply a coat of epoxy primer coating kit, Item 120, WP 1803 00 to shank of mounting bolts and loosely install bolts and countersunk washers in forward bellcrank support legs, while primer is still wet. Make sure countersunk side of washer faces bolthead.
32. Measure gap between tongue of forward bellcrank support and main transmission. Make sure gap is at least 0.001-inch.

WARNING**CSI**

Verification of minimum torque and safety of bolts are critical characteristics.

33. **QA TORQUE TWO SMALLER BOLTS ON LEFT SIDE OF FORWARD BELLCRANK SUPPORT TO 930 - 1030 INCH-POUNDS.** Wipe off excess primer using machinery wiping towel, Item 344, WP 1803 00. Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts to radius block, or to drilled bushing (Figure 12, Sheet 2, Detail B). Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued**WARNING****CSI**

Verification of minimum torque of bolt is a critical characteristic.

34. **QA TORQUE LARGE BOLT ON RIGHT SIDE OF FORWARD BELLCRANK SUPPORT TO 237 - 262 FOOT-POUNDS.** Wipe off excess primer using machinery wiping towel, Item 258, WP 1803 00. (Figure 12, Sheet 2, Detail C). Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).
35. Apply a bead of sealing compound, Item 258, WP 1803 00, around bolthead and washers.

WARNING**CSI**

Verification of dowel pin bushing retainer installation is critical characteristic.

36. **QA** Install and adjust eccentric bushing in forward bellcrank support tongue to fit over dowel pin on main transmission (Figure 12, Sheet 2, Detail D).
37. **QA** **W/O MWO 50-43** Install spring clip bushing retainer, 70400-08123-104, on support tongue as follows: ■
- a. Apply a bead of sealing compound, Item 258, WP 1803 00, around each spring clip bushing retainer bolt hole on top of forward bellcrank support tongue.
 - b. **QA** Position spring clip bushing retainer on forward bellcrank support tongue. Install bolts, countersunk washers, and nylon washers. Countersunk side of washer faces bolthead (Figure 12, Sheet 2, Detail D).

NOTE

Dowel pin retaining zee, 70400-08171-101, is used on transmissions with tall dowel pins. Retaining zee, 70400-08171-102, is used on transmissions with short dowel pins.

38. **QA** **MWO 50-43** Position dowel pin retaining zee on support tongue as follows: ■
- a. Apply a bead of sealing compound, Item 258, WP 1803 00, around each dowel pin retaining zee bolt hole on top of forward bellcrank support tongue.
 - b. **QA** Position dowel pin retaining zee on forward bellcrank support tongue. Install bolts, countersunk washers, and nylon washers. Countersunk side of washer faces bolthead (Figure 12, Sheet 2, Detail D).

NOTE

Lower washers and nylon washers must be installed wet.

39. Coat washers with epoxy primer coating kit, Item 120, WP 1803 00.
40. Apply a bead of sealing compound, Item 258, WP 1803 00, to one side of nylon washers.
41. **QA** Install nylon washers with sealing compound side towards main transmission, washers and nuts. **TORQUE NUTS TO 72 - 78 INCH-POUNDS.**

INSTALL - Continued

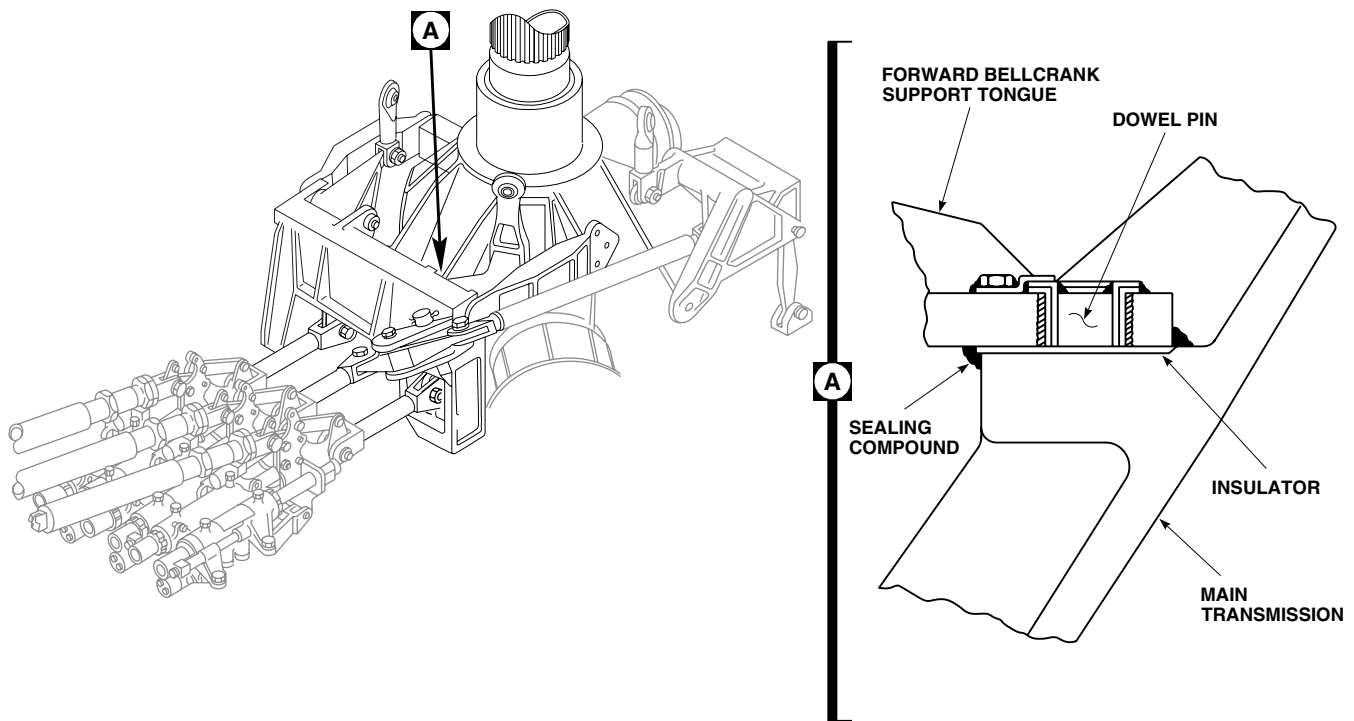
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Figure 11. Support Tongue Sealing.

NOTE

Do not use excessive sealing compound to form fillet around support tongue. Excessive sealing compound will hinder corrosion inspection.

42. Form a fillet with the squeezed out sealing compound around the joint between main transmission and forward bellcrank support tongue. Make sure sealing compound covers entire joint line but is no larger than 1/4-inch in diameter (Figure 11, Detail A).
43. Apply polyurethane coating, Item 231, WP 1803 00, to upper and lower surfaces of forward bellcrank support tongue mounting area (WP 0381 00).
44. Mask off adjacent area to prevent overspray, then apply 2 coats of corrosion preventive compound, Item 106, WP 1803 00, to forward bellcrank support tongue mounting area and also to water entrapment area behind mounting area. Allow to dry between coats.
45. Apply coat of sealing compound, Item 258, WP 1803 00, to dowel pin and entire tie rod mount pad surface on main transmission, including fillets around threaded inserts and base of dowel pin. Make sure there are no gaps in coating.
46. Position insulator on main transmission mounting pads. Apply coat of sealing compound, Item 258, WP 1803 00, to top surface of insulator and underside of tie rod. Make sure there are no gaps in coating.
47. Place left tie rod over main transmission dowel pin (Figure 12, Sheet 3, Detail E).

INSTALL - Continued

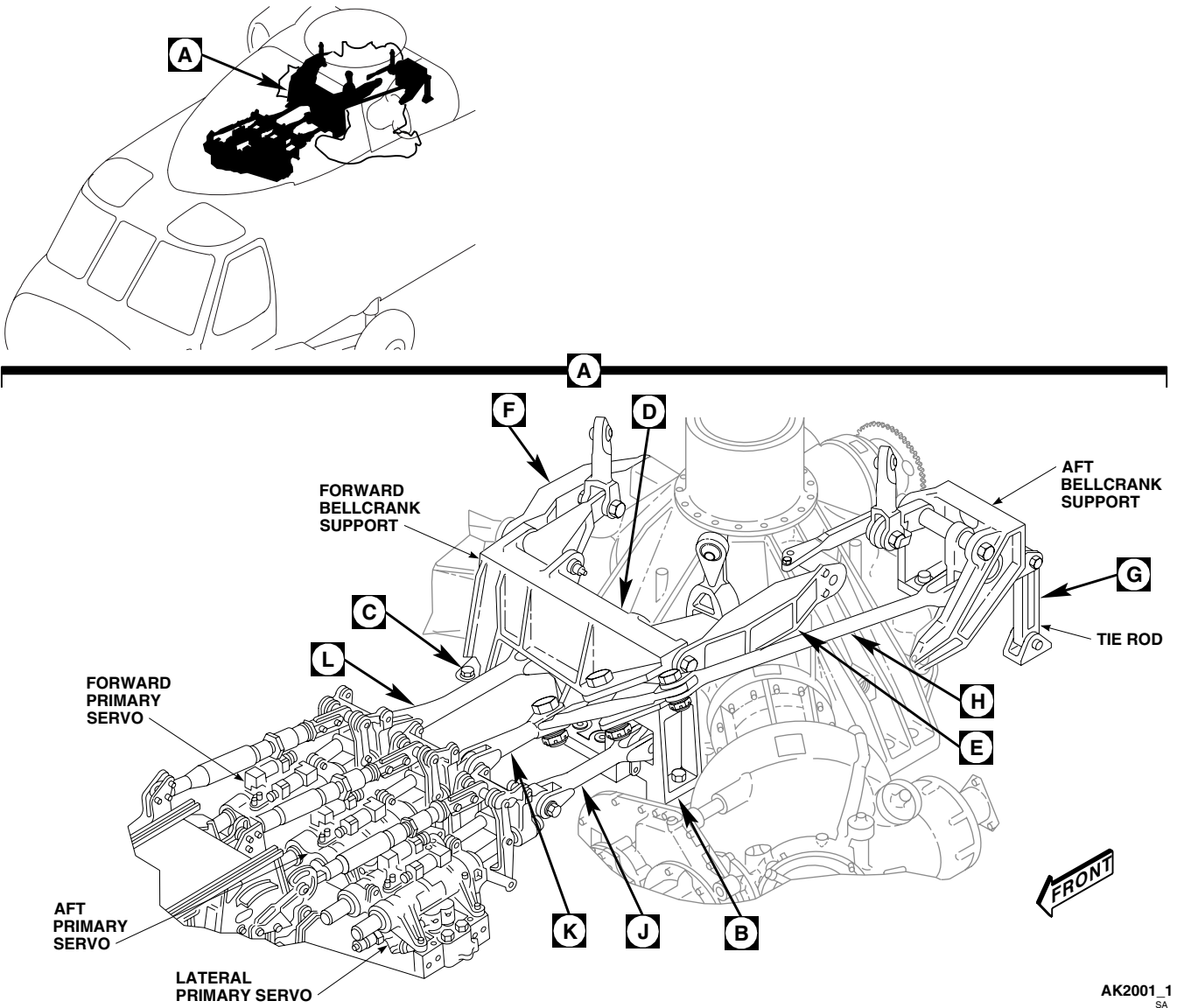


Figure 12. Install Bellcrank Supports and Connect Control Rods. (Sheet 1 of 5)

INSTALL - Continued

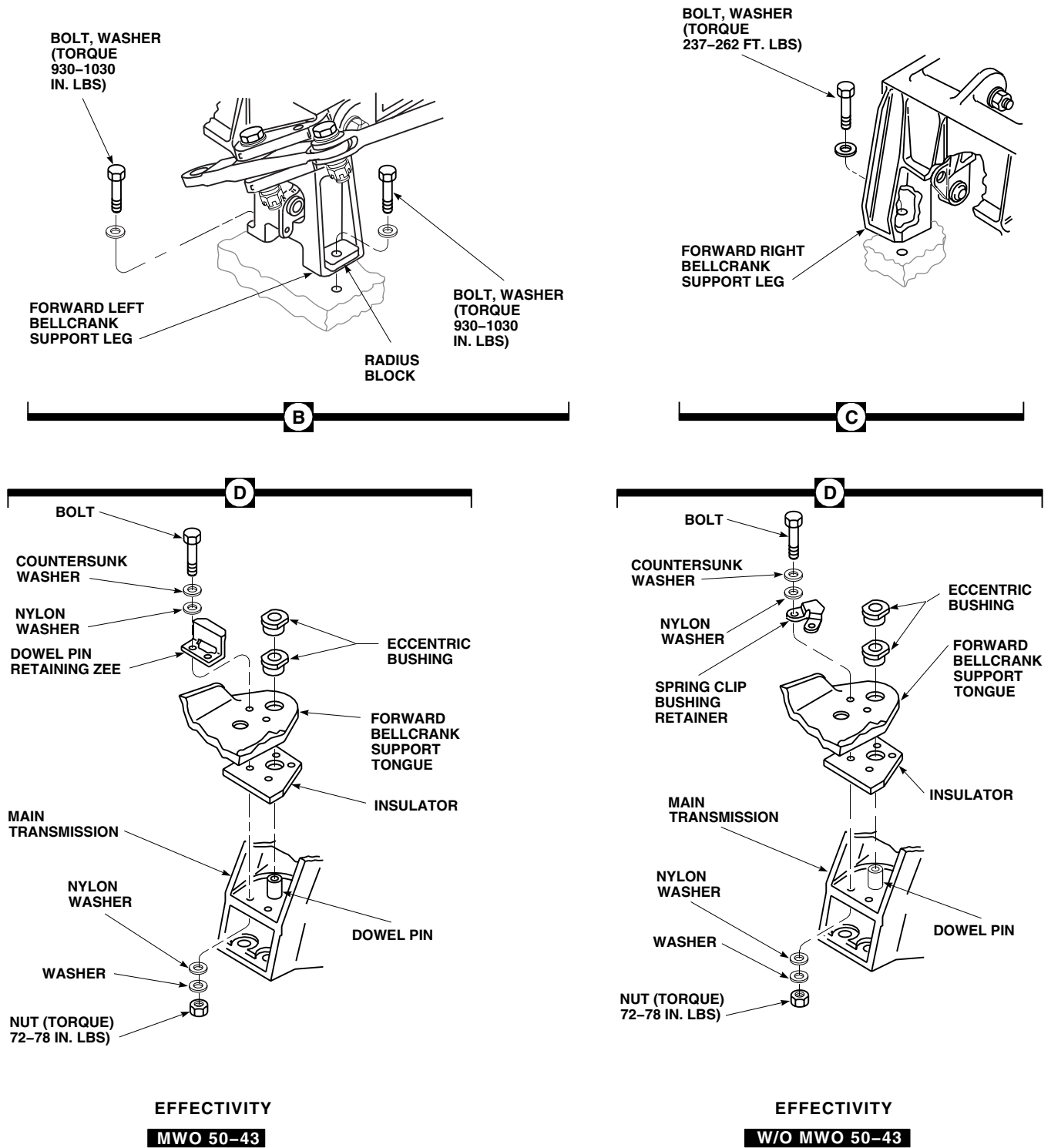


Figure 12. Install Bellcrank Supports and Connect Control Rods. (Sheet 2 of 5)

INSTALL - Continued

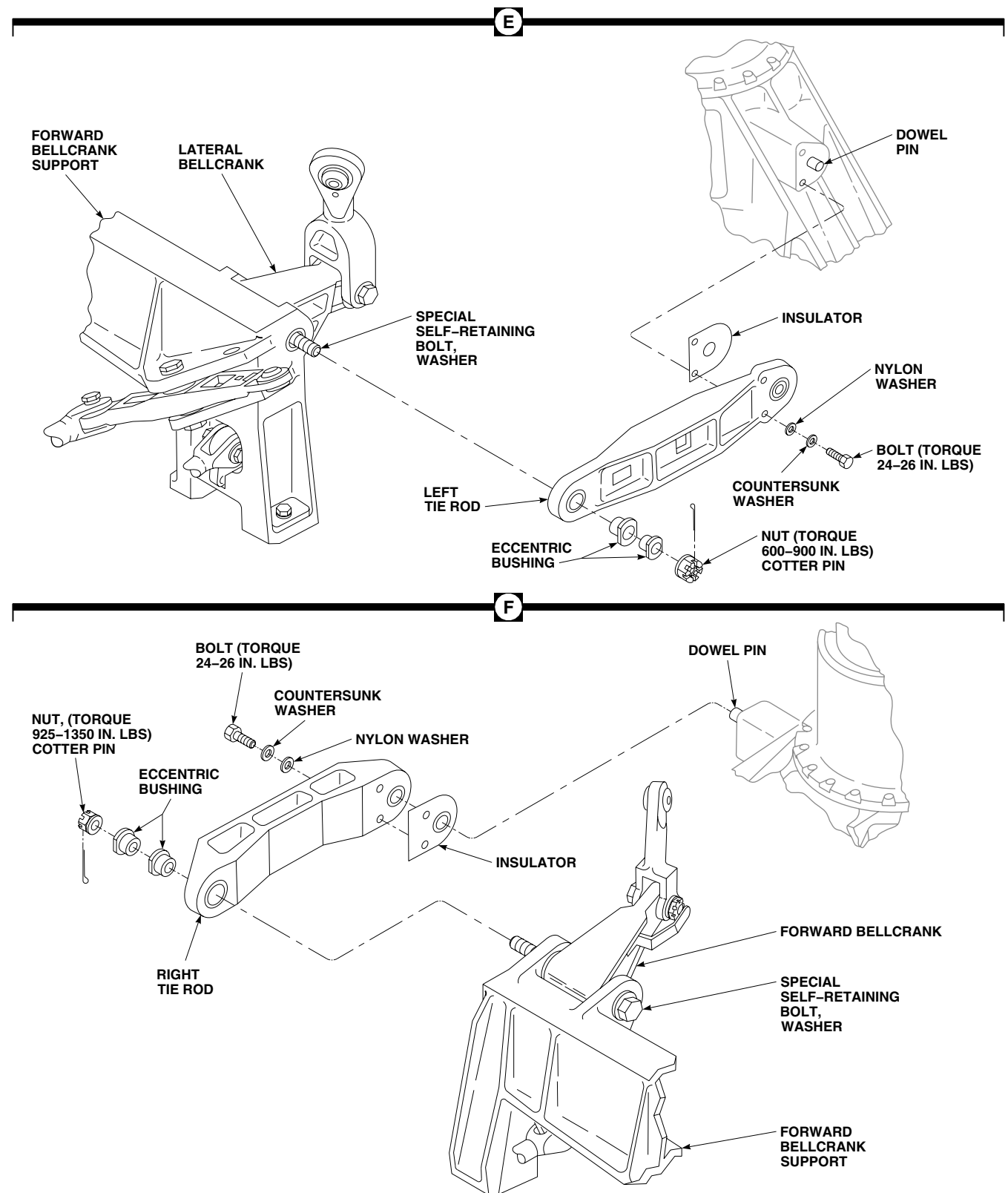
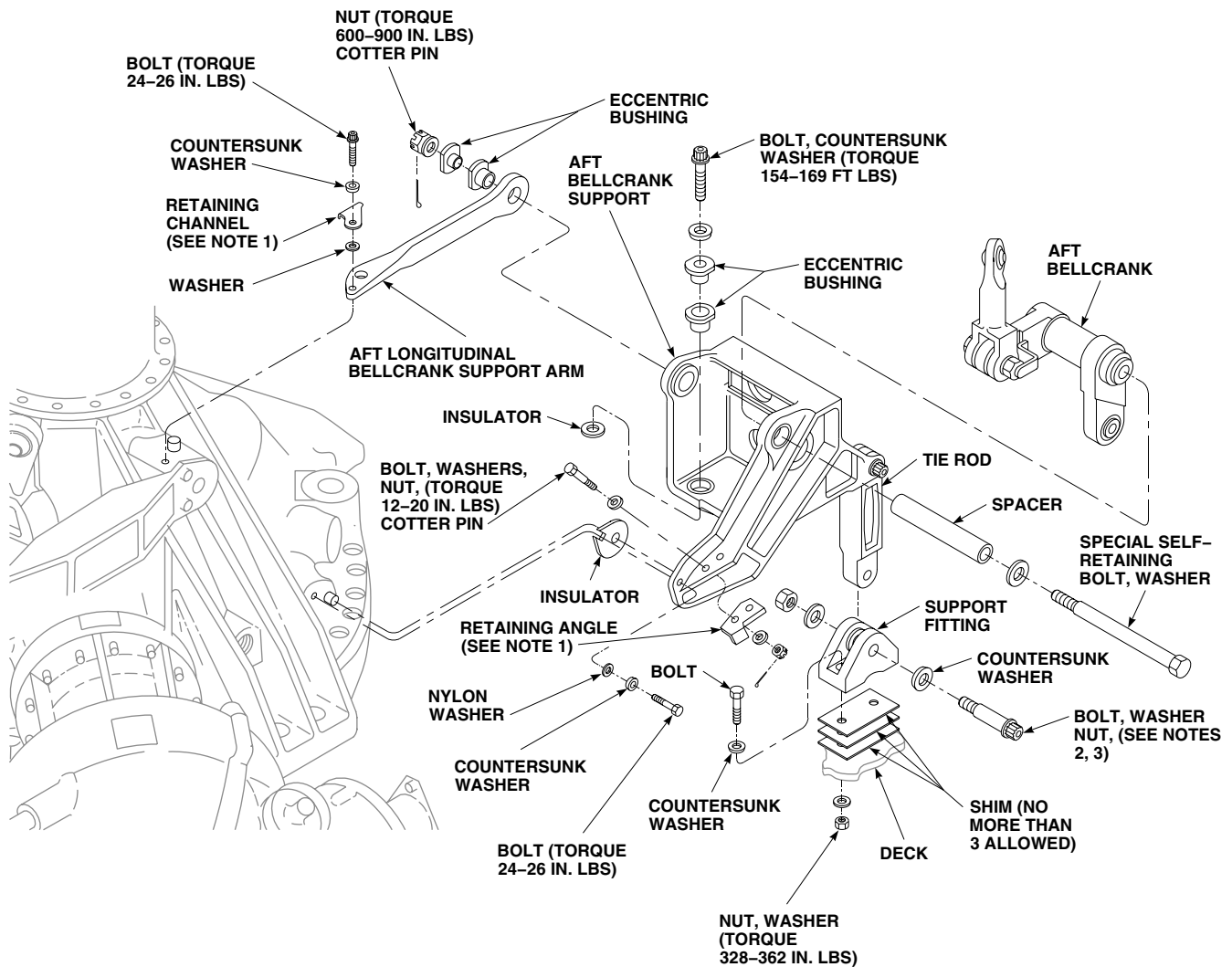
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Figure 12. Install Bellcrank Supports and Connect Control Rods. (Sheet 3 of 5)

INSTALL - Continued

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NOTES

1. **MWO 50-43**
2. ON HELICOPTERS WITH TIE ROD
70400-08111-041 OR 70400-08111-042,
TORQUE NUT TO 532-588 INCH-POUNDS.
3. ON HELICOPTERS WITH TIE ROD
70400-08111-043, TORQUE NUT TO
238-262 INCH-POUNDS.

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Figure 12. Install Bellcrank Supports and Connect Control Rods. (Sheet 4 of 5)

INSTALL - Continued

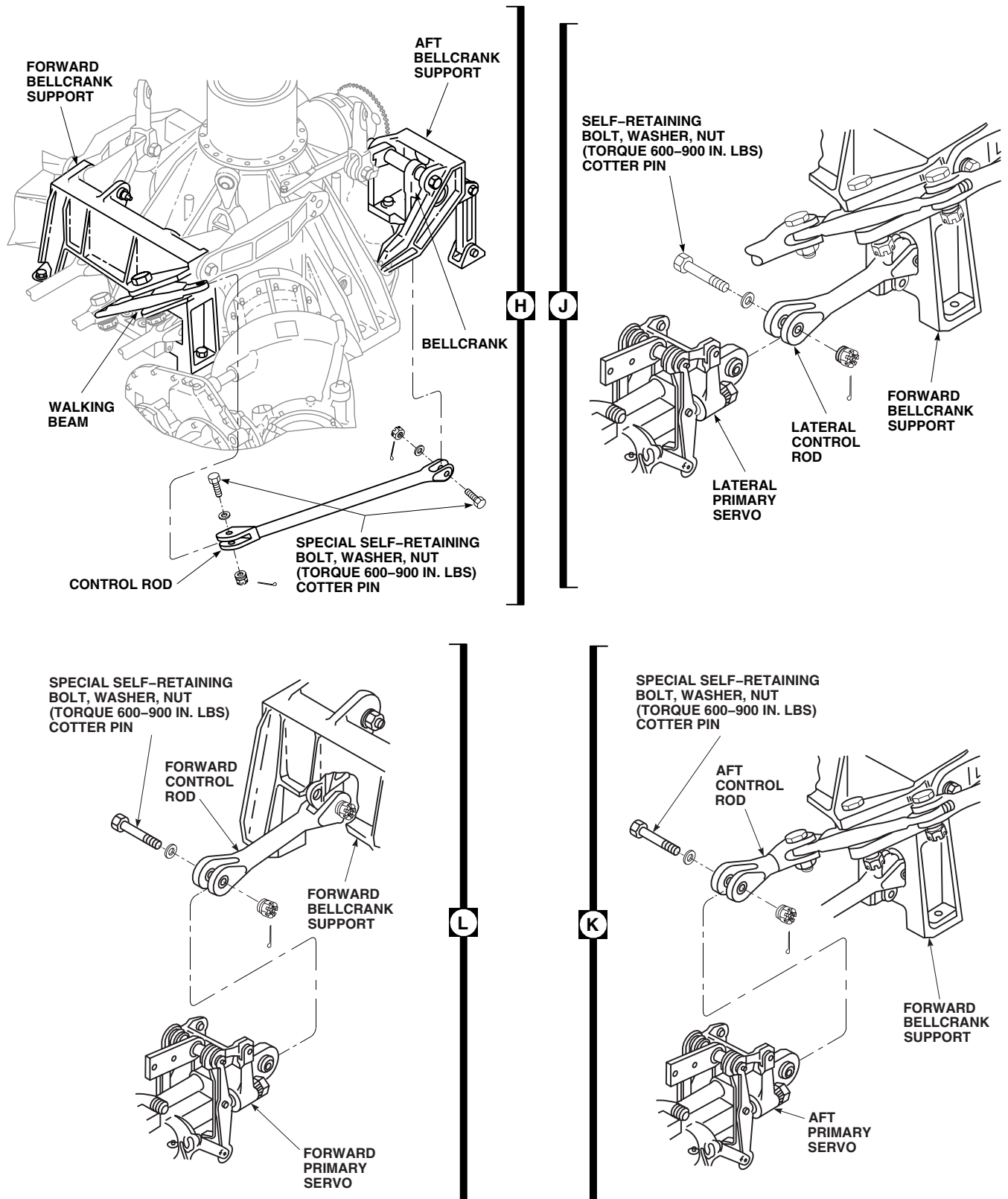
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Figure 12. Install Bellcrank Supports and Connect Control Rods. (Sheet 5 of 5)

INSTALL - Continued**WARNING****CSI**

Verification of proper hardware installation, torque, safety installation, and proof torque of special self-retaining bolts are critical characteristics.

NOTE

Add washers under bolthead as needed to make sure nut seats against tie rod and does not bottom out on bolt threads.

48. **QA** Install special self-retaining bolt and washers (maximum of 3 under bolthead) to provide correct grip length and cotter pin position.
49. Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.

NOTE

Hold tie rod in upward position while adjusting eccentric bushings with bolt.

50. **QA** Adjust eccentric bushings in left tie rod to line up with bolt. Install nut. **TORQUE NUT TO 600 - 900 INCH-POUNDS.** Install cotter pin.

NOTE

If unable to torque bolthead use a crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

51. **PROOF TORQUE BOLTHEAD TO 400 - 420 INCH-POUNDS.** If bolt rotates, adjust using washers. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).
52. Coat shank and boltheads with epoxy primer coating kit, Item 120, WP 1803 00. Allow primer to dry.
53. Lubricate bolt shank and head with corrosion preventive compound, Item 101, WP 1803 00. Coat threads of bolt with epoxy primer coating kit, Item 120, WP 1803 00.

NOTE

Bolts securing left tie rod to main transmission must be lockwired to each other and to support rod bolt, before applying sealing compound.

54. **QA** Install bolts and washers securing left tie rod to main transmission, while primer is still wet. Make sure insulator is installed and washers are positioned as shown (Figure 12, Sheet 3, Detail E). Make sure countersunk side of washer faces bolthead. **TORQUE BOLTS TO 24 - 26 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire boltheads together and to support rod bolt.
55. Apply a coat of sealing compound, Item 273, WP 1803 00, to dowel pin and entire mount pad surface on main transmission, including fillets around threaded inserts and base of dowel pin. Make sure there are no gaps in coating.
56. Position insulator on main transmission mounting pad. Apply coat of sealing compound, Item 258, WP 1803 00, to top surface of insulator and underside of tie rod. Make sure there are no gaps in coating.
57. Place right tie rod over main transmission dowel pin (Figure 12, Sheet 3, Detail F).

INSTALL - Continued**WARNING****CSI**

Verification of proper hardware installation, torque, safety installation, and proof torque of special self-retaining bolts are critical characteristics.

NOTE

Add washers under bolthead as needed, to make sure nut seats against tie rod and does not bottom out on bolt threads.

58. **QA** Install special self-retaining bolt and washers (maximum of 3 under bolthead) to provide correct grip length and cotter pin position.
59. Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.
60. **QA** Adjust eccentric bushings in right tie rod to line up with bolt. Install nut. **TORQUE NUT TO 925 - 1350 INCH-POUNDS.** Install cotter pin.

NOTE

If unable to torque bolthead use a crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

61. **QA PROOF TORQUE BOLTHEAD TO 630 - 650 INCH-POUNDS.** If bolt rotates, adjust using washers. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).
62. Coat shank and boltheads with epoxy primer coating kit, Item 120, WP 1803 00. Allow primer to dry.
63. **QA** Lubricate bolt shank and head with corrosion preventive compound, Item 101, WP 1803 00. Coat threads of bolt with epoxy primer coating kit, Item 120, WP 1803 00, and install bolt while primer is still wet.
64. **QA** Install bolts and washers securing right tie rod to main transmission. Make sure insulator is installed and washers are positioned as shown. Make sure countersunk side of washer faces bolthead. **TORQUE BOLTS TO 24 - 26 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, boltheads together (Figure 12, Sheet 3, Detail F).
65. Apply a bead of sealing compound, Item 258, WP 1803 00, around boltheads and washers. Form fillet with squeezed out sealing compound no larger than 0.250-inch diameter around rod and transmission mounting surface and boltheads. Wipe off excess primer.
66. If support rod is bolted to aft bellcrank support, unbolt it. Support contains eccentric bushings which must be lined up during installation.
67. Apply coat of sealing compound, Item 273, WP 1803 00, to dowel pin and entire vertical mount pad surface that mates with arm of aft bellcrank support, including fillets around threaded inserts and base of dowel pin. Make sure there are no gaps in coating.
68. Coat horizontal area on transmission that mates with aft bellcrank support with sealing compound, Item 258, WP 1803 00. Make sure there are no gaps in coating.
69. **QA** Place insulators on horizontal and vertical bellcrank support mounting areas of transmission. Apply coat of sealing compound, Item 258, WP 1803 00, to insulators and mating surfaces of bellcrank support. Make sure there are no gaps in coating. Place aft bellcrank support on main transmission (Figure 12, Sheet 4, Detail G).
70. Coat shank, threads, and area under bolthead with epoxy primer coating kit, Item 120, WP 1803 00.

INSTALL - Continued

71. **QA** Install bolt and washers securing leg of bellcrank support to side of main transmission, while primer is still wet. Position washers as shown. Make sure countersunk side faces bolthead. **TORQUE BOLT TO 24 - 26 INCH-POUNDS.**
72. Apply a bead of sealing compound, Item 273, WP 1803 00

WARNING**CSI**

Verification of dowel pin retaining angle installation is a critical characteristic.

73. Coat shank, threads, and area under bolthead with primer coating, Item 243, WP 1803 00.
74. **QA** **MWO 50-43** Position retaining angle on leg of bellcrank and install bolts, washers, and nuts. **TORQUE NUT TO 12 - 20 INCH-POUNDS.** Check for minimum 0.010-inch clearance between dowel pin and retaining angle. If clearance is less than 0.010-inch, reseal transmission dowel pin. Refer to RESEAT DOWEL PIN, in this work package. Install cotter pin. ■

WARNING**CSI**

Verification of minimum torque and safety installation of bolts are critical characteristics.

75. **QA** Line up and install eccentric bushings in aft bellcrank support. Install bolt and washer, while primer is still wet. **TORQUE BOLT TO 154 - 169 FOOT-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lock-wire this bolt to bolt installed in previous step.
76. Coat area on transmission that mates with support rod with primer coating, Item 243, WP 1803 00.
77. Place support rod over dowel pin on main transmission, while primer is still wet.

WARNING**CSI**

Verification of proper hardware installation, safety installation, and proof torque of special self-retaining bolts are critical characteristics.

78. **QA** Adjust eccentric bushings in support rod to line up boltholes.
79. Install special self-retaining bolt, spacer, and washers (maximum of 3 under bolthead) to provide correct grip length and cotter pin position.
80. Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.
81. **QA** Install nut. **TORQUE NUT TO 600 - 900 INCH-POUNDS.** Install cotter pin.

NOTE

If unable to torque bolthead use a crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

82. **QA PROOF TORQUE BOLT HEAD TO 400 - 420 INCH-POUNDS.** If bolt rotates, adjust using washers. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued

83. Coat shank, threads and area under bolthead with epoxy primer coating, Item 120, WP 1803 00.

WARNING**CSI**

Verification of dowel pin retaining channel installation is a critical characteristic.

84. **QA MWO 50-43** Position retaining channel on support rod. ■
85. **QA** Install bolt and washers securing support rod to main transmission, while primer is still wet. Position washers as shown. Make sure countersunk side of washer faces bolthead. **TORQUE BOLT TO 24 - 26 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolt to both bolts that hold left tie rod to main transmission (Figure 12, Sheet 4, Detail G).
86. Apply a bead of sealing compound, Item 258, WP 1803 00, around bolthead and washer. Form fillet with sealing compound no larger than 0.250-inch diameter. Wipe off excess primer.
87. **QA** Swing support fitting down and measure gap between fitting and deck. **PEEL SHIMS TO A THICKNESS OF 0.000 TO 0.010-INCH THICKER THAN MEASURED GAP.** No more than three shims may be used. Treat shims with corrosion resistant coating, Item 111, WP 1803 00.
88. Remove lower tie rod bolt.
89. Coat shank, threads, and area under boltheads with primer coating, Item 243, WP 1803 00.
90. **QA** Place shims between support fitting and deck. Install bolts, washers, and nuts, while primer is still wet. **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**

NOTE

Add washers under head of bolt as needed, to make sure nut seats against support and does not bottom out on bolt threads.

91. **QA** On helicopters with tie rod, 70400-08111-043, install bolt, washers, and nut securing tie rod to support fitting. **TORQUE NUT TO 238 - 262 INCH-POUNDS.**
92. **QA** On helicopters with tie rod, 70400-08111-041 or 70400-08111-042, install bolt, washers, and nut securing tie rod to support fitting. **TORQUE NUT TO 532 - 588 INCH-POUNDS.**
93. After all corrosion preventive coatings are dry, do this:
- Make sure all areas to be treated are clean and dry.
 - Apply two coats of corrosion preventive compound, Item 106, WP 1803 00, to flight control mounting pads on main transmission.
94. On transmissions 70351-08100-053 and 70351-08100-054, do this (WP 0381 00):
- Paint exposed end of pin white (WP 0381 00 and TM 55-1500-345-23).
 - Paint 1/8-inch wide black strip across end of pin, extending onto adjoining flight control supports.
 - Seal exposed end of transmission dowel pins with epoxy coating Kit, Item 118, WP 1803 00.
95. Place control rod between walking beam and aft bellcrank (Figure 12, Sheet 5, Detail H).

INSTALL - Continued**WARNING****CSI**

Verification of proper hardware installation, torque, safety installation, and proof torque of special self-retaining bolts are critical characteristics.

NOTE

Add washers under bolthead as needed, to make sure nut does not bottom out on bolt threads.

- 96. **QA** Install special self-retaining bolt and washers (maximum of 3 under bolthead) to provide correct grip length and cotter pin position.
- 97. Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.
- 98. **QA** Install nut. **TORQUE NUT TO 600 - 900 INCH-POUNDS.** Install cotter pin.

NOTE

If unable to torque bolthead use a crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

- 99. **QA PROOF TORQUE BOLTHEAD TO 400 - 420 INCH-POUNDS.** If bolt rotates, adjust using washers. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).
- 100. **QA** Bolt forward, aft, and lateral control rods to primary servos (Figure 12, Sheet 5, Detail J, Detail K, and Detail L).

WARNING**CSI**

Verification of proper hardware installation, safety installation, and proof torque of special self-retaining bolts are critical characteristics.

NOTE

Add washers under bolthead as needed, to make sure nut does not bottom out on bolt threads.

- 101. **QA** Install special self-retaining bolts and washers (maximum of 3 under bolthead) to provide correct grip length and cotter pin position.
- 102. Check bolts for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup.
- 103. **QA** Install nut. **TORQUE NUT TO 600 - 900 INCH-POUNDS.** Install cotter pin.

NOTE

If unable to torque bolthead use a crowfoot attachment to torque nut, without disturbing cotter pin, and without restraining bolthead.

- 104. **QA PROOF TORQUE BOLTHEADS TO 400 - 420 INCH-POUNDS.** If bolt rotates, adjust using washers. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

INSTALL - Continued

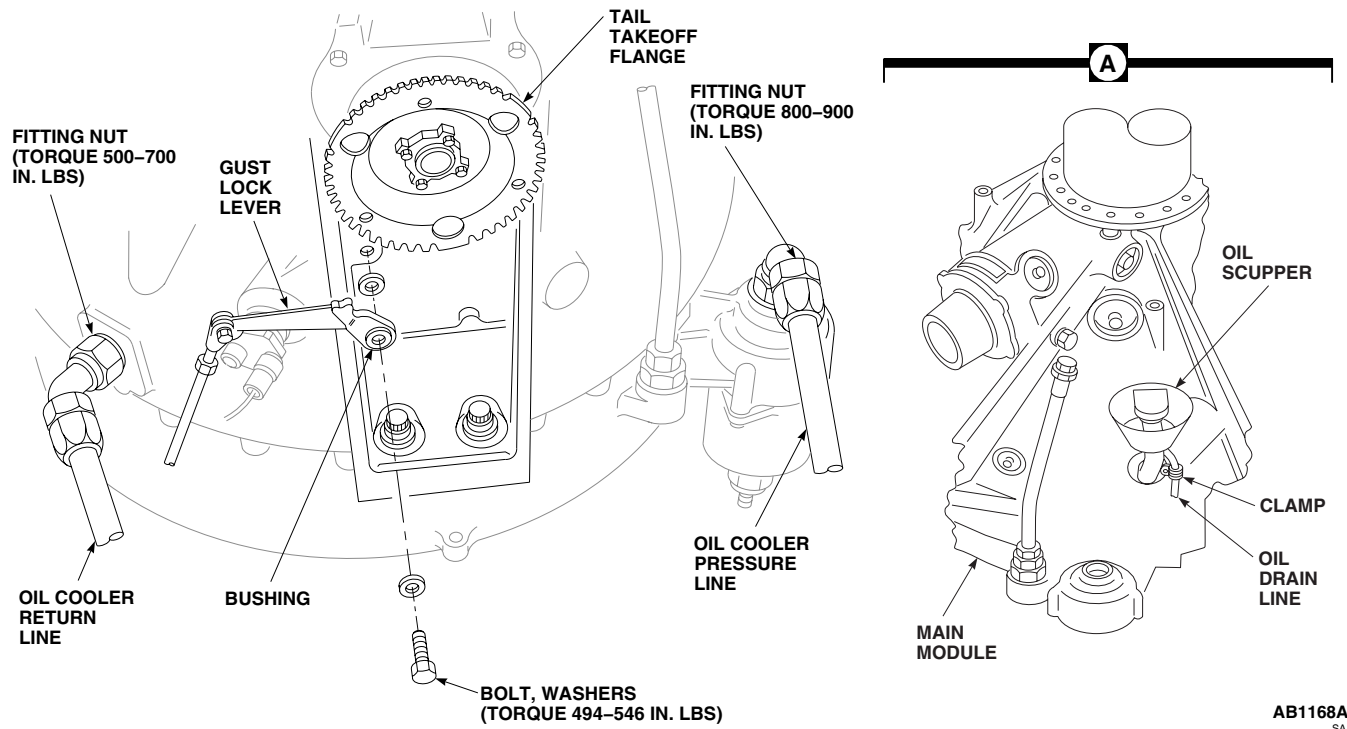


Figure 13. Connect Oil Lines to Transmission.

105. Install gust lock lever (WP 0637 00).
106. **QA** Connect oil drain line to scupper. Clamp oil drain line (Figure 13, Detail A).
107. **QA** Install main transmission oil cooler pressure and return lines (Figure 13).
108. **QA TORQUE FITTING NUT ON OIL COOLER PRESSURE LINE TO 800 - 900 INCH-POUNDS.**
109. **QA TORQUE FITTING NUT ON OIL COOLER RETURN LINE TO 500 - 700 INCH-POUNDS.**
110. Install section I tail rotor drive shaft (WP 0666 00).
111. Install bleed-air line (WP 1272 00).
112. Connect forward engine support to gimbal on input module (WP 0486 00).
113. Connect engine output drive shaft (WP 0486 00).
114. **QA** Connect hydraulic pressure and return lines to hydraulic pump module at quick-disconnects (Figure 14, Detail A).
115. Inspect and treat electrical connectors (WP 1703 00).
116. **QA** Connect electrical connector, P447 No. 1, or P432 No. 2, to hydraulic pump. Using nonelectrical wire, Item 354, WP 1803 00, lockwire connector.

INSTALL - Continued

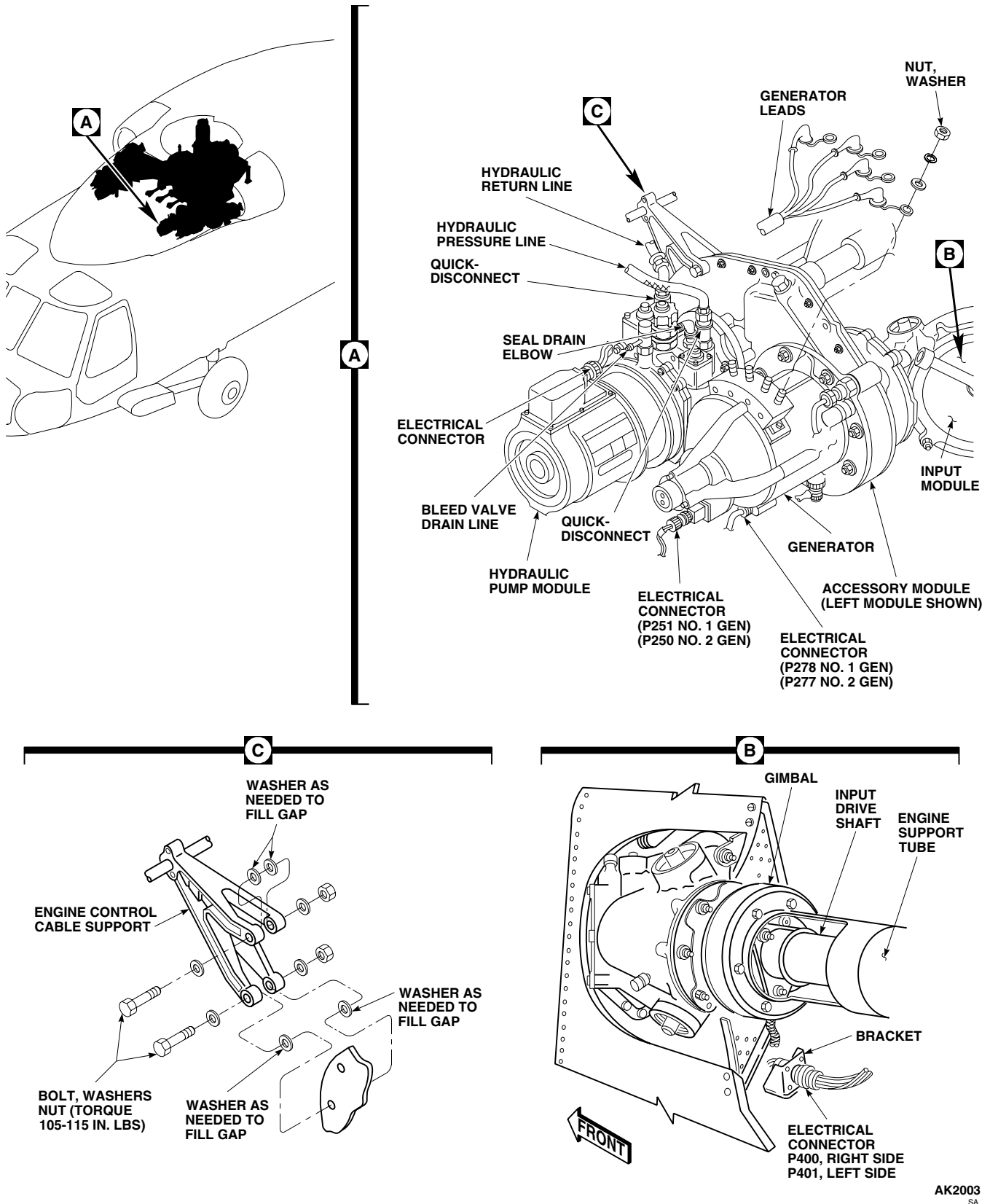


Figure 14. Connect Lines and Wiring.

INSTALL - Continued**NOTE**

Electrical connector on side of pump should be capped. It is not used in this installation.

- 117. **QA** Connect seal drain line to seal drain elbow.
- 118. **QA** Connect bleed valve drain line to bleed valve.

NOTE

When installing support on left module, place angle bracket under upper bolthead in place of washer.

- 119. Coat angle bracket and cable support boltholes on accessory module with primer coating, Item 243, WP 1803 00.
- 120. **QA** Position engine control cable supports on accessory modules. Support with clip or support, 70301-22107-101, goes on left module. Add washers as required to fill gap between bracket and module. Bolt bracket to module.
TORQUE TO 105 - 115 INCH-POUNDS (Figure 14, Detail C).
- 121. Apply sealing compound, Item 273, WP 1803 00, to boltheads, nuts, and exposed threads.
- 122. Reposition load demand cables, and reconnect collective bias tube rod ends to mixer (WP 0506 00).
- 123. Inspect and treat electrical connectors (WP 1703 00).

NOTE

- If tags or sleeves with T-numbers are missing from generator wires, see (WP 0839 00).
- Do not reinstall terminal covers.

- 124. **QA** Connect electrical connectors, P251 and P278 No. 1 generator, or P250 and P277 No. 2 generator, to generator.
- 125. **QA** Install rubber cover on each generator wire lead.
- 126. **QA** Connect wiring to generator terminals with nuts and washers. Remove tags from wires (Figure 14, Detail A).
- 127. Push rubber covers firmly over terminal connections.
- 128. **QA** Fasten wires with clamp to end of generator with screw and washer.
- 129. Inspect and treat electrical connectors (WP 1703 00).
- 130. **QA** Connect main transmission harness, P400, right side, P401, left side, to brackets under forward engine supports (Figure 14, Detail B).
- 131. Install swashplate (WP 0567 00).
- 132. Install main rotor head (WP 0547 00).
- 133. **QA** **UH-60L** Install fire extinguisher line on right firewall (WP 1199 00). ■
- 134. If needed, connect primary servo control rods/links to swashplate (WP 1125 00, WP 1126 00, WP 1127 00, WP 1128 00, and WP 1129 00).
- 135. Install air inlet fairings (WP 0301 00).
- 136. Install transmission covers (WP 0302 00).

INSTALL - Continued**NOTE**

Retirement time of main rotor shaft 70351-08131-044 through 70351-08131-046 is reduced from 1300 hours to 1000 hours if any type of external loads were installed on external stores support systems helicopter hard points. Upon addition of loads to hard points, main modules with main rotor shaft 70351-08131-044 through 70351-08131-046 installed must be reidentified with an "E" following main module serial number. There is no change in retirement life for main rotor 70351-08131-047 through 70351-08131-048.

137. Inspect and clean accessory, input, and main module oil return screens, chip detectors, and chip detector strainers. Check chip detector electrical circuits (WP 0085 00).
138. Install main rotor blades (WP 0576 00).
139. If main transmission was installed with main rotor head, connect main rotor head deice wiring to junction box (WP 1220 00).
140. Service main transmission (WP 1607 00).
141. Make sure area is clean and free of foreign material.
142. Close engine cowlings and main rotor pylon sliding cover.
143. Install oil cooler compartment access doors (WP 0308 00).
144. Perform a main rotor rig/check (WP 1037 00).
145. Do a functional test (WP 0615 00).
146. Install transmission drip pan (WP 0259 00, WP 0260 00, WP 0261 00, or WP 0261 00).

RESEAT DOWEL PIN**NOTE**

- This procedure applies to the dowel pin located at the forward leg of the aft bellcrank support. Reseating of any other transmission dowel pin is not authorized.
- Protect adjacent flight control surfaces, firewalls, hydraulic lines and wiring harnesses with pads.

1. **MWO 50-43** Remove bolts, washers, and nuts securing dowel pin retaining angle to aft bellcrank support leg. Remove retaining angle. 
2. Using 2 to 5 pound mallet and brass drift, seat dowel pin as follows:
 - a. On helicopters with aft bellcrank support, 70400-08117, seat dowel pin to the following dimensions:
 - (1) If dowel pin, 70351-08404-104, is installed, seat dowel pin to a maximum of 0.248-inch above surface of bellcrank support arm.
 - (2) **QA** If dowel pin, 70351-08404-102, 70351-08405-102, or NAS607-12-14P is installed, seat dowel pin until it is flush or slightly below surface of bellcrank support arm.
 - b. On helicopters with aft bellcrank support, 70400-08163, seat dowel pin to the following dimensions:
 - (1) If dowel pin, 70351-08404-104, is installed, seat dowel pin to a maximum of 0.128-inch above surface of bellcrank support arm.
 - (2) **QA** If dowel pin, 70351-08404-102, 70351-08405-102, or NAS607-12-14P is installed, seat dowel pin to a maximum of 0.122-inch below surface of bellcrank support arm.

RESEAT DOWEL PIN - Continued

3. **QA** **MWO 50-43** Install dowel pin retaining angle to bellcrank support arm with bolts, washers, and nuts.
TORQUE NUTS TO 12 - 20 INCH-POUNDS. Install cotter pin. ■
4. Check for minimum 0.010-inch clearance between dowel pin and retaining angle.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

MAIN TRANSMISSION TAIL TAKEOFF FLANGE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Inside Micrometer, GGG-C-105
Pin, 0.072-Inch Dia.
Powertrain Repairer's Toolkit, SC 5180-99-B13
Tail Drive Flange Wrench, 70700-20687-102
Torque Reactor, 70700-20688-041
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 0 - 600 ft. lbs.

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 55-1500-345-23
WP 0381 00
WP 0666 00
WP 1676 00
WP 1803 00

Material/Parts

Abrasive Cloth, Assortment, Item 1, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Primer Coating, Item 243, WP 1803 00

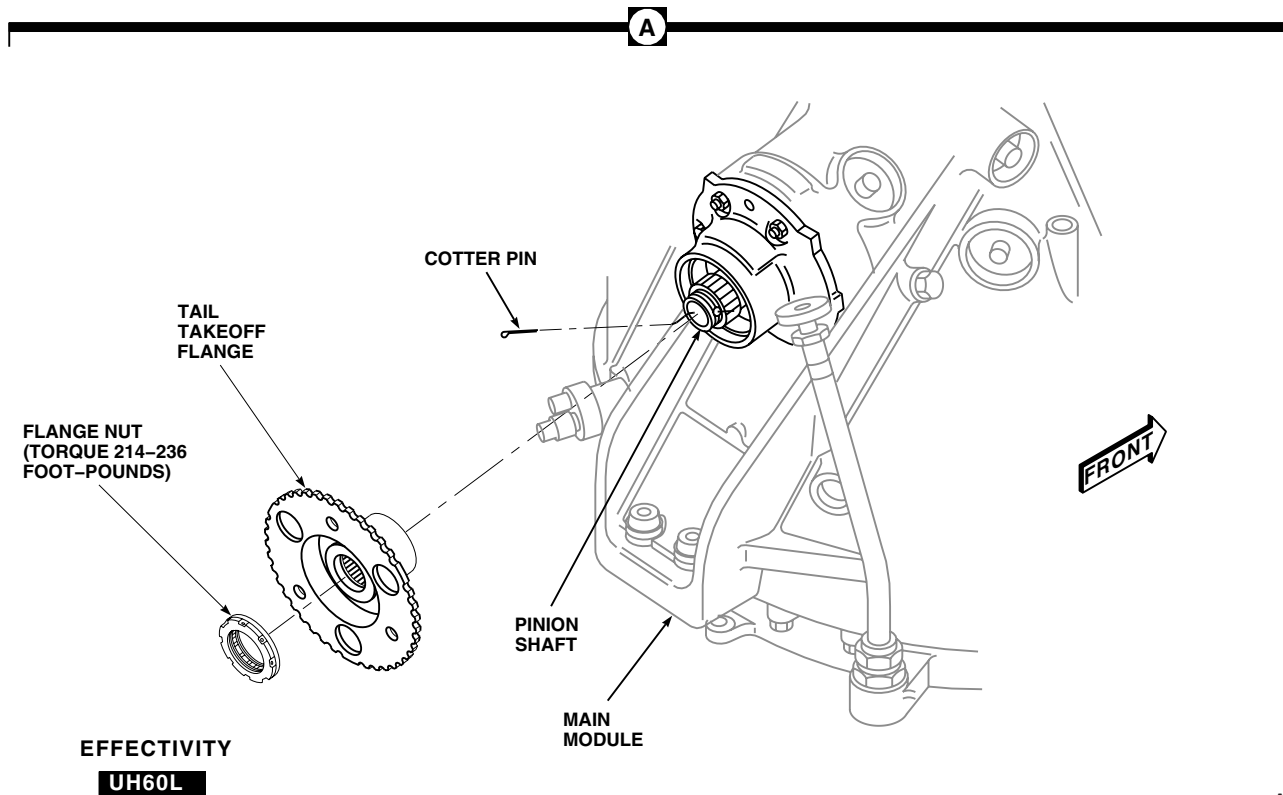
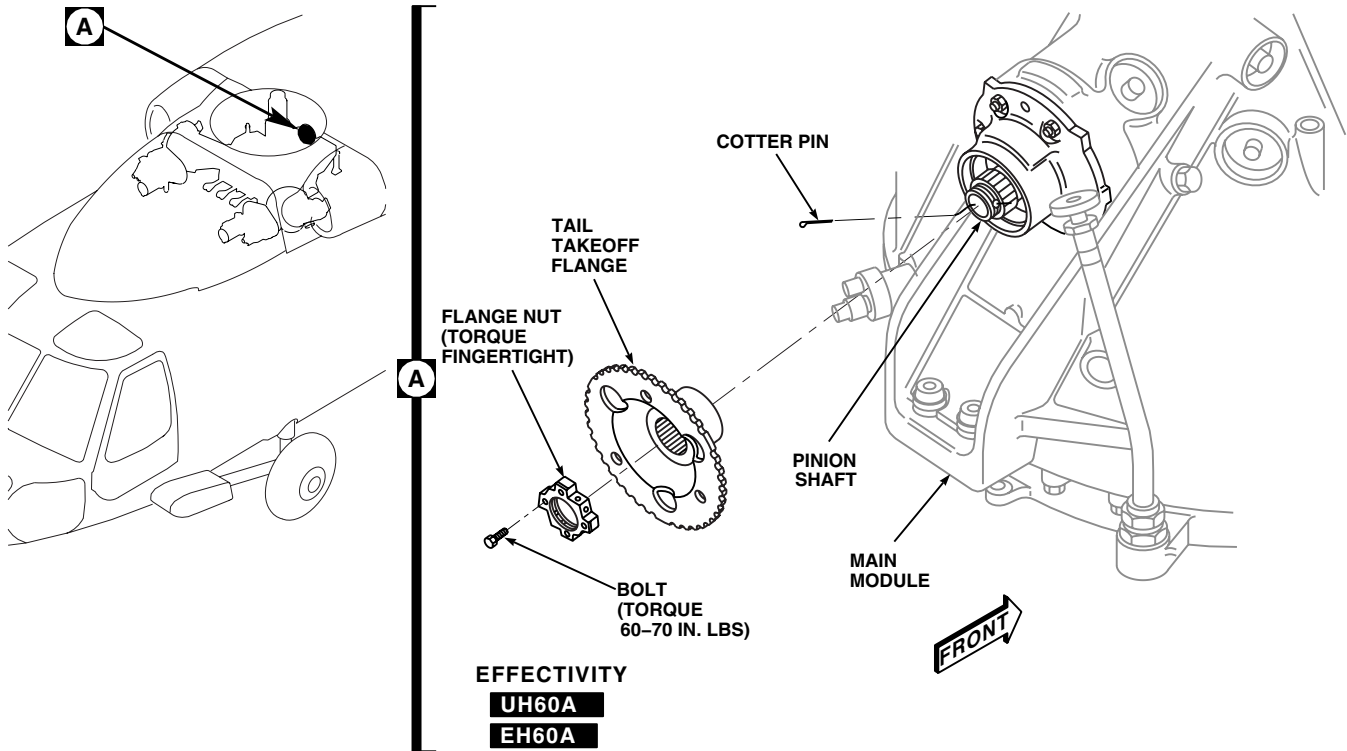
REMOVE

1. Tie down main and tail rotor blades to prevent blades from turning (WP 1676 00).
2. Release gust lock from inside cabin by pulling down on handle.
3. Open oil cooler compartment access doors.
4. Remove Section I tail rotor drive shaft. Keep coupling with shaft (WP 0666 00).
5. **UH60A EH60A** Remove tail takeoff flange as follows (Figure 1, Detail A):
 - a. Remove bolts from tail takeoff flange nut.
 - b. Remove cotter pin from flange nut. Unscrew and remove flange nut.
6. **UH60L** Remove tail takeoff flange as follows (Figure 1, Detail A).
 - a. Attach torque reactor to tail takeoff flange.
 - b. Loosen and remove nut attaching tail takeoff flange to pinion shaft.
 - c. Pull flange from pinion shaft.

INSPECT

1. Inspect chrome plated sleeve surface of flange for nicks or scratches (Figure 2). **NONE ALLOWED.** Replace flange if damage is found. Wear pattern caused by seal is not cause for replacement unless wear causes seal to leak.
2. Check present dimension of damaged area. Compare reading with minimum repair limits.
3. Check depth of damaged area to determine whether flange minimum dimension after repair can be maintained.
4. Inspect flange for scratches. Fine scratches need not be polished if not through paint.

INSPECT - Continued

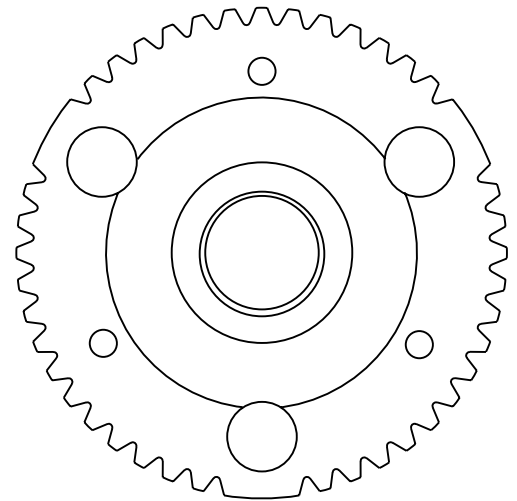


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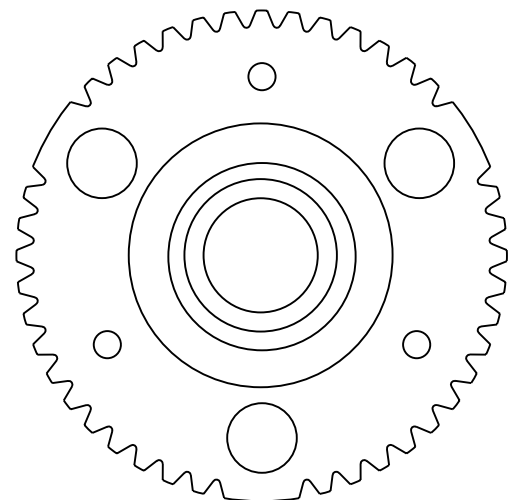
Figure 1. Remove/Install Main Transmission Tail Takeoff Flange.

EFFECTIVITY

EH60A



UH60L



ZONE	DIMENSION BEFORE REPAIR	MINIMUM DIMENSION AFTER REPAIR
A	0.175 (±0.005)	0.160
B	0.105 (±0.005)	0.090
C	0.125 (±0.010)	0.105

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INSPECT - Continued

5. Inspect splines for corrosion and wear. Evidence of corrosion of gear box splines is cause for removal of gear box. **NO WEAR INTO BASE METAL THROUGH PLATING IS PERMITTED (NO FRETTING OR SPALLING).**
6. Check internal spline dimension as follows:
 - a. Secure 0.072-inch diameter pin between internal splines.
 - b. Secure second 0.072-inch diameter pin between internal splines directly opposite first pin.
 - c. Measure inside diameter between two pins using inside micrometer. **INSIDE DIAMETER MUST NOT EXCEED 1.736-INCHES.** Replace flange if beyond limits.
 - d. Repeat check with pins located in four different positions.
7. Inspect flange nut for corrosion. **REPLACE NUT IF PITS ARE 0.030-INCH DEEP, IF PITTING COVERS MORE THAN 10% OF MATING SURFACE OR IF NUT THREADS SHOW SIGNS OF CORROSION.**
8. Remove corrosion from flange nut, using procedures in TM 1-1500-344-23.

REPAIR**NOTE**

- Rework of flange may be done on helicopter.
- Repair of minor damage such as nicks, scratches, or scoring shall be cleaned by polishing.
- Before starting repair on the flange, determine whether polishing out damage would result in a dimension below repairable limits. If damage repair is greater than permitted limits, replace flange (Figure 2).
- Polishing shall be done only in damaged area and not entire circumference, unless damage extends around it.

1. Remove damaged flange if necessary.
2. If damage can be repaired, blend smooth using abrasive cloth assortment, Item 1, WP 1803 00. Remove only what is necessary so flange will stay within limits. Blend radius on flange to +1/2-inch or greater.
3. Blend reworked area to remove all traces of abrasive marks and match surface smoothness with abrasive cloth, Item 82, WP 1803 00, or similar polishing cloth.
4. If flange splines have nicks, blend smooth using 150-grit abrasive cloth assortment, Item 1, WP 1803 00. **NICKS, DENTS, AND SCRATCHES MUST NOT EXCEED 0.010-INCH. NO RAISED AREAS ARE TO REMAIN AFTER POLISHING.**
5. Touch up paint on reworked areas per (WP 0381 00 and TM 55-1500-345-23). Do not paint around boltholes.

INSTALL

1. **QA** Slide takeoff flange on shaft (Figure 1, Detail A).
2. Put primer coating, Item 243, WP 1803 00, on inner side of nut.
3. **UH60A EH60A** Install flange nut as follows:
 - a. **QA INSTALL NUT AND TIGHTEN FINGERTIGHT. BACK OFF ENOUGH TO LINE UP COTTER PIN HOLE.** Insert cotter pin with head inside pinion shaft.
 - b. **QA INSTALL BOLTS ON FLANGE NUT AND TORQUE TO 60 - 70 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts.

INSTALL - Continued

4. **UH60L** Install flange nut as follows: 
 - a. Hand thread nut onto pinion shaft.
 - b. **QA** Using tail drive flange wrench and torque reactor, **TORQUE THE TAIL TAKEOFF RETAINING NUT AS FOLLOWS: THE TORQUE RANGE IS 214 - 236 FOOT-POUNDS. TORQUE TAIL TAKEOFF RETAINING NUT TO 214 FOOT-POUNDS.** Install new cotter pin with head inside pinion shaft. To align pin hole, turn nut in tightening direction. **DO NOT EXCEED MAX TORQUE. IF MAX TORQUE IS EXCEEDED AND NUT DOES NOT ALIGN, REMOVE NUT AND REPLACE IT WITH A NEW NUT AND REPEAT TORQUING PROCEDURE.**
5. Install Section I tail rotor drive shaft (WP 0666 00).
6. Make sure area is clean and free of foreign material.
7. Close oil cooler compartment access doors.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
MAIN TRANSMISSION OUTPUT SHAFT SEAL

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Ballpeen Hammer, 8-Ounce, GGG-H-868
 Barstock, 4 Inch Square
 Nylon Wedge
 Powertrain Repairer's Toolkit, SC-5180-99-B13
 Seal Installation Tool, Locally-Made, Figure 157,
 WP 1805 00
 Seal Puller Tool, Locally-Made, Figure 160,
 WP 1805 00

Material/Parts

Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Grease, Aircraft, Item 131, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 184,
 WP 1803 00

Lubricating Oil, Helicopter Transmission, Item 187,
 WP 1803 00
 Screws, No. 8 Self-Tapping
 Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0631 00
 WP 0666 00
 WP 1607 00
 WP 1803 00
 WP 1805 00

REMOVE**NOTE**

- Checking amount of seal leakage can prevent premature seal replacement before leakage is over recommended allowable limits. Leakage is gaged by measuring the drop in oil level as shown on dipstick. Total leakage must not be over 1/8-inch in any one flight. Allow at least 2 hours for oil to drain back before checking. If over 1/8-inch, replace seal.
- **UH60L** For extended flights over 3 hours, the allowable leakage is to be compatible with ten hour flight. Loss rate of 0.050-inch in 1 hour, 1/4-inch in 5 hours and 1/2-inch in 10 hours as indicated on dipstick. If loss of oil exceeds the above criteria, replace the seal. ■

1. Remove tail takeoff flange and Section I tail rotor drive shaft (WP 0631 00 and WP 0666 00).
2. Remove retaining ring (Figure 1).

NOTE

Self-tapping screws may be used as an alternate to remove oil seal if seal puller is not available. If seal puller is not available, proceed to step 8.

3. Locally make seal puller tool (WP 1805 00).
4. Locally make nylon wedge.

REMOVE - Continued

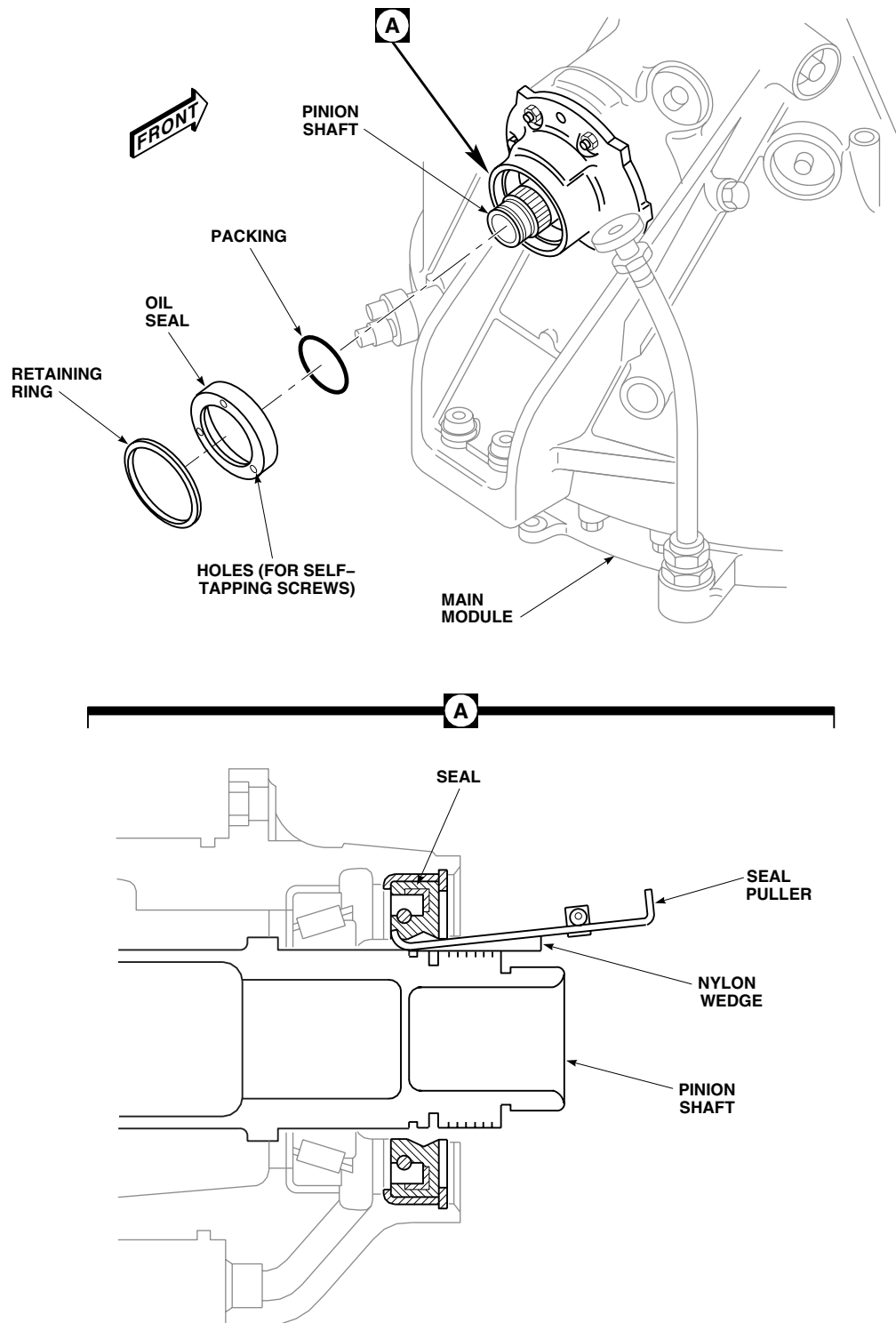
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Figure 1. Replace Main Transmission Output Shaft Seal.

REMOVE - Continued

5. Insert nylon wedge between seal puller and pinion shaft (Figure 1, Detail A) to protect gear box.

CAUTION

Damage to main module will result if too much force is applied while using hammer and seal puller. Gear box housing will crack. Use care when using hammer and seal puller.

6. Gently tap end of puller with an 8 ounce ballpeen hammer in an outward direction, away from gear box.
7. Moving seal puller and wedge in a clockwise direction, repeat step 5 and step 6 until seal is removed.
8. If seal puller is not available, remove oil seal from housing and pinion assembly as follows:

CAUTION

- Damage to main module will result if self-tapping screws completely penetrate oil seal. Screws will damage bearing inside main module. Install screws so that only outer face of oil seal is penetrated.
 - Damage to main module will result if metal chips or shavings enter gear box by drilling or punching holes in seal. Metal chips or shavings will contaminate gear box oil. Do not drill or punch holes in seal.
- a. Drill 5/32-inch holes in barstock (4 inches square) to align 1/8-inch holes in output shaft seal.
 - b. Insert 1/8 by 2 inch dia. long self-tapping screws through barstock and into output shaft seal holes.
 - c. Tighten screws until barstock plate is snugly seated against pinion shaft.
 - d. Tighten screws until output shaft seal is removed. **EXERCISE CAUTION NOT TO DAMAGE PINION SHAFT.**
9. Remove packing from pinion shaft. Discard packing.

INSTALL

1. Locally make seal installation tool (WP 1805 00).
2. **QA** Install packing on pinion shaft in groove behind splines.
3. Coat inside diameter of seal with aircraft grease, Item 131, WP 1803 00. Coat outside diameter of seal and seal mating surface (flange sleeve) with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
4. **QA** Locally make seal installation tool (WP 1805 00). Carefully install the oil seal by lightly tapping into housing and pinion assembly. Make sure seal is fully seated and beyond retaining ring groove. Be careful not to cock or distort the seal.
5. **QA** Apply epoxy primer coating kit, Item 120, WP 1803 00, to retaining ring and install in housing while wet.
6. Apply sealing compound, Item 273, WP 1803 00, to housing boss to prevent water entrapment and corrosion above retaining ring groove.
7. Touch up sealing compound with epoxy primer coating kit, Item 120, WP 1803 00.
8. Install takeoff flange and Section I tail rotor drive shaft (WP 0631 00 and WP 0666 00).
9. Service main transmission (WP 1607 00).

INSTALL - Continued

10. Run up helicopter and check for leaks.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****MAIN TRANSMISSION BREATHER PLUG**

This WP supersedes WP 0633 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC-5180-99-B13

Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00

References

WP 0646 00
WP 0653 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open oil cooler compartment access doors.
3. If breather plug is being replaced due to oil leakage, do this:
 - a. Check transmission for proper oil level.
 - b. Check transmission oil filler cap for evidence of debris or ice.
 - c. Check input module return tubes for proper torque (WP 0653 00).
 - d. Check transmission chip detectors for evidence of leaks (WP 0646 00).
4. Using nonmetallic scraper, remove sealing compound from breather plug.

NOTE

Do not remove screws from breather plug.

5. Remove retaining ring securing breather plug to transmission housing (Figure 1, Detail A).
6. Install bolt in center of breather plug.
7. Grasp bolt and remove breather plug from transmission housing.
8. Remove packing from breather plug. Discard packing.
9. Remove bolt from breather plug.

INSTALL

1. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00.
2. **QA** Install packing on breather plug (Figure 1, Detail A).

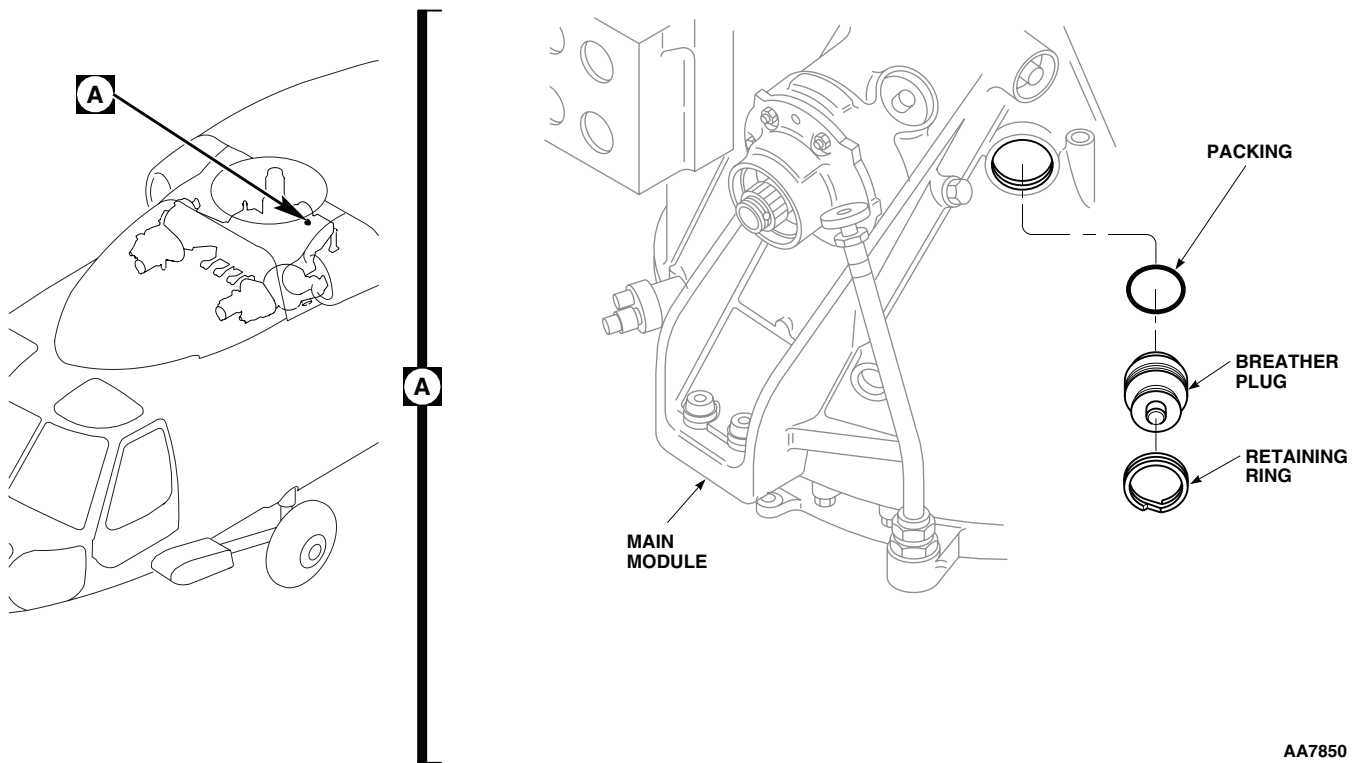
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Figure 1. Replace Breather Plug.

NOTE

Breather plug must be installed with a hole at bottom (6 o'clock) position to allow oil to drain.

3. **QA** Install breather plug in transmission with hole at (6 o'clock) position.
4. **QA** Install retaining ring into transmission housing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean breather plug and surrounding surface using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
6. Apply a bead of sealing compound, Item 273, WP 1803 00, around breather plug.
7. Close oil cooler compartment access doors.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

SWASHPLATE GUIDE AND SHAFT SEAL

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Depth Gage
Mallet, GGG-H-33
Nonmetallic Drift
Powertrain Repairer's Toolkit, SC 5180-99-CL-A13
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Assistants - (2)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
TM 1-1520-237-MTF
WP 0547 00
WP 0567 00
WP 0576 00
WP 1037 00
WP 1607 00
WP 1803 00

Material/Parts

Assembly Fluid, Item 54, WP 1803 00
Cloth, Abrasive, Item 78, WP 1803 00
Corrosion Resistant Coating, Item 111, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Grease, Aircraft, Item 131, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

REMOVE

1. Do a four-point rigging check (WP 1037 00).
2. Turn off all electrical power.
3. Remove main rotor blades (WP 0576 00).
4. Remove main rotor head (WP 0547 00).
5. Remove swashplate assembly (WP 0567 00).
6. Match mark swashplate guide housing and main module housing using chalk or grease pencil.

CAUTION

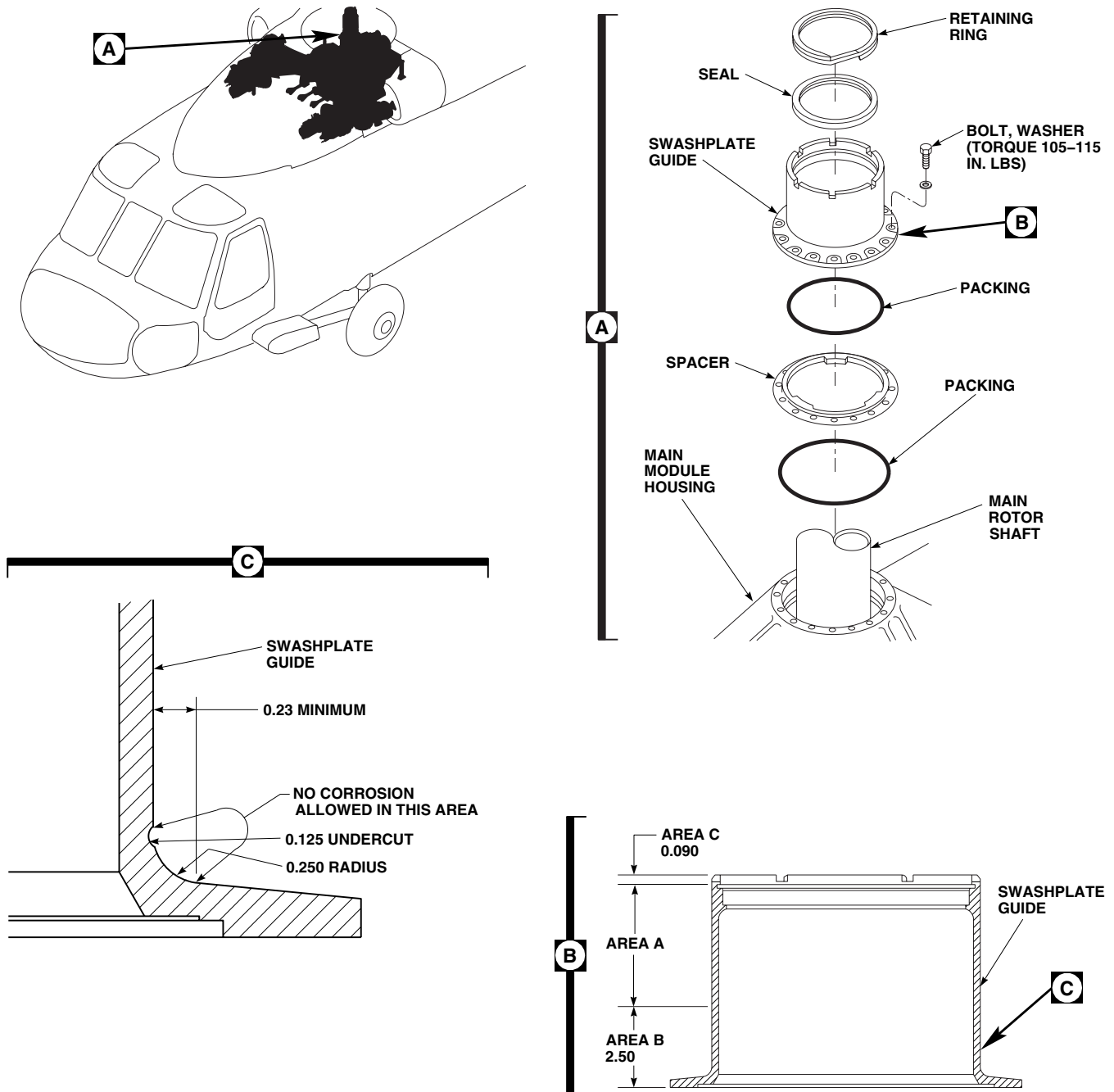
Damage to main module will result if foreign material enters housing. Use extreme care to keep foreign material out of main module.

7. Remove swashplate guide housing from main module and discard packing (Figure 1, Detail A).
8. Remove spacer and discard packing.
9. Remove retaining ring, turn housing over and tap out seal.

INSPECT/REPAIR

1. Inspect swashplate guide in AREA A and AREA B for wear and scoring (Figure 1, Detail B). **WEAR AND SCORING SHALL NOT PENETRATE INTO BASE METAL.**

INSPECT/REPAIR - Continued

**NOTE**

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

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Figure 1. Replace Swashplate Guide and Shaft Seal.

INSPECT/REPAIR - Continued

2. Inspect AREA A of swashplate guide surface for peeling, chipping, and flaking. **DAMAGE SHALL NOT EXCEED 0.250-INCH DIAMETER. NO MORE THAN 2 AREAS OF SURFACE DAMAGE WITHIN 2 INCHES OF EACH OTHER ARE ALLOWED. NO MORE THAN 20 TOTAL AREAS OF SURFACE DAMAGE.** Replace guide if damage is beyond limits or if damage penetrates into base metal.
3. If surface damage in AREA A is within limits, gently smooth edge of damage with abrasive cloth, Item 78, WP 1803 00.
4. Inspect AREA B of swashplate guide surface for peeling, chipping, or flaking. **NONE ALLOWED.** Replace guide if damage is found.
5. Inspect AREA C of swashplate guide for nicks, dents, or scratches. Minor damage may be blended smooth using abrasive cloth, Item 78, WP 1803 00. **DEPTH OF DAMAGE OR REWORK MUST NOT EXCEED 0.040-INCH.** Replace guide if damage or rework is beyond limits.

NOTE

Do all blending of corrosion with abrasive cloth, Item 78, WP 1803 00.

6. Inspect swashplate guide for corrosion.
 - a. **NO CORROSION ALLOWED IN 0.250-INCH RADIUS, OR 0.125-INCH UNDERCUT (Figure 1, Detail C).**
 - b. **SURFACE CORROSION IN AREA C MAY BE BLENDED OUT UP TO 0.020-INCH DEEP OVER NOT MORE THAN 20% OF SURFACE AREA.**
 - c. **SURFACE CORROSION ON FLANGE MAY BE BLENDED OUT UP TO 0.010-INCH DEEP OVER NOT MORE THAN 10% OF SURFACE AREA.**
 - d. Replace swashplate guide if damage or rework is beyond limits.
7. Apply corrosion resistant coating, Item 111, WP 1803 00, to all reworked areas (TM 1-1500-344-23).

INSTALL

1. **QA** Install seal using a rawhide mallet and nonmetallic drift. Tap seal evenly and carefully until retaining ring can be installed in slot on guide assembly.
2. **QA** Install retaining ring.
3. Install replacement packing on swashplate guide housing. Hold packing in place using assembly fluid, Item 54, WP 1803 00.
4. **QA** Install replacement packing in main module housing. Hold packing in place using assembly fluid, Item 54, WP 1803 00.
5. Apply a coat of epoxy primer coating kit, Item 120, WP 1803 00, to main module mating surface outboard of packing groove.
6. Install spacer.
7. Pack ID of seal with aircraft grease, Item 131, WP 1803 00.
8. Apply a light coat of aircraft grease, Item 131, WP 1803 00, to seal mating surface on shaft.

NOTE

If same swashplate guide is being installed, line up marks on swashplate guide and main module.

9. Line up match marks on swashplate guide housing and main module housing.

INSTALL - Continued

10. **QA** Apply a coat of epoxy primer coating kit, Item 120, WP 1803 00, to swashplate guide, underside the flange.
Install swashplate guide with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
11. Seal area around bolts, washers, nuts, and threads of bolts to nuts with sealing compound, Item 273, WP 1803 00.
12. Seal assembly with sealing compound, Item 273, WP 1803 00.
13. Remove excessive sealing compound.
14. Reinstall swashplate assembly (WP 0567 00).
15. Reinstall main rotor head (WP 0547 00).
16. Reinstall main rotor blades (WP 0576 00).
17. Service main transmission (WP 1607 00).
18. Make sure area is clean and free of foreign material.
19. Perform a main rotor rig/check (WP 1037 00).
20. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****MAIN ROTOR SHAFT PLUG**

This WP supersedes WP 0635 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 30 - 200 in. lbs.
Wood, 2 inch x 3 1/2-inch x 55 inch

Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Coating Kit, Rain Erosion, Item 94, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00

References

WP 0575 00
WP 1607 00
WP 1803 00

NOTE

Main rotor shaft plug is found only on helicopters without blade deice distributor and slipring installed.

REMOVE

1. Drain transmission (WP 1607 00).
2. Remove bifilar cover (WP 0575 00).
3. Remove nuts, washers, and main rotor shaft access cover from sump. Discard packing (Figure 1, Detail B).
4. Place a 2 x 3 1/2-inch x 55 inch piece of wood inside main rotor shaft. Tap end of wood with hammer until plug at top of shaft is unseated.
5. Remove main rotor shaft plug from inside shaft extension.

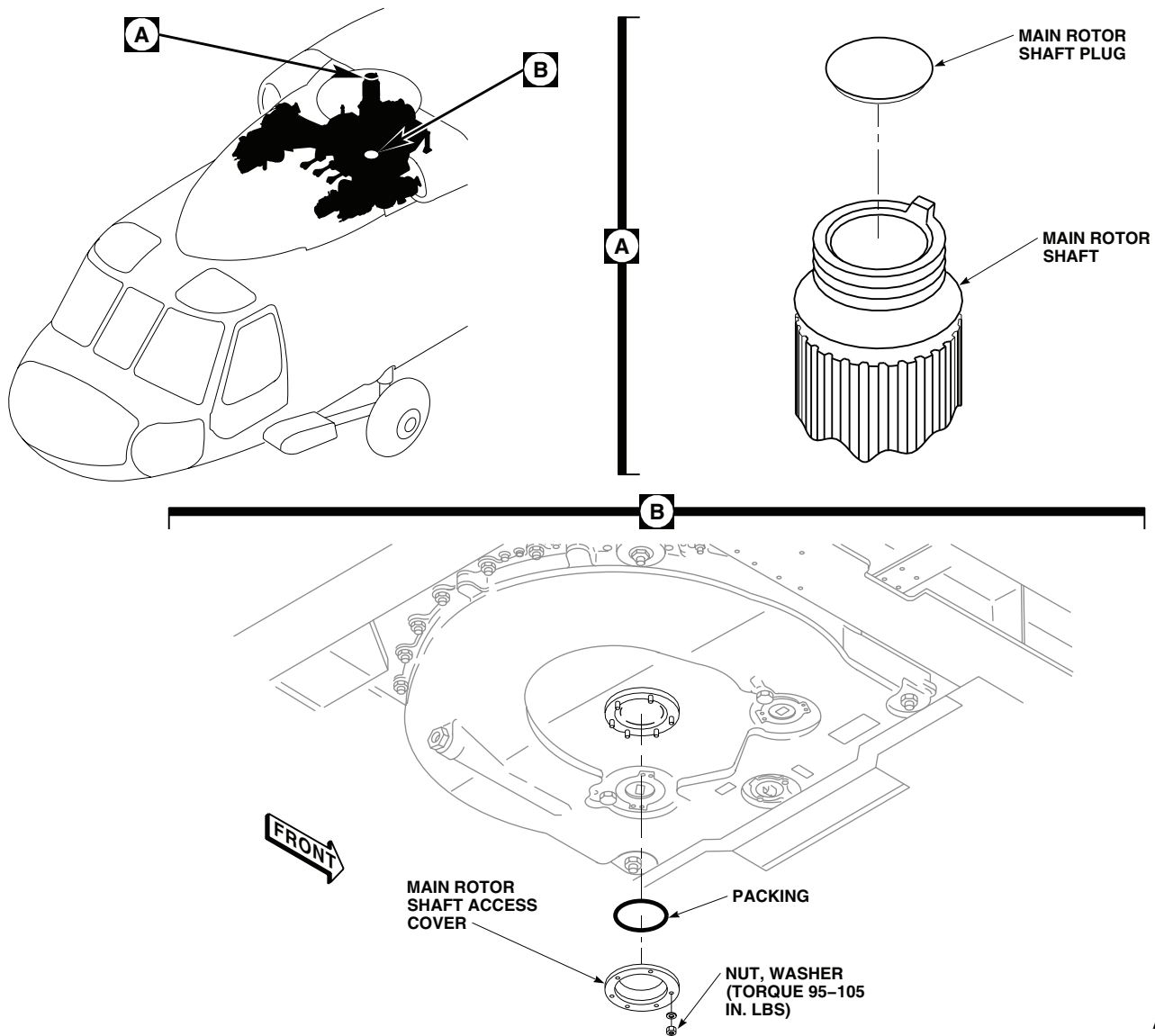
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Figure 1. Replace Main Rotor Shaft Plug.

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean areas to be bonded with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

INSTALL - Continued

2. Apply thin coat of rain erosion coating kit, Item 94, WP 1803 00, to bonding surfaces. Allow to cure at least 1 hour at 70°F(21°C) to 75°F(23°C).
3. Apply adhesive, Item 11, WP 1803 00, to bonding surfaces. Use weight (1 to 3 pounds) to secure shaft plug in place. Allow adhesive to cure for 2 hours at 110°F(43°C) to 130°F(54°C), or for 24 hours at 70°F(21°C) to 75°F(23°C) (Figure 1, Detail A).
4. Seal edges of plug with sealing compound, Item 273, WP 1803 00.
5. **QA** Install new packing, lubricated with aircraft turbine lubricating oil, Item 182, WP 1803 00, on access cover (Figure 1, Detail B).
6. **QA** Install main rotor shaft access cover on bottom of main module with nuts and washers. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
7. **QA** Install bifilar cover (WP 0575 00).
8. Service transmission (WP 1607 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****MAIN ROTOR SHAFT INSPECT/REPAIR**

This WP supersedes WP 0636 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-CL-B13

Epoxy Coating Kit, Item 118, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 12, WP 1803 00
Brush, Acid Swabbing, Item 57, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Abrasive, Item 78, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Preventive Compound, Item 103,
WP 1803 00
Corrosion Removing Compound, Item 108,
WP 1803 00

References

WP 0547 00
WP 0575 00
WP 0576 00
WP 0634 00
WP 0635 00
WP 1220 00
WP 1803 00

REMOVE

1. Engage gust lock.
2. If installed, remove deice distributor from main rotor shaft (WP 1220 00).

CAUTION

Damage to main rotor head assembly will result if metallic tools are used to remove sealant. Use nonmetallic scraper when removing sealant.

3. If installed, remove bifilar cover (WP 0575 00).
4. Remove upper main rotor shaft nut (WP 0547 00).
5. If installed, remove main rotor shaft plug (WP 0635 00).

INSPECT/REPAIR**CAUTION**

Damage to main gear box will result if foreign material enters gear box. To prevent damage, mask off main gear box bore where main rotor shaft passes through.

NOTE

If damage cannot be reached due to installation of shaft extension or swashplate guide, removal of main rotor blades, main rotor head, and swashplate guide may be required.

1. Remove main rotor blades (WP 0576 00).
2. Remove main rotor head (WP 0547 00).
3. Remove swashplate guide (WP 0634 00).

WARNING**CSI**

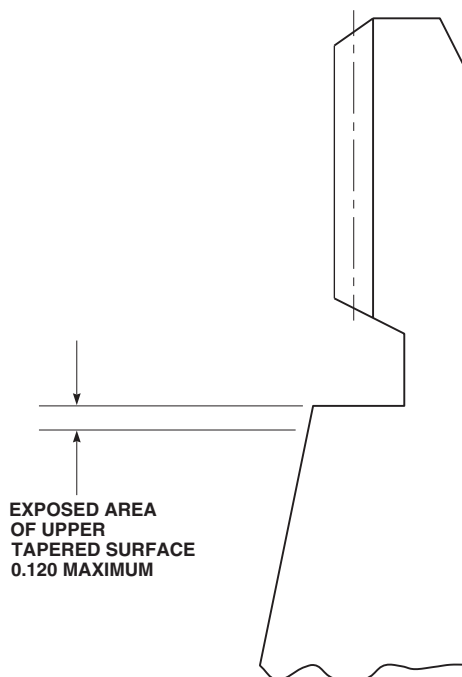
Nicks, dents, scratches, and tool or scuff marks within the critical characteristic area shall not exceed acceptance limits.

4. Inspect main rotor shaft surface for nicks, dents, scratches, and tool marks. **NONE ALLOWED.** Replace main module if damage is found.
5. Inspect main rotor shaft areas excluding lower split cone surface and upper tapered surface for corrosion. **NONE ALLOWED.** Replace main module if corrosion is found.
6. Inspect main rotor shaft lower split cone surface and upper tapered surface for corrosion and repair as follows (Figure 1):
 - a. Apply corrosion removing compound, Item 108, WP 1803 00, to corroded areas using acid swabbing brush, Item 57, WP 1803 00. Allow to dry for 30 seconds. Wipe dry with machinery wiping towel, Item 344, WP 1803 00.
 - b. Blend any corrosion remaining after application of corrosion removal and treatment compound per step 6.(a) using abrasive cloth, Item 82, WP 1803 00.
 - c. **REPAIR IN EXPOSED AREA OF UPPER TAPERED SURFACE MAY BE UP TO 100% OF AREA DEFINED IN FIGURE 1, UP TO 0.010-INCH DEEP.** Replace main module if repairs exceed limits.
 - d. **ON REMAINING UPPER TAPERED SURFACE AND LOWER SPLIT CONE SURFACE, NO SINGLE REPAIRED AREA SHALL EXCEED 1 SQUARE INCH. TOTAL REPAIRED AREA SHALL NOT EXCEED 20% OF SURFACE AND SHALL NOT EXCEED 0.010-INCH DEEP.** Replace main module if repairs exceed limits.
 - e. Apply thin coat of corrosion preventive compound, Item 103, WP 1803 00, to reworked areas.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Clean adhesive on area below main rotor shaft splines with technical acetone, Item 5, WP 1803 00.

INSPECT/REPAIR - Continued**NOTE**

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

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Figure 1. Inspect/Repair Main Rotor Shaft.

8. Inspect adhesive on area below main rotor shaft splines for damage extending into base metal. If damage extends into base metal, replace gear box.
9. If damage does not extend into base metal, do this.
 - a. Blend out epoxy coating kit damage using abrasive cloth, Item 78, WP 1803 00.
 - b. Clean repaired area using cheesecloth cloth, Item 85, WP 1803 00.
 - c. Apply adhesive, Item 12, WP 1803 00, to reworked areas. Allow adhesive to dry for 24 hours at room temperature.
 - d. Apply one coat of epoxy coating kit, Item 118, WP 1803 00.

INSTALL**WARNING****CSI**

Exposed surfaces of main rotor shaft are critical surfaces which must not be damaged during installation of components. After removal of protective covering, use extreme caution when installing components.

1. Install main rotor head (WP 0547 00).
2. Install main rotor blades (WP 0576 00).
3. If required, install main rotor blade deice distributor (WP 1220 00).

INSTALL - Continued

4. Install main rotor shaft plug if blade deice is not installed (WP 0635 00).
5. **QA** Replace bifilar cover if blade deice is not installed (WP 0575 00).
6. Disengage gust lock.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****GUST LOCK ROD AND LEVER**

This WP supersedes WP 0637 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill, 0.072-inch
Gloves
Goggles/Face Shield
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0259 00
WP 0260 00
WP 0261 00
WP 0261 01
WP 0638 00
WP 1803 00

REMOVE**WARNING**

Injury to personnel and damage to equipment will result if gust lock rod and lever are removed without rotor blades being secure. Make sure rotor blades are secured before removing gust lock rod and lever.

1. Turn off all electrical power.
2. Make sure rotor blades are tied down.
3. Open oil cooler compartment access doors.
4. Remove cotter pin, pin, washers and bushing and disconnect upper end of gust lock rod from gust lock lever (Figure 1, Detail A).
5. Remove transmission drip pan (WP 0259 00, WP 0260 00, WP 0261 00, or WP 0260 01).
6. Remove seal pad above gust lock handle.
7. Unfasten gust lock rod by drilling out pin and collar. Remove gust lock rod by pulling rod up and rotating it 90 degrees so handle will pass through slot in cabin ceiling.
8. Loosen jamnut and remove terminal from gust lock rod.
9. Remove nut and remove gust lock lever from lever bracket. Remove bushing.

REMOVE - Continued

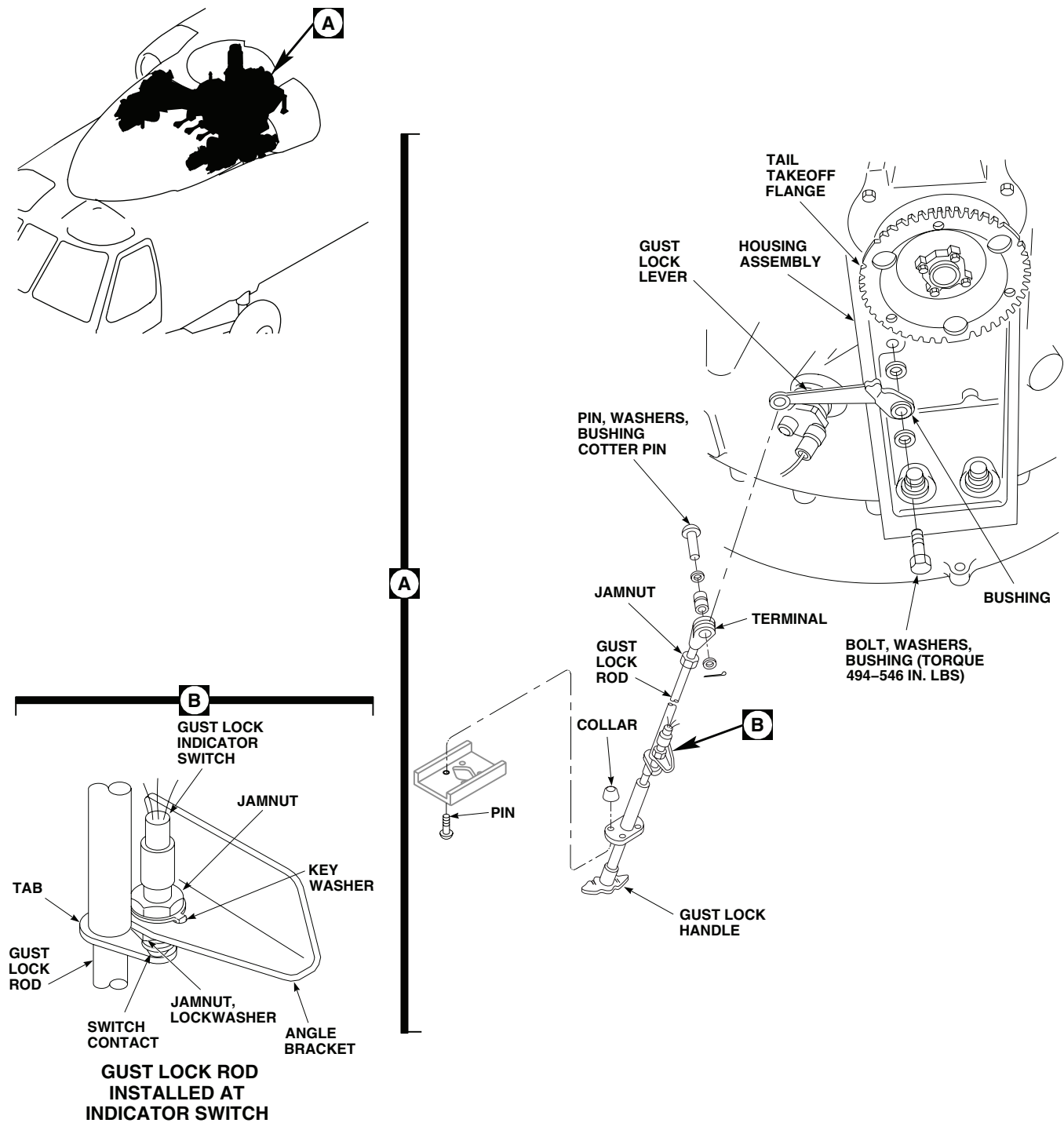


Figure 1. Replace Gust Lock Rod and Lever.

INSTALL

CAUTION

Damage to recess in frame or beam will occur if barrel-nut is incorrectly installed. To prevent damage to recess, make sure nut is installed with barrel (round side) toward bolthead.

NOTE

Make sure that only square barrel-nuts are used.

1. Test existing barrel-nut in housing assembly that attaches gust lock lever as follows:
 - a. Remove barrel-nut from housing assembly.
 - b. Place barrel-nut in soft-jawed vise.
 - c. Thread in bolt until two threads are showing beyond nut.
 - d. Back off bolt. **BREAKING TORQUE SHALL NOT BE LESS THAN 18 INCH-POUNDS.**
 - e. If breaking torque is 18 inch-pounds or more, bolt and barrel-nut may be reused. Reinstall barrel-nut in housing assembly.
 - f. If breaking torque is less than 18 inch-pounds, replace bolt and barrel-nut. Repeat steps 1. (b) through 1. (d).
2. **QA** Position gust lock lever, bushing, and washers against lever bracket and install bolt, washers, and nut. **TORQUE BOLT TO 494 - 546 INCH-POUNDS (Figure 1).**
3. **QA** Install jamnut and terminal on upper end of gust lock rod, handtight only.
4. **QA** Install gust lock rod by lowering handle through slot and turning 90 degrees so mounting plate holes line up with holes in structure and tab on rod lines up with switch. Attach lower end of gust lock rod in cabin ceiling with pins, and collars.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean seal pad mounting surface with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
6. Cement seal pad on cabin ceiling above gust lock handle with adhesive, Item 14, WP 1803 00.
7. **QA** Mesh tooth on tail takeoff flange with gust lock lever pawl. Adjust terminal so that when gust lock handle is engaged (in), holes in terminal and lever are lined up. Install pin and washer through terminal and lever. Make sure gust lock lever is fully engaged.
8. **QA TIGHTEN TERMINAL JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire jamnut.
9. **QA** Secure pin attaching terminal to gust lock lever with bushing, washers, and cotter pin.
10. Adjust gust lock (WP 0638 00).

INSTALL - Continued

11. Install transmission drip pan (WP 0259 00, WP 0260 00, WP 0261 00, or WP 0261 01).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

GUST LOCK INDICATOR SWITCH

INITIAL SETUP:**Tools and Special Tools**

Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)

Material/Parts

Tags

Wire, Nonelectrical, Item 354, WP 1803 00

References

TM 55-1500-323-24

WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Tie down rotor blades.
3. Open oil cooler compartment access doors.

CAUTION

Damage to gust lock will result if operated improperly. To prevent accidental gust lock operation and damage, gust lock handle button must be out when gust lock is fully engaged or disengaged.

4. Fully disengage gust lock. Measure and record the distance between the gust lock rod tab, and the gust lock indicator switch contact.
5. Tag wires and disconnect wire splice connections between gust lock indicator switch and harness assembly.
6. Remove jamnuts, washer, key washer, and indicator switch from angle bracket.

INSTALL

1. **QA** Assemble gust lock switch with upper jamnut and keywasher in position on angle bracket.
2. **QA** Add lockwasher and lower jamnut to indicator switch assembly, and position switch contact with gust lock rod tab to recorded measurement. **TIGHTEN JAMNUTS 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
3. Connect gust lock switch wiring to harness assembly using wire splice connections (TM 55-1500-323-24).
4. Turn on all electrical power.
5. Actuate gust lock in both engaged and disengaged positions, checking GUST LOCK indicator light on caution/advisory panel for proper operation.
6. Adjust gust lock switch if required. Refer to, ADJUST, in this work package. Using nonelectrical wire, Item 354, WP 1803 00, to lockwire jamnuts.
7. Turn off all electrical power.
8. Make sure area is clean and free of foreign material.
9. Close oil cooler compartment access doors.

ADJUST

1. Turn off all electrical power.
2. Open oil cooler compartment access doors.
3. Tie down rotor blades.
4. Turn on all electrical power.

CAUTION

Damage to gust lock will result if operated improperly. To prevent accidental gust lock operation and damage, gust lock handle button must be out when gust lock is fully engaged or disengaged.

5. Engage gust lock. Loosen jamnuts and position gust lock indicator switch up or down on angle bracket until GUST LOCK indicator on caution/advisory panel lights up when gust lock rod tab actuates indicator switch contact.
6. **QA** Tighten gust lock indicator switch in position with jamnuts. **TIGHTEN JAMNUTS 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
7. Actuate gust lock in both engaged and disengaged positions checking GUST LOCK indicator light on caution/advisory panel for proper operation.
8. **QA** Realign gust lock indicator switch if necessary. Using nonelectrical wire, Item 354, WP 1803 00, to lockwire jamnuts.
9. Turn off all electrical power.
10. Make sure area is clean and free of foreign material.
11. Close oil cooler compartment access doors.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

MAIN MODULE HOUSING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Beaker (10 ml), MIL-B-36268
Beaker (600 ml), MIL-B-36268
Beaker (2000 ml), MIL-B-36268
Container, Plastic, 1-Pint
Depth Gage, 643J
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13

Material/Parts

Acetic Acid, Item 4, WP 1803 00
Applicator, Disposable, Item 53, WP 1803 00
Chromium Trioxide, Item 65, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00
Corrosion Preventive Compound, Item 106,
WP 1803 00

Distilled Water, Reagent, Item 117, WP 1803 00
Epoxy Coating Kit, Item 118, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Ethylene Glycol, Technical, Item 123, WP 1803 00
Nitric Acid, Item 198, WP 1803 00
Polyurethane Coating, Item 231, WP 1803 00
Remover, Paint, Item 252, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Tape, Pressure Sensitive Adhesive, Item 323,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
WP 1803 00

GENERAL**WARNING****CSI**

Surfaces in the critical characteristic area shall be free of casting defects such as parting lines, sand pits, chills nicks, or dents. Area to be blended smooth to adjacent surface and minimum wall thickness of 0.28-inches is a critical characteristic.

NOTE

- Minor nicks, gouges, and scratches on surface of main module housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in main module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace main module.
- Minor nicks, gouges, scratches, and corrosion within snapping groove of jet or breather plug may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in main module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace main module.
- Repair of damaged area may be done with main module installed.
- All repairs shall be recorded on Aircraft Component Historical Record, DA Form 2408-16.

1. Clean any oil or grease from damaged area using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.

NOTE

Housings coated with epoxy coating kit are glossy silver. Housings coated with aircraft gray polyurethane paint are matte surface medium-gray.

2. Fine scratches which remove only paint may be blended smooth using abrasive cloth, Item 82, WP 1803 00. After smoothing out, touch up blended area using epoxy coating kit, Item 118, WP 1803 00, or polyurethane coating, Item 231, WP 1803 00, as required.

REPAIR LIMITS

1. Main module housing repair limits are as follows:
 - a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES AND ADJACENT RIBS OF MAIN MODULE IS 0.030-INCH (Figure 1, Sheet 1).**
 - b. **MAXIMUM DEPTH OF MATERIAL REMOVAL ON MOUNTING FEET AND MOUNTING FEET RIBS IS 0.030-INCH (Figure 1, Sheet 1, Detail A).**
 - (1) **NO MORE THAN TWO REPAIRS ALLOWED PER FOOT.**
 - (2) **REPAIRS SHALL NOT EXCEED 10% OF TOTAL SURFACE AREA OF FOOT.**
 - (3) **NO MATERIAL REMOVAL FROM BUSHING BORE ALLOWED.**

REPAIR LIMITS - Continued

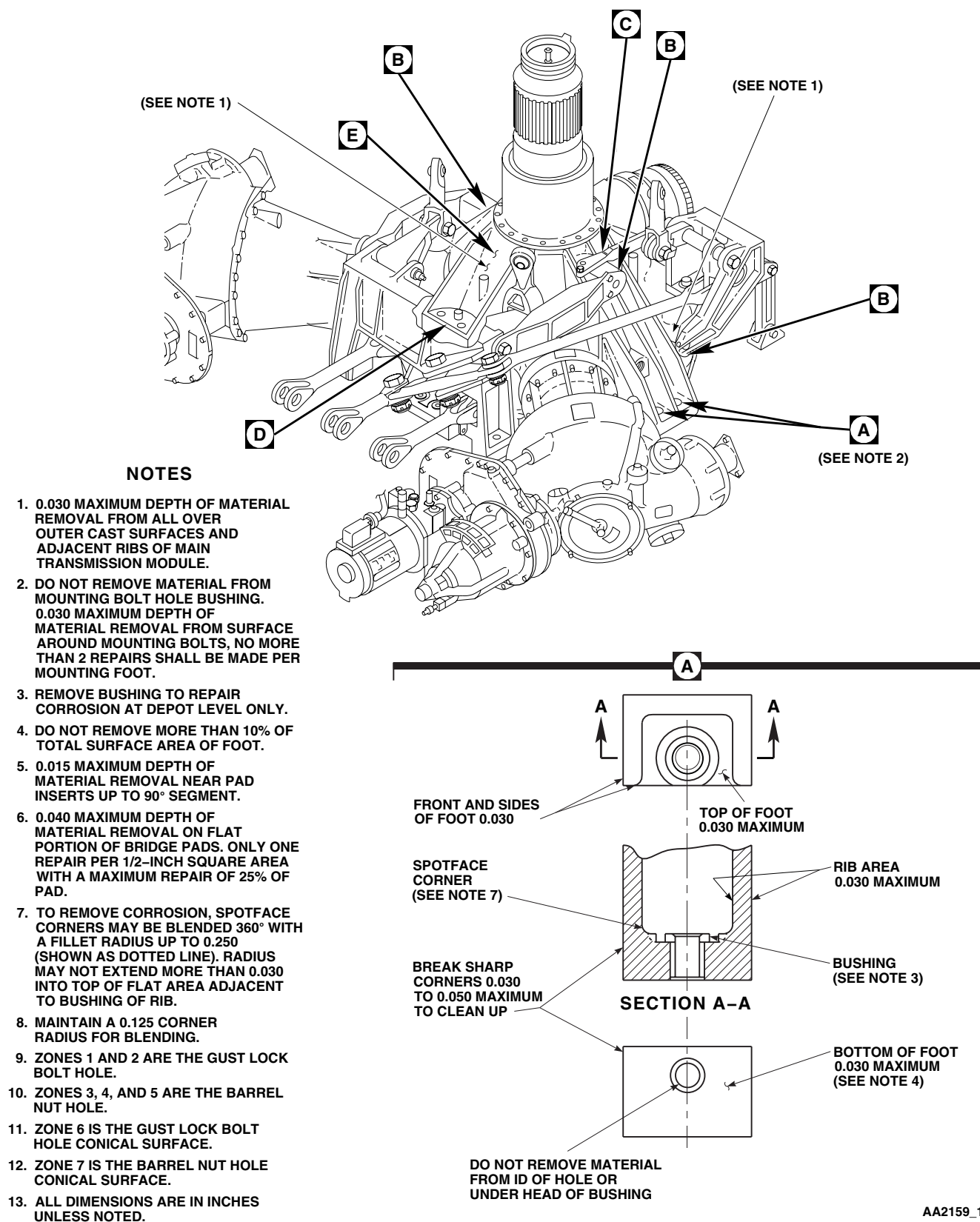


Figure 1. Repair Main Module Housing. (Sheet 1 of 4)

REPAIR LIMITS - Continued

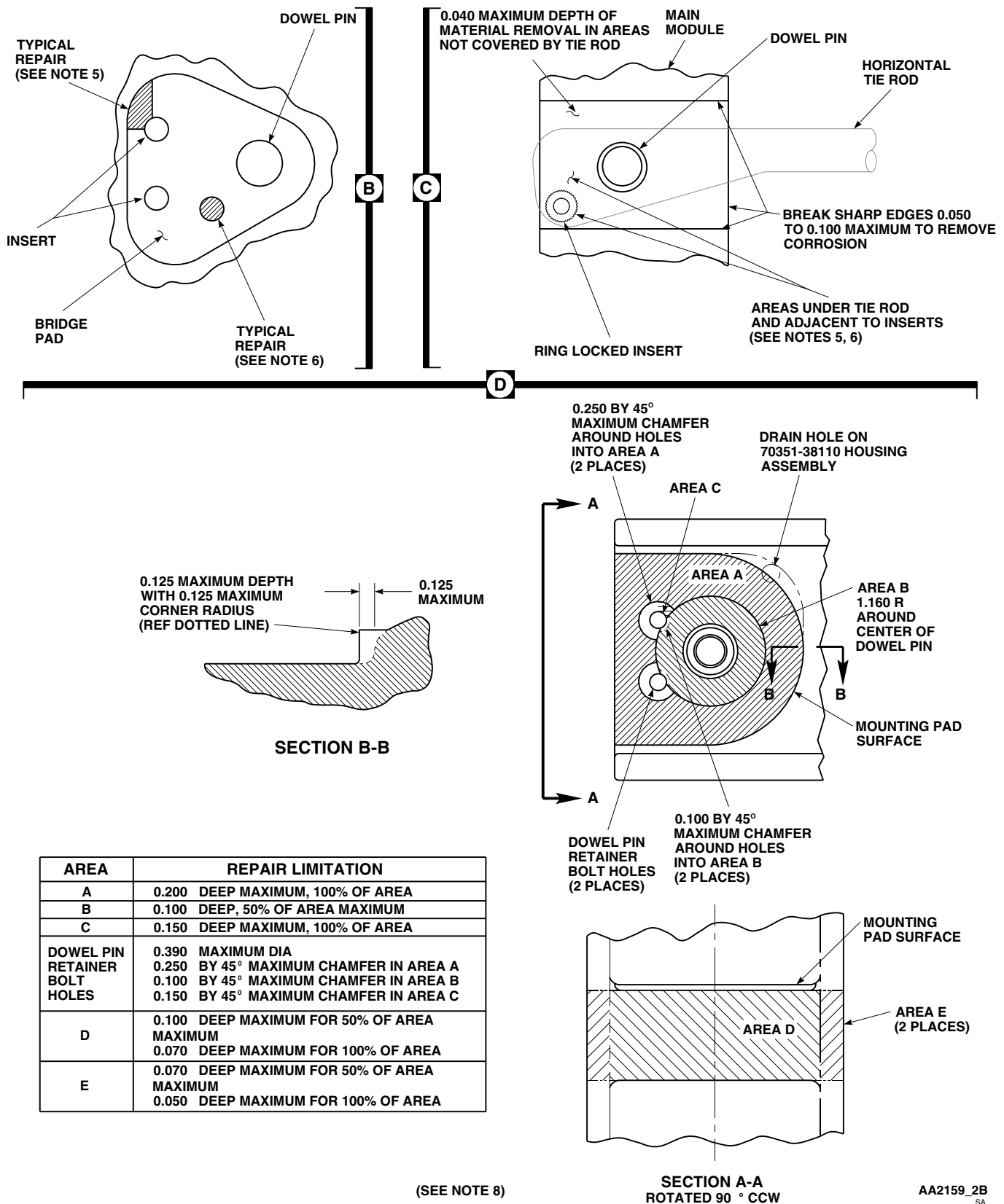


Figure 1. Repair Main Module Housing. (Sheet 2 of 4)

E

(SEE SECTION B-B)

(SEE SECTION C-C)

(SEE SECTION B-B)

(SEE SECTION C-C)

(SEE SECTION B-B)

(SEE SECTION C-C)

(SEE SECTION B-B)

(SEE SECTION C-C)

F

(SEE SECTION B-B)

(SEE SECTION C-C)

(SEE SECTION B-B)

SECTION B-B
TYPICAL JET ASSEMBLY

JET

(SEE VIEW A)

SECTION C-C
TYPICAL BREATHER PLUG ASSEMBLY

PLUG

(SEE VIEW B)

JET

B

0.030 MINIMUM FILLET RADIUS

A

BOTTOM OF SNAP RING GROOVE

UP TO 20% OF CIRCUMFERENCE OF SHADED AREA A AND AREA B MAY BE REMOVED

VIEW A

PACKING

BOTTOM OF SNAP RING GROOVE

B

A

UP TO 20% OF CIRCUMFERENCE OF SHADED AREA A AND AREA B MAY BE REMOVED

VIEW B

AA2159_3A
SA

0639 00-5

REPAIR LIMITS - Continued

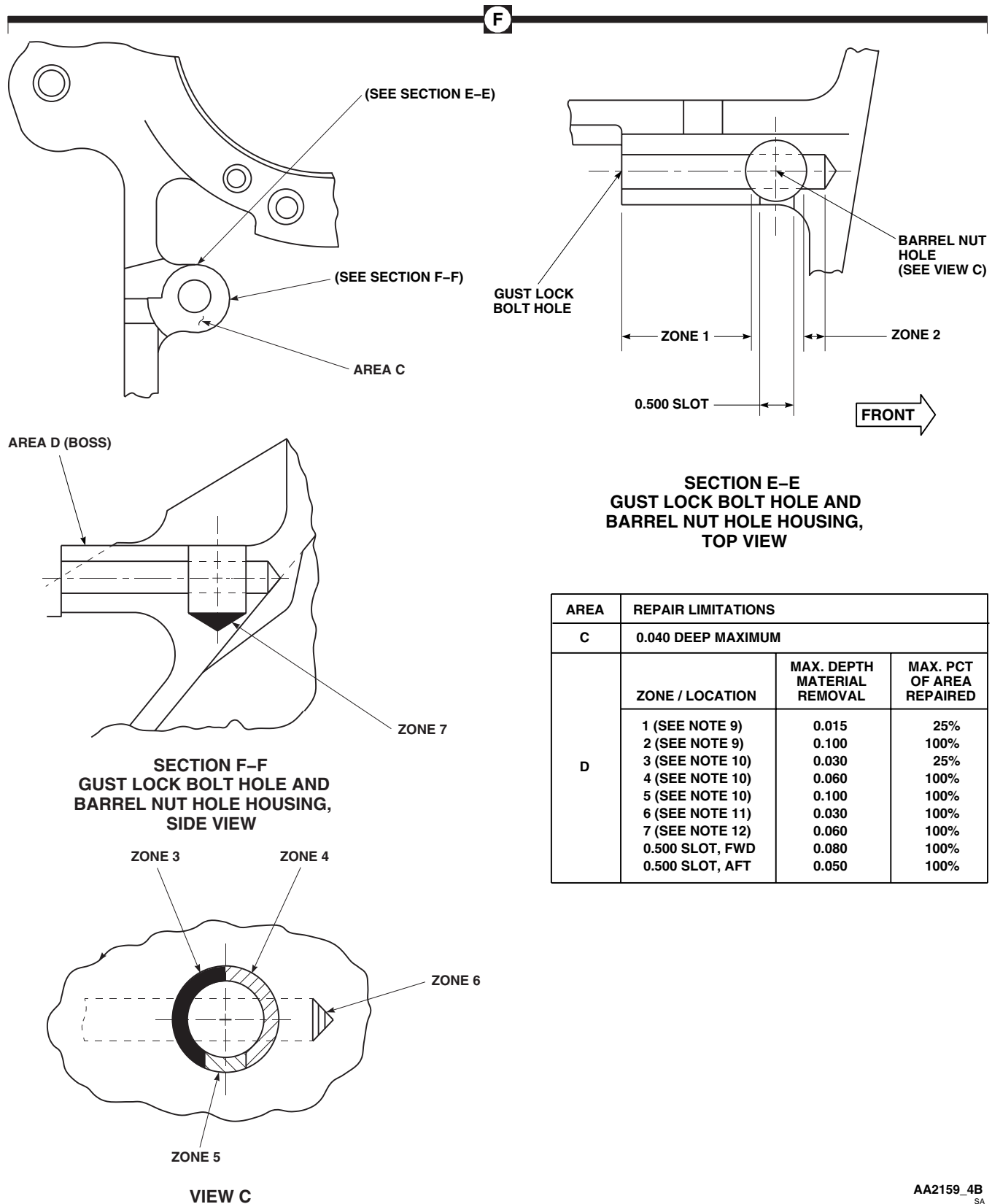


Figure 1. Repair Main Module Housing. (Sheet 4 of 4)

REPAIR LIMITS - Continued

- (4) **SPOTFACE CORNER MAY BE BLENDED TO 0.250-INCH RADIUS AS LONG AS REPAIR DOES NOT EXTEND MORE THAN 0.030-INCH INTO FLAT AREA ADJACENT TO BUSHING OR RIB.**
- (5) **SHARP CORNERS MAY BE BLENDED TO A 0.050-INCH RADIUS.**
- c. **MAXIMUM DEPTH OF MATERIAL REMOVAL ON FLAT PORTIONS OF BRIDGE PADS IS 0.040-INCH (Figure 1, Sheet 2, Detail B).**
 - (1) **NO MORE THAN ONE REPAIR PER 1/2-INCH SQUARE AREA WITH A MAXIMUM REPAIR OF 25% OF PAD.**
 - (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL NEAR PAD INSERTS (UP TO A 90 DEGREE SEGMENT) IS 0.015-INCH.**
- d. **FORWARD BELLCRANK SUPPORT MOUNTING PAD AREA MAY BE BLENDED TO LIMITS SHOWN IN Figure 1, Sheet 2, Detail D.**
- e. **SPOTFACE CORNER OF MOUNTING PAD MAY BE BLENDED TO LIMITS SHOWN IN Figure 1, Sheet 2, Section B-B.**
- f. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM HORIZONTAL TIEROD MOUNTING PAD AREA NOT COVERED BY TIEROD IS 0.040-INCH (Figure 1, Sheet 2, Detail C).**
 - (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM AREA UNDER TIEROD IS 0.040-INCH. NO MORE ONE REPAIR PER 1/2-INCH SQUARE AREA WITH A MAXIMUM REPAIR OF 25% OF PAD AREA.**
 - (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL NEAR RING LOCKED INSERT (UP TO A 90 DEGREE SEGMENT) IS 0.015-INCH.**
 - (3) **SHARP EDGES MAY BE BLENDED TO A 0.100-INCH RADIUS.**
- g. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM JET/BREATHING PLUG ASSEMBLIES IS 20% OF CIRCUMFERENCE OF SHADED AREA "A" AND AREA "B" (Figure 1, Sheet 3, Detail E).**
- h. **REFER TO Figure 1, Sheet 4, Detail F, FOR DAMAGE LIMITS TO GUST LOCK BOLTHOLE AND BARREL NUT HOLE.**

REPAIR PROCEDURES

1. Remove paint from damaged area using paint remover, Item 252, WP 1803 00.
2. Remove sealing compound from damaged area using nonmetallic scraper.
3. Carefully blend damaged area smooth using abrasive cloth, Item 83, WP 1803 00. Remove minimum amount of material required to blend out damaged area.
4. Using abrasive cloth, Item 82, WP 1803 00, blend reworked area to remove all traces of aluminum oxide cloth.
5. Remove displaced metal caused by abrasive blending by using magnesium etching procedure.

REPAIR PROCEDURES - Continued**CAUTION**

- Etchant measurements are to be as close to the values as can be measured in a milliliter beaker or vessel with equivalent mixing accuracy.
- Etchant formulas shall be prepared and stored in unbreakable containers and safely stored in designated areas not subjected to extreme heat or cold.
- Before preparing etchant, be sure storage area does not contain a premixed etchant formula. The correct stock bottle for this repair shall be marked MAGNESIUM ETCHANT.

6. Prepare magnesium etchant formula, Table 1:

Table 1. Magnesium Etchant Formula.

MATERIAL	AMOUNT
Ethylene Glycol, Technical Item 123, WP 1803 00	600.0 milliliters
Acetic Acid, Item 4, WP 1803 00	200.0 milliliters
Nitric Acid, Item 198, WP 1803 00	10.0 milliliters
Distilled Water, Reagent, Item 117, WP 1803 00	190.0 milliliters

7. Clean area to be etched with cleaning compound solvent, Item 71, WP 1803 00.
8. Mask area with pressure sensitive adhesive tape, Item 323, WP 1803 00.
9. Pour a small amount of etchant from stock bottle into 1 pint plastic container.

CAUTION

Damage to components will result if etchant is allowed to come in contact with areas not being etched. Extreme caution shall be used to protect those areas not being etched.

10. Apply etchant to reworked area using disposable applicator, Item 53, WP 1803 00. Swab area for about 10 to 15 seconds.
11. Neutralize etchant immediately by flooding area with a generous amount of clean water.
12. Dry area thoroughly with clean, dry, compressed air.
13. After use, do not return unused etchant to stock bottle. Dispose of used etchant in accordance with local instructions.
14. Identify contents of stock bottle as MAGNESIUM ETCHANT, before placing bottle in storage area.
15. Treat bare surface of housing with chromium trioxide pickle solution, Item 65, WP 1803 00 (TM 1-1500-344-23).
16. Touch up reworked areas with sealing compound, Item 273, WP 1803 00. After sealing compound has dried, apply one coat of epoxy primer coating kit, Item 120, WP 1803 00.

REPAIR PROCEDURES - Continued**NOTE**

Housings coated with epoxy coating kit are glossy silver; housings coated with polyurethane paint are matte surface medium gray.

17. Apply one coat of epoxy coating kit, Item 118, WP 1803 00, or one coat of polyurethane coating, Item 231, WP 1803 00, depending on original paint finish.
18. Apply one coat of corrosion preventive compound, Item 106, WP 1803 00. Allow to dry for 30 minutes and apply a second coat of corrosion preventive compound.
19. Check area for cleanness and foreign material.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM

INPUT MODULE HOUSING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Beaker (10 ml), MIL-B-36268
 Beaker (600 ml), MIL-B-36268
 Beaker (2000 ml), MIL-B-36268
 Container, Plastic, 1-Pint
 Depth Gage, 643J
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13

Material/Parts

Acetic Acid, Item 4, WP 1803 00
 Applicator, Disposable, Item 53, WP 1803 00
 Chromium Trioxide, Item 65, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00

Distilled Water, Reagent, Item 117, WP 1803 00
 Epoxy Coating Kit, Item 118, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Ethylene Glycol, Technical, Item 123, WP 1803 00
 Nitric Acid, Item 198, WP 1803 00
 Polyurethane Coating, Item 231, WP 1803 00
 Remover, Paint, Item 252, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 323,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
 WP 1803 00

GENERAL**NOTE**

- Minor nicks, gouges, and scratches on surface of input module housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in input module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace input module.
 - Minor nicks, gouges, scratches, and corrosion within snapping groove of jet or breather plug may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in input module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace input module.
 - Repair of damaged area may be done with input module installed.
 - All repairs shall be recorded on Aircraft Component Historical Record, on DA Form 2408-16.
1. Clean any oil or grease from damaged area using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.

GENERAL - Continued**NOTE**

Housings coated with epoxy coating are glossy silver. Housings coated with polyurethane coating are matte surface medium gray.

2. Fine scratches which remove only paint may be blended smooth using abrasive cloth, Item 82, WP 1803 00. After smoothing out, touch up blended area using epoxy coating kit, Item 118, WP 1803 00, or polyurethane coating, Item 231, WP 1803 00, as required.

REPAIR LIMITS

1. Input module housing limits are as follows:
 - a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES OF INPUT MODULE INPUT HOUSING IS 0.020-INCH.**
 - (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM PINION HOUSING IS 0.020-INCH (Figure 1, Detail A).**
 - (2) **NO MATERIAL REMOVAL ALLOWED ON PILOT DIAMETER.**
 - b. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM JET ASSEMBLY IS 20% OF SNAPPING GROOVE BOTTOM CIRCUMFERENCE (Figure 1, Detail B).**
 - c. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES OF INPUT MODULE CENTER HOUSING IS 0.040-INCH.**
 - d. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES OF INPUT MODULE OUTPUT HOUSING IS 0.030-INCH.**

NOTE

Output housing part number is stamped on housing.

- e. **OUTPUT HOUSING RIBS, 70351-08019-003, CAN BE BLENDED TO MINIMUM 0.420-INCH RIB THICKNESS. OUTPUT HOUSING RIBS, 70351-08019-001 AND 70351-08019-002, CAN BE BLENDED TO MINIMUM 0.170-INCH THICKNESS.**
 - (1) **NO MORE THAN TWO REPAIRS ALLOWED PER FOUR INCH SQUARE AREA.**
 - (2) **TOTAL LENGTH OF ALL BLENDED AREAS NOT TO EXCEED FIVE INCHES ALONG CIRCUMFERENCE (INCLUDING RIBS) AT ANY SECTION OF HOUSING.**

NOTE

Only 30% of the total number of holes in bolt pattern may be repaired.

- f. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTPUT HOUSING TO CENTER HOUSING MOUNTING FLANGE BOLTHOLES IS 0.015-INCH. MAXIMUM DIAMETER NOT TO EXCEED 0.440-INCH (Figure 1, Detail C).**
 - (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENER IS 0.015-INCH. REPAIR SHALL NOT COVER MORE THAN 25% OF SURFACE AREA UNDER FASTENER.**
 - (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
 - (3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOTFACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**

REPAIR LIMITS - Continued

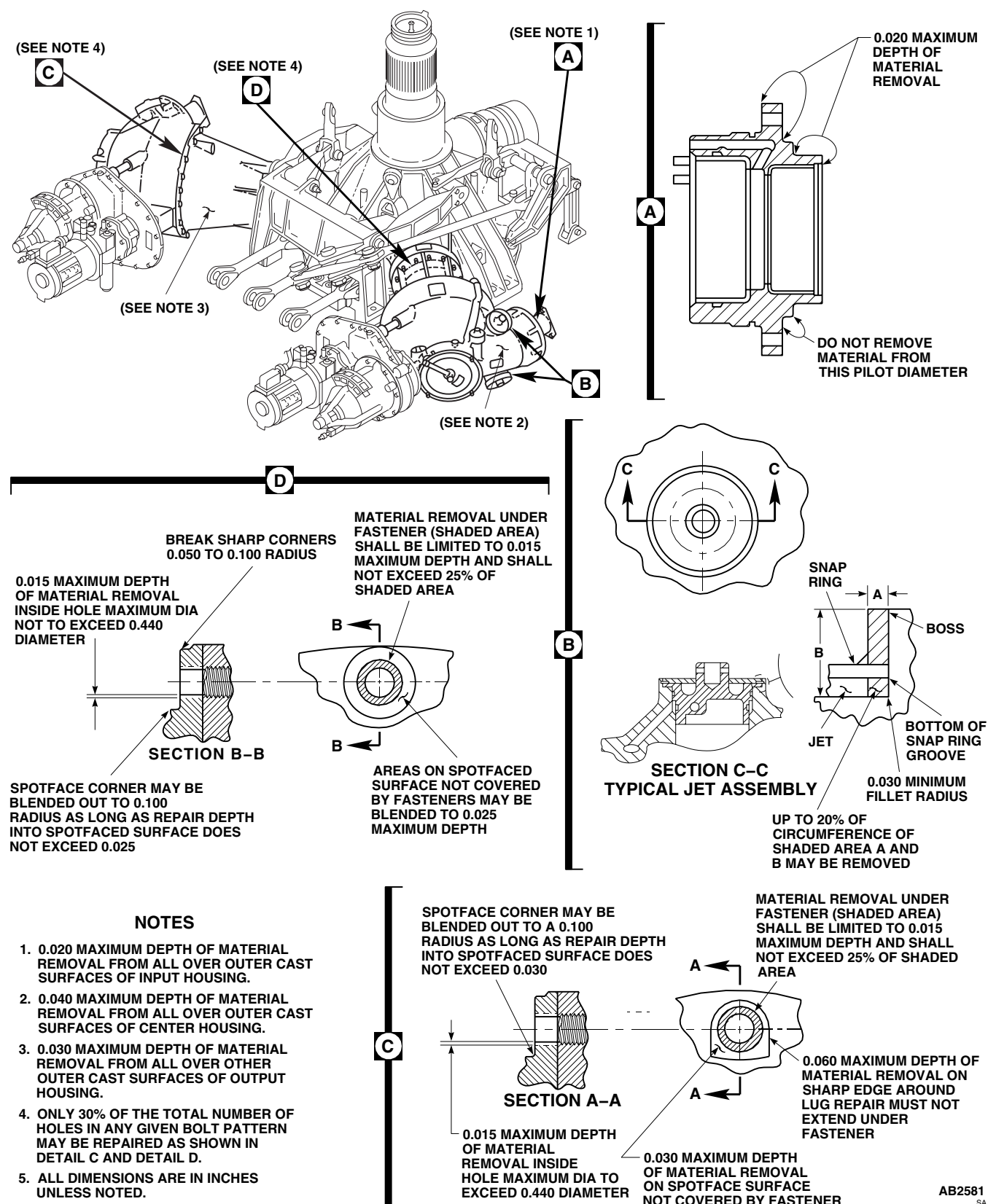


Figure 1. Repair Input Module Housing.

REPAIR LIMITS - Continued

- (4) **CORNER OF SPOTFACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOTFACE DOES NOT EXCEED 0.030-INCH.**

NOTE

Only 30% of the total number of holes in bolt pattern may be repaired.

- g. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTPUT HOUSING TO MAIN MODULE MOUNTING FLANGE BOLTHOLES IS 0.015-INCH. MAXIMUM DIAMETER NOT TO EXCEED 0.440-INCH (Figure 1, Detail D).**
- (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENER IS 0.015-INCH. REPAIR SHALL NOT COVER MORE THAN 25% OF SURFACE AREA UNDER FASTENER.**
- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE NOT COVERED BY FASTENER IS 0.025-INCH.**
- (3) **CORNER OF SPOTFACE MAY BE BLENDED TO 0.100 INCH RADIUS AS LONG AS DEPTH INTO SPOTFACE DOES NOT EXCEED 0.025-INCH.**
- (4) **SHARP CORNERS MAY BE BLENDED TO 0.100-INCH RADIUS.**

REPAIR PROCEDURES

1. Remove paint from damaged area using paint remover, Item 252, WP 1803 00.
2. Remove sealant from damaged area using nonmetallic scraper.
3. Carefully blend damaged area smooth using abrasive cloth, Item 83, WP 1803 00. Remove minimum amount of material required to blend out damaged area.
4. Using abrasive cloth, Item 82, WP 1803 00, blend reworked area to remove all traces of aluminum oxide cloth.
5. Remove displaced metal caused by abrasive blending by using magnesium etching procedure.

CAUTION

- Etchant measurements are to be as close to the values as can be measured in a milliliter beaker or vessel with equivalent mixing accuracy.
- Etchant formulas shall be prepared and stored in unbreakable plastic containers and safely stored in designated areas not subjected to extreme heat or cold.
- Before preparing etchant, be sure storage area does not contain a premixed etchant formula. The correct stock bottle for this repair shall be marked MAGNESIUM ETCHANT.

6. Prepare magnesium etchant formula Table 1:

Table 1. Magnesium Etchant Formula.

MATERIAL	AMOUNT
Ethylene Glycol, Technical Item 123, WP 1803 00	600.0 milliliters
Acetic Acid, Item 4, WP 1803 00	200.0 milliliters

REPAIR PROCEDURES - Continued**Table 1. Magnesium Etchant Formula. - Continued**

MATERIAL	AMOUNT
Nitric Acid, Item 198, WP 1803 00	10.0 milliliters
Reagent Distilled Water, Item 117, WP 1803 00	190.0 milliliters

7. Clean area to be etched with cleaning compound solvent, Item 71, WP 1803 00.
8. Mask area with pressure sensitive adhesive tape, Item 323, WP 1803 00.
9. Pour a small amount of etchant from stock bottle into plastic container.

CAUTION

Damage to components will result if etchant is allowed to come in contact with areas not being etched. Extreme caution shall be used to protect those areas not being etched.

10. Apply etchant to reworked area using disposable applicator, Item 53, WP 1803 00. Swab area for about 10 to 15 seconds.
11. Neutralize etchant immediately by flooding area with a generous amount of clean water.
12. Dry area thoroughly with clean, dry, compressed air.
13. After use, do not return unused etchant to stock bottle. Dispose of used etchant in accordance with local instructions.
14. Identify contents of stock bottle as MAGNESIUM ETCHANT, before placing bottle in storage area.
15. Treat bare surface of housing with chromium trioxide, Item 65, WP 1803 00 (TM 1-1500-344-23).
16. Touch up reworked areas with sealing compound, Item 273, WP 1803 00. After sealing compound has dried, apply one coat of epoxy primer coating kit, Item 120, WP 1803 00.

NOTE

Housings coated with epoxy coating are glossy silver; housings coated with polyurethane paint are matte surface medium gray.

17. Apply 1 coat of epoxy coating kit, Item 118, WP 1803 00, or 1 coat of polyurethane coating, Item 231, WP 1803 00, depending on original paint finish.
18. Apply 1 coat of corrosion preventive compound, Item 106, WP 1803 00. Allow to dry for 30 minutes and apply a second coat of corrosion preventive compound.
19. Check area for cleanness and foreign material.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM

ACCESSORY MODULE HOUSING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Beaker (10 ml), MIL-B-36268
 Beaker (600 ml), MIL-B-36268
 Beaker (2000 ml), MIL-B-36268
 Container, 1-Pint
 Depth Gage, 643J
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13

Material/Parts

Acetic Acid, Item 4, WP 1803 00
 Applicator, Disposable, Item 53, WP 1803 00
 Chromium Trioxide, Item 65, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00

Distilled Water, Reagent, Item 117, WP 1803 00
 Epoxy Coating Kit, Item 118, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Ethylene Glycol, Technical, Item 123, WP 1803 00
 Nitric Acid, Item 198, WP 1803 00
 Polyurethane Coating, Item 231, WP 1803 00
 Remover, Paint, Item 252, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 323,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
 WP 1803 00

GENERAL**NOTE**

- Minor nicks, gouges, and scratches on surface of accessory module housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in accessory module still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace accessory module.
- Repair of damaged area may be done with accessory module installed.
- All repairs shall be recorded on Aircraft Component Historical Record, DA Form 2408-16.

1. Clean any oil or grease from damaged area using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.

NOTE

Housings coated with epoxy coating are glossy silver. Housings coated with aircraft gray polyurethane paint are matte surface medium-gray.

2. Fine scratches which remove only paint may be blended smooth using abrasive cloth, Item 82, WP 1803 00. After smoothing out, touch up blended area using epoxy coating kit, Item 118, WP 1803 00, or polyurethane coating, Item 231, WP 1803 00, as required.

REPAIR LIMITS

1. Accessory module housing repair limits are as follows:

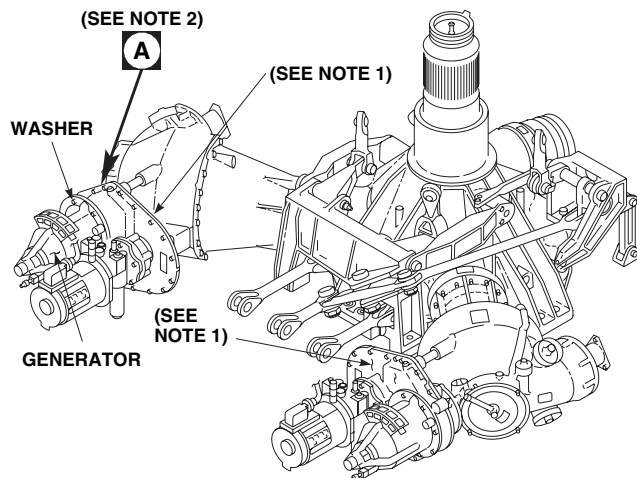
- a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES IS 0.060-INCH (Figure 1, Detail A).**

NOTE

- Only 30% of the total number of holes in bolt pattern may be repaired.
- Boltholes with bushings are not repairable.

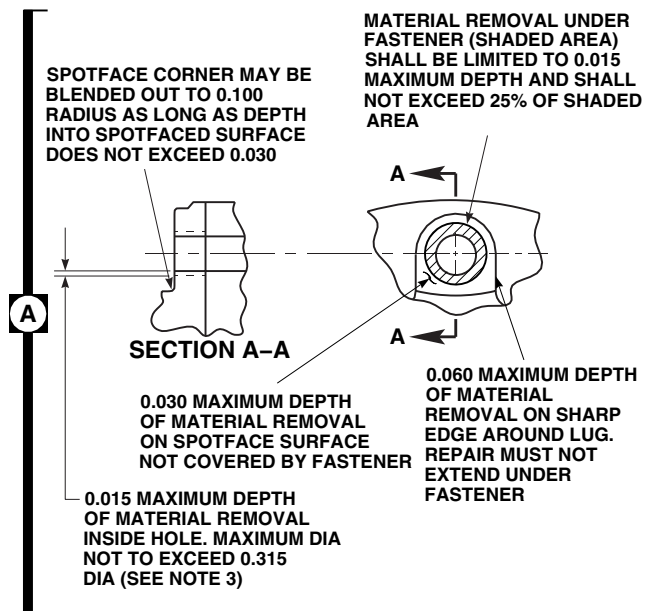
- b. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING FLANGE BOLTHOLES IS 0.015-INCH. MAXIMUM DIAMETER NOT TO EXCEED 0.315-INCH (Figure 1, Detail A).**

- (1) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENER IS 0.015-INCH. REPAIR SHALL NOT COVER MORE THAN 25% OF SURFACE AREA UNDER FASTENER.**
- (2) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- (3) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOTFACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**
- (4) **CORNER OF SPOTFACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOTFACE DOES NOT EXCEED 0.030-INCH.**



NOTES

1. 0.060 MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTER CAST SURFACES OF ACCESSORY MODULE.
2. ONLY 30% OF THE TOTAL NUMBER OF HOLES IN ANY GIVEN BOLT PATTERN MAY BE REPAIRED AS SHOWN IN DETAIL A.
3. HOLES 8, 9, 10, AND 11 (WITH BUSHINGS) ARE NOT REPAIRABLE IN THIS MANUAL.
4. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.



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Figure 1. Repair Accessory Module Housing.

REPAIR PROCEDURES

1. Remove paint from damaged area using paint remover, Item 252, WP 1803 00.
2. Remove sealant from damaged area using nonmetallic scraper.
3. Carefully blend damaged area smooth using abrasive cloth, Item 83, WP 1803 00. Remove minimum amount of material required to blend out damaged area.
4. Using abrasive cloth, Item 89, WP 1803 00, blend reworked area to remove all traces of aluminum oxide cloth.
5. Remove displaced metal caused by abrasive blending by using magnesium etching procedure.

CAUTION

- Etchant measurements are to be as close to the values as can be measured in a milliliter beaker or vessel with equivalent mixing accuracy.
- Etchant formulas shall be prepared and stored in unbreakable plastic containers and safely stored in designated areas not subjected to extreme heat or cold.
- Before preparing etchant, be sure storage area does not contain a premixed etchant formula. The correct stock bottle for this repair shall be marked MAGNESIUM ETCHANT.

6. Prepare magnesium etchant formula Table 1:

Table 1. Magnesium Etchant Formula.

MATERIAL	AMOUNT
Ethylene Glycol, Technical Item 123, WP 1803 00	600.0 milliliters
Acetic Acid, Item 4, WP 1803 00	200.0 milliliters
Nitric Acid, Item 198, WP 1803 00	10.0 milliliters
Reagent Distilled Water, Item 117, WP 1803 00	190.0 milliliters

7. Clean area to be etched with cleaning compound solvent, Item 71, WP 1803 00.
8. Mask area with pressure sensitive adhesive tape, Item 323, WP 1803 00.
9. Pour a small amount of etchant from stock bottle into plastic container.

CAUTION

Damage to components will result if etchant is allowed to come in contact with areas not being etched. Extreme caution shall be used to protect those areas not being etched.

10. Apply etchant to reworked area using disposable applicator, Item 53, WP 1803 00. Swab area for about 10 to 15 seconds.
11. Neutralize etchant immediately by flooding area with a generous amount of clean water.
12. Dry area thoroughly with clean, dry, compressed air.

REPAIR PROCEDURES - Continued

13. After use, do not return unused etchant to stock bottle. Dispose of used etchant in accordance with local instructions.
14. Identify contents of stock bottle as MAGNESIUM ETCHANT, before placing bottle in storage area.
15. Treat bare surface of housing with chromium trioxide, Item 65, WP 1803 00 (TM 1-1500-344-23).
16. Touch up reworked areas with sealing compound, Item 273, WP 1803 00. After sealing compound has dried, apply one coat of epoxy primer coating kit, Item 120, WP 1803 00.

NOTE

Housings coated with epoxy coating are glossy silver; housings coated with polyurethane paint are matte surface medium gray.

17. Apply one coat of epoxy coating kit, Item 118, WP 1803 00, or one coat of polyurethane coating, Item 231, WP 1803 00, depending on original paint finish.
18. Apply one coat of corrosion preventive compound, Item 106, WP 1803 00. Allow to dry for 30 minutes and apply a second coat of corrosion preventive compound.
19. Check area for cleanness and foreign material.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****MAIN, INPUT, AND ACCESSORY MODULE HOUSING WATER ENTRAPMENT AREA CORROSION PROTECTION**

This WP supersedes WP 0642 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-B01
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-CL-B13

Polyurethane Coating, Item 235, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Corrosion Preventive Compound, Item 105,
WP 1803 00

References

WP 1803 00

CORROSION PROTECTION OF WATER ENTRAPMENT AREAS

1. Clean water entrapment areas (Figure 1) with corrosion preventive compound, Item 105, WP 1803 00, and allow to dry.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Repeat cleaning procedure using technical acetone, Item 5, WP 1803 00, and allow to dry.
3. Apply a coat of polyurethane coating, Item 235, WP 1803 00, to cleaned areas.

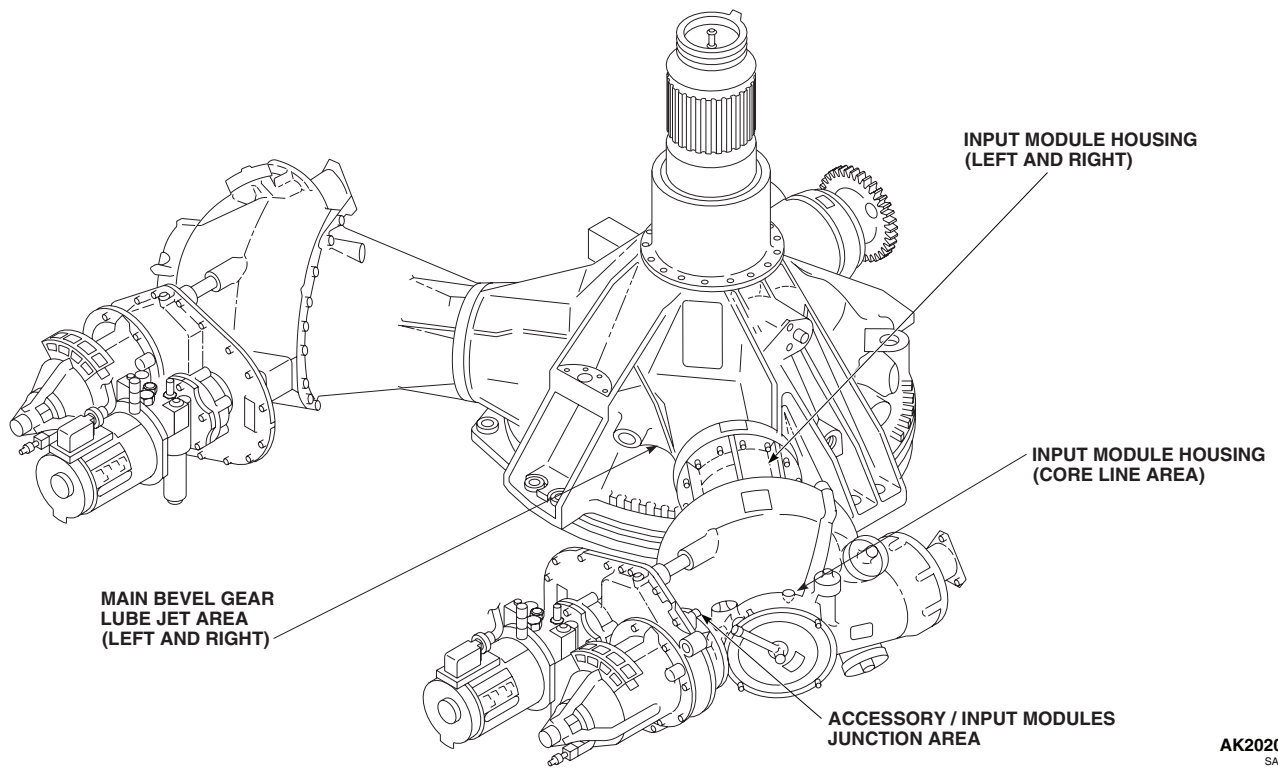
CORROSION PROTECTION OF WATER ENTRAPMENT AREAS - Continued

Figure 1. Water Entrapment Areas.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**ACCESSORY MODULE CHIP DETECTOR**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Electrical Repairer's Toolkit, SC 5180-99-B06

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cleaning Compound Solvent, Item 71, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 184,
 WP 1803 00
 Lubricating Oil, Helicopter Transmission, Item 187,
 WP 1803 00

References

WP 0085 00
 WP 0617 00
 WP 0630 00
 WP 1703 00
 WP 1803 00

REMOVE**NOTE**

Instructions are the same for left or right chip detector.

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover.
3. Disconnect chip detector electrical connector, P406, left side or electrical connector, P405, right side from main transmission wiring harness (Figure 1).

WARNING

Injury to personnel will result if precautions are not taken when removing chip detector. To prevent injury, personnel must wear eye protection and make sure transmission oil is cool before removing chip detector.

4. Remove chip detector from self-closing valve in accessory module. Discard packings (Figure 1, Detail A).
5. Inspect chip detector element for metal particles (WP 0617 00).
6. Remove self closing valve and packing from accessory module. Discard packing.
7. Clean chip detector magnetic plug in cleaning compound solvent, Item 71, WP 1803 00. Clean self-closing valve, if removed, in solvent.

INSTALL

1. Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
2. **QA** Install packing on self-closing valve and install in accessory module (Figure 1, Detail A). **TIGHTEN SELF-CLOSING VALVE 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

INSTALL - Continued

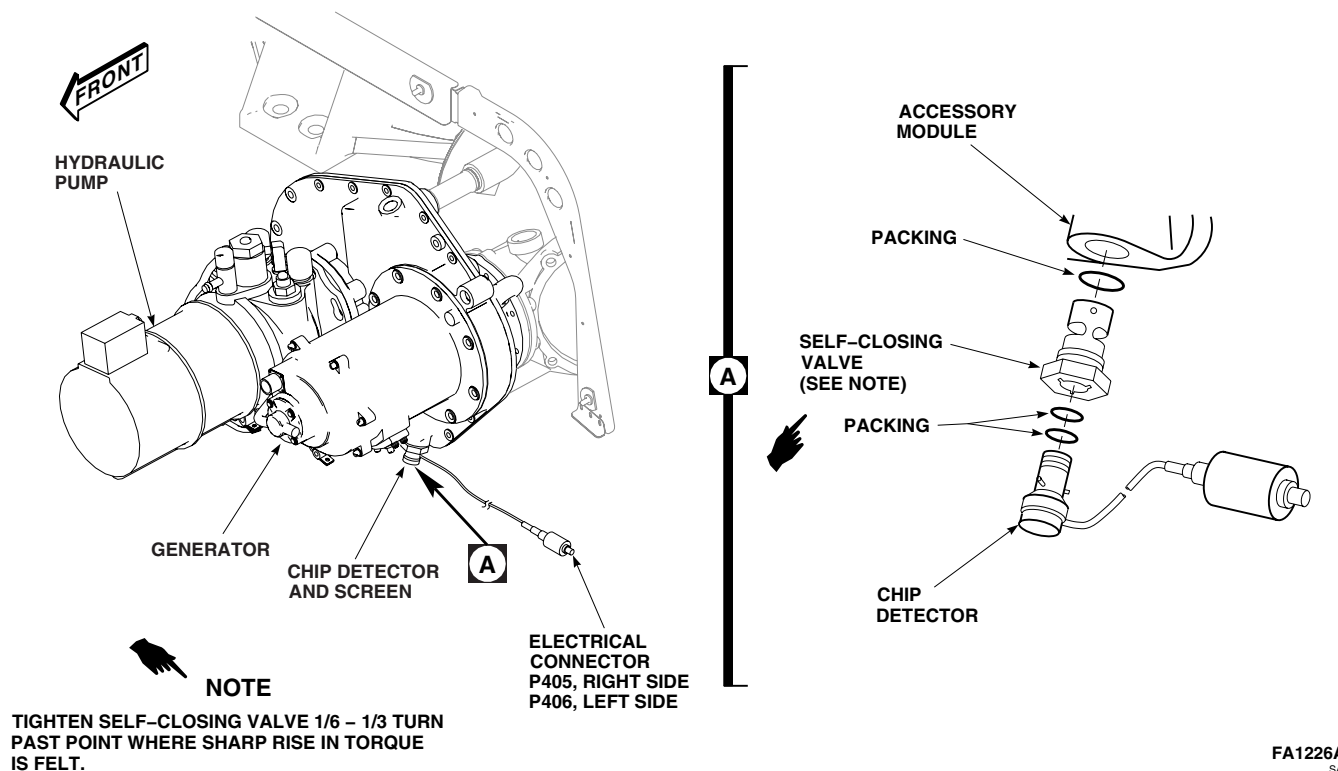


Figure 1. Replace Accessory Module Chip Detector.

3. Check chip detector electrical circuit (WP 0085 00).
4. Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
5. **QA** Install chip detector and packings into self-closing valve in accessory module (Figure 1, Detail A).
6. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire chip detector and self-closing valve.
7. Inspect and treat electrical connectors (WP 1703 00).
8. **QA** Connect chip detector electrical connector, P406, left side, or P405, right side, to main transmission wiring harness (Figure 1). See WP 0630 00 for wiring harness routing.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

ACCESSORY MODULE OIL SCREEN

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-A01
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-CL-A13

Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00
Sealing Compound, Item 273, WP 1803 00

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0617 00
WP 1803 00

REMOVE

1. Remove sealing compound from both ends of tube with nonmetallic scraper.

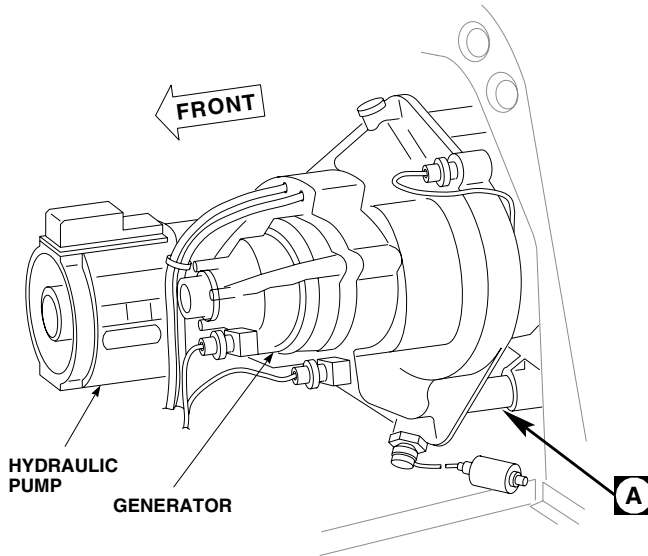
WARNING

Injury to personnel will result if precautions are not taken when removing oil screen.
To prevent injury, personnel must wear eye protection and make sure transmission oil
is cool before removing oil screen.

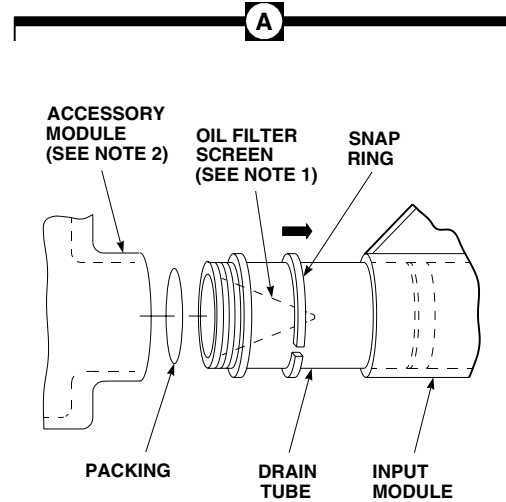
2. Unlock snapping on drain tube on input module side between accessory and input modules and slide to front along drain tube (Figure 1, Detail A).
3. Push tube into input module to expose cone shaped oil filter screen.
4. Remove oil screen and inspect for metal particles (WP 0617 00).
5. Clean oil screen with cleaning compound solvent, Item 71, WP 1803 00.

INSTALL

1. **QA** Insert oil screen into drain tube (Figure 1, Detail A).
2. Apply epoxy primer coating kit, Item 120, WP 1803 00, to snapping.
3. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
4. **QA** Seat drain tube with packing in accessory module. Slide snapping to rear along drain tube and lock in groove (Figure 1, Detail A).
5. Apply sealing compound, Item 273, WP 1803 00, to drain tube at both ends to prevent water entrapment and corrosion above snapping groove.
6. Make sure area is clean and free of foreign material.

INSTALL - Continued**NOTES**

1. THERE ARE (4 EACH) OIL FILTER SCREENS.
2. LEFT AND RIGHT MODULES ARE THE SAME.



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Figure 1. Replace Accessory Module Oil Screen.

7. Close main rotor pylon sliding cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**INPUT MODULE CHIP DETECTOR**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Borescope Kit, 6650-T700-003
 Container, 2-Gallon
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Torque Wrench, 30 - 200 in. lbs.

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 184,
 WP 1803 00
 Lubricating Oil, Helicopter Transmission, Item 187,
 WP 1803 00
 Tape, Lacing and Tying, Item 321, WP 1803 00

References

WP 0085 00
 WP 0259 00
 WP 0260 00
 WP 0261 00
 WP 0617 00
 WP 0630 00
 WP 1607 00
 WP 1703 00
 WP 1803 00

REMOVE**NOTE**

Instructions are the same for left or right chip detector.

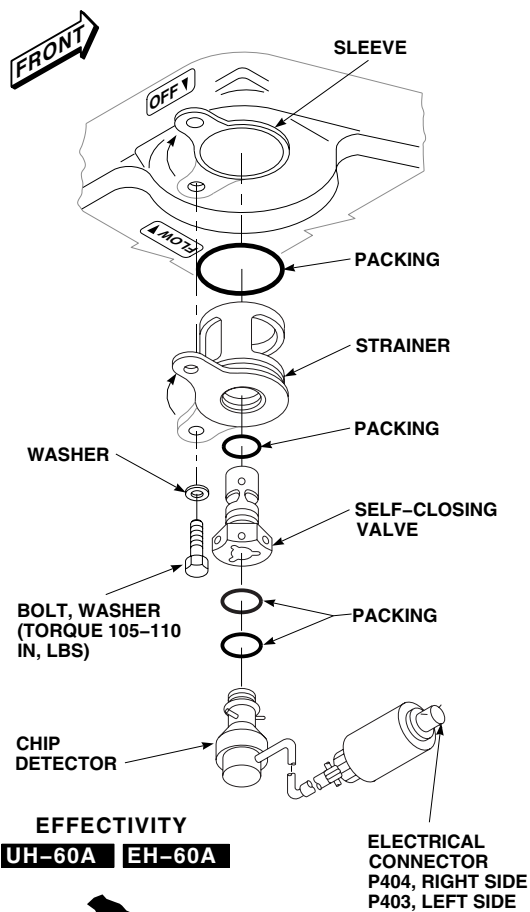
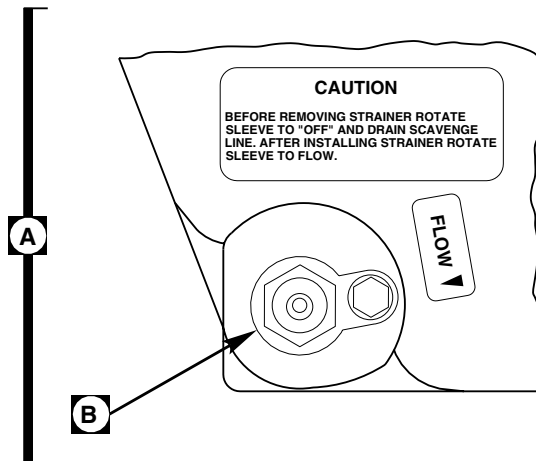
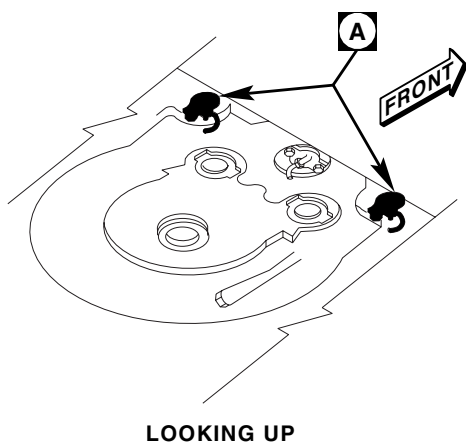
1. Turn off all electrical power.
2. Remove transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).

WARNING

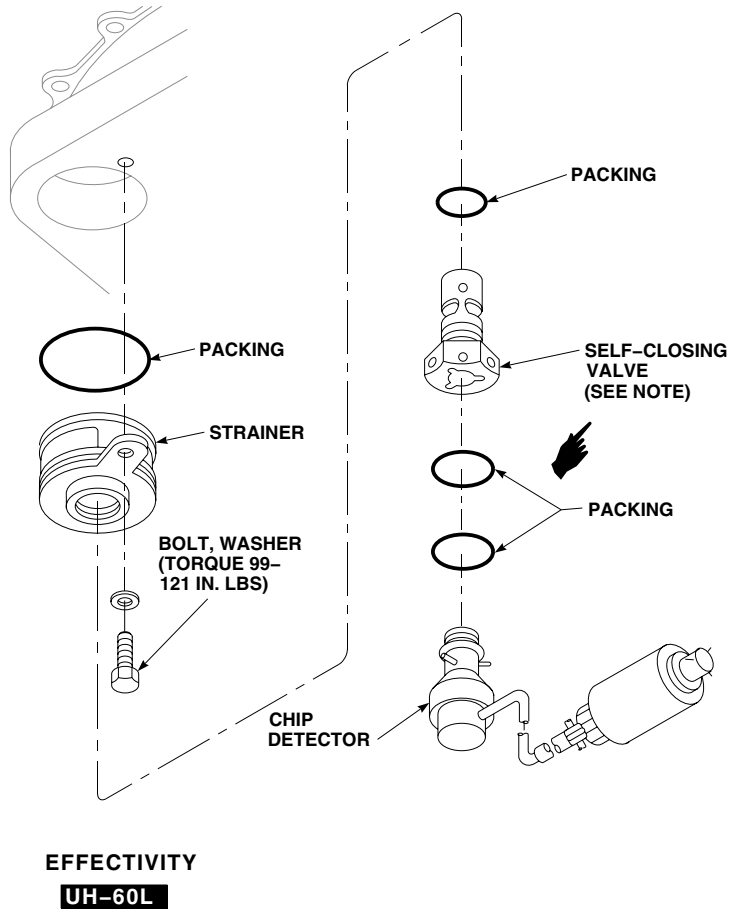
Injury to personnel will result if precautions are not taken when removing chip detector and strainer. To prevent injury, personnel must wear eye protection and make sure transmission oil is cool before removing chip detector and strainer.

3. Remove lacing tape and clamps securing chip detector wiring harness to sump. Keep all attaching hardware (WP 0630 00).
4. Disconnect chip detector electrical connector, P403, left side, or P404, right side, from main transmission wiring harness (Figure 1, Detail B).
5. **UH-60A EH-60A** Remove chip detector, self-closing valve, and strainer as follows:
 - a. Remove bolt holding strainer and sleeve to sump. Turn strainer and sleeve OFF. Remove chip detector from self-closing valve by pushing in, turning, and pulling out.

REMOVE - Continued



NOTE
TIGHTEN SELF-CLOSING VALVE 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.



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Figure 1. Replace Input Module Chip Detector.

REMOVE - Continued**NOTE**

When removing strainer, about 1 quart of lubrication oil will drain out.

- b. Place a 2 gallon container, or larger, underneath strainer to catch oil.
- c. Remove strainer and packing leaving sleeve in sump. Discard packing.
- d. Remove self-closing valve and packing from strainer. Remove packing from self-closing valve. Discard packing.
6. **UH-60L** Remove chip detector, self-closing valve, and strainer as follows:
 - a. Place a 2 gallon container, underneath chip detector to catch oil.
 - b. Remove chip detector and packings from self-closing valve by pushing in, turning and pulling out. Remove packing from chip detector. Discard packing.

NOTE

About 1 gallon of lubrication oil will drain out when self-closing valve is removed.

- c. Remove self-closing valve and packing from strainer. Remove packing from self-closing valve. Discard packing.
- d. Remove bolt and strainer from sump.
7. Inspect chip detector, self-closing valve, and strainer for metal particles (WP 0617 00).
8. **UH-60A EH-60A** Inspect openings in chip detector sleeve and sump cored hole, using borescope kit. **MAXIMUM OPENING OVERLAP IS 0.030-INCH.** Replace gearbox if limit is exceeded.
9. **UH-60A EH-60A** Inspect the scavenge passage between chip detector sleeve window and main gear box lube pump using borescope kit and a light source. **IF SCAVENGE PASSAGE IS NOT FREE OF OBSTRUCTIONS, REPLACE GEAR BOX.**
10. Clean chip detector, self closing valve, and strainer in cleaning compound solvent, Item 71, WP 1803 00.

INSTALL

1. **UH-60A EH-60A** Install chip detector, self-closing valve, and strainer as follows:
 - a. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on self-closing valve (Figure 1, Detail B).
 - b. **QA** Install self-closing valve and packing, in strainer. **TIGHTEN SELF-CLOSING VALVE 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
 - c. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on strainer.
 - d. **QA** Install strainer and packing, in sleeve in sump.
 - e. Check chip detector electrical circuit (WP 0085 00).
 - f. **QA** Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on chip detector.
 - g. **QA** Install chip detector and packings in self-closing valve, push in and turn. Turn sleeve to FLOW.
 - h. **QA** Install bolt and washer. **TORQUE BOLT TO 105 - 110 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolt to self-closing valve.

INSTALL - Continued

2. **UH-60L** Install chip detector, self-closing valve, and strainer as follows:
 - a. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on strainer (Figure 1, Detail B).
 - b. **QA** Install strainer in sump.
 - c. **QA** Install strainer retaining bolt and washer in sump. **TORQUE BOLT TO 99 - 121 INCH-POUNDS.**
 - d. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on self-closing valve.
 - e. **QA** Install self-closing valve in strainer. Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolt to self-closing valve.
 - f. Check chip detector electrical circuit (WP 0085 00).
 - g. **QA** Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on chip detector.
 - h. **QA** Install chip detector in self-closing valve, push in and turn.
3. Inspect and treat electrical connectors (WP 1703 00).
4. **QA** Connect electrical connector, P403, left side, or P404, right side, to main transmission harness. Using nonelectrical wire, Item 357, WP 1803 00, lockwire connector. See WP 0630 00 for wiring harness routing.
5. **QA** Install screws, clamps, with lacing and tying tape, Item 321, WP 1803 00, to secure chip detector wire harness to sump.
6. Check oil for proper level. Service main transmission if necessary (WP 1607 00).
7. Make sure area is clean and free of foreign material.
8. Install transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

MAIN MODULE SUMP CHIP DETECTOR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-A01
Container, 2-Gallon
Powertrain Repairer's Toolkit, SC 5180-99-CL-A13
Torque Wrench, 30 - 200 in. lbs.

References

WP 0085 00
WP 0259 00
WP 0260 00
WP 0261 00
WP 0616 00
WP 0617 00
WP 0630 00
WP 1607 00
WP 1703 00
WP 1803 00

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**CAUTION**

A sustained loss of transmission oil pressure of 10 psi or more, in conjunction with the presence of numerous aluminum foil-like debris in the chip detector strainer, is indicative of a planetary carrier crack. A portion of such crack could be visible by draining the transmission oil, removing one of the transmission oil pumps, and inspecting the underside of the carrier through the oil pump opening in the transmission sump housing. This will require walking the rotor system to rotate the carrier assembly

1. Turn off all electrical power.
2. Remove transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).

WARNING

Injury to personnel will result if precautions are not taken when removing strainer. To prevent injury, personnel must wear eye protection and make sure transmission oil is cool before removing strainer.

3. Remove clamps and lacing tape securing chip detector wiring to sump (WP 0630 00).
4. Disconnect strainer electrical connector, P412, from main transmission harness (Figure 1, Detail B).
5. Remove bolts and washers from sump strainer (chip detector and oil temperature indicator). Turn sleeve in sump to OFF position as indicated by arrow (Figure 1, Detail A).

REMOVE - Continued

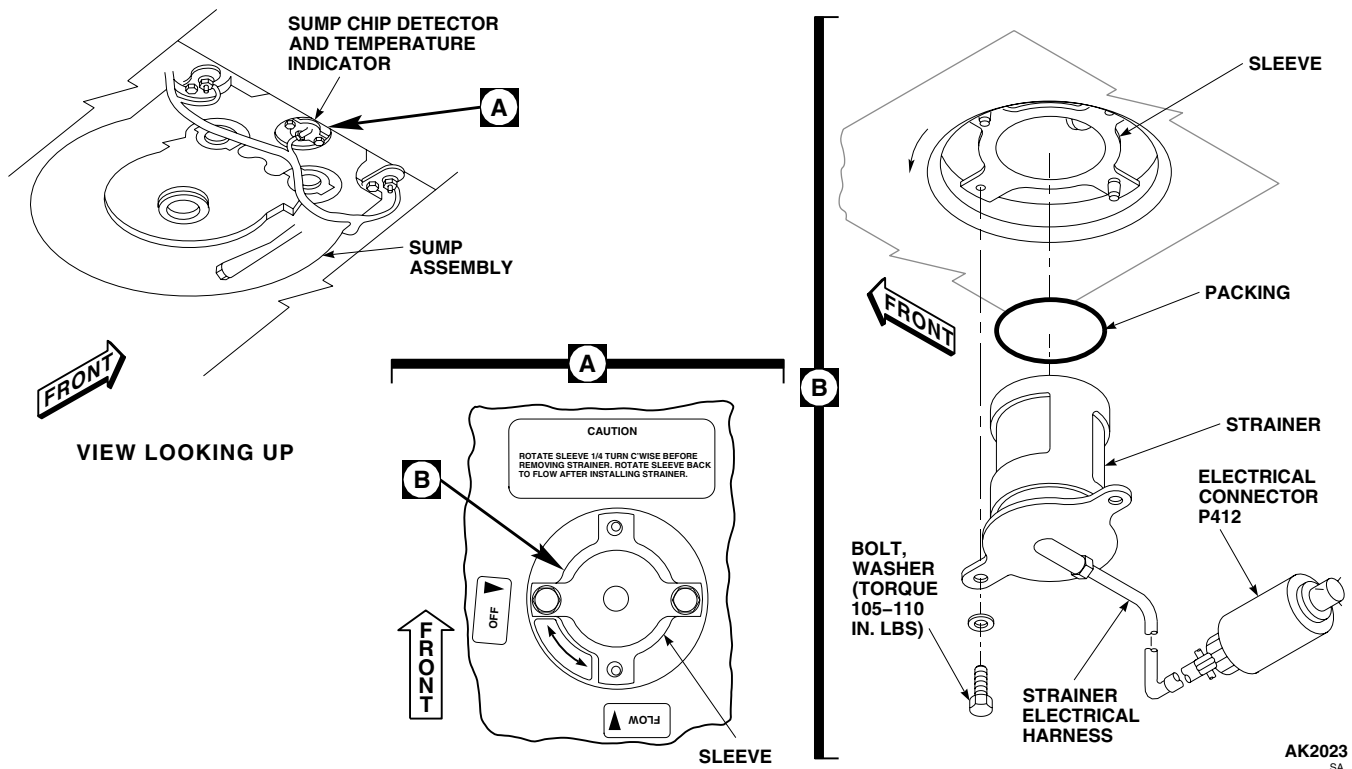


Figure 1. Replace Main Module Sump Chip Detector.

NOTE

When removing strainer, about 1 pint of oil will drain out.

6. Place a 2 gallon container underneath screen when removing it.
7. Remove strainer from sleeve in sump. Remove and discard packing.
8. Inspect chip detector and strainer for metal particles (WP 0617 00).
9. Clean chip detector magnetic element and strainer, if necessary, with cleaning compound solvent, Item 71, WP 1803 00.
10. Check chip detector electrical circuit (WP 0085 00).

INSTALL

1. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
2. **QA** Install packing in groove on strainer. Insert strainer into sleeve in sump (Figure 1, Detail B).
3. **QA** Turn sleeve in sump to FLOW position as indicated by arrow, and install bolts and washers (Figure 1, Detail A). **TORQUE BOLTS TO 105 - 110 INCH-POUNDS (Figure 1, Detail B).**
4. Inspect and treat electrical connectors (WP 1703 00).
5. **QA** Connect strainer electrical connector, P412, to main transmission harness. Secure wiring with clamps and lacing tape. See WP 0630 00 for wiring harness routing.

INSTALL - Continued

6. Check oil filter for bypass condition. If in bypass, refer to WP 0616 00.
7. Service main transmission if necessary (WP 1607 00).
8. Make sure area is clean and free of foreign material.
9. Install transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).
10. After main transmission chip detectors, oil strainer screens, and filters have been checked, do this:
 - a. Refer to WP 0617 00 to determine serviceability of the main transmission.
 - b. Either continue gear box in service, make a serviceability check (WP 0616 00), or replace module, as indicated in WP 0617 00.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**MAIN TRANSMISSION OIL FILTER/SLEEVE**

This WP supersedes WP 0647 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-A01
 Container, 1-Quart
 Main Transmission Oil Filter Flow Valve Positioner,
 Locally-Made, Figure 264, WP 1805 00
 Nonmetallic Drift
 Powertrain Repairer's Toolkit, SC 5180-99-CL-A13
 Torque Wrench, 30 - 200 in. lbs.

Sealing Compound, Item 273, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0259 00
 WP 0260 00
 WP 0261 00
 WP 1607 00
 WP 1803 00
 WP 1805 00

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 184,
 WP 1803 00
 Lubricating Oil, Helicopter Transmission, Item 187,
 WP 1803 00

REMOVE**NOTE**

- **UH-60L** Make sure helicopter is level or slightly nose down when replacing oil filter. ■
- **UH-60L** Filter assembly is installed on sump with three bolts. ■
- **UH-60A** Transmission oil must be drained before replacing oil filter/sleeve. ■
- **UH-60L** Transmission oil does not need to be drained before replacing oil filter/sleeve. ■

1. Remove transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).
2. Remove bolts and washers holding filter assembly to main gear box sump (Figure 1, Sheet 1, Detail A).
3. **UH-60A EH-60A** Locally make main transmission oil filter flow valve positioner (WP 1805 00). Turn the gearbox sleeve assembly clockwise to the OFF position. ■

REMOVE - Continued**NOTE**

- **UH-60A EH-60A** Filter will turn with sleeve. ■
- **UH-60L** No filter sleeve installed in sump. ■

4. **UH-60A EH-60A** Turn filter assembly 1/4 turn clockwise. ■
5. **UH-60L** Turn filter assembly 1/2-turn. ■
6. Pull filter assembly out of sleeve and catch any oil with 1 quart container.

NOTE

UH-60L Valve may or may not remain in main gear box sump. If valve remains in main gear box sump, it is not necessary to remove valve. ■

7. **UH-60A EH-60A** Depress spring clips, if installed, and lift valve from filter bowl (Figure 1, Sheet 1, Detail B). ■
8. Remove first and second stage filter elements and packings. Discard packings.
9. If required, remove pressure indicator as follows:
 - a. Remove pressure indicator from filter bowl (Figure 1, Sheet 1, Detail B, Figure 1, Sheet 2, Detail B, or Detail C).
 - b. Remove and discard packings and packing retainers.
10. Remove sleeve assembly as follows:
 - a. Turn sleeve assembly so that dowel pin lines up with hole in sleeve assembly (Figure 1, Sheet 3, Detail D).
 - b. Remove dowel pin by pushing through housing from outside using a nonmetallic drift. Use care not to damage housing.
 - c. Remove sleeve assembly. Remove and discard packing.
 - d. Visually inspect sleeve assembly for thickness (0.1245 to 0.1250-inch) and damage.

INSTALL

1. Install sleeve assembly as follows:
 - a. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, and install new packing on sleeve assembly (Figure 1, Sheet 3, Detail D).
 - b. **QA** Install sleeve assembly. Align hole in sleeve assembly with hole in housing.
 - c. Select new dowel pin with an OD of 0.1256 to 0.1261-inch diameter.
 - d. **QA** Coat dowel pin with sealing compound, Item 273, WP 1803 00. Install dowel pin using a nonmetallic drift until pin is 0.010-inch below to flush with housing. Clean excess sealing compound from inside sleeve assembly.

NOTE

Make sure that dowel pin does not protrude beyond ID of sleeve assembly.

- e. Coat outside end of dowel pin with sealing compound, Item 273, WP 1803 00, to seal dowel pin and housing hole.
2. Clean filter bowl in cleaning compound solvent, Item 71, WP 1803 00.

INSTALL - Continued

3. If required, install pressure indicator as follows:
 - a. **QA** Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00. Install packings and packing retainer, if required, on pressure indicator.
 - b. **QA** Install pressure indicator in filter bowl (Figure 1, Sheet 1, Detail B or Figure 1, Sheet 2, Detail C).
TIGHTEN PRESSURE INDICATOR 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.
 - c. **QA** If required, using nonelectrical wire, Item 357, WP 1803 00, lockwire pressure indicator to filter bowl.
4. To reset bypass indicator button on filter bowl, 9382-4190, do this:
 - a. Press indicator spring inside bowl down with 6 inch long 1/4-inch drive extension and hold.
 - b. Push indicator up and hold.
 - c. Release indicator spring, then release indicator.
5. To reset bypass indicator button on filter bowls, AD-AO89BEP-2 and 426729, do this:
 - a. Turn filter bowl upside down.
 - b. Press indicator button into filter bowl.

CAUTION

Damage to main transmission will result if incorrect filter element is installed. Incorrect filter element will allow chips to enter main transmission. Make sure the correct filter elements are used when installing oil filter assemblies.

NOTE

First and second stage oil filter elements are throwaway type and must be replaced.

6. **UH-60A EH-60A** With oil filter assembly, 70351-08143-101, do the following (Figure 1, Sheet 1, Detail B):
 - a. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in first stage filter element.
 - b. **QA** Install first stage filter element in filter bowl.
 - c. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in second stage filter element.

CAUTION

The second stage filter element must be installed in the proper orientation as illustrated in Figure 1, Sheet 1, Detail B. Installing the filter upside down can produce main transmission oil pressure fluctuations.

- d. **QA** Install second stage filter in first stage filter element.
- e. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on valve.

INSTALL - Continued**NOTE**

If installed, make sure that tips of spring clips are seated in filter bowl groove.

- f. Depress spring clips if installed and place valve in filter bowl.
- g. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on filter bowl.
- h. **QA** Push filter assembly into filter sleeve.

CAUTION

Damage to the main transmission will result if the arrow on the filter sleeve does not line up with the FLOW position arrow.

- i. **QA** Locally make main transmission oil filter flow valve positioner (Figure 264, WP 1805 00). Turn the gearbox sleeve assembly counterclockwise, until arrow on filter sleeve lines up with FLOW position arrow.
 - j. **QA** Install bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
7. **UH-60A EH-60A** With oil filter assembly, 70351-08143-104, do the following (Figure 1, Sheet 2, Detail B):
- a. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in first stage filter element.
 - b. Install first stage filter element in filter bowl.
 - c. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in second stage filter element.

CAUTION

The second stage filter element must be installed in the proper orientation as illustrated in Figure 1, Sheet 2, Detail B. Installing the filter upside down can produce main transmission oil pressure fluctuations.

- d. Install second stage filter element in first stage filter element.
- e. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on valve.
- f. Install valve in filter bowl.
- g. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on filter bowl.
- h. Install filter assembly into filter sleeve in main gear box sump (Figure 1, Sheet 1, Detail A).

CAUTION

Damage to the main transmission will result if the arrow on the filter sleeve does not line up with the FLOW position arrow.

- i. Locally make main transmission oil filter flow valve positioner (Figure 264 , WP 1805 00). Turn the gearbox sleeve assembly counterclockwise, until arrow on filter sleeve lines up with FLOW position arrow.
- j. Install bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

INSTALL - Continued

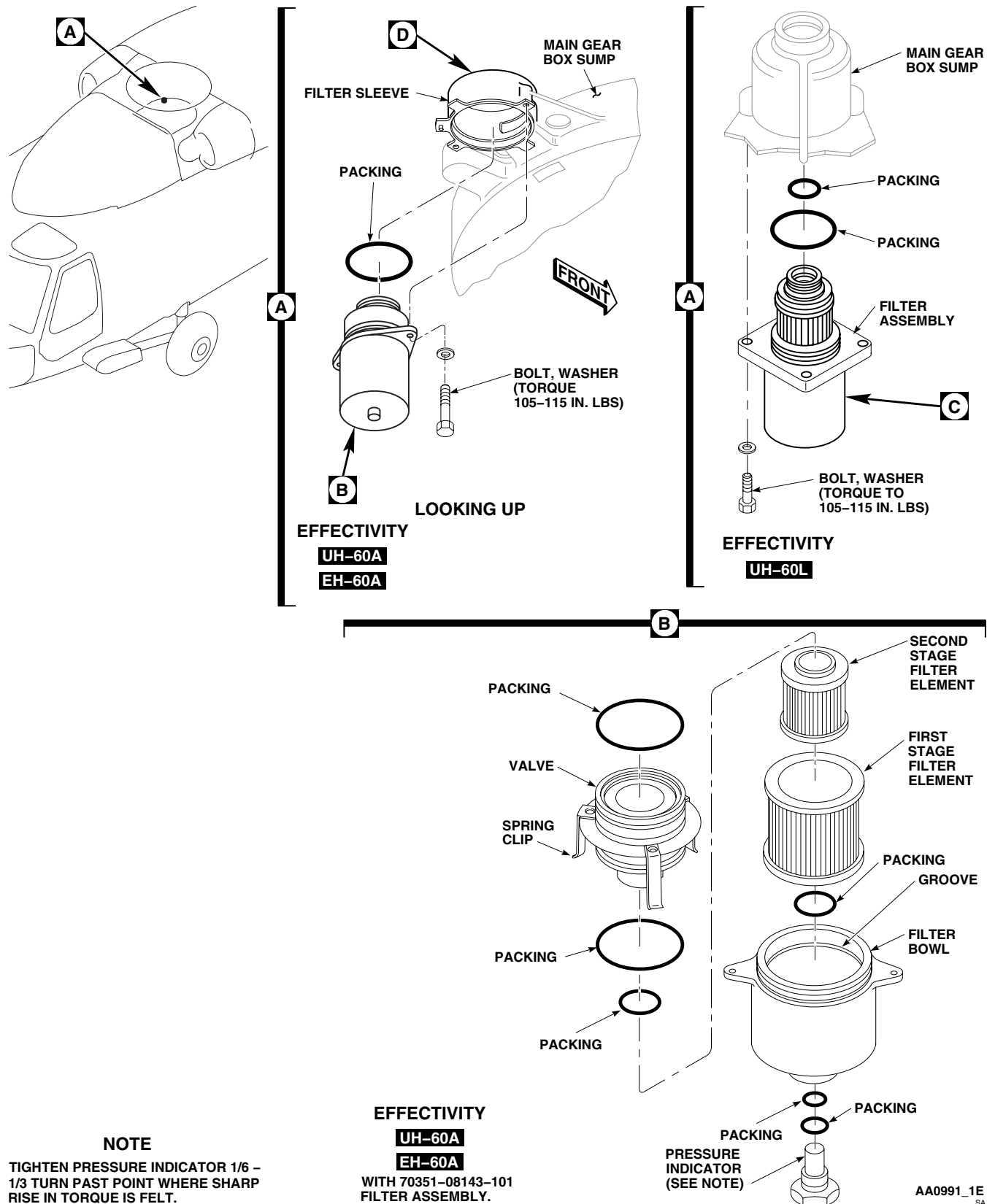
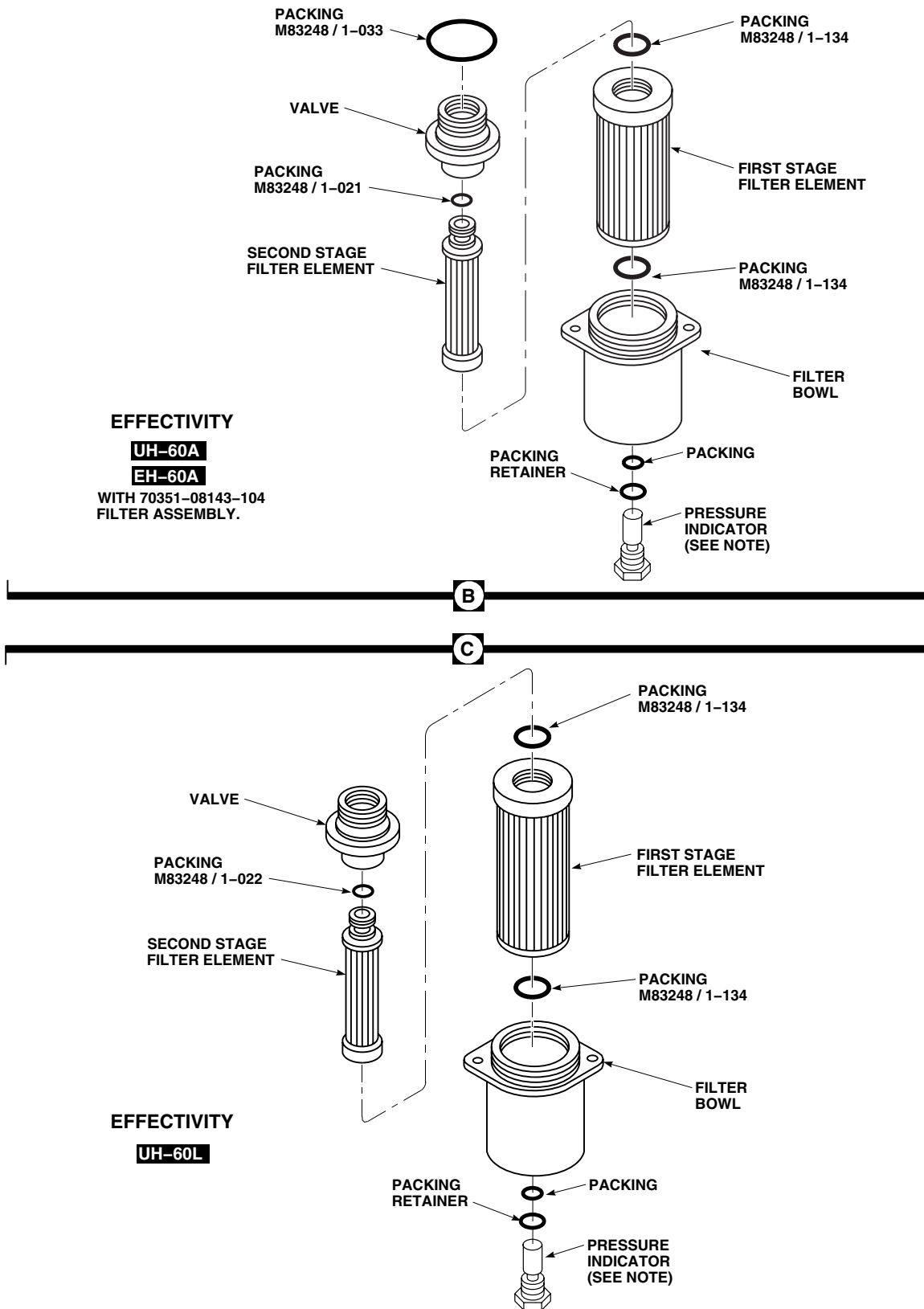


Figure 1. Replace Main Transmission Oil Filter/Sleeve. (Sheet 1 of 3)

INSTALL - Continued



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SA

Figure 1. Replace Main Transmission Oil Filter/Sleeve. (Sheet 2 of 3)

INSTALL - Continued

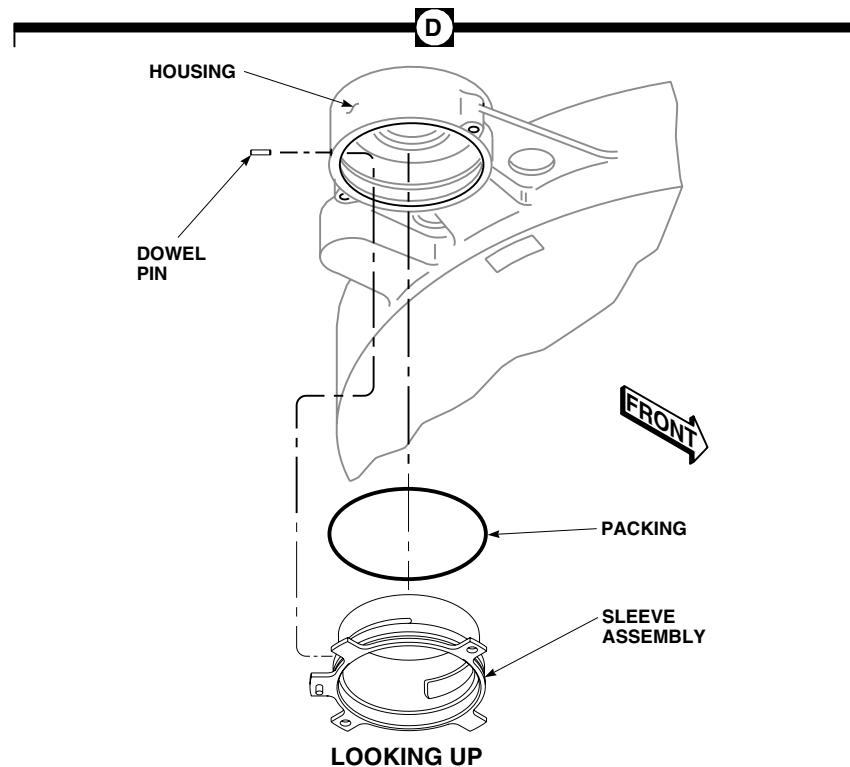
AA0991_3A
SA

Figure 1. Replace Main Transmission Oil Filter/Sleeve. (Sheet 3 of 3)

CAUTION

Damage to main transmission will result if incorrect filter element is installed. Incorrect filter element will allow chips to enter main transmission. Make sure the correct filter elements are used when installing oil filter assemblies.

8. **UH-60L** If valve remained with filter assembly, do the following (Figure 1, Sheet 2, Detail C):
 - a. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in first stage filter element.
 - b. **QA** Install first stage filter element in filter bowl.
 - c. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in second stage filter element.

CAUTION

The second stage filter element must be installed in the proper orientation as illustrated in Figure 1, Sheet 2, Detail C. Installing the filter upside down can produce main transmission oil pressure fluctuations.

- d. **QA** Install second stage filter element in first stage filter element.

INSTALL - Continued

- e. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on filter bowl.
- f. **QA** Install valve on filter assembly.
- g. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in valve.
- h. **QA** Push filter assembly into main gear box sump (Figure 1, Sheet 1, Detail A).
- i. **QA** Install bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**

CAUTION

Damage to main transmission will result if incorrect filter element is installed. Incorrect filter element will allow chips to enter main transmission. Make sure the correct filter elements are used when installing oil filter assemblies.

- 9. **UH-60L** If valve remained in main gear box sump, do the following (Figure 1, Sheet 2, Detail C):
 - a. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in second stage filter element.
 - b. **QA** Install second stage filter element on valve in main gear box sump.
 - c. **QA** Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install in first stage filter element.
 - d. **QA** Install first stage filter element on valve in sump.
 - e. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on filter bowl.
 - f. **QA** Install filter bowl on main gear box sump with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS (Figure 1, Sheet 1, Detail A).**
- 10. Check oil level on dipstick and service if necessary (WP 1607 00).
- 11. Make sure area is clean and free of foreign material.
- 12. Run up helicopter and check for leaks.
- 13. Install drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

MAIN TRANSMISSION OIL PUMP

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-A01
Bolt, NAS563 (2 inch threaded shank, minimum)
Container, 1-Quart
Container, 10-Gallon
Drain Tube, D730L
Installation Tool, 1420-BH-007, Locally-Made, Figure 45, WP 1805 00
Powertrain Repairer's Toolkit, SC 5180-99-B13
Rawhide Mallet
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Assembly Fluid, Item 54, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184, WP 1803 00

Lubricating Oil, Helicopter Transmission, Item 187, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0259 00
WP 0260 00
WP 0261 00
WP 0615 00
WP 1607 00
WP 1803 00
WP 1805 00

NOTE

The following procedures apply to both oil pumps.

REMOVE

1. Remove transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).
2. Remove plug from sump drain (Figure 1).
3. Connect drain tube to sump drain. Drain oil into 10 gallon container.
4. Remove drain tube and replace plug. Using nonelectrical wire, Item 357, WP 1803 00, lockwire plug to sump.
5. Remove nuts and washers holding pump (Figure 1, Detail A).

NOTE

- When removing pump, have 1 quart empty container ready to catch lubricating oil.
 - Two bolts, NAS563, are used as jacking bolts and, when tightened, pushes pump away from gear box housing.
6. Install two bolts, NAS563, with at least a 2 inch thread (NAS563-65 or longer) in threaded holes of pump base. Tighten bolts evenly to draw pump away from sump. Remove pump and packings. Discard packings (Figure 1, Detail A).

REMOVE - Continued

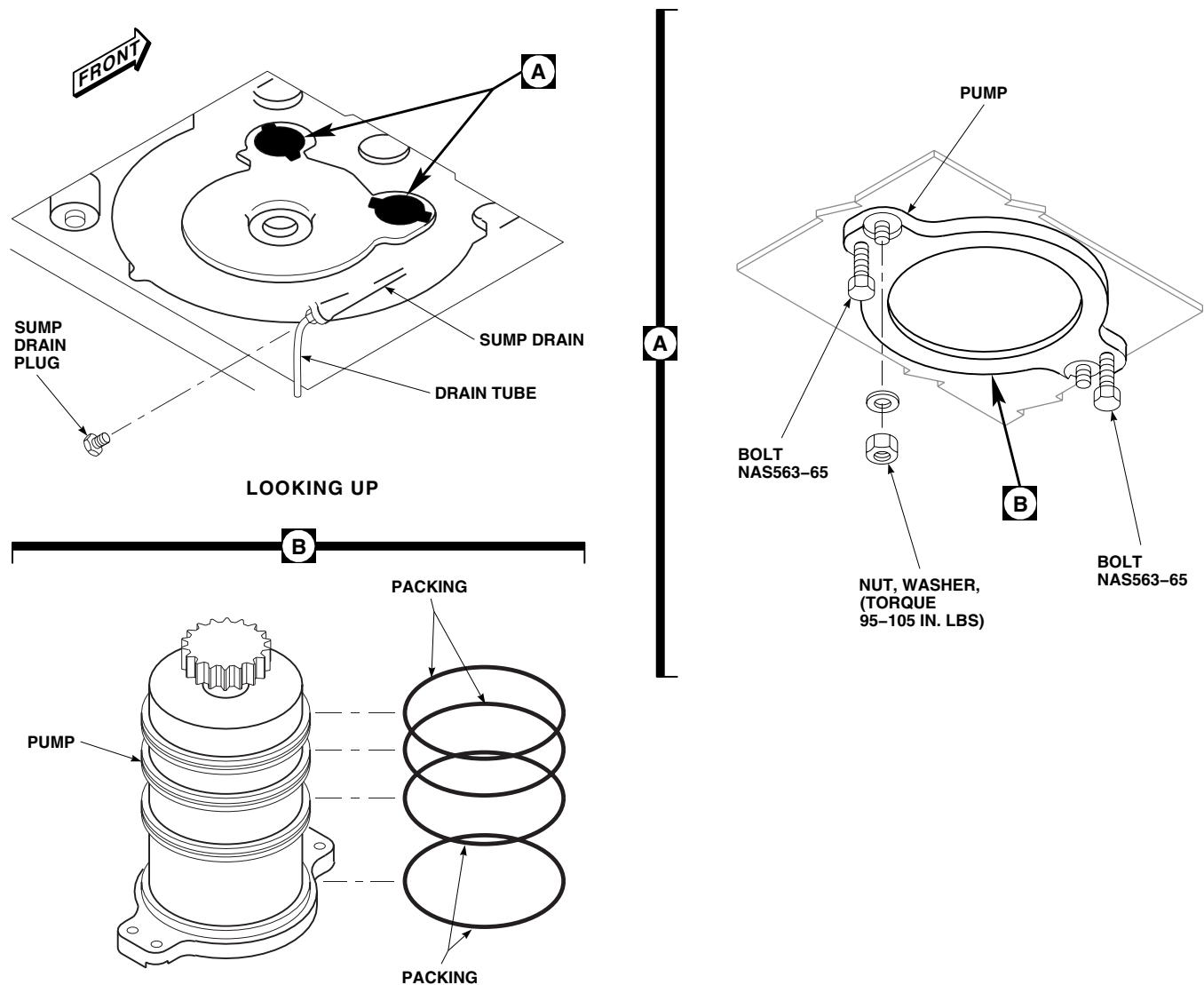
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Figure 1. Replace Main Transmission Oil Pump.

INSTALL

1. Locally make oil pump installation tool (WP 1805 00).
2. Lubricate packings with assembly fluid, Item 54, WP 1803 00, aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
3. **QA** Install packings on pump (Figure 1, Detail B).

INSTALL - Continued**CAUTION**

Damage to pump housing will result if attaching nuts are used to draw pump into sump. Turn rotor head to align splines and install pump.

NOTE

Main gear box lubrication pumps can be installed in either direction.

4. **QA** Push pump into main gear box sump. Seat pump using locally made installation tool and rawhide mallet. Tap around tube with mallet, being careful not to cock pump or damage packings.
5. **QA** Install washers and nuts on pump. **TORQUE NUTS TO 95 - 105 INCH-POUNDS (Figure 1, Detail A).**
6. Service main transmission (WP 1607 00).
7. Make sure area is clean and free of foreign material.
8. Do an operational check (WP 0615 00).
9. Install transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****MAIN TRANSMISSION OIL PRESSURE TRANSMITTER****This WP supersedes WP 0649 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 1-Quart
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00

References

WP 0615 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Open oil cooler compartment access doors.
3. Disconnect electrical connector, P444, from oil pressure transmitter (Figure 1, Detail A).

CAUTION

Damage to threaded port will result if excessive torque is used to remove oil pressure transmitter. Make sure all sealant is removed prior to removing oil pressure transmitter.

4. Remove sealing compound from area around mating flange using nonmetallic scraper.
5. Remove oil pressure transmitter from main gear box (Figure 1, Detail A).
6. Position 1 quart container to catch any residual oil.

INSTALL

1. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
2. **QA** Install packing on oil pressure transmitter.

CAUTION

Damage to oil pressure transmitter will result if over torqued. Oil pressure readings will be effected. Do not over torque oil pressure transmitter.

3. **QA** Install oil pressure transmitter in main gear box. **TORQUE TO 120-130 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire transmitter (Figure 1, Detail A).

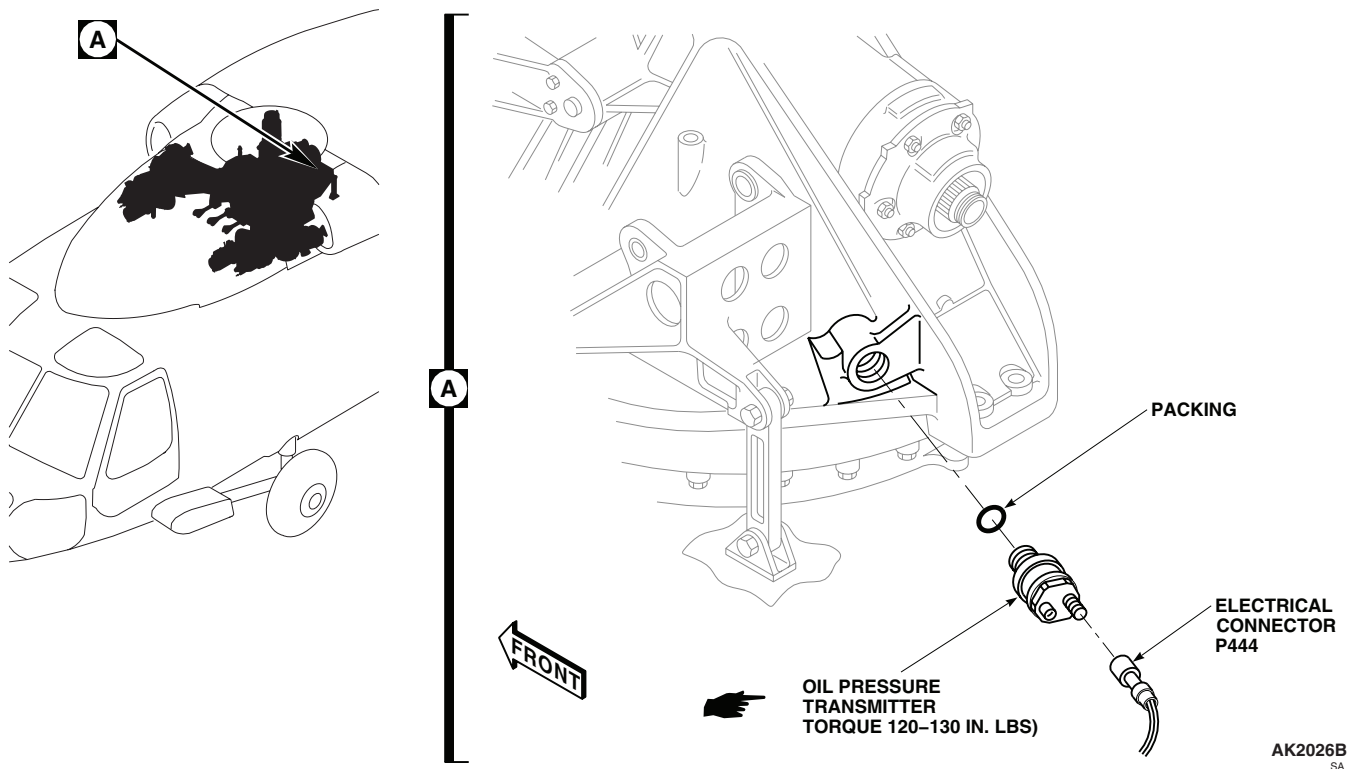
INSTALL - Continued

Figure 1. Replace Main Transmission Oil Pressure Transmitter.

4. Seal area around mating flange with sealing compound, Item 273, WP 1803 00.
5. Inspect and treat electrical connector (WP 1703 00).
6. **QA** Install electrical connector, P444, into transmitter.
7. Make sure area is clean and free of foreign material.
8. Close oil cooler compartment access doors.
9. Do a functional check and check for leaks (WP 0615 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

MAIN TRANSMISSION OIL PUMP PRESSURE REGULATING VALVES

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 1-Quart
Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00
Sealing Compound, Item 287, WP 1803 00
Wire, Nonelectrical, Item 354, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

References

WP 0259 00
WP 0260 00
WP 0261 00
WP 0615 00
WP 0617 00
WP 0618 00
WP 1607 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).
3. Drain main transmission (WP 1607 00).
4. Unscrew pump pressure regulating valves (Figure 1).
5. Position 1 quart container to catch any residual oil.

INSTALL

1. Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
2. Install new packings on new pressure regulating valve.
3. Screw pressure regulating valve into module until it is bottomed. Using nonelectrical wire, Item 357, WP 1803 00, lockwire valve to module (Figure 1).
4. Seal area around pressure relief valve to sump interface with sealing compound, Item 287, WP 1803 00.
5. Make sure area is clean and free of foreign material.
6. Service main transmission (WP 1607 00).
7. Do a functional check (WP 0615 00) and check for leaks.
8. Install transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).

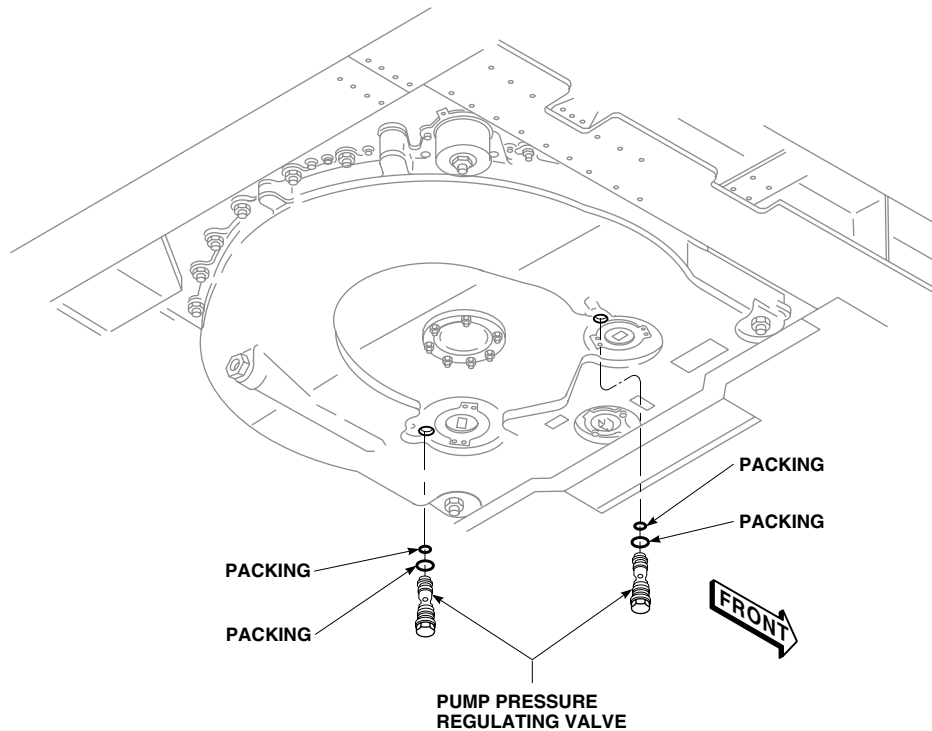
INSTALL - ContinuedAK2027
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Figure 1. Replace Oil Pump Pressure Regulating Valves.

ADJUST**NOTE**

- Normal oil pressure, stabilized after warmup, is 50 to 55 psi.
- Adjust oil pressure while engine is shut down.

1. Remove transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).
2. If oil pressure, after warmup, is above normal operating pressure, do this:
 - a. Using broad-tipped screwdriver turn both pressure regulator screws counter clockwise an equal amount (Figure 2, Detail A).

NOTE

One complete counterclockwise turn of both screws will lower oil pressure 3 to 4 psi.

- b. After adjustment, screws may be adjusted up to 1/4 turn counterclockwise maximum, to align hole with lockwire slots.
 - c. **QA** Using nonelectrical wire, Item 354, WP 1803 00, lockwire adjustment screws.
 - d. Ground run engine 100% Nr and record oil pressure (WP 0615 00).
 - e. If oil pressure will not stabilize at 50 to 55 psi after valve adjustment, replace both oil pressure regulator valves.
3. If oil pressure, after warmup, is below normal operating pressure, do this:

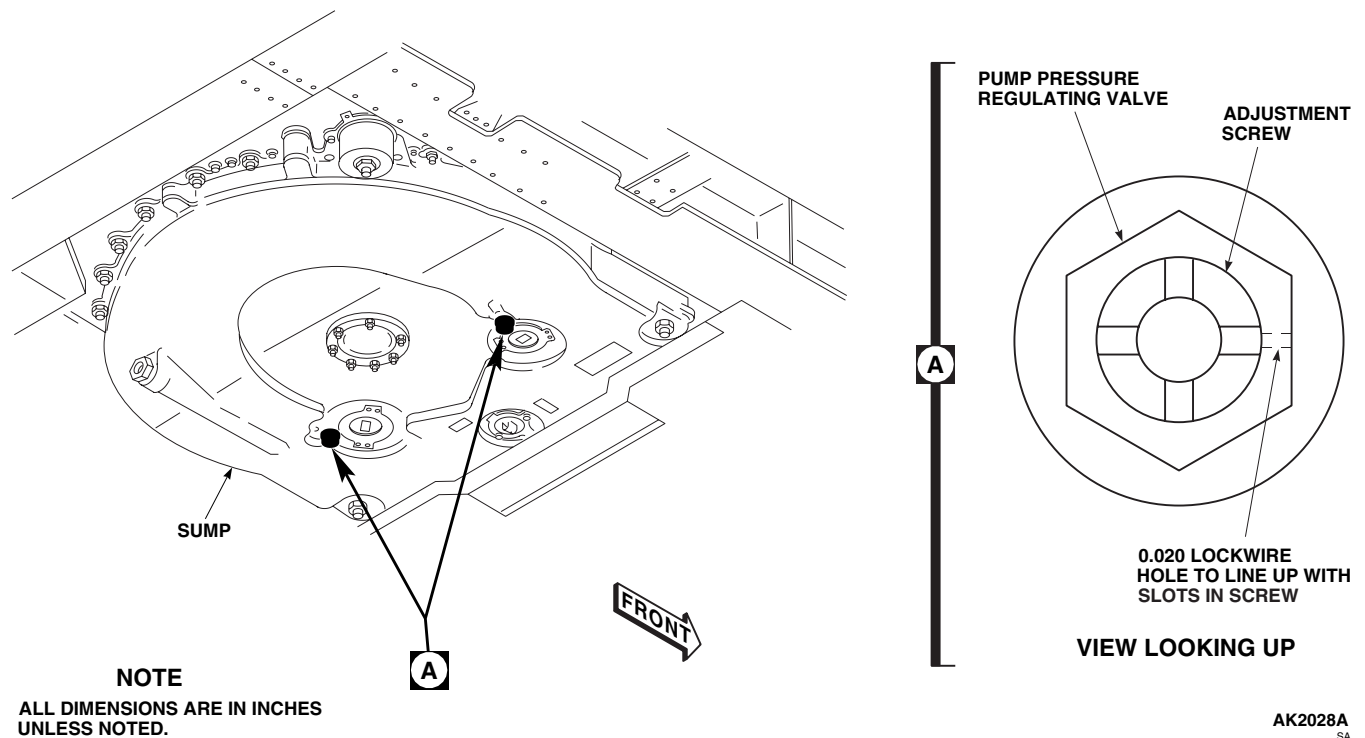
ADJUST - Continued

Figure 2. Replace Oil Pump Pressure Regulating Valves.

- a. **QA** Check chip detectors, strainers, and screens for contamination in main, input, and accessory modules (WP 0617 00).
- b. Check oil filter for contamination (WP 0618 00).

NOTE

Adjust oil pressure regulating valves while the engine is shut down.

4. If no contamination is found, turn both pressure regulator screws clockwise an equal amount using broad-tipped screwdriver.

NOTE

One complete clockwise turn of both screws will raise oil pressure 3 to 4 psi.

5. After adjustment, screws may be adjusted up to 1/4 turn counterclockwise maximum, to align hole with lockwire slots.
6. **QA** Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws
7. Ground run engine 100% Nr and record oil pressure (WP 0615 00).
8. If oil pressure will not stabilize at 50 to 55 psi after valve adjustment, replace both oil pressure regulator valves.
9. Install transmission drip pan (WP 0259 00, WP 0260 00, or WP 0261 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
MAIN TRANSMISSION OIL GAGE TUBE ASSEMBLY

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-A01
Powertrain Repairer's Toolkit, SC 5180-99-CL-A13
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 300 - 2500 in. lbs.

Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00

References

WP 1803 00

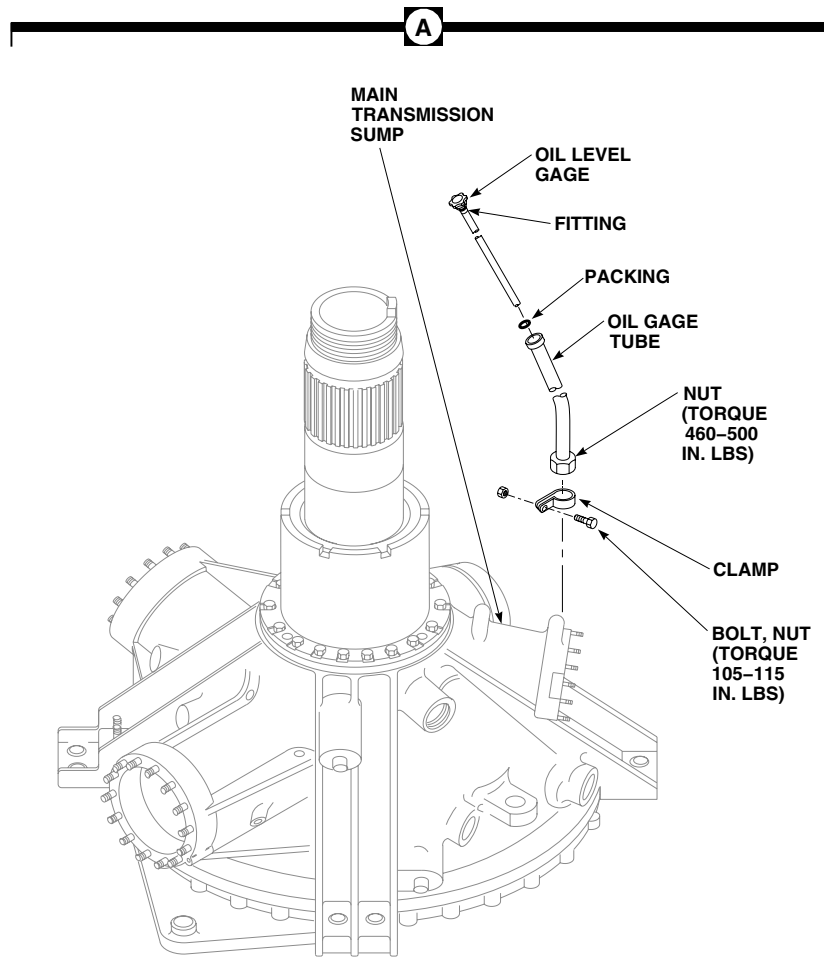
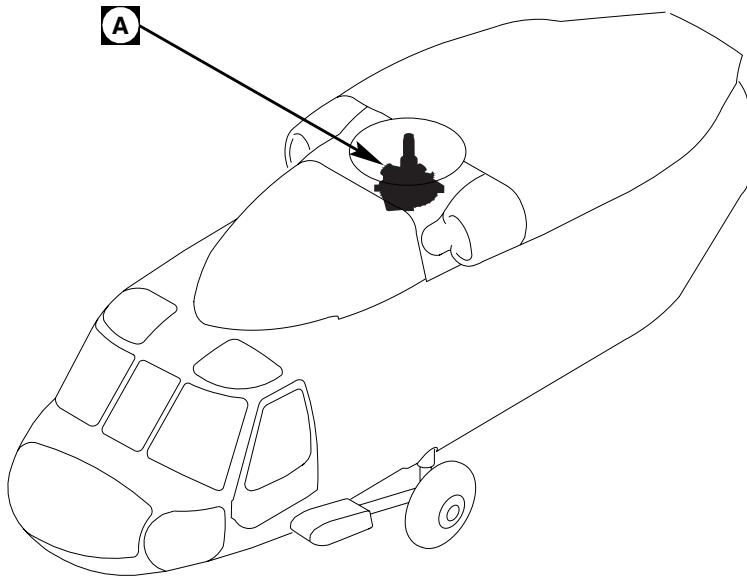
REMOVE

1. Turn off all electrical power.
2. Open oil cooler compartment access doors.
3. Push down and twist oil level gage to remove from oil gage tube assembly (Figure 1, Detail A).
4. Remove fitting and packing from oil gage tube assembly (Figure 1, Detail A). Discard packing.
5. Clean connection area at base of oil gage tube and main transmission sump using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00.
6. Disassemble nut, bolt, and clamp from bracket on transmission.
7. Loosen nut from transmission sump connection and remove oil gage tube assembly. Plug hole in sump.
8. Keep oil level gage, clamp, and hardware for installation.

INSTALL

1. Turn off all electrical power.
2. Remove plug from transmission sump connection and clean threads using machinery towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00. Use care not to contaminate oil reservoir.
3. **QA** Position and install oil gage tube onto transmission sump connection.
4. **QA** Align and assemble clamp to bracket using bolt and nut. **TORQUE NUT TO 105 - 115 INCH-POUNDS.**
5. **QA** Tighten transmission connection. **TORQUE TO 460 - 500 INCH-POUNDS.**
6. **UH60L** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00. ■
7. **QA** Install packing on oil level gage fitting.

INSTALL - Continued



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Figure 1. Replace Main Transmission Oil Gage Tube Assembly.

INSTALL - Continued**NOTE**

Nonelectrical wire not required for oil level gage fitting.

8. **QA** Install oil level gage fitting with packing into oil gage tube assembly.
9. **QA** Install oil level gage into oil gage tube assembly. Push down and twist oil level gage to lock into fitting.
10. Make sure area is clean and free of foreign material.
11. Close oil cooler compartment access doors.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

MAIN TRANSMISSION OIL TEMPERATURE SWITCH

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powertrain Repairer's Toolkit, SC 5180-99-B13

Sealing Compound, Item 273, WP 1803 00

Wire, Nonelectrical, Item 357, WP 1803 00

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1803 00
WP 1703 00
WP 0615 00

REMOVE

1. Turn off all electrical power.
2. Open oil cooler compartment access doors.
3. Disconnect electrical connector, P408, from oil temperature switch (Figure 1, Detail A).
4. Remove oil temperature switch from main module.

INSTALL

1. Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
2. **QA** Install packing on oil temperature switch (Figure 1, Detail A).
3. **QA** Install oil temperature switch in main gear box. Using nonelectrical wire, Item 357, WP 1803 00, lockwire switch.
4. Seal area around mating flange with sealing compound, Item 273, WP 1803 00.
5. Inspect and treat electrical connector (WP 1703 00).
6. **QA** Connect electrical connector, P408, to oil temperature switch.
7. Make sure area is clean and free of foreign material.
8. Close oil cooler compartment access doors.
9. Do a functional check and check for leaks (WP 0615 00).

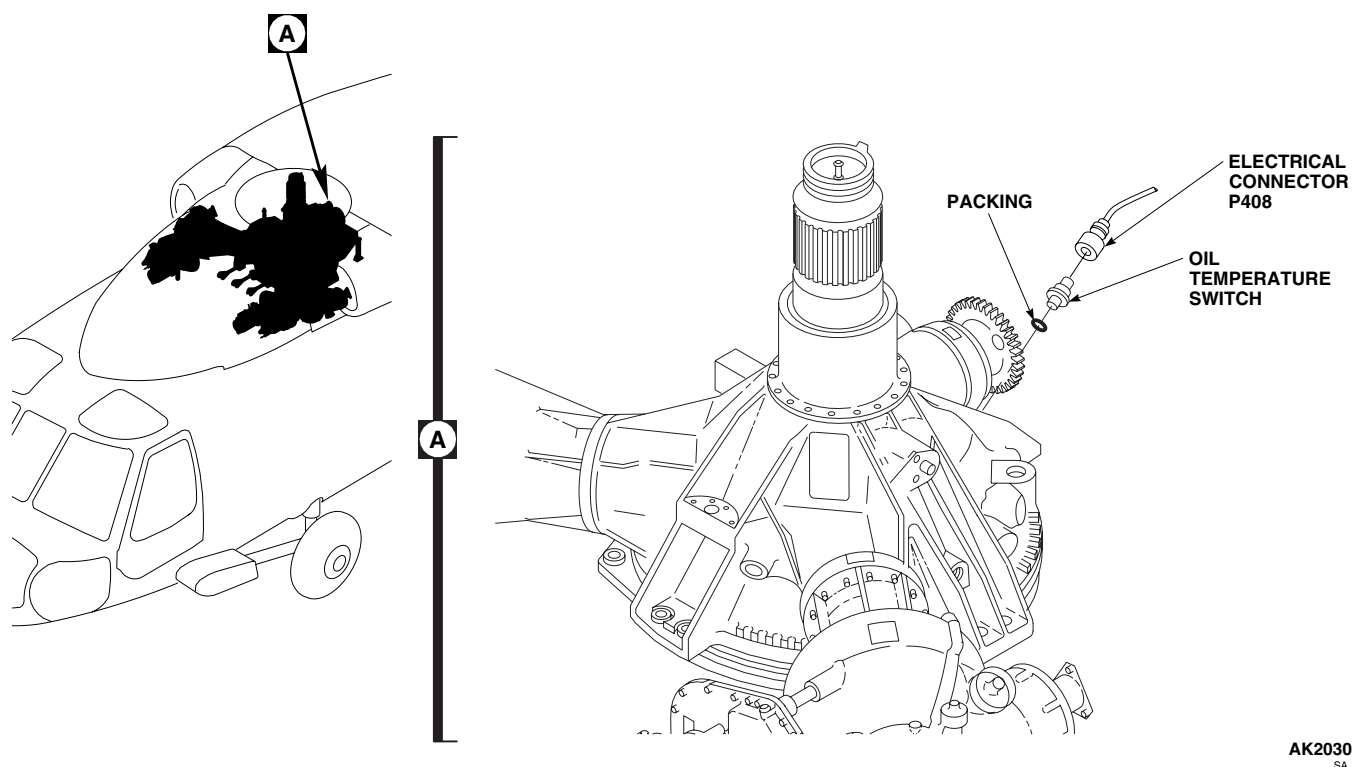
INSTALL - Continued

Figure 1. Replace Main Transmission Oil Temperature Switch.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**INPUT MODULE**

This WP supersedes WP 0653 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Jacking Bolts, 1/4-28
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Sling, Aircraft Maintenance, 70700-20320-045
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Chromium Trioxide Item 65, WP 1803 00
 Cleaning Compound Solvent, Item 71, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 184,
 WP 1803 00
 Lubricating Oil, Helicopter Transmission, Item 187,
 WP 1803 00
 Petrolatum, Item 216, WP 1803 00
 Polyurethane Coating, Item 231, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
 TM 55-1500-345-23
 WP 0085 00
 WP 0101 00
 WP 0301 00
 WP 0381 00
 WP 0484 00
 WP 0486 00
 WP 0615 00
 WP 0619 00
 WP 0630 00
 WP 0643 00
 WP 0644 00
 WP 0645 00
 WP 0646 00
 WP 0656 00
 WP 0706 00
 WP 0839 00
 WP 1590 00
 WP 1703 00
 WP 1803 00

REMOVE**NOTE**

- Input module with accessory module installed, but without pump and generator installed weighs 140 pounds.
- When replacing input module make sure to remove and keep gimbal and input flange for reuse.
- The following procedure applies to both left and right input modules, except as noted.

1. If maintenance crane is going to be used, assemble and erect crane (WP 1590 00).

REMOVE - Continued

2. Turn off all electrical power.
3. Open main rotor pylon sliding cover.
4. Open engine cowling.
5. Remove engine air inlets (WP 0484 00).
6. Remove air inlet fairing (WP 0301 00).
7. Disconnect engine output shaft from input module flange (WP 0486 00).
8. Disconnect forward engine support tube from gimbal on input module (WP 0486 00).

NOTE

When removing right input module, both hydraulic pump and generator must be removed first, in order to clear refill handpump. When removing left module, only hydraulic pump must be removed first.

9. Remove hydraulic pump module (WP 0706 00).
10. Remove generator (WP 0839 00).
11. Keep internal oil tubes between generator and accessory module with accessory module.
12. Disconnect chip detector electrical connector, P406, left side or electrical connector, P405, right side from main transmission wiring harness (Figure 1).
13. Disconnect electrical connector, P443, from Nr sensor on right accessory module (Figure 1).
14. **UH-60L** Disconnect electrical connector, P407, from Nr sensor on left accessory module (Figure 1).
15. Disconnect low oil pressure electrical connector, P409, from left module (Figure 1).
16. Disconnect engine control cable supports from accessory modules. Keep bolts, washers, and nuts with supports.
17. Attach hoisting sling straps to input module. Place one strap around accessory module mounting flange and connect other strap to module lifting bracket (Figure 2, Sheet 1, Detail A).

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

18. Attach sling to maintenance crane or a hoist with enough capacity to lift input module.
19. Adjust length of hoisting sling straps for even distribution of weight before lifting.
20. Raise hoist and take up slack.
21. Remove nuts and washers attaching input module to main module.

REMOVE - Continued

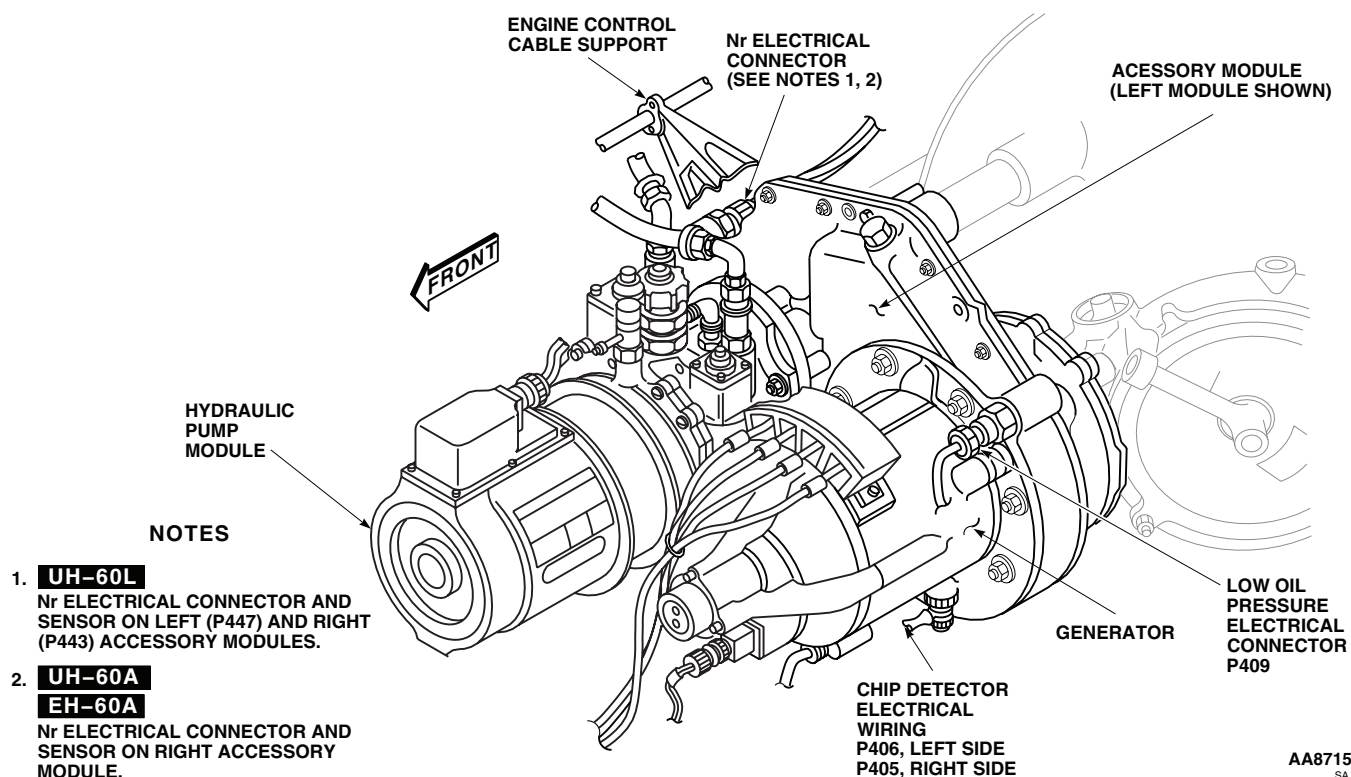


Figure 1. Remove Pump Module and Generator.

CAUTION

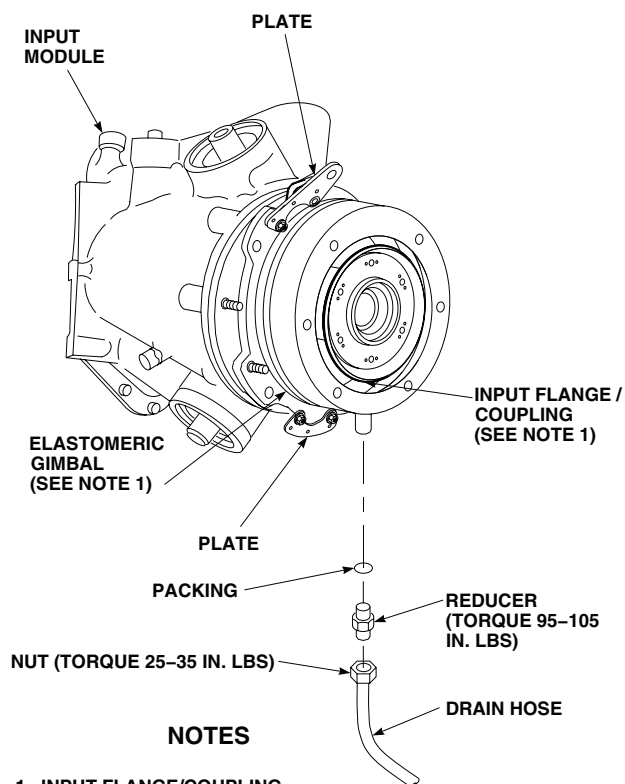
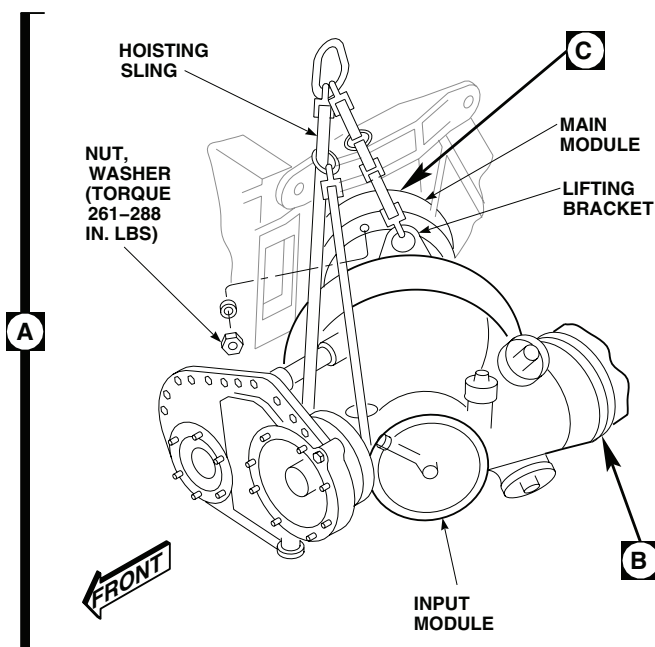
Damage to quill shaft will result if quill shaft is not removed from input module, or main module while removing input module. Quill shaft may fall damaging splines. Remove quill shaft while removing input module.

NOTE

Care must be taken not to burr studs while removing input module.

22. Carefully pull input module from main module.
23. Lift input module clear of helicopter and place on stand. If input module does not pull out of main module easily, it may be jacked out. Use 1/4-28 bolts in the three jacking holes in input module. Slowly and equally thread in bolts to push input module away from main module. Be careful not to cock module. Make sure oil return tube stays with main gear box module. Cover port of oil return tube. Remove hoisting sling.
24. Remove sealing compound using nonmetallic scraper.

A line drawing of a car from a side-rear perspective. The rear window and the area immediately behind it are filled with a solid black silhouette, representing a large object or person. A black arrow points from a small black square containing the white letter 'A' to this blacked-out area.



1. INPUT FLANGE/COUPLING
70351-08255-102, AND GIMBAL
70361-08015-102.
2. INPUT FLANGE 70351-08206-102, AND
GIMBAL 70361-08001-101,
70361-08001-102, 70361-08001-103.

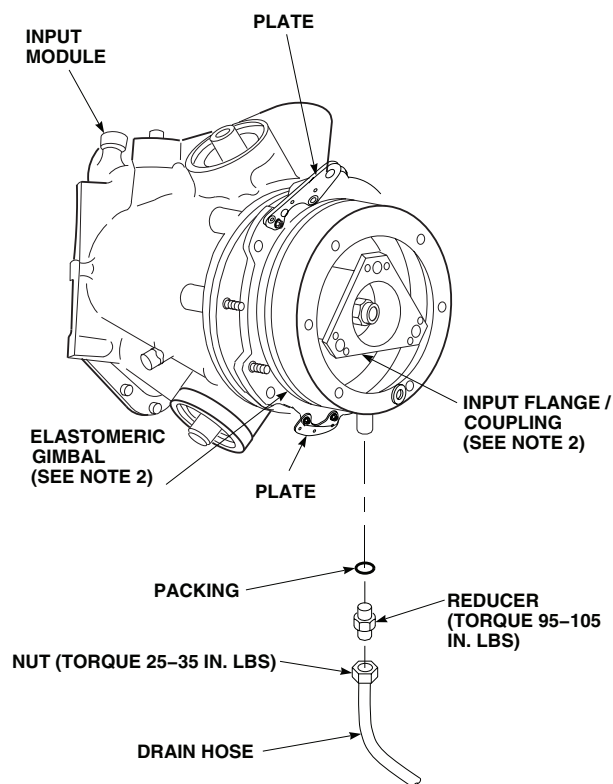
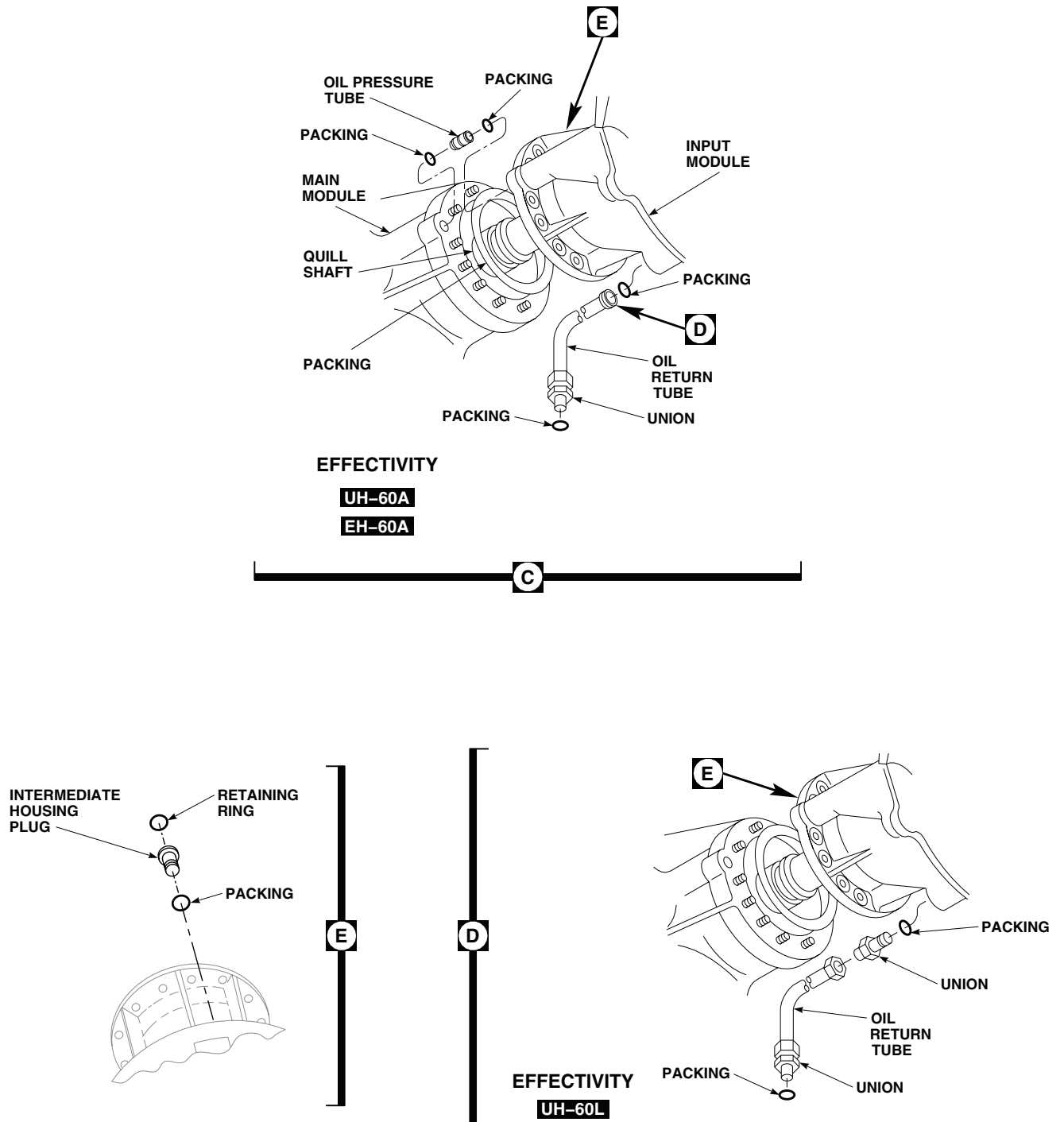
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Figure 2. Replace Input Module. (Sheet 1 of 2)

REMOVE - Continued



AA7854_2A
SA

Figure 2. Replace Input Module. (Sheet 2 of 2)

REMOVE - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

25. Clean primer off mating surface and hardware using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
26. Inspect input pinion for axial movement (WP 0619 00).

INPUT MODULE BUILDUP**NOTE**

If a replacement input module is going to be installed, remove the following parts from the old module.

1. Remove air inlet fairing bracket from input module (Figure 2, Sheet 1, Detail B).
2. Remove lifting bracket (Figure 2, Sheet 1, Detail A).
3. Disconnect elastomeric gimbal drain hose from reducer, install protective cap on hose (Figure 2, Sheet 1, Detail B).
4. Remove reducer and packing from elastomeric gimbal, install protective plug in gimbal. Discard packing.

NOTE

When removing elastomeric gimbal, do not remove two nuts on short studs. These nuts are part of input module and removal is not required to remove gimbal.

5. Remove elastomeric gimbal and attaching plates. Remove six nuts in equal steps, otherwise they will not clear studs before hitting housing. If new input module is not readily available for installation, cover opening in main module.
6. Remove retaining ring and intermediate housing plug. Remove and discard packing from plug (Figure 2, Sheet 2, Detail E).
7. Remove accessory module (WP 0656 00).
8. Clean quill shaft using clean machinery wiping towel, Item 344, WP 1803 00, and cleaning compound solvent, Item 71, WP 1803 00.
9. Visually inspect quill shaft and splines for damage. **NONE ALLOWED.** Replace quill shaft if damaged.
10. Lubricate shaft with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
11. **QA** Install quill shaft in input module.
12. Install accessory module (WP 0656 00).
13. If required, install lifting bracket on input module split half at 12 o'clock position. **TORQUE NUTS TO 261 - 288 INCH-POUNDS (Figure 2, Sheet 1, Detail A).**

INPUT MODULE BUILDUP - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

14. Clean mounting surfaces on gimbal and input module with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
15. Apply primer coating, Item 243, WP 1803 00, to gimbal mounting surface on input module.

CAUTION

- Disc pack coupling will be damaged if not connected to correct gimbal.
- Make sure to use the correct gimbal with input flange/coupling being used. Elastomeric gimbal 70361-08015-102 must be used with input flange/coupling 70351-08255-102 and engine output shaft 70361-08014-041. Elastomeric gimbal 70361-08001-103, must be used with input flange 70351-08001-102, or 70361-08001-103, must be used with input flange 70351-08206-102 and engine output shaft 70361-08004-042, or 70361-08004-043 and disk pack 70361-08002-041.

NOTE

- When installing elastomeric gimbal, do not install washers under nuts which secure plates.
 - Gimbal drain port must be located at 6 o'clock position when installed.
16. **QA** Install elastomeric gimbal and plates on studs only far enough to allow nuts to be started. Start all nuts together and run them down two or three threads at a time.
 - a. **FOR GIMBAL 70361-08001-101, 70361-08001-102, OR 70361-08001-103, TORQUE NUTS TO 261 - 288 INCH-POUNDS (Figure 2, Sheet 1, Detail B).**
 - b. **FOR GIMBAL 70361-08015-102, TORQUE NUTS TO 145-165 INCH-POUNDS (Figure 2, Sheet 1, Detail B).**
 17. Lubricate packing with petrolatum, Item 216, WP 1803 00.
 18. **QA** Install packing onto reducer. Install reducer in elastomeric gimbal. **TORQUE REDUCER TO 95 - 105 INCH-POUNDS.**
 19. **QA** Install drain hose onto reducer. **TORQUE NUT TO 25 - 35 INCH-POUNDS.**
 20. Apply epoxy primer coating kit, Item 120, WP 1803 00, to all exposed washers, nuts, and studs securing gimbal to module.
 21. Cover gimbal mounting hardware with sealing compound, Item 273, WP 1803 00.
 22. Apply epoxy primer coating kit, Item 120, WP 1803 00, over sealing compound and apply one coat of polyurethane coating, Item 231, WP 1803 00, over primer.
 23. **QA** Install air inlet fairing bracket on plates.

INPUT MODULE BUILDUP - Continued**CAUTION**

Damage to input module will result if intermediate housing shipping plug is not removed. Make sure intermediate housing shipping plug is removed.

24. Remove intermediate housing shipping plug from replacement module.
25. **QA** Coat packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00. Install packing on intermediate housing plug.
26. **QA** Install intermediate housing plug into replacement input module (Figure 2, Sheet 2, Detail E).
27. Apply epoxy primer coating kit, Item 120, WP 1803 00, to retaining ring. Allow to dry.
28. **QA** Apply epoxy primer coating kit, Item 120, WP 1803 00, to retaining ring groove in intermediate housing. Install retaining ring while primer is still wet. Apply primer to face of installed retaining ring. Allow to dry.
29. Apply sealing compound, Item 276, WP 1803 00, around area of retaining ring
30. Apply epoxy primer coating kit, Item 120, WP 1803 00, over sealing compound and apply one coat of polyurethane coating, Item 231, WP 1803 00, over primer.

INSTALL**CAUTION**

Damage to input module will result if all shipping plugs are not removed. Make sure shipping plugs are removed from all port openings.

1. Attach straps of hoisting sling, to input module. Position one strap around accessory module attachment flange. Connect other strap to module lifting bracket (Figure 2, Sheet 1, Detail A).
2. Attach sling to hoist or maintenance crane. Adjust lengths of hoisting sling straps for even distribution of weight.
3. Remove oil return tube from main module sump (Figure 2, Sheet 2, Detail C and Detail D).
4. Remove packings from both ends of oil return tube. Discard packing.
5. Coat replacement packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
6. **QA** Install packings on oil return tube.
7. **QA** Install oil return tube in main module sump.

NOTE

Care must be taken not to burr mounting studs while installing input module.

8. **QA** Coat packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and place packing in groove on input module (Figure 2, Sheet 2, Detail C and Detail D).
9. **QA** Lift module into place. Make sure oil pressure tube and return tube packings are in position and seated.

INSTALL - Continued**NOTE**

To tighten or loosen nut on oil return tube, hold hex on fitting in main module sump using 1 3/8-inch open-end wrench. End of fitting in sump is not threaded.

10. Coat area between Rosan lockring and mating housing counterbore with sealing compound, Item 273, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

11. Clean mounting surface on main and input modules with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
12. Touch up bare magnesium surfaces with chromium trioxide pickle solution, Item 65, WP 1803 00 (TM 1-1500-344-23).

WARNING

Injury to personnel and damage to equipment will result if transmission oil is contaminated. To avoid contaminating transmission oil system, do not apply primer to bore diameters that are sealing surfaces for packings.

13. Apply epoxy primer coating kit, Item 120, WP 1803 00, to input module mounting surface and fastener holes. Do not apply primer to packing or packing groove.

NOTE

- Install input module while primer is still wet.
- When changing input module to opposite side, make sure that boltholes of gimbal and forward support line up.

14. **QA** Guide quill shaft into main module while installing input module onto mounting flange. If necessary, turn input flange to line up splines. Also guide oil return tube into input module. If necessary, loosen nut on oil return tube to line up.
15. **QA** Install nuts and washers. **TORQUE NUTS TO 261 - 288 INCH-POUNDS (Figure 2, Sheet 1, Detail A).**
16. Apply epoxy primer coating kit, Item 120, WP 1803 00, to all exposed washers, nuts, and studs securing input module to main module.
17. Seal area around joint of washer and nut, threads of bolts to nut, and seam between input module and main module with sealing compound, Item 273, WP 1803 00.
18. Remove extra sealing compound.
19. Apply epoxy primer coating kit, Item 120, WP 1803 00, over sealing compound, and apply one coat of polyurethane coating, Item 231, WP 1803 00, over primer (WP 0381 00 and TM 55-1500-345-23).
20. Tighten nut on oil return tube. Hold hex on union while tightening.
21. Remove hoisting sling.

INSTALL - Continued

22. Inspect and clean accessory, input, and main module oil return screens, chip detectors, and chip detector strainers (WP 0643 00, WP 0644 00, WP 0645 00, and WP 0646 00).
23. Check chip detector electrical circuit (WP 0085 00).
24. Connect forward engine support tube to gimbal on input module (WP 0486 00).
25. Connect engine output drive shaft to input module flange (WP 0486 00).
26. Install generator (WP 0839 00).
27. Install pump module (WP 0706 00).
28. Inspect and treat electrical connectors (WP 1703 00).

NOTE

See WP 0630 00 for wiring harness routing.

29. **QA** Connect chip detector electrical connector P406 left side or P405 right side, to main transmission wiring harness (Figure 1).
30. **QA** **UH-60L** Connect Nr electrical connectors to back side of Nr sensors on left and right accessory modules. Using nonelectrical wire, Item 357, WP 1803 00, lockwire connectors (Figure 3, Detail A). ■
31. **QA** **UH-60A EH-60A** Connect Nr electrical connector to back side of Nr sensor on right accessory module. Using nonelectrical wire, Item 357, WP 1803 00, lockwire connector. ■
32. **QA** Connect low oil pressure electrical connector, P409, to low oil pressure switch on left module over generator. Using nonelectrical wire, Item 357, WP 1803 00, lockwire connector.
33. Coat cable support boltholes on accessory module with primer coating, Item 243, WP 1803 00.
34. **QA** Position engine control cable support on accessory module. Support, 70301-22107-101, with clip, 70301-02119-041, is installed on left accessory module. Add washers as required to fill gap between support and module. Install bolts, washers, and nuts attaching cable support to accessory module. **TORQUE NUTS TO 105 - 115 INCH-POUNDS (Figure 3, Detail B).**
35. Apply one coat of polyurethane coating, Item 231, WP 1803 00, to any water entrapment areas of input module.
36. Install air inlet fairing (WP 0301 00).
37. Install engine air inlets (WP 0484 00).
38. Service main transmission (WP 1607 00).
39. Make sure area is clean and free of foreign material.
40. Close engine cowlings and main rotor pylon sliding cover.
41. Do a functional test and check for leaks (WP 0615 00).
42. Do an operational check of ac electrical system and hydraulic pump module (WP 0101 00).

INSTALL - Continued

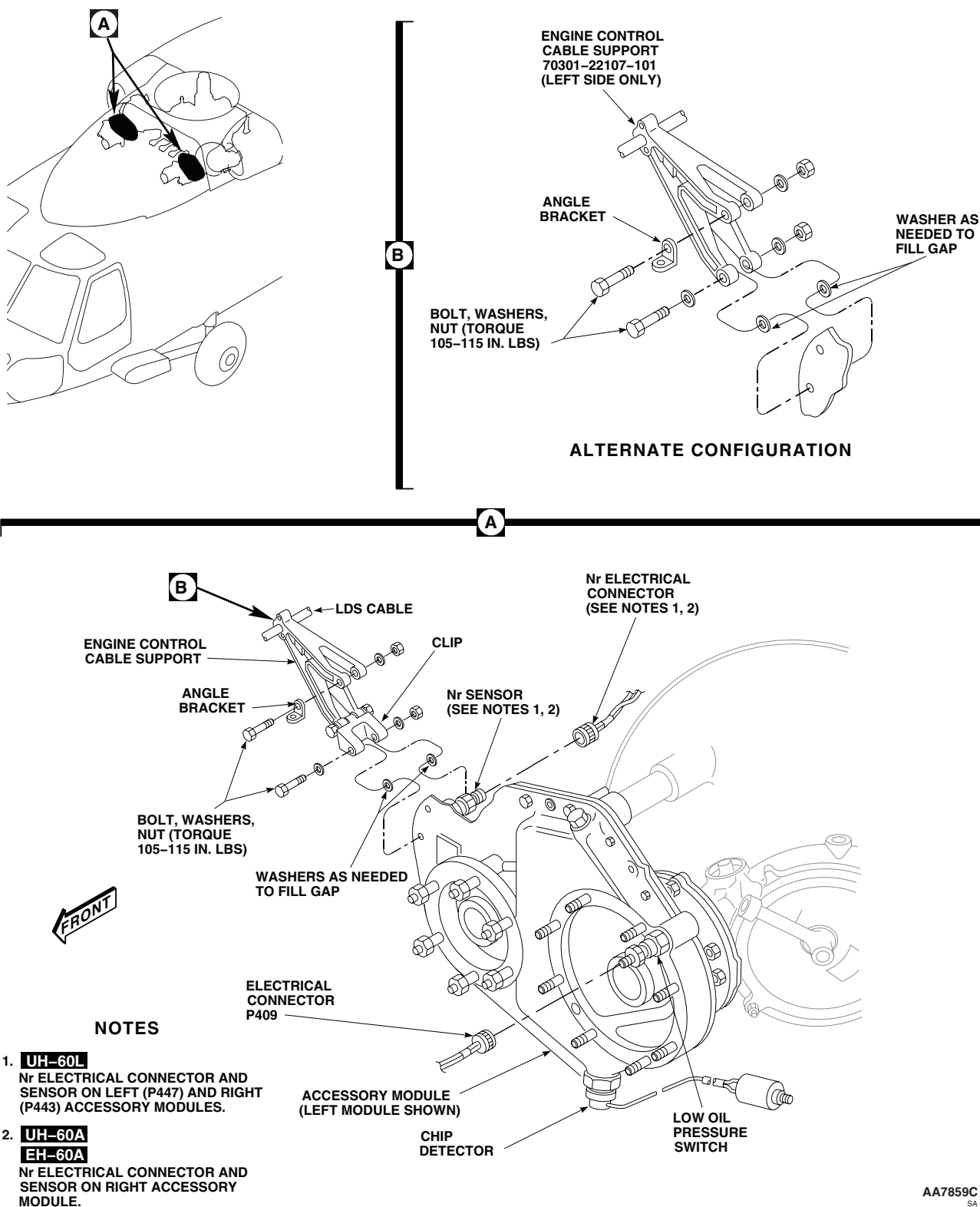
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Figure 3. Install Pump Module and Generator.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****INPUT MODULE SEAL**

This WP supersedes WP 0654 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Flange Puller, Locally-Made, Figure 50, WP 1805 00
Powertrain Repairer's Toolkit, SC 5180-99-B13
Scribes
Seal Installation Tool, Locally-Made, Figure 157,
WP 1805 00
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.
Torque Wrench, Dial Indicating

Material/Parts

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00

Primer Coating, Item 243, WP 1803 00
Screws, MS35206-245 or Equivalent
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0481 00
WP 0615 00
WP 0619 00
WP 0654 01
WP 0655 00
WP 1803 00
WP 1805 00

REMOVE

1. Remove input module gimbal (WP 0654 01).

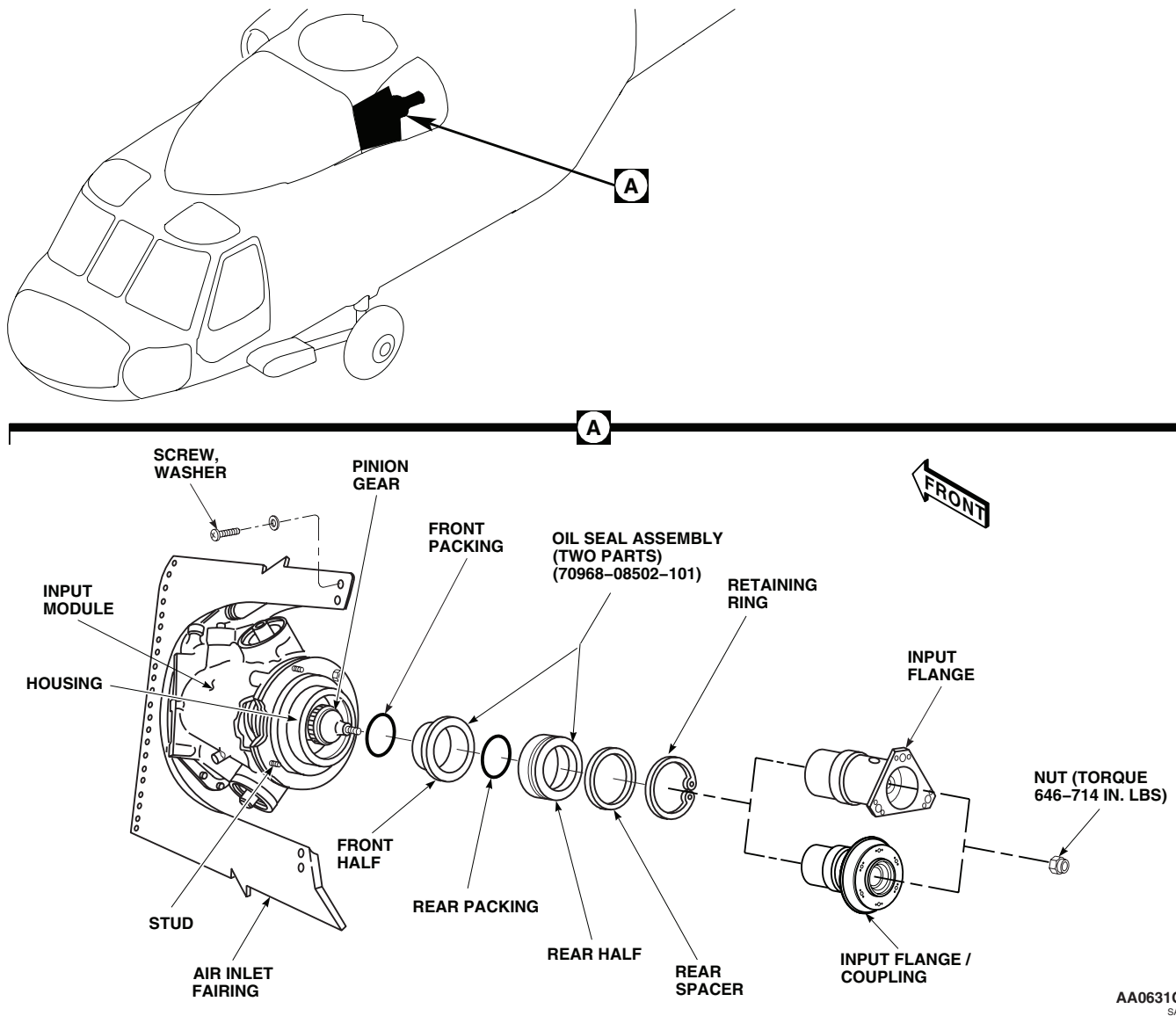
REMOVE - Continued

Figure 1. Replace Input Module Seal.

2. Remove input flange (WP 0655 00).
3. Remove retaining ring (Figure 1).

NOTE

Seal has two pieces. When replacing seal, both halves are replaced.

4. Remove rear spacer.
5. Thread three screws, MS35206-245, or equivalent, into three holes in rear half of seal. Using screws, pull out rear half of seal from input module pinion and housing assembly.
6. Remove front half of seal from pinion gear as follows.

REMOVE - Continued

- a. Using 2 scribes or picks, find flats on seal and gently work seal off packing.
- b. Remove front half of seal from pinion gear.
7. If front half of seal does not have flats or is difficult to remove, perform alternate method to remove seal as follows:
 - a. Wrap nonelectrical wire, Item 357, WP 1803 00, around lip of seal and twist leaving enough nonelectrical wire to grip with pliers.
 - b. Wrap a second piece of nonelectrical wire, Item 357, WP 1803 00, around lip of seal 180 degrees from the first wire and twist leaving enough nonelectrical wire to grip with pliers.
 - c. Using two pairs of pliers, grip both wires and pull removing front half of seal from pinion gear.
8. Remove packings from pinion gear. Discard packing.

INSTALL

1. Locally make seal installation tool (WP 1805 00).
2. On input flange, 70351-08206-041, install bolts into anchor nuts on input flange. If bolts can be fully threaded in by hand, replace input flange nuts.
3. **QA** Coat front packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on pinion gear shaft. Make sure packing is installed in groove.

CAUTION

Damage to seal will result if seal mating surfaces come in contact with skin or if mating surfaces are not lubricated during assembly. Oil from skin will damage carbon seal material. Do not touch mating surfaces of seal.

4. **QA** Lubricate seal packings and mating surfaces of both halves of seal, 70968-08502-101, with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00. Only lubricate area on both halves of seal that comes in contact with packing.
5. **QA** Install front half of seal on pinion gear shaft with flange toward rear. Seat seal half against bearing (Figure 1).
6. **QA** Coat rear packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on pinion gear shaft. Make sure packing is installed in groove.
7. **QA** Install rear half of seal in housing, with metallic surface toward rear (Figure 1). Seat seal assembly by gently tapping into position using locally made seal installation tool (WP 1805 00). Make sure seal is seated.
8. **QA** Install rear spacer.
9. **QA** Install retaining ring.

INSTALL - Continued**WARNING****CSI**

All surfaces of input flange are critical characteristics which must not be damaged during installation. After removal of protective covering, use extreme caution when installing.

CAUTION

Damage to input module internal bearing retaining clips will result if input module flange is seated by tapping with a mallet. Use only slight hand pressure and torque on flange retaining nut to seat flange.

10. Install input flange (WP 0655 00).
11. Install input module gimbal (WP 0654 01).
12. Install engine (WP 0481 00).
13. Do a functional test of transmission system (WP 0615 00).

REPLACE FORWARD SUPPORT TUBE ATTACHMENT NUTPLATES**CAUTION**

Use extreme care while drilling out rivets so as not to damage input gimbal, flange, bearing and seal.

NOTE

This procedure applies P/N 70361-080001-101, 70361-08001-102, 70361-080001-103 and 70361-080015-102. Removal of the gimbal is not required for nutplate replacement.

1. Remove rivets securing nutplate(s) to gimbal as required (TM 1-1500-204-23).
2. **QA** Inspect rivet holes in gimbal for elongation or deformation, and for any damage to the gimbal as the result of rivet removal. None allowed. Replace gimbal as required.
3. Inspect input module flange/coupling, oil seal assembly and input module housing for damage as a result of rivet removal. Refer to (WP 0619 00, WP 0654 01 or WP 0655 00) for damage limits.
4. Replace nutplate(s) (TM 1-1500-204-23).
5. **QA** Remove all FOD from input module gimbal and surrounding areas.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****INPUT MODULE GIMBAL**

This WP supersedes WP 654 01, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Flange Puller, Locally-Made, Figure 50, WP 1805 00
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 150 - 1000 in. lbs.
Wrench, Crowfoot, 7/16, 3/8 Drive, FC014A

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Nonmetallic Scraper
Polyurethane Coating, Item 231, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0381 00
WP 0481 00
WP 0615 00
WP 0619 00
WP 0654 00
WP 1803 00

REMOVE

1. Disconnect and remove engine (WP 0481 00).
2. Inspect input pinion for axial movement (WP 0619 00).
3. Disconnect bracket from air inlet fairing (Figure 1, Detail A).
4. Disconnect gimbal drain line from gimbal.
5. Using nonmetallic scraper, remove sealing compound between gimbal and input module and from nuts securing gimbal to input module.
6. Remove nuts securing gimbal to input module with crowfoot. Loosen nuts in equal steps, otherwise they will not clear studs before hitting housing. Remove gimbal from input module.

INSTALL

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean mounting surface on gimbal and input module with machinery towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
2. Apply primer coating, Item 243, WP 1803 00, to mating surface of input module.

NOTE

Gimbal drain port must be located at 6 o'clock position when installed.

3. Install gimbal and plate onto input module while primer is still wet. Install only far enough to allow nuts to be started. Start all nuts and run them down two or three threads at a time. Torque gimbal nuts as follows:
 - a. **FOR GIMBAL 70361-08001-101, 70361-08001-102, OR 70361-08001-103, TORQUE NUTS TO 261 - 288 INCH - POUNDS (Figure 1, Detail A).**
 - b. **FOR GIMBAL 70361-08015-102, TORQUE NUTS TO 145 - 165 INCH - POUNDS (Figure 1, Detail A).**
4. Apply epoxy primer coating kit, Item 120, WP 1803 00, to all exposed washers, nuts, and studs securing gimbal to module.
5. Cover gimbal mounting hardware with sealing compound, Item 273, WP 1803 00.
6. Apply epoxy primer coating kit, Item 120, WP 1803 00, over sealing compound and apply one coat of polyurethane coating, Item 231, WP 1803 00, over primer (WP 0381 00 and TM 55-1500-345-23).
7. Connect gimbal drain line.
8. Connect air inlet fairing to bracket.
9. Install engine (WP 0481 00).
10. Do a functional test of transmission system (WP 0615 00).

INSTALL - Continued

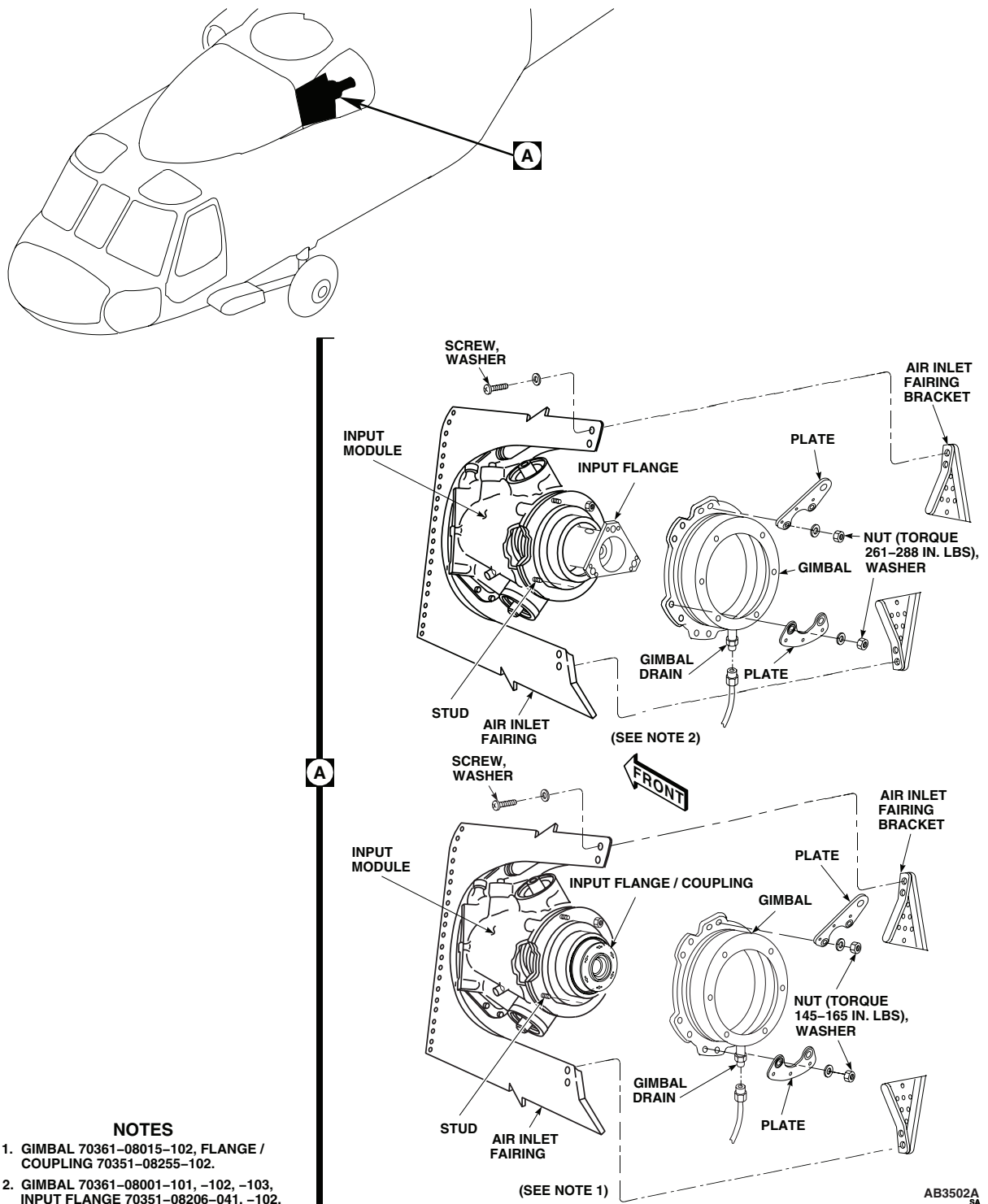


Figure 1. Replace Input Module Gimbal.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

DRIVE SYSTEM

INPUT MODULE INPUT FLANGE

This WP supersedes WP 0655 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Flange Puller, Locally-Made, Figure 50, WP 1805 00
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Torque Reactor Plate, Locally-Made, Figure 181, WP 1805 00
 Torque Reactor Plate, Locally-Made, Figure 262, WP 1805 00
 Torque Reactor Plate, Locally-Made, Figure 271, WP 1805 00
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Bolt, AN4-12A (3 Required)
 Cloth, Abrasive, Item 81, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00

Sealing compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0381 00
 WP 0481 00
 WP 0486 00
 WP 0619 00
 WP 0653 00
 WP 0654 01
 WP 1607 00
 WP 1803 00
 WP 1805 00

REMOVE

WARNING

CSI

All surfaces of input flange are critical characteristics which must not be damaged during installation.

NOTE

Input module flange, 70351-08206-102, is the preferred flange and should be installed if available. If the -102 flange is not available, input module flange, 70351-08206-041, may be used until the -102 flange becomes available.

1. Disconnect and remove engine (WP 0481 00).
2. Inspect input pinion for axial movement (WP 0619 00).
3. Locally make torque reactor plate Figure 181, (WP 1805 00), or torque reactor plate Figure 262, (WP 1805 00).
4. Hold flange with torque reactor and remove nut securing input flange to input module. Remove flange (Figure 1). If flange is difficult to remove, do this:
 - a. Locally make flange puller (Figure 50, WP 1805 00).
 - b. Carefully attach flange puller to flange using flange bolts.

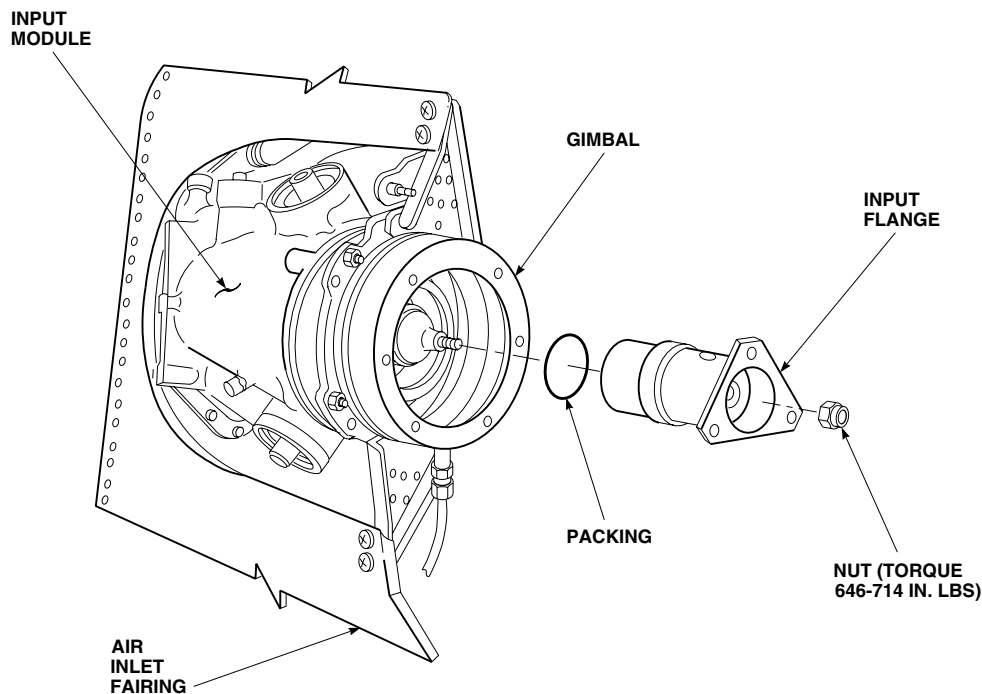
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Figure 1. Replace Input Flange (P/N 70351-08206-102).

- c. Slowly screw in main puller bolt until flange is pulled free from input shaft.
- d. Discard packing.
5. On helicopters with input flange/coupling 70351-08255-102, remove flange/coupling as follows:
 - a. Remove gimbal (WP 0654 01).
 - b. Using locally made torque reator plate, WP 1805 00, hold flange/coupling, and remove nut securing flange/coupling to input module.

CAUTION

Flange/coupling will be damaged if hit or tapped. Do not hit or tap flange/coupling to remove.

- c. Remove input flange/coupling (Figure 2) as follows:
 - (1) Using locally made flange puller, Figure 50, WP 1805 00, carefully attach flange puller to flange/coupling using three AN4-12A bolts.
 - (2) Discard packing.

INSPECT

1. Inspect input flange/coupling 70351-08255-102 as follows (Figure 3, Sheet 1 and Figure 3, Sheet 2):

INSPECT - Continued

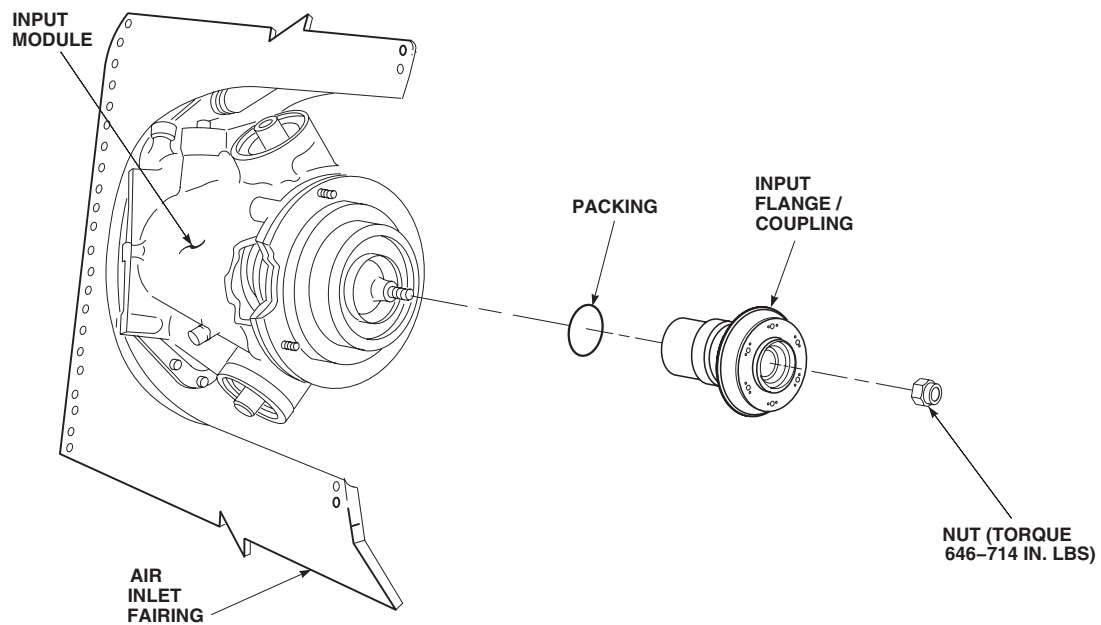
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Figure 2. Replace Input Flange/Coupling (P/N 70351-08255-102).

NOTE

Flange/coupling is made of titanium alloy, so bare metal may have some discoloration and variation in appearance (non-uniform color, from straw to blue, staining, handling marks, etc.) Flange/coupling is a welded construction and weld seams are visible. Discoloration as described, or visible weld seams are not reason for rejection.

- a. Using magnifying glass, visually inspect flange/coupling for cracks. **NONE ALLOWED.** Replace flange/coupling if cracks are found.
- b. Coupling is coated with green epoxy primer on surfaces that are external when coupling is assembled into helicopter. Chipping and flaking of green epoxy primer is acceptable and may be touched up. The coupling has a ring of material on external surface above internal spline for balance correction. This ring will typically have an area where material was removed and coated with green epoxy primer during manufacturing.
- c. Using magnifying glass, visually inspect flange/coupling diaphragm profile area for dents or wavy appearance. **NONE ALLOWED.** Replace flange/coupling if dents or wavy appearance or any damage is found in diaphragm profile area.
- d. Using magnifying glass, visually inspect flange/coupling diaphragm profile surface for scratches or nicks into base metal. **NONE ALLOWED.** Replace flange/coupling with scratches or nicks into base metal of diaphragm profile surface.

INSPECT - Continued

NOTE

There are four vent slots equally spaced in the diaphragm rim. These are normal and not cause for rejection.

- e. Scratches and nicks to a maximum of 0.015-inch in diaphragm rim, hub, or flanges (all surfaces other than diaphragm profile) are acceptable. Replace flange/coupling with scratches or nicks deeper than 0.015-inch in diaphragm rim, hub, or flange.
 - f. Inspect bolted mounting flange face and axial stop face for fretting. Light fretting, up to 0.0005-inch deep, is acceptable and may be polished out using abrasive cloth, Item 82, WP 1803 00.
 - g. Inspect for raised edges or otherwise proud material on mounting flange face, flange pilot (OD), spline end pilot (ID) and spline end face. Blend and polish raised areas using abrahsive cloth, Item 81, WP 1803 00, abrahsive cloth, Item 82, WP 1803 00. **NO BLENDING OR POLISHING IS ALLOWED IN THE DIAPHRAGM PROFILE AREA.**
 - h. Internal spline area is coated with Ever-lube #620C dry film lubricant. Light marking of the coating on the spline teeth is acceptable.
 - i. Inspect flange/coupling nutplates for missing rivits, rivets above surface of flange face, or missing lock nuts. **NONE ALLOWED.** Replace flange/coupling with missing rivets or lock nuts or rivets above surface of flange face.
 - j. Check running torque on nutplate lock nuts. Replace flange/coupling if running torque is below 3 inch-pounds on any lock nut.
 - k. Inspect axial stop face for nicks or scratches. Nicks or scratches up to 0.010-inch over 10% of area is allowed. Remove only raised edges.
 - l. Measure ID of flange pilot. If ID is less than 0.5287-inches, replace flange/coupling.
 - m. Measure OD of flange pilot. If OD is less than 2.4762-inches, replace flange/coupling.
 - n. Inspect flange pilot OD for nicks and scratches up to 0.005-inch deep over less than 30% of entire circumference allowed. No more than 10% allowed in any one 90 degree quadrant section.
 - o. Measure ID of pilot face. If ID exceeds 2.3664-inches, replace flange/coupling.
 - p. Measure OD of pilot face. If OD exceeds 2.580-inches, replace OD flange/coupling.
 - q. Measure wear of spline. No more than 0.002-inch wear step in spline allowed. If spline wear step is greater than 0.002-inches, replace flange/coupling.
- 2. Inspect input flange 70351-08206-102 or 70351-08206-041 as follows:
 - a. Check dimension of damaged areas. Compare with minimum repair limits (Figure 2).
 - 3. Check to determine whether flange minimum dimension after repair can be maintained.
 - 4. Inspect input module flange for fretting in washer contact area. **MAXIMUM FRETTING DEPTH IS 0.0005-INCH.** Replace input module flange if limit is exceeded.

INSPECT - Continued

NOTE

If input module flange, 70351-08206-041 with nutplates is used, replace with input module flange, 70351-08206-102 if available, if input module flange, 70351-08206-041 is used, after torque has stabilized, place a drop of seal compound, Item 273, WP 1803 00, to threads of bolts adjacent to nutplates. Make sure compound seeps into nutplates.

- On input flange, 70351-08206-041, check flange nutplate rivets for proper installation, looseness, and fretting on heads. Replace input flange if rivets are loose, not flush with flange surface, or fretting is found on heads.

REPAIR

NOTE

- Repair is for input flange 70351-08206-102 and 70351-08206-041 only.
- Before starting repair on flange, determine whether polishing out damage would result in a diameter below reparable limits (Figure 4). If damage repair is greater than permitted limits, replace flange.
- No balance checks are required after reworking flange in accordance with this procedure.
- Repair of flange may be done with flange installed on helicopter.

- Remove damaged flange if necessary.

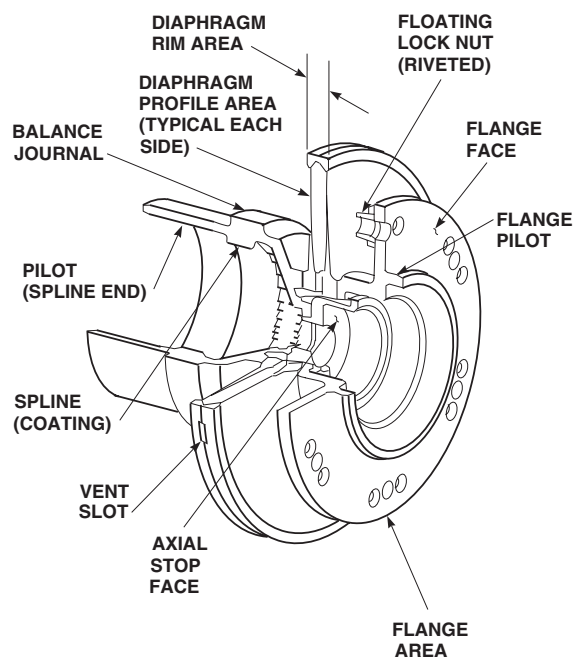
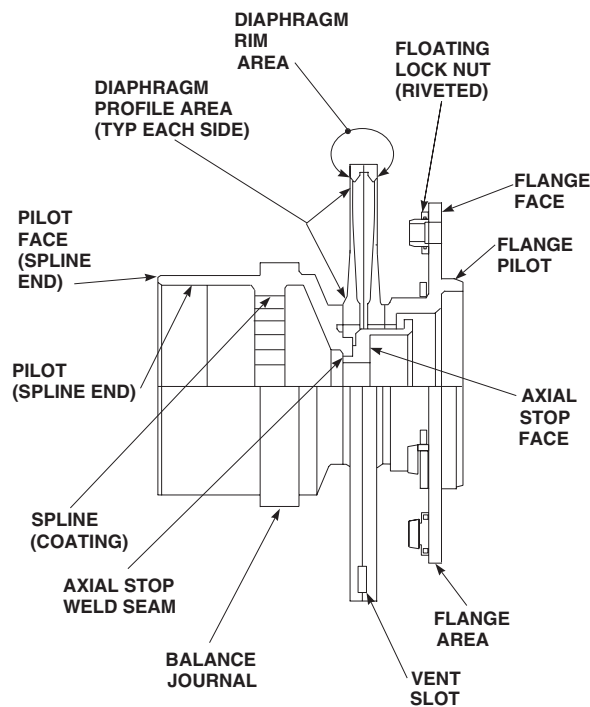
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Figure 3. Inspect Input Flange, 70351-08255-102. (Sheet 1 of 2)

REPAIR - Continued

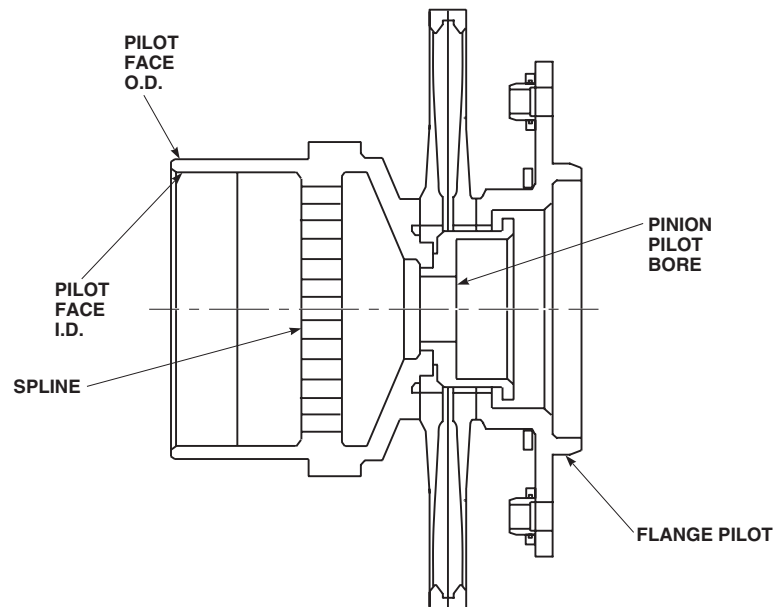
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Figure 3. Inspect Input Flange, 70351-08255-102. (Sheet 2 of 2)

2. Polish out minor nicks, scratches, and scoring using abrasive cloth, Item 82, WP 1803 00.
3. Blend out damage using abrasive cloth, Item 81, WP 1803 00. Remove only what is necessary so flange will stay within limits. Blend damaged area to a minimum 1/2-inch radius.
4. Polish reworked area to remove all traces of abrasion marks. Match surface smoothness using abrasive cloth, Item 82, WP 1803 00.

REPAIR - Continued

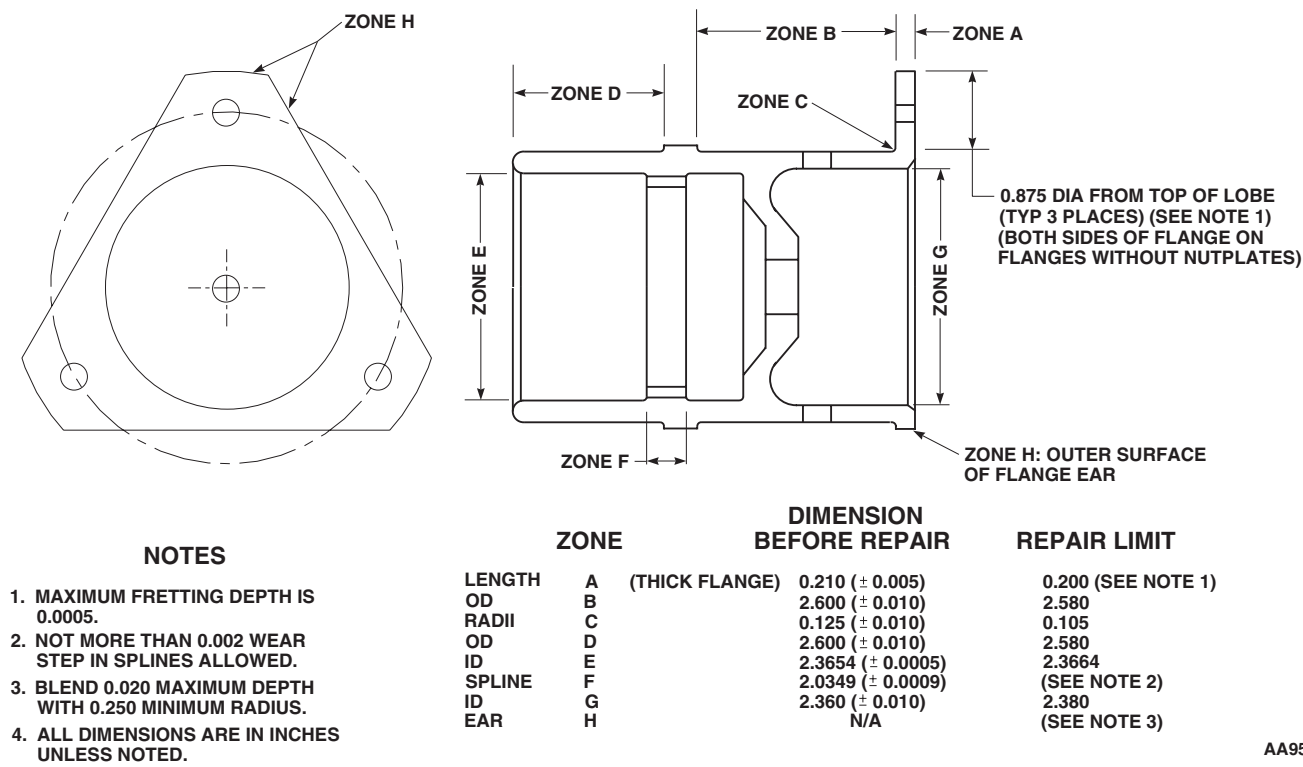


Figure 4. Repair Input Flange.

INSTALL**WARNING****CSI**

All surfaces of input flange are critical characteristics which must not be damaged during installation.

CAUTION

- Damage to input module internal bearing retaining clips will result if input module flange is seated by tapping with a mallet. Use only slight hand pressure and torque on flange retaining nut to seat flange.
- Disc pack coupling will be damaged if not connected to correct gimble.
- Input module flange/coupling, 70351-08255-102, must be installed with elastomeric gimbal, 70361-08015-102 (WP 0653 00) and engine output drive shaft, 70361-08014-041 (WP 0486 00). Input module flange, 70351-08206-102, or 70351-08206-041, with gimbal 70361-08001-101, 70361-08001-102, or 70361-08001-103, may be used with driveshaft 70361-08004-042 or 70361-08004-043.

NOTE

- When using input module flange/coupling, 70351-08255-102, verify 70351-08255-102 flange/coupling is properly marked, "For H-60/S-70 only". Dash numbers other than -102 are not interchangeable.
- If input module flange, 70351-08206-041 with nutplates is used, replace with input module flange, 70351-08206-102 if available, if input module flange, 70351-08206-041 is used, after torque has stabilized, place a drop of seal compound, Item 273, WP 1803 00, to threads of bolt adjacent to nutplates. Make sure compound seeps into nutplates.

1. **QA** Install packing on pinion gear (Figure 1).
2. Slide input flange into seal until seated.
3. Using torque wrench, check drag torque of retaining nut. **MINIMUM DRAG TORQUE IS 18 INCH-POUNDS.** Replace nut if beyond limits.

WARNING**CSI**

Verification of minimum torque is a critical characteristic.

4. **QA** Install nut. **TORQUE NUT TO 646 - 714 INCH-POUNDS.** Apply torque stripe (WP 0381 00).
5. Inspect input pinion (WP 0619 00).
6. Service main transmission (WP 1607 00).
7. Install engine (WP 0481 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**ACCESSORY MODULE**

This WP supersedes WP 0656 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Jacking Bolts, 1/4-28
 Mallet, GGG-H-33
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Protective Plugs
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 750 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cleaning Compound Solvent, Item 71, WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Epoxy Coating Kit, Item 118, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 184,
 WP 1803 00
 Lubricating Oil, Helicopter Transmission, Item 187,
 WP 1803 00
 Polyurethane Coating, Item 231, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
 TM 1-1500-344-23
 TM 55-1500-345-23
 WP 0085 00
 WP 0381 00
 WP 0615 00
 WP 0617 00
 WP 0618 00
 WP 0630 00
 WP 0641 00
 WP 0643 00
 WP 0644 00
 WP 0645 00
 WP 0646 00
 WP 0658 00
 WP 0706 00
 WP 0839 00
 WP 1603 00
 WP 1607 00
 WP 1703 00
 WP 1803 00

REMOVE**NOTE**

Accessory module without the pump module and generator on weighs about 28 pounds.

1. Turn off all electrical power.
2. Open main rotor pylon sliding cover.
3. Remove hydraulic pump module (WP 0706 00).
4. Remove generator (WP 0839 00).

REMOVE - Continued

5. Disconnect chip detector electrical connector, P406, left side, or P405, right side, from main transmission wiring harness (Figure 1, Detail A).
6. Disconnect Nr electrical connector, P443, from right accessory module.
7. **UH-60L** Disconnect Nr electrical connector, P407, from left accessory module. ■
8. Disconnect low oil pressure electrical connector, P409, from left module. Right module does not have an electrical connector in this location.
9. Remove bolts, washers, and nuts securing engine control cable support to accessory module. Keep attaching hardware with support.
10. Remove sealing compound from oil return tubes with nonmetallic scraper.
11. Remove retaining rings from upper and lower return tubes on accessory module.
12. Remove nuts from flange of module.
13. Lift off accessory module. Oil return tube will slide free of module and stay with input module. If accessory module does not pull off input module easily, it may be jacked out. Use a 1/4-28 bolt in jacking hole in input module flange. Slowly thread in bolt and at same time gently tap opposite side of accessory module with a soft mallet. Be careful not to cock module.
14. Check shim on accessory module flange for separation and torn areas. **SHIM MAY BE REUSED AS LONG AS IT DOES NOT HAVE ANY MAJOR DAMAGE SUCH AS CUTS, RAISED OR DEPRESSED AREAS, OR ANY OTHER DAMAGE THAT PREVENTS ACCESSORY MODULE FROM MATING PROPERLY WITH INPUT MODULE FLANGE.** Replace shim if damaged beyond limits (WP 0658 00).
15. Check cone oil screens for metal particle contamination. Compare any particles found with accept/reject table (WP 0617 00).
16. Carefully package and return any foreign material with module. If necessary, replace cone oil screens.
17. Keep cone oil screen with input module.
18. Remove oil tubes between accessory module and input module (Figure 2). Keep oil tubes with input module.
19. Remove both bolts, washers, and nuts from cable support attachment area on accessory module to be installed. Install these bolts, washers, and nuts in module just removed (Figure 1, Detail A). **TORQUE NUTS TO 114 - 126 INCH-POUNDS.**
20. Remove sealing compound using nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

21. Clean primer off of mating surface and hardware using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
22. If new accessory module is not readily available for installation, cover open input module.

INSPECT

Visually inspect module housing for nicks, gouges, scratches, and corrosion. Repair or replace accessory module housing (WP 0641 00).

INSPECT - Continued

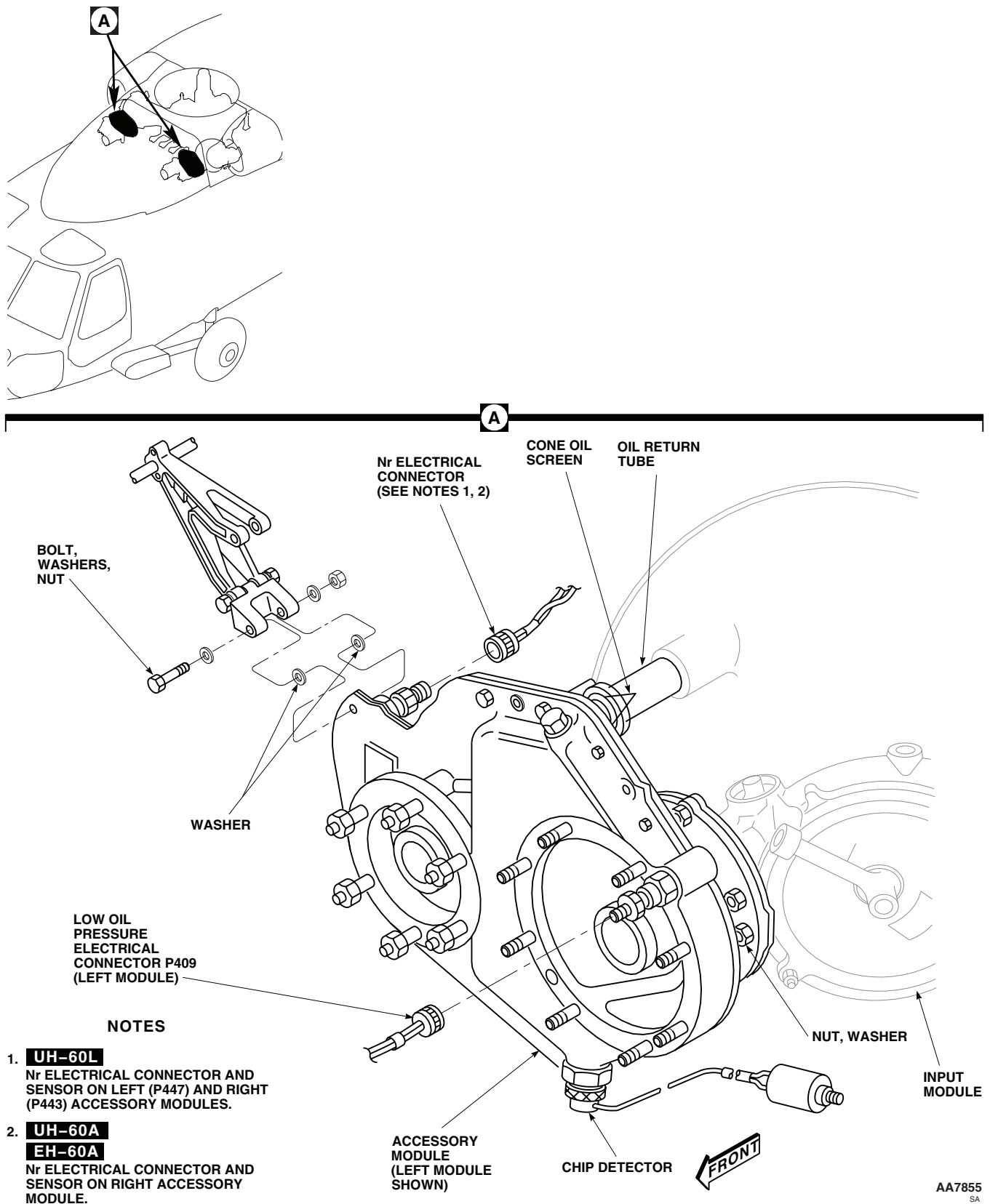


Figure 1. Replace Accessory Module.

INSPECT - Continued

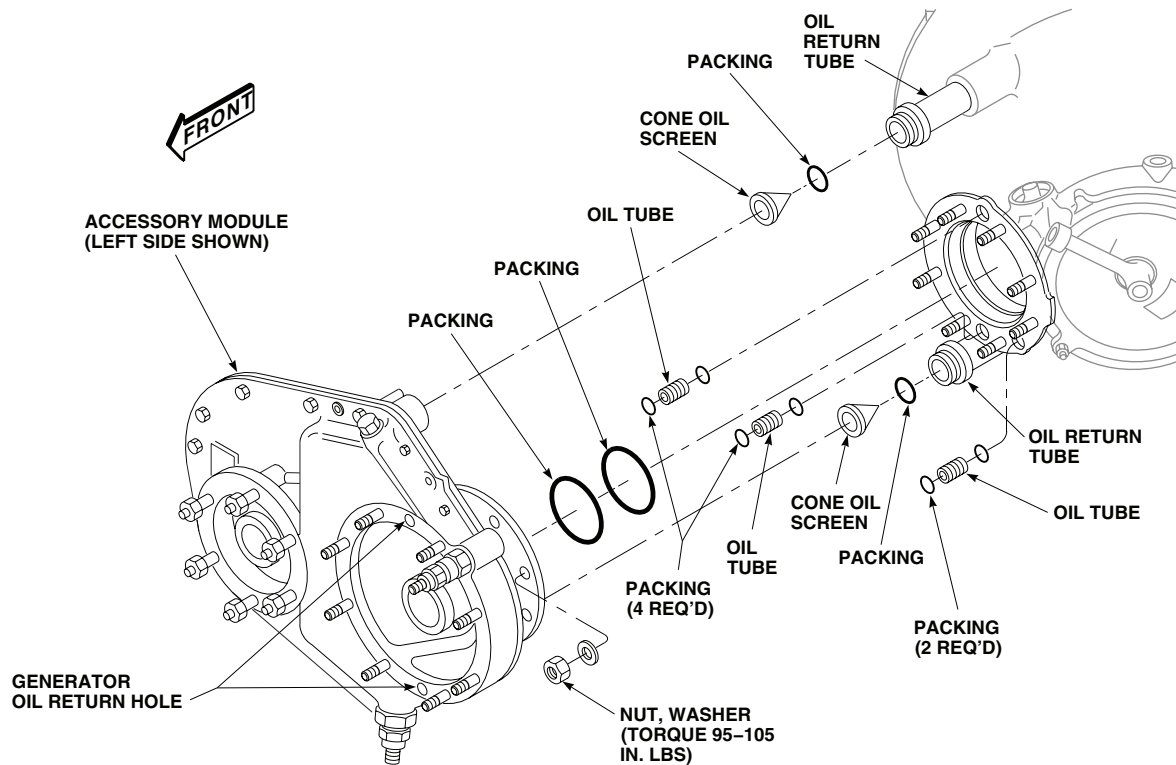
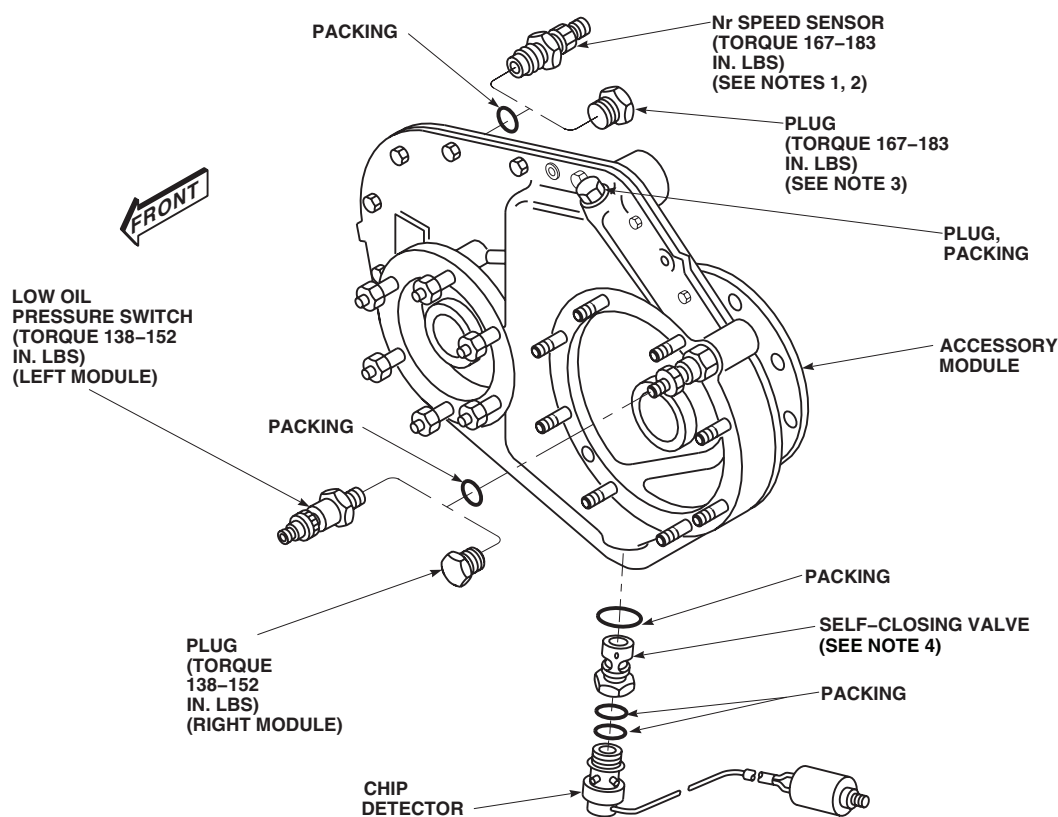
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Figure 2. Replace Accessory Module.

ACCESSORY MODULE BUILDUP

1. Remove low pressure switch and packing from left accessory module. Keep pressure switch. Discard packing (Figure 3).
2. Remove plug and packing from right accessory module. Keep plug. Discard packing.
3. Install protective plug in hole.
4. Remove Nr speed sensor and packing from right accessory module. Keep speed sensor. Discard packing.
5. **UH-60L** Remove Nr speed sensor and packing from left accessory module. Keep speed sensor. ■
6. **QA** Install protective plug in hole.
7. Remove chip detector, sleeve, and packings from bottom of module. Check chip detector for metal particle contamination. Carefully package and return any foreign material with the module. Discard packings.
8. Install protective plug in hole.
9. Clean chip detector using cleaning compound solvent, Item 71, WP 1803 00.
10. Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, or helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
11. **QA** Remove protective plug. Install packing on sleeve. Install sleeve in bottom of self closing valve. **TIGHTEN SELF-CLOSING VALVE 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT (Figure 3).**
12. Check chip detector electrical circuit (WP 0085 00).

ACCESSORY MODULE BUILDUP - Continued



NOTES

1. **UH-60A**
EH-60A RIGHT MODULE.
2. **UH-60L**
LEFT AND RIGHT MODULE.
3. **UH-60A**
EH-60A LEFT MODULE.
4. **TIGHTEN SELF-CLOSING VALVE 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

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Figure 3. Replace Accessory Module Components.

13. Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, or helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
14. **QA** Install packings on chip detector. Install chip detector in sleeve.
15. **QA UH-60A EH-60A** Remove protective plug. Install packing on Nr speed sensor. Install Nr speed sensor in rear of right accessory module. **TORQUE Nr SPEED SENSOR TO 167 - 183 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire sensor. Install packing on plug. Install plug in rear of left accessory module. **TORQUE PLUG TO 167 - 183 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire plug. ■
16. **QA UH-60L** Install packing on Nr speed sensor. Install Nr speed sensor in rear of left or right accessory module. **TORQUE Nr SPEED SENSOR TO 167 - 183 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire sensor. ■

ACCESSORY MODULE BUILDUP - Continued

17. **QA** Remove protective plug. Install packing on low oil pressure switch. Install low oil pressure switch in front side of left accessory module. **TORQUE LOW OIL PRESSURE SWITCH TO 138 - 152 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire switch.
18. **QA** Install packing on plug. Install plug in front side of right accessory module. **TORQUE PLUG TO 138 - 152 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire plug.

INSTALL

1. Remove cone oil screens and packings from oil return tubes (Figure 2).
2. Clean screens with cleaning compound solvent, Item 71, WP 1803 00.
3. Lubricate packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, or helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
4. **QA** Install cone oil screens and packings in tubes, with point of screen in tube.

NOTE

Oil return ports for generator may or may not be drilled in accessory gear box. Oil tube bores will always be drilled in accessory gear box to accept oil tubes and packings. Make sure oil tubes and packings are installed between input and accessory gear box, whether oil return ports are drilled or not, to prevent leakage from input module oil return ports.

5. **QA** Place packings on oil pressure tubes.
6. **QA** Install oil tubes in input module.
7. **QA** Place packings on accessory module.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

8. Clean mounting surfaces on input and accessory module with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
9. Coat area between Rosan lockring and mating housing counterbore with sealing compound, Item 273, WP 1803 00.
10. Touch up magnesium surfaces with corrosion resistant coating, Item 111, WP 1803 00 (TM 1-1500-344-23).

WARNING

Injury to personnel and damage to equipment will result if transmission oil is contaminated. To avoid contaminating transmission oil system, do not apply primer to bore diameters that are sealing surfaces for packings.

NOTE

Install accessory module while primer is wet.

11. Apply primer coating, Item 243, WP 1803 00, to accessory module mounting surface and fastener holes on input module. Do not apply primer to packing or packing grooves.

INSTALL - Continued**CAUTION**

Damage to accessory module will result if it is not carefully installed on input module.
Be careful not to drop module or burr stud threads when installing accessory module.

12. **QA** Place accessory module on input module mounting plate. Make sure oil return tubes and oil tubes are properly seated.
13. **QA** Install washers and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS (Figure 2).**

NOTE

Housings coated with epoxy coating kit, Item 118, WP 1803 00, are glossy silver;
housings coated with polyurethane coating, Item 231, WP 1803 00, are matte surface
medium-gray.

14. Apply epoxy primer coating kit, Item 120, WP 1803 00, to all exposed washers, nuts, and studs securing accessory module to input module.
15. Touch up paint on exposed boltheads, nuts, and threads, as necessary (WP 0381 00 and TM 55-1500-345-23).
16. Seal area around joint of washer and nut, thread of studs to nut and seam between accessory module and main module with sealing compound, Item 273, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

17. Remove excess sealing compound using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
18. Apply epoxy primer coating kit, Item 120, WP 1803 00, over sealing compound and apply one coat of epoxy coating kit, Item 118, WP 1803 00, or polyurethane coating, Item 231, WP 1803 00, depending on original paint finish.
19. Install generator (WP 0839 00).
20. Install hydraulic pump module (WP 0706 00).
21. Inspect and clean accessory, input, and main module oil return screens, chip detectors, and chip detector strainers (WP 0643 00, WP 0644 00, WP 0645 00, and WP 0646 00).
22. Check chip detector electrical circuit (WP 0085 00).
23. Inspect and clean main oil filter (WP 0618 00).
24. Inspect and treat electrical connectors (WP 1703 00).

NOTE

See WP 0630 00 for wiring harness routing.

25. **QA** Connect chip detector electrical connector, P406, left side or electrical connector, P405, right side to main transmission wiring harness (Figure 1, Detail A).
26. **QA** Connect Nr electrical connector, P443, to back side of right module. Using nonelectrical wire, Item 357, WP 1803 00, lockwire connector (Figure 4, Detail A).

INSTALL - Continued

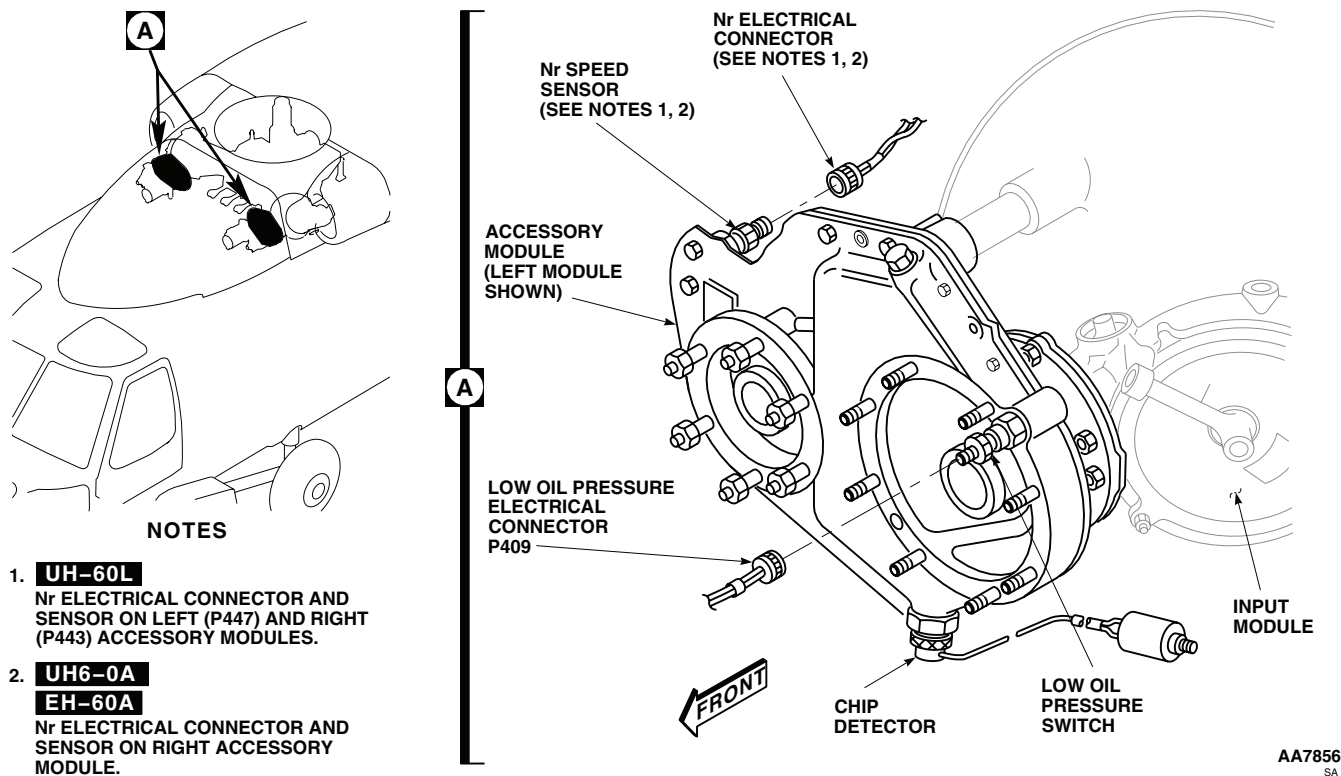


Figure 4. Install Accessory Module Electrical Connectors.

27. **QA UH-60L** Connect Nr electrical connector, P407, to back side of left module. Using nonelectrical wire, Item 357, WP 1803 00, lockwire connector (Figure 4, Detail A).
28. **QA** Connect low oil pressure electrical connector, P409, to low oil pressure switch on left module over generator. Using nonelectrical wire, Item 357, WP 1803 00, lockwire connector if connector is not self locking.
29. Coat cable support boltholes on accessory module with primer coating, Item 243, WP 1803 00.
30. **QA** Position engine control cable support on accessory module. Support, 70301-22107-101, with clip, 70301-02119-041, is installed on left accessory module. Add washers as required to fill gap between support and module. Install bolts, washers, and nuts attaching cable support to accessory module. **TORQUE NUTS TO 105 - 115 INCH-POUNDS (Figure 5).**
31. Do a functional test (WP 0615 00).
32. Service main transmission (WP 1607 00).
33. Check hydraulic fluid level in pump module. Service as necessary (WP 1603 00).
34. Make sure area is clean and free of foreign material.
35. Close sliding cover.

INSTALL - Continued

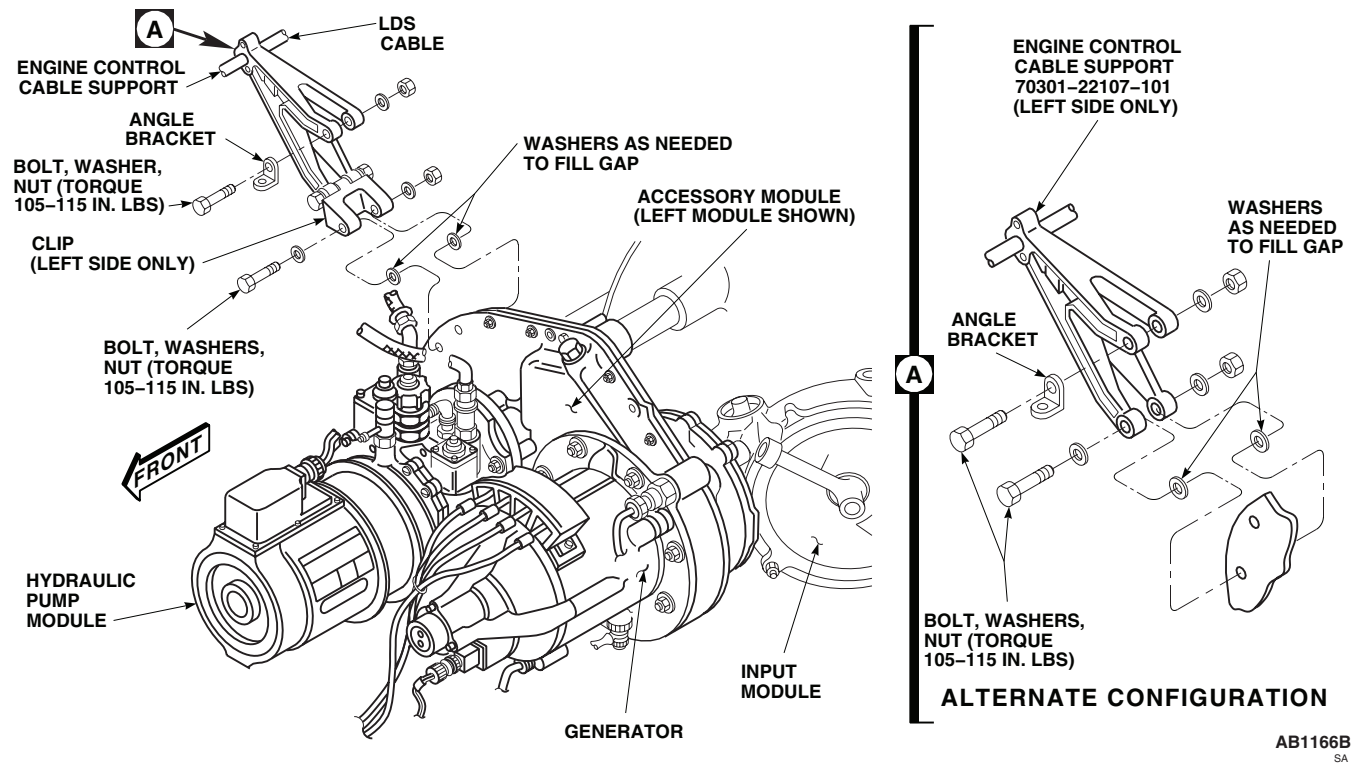


Figure 5. Replace Accessory Module.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****ACCESSORY MODULE SPLINE ADAPTER**

This WP supersedes WP 0657 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Mallet
Nonmetallic Drift
Spline Adapter Removal Tool, Locally-Made,
Figure 266, WP 1805 00

References

WP 0839 00
WP 1805 00

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Remove generator (WP 0839 00).
2. Install spline adapter removal tool (WP 1805 00) into spline adapter, making certain splines engage each other (Figure 1).
3. Tighten nut of spline adapter removal tool and remove spline adapter. Remove spline adapter removal tool by loosening nut and removing from spline adapter.

INSPECT

1. Inspect spline adapter for wear, cracks, or discoloration due to heat. **NONE ALLOWED.** Replace worn, cracked, or discolored spline adapter.
2. Inspect spline adapter for missing pieces. **DIMENSION OF MISSING PIECES SHALL NOT EXCEED 0.10-INCH. IF ANY PIECE OF SPLINE ADAPTER WITH DIMENSION GREATER THAN 0.010 INCH IS MISSING, CHECK INPUT MODULE OIL RETURN TUBES AND CHIP DETECTOR SCREENS FOR MISSING PIECES. IF MISSING PIECES CANNOT BE LOCATED, THE ACCESSORY MODULE AND SPLINE ADAPTER MUST BE REPLACED.**
3. Inspect for missing pieces of spline adapter in oil tubes and chip detectors.

INSTALL

1. Check accessory module gear box splines to be sure they are free and clean of all foreign matter.
2. **QA** Carefully tap spline adapter (tapered end first) into accessory module gear box splines, using mallet and nonmetallic drift, until seated (Figure 1).
3. Install generator (WP 0839 00).

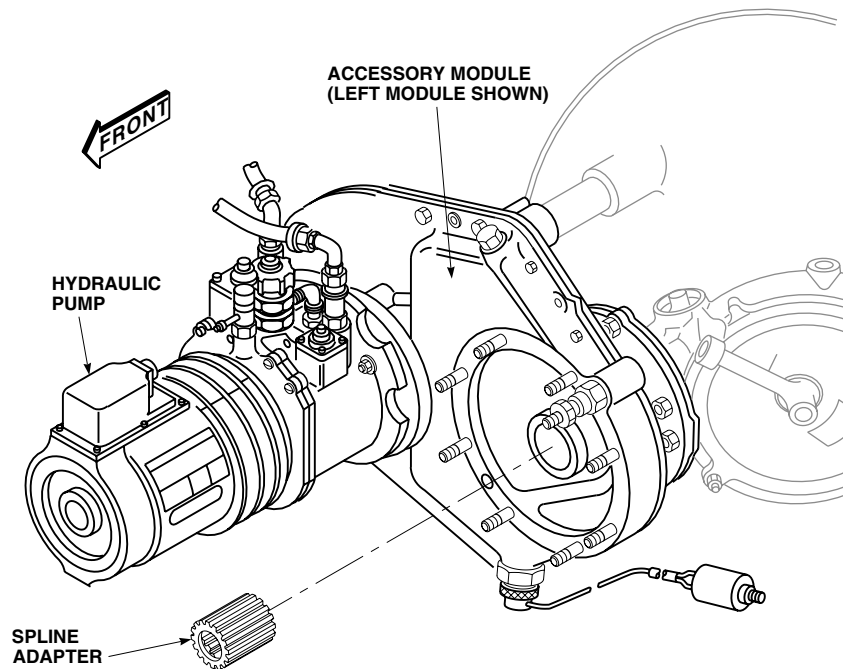
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Figure 1. Replace Accessory Module Spline Adapter.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****ACCESSORY MODULE SHIM**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Micrometer, 0 to 1 inch
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1803 00

Material/Parts

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Shim, 70351-08076-101

REMOVE

1. Insert a nonmetallic scraper between accessory module flange and shim. Using nonmetallic scraper as a wedge, gently pry shim away from flange until head of pin is above shim surface (Figure 1).
2. Grasp head of pin with pliers and pull pin out.
3. Repeat procedure for remaining pin and remove shim.

INSTALL

1. Measure and record shim thickness in three places 120 degrees apart. Record average of three measurements. If unable to obtain shim measurement due to excessive shim damage, submit CAT II EIR per PAM 738-751.
2. Peel new shim to average thickness ± 0.001 -inch of average measurement recorded in step 1.
3. **QA** Apply epoxy primer coating kit, Item 120, WP 1803 00, to accessory module flange. Position shim on flange and secure with pins (Figure 1).

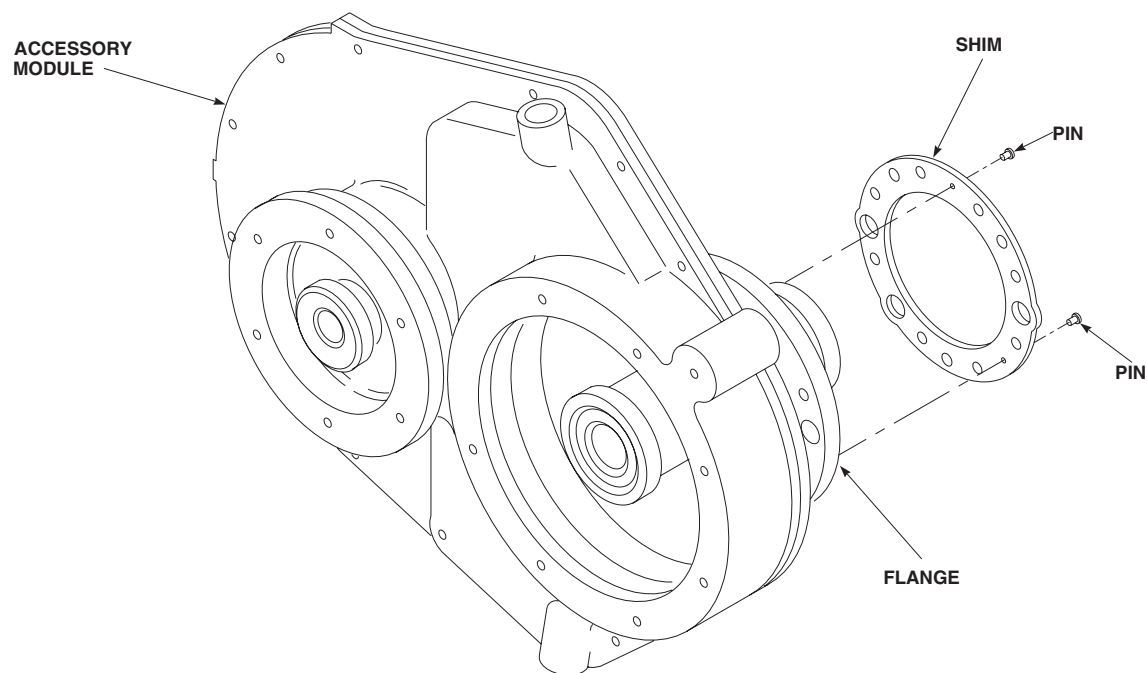
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Figure 1. Replace Shim.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**INTERMEDIATE GEAR BOX**

This WP supersedes WP 0659 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Airframe Repairer's Toolkit, SC 5180-99-B02
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Nonmetallic Scraper
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.
 Torque Wrench, 0 - 600 ft. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Brush, Acid Swabbing, Item 57, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Molybdenum Disulfide, Item 197, WP 1803 00
 Nonpermanent Marker
 Polyurethane Coating, Item 231, WP 1803 00
 Polyurethane Coating, Item 235, WP 1803 00
 Primer, Adhesive, Item 241, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00

Shim, 1, Locally-Made, Figure 199, WP 1805 00
 70219-06000-106, 107, 108, 110, 11
 Tape, Pressure Sensitive, Adhesive, Item 326,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wooden Blocks to Support Drive Shaft

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
 TM 1-1520-237-MTF
 TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0085 00
 WP 0335 00
 WP 0381 00
 WP 0665 00
 WP 1608 00
 WP 1681 00
 WP 1803 00
 WP 1805 00

REMOVE**NOTE**

Intermediate gear box weighs 32 pounds.

1. Turn off all electrical power.
2. Remove intermediate gear box fairing (WP 0335 00).
3. Disconnect electrical connector, P613, from chip detector/temperature sensor (Figure 1, Detail A).
4. Firmly support rear end of section III tail rotor drive shaft and lower end of section IV tail rotor drive shaft with machinery wiping towel, Item 344, WP 1803 00, or block with wood.
5. Secure tail rotor blades to pylon to keep from freewheeling.
6. Engage gust lock.
7. Remove bolts, spherical washers, shims, and nuts securing drive shafts to input and output flanges of intermediate gear box (Figure 4, Detail A).

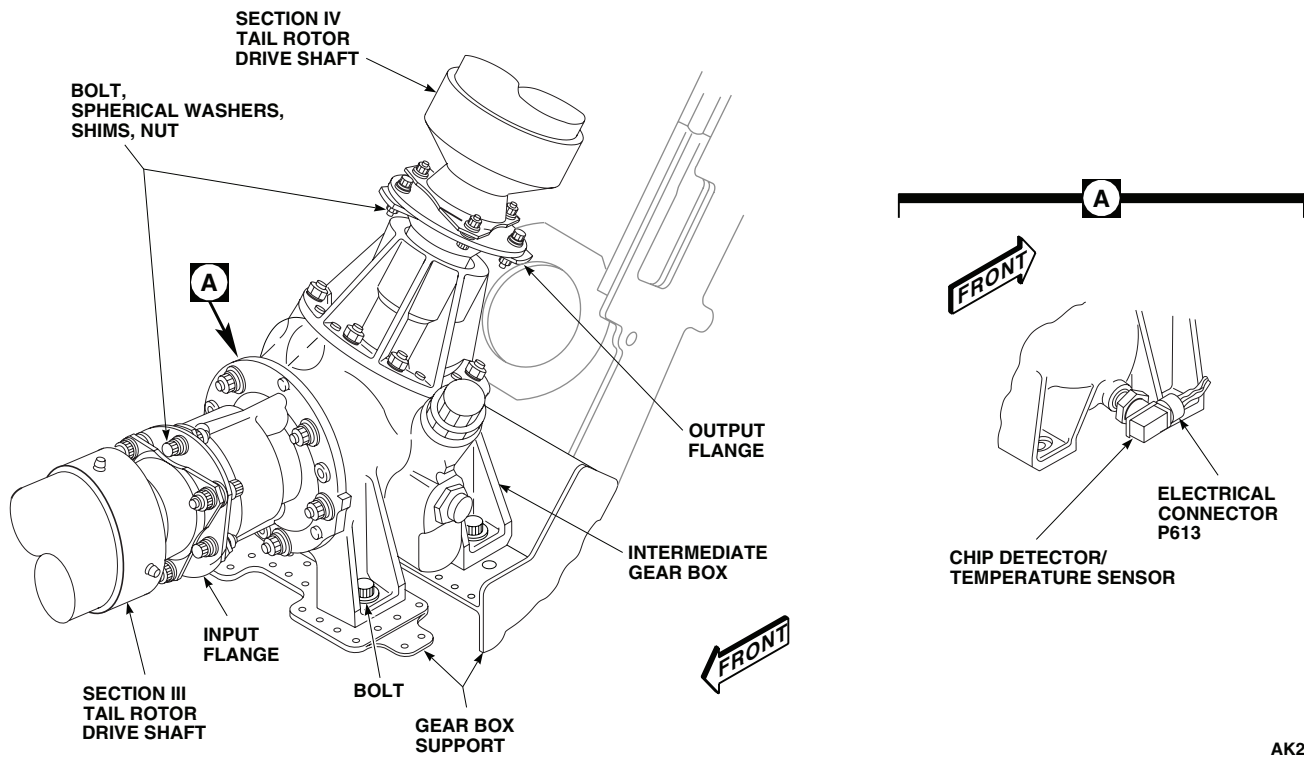
REMOVE - Continued

Figure 1. Replace Intermediate Gear Box.

8. Remove bolts securing intermediate gear box to supports (Figure 3).
9. Remove intermediate gear box.

INSPECT**CAUTION**

Damage to barrel-nuts will result if mounting bolts are forced into them. Threads will be stripped. Do not attempt to force mounting bolts into barrel-nuts.

NOTE

Make sure that only square barrel-nuts are used.

1. Thread bolts into forward gear box mounting holes until gap between mounting surface and underside of bolthead is 0.669 to 0.741-inch.
2. Thread bolts into rear gear box mounting holes until gap between mounting surface and underside of bolthead is 0.619 to 0.691-inch.
3. Using a torque wrench, back out bolt. **BREAKAWAY TORQUE SHALL BE AT LEAST 9.5 INCH-POUNDS. IF BREAKAWAY TORQUE IS LESS THAN 9.5 INCH-POUNDS, BARREL NUTS MUST BE REPLACED.** Fold tail pylon (WP 1681 00) to gain access to barrel nuts.

INSTALL

1. Check intermediate gear box for mounting gap as follows:

INSTALL - Continued

- a. Remove any sealant from gear box mounting surfaces and from airframe gear box supports using nonmetallic scraper.
 - b. Place intermediate gear box on supports and install mounting bolts. Tighten bolts as required until bolthead makes light contact with gear box mounting feet. **DO NOT EXCEED 100 INCH-POUNDS.**
 - c. Check for gaps between intermediate gear box mounting surfaces using feeler gage. **MAXIMUM GAP IS 0.030-INCH.**
2. If gap exceeds 0.030-inch under any mounting foot, install shim to bring gap within limits as follows:
- a. Remove bolts and remove intermediate gear box from airframe.

NOTE

If a shim was previously installed on a mounting pad which now requires a thicker shim, existing shim must be removed.

- b. Remove pin rivets securing shim to airframe (TM 1-1500-204-23).
- c. Remove original shim and spacer from mounting pad using a nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- d. Remove old sealing compound from mounting pad using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

NOTE

- Use only one gear box shim, one gear box spacer, and two rivets on each mounting pad.
 - Replacement gear box shim thickness must be original shim thickness plus thickness required for correction.
- e. Locally make replacement gear box shim and gear box spacer (Figure 199 and Figure 200, WP 1805 00).
 - f. Position replacement gear box shim and gear box spacer, 70219-06000-109, on airframe mounting pad (Figure 2, Detail A).
 - g. Mark location of fastener holes on shim using nonpermanent marker.
 - h. Drill holes in shim using 11/64-inch diameter drill.
 - i. Apply adhesive primer, Item 241, WP 1803 00, to all shim and spacer surfaces. Allow to dry.
 - j. Apply a brush coat of sealing compound, Item 273, WP 1803 00, to mounting pad using acid swabbing brush, Item 57, WP 1803 00.
 - k. Install gear box shim and gear box spacer, 70219-06000-109, on mounting pad using pin rivets (TM 1-1500-204-23).
3. **QA** Install intermediate gear box on airframe mounting surface. **TORQUE BOLTS TO 385 - 425 INCH-POUNDS (Figure 3).**

INSTALL - Continued

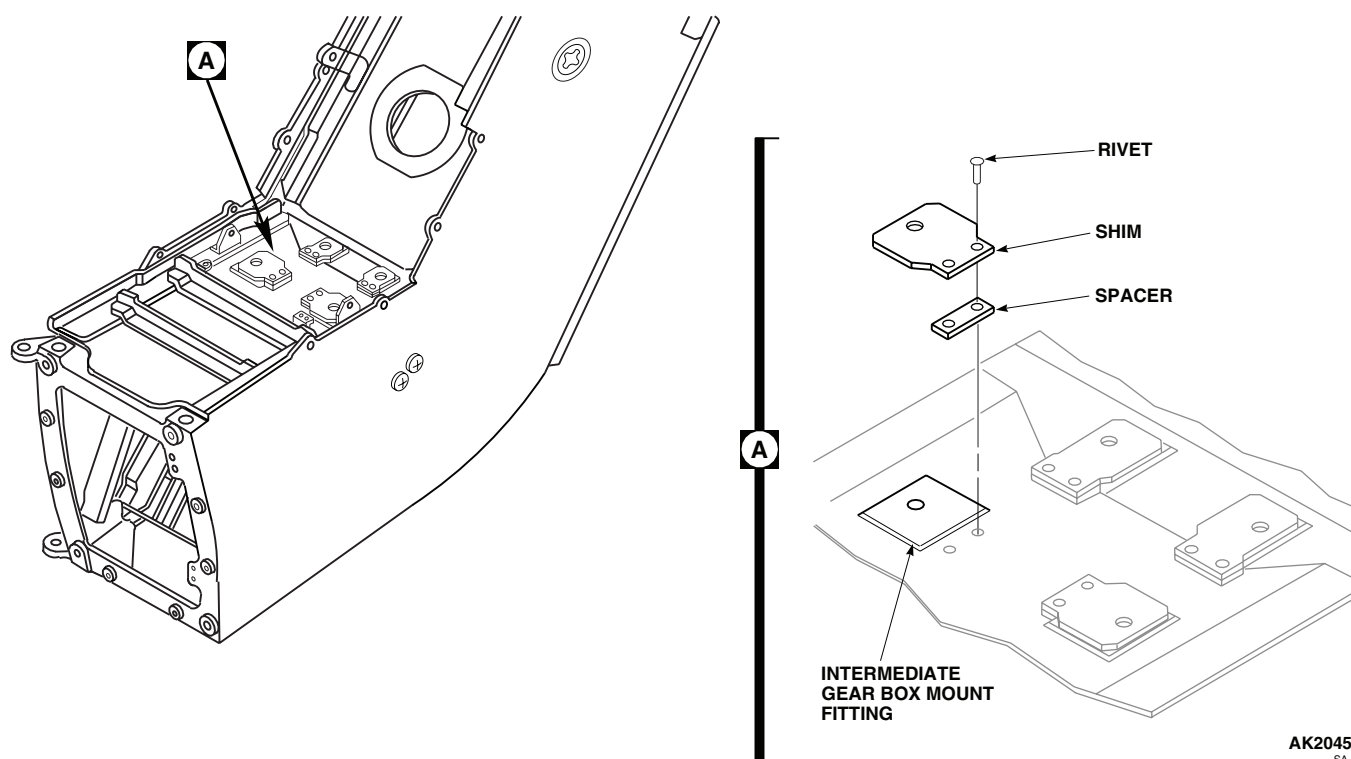


Figure 2. Replace Intermediate Gear Box.

4. **QA** Measure gap between gear box mounting feet and mounting pads using feeler gage. **NO GAP ALLOWED.** If gap exists, do the following:
 - a. Remove bolts and remove intermediate gear box from airframe supports.
 - b. Fluorescent penetrant inspect gear box mounting legs (TM 1-1520-265-23). **NO CRACKS ALLOWED.** If any cracks exist in gear box mounting feet, replace gear box.
 - c. If no cracks are found repeat step 2 through step 7.
5. Remove intermediate gear box from airframe supports.
6. Coat intermediate gear box airframe mounting surface with molybdenum disulfide, Item 197, WP 1803 00.
7. Coat gear box mounting feet with sealing compound, Item 273, WP 1803 00.
8. Apply a small bead of sealing compound, Item 273, WP 1803 00, to inner rim of each mounting bolthole on top side of mounting feet.

WARNING**CSI**

Verification of minimum torque of bolts is a critical characteristic.

9. **QA** Place intermediate gear box on supports. Apply a coating of epoxy primer coating kit, Item 120, WP 1803 00, to mounting bolts. **TORQUE TO 385 - 425 INCH-POUNDS (Figure 3).**
10. Form a fillet with squeezed out sealing compound around base of mounting feet.

INSTALL - Continued

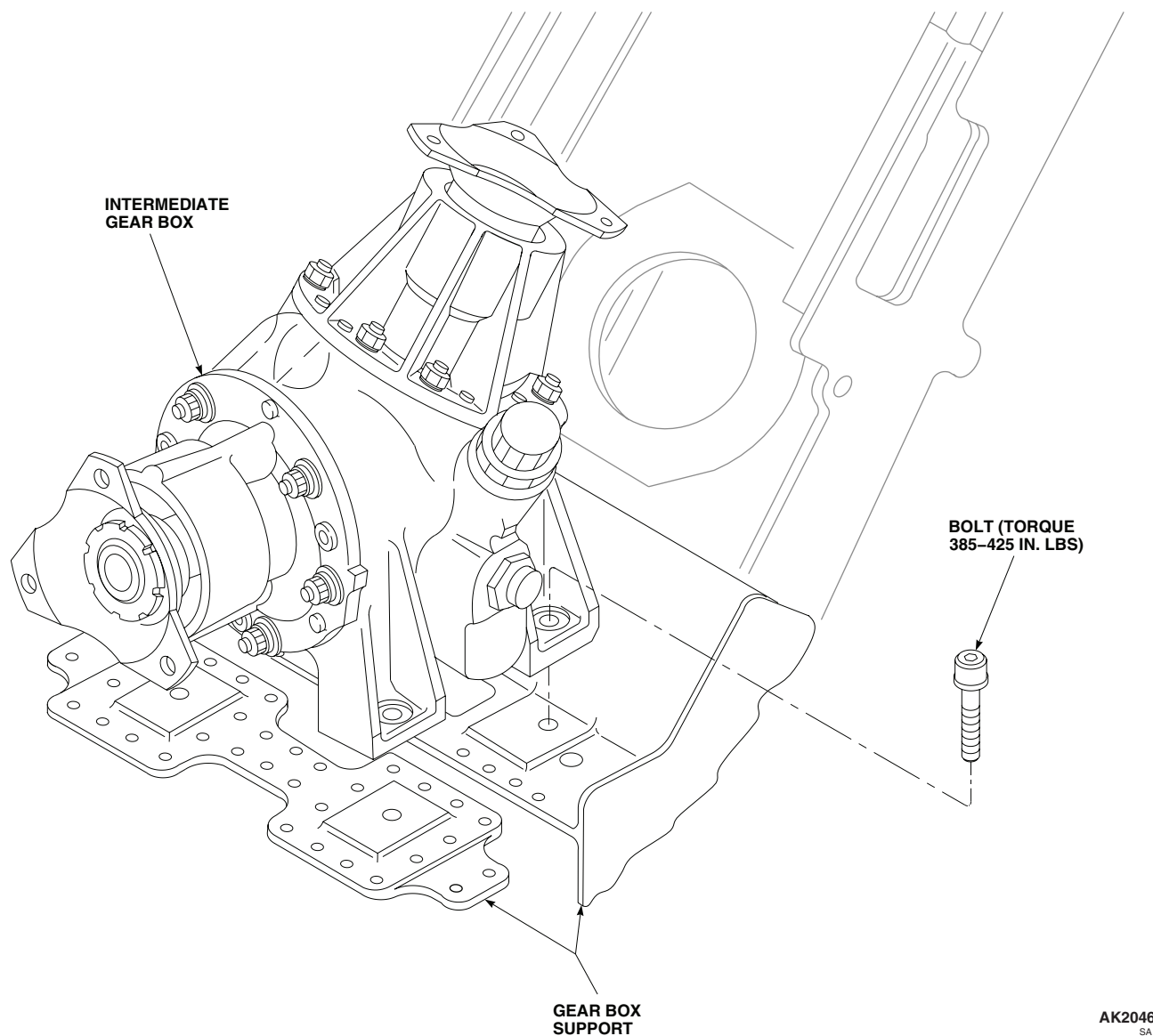


Figure 3. Replace Intermediate Gear Box.

11. Mask off adjacent areas using pressure sensitive adhesive tape, Item 326, WP 1803 00, to prevent overspray.
12. Apply corrosion preventive compound, Item 106, WP 1803 00, to entire area of mounting feet.
13. Allow to dry for 10 to 30 minutes and apply a second coat of corrosion preventive compound, Item 106, WP 1803 00.
14. Inspect every three days and reapply two coats of corrosion preventive compound, Item 106, WP 1803 00, as necessary until bolt torque stabilizes.
15. Remove supporting wood blocks from rear end of section III tail rotor drive shaft.

INSTALL - Continued**NOTE**

Do not turn shaft from original position where largest gap was measured, as gap may change. If shaft is turned, recheck gap.

16. Connect tail rotor drive shaft to input flange of intermediate gear box as follows (Figure 4, Detail A and Detail C):
 - a. Hold shaft in position and measure gap at three lobes, using feeler gage.
 - b. Determine average gap by adding readings and dividing by 3, refer to Table 1.

Table 1. EXAMPLE.

GAP	A = 0.125
	B = 0.142
	C = 0.137
Total	0.404
0.404 ÷ 3 = 0.1346-inch average gap.	
Round off to 0.135-inch	

NOTE

Do not interchange shims from one bolt stackup to another.

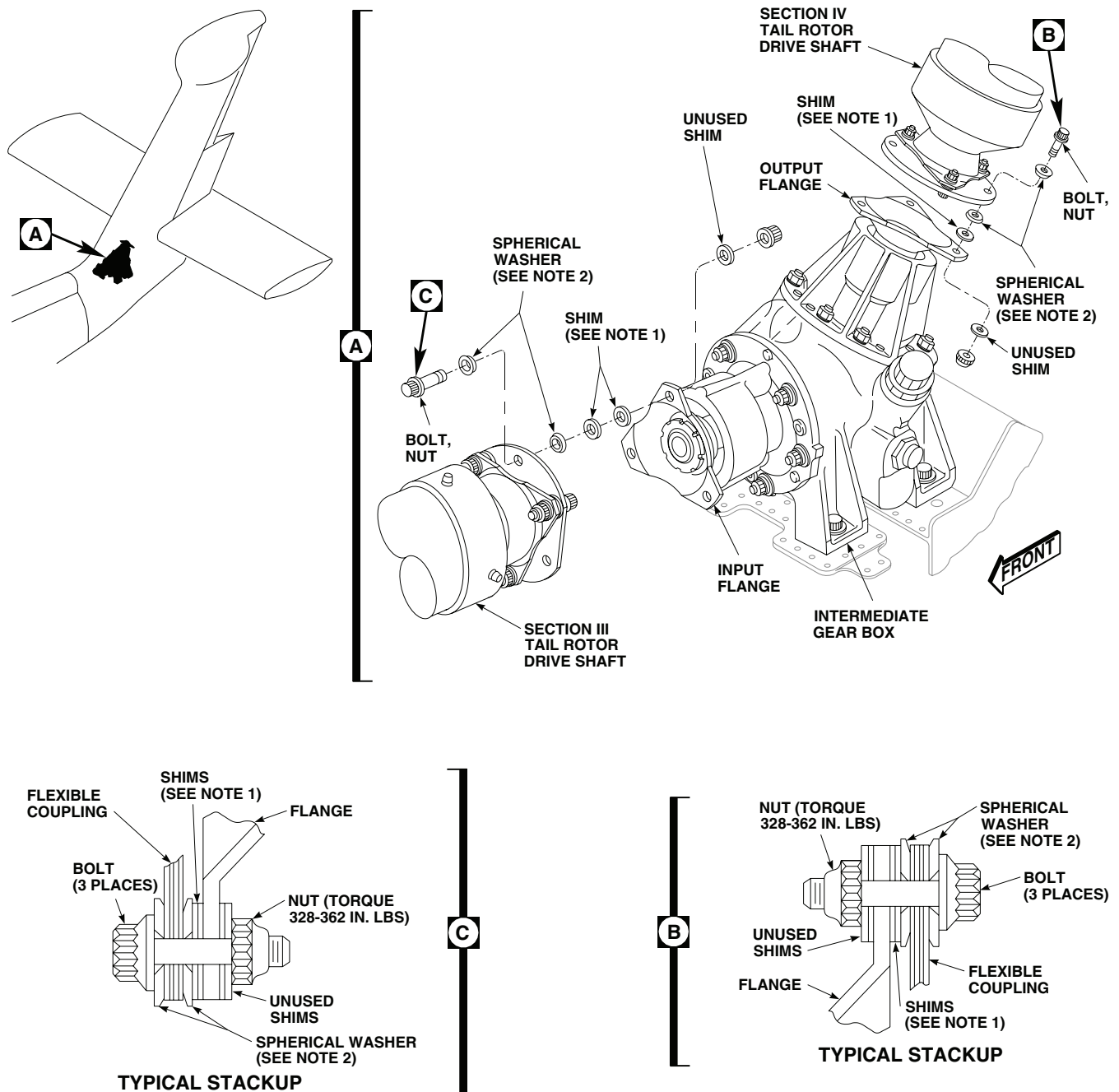
17. Select spherical washer and shim combination closest to average gap. Make three stackups.

WARNING**CSI**

Verification of minimum torque of nuts is a critical characteristics.

18. Spray spherical washers with corrosion preventive compound, Item 106, WP 1803 00.
19. **QA** Install appropriate stackup on bolt. Insert spherical washers with spherical side facing coupling on both sides of coupling, and install washers, unused shims, and nut. Make sure nut is on gear box flange. **TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
20. Remove supporting block from lower end of section IV tail rotor drive shaft.
21. **QA** Connect section IV tail rotor drive shaft to output flange of intermediate gear box. Procedure for output flange connection is same as for input flange connection. Repeat step 21 through step 24 (Figure 4, Detail A and Detail B).
22. Check chip detector electrical circuit (WP 0085 00).
23. Install chip detector/temperature sensor (WP 0665 00).

INSTALL - Continued



NOTES

1. SHIM THICKNESS:
70361-05009-105, 0.020
70361-05009-106, 0.030
70361-05009-107, 0.035
70361-05009-108, 0.040
ONE OF EACH SHIM WITH BOLT STACKUP.
2. SPHERICAL WASHER THICKNESS
70361-05010-102, 0.080
TO 0.083.
3. ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

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Figure 4. Replace Intermediate Gear Box.

INSTALL - Continued**WARNING**

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

CAUTION

Damage to gear box will result if gear box is overfilled. Do not overfill gear box.

24. Service intermediate gear box (WP 1608 00).
25. Make sure area is clean and free of foreign material.
26. Install intermediate gear box fairings (WP 0335 00).
27. Check DA Form 2408-13-1 to make sure tail rotor drive shaft and intermediate gear box mount bolts torque checks are shown as due 9 to 11 flight hours after installation.
28. After torque has been stabilized, do this:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- a. Remove corrosion preventive compound using technical acetone, Item 5, WP 1803 00.
 - b. Apply sealing compound, Item 273, WP 1803 00, to joints around boltheads, nuts, and washers. Allow to dry.
 - c. Apply epoxy primer coating kit, Item 120, WP 1803 00, to joints around boltheads, nuts, and washers.
 - d. Apply one coat of polyurethane coating, Item 231, WP 1803 00, to joints around boltheads, nuts, and washers (WP 0381 00 and TM 55-1500-345-23).
29. Apply one coat of polyurethane coating, Item 235, WP 1803 00, to any water entrapment areas of gear box.
 30. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

INTERMEDIATE GEAR BOX INPUT SEAL AND FLANGE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Ballpeen Hammer, GGG-H-86
Container, 2-Quart
Drain Hose, H538A-3
Nylon Wedge
Powertrain Repairer's Toolkit, SC 5180-99-B13
Seal Installation Tool, Locally-Made, Figure 157,
WP 1805 00
Seal Puller, Locally-Made, Figure 160, WP 1805 00
Torque Reactor, 70700-20688-041
Torque Wrench, 0 - 600 ft. lbs.
Wrench, 70700-20687-041

Material/Parts

Corrosion Preventive Compound, Item 106,
WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Grease, Aircraft, Item 131, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00

Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00
Screws, Self-Tapping
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
WP 0335 00
WP 0621 00
WP 0668 00
WP 1608 00
WP 1803 00
WP 1805 00

REMOVE

1. Turn off all electrical power.
2. Remove fairing from around intermediate gear box (WP 0335 00).
3. Remove section III tail rotor drive shaft (WP 0668 00).
4. Remove chip detector/temperature sensor. Connect drain hose, H538A-3, and drain oil into a 2 quart container.
5. Remove drain hose and reinstall chip detector/temperature sensor.
6. Using torque reactor and wrench remove flange nut (Figure 1).
7. Pull flange off shaft.
8. Remove retaining ring.

NOTE

Self-tapping screws may be used as an alternate to remove oil seal if seal puller is not available. If seal puller is not available, proceed to step 14.

9. Locally make nylon wedge.
10. Locally make seal puller (WP 1805 00).

REMOVE - Continued

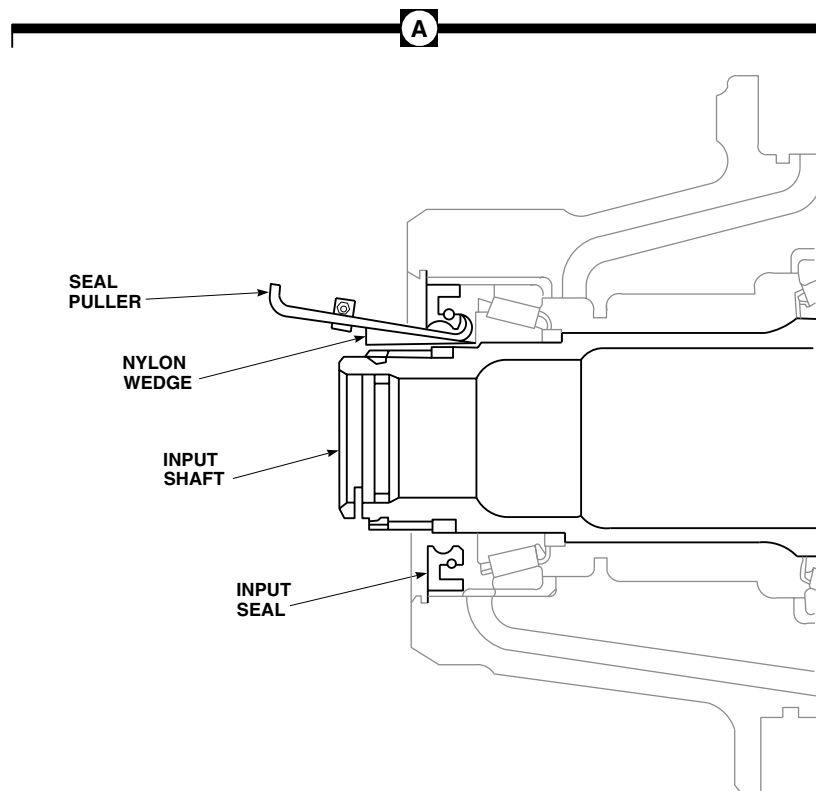
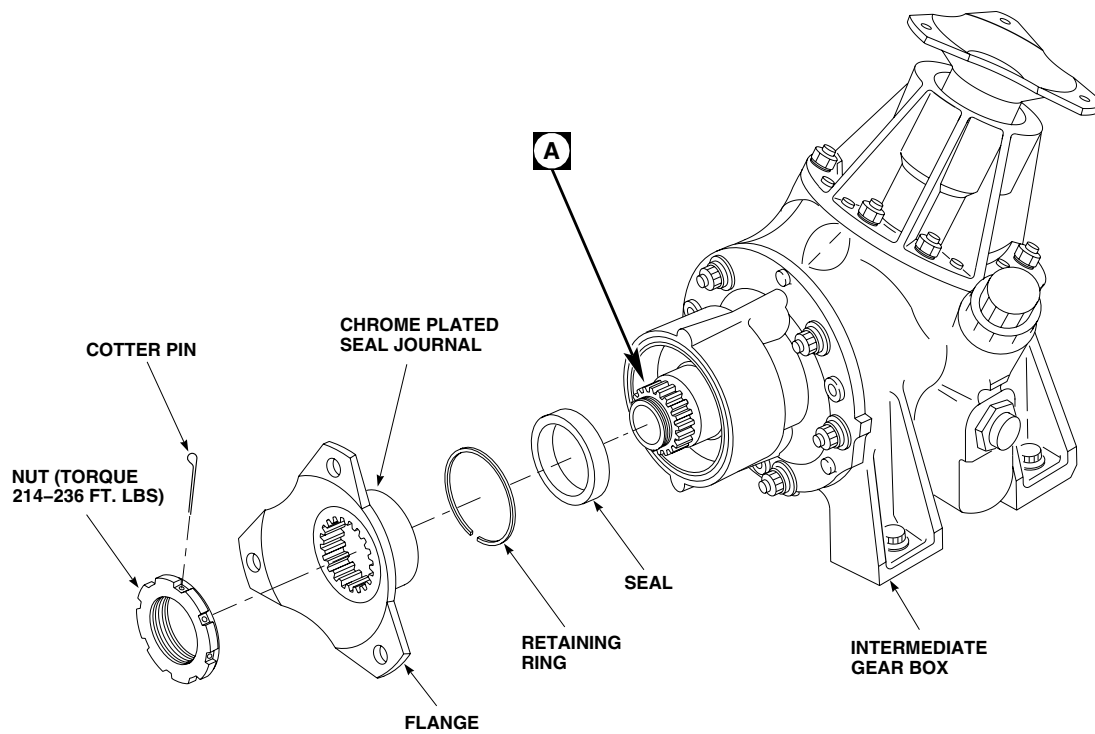
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Figure 1. Replace Input Seal and Flange.

REMOVE - Continued

11. Insert nylon wedge between seal puller and shaft to protect gear box (Figure 1, Detail A).

CAUTION

Damage to intermediate gear box will result, if too much force is applied while using hammer and seal puller. Gear box housing will crack. Use care when using hammer and seal puller.

12. Gently tap end of seal puller with an 8-ounce ballpeen hammer in an outward direction, away from gear box.
13. Moving seal puller and wedge in a clockwise direction, repeat steps 11 and 12 until seal is removed.

WARNING

- Damage to intermediate gear box will result if self-tapping screws completely penetrate input seal. Screws will damage bearing inside gear box. Install screws so that only outer face of input seal is penetrated.
- Damage to intermediate gear box will result if metal chips or shavings enter gear box by drilling or punching holes in seal. Metal chips or shavings will contaminate gear box oil. Do not drill or punch holes in seal.

14. If seal puller is not available, remove oil seal as follows:
 - a. Thread three self-tapping screws into holes in seal.
 - b. Carefully pull out seal.

INSPECT

1. Inspect flange (WP 0621 00).
2. Inspect flange nut for corrosion. **REPLACE NUT IF PITS ARE 0.030 INCH DEEP, IF PITTING COVERS MORE THAN 10% OF MATING SURFACE, OR IF NUT THREADS SHOW SIGNS OF CORROSION.**
3. Inspect splines for corrosion and wear. Evidence of corrosion of gear box splines is cause for removal of gear box. **NO WEAR INTO BASE METAL THROUGH PLATING IS PERMITTED (NO FRETTING OR SPALLING).**

REPAIR

1. Remove corrosion from flange nut, using procedures in TM 1-1500-344-23.

NOTE

Do not paint threads or mating surface of flange nut.

2. Apply epoxy primer coating kit, Item 120, WP 1803 00, to flats and exposed face of flange nut.

INSTALL

1. Locally make seal installation tool (WP 1805 00).
2. Coat inside diameter of seal with aircraft grease, Item 131, WP 1803 00. Coat outside diameter of seal and seal mating surface (flange sleeve) with aircraft turbine lubricating oil, Item 182, WP 1803 00, or helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.

INSTALL - Continued**CAUTION**

Input seal and gear box will be damaged if seal is incorrectly installed. Seal will leak. Make sure seal is installed flush to 0.020-inch below retaining ring groove prior to installing retaining ring.

3. **QA** Install new seal by gently tapping into housing using locally made seal installation tool (WP 1805 00). Be careful not to cock or distort seal (Figure 1).
4. Apply epoxy primer coating kit, Item 120, WP 1803 00, to retaining ring and install in housing while wet.
5. Apply sealing compound, Item 273, WP 1803 00, to housing boss to prevent water entrapment and corrosion above retaining ring groove.
6. Touch up sealing compound with epoxy primer coating kit, Item 120, WP 1803 00.
7. **QA** Slide flange on shaft (Figure 1).
8. Apply corrosion preventive compound, Item 106, WP 1803 00, over entire nut.
9. **QA** Make sure bevel side of nut is toward seal and that nut can be fully run on by hand. Using torque reactor and wrench, **TORQUE NUT AS FOLLOWS: THE TORQUE RANGE IS 214 - 236 FOOT-POUNDS. TORQUE RETAINING NUT TO 214 FOOT-POUNDS.** Install new cotter pin. To align pin hole, turn nut in tightening direction. **DO NOT EXCEED MAX TORQUE. IF MAX TORQUE IS EXCEEDED AND NUT DOES NOT ALIGN, REMOVE NUT AND REPLACE IT WITH A NEW NUT AND REPEAT TORQUING PROCEDURE.**
10. **QA** Connect section III tail rotor drive shaft (WP 0668 00).

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

11. Service intermediate gear box (WP 1608 00).
12. Run up helicopter until gear box oil warms up. Shut down and check for leaks.
13. Make sure area is clean and free of foreign material.
14. Install tail gear box fairing (WP 0335 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

INTERMEDIATE GEAR BOX OUTPUT SEAL AND FLANGE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Ballpeen Hammer, GGG-H-86
Nylon Wedge
Powertrain Repairer's Toolkit, SC 5180-99-B13
Seal Installation Tool, Locally-Made, Figure 157,
WP 1805 00
Seal Puller, Locally-Made, Figure 160, WP 1805 00
Torque Reactor, 70700-20688-041
Torque Wrench, 0 - 600 ft. lbs.
Wrench, 70700-20687-041

Material/Parts

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Grease, Aircraft, Item 131, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00

Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00
Screws, Self-Tapping
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
WP 0335 00
WP 0621 00
WP 0669 00
WP 1608 00
WP 1803 00
WP 1805 00

REMOVE

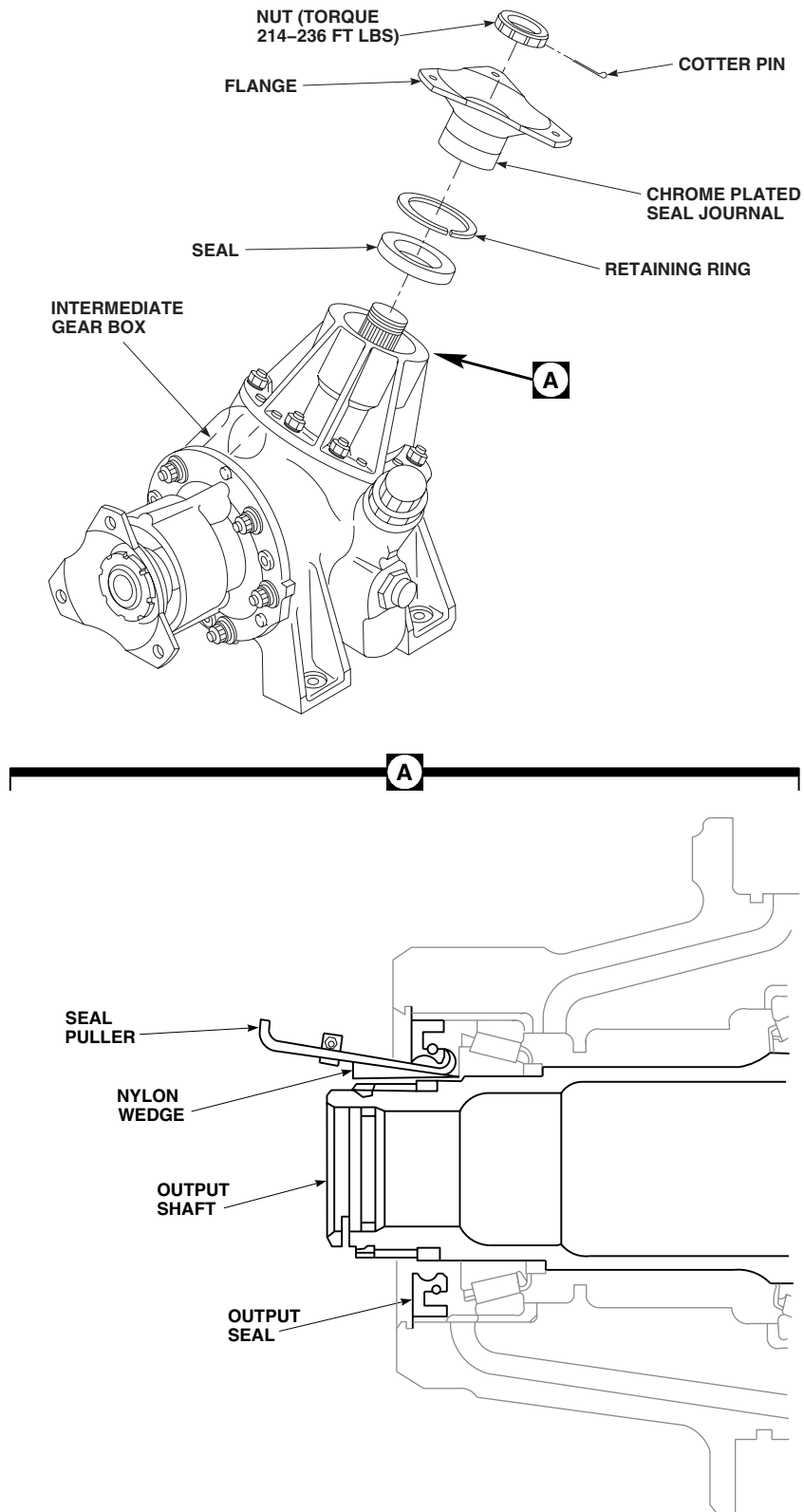
1. Turn off all electrical power.
2. Remove fairing from around intermediate gear box (WP 0335 00).
3. Remove section IV tail rotor drive shaft (WP 0669 00).
4. Using torque reactor and wrench remove nut (Figure 1).
5. Pull flange off shaft.
6. Remove retaining ring.

NOTE

Self-tapping screws may be used as an alternate to remove oil seal if seal puller is not available. If seal puller is not available, proceed to step 12.

7. Locally make nylon wedge.
8. Locally make seal puller (WP 1805 00).
9. Insert nylon wedge between seal puller and shaft to protect gear box (Figure 1).

REMOVE - Continued



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Figure 1. Replace Intermediate Gear Box Output Seal and Flange.

REMOVE - Continued**CAUTION**

Damage to intermediate gear box will result, if too much force is applied while using hammer and seal puller. Gear box housing will crack. Use care when using hammer and seal puller.

10. Gently tap end of seal puller with an 8 ounce ballpeen hammer in an outward direction, away from gear box.
11. Moving seal puller and wedge in a clockwise direction, repeat step 9 and step 10 until seal is removed.

WARNING

- Damage to intermediate gear box will result if self-tapping screws completely penetrate input seal. Screws will damage bearing inside gear box. Install screws so that only outer face of input seal is penetrated.
 - Damage to intermediate gear box will result if metal chips or shavings enter gear box by drilling or punching holes in seal. Metal chips or shavings will contaminate gear box oil. Do not drill or punch holes in seal.
12. If seal puller is not available, remove oil seal as follows:
 - a. Thread three self-tapping screws into holes in seal.
 - b. Carefully pull out seal.

INSPECT

1. Inspect flange (WP 0621 00).
2. Inspect flange nut for corrosion. **REPLACE NUT IF PITS ARE 0.030 INCH DEEP, IF PITTING COVERS MORE THAN 10% OF MATING SURFACE, OR IF NUT THREADS SHOW SIGNS OF CORROSION.**
3. Inspect splines for corrosion and wear. Evidence of corrosion of gear box splines is cause for removal of gear box. **NO WEAR INTO BASE METAL THROUGH PLATING IS PERMITTED (NO FRETTING OR SPALLING).**

REPAIR

1. Remove corrosion from flange nut (TM 1-1500-344-23).
2. Apply epoxy primer coating kit, Item 120, WP 1803 00, to flats and exposed face of flange nut. Do not paint threads or mating surface of flange nut.

INSTALL

1. Locally make seal installation tool (WP 1805 00).
2. Coat inside diameter of seal with aircraft grease, Item 131, WP 1803 00. Coat outside diameter of seal and seal mating surface (flange sleeve) with aircraft turbine lubricating oil, Item 182, WP 1803 00, or helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.

CAUTION

Input seal and gear box will be damaged if seal is incorrectly installed. Seal will leak. Make sure seal is installed flush to 0.020 inch below retaining ring groove prior to installing retaining ring.

3. **QA** Install seal by gently tapping into housing using locally made seal installation tool (WP 1805 00). Be careful not to cock or distort seal (Figure 1).

INSTALL - Continued

4. **QA** Apply epoxy primer coating kit, Item 120, WP 1803 00, to retaining ring and install in housing while wet.
5. Apply sealing compound, Item 273, WP 1803 00, to housing boss to prevent water entrapment and corrosion above retaining ring groove.
6. Touch up sealing compound with epoxy primer coating kit, Item 120, WP 1803 00.
7. **QA** Slide flange on shaft (Figure 1).
8. **QA** Make sure bevel side of nut is toward seal and that nut can be fully run on by hand. Using torque reactor and wrench, **TORQUE NUT AS FOLLOWS: THE TORQUE RANGE IS 214 - 236 FOOT-POUNDS. TORQUE RETAINING NUT TO 214 FOOT-POUNDS.** Install new cotter pin. To align pin hole, turn nut in tightening direction. **DO NOT EXCEED MAX TORQUE. IF MAX TORQUE IS EXCEEDED AND NUT DOES NOT ALIGN, REMOVE NUT AND REPLACE IT WITH A NEW NUT AND REPEAT TORQUING PROCEDURE.**
9. Install section IV tail rotor drive shaft (WP 0669 00).

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

10. Service intermediate gear box (WP 1608 00).
11. Run up helicopter until gear box oil warms up. Shut down and check for leaks.
12. Make sure area is clean and free of foreign material.
13. Install intermediate gear box fairing (WP 0335 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

INTERMEDIATE GEAR BOX SIGHT PLUG

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 2-Quart
Drain Hose, H538A-3

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00

References

WP 0335 00
WP 1608 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove fairing from around intermediate gear box (WP 0335 00).
3. Remove chip detector.
4. Install drain hose, H538A-3, to self-closing valve, and drain enough oil into a 2 quart container, to lower level below bottom of sight plug.
5. Remove drain hose.
6. Install chip detector.
7. Unscrew sight plug. Discard packing (Figure 1).

INSTALL

1. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on sight plug.
2. **QA** Screw sight plug into intermediate gear box. **TIGHTEN UNTIL A SHARP RISE IN TORQUE IS FELT. THEN TIGHTEN 1/6 - 1/3 TURN MORE.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire plug to gear box housing (Figure 1).

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

3. Service intermediate gear box (WP 1608 00).
4. Make sure area is clean and free of foreign material.

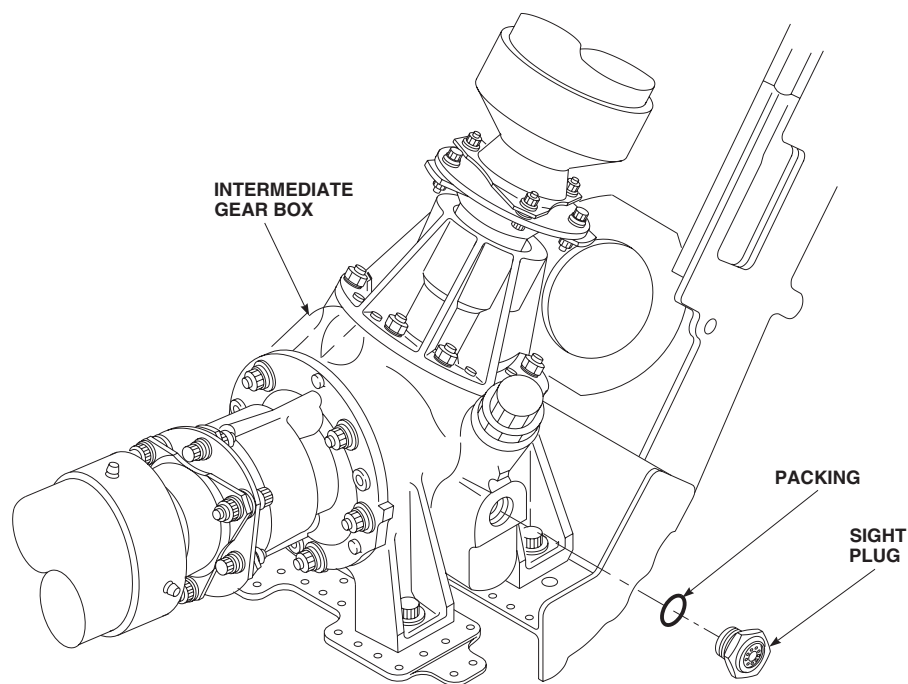
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Figure 1. Replace Intermediate Gear Box Sight Plug.

5. Install intermediate gear box fairing (WP 0335 00).
6. Run up helicopter until gear box oil warms up. Shut down and check for leaks.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

INTERMEDIATE GEAR BOX OIL FILLER PLUG

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0335 00
WP 1803 00

REMOVE

1. Remove fairing from gear box (WP 0335 00).

NOTE

Do not interchange filler plug of intermediate and tail rotor gearboxes. Tail rotor gearbox has a vented filler plug, P/N F401STV-2, installed. Intermediate gearbox has a non-vented filler plug, P/N F401STV-1, installed.

2. Unscrew filler plug, with cap attached, from gear box (Figure 1).

CAUTION

Damage to intermediate gear box will result if sealing compound enters gear box. Use extreme care when scraping sealing compound from filler plug boss.

3. Scrape sealing compound from filler plug boss with nonmetallic scraper.

INSTALL

1. Install packing lubricated with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, on filler plug.
2. Install filler plug on gear box. **TORQUE PLUG TO 120 - 190 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire filler plug to gear box (Figure 1).
3. Seal area around base of filler plug with sealing compound, Item 273, WP 1803 00.
4. Run up helicopter and check for leaks.
5. Install intermediate gear box fairing (WP 0335 00).

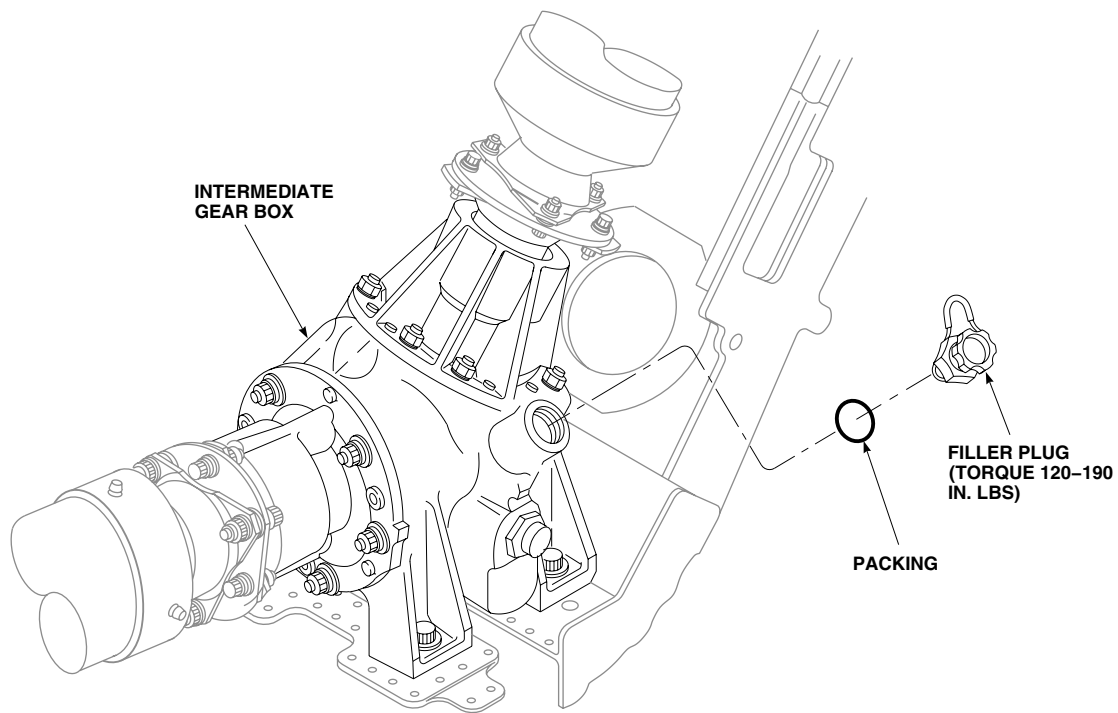
INSTALL - Continued**AK2051**
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Figure 1. Replace Intermediate Gear Box Oil Filler Plug.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM

INTERMEDIATE GEAR BOX HOUSING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Depth Gage, Dial Indicating, 643J
 Fluorescent Penetrant Inspection Kit, XMA101
 Powertrain Repairer's Toolkit, SC 5180-99-B13

Remover, Paint, Item 252, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Chromium Trioxide, Item 65, WP 1803 00
 Cleaning Compound Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00
 Epoxy Coating Kit, Item 118, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Polyurethane Coating, Item 231, WP 1803 00

References

TM 1-1500-344-23
 TM 1-1520-265-23
 WP 0639 00
 WP 1803 00

GENERAL REPAIRS
NOTE

- Minor nicks, gouges, scratches, or corrosion on surface of gear box housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in gear box still being within repair limits. Make sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace gear box.
 - Repair of damaged area may be done with gear box installed.
 - The following repair procedures apply to input, center, and output housings (Figure 1).
 - All repairs shall be recorded on Aircraft Component Historical Record, DA Form 2408-16.
1. Clean any oil or grease from damaged area using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
 2. Before blending, visually check damaged area for cracks. **NO CRACKS ALLOWED.** Replace cracked gear box.
 3. Fine scratches which remove only paint may be blended smooth using abrasive cloth, Item 82, WP 1803 00. After smoothing out, touch up with epoxy coating kit, Item 118, WP 1803 00, or polyurethane coating, Item 231, WP 1803 00, depending on original paint finish.
 4. Nicks, gouges, scratches, or corrosion on surface of all gear box housings may be blended out to a 1/2-inch radius. Only two repairs are allowed in any 4 square inch area. Refer to repair limits for depths of repair.

GENERAL REPAIRS - Continued

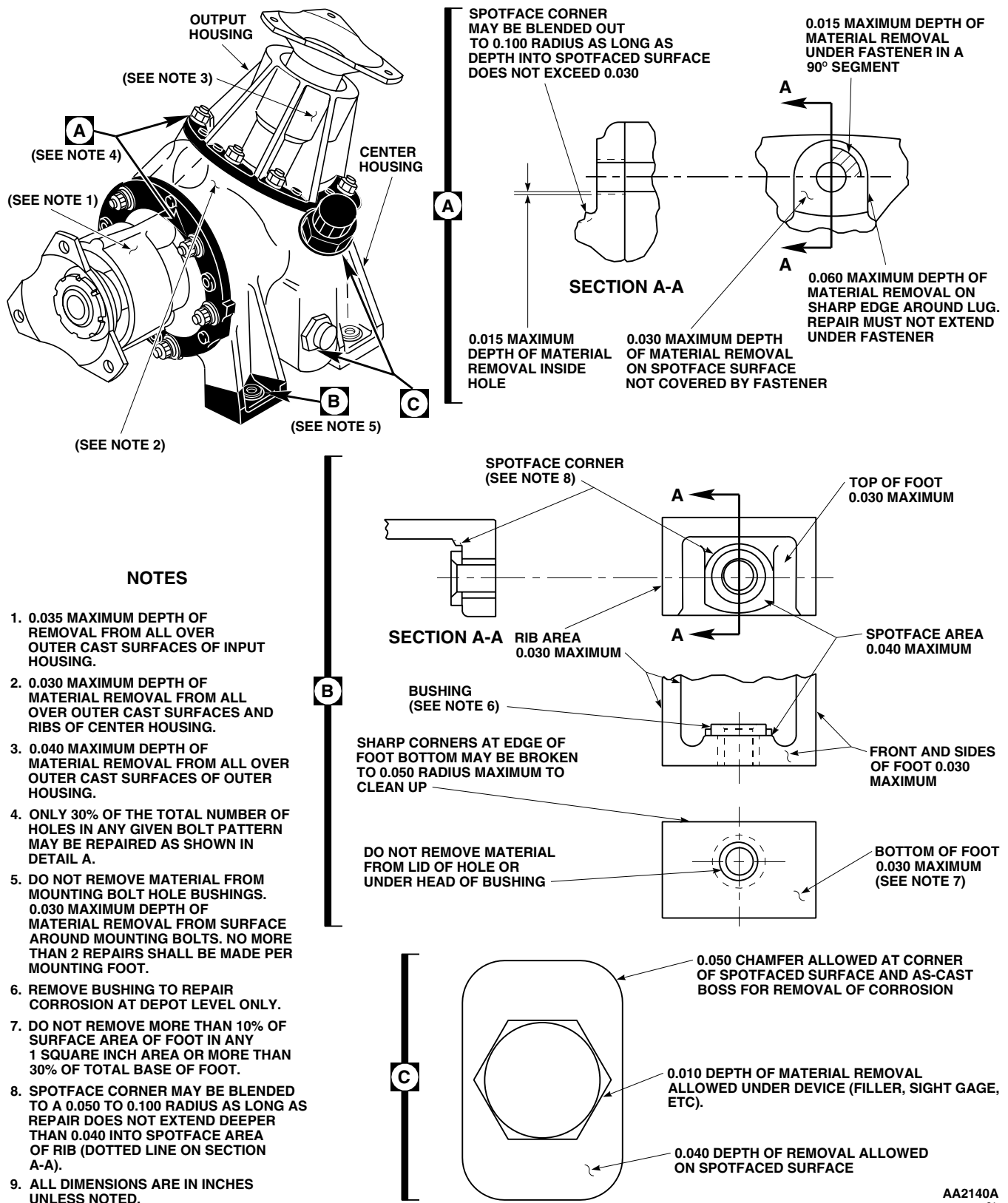


Figure 1. Repair Intermediate Gear Box Housing.

REPAIR LIMITS

1. Input housing limits are as follows:

- a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACE IS 0.035-INCH.**

NOTE

Only 30% of the total number of holes in bolt pattern may be repaired.

- b. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM INSIDE OF BOLTHOLES IS 0.015-INCH (Figure 1, Detail A).**
- c. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENERS (BUT NOT COVERING MORE THAN A 90 DEGREE SEGMENT) IS 0.015-INCH.**
- d. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- e. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOTFACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**
- f. **CORNER OF SPOTFACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOTFACE DOES NOT EXCEED 0.030-INCH.**

2. Center housing limits are as follows:

- a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACE IS 0.030-INCH.**
- b. **MAXIMUM DEPTH OF MATERIAL REMOVAL ON MOUNTING FEET AND MOUNTING FEET RIBS IS 0.030-INCH (Figure 1, Detail B).**
 - (1) **NO MORE THAN TWO REPAIRS ALLOWED PER FOOT.**
 - (2) **TOTAL REPAIR SHALL NOT EXCEED 10% OF SURFACE AREA OF FOOT IN ANY 1 SQUARE INCH AREA OR MORE THAN 30% OF TOTAL BASE OF FOOT.**
 - (3) **NO MATERIAL REMOVAL FROM BUSHING BORE ALLOWED.**
 - (4) **SPOTFACE CORNER MAY BE BLENDED TO 0.050 TO 0.100-INCH RADIUS AS LONG AS REPAIR DOES NOT GO DEEPER THAN 0.040-INCH INTO SPOTFACE.**
 - (5) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE OF MOUNTING FOOT IS 0.040-INCH.**
- c. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OPEN SPOTFACE SURFACE NOT COVERED BY SIGHT GAGE, CHIP DETECTOR, OR FILLER PLUG IS 0.040-INCH (Figure 1, Detail C).**
- d. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE UNDER SIGHT GAGE, CHIP DETECTOR, OR FILLER PLUG IS 0.010-INCH.**
- e. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE CORNER AROUND SIGHT GAGE, CHIP DETECTOR, OR FILLER PLUG IS 0.040-INCH WHEN BLENDED TO A 0.050-INCH RADIUS.**

3. Output housing limits are as follows:

- a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES IS 0.040-INCH.**

REPAIR LIMITS - Continued**NOTE**

Only 30% of the total number of holes in bolt pattern may be repaired.

- b. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM INSIDE OF BOLTHOLES IS 0.015-INCH (Figure 1, Detail A).**
- c. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENERS (BUT NOT COVERING MORE THAN A 90 DEGREE SEGMENT) IS 0.015-INCH.**
- d. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- e. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOTFACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**
- f. **CORNER OF SPOTFACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOTFACE DOES NOT EXCEED 0.030-INCH.**

REPAIR PROCEDURES

- 1. Before starting rework, remove paint from damaged area using paint remover, Item 252, WP 1803 00. Blend damaged area smooth using aluminum oxide abrasive cloth, Item 83, WP 1803 00. Remove only what is necessary to blend out damaged area.
- 2. Using abrasive cloth, Item 82, WP 1803 00, blend reworked area to remove all traces of aluminum oxide abrasive cloth.
- 3. Remove displaced metal caused by blending with magnesium etchant (WP 0639 00).
- 4. Treat bare surface of housing with chromium trioxide, Item 65, WP 1803 00 (TM 1-1500-344-23).
- 5. Fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.**

NOTE

Housings coated with epoxy coating kit, Item 118, WP 1803 00, are glossy silver;
housing coated with polyurethane coating, Item 231, WP 1803 00, are matte surface
medium-gray.

- 6. Touch up reworked area with sealing compound, Item 273, WP 1803 00. After sealing compound has dried, apply one coat of epoxy primer coating kit, Item 120, WP 1803 00, or polyurethane coating, Item 231, WP 1803 00, depending on original paint finish.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
INTERMEDIATE GEAR BOX CHIP DETECTOR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Container, 2-Quart
 Drain Hose, H538A-3
 Powertrain Repairer's Toolkit, SC 5180-99-B13

Wire, Nonelectrical, Item 355, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 184,
 WP 1803 00
 Lubricating Oil, Helicopter Transmission, Item 187,
 WP 1803 00

References

WP 0085 00
 WP 0335 00
 WP 0622 00
 WP 1608 00
 WP 1703 00
 WP 1803 00

REMOVE

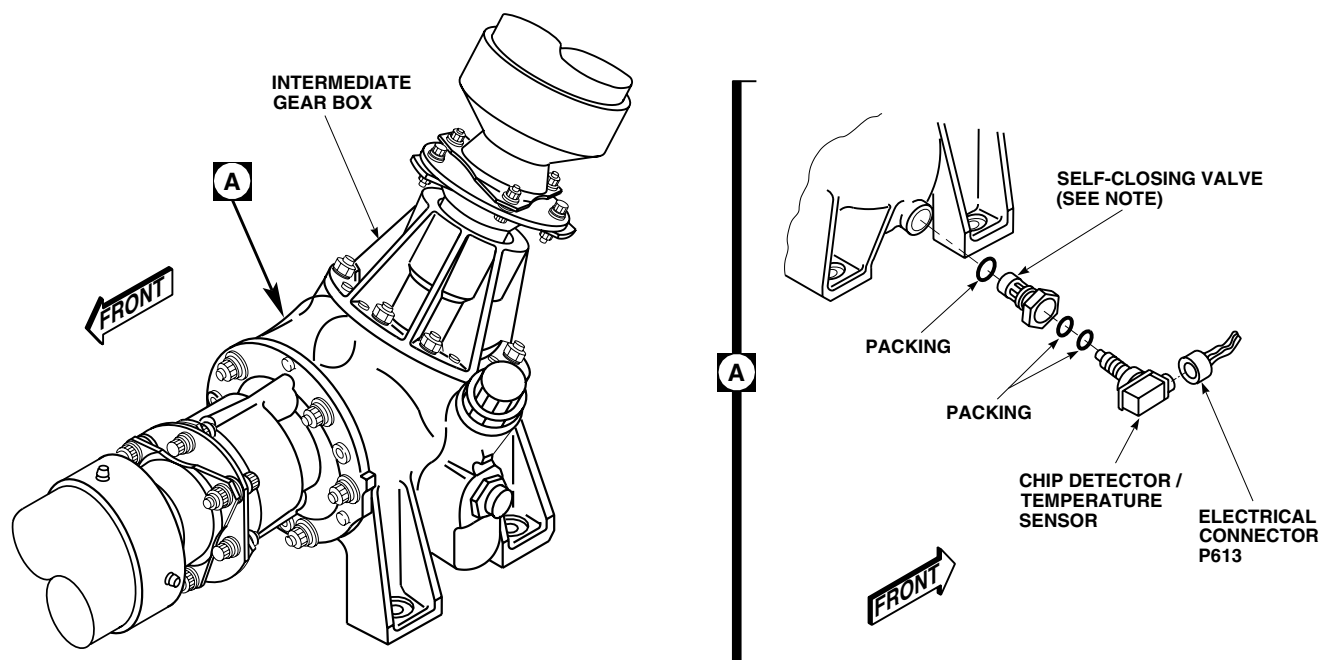
1. Turn off all electrical power.
2. Pull out CHIP DET circuit breaker (copilot's circuit breaker panel).
3. Remove intermediate gear box fairing (WP 0335 00).
4. Disconnect electrical connector, P613, from chip detector (Figure 1, Detail A).
5. Remove chip detector. Discard packings.

INSPECT

1. Check inside of self-closing valve for contact pin located in center of valve and for leakage of oil. If contact pin is loose or missing or if leakage is found, locate pin and washer and replace self-closing valve.
2. If pin or washer cannot be located, perform serviceability check (WP 0622 00).
3. If self-closing valve is to be removed attach drain hose, H538A-3, to self-closing valve and drain oil into a 2 quart container. Remove self-closing valve and packing.

INSTALL

1. Turn off all electrical power.
2. Lubricate all packings with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
3. **QA** Install packing on self-closing valve (Figure 1, Detail A).
4. **QA** Install self-closing valve in intermediate gear box. **TIGHTEN SELF-CLOSING VALVE 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
5. Check chip detector electrical circuit (WP 0085 00).

INSTALL - Continued**NOTE**

TIGHTEN SELF-CLOSING VALVE 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

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Figure 1. Replace Chip Detector.

6. **QA** Install packings on chip detector. Install chip detector.
7. **QA** Using nonelectrical wire, Item 355, WP 1803 00, lockwire self-closing valve.
8. Inspect and treat electrical connector (WP 1703 00).
9. **QA** Connect electrical connector, P613, to chip detector.

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

10. Service intermediate gear box (WP 1608 00).
11. Make sure area is clean and free of foreign material.
12. Install intermediate gear box fairing (WP 0335 00).
13. Do an operational check of both chip detector electrical circuits (WP 0085 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
TAIL ROTOR DRIVE SHAFT SECTION I

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Micrometer
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 300 - 2500 in. lbs.

Epoxy Primer Coating Kit, Item 120, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cloth, Abrasive, Item 76, WP 1803 00
 Corrosion Preventive Compound, Item 105,
 WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Corrosion Removing Compound, Item 108,
 WP 1803 00

References

WP 0626 00
 WP 0666 00
 WP 0670 00
 WP 0680 00
 WP 1693 00
 WP 1803 00

REMOVE**WARNING**

Injury to personnel and damage to equipment will result if tail rotor is not secured when drive shaft is disconnected. Secure tail rotor to prevent shafts and rotor from turning.

CAUTION

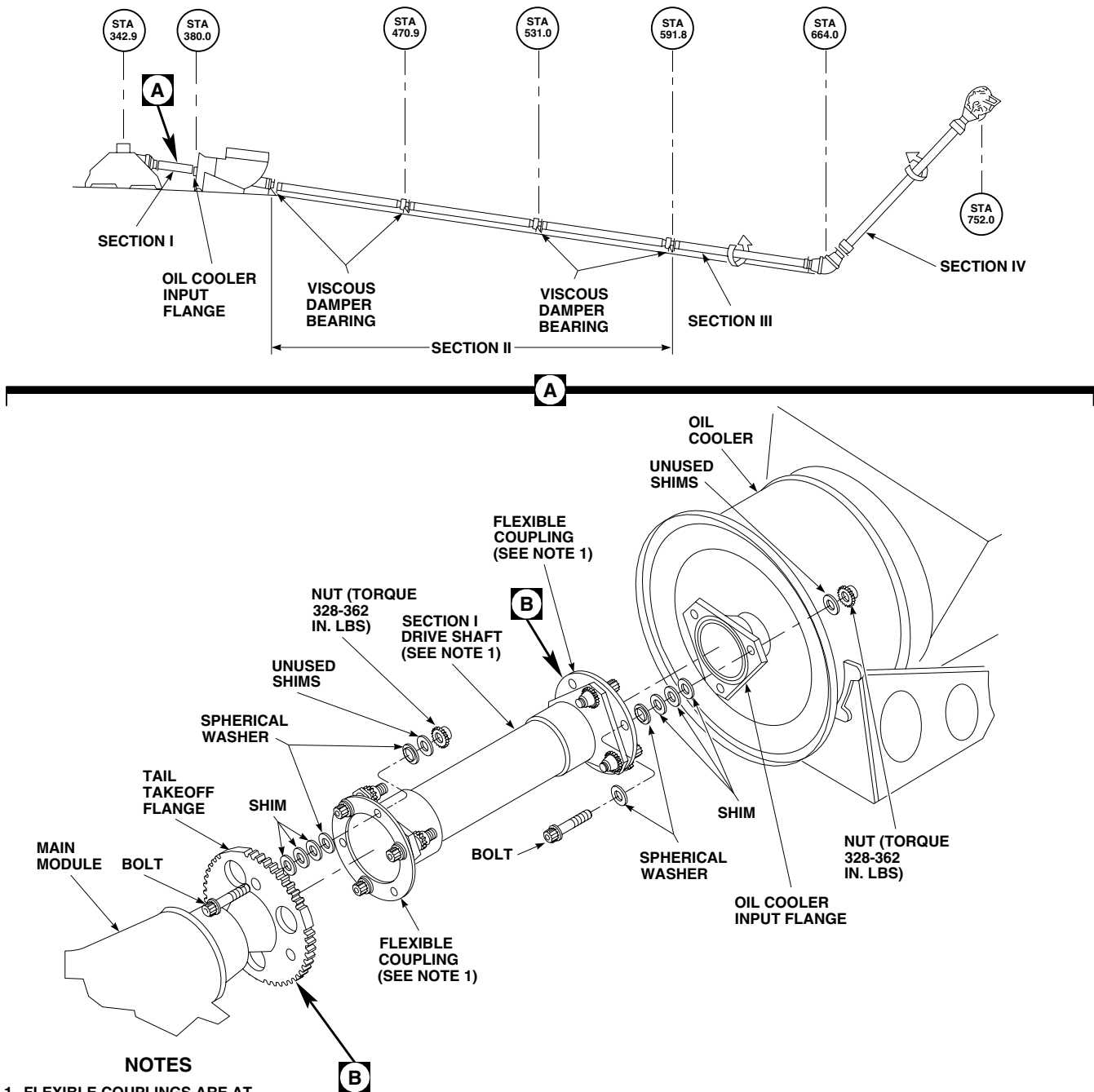
Damage to tail rotor drive shaft will result if handled improperly. Handle drive shaft carefully so it will not get dented, scratched, or bent.

NOTE

- Drive shaft coupling bolts and self-locking nuts can be reused provided they show no visual evidence of damage.
- Note location of retaining hardware so you can reinstall in same place.

1. Turn off all electrical power.
2. Open oil cooler compartment access panel doors.
3. Support drive shaft and remove bolts, spherical washers, shims, and nuts connecting flexible couplings to tail takeoff flange and oil cooler input flange. Remove shaft and flexible coupling assembly (Figure 1, Sheet 1).

REMOVE - Continued



NOTES

1. FLEXIBLE COUPLINGS ARE ATTACHED TO SECTION 1 DRIVE SHAFT PRIOR TO SHAFT AND COUPLING INSTALLATION.
2. SHIM THICKNESSES:
 70361-05009-105, 0.020
 70361-05009-106, 0.030
 70361-05009-107, 0.035
 70361-05009-108, 0.040
 ONE OF EACH SHIM WITH BOLT STACKUP.
3. SPHERICAL WASHER THICKNESS 0.080 TO 0.083.
4. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Replace Tail Rotor Drive Shaft Section I. (Sheet 1 of 2)

REMOVE - Continued

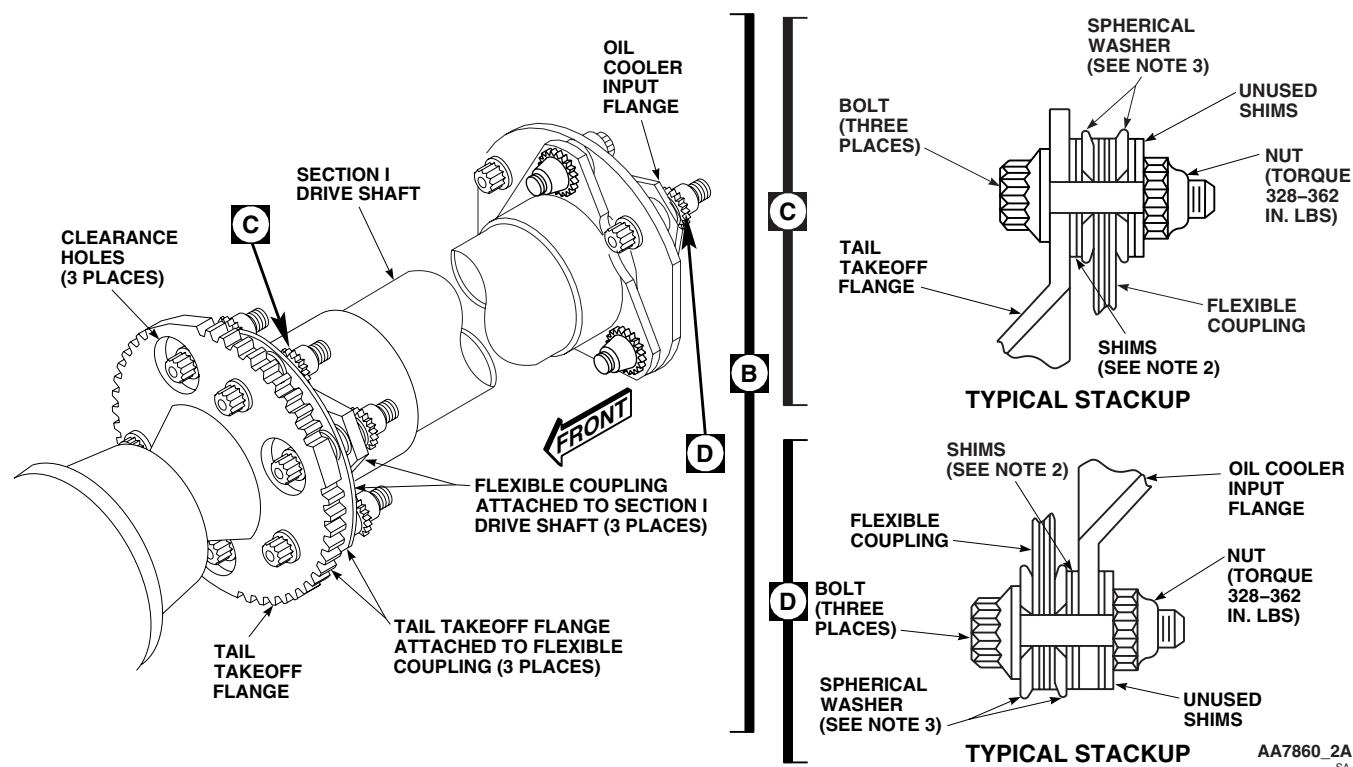


Figure 1. Replace Tail Rotor Drive Shaft Section I. (Sheet 2 of 2)

INSPECT

NOTE

Corrosion preventive compound may turn reddish brown over time. This is normal and should not be mistaken for corrosion.

1. Visually inspect flexible coupling spherical washers for evidence of corrosion, damage, or surface roughness.
NONE ALLOWED.
2. Inspect section I tail rotor drive shaft flange boltholes for elongation. **NONE ALLOWED.** Replace tail rotor drive shaft if boltholes are elongated (WP 0666 00).
3. If washer, 70361-05010-103, is unavailable, washer, 70361-05010-102, may be reworked as follows:
 - a. Clean washer using corrosion removing compound, Item 108, WP 1803 00.
 - b. If corrosion is still present, remove corrosion using abrasive cloth, Item 76, WP 1803 00. Make sure all corrosion is removed.
 - c. **MINIMUM THICKNESS AFTER BLENDING IS 0.080 INCH THICK.** Replace washers if minimum thickness is under this limit.
 - d. Apply corrosion preventive compound, Item 105, WP 1803 00, to washer surfaces.
 - e. Apply corrosion preventive compound, Item 106, WP 1803 00, to hardware assembly and allow to dry.

INSTALL**WARNING**

Injury to personnel and damage to equipment will result if Section I drive shaft is bolted incorrectly to tail takeoff flange. Three drive shaft bolts will be connected to flexible coupling only, and will protrude through three large holes in tail takeoff flange. Three tail takeoff flange bolts will be connected to flexible coupling only. Make sure Section I drive shaft flange is not bolted directly through flexible coupling to tail takeoff flange.

NOTE

- When reinstalling bolts in flexible couplings, be sure boltheads are against tail takeoff flange.
- Do not lockwire bolts with drilled heads installed on flexible couplings.

1. **QA** Install flexible couplings on Section I drive shaft (WP 0670 00).
2. Support Section I tail rotor drive shaft in position (Figure 1, Sheet 1).

CAUTION

Damage to drive shaft will result if handled improperly. Handle drive shaft carefully so it will not get dented, scratched, or bent. To prevent possible damage to drive shaft, place shaft on padded area.

NOTE

When installing bolts in flexible couplings, make sure a spherical washer is installed between the bolthead and coupling. Make sure the spherical side of washers are against the flexible coupling.

3. Install bolts at front end of shaft with spherical washers and nuts. Install washers on each side of flexible coupling. Tighten nuts only enough to prevent axial movement (Figure 1, Sheet 1).
4. Support opposite end of shaft (rear end) with bolts and spherical washers. Do not install nuts at this time.

NOTE

Do not turn shaft from original position, as gap may change. If shaft is moved, recheck gap.

5. Determine shim thickness at rear end of shaft as follows:
 - a. Hold shaft in installed position with spherical washers on each side of flexible coupling. Measure gap at three attachment points, using a feeler gage between washers and flange.
 - b. Determine average gap by adding readings and dividing by 3.
 - c. Divide average gap by 2 to determine thickness (shims are used on both ends of shaft).
 - d. **QA** Select shim combination closest to shim thickness calculated in step 5. c. Make three stackups for each end of shaft. **MAXIMUM GAP IS 0.125-INCH AT EACH END OF SHAFT.**

Table 1. Example.

GAP	A = 0.125
-----	-----------

INSTALL - Continued**Table 1. Example. - Continued**

	B = 0.142
	C = 0.137
Total	0.404
$0.404 \div 3 = 0.1346$ -inch average gap.	
$0.1346 \div 2 = 0.0673$ -inch average shim required on each end.	
Round off to 0.068-inch	

NOTE

- Shim thickness must be the same for all three bolt locations at each end of shaft. Shim combination at one end of shaft does not have to be the same as shim combination at other end of shaft as long as total shim combination adds up closest to average gap.
 - Drive shaft attaching nuts must be torqued within 15 minutes after applying epoxy primer to bolts.
6. Apply epoxy primer coating kit, Item 120, WP 1803 00, to bolt grip lengths and under bolt heads. Do not apply primer to threads.
 7. **QA** Install one stackup on each bolt. Insert spherical washers on each side of coupling. Make sure spherical side of washer faces coupling. Place unused shims under nuts.
 8. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** Replace nuts not within limits.

WARNING**CSI**

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

9. **QA TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
10. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure a tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (WP 1693 00).

CAUTION

Corrosion preventive compounds can penetrate bearing seals and dissolve lubrication grease. Do not spray corrosion preventive compounds in vicinity of drive shaft bearings. Shield bearings from spray.

11. Spray all flexible coupling spherical washers with corrosion preventive compound, Item 106, WP 1803 00.

INSTALL - Continued**WARNING**

Injury to personnel and damage to equipment will result if spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and spline wear inspection is performed (WP 0626 00).

NOTE

Spline wear inspection is not required when tail rotor drive shaft is disconnected or replaced aft of STA 531.0.

12. On helicopters with oil cooler fan, 70361-03005-103 through 70361-03005-106, installed, perform oil cooler spline wear inspection if required (WP 0626 00).

WARNING

Injury to personnel and damage to equipment will result if vibration check is not performed every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive vibration indicates improper installation of oil cooler assembly which will cause oil cooler axial fan drive shaft splines and fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed (WP 0680 00).

13. Perform oil cooler vibration check (WP 0680 00).
14. Check area for cleanness and foreign material.
15. Close oil cooler compartment access doors.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**TAIL ROTOR DRIVE SHAFT SECTION II**

This WP supersedes WP 0667 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Micrometer
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Cloth, Abrasive, Item 76, WP 1803 00
 Corrosion Preventive Compound, Item 105,
 WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Corrosion Removing Compound, Item 108,
 WP 1803 00

Epoxy Primer Coating Kit, Item 120, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0303 00
 WP 0369 00
 WP 0626 00
 WP 0670 00
 WP 0671 00
 WP 0680 00
 WP 1693 00
 WP 1803 00

REMOVE**WARNING**

Injury to personnel and damage to equipment will result if tail rotor is not secured when tail rotor drive shaft is disconnected. Secure tail rotor to prevent shafts and rotor from turning.

CAUTION

Damage to drive shafts will result if handled improperly. Handle drive shafts carefully so they will not get dented, scratched, or bent.

NOTE

Drive shaft coupling bolts, self-locking nuts, and viscous damper housing attachment bolts and nuts can be reused provided they show no visual evidence of damage.

1. Open APU access doors and remove APU panel (STA 485.0 through STA 525.0) as required to gain access to drive shafts (WP 0303 00).
2. Open tail rotor drive shaft covers as required to gain access to drive shafts.

NOTE

Remove/Install procedures for the three section II drive shafts are identical.

3. Identify section II drive shafts as front, middle, and rear section II drive shaft.

REMOVE - Continued**NOTE**

- Keep hardware together and in proper location.
 - Forward end flex coupling comes off with drive shaft. Rear end coupling remains on helicopter.
4. Support drive shaft and disconnect flexible coupling at forward end by removing nuts, washers, shims, and bolts (Figure 1, Detail B).
 5. Disconnect rear end of shaft from flexible coupling by removing nuts, washers, shims, and bolts.
 6. Remove bolts, washers, and nuts securing viscous damper to airframe support. Remove drive shaft (Figure 1, Detail A).

INSPECT

1. Inspect airframe supports for visible cracks and distortion. **NONE ALLOWED.**
2. Inspect airframe support boltholes for elongation as follows:
 - a. **IF ONE OF THE FOUR BOLTHOLES IS ELONGATED MORE THAN 0.010-INCH, (GREATER THAN 0.275-INCH DIAMETER), REPLACE AIRFRAME SUPPORT(WP 0369 00).**
 - b. **IF TWO OR MORE BOLTHOLES ARE ELONGATED MORE THAN 0.006-INCH, (GREATER THAN 0.271-INCH DIAMETER), REPLACE AIRFRAME SUPPORT(WP 0369 00).**
3. Inspect tail rotor drive shaft flexible coupling (WP 0670 00).

NOTE

Corrosion preventive compound may turn reddish brown over time. This is normal and should not be mistaken for corrosion.

4. Visually inspect flexible coupling spherical washers for evidence of corrosion damage or surface roughness. **NONE ALLOWED.**
5. Inspect section II tail rotor drive shaft flange boltholes for elongation. **NONE ALLOWED.** If forward flange boltholes are elongated, replace section II tail rotor drive shaft (WP 0667 00). If aft flange boltholes are elongated, replace flange (WP 0671 00).
6. If washer, 70361-05010-103, is unavailable, washer, 70361-05010-102, may be reworked as follows:
 - a. Clean washer using corrosion removing compound, Item 108, WP 1803 00.
 - b. If corrosion is still present, remove corrosion using abrasive cloth, Item 76, WP 1803 00. Make sure all corrosion is removed.
 - c. **MINIMUM THICKNESS AFTER BLENDING IS 0.080-INCH THICK.** Replace washers if minimum thickness is under this limit.
 - d. Apply corrosion preventive compound, Item 105, WP 1803 00, to washer surfaces.
 - e. Apply corrosion preventive compound, Item 106, WP 1803 00, to hardware assembly and allow to dry.

INSPECT - Continued

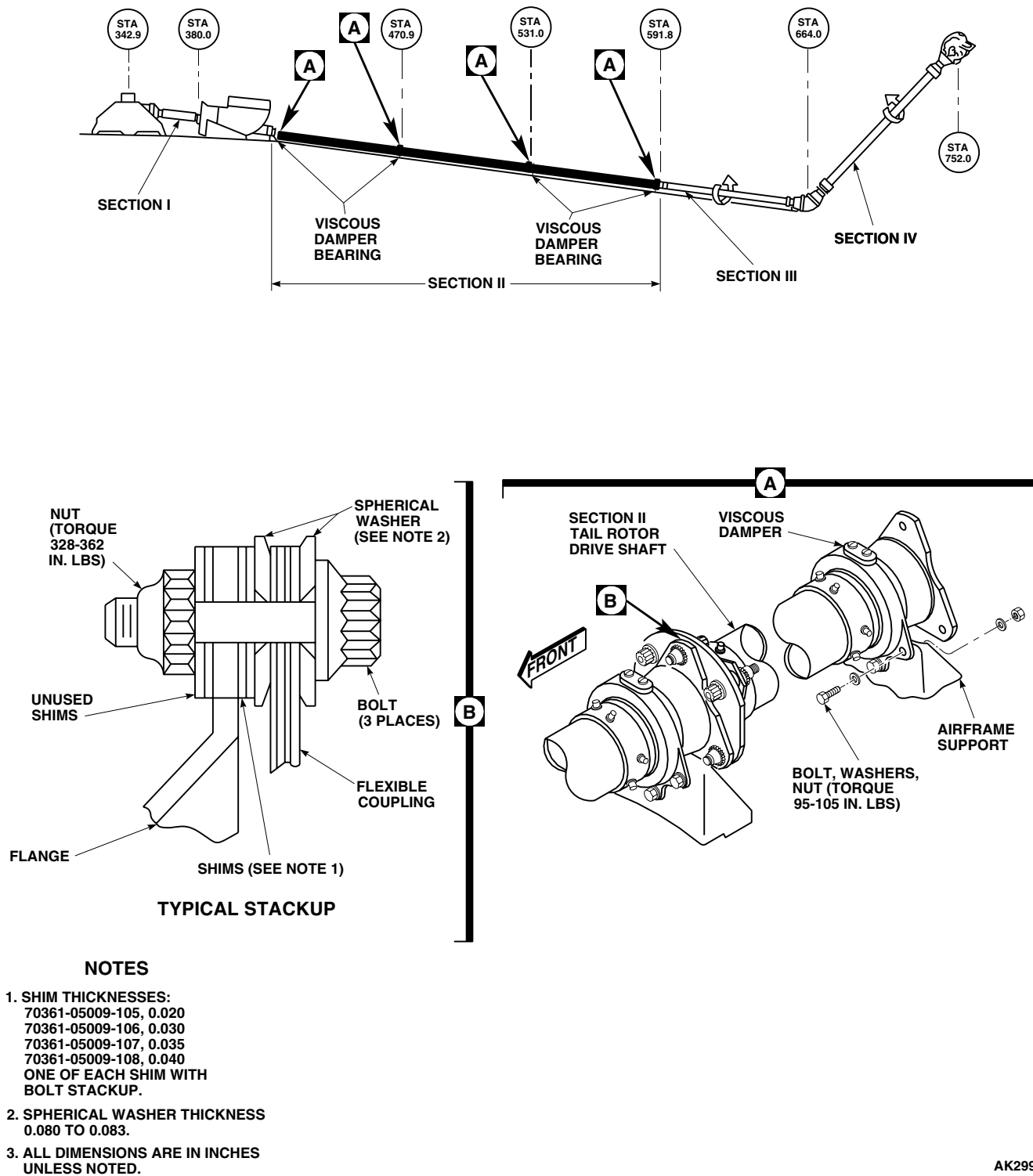


Figure 1. Replace Tail Rotor Drive Shaft Section II.

INSTALL

WARNING

Injury to personnel and damage to equipment will result if Section II drive shaft is bolted incorrectly to adjacent flange. Three adjacent flange bolts will be connected to flexible coupling only. Three drive shaft bolts will be connected to flexible coupling only. Make sure Section II drive shaft flange is not bolted directly through flexible coupling to adjacent flange.

CAUTION

Damage to drive shaft will result if handled improperly. Handle drive shafts carefully so they will not get dented, scratched, or bent. To prevent possible damage to drive shaft, place shaft on padded area.

NOTE

- When installing bolts in flexible couplings, make sure a spherical washer is installed between the bolthead and coupling. Make sure the spherical side of washers are against the flexible coupling.
- Do not lockwire bolts with drilled heads that are installed on flexible couplings.
- Front, middle, and rear Section II drive shafts are installed the same. For any shaft removed or disconnected, install both ends using the following procedures.

1. **QA** Remove flexible couplings from old shaft and install on new shaft (WP 0670 00).
2. Support front end of shaft. Attach viscous damper to airframe support with bolts, washers, and nuts (Figure 1, Detail A).
3. **QA** Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 3.5 INCH-POUNDS.** Replace nuts not within limits. **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
4. Center bearing in viscous damper support.

NOTE

- When installing spherical washers, make sure spherical side of washer is against flexible coupling.
 - When measuring gap for shims, do not turn shaft from original position, as gap may change. If shaft is moved, recheck gap.
5. Install bolts at front end of shaft with spherical washers on each side of flexible coupling. Measure gap at three attachment points, using a feeler gage.
 6. Determine average gap by adding readings and dividing by 3, Table 1.
 7. **QA** Select shim combination closest to average gap. Make three stackups. **MAXIMUM GAP IS 0.125-INCH.**

INSTALL - Continued**NOTE**

- Shim thickness must be same for all three bolt locations at each end of shaft. Shim combination at one end of shaft does not have to be the same as shim combination at other end of shaft as long as total shim combination adds up closest to average gap.
 - Drive shaft attaching nuts must be torqued within 15 minutes after applying primer to bolts.
8. Apply epoxy primer coating kit, Item 120, WP 1803 00, to bolt grip lengths and under bolt heads. Do not apply primer to threads.
 9. **QA** Install one stackup on each bolt. Insert spherical washers on each side of coupling. Make sure spherical side of washer faces coupling. Place unused shims under nuts (Figure 1, Detail B).

Table 1. EXAMPLE.

GAP	A = 0.062
	B = 0.071
	C = 0.068
Total	0.201
0.201 ÷ 3 = 0.067-inch average gap.	

10. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** Replace nuts not within limits.

WARNING**CSI**

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

11. **QA TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
12. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure a tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (WP 1693 00).

CAUTION

Damage to bearings will result if contact occurs with corrosion preventive compounds. Corrosion preventive compounds will penetrate bearing seals and dissolve lubrication grease. Do not spray corrosion preventive compounds in vicinity of drive shaft bearings. Shield bearings from spray.

13. Spray all flex coupling spherical washers with corrosion preventive compound, Type I, Item 106, WP 1803 00.

INSTALL - Continued**WARNING**

Injury to personnel and damage to equipment will result if spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and spline wear inspection is performed (WP 0626 00).

NOTE

Spline wear inspection is not required when tail rotor drive shaft is disconnected or replaced aft of STA 531.0.

14. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed, perform oil cooler spline wear inspection if required (WP 0626 00).

WARNING

Injury to personnel and damage to equipment will result if vibration check is not performed every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive vibration indicates improper installation of oil cooler assembly which will cause oil cooler axial fan drive shaft splines and fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed (WP 0680 00).

NOTE

Vibration check is not required when tail rotor drive shaft is disconnected or replaced aft of STA 531.0.

15. Perform oil cooler vibration check if required (WP 0680 00).
16. Make sure area is clean and free of foreign material.
17. Install APU panel and close APU access doors (STA 485.0 through STA 525.0) as required (WP 0303 00).
18. Close tail rotor drive shaft cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

TAIL ROTOR DRIVE SHAFT SECTION III

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Micrometer
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Corrosion Removing Compound, Item 108,

WP 1803 00

Epoxy Primer Coating Kit, Item 120, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cloth, Abrasive, Item 76, WP 1803 00
Corrosion Preventive Compound, Item 105,
WP 1803 00
Corrosion Preventive Compound, Item 106,
WP 1803 00

References

WP 0668 00

WP 0670 00

WP 1693 00

WP 1803 00

REMOVE**WARNING**

Injury to personnel and damage to equipment will result if tail rotor is not secured when drive shaft is disconnected. Secure tail rotor to prevent shafts and rotor from turning.

CAUTION

Damage to drive shaft will result if handled improperly. Handle drive shaft carefully so it will not get dented, scratched, or bent.

NOTE

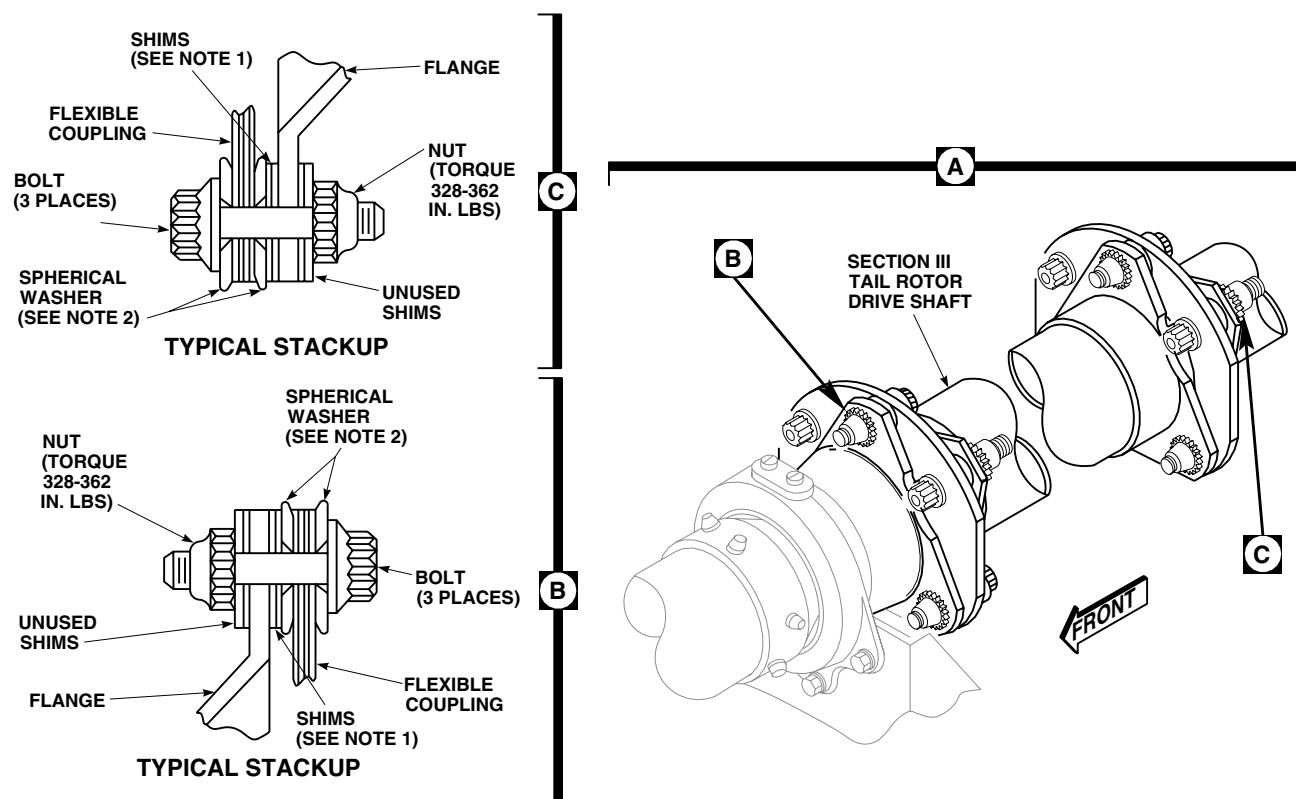
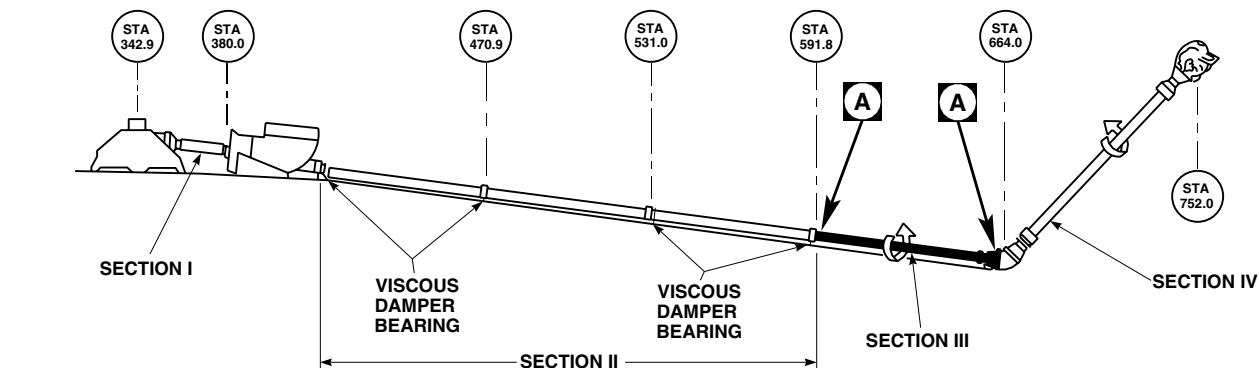
Drive shaft coupling bolts, self-locking nuts, and viscous damper housing attachment bolts and nuts can be reused provided they show no visual evidence of damage.

1. Open drive shaft cover over Section III tail rotor drive shaft.
2. Support rear end of Section III tail rotor drive shaft and remove bolts, spherical washers, shims, and nuts connecting flexible coupling on shaft to input flange on intermediate gear box (Figure 1, Detail A and Detail B).
3. Support front end of Section III tail rotor drive shaft and remove bolts, spherical washers, shims, and nuts connecting flexible coupling on shaft to aft Section II tail rotor drive shaft (Figure 1, Detail A and Detail B). Remove shaft with flexible couplings.

INSPECT

1. Inspect airframe supports for visible cracks and distortion. **NONE ALLOWED.**
2. Inspect tail rotor drive shaft flexible coupling (WP 0670 00).

INSPECT - Continued



NOTES

- SHIM THICKNESSES:
70361-05009-105, 0.020
70361-05009-106, 0.030
70361-05009-107, 0.035
70361-05009-108, 0.040
ONE OF EACH SHIM WITH BOLT STACKUP.
- SPHERICAL WASHER THICKNESS 0.080 TO 0.083.
- ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Replace Tail Rotor Drive Shaft Section III.

INSPECT - Continued**NOTE**

Corrosion preventive compound may turn reddish brown over time. This is normal and should not be mistaken for corrosion.

3. Visually inspect flexible coupling spherical washers for evidence of corrosion damage or surface roughness. **NONE ALLOWED.**
4. Inspect section III tail rotor drive shaft flange bolt holes for elongation. **NONE ALLOWED.** Replace tail rotor drive shaft if boltholes are elongated. Refer to, REMOVE, in this work package.
5. If washer, 70361-05010-103, is unavailable, washer, 70361-05010-102, may be reworked as follows:
 - a. Clean washer using corrosion removing compound, Item 108, WP 1803 00.
 - b. If corrosion is still present, remove corrosion using abrasive cloth, Item 76, WP 1803 00. Make sure all corrosion is removed.
 - c. **MINIMUM THICKNESS AFTER BLENDING IS 0.080-INCH THICK.** Replace washers if minimum thickness is under this limit.
 - d. Apply corrosion preventive compound, Item 105, WP 1803 00, to washer surfaces.
 - e. Apply corrosion preventive compound, Item 106, WP 1803 00, to hardware assembly and allow to dry.

INSTALL**WARNING**

Injury to personnel and damage to equipment will result if Section III drive shaft is bolted incorrectly to adjacent flange. Three adjacent flange bolts will be connected to flexible coupling only. Three drive shaft bolts will be connected to flexible coupling only. Make sure Section III drive shaft flange is not bolted directly through flexible coupling to adjacent flange.

CAUTION

Damage to drive shaft will result if handled improperly. Handle drive shafts carefully so they will not get dented, scratched, or bent. To prevent possible damage to drive shaft, place shaft on padded area.

NOTE

- When installing bolts in flexible couplings, make sure a spherical washer is installed between the bolthead and coupling. Make sure the spherical side of washers are against the flexible coupling.
 - Do not lockwire bolts with drilled heads that are installed on flexible couplings.
1. **QA** Remove flexible couplings from old shaft and install on new shaft (WP 0670 00).
 2. Support Section III tail rotor drive shaft in position (Figure 1).

NOTE

When installing spherical washers, make sure spherical side of washers are against flexible coupling.

3. Install bolts at forward end of shaft with spherical washers on each side of flexible coupling. Install nuts and tighten only enough to prevent axial movement (Figure 1, Detail B).

INSTALL - Continued**NOTE**

Do not turn shaft from original position, as gap may change. If shaft is moved, recheck gap.

4. Determine shim thickness at opposite end of shaft as follows:
 - a. Support shaft in installed position with 3 bolts and spherical washers on each side of flexible coupling. Do not install nuts at this time.
 - b. Measure gap at three attachment points, using a feeler gage.
 - c. Determine average gap by adding readings and dividing by 3.
 - d. Divide average gap by 2 to determine best shim combination measurement for both ends of shaft, refer to Table 1.

Table 1. EXAMPLE.

GAP	A = 0.125
	B = 0.142
	C = 0.137
Total	0.404
$0.404 \div 3 = 0.1346$ -inch average gap.	
Round off to 0.135-inch.	
$0.135 \div 2 = 0.068$ -inch gap at both ends of shaft.	

5. **QA** Select shim combination resulting from computation above. Make three stackup for each end of shaft. **MAXIMUM GAP IS 0.125-INCH AT EACH END OF SHAFT.**

NOTE

- Shim thickness must be the same for all three bolt locations at each end of shaft. Shim combination at one end of shaft does not have to be the same as shim combination at other end of shaft as long as total shim combination adds up closest to average gap.
 - Drive shaft attaching nuts must be torqued within 15 minutes after applying primer to bolts.
6. Apply epoxy primer, Item 120, WP 1803 00, to bolt grip lengths and under bolt heads. Do not apply primer to threads.
 7. **QA** Install one stackup on each bolt. Insert spherical washers on each side of coupling. Make sure spherical side of washer faces coupling (Figure 1, Detail B and Detail C). Place unused shims under nuts.
 8. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** Replace nuts not within limits.

INSTALL - Continued**WARNING****CSI**

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

9. QA TORQUE NUTS TO 328 - 362 INCH-POUNDS.

10. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure a tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (WP 1693 00).

CAUTION

Damage to bearings will result if contact occurs with corrosion preventive compounds. Corrosion preventive compounds will penetrate bearing seals and dissolve lubrication grease. Do not spray corrosion preventive compounds in vicinity of drive shaft bearings. Shield bearings from spray.

11. Spray all flexible coupling spherical washers with corrosion preventive compound, Item 106, WP 1803 00.
12. Make sure area is clean and free of foreign material.
13. Close tail rotor drive shaft cover.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****TAIL ROTOR DRIVE SHAFT SECTION IV****This WP supersedes WP 0669 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Micrometer
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Epoxy Primer Coating Kit, Item 120, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cloth, Abrasive, Item 76, WP 1803 00
Corrosion Preventive Compound, Item 105,
WP 1803 00
Corrosion Preventive Compound, Item 106,
WP 1803 00
Corrosion Removing Compound, Item 108,
WP 1803 00

References

WP 0333 00
WP 0335 00
WP 0669 00
WP 0670 00
WP 1693 00
WP 1803 00

REMOVE**WARNING**

Injury to personnel and damage to equipment will result if tail rotor is not secured when drive shaft is disconnected. Secure tail rotor to prevent shafts and rotor from turning.

CAUTION

Damage to drive shaft will result if handled improperly. Handle drive shaft carefully so it will not get dented, scratched, or bent.

NOTE

Drive shaft coupling bolts, self-locking nuts, and viscous damper housing attachment bolts and nuts can be reused provided they show no visual evidence of damage.

1. Turn off all electrical power.
2. Remove intermediate gear box fairing (WP 0335 00).
3. Remove tail gear box fairing (WP 0333 00).
4. Open section IV tail rotor drive shaft cover.
5. Remove bolts connecting flexible coupling on Section IV to output flange on intermediate gear box (Figure 1, Detail A).
6. Remove bolts connecting flexible coupling on Section IV to input flange on tail gear box (Figure 1, Detail B).

REMOVE - Continued

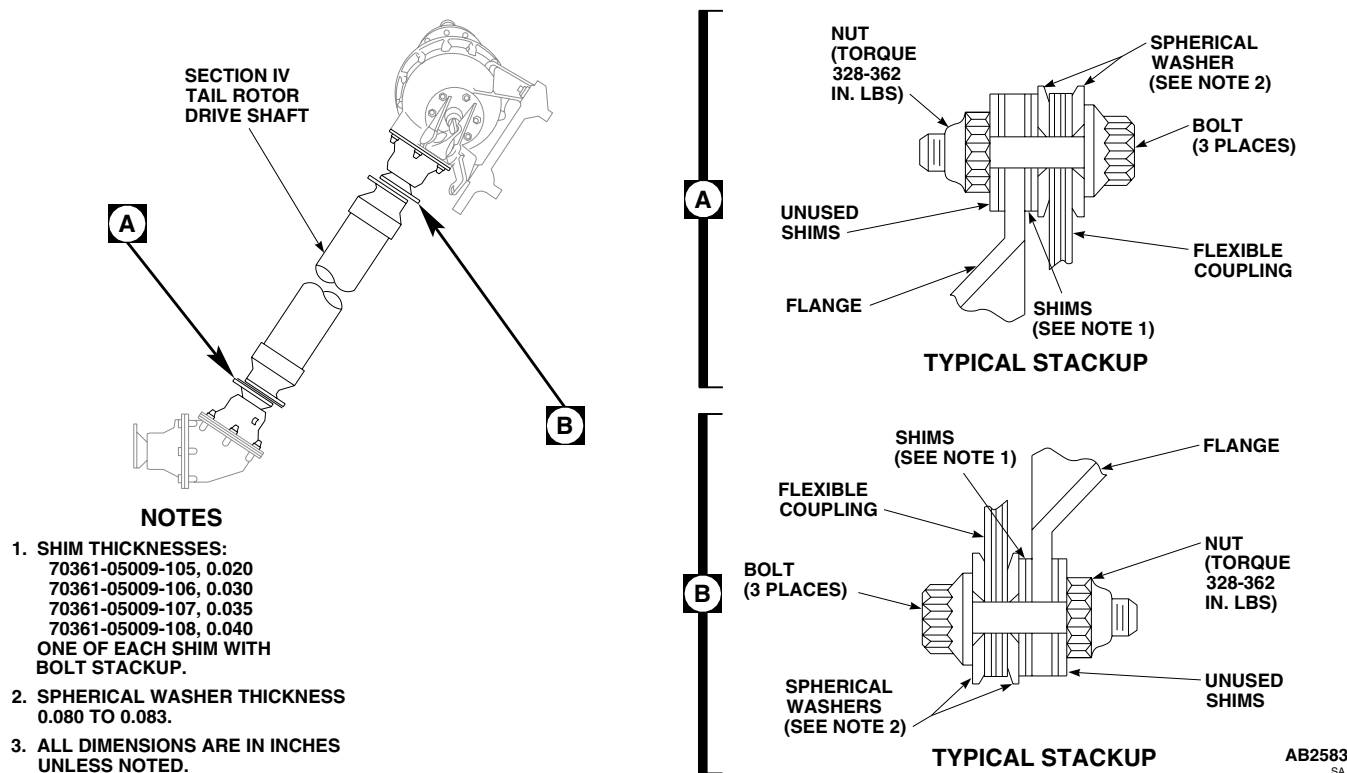


Figure 1. Replace Section IV Tail Rotor Drive Shaft.

7. Remove Section IV tail rotor drive shaft with flexible coupling from pylon.

INSPECT

1. Inspect airframe supports for visible cracks and distortion. **NONE ALLOWED.**
2. Inspect tail rotor drive shaft flexible coupling (WP 0670 00).

NOTE

Corrosion preventive compound may turn reddish brown over time. This is normal and should not be mistaken for corrosion.

3. Visually inspect flexible coupling spherical washers for evidence of corrosion damage or surface roughness. **NONE ALLOWED.**
4. Inspect section IV tail rotor drive shaft flange boltholes for elongation. **NONE ALLOWED.** Replace tail rotor drive shaft if boltholes are elongated (WP 0669 00).
5. If washer, 70361-05010-103, is unavailable, washer, 70361-05010-102, may be reworked as follows:
 - a. Clean washer using corrosion removing compound, Item 108, WP 1803 00.
 - b. If corrosion is still present, remove corrosion using abrasive cloth, Item 76, WP 1803 00. Make sure all corrosion is removed.
 - c. **MINIMUM THICKNESS AFTER BLENDING IS 0.080-INCH THICK.** Replace washers if minimum thickness is under this limit.

INSPECT - Continued

- d. Apply corrosion preventive compound, Item 105, WP 1803 00, to washer surfaces.
- e. Apply corrosion preventive compound, Item 106, WP 1803 00, to hardware assembly and allow to dry.

INSTALL

WARNING

Injury to personnel and damage to equipment will result if Section IV drive shaft is bolted incorrectly to adjacent flange. Three adjacent flange bolts will be connected to flexible coupling only. Three drive shaft bolts will be connected to flexible coupling only. Make sure Section IV drive shaft flange is not bolted directly through flexible coupling to adjacent flange.

CAUTION

Damage to drive shaft will result if handled improperly. Handle drive shafts carefully so they will not get dented, scratched, or bent. To prevent possible damage to drive shaft, place shaft on padded area.

NOTE

When installing bolts in flexible couplings, make sure a spherical washer is installed between the bolthead and coupling. Make sure the spherical side of washers are against the flexible coupling.

1. **QA** Remove flexible couplings from old shaft and install on new shaft (WP 0670 00).
2. Support Section IV tail rotor drive shaft in position (Figure 1).
3. Install bolts at intermediate gear box flange end of shaft, with spherical washers on each side of flexible coupling (Figure 1, Detail A).
4. Install nuts and tighten only enough to prevent axial movement.
5. Have assistant support shaft to keep flexible coupling from bending under weight of shaft.

NOTE

Do not turn shaft from original position, as gap may change. If shaft is moved, recheck gap.

6. Determine shim thickness at opposite end of shaft as follows:
 - a. Hold shaft in installed position. Install bolts at tail gear box end of shaft, with spherical washers on each side of flexible coupling, and boltheads against flexible coupling. Measure gap at three attachment points, and flange (Figure 1, Detail B).
 - b. Determine average gap by adding readings and dividing by 3.
 - c. Divide average gap by 2 to determine best shim combination measurement for both ends of shaft.

Table 1. EXAMPLE.

GAP	A = 0.125
	B = 0.142
	C = 0.137

INSTALL - Continued**Table 1. EXAMPLE. - Continued**

Total	0.404
$0.404 \div 3 = 0.1346$ -inch average gap.	
Round off to 0.135-inch.	
$0.135 \div 2 = 0.068$ -inch gap at both ends of shaft.	

- d. **QA** Select shim combination resulting from computation above. Make three stackup for each end of shaft. **MAXIMUM GAP IS 0.125 - INCH AT EACH END OF SHAFT.**

NOTE

- Shim thickness must be the same for all three bolt locations at each end of shaft. Shim combination at one end of shaft does not have to be the same as shim combination at other end of shaft as long as total shim combination adds up closest to average gap.
 - Drive shaft attaching nuts must be torqued within 15 minutes after applying primer to bolts.
7. Apply epoxy primer coating kit, Item 120, WP 1803 00, to bolt grip lengths and under bolt heads. Do not apply primer to threads.
 8. **QA** Install one stackup on each bolt. Insert spherical washers on each side of coupling. Make sure spherical side of washer faces coupling. Place unused shims under nut (Figure 1, Detail A and Detail B).
 9. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** Replace nuts not within limits.

WARNING**CSI**

Verification of minimum torque and torque stabilization check of nuts is a critical characteristic.

10. **QA TORQUE NUTS TO 328 - 362 INCH-POUNDS.**
11. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure a tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (WP 1693 00).
12. Spray flex coupling spherical washers with corrosion preventive compound, Item 106, WP 1803 00.
13. Make sure area is clean and free of foreign material.
14. Close tail rotor pylon drive shaft cover.
15. Install tail gear box fairing (WP 0333 00).
16. Install intermediate gear box fairing (WP 0335 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****TAIL ROTOR DRIVE SHAFT COUPLING**

This WP supersedes WP 0670 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Borescope Kit, 6650-T700-003
Depth Gage, Dial Indicating, 643J
Gloves
Goggles/Face Shield
Micrometer GGG-C-105
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Abrasive, Item 76, WP 1803 00

Cloth, Abrasive, Item 82, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-6625-724-13&P
WP 0626 00
WP 1693 00
WP 1803 00

REMOVE

1. Support end of drive shaft.

NOTE

- There is no longer a requirement that the order of discs alternate between those marked "A" and those marked "B". The coupling may contain any order of "A" and "B" discs. Maintain the original orientation and order of discs when disassembling and inspecting coupling.
 - If any one disc making up tail rotor drive shaft coupling stackup is damaged, entire flexible coupling must be replaced. Tail rotor drive shaft coupling discs are a matched set. Discs cannot be replaced individually. If any one disc is damaged beyond limits, the entire coupling must be replaced. Two or more discs between bolts on coupling flanges may separate or buckle. This is acceptable unless there is a sharp bend or crease in disc or in lamination spread (gap between 2 individual discs) is over 0.030-inch. To determine spread, measure gap between the disc laminations with a feeler gage.
2. Remove bolts, spherical washers, and nuts securing tail rotor drive shaft coupling to drive shaft. Remove coupling (Figure 1).

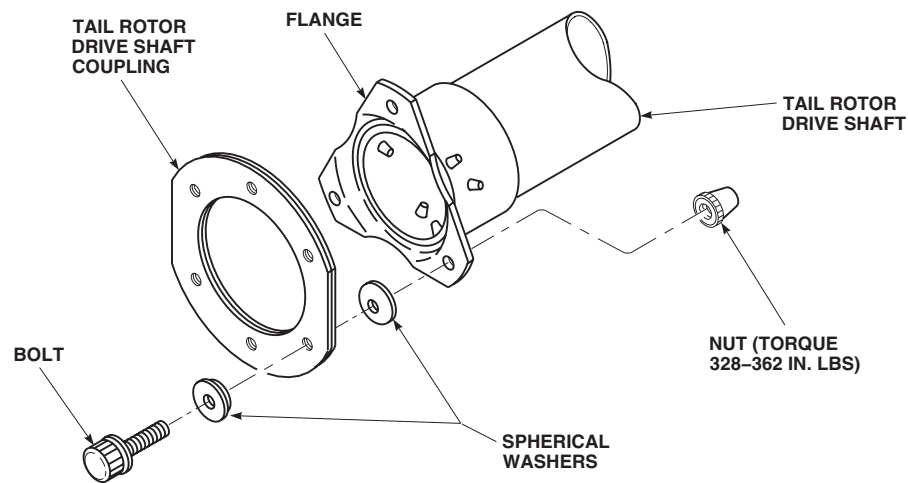
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Figure 1. Replace Tail Rotor Drive Shaft Coupling.

INSPECT**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Damage to bearings will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids will penetrate bearing seals and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of drive shaft bearings. Apply solvents only by wiping.

NOTE

- No more than half the number of discs may be blended in the same assembled area.
- Fretting of coupling discs occurs in area under spherical washer.
- Red oxide stains on coupling discs are acceptable.
- When inspecting tail rotor drive shaft couplings, pay particular attention to edges around the circumference of the spherical washers.
- If tail rotor drive shaft coupling is damaged, drive shaft may also be damaged. Before installing new coupling, visually inspect drive shaft.

1. Clean the coupling using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
2. Inspect tail rotor drive shaft couplings for cracks, using borescope kit, or magnifying glass with adequate lighting.
NO CRACKS ALLOWED. Replace coupling if cracks are found.

INSPECT - Continued

3. Inspect tail rotor drive shaft coupling for corrosion pitting using dial indicating depth gauge. **CORROSION PITTING SHALL NOT EXCEED 0.0005 - INCH DEEP.**
4. Blend out corrosion pitting that are no deeper than 0.0005-inch deep using abrasive cloth, Item 82, WP 1803 00. Replace tail rotor drive shaft coupling if limit is exceeded.
5. Inspect for dents and nicks on edges of individual disks. **NICKS AND DENTS SHALL NOT EXCEED 0.020-INCH DEEP.**
6. Blend out nicks or dents that are no deeper than 0.020-inch with abrasive cloth, Item 76, WP 1803 00. Smooth area with abrasive cloth, Item 82, WP 1803 00. Replace tail rotor drive shaft coupling if limit is exceeded.
7. Inspect for fretting using dial indicating depth gauge. **FRETTING SHALL NOT EXCEED 0.0005 - INCH DEEP.**
8. Remove fretting up to 0.0005-inch deep using abrasive cloth, Item 82, WP 1803 00. Replace tail rotor drive shaft coupling if limit is exceeded.
9. Wipe away any grease or dirt from tail rotor drive shaft coupling.
10. Inspect coupling for buckling. **NO GAP LARGER THAN 0.030 - INCH PERMITTED BETWEEN LAMINATES.**
11. **IF GAP BETWEEN ANY TWO LAMINATES IS GREATER THAN 0.030 - INCH, BREAK TORQUE ON SIX NUTS, THEN RETORQUE NUTS IN 50 INCH-POUND INCREMENTS, IN SEQUENCE, UNTIL EACH NUT IS TORQUED TO 328 - 362 INCH-POUNDS. IF GAP STILL EXCEEDS 0.030 - INCH, REPLACE COUPLING.**

INSTALL

1. **QA** Check disc stackup with micrometer before installing tail rotor drive shaft coupling. **DISC STACKUP MUST MEASURE 0.140 to 0.151-INCH THICK. TAIL ROTOR DRIVE SHAFT COUPLING MUST CONTAIN 12 ± 2 DISCS.** Replace coupling if not within limits.

NOTE

Tail rotor drive shaft coupling nuts must be torqued within 15 minutes after applying primer to bolts.

2. Apply epoxy primer coating kit, Item 120, WP 1803 00, to bolt grip lengths and under bolt heads. Do not apply primer to threads.
3. Install tail rotor drive shaft coupling on drive shaft with bolts, spherical washers, and nuts. Spherical side of washers must be against tail rotor drive shaft coupling (Figure 1).
4. Using torque wrench, check drag torque on nuts. **MINIMUM DRAG TORQUE IS 9.5 INCH-POUNDS.** Replace nuts not within limits.

WARNING**CSI**

Verification of minimum torque of nuts is a critical characteristic.

5. **QA TORQUE NUT TO 328 - 362 INCH-POUNDS.**
6. **QA** Connect tail rotor drive shaft.

INSTALL - Continued**WARNING**

- Injury to personnel and damage to equipment will result if vibration check is not performed every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive vibration indicates improper installation of oil cooler assembly which will cause oil cooler axial fan drive shaft splines and fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed in accordance with TM 1-6625-724-13&P.
- Injury to personnel and damage to equipment will result if spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and spline wear inspection is performed in accordance with WP 0626 00.

7. Perform vibration check if tail rotor drive shaft was disconnected or replaced from transmission to STA 531.0 (TM 1-6625-724-13&P).
8. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure tail rotor drive shaft torque check is shown as due 9 to 11 flight hours after installation (WP 1693 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

TAIL ROTOR DRIVE SHAFT SUPPORT ASSEMBLY

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13
Puller Kit, Mechanical, GGG-P-781
Torque Reactor
Torque Wrench, 0 - 600 ft. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0626 00
WP 0666 00
WP 0667 00
WP 0668 00
WP 0669 00
WP 0674 00
WP 0680 00
WP 1803 00

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

REMOVE**CAUTION**

Damage to tail rotor drive shaft will result if handled improperly. Handle drive shafts carefully so they will not get dented, scratched or bent. To prevent possible damage to drive shaft, place shaft on padded area.

1. Remove tail rotor drive shafts as required (WP 0666 00, WP 0667 00, WP 0668 00, or WP 0669 00).
2. Remove nut from drive shaft using wrench and torque reactor (Figure 1).

NOTE

Make sure index marks are present on drive shaft and flange. If no index marks are present, apply index marks to drive shaft and flange using permanent ink marker.

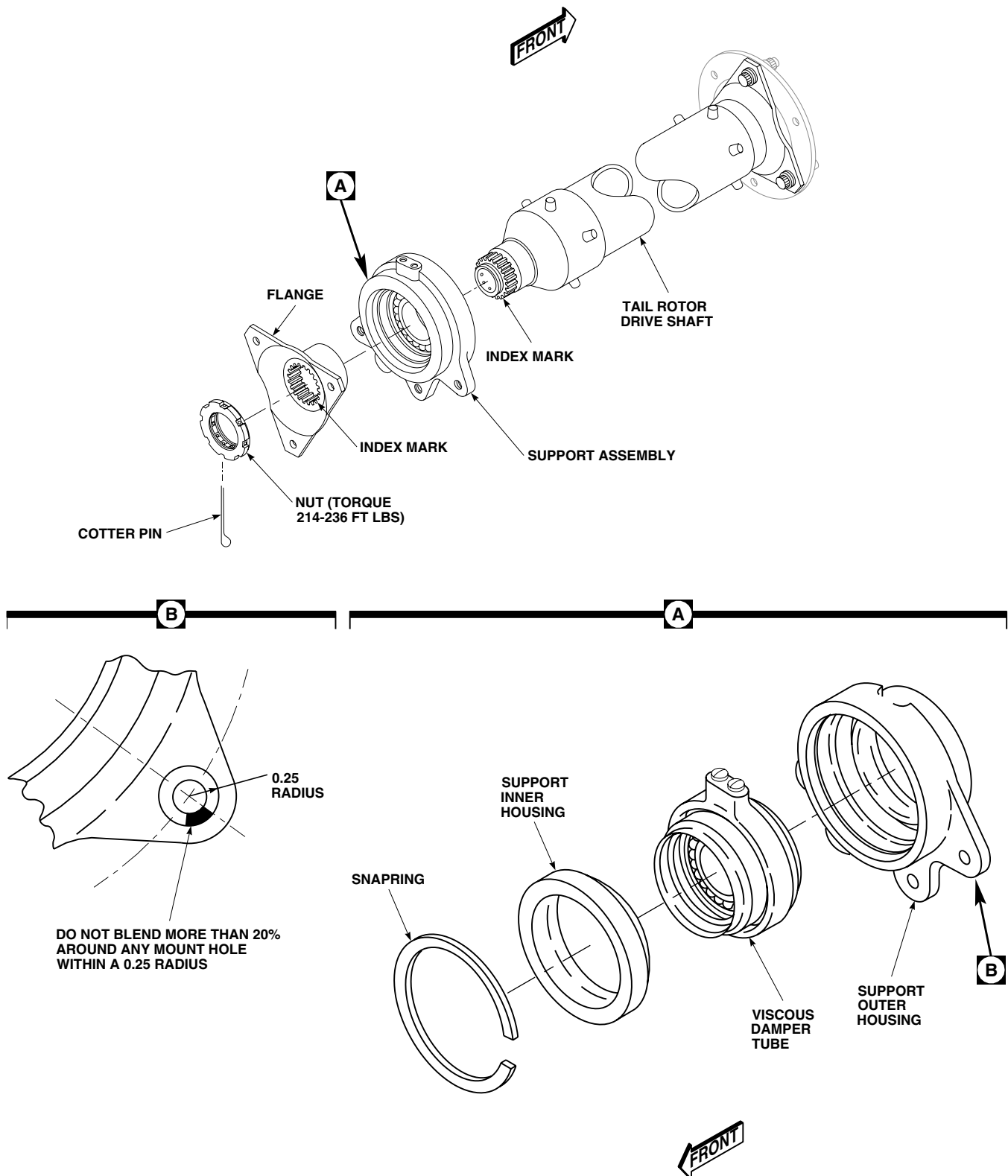
3. Remove flange from shaft by pulling straight off.
4. Remove support assembly from shaft using puller.

DISASSEMBLE**NOTE**

Viscous damper bearing tube assembly allows movement of bearing liner relative to support outer housing. However, movement of viscous damper tube within support outer housing is not allowed. Radial and axial movement of bearing inside liner is allowed to accommodate shaft deflections during flight.

1. Remove sealing compound from around edges of viscous damper tube filler neck and from around mating surfaces of inner housing, outer housing, and snapping using nonmetallic scraper.

DISASSEMBLE - Continued

**NOTE**

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

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Figure 1. Replace Tail Rotor Drive Shaft Support Assembly.

DISASSEMBLE - Continued

2. Remove snapping from support inner housing (Figure 1, Detail A).
3. Manually press out support inner housing.
4. Remove viscous damper tube assembly.

INSPECT/REPAIR

1. Inspect tail rotor drive shaft for damage (WP 0674 00).
2. Inspect flange nut for corrosion.
 - a. Minor corrosion may be blended out using abrasive cloth, Item 82, WP 1803 00, or abrasive mat, Item 2, WP 1803 00 (grits 320 or finer).
 - b. **REPLACE NUT IF PITS ARE 0.030-INCH DEEP, IF PITTING COVERS MORE THAN 10% OF MATING SURFACE OR IF NUT THREADS SHOW SIGNS OF CORROSION.**

NOTE

Do not paint threads or mating surface of flange nut.

- c. Apply epoxy primer coating kit, Item 120, WP 1803 00, to flats and exposed face of flange nut.
3. Inspect support assembly outer housing for damage or corrosion.

NOTE

All blended areas to be smooth to adjacent surfaces. No sharp edges allowed.

- a. Damage or corrosion on mounting lugs may be blended out up to 0.030-inch deep over not more than 20% of any one lug, and not more than 20% around any hole with abrasive cloth, Item 82, WP 1803 00 (Figure 1, Detail B).
 - b. Damage or corrosion on outer housing may be blended out up to 0.030-inch deep over not more than 20% of total surface area with abrasive cloth, Item 82, WP 1803 00.
 - c. Replace outer housing if damage or corrosion is beyond limits.
4. Inspect support assembly inner housing for damage or corrosion.

NOTE

All blended areas to be smooth to adjacent surfaces. No sharp edges allowed.

- a. Damage or corrosion on inner housing may be blended out up to 0.030-inch deep over not more than 20% of total surface area with abrasive cloth, Item 82, WP 1803 00.
 - b. Replace inner housing if damage or corrosion is beyond limits.

ASSEMBLE

1. **QA** Install viscous damper tube in support outer housing. Make sure tube is seated.
2. **QA** Install support inner housing with snapping.
3. Apply 1/16 to 1/8-inch bead of sealing compound, Item 273, WP 1803 00, around edges of viscous damper tube filler neck and around mating surfaces of inner housing, outer housing, and snapping.

INSTALL

1. **QA** Press support assembly on drive shaft. Make sure support is installed in direction shown (Figure 1).

INSTALL - Continued**NOTE**

Make sure rear section of support assembly is against tapered part of drive shaft.

2. **QA** Install flange on drive shaft. Make sure flange and shaft index marks line up.
3. **QA** Make sure bevel side of nut is toward seal and that nut can be fully run on by hand. Using torque reactor and wrench, **TORQUE NUT AS FOLLOWS: THE TORQUE RANGE IS 214 - 236 FOOT-POUNDS. TORQUE RETAINING NUT TO 214 FOOT-POUNDS.** Install new cotter pin. To align pin hole, turn nut in tightening direction. **DO NOT EXCEED MAX TORQUE. IF MAX TORQUE IS EXCEEDED AND NUT DOES NOT ALIGN, REMOVE NUT AND REPLACE IT WITH A NEW NUT AND REPEAT TORQUING PROCEDURE.**

WARNING

Injury to personnel and damage to equipment will result if vibration check is not performed every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive vibration indicates improper installation of oil cooler assembly which will cause oil cooler axial fan drive shaft splines and fan exit flange splines to wear, resulting in loss of tail rotor drive.

4. Perform vibration check if tail rotor drive shaft was disconnected or replaced from transmission to STA 531.0 (WP 0680 00).

WARNING

Injury to personnel and damage to equipment will result if spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed.

5. Perform spline wear inspection on oil cooler fans, 70361-03005-103 through 70361-03005-106, if tail rotor drive shaft was disconnected or replaced from transmission to STA 531.0 (WP 0626 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
TAIL ROTOR DRIVE SHAFT SUPPORT BEARING

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Drift
Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

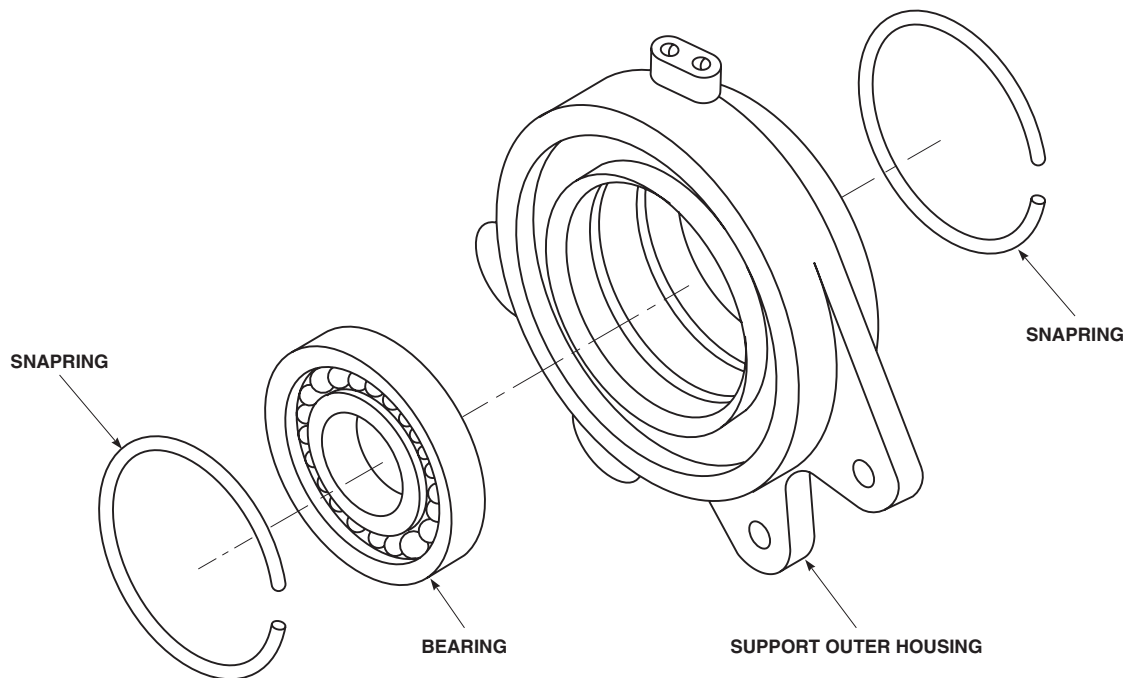
Cloth, Abrasive, Item 82, WP 1803 00
Corrosion Preventive Compound, Item 105,
WP 1803 00

References

WP 0671 00
WP 1803 00

REMOVE

1. Remove support assembly (WP 0671 00).
2. Remove snaprings (Figure 1).
3. Manually press out bearing. If necessary, tap out gently with a nonmetallic drift.



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Figure 1. Replace Tail Rotor Drive Shaft Support Bearing.

INSPECT

1. Inspect bearing liner on viscous damper tube for scratches with raised edges and burrs. **MAXIMUM DEPTH OF SCRATCHES IS 0.015-INCH.**
2. Blend out minor scratches using abrasive cloth, Item 82, WP 1803 00.

INSTALL**WARNING****CSI**

Presence of lubricant is a critical characteristic

CAUTION

Damage to bearings will result if contact occurs with corrosion preventive compound. Corrosion preventive compound can dissolve lubrication grease in bearing. Do not allow corrosion preventive compound to contact ball bearings. Apply corrosion preventive compound to bearing outer race only.

1. Apply light coat of corrosion preventive compound, Item 105, WP 1803 00, to bearing outer race.
2. **QA** Manually press bearing into support outer housing. If necessary, gently tap in place with a nonmetallic drift (Figure 1).

NOTE

Viscous damper bearing tube assembly allows movement of bearing liner relative to support outer housing. However, movement of viscous damper tube within support outer housing is not allowed. Radial and axial movement of bearing inside liner is allowed to accommodate shaft deflections during flight.

3. **QA** Install snaprings.
4. Install support assembly on tail rotor drive shaft (WP 0671 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****VISCOUS DAMPER TUBE****This WP supersedes WP 0673 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Drift
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0625 00
WP 0671 00
WP 1611 00
WP 1803 00

Material/Parts

Alcohol, Denatured, Item 48, WP 1803 00
Corrosion Preventive Compound, Item 105,
WP 1803 00
Sealing Compound, Item 273, WP 1803 00

REMOVE**NOTE**

For damper leakage limitations, refer to WP 0625 00.

1. Remove support assembly (WP 0671 00).
2. Remove sealing compound from around edges of viscous damper tube filler neck and from around mating surfaces of inner housing, outer housing, and snapping using nonmetallic scraper.
3. Remove snapping from support inner housing (Figure 1).
4. Manually press out support inner housing.
5. Remove viscous damper tube.
6. Remove snaprings holding bearing in place.
7. Manually press out bearing. If necessary, gently tap out bearing using a nonmetallic drift.

CLEAN

1. Remove viscous damper tube from packing and clean as follows:
 - a. Flush with hot water.
 - b. Rinse with denatured alcohol, Item 48, WP 1803 00.
2. Visually inspect viscous damper tube for presence of glue spots, mold, or other discoloration that mimic the appearance of an air bubble or obscure view of air bubble. If present and cannot be rectified by cleaning, replace viscous damper tube.

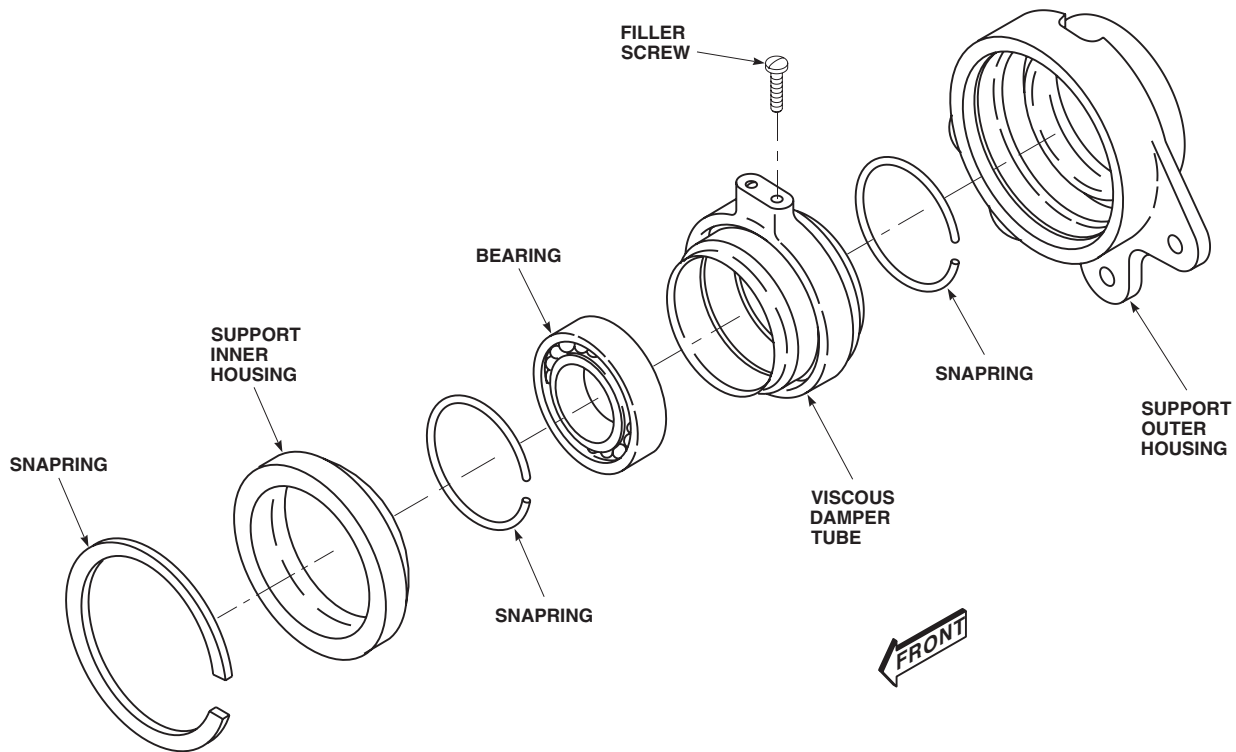
CLEAN - ContinuedAK2061
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Figure 1. Replace Viscous Damper Tube.

INSTALL**NOTE**

Viscous damper bearing tube assembly allows movement of bearing liner relative to support outer housing. However, movement of viscous damper tube within support outer housing is not allowed. Radial and axial movement of bearing inside liner is allowed to accommodate shaft deflections during flight.

1. **QA** Install one snapring in viscous damper tube (Figure 1).

WARNING**CSI**

Presence of lubricant is a critical characteristic

CAUTION

Damage to bearings will result if contact occurs with corrosion preventive compound. Corrosion preventive compound can dissolve lubrication grease in bearing. Do not allow corrosion preventive compound to contact ball bearings. Apply corrosion preventive compound to outer race only.

2. Apply light coat of corrosion preventive compound, Item 105, WP 1803 00, to bearing outer race.
3. **QA** Manually press in bearing. If necessary, tap gently with a nonmetallic drift.

INSTALL - Continued

4. **QA** Install other snapring.
5. **QA** Install viscous damper tube on support outer housing. Make sure tube is seated.
6. **QA** Install support inner housing with beveled edge towards tube.
7. **QA** Install snapring.
8. Apply 1/16 to 1/8-inch bead of sealing compound, Item 273, WP 1803 00, around edges of viscous damper tube filler neck and around mating surfaces of inner housing, outer housing, and snapring.
9. Service viscous damper (WP 1611 00).
10. Install support assembly on tail rotor drive shaft (WP 0671 00).
11. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1, to make sure viscous damper is inspected for leakage after the next flight. If air bubbles are more than 1 inch long replace damper tube.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
TAIL ROTOR DRIVE SHAFT REPAIR

INITIAL SETUP:
Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Caliper Set, GGG-C-105
 Depth Gage, Dial Indicating, 643J
 Powertrain Repairer's Toolkit, SC 5180-99-B13

References

TM 1-1500-345-23
 WP 0381 00
 WP 0666 00
 WP 0667 00
 WP 0668 00
 WP 0669 00
 WP 1803 00

Material/Parts

Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Abrasive, Item 78, WP 1803 00
 Primer Coating, Item 243, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

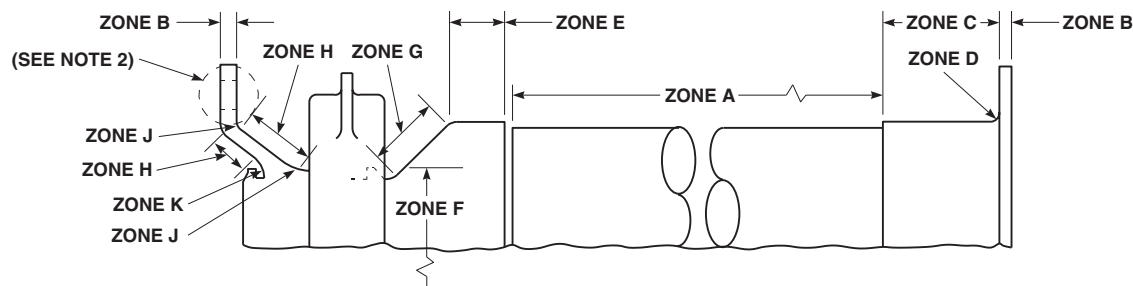
REPAIR
NOTE

- Before starting repair of shaft, determine whether polishing out damage would result in a diameter below reparable limits. If damage repair is greater than permitted limits, replace shaft/flange.
- Rework of drive shafts may be done on helicopter.
- No balance checks are required after reworking shaft in accordance with this task.

1. Remove damaged section of tail rotor drive shaft if required (WP 0666 00, WP 0667 00, WP 0668 00, or WP 0669 00).
2. Fine scratches which remove only paint may be blended smooth using abrasive cloth, Item 82, WP 1803 00. After blending, touch up with matching paint (WP 0381 00 and TM 1-1520-345-23).
3. Check present dimension of damaged area using dial indicating depth gage and calipers. Compare reading with minimum repair limits (Figure 1, Sheet 1 and Figure 1, Sheet 2).
4. Check depth of damaged area with depth gage to determine whether drive shaft minimum dimension after repair can be maintained.
5. If damage can be repaired, blend smooth using abrasive cloth, Item 78, WP 1803 00. Remove only what is necessary so shaft will stay within limits.
6. Light corrosion on rivets or on visible part of shaft ID may be removed using abrasive cloth, Item 78, WP 1803 00. Remove only what is necessary so shaft will stay within limits.
7. Blend reworked area to remove all traces of abrasive marks and match surface smoothness using abrasive cloth, Item 82, WP 1803 00, or appropriate polishing cloth.
8. Clean polished area. Apply primer coating, Item 243, WP 1803 00, to area. Touch up area with matching paint

REPAIR - Continued

TYPICAL ILLUSTRATION FOR DRIVE SHAFT SECTIONS I, II, AND III



COMPONENT	ZONE	DIMENSION BEFORE REPAIR	REPAIR LIMIT
TUBE OD	A	3.500 (+0.005 TO 0.000) DIA	3.485
TUBE ID	A	3.312 (+0.010) DIA	3.322
FLANGE LUG	B	0.175 (+0.005)	0.160
OD	C	3.690 (+0.010) DIA	3.670 DIA
RADIUS	D	0.312 (+0.010) RADIUS	(SEE NOTE 3)
STUB SHAFT OD	E	3.690 (+0.010) DIA	3.670 DIA
OD	F	1.9686 (+0.007 TO 0.000) DIA	1.9676 MINIMUM DIA
	G	0.125 (+0.010) (WEB THICKNESS)	BLEND TO 0.250 RADIUS
COUPLING FLANGE			
CONE	H	0.105 (+0.005)	0.010 MAXIMUM DEPTH
RADII	J	0.125 (+0.010) RADIUS	BLEND TO 0.250 RADIUS
			(SEE NOTE 3)
RADII	K	0.190 RADIUS	0.010 MAXIMUM DEPTH
		0.45	BLEND TO 0.250 RADIUS
			(SEE NOTE 3)
OUTER SURFACE OF FLANGE EARS (BOTH ENDS)			0.010 MAXIMUM DEPTH
			BLEND TO 0.250 RADIUS
			(SEE NOTE 3)

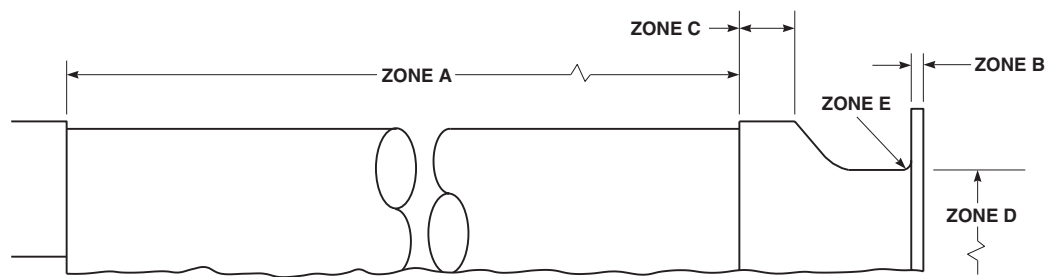
NOTES

- REPAIR LIMITATIONS
 - REPAIR OF MINOR DAMAGE SUCH AS LIMITED NICKS, SCRATCHES, OR SCORING SHALL BE CLEANED UP BY POLISHING.
 - POLISHING SHALL BE DONE ONLY IN THE DAMAGED AREA AND NOT THE ENTIRE CIRCUMFERENCE UNLESS THE DAMAGE EXTENDS AROUND IT.
 - THE BLEND RADIUS ON SHAFT OD SHALL BE 0.500 OR GREATER.
 - FINE SCRATCHES NEED NOT BE POLISHED IF NOT THROUGH PAINT.
 - AREAS WHERE PAINT HAS BEEN REMOVED ARE TO BE POLISHED WITH CROCUS CLOTH AND REPAINTED.
- DO NOT PAINT AROUND BOLT HOLES.
- BLEND RADIUS SHALL NOT BE LESS THAN 0.010 BELOW THE MINIMUM TOLERANCES SHOWN IN "DIMENSION BEFORE REPAIR" COLUMN.
- ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

Figure 1. Repair Tail Rotor Drive Shaft. (Sheet 1 of 2)

REPAIR - Continued

ILLUSTRATION FOR PYLON DRIVE SHAFT SECTION IV



COMPONENT	ZONE	DIMENSION BEFORE REPAIR	REPAIR LIMIT
TUBE OD	A	5.250 (+0.005 TO 0.000) DIA	5.235 DIA
TUBE ID	A	5.042 (+0.010) DIA	5.052 DIA
FLANGE LUG OD OD RADIUS	B C D E	0.175 (+0.005) 5.450 (+0.010) DIA 3.500 (+0.010) DIA 0.250 (+0.010) RADIUS	0.160 5.430 DIA 3.480 DIA (SEE NOTE 3)

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Figure 1. Repair Tail Rotor Drive Shaft. (Sheet 2 of 2)

(WP 0381 00 and TM 1-1520-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
OIL COOLER RADIATOR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01	WP 0304 00
Container, 1-Quart	WP 0308 00
Powertrain Repairer's Toolkit, SC 5180-99-B13	WP 0309 00
Torque Wrench, 30 - 200 in. lbs.	WP 0310 00
	WP 0381 00

Material/Parts

Cloth, Abrasive, Item 82, WP 1803 00	WP 0617 00
Protective Caps and Plugs	WP 0676 00
	WP 0677 00
	WP 0700 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)	WP 1607 00
UH-60 Helicopter Repairer MOS 15T (1)	WP 1803 00

References

TM 11-1520-237-23
 TM 55-1500-345-23

REMOVE

1. Turn off all electrical power.
2. Turn main rotor blades as required to gain access to oil cooler radiator.
3. Engage gust lock.
4. Remove oil cooler access doors (WP 0308 00).
5. Remove APU screens (WP 0304 00).
6. Remove IR jammer fairing (WP 0309 00).
7. Remove IR jammer (TM 11-1520-237-23).
8. Remove IR jammer panel (WP 0310 00).
9. Remove pneumatic starter line (WP 0700 00).
10. Place 1 quart container underneath oil hoses and radiator to catch any spilled oil.
11. Disconnect inlet and outlet oil hoses from radiator. Cap hoses and fittings (WP 0677 00).

NOTE

Radiator weighs about 31 pounds.

12. Remove screws and washers securing front support beam to airframe. Remove front support beam from airframe. If necessary, remove outlet fitting on radiator to remove beam, then reinstall fitting (Figure 1).
13. Remove thermostatic valve (WP 0676 00).

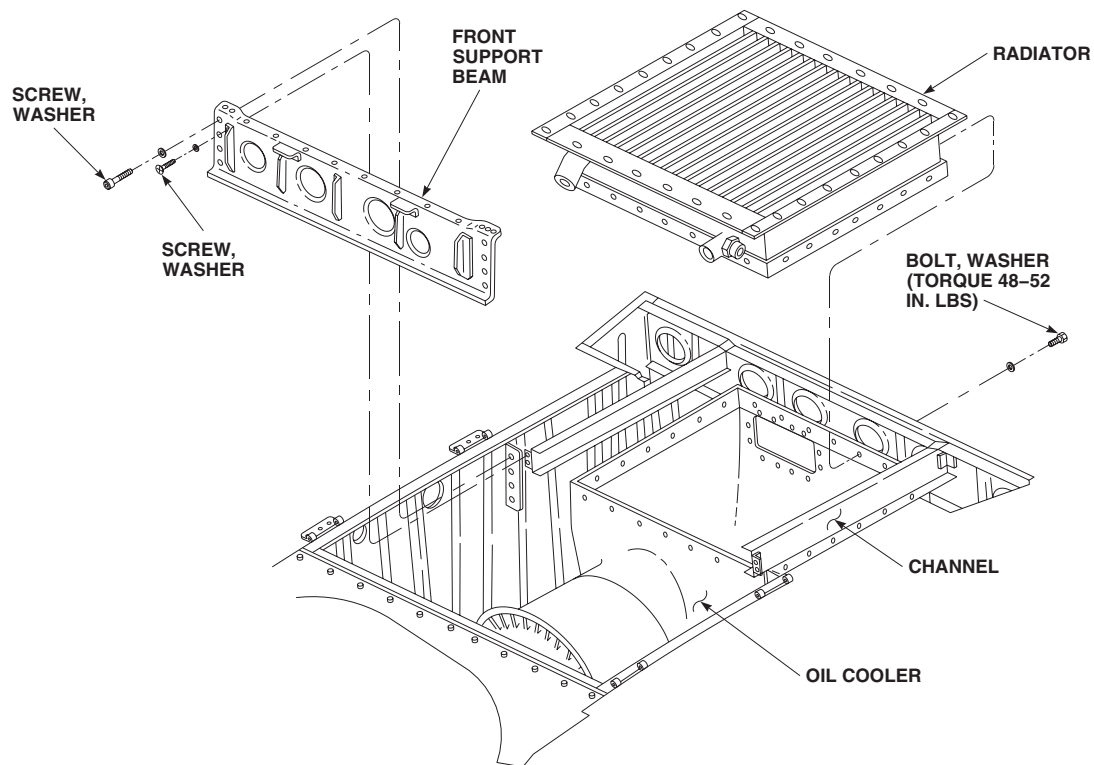
REMOVE - ContinuedAK2063
SA

Figure 1. Replace Oil Cooler Radiator.

14. Remove bolts and washers securing radiator to oil cooler. Remove radiator from oil cooler.
15. Drain any remaining oil from radiator.
16. If radiator was removed due to clogging, contamination, or blockage, do this:
 - a. Flush inlet and outlet oil lines.
 - b. Inspect main transmission chip detectors (all) and main module oil strainer (WP 0617 00).
 - c. Send radiator to depot for cleaning or repair.

INSPECT

1. Nicks and gouges up to 0.100-inch deep (measured from edge of mounting flange) are to be blended smooth with a blend radius of not less than 1/8-inch. An edge distance of not less than 0.120-inch must be maintained at all attachment holes.
2. Fine scratches that remove only paint may be blended smooth by using abrasive cloth, Item 82, WP 1803 00. Touch up with matching paint (WP 0381 00 and TM 55-1500-345-23).
3. Fine scratches not through paint need not be polished.

INSPECT - Continued**CAUTION**

Damage to radiator core will result if attempts are made to straighten core fins. Do not straighten core fins. Core fins may break and cause leakage.

4. Minor damage to cooler core fins is acceptable, provided there is no leakage or transmission cooling problems reported. **MINOR DAMAGE IS DAMAGE THAT DOES NOT EXCEED THE FIRST TIER IN DEPTH OR A MAXIMUM OF 2% OF EACH SIDE SURFACE AREA.**

INSTALL

1. **QA** Place radiator on oil cooler. Install bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS (Figure 1).**
2. **QA** Install front support beam to airframe using screws and washers. If necessary, remove outlet fitting to install brace, then reinstall fitting.
3. Install thermostatic valve (WP 0676 00).
4. Install packing on return fitting (WP 0677 00).
5. Install packing on outlet fitting (WP 0677 00).
6. **QA** Connect inlet and outlet oil hoses to fittings on radiator (WP 0677 00).
7. Install pneumatic starter line (WP 0700 00).
8. Make sure area is clean and free of foreign material.
9. Install IR jammer panel (WP 0310 00).
10. Install IR jammer (TM 11-1520-237-23).
11. Install IR jammer fairing (WP 0309 00).
12. Install APU screens (WP 0304 00).
13. Install oil cooler compartment access doors (WP 0308 00).
14. Close APU access doors.
15. Service main transmission (WP 1607 00).
16. Do an operational check as follows:
 - a. With qualified person at controls, run up helicopter.
 - b. Bring main transmission up to operating temperature and run about 10 minutes.
 - c. Temperature should remain constant.
 - d. Shut down. Check oil cooler and radiator for leaks.
 - e. Recheck oil level.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
OIL COOLER RADIATOR THERMOSTATIC VALVE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 1-Quart
Pail
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0308 00
WP 1803 00

Material/Parts

Wire, Nonelectrical, Item 357, WP 1803 00

REMOVE

1. Turn main rotor blades as required to access oil cooler fan and radiator.
2. Engage gust lock.
3. Remove oil cooler access doors (WP 0308 00).
4. Remove forward screws holding IR fairing and APU screens panel to support brace (Figure 1).
5. Remove screws holding support brace to main rotor pylon.
6. Slide support brace forward over oil cooler inlet and outlet oil lines.
7. Place 1 quart container under thermostatic valve to catch any leaking oil.
8. Remove thermostatic valve (Figure 1, Detail A).

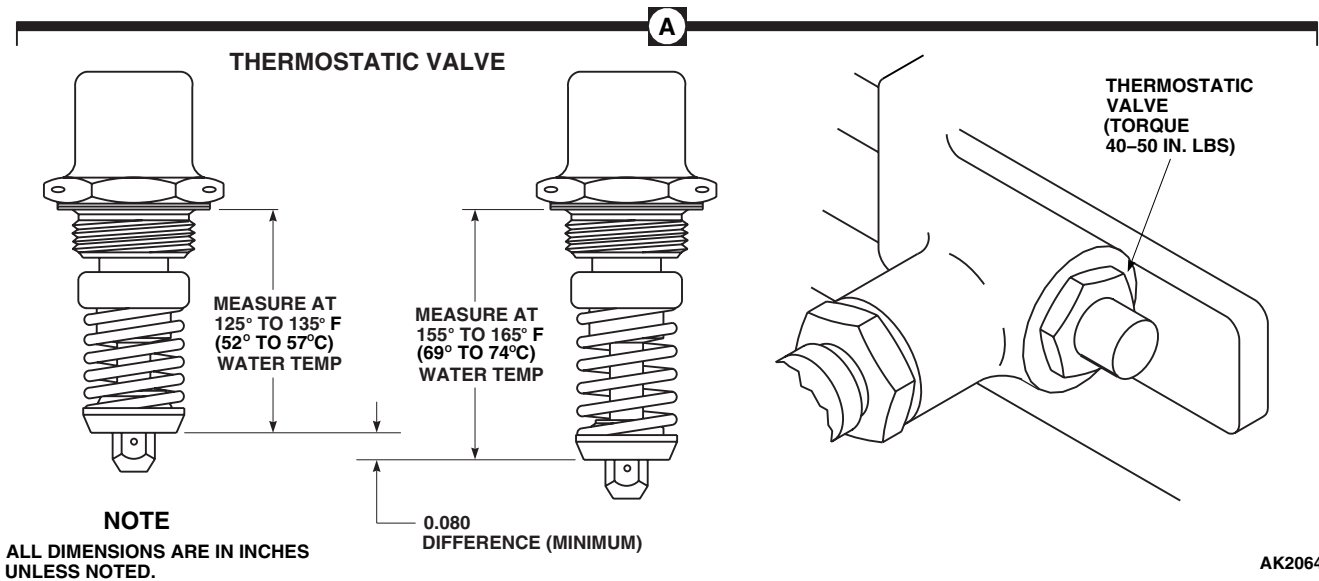
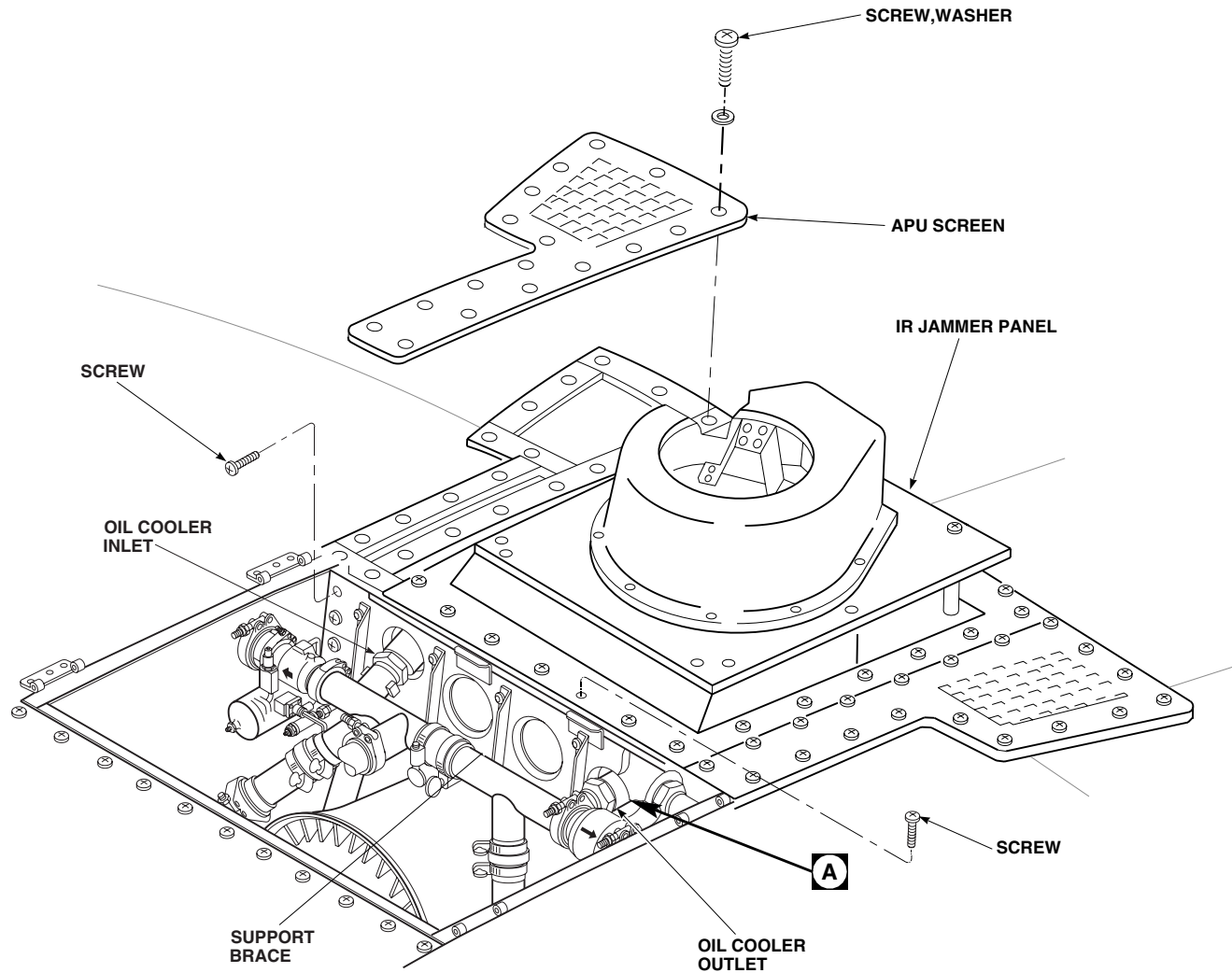
TEST

1. Heat small pail of water to 125°F(52°C) to 135°F(57°C).
2. Place thermostatic valve in heated water for 5 minutes.
3. Remove thermostatic valve from water and measure quickly.
4. Heat pail of water to 155°F(69°C) to 165°F(74°C).
5. Place thermostatic valve in water for 5 minutes.
6. Remove thermostatic valve from water and measure quickly.
7. **MINIMUM DIFFERENCE BETWEEN TWO MEASUREMENTS IS 0.080-INCH.** If thermostatic valve moves less than this, replace valve (Figure 1, Detail A).

INSTALL

1. Check valve seat on radiator for scoring or wear that would cause leakage.
2. **QA** Install thermostatic valve and gasket (Figure 1). **TORQUE VALVE TO 40 - 50 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire valve.

INSTALL - Continued



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Figure 1. Replace Oil Cooler Radiator Thermostatic Valve.

INSTALL - Continued

3. **QA** Place support brace on main rotor pylon and install screws (Figure 1).
4. Make sure area is clean and free of foreign material.
5. **QA** Install screws holding fairing and screened panel to support brace.
6. Install oil cooler access doors (WP 0308 00).
7. Do a maintenance operational check on thermostatic valve as follows:
 - a. With qualified person at controls, run up helicopter.
 - b. Run helicopter at normal operating temperature for about 10 minutes.
 - c. Check for normal oil temperature indication.
 - d. Shut down, check oil level, and inspect for leaks.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

OIL COOLER RADIATOR LINE AND ELBOWS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-A01
Container, 1-Quart
Powertrain Repairer's Toolkit, SC 5180-99-CL-A13
Torque Wrench, 300 - 2500 in. lbs.

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1607 00
WP 1803 00

Material/Parts

Petrolatum, Item 216, WP 1803 00
Protective Caps and Plugs

REMOVE

1. Turn off all electrical power.
2. Open oil cooler compartment access doors.
3. Remove screws, washers, and nuts securing clamps to lines and brackets. Remove clamps (Figure 1, Detail B).
4. Place a 1 quart container under oil line to catch oil trapped in line.
5. Disconnect and remove inlet and outlet oil lines (Figure 1, Detail A).
6. Remove elbows from radiator. Cap open fittings and lines.

INSTALL

1. Turn off all electrical power.
2. Coat threads of elbows and both packings with petrolatum, Item 216, WP 1803 00.

CAUTION

Damage to starter control valve will result if contact occurs between starter control valve and oil cooler inlet elbow. Make sure there is proper clearance between start control valve and oil cooler inlet elbow.

3. **QA** Install nut and packing on 45 degree elbow. Install elbow in inlet fitting (Figure 1, Detail A).
4. **QA** Install nut and packing on 90 degree elbow. Install elbow in outlet fitting.
5. **TIGHTEN NUT ON ELBOWS TO WHERE A SHARP RISE IN TORQUE IS FELT, THEN TIGHTEN 1/6 - 1/3 TURN MORE.**

INSTALL - Continued

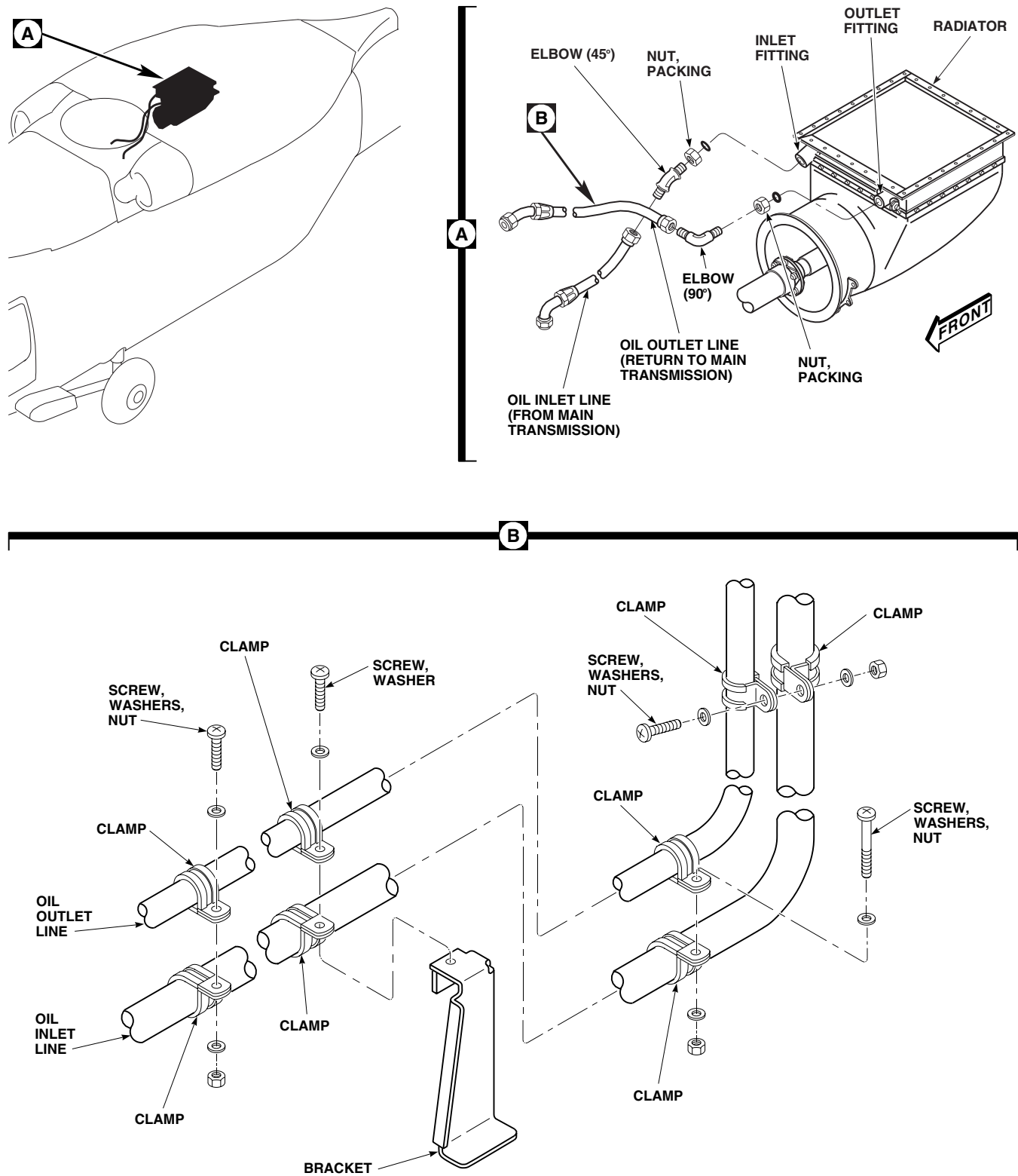


Figure 1. Replace Oil Cooler Radiator Lines and Elbows.

INSTALL - Continued**WARNING****CSI**

Verification of clearance between hoses is a critical characteristic.

6. **QA** Install inlet and outlet oil lines. **MAKE SURE THERE IS A 0.120 INCH MINIMUM CLEARANCE BETWEEN HOSES OVER THE ENTIRE LENGTH.**
7. **QA TORQUE FITTING NUTS ON INLET LINE TO 850 - 1020 INCH-POUNDS.**
8. **QA TORQUE FITTING NUTS ON OUTLET LINE TO 700 - 840 INCH-POUNDS.**
9. **QA** Install clamps on oil lines with screws, washers, and nuts (Figure 1, Detail B).
10. **QA** Install clamps on oil lines and attach to bracket with screw and washer.
11. Service main transmission (WP 1607 00).
12. Make sure area is clean and free of foreign material.
13. Close oil cooler compartment access doors.
14. Close engine cowlings.
15. Do a maintenance operational check and check for leaks.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**OIL COOLER**

This WP supersedes WP 0678 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Borescope Kit
 Caliper Set, GGG-C-105
 Dial Indicator, MIL-I-18422
 Micrometer, GGG-C-105
 Oil Cooler Adapter, 70700-20425-041
 Pin, 0.096-Inch Diameter
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Protective Caps and Plugs
 Puller Kit, Mechanical
 Spring Scale, 643J
 Torque Reactor, 70700-20688-041
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.
 Torque Wrench, 0 - 600 ft. lbs.

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
 Adhesive, Item 14, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 78, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Pad, Scouring, Item 205, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Shim, 70361-03013-104
 Shim, 70361-03013-105
 Shim, 70361-03013-106

Strap, Tiedown, Electrical, Item 310, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
 TM 1-1500-344-23
 TM 11-1520-237-23
 TM 55-1500-345-23
 WP 0304 00
 WP 0308 00
 WP 0309 00
 WP 0310 00
 WP 0311 00
 WP 0369 00
 WP 0379 00
 WP 0381 00
 WP 0615 00
 WP 0626 00
 WP 0646 00
 WP 0666 00
 WP 0667 00
 WP 0671 00
 WP 0675 00
 WP 0677 00
 WP 0679 00
 WP 0680 00
 WP 0700 00
 WP 1590 00
 WP 1607 00
 WP 1803 00

REPLACE OIL COOLER**Remove****NOTE**

Oil cooler weighs 70 pounds.

1. If maintenance crane is going to be used, assemble and erect crane (WP 1590 00).
2. Remove oil cooler compartment access doors (WP 0308 00).
3. Remove APU screens (WP 0304 00).
4. Remove IR jammer fairing (WP 0309 00).
5. Remove IR jammer (TM 11-1520-237-23).
6. Remove IR jammer panel (WP 0310 00).
7. Remove pneumatic starter lines (WP 0700 00).
8. Disconnect inlet and outlet oil lines at radiator (WP 0677 00).
9. Open APU access doors.
10. Remove screws and washers that secure front support beam to airframe and remove front support beam.

WARNING

Injury to personnel and damage to equipment will result if tail rotor is not secured when drive shaft is disconnected. Secure tail rotor to prevent shafts and rotor from turning.

11. Remove fairing to gain access to Section I tail rotor drive shaft (WP 0311 00).
12. Remove Section I tail rotor drive shaft (WP 0666 00).
13. Open drive shaft cover above Section II tail rotor drive shaft.
14. Support front end of Section II tail rotor drive shaft.
15. Remove bolts, spherical washers, shims, and nuts connecting Section II tail rotor drive shaft to oil cooler drive shaft flange (Figure 1).
16. Attach straps of hoisting sling to oil cooler (Figure 2).

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

17. Attach sling to hoist with enough capacity to lift oil cooler.
18. Adjust length of hoisting sling straps.
19. Raise hoist and remove slack from sling.

REPLACE OIL COOLER - Continued

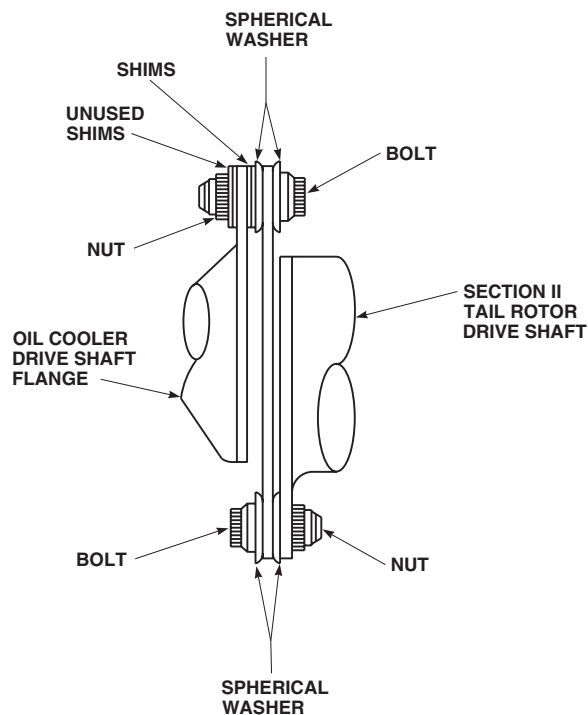
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Figure 1. Remove Oil Cooler.

20. Remove bolts and nuts attaching oil cooler to front and rear support fittings.
21. Lift oil cooler fan and radiator clear of helicopter.
22. Lower oil cooler onto adapter (Figure 3).
23. Secure oil cooler to adapter.
24. Remove hoisting sling.
25. If oil cooler is removed due to clogging, contamination, or blockage, do this:
 - a. Flush inlet and outlet oil lines.
 - b. Inspect main transmission chip detectors (all) and main module oil strainer (WP 0646 00).
 - c. Send radiator to depot level for cleaning/repair.

Inspect

1. Inspect airframe oil cooler supports for visible cracks and distortions. **NONE ALLOWED.** Inspect rear support boltholes for elongation as follows:
 - a. **IF ONE OF FOUR BOLTHOLES ARE ELONGATED MORE THAN 0.010-INCH, (GREATER THAN 0.275-INCH DIAMETER), REPLACE AIRFRAME SUPPORT (WP 0369 00).**
 - b. **IF TWO OR MORE BOLTHOLES ARE ELONGATED MORE THAN 0.006-INCH, (GREATER THAN 0.271-INCH DIAMETER), REPLACE AIRFRAME SUPPORT (WP 0369 00).**

REPLACE OIL COOLER - Continued

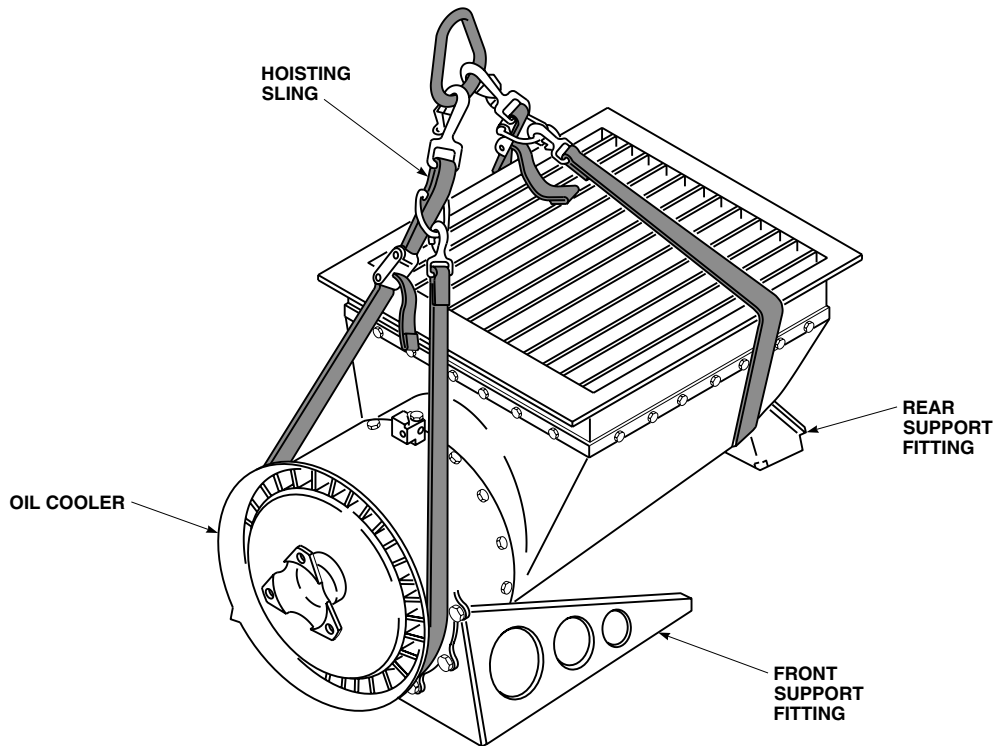
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Figure 2. Attach Hoisting Sling to Oil Cooler.

NOTE

The following procedure not required on new oil coolers being installed.

2. Perform axial fan input flange retaining nut torque check as follows:
 - a. Remove cotter pin from axial fan input flange retaining nut and install torque reactor.

WARNING**CSI**

Verification of 214 foot-pounds minimum torque is a critical characteristic.

NOTE

Torque reactor will be attached to axial fan input flange to react against torquing of axial fan input flange retaining nut.

- b. Check torque of axial fan input flange retaining nut using torque wrench and torque reactor. Check torque in a tightening direction, while holding axial fan shaft with torque reactor. **IF RETAINING NUT TORQUE IS LESS THAN 180 FOOT-POUNDS, REPLACE AXIAL FAN (WP 0678 00).**
- c. **IF RETAINING NUT TORQUE IS 180 - 213 FOOT-POUNDS, TORQUE RETAINING NUT TO 214 - 236 FOOT-POUNDS.** Remove torque reactor and install cotter pin.
- d. **IF RETAINING NUT TORQUE IS 214 - 236 FOOT-POUNDS, TORQUE CHECK IS COMPLETE.** Install new cotter pin.

REPLACE OIL COOLER - Continued

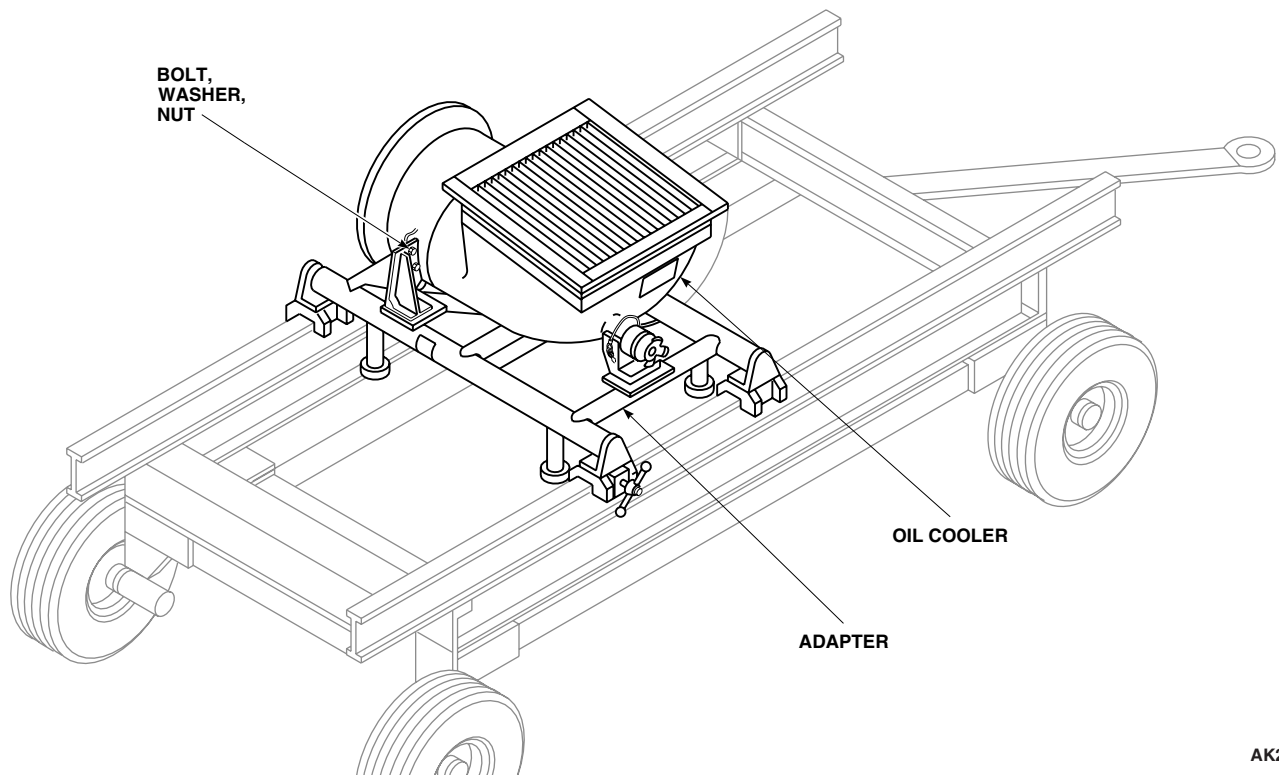


Figure 3. Lower Oil Cooler to Adapter.

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e. **ON HELICOPTERS WITH OIL COOLER FAN 70361-03005-107, IF RETAINING NUT TORQUE IS LESS THAN 180 FOOT-POUNDS, DO THE FOLLOWING:**

- (1) Remove retaining nut and input flange.

NOTE

Wear is defined as a step or slope between contacting surfaces and noncontacting surfaces of spline teeth.

- (2) Visually inspect splines on both flange and axial fan shaft for wear. If any wear is detectable by touch, replace axial fan.
 - (3) If no wear is detectable, align balance index marks on shaft and input flange. Install input flange and retaining nut. **TORQUE NUT TO 214 - 236 FOOT-POUNDS.**
 - (4) Remove torque reactor and install cotter pin.
 - (5) Repeat check of input flange retaining nut torque in 9-11 flight hours. If torque is less than 214 foot-pounds at second check, replace axial fan.
3. Inspect oil cooler drive shafts flange for elongation. **NONE ALLOWED.** If forward flange boltholes are elongated, replace section II tail rotor drive shaft. (WP 0667 00). If aft flange boltholes are elongated, replace flange (WP 0671 00).
 4. Apply epoxy primer coating kit, Item 120, WP 1803 00, to exposed surfaces of retaining nut.
 5. Apply corrosion preventive compound, Item 106, WP 1803 00, to retaining nut and area around nut. Do not apply compound to flange mating surfaces.

REPLACE OIL COOLER - Continued**Install****WARNING**

Injury to personnel and damage to equipment will result if oil cooler drive shaft runout check is not performed before oil cooler is installed on helicopter. Failure to properly perform runout check will result in loss of tail rotor drive.

1. If cooler viscous damper bearing support is being replaced, an oil cooler drive shaft runout check must be performed. Refer to, OIL COOLER DRIVE SHAFT RUNOUT CHECK, in this work package.

WARNING

Injury to personnel and damage to equipment will result if oil cooler drive shaft bolts are not properly torqued before the radiator is installed on the oil cooler duct. Ensure oil cooler drive shaft mounting bolts are properly torqued.

2. Install radiator on oil cooler duct (WP 0675 00).
3. Attach straps of hoisting sling to oil cooler (Figure 2).
4. Adjust rear hoisting sling strap so it does not interfere with beams on main rotor pylon.
5. Attach sling to hoist or maintenance crane.
6. Adjust length of hoisting sling straps.
7. Lift oil cooler into place (Figure 4).
8. **QA** Install bolts and washers on front support to line up oil cooler. **TORQUE BOLTS ON FRONT SUPPORT TO 105 - 115 INCH-POUNDS.** Do not install bolts in rear support fittings now.
9. Remove sling.
10. **QA** Make sure mounting flange on radiator assembly is flush with airframe, and tighten flange attachment hardware.
11. Center bearing in rear viscous damper support.

WARNING**CSI**

Verification of proper preload is a critical characteristic.

NOTE

Aircraft fuel tank weight can affect the installation preload of the oil cooler.

12. Do preload check of oil cooler drive shaft as follows:
 - a. Align oil cooler drive shaft viscous damper bearing support with rear support boltholes using spring scale and electrical tiedown strap, Item 310, WP 1803 00. **FORCE REQUIRED TO ALIGN BOLTHOLES SHALL NOT EXCEED 10 POUNDS.**

WARNING**CSI**

Verification of alignment is a critical characteristic.

- b. Verify alignment by inserting a bolt through bearing support and rear support on each side of rear support.

REPLACE OIL COOLER - Continued

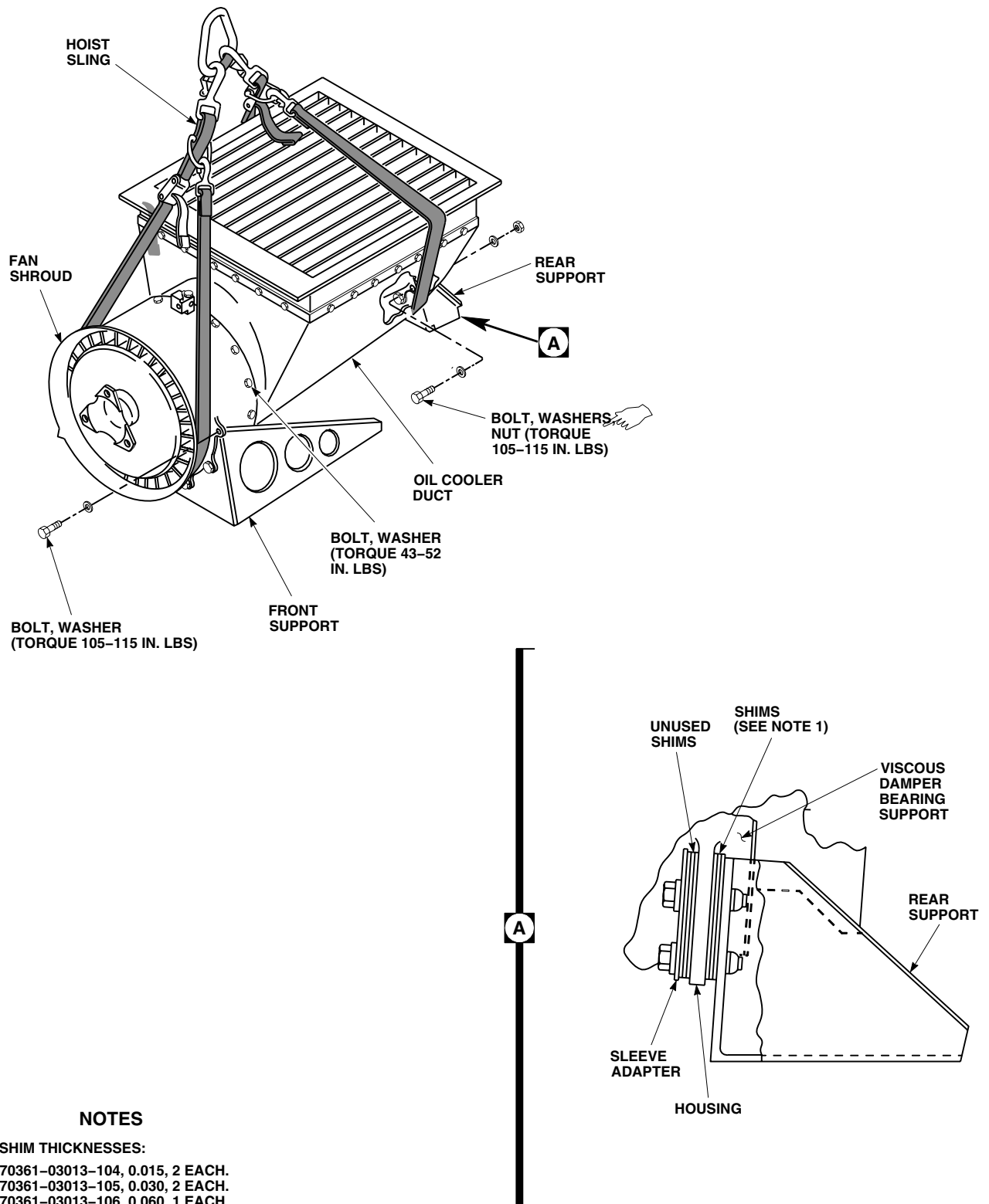


Figure 4. Install Oil Cooler.

REPLACE OIL COOLER - Continued

13. If more than 10 pounds of force is needed to align oil cooler drive shaft viscous damper bearing support, do this:
 - a. Remove oil cooler. Refer to, REPLACE OIL COOLER, in this work package. Do not remove hoisting sling.
 - b. Loosen all bolts attaching fan to oil cooler duct.
 - c. Loosen all bolts attaching radiator to oil cooler duct.
 - d. Lift oil cooler into place.
 - e. Repeat step 8 through step 12.
 - f. If preload is still beyond limits, defuel aircraft and repeat step a. through step e.
 - g. If preload is still more than 10 pounds and aircraft has been defueled, replace oil cooler.

WARNING**CSI**

Verification of proper preload is a critical characteristic.

- h. If preload is less than 10 pounds install 3 bolts on each side of radiator connection and 4 bolts around fan connection.

NOTE

- Oil cooler assembly may be removed from front and rear support to provide easier access to secure radiator and fan. Recheck preload on reinstallation.
- Make sure oil cooler duct does not interfere with rotation of drive shaft.

- i. **TORQUE BOLTS ATTACHING RADIATOR TO OIL COOLER DUCT TO 48 - 52 INCH-POUNDS.**

- j. **TORQUE BOLTS ATTACHING FAN TO OIL COOLER DUCT TO 43 - 52 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire accelerometer bracket to adjacent bolt and bracket mounting bolt to other adjacent bolt.

14. Measure gap between rear support and viscous damper bearing support (Figure 4).
15. Remove bolts installed to verify alignment.

NOTE

A total of 5 shims are used to assemble rear support, viscous damper bearing support, and sleeve adapter. Use two 70361-03013-104 shims, two 70361-03013-105 shims, and one 70361-03013-106 shim for this assembly.

16. Select combination of 5 shims (to fill measured gap) and place between rear support and viscous damper bearing support. Place any unused shims between sleeve adapter and viscous damper bearing support.
17. **QA** Install sleeve adapter and viscous damper bearing support to rear support with bolts, washers, and nuts.
TORQUE NUTS TO 105 - 115 INCH-POUNDS.
18. Install Section I tail rotor drive shaft (WP 0666 00).
19. **QA** Connect front end of Section II tail rotor drive shaft to oil cooler drive shaft flange (WP 0667 00).
20. **QA** Install front support brace using screws and washers.

REPLACE OIL COOLER - Continued**WARNING****CSI**

Verification of 0.12-inches minimum between flexible hoses is a critical characteristic.

21. Connect inlet and outlet oil hoses to fittings of oil cooler radiator (WP 0677 00).
22. Make sure area is clean and free of foreign material.
23. Install IR jammer panel (WP 0310 00).
24. Install IR jammer (TM 11-1520-237-23).
25. Install IR jammer fairing (WP 0309 00).
26. Install APU screens (WP 0304 00).
27. Install pneumatic starter lines (WP 0700 00).
28. Make sure area is clean and free of foreign material.
29. **QA** Install oil cooler compartment access doors (WP 0308 00).
30. Close APU access doors.
31. Service main transmission (WP 1607 00).

NOTE

The spline wear indicator system is not used on oil cooler fan, 70361-03005-107.

32. Reinstall oil cooler spline wear indicator (WP 0679 00).

WARNING

Injury to personnel and damage to equipment will result if spline wear inspection is not performed on oil cooler fans, 70361-03005-103 through 70361-03005-106, every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive spline wear will result in loss of tail rotor drive. Make sure spline wear indicator is installed and spline wear inspection is performed in accordance with WP 0626 00.

33. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed, perform spline wear inspection (WP 0626 00).
34. Perform functional check of main gear box (WP 0615 00).
35. Do an operational check on oil cooler and check for leaks.

WARNING

Injury to personnel and damage to equipment will result if vibration check is not performed every time tail rotor drive shaft is disconnected or replaced from transmission to STA 531.0. Excessive vibration indicates improper installation of oil cooler assembly which will cause oil cooler axial fan drive shaft splines and fan exit flange splines to wear, resulting in loss of tail rotor drive. Make sure vibration check is performed (WP 0680 00).

36. Perform vibration check on oil cooler assembly (WP 0680 00).

REPLACE OIL COOLER FAN, SHAFT, AND DUCT**Remove****NOTE**

Rework of oil cooler drive shaft may be performed on helicopter.

1. Remove oil cooler. Refer to, REPLACE OIL COOLER, in this work package.
2. Remove radiator from oil cooler duct (WP 0675 00).
3. Remove cotter pin from oil cooler drive shaft flange nut near viscous damper bearing (Figure 5).
4. Remove flange nut from oil cooler drive shaft using wrench and torque reactor.

NOTE

Check where index marks on flange and drive shaft are to aid in installation.

5. Remove drive shaft flange from oil cooler drive shaft by pulling straight off.
6. Remove viscous damper bearing support from oil cooler drive shaft using puller.
7. Remove sleeve adapter from oil cooler drive shaft.
8. Remove accelerometer bracket, bolts and washers from around fan shroud. Remove fan and oil cooler drive shaft from oil cooler duct.

WARNING**CSI**

Verification of 214 foot-pounds minimum torque is a critical characteristic.

9. Remove bolts attaching oil cooler drive shaft to fan exit flange.

WARNING

Injury to personnel and damage to equipment will result if fan exit flange retaining nut is not torqued properly on oil cooler fans, 70361-03005-103 through 70361-03005-106. Loosening or removal of retaining nut may cause improper torque to be present and is not allowed. On oil cooler fans, 70361-03005-103 through 70361-03005-106, do not loosen or remove fan exit flange retaining nut.

10. On oil cooler fans, 70361-03005-103 through 70361-03005-106, do torque check on fan exit flange retaining nut (WP 0626 00). Do not install cotter pin now.

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued

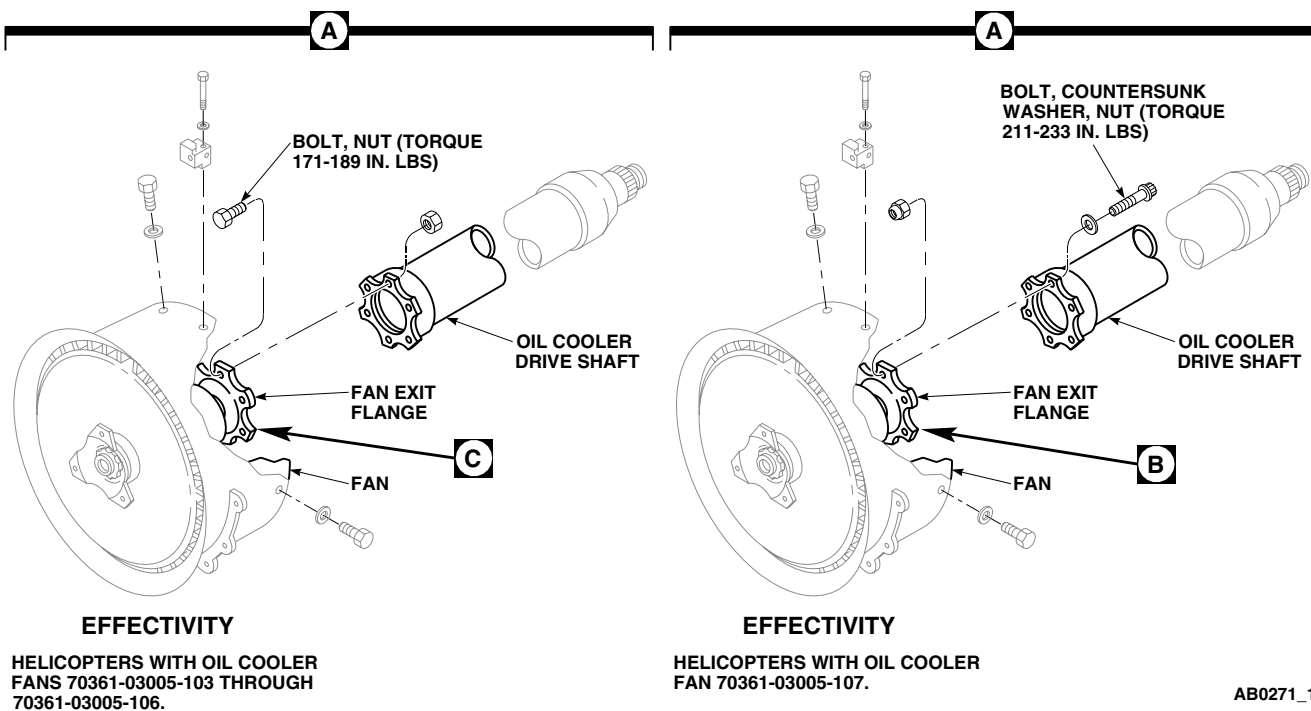
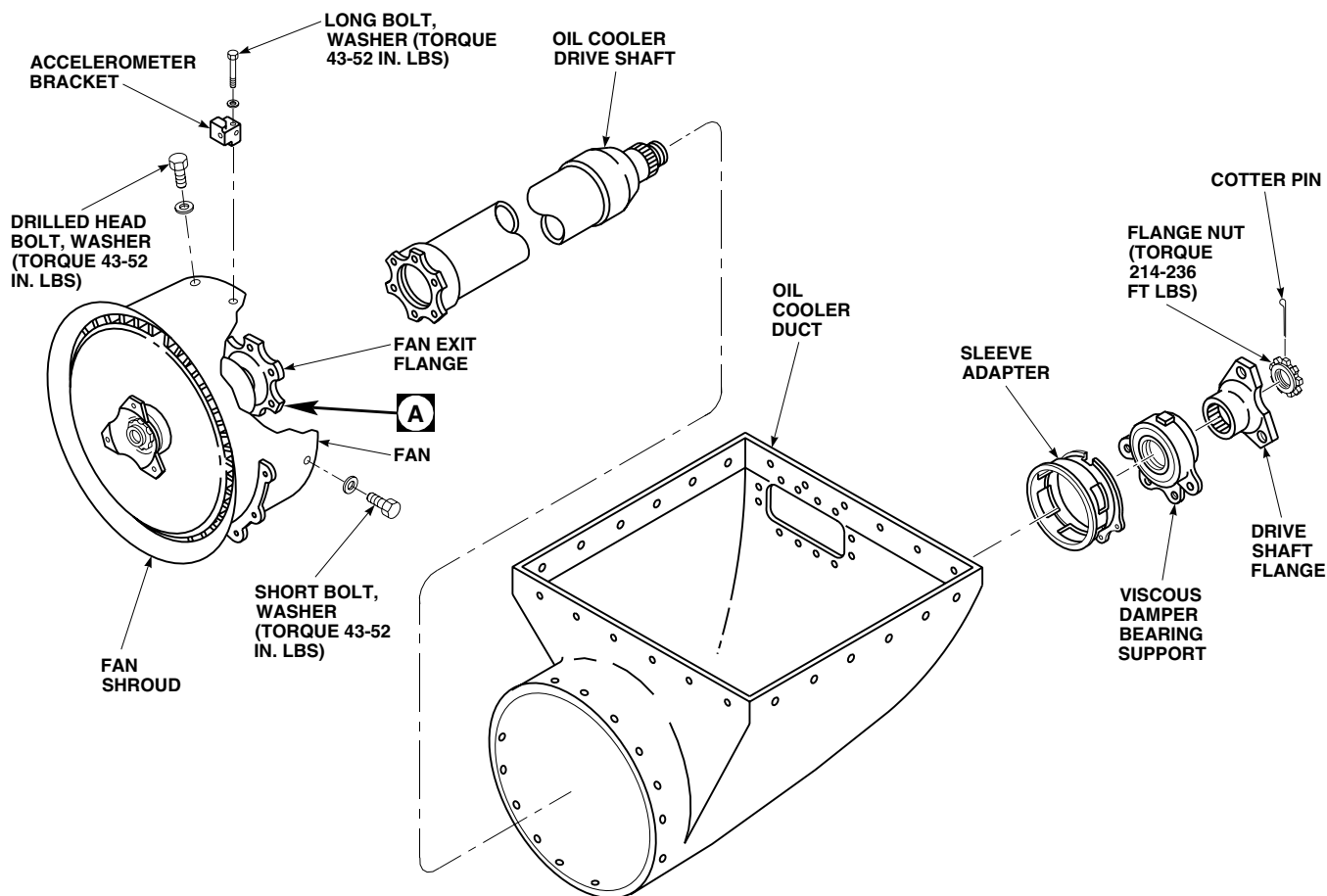


Figure 5. Replace Oil Cooler Fan, Shaft, and Duct. (Sheet 1 of 2)

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued

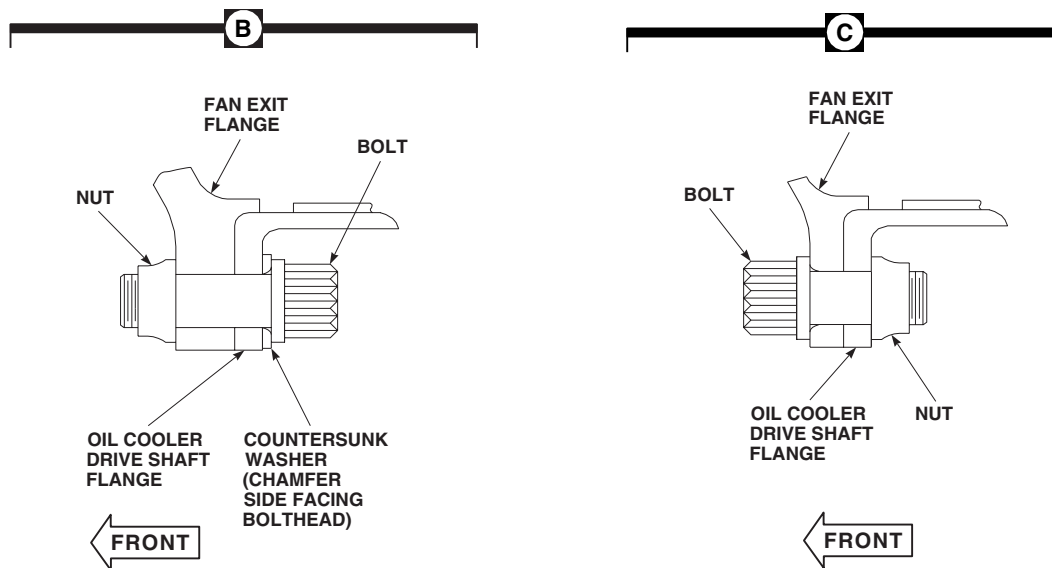
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Figure 5. Replace Oil Cooler Fan, Shaft, and Duct. (Sheet 2 of 2)

Inspect/Repair

CAUTION

Damage to bearings will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate bearing seals and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of drive shaft bearings. Apply solvents only by wiping.

NOTE

No balance checks are required after reworking in accordance with this task.

1. Clean oil cooler drive shaft thoroughly with machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
2. Visually inspect oil cooler drive shaft for cracks. **NONE ALLOWED.** Replace drive shaft if cracks are found.
3. Visually inspect drive shaft for corrosion between oil cooler drive shaft tube and stub shaft, and drive shaft tube and flange. Corrosion between drive shaft tube and stub shaft, and drive shaft tube and flange is identified by flaking or lifting of paint, cracks, or presence of white powder on shaft. **NO CORROSION ALLOWED.** Replace drive shaft if corrosion is found.
4. Blend fine scratches which remove only paint using abrasive cloth, Item 82, WP 1803 00. After blending, touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
5. Using depth gage and calipers, check dimension of damaged area before rework.

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued

6. Compare readings with minimum repair limits (Figure 6). **DRIVE SHAFT WILL NOT BE DAMAGED BEYOND MINIMUM REPAIR LIMITS.** Replace drive shaft which is under limit.
7. If oil cooler drive shaft minimum dimension after repair can be maintained, repair as follows:
 - a. Blend damage smooth using abrasive cloth, Item 78, WP 1803 00.
 - b. Blend reworked area to remove all traces of abrasive marks and match surface smoothness using abrasive cloth, Item 82, WP 1803 00, or polishing cloth.

CAUTION

Damage to bearing will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate bearing seals and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of drive shaft bearings. Apply solvents only by wiping.

- c. Clean oil cooler drive shaft thoroughly with machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
- d. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23). Do not apply paint to boltholes or areas 1 inch around boltholes.

NOTE

Measurement of shaft is only required if wear is visually detected.

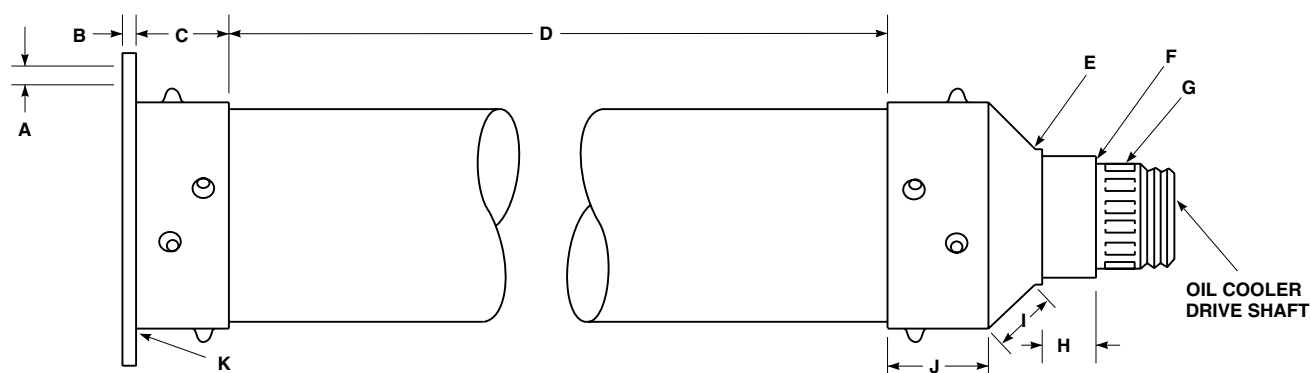
8. Visually inspect stub shaft splines for wear, if wear is detected, inspect as follows:
 - a. Secure 0.096-inch diameter pin between two splines.
 - b. Secure second 0.096-inch diameter pin between two splines directly opposite first pin.
 - c. Measure outside diameter of shaft splines, using micrometer, over pins. **OUTSIDE DIAMETER MUST NOT BE LESS THAN 1.9401-INCHES.** Replace shaft if worn beyond limits.
 - d. Repeat check with pins located in four different positions.
9. Inspect threads on end of stub shaft for crossed, stripped or flattened threads. **NO DAMAGE ALLOWED.** Replace shaft if threads are damaged.
10. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, inspect axial fan shaft ID as follows:
 - a. Clean axial fan shaft ID of grease and dirt using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.

WARNING**CSI**

Existence of electroless nickel plate all over 21E47-1 axial fan shaft, except bearing journals, is a critical characteristic. Existence of cadmium plate and epoxy primer all over 21E47-1 axial fan shaft, except bearing journals, is a critical characteristic.

- b. Inspect axial fan shaft ID for missing primer, corrosion, pits, and flaking plating, using borescope kit. **NO CORROSION, PITS, OR FLAKING PLATING ALLOWED.**
- c. Inspect axial fan shaft ID for cracks particularly in area of cotter pin holes. **NO CRACKS ALLOWED.**

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued



COMPONENT	ZONE	DIMENSION BEFORE REPAIR	REPAIR LIMIT
FLANGE HOLE	A	0.322 (+0.005 TO 0.000) DIA	0.327 MAXIMUM DIA 0.0015 OUT-OF-ROUND
LUG OD	B C	0.175 (± 0.005) 3.710 (± 0.010)	0.160 MINIMUM THICKNESS 3.690 MINIMUM DIA
TUBE	D	3.500 (+0.005 TO 0.000) DIA	3.490 MINIMUM DIA
STUB SHAFT RADIUS RADIUS SPLINES	E F G	0.250 (± 0.005) 0.032 (± 0.007) 1.9446 (± 0.0012)	0.230 MINIMUM RADIUS 0.020 MINIMUM RADIUS 1.9401 MINIMUM DIA (SEE NOTE 4)
OD	H	1.9686 (+0.007 TO 0.000) DIA	1.9682 MINIMUM DIA
	I	0.125 (± 0.010) (WEB THICKNESS)	0.010 MAXIMUM DEPTH BLEND TO 0.250 RADIUS
OD	J	3.690 (± 0.010) DIA	3.670 MINIMUM DIA
FLANGE RADIUS	K	0.125 (± 0.010)	0.105 MINIMUM RADIUS (SEE NOTE 3)

NOTES

- REPAIR LIMITATIONS
 - REPAIR OF MINOR DAMAGE SUCH AS LIMITED NICKS, SCRATCHES, OR SCORING SHALL BE CLEANED UP BY POLISHING.
 - POLISHING SHALL BE DONE ONLY IN THE DAMAGED AREA AND NOT THE ENTIRE CIRCUMFERENCE UNLESS THE DAMAGE EXTENDS AROUND IT.
 - THE BLEND RADIUS ON SHAFT OD SHALL BE 0.500 OR GREATER.
 - FINE SCRATCHES NEED NOT BE POLISHED IF NOT THROUGH PAINT.
 - AREAS WHERE PAINT HAS BEEN REMOVED ARE TO BE POLISHED WITH CROCUS CLOTH AND REPAINTED.
- DO NOT PAINT AROUND BOLTHOLES.
- BLEND RADIUS SHALL NOT BE LESS THAN 0.010 BELOW THE MINIMUM TOLERANCES SHOWN IN "DIMENSION BEFORE REPAIR" COLUMN.
- MINIMUM DIA OF SPLINES TO BE MEASURED OVER 2 0.096 DIA PINS.
- ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 6. Repair Oil Cooler Drive Shaft.

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued**NOTE**

Minor erosion may be noticed on fan blades. This is normal and not a cause for rejection.

11. Inspect fan blades for nicks, gouges, and cracks. **NO CRACKS ALLOWED IN BLADE/HUB ATTACHMENT. NICKS AND GOUGES SHALL NOT EXCEED 0.032-INCH DEEP.**
12. Blend nicks and gouges in fan blades that do not exceed 0.032-inch deep using abrasive cloth, Item 83, WP 1803 00, to a blend radius of 1/8-inch or greater.
13. Touch up blended area of fan blades with corrosion resistant coating, Item 111, WP 1803 00 (TM 1-1500-344-23).
14. Inspect bellmouth of fan shroud for nicks and gouges. **NICKS AND GOUGES SHALL NOT EXCEED 0.062-INCH DEEP.**
15. Blend nicks and gouges in bellmouth that do not exceed 0.062-inch deep using abrasive cloth, Item 83, WP 1803 00, to a blend radius of 1/8-inch or greater.
16. Prime blended area of bellmouth with primer coating, Item 243, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
17. Touch up paint on bellmouth with matching paint (WP 0381 00 and TM 55-1500-345-23).
18. Repair surface damage/corrosion on oil cooler fan external housing as follows:

CAUTION

Damage to bearings and bonded joints will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate bearing seals and bonding joints dissolving lubrication grease and adhesives. Do not spray on cleaning solvents or fluids in vicinity of oil cooler fan bearings or bonded joints. Apply solvents only by wiping.

NOTE

- Oil cooler fan housing rework may be performed on helicopter.
 - No balance checks are required after rework on oil cooler fan housing using the following procedures.
- a. Clean oil and grease from oil cooler fan housing using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
 - b. Blend fine scratches which remove only paint using abrasive cloth, Item 82, WP 1803 00.
 - c. Check depth of damage area. Blend damage area if damage can be fully repaired and blended without blending deeper than 0.010-inch. Blend repair smooth using abrasive cloth, Item 82, WP 1803 00, or abrasive mat, Item 2, WP 1803 00. Remove only what is required to remove damage.
 - d. Prime blended area of oil cooler fan housing using primer coating, Item 243, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
 - e. Touch up paint on oil cooler fan housing with matching paint (WP 0381 00 and TM 55-1500-345-23).
19. On oil cooler fans, 70361-03005-107, repair surface damage/corrosion on oil cooler one piece shaft exit flange as follows:

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued**CAUTION**

Damage to bearings will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate bearing seals dissolving lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of oil cooler fan bearings. Apply solvents only by wiping.

NOTE

- Oil cooler fan must be separated from oil cooler duct and radiator.
 - No balance checks are required after rework on oil cooler fan one piece shaft exit flange using the following procedures.
 - Before starting repair of oil cooler fan one piece shaft exit flange, determine whether polishing out damage would result in dimension beyond repairable limits.
- a. Clean oil and grease from oil cooler fan one piece shaft exit flange using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
 - b. Blend fine scratches using abrasive cloth, Item 82, WP 1803 00.
 - c. Check depth of damage area. Blend damage area if damage can be fully repaired and blended without exceeding limits (Figure 7). Blend repair smooth using abrasive cloth, Item 82, WP 1803 00, or abrasive mat, Item 2, WP 1803 00. Remove only what is required to remove damage.
20. On oil cooler fans, 70361-03005-103, 70361-03005-106, and 70361-03005-107, repair surface damage/corrosion on oil cooler input flange as follows:

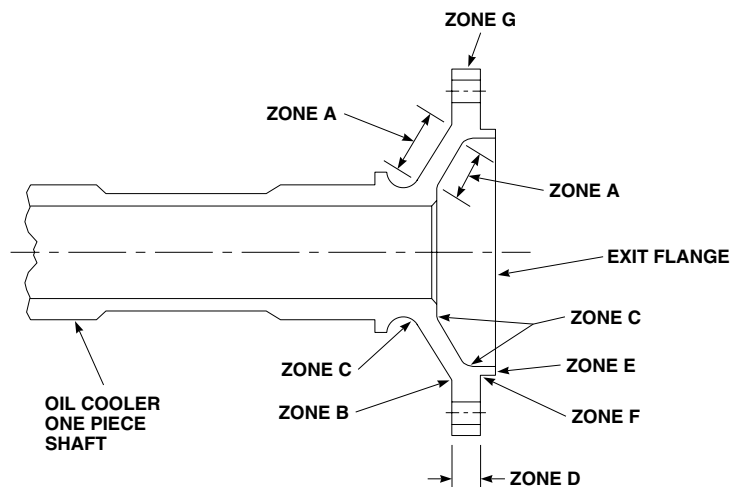
CAUTION

Damage to bearings will result if contact occurs with cleaning solvents and fluids. Cleaning solvents and fluids can penetrate bearing seals dissolving lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of oil cooler fan bearings. Apply solvents only by wiping.

NOTE

- Oil cooler fan must be separated from oil cooler duct and radiator.
 - No balance checks are required after rework on oil cooler fan input flange using the following procedures.
 - Before starting repair of oil cooler fan input flange, determine whether polishing out damage would result in dimension beyond repairable limits.
- a. Clean oil and grease from oil cooler fan input flange using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
 - b. Blend fine scratches using abrasive cloth, Item 82, WP 1803 00.
 - c. Check depth of damage area. Blend damage area if damage can be fully repaired and blended without exceeding limits (Figure 8). Blend repair smooth using abrasive cloth, Item 82, WP 1803 00, or abrasive mat, Item 2, WP 1803 00. Remove only what is required to remove damage.
 - d. Prime blended area of oil cooler fan input flange using primer coating, Item 243, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
 - e. Touch up paint on oil cooler fan input flange with matching paint (WP 0381 00 and TM 55-1500-345-23).

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued



COMPONENT	ZONE	DIMENSION BEFORE REPAIR	REPAIR LIMIT
WEB	A	0.310/0.280	0.270 MIN
RADIUS	B	0.125 R (± 0.010)	(SEE NOTE 3)
RADIUS	C	0.250 R (± 0.010)	(SEE NOTE 3)
LUG	D	0.400 (± 0.010)	0.380 (SEE NOTE 2)
FACE	E	N/A	(SEE NOTE 4)
RADIUS	F	0.060 (± 0.010)	(SEE NOTE 3)
OUTER SURFACE OF FLANGE EAR	G	N/A	(SEE NOTE 4)

EFFECTIVITY

HELICOPTERS WITH OIL COOLER FAN
70361-03005-107.

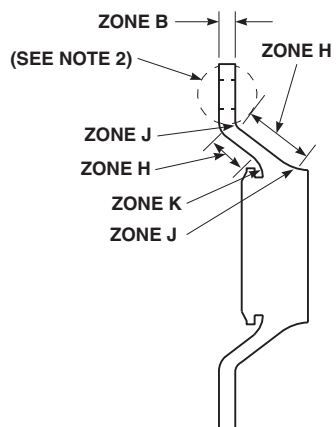
NOTES

- REPAIR LIMITATIONS
 - REPAIR OF MINOR DAMAGE SUCH AS LIMITED NICKS, SCRATCHES, OR SCORING SHALL BE CLEANED UP BY POLISHING.
 - POLISHING SHALL BE DONE ONLY IN THE DAMAGED AREA AND NOT THE ENTIRE CIRCUMFERENCE UNLESS THE DAMAGE EXTENDS AROUND IT.
 - THE BLEND RADIUS ON FLANGES SHALL BE 0.500 OR GREATER.
- MAXIMUM FRETTING DEPTH IS 0.0005. POLISH OUT MINOR NICKS, SCRATCHES, FRETTING AND SCORING USING CROCUS CLOTH.
- BLEND RADIUS SHALL NOT BE LESS THAN 0.010 BELOW THE MINIMUM TOLERANCES SHOWN IN THE "DIMENSION BEFORE REPAIR" COLUMN.
- BLEND UP TO 0.020 DEEP IN NO MORE THAN 10% OF THIS TOTAL SURFACE.
- ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 7. Repair Oil Cooler Fan One Piece Shaft Exit Flange.

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued



INPUT FLANGE

COMPONENT	ZONE	DIMENSION BEFORE REPAIR	REPAIR LIMIT
FLANGE LUG	B	0.175 (+0.005)	0.160
COUPLING FLANGE CONE	H	0.105 (+0.005)	0.010 MAXIMUM DEPTH BLEND TO 0.250 RADIUS (SEE NOTE 3)
RADII	J	0.125 (+0.010) RADIUS	
RADII	K	0.190 0.45 RADIUS	0.010 MAXIMUM DEPTH BLEND TO 0.250 RADIUS (SEE NOTE 3)
OUTER SURFACE OF FLANGE EARS (BOTH ENDS)			0.010 MAXIMUM DEPTH BLEND TO 0.250 RADIUS (SEE NOTE 3)

NOTES

1. REPAIR LIMITATIONS
 - (a) REPAIR OF MINOR DAMAGE SUCH AS LIMITED NICKS, SCRATCHES, OR SCORING SHALL BE CLEANED UP BY POLISHING.
 - (b) POLISHING SHALL BE DONE ONLY IN THE DAMAGED AREA AND NOT THE ENTIRE CIRCUMFERENCE UNLESS THE DAMAGE EXTENDS AROUND IT.
 - (c) FINE SCRATCHES NEED NOT BE POLISHED IF NOT THROUGH PAINT.
 - (d) AREAS WHERE PAINT HAS BEEN REMOVED ARE TO BE POLISHED WITH CROCUS CLOTH AND REPAINTED.
2. DO NOT PAINT AROUND BOLT HOLES.
3. BLEND RADIUS SHALL NOT BE LESS THAN 0.010 BELOW THE MINIMUM TOLERANCES SHOWN IN "DIMENSION BEFORE REPAIR" COLUMN.
4. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 8. Repair Oil Cooler Fan Input Flange.

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued

21. Inspect transition duct as follows:

- a. Clean transition duct of grease and dirt using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
- b. Inspect duct for damage. Repair duct as required (WP 0379 00).
- c. Inspect vanes inside duct for damage. **NO DAMAGE ALLOWED.** Replace transition duct if vanes are damaged.
- d. Inspect all nutplates for damage and security. Repair nutplates as required (TM 1-1500-204-23).

22. Rework the oil cooler transition duct, 70361-03006-044, to allow use of the slotted sleeve adapter as follows:

NOTE

Rework of the oil cooler transition duct may be performed on helicopter.

- a. Trim duct to 24.65 - inch (± 0.03) (Figure 9). Oil cooler transition duct, 70361-03006-045, may be ordered to avoid having to trim the transition duct currently in aircraft. This duct is already trimmed to accommodate the slotted sleeve adapter.
- b. Clean end of duct using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00. Abrade end of duct using scouring pad, Item 205, WP 1803 00, then wipe again using machinery towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
- c. Bond only in a well ventilated area. Bond ends of sealing ring to each other and then entire sealing ring to transition duct with adhesive, Item 14, WP 1803 00, (Figure 9). Apply adhesive, Item 14, WP 1803 00, in a thin uniformed coating to both surfaces to be bonded. The adhesive should be allowed to evaporate until minimum needed tackiness remains.

Install**WARNING**

- Injury to personnel and damage to equipment will result if fan exit flange retaining nut is not torqued properly. Loosening or removal of retaining nut may cause improper torque to be present and is not allowed. Only inspect and repair surfaces that do not require removal or loosening of fan exit flange retaining nut.
- Verify proper installation of attaching hardware, especially countersunk washers. Oil cooler fan, 70361-03005-107, requires different hardware.

1. On oil cooler fans, 70361-03005-103 through 70361-03005-106, inspect fan exit flange and retaining nut (WP 0626 00).
2. Perform breakaway torque check on fan to drive shaft retaining nuts as follows:

NOTE

Do not install bolts and nuts on components when performing this torque check.

- a. Thread bolts into each nut until two threads stick out past end of nut.
 - b. Using torque wrench, back out bolt. **MINIMUM BREAKAWAY TORQUE IS 6.5 INCH-POUNDS.**
3. If breakaway torque is not within limits, do this:
- a. Inspect bolts and nuts for evidence of corrosion or wear. **NONE ALLOWED.** Replace bolts or nuts if corrosion or wear is found.
 - b. Repeat step 2.

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued

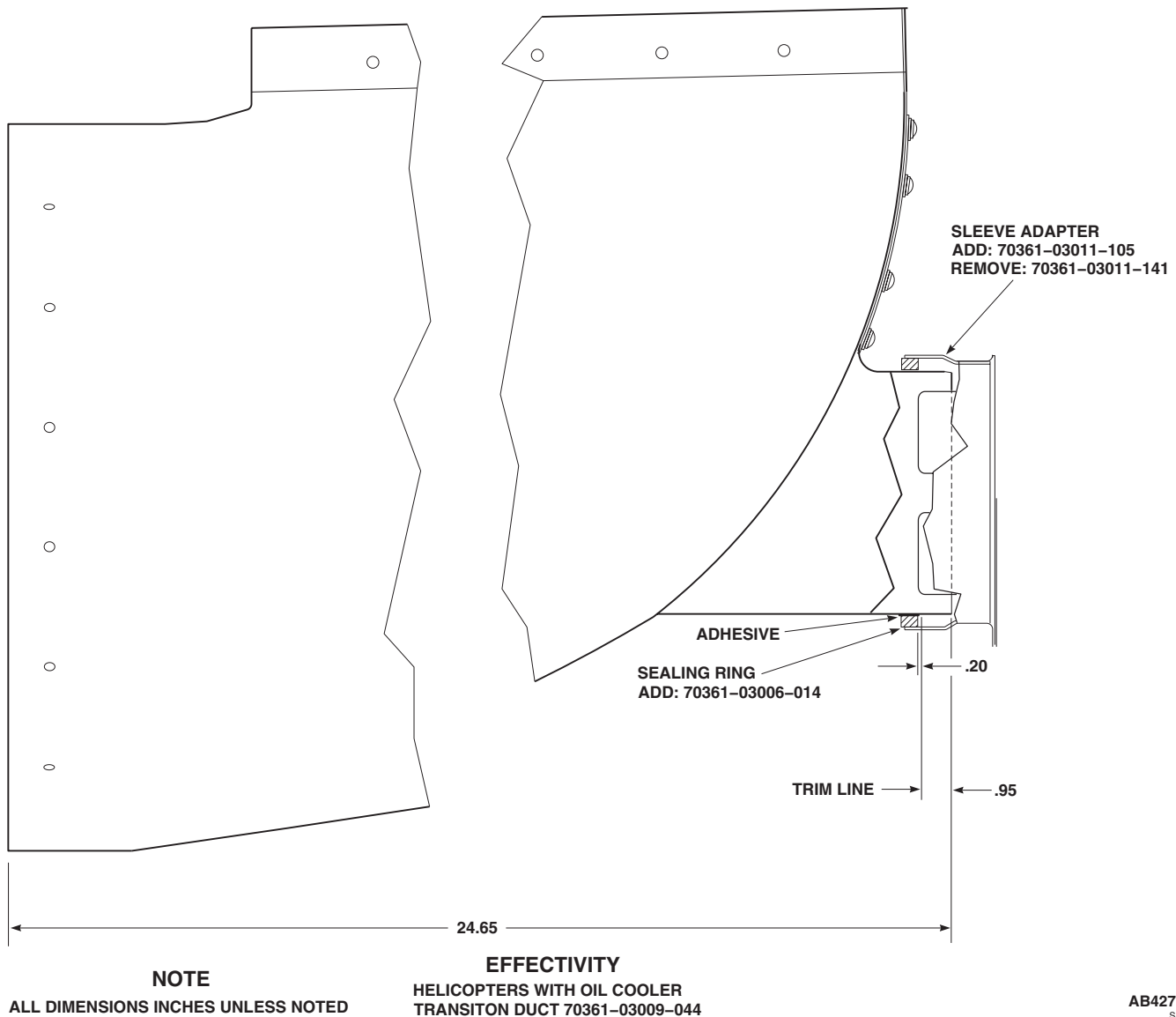


Figure 9. Rework Of Oil Cooler Transition Duct.

NOTE

To prevent improper torquing of the exit flange to the oil cooler drive shaft, verify proper attaching hardware and proper installation of countersunk washers. Make sure that the countersunk washer is installed with the countersunk side facing the bolthead as shown in Figure 5. The 70361-03005-107 oil cooler fans have longer attaching bolts and countersunk washers. After assembly, verify that the bolt thread extends past self locking nut.

4. On helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, do this (Figure 5, Detail A):
 - a. Connect fan to drive shaft with bolts and nuts (bolthead facing front).
 - b. **QA TORQUE NUTS TO 171 - 189 INCH-POUNDS.**

REPLACE OIL COOLER FAN, SHAFT, AND DUCT - Continued**NOTE**

To prevent improper torquing of the exit flange to the oil cooler drive shaft, verify proper attaching hardware and proper installation of countersunk washers. Make sure that the countersunk washer is installed with the countersunk side facing the bolthead as shown in Figure 5. The 70361-03005-107 oil cooler fans have longer attaching bolts and countersunk washers. After assembly, verify that the bolt thread extends past self locking nut.

5. On helicopters with oil cooler fan, 70361-03005-107, do this (Figure 5, Detail A):
 - a. Connect fan to drive shaft with bolts, countersunk washers, and nuts (bolthead facing rear).
 - b. **QA TORQUE NUTS TO 211 - 233 INCH-POUNDS.**

NOTE

When installing fan shroud onto oil cooler duct, make sure drilled head bolts are installed on both sides of accelerometer bracket position.

6. **QA** Install fan and drive shaft into oil cooler duct using bolts and washers. **TORQUE BOLTS TO 43 - 52 INCH-POUNDS (Figure 5, Sheet 1).**
7. **QA** Install accelerometer bracket with long bolt and washer (Figure 5). **TORQUE BOLT TO 43 - 52 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire accelerometer bracket to adjacent bolt and long bolt to other adjacent bolt.
8. Do an oil cooler drive shaft runout check. Refer to, OIL COOLER DRIVE SHAFT RUNOUT CHECK, in this work package.
9. **QA** Install sleeve adapter on oil cooler drive shaft.
10. **QA** Using an arbor press, press viscous damper bearing support onto drive shaft.
11. Line up index mark on flange and index mark on drive shaft.
12. Slide drive shaft flange on oil cooler drive shaft.

WARNING**CSI**

Verification of minimum torque of nut is a critical characteristic.

13. **QA** Make sure bevel side of nut is toward seal and that nut can be fully run on by hand. Using torque reactor and wrench, **TORQUE NUT AS FOLLOWS: THE TORQUE RANGE IS 214 - 236 FOOT-POUNDS. TORQUE RETAINING NUT TO 214 FOOT-POUNDS.** Install new cotter pin. To align pin hole, turn nut in tightening direction. **DO NOT EXCEED MAX TORQUE. IF MAX TORQUE IS EXCEEDED AND NUT DOES NOT ALIGN, REMOVE NUT AND REPLACE IT WITH A NEW NUT AND REPEAT TORQUING PROCEDURE.**
14. Install oil cooler. Refer to, REPLACE OIL COOLER, in this work package.

OIL COOLER DRIVE SHAFT RUNOUT CHECK

1. Remove oil cooler. Refer to, REPLACE OIL COOLER, in this work package.
2. Remove bolts securing radiator to oil cooler duct. Remove radiator.

OIL COOLER DRIVE SHAFT RUNOUT CHECK - Continued

3. Remove oil cooler drive shaft nut, flange, and viscous damper bearing. Refer to, INSPECT, in this work package.
4. Attach dial indicator to adapter near rear of oil cooler. Make sure dial indicator will not be disturbed during runout check.
5. Place dial indicator pointer on oil cooler drive shaft viscous damper bearing journal surface. Make sure dial indicator pointer is perpendicular to drive shaft centerline.

NOTE

A clicking or rattling noise when oil cooler fan is rotated is not an indication of possible bearing failure. Actual operating loads are not being simulated by rotating fan. Bearing can only be checked by performing the following procedures.

6. Rotate drive shaft by hand through front axial fan flange. Make sure no force is placed on front flange that will affect runout reading. **RUNOUT SHALL NOT EXCEED 0.010-INCH.**
7. If runout is greater than 0.010-inch, do this:
 - a. Loosen bolts and nuts attaching oil cooler drive shaft to fan exit flange.
 - b. Tighten bolts and nuts until drive shaft is secure. **DO NOT TORQUE BOLTS NOW.**
 - c. Repeat runout check. Note point where greatest change in dial indicator reading occurs. Torque nut opposite point of greatest change as follows:
 - (1) **ON OIL COOLER FANS, 70361-03005-103 THROUGH 70361-03005-106, TORQUE NUTS TO 171 - 189 INCH-POUNDS.**
 - (2) **ON OIL COOLER FAN, 70361-03005-107, TORQUE NUTS TO 211 - 233 INCH-POUNDS.**
 - d. Repeat step 7. c. until all nuts attaching oil cooler drive shaft to fan exit flange are torqued.
 - e. Repeat runout check. **RUNOUT SHALL NOT EXCEED 0.010-INCH.**
8. If runout is still greater than 0.010-inch, do this:
 - a. Remove bolts and nuts attaching oil cooler drive shaft to fan exit flange.
 - b. Turn oil cooler drive shaft one bolthole with respect to fan exit flange.
 - c. Install oil cooler drive shaft to fan exit flange with bolts, washers, and nuts. Tighten bolts and nuts until drive shaft is secure. **DO NOT TORQUE BOLTS NOW.**
 - d. Repeat step 7. c. and step 7. d.
 - e. Repeat runout check. **RUNOUT SHALL NOT EXCEED 0.010-INCH.** Repeat step 8. a. through step 8. d. until runout is within limits.
 - f. If all combinations of boltholes have been used and runout still exceeds 0.010-inch, replace oil cooler drive shaft. Repeat step 3. through step 8. for new oil cooler drive shaft.
 - g. If new oil cooler drive shaft runout still exceeds 0.010-inch, replace oil cooler axial fan. Refer to, INSPECT, in this work package.
9. Install radiator on oil cooler duct (WP 0675 00).
10. Install oil cooler. Refer to, REPLACE OIL COOLER, in this work package.

TORQUE CHECK OIL COOLER DRIVE SHAFT FLANGE RETAINING NUT

1. Remove oil cooler. Refer to, REPLACE OIL COOLER, in this work package.
2. Remove cotter pin from oil cooler drive shaft flange retaining nut.

TORQUE CHECK OIL COOLER DRIVE SHAFT FLANGE RETAINING NUT - Continued

3. Attach torque reactor to oil cooler drive shaft flange. **CHECK OIL COOLER DRIVE SHAFT FLANGE RETAINING NUT TORQUE IN A TIGHTENING DIRECTION. IF RETAINING NUT TORQUE IS LESS THAN 180 FOOT-POUNDS, DO THE FOLLOWING:**
 - a. Remove retaining nut and oil cooler drive shaft flange.
 - b. Visually inspect splines on both flange and axial fan shaft for wear. Wear is defined as a step or slope, or pitting on the spline teeth.
 - c. If any wear is detectable by touch, replace the fan.
 - d. **QA** If no wear is detected, align balance index marks on shaft and flange. Install flange and retaining nut.
TORQUE NUT TO 214 - 236 FOOT-POUNDS.

WARNING**CSI**

Verification of minimum torque of nut is a critical characteristic.

NOTE

If retaining nut torque is 180 - 213 foot-pounds, retorque to 214 - 236 foot-pounds.
Install cotter pin.

4. **QA** If retaining nut torque is 214 - 236 foot-pounds torque check is complete. Install cotter pin.
5. Apply epoxy primer coating kit, Item 120, WP 1803 00, to retaining nut and corrosion preventive compound, Item 106, WP 1803 00, to retaining nut and surrounding areas.
6. Install oil cooler. Refer to, REPLACE OIL COOLER, in this work package.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM****OIL COOLER SPLINE WEAR INDICATOR SYSTEM****This WP supersedes WP 0679 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-CL-A01
Gloves
Goggles/Face Shield
Spring Scale
Torque Wrench, 30 - 200 in. lbs.

Polyurethane, Coating, Item 235, WP 1803 00
Strap, Tiedown, Electrical, Item 310, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 325,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 14, WP 1803 00
Brush, Artists, Item 58, WP 1803 00
Epoxy Coating Kit, Item 118, WP 1803 00
Polyurethane, Coating, Item 229, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

WP 0626 00
WP 1803 00

INSTALLATION**NOTE**

This procedure applies only to helicopters with oil cooler fans, 70361-03005-103 through 70361-03005-106, installed.

1. Open tail rotor drive shaft cover on tail cone.
2. Remove panels and access doors above oil cooler.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Cleaning solvents and fluids can penetrate bearing seals and dissolve lubrication grease. Do not spray on cleaning solvents or fluids in vicinity of drive shaft bearings. Apply solvents only by wiping.

3. Clean area on tail rotor drive shaft near No. 3 viscous damper at STA 531.40 using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00 (Figure 1). Wipe dry with clean machinery towel.

INSTALLATION - Continued

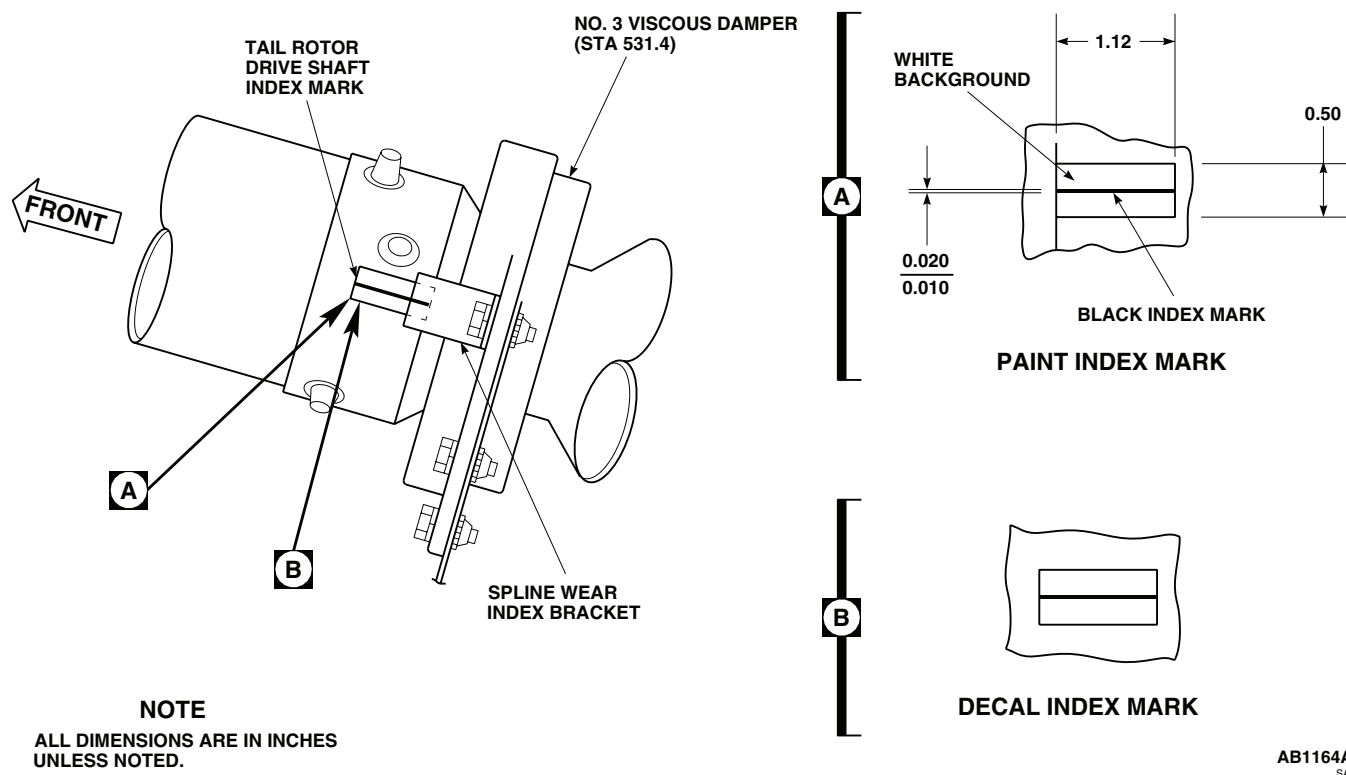


Figure 1. Spline Wear Indicator System Installation.

NOTE

- Location of decal on circumference of tail rotor drive shaft is not important.
- Black index mark line on decal is to be installed parallel with centerline of tail rotor drive shaft.

4. To install decal index mark on tail rotor drive shaft, do the following:

- QA** Bond decal to cleaned area on tail rotor drive shaft using adhesive, Item 14, WP 1803 00 (Figure 1, Detail B). Allow to dry.
- Apply a coat of polyurethane coating, Item 235, WP 1803 00, using artists brush, Item 58, WP 1803 00, over decal. Extend coat 1/4-inch maximum beyond edges of decal.

NOTE

- If index mark decal for tail rotor drive shaft and wear indicator decal for axial fan housing are available, install in place of paint.
- Location of paint index mark on circumference of tail rotor drive shaft is not important.
- Black index mark line is to be applied parallel with centerline of tail rotor drive shaft.

5. To apply paint index mark to tail rotor drive shaft, do the following:

- Mask cleaned area on tail rotor drive shaft using pressure sensitive adhesive tape, Item 325, WP 1803 00, to dimensions shown in Figure 1, Detail A.

INSTALLATION - Continued

- b. Paint background inside masked area using polyurethane coating, Item 229, WP 1803 00, and artists brush, Item 58, WP 1803 00. Allow to dry.
 - c. Mask background area using pressure sensitive adhesive tape, Item 325, WP 1803 00, to dimension shown in Figure 1, Detail A, for black index mark.
 - d. **QA** Paint index mark on background using epoxy coating kit, Item 118, WP 1803 00, and artists brush, Item 58, WP 1803 00. Allow to dry.
 - e. Apply polyurethane coating, Item 235, WP 1803 00, using artists brush, Item 58, WP 1803 00, over index mark and background. Allow to dry.
 - f. Remove masking tape.
6. Install spline wear index bracket to left side of No. 3 viscous damper at STA 531.40 as follows:
 - a. Remove upper left bolt and nut from No. 3 viscous damper bracket (Figure 1).
 - b. **QA** Install spline wear index bracket to No. 3 viscous damper bracket using new bolt, washer, and nut.
TORQUE NUT TO 95 - 105 INCH-POUNDS.
 7. Rotate tail rotor blade in direction of normal rotor rotation until index mark on tail rotor drive shaft aligns with spline wear index bracket slot on No. 3 viscous damper (Figure 1). Use 6 inch ruler or similar straight edge to line up bracket slot with black index mark line.

WARNING

- Spline wear inspection cannot be accurately performed if alignment of index mark on oil cooler drive shaft and spline wear index bracket on No. 3 viscous damper is disturbed while installing index marks on oil cooler axial fan. False spline wear readings will result in loss of tail rotor drive. Make sure index mark on tail rotor drive shaft and spline wear index bracket slot remain aligned.
 - Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.
8. Clean area on oil cooler axial fan housing at approximately 6 o'clock position using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00 (Figure 2, Section A-A). Wipe dry with clean machinery towel.

WARNING

- Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.
9. Clean leading edge, top edge and front face of axial fan blade at approximately 6 o'clock position using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00 (Figure 2). Wipe dry with clean machinery towel.
 10. **QA** Paint leading edge, top edge and front face of axial fan blade using epoxy coating kit, Item 118, WP 1803 00, using artists brush, Item 58, WP 1803 00 (Figure 2). Allow to dry.

INSTALLATION - Continued

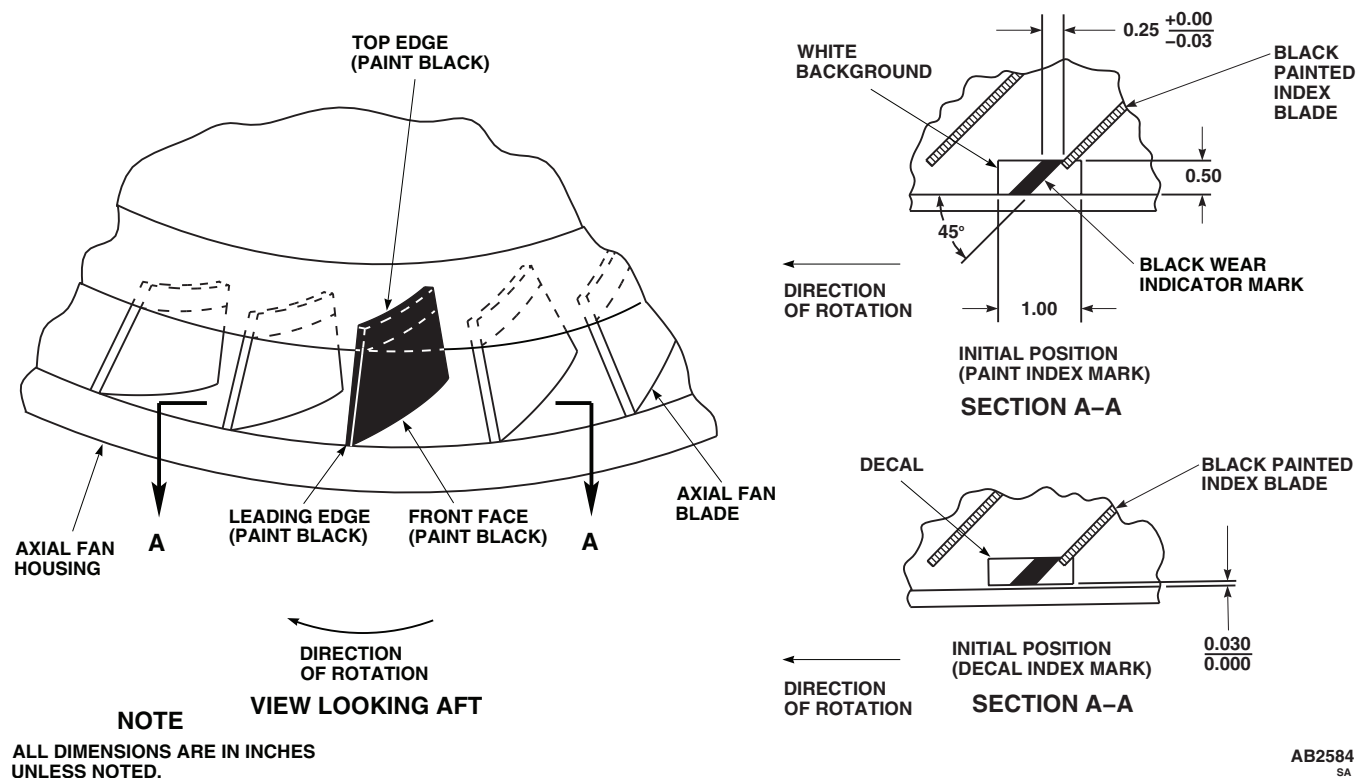


Figure 2. Spline Wear Indicator System Installation.

11. Apply two coats of polyurethane coating, Item 235, WP 1803 00, using artists brush, Item 58, WP 1803 00, over painted area on fan blade. Allow to dry.

NOTE

If index mark decal for oil cooler axial fan housing is available, install decal in place of paint. Proceed to step 12.

12. To apply paint index mark to oil cooler axial fan housing, do the following:
 - a. Mask cleaned area on axial fan housing using pressure sensitive adhesive tape, Item 325, WP 1803 00, to dimensions shown in (Figure 2, Section A-A).
 - b. Paint background inside masked area using polyurethane coating, Item 229, WP 1803 00, and artists brush, Item 58, WP 1803 00. Allow to dry.
 - c. Engage gust lock.
 - d. Apply a 20 pound load for 5 seconds to tail rotor blade trailing edge using electrical tiedown strap, Item 310, WP 1803 00, and spring scale. Make sure tail rotor blade is rotated in direction of normal rotation.
 - e. Release gust lock with load applied.
 - f. Slowly rotate tail rotor blade in direction of normal rotation until index mark on tail rotor drive shaft aligns with spline wear index bracket slot on No. 3 viscous damper. Use 6 inch ruler or similar straight edge to line up bracket slot with black index mark line.
 - g. If index mark on tail rotor drive shaft rotates past spline wear index bracket slot on No. 3 viscous damper, continue to rotate blade in direction of normal rotation until index mark is aligned with slot.

INSTALLATION - Continued

- h. If tail rotor is inadvertently rotated in direction opposite normal rotation, repeat step 12. c. through step 12. g.

WARNING

Spline wear inspection cannot be accurately performed if alignment of index mark on oil cooler drive shaft and spline wear index bracket on No. 3 viscous damper is disturbed while installing index marks on oil cooler axial fan. False spline wear readings will result in loss of tail rotor drive. Make sure index mark on tail rotor drive shaft and spline wear index bracket slot remain aligned.

NOTE

Black painted axial fan index blade must be at approximately 6 o'clock position.

- i. Mask background area on axial fan housing using pressure sensitive adhesive tape, Item 325, WP 1803 00, to dimensions shown in (Figure 2, Section A-A). Make sure that masking tape is properly aligned with black painted index blade in INITIAL POSITION.
 - j. Paint wear indicator mark inside masked area using epoxy coating kit, Item 118, WP 1803 00, and artists brush, Item 58, WP 1803 00. Allow to dry.
 - k. Remove masking tape.
 - l. Repeat step 12. f. through step 12. h.
 - m. Check position of black painted index fan blade. **BLACK PAINTED INDEX BLADE MUST BE AT INITIAL POSITION** (Figure 2).
 - n. If black painted index blade is at INITIAL POSITION, apply two coats of polyurethane coating, Item 235, WP 1803 00, using artists brush, Item 58, WP 1803 00, to black wear indicator mark and white background.
 - o. If black painted index blade is not at INITIAL POSITION, do the following:
 - (1) Remove painted index mark from axial fan housing.
 - (2) Repeat step 12. a. through step 12. o.
13. To apply decal index mark to oil cooler axial fan housing, do the following:
- a. Engage gust lock.
 - b. Apply a 20 pound load for 5 seconds to tail rotor blade trailing edge using electrical tiedown strap, Item 310, WP 1803 00, and spring scale. Make sure tail rotor blade is rotated in direction of normal rotation.
 - c. Release gust lock with load applied.
 - d. Slowly rotate tail rotor blade in direction of normal rotation until index mark on tail rotor drive shaft aligns with spline wear index bracket slot on No. 3 viscous damper. Use 6 inch ruler or similar straight edge to line up bracket slot with black index mark line.
 - e. If index mark on tail rotor drive shaft rotates past spline wear index bracket slot on No. 3 viscous damper, continue to rotate blade in direction of normal rotation until index mark is aligned with slot.
 - f. If tail rotor is inadvertently rotated in direction opposite normal rotation, repeat step 13. a. through step 13. e.

INSTALLATION - Continued**WARNING**

Spline wear inspection cannot be accurately performed if alignment of index mark on oil cooler drive shaft and spline wear index bracket on No. 3 viscous damper is disturbed while installing index marks on oil cooler axial fan. False spline wear readings will result in loss of tail rotor drive. Make sure index mark on tail rotor drive shaft and spline wear index bracket slot remain aligned.

NOTE

Black painted axial fan index blade must be at approximately 6 o'clock position.

- g. **QA** Bond decal index mark to cleaned area on axial fan housing using adhesive, Item 14, WP 1803 00. Make sure decal is properly aligned with black painted index blade in INITIAL POSITION (Figure 2, Section A-A). Allow to dry.
- h. Repeat step 13. d. through step 13. f.
- i. **QA** Check position of black painted index fan blade. **BLACK PAINTED INDEX BLADE MUST BE AT INITIAL POSITION (Figure 2).**
- j. If black painted index blade is at INITIAL POSITION, apply polyurethane coating, Item 235, WP 1803 00, to decal. Extend coat 1/4-inch maximum beyond edges of decal.
- k. If black painted index blade is not at INITIAL POSITION, do the following:
 - (1) Remove decal from axial fan housing.
 - (2) Repeat step 13. a. through step 13. k.
- 14. Perform spline wear inspection (WP 0626 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

DRIVE SYSTEM

OIL COOLER VIBRATION CHECK

This WP supersedes WP 0680 00, dated 17 April 2006.

INITIAL SETUP:

NA

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END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM**TAIL GEAR BOX**

This WP supersedes WP 0681 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Airframe Repairer's Toolkit, SC 5180-99-B02
 Drill, No. 29
 Gloves
 Goggles/Face Shield
 Fluorescent Penetrant Inspection Kit, XMA101
 Nonmetallic Scraper
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Antiseize Compound, Item 49, WP 1803 00
 Brush, Acid Swabbing, Item 57, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Epoxy Coating Kit, Item 118, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Molybdenum Disulfide, Item 197, WP 1803 00
 Nonpermanent Marker
 Primer, Ahesive, Item 241, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Shim, Locally-Made, Figure 199, WP 1805 00
 Spacer, Locally-Made, Figure 200, WP 1805 00
 Tags
 Tape, Pressure Sensitive, Adhesive, Item 326,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 Aircraft Structural Repairer MOS 15G (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
 TM 1-1520-237-MTF
 TM 1-1520-265-23
 WP 0333 00
 WP 0585 00
 WP 0669 00
 WP 0683 00
 WP 0686 00
 WP 0688 00
 WP 0910 00
 WP 0913 00
 WP 1039 00
 WP 1145 00
 WP 1146 00
 WP 1148 00
 WP 1152 00
 WP 1153 00
 WP 1155 00
 WP 1156 00
 WP 1247 00
 WP 1591 00
 WP 1609 00
 WP 1693 00
 WP 1695 00
 WP 1703 00
 WP 1803 00
 WP 1805 00

REMOVE**NOTE**

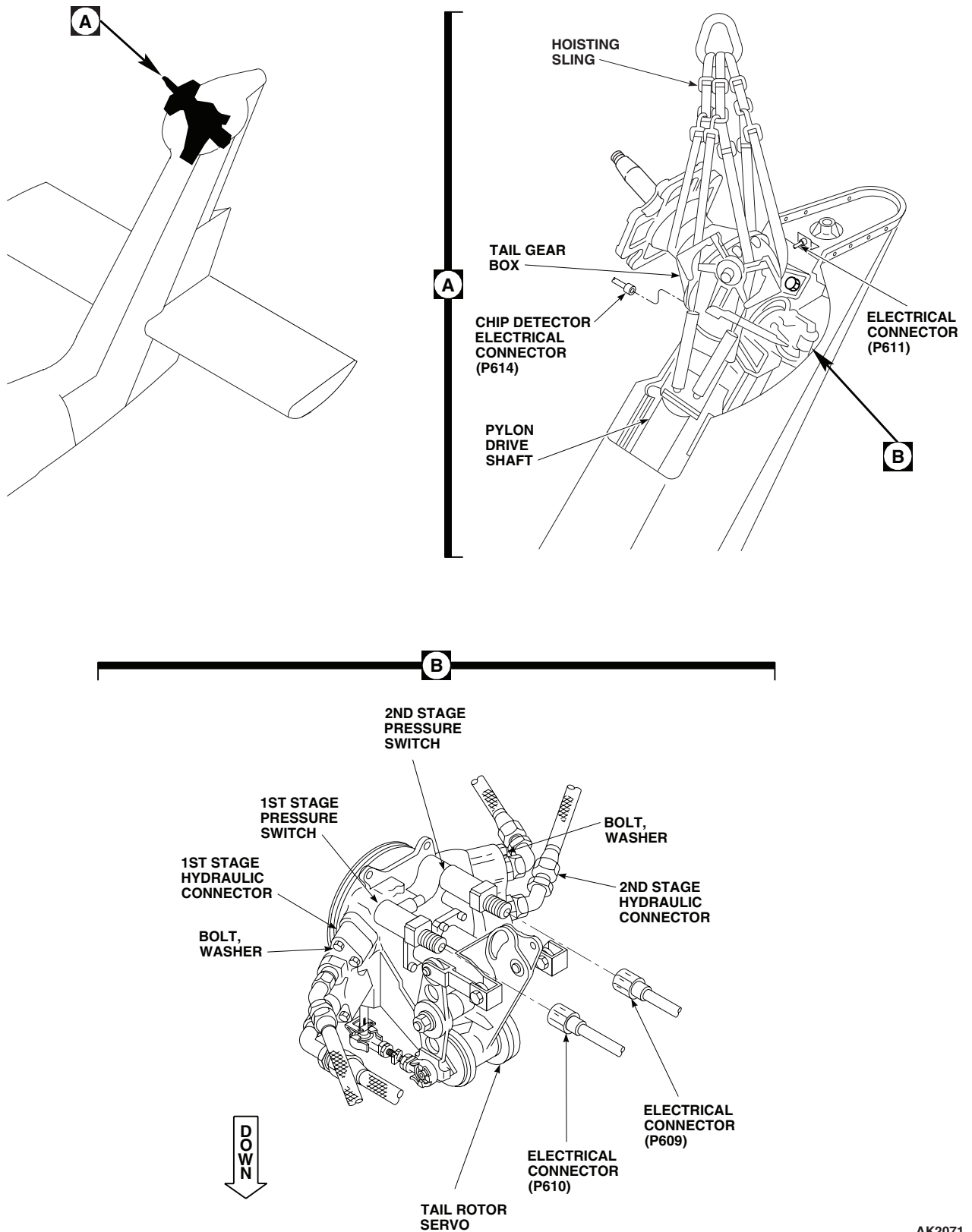
- Tail gear box with servo installed weighs 125 pounds.
- If gear box is being replaced because of contamination, make sure all foreign material from chip detector and oil is carefully packaged and returned with it.

1. Turn off all electrical and hydraulic power.
2. Engage gust lock.
3. Disconnect and tag wiring and grounding straps from anticollision and position lights (WP 0910 00 and WP 0913 00).
4. Remove tail rotor gear box fairing (WP 0333 00).
5. If tail rotor maintenance crane is going to be used, erect crane (WP 1591 00).
6. Remove tail rotor assembly (WP 0585 00).
7. Open tail pylon drive shaft cover.
8. Support pylon tail rotor drive shaft and disconnect from tail gear box input flange. Leave flexible coupling with shaft (WP 0669 00).
9. Disconnect electrical connectors, P610 and P609, from 1st and 2nd stage pressure switches (Figure 1, Detail B).
10. Remove bolts and disconnect 1st and 2nd stage hydraulic connectors from tail rotor servo.
11. Disconnect electrical connector, P611, on top of pylon structure (Figure 1, Detail A).
12. Disconnect tail rotor control cables (WP 1145 00).
13. Disconnect chip detector electrical connector, P614, from chip detector/temperature sensor.
14. Attach hoisting sling to tail gear box.

WARNING

- Hoist cable may break, causing injury to personnel or damage to equipment, if hoist is used to free a sticking component being removed from helicopter. Do not use too much force when removing a component.
 - Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.
15. Attach sling to maintenance crane or overhead hoist with enough capacity to lift gear box. Adjust length of sling straps. Take up any slack.
 16. If blade deice kit is installed, remove rotor slipring (WP 0683 00 and WP 1247 00).
 17. Remove bolts securing tail gear box to tail gear box support.
 18. Lift gear box from pylon and place on work surface. Remove sling.
 19. If new gear box is being installed, remove spring cylinders, quadrant, quadrant support, spring cylinder support, pulleys and tail rotor servo assembly from gear box (WP 1155 00, WP 1152 00, WP 1153 00, WP 1156 00, WP 1146 00, and WP 1148 00).

REMOVE - Continued



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Figure 1. Remove Tail Gear Box.

INSPECT**NOTE**

When inspecting tail rotor gear box fitting pay particular attention to joining area of fitting to front spar caps, in area of high strength fasteners.

1. Inspect tail rotor gear box fitting for surface cracks around lower barrel-nut holes. **NO CRACKS ALLOWED.**
2. Inspect inside surfaces of fitting where attached to front spar cap angles, using inspection mirror and flashlight through access hole in front spar web. **NO CRACKS ALLOWED.** A one time flight to depot is allowed.

INSTALL

1. If removed, install tail rotor servo assembly, spring cylinder support, quadrant support, pulleys, quadrant and spring cylinders on replacement tail gear box (WP 1155 00, WP 1152 00, WP 1153 00, WP 1156 00, WP 1146 00, and WP 1148 00).

CAUTION

Damage to barrel-nut fitting will result if barrel-nut is incorrectly installed. To prevent damage to fitting make sure nut is installed with barrel (round side) towards bolthead.

NOTE

Make sure that only square barrel-nuts are used.

2. Check break away torque of tail gear box barrel-nuts in support as follows (Figure 3):
 - a. Install bolt in each nut until two threads protrude from nut.
 - b. Using a torque wrench, back out each bolt. **BREAKAWAY TORQUE SHALL BE AT LEAST 14 INCH-POUNDS.**
 - c. If breakaway torque is less than 14 inch-pounds, replace barrel-nut.
3. Remove any protective sealant from gear box mounting surfaces and from airframe mounting pads with nonmetallic scraper.
4. Attach hoisting sling to gear box.
5. Lift tail gear box into place on airframe mounting pads. Make sure wiring harness going under gear box is in position and not crushed by gear box.
6. Install gear box to airframe with mounting bolts. Tighten bolts as required until bolthead makes light contact with gear box mounting feet. **DO NOT EXCEED 100 INCH-POUNDS.**
7. Check for gaps between tail gear box mounting feet and mounting surfaces using feeler gauge. **MAXIMUM GAP IS 0.030-INCH.**
8. If gap exceeds 0.030-inch under any mounting foot, install shim to bring gap within limits as follows:
 - a. Remove tail gear box from tail pylon.

NOTE

If a shim was previously installed on a mounting pad which now requires a thicker shim, existing shim must be removed (Figure 2, Detail A).

- b. Remove rivets securing shim to airframe (TM 1-1500-204-23).
- c. Remove original gear box shim and gear box spacer from mounting pad using a nonmetallic scraper.

INSTALL - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- d. Remove old sealing compound from mounting pad using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

NOTE

- Use only one shim, one spacer and two rivets on each mounting pad.
 - Replacement shim thickness must be original shim thickness plus thickness required for correction.
- e. Locally make replacement gear box shim and gear box spacer (WP 1805 00).
 - f. Position replacement gear box shim and gear box spacer on airframe mounting pad.
 - g. Mark location of fastener holes on shim using nonpermanent marker.
 - h. Drill holes in shim using No. 29 drill.
 - i. Apply adhesive primer, Item 241, WP 1803 00, to all gear box shim and gear box spacer surfaces. Allow to dry.
 - j. Apply a brush coat of sealing compound, Item 273, WP 1803 00, to mounting pad using acid brush, Item 57, WP 1803 00.
 - k. Install shim and spacer, 70219-06000-101, on forward mounting pads as required.
 - l. Install shim and spacer, 70219-06000-102, on aft mounting pads as required.
 - m. Secure shim and spacer to airframe using rivets (TM 1-1500-204-23).
 - n. Lift gear box into place on airframe mounting pads. Make sure wiring harness going under gear box is in position and not crushed by gear box.
9. **QA** Install mounting bolts and secure gear box to airframe. **TORQUE BOLTS TO 656 - 724 INCH-POUNDS (Figure 3).**
 10. **QA** Using feeler gauge, measure gap between gear box mounting feet and mounting pad. **NO GAP ALLOWED.** If any gaps exist do the following:
 - a. Remove tail gear box from tail pylon.
 - b. Inspect tail gear box housing (WP 0686 00).
 - c. Fluorescent penetrant inspect gear box mounting legs (TM 1-1520-265-23). **NO CRACKS ALLOWED.** If any cracks exist in gear box mounting legs, replace gear box.
 - d. If no cracks are found repeat step 6 through step 10.
 11. Remove tail gear box from airframe supports.
 12. Coat tail gear box airframe mounting surface with molybdenum disulfide, Item 197, WP 1803 00.
 13. Coat tail gear box mounting feet with sealing compound, Item 273, WP 1803 00.
 14. Lift gear box into place. Make sure wiring harness going under gear box is in position and not crushed by gear box.

INSTALL - Continued

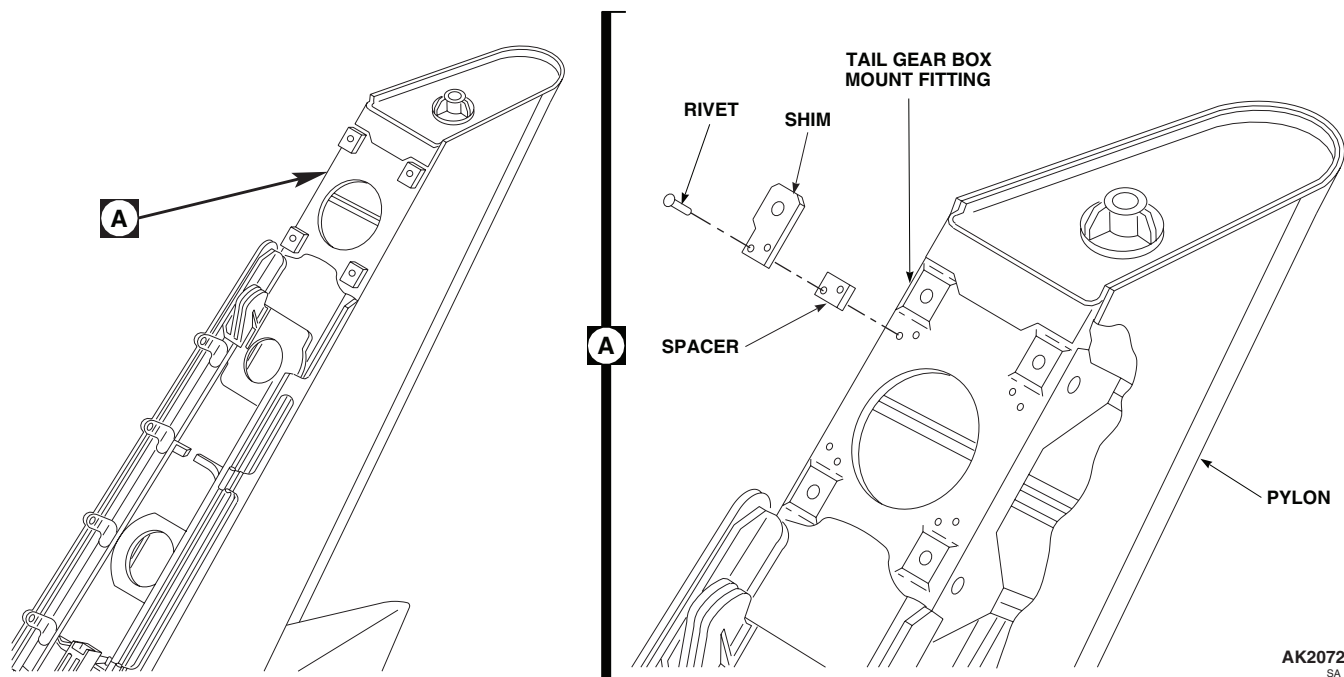


Figure 2. Shim Tail Gear Box.

WARNING**CSI**

Verification of minimum torque of bolts is a critical characteristic.

15. Apply a light coat of antiseize compound, Item 49, WP 1803 00, to mounting bolts.
16. Apply epoxy primer coating kit, Item 120, WP 1803 00, to shoulders of bolt and to areas of gear box around bolt-holes.
17. **QA** Install bolts while primer is still wet. It may be easier to install top two bolts first. **TORQUE BOLTS TO 656 - 724 INCH-POUNDS.**
18. Remove sling.
19. Form a fillet with squeezed out sealing compound around base of mounting feet.
20. Mask off adjacent areas using pressure sensitive adhesive tape, Item 326, WP 1803 00, to prevent overspray.
21. Apply corrosion preventive compound, Item 106, WP 1803 00, to entire area of mounting feet.
22. Allow to dry for 10 to 30 minutes and apply a second coat of corrosion preventive compound, Item 106, WP 1803 00.
23. Inspect every three days and reapply two coats of corrosion preventive compound, Item 106, WP 1803 00, as necessary until bolt torque stabilizes.

INSTALL - Continued

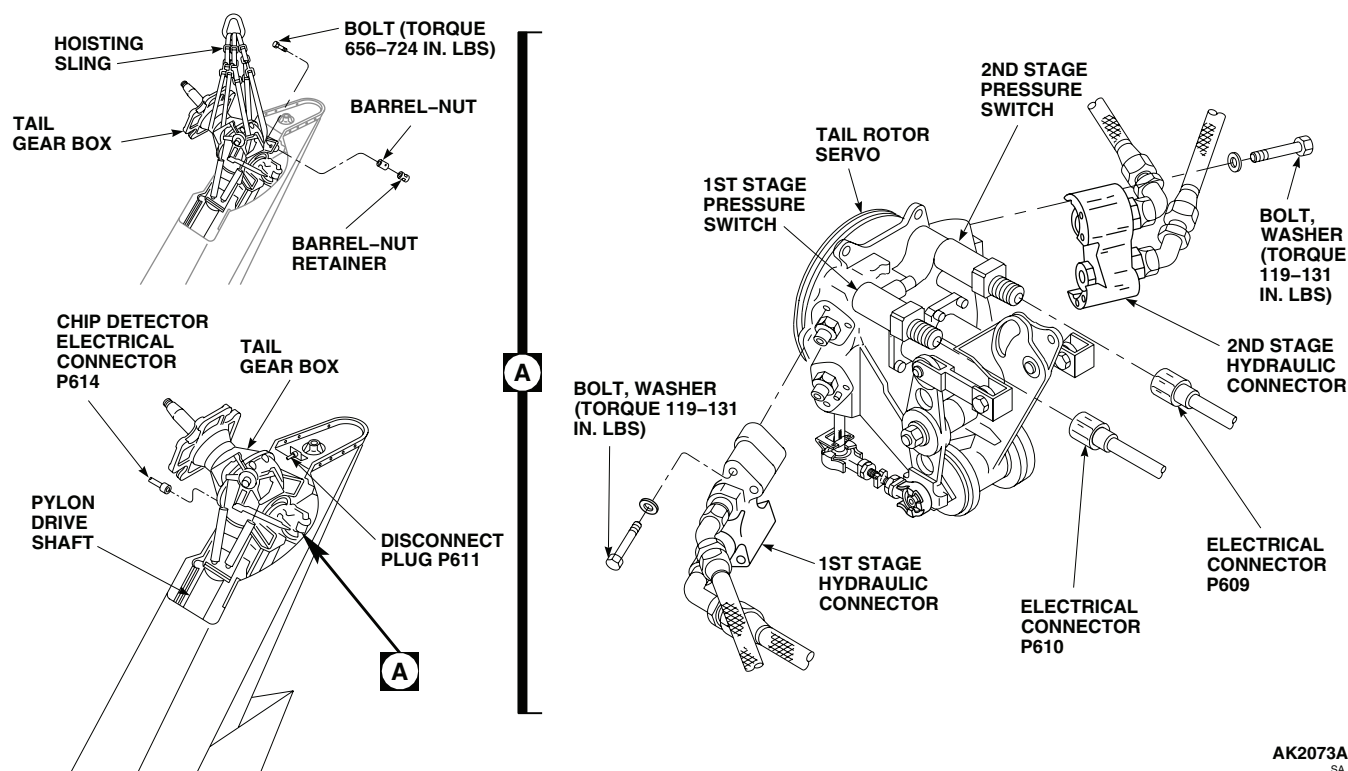


Figure 3. Install Tail Gear Box.

24. **QA** If blade deice kit was installed and inboard retention plate removed, install tail rotor slipping rotor to retention plate and retention plate to tail gear box (WP 0683 00 and WP 1247 00).

WARNING

Injury to personnel and damage to equipment will result if inboard retention plate is installed without nonelectrical wire on bolts and jamnuts. Make sure bolts and jamnuts are lockwired prior to installation of tail rotor blades.

25. **QA** Check inboard retention plate bolts and jamnuts for nonelectrical wire. If required, using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts and jamnuts.
26. Check pylon drive shaft to tail gear box coupling alignment as follows:
- Support pylon drive shaft and connect shaft to gear box flange using bolts and spherical washers. Do not install nuts or shims.
 - Measure and record gap between flange and spherical washer at the three attachment points. Do not rotate shaft when measuring gaps.
 - QA** Calculate difference between maximum and minimum gaps and record. **MAXIMUM GAP DIFFERENCE MUST NOT EXCEED 0.017-INCH.**
 - If gap difference exceeds limit, remove gear box and recheck tail gear box mounting.
 - Support drive shaft and remove bolts and washers.
27. Connect pylon tail rotor drive shaft to tail gear box (WP 0669 00).

INSTALL - Continued

28. Connect tail rotor control cables (WP 1145 00).
29. Install tail gear box chip detector/temperature sensor (WP 0688 00).
30. Inspect and treat electrical connector (WP 1703 00).
31. **QA** Connect chip detector electrical connector, P614, to chip detector/temperature sensor (Figure 3).
32. Connect first and second stage hydraulic connectors to tail rotor servo. Install bolts and washers. **TORQUE BOLTS TO 119 - 131 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts (Figure 3, Detail A) (WP 1148 00).
33. **QA** Connect electrical connectors, P610, to first stage pressure switch and P609 to second stage pressure switch.
34. **QA** Connect electrical connector, P611, on top of pylon structure (Figure 3).
35. Install tail rotor (WP 0585 00).
36. Do a tail rotor rig check (WP 1039 00).

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

37. Service tail gear box (WP 1609 00).
38. Inspect tail rotor servo valve input linkage (WP 1148 00).
39. Make sure area is clean and free of foreign material.
40. **QA** Install tail rotor gear box fairing around tail gear box (WP 0333 00).
41. Connect wiring and grounding straps to anticollision light and position light (WP 0910 00 and WP 0913 00).

WARNING**CSI**

Verification of indicated torque during initial assembly or installation and subsequent stabilization retorques is a critical characteristic.

42. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1 to make sure tail rotor gear box mounting bolt torque check, section IV tail rotor drive shaft, and tail rotor cable tension check are shown as due 9 to 11 flight hours after installation (WP 1693 00 and WP 1695 00).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

43. After torque has been stabilized, remove corrosion preventive compound, using machinery towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Apply sealing compound, Item 273, WP 1803 00, to joints around boltheads, nuts, and washers.
44. Apply epoxy primer coating kit, Item 120, WP 1803 00, over sealant. Apply epoxy coating kit, Item 118, WP 1803 00, over primer.

INSTALL - Continued

45. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

TAIL GEAR BOX INPUT SEAL, PLUG SEAL AND FLANGE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Ballpeen Hammer, GGG-H-86
Container, 2-Quart
Drain Hose, H538A-3
Mallet, GGG-H-33
Nylon Wedge
Powertrain Repairer's Toolkit, SC 5180-99-B13
Seal Installation Tool, Locally-Made, Figure 157,
WP 1805 00
Seal Puller, Locally-Made, Figure 160, WP 1805 00
Torque Reactor, 70700-20688-041
Torque Wrench, 0 - 600 ft. lbs.
Wrench, 70700-20687-101

Material/Parts

Bolt, 10-32
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Grease, Aircraft, Item 131, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00

Lubricating Oil, Aircraft Turbine, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00
Screws, No. 8 Self-Tapping
Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
WP 0333 00
WP 0621 00
WP 0669 00
WP 1609 00
WP 1703 00
WP 1803 00
WP 1805 00

REMOVE

1. Turn off all electrical power.
2. Remove tail rotor gear box fairing (WP 0333 00).
3. Remove section IV tail rotor drive shaft (WP 0669 00).
4. Disconnect electrical connector, P614, from chip detector. Remove chip detector. Connect drain hose, H538A-3 and drain oil into a 2-quart container. Remove drain hose and reinstall chip detector.
5. Remove cotter pin and using torque reactor and wrench, remove flange nut (Figure 1, Detail A).
6. Remove input shaft plug by inserting any 10-32 bolt into plug and wiggle plug from shaft.
7. Pull flange off shaft. If necessary, tap flange gently and evenly with a rawhide/rubber mallet.
8. Remove retaining ring holding in seal.

NOTE

Self-tapping screws may be used as an alternate to remove oil seal if seal puller is not available. If seal puller is not available, proceed to step 13.

9. Locally make nylon wedge.
10. Locally make seal puller (WP 1805 00).

REMOVE - Continued

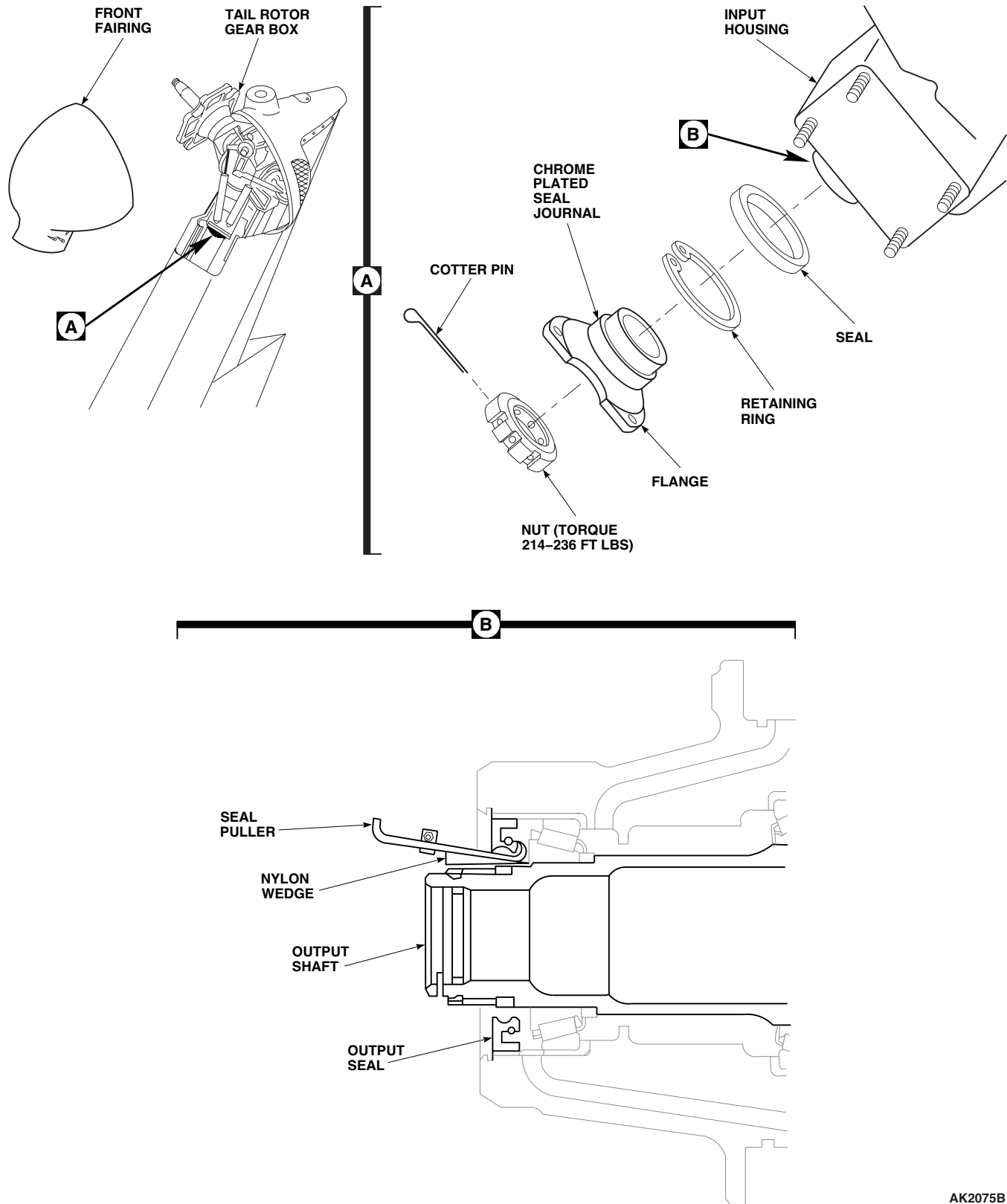
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Figure 1. Replace Input Seal and Flange.

REMOVE - Continued

11. Insert nylon wedge between seal puller and shaft to protect gear box (Figure 1, Detail B).

CAUTION

Damage to tail gear box will result, if too much force is applied while using hammer and seal puller. Gear box housing will crack. Use care when using hammer and seal puller.

12. Gently tap end of seal puller with an 8-ounce ballpeen hammer in an outward direction, away from gear box.
13. Moving seal puller and wedge in a clockwise direction, repeat steps 10 and 11 until seal is removed.

WARNING

- Damage to tail gear box will result if self-tapping screws completely penetrate input seal. Screws will damage bearing inside gear box. Install screws so that only outer face of input seal is penetrated.
 - Damage to tail gear box will result if metal chips or shavings enter gear box by drilling or punching holes in seal. Metal chips or shavings will contaminate gear box oil. Do not drill or punch holes in seal.
14. If seal puller is not available, remove oil seal as follows:
 - a. Thread three self-tapping screws into holes in seal.
 - b. Carefully pull out seal.

INSPECT/REPAIR

1. Repair of tail gear box flange is identical to intermediate gear box flange repair. See WP 0621 00 for procedure to repair tail gear box flange.
2. Inspect flange nut for corrosion. **REPLACE NUT IF PITS ARE 0.030 INCH DEEP, IF PITTING COVERS MORE THAN 10% OF MATING SURFACE OR IF NUT THREADS SHOW SIGNS OF CORROSION.**
3. Remove corrosion from flange nut (TM 1-1500-344-23).
4. Inspect inside diameter for corrosion.
5. Remove corrosion, if any, from shaft hole according to (TM 1-1500-344-23) and step 8..
6. Inspect splines for corrosion or wear. Evidence of corrosion of gear box splines is cause for removal of gear box. **NO WEAR INTO BASE METAL IS PERMITTED (NO FRETTING OR SPALLING).**

NOTE

Do not paint threads or mating surface of flange nut.

7. Apply epoxy primer coating kit, Item 120, WP 1803 00, to flats and exposed face of flange nut. Do not paint threads or mating surface of flange nut.
8. Visually inspect input shaft for damage such as nicks, scratches, gouges, and corrosion. Using burnisher or equivalent, blend out damaged area. **MAXIMUM ALLOWABLE DEPTH AFTER REPAIR IS 0.005-INCH.**

INSTALL

1. Install new seal on plug and lubricate with aircraft turbine lubricating oil, Item 182, WP 1803 00, aircraft turbine lubricating oil, Item 184, WP 1803 00, or helicopter transmission lubricating oil, Item 187, WP 1803 00,
2. Insert plug into shaft

INSTALL - Continued

3. Locally make seal installation tool (WP 1805 00).
4. Coat inside diameter of seal with aircraft grease, Item 131, WP 1803 00. Coat outside diameter of seal and seal mating surface (flange sleeve) with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.

CAUTION

Input housing will be damaged if seal is incorrectly installed in housing. Make sure seal is flush with retaining ring groove, or no more than 0.020-inch below retaining ring groove.

5. Locally make seal installation tool (WP 1805 00). Install new seal by gently tapping into housing. Be careful not to cock or distort seal.
6. **QA** Apply epoxy primer coating kit, Item 120, WP 1803 00, to retaining ring and install in housing while wet.
7. Apply sealing compound, Item 273, WP 1803 00, at corner of plug.
8. Apply epoxy primer coating kit, Item 120, WP 1803 00, to exposed area of plug.
9. Apply sealing compound, Item 273, WP 1803 00, to housing boss to prevent water entrapment and corrosion above retaining ring groove.
10. Touch up sealing compound with epoxy primer coating kit, Item 120, WP 1803 00.
11. **QA** Slide flange on shaft (Figure 1, Detail A).

NOTE

Be careful not to cock flange. If necessary, tap flange gently and evenly with a rawhide/rubber mallet.

12. **QA** Make sure bevel side of nut is toward seal and that nut can be fully run on by hand. Using torque reactor and wrench, **TORQUE NUT AS FOLLOWS: THE TORQUE RANGE IS 214 - 236 FOOT-POUNDS. TORQUE RETAINING NUT TO 214 FOOT-POUNDS.** Install new cotter pin. To align pin hole, turn nut in tightening direction. **DO NOT EXCEED MAX TORQUE. IF MAX TORQUE IS EXCEEDED AND NUT DOES NOT ALIGN, REMOVE NUT AND REPLACE IT WITH A NEW NUT AND REPEAT TORQUING PROCEDURE.**
13. Install section IV tail rotor drive shaft (WP 0669 00).

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

14. Service tail gear box (WP 1609 00).
15. Inspect and treat electrical connector (WP 1703 00).
16. **QA** Connect electrical connector, P614, to chip detector/temperature sensor.
17. Make sure area is clean and free of foreign material.
18. Install tail gear box fairing (WP 0333 00).
19. Run up helicopter until gear box oil warms up. Shut down and check for leaks.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

DRIVE SYSTEM

INNER RETENTION PLATE AND SEALS

This WP supersedes WP 0683 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Allen Wrench, 5/32-inch
 Ballpeen Hammer, GGG-H-86
 Depth Gage, Dial Indicating, 643J
 Fluorescent Penetrant Inspection Kit, XMA101
 Mallet, GGG-H-33
 Micrometer, GGG-H-41
 Nonmetallic Drift
 Nylon Wedge, Locally-Made, Figure 268, WP 1805 00
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Seal Installation Tool, Locally-Made, Figure 157, WP 1805 00
 Seal Puller, Locally-Made, Figure 160, WP 1805 00
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Lubricating Oil, Aircraft Turbine, Item 184, WP 1803 00
 Lubricating Oil, Helicopter Transmission, Item 187, WP 1803 00
 Marker, Nonpermanent
 Screws, No. 8 Self-Tapping
 Sealing Compound, Item 273, WP 1803 00
 Sealing Compound, Item 282, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 325, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Alcohol, Denatured, Item 48, WP 1803 00
 Assembly Fluid, Item 54, WP 1803 00
 Cloth, Abrasive, Item 79, WP 1803 00
 Cloth, Abrasive, Item 77, WP 1803 00
 Damping Fluid, Item 113, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Grease, Aircraft, Item 131, WP 1803 00
 Grease, Aircraft, Item 132, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182, WP 1803 00

References

TM 1-1520-265-23
 TM 1-1500-345-23
 WP 0333 00
 WP 0381 00
 WP 0585 00
 WP 1246 00
 WP 1609 00
 WP 1694 00
 WP 1803 00
 WP 1805 00

REMOVE

WARNING

CSI

Exposed surfaces of bevel gear are critical surfaces which must not be damaged during and after removal of inner retention plate. Use extreme caution when removing and provide protective covering over critical surfaces after removal of inner retention plate.

1. Remove tail rotor gear box fairing (WP 0333 00).
2. Remove tail rotor assembly (WP 0585 00).

REMOVE - Continued

3. Loosen bolts from seal housing (Figure 1, Sheet 1).

NOTE

It is not necessary to completely remove bolts. Bolts may be loosened enough to unthread nut with bolts and seal housing still attached.

4. Remove retaining ring, pitch change scraper seal, nut and washer.

NOTE

If cones do not loosen after tapping with mallet, apply damping fluid, Item 113, WP 1803 00, or equivalent, wait about 30 minutes and tap retention plate again. If necessary, repeat procedure.

5. Remove outer cones. Tap retention plate with a mallet to loosen cones.
6. Check outer cones for damage and fretting.
7. Slide inner retention plate off. Cover all exposed surfaces of bevel gear with pressure sensitive adhesive tape, Item 325, WP 1803 00.
8. If blade deice kit is installed, remove tail rotor slipring rotor (WP 1246 00).
9. Remove retaining ring holding output seal.

NOTE

Self-tapping screws may be used as an alternate to remove oil seal if seal puller is not available. If seal puller is not available, proceed to step 15.

10. Locally make seal puller (WP 1805 00).
11. Locally make nylon wedge (WP 1805 00).
12. Insert nylon wedge between seal puller and shaft to protect gear box (Figure 1, Sheet 1, Detail D).

CAUTION

Damage to tail rotor gear box will result, if too much force is applied while using hammer and seal puller. Gear box housing will crack. Use care when using hammer and seal puller.

13. Gently tap end of seal puller with an 8-ounce ballpeen hammer in an outward direction, away from gear box.
14. Moving seal puller and wedge in a clockwise direction, repeat step 12. and step 13. until seal is removed.

CAUTION

- Damage to tail rotor gear box will result if self-tapping screws completely penetrate output seal. Screws will damage bearing inside gear box. Install screws so that only outer face of output seal is penetrated.
- Damage to tail rotor gear box will result if metal chips or shavings enter gear box by drilling or punching holes in seal. Metal chips or shavings will contaminate gear box oil. Do not drill or punch holes in seal.

15. If seal puller is not available, remove oil seal as follows:
 - a. Thread three self-tapping screws into holes in seal.
 - b. Carefully pull out seal.

REMOVE - Continued

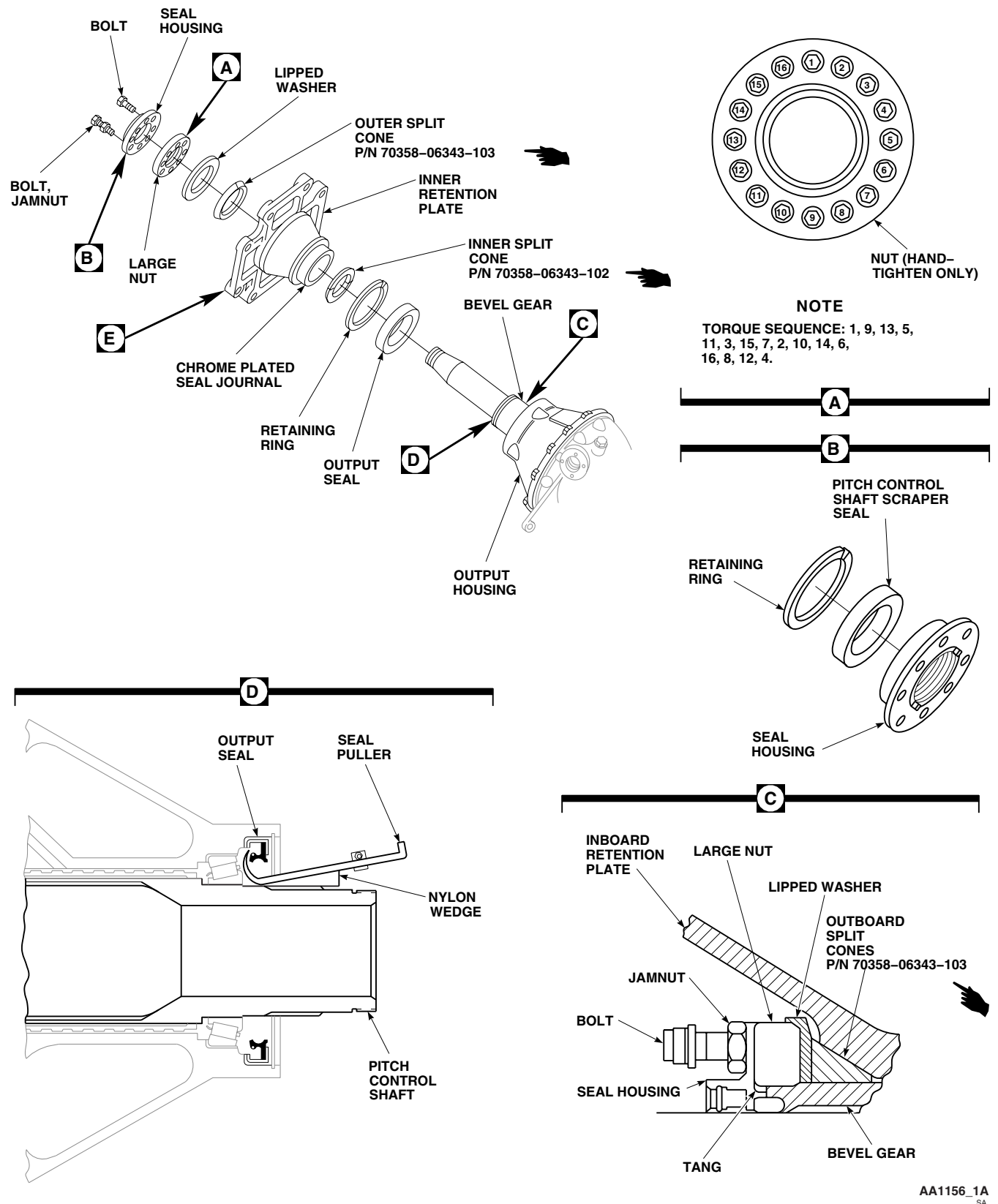
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Figure 1. Replace Inner Retention Plate and Seals. (Sheet 1 of 2)

REMOVE - Continued

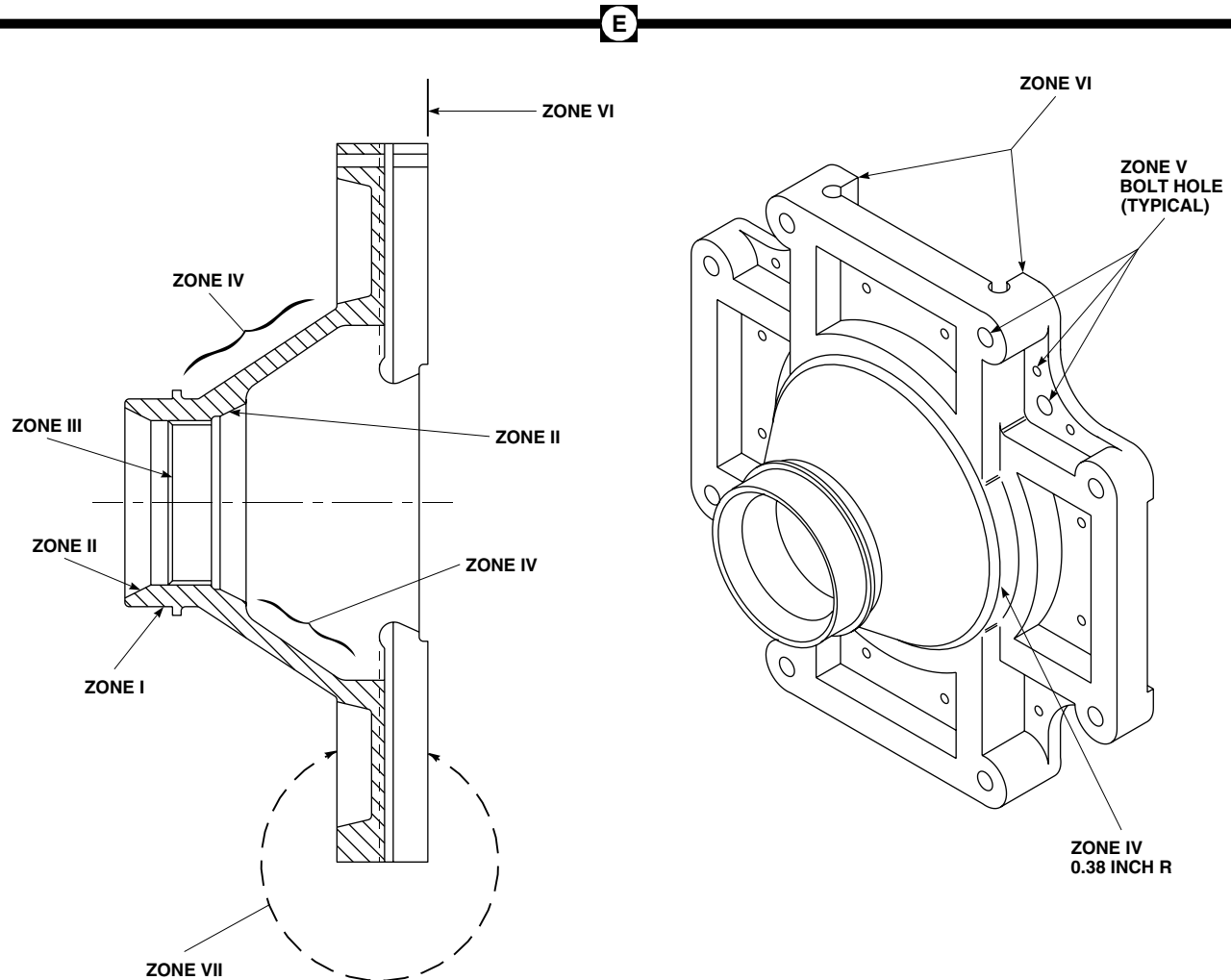
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Figure 1. Replace Inner Retention Plate and Seals. (Sheet 2 of 2)

NOTE

Pitch control shaft scraper seal should be replaced whenever the output seal is replaced.

16. Remove retaining ring from seal housing. Turn seal housing upside down (boltheads down) and drive out pitch control shaft scraper seal, using a nonmetallic drift (Figure 1, Sheet 1, Detail B).
17. Check inner cones for damage and fretting. Remove cones only if they are being replaced.

INSPECT/REPAIR

1. Visually inspect inboard retention plate for cracks. **NO CRACKS ALLOWED.** Replace retention plate if cracks are found.

INSPECT/REPAIR - Continued

2. If inboard retention plate was removed due to leakage, inspect inboard retention plate Zone I as follows (Figure 1, Sheet 2, Detail E):

NOTE

Seal contact area is identified by a polished circumferential area or slight groove.

- a. Check Zone I seal contact area for grooving. **MAXIMUM GROOVE DEPTH IS 0.003-INCH (0.006-INCH ON CHROME PLATED SEAL JOURNAL OUTSIDE DIAMETER)**. Replace inboard retention plate if grooving is beyond limits.
 - b. Nicks and scratches in Zone I seal contact area that do not penetrate into base metal may be blended out. **NO REPAIR INTO BASE METAL ALLOWED IN SEAL CONTACT AREA**. Replace inboard retention plate if damage or repair in seal contact area penetrates into base metal.
 - c. Nicks and scratches in Zone I areas other than seal contact area may be blended out up to 0.010-inch. Replace inboard retention plate if damage is beyond limits.
3. Inspect Zone II tapered cone seats for damage. **NO FRACTURES, CRACKS, OR PITS ALLOWED**. Minor fretting and scratches on cone seats may be blended out up to 0.002-inch using abrasive cloth, Item 77, WP 1803 00. **REPAIR AREA NOT TO EXCEED 10% OF TOTAL SURFACE AREA**. Replace inboard retention plate if tapered cone seats are damaged beyond limits.
4. Inspect Zone III internal spline area for damage.
- a. **NO DAMAGE MORE THAN UNIFORM WEAR FROM MATING SPLINE ALLOWED. WEAR STEP ON SPLINE IS LIMITED TO 0.002-INCH.**
 - b. **MINOR SCRATCHES UP TO 0.002-INCH DEEP ARE ACCEPTABLE.**
 - c. Replace inboard retention plate if internal splines are damaged beyond limits.
5. Inspect nylon shims as follows:
- a. Remove screws securing nylon shim to inboard retention plate. Remove shims.
 - b. Inspect nylon shims 1 inch in from edge of each corner using micrometer. **SHIM THICKNESS MUST BE 0.123 to 0.126-INCH**. Replace shims not within limits or if surface is not flat.
6. Inspect threaded inserts as follows:
- a. Thread screws into inserts and check drag torque required to install screws. **DRAG TORQUE SHALL BE 2 - 18 INCH-POUNDS.**
 - b. If drag torque is not within limits, replace screw and clean insert. Recheck drag torque. If drag torque is still not within limits, replace insert as follows:
 - (1) Remove insert using 5/32 Allen wrench.
 - (2) Clean tapped hole in retention plate with denatured alcohol, Item 48, WP 1803 00. Blow dry cleaned area using clean, dry compressed air.
 - (3) Inspect tapped hole in retention plate for thread yielding. Replace retention plate if more than 10% of thread had yielded.
 - (4) Clean outer threads of insert with denatured alcohol, Item 48, WP 1803 00. Blow dry cleaned area using clean, dry compressed air.
 - (5) Apply a thin coat of sealing compound, Item 282, WP 1803 00, to first two threads of insert.
 - (6) Using 5/32 Allen wrench, install insert flush to 0.025-inch below surface within tapped hole, to a driving torque of 25 - 300 inch-pounds.

INSPECT/REPAIR - Continued

7. Position nylon shims on retention plate. Thread screws into inserts and note drag torque required to install screws.
TORQUE SCREWS TO A VALUE 4 - 10 INCH-POUNDS GREATER THAN DRAG TORQUE.
8. Visually inspect inboard retention plate for nicks, gouges, fretting, and scoring.
9. Remove raised material and blend out damage using abrasive wheel, Item 350, WP 1803 00. **REPLACE RETENTION PLATE IF (Figure 1, Sheet 2, Detail E):**
 - a. **CORNER DAMAGE IN ZONE VII AFTER BLENDING IS GREATER THAN 0.100-INCH DEEP, OR 0.040-INCH DEEP IN BOLTHOLE AREAS. MULTIPLE AREAS OF DAMAGE MUST BE AT LEAST 1 INCH APART.**
 - b. **SURFACE DAMAGE AFTER BLENDING IS GREATER THAN 0.020 -INCH DEEP IN OPEN AREAS, OR GREATER THAN 0.020-INCH DEEP IN BOLTHOLE AREAS. MULTIPLE AREAS OF DAMAGE MUST BE AT LEAST 1 INCH APART.**
 - c. **SURFACE DAMAGE IN ZONE VI MATING AREAS AFTER BLENDING IS GREATER THAN 0.020-INCH DEEP, OR COVERS MORE THAN 10% OF MATING AREA. BLEND TO REMOVE ANY DAMAGE USING A 0.050-INCH BLEND RADIUS.**
 - d. **FRETTING IN ZONE VI AREAS THAT MATE WITH OUTBOARD RETENTION PLATE AFTER BLENDING IS GREATER THAN 0.010-INCH DEEP, OR COVERS MORE THAN 10% OF MATING AREA. BLEND TO REMOVE ANY DAMAGE USING A 0.050-INCH BLEND RADIUS.**
 - e. **BOLTHOLE BORE DAMAGE IN ZONE V EXCEEDS 0.010-INCH DEEP OR COVERS MORE THAN 10% OF BORE SURFACE AREA.**

CAUTION

Damage to inboard retention plate will result if care is not taken when inspecting or reworking Zone IV areas. Zone IV areas are critical areas of the inboard retention plate. Use extreme care when inspecting or reworking all Zone IV areas.

- f. **DAMAGE OR REWORK IN ZONE IV AREAS (INCLUDING 0.38-INCH RADIUS) IS DEEPER THAN 0.0003-INCH. MULTIPLE AREAS OF DAMAGE MUST BE AT LEAST 1 INCH APART.**
10. Touch up paint on repaired areas (WP 0381 00 and TM 1-1520-345-23).
11. Inspect split cones for scratches or pits and repair using abrasive cloth, Item 85, WP 1803 00, as follows:
 - a. **BLEND SCRATCHES OR PITS TO FORM A SMOOTH TRANSITION TO SURROUNDING AREA UP TO 0.005-INCH DEEP NO MORE THAN 10% OF TOTAL SURFACE AREA.**
 - b. Replace split cones if repairs are more than limits.
12. Perform fluorescent dye penetrant inspection of repaired cones (TM 1-1520-265-23).
13. Visually inspect exposed surfaces of bevel gear as follows:

WARNING**CSI**

Nicks, dents, scratches, and tool or scuff marks shall not exceed limits and existence of silver plating on external splines are critical characteristics.

- a. Inspect splines on bevel gear for wear and damage. **SCRATCHES UP TO 0.0003-INCH DEEP AND UNIFORM WEAR UP TO 0.0003-INCH FROM MATING SPLINE IS PERMITTED AS LONG AS DAMAGE OR WEAR DOES NOT PENETRATE THROUGH SILVER PLATE INTO BASE METAL.** Replace tail gear box if bevel gear splines are damaged or worn beyond limits.

INSPECT/REPAIR - Continued**WARNING****CSI**

Nicks, dents, scratches, and tool or scuff marks shall not exceed limits and existence of silver plating on split cone journal are critical characteristics.

- b. Inspect split cone journals on bevel gear for damage. **SCRATCHES UP TO 0.0003-INCH DEEP AND UNIFORM WEAR UP TO 0.0003-INCH IS PERMITTED AS LONG AS DAMAGE OR WEAR DOES NOT PENETRATE THROUGH SILVER PLATE INTO BASE METAL. NO CRACKS, FRACTURES, OR FRETTING THROUGH SILVER PLATE ALLOWED.** Replace tail gear box if split cone journals on bevel gear are damaged beyond limits.
- c. Inspect threads on bevel gear for damage. **NO STRIPPED, CROSSED, OR FLATTENED THREADS ALLOWED.** Replace tail gear box if threads on bevel gear are damaged.
- 14. Visually inspect the large nut for nicks and scratches. **REPLACE NUT IF NICKS OR SCRATCHES GREATER THAN 0.020-INCH DEEP ARE FOUND. NO CROSSED, STRIPPED, FLATTENED, FRACTURED, OR CRACKED THREADS ALLOWED.**
- 15. Visually inspect lipped washer for nicks and scratches. **REPLACE LIPPED WASHER IF DAMAGE IS DEEPER THAN 0.015-INCH OR REPAIR AREA IS GREATER THAN 20% OF SURFACE AREA.**

INSTALL**WARNING****CSI**

- Exposed surfaces of bevel gear shaft are critical surfaces which must not be damaged during installation of inner retention plate. After removal of protective covering, use extreme caution when installing inner retention plate.
- Verification of cone seat is a critical characteristic.
- Verification of P/N 70358-04363-103 split cone installation in outboard position, P/N 70358-06343-102 split cone installation in inboard position is critical. **IMPROPER INSTALLATION OF SPLIT CONES WILL LEAD TO PREMATURE FAILURE AND POSSIBLE LOSS OF TAIL ROTOR THRUST.**

CAUTION

Damage to gear box and split cones will result if split cones are not cleaned prior to installation. Split cones will not seat properly. Make sure split cones are clean and free of lubricant before installing split cones on gear box.

NOTE

Split cones are installed and removed in matched sets. Make sure serial numbers are the same for each cone in a pair.

1. If inner split cones were removed, thoroughly clean lubricant from cones using machinery wiping towel, Item 344, WP 1803 00.
2. Fully clean all exposed surfaces of the bevel gear with a clean machinery wiping towel, Item 344, WP 1803 00.

INSTALL - Continued

3. Apply assembly fluid, Item 54, WP 1803 00, to inner surfaces of split cones and install with beveled side facing outboard. Make sure gaps between inner cones are equal.
4. If blade deice kit is installed, attach tail rotor slipping rotor to inboard retention plate (WP 1246 00).
5. Coat pitch control shaft scraper seal lightly with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.

CAUTION

Tail gear box and scraper seal will be damaged if seal is improperly installed. Make sure seal is not cocked during installation. Make sure seal is properly seated beyond retaining ring groove before installing retaining ring.

6. **QA** Install pitch control shaft scraper seal in seal housing, using locally made nonmetallic drift (Figure 1, Sheet 1, Detail B).
7. **QA** Install retaining ring in groove in seal housing.
8. Coat inside surface of output seal with a thin coat of aircraft grease, Item 131, WP 1803 00.
9. Coat outside surface of output seal and retention plate liner (liner to seal mating surface) with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.

CAUTION

Damage to tail gear box and scraper seal will result if seal is improperly installed. Make sure seal is not cocked during installation. Make sure seal is properly seated beyond retaining ring groove before installing retaining ring.

10. **QA** Locally make seal installation tool (WP 1805 00). Install new seal by gently tapping into housing. Be careful not to cock or distort seal.
11. **QA** Apply epoxy primer coating kit, Item 120, WP 1803 00, to retaining ring and install in housing while wet.
12. Apply sealing compound, Item 273, WP 1803 00, to housing boss to prevent water entrapment and corrosion above retaining ring groove.
13. Touch up sealing compound with epoxy primer coating kit, Item 120, WP 1803 00.
14. Apply aircraft grease, Item 132, WP 1803 00, to splines on bevel gear and inboard retention plate.

WARNING**CSI**

Verification of cone seat is a critical characteristic.

15. **QA** Install retention plate (Figure 1).

CAUTION

Retention plate and split cones will be damaged if split cones are not clean when installed. Split cones will not seat properly. Make sure split cones are clean and free of lubricant before installing on gear box.

16. Clean lubricant from outer split cones with machinery wiping towel, Item 344, WP 1803 00.

INSTALL - Continued**NOTE**

Make sure outer split cone gaps are positioned 90 degrees from gaps of inner split cones.

17. Apply assembly fluid, Item 54, WP 1803 00, to inner surfaces of split cones and install with beveled side facing inboard. Make sure cones are properly seated, and gaps between cones are the same size. Tap lightly using nonmetallic drift if necessary to seat cones.
18. Before assembling lipped washer, check to make sure that large nut face seats with both sides of washer face.
19. **QA** Install lipped washer on bevel gear with lip outboard from gear box.

CAUTION

Damage to bevel gear threads will result if nut is installed incorrectly. When installing nut on bevel gear, make sure beveled side of nut faces inboard and that nut fits flush with lipped washer.

20. **QA** Install nut on bevel gear with beveled side of nut facing inboard and flush with lipped washer. **TORQUE NUT HANDTIGHT WITH MINIMUM AMOUNT OF TORQUE THAT WILL ACHIEVE SEAL TANG ENGAGEMENT. DO NOT LOOSEN NUT.**
21. **QA** Install seal housing, pitch change scraper seal and retaining ring on bevel gear and engage seal tangs with slots in bevel gear.

NOTE

Make sure seal housing does not rotate when aligning boltholes.

22. Rotate nut in the tightening direction until boltholes line up with holes in seal housing.
23. Install 2 bolts with jam nuts 180 degrees apart and located as close as possible to tangs. Install 14 additional bolts securing seal housing to nut. Install all bolts handtight.

WARNING**CSI**

Verification of torque stabilization is a critical characteristic.

24. **USING INK MARKER, MARK ONE OF THE SIXTEEN BOLTS AS A REFERENCE POINT. TORQUE BOLTS TO 19 - 21 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4 (Figure 1, Sheet 1, Detail A).**
25. **TORQUE BOLTS TO 38 - 42 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4 (Figure 1, Sheet 1, Detail A). REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 40 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**
26. Remove bolts securing seal housing to large nut. Remove seal housing.

INSTALL - Continued**NOTE**

- Tangs may not be fully engaged in slots until cones are properly seated.
- Make sure nut is fully seated to ensure tangs are engaged in slots in bevel gear.
- To ensure proper seating, be sure nut seats flat against lipped washer on both sides.

27. Retighten nut on bevel gear handtight and make sure nut is properly seated against lipped washer (Figure 1, Sheet 1, Detail C). Repeat step 21. and step 22.
28. Repeat step 23.

WARNING**CSI**

Verification of torque stabilization is a critical characteristic.

29. **USING INK MARKER, MARK ONE OF THE SIXTEEN BOLTS AS A REFERENCE POINT. TORQUE BOLTS TO 19 - 21 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4 (Figure 1, Sheet 1, Detail A). REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 20 INCH-POUNDS. DO NOT TIGHTEN JAM-NUTS AT THIS TIME. .REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 20 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**
30. **TORQUE BOLTS TO 38 - 42 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4 (Figure 1, Sheet 1, Detail A). REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 40 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**
31. **TORQUE BOLTS TO 57 - 63 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4 (Figure 1, Sheet 1, Detail A). REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 60 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**
32. **TORQUE BOLTS TO 86 - 94 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4 (Figure 1, Sheet 1, Detail A). REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 90 INCH-POUNDS. DO NOT TIGHTEN JAMNUTS AT THIS TIME.**

NOTE

Upon completion of the final torque sequence at 114 - 126 inch-pounds, the bolted connection may be heated up from friction. If possible, repeat the 114 - 126 inch-pound torque sequence after a 30 minute cool down period.

33. **QA TORQUE BOLTS TO 114 - 126 INCH-POUNDS IN THE FOLLOWING SEQUENCE: 1, 9, 13, 5, 11, 3, 15, 7, 2, 10, 14, 6, 16, 8, 12, AND 4 (Figure 1, Sheet 1, Detail A). REPEAT TORQUE SEQUENCE UNTIL ALL BOLTS HAVE STABILIZED AND NO MOVEMENT IS FELT AT 120 INCH-POUNDS.**

INSTALL - Continued**WARNING**

Use of safety cable is not allowed in this application. Use only nonelectrical wire as specified.

CAUTION

Damage to retention plate will result if tools come in contact while tightening jamnuts. Do not allow tools to come in contact with retention plate when tightening jamnuts.

34. **QA** Tighten jamnuts until a sharp rise in torque is felt; then tighten 1/6 - 1/3 turn more. Using nonelectrical wire, Item 357, WP 1803 00, lockwire all bolts and jamnuts.
35. Install tail rotor assembly (WP 0585 00).

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

36. Service tail gearbox (WP 1609 00).
37. Make sure area is clean and free of foreign material.
38. Check Aircraft Inspection and Maintenance Record, DA Form 2408-13-1, to make sure retention plate bolt torque check is shown due 9 to 11 flight hours after installation (WP 1694 00).
39. Run up helicopter until gear box oil warms up. Shut down and check for leaks.
40. Install tail rotor gear box fairing (WP 0333 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

DRIVE SYSTEM

TAIL GEAR BOX SIGHT PLUG

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Container, 2-Quart
 Drain Hose, H538A-3
 Powertrain Repairer's Toolkit, SC 5180-99-B13

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Lubricating Oil, General Purpose, Item 184,
 WP 1803 00
 Lubricating Oil, Helicopter Transmission, Item 187,
 WP 1803 00

References

WP 0333 00
 WP 0688 00
 WP 1609 00
 WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove tail rotor gear box fairing (WP 0333 00).
3. Remove chip detector/temperature sensor (WP 0688 00).
4. Install drain hose, H538A-3, to self-closing valve, and drain enough oil into a 2 quart container to lower level below bottom of sight plug.
5. Remove drain hose.
6. Install chip detector/temperature sensor (WP 0688 00).
7. Unscrew sight plug. Discard packing (Figure 1, Detail A).

INSTALL

1. **QA** Lubricate packing with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, and install on sight plug.
2. **QA** Screw sight plug into gear box. Tighten sight plug **UNTIL SHARP RISE IN TORQUE IS FELT, THEN TIGHTEN 1/6 - 1/3 TURN MORE.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire plug to gear box housing.

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

3. Service tail gear box (WP 1609 00).

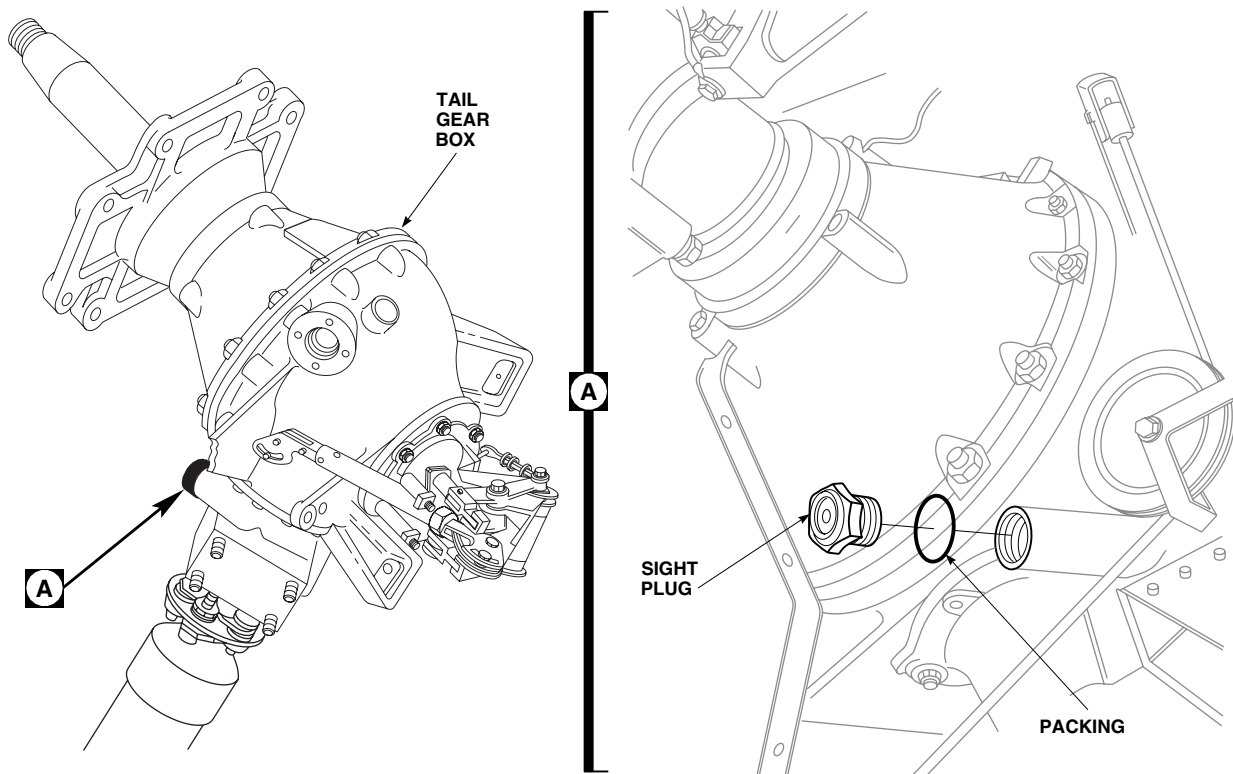
INSTALL - Continued

Figure 1. Replace Sight Plug.

4. Make sure area is clean and free of foreign material.
5. Install tail rotor gear box fairing (WP 0333 00).
6. Run up helicopter until gear box oil warms up. Shut down and check for leaks.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

TAIL GEAR BOX OIL FILLER PLUG

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Nonmetallic Scraper
Powertrain Repairer's Toolkit, SC 5180-99-B13
Torque Wrench, 30 - 200 in. lbs.

Sealing Compound, Item 273, WP 1803 00

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, General Purpose, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00

References

WP 0333 00

WP 1803 00

REMOVE

1. Remove tail rotor gear box fairing (WP 0333 00).
2. Unscrew filler plug, with cap attached. Remove from gear box (Figure 1). Discard packing.

CAUTION

Damage to tail gear box will result if sealing compound is allowed to enter gear box during cleaning. Use extreme care when scraping sealing compound from filler plug boss.

3. Scrape sealing compound from filler plug boss with nonmetallic scraper.

INSTALL

1. **QA** Install packing lubricated with aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00, on filler plug (Figure 1).
2. **QA** Install filler plug on gear box. **TORQUE PLUG TO 120 - 160 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire filler plug to gear box.
3. Seal area around base of filler plug with sealing compound, Item 273, WP 1803 00.
4. Run up helicopter, and check for leaks.
5. Install tail rotor gear box fairing (WP 0333 00).

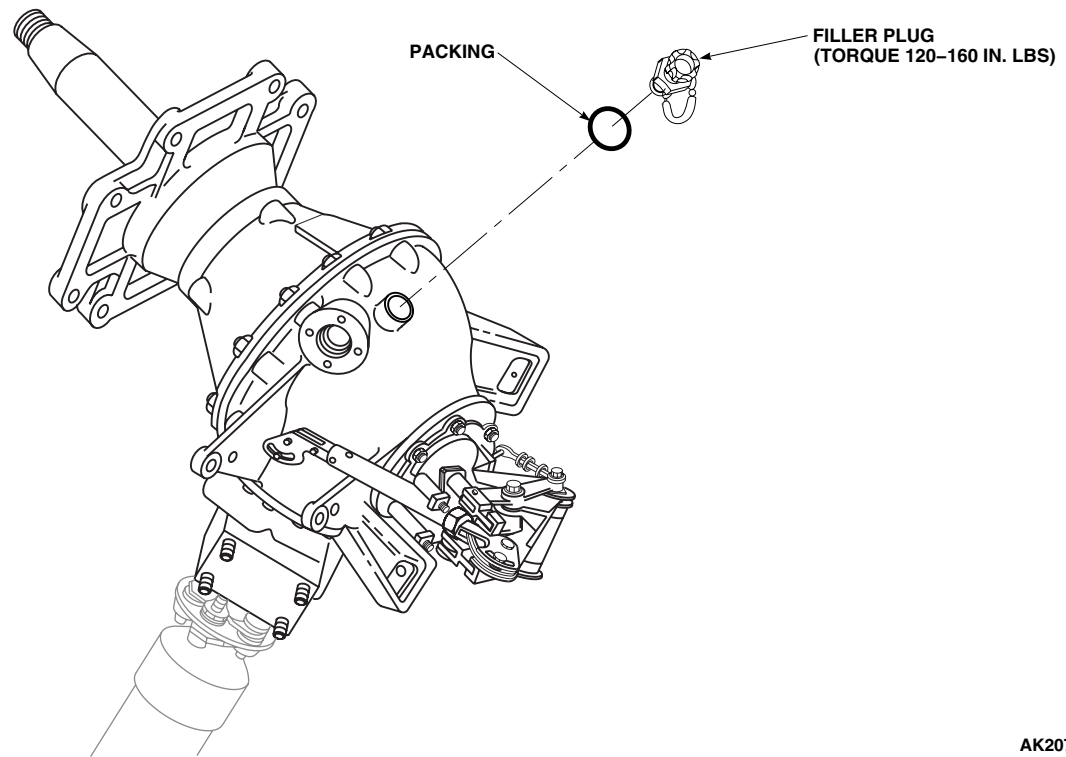
INSTALL - Continued

Figure 1. Replace Oil Filler Plug.

END OF WORK PACKAGE

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AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
TAIL GEAR BOX HOUSING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Airframe Repairer's Toolkit, SC 5180-99-B02
 Depth Gage, 643J
 Fluorescent Penetrant Inspection Kit, XMA101
 Powertrain Repairer's Toolkit, SC 5180-99-B13

Remover, Paint, Item 252, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 Aircraft Structural Repairer MOS 15G (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Chromium Trioxide, Item 65, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00
 Epoxy Coating Kit, Item 118, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Polyurethane, Coating, Item 231, WP 1803 00

References

TM 1-1500-344-23
 TM 1-1500-345-23
 TM 1-1520-265-23
 WP 0381 00
 WP 0639 00
 WP 1803 00

GENERAL REPAIRS**NOTE**

- Minor nicks, gouges, scratches, or corrosion on surface of gear box housing may be repaired. Before starting repair, measure depth of damage to determine whether blending out damage will result in gear box still being within repair limits. Be sure to measure depth of damage from an area which has not been previously repaired. If limits will be exceeded, replace gear box.
- Repair of damaged area may be done with gear box installed.
- The following repair procedures apply to input, center, and output housings (Figure 1).

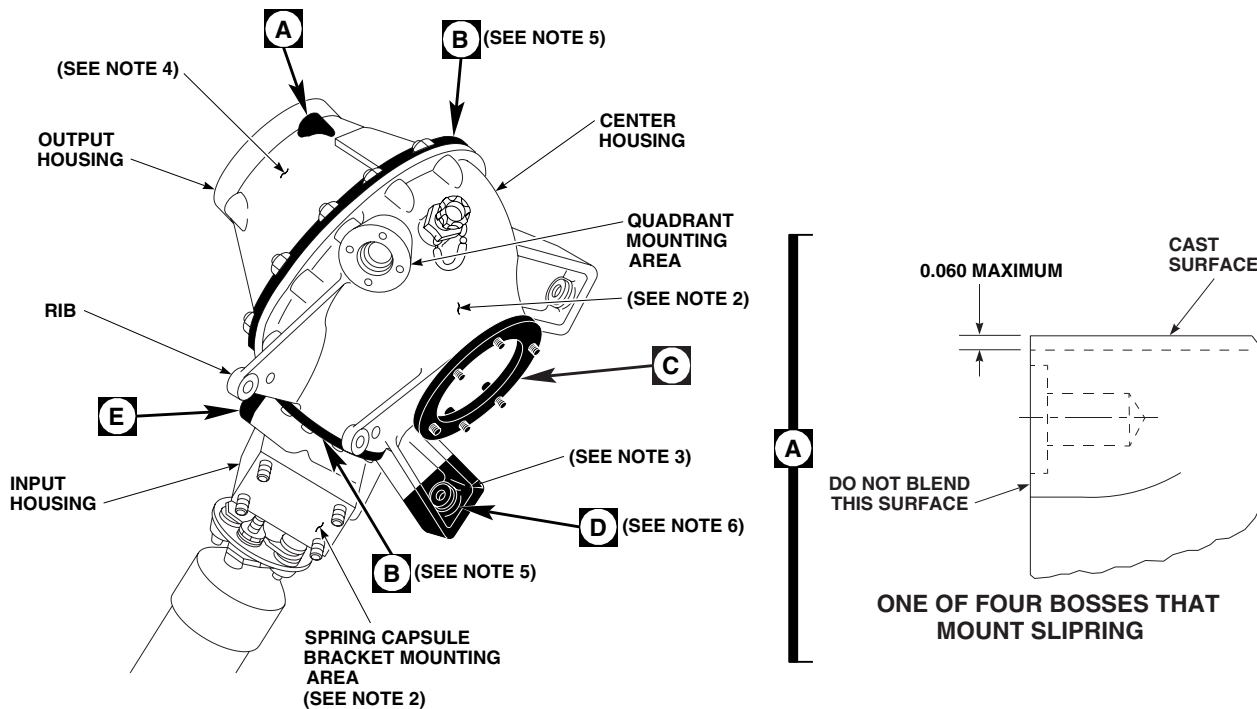
1. Clean any oil or grease from damaged area using cleaning compound solvent, Item 71, WP 1803 00.

NOTE

Cracks will appear as radial lines extending outward from center of gear box bushing and/or extending down the side of mounting foot.

2. Before blending, visually check damaged area for cracks. **NO CRACKS ALLOWED.** Replace cracked gear box.
3. Fine scratches which remove only paint may be blended smooth using abrasive cloth, Item 82, WP 1803 00. After smoothing out, touch up with epoxy coating kit, Item 118, WP 1803 00.
4. Nicks, gouges, scratches, or corrosion on surface of gear box housing may be blended out to a 1/2-inch radius. Only two repairs are allowed in any 4 square inch area. Refer to, REPAIR LIMITS, in this work package for depths of repair.

GENERAL REPAIRS - Continued



NOTES

1. 0.040 MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES OF INPUT HOUSING.
2. 0.025 MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES, OTHER THAN MOUNTING FEET AND ADJACENT RIBS, OF CENTER HOUSING.
3. 0.015 MAXIMUM DEPTH OF MATERIAL REMOVAL IN AREA OF MOUNTING FEET AND ADJACENT RIBS OF CENTER HOUSING.
4. 0.040 MAXIMUM DEPTH OF MATERIAL REMOVAL ALL OVER OUTER CAST SURFACES, OTHER THAN BOSSES THAT MOUNT SLIPRING, OF OUTPUT HOUSING.
5. ONLY 30% OF THE TOTAL NUMBER OF HOLES IN ANY GIVEN BOLT PATTERN MAY BE REPAIRED AS SHOWN IN DETAIL B.
6. DO NOT REMOVE MATERIAL FROM MOUNTING BOLT HOLE BUSHINGS. 0.015 MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE AROUND MOUNTING BOLTS. NO MORE THAN 2 REPAIRS SHALL BE MADE PER MOUNTING FOOT.
7. SPOTFACE CORNER MAY BE BLENDED TO A 0.050 TO 0.100 RADIUS AS LONG AS REPAIR DOES NOT EXTEND DEEPER THAN 0.040 INTO SPOTFACE AREA OF RIB (DOTTED LINE ON SECTION A-A).
8. DEPTH OF REPAIR NOT TO EXCEED 5.124 DIA.
9. REWORK BORE OVERSIZE TO 5.122 / 5.124 DIA TO REMOVE CORROSION MAINTAINING CONCENTRICITY OF 0.002 DIA.
10. REMOVE BUSHING TO REPAIR CORROSION AT DEPOT LEVEL ONLY.
11. DO NOT REMOVE MORE THAN 10% OF SURFACE AREA OF FOOT IN ANY 1 SQUARE INCH AREA OR MORE THAN 30% OF TOTAL BASE OF FOOT.
12. BLEND AREAS OF CORROSION USING 0.250 MINIMUM RADIUS. MAXIMUM DEPTH IS 0.030. MAXIMUM AREA IS 20% AT EACH LUG.
13. BLEND AREAS OF CORROSION USING 0.250 MINIMUM RADIUS. MAXIMUM DEPTH IS 0.060 FROM SURFACE OF LUG. BLENDED AREA SHALL NOT EXCEED 50% BETWEEN LUGS.
14. BLEND WITH A MINIMUM OF 0.375 RADIUS MAINTAINING A 0.650 DIMENSION AND NOT EXCEEDING 5.220 DIA AS SHOWN. DEPTH OF BLENDED AREAS IS NOT TO EXTEND BEYOND THE AREA SPECIFIED.
15. BLEND AREAS OF CORROSION USING 0.375 MINIMUM RADIUS. DEPTH OF BLENDED AREA IS NOT TO EXTEND BEYOND CHAMFER. DIA OF CHAMFER IS NOT TO EXCEED 5.320 DIA AND DEPTH OF CHAMFER NOT TO EXCEED 0.350.
16. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

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Figure 1. Repair Housing. (Sheet 1 of 3)

GENERAL REPAIRS - Continued

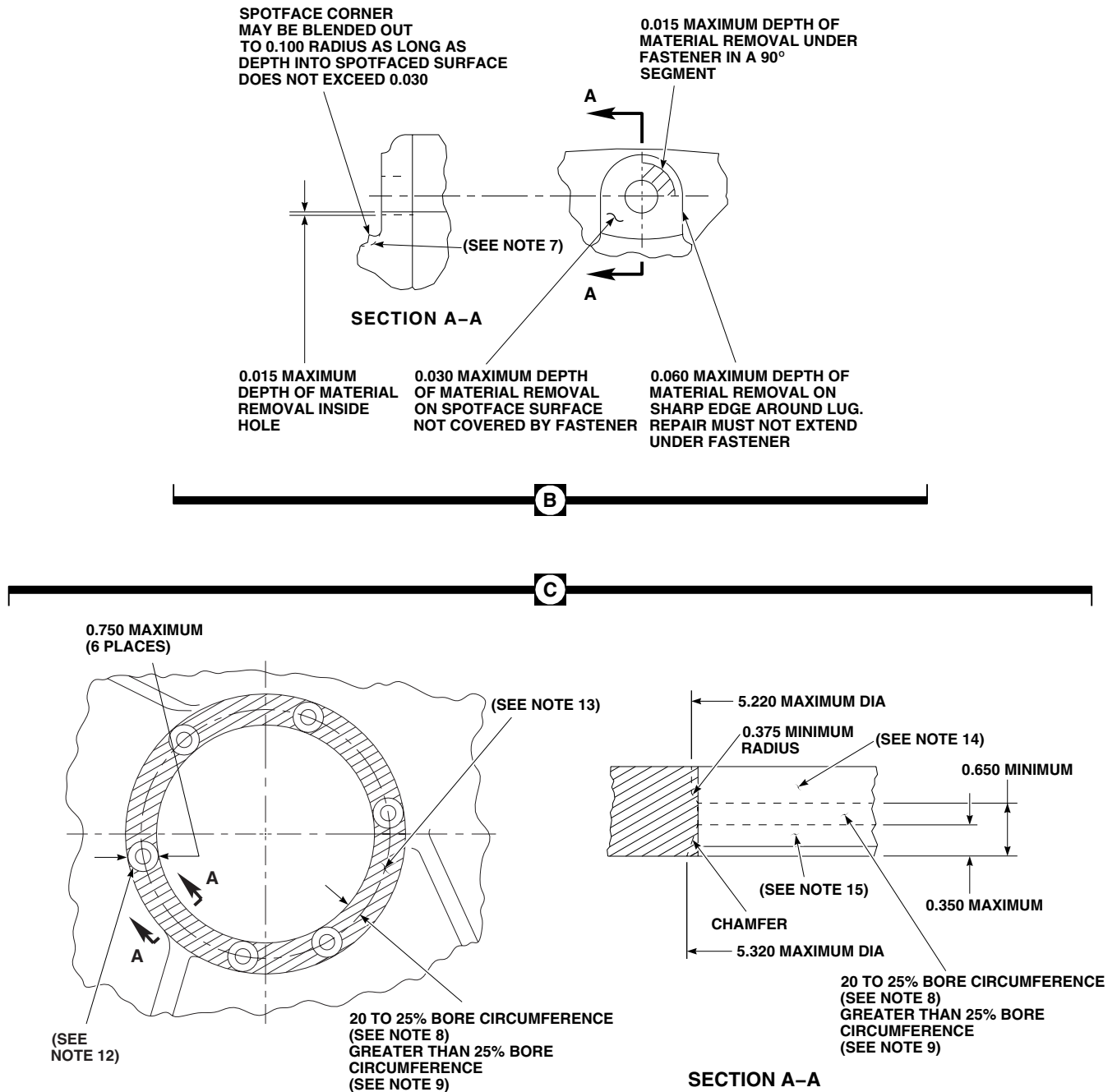
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Figure 1. Repair Housing. (Sheet 2 of 3)

GENERAL REPAIRS - Continued

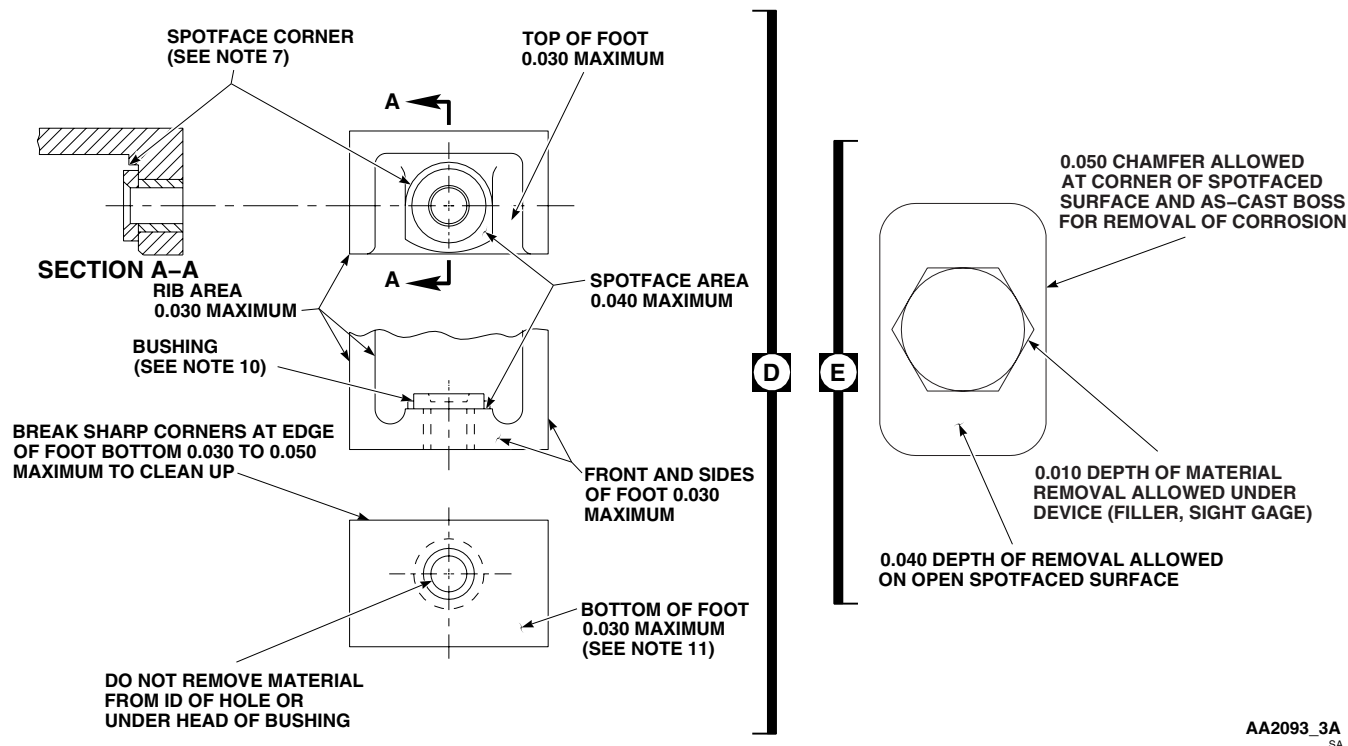


Figure 1. Repair Housing. (Sheet 3 of 3)

REPAIR LIMITS

1. Input housing limits are as follows:

- a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM ALL OVER OUTER CAST SURFACES IS 0.040-INCH.**

NOTE

Only 30% of the total number of holes in bolt pattern may be repaired.

- b. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM INSIDE OF BOLTHOLES IS 0.015-INCH (Figure 1, Sheet 1, Detail B).**
- c. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENERS (BUT NOT COVERING MORE THAN A 90 DEGREES SEGMENT) IS 0.015-INCH.**
- d. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- e. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOTFACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**
- f. **CORNER OF SPOTFACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOTFACE DOES NOT EXCEED 0.030-INCH.**
- g. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OPEN SPOTFACE SURFACE NOT COVERED BY SIGHT GAGE OR CHIP DETECTOR IS 0.040-INCH (Figure 1, Sheet 3, Detail E).**

REPAIR LIMITS - Continued

- h. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE UNDER SIGHT GAGE OR CHIP DETECTOR IS 0.010-INCH.**
 - i. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE CORNER AROUND SIGHT GAGE OR CHIP DETECTOR IS 0.040-INCH WHEN BLENDED TO A 0.050-INCH RADIUS.**
 - j. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPRING CAPSULE BRACKET MOUNTING AREA IS 0.040-INCH.**
2. Center housing limits are as follows:
- a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTER CAST SURFACES (NOT INCLUDING MOUNTING FEET AND MOUNTING FEET RIBS) IS 0.025-INCH.**
 - b. **MAXIMUM DEPTH OF MATERIAL REMOVAL ON MOUNTING FEET AND MOUNTING FEET RIBS IS 0.030-INCH (Figure 1, Sheet 2, Detail D).**
 - (1) **NO MORE THAN TWO REPAIRS ALLOWED PER FOOT.**
 - (2) **TOTAL REPAIR SHALL NOT EXCEED 10% OF SURFACE AREA OF FOOT IN ANY 1 SQUARE INCH AREA OR MORE THAN 30% OF TOTAL BASE OF FOOT.**
 - (3) **NO MATERIAL REMOVAL FROM BUSHING BORE ALLOWED.**
 - (4) **SPOTFACE CORNER MAY BE BLENDED TO 0.050 TO 0.100-INCH RADIUS AS LONG AS REPAIR DOES NOT PENETRATE DEEPER THAN 0.040-INCH INTO SPOTFACE.**
 - (5) **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE OF MOUNTING FOOT IS 0.040-INCH.**
 - c. **MAXIMUM DEPTH OF MATERIAL REMOVAL ON OPEN SPOTFACE AREA NOT COVERED BY FILLER PLUG IS 0.040-INCH.**
 - d. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SURFACE UNDER FILLER PLUG IS 0.010-INCH.**
 - e. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE CORNER AROUND FILLER PLUG IS 0.040-INCH WHEN BLENDED TO A 0.050-INCH RADIUS.**
 - f. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM QUADRANT MOUNTING AREA IS 0.025-INCH.**
 - g. Refer to Figure 1, Sheet 2, Detail C, for repair limits on tail rotor servo mounting area.
3. Output housing limits are as follows:
- a. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM OUTER CAST SURFACE (EXCEPT SLIPRING MOUNT BOSSES) IS 0.040-INCH.**

NOTE

Only 30% of the total number of holes in bolt pattern may be repaired.

- b. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM INSIDE OF BOLTHOLES IS 0.015-INCH (Figure 1, Sheet 1, Detail B).**
- c. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM MOUNTING SURFACE UNDER FASTENERS (BUT NOT COVERING MORE THAN A 90 DEGREES SEGMENT) IS 0.015-INCH.**
- d. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SPOTFACE SURFACE NOT COVERED BY FASTENER IS 0.030-INCH.**
- e. **MAXIMUM DEPTH OF MATERIAL REMOVAL FROM SHARP EDGE AROUND SPOTFACE SURFACE (NOT INCLUDING AREA UNDER FASTENER) IS 0.060-INCH.**

REPAIR LIMITS - Continued

- f. **CORNER OF SPOTFACE MAY BE BLENDED TO 0.100-INCH RADIUS AS LONG AS DEPTH INTO SPOTFACE DOES NOT EXCEED 0.030-INCH.**
- g. **MAXIMUM DEPTH OF REPAIR TO DEICE SLIPRING MOUNTING BOSS CAST SURFACE IS 0.060-INCH (Figure 1, Sheet 1, Detail A).**
- h. **NO MATERIAL REMOVAL FROM SLIPRING MOUNTING SURFACE ALLOWED.**

REPAIR PROCEDURES

1. Before starting rework, remove paint from damaged area using paint remover, Item 252, WP 1803 00. Blend damaged area smooth using abrasive cloth, Item 83, WP 1803 00. Remove only what is necessary to blend out damaged area.
2. Using abrasive cloth, Item 82, WP 1803 00, blend reworked area to remove all traces of aluminum abrasive oxide cloth.
3. Remove displaced metal caused by blending with magnesium etchant (WP 0639 00).
4. Treat bare surface of housing with chromium trioxide, Item 65, WP 1803 00 (TM 1-1500-344-23).
5. Fluorescent penetrant inspect, Type I, Method C, reworked area for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace gear box if cracks are found.

NOTE

Housings coated with epoxy coating kit, Item 118, WP 1803 00, are glossy silver;
housings coated with polyurethane coating, Item 231, WP 1803 00, are matte surface
medium-gray.

6. Touch up reworked area with sealing compound, Item 273, WP 1803 00. After sealing compound has dried apply one coat of epoxy primer coating kit, Item 120, WP 1803 00, then apply one coat of epoxy coating kit, Item 118, WP 1803 00, or one coat of polyurethane coating, Item 231, WP 1803 00, depending on original paint finish (WP 0381 00 and TM 1-1520-345-23).
7. Record brief description of tail gear box repairs on Aircraft Component Historical Record, DA Form 2408-16.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

TAIL GEAR BOX PULLEY GUARDS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Hand Reamer Set, GGG-R-180

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Primer Coating Item 243, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

References

TM 1-1500-204-23
WP 1146 00
WP 1803 00

REMOVE**NOTE**

Removal procedures for both pulley guards are the same.

1. Remove pulley (WP 1146 00).
2. Drill out rivet (TM 1-1500-204-23). Remove rivet, washer, and spacer (Figure 1, Detail A).

NOTE

There is a 1/8-inch gap between bushing ends in center of tail rotor gear box boss.

3. Tap bushing and guard from gear box with drift. Be careful not to scar hole in gear box boss.
4. Tap out remaining bushing.

INSTALL

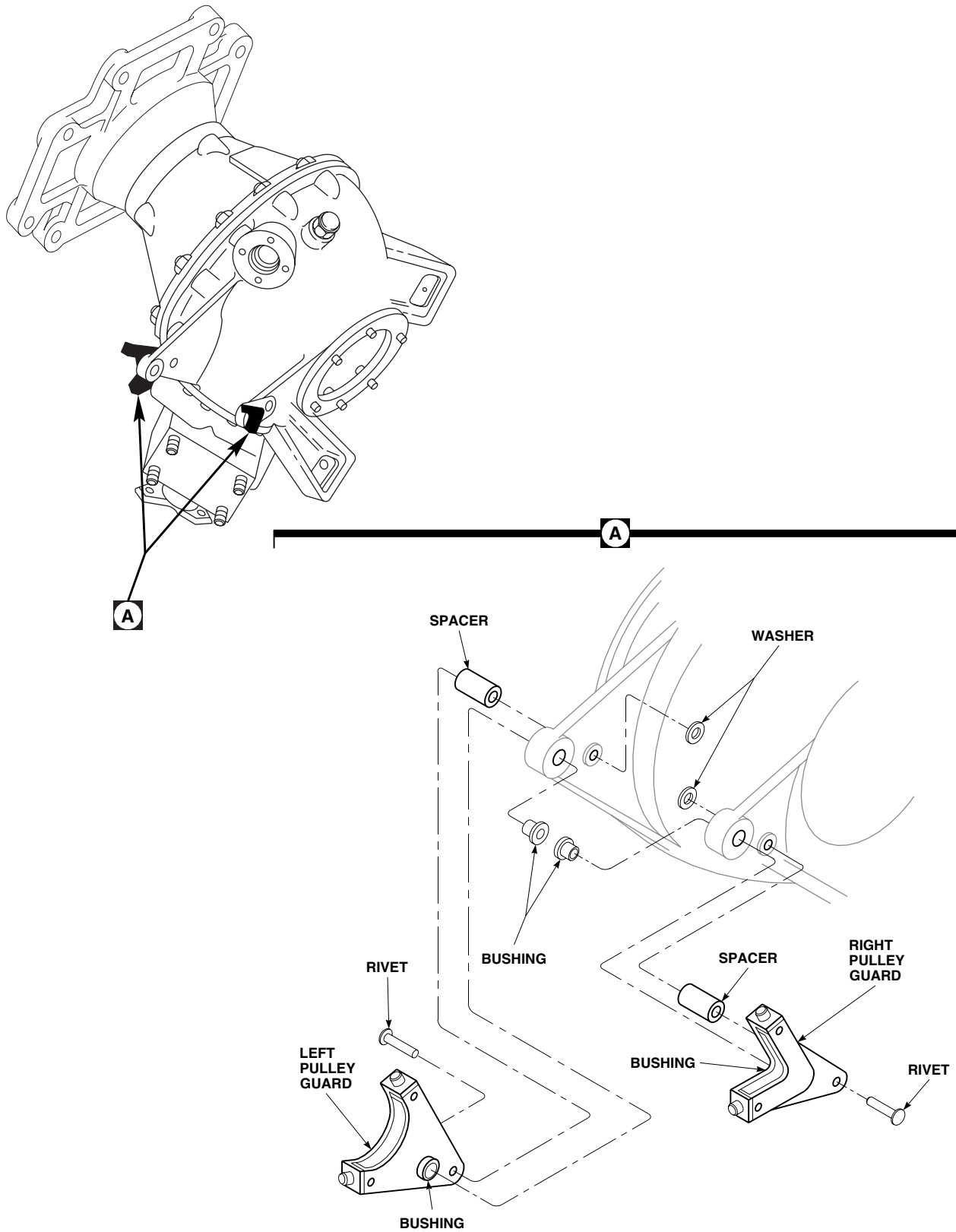
1. If necessary, ream large hole in pulley guard to 0.437 to 0.438-inch, using a hand reamer set (TM 1-1500-204-23).
2. Coat area under shoulder of bushing with primer coating, Item 243, WP 1803 00.
3. **QA** Press bushing into guard while primer is still wet.
4. Coat outer surface of bushing and area where guard will contact housing with primer coating, Item 243, WP 1803 00.

CAUTION

Damage to control cable will result if wrong pulley guard is installed. Right and left pulley guards are not interchangeable. Prior to installing pulley guard, make sure that left pulley guard is Part No. 70400-06670-042, 70400-06670-045, or 70400-06670-047, and right pulley guard is Part No. 70400-06670-046 or 70400-06670-048 (viewed looking aft).

5. **QA** Press bushing and guard into housing while primer is still wet.
6. Coat outer surface and area under shoulder of second bushing with primer coating, Item 243, WP 1803 00.

INSTALL - Continued



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Figure 1. Replace Pulley Guards.

INSTALL - Continued

7. **QA** Press bushing into housing while primer is still wet.
8. Install rivet, spacer, and washer in housing and guard (TM 1-1500-204-23).
9. Seal areas around shoulders of bushings and end of rivet with washer with sealing compound, Item 273, WP 1803 00.
10. Line ream bushings to 0.312 to 0.313-inch.
11. Install pulley (WP 1146 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**DRIVE SYSTEM**

TAIL GEAR BOX CHIP DETECTOR/TEMPERATURE SENSOR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 2-Quart
Drain Hose, H538-A-3

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Lubricating Oil, General Purpose, Item 184,
WP 1803 00
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00

References

WP 0085 00
WP 0333 00
WP 0628 00
WP 1609 00
WP 1703 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Pull out CHIP DET circuit breaker (copilot's circuit breaker panel).
3. Remove tail rotor gear box fairing (WP 0333 00).
4. Disconnect electrical connector, P614, from chip detector/temperature sensor (Figure 1, Detail A).
5. Remove chip detector/temperature sensor.

NOTE

Leakage of lubricating oil through center of self-closing valve indicates loose internal hardware. This disables temperature sensor portion of chip detector.

6. Visually inspect self-closing valve for leakage, from center of valve. **NONE ALLOWED.** Replace valve if any leakage is evident.
7. Check inside of self-closing valve for security of contact pin located in center of valve. If contact pin is loose or missing, locate pin and washer and replace self-closing valve.
8. If pin or washer cannot be located, perform serviceability check (WP 0628 00).
9. If self-closing valve is going to be removed, attach drain hose, H538A-3, to valve and drain oil into a 2 quart container.
10. Remove self-closing valve.

INSTALL

1. Lubricate all packings using aircraft turbine lubricating oil, Item 182, WP 1803 00, helicopter transmission lubricating oil, Item 187, WP 1803 00, or aircraft turbine lubricating oil, Item 184, WP 1803 00.
2. **QA** Install packing on self-closing valve (Figure 1, Detail A).

INSTALL - Continued

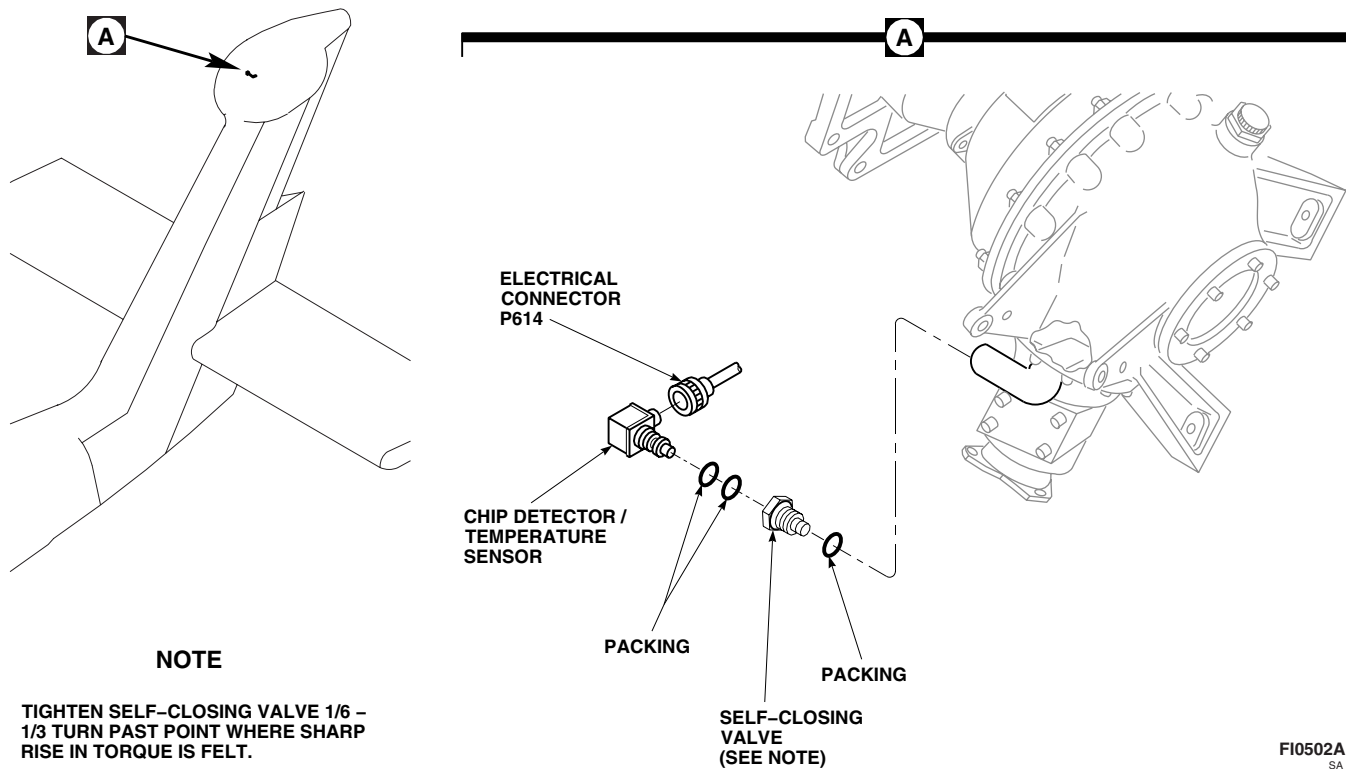


Figure 1. Replace Tail Gear Box Chip Detector/Temperature Sensor.

3. **QA** Install self-closing valve in tail gear box. **TIGHTEN SELF-CLOSING VALVE 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
4. Check both chip detector/temperature sensor electrical circuits (WP 0085 00).
5. **QA** Install packings on chip detector/temperature sensor.
6. **QA** Install chip detector/temperature sensor.
7. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire self-closing valve to housing.
8. Inspect and treat electrical connector (WP 1703 00).
9. **QA** Connect electrical connector, P614, to chip detector/temperature sensor.

WARNING

Injury to personnel and damage to equipment will result if gear box is operated with low oil level. Operation with low oil level or without oil will cause loss of tail rotor drive. Lightly shake pylon to observe oil sloshing when checking oil level.

10. Service tail gear box (WP 1609 00).
11. Make sure area is clean and free of foreign material.
12. Install tail rotor gear box fairing (WP 0333 00).
13. Perform an operational check of both chip detector and temperature sensor electrical circuits (WP 0085 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
TAIL DRIVE SHAFT COUPLING ALIGNMENT CHECK

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Alignment Bar and Bracket, Locally-Made, Figure 267,
WP 1805 00
Dial Indicator, MIL-I-18422
Mirror
Torque Wrench, 150 - 1000 in. lbs.

References

WP 0659 00
WP 0666 00
WP 0667 00
WP 0668 00
WP 0669 00
WP 0681 00
WP 1805 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

CHECK**NOTE**

Coupling alignment checks are required by conditional inspections or special maintenance. Conditional inspections which reveal a cracked laminate or degraded bladder, trigger an alignment check. Hard landings may require alignment checks depending on severity. Special maintenance such as shimming the forward mount of the oil cooler bracket, shimming tail or intermediate gear boxes, or replacing a drive shaft mount bracket are examples of reasons for an alignment check.

1. Record the following data for each coupling on the Inspection Worksheet (Figure 2).
 - a. Measured Total Indicator Reading (TIR).
 - b. Arm length of alignment bar and bracket from center of coupling to center of dial indicator cam (Distance "D").

NOTE

Make sure that shim and washer stackup are not disturbed.

2. Perform the following at each flexible coupling connection:
 - a. Locally make alignment bar and bracket (WP 1805 00)
 - b. Loosen nut (do not remove) on mating shaft flange.

NOTE

- Bracket may be placed on any of the 3 bolted connections of the flexible coupling and mating shaft flange.
 - Bracket must not be placed on the bolted connection of the flexible coupling and the shaft assembly to be measured.
- c. Position alignment bar and bracket on shaft. Position bolthead and spherical washer on flexible coupling flange (Figure 1, Detail A). If necessary, loosen nut (do not remove) to adjust bracket/bolthead clearance.

CHECK - Continued

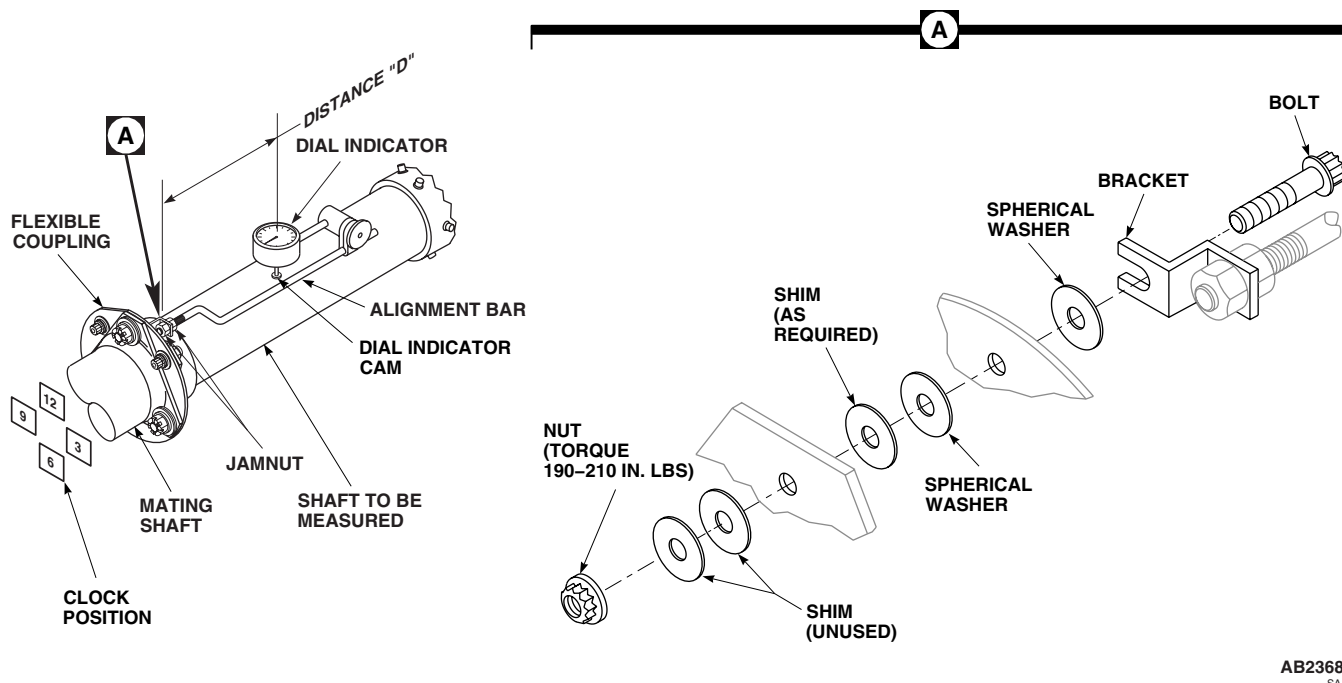


Figure 1. Dial Indicator Setup.

NOTE

Make sure that bracket is positioned so that dial indicator and bracket are allowed to rotate 360 degrees when drive shaft is rotated.

- d. Tighten nut and secure alignment bar and bracket to shaft. **TORQUE NUT TO 190 - 210 INCH-POUNDS.**
- e. Attach dial indicator to alignment bar and bracket. Adjust dial indicator so that dial indicator cam rests against drive shaft approximately 6 to 8 inches from the center of the flexible coupling (Figure 1).
- f. Determine the distance between centerline of flexible coupling and centerline of dial indicator cam (Distance "D"). Record Distance "D" measurement to Inspection Worksheet.

NOTE

Use mirror to observe readings to maintain complete reading of dial indicator through 360 degree drive shaft rotation.

- g. Zero indicator dial by slowly rotating drive shaft 360 degrees and observing lowest reading. Return dial indicator to lowest observed reading and zero dial indicator.

NOTE

Use mirror to observe readings to maintain complete reading of dial indicator through 360 degree drive shaft rotation.

- h. To obtain Measured TIR, rotate drive shaft 360 degrees and record maximum observed total indicator reading

CHECK - Continued**INSPECTION WORKSHEET
(REPRODUCE LOCALLY)**

1. A/C BUNO _____ A/C HRS. _____

2. TOTAL INDICATED RUNOUT DATA, (READ IN FOUR COLUMNS):

	DISTANCE "D"	MEASURED TIR	CLOCK POSITION	MAX ALLOWABLE TIR
(A) STA 363.0				
(B) STA 376.0				
(C) STA 410.0				
(D) STA 470.0				
(E) STA 531.0				
(F) STA 592.0				
(G) STA 633.0				
(H) STA 644.0				
(I) STA 655.0				
(J) STA 725.0				

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Figure 2. Inspection Worksheet.

(TIR) and clock range (3 to 9 o'clock or 6 to 12 o'clock position) of maximum observed total indicator reading (TIR). Record Measured TIR and clock position on Inspection Worksheet.

- i. Calculate the Maximum Allowable TIR for appropriate drive shaft using the following equation: Max. Allowable TIR = $0.0088 \times \text{Distance "D"}$. Record Maximum Allowable TIR on Inspection Worksheet.
- j. If Measured TIR from step 2. h. is less than or equal too Maximum Allowable TIR from step 2. i. alignment is acceptable.

Table 1. Example.

Distance "D"	7.0 inch
Measured Tier	0.010-inch
Clock Position	12-o'clock
0.0088 x 7.0 = 0.060-inch Maximum Allowable TIR.	
Measured TIR of 0.010-inch is less than Maximum Allowable TIR of 0.060-inch. Alignment is acceptable.	

- k. If Measured TIR from step 2. h. is greater than Maximum Allowable TIR from step 2. i., **ALIGNMENT IS NOT ACCEPTABLE**. Perform the following:

- (1) Check airframe mount bracket position.

CHECK - Continued

- (2) If misalignment is at intermediate or tail gearbox connection points remove sealing compound, loosen mounting bolts, and check gear box mounting pads for gaps under gearbox mounting feet (WP 0659 00 and/or WP 0681 00 00).
 - (3) To correct angular alignment determine the clock position recorded in step 2. h. If Measured TIR falls in the horizontal clock position (3 to 9 o'clock position) angular alignment can only be corrected by AMCOM Maintenance Engineering.
 - (4) If Measured TIR falls into the vertical clock positions (6 to 12 o'clock position) angular alignment can be corrected by shimming the intermediate gear box (WP 0659 00) and/or tail gear box (WP 0681 00).
 - (5) If support bracket was repaired or intermediate gear box and/or tail gear box reshimmed, recheck Measured TIR of drive shafts forward and aft of repaired support bracket or reshimmed gear box.
- l. Loosen nut (do not remove) on mating shaft flange. Remove alignment bar and bracket from flexible coupling.

NOTE

Make sure shim and washer stackup is not disturbed.

- m. Tighten and torque nut on applicable drive shaft section (WP 0666 00, WP 0667 00, WP 0668 00, and/or WP 0669 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
DRIVE SYSTEM
TAIL DRIVE SHAFT COUPLING SHIMMING CHECK

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Straight Edge

References

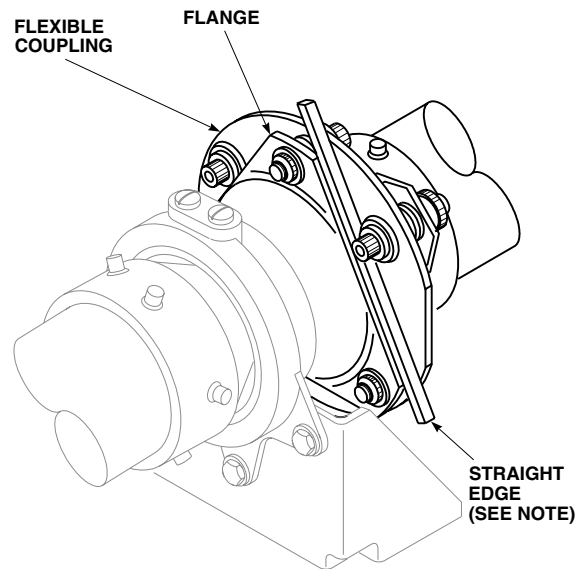
WP 0666 00
WP 0667 00
WP 0668 00
WP 0669 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

CHECK**NOTE**

- Shims used for tail drive shafts are placed between the mating flange spherical washer. Unused shims are placed between other spherical washer and nut.
 - Sections I, III, and IV drive shafts have shims at both ends of shaft assemblies. These shaft use equal amounts of shims at both ends.
 - Section II drive shaft has shims at forward end only.
 - Disconnect coupling pylon shaft assembly has shims at aft end only.
1. Check for proper shimming of all tail drive shafts, Sections I through IV. Make sure the same amount of shims are used between spherical washer and adjacent flange. There should be the same amount of unused shims between spherical washer and nut.
 2. Make sure all spherical washers have spherical side facing flexible coupling.
 3. **MAKE SURE THERE IS NOT MORE THAN 0.015-INCH DEFLECTION IN FLEXIBLE COUPLINGS (Figure 1).** This can be determined by using feeler gage, and a straight edge placed along the flange surface and flexible coupling.
 4. If deflection is more than 0.015-inch in flexible couplings reshim tail drive shafts, Sections I through IV (WP 0666 00, WP 0667 00, WP 0668 00, and WP 0669 00).

CHECK - Continued**NOTE**

INSERT FEELER GAGE
BETWEEN FLEXIBLE COUPLING
AND STRAIGHT EDGE. MAXIMUM
DEFLECTION NOT TO EXCEED
0.015-INCH.

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Figure 1. Tail Drive Shaft Coupling Shimming Check.

END OF WORK PACKAGE

By Order of the Secretary of the Army:

Official:

A handwritten signature in black ink, appearing to read "Joyce E. Morrow". The signature is fluid and cursive, with the first name "Joyce" and last name "Morrow" clearly distinguishable.

JOYCE E. MORROW

*Administrative Assistant to the
Secretary of the Army*

0608618

PETER J. SCHOOMAKER
*General, United States Army
Chief of Staff*

DISTRIBUTION:

To be distributed in accordance with Initial Distribution Number (IDN) 311290, requirements for TM 1-1520-237-23-7.

These are the instructions for sending an electronic 2028

The following format must be used if submitting an electronic 2028. The subject line must be exactly the same and all fields must be included; however only the following fields are mandatory: 1, 3, 4, 5, 6, 7, 8, 9, 10, 13, 15, 16, 17, and 27.

From: "Whomever" <whomever@wherever.army.mil>

To: 2028@redstone.army.mil

Subject: DA Form 2028

1. From: Joe Smith
2. Unit: home
3. Address: 4300 Park
4. City: Hometown
5. St: MO
6. Zip: 77777
7. Date Sent: 19--OCT--93
8. Pub no: 55--2840--229--23
9. Pub Title: TM
10. Publication Date: 04--JUL--85
11. Change Number: 7
12. Submitter Rank: MSG
13. Submitter FName: Joe
14. Submitter MName: T
15. Submitter LName: Smith
16. Submitter Phone: 123-123-1234
17. Problem: 1
18. Page: 2
19. Paragraph: 3
20. Line: 4
21. NSN: 5
22. Reference: 6
23. Figure: 7
24. Table: 8
25. Item: 9
26. Total: 123
27. Text:

This is the text for the problem below line 27.

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, 35898				FROM: (Activity and location) (Include ZIP Code) MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565			DATE 8/30/02	
PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS								
PUBLICATION NUMBER				DATE		TITLE		
PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION
PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)								
EXAMPLE								
TYPED NAME, GRADE OR TITLE MSG, Jane Q. Doe, SFC				TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION 788-1234			SIGNATURE	

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.						Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/Supply Manuals (SC/SM)	DATE
TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898 - 5000						FROM: (Activity and location)(Include ZIP Code)	
PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS							
PUBLICATION/FORM NUMBER						DATE	TITLE
ITEM NO.	PAGE NO.	PARA- GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON	
* Reference to line numbers within the paragraph or subparagraph.							
TYPED NAME, GRADE OR TITLE					TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION		SIGNATURE

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898				FROM: (Activity and location) (Include ZIP Code)			DATE	
PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS								
PUBLICATION NUMBER				DATE		TITLE		
PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION
PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)								
TYPED NAME, GRADE OR TITLE				TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION			SIGNATURE	

The Metric System and Equivalents

Linear Measure

1 centimeter = 10 millimeters = .39 inch
 1 decimeter = 10 centimeters = 3.94 inches
 1 meter = 10 decimeters = 39.37 inches
 1 dekameter = 10 meters = 32.8 feet
 1 hectometer = 10 dekameters = 328.08 feet
 1 kilometer = 10 hectometers = 3,280.8 feet

Weights

1 centigram = 10 milligrams = .15 grain
 1 decigram = 10 centigrams = 1.54 grains
 1 gram = 10 decigrams = .035 ounce
 1 decagram = 10 grams = .35 ounce
 1 hectogram = 10 decagrams = 3.52 ounces
 1 kilogram = 10 hectograms = 2.2 pounds
 1 quintal = 100 kilograms = 220.46 pounds
 1 metric ton = 10 quintals = 1.1 short tons

Liquid Measure

1 centiliter = 10 milliliters = .34 fl. ounce
 1 deciliter = 10 centiliters = 3.38 fl. ounces
 1 liter = 10 deciliters = 33.81 fl. ounces
 1 dekaliter = 10 liters = 2.64 gallons
 1 hectoliter = 10 dekaliters = 26.42 gallons
 1 kiloliter = 10 hectoliters = 264.18 gallons

Square Measure

1 sq. centimeter = 100 sq. millimeters = .155 sq. inch
 1 sq. decimeter = 100 sq. centimeters = 15.5 sq. inches
 1 sq. meter (centare) = 100 sq. decimeters = 10.76 sq. feet
 1 sq. dekameter (are) = 100 sq. meters = 1,076.4 sq. feet
 1 sq. hectometer (hectare) = 100 sq. dekameters = 2.47 acres
 1 sq. kilometer = 100 sq. hectometers = .386 sq. mile

Cubic Measure

1 cu. centimeter = 1000 cu. millimeters = .06 cu. inch
 1 cu. decimeter = 1000 cu. centimeters = 61.02 cu. inches
 1 cu. meter = 1000 cu. decimeters = 35.31 cu. feet

Approximate Conversion Factors

<i>To change</i>	<i>To</i>	<i>Multiply by</i>	<i>To change</i>	<i>To</i>	<i>Multiply by</i>
inches	centimeters	2.540	ounce-inches	Newton-meters	.007062
feet	meters	.305	centimeters	inches	.394
yards	meters	.914	meters	feet	3.280
miles	kilometers	1.609	meters	yards	1.094
square inches	square centimeters	6.451	kilometers	miles	.621
square feet	square meters	.093	square centimeters	square inches	.155
square yards	square meters	.836	square meters	square feet	10.764
square miles	square kilometers	2.590	square meters	square yards	1.196
acres	square hectometers	.405	square kilometers	square miles	.386
cubic feet	cubic meters	.028	square hectometers	acres	2.471
cubic yards	cubic meters	.765	cubic meters	cubic feet	35.315
fluid ounces	milliliters	29.573	cubic meters	cubic yards	1.308
pints	liters	.473	milliliters	fluid ounces	.034
quarts	liters	.946	liters	pints	2.113
gallons	liters	3.785	liters	quarts	1.057
ounces	grams	28.349	liters	gallons	.264
pounds	kilograms	.454	grams	ounces	.035
short tons	metric tons	.907	kilograms	pounds	2.205
pound-feet	Newton-meters	1.356	metric tons	short tons	1.102
pound-inches	Newton-meters	.11296			

Temperature (Exact)

F	Fahrenheit temperature	5/9 (after subtracting 32)	Celsius temperature	°C
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